



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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34-00-00-001

NAVIGATION, GENERAL

Introduction

The navigation system has different sub-systems to supply other systems with the data that follows:

- Pitot and static air pressures
- Pitch and bank attitude and stabilized heading
- Microwave Landing System (MLS) azimuth and elevation
- Weather, ground proximity, location and threat status of other aircraft, and Above Ground Level (AGL) altitude
- Radio navigation
- Flight management.

General Description

[Refer to Figure 1.](#)

The navigation system has the sub-systems that follow:

- 34-10-00 Air Data System
- 34-20-00 Attitude and Heading System
- 34-31-00 Microwave Landing System
- 34-40-00 External Protection System

- 34-51-00 VHF Navigation
- 34-52-00 Automatic Direction Finding (ADF)
- 34-53-00 Distance Measuring Equipment (DME)
- 34-54-00 Air Traffic Control Transponder
- 34-55-00 Global Positioning System
- 24-60-00 Flight Management System (FMS).

Air Data System

The Air Data System (ADS) supplies air data to the Electronic Instrument System (EIS) for indication and to other aircraft systems.

The air data parameters that are shown to the pilots and used for flight guidance are critical parameters. For this reason, the ADS has the three isolated air data systems, two primary and one standby channel.

Attitude and Heading System

The Attitude and Heading System (AHRS) supplies attitude and heading data to the Electronic Instrument System (EIS) indication and to other aircraft systems.

The attitude and heading parameters that are shown to the pilots and used for flight guidance are critical parameters. For this reason, the AHRS has three isolated attitude and heading data systems, two primary and one standby channel.



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Microwave Landing System

The Microwave Landing System (MLS) receives radio signals to give the navigation data azimuth and elevation that gives the approach flight path to landing on a runway.

External Protection System

The External Protection System (EPS) has different sub-systems to give visual and aural indications that relate to the conditions that follow:

- Weather
- Ground proximity
- Location and threat status of other aircraft
- Above Ground Level (AGL) altitude.

It also gives other aircraft and ground stations with data relating to the aircraft identification and location.

The external protection system has the sub-systems that follow:

- Weather Radar Systems
- Ground Proximity Warning System (GPWS)
- Traffic Alerting and Collision Avoidance System II (TCAS II)
- Radio Altimeter.

Very High Frequency Navigation

The Very High Frequency Navigation (VHF NAV) system receives radio signals to give the navigation data that follows:

- VHF Omni-range (VOR)
- Instrument Landing System (ILS).

VHF Omni-range (VOR): The Very High Frequency Omni-range (VOR) ground stations form part of the Victor airways navigation system. The VOR gives guidance on any track to or from the station.

Instrument Landing System (ILS): The Instrument Landing System (ILS) is an integrated system that gives the approach flight path to landing on a runway.

Automatic Direction Finding

The Automatic Direction Finding (ADF) system shows the magnetic bearing to a fixed Non-Directional Beacon (NDB).

Distance Measuring Equipment

The airborne Distance Measuring Equipment (DME) measures the line of sight slant range from the aircraft to the DME ground station. It uses the slant distance data to calculate the ground speed and the time-to-go parameters.

Mode S Transponder

The Mode S Transponder system transmits a signal to identify the aircraft and its altitude parameters to Air Traffic Control (ATC) ground stations and to other aircraft equipped with Traffic Alert and Collision Avoidance System (TCAS):



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Global Positioning System

The Flight Management System (FMS), Global Positioning System (GPS), and VORTAC Position Unit (VPU) are completely self-contained within a single Multi-Functional Control Display Unit (MCDU). It is located in the forward left side of the flight compartment console.

Flight Mangement System (FMS)

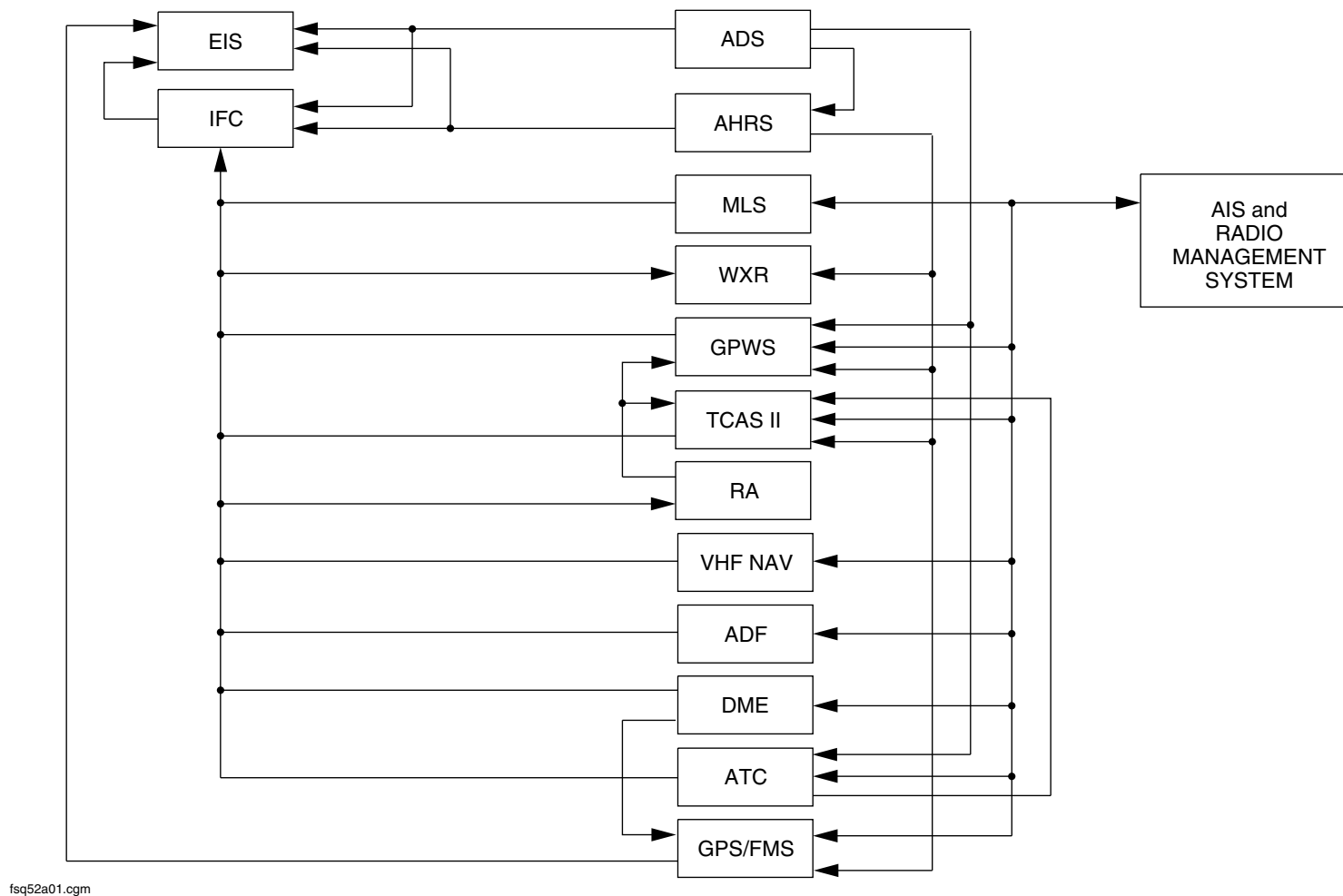
The FMS along with other interfacing equipment in the aircraft provides the following functions:

- Calculating the aircraft's position and velocity
- Guidance for the Automatic Flight Control System (AFCS)
- Managing the aircraft's flight path from origin to destination
- Tuning communication and navigation radios
- Make fuel management computations

The FMS assists the pilots by automating many routine tasks and manual computations.



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NAVIGATION SYSTEM BLOCK DIAGRAM
Figure 1



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FLIGHT ENVIRONMENT DATA

Introduction

The Air Data System (ADS) supplies air data to the Electronic Instrument System (EIS) for indication and to other aircraft systems.

General Description

The air data parameters that are shown to the pilots and used for flight guidance are critical parameters. For this reason, the ADS has the three isolated air data systems, two primary and one standby channel.

[Refer to Figures 1 and 2.](#)

Each primary channel has one Air Data Unit (ADU), pitot-static probe, and temperature probe to do the functions that follow:

- Measure static and pitot pressures
- Receive air temperature data
- Receive barometric corrections
- Calculate air data parameters
- Supplies air data parameters to other systems through ARINC 429 buses to other systems.

The static air pressure sensed at each ADU is the average of the static pressures from the two pitot-static probes. Each ADU also

senses pitot pressure from its related pitot-static probe and temperature from its related temperature probe.

The Index Control Panels (ICPs) are used to set the barometric correction for its related ADU.

[Refer to Figure 3.](#)

The standby air data instruments function independently of the primary channel Air Data Units (ADU1, ADU2) to show the data that follows:

- Altitude
- Air speed
- Barometric correction settings.

The Air Data System (ADS) has the components that follow:

- Unit, Air Data (34-11-01)
- Probes, Pitot-Static (34-11-06)
- Probes, Static Temperature (34-11-11)
- Indicator, standby altitude (34-12-01)
- Indicator, airspeed-standby (34-12-06)
- Probe, standby pitot-static (34-12-11).

Air Data Unit

[Refer to Figure 4.](#)

The Air Data Unit (ADU) weighs 6.12 lb (2.78 kg). It has subassemblies mechanically packaged into a 2 in. (51 mm) high, 6 in. (152 mm) wide by 7.9 in. (201 mm) long case.



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It has quick disconnect pitot and static pneumatic connectors.

Both ADUs are installed in the flight compartment on the aft bulkhead using four mounting screws and have the pneumatic connectors oriented downwards to prevent entry of moisture into the units.

Probes, Pitot–Static

[Refer to Figure 5.](#)

The two primary heated Pitot–Static probes are installed symmetrically on the left and right sides of the fuselage.

The base plates have electrical and pressure connectors. Locating pins on the plate give an accurate angular orientation of the probe with the aircraft axis.

The primary probes are oriented 9 degrees downward relative to the horizontal reference axis of the aircraft. The angle of the probe tube relative to the mast gives a 5 degree angle in or out between the tube line and the tangential plane of the fuselage skin.

Probe, Static Temperature

[Refer to Figure 6.](#)

The static temperature probe weighs 0.25 pounds and is attached flush to the fuselage below the left wing root by four screws.

Standby Altitude Indicator

[Refer to Figure 7.](#)

The standby altitude indicator weighs 2.2 lb (1 kg). It is located in the instrument panel between the pilot's Multi–Function Display (MFD1)

and the Engine Display (ED) below the standby airspeed indicator in view of the two pilots.

Standby Airspeed Indicator

[Refer to Figure 8.](#)

The standby altitude indicator weighs 1.1 lb (0.5 kg). It is located in the instrument panel between the pilot's Multi–Function Display (MFD1) and the Engine Display (ED) above the standby altitude indicator in view of the two pilots.

Probe, Standby Pitot–Static

[Refer to Figure 9.](#)

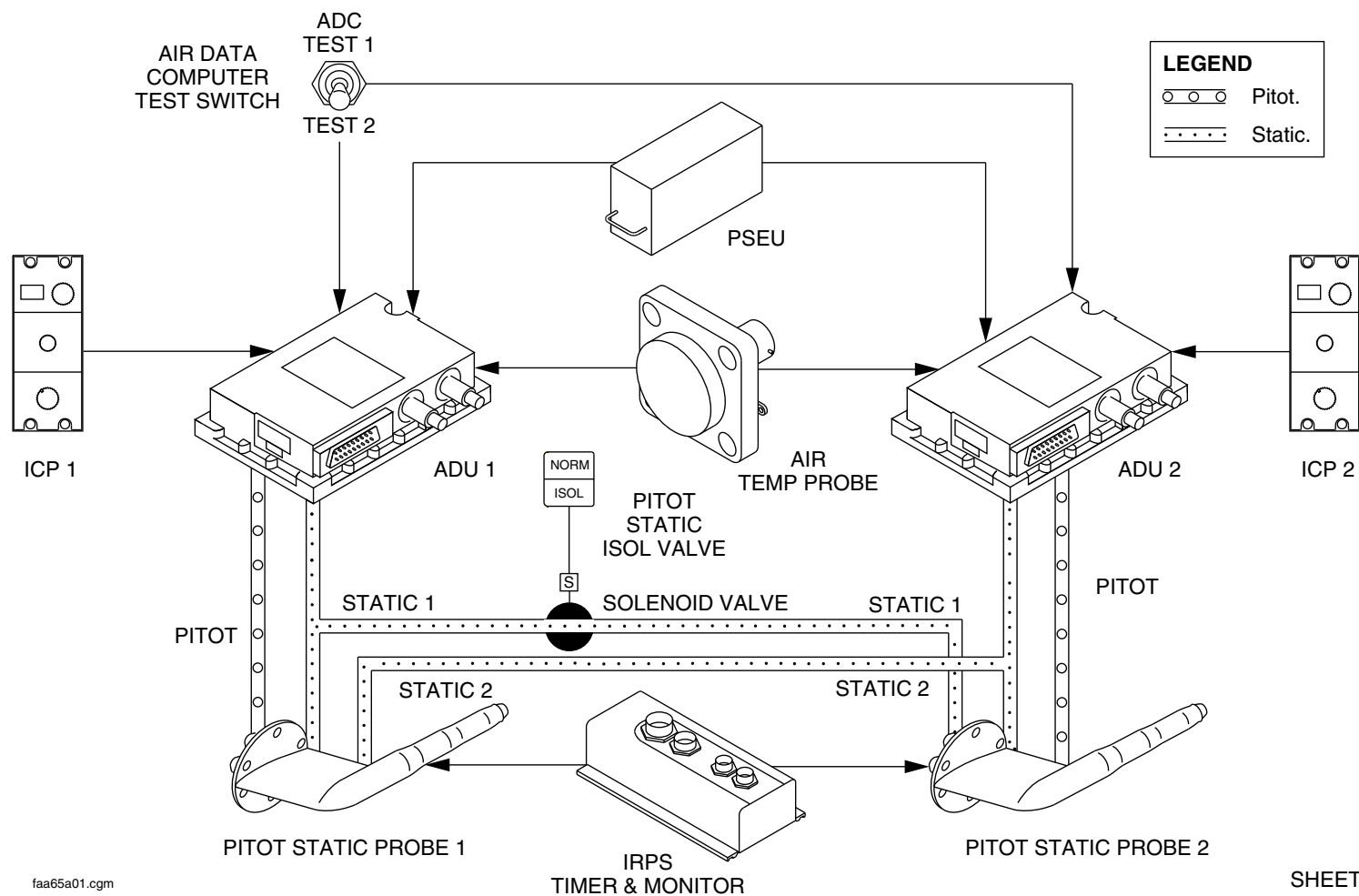
The standby pitot–static probe is installed on the right side of the forward fuselage section.

The base plates have electrical and pressure connectors. Locating pins on the plate give an accurate angular orientation of the probe with the aircraft axis.



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AIR DATA SYSTEM BLOCK DIAGRAM, INPUTS
Figure 1

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The diagram illustrates the avionics architecture for a twin-engine aircraft, showing the interconnections between various systems. The architecture is divided into two main sections: **PILOT EFIS DISPLAY** and **COPILOT EFIS DISPLAY**.

PILOT EFIS DISPLAY: Includes **PFD 1** (Primary Flight Display), **MFD 1** (Multi-Function Display), and **ED** (Engine Display).

COPILOT EFIS DISPLAY: Includes **PFD 2** (Primary Flight Display), **MFD 2** (Multi-Function Display), and **ED** (Engine Display).

Central Avionics and Control Units:

- FADEC 1 FADEC 2:** Fuel/Air/Fuel/Engine Control Unit.
- CABIN PRESSURE CONTROLLER:** Controls cabin pressure.
- CABIN INDICATION MODULE:** Provides cabin pressure indications.
- AVIONICS (LEFT ESSENTIAL), A7:** Essential avionics for the left engine.
- AVIONICS (RIGHT MAIN), A6:** Main avionics for the right engine.
- ADU 1 ADU 2:** Air Data Units.
- IFC 1 IFC 2:** Instrument Flight Controller.
- FMS 1 FMS 2:** Flight Management System.
- AHRS 1 AHRS 2:** Attitude Heading Reference System.
- ATC 1 ATC 2:** Traffic Alert and Collision Avoidance System.
- FCECU:** Fuel Control Electronic Control Unit.

Data Flow and Connections:

- The **PILOT EFIS DISPLAY** and **COPILOT EFIS DISPLAY** are connected to the **FADEC 1 FADEC 2** and **IFC 1 IFC 2** units.
- The **FADEC 1 FADEC 2** unit is connected to the **AVIONICS (LEFT ESSENTIAL), A7** and **AVIONICS (RIGHT MAIN), A6** units.
- The **CABIN PRESSURE CONTROLLER** is connected to the **AVIONICS (LEFT ESSENTIAL), A7** and **AVIONICS (RIGHT MAIN), A6** units.
- The **CABIN INDICATION MODULE** is connected to the **AVIONICS (LEFT ESSENTIAL), A7** and **AVIONICS (RIGHT MAIN), A6** units.
- The **AVIONICS (LEFT ESSENTIAL), A7** and **AVIONICS (RIGHT MAIN), A6** units are connected to the **ADU 1 ADU 2** units.
- The **ADU 1 ADU 2** units are connected to the **IFC 1 IFC 2** and **FMS 1 FMS 2** units.
- The **IFC 1 IFC 2** and **FMS 1 FMS 2** units are connected to the **AHRS 1 AHRS 2** and **ATC 1 ATC 2** units.
- The **AHRS 1 AHRS 2** and **ATC 1 ATC 2** units are connected to the **FCECU**.

Sheet Information:

- SHEET 2:** The diagram is labeled as **SHEET 2** in the bottom right corner.
- faa66a01.cqm:** The file name **faa66a01.cqm** is located in the bottom left corner.

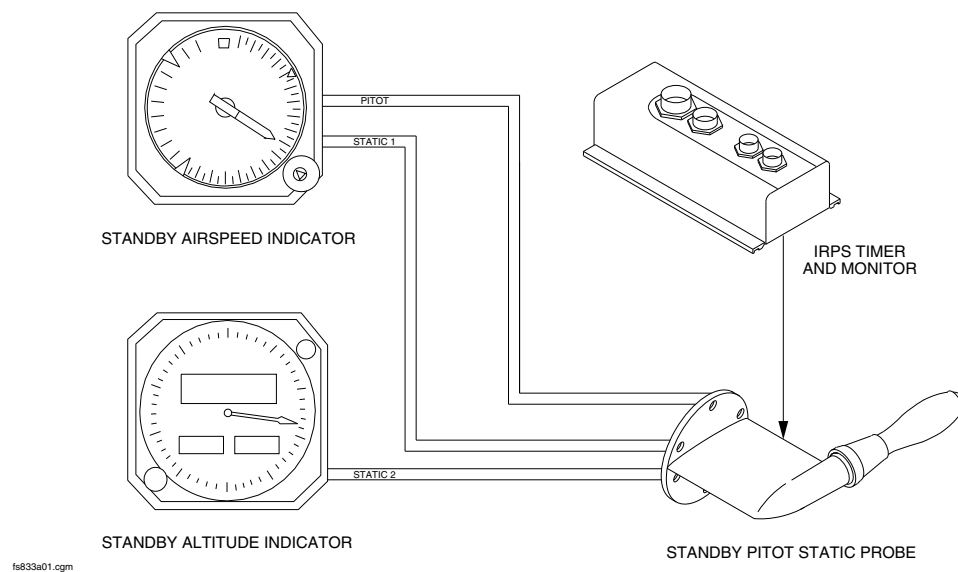
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STANDBY AIR DATA INSTRUMENTS BLOCK DIAGRAM
Figure 3

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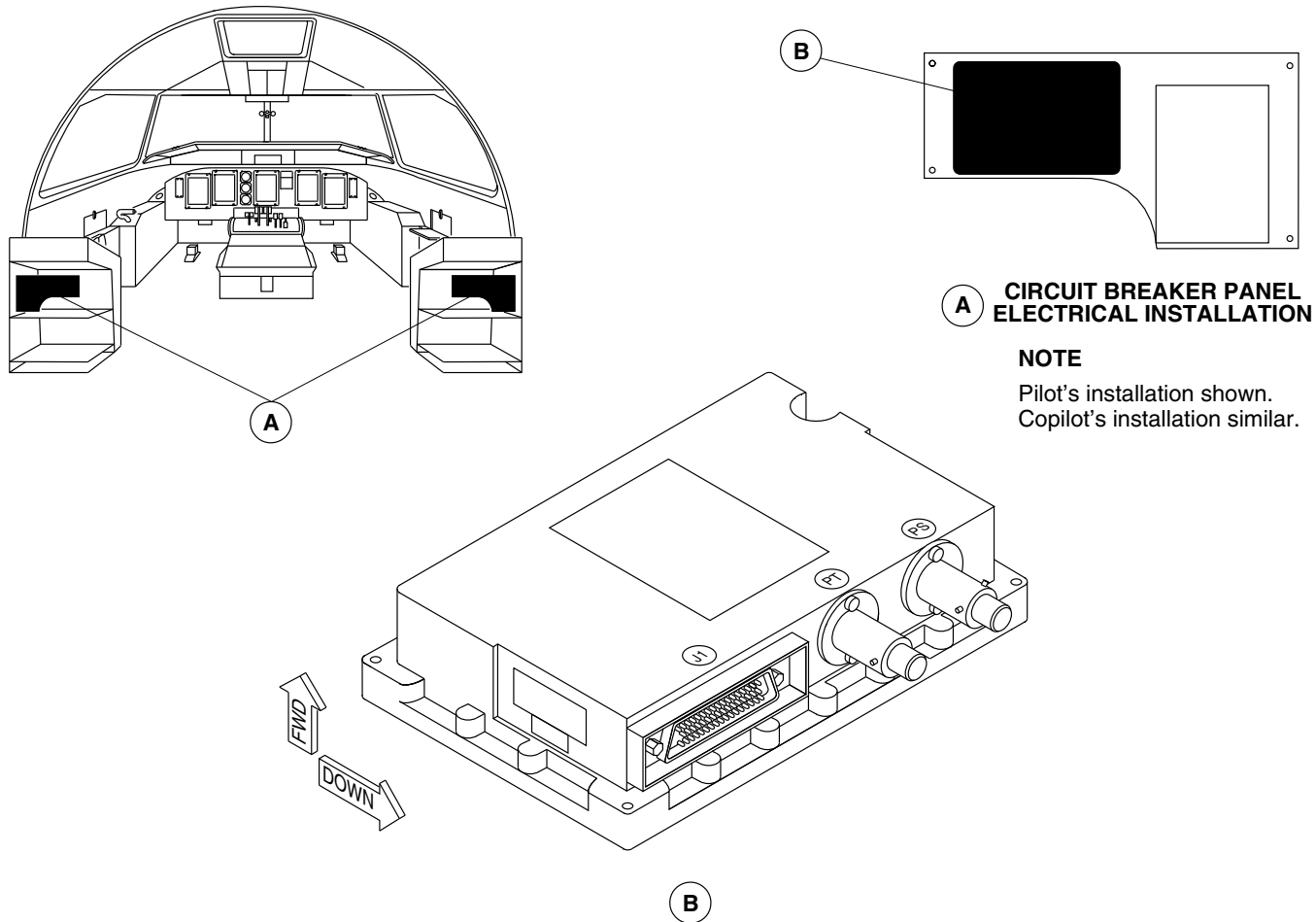
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AIR DATA UNIT LOCATOR
Figure 4

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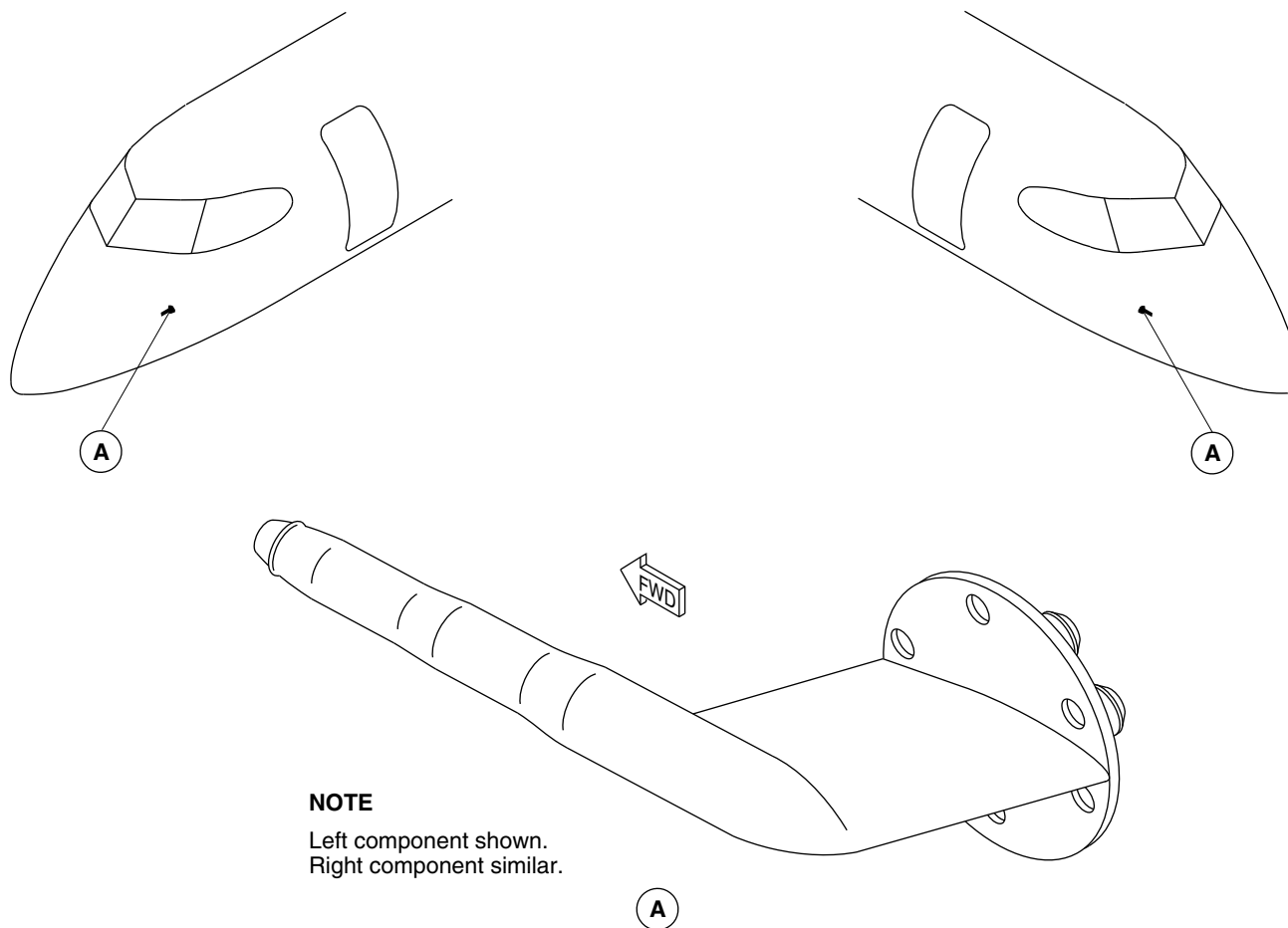
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NOTE

Left component shown.
Right component similar.

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PITOT-STATIC PROBES LOCATOR
Figure 5

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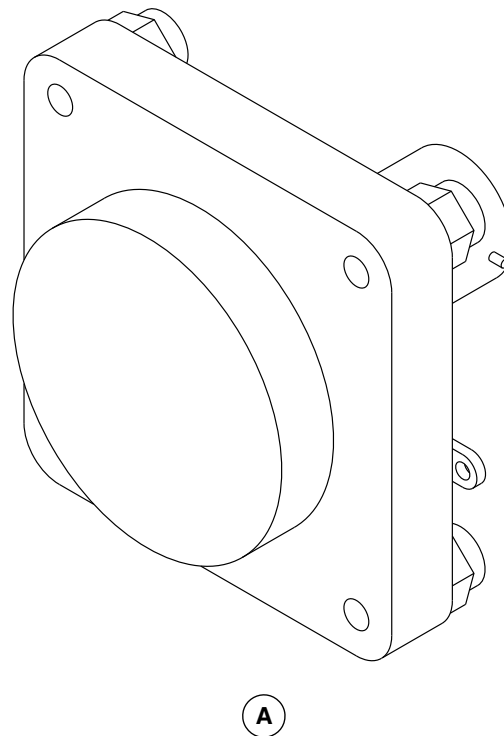
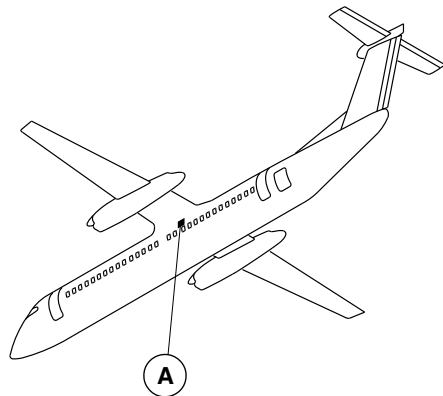
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STATIC TEMPERATURE PROBE LOCATOR
Figure 6

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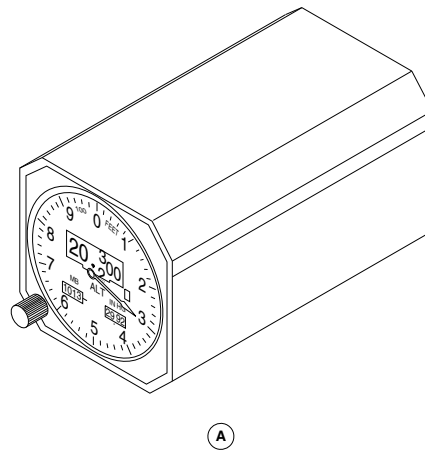
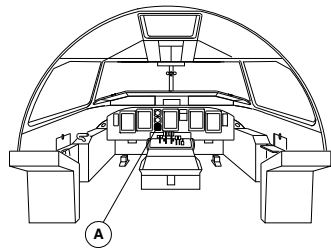
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STANDBY ALTITUDE INDICATOR LOCATOR
Figure 7

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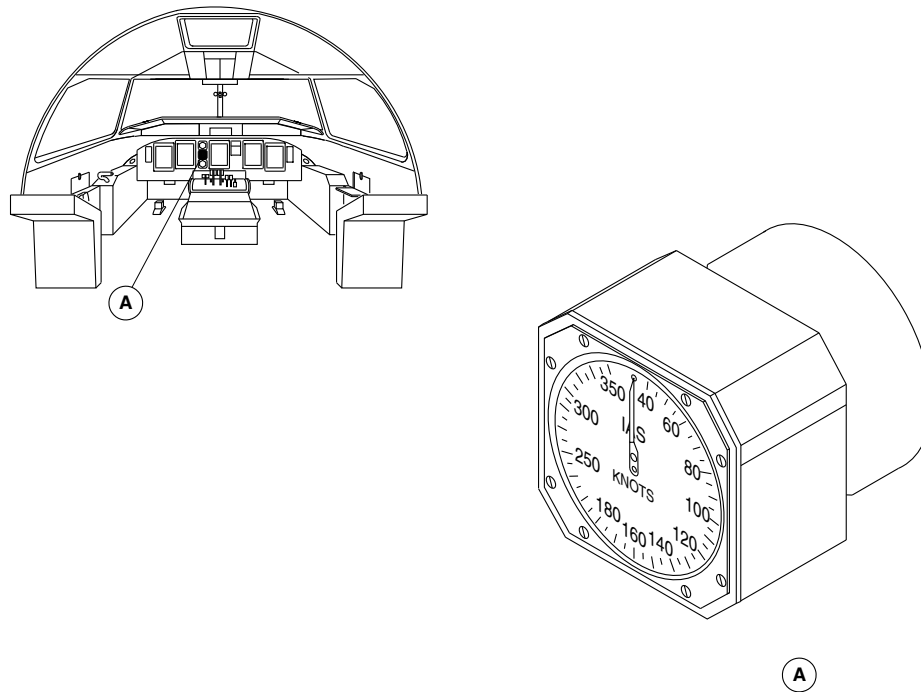
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STANDBY AIRSPEED INDICATOR, LOCATOR
Figure 8

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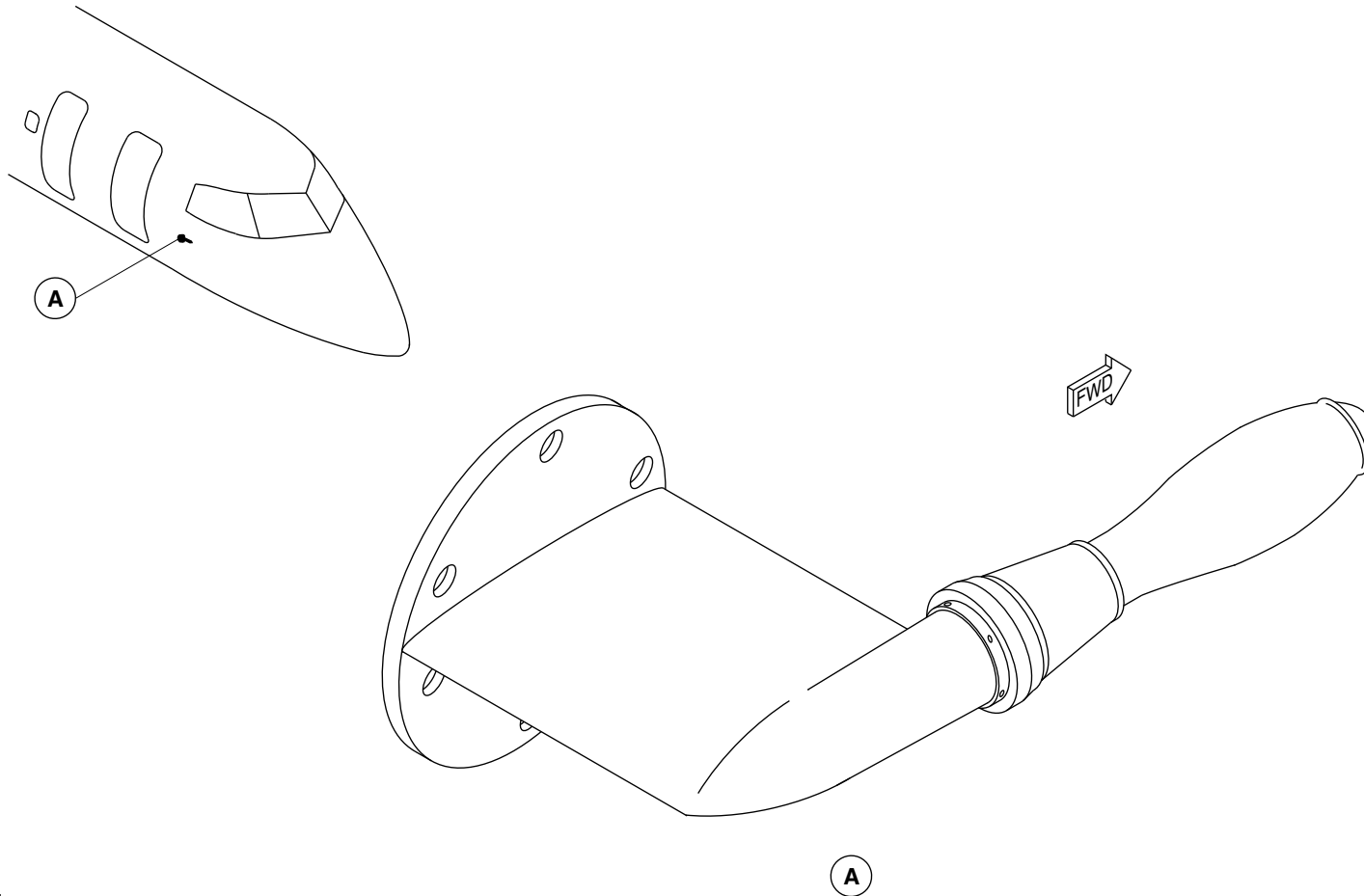
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STANDBY PITOT-STATIC PROBE, LOCATOR
Figure 9

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AIR DATA SYSTEM

Introduction

The primary Air Data System (ADS) senses and converts static pressure, pitot (impact) pressure, and static air temperature to electronic data for use by the systems that follow:

- Cabin Pressure Control
- Flight Guidance
- Flight Controls
- Indication
- Vmo/Mmo Warning
- Navigation
- Powerplant Operation.

General Description

Refer to Figures 1 and 2.

The two primary channel Air Data Computers (ADC1, ADC2) operate independently to calculate the data that follows:

- Altitude
- Air speed
- Vertical speed

- Temperature
- Barometric corrections.

The ADC operates in the modes that follow:

- Operational
- Initiated Test
- Maintenance.

The primary Air Data System (ADS) has the components that follow:

- Computer, Air Data (34-11-01)
- Probes, Pitot-Static (34-11-06)
- Probe, Static Temperature (34-11-11)
- Valve, pitot-static isolation (34-xx-xx)
- Panel, pitot-static isolation valve (34-xx-xx).

The Air Data Computers (ADC1, ADC2) supply data to the systems that follow:

- Cabin Pressure Controller, (CPC)
- Cabin Indication Module (CIM)
- Flight Guidance Module (FGM1, FGM2)
- Flight Controls Electronic Control Unit (FCECU1, FCECU2)
- Stall Protection Modules (SPM1, SPM2)
- Input/Output Processors (IOP1, IOP2)
- Primary Flight Displays (PFD1, PFD2)
- Mult-Functional Displays (MFD1, MFD2)



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- Attitude and Heading Reference Units (AHRS1, AHRS2)
- ATC Transponder (ATC1, ATC2)
- Flight Management System (FMS1, FMS2)
- Full Authority Digital Engine Control (FADEC1, FADEC2).

Detailed Description

Refer to Figure 3.

The Air Data Computers (ADC1, ADC2) operate in the modes that follow:

- Power-On Self Test (POST)
- Line operational
- Initiated test
- Maintenance mode.

Power-On Self Test (POST): This mode automatically continues for 2 seconds after power up.

The ADCs does the self-tests that include:

- CPU self-tests
- Memory check sum
- Test of peripheral circuits
- Test of I/O ports

- Indication of test in progress and successful completion.

NOTE

The air data output parameters are invalid during the power-up tests.

The ADC calculates a failure warning if it senses a hardware failure.

When the ADC POST mode is completed, all ADC parameters are available.

In the line operational mode, the Air Data Computers (ADC1, ADC2) do the primary channel calculations as follows:

- Measure static and total pressures
- Measure outside air temperature
- Calculate barometric correction
- Sense probe heater failure data
- Calculate air data parameters for flight compartment indication and used by the air-conditioning, flight guidance, flight control, navigation and engine systems.

The Air Data Computers (ADC1, ADC2) calculate the air data parameters that follow:

- Static pressure (ps)
- Impact pressure (dp)
- Barometric correction in hPa and in InHg (Bcor)
- Barometric altitude corrected by barometric correction (Hc)
- Altitude rate (Vz)
- Calibrated airspeed (CAS)



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- True airspeed (TAS)
- Maximum velocity in operation (VMO)
- Mach number (M)
- Static Air Temperature (SAT).

Pitot pressure from each pitot-static probe is supplied through tubing to its related ADC and averaged static pressure is supplied through tubing and a shutoff valve to the two ADCs.

A bird strike at the right side of the forward fuselage section can cause damage to the copilot's primary pitot-static probe and to the secondary pitot-static probe. All air data indications can possibly become incorrect because the ADC1 static tubing is connected through a shutoff valve to pitot-static probe No. 2.

The shutoff valve is a usually open valve that is electrically controlled. The NORM/ISOL pushbutton switch on the pitot static isolation control panel is pushed to close the shutoff valve.

The pitot-static probe tubes have an encased heating element with a self regulating power consumption feature to prevent ice accumulation on the pressure probes and help eliminate erroneous total and static measurements. Each pitot-static probe heater is controlled by a PITOT STATIC toggle switch on the ice protection panel.

The static temperature probe has two resistor sensors. Each sensor is connected to one ADC in order to supply the aircraft system with two independent temperature values. The Indicated Outside Air Temperature (IOAT) measured by the two sensors is affected by airspeed. A recovery factor is set in each ADC and is used to calculate and compare, using the IOAT, the Static Air Temperature (SAT) and Total Air Temperature (TAT).

The ADC receives barometric correction data from a rotary switch on its related Index Control Panel (ICP) to calculate the barometric correction and the barometric altitude.

The barometric correction value is shown on the PFD altitude digital readout in Millibars (MB) or Inches of Mercury (In Hg).

On aircraft with the Millibars (MB) barometric correction value (823SO00017): The MB barometric correction indication is set by ADC pin programming logic.

When the ADC is not energized, the barometric selection value is saved in ADC Non Volatile Memory (NVM).

The trend of the altitude calculated by the ADCs can be incorrect when the aircraft is dynamically pitched. For indication, the barometric corrected altitude data from the ADC are dampened by the Attitude and Heading Reference Units (AHRS) to show barometric inertial altitude. If the AHRU malfunctions or an opposite EFIS ATT/HDG SOURCE reversion is set, barometric corrected altitude is shown on the related PFD.

NOTE

Note: The EIS has no indications to tell the pilots that barometric corrected altitude is shown.

The barometric altitude rate data from the ADCs are supplied through the AHRUs to make the indications instantaneous. If the related ADC malfunctions, the altitude rate data from the other ADC is automatically used to keep the IVSI indication in view.

The calibrated airspeed is supplied to the display units where it is shown as indicated airspeed.



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The Air Data Computers (ADCs) calculate Non Computed Data (NCD) if a primary parameter, static pressure (ps), impact pressure (dp), or Outside Air Temperature (OAT), is sensed out of range.

NOTE

Note: The airspeed indication is calculated as NCD when the airspeed is less than 30 kt, but, when the aircraft is stationary and on the ground, a Weight-On-Wheels (WOW) signal from the proximity sensor electronics unit will cause the Electronic Instruments System (EIS) to show 30 KIAS and 0 KTAS.

Initiated Test Mode: The ADC pre-flight test switch is used to check Vmo Warning Tone Generator (WTG) and the ADC outputs.

Maintenance Mode: The Built In Test Equipment (BITE) uses the Central Diagnostic System (CDS) to give the condition of the component. It saves faults in a Non Volatile Memory (NVM) for reporting to line and shop maintenance.

The BITE allows aircraft maintenance personnel to:

- Do fault isolation and return to service testing after completing maintenance actions
- Access failure reports from last or previous flight legs
- Get the avionics status report
- Get the part number of an LRU.

The BITE mode monitors the condition of the component as follows:

- Power-On Self Test (POST)
- Continuous monitoring.

Power-On Self Test (POST): The POST checks the condition of the component at power-up or after a long power interruption of more than 200 milliseconds.

Continuous Monitoring: The continuous monitoring checks the status of the component in flight. It records faults in a Non Volatile Memory (NVM) for later troubleshooting using the Central Diagnostic System (CDS).

The maintenance mode is selected from one or the other Audio and Radio Control Display Units (ARCDUs) during maintenance operations. The ADCs perform the tests when they receive a maintenance discrete command from IOM1 while the air speed is less than 50 kt (93 km/hr) and the aircraft is on the ground.

[Refer to Figure 4.](#)

The EFIS ADC SOURCE reversion selector on the ESID Control Panel (ESCP) is set to the NORM position to show ADC data on the related Primary Flight Displays (PFD1, PFD2).

NOTE

Note: The ADC data is supplied through its related AHRU.

[Refer to Figure 5.](#)

The Electronic Instrument System (EIS) shows the Air Data Computer (ADC1, ADC2) parameters that follow:

- Indicated Air Speed (IAS1, IAS2)
- Altitude (ALT1, ALT2)
- True Airspeed (TAS1, TAS2)
- Minimum Descent Altitude (MDA)
- Static Air Temperature (SAT).



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Indicated Air Speed (IAS): The Indicated Air Speed (IAS) indicator shows the IAS of the selected Air Data Computer (ADC1, ADC2). It has a vertical scale with an IAS reference line and digital readout. The IAS vertical scale and reference line give an analogue indication.

The Primary Flight Display shows the Indicated Air Speed (IAS) indicator components that follow:

- IAS scale
- IAS digital readout
- IAS reference line
- VMO speed cue
- Predicted airspeed trend
- IAS mismatch flag
- IAS failure flag.

[Refer to Figure 6.](#)

IAS Scale: The IAS scale has a speed range from 30 to 500 kts. It is a black filled rectangular vertical tape with scale with marks every 10 kts and numbers every 20 kts. The tape moves up for decreasing airspeeds and down for increasing airspeeds to show the IAS value.

IAS Digital Readout: The IAS value is shown through a window in the middle of the IAS indicator. The IAS indicator also shows range of values 42 kts. more and less than the value shown in the window.

The IAS digital readout has white numbers that change to red when the IAS is more than the Maximum Operating Speed (VMO). The display changes back to white when the IAS decreases 2 at less than the VMO.

The numbers also change to red when the IAS is less than the low speed cue. The display changes back to white when the IAS increases 2 kts. more than the low speed cue.

IAS Reference Line: The IAS indicator has a white line adjacent to the window to point to the current IAS.

VMO Speed Cue: The VMO speed cue gives indication of the Maximum Operating Speed (VMO) along the IAS scale from 30 to 400 kts. It is a red and black band that starts with a red edge at the VMO value and extends to the top of the scale.

The VMO speed cue is out of view when the IAS is 42 kts. less than VMO.

NOTE

Note: When VMO is invalid, the ADC will also send an invalid IAS signal to make the PFD show an IAS failure flag.

Predicted Airspeed Trend: The predicted airspeed trend gives an analog indication of the predicted airspeed in 10 seconds while the aircraft is airborne. It is a white vertical line with an arrow that starts at IAS reference line and points to the predicted airspeed.

If the airspeed trend is more than the shown IAS scale, part of the indication goes out of view.

When the difference between the actual IAS and the predicted airspeed value is less than 1 kt or when the IAS parameter is not correct, the indication goes out of view.

[Refer to Figure 7.](#)

IAS Mismatch Flag: The Flight Data Processing System (FDPS) senses signal differences between the IAS1 and the IAS2. The two IAS indicators show a yellow IAS flag above its IAS digital readout and the two Flight Mode Annunciators (FMA) show a yellow IAS



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MISMATCH message in the centre row of the centre column. The indications flash for five seconds when they come into view and then go steady.

The airspeed mismatch threshold is equal to 10 kt.

[Refer to Figure 8.](#)

IAS Failure Flag: When the airspeed parameter malfunctions, an IAS failure flag comes into view as a red IAS FAIL message in a white rectangle.

Speed bugs are shown along the edge of the IAS scale to reference V(V1, VR, V2) speeds.

[Refer to Figure 9.](#)

The SPEED BUG SEL pushbutton switch located on the Index Control Panel (ICP1, ICP2) is pushed to set the speed bugs that follow:

- V1, take off decision speed
- VR, rotation speed
- V2, take off safety speed
- No. 1 (Solid)
- No. 2 (Outline).

The SPEED BUG SEL pushbutton switch is pushed to set speed bugs V1, VR, and V2 when the aircraft is on the ground and speed bugs No. 1 and No.2 when the aircraft is airborne.

The SPEED BUG SEL pushbutton switch is pushed to change speed bug V1 or No. 1. It is pushed again to set V1 or No. 1 and to change VR or No. 2. The SPEED BUG SEL pushbutton switch is pushed again to set V2. Each speedbug is also set five seconds after the last

SPEED BUG SEL pushbutton switch or SPEED BUG rotary selection.

The SPEED BUG rotary selector is turned to change the speed bug value.

[Refer to Figure 10.](#)

The cyan bug moves along the IAS scale from minimum 50 kt value to 400 kt, and is parked at this position for higher values. When the index value is lower than 50 kt or when not correct, it is removed from the indication.

Each index bug has a digital readout shown in cyan characters to give a digital value for the Index Control Panel (ICP1, ICP2) selection. It shows the selection in 1kt increments from 50 kts to 400 kts. The indication is out of view when the selection is less than 50 kts. Each digital value has a bug reminder.

NOTE

Note: The speed bug is not an autopilot target and is only used as an advisory indication.

[Refer to Figure 11.](#)

The BARO SET knob located on the Index Control Panel (ICP1, ICP2) is turned to supply the related Air Data Computers (ADC1, ADC2) with a barometric correction.

[Refer to Figure 12.](#)

The barometric correction range is 745 to 1050 mbar (22 to 31 In Hg). The Related Primary Flight Displays (PFD1 and PFD2) show the barometric correction indication below the altitude indicator with four digits shown in cyan followed by a Millibars (MB) label in white fonts. It is shown in Millibars (MB) or in Hg, as set by pin programming logic on the ADC. If a power interruption occurs, the



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barometric selection value is saved by the ADC for use when restarting.

[Refer to Figure 13.](#)

Altitude Indicator: The altitude indicator shows the barometric altitude of the selected Air Data Computer (ADC1, ADC2). It has a vertical scale with an IAS reference line and digital readout. The IAS vertical scale and reference line give an analogue indication.

The Primary Flight Display shows the altitude indicator parameters that follow:

- Altitude scale
- Altitude digital readout
- Altitude reference line
- Altitude mismatch flag
- Altitude failure flag.

Altitude scale: The altitude scale has a range from –990 ft to 50000 ft. It is a black filled rectangular vertical tape with scale with marks every 100 ft and numbers every 500 ft. The tape moves up for decreasing altitudes and down for increasing altitudes.

For negative altitudes, A white NEG label is shown vertically between the 0 ft and –500 ft, –500 and –1000 ft, and –1000 ft and –1500 ft marks.

Altitude digital readout: The altitude value is shown through a T shaped window in the middle of the altitude indicator as a rolling drum indication in 20 ft increments. The altitude indicator also shows a range of values 550 ft more and less than the value shown in the window.

Altitude reference line: The altitude indicator has a white line adjacent to the window to point to the current altitude.

[Refer to Figure 14.](#)

Altitude Mismatch Flag: The Flight Data Processing System (FDPS) senses signal differences between the ALT1 and the ALT2. The two altitude indicators show a yellow ALT flag above its altitude digital readout and the two Flight Mode Annunciators (FMA) show a yellow ALT MISMATCH message in the centre row of the centre column. The indications flash for five seconds when they come into view and then go steady.

The altitude mismatch threshold increases from 60 ft (18.3 m) on the ground to 180 ft (55 m) at 27,000 ft (8230 m).

[Refer to Figure 15.](#)

Altitude Failure Flag: When the altitude indicator parameter malfunctions, an altitude indicator failure flag comes into view as a red ALT FAIL message in a white rectangle.

[Refer to Figure 16.](#)

On aircraft with the metric secondary altitude indications (839CH00016 OR SB84–31–74 OR SB84–31–93 OR SB84–31–97): When the PFD ALTIMETER UNITS toggle on the PFD altimeter unit control panel is set to the FT+M position, the altitude indicator will continue to show altitude in feet and it will also show a metric indication in meters.



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On aircraft with SB84–31–44 OR SB84–31–84 incorporated, the PFD altimeter unit control panel is not installed.

[Refer to Figure 17.](#)

The metric indication shows the aircraft's current altitude in 5 m increments from negative 500 m to 15900 m. It has white numbers in white box with an M label.

When the metric altitude indication malfunctions, the 5 numbers are replaced with 5 white dashes.

NOTE

The metric altitude indication is supplied through its related IOP. If it malfunctions, the indication is supplied through the opposite (IOP).

When the metric altitude indication is in view, the pre-selected altitude readout is also shown in meters. The range of the altitude readout is between 0 and 30500 m.

NOTE

Note: The metric altitude readout indication changes in 100 ft increments but is rounded up to match the 5 m metric altitude indication.

[Refer to Figure 18.](#)

Minimum Descent Altitude (MDA): The MDA is a specified altitude that is referenced to sea level for non-precision approaches. A descent below the MDA cannot be made until a visual reference to continue the approach to land is made.

The DH/MDA rotary switch on the Index Control Panel (ICP1, ICP2) is set to MDA to let the DH/MDA knob change the MDA value. The DH/MDA knob is then turned to set a MDA for minimum descent altitude calculations.

[Refer to Figure 19.](#)

The MDA indication has the components that follow:

- Digital readouts
- Index bug
- Alert annunciation.

MDA Digital Readouts: The bottom right part of the attitude indicator shows the selection as cyan numbers followed by a white MDA label. It shows the altitude selection in ten-foot increments from 0 through 19990 ft. When MDA setting is less than 0 ft, all of the indication goes out of view.

MDA Index Bug: When the MDA is set to a value more than zero, a cyan analogue MDA index bug on the altitude indicator also comes into view to show the selected MDA value.

When aircraft is at the selected MDA, the MDA index bug changes to a yellow indication that flashes for the first 3 seconds then goes steady.

MDA Alert Annunciation: An MDA alert annunciation shows that the aircraft is going down to the Minimum Descent Altitude (MDA).

This indication is armed while the aircraft's barometric corrected altitude is 100 ft more than the set MDA and ground. A Weight On Wheels (WOW) condition causes the indication to disarm to make sure that the MDA alert annunciation is out of view during take off and after landing.

When the MDA alert annunciation is armed and the aircraft altitude is less than the set MDA value, a yellow MDA alert annunciation that flashes for the first 3 seconds then goes steady is shown in the attitude sphere.



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The indication stays in view while the barometric corrected altitude is less than 100 ft above the set MDA and ground to keep the MDA alert annunciation in view during a flight level hold at the set MDA.

Five white dashes replace the numbers and the MDA alert annunciation goes out of view when a Primary Flight Display (PFD1, PFD2) does not receive valid data from its related Input/Output Processor (IOP1, IOP2).

[Refer to Figure 20.](#)

When the EFIS ADC SOURCE reversion selector on the ESID Control Panel (ESCP) is set to the 1 or 2 position, the two Primary Flight Displays (PFD1, PFD2) will show barometric inertial altitude from the selected ADC (ADC1 or ADC2) through its related AHRU (AHRU1 and AHRU2).

NOTE

Note: When the EFIS ATT/HDG SOURCE reversion selector on the ESID Control Panel (ESCP) is set to the 1 or 2 position, the related PFD will show barometric inertial altitude from the selected ADC and AHRU and the other PFD will show barometric corrected altitude data from its related ADC.

[Refer to Figure 21.](#)

When the two Primary Flight Displays (PFD1, PFD2) are showing the same air data source, an air data source annunciator is shown in yellow fonts.

[Refer to Figure 22.](#)

The MFD1 and MFD2 reversion switches located on the ESID Control Panel (ESCP) are used to show the navigation page on the multi-functional displays.

The navigation page is usually shown on the upper part of MFD1 and permanent data is shown on the bottom.

[Refer to Figure 23.](#)

The navigation page shows the TAS digital value in white numbers from 9 to 999 kt in 1 kt increments with a TAS label.

[Refer to Figure 24.](#)

When the TAS data malfunctions, the 3 digits are replaced by 3 white dashes.

[Refer to Figure 25.](#)

The Engine Display (ED) shows the Static Air Temperature (SAT) digital display in white digits with a + or – symbol and a °C label to the right of the digits.

The ED shows SAT from ADC1. When an EFIS ADC SOURCE is set to 2, the ED will show the SAT parameter from ADC2.

[Refer to Figure 26.](#)

When the SAT data malfunctions, the 2 digits are replaced by 2 white dashes.

[Refer to Figure 27.](#)

The NORM/ISOL annunciator switch on the pitot static isolation control panel is pushed to isolate ADC1 from the pitot–static probe No. 2 static source.



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The green NORM indication goes out and the amber ISOL comes on to show that the shutoff valve has moved to the closed position.

Refer to Figure 28.

The ADC pre-flight test switch on the pilot's side console is set to TEST 1 or TEST 2 to check Vmo Warning Tone Generator (WTG) and the related ADC interfaces.

The Vmo warning sounds 5 seconds after making the selection for aural warning confirmation.

The specific values of air data parameters are shown in the table that follows:

Table 1: Table 1. Air Data Parameters and Output Values

PARAMETER	OUTPUT VALUES
Static Pressure	571.8125 mbar
Impact Pressure	137.875 mbar
Standard Altitude	15,000 ft
Baro correction	990 mbar 29.23 In Hg
Baro corrected altitude	14360 ft (4377 km/h)
Vertical speed	0 ft/min
Calibrated Airspeed	285 KIAS (528 km/h)

PARAMETER	OUTPUT VALUES
Mach	0.564
True Air Speed	353.125 kt (654.41 km/h)
VMO	280 KIAS (519 km/h)
Static Air Temp.	-15 °C
Total Air Temp	+1.5 °C

The PITCH TRIM, ELEVATOR FEEL, and AVIONICS caution lights also come on.

The aircraft left essential and right main busses supply 28 Vdc electrical power through 2 A circuit breakers to the Air Data Computers (ADC1, ADC2). The circuit breakers are located on the 28 Vdc avionics circuit breaker panel in position A7 and A6.

The temperature probe has 2 sensing elements. Each element receives an excitation current from its related ADC to measure its resistance. It is less than 0.5 mA to limit the self-heating error.

Each ADC receives total pressure (pt) supplied by its related pitot-static probe and a static pressure (ps) value obtained as a pneumatic average between static pressures from the two pitot-static probes.

In addition to the data received from the pneumatic interfaces, each ADC receives discrete data from the systems that follow:

- An electrical signal from its related temperature sensor to calculate the Indicated Outside Air Temperature (IOAT)



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- Barometric correction data from the related Index Control Panel (ICP)
- Test discrete command from a 3 position ADC TEST switch on the pilot's side console
- Initiated Built-In-Test (IBIT) discrete command from the Centralized Diagnostic System (CDS)
- Pitot-static probe heating failure discrete from the Ice and Rain Protection System (IRPS)
- Air data source manual reversion command from the ESID Control Panel (ESCP)
- Weight-On-Wheels discrete from the Proximity Sensor Electronics Unit (PSEU).

The air data parameters that follow are controlled using ARINC 429 low speed busses (12.0 to 14.5 kilobytes per second):

Table 2: Table 2. ADC Interfaces

PARAMETER	SOURCE
Static Pressure	FADEC1, FADEC2
Impact Pressure	FADEC1, FADEC2
Pressure Altitude	FGM1, FGM2 IOP1, IOP2 ATC1, ATC2 FMS CPC CIM

PARAMETER	SOURCE
Barometric Correction	PFD1, PFD2 MFD1, PFD2 FDR CPC
Barometric Corrected Altitude	PFD1, PFD2 MFD1, MFD2 FGM1, FGM2 AHRS1, AHRS2 ATC1, ATC2 FMS
Altitude Rate	FMS CPC CIM
Calibrated Air Speed	PFD1, PFD2 MFD1, MFD2 FGM1, FGM2 SPM1, SPM2 IOP1, IOP2 FMS FADEC1, FADEC2 FCECU1, FCECU2
Maximum Velocity in Operation	PFD1, PFD2 MFD1, MFD2 FGM1, FGM2 FMS
VMO Warning	IOP1, IOP2
Mach	SPM1, SPM2 FMS



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PARAMETER	SOURCE
True Air Speed	PFD1, PFD2 MFD1, MFD2 FGM1, FGM2 SPM1, SPM2 AHRS1, AHRS2 IOP1, IOP2 FMS FADEC1, FADEC2 CIM
Static Air Temperature	MFD1, MFD2 ED IOP1, IOP2 FMS FADEC1, FADEC2
Total Air Temperature	IOP1, IOP2 FMS

Each ADC supplies air data parameters through 5 identical ARINC 429 low speed busses for use by critical systems and other avionics subsystems. For non-avionics non-critical systems, air data parameters are available on two General Purpose Data Busses (GPDB) from the IOP modules.

Each ADC also transmits to each IOM a Vmo/Mmo warning discrete. The components interfaced with ADC ARINC 429 outputs are listed as follows:

Table 3: ADC1

ARINC 1	ARINC 2	ARINC 3	ARINC 4	ARINC 5
MFD1 PFD1	FGM1 SPM1 FCECU1 AHRS1 IOP1	FADEC1 Channel A FADEC2 Channel A	MFD2 PFD2 FGM2 SPM2 AHRS2	IOP2 ATC1 ATC2 CPC CIM

Table 4: ADC2

ARINC 1	ARINC 2	ARINC 3	ARINC 4	ARINC 5
MFD2 PFD2	FGM2 SPM2 AHRS2 FCECU2 IOP2	FADEC1, Channel B FADEC2, Channel B	MFD1 PFD1 FGM1 SPM1 AHRS2	IOP1 ATC1 ATC2 CPC CIM

The Input/Output Processor (IOP1, IOP2), Stall Protection Modules (SPM1,SPM2), and Flight Guidance Modules (FGM1, FGM2) are located inside its related Integrated Flight Cabinet (IFC1, IFC2).

[Refer to Figure 29.](#)

An electrically controlled solenoid shutoff valve is used to isolate ADC1 from the pitot-static probe No. 2 static source.

An annunciator switch located near the pilot in the flight compartment give manual control of the shutoff valve.

Electrical power is supplied from the 28 Vdc left essential bus through a flight compartment annunciator switch and relay to let the same electrical power source energize the shutoff valve.



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When the NORM/ISOL annunciator switch is pushed, the 28 Vdc left essential bus energizes a relay to let the 28 Vdc left essential bus supply electrical power through relay contacts to the shutoff valve. The shutoff valve is stays closed while it is energized.

When the solenoid valve is in the open position, it supplies different signals through the Advisory Control Computer (ACU) to make the NORM lights in the annunciator switch come on and the ISOL lights go out.

If the solenoid valve is in the closed position, it supplies different signals through the Advisory Control Computer (ACU) to make the ISOL lights in the annunciator switch come on and the NORM lights go out.

Computer, Air Data

[Refer to Figure 30.](#)

The ADC measures static pressure (ps) and total pressure (pt) supplied from the primary pressure probes and compensates these pressures for probe position errors. The ADC also measures the temperature probe resistance to calculate Outside Air Temperature (OAT). From these primary parameters and with the acquired barometric setting information transmitted by the Index Control Panel (ICP), the ADC does the air data calculations.

The ADC is a Line Replaceable Unit (LRU) that has the parts that follow:

- An electrical board with two silicon pressure sensors P90,a CPU, analog and discrete inputs/outputs, ARINC 429 interface, power supply and electrical interface with a connector and EMC protection

- A mechanical case that secures the pressure ports

The Air Data Computer (ADC) weighs 6.12 lb (2.78 kg). It has subassemblies mechanically packaged into a 2 in. (51 mm) high, 6 in. (152 mm) wide by 7.9 in. (201 mm) long case.

It has quick disconnect pitot and static pneumatic connectors.

The two ADCs are installed in the flight compartment on the aft bulkhead using four mounting screws and have the pneumatic connectors oriented downwards to prevent entry of moisture into the units.

[Refer to Figure 31.](#)

The ADC transmits the calculated air data parameters on 5 ARINC 429 low speed busses.

The validity of these parameters is dependent on the results of internal monitoring performed during power-up tests and continuous BIT. Interfaces with the Centralized Diagnostic System (CDS) lets any failure sensed by the ADC monitoring to be recorded during operations and lets control of a BIT in order to get ADC status.

A test mode enables confirmation of correct interface of ADC with the flight compartment displays.

Probes, Pitot-Static

[Refer to Figure 32.](#)

The two primary pitot-static probes are designed to accurately sense pitot and static pressures over the complete operating range of the aircraft.

An opening located at the nose of the probe senses pitot pressure. The shape of the opening is designed to reduce the sensitivity due to



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

inclination of the tube axis to the effected varying airflow angles. This results in a total pressure measurement constant through the entire AOA and sideslip envelope. Static pressure ports are located on the side of each probe tube.

The primary probes have one total pressure connector and two static pressure connectors (for pneumatic connection with each ADC).

To prevent ice accumulation on pressure ports, probes have a single coaxial-type electrical heating element with a self-regulating power consumption feature.

Moisture accumulation in the tube is also eliminated by the use of drain holes for the total pressure circuit. The static pressure circuit is self-drained due to the locations of the static ports on the tube.

The two primary heated pitot-static probes are installed symmetrically on the left and right sides of the fuselage.

The base plates have electrical and pressure connectors. Locating pins on the plate give an accurate angular orientation of the probe with the aircraft axis.

The primary probes are oriented 9 degrees downward in respect to the horizontal reference axis of the aircraft. The angle of the probe tube in respect with the mast gives a 5 degree angle in or out between the tube line and the tangential plane of the fuselage skin.

These probes are connected to aluminum tubing by flexible hoses. Two T-pipes, located roughly equal distance from the two primary probes, gives the pneumatic averaging of left and right static pressures delivered to each ADC. Tubing installation provides moisture drainage due to a tubing slope always greater than 3.5 degrees with drains holes installed at the lower point of each pipe. Six drains are installed.

Probe, Static Temperature

[Refer to Figure 33.](#)

The static temperature probe measures the surrounding aircraft Indicated Outside Air Temperature (IOAT). It is a flush-attached electrical resistance transmitter with temperature resistance properties.

The IOAT measured by the two sensors is affected by airspeed, a recovery factor, (a value programmed into the ADC to change IOAT data to SAT) is stored in each ADC and is used to calculate from IOAT the Static Air Temperature (SAT) and the True Air Temperature (TAT).

The probe has two isolated platinum temperature sensing elements that have an 0 °C base resistance of 500 ohms, and a thermal coefficient of resistance of 0.00385 ohms/degree C.

The accuracy of the resistance over the -50 to +50 °C range is ± 1 ohm. This resistance error is equivalent to a temperature accuracy of 0.52 °C within this range.

The static temperature probe weighs 0.25 pounds and is attached flush to the fuselage below the left wing root by four screws.

Valve, pitot-static isolation

[Refer to Figure 34.](#)

The pitot-static isolation valve provides physical separation between the two pitot-static detection systems. The valve is actuated with the NORM/ISOL annunciator switch.



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Panel, pitot-static isolation valve

Refer to Figure 35.

The pitot-static isolation valve panel has a NORM/ISOL annunciator switch. It is pushed to energize the shutoff valve.

The NORM/ISOL annunciator switch is a guarded annunciator switch. A transparent switch cover prevents accidental selection of the switch.

The pitot-static isolation valve panel is attached to the bottom left part of the instrument panel with two mounting screws. A bonding wire connected to the chassis makes a ground continuity connection between the control panel and the aircraft structure.

Water accumulators, pitot/static system

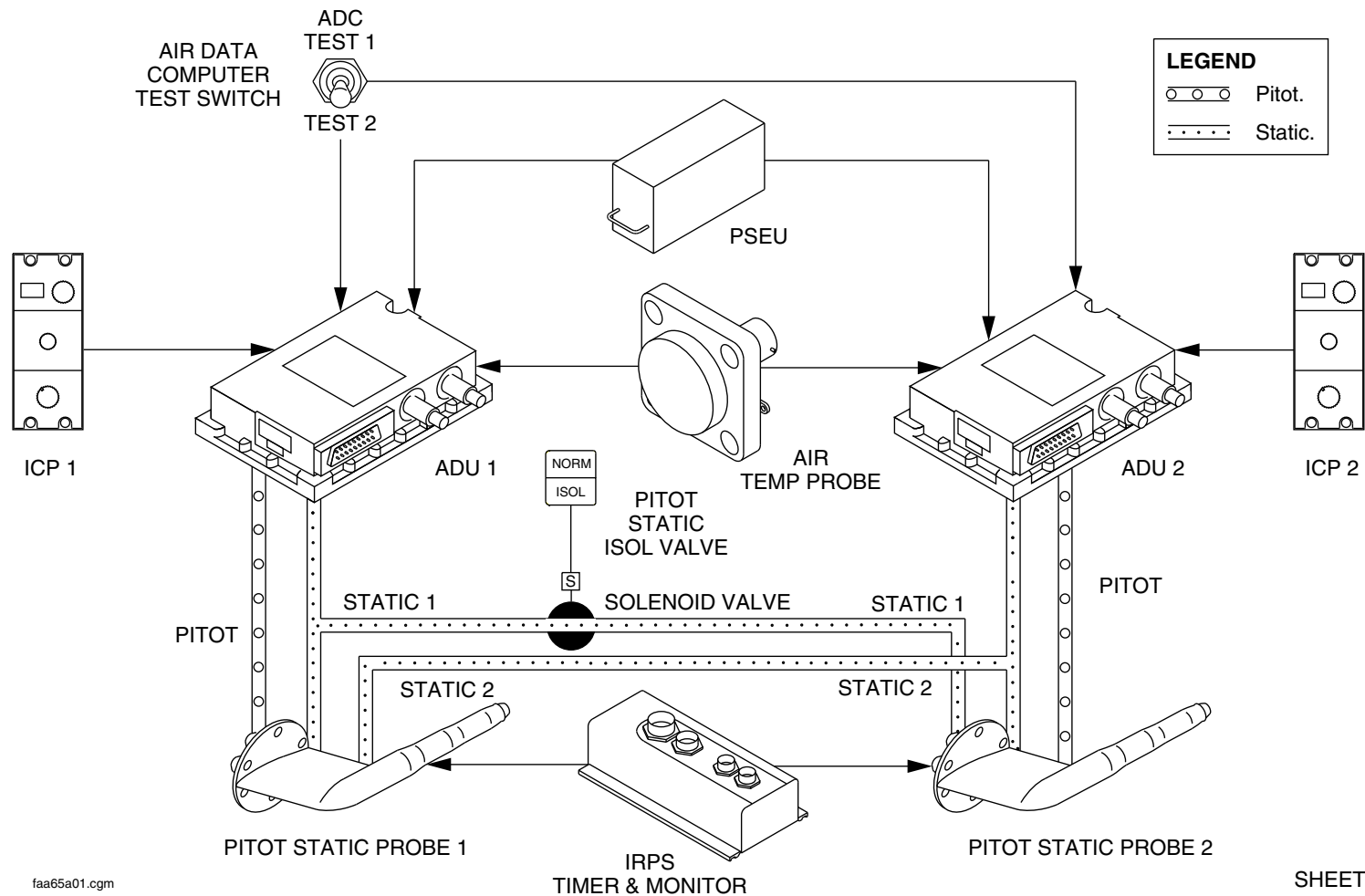
Refer to Figure 36.

For the aircraft with the SB84-34-59 or ModSum 4-113488 incorporated, there are eight pitot-static water accumulators installed, which includes six primary pitot-static water accumulators and two standby pitot/static water accumulators.

For the aircraft with the SB84-34-156 or ModSum 4-113815 incorporated, there are nine pitot-static water accumulators installed, which includes six primary pitot-static water accumulators, two standby pitot-static water accumulators and one additional standby pitot water accumulator.



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AIR DATA SYSTEM BLOCK DIAGRAM, INPUTS
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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PILOT EFIS DISPLAY

COPILOT EFIS DISPLAY

FADEC 1
FADEC 2

CABIN PRESSURE
CONTROLLER

CABIN INDICATION
MODULE

ADU 1
ADU 2

AVIONICS (LEFT ESSENTIAL), A7

AVIONICS (RIGHT MAIN), A6

ED

IFC 1
IFC 2

FMS 1
FMS 2

AHRS 1
AHRS 2

ATC 1
ATC 2

FCECU

SHEET 2

faa66a01.cqm

PRIMARY AIR DATA SYSTEM BLOCK DIAGRAM
Figure 2

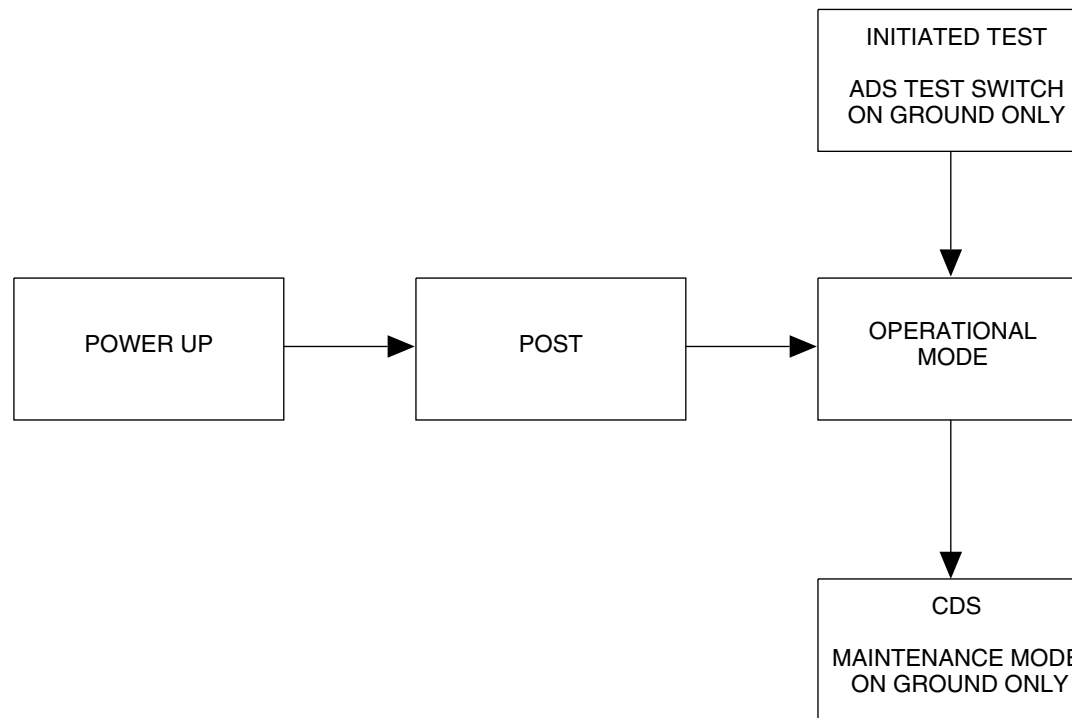
PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
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fss83a01.cgm

ADC MODES OF OPERATION
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

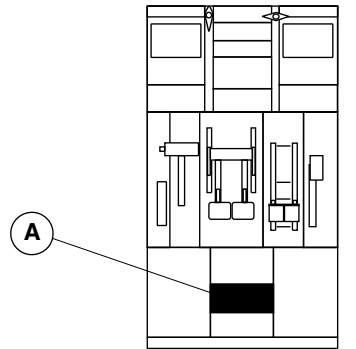
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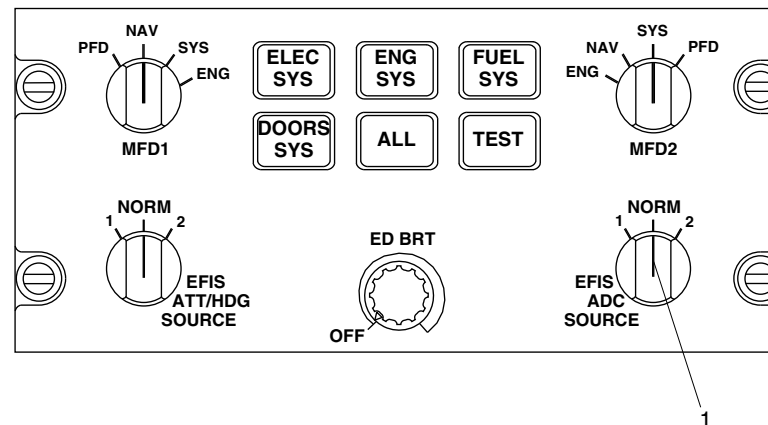
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CENTRE CONSOLE

LEGEND

1. EFIS ADC Source Reversion Selector.



A

fsa73a11.cgm

ESCP EFIS ADC SOURCE REVERSION PAGE 1
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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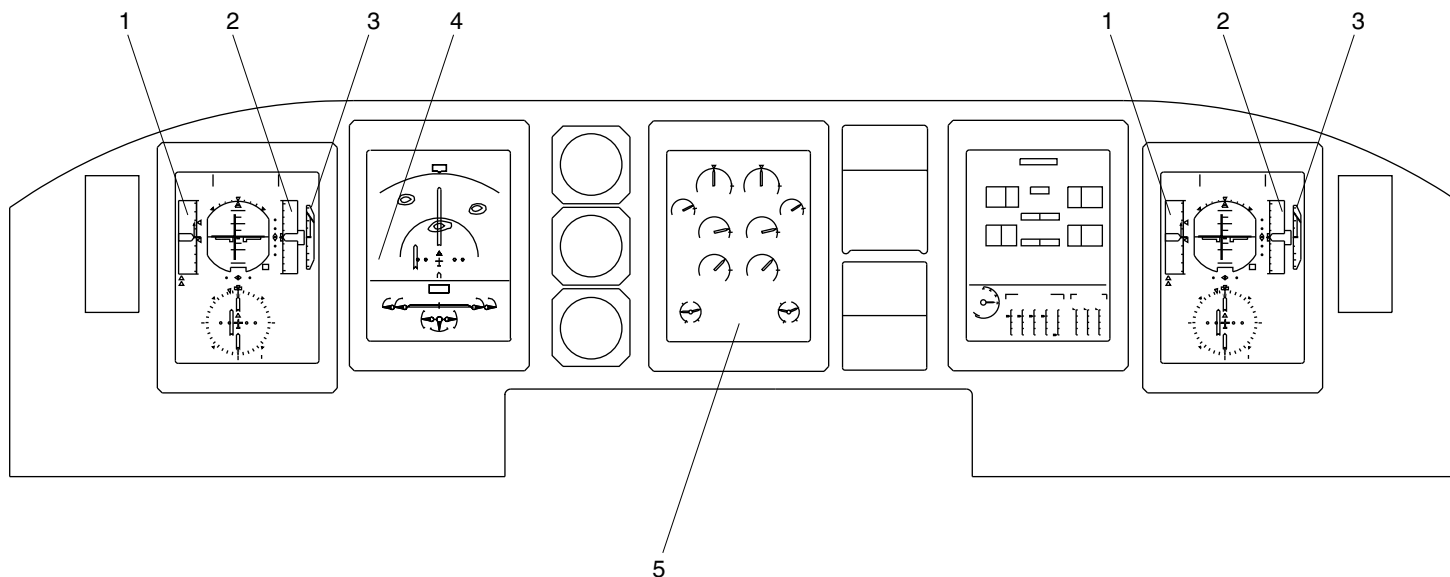


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LEGEND

1. Air Speed Indicator.
2. Altimeter.
3. IVSI.
4. True Airspeed Indication.
5. Static Air Temperature Indication.



faa97a01.cgm

ELECTRONIC INSTRUMENT SYSTEM (EIS) INDICATIONS
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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34-11-00

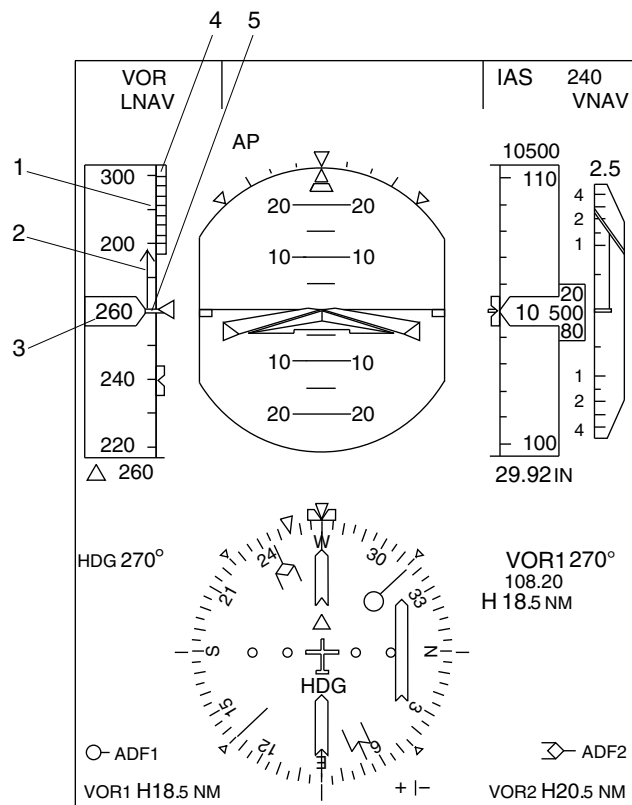
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A schematic diagram of a vehicle dashboard. The dashboard features a central instrument cluster with three circular gauges on the left and a rectangular display on the right. To the left of the instrument cluster is a large black rectangular area, and to the right is a large black rectangular area. A callout line labeled 'A' points to the instrument cluster area.

LEGEND

1. IAS Scale.
2. Predicted Airspeed Trend.
3. IAS Digital Readout.
4. VMO Speed Cue.
5. IAS Reference Line.



faa61a01.cgm

PFD INDICATED AIR SPEED INDICATIONS
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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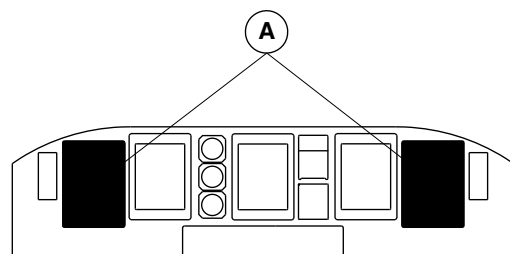
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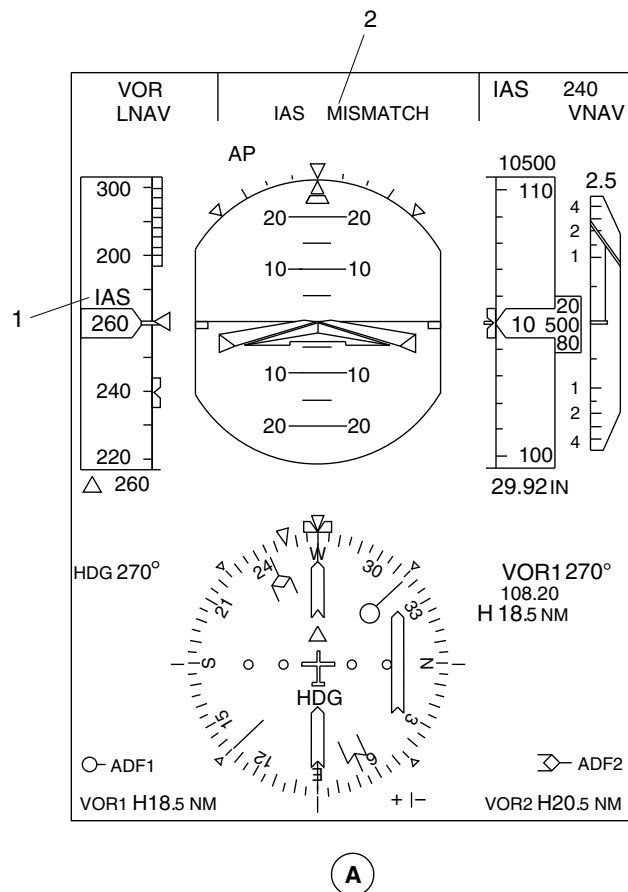
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Mismatch Flag.
2. IAS Mismatch Message.



faa51a01.cgm

PFD INDICATED AIR SPEED MISMATCH INDICATION
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
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A schematic diagram of a vehicle dashboard. It features a central instrument cluster with three circular gauges on the left and a rectangular display on the right. Flanking the central cluster are two large rectangular areas, likely for air vents or infotainment screens. A callout line labeled 'A' points to the top center of the dashboard, above the instrument cluster.

Figure 1 displays a simulated instrument panel with the following components:

- ILS Indicator (Top Left):** A vertical scale labeled "IAS" and "FAIL". A pointer indicates a value of 1.
- Heading Indicator (Top Center):** A circular scale with markings from 20 to 300. The heading is set to 270° (labeled "HDG 270°"). The scale also shows "200 DH" and "300" (likely 300 feet).
- Altitude Indicator (Top Right):** A vertical scale labeled "ALT" and "FAIL". The scale ranges from 0 to 2.5. The altitude is set to 108.15 (labeled "108.15").
- Heading Indicator (Bottom Center):** A circular scale with markings from 0 to 360. The heading is set to 270° (labeled "HDG 270°"). The scale also shows "200 DH" and "300" (likely 300 feet).
- Altitude Indicator (Bottom Right):** A vertical scale labeled "ALT" and "FAIL". The scale ranges from 0 to 2.5. The altitude is set to 108.15 (labeled "108.15").
- ILS1 H18.6 NM (Bottom Left):** A label indicating the ILS1 frequency and distance.
- ILS1 270° 108.15 H 18.5 NM (Bottom Center):** A label indicating the ILS1 frequency, heading, and distance.
- VOR2 H20.5 NM (Bottom Right):** A label indicating the VOR2 frequency and distance.

1. IAS Failure Flag.

PFD INDICATED AIR SPEED FAILURE FLAG
Figure 8

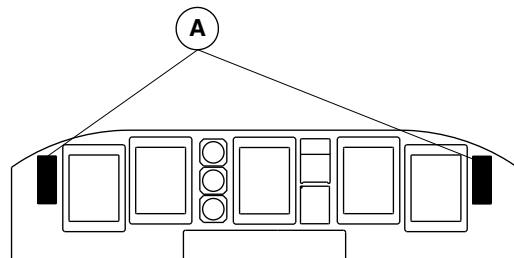
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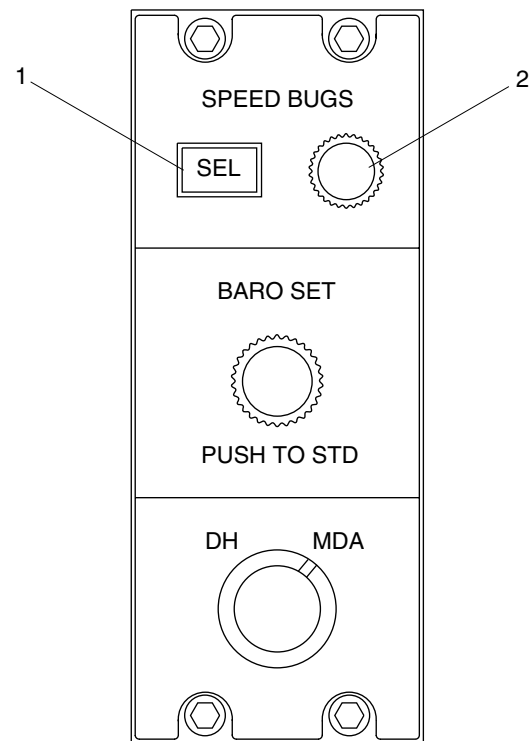
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Speed Bug Index SEL Pushbutton Switch.
2. Speed Bug Rotary Selector.



fa977a01.cgm

INDEX CONTROL PANEL (ICP), SPEED BUG SELECTION
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

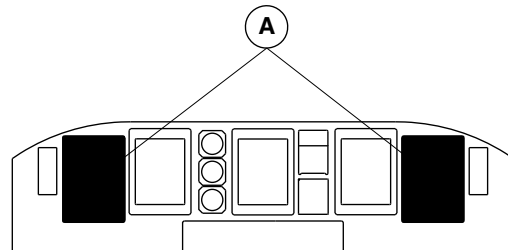
34-11-00

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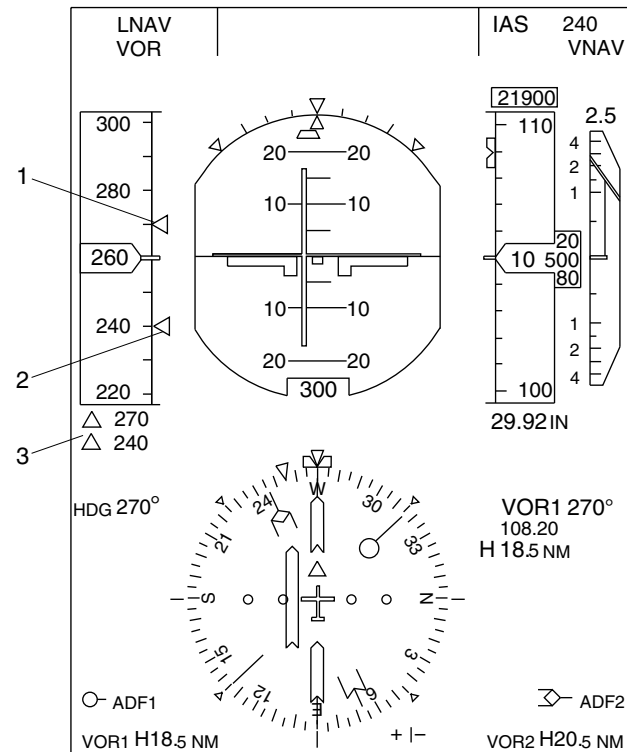
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. IAS Index Bug No. 1.
2. IAS Index Bug No. 2.
3. IAS Index Bug Digital Readouts.



A

faa59a01.cgm

PFD SPEED BUG INDICATIONS
Figure 10

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

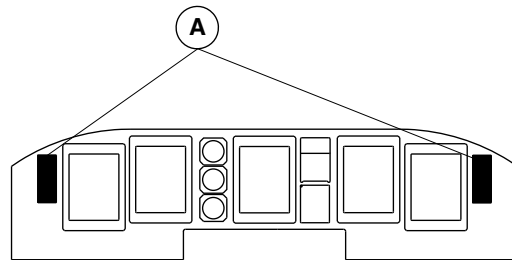
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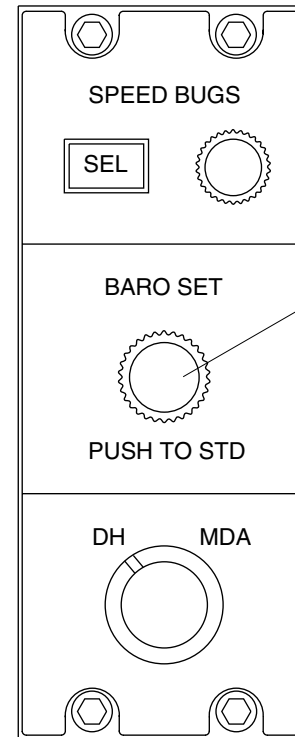
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. BARO SET Knob / PUSH TO STD Switch.



A

fa976a01.cgm

INDEX CONTROL PANEL (ICP), BAROMETRIC CORRECTION SELECTION
Figure 11

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

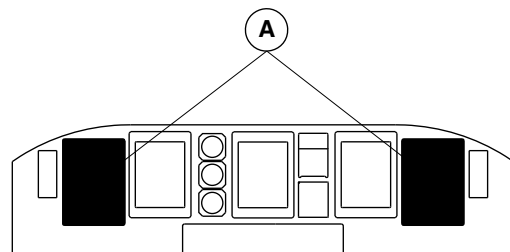
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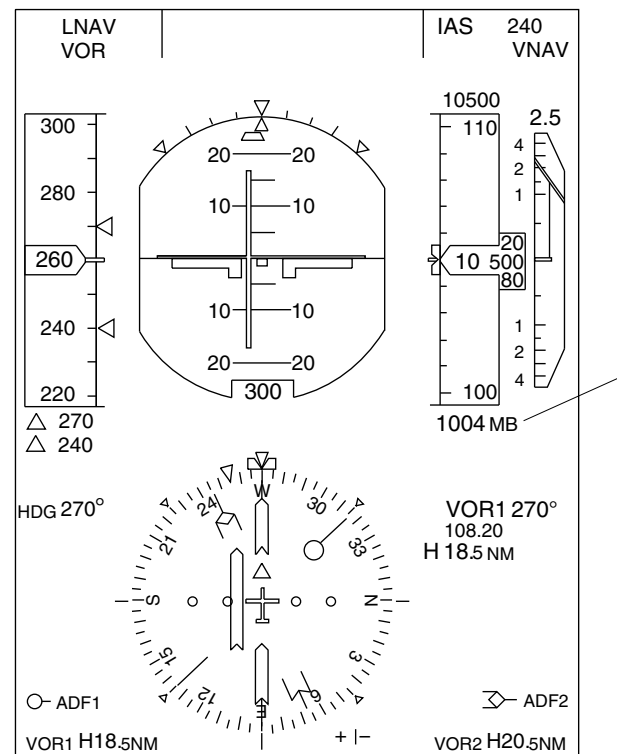
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Baro Setting Digital Readout.



A

faa64a01.cgm

PFD BAROMETRIC CORRECTION INDICATION
Figure 12

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

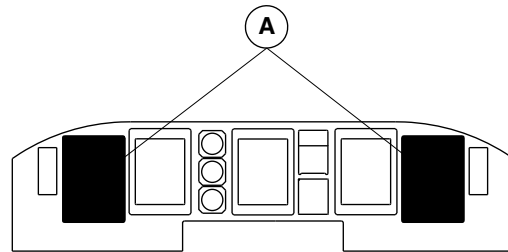
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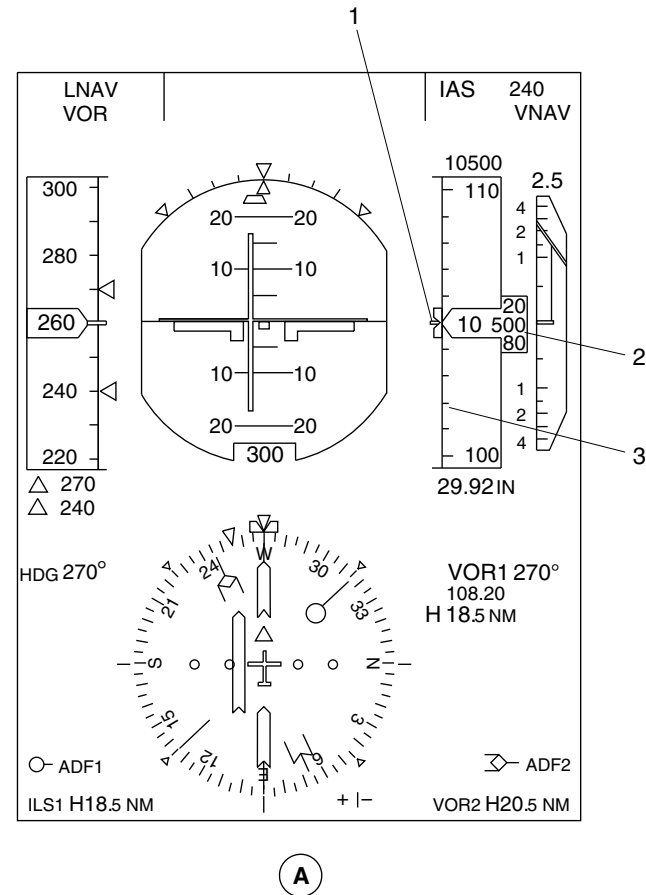
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Altitude Reference Line.
2. Altitude Digital Readout.
3. Altitude Scale.



faa63a01.cgm

PFD ALTITUDE INDICATOR INDICATION
Figure 13

PSM 1-84-2A
EFFECTIVITY:
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Config 001

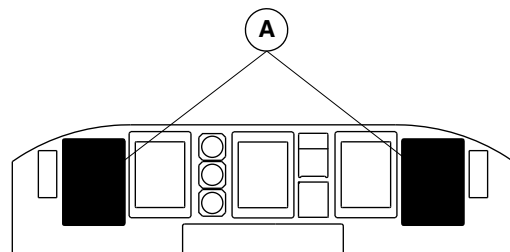
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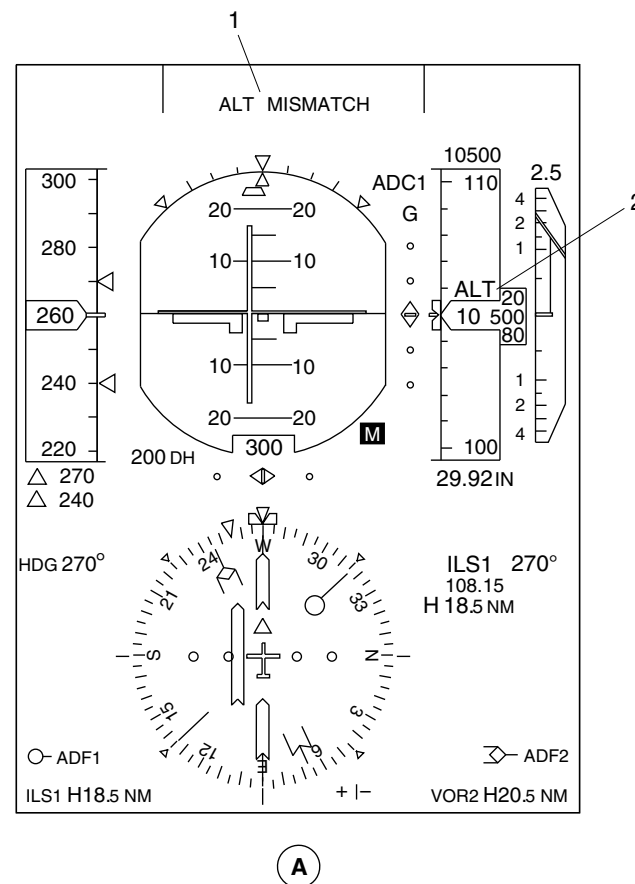
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Mismatch Message.
2. Altitude Mismatch Flag.



faa52a01.cgm

PFD ALTITUDE INDICATOR MISMATCH INDICATION
Figure 14

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

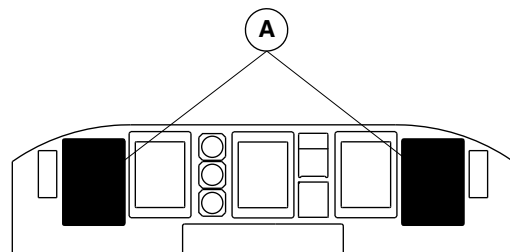
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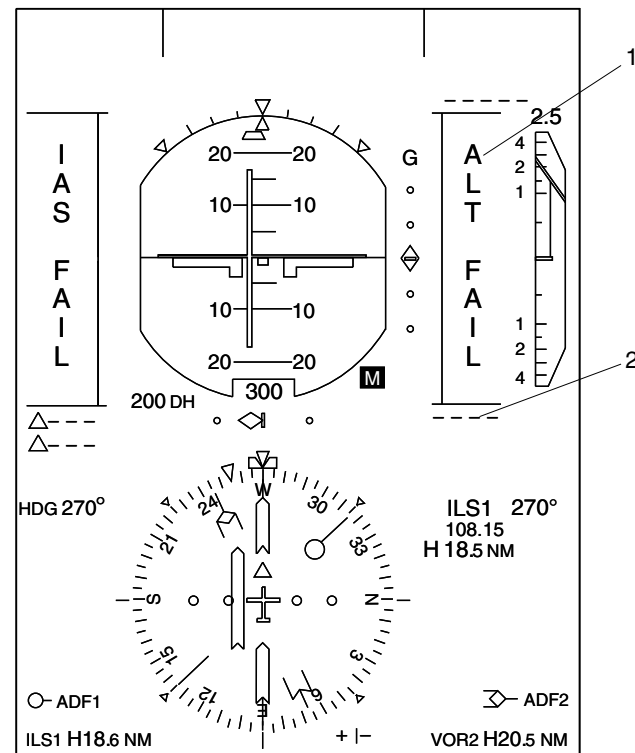
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Altimeter Failure Flag.
2. Baro Setting Fail Annunciation.



A

faa62a01.cgm

PFD ALTITUDE INDICATOR FAILURE FLAG
Figure 15

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

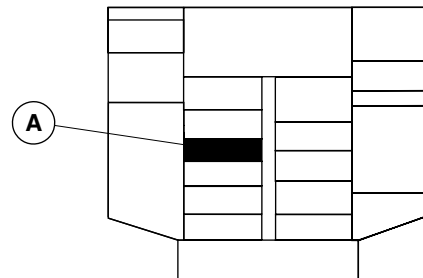
34-11-00

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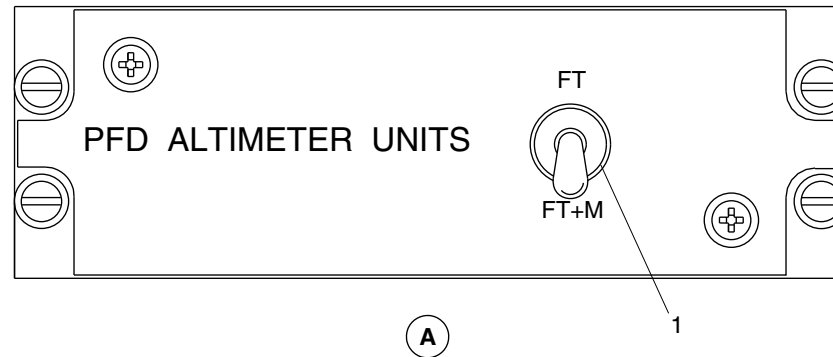


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE



LEGEND

1. PFD ALTIMETER UNITS Toggle Switch.

fsr97a01.cgm

(839CH00016): PFD METRIC SECONDARY ALTITUDE INDICATION SELECTION
Figure 16

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

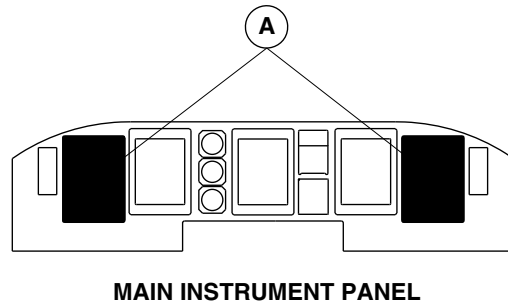
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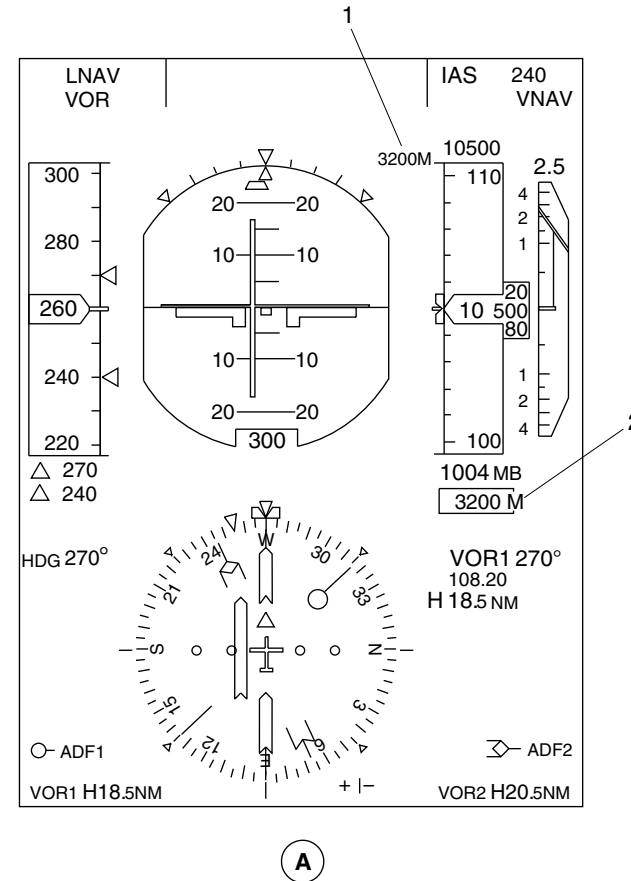
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Metric Altitude Digital Readout.
2. Metric Preselect Altitude Digital Readout.



far95a01.cgm

(839CH00016): PFD METRIC ALTITUDE INDICATION
Figure 17

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

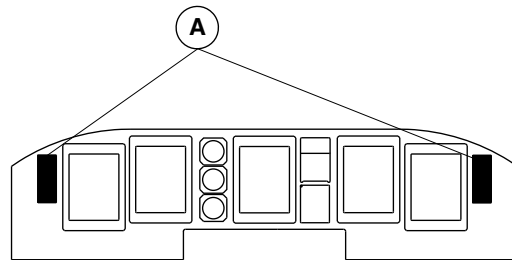
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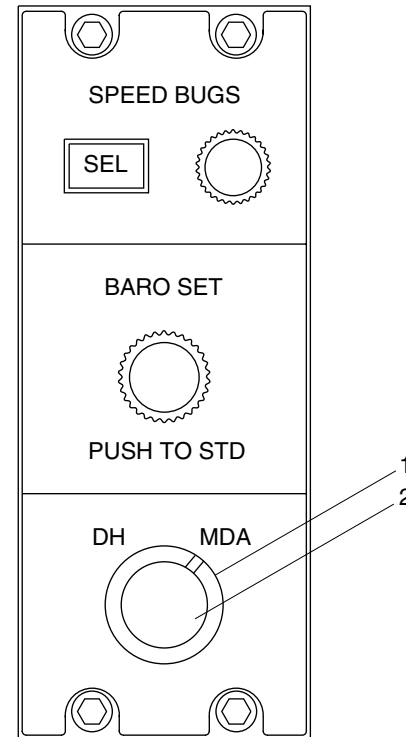
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. DH/MDA Selector Switch.
2. DH/MDA Knob.



A

far44a01.cgm

ICP MDA SELECTION
Figure 18

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

34-11-00

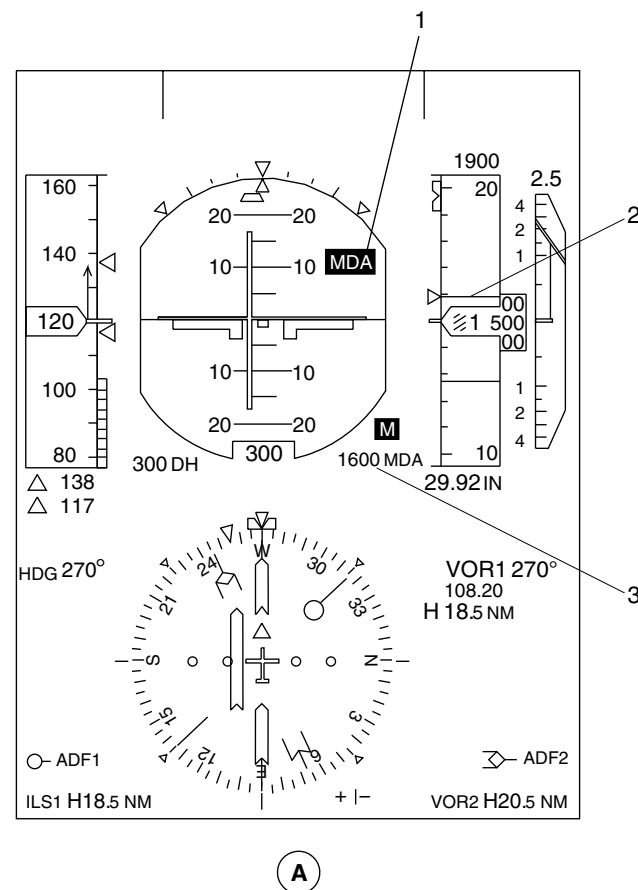
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A

MAIN INSTRUMENT PANEL

1. MDA Alert Annunciator.
2. MDA Index Bug.
3. MDA Digital Readout.



EIS MDA INDICATION
Figure 19

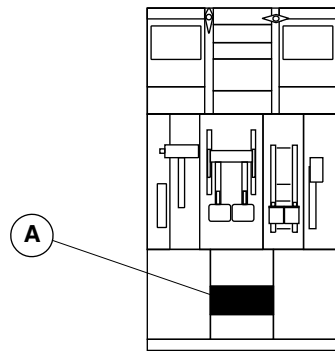
34-11-00

Print Date: 2025-04-22



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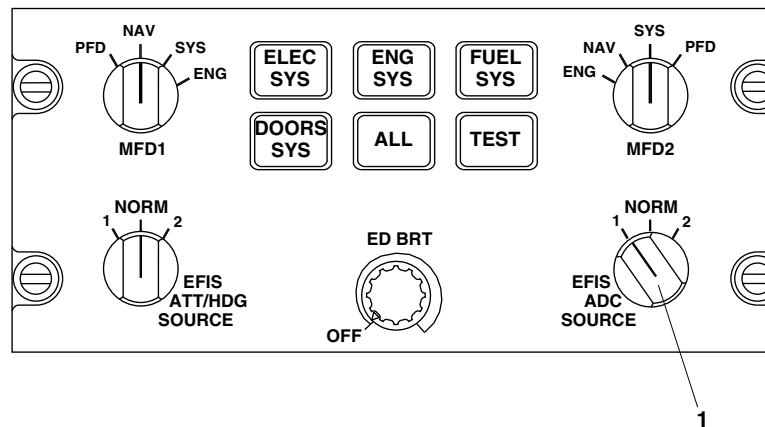
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

LEGEND

1. EFIS ADC Source Reversion Selector.



fsb04a01.cgm

ESCP EFIS ADC SOURCE REVERSION PAGE 2
Figure 20

PSM 1-84-2A
EFFECTIVITY:
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Config 001

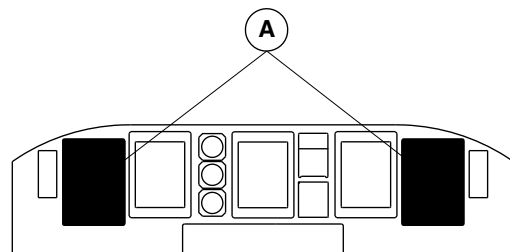
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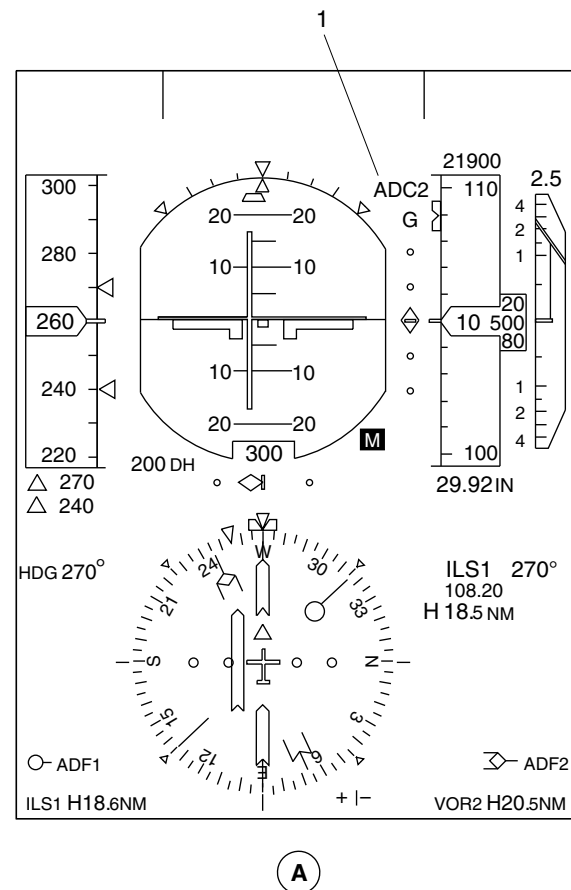
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Air Data Source Annunciator.



fab05a01.cgm

PFD EFIS ADC SOURCE REVERSION INDICATION
Figure 21

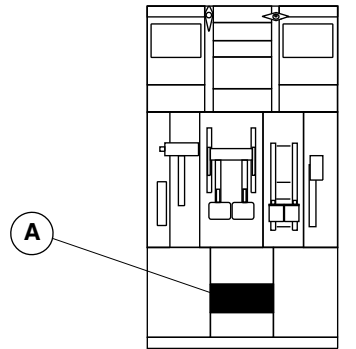
PSM 1-84-2A
EFFECTIVITY:
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Config 001

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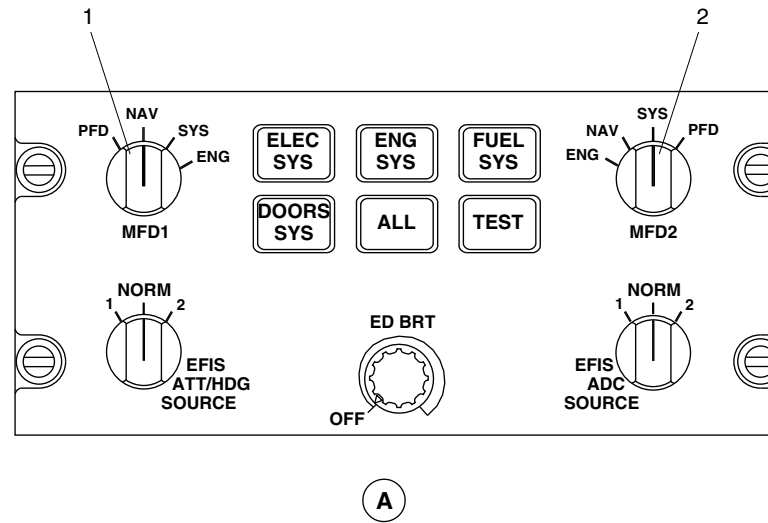
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

LEGEND

1. MFD1 Reversion Selector.
2. MFD2 Reversion Selector.



fs750a01.cgm

ESCP MFD REVERSION SELECTION
Figure 22

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

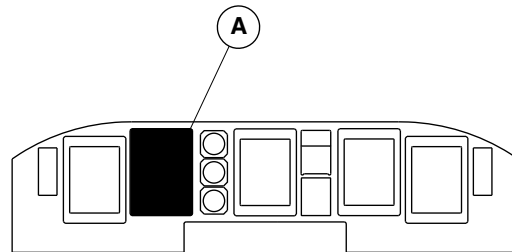
34-11-00

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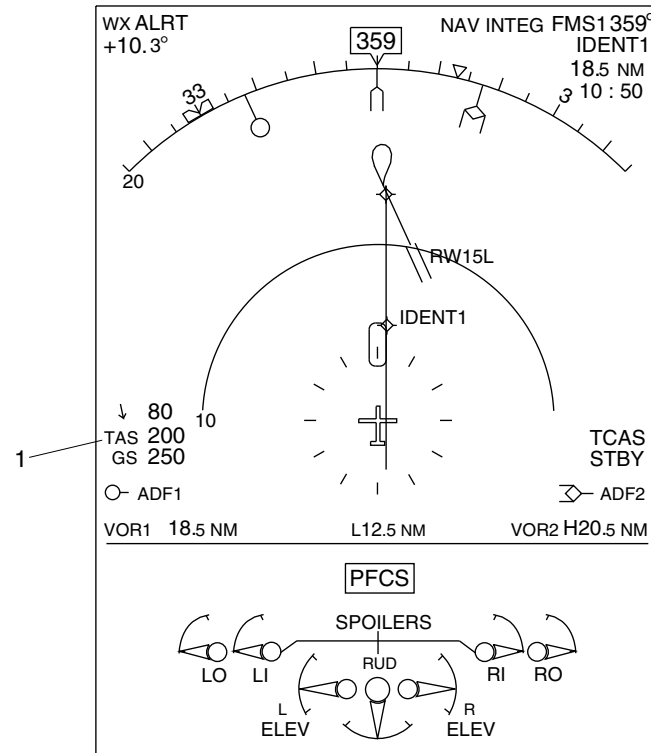
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. True Air Speed Digital Value.



A

fab07a01.cgm

PFD TRUE AIR SPEED (TAS) INDICATION
Figure 23

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

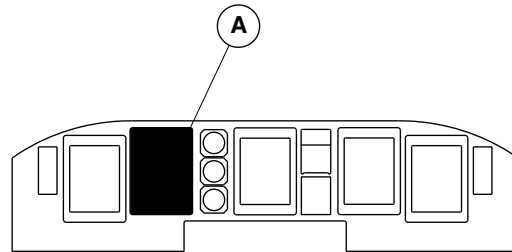
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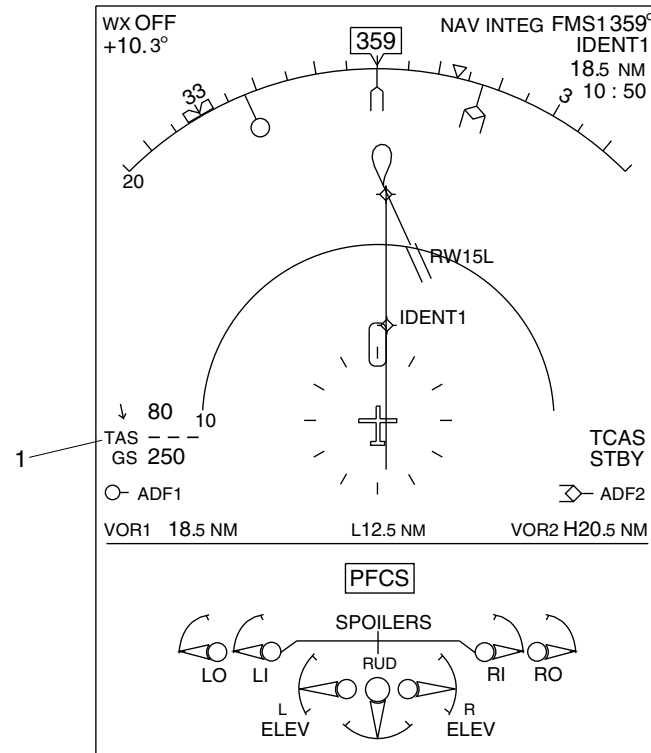
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. True Air Speed Fail Annunciation.



A

fab06a01.cgm

PFD TRUE AIR SPEED (TAS) MALFUNCTION
Figure 24

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

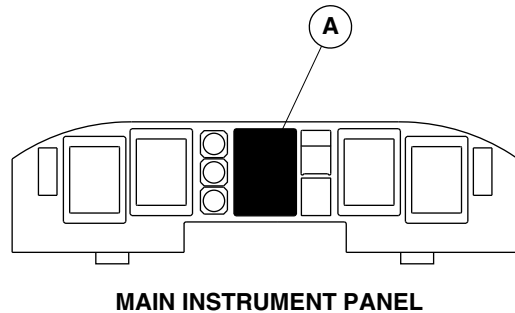
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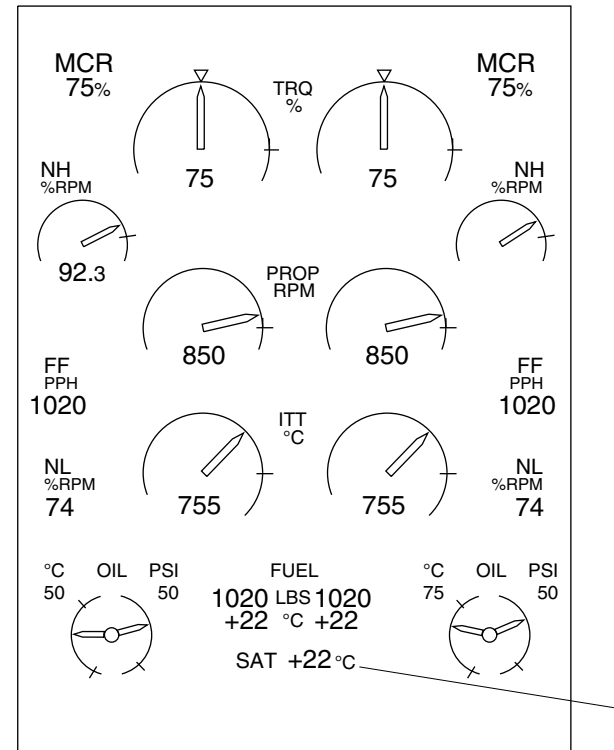
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Static Air Temperature Indication.



A

fsb02a01.cgm

ED STATIC AIR TEMPERATURE (SAT) INDICATION
Figure 25

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

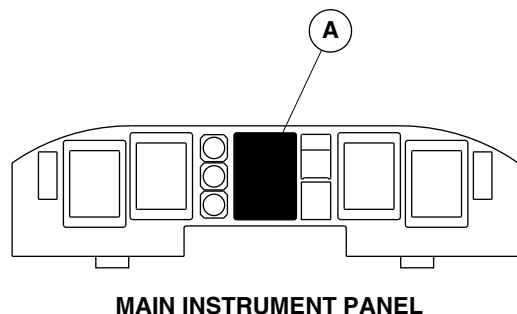
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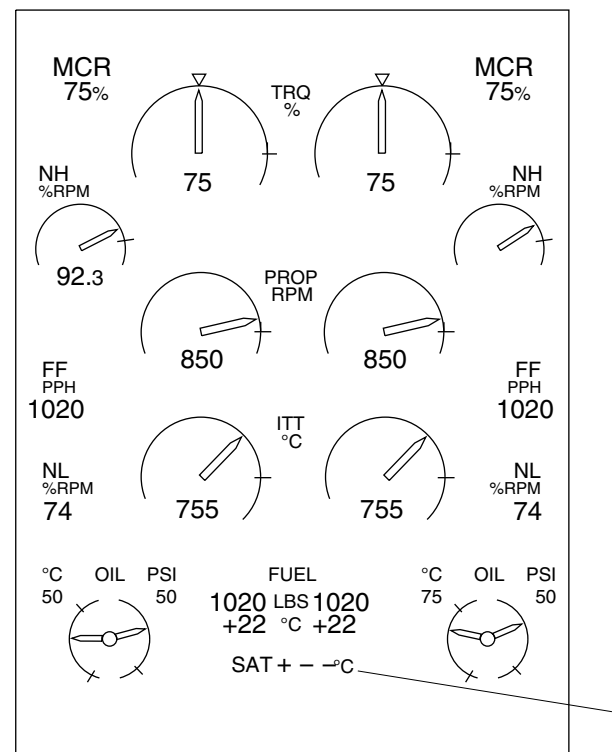
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Static Air Temperature Fail Annunciation.



A

fsb03a01.cgm

ED STATIC AIR TEMPERATURE (SAT) MALFUNCTION
Figure 26

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-11-00
Config 001

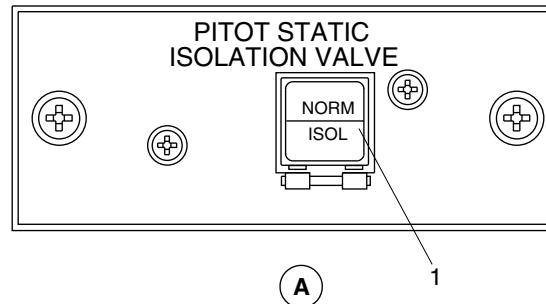
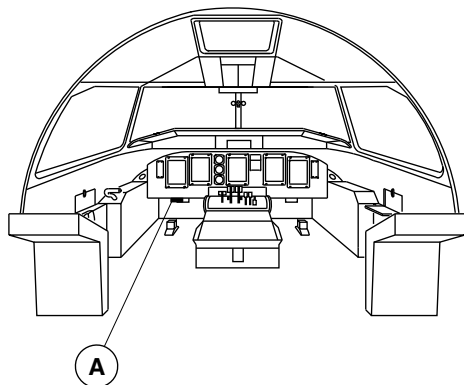
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Pitot Static Isolation Valve Pushbutton Switch.

fsr94a01.cgm

PITOT STATIC ISOLATION SELECTION
Figure 27

PSM 1-84-2A
EFFECTIVITY:
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Config 001

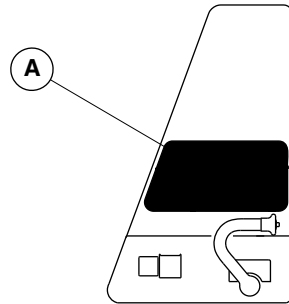
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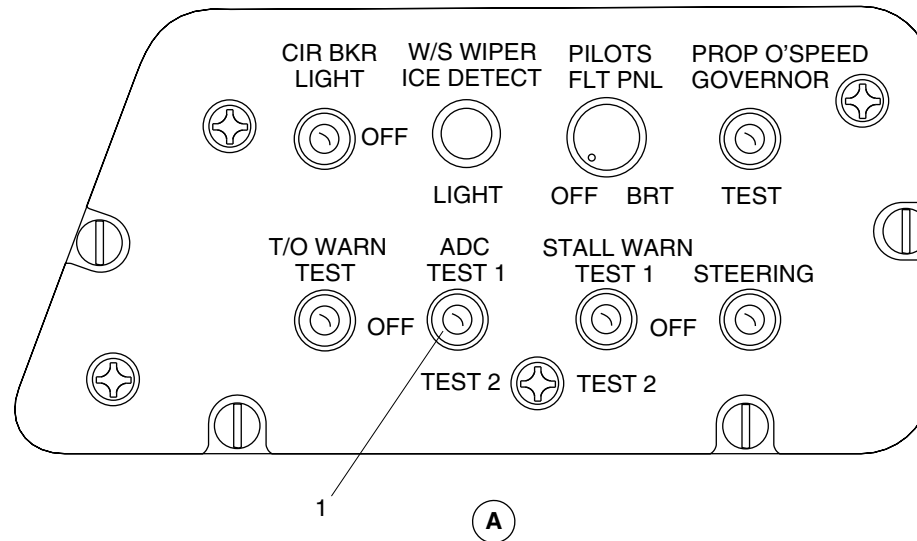
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEFT SIDE CONSOLE

LEGEND

1. Air Data Computer Test Toggle Switch.



fs981a01.cgm

ADC PRE-FLIGHT TEST SWITCH
Figure 28

PSM 1-84-2A
EFFECTIVITY:
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Config 001

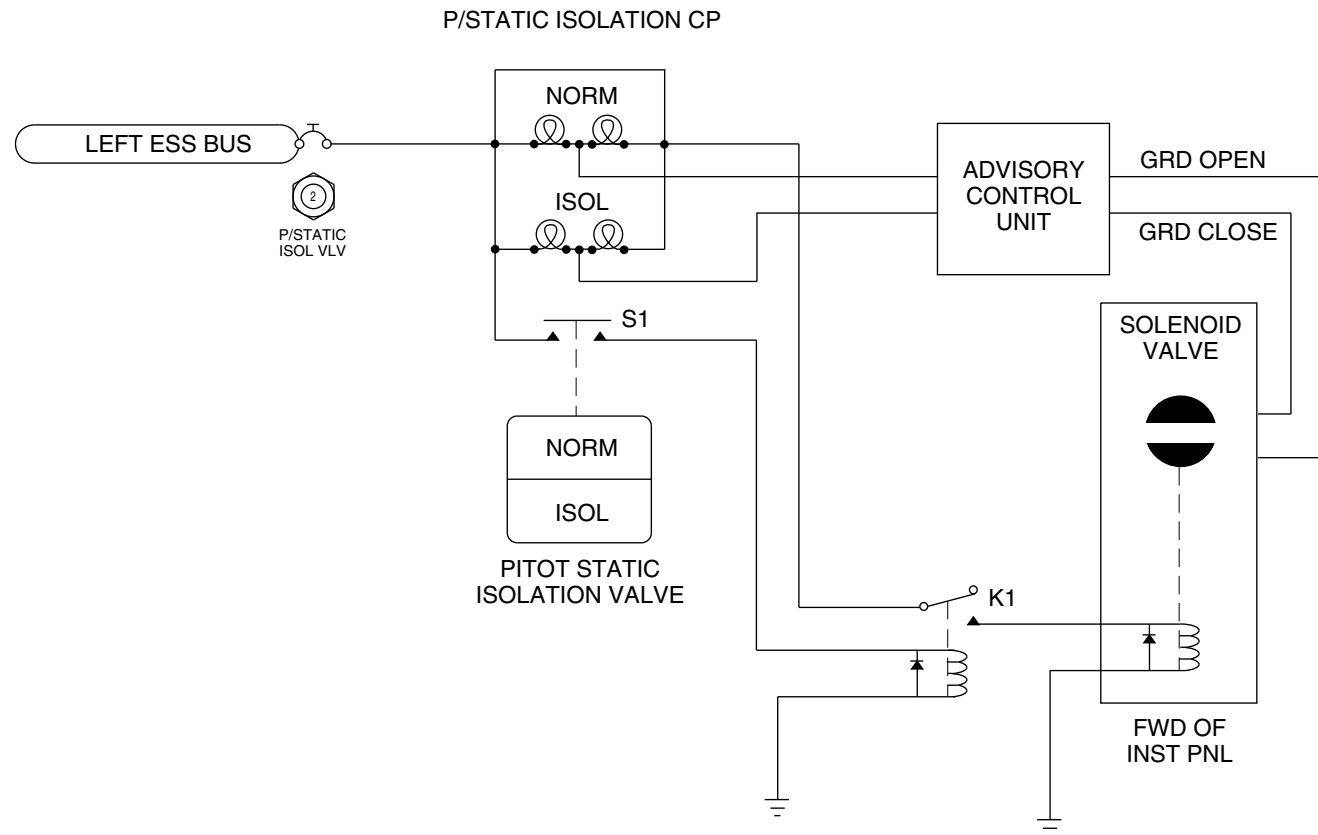
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fsu82a01.cgm

SOLENOID SHUTOFF VALVE SIMPLIFIED SCHEMATIC
Figure 29

PSM 1-84-2A
EFFECTIVITY:
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Config 001

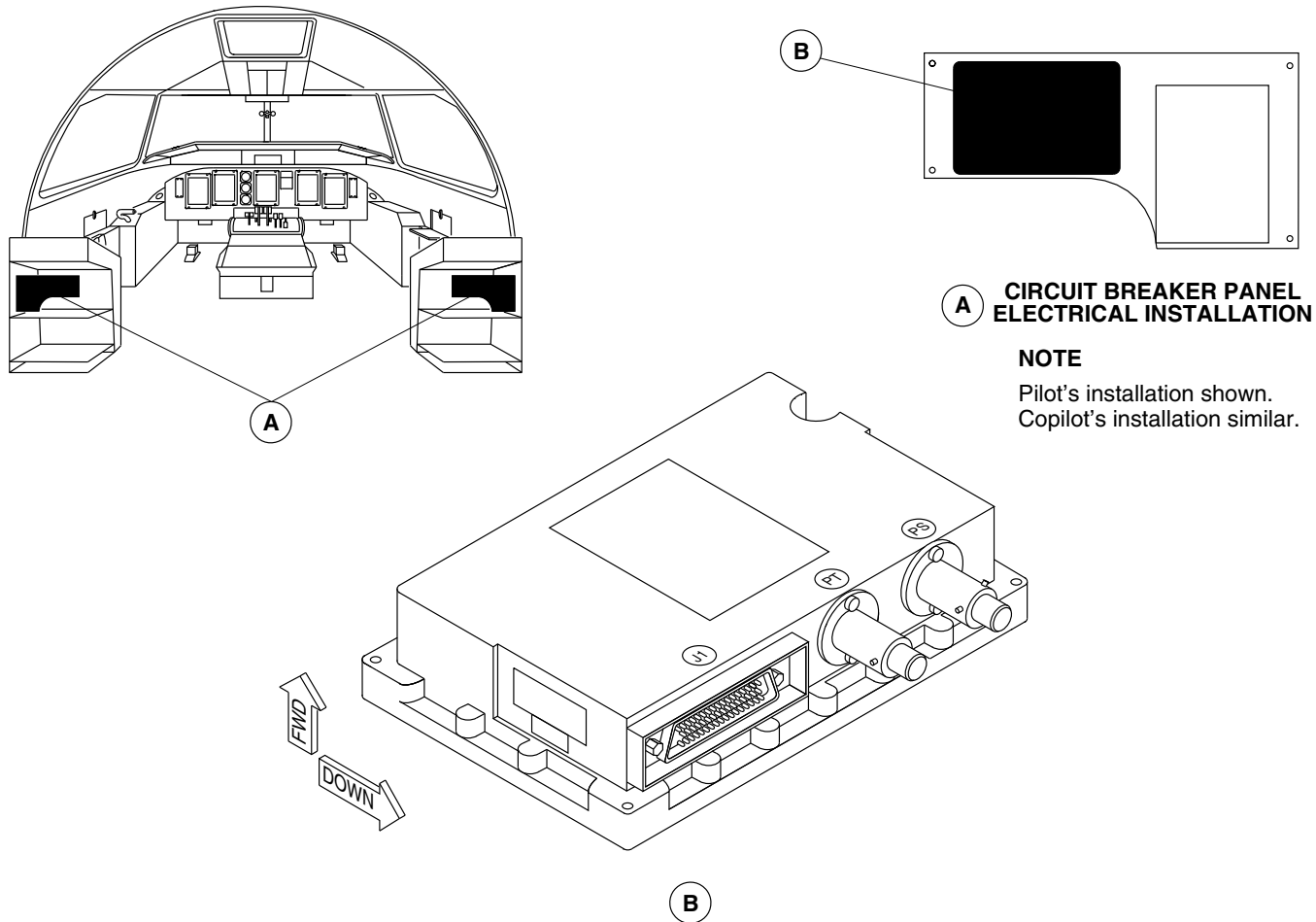
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fsa21a01.cgm

AIR DATA UNIT LOCATOR
Figure 30

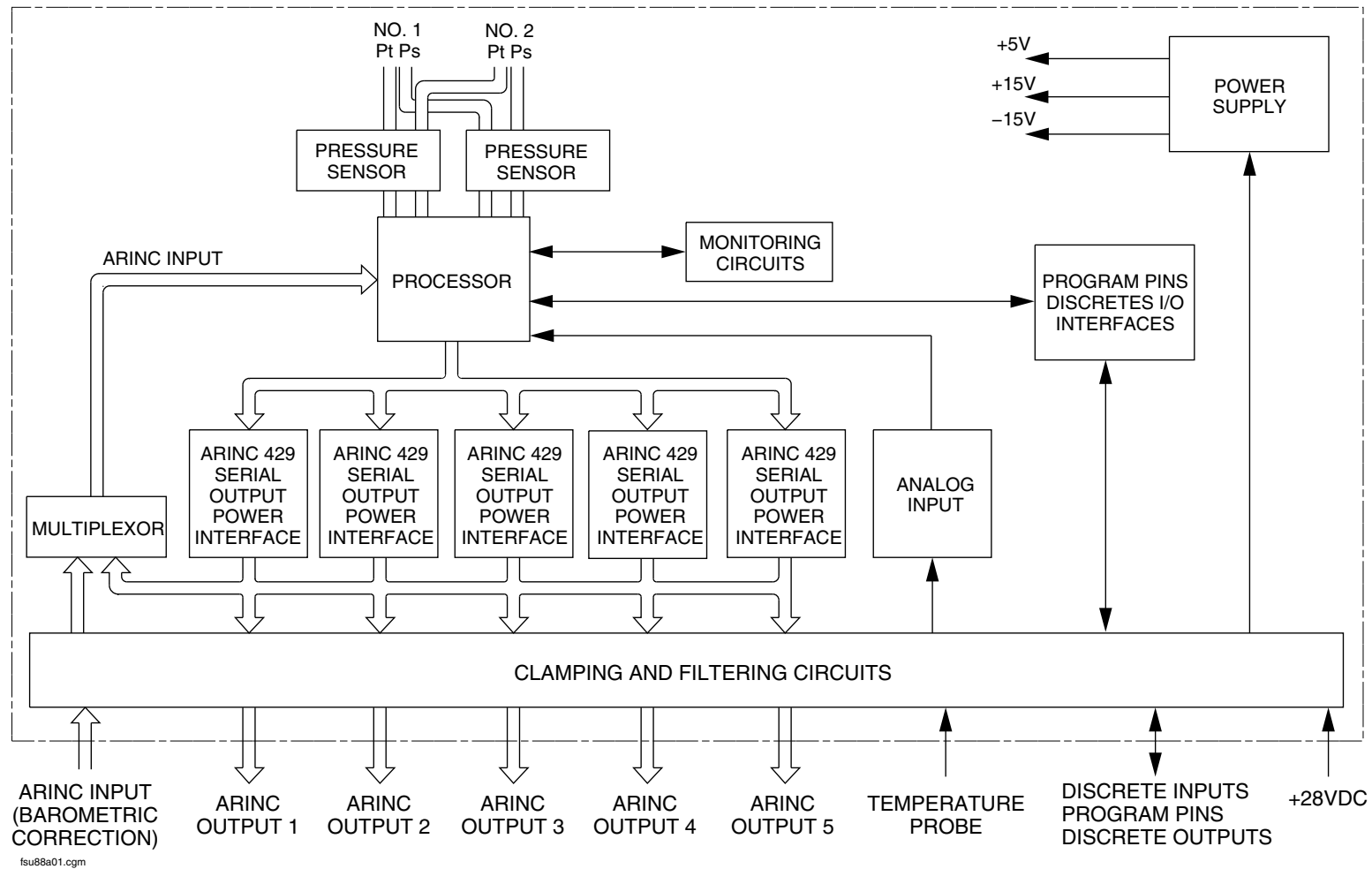
PSM 1-84-2A
EFFECTIVITY:
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Config 001

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



AIR DATA COMPUTER INTERNAL BLOCK DIAGRAM
Figure 31

PSM 1-84-2A
EFFECTIVITY:
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Config 001

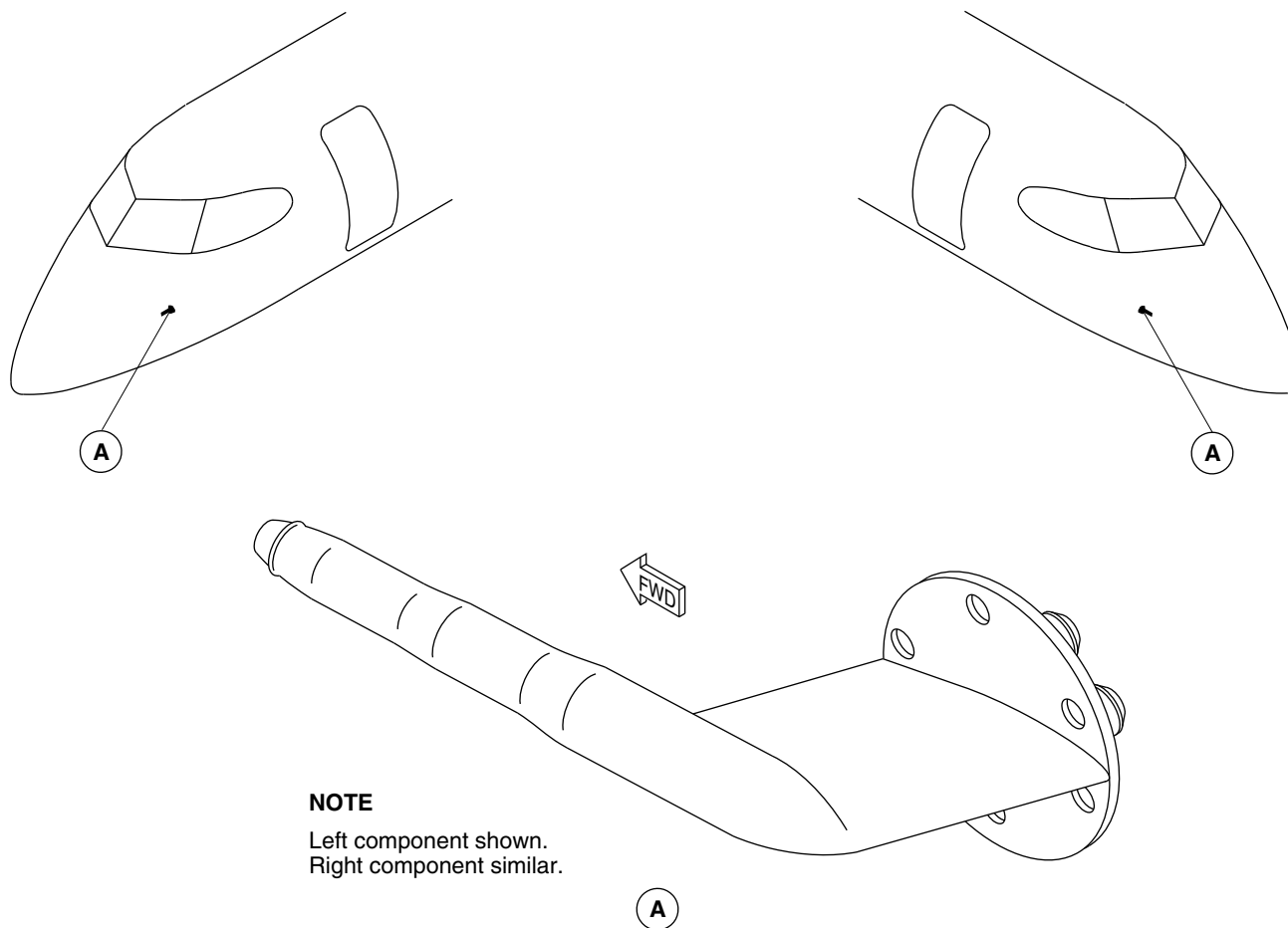
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

Left component shown.
Right component similar.

fs732a01.cgm

PITOT-STATIC PROBES LOCATOR
Figure 32

PSM 1-84-2A
EFFECTIVITY:
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Config 001

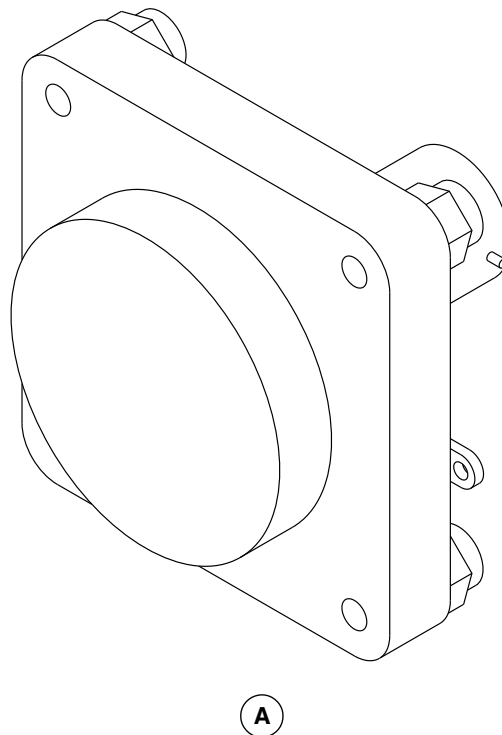
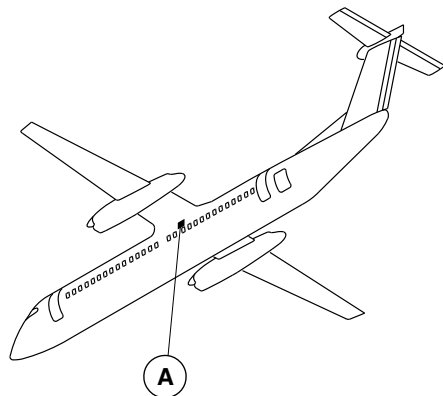
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fsa96a01.cgm

STATIC TEMPERATURE PROBE LOCATOR
Figure 33

PSM 1-84-2A
EFFECTIVITY:
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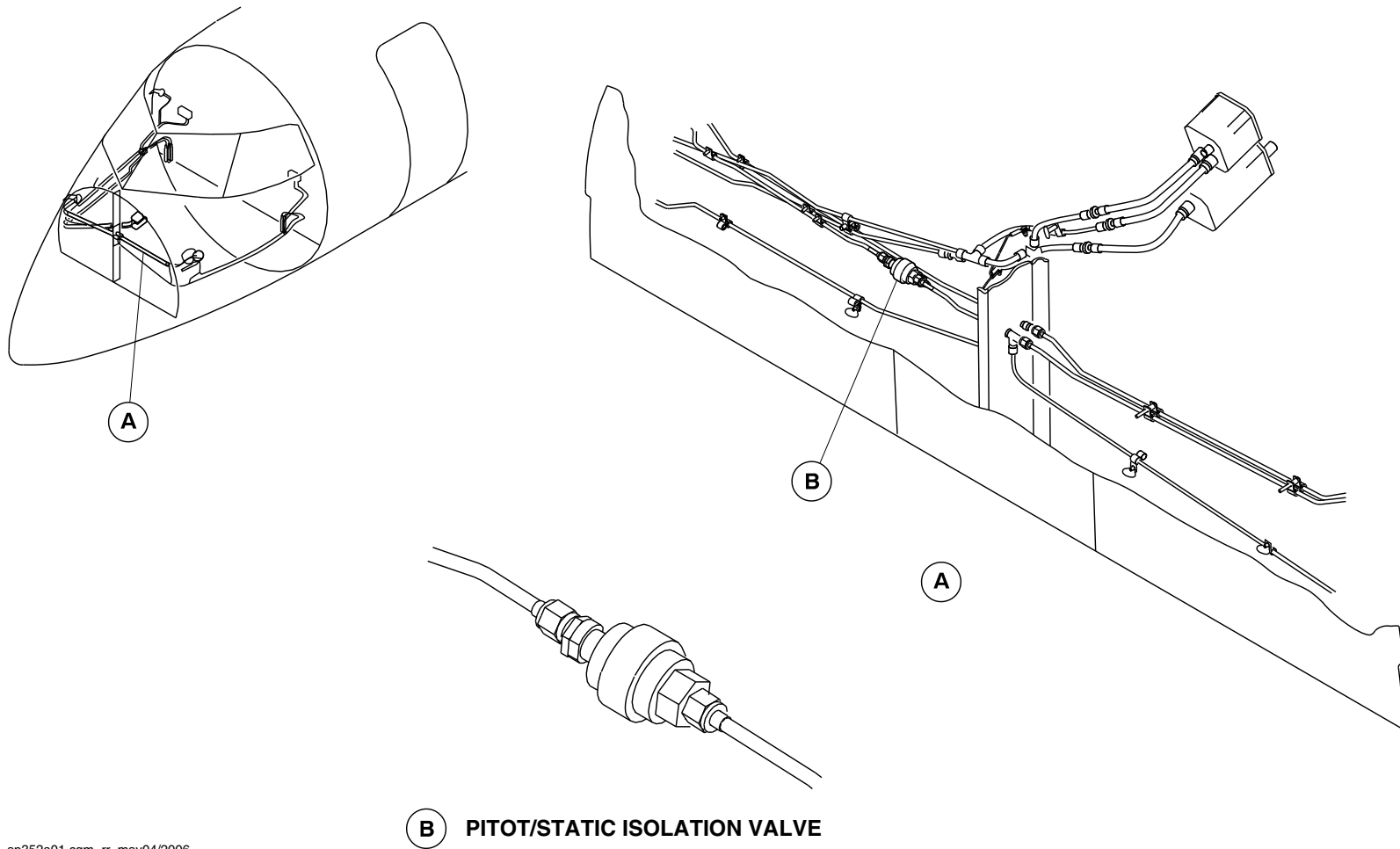
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ap352a01.cgm, rr, may04/2006

Pitot/Static Isolation Valve
Figure 34

PSM 1-84-2A
EFFECTIVITY:
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Config 001

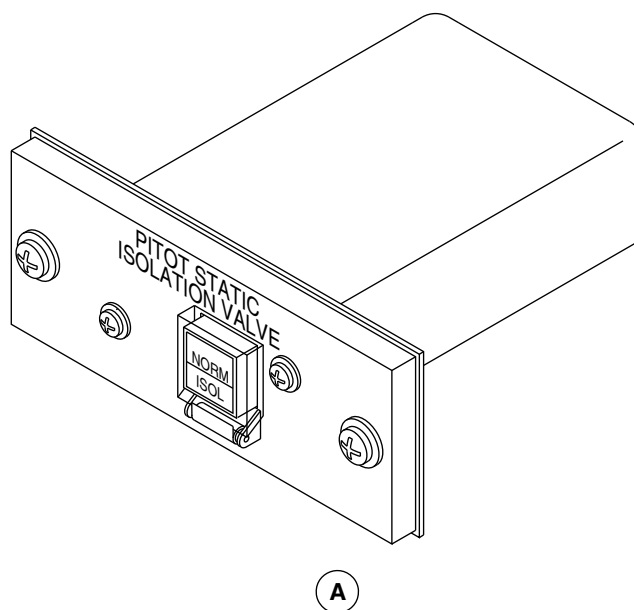
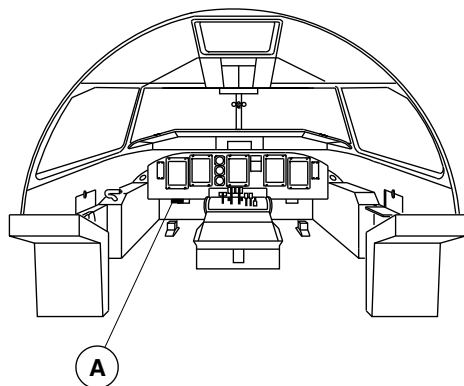
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fsr94a02.cgm

PITOT-STATIC ISOLATION VALVE PANEL LOCATOR
Figure 35

PSM 1-84-2A
EFFECTIVITY:
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Config 001

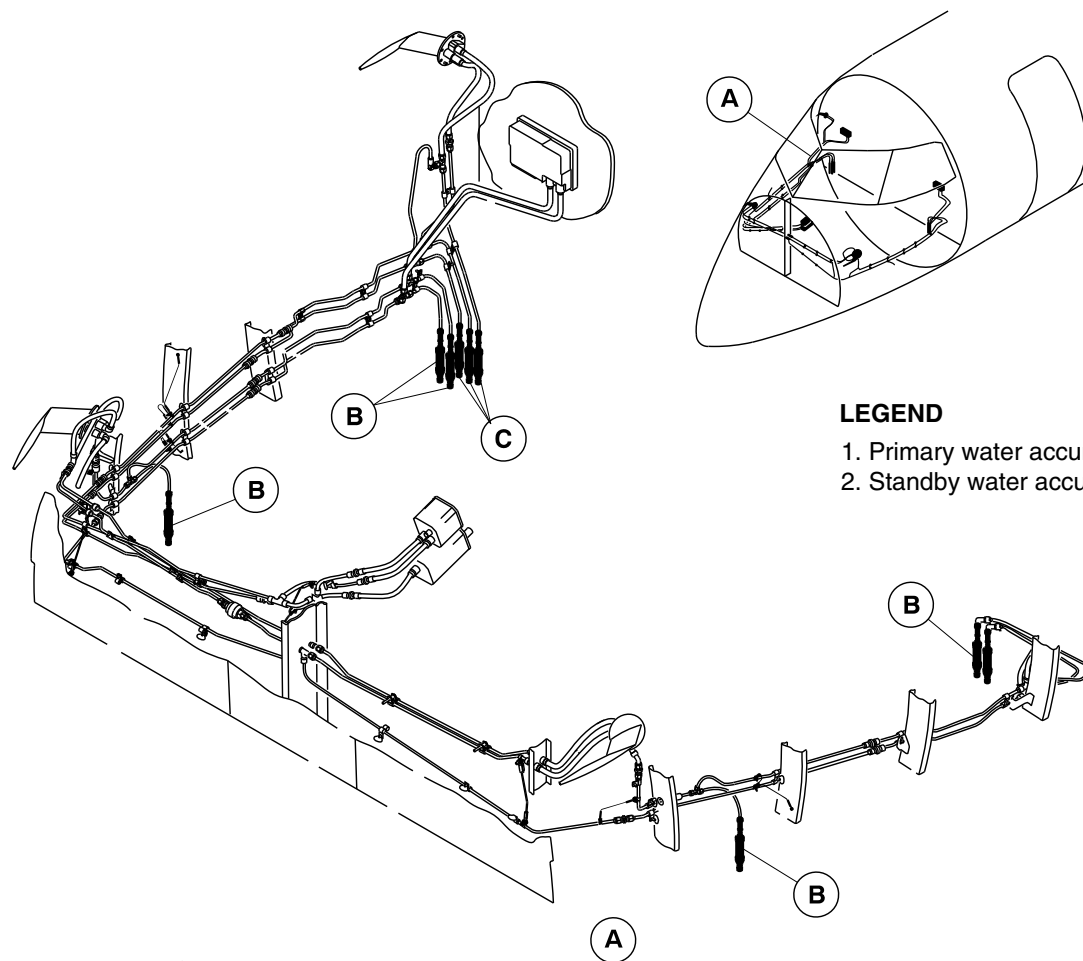
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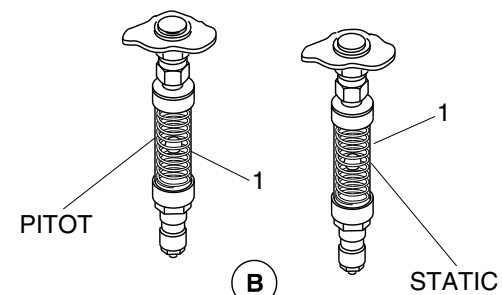
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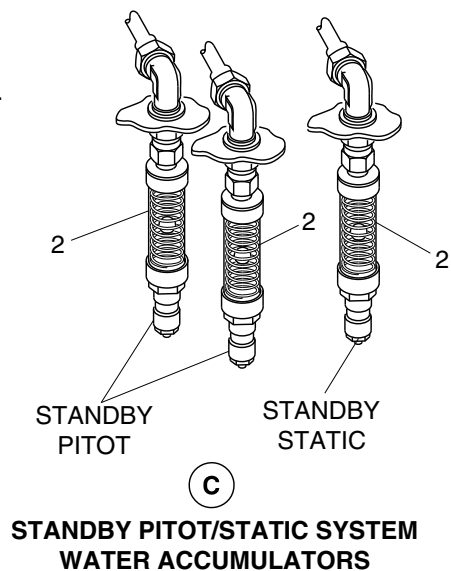


LEGEND

- 1. Primary water accumulator.
- 2. Standby water accumulator.



PRIMARY PITOT/STATIC SYSTEM WATER ACCUMULATORS



STANDBY PITOT/STATIC SYSTEM WATER ACCUMULATORS

Water Accumulators, Pitot/Static System
Figure 36

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

34-12-00-001

STANDBY AIR DATA INSTRUMENTS

Introduction

The standby air data system senses and changes static pressure and pitot (impact) pressure to give altitude and airspeed indication independently of the primary air data system.

General Description

Refer to Figure 1.

The standby air data instruments function independently of the primary channel Air Data Units (ADU1, ADU2) to show the data that follows:

- Altitude
- Air speed
- Barometric correction settings.

The standby air data system has the components that follow:

- Indicator, standby altitude (34-12-01)
- Indicator, airspeed-standby (34-12-06)
- Probe, standby pitot-static (34-12-11).

The standby pitot-static probe has one pitot pressure port and two static pressure port to supply air data to the standby air data instruments. The standby altitude indicator receives air data from one

static port and the standby airspeed indicator receives air data from the pitot and static ports.

Detailed Description

The standby altitude indicator and the standby airspeed indicator are mechanical devices.

The standby altitude indicator uses static air pressure to calculate altitude and the standby airspeed indicator uses the difference between pitot and static pressures to calculate airspeed.

The forward part of the probe is an open tube with a knife edge around a circular opening. The standby pitot-static probe has a location with minimal distorted airflow and is pointed directly into the airstream.

The standby altitude and airspeed indicators are connected to the standby pitot-static probe. It has an integral electrical heating element to prevent ice accumulation that causes irregular measurements. The electrical power to the heater is controlled by the Timer Monitor Unit (TMU) of ice and rain protection system.

The aircraft right essential busses supply 28 Vdc electrical power through a 1 ampere circuit breaker to a vibrator in the standby altitude indicator. The circuit breaker is located on the 28 Vdc right circuit breaker panel in position G2.

The aircraft right essential busses supply 28 Vdc electrical power through 25 ampere circuit breaker to the standby pitot-static probe. The circuit breaker is located on the 28 Vdc right circuit breaker panel in position L7.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Standby Altitude Indicator

[Refer to Figure 2.](#)

The standby altitude indicator has a high-precision aneroid capsule that expands or compresses with changes in atmospheric pressure. A multiplying gear train mechanism is used to change the capsule variations into angular movements of the altitude pointer. A vibrator in the indicator shakes the mechanism to decrease the friction and to prevent the needle pointer from sticking.

[Refer to Figure 3.](#)

The standby altitude indicator shows the altitude parameters that follow:

- Dial scale and needle pointer
- Digital readout
- Barometric set knob
- Barometric correction indication, millibars
- Barometric correction indication, inches of mercury
- Failure flag.

Dial Scale and Needle Pointer: The dial scale with a needle pointer gives analog indicated barometric corrected altitude. The scale has a circle with short markings shown every 20 ft and long markings every 100. The dial scale shows thousands of feet per turn with each 100 ft. mark shown with a digital value.

Digital Readout: The digital indicator is a T-shaped window inside the circular dial scale. It shows the aircraft's current altitude in 20 ft (6.1 m) increments from negative 1000 ft (304 m) up to 35,000 ft (30474 m) as a rolling drum indication. A black and a white hatched

symbol comes into view in the left part of the indication when the altitude is less than 10,000 ft (3048 m). It shows a NEG label when the barometric altitude is less than 0 ft.

Barometric Set Knob: A barometric set knob located in the lower left part of the standby altitude indicator is turned to set the barometric correction.

Barometric Correction Indication: The standby altitude indicator shows the barometric set knob selection. The left part of the indication shows the barometric selection from 0749 to 1051 millibars and the right part shows the same value in inches of mercury from 22.01 to 31.03.

Failure Flag: When the vibrator malfunctions, an orange flag comes into view.

The standby altitude indicator weighs 2.2 lb (1 kg). It is located in the instrument panel between the pilot's Multi-Function Display (MFD1) and the Engine Display (ED) below the standby airspeed indicator in view of the two pilots.

Standby Airspeed Indicator

[Refer to Figure 4.](#)

The standby airspeed indicator has a high-precision aneroid capsule that expands or compresses with changes in pitot and static pressure. A multiplying gear train mechanism is used to change the capsule variations into angular movements of the airspeed pointer.

[Refer to Figure 5.](#)

The standby airspeed indicator shows the altitude parameters that follow:

- Dial scale and needle pointer



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- Yellow speed bug
- White speed bugs.

Dial Scale and Needle Pointer: The dial scale with a needle pointer gives analog indicated airspeed. The dial scale has a speed range from 40 to 350 kt. with marks at 5 kt intervals. The 40, 60, 80, 100, 120, 140, 160, 180, 200, 250, 300, and 350 marks are shown with digital values.

Yellow Speed Bug: A speed bug rotary selector located in the lower right part of the standby airspeed indicator is turned to set the speed bug value. The speed bug is a yellow triangular indicator. It moves along the dial scale from minimum 40 kt value to 350 kt.

White Speed Bugs: Two removable white bugs slide along the edge of the indicator's bezel. Their position is manually adjustable.

The standby altitude indicator weighs 1.1 lb (0.5 kg). It is located in the instrument panel between the pilot's Multi-Function Display (MFD1) and the Engine Display (ED) above the standby altitude indicator in view of the two pilots.

Probe, Standby Pitot-Static

[Refer to Figure 6.](#)

The fuselage-mounted standby pitot-static probe accurately senses pitot and static pressures over the complete operating range of the aircraft.

An opening located at the nose of the probe senses pitot pressure. The shape of the opening is designed to reduce the sensitivity due to inclination of the tube axis to the effected varying airflow angles. This results in a total pressure measurement constant through the entire

Angle Of Attack (AOA) and sideslip envelope. Static pressure ports are located on the side of the probe tube.

The probe is a swept low drag mast and a specially contoured tube that are brazed together. The special shape of the tube gives Static Source Error Compensation (SSEC) resulting from the specific probe location on the aircraft fuselage and from specific features of the airflow at this location.

The standby pitot-static probe has one total pressure connector and two static pressure connectors for pneumatic connections to the standby altimeter and airspeed indicators.

To prevent ice accumulation on pressure ports, probes have a single coaxial-type electrical heating element with a self-regulating power consumption feature.

Moisture accumulation in the tube is also eliminated by the use of drain holes for the total pressure circuit. The static pressure circuit is self-drained due to the locations of the static ports on the tube.

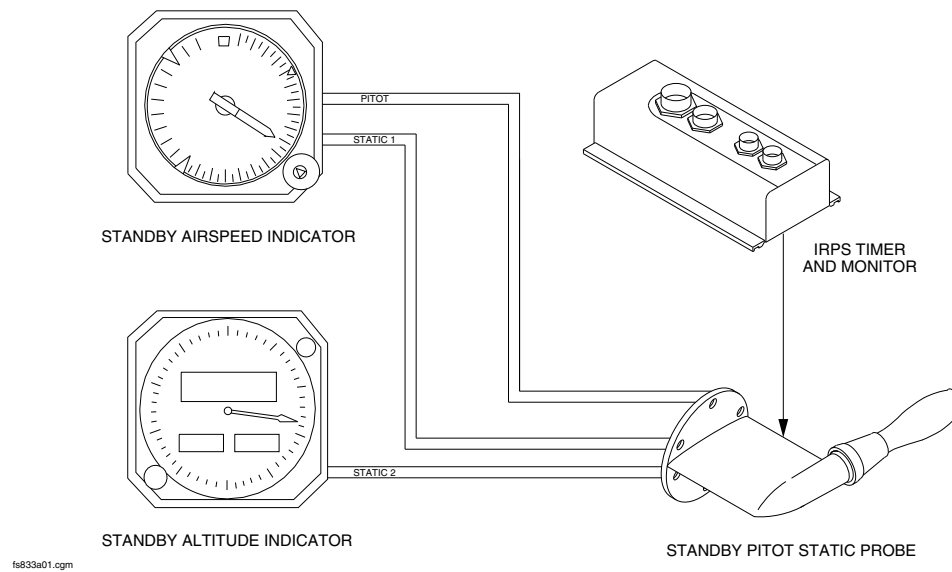
The standby pitot/static probe is installed on the right side of the forward fuselage section.

The base plates have electrical and pressure connectors. Locating pins on the plate give an accurate angular orientation of the probe with the aircraft axis.



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STANDBY AIR DATA INSTRUMENTS BLOCK DIAGRAM
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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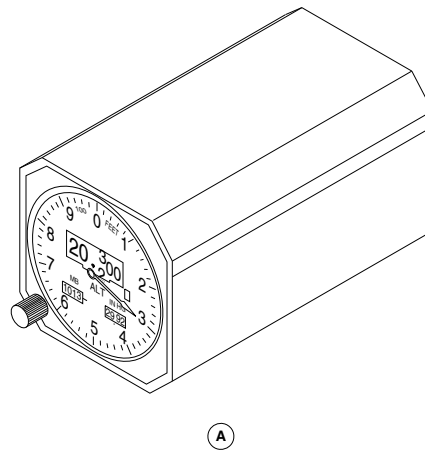
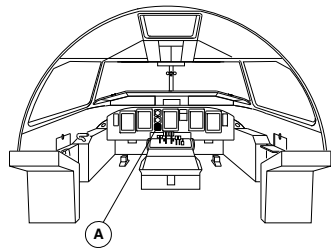
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ts514a01.cgm

STANDBY ALTITUDE INDICATOR LOCATION
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

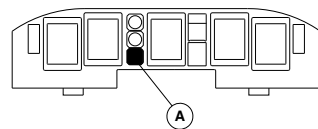
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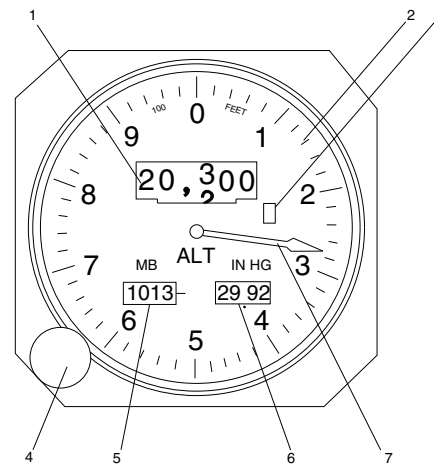
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Digital Readout.
2. Dial Scale.
3. Failure Flag.
4. Barometric Set Knob.
5. Barometric Correction Indication, Millibars.
6. Barometric Correction Indication, Inches of Mercury.
7. Needle Pointer.



ts653a01.cgm

STANDBY ALTITUDE INDICATOR, DETAIL
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

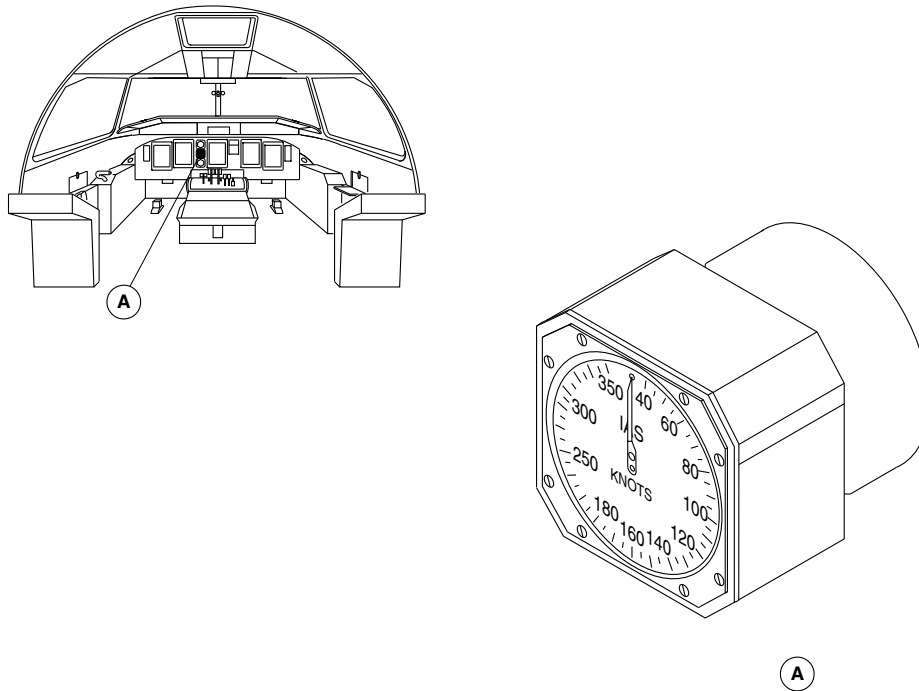
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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STANDBY AIRSPEED INDICATOR, LOCATOR
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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Config 001

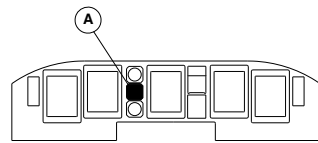
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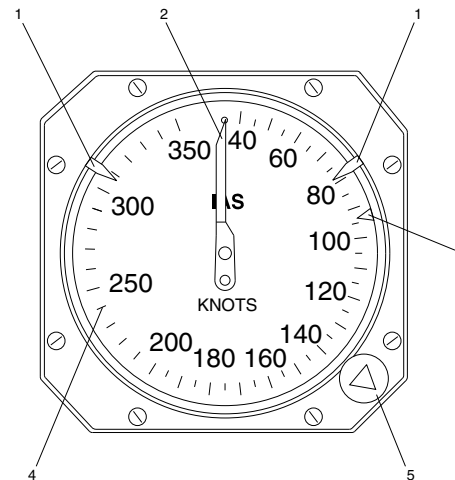
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. White Speed Bug.
2. Needle Pointer.
3. Yellow Speed Index Bug.
4. Dial Scale.
5. Speed Bug Rotary Selector.



fsb46a01.cgm

STANDBY AIRSPEED INDICATOR, DETAIL
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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Config 001

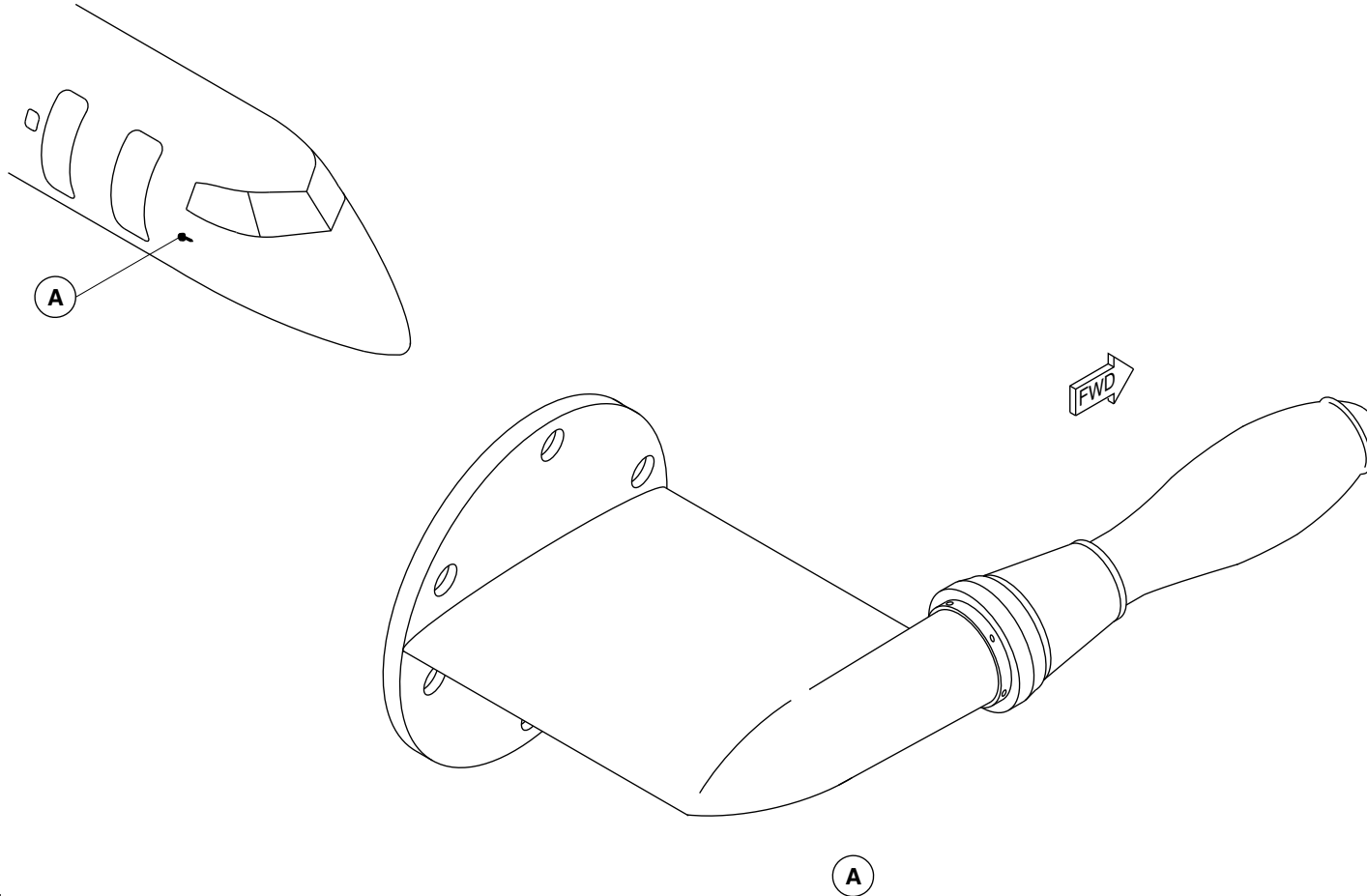
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STANDBY PITOT-STATIC PROBE, LOCATOR
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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INTEGRATED STANDBY INSTRUMENT (ISI)

Introduction

The Integrated Standby Instrument (ISI) replaces the current mechanical standby instruments and performs their function within a single electronic unit. The ISI system provides the following information to the pilots:

- Attitude
- Altitude
- Airspeed

The ISI system operates independently and does not interface with any other systems.

General Description

[Refer to Figure 1.](#)

The ISI is located on the instrument panel between Multi-Function Display 1 (MFD1) and the Engine Display (ED). It replaces the following mechanical standby instruments:

- Standby Attitude Direction Indicator
- Standby Altimeter
- Standby Airspeed Indicator

The ISI displays all three standby instruments on a single High Resolution Active Matrix Liquid Crystal Display (LCD) using Vertical Tape Symbolology for the Indicated Airspeed (IAS) and the Altimeter. It also provides standby display for:

- Vmo (Maximum Operating Speed)
- Side-slip

Detailed Description

[Refer to Figure 2.](#)

As a standby instrument, the external requirements are limited to:

- Standby Pitot Probe
- Standby Static Probe
- Battery Bus
- Bezel Lighting

The ISI is connected to the standby pitot-static probe. The probe is in a location with minimal distorted airflow and is pointed directly into the airstream. It has an integral electrical heating element to prevent ice accumulation. The electrical power to the heater is controlled by the Timer Monitor Unit (TMU) of ice and rain protection system.

The ISI utilizes internal sensors and outside static air pressure to calculate altitude and airspeed. An Inertial Measurement Unit computes rotations and linear accelerations along the aircraft axes for the attitude function.

The aircraft right essential bus supplies 28 VDC electrical power through a 25 ampere circuit breaker to power the standby pitot-static probe. The circuit breaker is located on the 28 VDC right circuit breaker panel in position L7.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The aircraft left battery bus supplies 28 VDC electrical power through a 3 ampere circuit breaker to power the ISI. The circuit breaker is located on the 28 VDC left circuit breaker panel in position H1.

The bezel lighting (5 VDC) is embedded in the front face structure (pushbuttons) of the ISI and is controlled from the pilot's side console.

The Integrated Standby Instrument (ISI) operational features are located on the front face of the instrument and consist of:

- Control Panel Push Buttons
- Standby Attitude Display
- Standby Airspeed Display
- Standby Altitude Display

Control Panel

[Refer to Figure 3.](#)

The ISI front face is comprised of four illuminated pushbuttons and one rotary knob surrounding the LCD. The adjustments include:

- STD Pushbutton
- “+” Pushbutton
- “-” Pushbutton
- CAGE Pushbutton
- BARO Rotary Knob

The “STD” (Standard) pushbutton is used to reset barometric correction to the standard value.

The “+” pushbutton manually increases LCD brightness.

The “-” pushbutton manually decreases LCD brightness.

The “BARO” (Barometric) rotary knob enables selection of baro correction in 'mb' (millibars) or 'in Hg' (inches of mercury) and is used to control the barometric correction scale. A clockwise rotation increases the barometric value while a counterclockwise rotation decreases it.

The “CAGE” pushbutton resets the horizontal function to zero when depressed for more than two seconds. It will also cause a CAGE warning flag to appear in the top right of display. It is used to cancel errors induced in the attitude display following unusual aircraft maneuvering or an upset.

Standby Attitude Display

[Refer to Figure 4.](#)

The standby attitude function supplies an alternate display of airplane attitude utilizing three internal quartz gyros sensors and two silicon accelerometers. It is powered by the battery bus.

The Standby Attitude Display shows the attitude information that follows:

- Airplane Symbol (white)
- Index Pointer and Roll Scale (white)
- Side-slip Indicator
- Pitch scale (white)
- Attitude Sphere and Horizon Line
- ATT Flag



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The airplane symbol is a fixed airplane reference against the attitude sphere and shows the amount of airplane pitch and roll. An optional single cue airplane symbol replaces the standard airplane symbol.

The amount of roll is determined by an incremental roll scale (0°, 10°, 20°, 30°, 45° and 60°) and moves relative to a fixed index pointer.

The side-slip indicator shows lateral acceleration left or right to a maximum of 0.14g.

The pitch scale is shown on the attitude sphere and is incremented (0°, 5°, 10°, 15°, 20°, 25°, 30°, 40°, 60° and 90°). In the event of excessive pitch angles (greater than 40° pitch up or 30° pitch down), red chevrons will come into view and point to horizon.

The attitude sphere has a sky permanent sector in blue above a ground permanent sector in brown. A white horizon line divides these two sectors. The sphere moves to show pitch and roll.

An 'ATT' (Attitude) flag indicates an invalid display due to failure of the attitude function as detected by internal monitors.

Standby Airspeed Display

Refer to Figure 5.

The standby airspeed function supplies an alternate display of aircraft indicated airspeed in vertical tape format. It receives data from static and pitot pressure sensors utilizing static and pitot pressure supplied from the Standby Pitot Static Probe. Data is converted into digital format allowing the ISI to compute the corresponding conventional indicated airspeed. The STBY system is independent of the Air Data Computer (ADC) primary system.

The Standby Airspeed Display shows the airspeed information that follows:

- Vmo (red limit)
- Airspeed Pointer and Index (white)
- Airspeed Scale (white)
- IAS Failure Flag

The Vmo (Maximum Operating Speed) is displayed as a red tape in the left airspeed display area.

The airspeed pointer and index indicates airspeed. The airspeed ranges from 40 knots to 520 knots.

The airspeed scale is graduated every 5 knots between 40 knots and 250 knots. Above 250 knots, the scale is graduated every 10 knots up to 520 knots.

The IAS (Indicated Air Speed) failure flag replaces the airspeed tape and pointer in the event of an airspeed function failure as detected by internal monitors.

Standby Altitude Display

Refer to Figure 6.

The standby barometric altimeter function supplies an alternate display of barometrical corrected altitude in vertical tape format. The ISI standby altimeter function receives data from the static pressure sensor utilizing static pressure supplied from the Standby Pitot static Probe. Data is converted into digital format allowing the ISI to compute the static pressure.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The Standby Altitude Display shows the altitude information that follows:

- Barometric Correction (hPa)
- Altitude Scale (white)
- Altitude Counter (white)
- Baro Correction (in Hg)
- ALT Flag
- SSEC Flag

The barometric correction displayed in the upper right quadrant of the LCD is set by the BARO rotary knob. The barometric correction ranges from 740 hPa to 1100 hPa.

The altitude scale is located on the right side of the display. It is comprised of a moving tape incremented every 100 feet and identified every 500 feet. The altitude tape range is –2000 feet to +50,000 feet .

The altitude counter appears as a digital readout right of center. It displays current altitude digitally in 20 foot increments. White hatch marks are displayed below 10,000 feet. An 'N' is displayed in place of the ten thousands digit for negative altitudes.

The barometric correction displayed in the lower right quadrant of the LCD is set by the BARO rotary knob. The barometric correction ranges from 21.85 in Hg to 32.48 in Hg. An optional ALT/METRIC switch on the overhead panel allows the barometric correction readout to be displayed in either meters or feet.

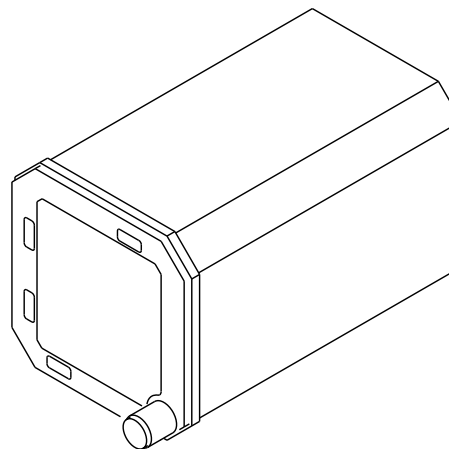
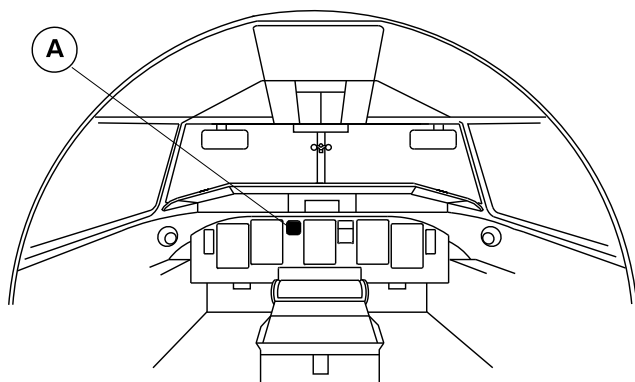
An 'ALT' (Altitude) flag replaces the altitude tape in event of failure of the altitude function as detected by internal monitors.

An 'SSEC' (Static Source Error Correction) flag is displayed when SSEC corrections are no longer available for altitude computation.



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Integrated Standby Instrument (ISI)
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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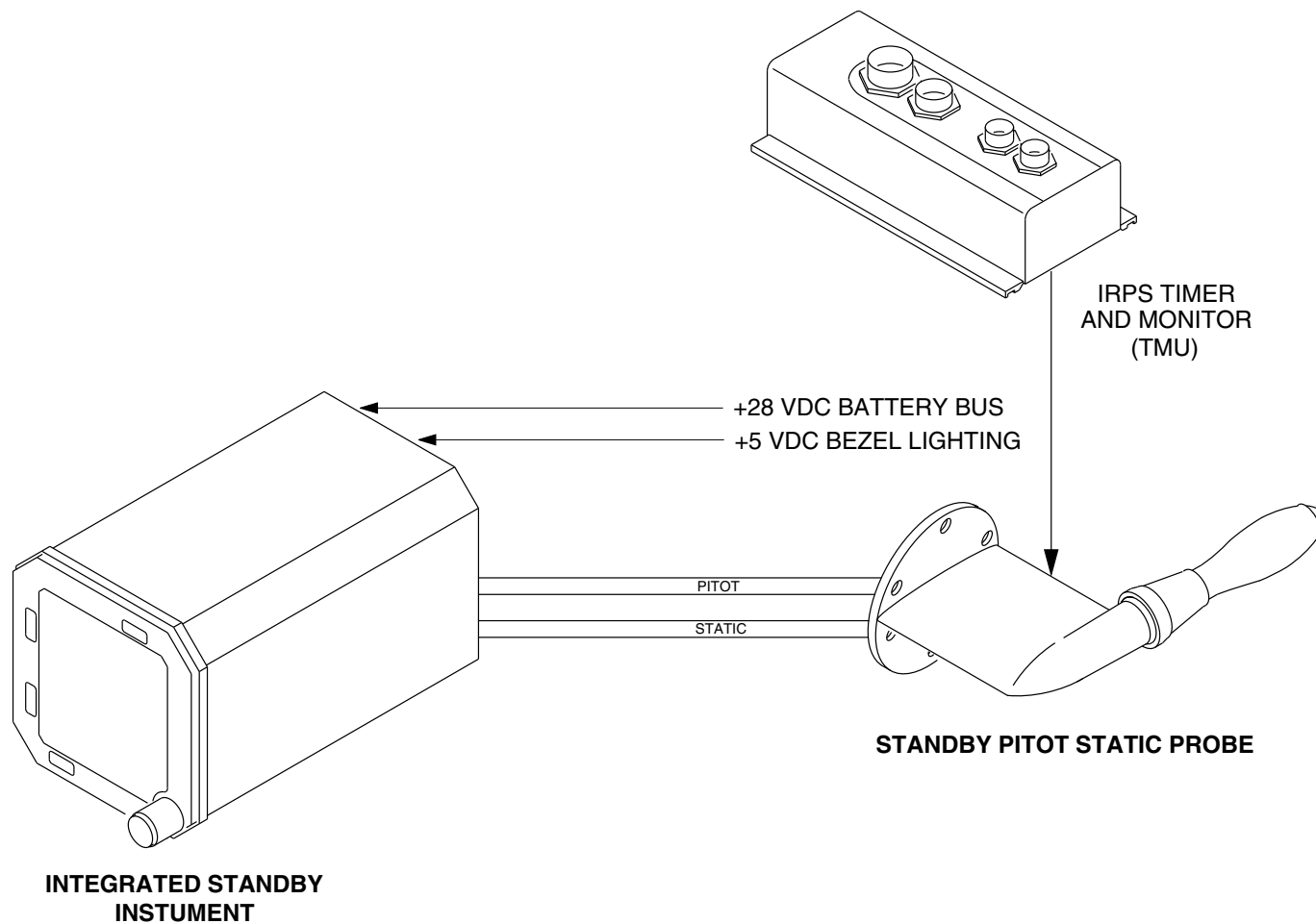
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



brar33a01.dg, gw, 15/04/03

Integrated Standby Instrument (ISI) Block Diagram
Figure 2

PSM 1-84-2A
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Config 001

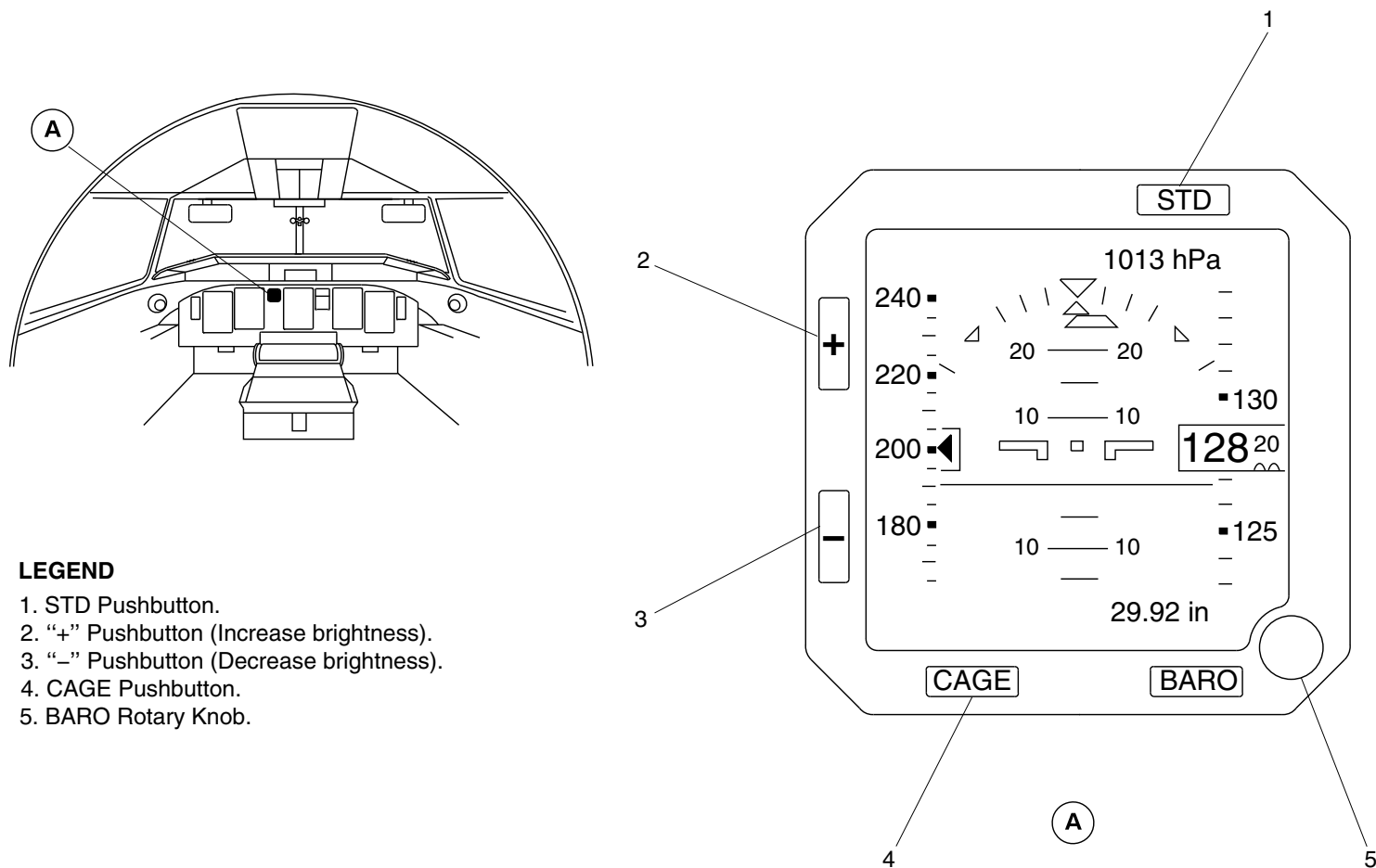
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. STD Pushbutton.
2. "+" Pushbutton (Increase brightness).
3. "-" Pushbutton (Decrease brightness).
4. CAGE Pushbutton.
5. BARO Rotary Knob.

brar34a01.dg, gw, 15/05/03

Integrated Standby Instrument (ISI) Control Panel
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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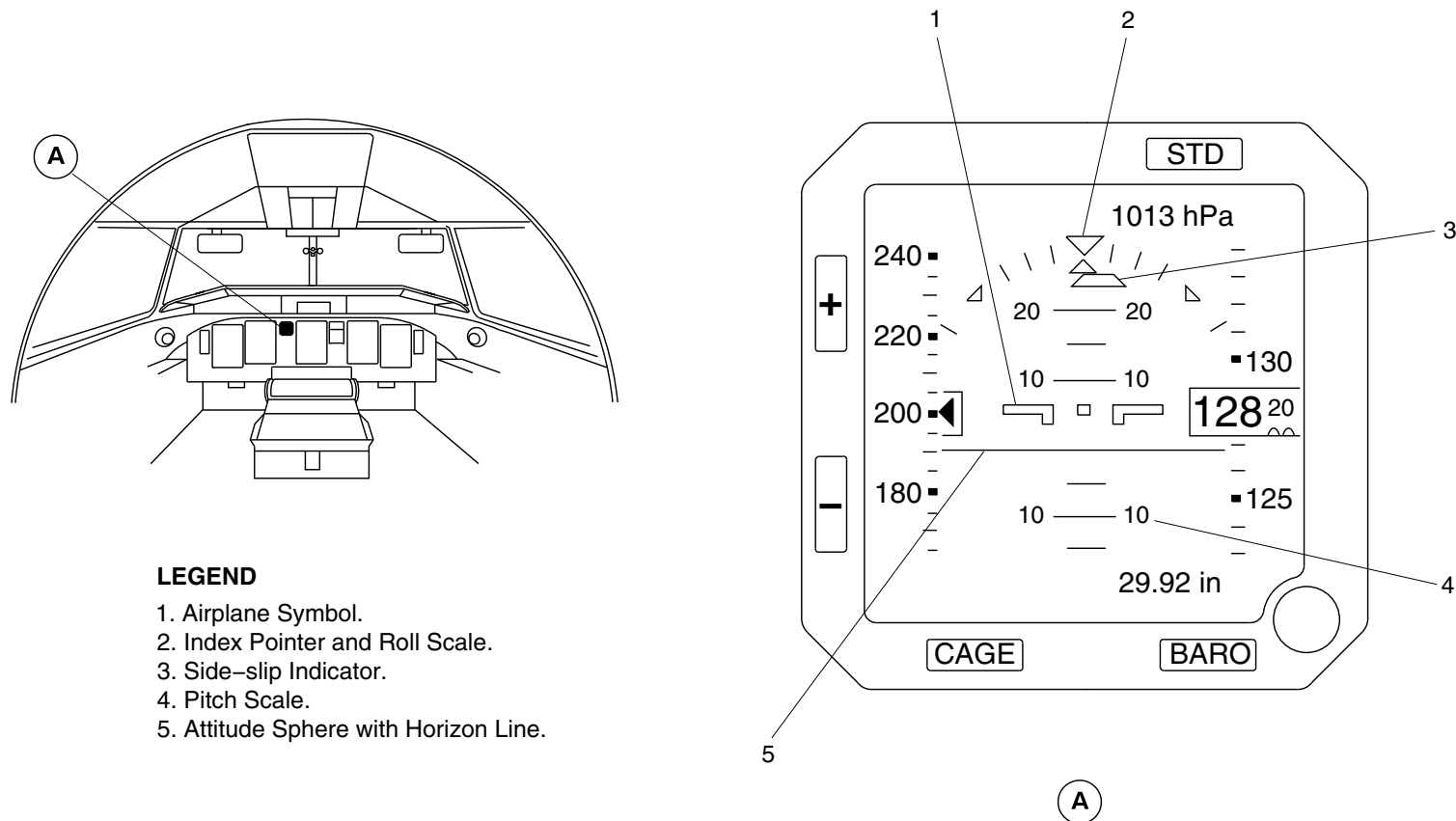
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Integrated Standby Instrument (ISI) Attitude Display
Figure 4

PSM 1-84-2A
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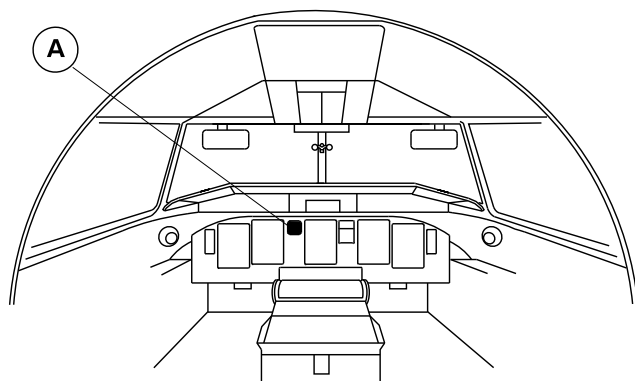
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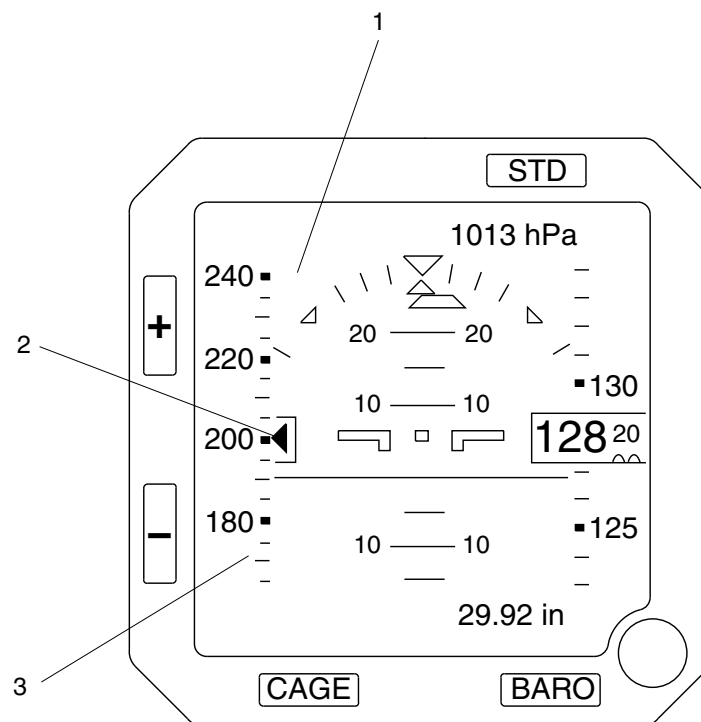
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LEGEND

1. V_{MO} (red limit).
2. Airspeed Pointer and Index.
3. Airspeed Scale (white).



brar36a01.dg, gw, 15/05/03

Integrated Standby Instrument (ISI) Airspeed Display
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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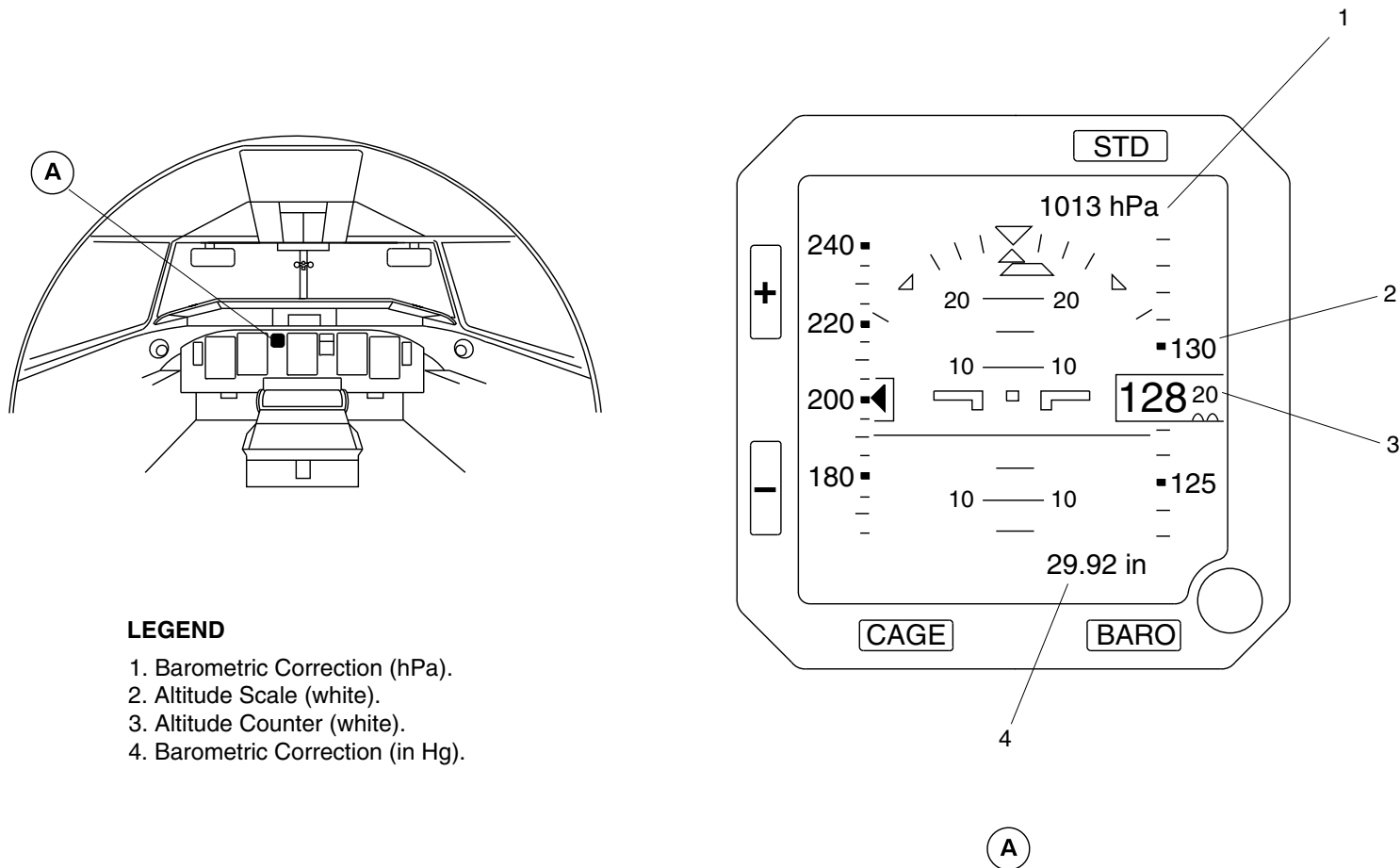
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LEGEND

1. Barometric Correction (hPa).
2. Altitude Scale (white).
3. Altitude Counter (white).
4. Barometric Correction (in Hg).

brar37a01.dg, gw, 15/05/03

Integrated Standby Instrument (ISI) Altitude Display
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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ATTITUDE AND DIRECTION

Introduction

The Attitude and Heading System (AHS) supplies attitude and heading data to the Electronic Instrument System (EIS) indication and to other aircraft systems.

General Description

The attitude and heading parameters that are shown to the pilots and used for flight guidance are critical parameters. For this reason, the AHS has three isolated attitude and heading data systems, two primary and one standby channel.

[Refer to Figure 1.](#)

Each primary channel has one Attitude and Heading Reference Unit (AHRU), flux valve, and AHS control panel to the parameters that follow:

- Attitude
- Angular rates
- Lateral accelerations
- Vertical speed
- Magnetic heading.

The Attitude Heading Reference Units (AHRU1, AHRU2) function in the modes that follow:

- Power-On Self Test (POST)
- Operational
- Maintenance
- Magnetic Compensation
- Magnetic Unit Calibration.

The standby attitude and heading instruments function independently of the primary channel Attitude and Heading Reference Units (AHRU1, AHR2) to show the data that follows:

- pitch and roll attitude
- magnetic heading.

The Attitude and Heading Reference System (AHS) has the components that follow:

- Units, Attitude and Heading Reference (34-21-01)
- Valves, Flux (34-21-11)
- Panel, AHS Control (34-21-16).
- Module, Remote Memory (34-21-21)
- Indicator, Standby Horizon (34-22-01)
- Compass, Standby (33-22-06).



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Attitude and Heading Reference Units

Refer to Figure 2.

The front panel has the aircraft interface connector and the HIRF and lightning protection filter. It gives an interface with the Input/Output and power supply modules.

The AHRU weighs 7.7 lb (3.49 kg). It is 4.2 in. (106.7 mm) wide, 4.5 in. (114 mm) high and 11.4 in. (290 mm) long.

The Attitude Heading Reference Units (AHRU1, AHRU2) are attached to a shock-mount with four screws.

The two AHRUs have shock mounts that attach to the wing plane below the floor within the Center of Gravity (CG) limits. The AHRU is installed in a mounting tray in a lateral orientation.

Shims are installed below the AHRU mounting trays to give the correct physical alignment with aircraft attitude. The seat rails are the longitudinal and lateral references.

Flux Valves

Refer to Figure 3.

The flux valve is a metallic, non-magnetic, half-spherical component. It weighs 0.9 lb (0.41 kg).

Flux valves (MDU 1, MDU 2) are installed in the wing tips to minimize magnetic disturbances. The magnetic field at the wing tip is the most stable.

All flux valves (MDU 1, MDU 2) mounting hardware and fasteners located within an 18-inch radius are made of low magnetic permeability material.

The reference axes of the magnetic detector align with the reference axes of the aircraft.

AHRS Control Panel

Refer to Figure 4.

The AHRS Control Panel (AHCP) weighs 0.5 lb (0.23 kg). The AHRS Control Panel (AHCP) is attached to the center-console with four DZUS fasteners.

Remote Memory Module

Refer to Figure 5.

The Remote Memory Modules (RMMs) connect to their related AHRUs. Two screws secure it in place. The RMMs physically attach to the aircraft by a lanyard. This makes sure that it only connects to its related Attitude Heading Reference Unit (AHRU1, AHRU2). The RMM stays in the aircraft when the Attitude Heading Reference Units (AHRU 1, AHRU 2) are removed for maintenance.

Standby Horizon Indicator

Refer to Figure 6.

The standby horizon indicator is located on the instrument panel between Multi-Functional Display 1 (MFD 1) and the Engine Display (ED) above the standby airspeed indicator in view of the two pilots. The standby horizon indication is mechanically adapted to the 19.5 degrees of instrument panel tilt.



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Standby Compass

[Refer to Figure 7.](#)

The standby compass is installed with non-magnetic screws to the centre structural support of the windshield directly below the Caution and Warning Panel (CWP).

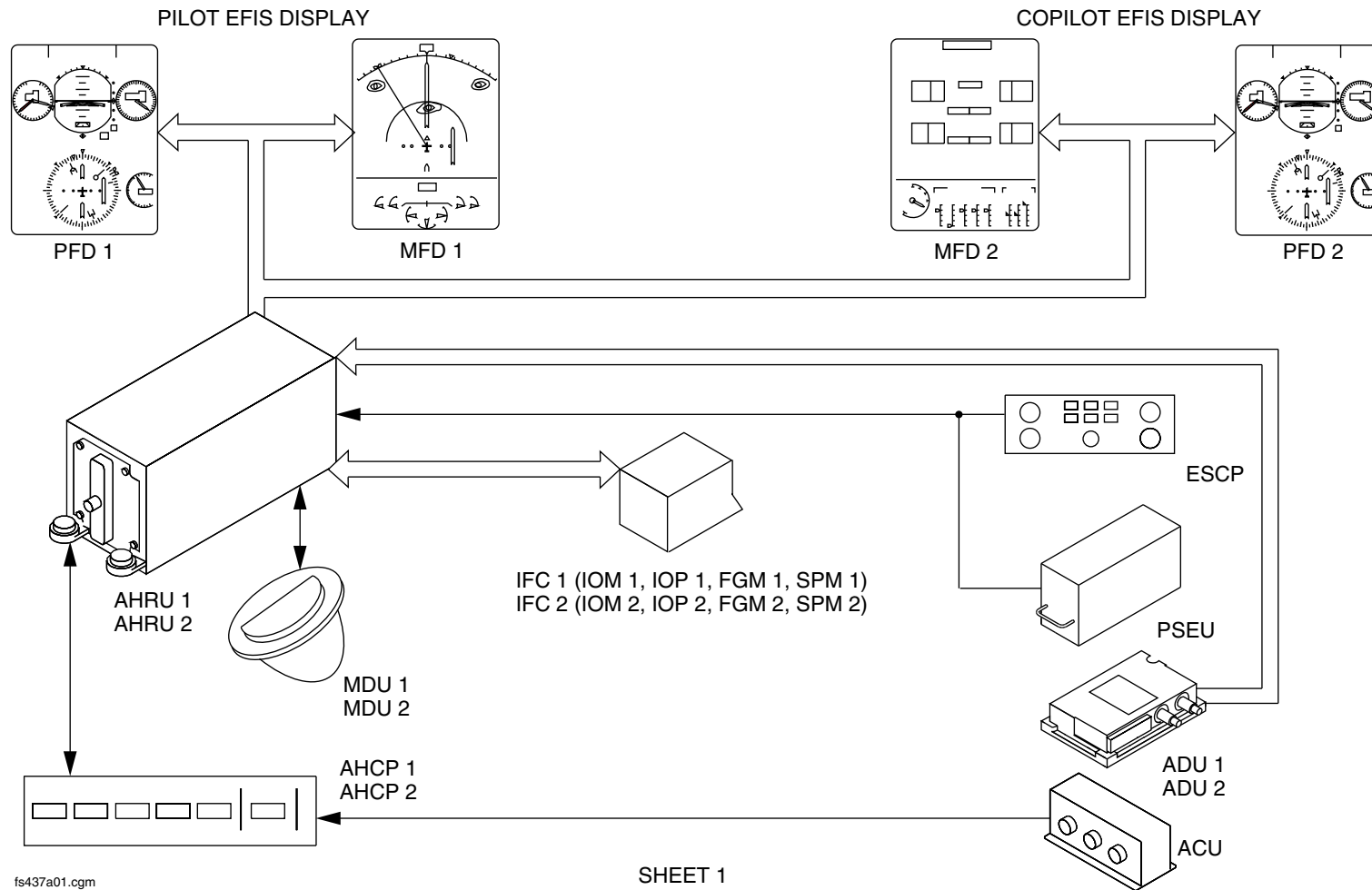
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ATTITUDE AND HEADING REFERENCE SYSTEM BLOCK DIAGRAM
Figure 1

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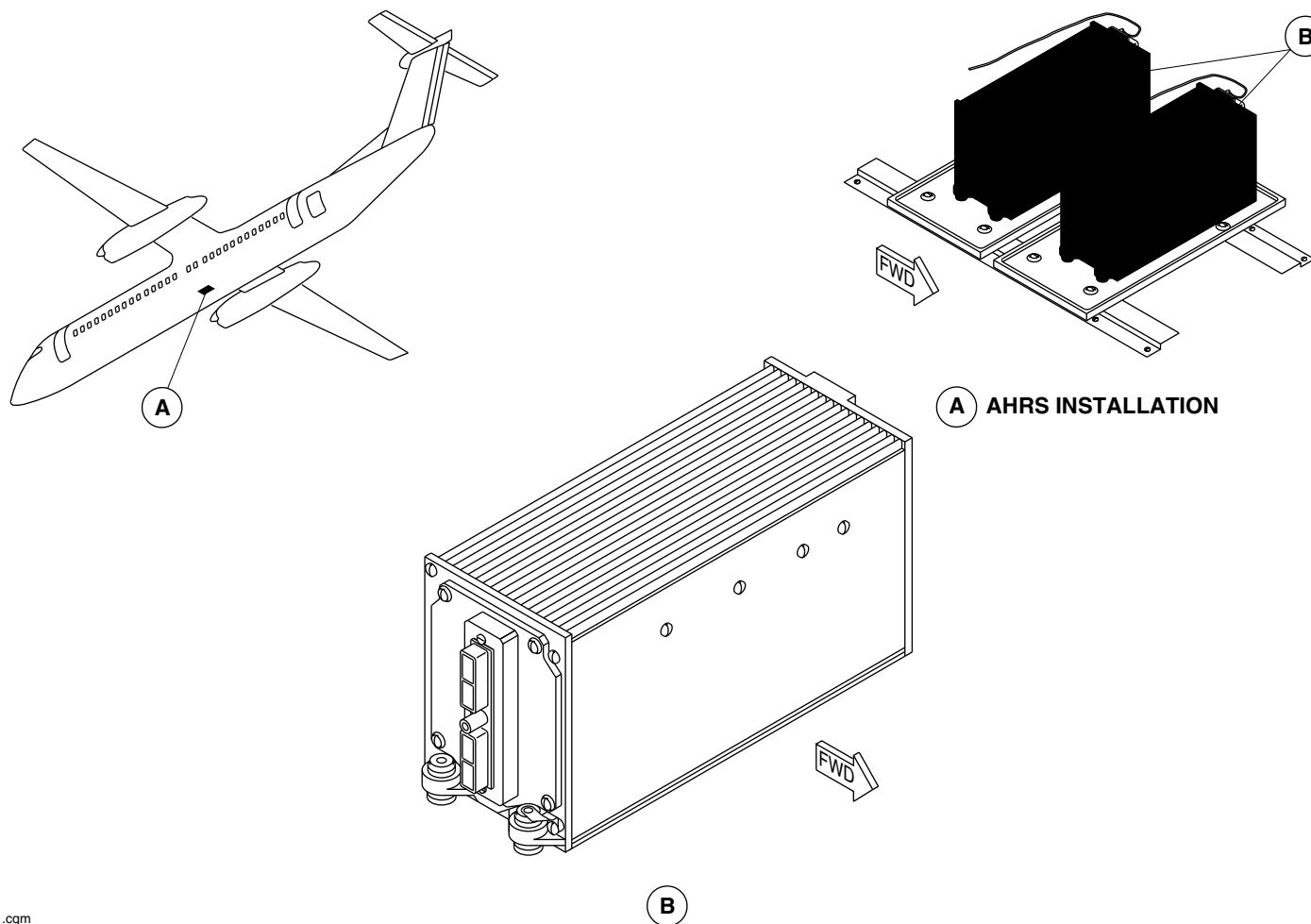
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fsd03a01.cgm

ATTITUDE AND HEADING REFERENCE UNITS LOCATOR
Figure 2

PSM 1-84-2A
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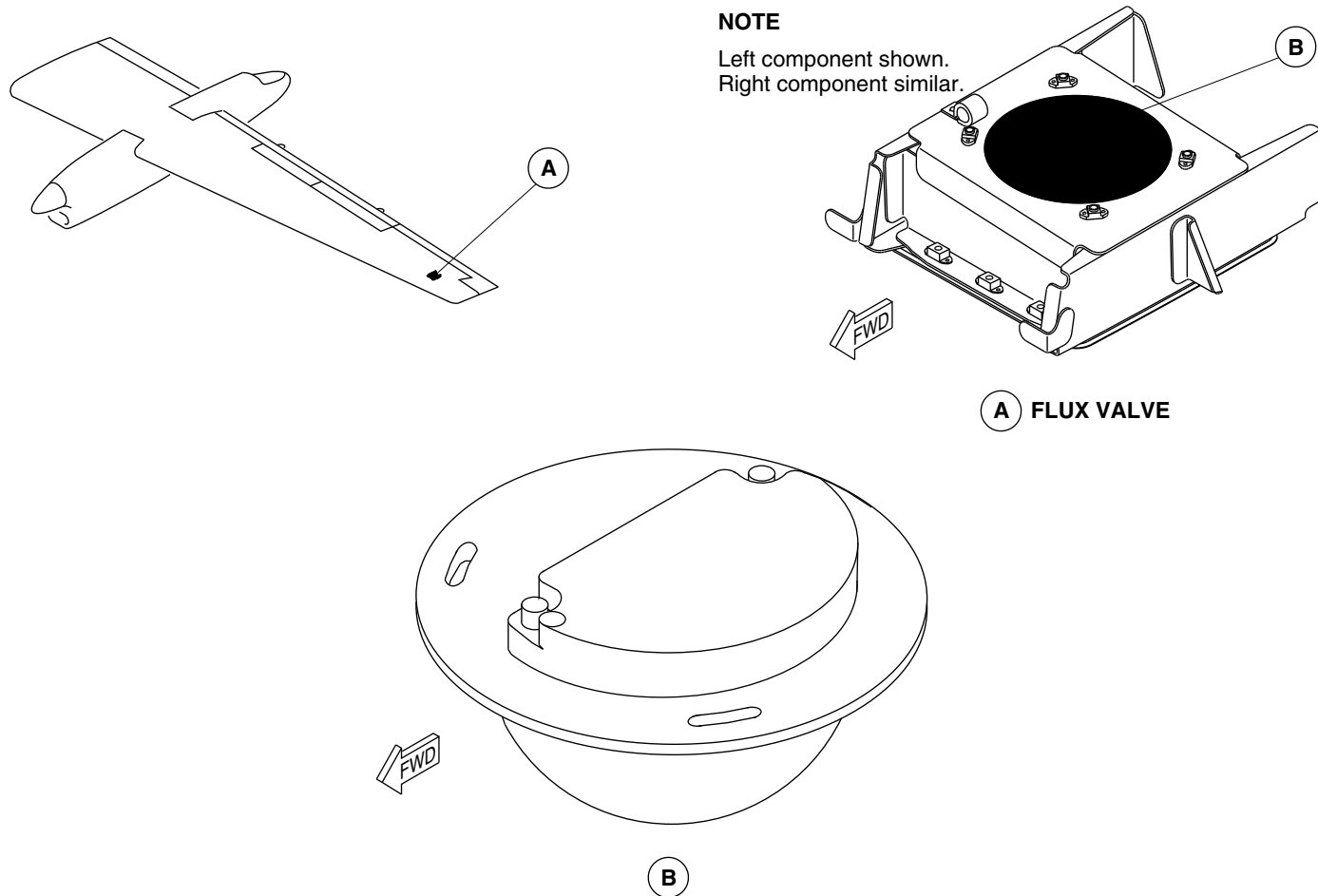
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fsb92a01.cgm

FLUX VALVES LOCATOR
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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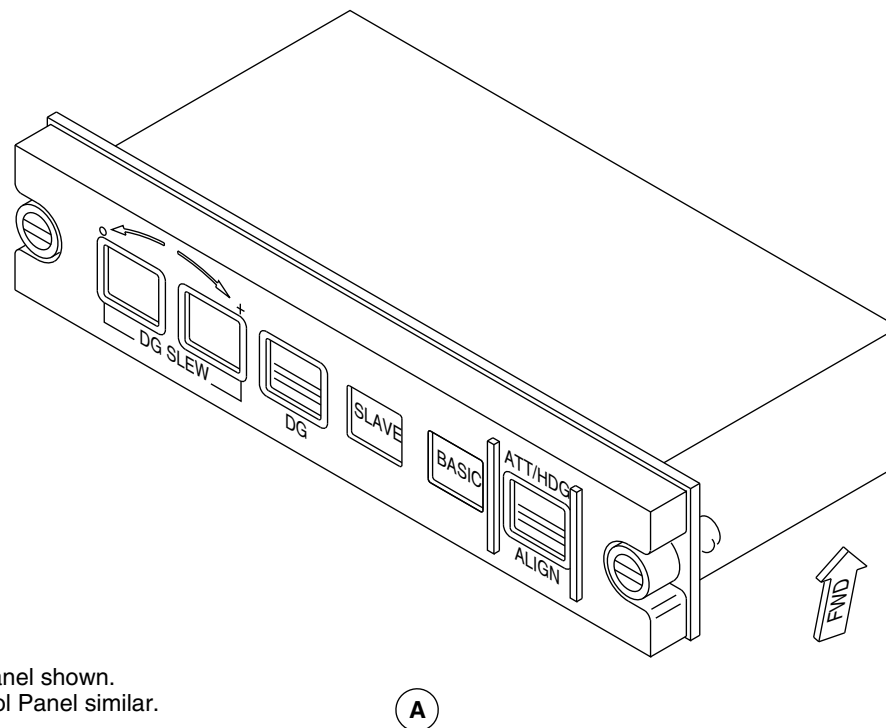
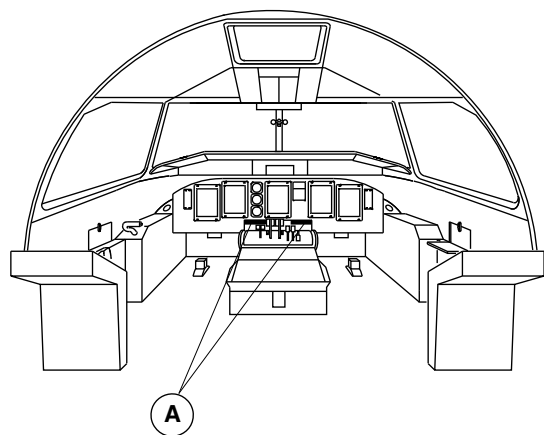
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NOTE

Pilot Control Panel shown.
Co-Pilot Control Panel similar.

fsd02a01.cgm

ATTITUDE AND HEADING REFERENCE UNIT CONTROL PANELS LOCATOR
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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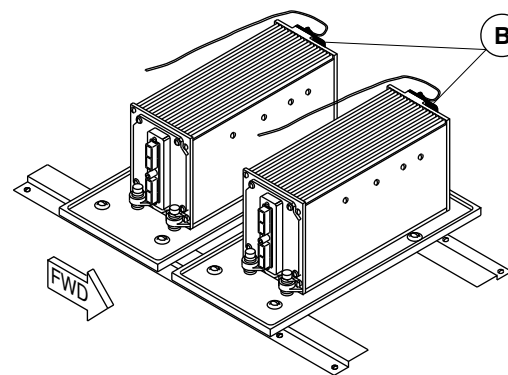
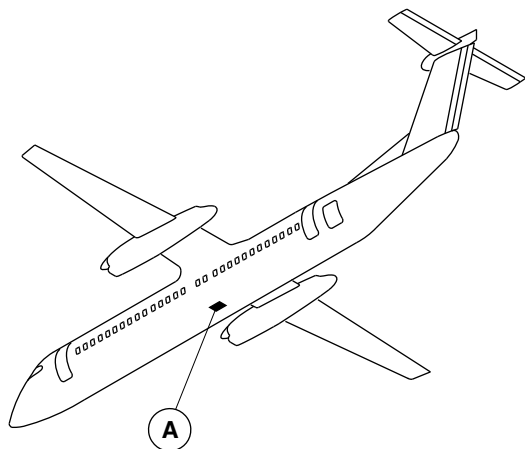
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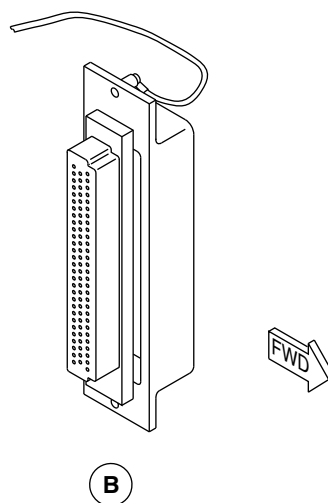


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A AHRS INSTALLATION



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REMOTE MEMORY MODULES LOCATOR
Figure 5

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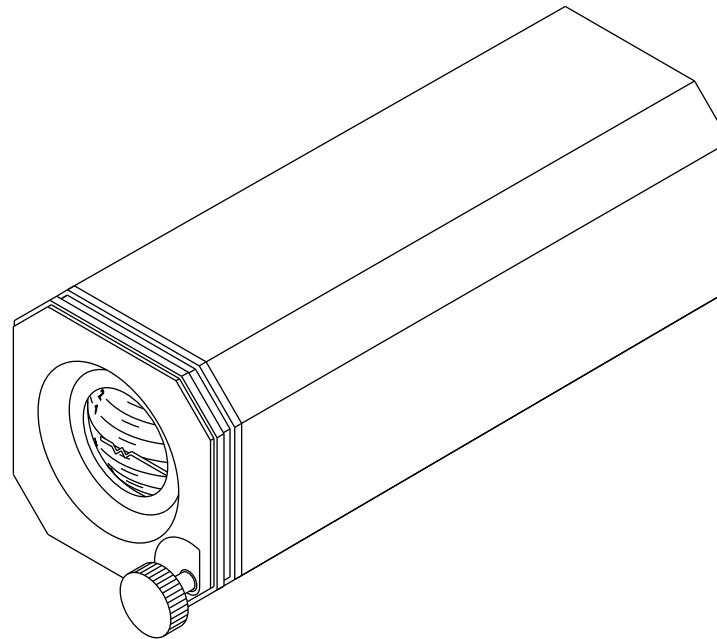
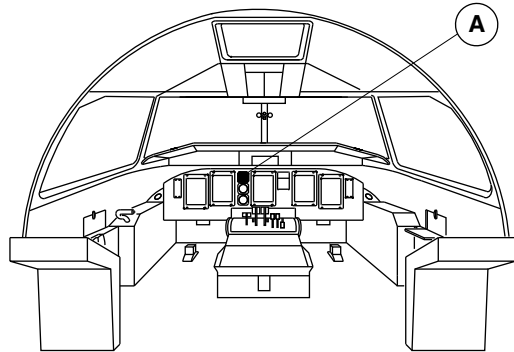
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fs648a01.cgm

STANDBY HORIZON INDICATOR LOCATOR
Figure 6

PSM 1-84-2A
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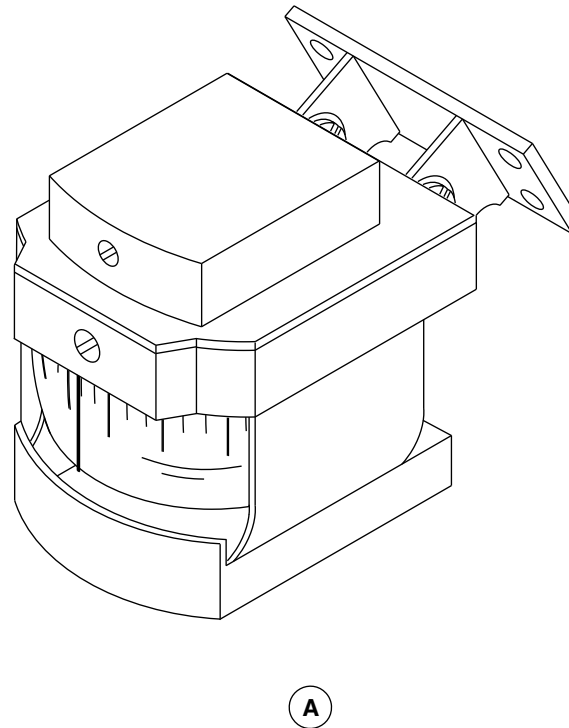
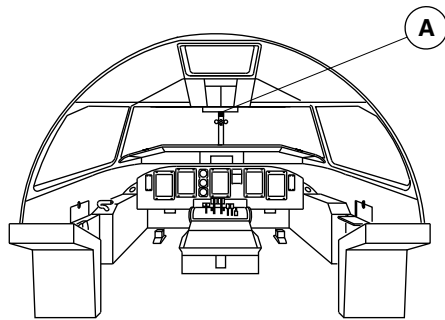
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fsc96a01.cgm

STANDBY COMPASS LOCATOR
Figure 7

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ATTITUDE AND HEADING REFERENCE SYSTEM (AHRS)

Introduction

The Attitude and Heading Reference System supplies primary attitude and heading information to the systems that follow:

- Electronic Flight Instrumentation System (EFIS)
- Auto Flight Control System (AFCS)
- Other aircraft systems.

General Description

[Refer to Figures 1 and 2.](#)

The two primary channel Attitude and Heading Reference Systems (AHRS 1, AHRS 2) channels supply the attitude and heading information that follows:

- Attitude
- Angular rates
- Lateral accelerations
- Vertical speed
- Magnetic heading.

The Attitude Heading Reference Units (AHRU 1, AHRU 2) function in the modes that follow:

- Power-On Self Test (POST)
- Operational
- Maintenance
- Magnetic compensation
- Flux Valve calibration.

Each Attitude and Heading Reference System (AHRS 1, AHRS 2) has the components that follow:

- Units, Attitude and Heading Reference (34-21-01)
- Valves, Flux (34-21-11)
- Panel, AHRS Control (34-21-16).
- Module, Remote Memory (34-21-21).

Each Attitude and Heading Reference System (AHRS 1, AHRS 2) receives digital air data parameters from the two Air Data Units (ADU 1, ADU 2) and discrete inputs from the following systems:

- ESID Control Panel (ESCP 1, ESCP 2)
- Proximity Sensor Electronics Unit (PSEU)
- Central Diagnostic System (CDS).

The Attitude and Heading Reference Unit (AHRU) supplies specific attitude and heading parameters directly to the systems that follow:

- Auto Flight Control System (AFCS)
- Stall Protection System (SPS)
- Electronic Flight Instrumentation System (EIS).

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The AHRU also supplies specific attitude and heading data to the systems that follow through the two Integrated Flight Cabinets (IFC 1, IFC 2):

- Flight Data Recorder (FDR)
- Flight Data Processing System (FDPS)
- Weather Radar (WXR)
- Ground Proximity Warning System (GPWS)
- Traffic Collision Avoidance System (TCAS)
- Central Diagnostic System (CDS).

The General Purpose Data Busses (GPDB) are used to supply attitude and heading information to the systems that follow:

- Flight Control System (FCS)
- Flight Management System (FMS).

A connection to the Portable Maintenance Access Terminal (PMAT) is used to download failure data.

Detailed Description

[Refer to Figure 3.](#)

The Attitude Heading Reference Units (AHRU 1, AHRU 2) function in the modes that follow:

- Power-On Self Test (POST)
- Operational
- Maintenance
- Magnetic compensation

- Flux Valve calibration.

Power-On Self Test (POST): The Power-On Self Test (POST) mode automatically starts and continues for 5 seconds after power-up, during which time the AHRUs do the self-tests that follow:

- Memory check sum
- Test of peripheral circuits
- Test of I/O ports
- Indication of test in progress and successful completion.

During the self-test, the output data from AHRU 1, AHRU 2 cannot be used.

If the self-test senses a malfunction, it is sent to the Central Diagnostic System (CDS) for later analysis during maintenance functions.

Operational Mode: The operational mode has the modes that follow:

- Alignment
- Normal
- Basic
- Directional Gyro.

Alignment Mode: The alignment mode automatically starts at the end of the Power-On Self Test (POST) mode if no malfunctions are found.

AHRU 1 and AHRU 2 calculate the parameters during the alignment mode that follow:

- Attitude and Vertical speed using Air Data Unit (ADU 1, ADU 2) altitude reference



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- Heading using flux valve measurements
- Attitude using accelerations representative of gravity.

A re-alignment mode may be set using the AHRU Control Panels (AHCP 1, AHCP 2) when the aircraft is on the ground or in flight. The aircraft must be in a straight and level attitude when a re-alignment mode selection is made in flight.

AHRU 1, AHRU 2 parameters are available 20 seconds after the re-alignment mode starts.

When the re-alignment mode is completed, the AHRU operates in the primary modes that follow:

- Normal mode for attitude
- Slaved mode for heading.

Normal Mode: The AHRU will start the normal mode automatically following the POST, if the air data parameters are available that follow:

- True Air Speed (TAS)
- Barometric altitude (Zb).

Each Attitude and Heading Reference Unit (AHRU) receives air data parameters from both Air Data Units (ADU 1, ADU 2) through two data busses.

Normally, the AHRU uses its related ADU's parameters. If the related parameter is not correct, a manual ADU reversion is initiated. In this case, the other ADU is used to supply True Air Speed (TAS) and barometric altitude (Zb) parameters.

The Attitude Heading Reference Unit (AHRU) monitors its on side ADU input. If it fails completely, the AHRU automatically reverts to use the other ADU.

True Air Speed (TAS) from the ADU is used to compensate for acceleration induced attitude error. Barometric altitude is used to calculate vertical speed data for long term reference. In the short term, acceleration measurements provide vertical speed variations.

In the normal mode, if the TAS parameter is not available, the basic mode is automatically started and the AHRU functions similarly to a Vertical Gyro (VG). A green light on the AHRU Control Panel will come on to indicate the Basic mode.

Slaved Mode: When functioning in the slaved mode for heading, the AHRU slaves its inertial heading with analog magnetic heading information calculated from analog signals from the flux valve. It uses data from the MDU to calculate the magnetic heading and compensate for heading error caused by directional gyro drift.

The AHRU monitors the MDU. It causes the EIS to show a heading failure flag when the magnetic heading input from the MDU malfunctions. The apparent gravity during dynamic flight phases affects the MDU measurements. Due to the pendulous nature of the MDU, the AHRU internal logic automatically limits heading performance during accelerations.

When a heading error occurs while in the slave mode, the error may be removed by a fast selection of the DG mode and then back to the SLAVED mode. The DG push button switch on the AHRU Control Panel is pushed to start the DG mode and then is pushed again to return the system to the heading mode.

A fast selection and de-selection of the DG mode immediately causes the system to reinitialize to the MDU heading and has no effect on attitude.



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In the slaved mode, if an MDU malfunctions, the PFD will show a heading failure flag and a red SLAVE annunciation on the AHRs Control Panel will come on. When the AHRU is set to DG mode, the MDU data is removed from the heading calculation and a DG heading indication is shown on the PFD. The SLAVE annunciation remains on.

The AHRU is set to the DG mode when the flux valve malfunctions or degrades when flying close to the magnetic poles.

When the DG mode is set, the Attitude and Heading Reference Units (AHRU 1, AHRU 2) do not receive magnetic data from the flux valves.

The heading is manually corrected by a plus or minus selection using the Attitude and Heading Control Panels (AHCP 1, AHCP 2). Two rates of correction speed are available. The low speed functions at 1 degree per second, and fast speed at 10 degrees per second.

Maintenance Mode: The Built In Test Equipment (BITE) uses the Central Diagnostic System (CDS) to give the condition of the component. It saves faults in a Non Volatile Memory (NVM) for reporting to line and shop maintenance.

The maintenance mode is selected from one or the other ARCDU during maintenance operations. The AHRU does a test when it receives a maintenance discrete command through the IOM 1 while the air speed is less than 50 knots (93 km/h) and the aircraft is on the ground.

The Central Diagnostic System (CDS) sends a command directly to each primary Attitude Heading Reference Unit (AHRU 1, AHRU 2) to do a Power-On-Self-test (POST) that completes in less than 10 seconds. All sensed failures are sent in real time to the Central Diagnostic System (CDS).

The Built In Test Equipment (BITE) modes monitor the condition of the component as follows:

- Power-On Self Test (POST)
- Continuous Monitoring.

Power-On Self Test (POST): The Power-On Self Test (POST) checks the condition of the component at power-up or after a long power interruption, more than 200 milliseconds.

Continuous Monitoring: The continuous monitoring checks the status of the component in flight. It records faults in a Non Volatile Memory (NVM) for later troubleshooting using the Central Diagnostic System (CDS).

When powered, the Attitude Heading Reference Unit (AHRU 1, AHRU 2) does the different types of monitoring that follow:

- Processors and memories (program memory, calibration sensor memory, RAM memory)
- Power supply monitoring
- Sensor, FOG and accelerometer, failure monitoring and measurement range monitoring
- Internal temperature monitoring
- Flux-valve measurement monitoring, magnetic field module
- ARINC inputs and outputs.

Magnetic Compensation Mode: The magnetic compensation mode functions on the ground only. It senses magnetic heading error induced by the flux valve and aircraft heading.



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The Attitude and Heading Reference Unit (AHRU) compares the magnetic heading field measurements supplied by its related flux valve and the magnetic reference values. The magnetic reference value is supplied to AHRU 1 and AHRU 2 through the Portable Maintenance Access Terminal (PMAT).

Flux Valve Calibration Mode: The flux valve calibration mode is used to set and store magnetic heading corrections.

The conditions that follow start the flux valve calibration mode:

- The aircraft is on the ground
- The ALIGN and DG push buttons switches on the AHRS Control Panel (AHCP) are pushed at the same time for three seconds.

NOTE

If the calibration mode is set, a Weight–Off–Wheels (WOFW) state causes the flux valve calibration mode to automatically stop.

The red SLAVE annunciator light on the AHRS Control Panel (AHCP) flashes throughout the calibration mode. In addition, the Electronic Horizontal Situation Indicator (EHSI) area of the Primary Flight Display (PFD) shows an amber HDG CALIBRATION message.

[Refer to Figure 4.](#)

The EFIS ATT/HDG SOURCE reversion selector is set to the NORM position on the ESID Control Panel (ESCP) to show data from their

related Attitude and Heading Reference Units (AHRU 1, AHRU 2) on the Primary Flight Displays (PFD 1, PFD 2).

[Refer to Figure 5.](#)

The Electronic Instrument System (EIS) shows the Attitude and Heading Reference Units (AHRU 1, AHRU 2) parameters that follow:

- Electronic Attitude Direction Indicator (EADI 1, EADI 2)
- Electronic Horizontal Situation Indicator (EHSI 1, EHSI 2)
- Inertial Vertical Speed Indicator (IVSI 1, IVSI 2).

Electronic Attitude Direction Indicator (EADI 1, EADI 2): The Electronic Attitude Direction Indicator (EADI 1, EADI 2) is used to show the aircraft's pitch, bank, and slip/skid against a representation of the earth and sky.

[Refer to Figure 6.](#)

The Primary Flight Display (PFD) shows the Electronic Attitude Direction Indicator (EADI) components that follow:

- Attitude sphere
- Pitch scale
- Roll pointer and scale
- Airplane symbol
- Slip/skid indicator
- Mismatch flags
- Attitude failure flag.

Attitude Sphere: The attitude sphere has a sky permanent sector in blue above a ground permanent sector in brown and a white horizon



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line to divide the sectors. The attitude sphere moves to show pitch and roll.

Pitch Scale: The pitch scale is shown on the attitude sphere in 5 degree increments from 0 to 30 degrees. There are numbers shown on the two sides of the pitch scale at 10, 20, 30, 40, 60, and 90 degree increments for the pitch up and pitch down attitudes.

The pitch scale has red chevrons that point towards the pitch scale to help recover from an unusual attitude.

Roll pointer and scale: The roll scale is shown on the attitude sphere in 0, 10, 20, 30, 45, and 60 degree increments at the top of the attitude sphere. A 60 degree roll mark comes into view when the roll is more than 30 degrees.

The roll pointer is located at the top of the attitude sphere, above the slip/skid indicator to show the amount of aircraft roll. It moves with the horizon line to measure aircraft roll.

Airplane Symbol: The attitude indication has an attitude sphere that moves relative to a fixed airplane symbol to show the amount of aircraft pitch.

Slip/Skid Indicator: The Slip/Skid Indicator shows the lateral acceleration of the aircraft and is a trapezoid symbol. It turns with the roll pointer and moves to left or right to show the intensity of lateral acceleration.

The maximum deflection indication shows a 0.14 gravity (g) lateral acceleration. The slip skid Indicator is shown to the left when the aircraft is slipping to the right and to the right when the aircraft is slipping to the left.

The Slip/Skid Indicator goes out of view when the aircraft roll is more than 60 degrees or the related AHRU has malfunctioned.

[Refer to Figure 7.](#)

Mismatch Flags: The Flight Data Processing System (FDPS) senses signal differences between the AHRU 1 and AHRU 2 pitch and roll parameters. Both Flight Mode Annunciators (FMA) located at the top of the Primary Flight Displays (PFD 1, PFD 2) show yellow PITCH MISMATCH or ROLL MISMATCH messages in the center row of the center column. If a pitch and roll mismatch condition occur at the same time, the pitch mismatch message will be shown. It has a higher indication priority than the roll mismatch message. The indications flash for five seconds when they come into view and then go steady.

The attitude mismatch threshold is 3 degrees.

[Refer to Figure 8.](#)

Attitude Failure Flag: When the attitude parameters malfunction, the roll scale, attitude sphere, and airplane symbol are replaced by a flag that comes into view as a red ATT FAIL message inside a white circle.

Electronic Horizontal Situation Indicator (EHSI): The Electronic Horizontal Situation Indicator (EHSI 1, EHSI 2) is used to show the aircraft heading and the slaving error.

[Refer to Figure 9.](#)

The Primary Flight Display (PFD) shows the Electronic Horizontal Situation Indicator EHSI components that follow:

- Rotating heading dial
- Fixed markings
- Actual heading marker and aircraft symbol
- Slaving error annunciator



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- Heading mismatch flag
- Heading failure flag.

Rotating heading dial: The rotating heading dial is a full compass rose indication with markings at 5 degree intervals and numbers at 30 degree intervals except North, East, South, and West. The North, East, South, and West heading are shown as N, E, S, and W.

Fixed markings: The fixed markers are shown around the rotating heading dial at 45 degree interval.

Actual heading marker and aircraft symbol: A fixed white triangle actual heading marker is shown at the top of the EHSI with a fixed aircraft symbol in the center of the rotating heading dial to give a lubber line indication.

Slaving Error Annunciator: The compass slaving error annunciator shows the difference between the Attitude and Heading Reference Unit (AHRU 1, AHRU 2) heading and its related flux valve (MDU 1, MDU 2) heading.

A white vertical pointer that moves between a green + and – symbol is shown near the rotating heading dial indication. The maximum indication is plus or minus 4 degrees.

When the data from the AHRU is not correct, the pointer and scale are removed.

[Refer to Figure 10.](#)

Mismatch Flags: The Flight Data Processing System (FDPS) senses signal differences between the AHRU 1 and AHRU 2 heading parameters. Both Flight Mode Annunciators (FMA) located at the top of the Primary Flight Displays (PFD 1, PFD 2) show yellow HDG MISMATCH messages in the center row of the center column. If an attitude and heading mismatch condition occurs at the same time,

the related attitude mismatch message will be shown. It has a higher indication priority than the heading mismatch message. The indications flash for five seconds when they come into view and then go steady.

The heading mismatch threshold is 8 degrees.

The AHRU deactivates the flux slaving when the aircraft roll angle is greater than 7 degrees, pitch angle is greater than 15 degrees or during specified horizontal acceleration. For this reason certain aircraft maneuvers, such as extended, continuous turns may cause a “HDG MISMATCH” message to be displayed on the PFD. This is considered normal as long as the “HDG MISMATCH” message is removed within 5 minutes of returning to straight and level flight.

The AHRU deactivates the flux valve slaving on the occurrences of any of the following conditions (with hysteresis):

- The roll angle is above 7 degrees and the roll angle must return to below 6 degrees to clear this condition.
- The pitch angle is above 15 degrees and it must return to below 14 degrees to clear this condition.
- Any individual component of horizontal acceleration is above 70 mg and must return to below 60 mg.
- Filtered intensity (time constant = 5 seconds) of horizontal acceleration is above 23 mg and must return to below 20 mg.

[Refer to Figure 11.](#)

Heading Failure Flag: When the heading parameters malfunction, the rotating heading dial, aircraft symbol, and slaving error annunciator are replaced by a flag that comes into view as a red HDG FAIL message inside a white circle. The fixed markings remain in view.



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Refer to Figure 12.

Inertial Vertical Speed Indicator (IVSI 1, IVSI 2): The Inertial Vertical Speed Indicator (IVSI 1, IVSI 2) shows the aircraft inertial vertical speed.

The Primary Flight Display (PFD) shows the Inertial Vertical Speed Indicator (IVSI) components that follow:

- Scale
- Pointer needle
- Digital readout
- Failure flag.

Scale: The IVSI 1 scale shows vertical speed from –4500 to +4500 feet per minute. It has a compressed scale that is in 100 feet per second increments for values between –1000 feet per minute and +1000 feet per minute. The scale expands to show increments at 500 feet per minute between –1000 and –4500 feet per minute and between +1000 and +4500 feet per minute. The markings for 1000, 2000, and 4000 feet per minute are shown as 1, 2, and 4.

Pointer Needle: The Pointer Needle points to the current inertial vertical speed of the aircraft along a fixed scale from –4500 to +4500 feet per minute.

Digital Readout: The digital readout can show the inertial vertical speed value in white numbers from –9900 to 9900 feet per minute. It shows a compressed value that is in 50 feet per minute increments from –1000 through +1000. It changes to an expanded value that is

in 100 feet per minute increments between –1000 and –9900 feet per minute and between +1000 and +9900 feet per minute.

Refer to Figure 13.

IVSI Failure Flag: When the IVSI parameters malfunction, the scale, pointer needle, and digital readout are replaced by a flag that comes into view as a red IVSI FAIL message inside a white rectangle.

Refer to Figure 14.

The ATT/HDG ALIGN annunciator switch located on the AHRS Control Panel (AHCP) is pushed to manually start the alignment mode.

Refer to Figure 15.

An amber bar in the ATT/HDG ALIGN annunciator switch comes to show the alignment mode and the two Electronic Attitude Direction Indicators (EADI 1, EADI 2) show an ALIGNING message. The message comes into view in yellow fonts that flash for five seconds and then change to steady.

The alignment mode may be set when the aircraft is on the ground or in-flight.

NOTE

The aircraft must be in a straight and level attitude when making an alignment mode selection in-flight.

The alignment mode continues for 60 seconds on the ground and 90 seconds in-flight. The Attitude Heading Reference System (AHRS 1, AHRS 2) parameters are available 20 seconds after starting the alignment mode.

The normal mode automatically starts at the end of the alignment mode if the external inputs are available. It calculates attitude



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parameters when external air data inputs are available and slaved heading when the external MDU input is available.

[Refer to Figure 16.](#)

If the TAS parameter is not available, a green light on the AHCP will come on to show the automatic reversion to the basic mode.

[Refer to Figure 17.](#)

The SLAVE light on the AHCP comes on to show a flux valve malfunction.

[Refer to Figure 18.](#)

The DG switch on the AHCP is pushed to manually set the DG Mode. A green bar in the annunciator switch comes on.

The + or – DG SLEW switches located on the Attitude and Heading Control Panel (AHCP 1, AHCP 2) are pushed to correct the rotating heading dial for gyro drift. The rotating heading dial turns in a clockwise direction while the + DG SLEW switch is pushed, and it moves in a counterclockwise when the – DG SLEW switch is pushed. If one slew switch or the other is initially pushed and held, the rotating heading dial turns at the low speed rate (1 degree per second), and if held for more than 3 seconds, it will then turn at the high-speed rate (10 degrees per second).

[Refer to Figure 19.](#)

The EHSI shows a DG heading source annunciation and the slaving error indication goes out of view to show the selection.

[Refer to Figure 20.](#)

The flux valve calibration mode is set when the ALIGN and DG SLEW switches on the AHRS Control Panel (AHCP 1, AHCP 2) are

pushed for three seconds at the same time while the aircraft is on the ground.

The flux valve calibration mode automatically cancels at take-off when the Weight On Wheels (WOW) condition changes from ground to flight.

[Refer to Figure 21.](#)

The PFD and MFD EHSIs show a HDG CALIBRATION message. The MFD actual heading digital value and reference indication changes to show the heading value with four numbers. The red SLAVE annunciator light on the AHCP flashes during the calibration mode.

The flux valve calibration mode starts with an alignment phase. The ALIGN light comes on and the alignment phase continues for 5 minutes.

Heading error measurements are made when the ALIGN annunciator light goes out.

In the measurement phase, the aircraft is positioned on a northerly heading. Sight measurements are made at 45 degree intervals through 360 degrees. The AHRU is supplied with the measured reference heading data for each interval and the DG SLEW push buttons switches are used to match the rotating heading dial indication on the EHSI with the sight measurement. The DG SLEW switches are pushed at the same time to put the heading value into the AHRU.

The red SLAVE annunciator light flash rate increases to indicate a difference between the measured value and the value supplied by flux valve. The red SLAVE annunciator light goes back to the normal flash rate when the update is completed. It continues for less than 60 seconds.



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The aircraft is positioned on a 30 degree heading. Eight more sight measurements are made at 45 degree intervals. The Attitude and Heading Reference Unit (AHRU) is supplied with the measured reference heading data for each interval.

The red SLAVE annunciator light stays on to show a satisfactory completion of the last measurement. The ALIGN and DG SLEW switches located on the AHCP are pushed at the same time for 3 seconds to complete the procedure.

A new alignment mode automatically initiates for 60 seconds.

The Attitude and Heading Reference System (AHRS) circuit breakers are cycled to stop the calibration procedure.

[Refer to Figure 22.](#)

When the EFIS ATT/HDG SOURCE reversion selector is set to 1 or 2 on the ESID Control Panel (ESCP), both Primary Flight Displays (PFD 1, PFD 2) will show data from the selected Attitude and Heading Reference Unit (AHRU 1, AHRU 2).

[Refer to Figure 23.](#)

When the two PFDs are showing the same attitude and heading source, an attitude and heading source annunciator is shown in yellow fonts that follow:

- ATT 1, ATT 2
- HDG 1, HDG 2
- DG 1, DG 2.

[Refer to Figure 24.](#)

The MFD 1 and MFD 2 Reversion switches located on the ESID Control Panel (ESCP) are used to select the Navigation Page.

[Refer to Figure 25.](#)

The Navigation Page is usually shown on the upper part of MFD1 and permanent data is shown on the bottom. The Navigation Page comes into view in the map FMS mode format. The map mode shows the same heading parameters as its related Primary Flight Display (PFD 1, PFD 2) indication on a 90 degree heading arc.

[Refer to Figure 26.](#)

The FORMAT pushbutton switch located on the EFIS Control Panel (EFCP) is pushed for more than one second (long push) to select the plan FMS mode format.

[Refer to Figure 27.](#)

The plan FMS mode format has a white aircraft symbol that turns about its center in the center of a fixed full map indication. When the heading parameter is not available, it is removed from view.

The Left Main and Essential Busses supply 28 Vdc electrical power through four 5 A circuit breakers to the AHRS 1 components. The circuit breakers are located in positions A4, A9, B4, and B9 on the avionics circuit breaker panel.

The right main and essential buses supply 28 Vdc power through four 5 A circuit breakers to the AHRS 2 components. The circuit breakers are located in positions A3, A10, B3, and B10 on the avionics circuit breaker panel.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Table 1: Table 1. Electrical Power Interfaces

COMPONENT	LEFT MAIN BUS	LEFT ESS BUS	RIGHT MAIN BUS	RIGHT ESS BUS
AHRU 1	x	x		
AHCP 1	x	x		
AHRU 2			x	x
AHCP 2			x	x

The AHRUs receive digital air data parameters from the two ADUs and discrete inputs from the systems that follow:

Table 2: Table 2. AHRU Input Interfaces

PARAMETER	SOURCE
ATT/HDG Source Reversion	ESCP 1, ESCP 2
Weight On Wheels	PSEU
Mode Commands	AHCP 1, AHCP 2
IBIT	CDS

The Attitude and Heading Reference Unit (AHRU) supplies specific attitude and heading parameters directly to the systems that follow:

Table 3: Table 3. AHRU Output Interfaces

PARAMETER	SOURCE
Bank Angle	AFCS 1, AFCS 2 SPS EFIS 1, EFIS 2
Pitch Angle	AFCS 1, AFCS 2 SPS EFIS 1, EFIS 2
Body Linear Accelerations, x	AFCS 1, AFCS 2
Body Linear Accelerations, y	AFCS 1, AFCS 2
Body Linear Accelerations, z	AFCS 1, AFCS 2 SPS
Body Linear Accelerations, j	EFIS 1, EFIS 2
Body Angular Rates	AFCS 1, AFCS 2 SPS
Heading	AFCS 1, AFCS 2 EFIS 1, EFIS 2
Compass Synchro Error	EFIS 1, EFIS 2
Baro Inertial Vertical Speed	AFCS 1, AFCS 2 EFIS 1, EFIS 2

It supplies specific attitude and heading data to the systems that follow through the Integrated Flight Cabinets (IFC):



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Table 4: Table 4. AHRU Output Interfaces, IFCs

PARAMETER	SOURCE
Bank Angle	FDR FDPS 1, FDP 2 WXR ATC 1, ATC 2
Pitch Angle	FDR FDPS 1, FDPS 2 WXR ATC 1, ATC 2
Body Linear Accelerations, x	FDR FDPS 1, FDPS 2
Body Linear Accelerations, y	FDR FDPS 1, FDPS 2
Body Linear Accelerations, z	FDR FDPS 1 FDPS 2 FMS FCECU 1, FCECU 2
Body Angular Rates	FDPS 1, FDPS 2 FMS
Heading	FDR FDPS 1, FDPS 2

A General Purpose Data Busses (GPDB) supply attitude and heading information to the systems that follow:

Table 5: Table 5. AHRU Output Interfaces, GPDB

PARAMETER	SOURCE
Body Linear Acceleration, z	FMS 1, FMS 2 FCECU 1, FCECU 2
Body Angular Rates	FMS 1, FMS 2

The AHRU uses four ARINC 429 High-Speed data busses to supply specific attitude and heading parameters to the systems that follow:

Table 6: AHRU 1

ARINC 1	ARINC 2	ARINC 3	ARINC 4
MFD 1 PFD 1	FGM 1 SPM 1 CDS FDR FDPS 1	PFD 2 MFD 2 FGM 2 SPM 2	FDPS 2

Table 7: AHRU 2

ARINC 1	ARINC 2	ARINC 3	ARINC 4
PFD 2 MFD 2	FGM 2 SPM 2 FDPS 2	PFD 1 MFD 1 FGM 1 SPM 1	FDPS 2 CDS FDR FDPS 1

The Input/Output Processor (IOP 1, IOP 2), Stall Protection Modules (SPM 1, SPM 2), and Flight Guidance Modules (FGM 1, FGM 2) are located inside its related Integrated Flight Cabinet (IFC 1, IFC 2).



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Units, Attitude and Heading Reference

Refer to Figure 28.

The Attitude Heading Reference Unit (AHRU) uses fibre optic gyros and accelerometers to measure angular rates and accelerations in three body axes.

It weighs 7.7 lb (3.49 kg). It is 4.2 in. (106.7 mm) wide, 4.5 in. (114 mm) high and 11.4 in. (290 mm) long.

The case has cooling fins to dissipate heat.

The Attitude Heading Reference Unit (AHRU 1, AHRU 2) is attached to a shock-mount with four screws.

The two AHRUs have shock mounts that attach to the wing plane below the floor within the Center of Gravity (CG) limits. The AHRU is installed in a mounting tray in a lateral orientation. This prevents possible damage to the two Attitude Heading Reference Units (AHRU 1, AHRU 2) in case of engine rotor-burst. The installation is specified by the configuration software.

Shims are installed below the AHRU mounting trays to give the correct physical alignment with aircraft attitude. The seat rails are the longitudinal and lateral references.

Refer to Figure 29.

The Attitude and Heading Reference Unit (AHRU) has the three electronic modules that follow, installed in a rigid tubular structure:

- Sensor
- Input/Output
- Power Supply.

Sensor Module: The Sensor Module has the components that follow:

- Three-axis Fibre Optic Gyros (FOG)
- Three accelerometers
- Main processor board
- Flux-valve electronic interface board

The Fibre Optic Gyros (FOG) transmit data by a micro controller to the main processor through a data bus. The electronic circuitry within the sensor module gives the functions that follow:

- Laser diode temperature regulation
- Fibre Optic Gyros (FOG) modulation
- Signal detection and measurement.

The main processor board does all the inertial software processing. It uses a microprocessor, EPROM, RAM and an EEPROM.

The main software does the functions that follow:

- Calculate attitude, heading and vertical speed
- Give angular rates along carrier axes
- Give linear accelerations along carrier axes
- Update attitude and inertial/magnetic sensor reference hybridization
- Update attitude and inertia/airspeed hybridization
- Update of altitude and vertical speed and baro-inertial hybridization
- Interfacing
- Test



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- Magnetic compensation.

The flux–valve interface board calculates the functions that follow:

- Flux valve excitation
- Multiplexes the analog data
- Analog–to–Digital conversions
- Determines temperature
- Filters and computes accelerometer outputs.

Input/Output Module: The Input/Output module receives air data parameters from the two Air Data Units (ADU 1, ADU 2) through data busses. It transmits on four output buses and uses the bi–directional RS232 line. It connects to the main processing board through a 16–bit data bus. The Input/Output module uses ARINC 429, RS232, and discrete interfaces.

The power supply module changes the dc Generation System 28 Vdc to the voltages that follow:

- +15 Vdc
- –15 Vdc
- +5 Vdc.

The motherboard is attached to the rear panel and supplies the connection between the three electronic modules. It connects to the external Remote Memory Module (RMM) connector.

The front panel has the aircraft interface connector and the HIRF and lightning protection filter. It gives an interface with the Input/Output and power supply modules.

[Refer to Figure 30.](#)

The attitude is updated with the angular rate measurements to project the acceleration supplied by the accelerometers in the geographical axes. The acceleration is integrated to estimate the inertial speed. The difference between this inertial speed and the external reference speed (TAS) supplied by ADC is used to correct the attitude.

[Refer to Figure 31.](#)

The gyromagnetic heading is updated with angular rate measurements. It is then corrected with a magnetic compensation model stored in EEPROM. The corrected measures of the earth magnetic field components supplied by the flux valve are projected with the gyromagnetic heading in the geographical axes. The calculated East component lets the AHRU estimate the heading error. This error is used to correct the gyromagnetic heading and is filtered for indication as the slaving error annunciator.

The correction is not calculated when the DG mode is set.

[Refer to Figure 32.](#)

The vertical speed is updated by adding vertical acceleration data (projection of accelerometer measurements with computed attitude). A second addition lets the AHRU estimate the inertial altitude. This altitude is compared with barometric altitude supplied by the ADC. The filtered error is then used to correct the barometric inertial vertical speed.

When barometric altitude is no longer supplied, the vertical speed parameter becomes invalid after 20 seconds.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Valves, Flux

[Refer to Figure 33.](#)

The flux valve is a two-axis sensor. It measures the horizontal component of the earth's magnetic field.

The flux valve is a metallic, non-magnetic, half-spherical component. It weighs 0.9 lb (0.41 kg).

One flux valve is installed in each wing tip to minimize magnetic disturbances. The magnetic field at the wing tip is most stable.

The flux valves have reference axes that are aligned with the reference axes of the aircraft.

All flux valves mounting hardware and fasteners located within an 18-inch radius are made of low magnetic permeability material.

The flux valve output has two voltages that are proportional to the sine and cosine of the magnetic heading angle. Two coils that are positioned at 90 degrees to each other in the horizontal plane give the sine and cosine signals.

A large pendulum weight is suspended in a double gimbal arrangement to hold the plane of the coils in the horizontal position. The weight is immersed in fluid to give viscous damping.

The two sets of coils have an excitation coil and a sensing coil. The AHRU supplies 400 Hz excitation to the coil. The excitation signal alternately drives the core material in the coils into and out of saturation. The sense windings on each coil sense the transition between saturation levels that will have an equal duty cycle without any external magnetic field present. The presence of an external magnetic field causes the core to stay in saturation longer, and enhances the duty cycle of the sense coils in the direction of the

external field. Electronics within the Attitude Heading Reference Unit (AHRU 1, AHRU 2) receives the sense-winding signal.

The flux valve is attached to a support plate. Three elliptical slots and a white index mark on the support plate lets the flux valve align with the longitudinal aircraft reference axis. Its accuracy is better than ± 3 degrees.

Panel, AHRS Control

[Refer to Figure 34.](#)

The AHCP is used to set the Attitude Heading Reference System (AHRS) modes and show its status.

The AHCP weighs 0.5 lb (0.23 kg). It is attached to the forward part of the center-console with four DZUS fasteners.

The AHRS Control Panel (AHCP) has the assemblies that follow:

- Front Face
- Interface Board Module
- Outer Box.

The AHRS Control Panel (AHCP 1, AHCP 2) has the annunciators and switches that follow:

- ATT/HDG ALIGN annunciator switch
- DG annunciator switches
- DG SLEW push buttons switches
- BASIC annunciator
- SLAVE annunciator.



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Module, Remote Memory

Refer to Figure 35.

The Remote Memory Module (RMM) stores the AHRS configuration parameters for the aircraft configuration that follows:

- Magnetic compensation (usually different for each aircraft)
- Location of the AHRU in the aircraft
- Mechanical misalignment of AHRU reference axes relating to the aircraft reference axes
- The permanent orientation of the AHRU in the aircraft.

The AHRU supplies the magnetic compensation parameters to the RMM through a bi-directional RS232 serial data bus. An EEPROM in the RMM stores the magnetic compensation parameters.

The position of a single jumper at the RMM gives the pilot or copilot location in the aircraft.

The AHRU mechanical alignment make sure that the needed system accuracy without additional software compensation. The RMM installation is initially set with no compensation.

The AHRU energizes the RMM. It has no other interface.

The RMM connects to its related AHRU and is secured with two screws. The RMM is also attached to the aircraft by a lanyard to make sure that it can only be connected to its related AHRU. The RMM stays in the aircraft when the AHRU is removed for maintenance.

Training Information Points

The performance values that follow is applicable for 95 percent of the time when the external references used by the AHRU are available.

In straight and level flight, the Attitude and Heading Reference System (AHRS) gives attitude accuracy that is better than ± 0.5 degree.

Attitude accuracy degradation may occur after a maneuver such as a turn, acceleration, or deceleration. If the maneuver is completed in less than 3 minutes, the developed attitude error is less than ± 1.5 degrees. It goes back to nominal accuracy within 2 minutes after the completion of the manoeuvre.

Attitude error is less than 1.5 degree for 95 percent of the time when a true airspeed reference is not available.

In straight and level flight, the table that follows shows the heading performance as a function of the magnetic field dip angle of the earth:

DIP ANGLE (I) IN DEGREES	MAGNETIC HEADING ERROR IN DEGREES
Less than 71	1.5
Between 71 and 73	2.0
Between 74 and 78	2.7
Between 79 and 82	4.0

The DG SLEW mode is selected to operate the Attitude and Heading Reference System (AHRS) above an 82 degree dip angle.



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During a manoeuvre such as a turn, acceleration, or deceleration or uncoordinated flight, the flux valve (MDU 1, MDU 2) data is not used to prevent use of erroneous magnetic heading.

PSM 1-84-2A
EFFECTIVITY:
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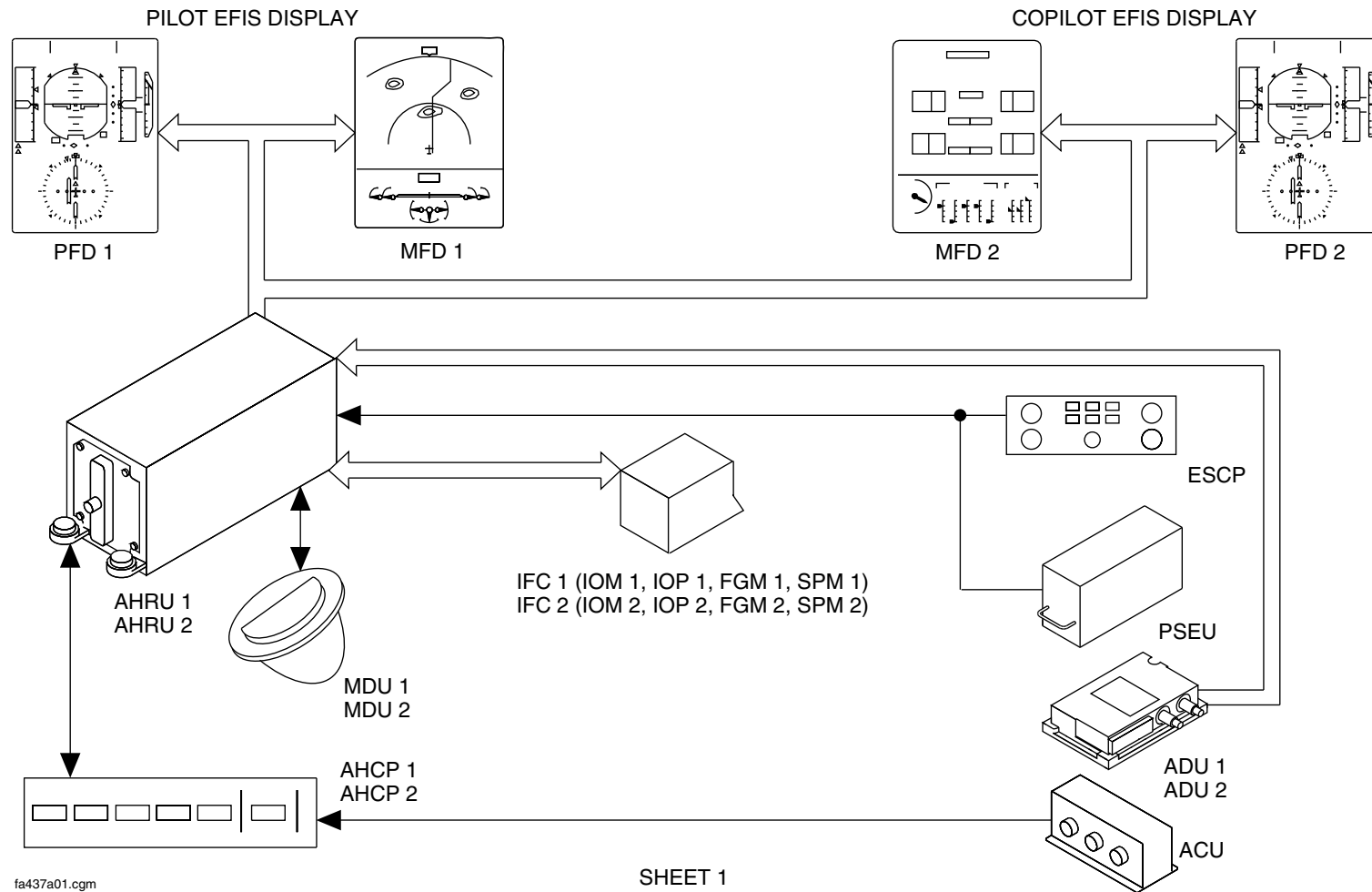
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



ATTITUDE AND HEADING REFERENCE SYSTEM BLOCK DIAGRAM
Figure 1

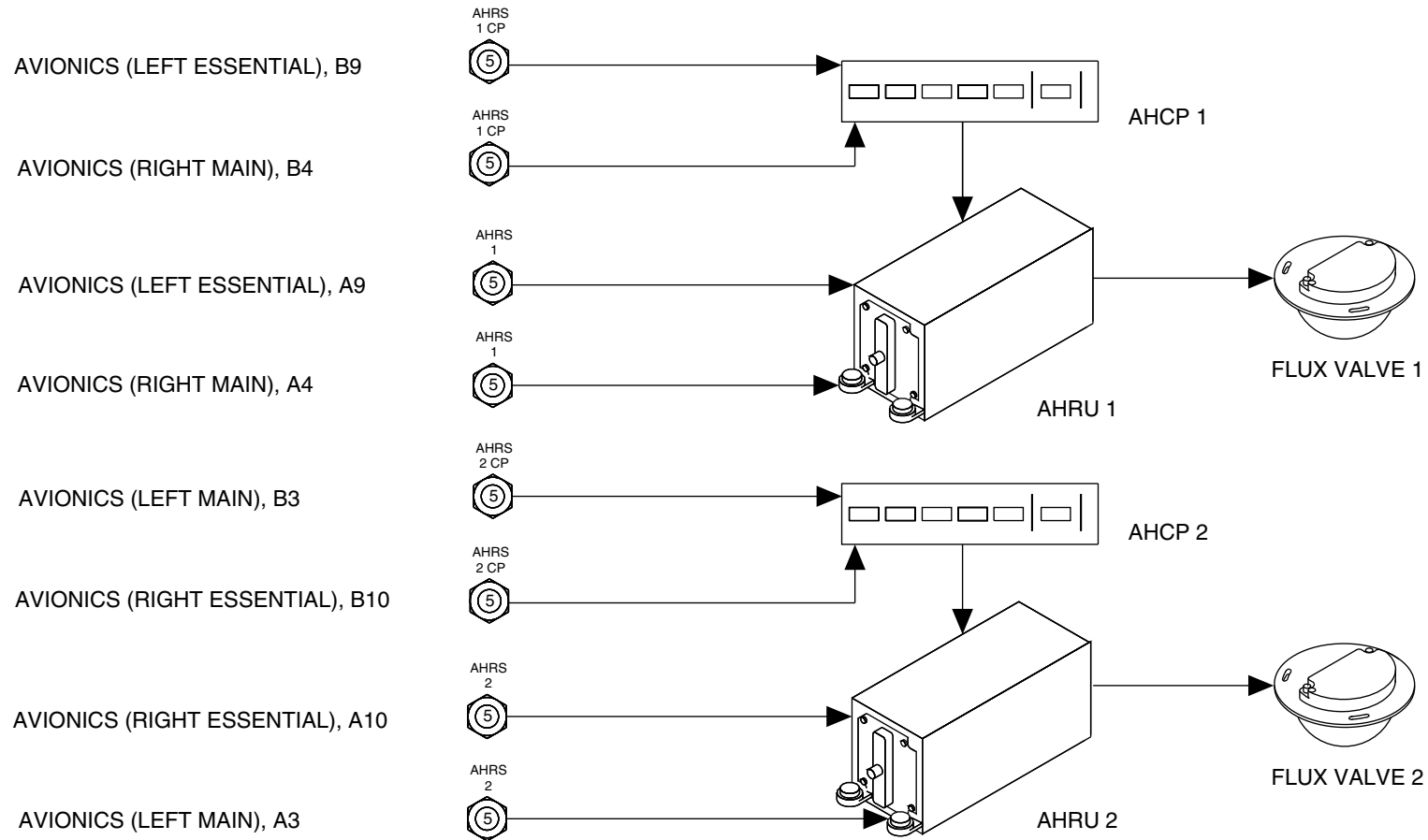
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SHEET 2

ATTITUDE AND HEADING REFERENCE SYSTEM BLOCK DIAGRAM, POWER
Figure 2

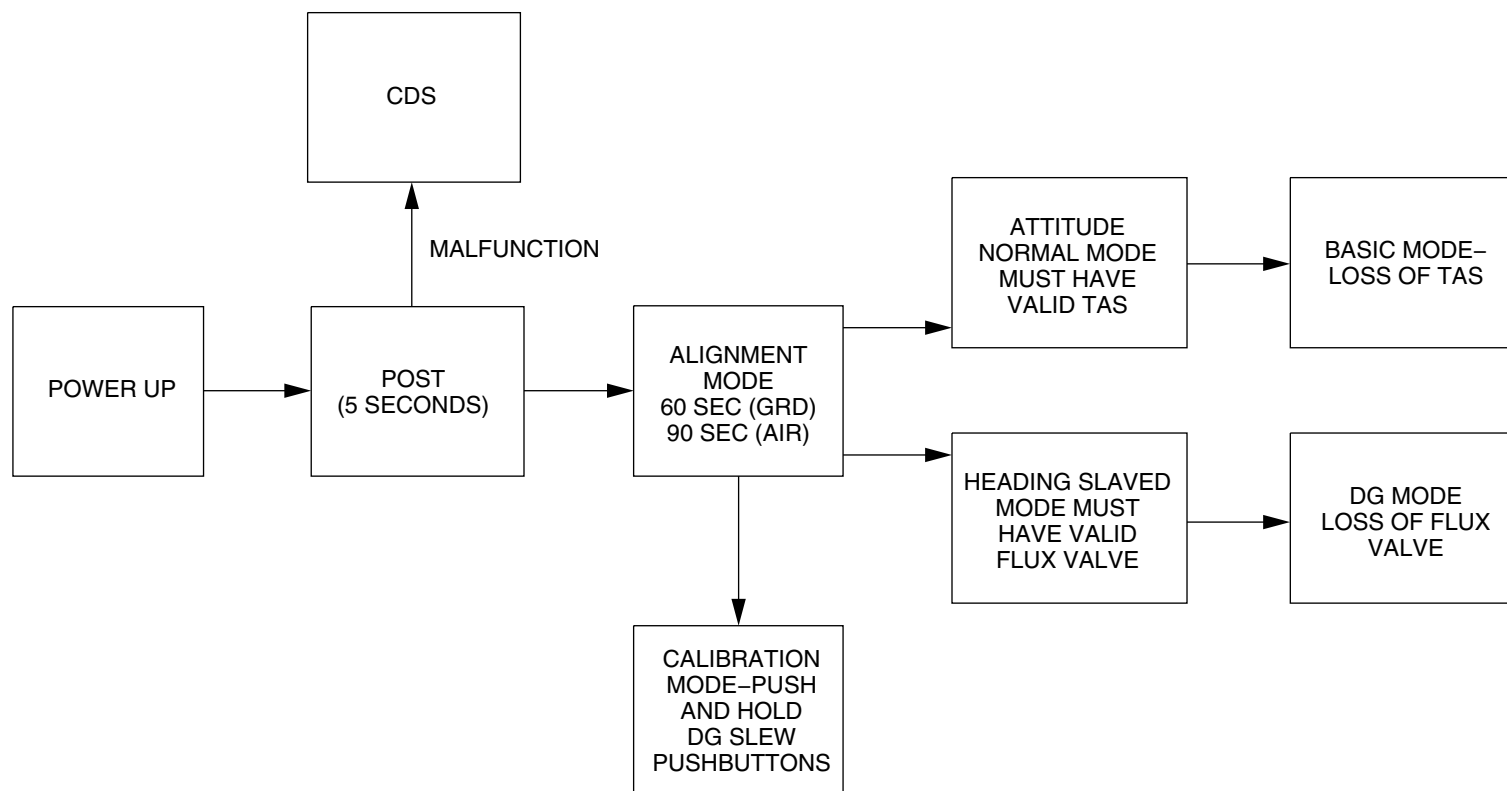
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AHRS MODES OF OPERATION
Figure 3

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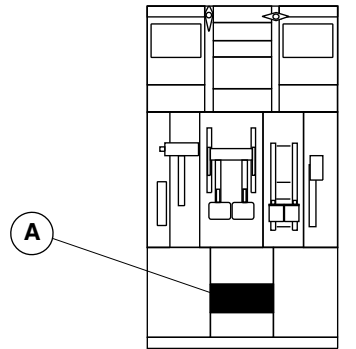
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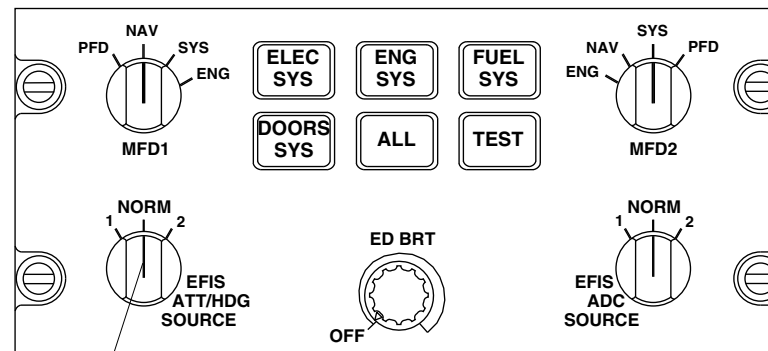
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CENTRE CONSOLE

LEGEND

1. EFIS Attitude and Heading Source Reversion Selector.



1

A

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ESCP, EFIS AHRS SOURCE REVERSION PAGE 1
Figure 4

PSM 1-84-2A
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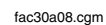
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A schematic diagram of a vehicle dashboard. It features a central instrument cluster with three circular gauges. To the left and right of the gauges are rectangular display screens. On the far left and right are black rectangular areas representing side mirrors. A callout bubble labeled 'A' points to the top center of the dashboard, above the instrument cluster.

1. EHSI.
2. EADI.
3. IVSI.



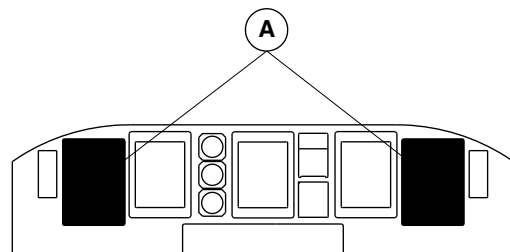
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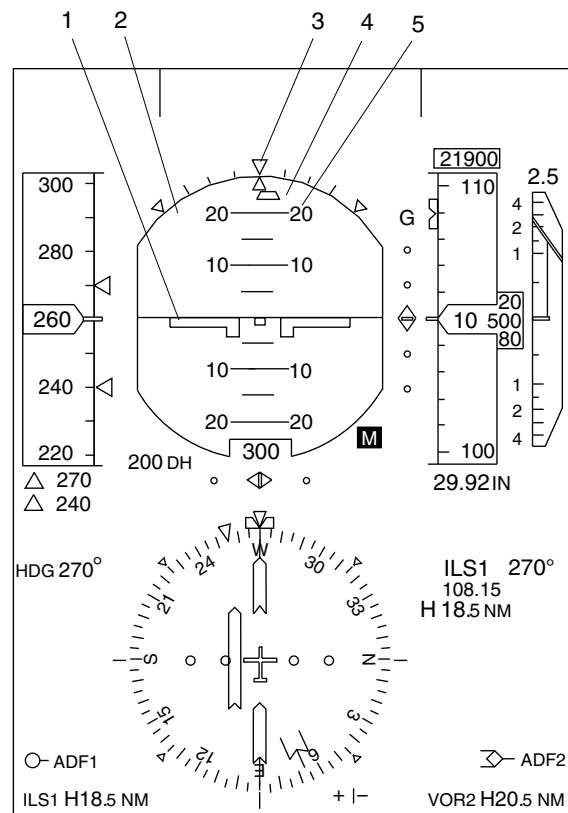
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MAIN INSTRUMENT PANEL

LEGEND

1. Airplane Symbol.
2. Attitude Sphere.
3. Roll Pointer and Scale.
4. Slip/Skid Indicator.
5. Pitch Scale.



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PFD, ATTITUDE INDICATIONS
Figure 6

PSM 1-84-2A
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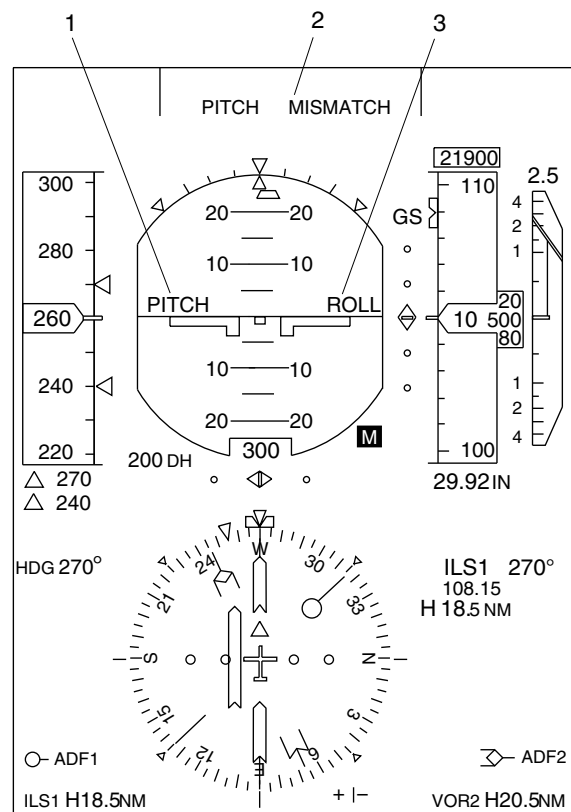
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A schematic diagram of a vehicle dashboard. The dashboard features a central instrument cluster with three circular gauges on the left and a rectangular display on the right. To the left of the instrument cluster is a large black rectangular area, and to the right is a large black rectangular area. A callout line labeled 'A' points to the instrument cluster area.

LEGEND

1. Pitch Mismatch Flag.
2. Mismatch Message.
3. Roll Mismatch Flag.



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PFD, ATTITUDE MISMATCH INDICATION
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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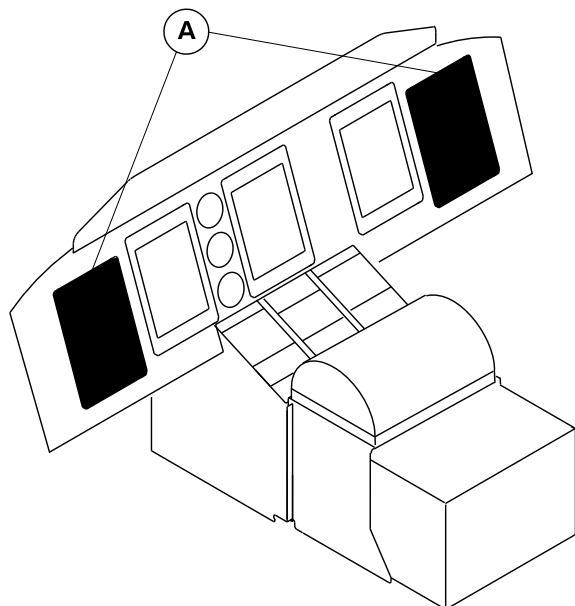
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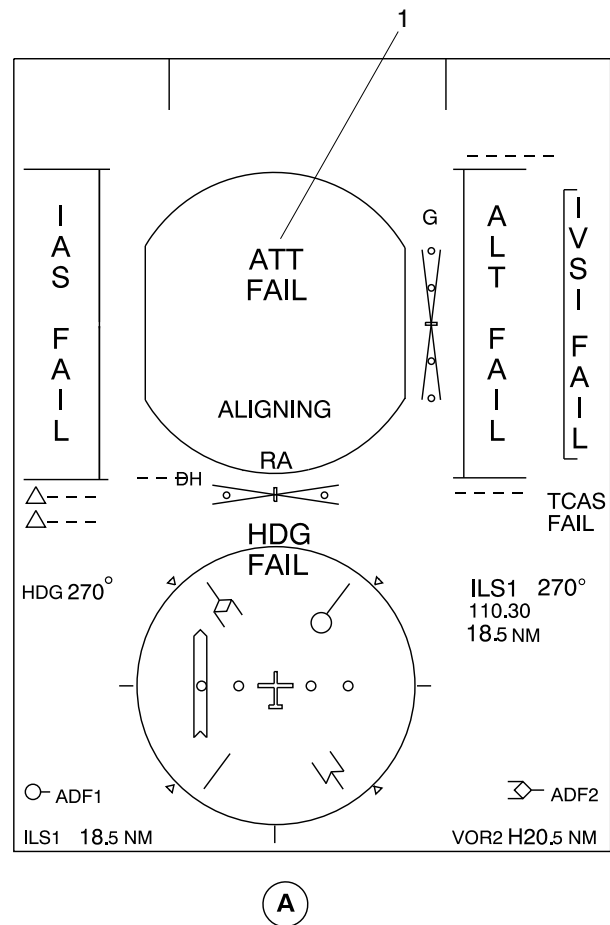
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LEGEND

1. Attitude failure flag.



fac30a04.dg, av, jun11/2011

PFD, Attitude Failure Flag
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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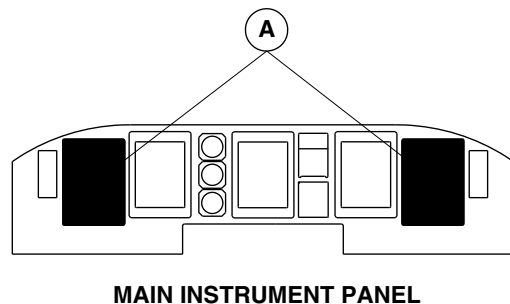
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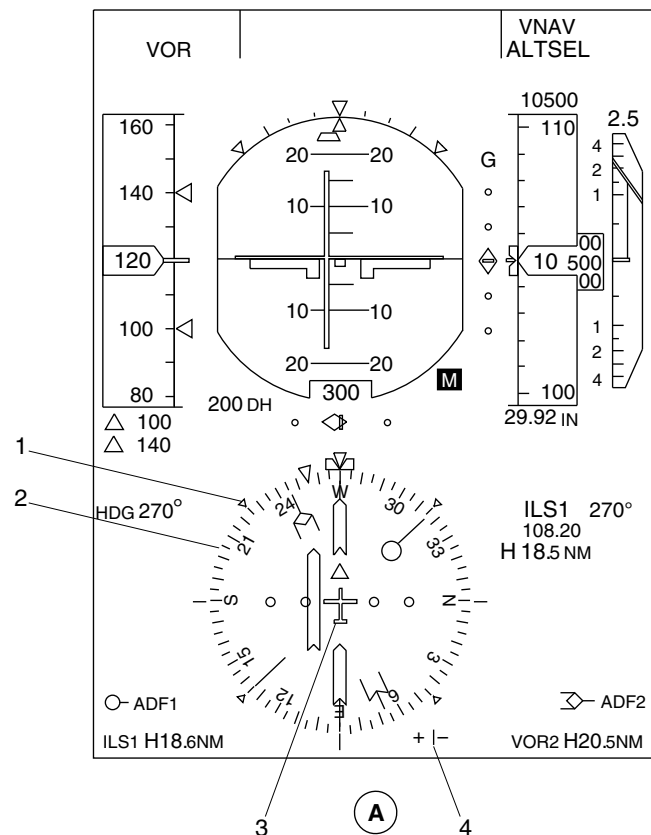
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LEGEND

1. Fixed Markings.
2. Rotating Heading Dial.
3. Actual Heading Marker and Aircraft Symbol.
4. Slaving Error Annunciator.



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PFD HEADING INDICATIONS
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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Config 001

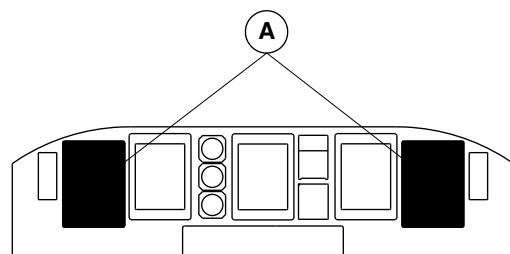
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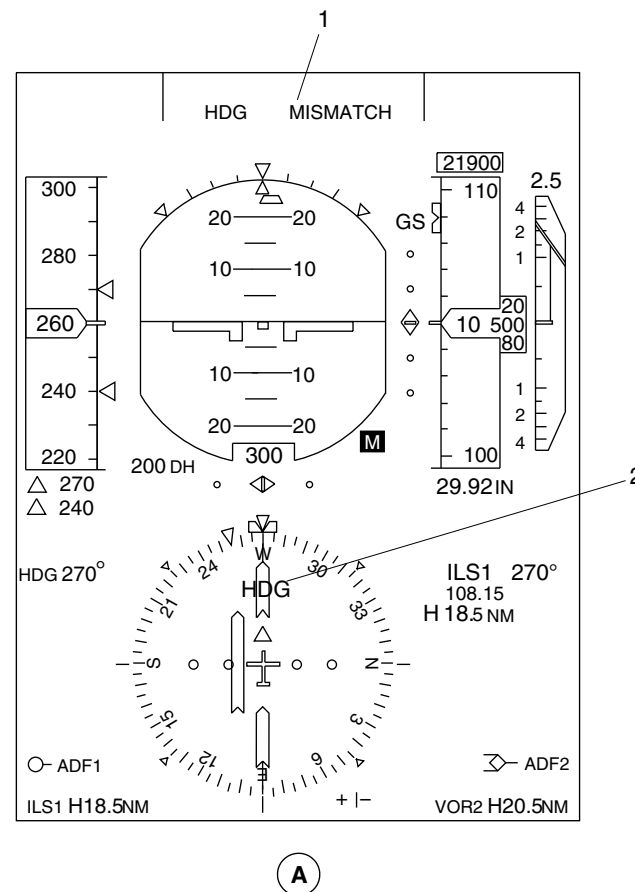
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MAIN INSTRUMENT PANEL

LEGEND

1. Mismatch Message.
2. Heading Mismatch Flag.



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PFD, HEADING MISMATCH INDICATION
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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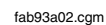
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A schematic diagram of the main instrument panel. It features a central cluster of gauges and displays, flanked by two large black rectangular areas representing the A-pillars. A circle labeled 'A' is positioned above the center, with two lines extending downwards to the top corners of the instrument panel, indicating the location of the A-pillars.

MAIN INSTRUMENT PANEL

1. Heading Failure Flag.



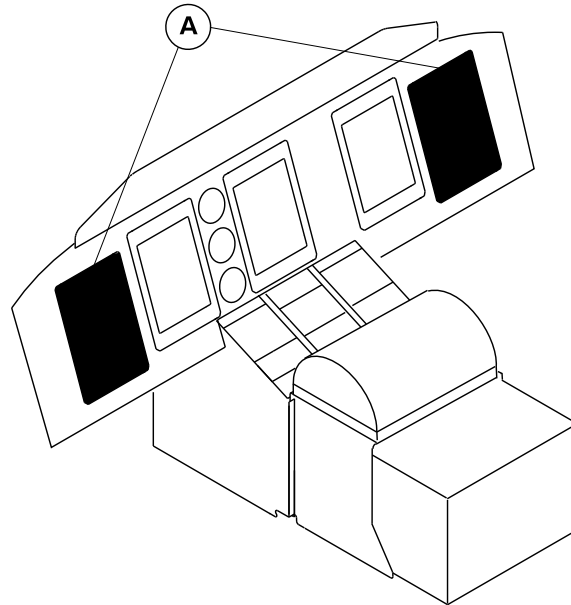
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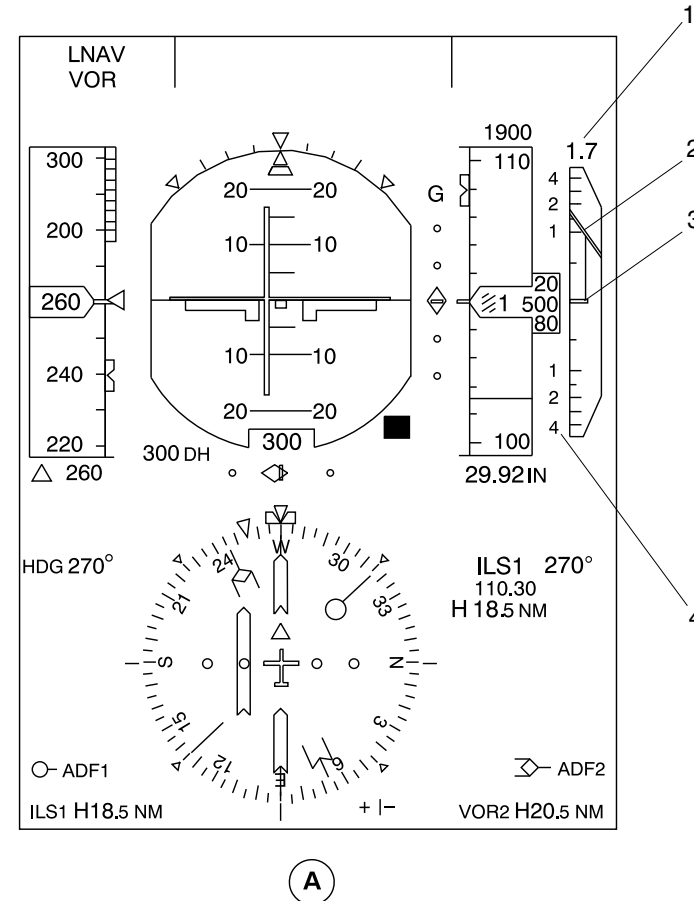
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. IVSI digital readout.
2. IVSI pointer.
3. IVSI zero reference line.
4. IVSI scale.



fac30a06.dg, av, jun11/2011

PFD, Inertial Vertical Speed Indicator Indications
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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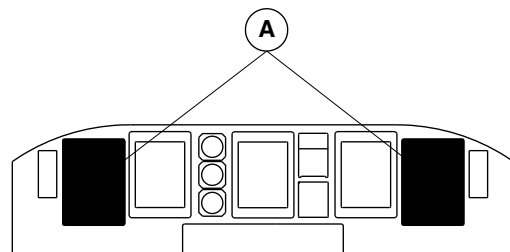
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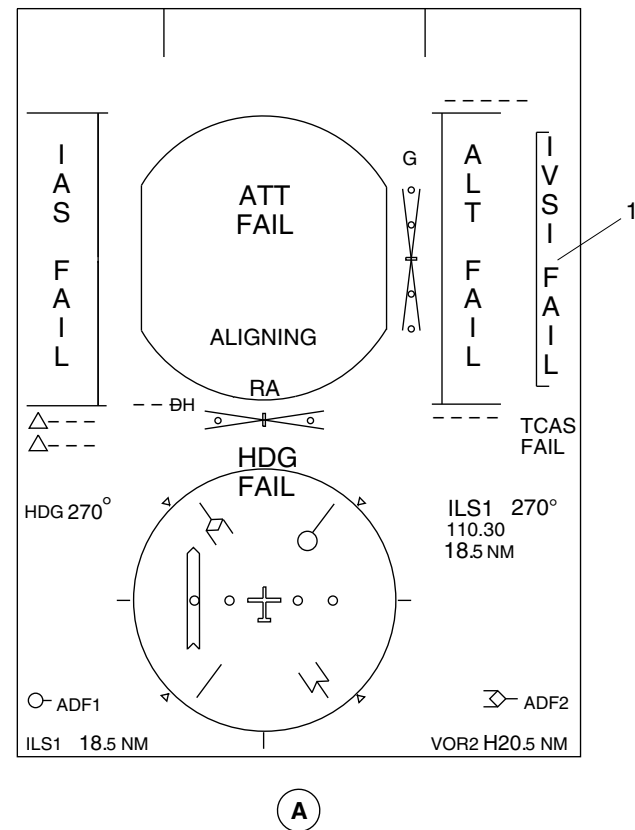
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. IVSI Failure Flag.



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PFD, INERTIAL VERTICAL SPEED INDICATOR FAILURE FLAG
Figure 13

PSM 1-84-2A
EFFECTIVITY:
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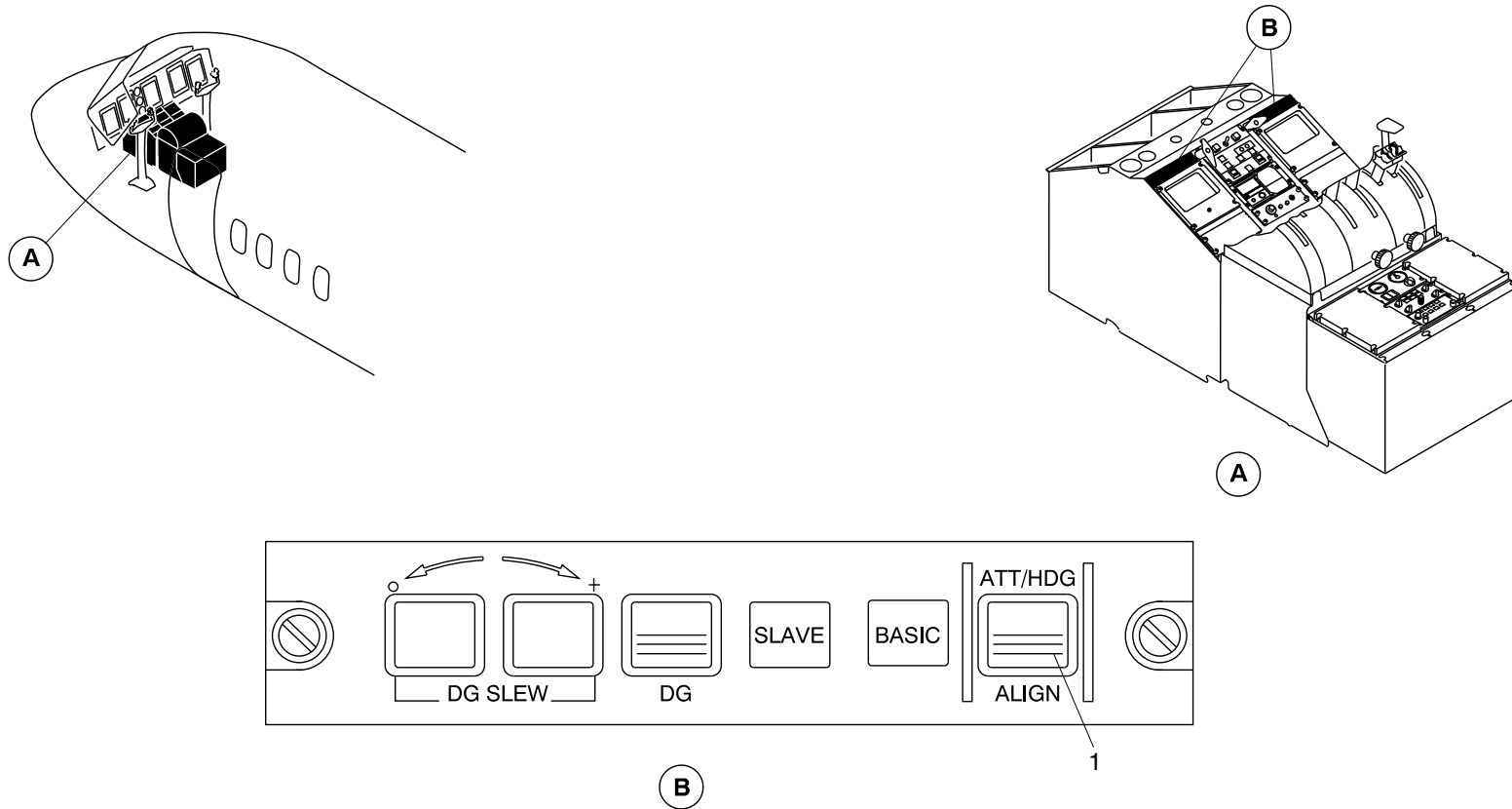
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. ALIGN annunciator switch.

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AHCP, Alignment Mode Indications
Figure 14

PSM 1-84-2A
EFFECTIVITY:
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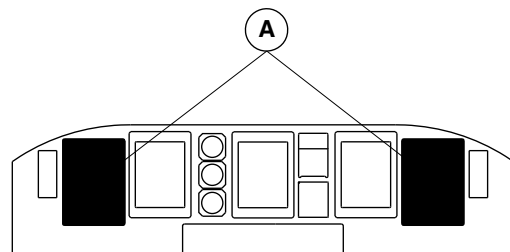
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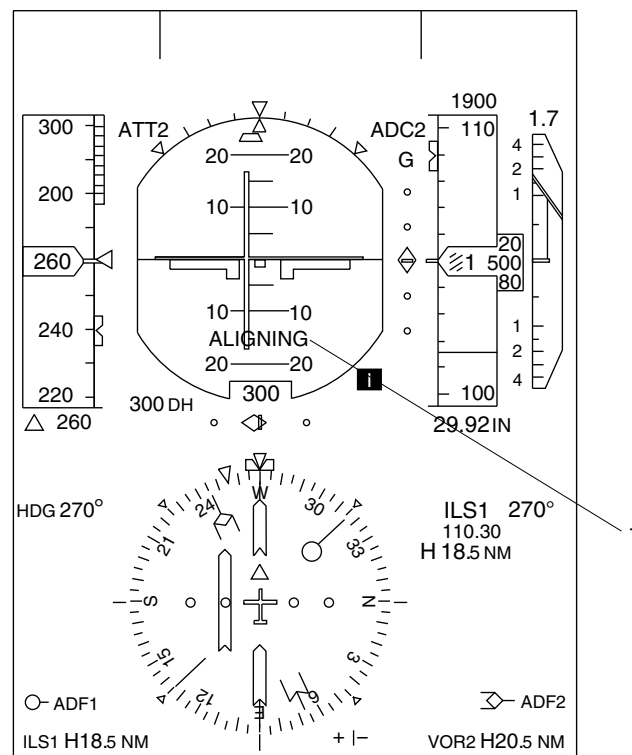
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. AHRS Aligning Annunciator.



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PFD, ALIGNMENT MODE INDICATIONS
Figure 15

PSM 1-84-2A
EFFECTIVITY:
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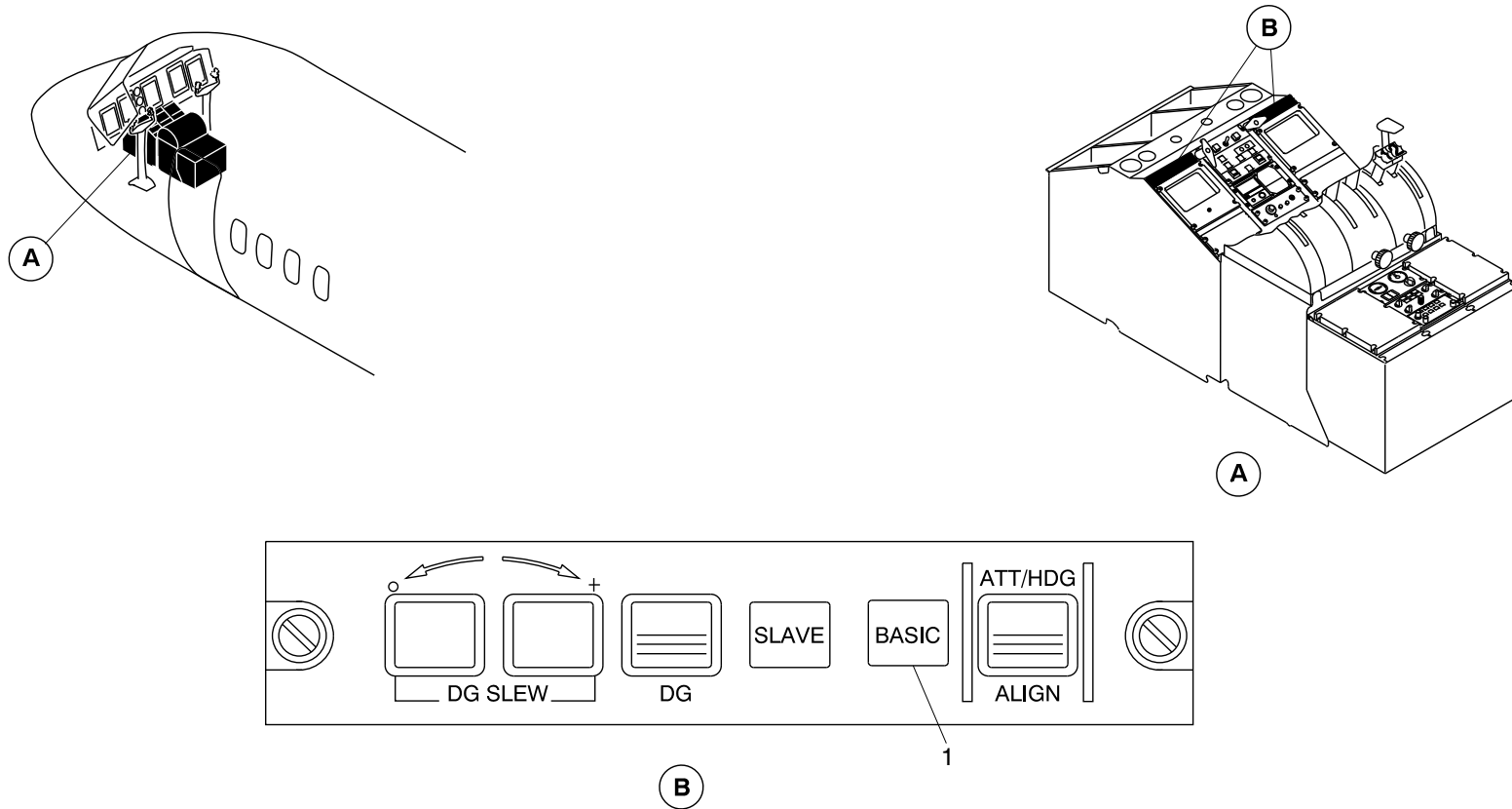
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. BASIC annunciator light.

fsc99a02.dg, av, jun11/2011

AHCP, Basic Mode Indication
Figure 16

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-21-00
Config 001

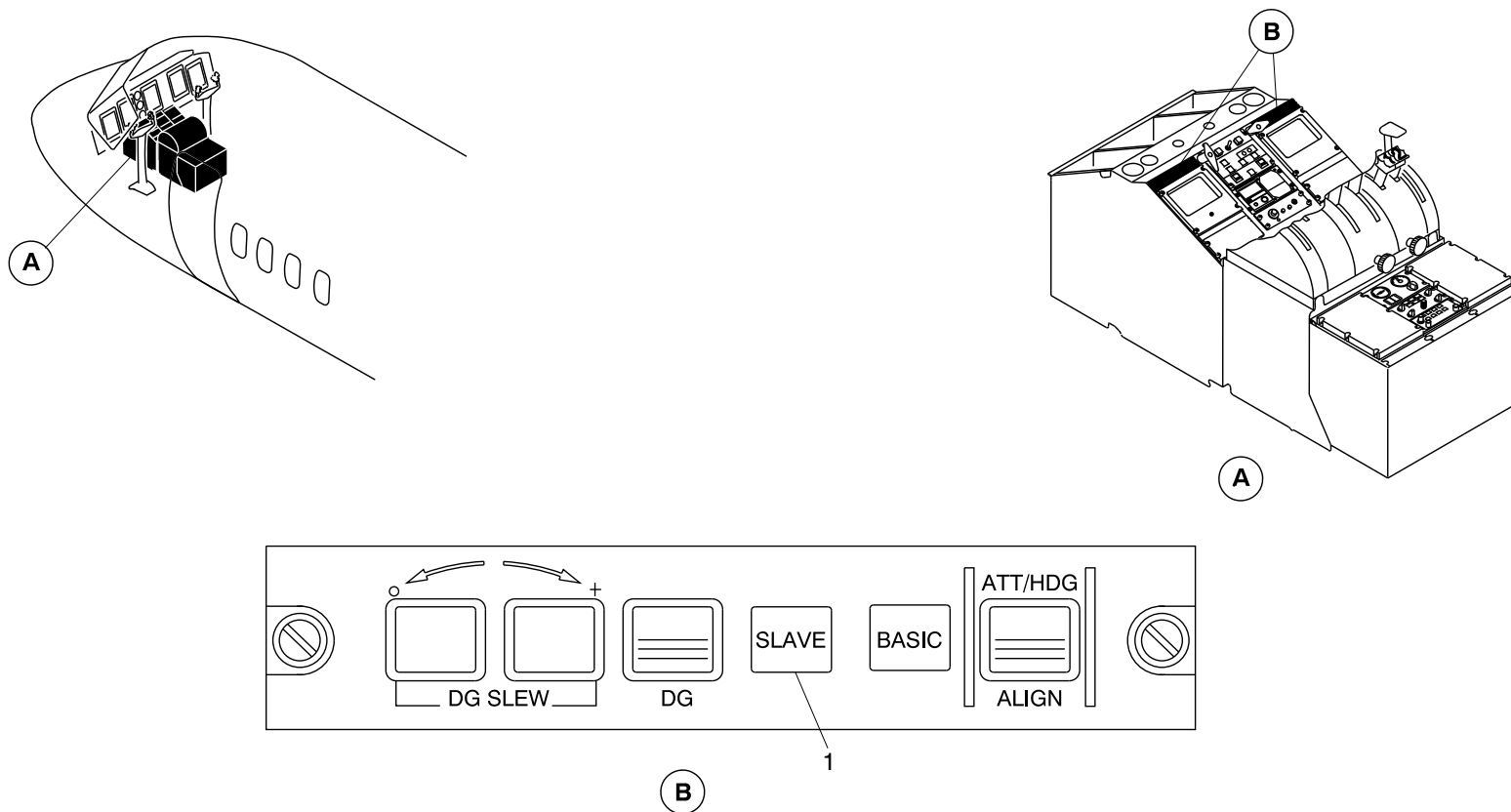
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. SLAVE annunciator light.

fsc99a03.dg, av, jun11/2011

AHCP, Slave Indication
Figure 17

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-21-00
Config 001

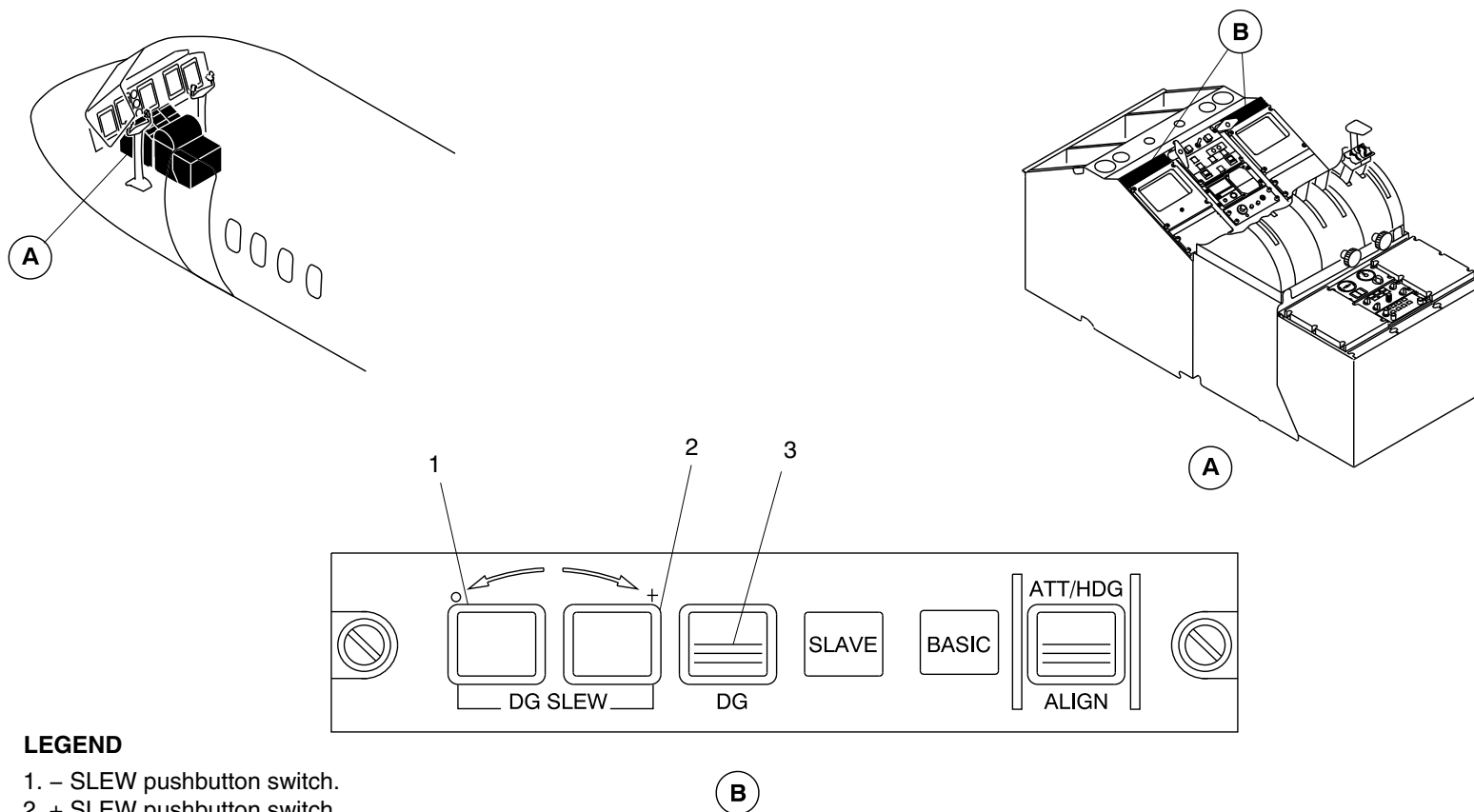
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. – SLEW pushbutton switch.
- 2. + SLEW pushbutton switch.
- 3. DG annunciator switch.

fsc99a04.dg, av, jun11/2011

AHCP, DG Mode Selection
Figure 18

PSM 1-84-2A
EFFECTIVITY:
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A schematic diagram of a vehicle dashboard. The dashboard features a central instrument cluster with three circular gauges on the left and a rectangular display on the right. To the left of the instrument cluster is a large black rectangular area, and to the right is a large black rectangular area. A callout line labeled 'A' points to the instrument cluster area.

LEGEND

110
140
120
100
80

△ 138
△ 117

200 DH

300

M

10500
110
2.5
4
2
1
100

DG

HDG 270°

ADF1

ADF2

VOR2 H20.5 NM

ILS1 H18.5 NM

ILS1 270°
110.30
H 18.5 NM

PFD, DG MODE INDICATION
Figure 19

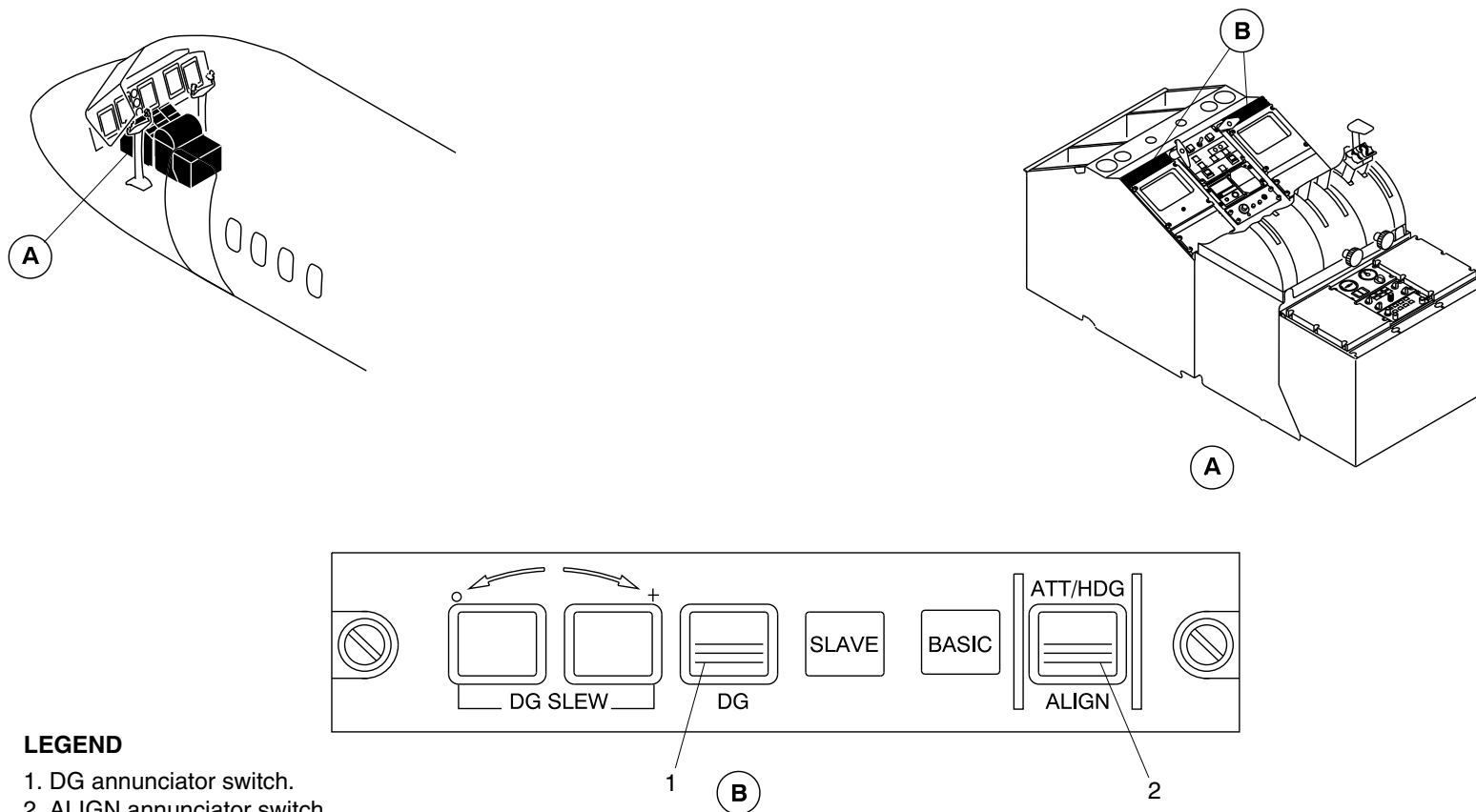
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LEGEND

1. DG annunciator switch.
2. ALIGN annunciator switch.

fsc99a05.dg, av, jun11/2011

AHCP, Calibration Mode Selection
Figure 20

PSM 1-84-2A
EFFECTIVITY:
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A schematic diagram of a vehicle dashboard. The dashboard features a central instrument cluster with three circular gauges on the left and a rectangular display on the right. To the left of the instrument cluster is a large black rectangular area, and to the right is a large black rectangular area. A callout line labeled 'A' points to the instrument cluster area.

LEGEND

-
- The instrument panel displays the following instruments and information:
- Altimeter (Top Left):** Shows altitude in feet. The scale ranges from 80 to 110. The current reading is 120. The altimeter is set to 10500.
 - Heading Indicator (Top Center):** A circular instrument showing heading. The scale ranges from 0 to 360. The current heading is 270. The heading is set to 270.
 - Attitude Indicator (Top Right):** Shows pitch and roll. The pitch scale ranges from 0 to 2.5. The current pitch is 1.5. The attitude is set to 10500.
 - Speedometer (Bottom Left):** Shows airspeed. The scale ranges from 0 to 100. The current speed is 100. The speed is set to 100.
 - Heading Indicator (Bottom Center):** A circular instrument showing heading. The scale ranges from 0 to 360. The current heading is 270. The heading is set to 270.
 - Heading Indicator (Bottom Right):** A circular instrument showing heading. The scale ranges from 0 to 360. The current heading is 270. The heading is set to 270.
 - Heading Indicator (Bottom Left):** A circular instrument showing heading. The scale ranges from 0 to 360. The current heading is 270. The heading is set to 270.
 - Heading Indicator (Bottom Right):** A circular instrument showing heading. The scale ranges from 0 to 360. The current heading is 270. The heading is set to 270.

PFD, CALIBRATION MODE INDICATION
Figure 21

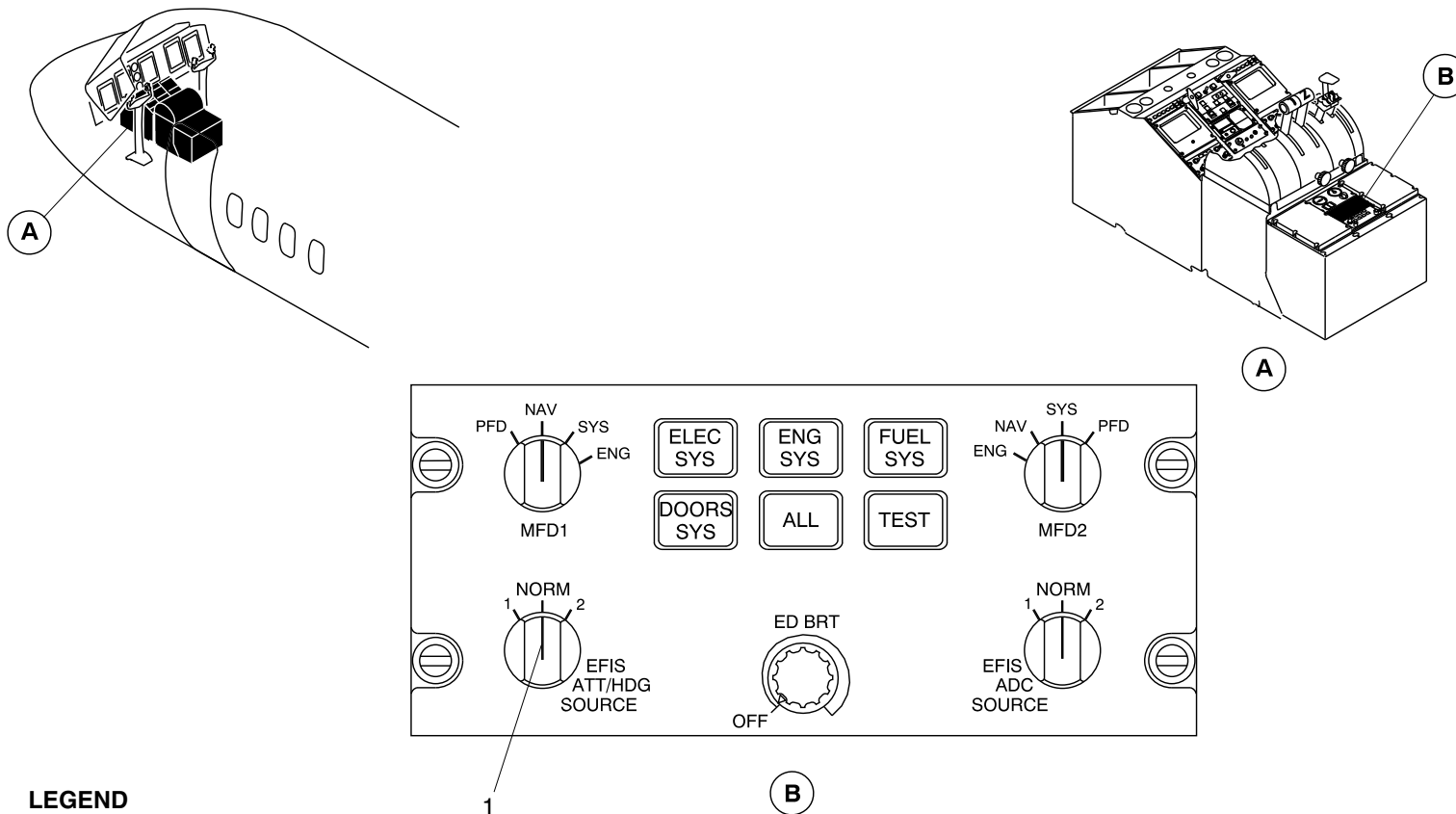
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LEGEND

1. EFIS attitude and heading source reversion selector.

fsc95a01.dg, av, jul11/2011

ESCP, EFIS AHRs Source Reversion Selection
Figure 22

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-21-00
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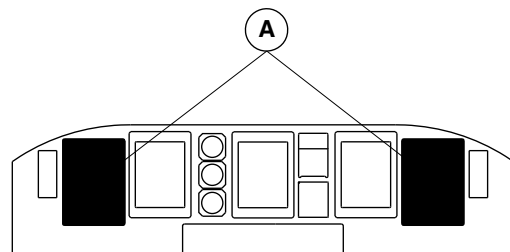
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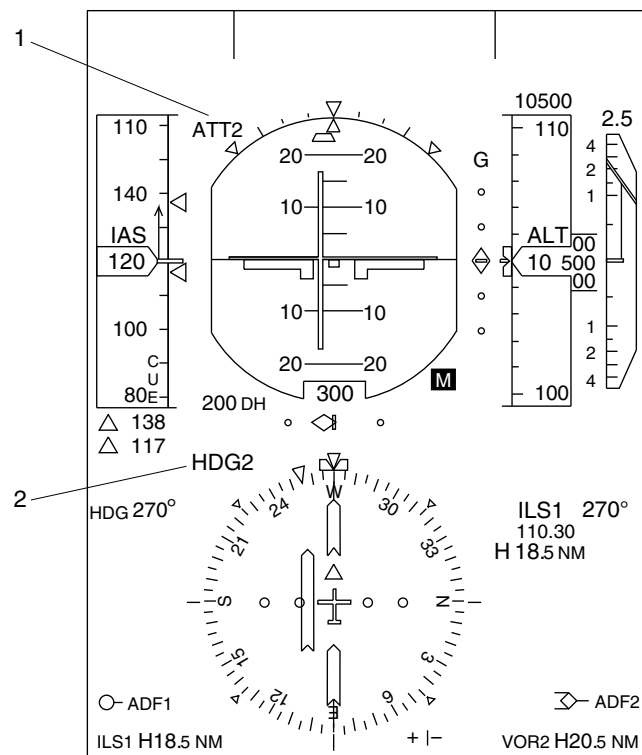
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MAIN INSTRUMENT PANEL

LEGEND

1. Attitude Source Annunciation.
2. Heading Source Annunciation.



fac30a03.cgm

PFD, EFIS ATT/HDG SOURCE REVERSION INDICATION
Figure 23

PSM 1-84-2A
EFFECTIVITY:
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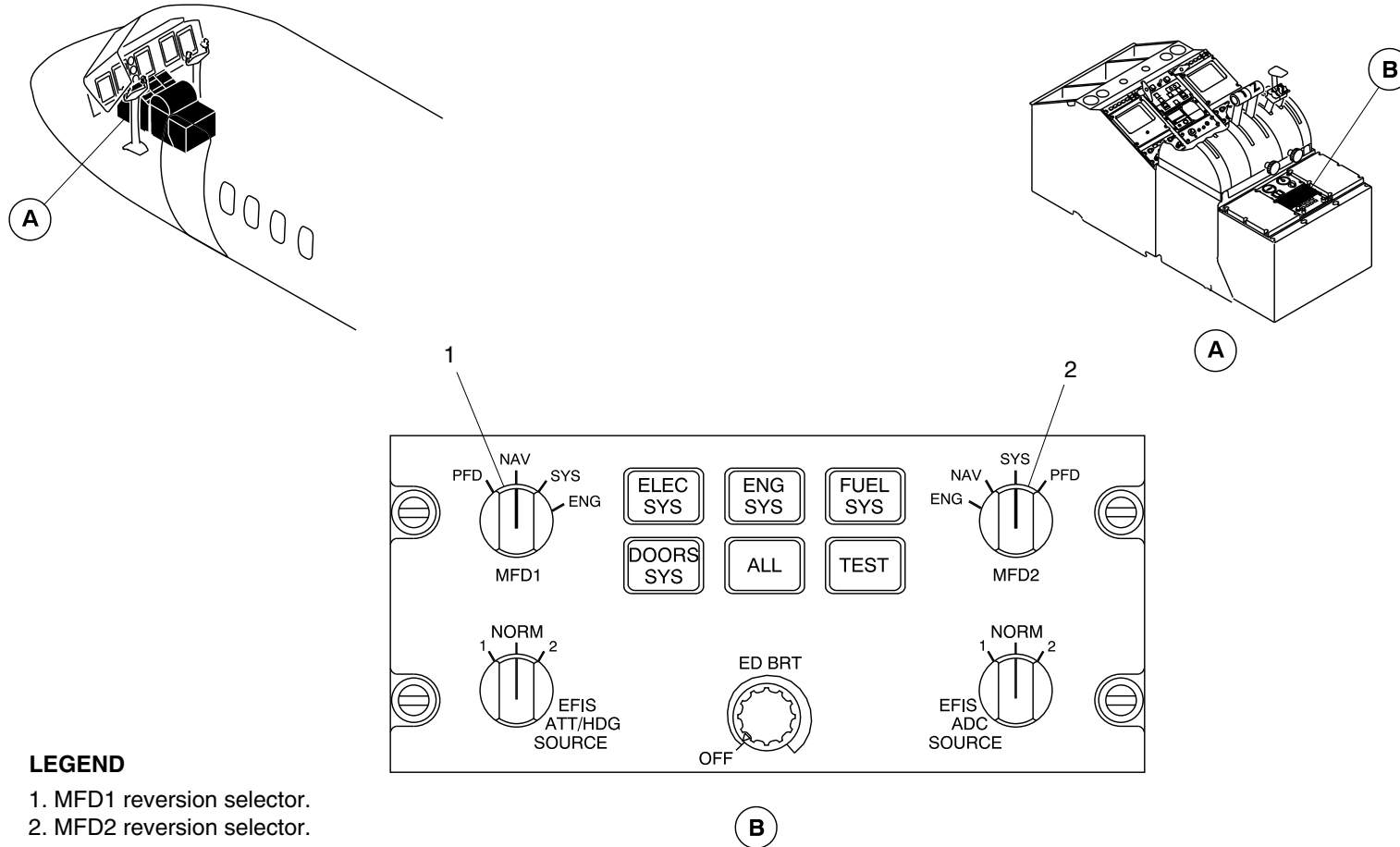
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fs750a01.dg, av, jul11/2011

ESCP, MFD Reversion Selection
Figure 24

PSM 1-84-2A
EFFECTIVITY:
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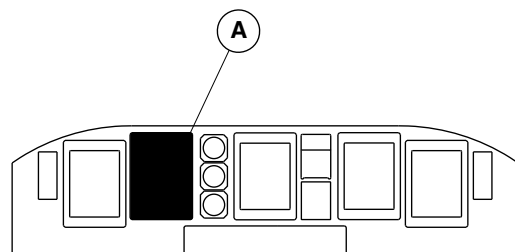
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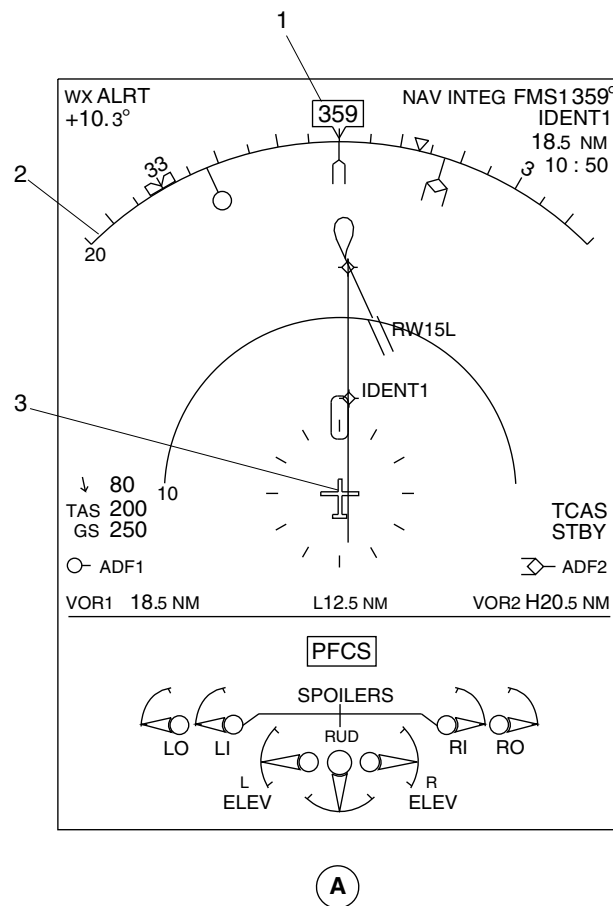
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MAIN INSTRUMENT PANEL

LEGEND

1. Actual Heading Digital Value and Reference.
2. Heading Scale in Arc Format.
3. Aircraft Symbol.



fs752a02.cgm

MFD, ARC FORMAT INDICATION
Figure 25

PSM 1-84-2A
EFFECTIVITY:
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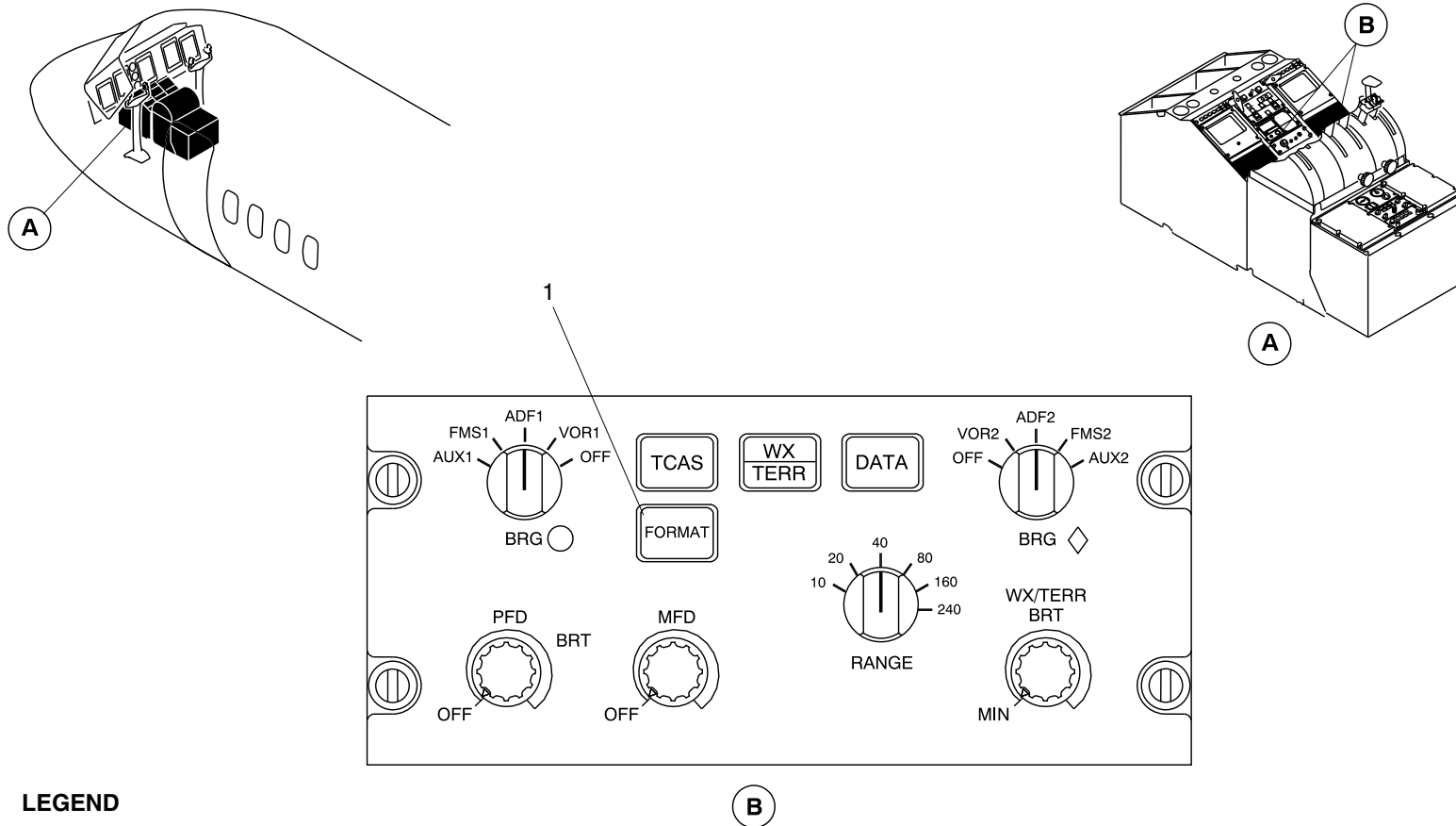
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LEGEND

1. FORMAT pushbutton switch.

fa786a01.dg, av, jul11/2011

ESCP, Plan Format Selection
Figure 26

PSM 1-84-2A
EFFECTIVITY:
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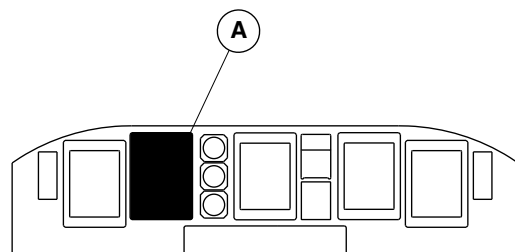
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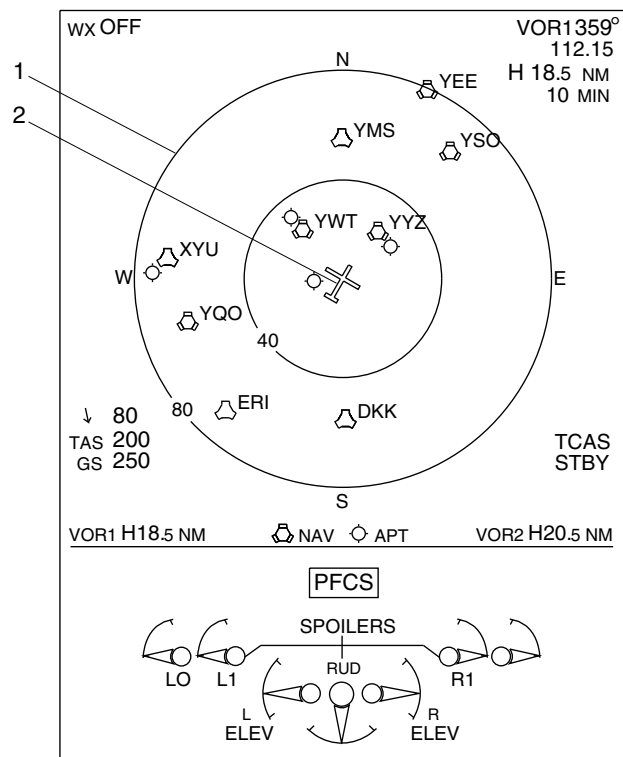
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MAIN INSTRUMENT PANEL

LEGEND

1. Fixed Heading Indication.
2. Aircraft Symbol.



A

fa759a02.cgm

MFD, PLAN FORMAT INDICATION
Figure 27

PSM 1-84-2A
EFFECTIVITY:
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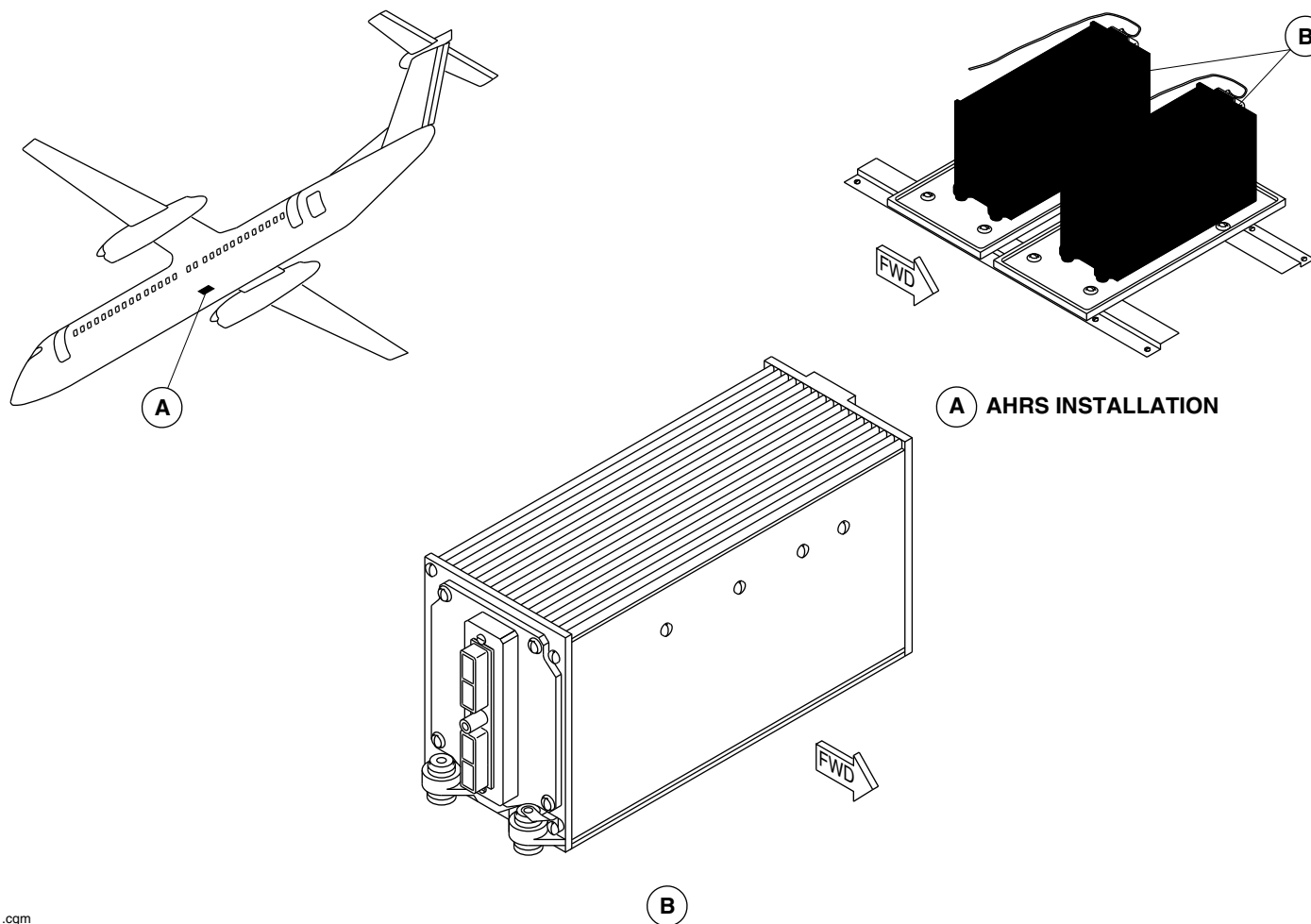
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fsd03a01.cgm

ATTITUDE AND HEADING REFERENCE UNITS LOCATOR
Figure 28

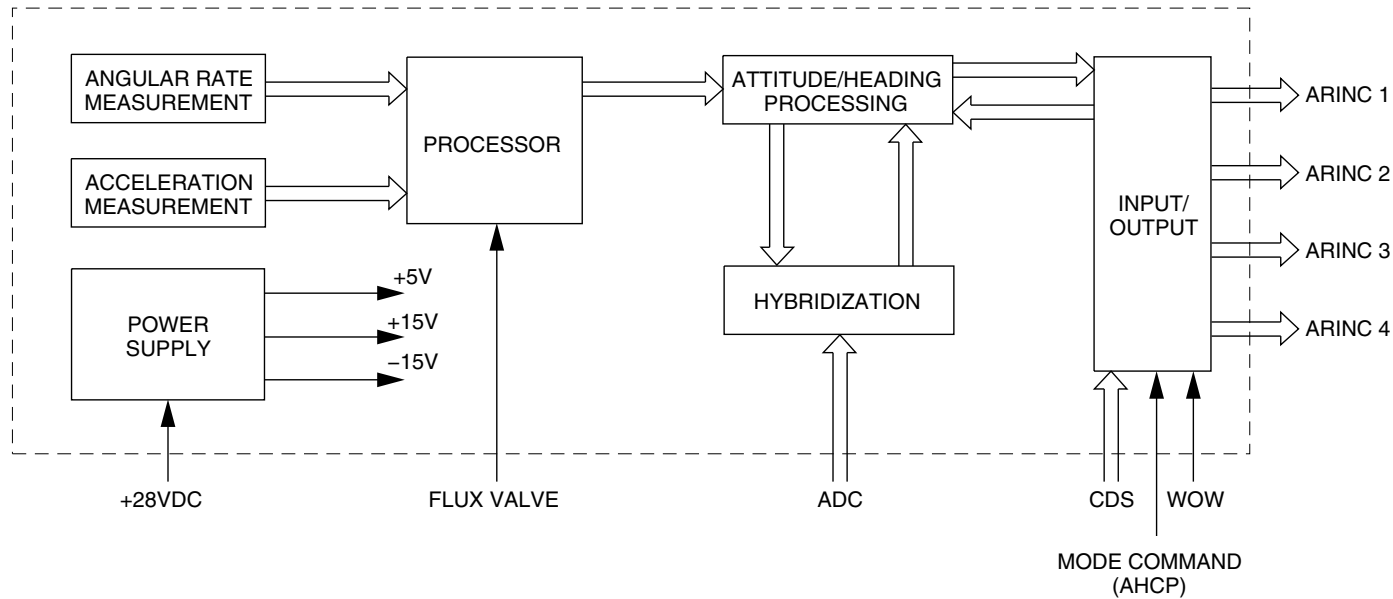
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fsu89a01.cgm

ATTITUDE AND HEADING REFERENCE INTERNAL BLOCK DIAGRAM
Figure 29

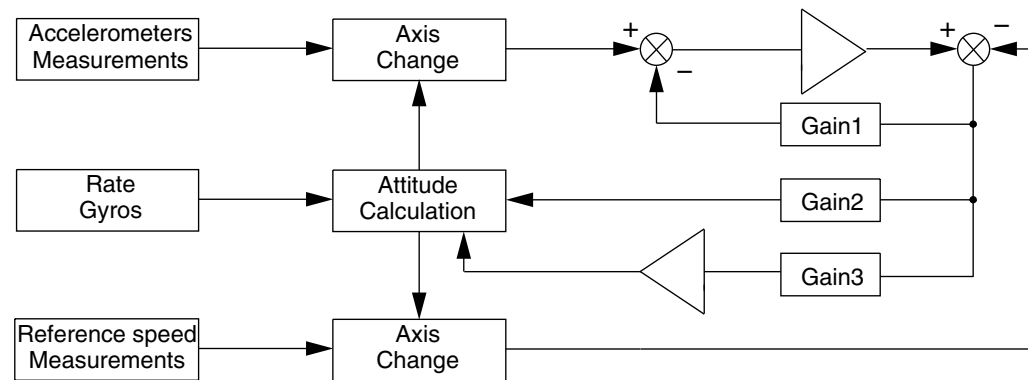
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fss70a01.cgm

ATTITUDE CALCULATION PRINCIPLE
Figure 30

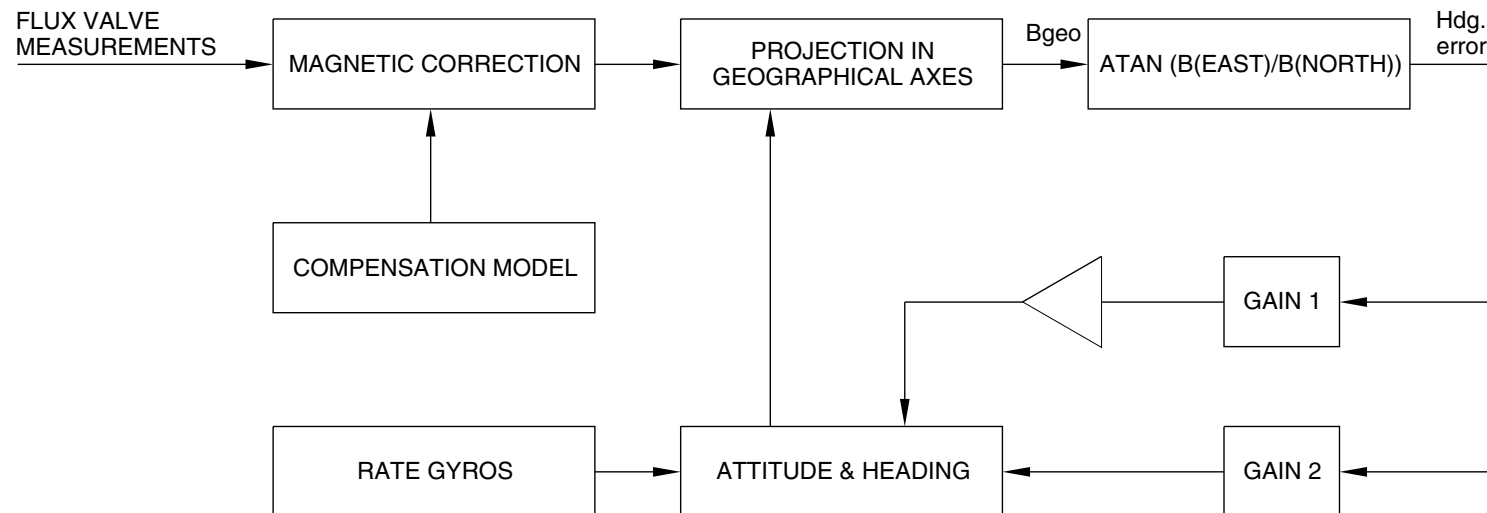
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fss81a01.cgm

HEADING CALCULATION PRINCIPLE
Figure 31

PSM 1-84-2A
EFFECTIVITY:
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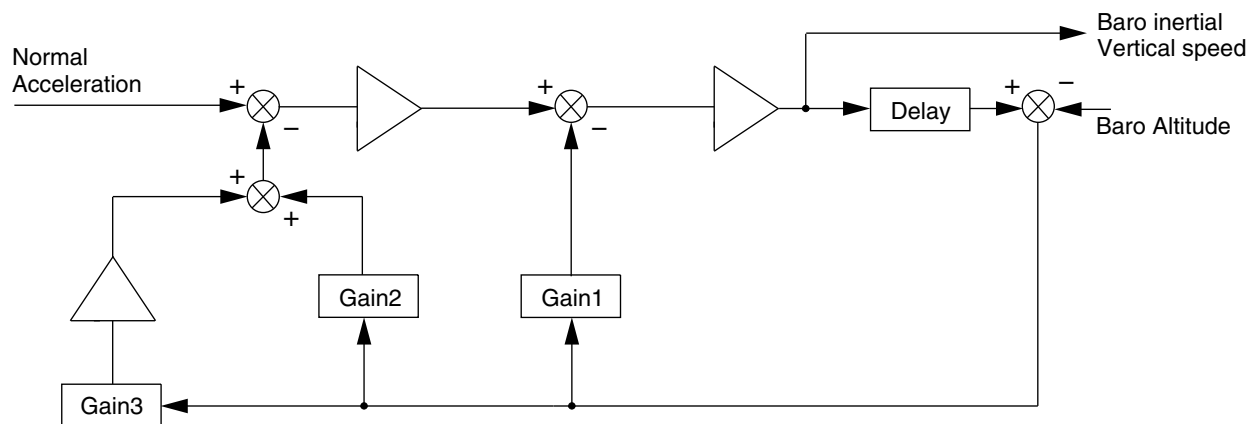
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fss71a01.cgm

VERTICAL SPEED CALCULATION PRINCIPLE
Figure 32

PSM 1-84-2A
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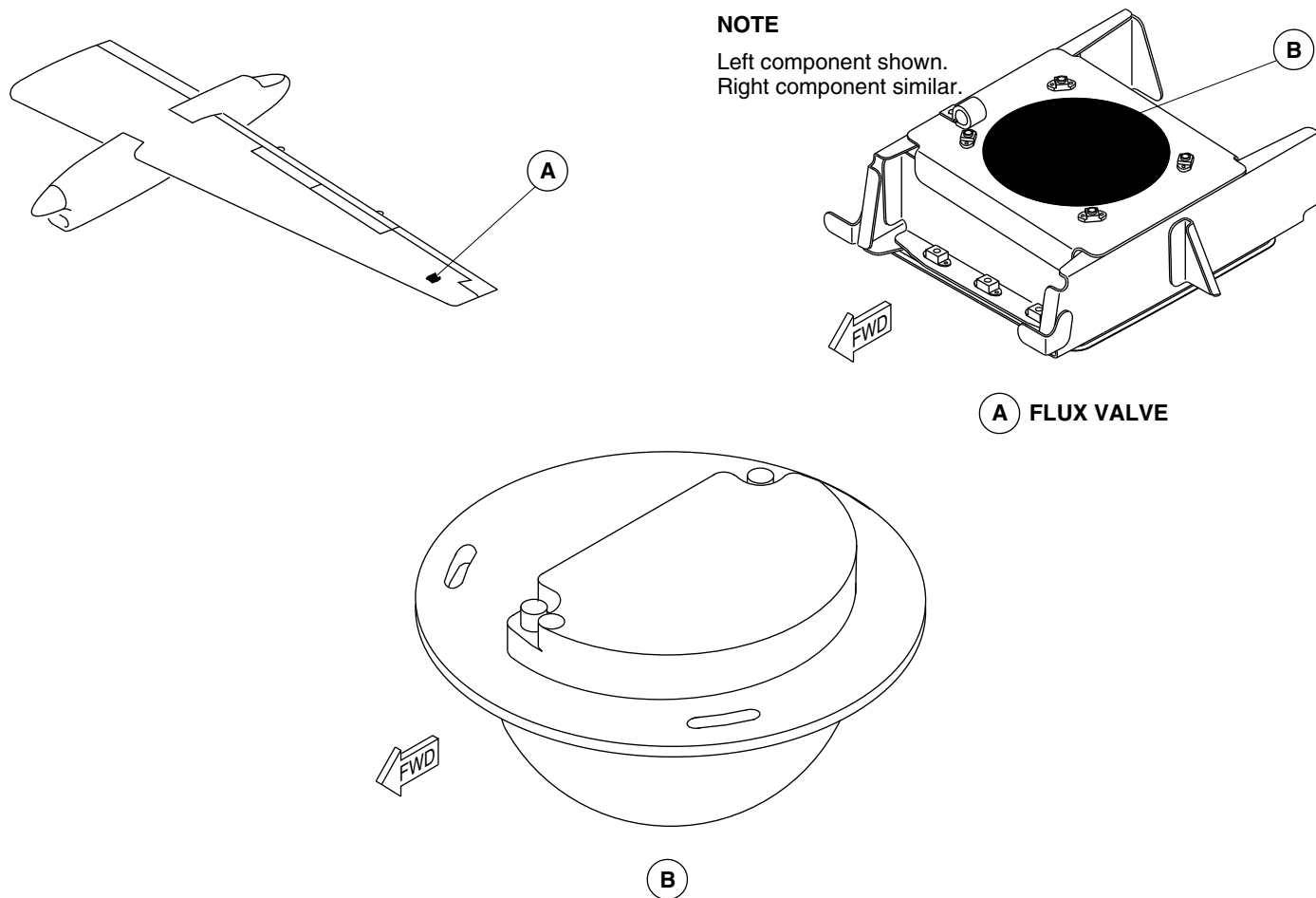
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fsb92a01.cgm

FLUX VALVES LOCATOR
Figure 33

PSM 1-84-2A
EFFECTIVITY:
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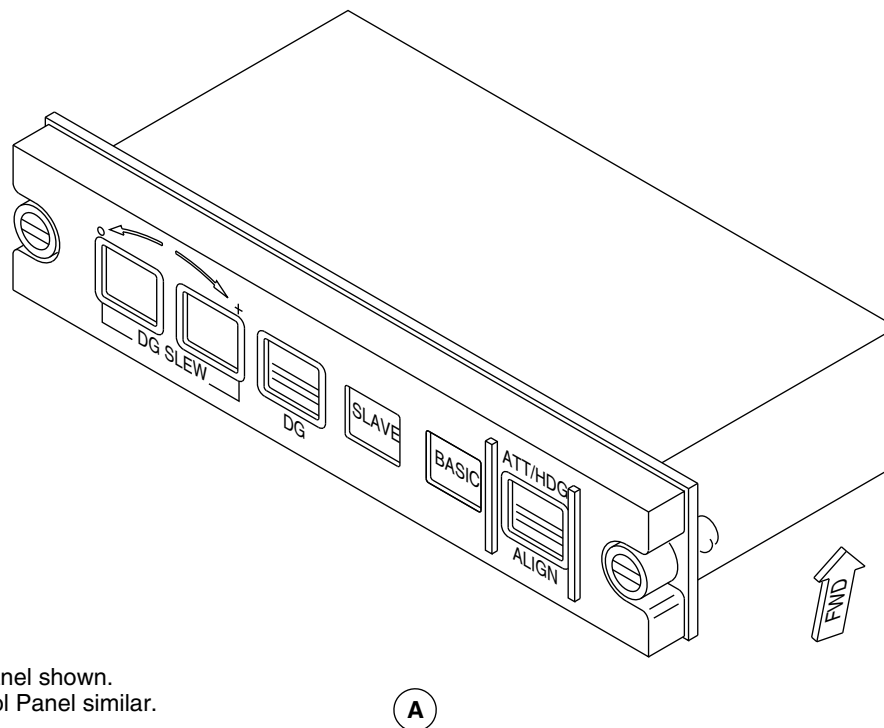
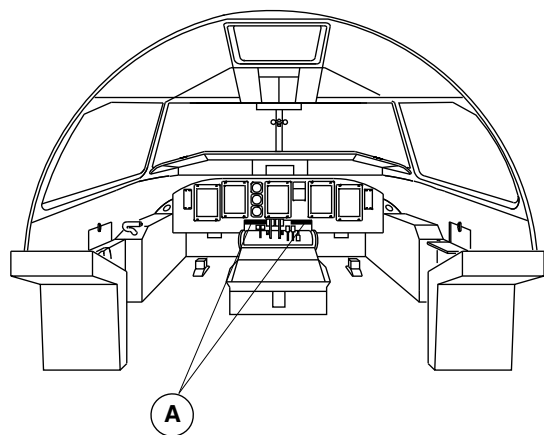
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NOTE

Pilot Control Panel shown.
Co-Pilot Control Panel similar.

fsd02a01.cgm

ATTITUDE AND HEADING REFERENCE UNIT CONTROL PANELS LOCATOR
Figure 34

PSM 1-84-2A
EFFECTIVITY:
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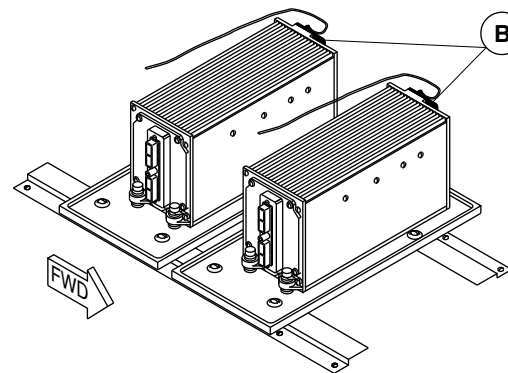
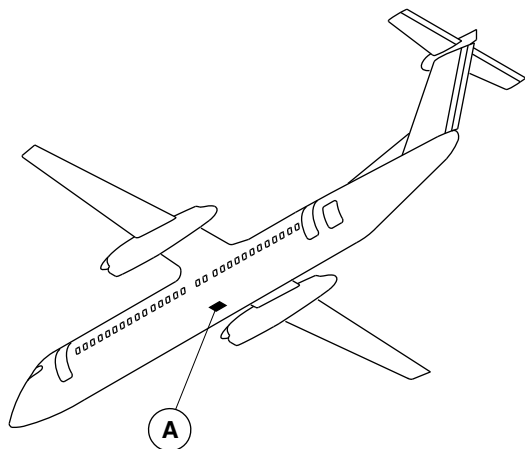
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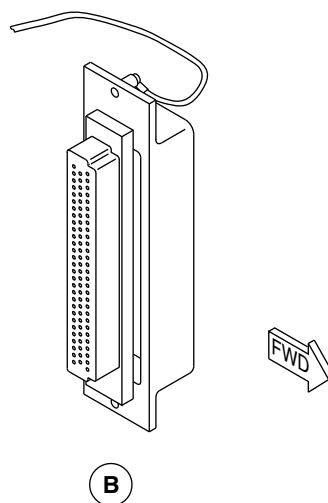


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A AHRS INSTALLATION



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REMOTE MEMORY MODULES LOCATOR
Figure 35

PSM 1-84-2A
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**ON A/C ALL

34-22-00-001

STANDBY ATTITUDE AND HEADING INSTRUMENTS

Introduction

The standby attitude and heading reference system shows attitude and heading information.

General Description

The standby instruments give pitch, roll, and heading indications.

A standby horizon is used to give pitch and roll attitude indications from an electro-mechanical gyroscope.

A standby magnetic compass is used to give a magnetic heading indication.

The standby units operate independently and do not interface with any other systems.

The Standby Attitude and Heading Instrument system has the components that follow:

- Indicator, Standby Horizon (34-22-01)
- Compass, Standby (33-22-06).

Detailed Description

The standby horizon indicator has a vertical gyroscope that spins to remain fixed in space. It remains stable with respect to the

movements of the aircraft to give an attitude indication. The standby horizon indicator has an attitude sphere with a pitch scale and roll index that moves relative to a fixed airplane symbol and roll scale.

[Refer to Figure 1.](#)

The standby horizon indicator shows the components that follow:

- Attitude sphere
- Pitch scale
- Index pointer and roll scale
- Airplane symbol
- Attitude failure flag.

Attitude Sphere: The attitude sphere has a sky permanent sector in blue above a ground permanent sector in brown and a white horizon line to divide the sectors. The attitude sphere moves to show pitch and roll.

Pitch Scale: The pitch scale is shown on the attitude sphere in 5 degree increments from 0 to ± 80 degrees.

Index Pointer and Roll Scale: A blue index pointer moves against a white scale that has 10 degree graduations between ± 60 degrees.

Airplane Symbol: The orange airplane symbol is a fixed aircraft reference against the attitude sphere shows the amount of aircraft pitch and roll.

Attitude Failure Flag: The instrument monitors its gyroscope for speed and its power supply for overcurrent and overheating. The indicator fail warning flag comes into view when the gyroscope speed is less than 18,000 revolutions per minute (rpm) or when a power supply malfunction is sensed. The fail warning indication is an orange flag with white cross-hatches.

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The fast erection mechanism is pulled and slowly released to stabilize the gyroscope in the vertical position after power up. The knob is located at the lower right hand corner of the instrument.

The left and right essential busses supply 28 VDC electrical power through two 3 A circuit breakers to the standby horizon indicator. The circuit breakers are located in positions H2 on the left circuit breaker panel and A11 on the avionics circuit breaker panel.

The standby horizon indicator uses 1.5 A at power up and it decreases to less than 0.3 A after 3 minutes.

[Refer to Figure 2.](#)

The standby compass shows the heading of the aircraft. It uses a compass card that is free to turn against a fixed lubber line. The compass card is marked in 10 degree graduations with a digital value every 30 degrees.

Indicator, Standby Horizon

[Refer to Figure 3.](#)

The standby horizon indicator has a gimbal mechanism that supports the vertical gyroscope. A three-phase asynchronous motor supplies the rotating movement. A static inverter powered by 28 VDC supplies the motor with 400 Hz VAC.

A mechanical erection mechanism keeps the gyroscope in the vertical axis position. It stops when a significant horizontal acceleration is sensed.

The standby horizon indicator has a glass bezel on its front panel and a black case to protect the mechanisms and electronic circuitry in it. A fast erection is located at the lower part of the front panel to erect the gyroscope and to adjust aircraft symbol.

The standby horizon indicator is located on the instrument panel between Multi-Functional Display 1 (MFD 1) and the Engine Display (ED) above the standby airspeed indicator in view of the two pilots. The standby horizon indication is mechanically adapted to the 19.5 degrees of instrument panel tilt.

Compass, Standby

[Refer to Figure 4.](#)

The standby compass gives a third source of magnetic heading. It is used as a reference when the two primary systems malfunction or when the two primary systems are calculating different heading values.

The compass has the assemblies that follow:

- Bowl
- Compensator unit
- Support plate.

Bowl: The semi-spherical bowl is transparent and filled with a stabilizing fluid in which the compass card can rotate. The compass card has directional magnets and graduations to show magnetic heading.

Compensator Unit: The deviation compensation unit is attached to the baseplate above the bowl. It has adjustable north/south and east/west magnets to compensate for the magnetic indication errors caused by magnetism around it. A related screw is turned to adjust the magnets. A white mark is shown above north/south compensator and a red mark is shown above the east/west compensator to show the "zero effect" positions. The compensation can be adjusted to 45 ±15 degrees. A cover protects the adjustment screws.



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Support Plate: The bowl is attached to the support plate. A scale in degrees is engraved on the support plate near the heading mark. The bowl can be turned as part of the compensation procedure to change the heading indication. A screw is tightened in the support plate to hold the bowl in place after compensation.

The standby compass is installed with non-magnetic screws to the centre structural support of the windshield directly below the Caution and Warning Panel (CWP).

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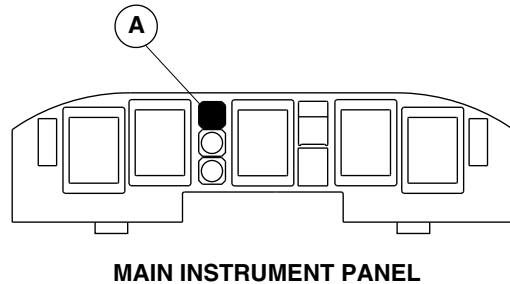
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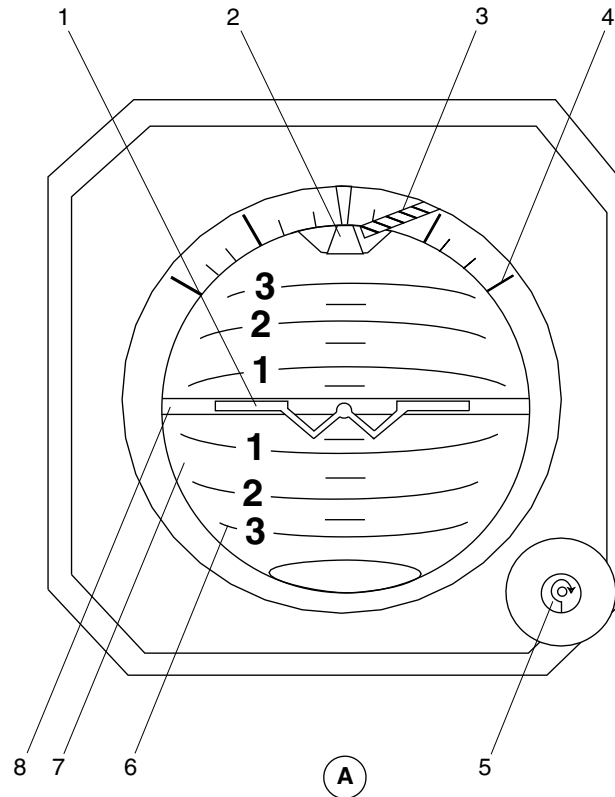
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MAIN INSTRUMENT PANEL

LEGEND

1. Aircraft Symbol.
2. Index Pointer.
3. Attitude Failure Flag.
4. Roll Scale.
5. Fast Erect Knob.
6. Pitch Scale.
7. Attitude Sphere.
8. Horizon Line.



fs649a01.cgm

STANDBY HORIZON INDICATOR
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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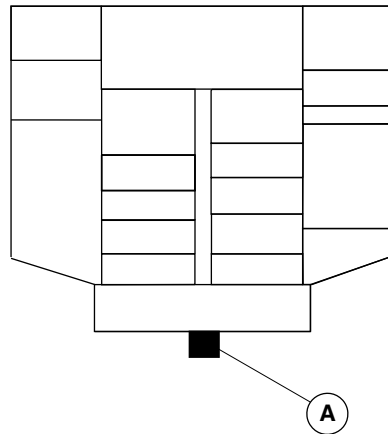
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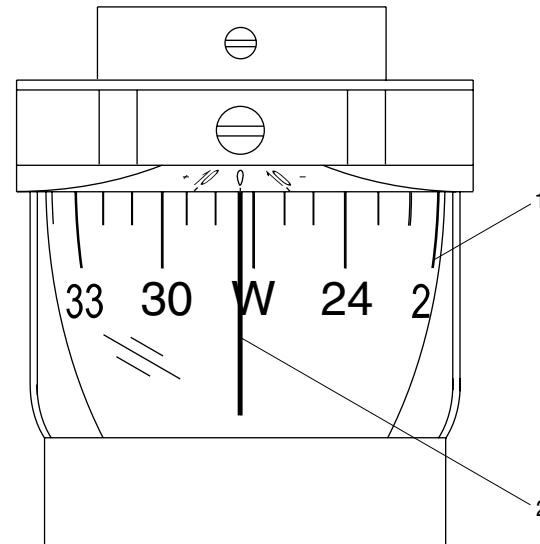
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LEGEND

- 1. Heading Scale.
- 2. Lubber Line.



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STANDBY COMPASS
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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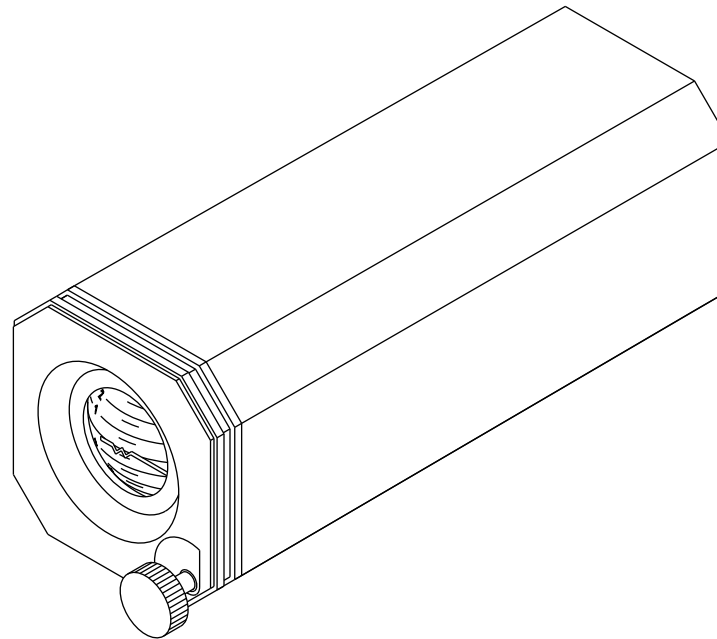
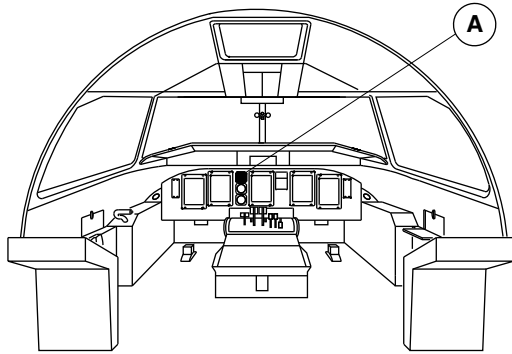
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STANDBY HORIZON INDICATOR LOCATOR
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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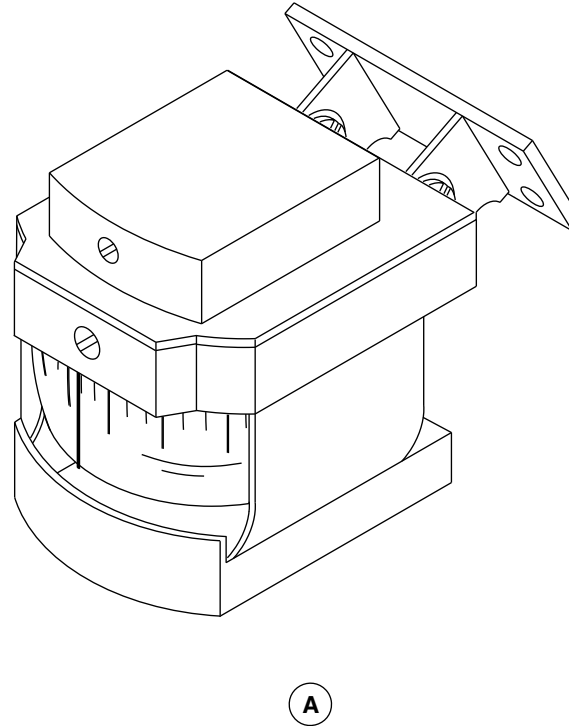
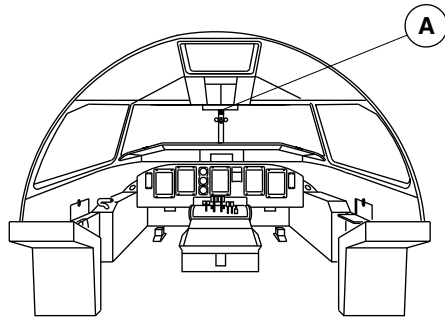
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STANDBY COMPASS LOCATOR
Figure 4

PSM 1-84-2A
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**ON A/C ALL

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MICROWAVE LANDING SYSTEM

General

The microwave landing system (MLS) is an airborne navigation system. It receives radio signals from a ground station that contains azimuth and elevation data for an approach flight path to a runway landing.

General Description

[Refer to Figure 1.](#)

The MLS receives and processes radio signals from a ground-based MLS station that gives approach flight path data transmitted in the 5 GHz range. The flight displays show this data in the form of azimuth (lateral) and elevation (glidepath/vertical) information.

The MLS is tuned with one of the two audio and radio control display units (ARCDU) installed in the flight deck center pedestal. MLS audio is selected with the ARCDUs for the flight crew.

When the MLS is set as the navigation source, the MLS automatically tunes the DME transceivers to supply distance information during approach. Distance is shown on the primary flight displays (PFDs) and, if selected to display the navigational page, on the multifunctional displays (MFDs).

System outputs are supplied for use by flight guidance modules (FGM) 1 and 2 for autopilot and flight director use. These modules are located in integrated flight cabinets (IFC) 1 and 2.

The microwave landing system has the components that follow:

- angle receivers (34-31-01)
- antennas (34-31-11)
- power divider (34-31-16).

Detailed Description

An MLS ground-based station supplies narrow, fan-shaped beams that are swept across a coverage area. An azimuth scanning beam is narrow in the horizontal plane and wide in the vertical plane and the elevation scanning beam is narrow in the vertical plane and wide in the horizontal plane. The approach sector is then fully covered as the pattern of the two beams is scanned between horizontal (azimuth) and vertical (elevation) angular limits.

Each angle transmission has a TO scan followed by a FRO scan. The TO scan of the antenna is in the direction of increasing angles (clockwise for azimuth and upward for elevation). The elapsed time between reception at the aircraft of the TO and FRO scans is a linear function of the angle of the receiving antenna location. The aircraft position, relative to the runway centerline, is then calculated according to the difference in time between when the aircraft receives the TO and FRO scans.

The scan rate for the azimuth and elevation beams is 20,000 degrees per second and is sensed by the receivers as unmodulated pulses of C-band RF energy. The MLS ground stations transmit the approach azimuth guidance out about 20 nautical miles in an arc of



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–40° to +40° around the runway centerline. The elevation vertical guidance is transmitted up to a height of 20,000 feet from 0.9° to 15°.

Basic and auxiliary data is also transmitted with differential phase-shift keying (DPSK) modulation to supply information related to the operation of the ground station. This data includes:

- minimum glidepath
- beamwidth
- operational status
- proportional coverage sectors
- antenna locations
- offsets
- heights.

Distance information is transmitted from the approach azimuth site to supply slant-range guidance. The transmitter frequency is paired with the MLS channel number so when an MLS channel is selected, it will automatically tune the correct DME frequency at the same time.

The angle receivers receive and process the DPSK C-band azimuth and elevation signals transmitted from an MLS ground-based transmitter. The signals received are in the frequency band from 5031.00 MHz to 5090.70 MHz for a total of 200 frequencies designated as channels 500 through 699.

Outputs are supplied for deviation presentations on the PFDs and/or MFDs, audio distribution and DME channel information. The system

decodes and stores information that is in the transmitted DPSK data words and supplies flag warnings when:

- there is a lack of confidence in the received signal for any function,
- there is an airborne system failure, or
- the pilot-selected path is less than the minimum path given by the basic data.

Three MLS antennas are installed on the aircraft, two below the windshield and one on the aft underside of the fuselage. The two under the windshield (left and right) are dedicated to MLS 1 and MLS 2, respectively. The third antenna routes the signal to a power divider before it is input to the individual receivers. During the transmitted preamble time-slot, the antenna that receives the strongest signal is selected by the angle receiver, which then demodulates the DPSK preamble. After recognition of the function identification code as an azimuth or approach elevation function, the receiver processes the beam information to determine the TO and FRO beam centers.

The receiver does an MLS signal validation on an individual basis over a number of frames. There is a monitor in the system and an automatic test signal is injected periodically to sense receiver failures. The flight crew can also start a manual system self-test.

Interface with the aircrew is through the two ARCDUs and two EFIS control panels (EFCP) located in the center console. Manual channel selection is done over a low-speed ARINC 429 databus. Either pilot can select an MLS channel from the MLS page. The MLS page is accessed on page 2 of the ARCDUs main page. The MLS line select key (LSK) is pushed followed by the EXP button. The MLS page allows the aircrew to select various MLS functions.

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EFFECTIVITY:

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The MLS can be set as the navigation source with either of the two NAV SOURCE rotary switches located on the flight guidance control panel (FGCP). When selected, a cyan legend MLS1 or MLS2 is shown on the applicable PFD or MFD. If the same MLS system is set as the navigation source on both PFDs, the annunciator changes to amber.

The dual MLS system operates on 28 Vdc electrical power. MLS 1 receives power through a 3-ampere circuit breaker on the left main bus and MLS 2 receives power through a 3-ampere circuit breaker on the right main bus. The circuit breakers are on the avionics circuit breaker panel and are located at positions K3 for MLS 1 and K4 for MLS 2.

System Control and Tuning

Refer to Figures 2 and 3.

The ARCDUs, located on the center pedestal, are used to:

- tune (channel selection) the on-side MLS,
- select manual self-test, and
- select MLS mode.

The MLS window is accessed through page 2 of the ARCDU main page. The window shows:

- the vertically oriented legend for MLS 1 or MLS 2,
- MODE selection (AUTO or MAN),
- the active channel selection,
- the active azimuth (in degrees) with the legend AZ above, and

- the active glideslope (in degrees) with the legend GS above.

The LSK next to the MLS window in either main page 2 or the MLS expanded page does the functions that follow:

- On first activation, allows tuning with the tune control knob of MLS channels 500 to 699.
- On second activation, allows manual selection of the azimuth (AZ) from 0° to 359°.
- On third activation, allows manual selection of the glideslope (GS) from 2.0° to 19.9°.
- On fourth activation, exits the MLS selection mode.

MLS Mode Selection: Mode selection is done on the MLS expanded window. Selection of the adjacent LSK will toggle the MLS mode between AUTO and MANUAL. In AUTO mode, the MLS will automatically accept the azimuth and elevation data encoded in the received signal. The only parameter that the flight crew can select is the channel number. Selection of the MANUAL mode allows the flight crew to select the azimuth and elevation angles as well as the channel.

MLS Channel Selection: To tune in an MLS channel, the LSK next to the MLS window is pushed once and the flight crew then turns the inner and outer concentric controls of the ARCDU tuning knob. The MLS channels number from 500 to 699. The outer knob controls the two left channel digits and the inner knob controls the third channel digit. The LSK is pushed again to make the selected channel active.

NOTE

There is no storage of preset channel selection for the MLS as with other tuning displays of navigation systems.



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Crew Initiated Self-test: The self-test is accessed on the MLS window page. When the LSK next to the TEST legend on the right side of the ARCDU screen is pushed, a 12-second test of the MLS starts. If the system passes the test, a green OK will be shown on the ARCDU for 5 seconds. If a problem is found, a red FAIL will be shown until a second test is run. While in test mode, the white TEST legend on the ARCDU display will become ON TEST in black on a green background.

If self-test mode is selected while the MLS is the navigation source, red lateral and vertical flags (crosses) will appear over the respective azimuth and elevation/glidepath displays on the PFD. After 4 seconds, the flags will be removed and a 4-second indication of FLY LEFT and FLY UP deviation bar/pointer will appear followed by a 4-second indication of FLY RIGHT and FLY DOWN.

Audio Selection: MLS audio can be selected when the VOR/MLS 1 or 2 pushbutton is pushed on either ARCDU. This supplies station Morse code MLS identification audio to the flight crew headsets. If the ARCDU SPKR pushbutton is pushed, the MLS audio is routed to the two cockpit loudspeakers. The MLS audio is also available at the observer station by selection of the combined VOR/MLS 1 or 2 switch on the observer's audio control panel.

Indication

Refer to Figure 4.

Azimuth (Course) Selection and Deviation: Azimuth selection is done with the on-side COURSE control located on the outer side of the FGCP. A cyan lateral deviation bar will move independently left or right of the course pointer (2 dots = 150 microamperes) against a white scale that moves with the course pointer. On course is indicated when the lateral deviation bar and course pointer are

displayed as a single unit. Maximum deflection is shown by a deviation of 180 microamperes and the deviation bar is parked. A digital readout of the selected course is shown on the left side of the PFD in cyan with the legend CRS and degree symbol (°) shown in white.

Elevation/Glidepath: The white elevation/glidepath scale and cyan, diamond-shaped pointer are shown to the right of the ADI sphere. A cyan symbol E is located above the scale.

Distance: A digital distance readout is shown on the right side of the EHSI on the PFD and MFD. The associated DME transceiver is automatically tuned as a function of the MLS channel selection. The readout is shown in white with a resolution of 0.1 NM below 99.9 NM. If the distance is not valid, three white dashes will replace the digital readout. The DME receiver can tune to three stations:

- DME channel 1 is dedicated to the pilot's radio-nav source (VOR1, ILS1 and MLS1).
- DME channel 2 is dedicated to the pilot's radio-nav source (VOR2, ILS2 and MLS2).
- DME channel 3 is dedicated to the Flight Management System (FMS) use only.

Failure Indication and Annunciation

If the MLS receiver fails or the ARINC 429 output is No Computed Data, the associated azimuth and elevation/glidepath deviations will be removed from the PFD.

Azimuth Deviation Failure: Azimuth deviation failure is shown by a red cross that overlays the course deviation scale. If the course is invalid, three white dashes will replace the digital readout. If a heading failure occurs, the course bar will be removed, but the



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azimuth deviation scale will stay in view and the azimuth deviation bar will stay active. The failure indication appears when one of the conditions that follow occurs:

- Deviation data from the MLS is invalid.
- The MLS is the selected navigation source but the selected course is not valid.
- The aircraft is out of range of the tuned ground station.
- The navigation source selection data are invalid.

Elevation/Glidepath Deviation Failure: A red cross over the length of the elevation/glidepath display is shown for any invalid condition. Full scale deflection is 150 microamperes. For a deviation signal of 180 microamperes, the diamond-shaped pointer will become a fixed arrow and show the direction to fly.

Angle Receiver

[Refer to Figure 5.](#)

The MLS angle receivers are located beneath the cabin floor, in the forward section between stations X71.20 and X94.80. They are rigidly mounted on trays that are oriented along the direction of flight. A single recessed DPX rack-mounted rear-panel connector and two rear-panel connectors are fitted for connection to the aircraft cable/antenna installation. Low-loss coaxial cables connect each receiver to a front antenna and also to one of the output connectors of the antenna power divider located beside the receivers.

Each angle receiver consists of the subassemblies that follow:

- a radio frequency (RF) assembly
- a multifunction module

- a processor module
- an interface module
- a power supply module

The radio frequency (RF) assembly contains a C-band module, 200 channel synthesizer, and all the intermediate frequency (IF) sections on a single print-circuit board. It also includes a Built-In Test Equipment (BITE) module and a bias module.

The multifunction module consists of the functional circuits. These circuits include the DPSK demodulator, the multifunction processor, the interprocessor communications interface, the self-test generator, monitor and the ARINC 429 interface.

The processor module computes the azimuth and elevation/glidepath deviation values, initiates automatic self-test routines when outside the coverage area, transfers assembled ARINC 429 words to the multifunction module through the dual-port RAM, and selects the antenna receiving the strongest signal and determines the paired DME frequency. The module determines which antenna input will be used by continuously switching at a 1 Hz rate between the forward and aft antennas. The processor continues to switch until a decodable azimuth or high-rate azimuth is sensed. Once it is sensed, the processor will lock onto the antenna that produces the larger signal.

The interface module supplies a 600-ohm Morse code output to the audio integrating system and subsequent flight deck distribution.

The power supply module converts the aircraft's 28 Vdc to regulated +15, -15 and +5 Vdc for use by the receiver modules.



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MLS Antennas

Refer to Figures 6 and 7.

Three C-band, vertically polarized, omnidirectional antennas are installed on the aircraft. Two are antennas located on the nose just below the windshield. The left antenna is connected to the MLS 1 receiver and the right antenna is connected to the MLS 2 receiver. The third antenna is located on the aft underside of the fuselage at station X813 and Y2.60. The two receivers share this antenna and it is connected through the power divider.

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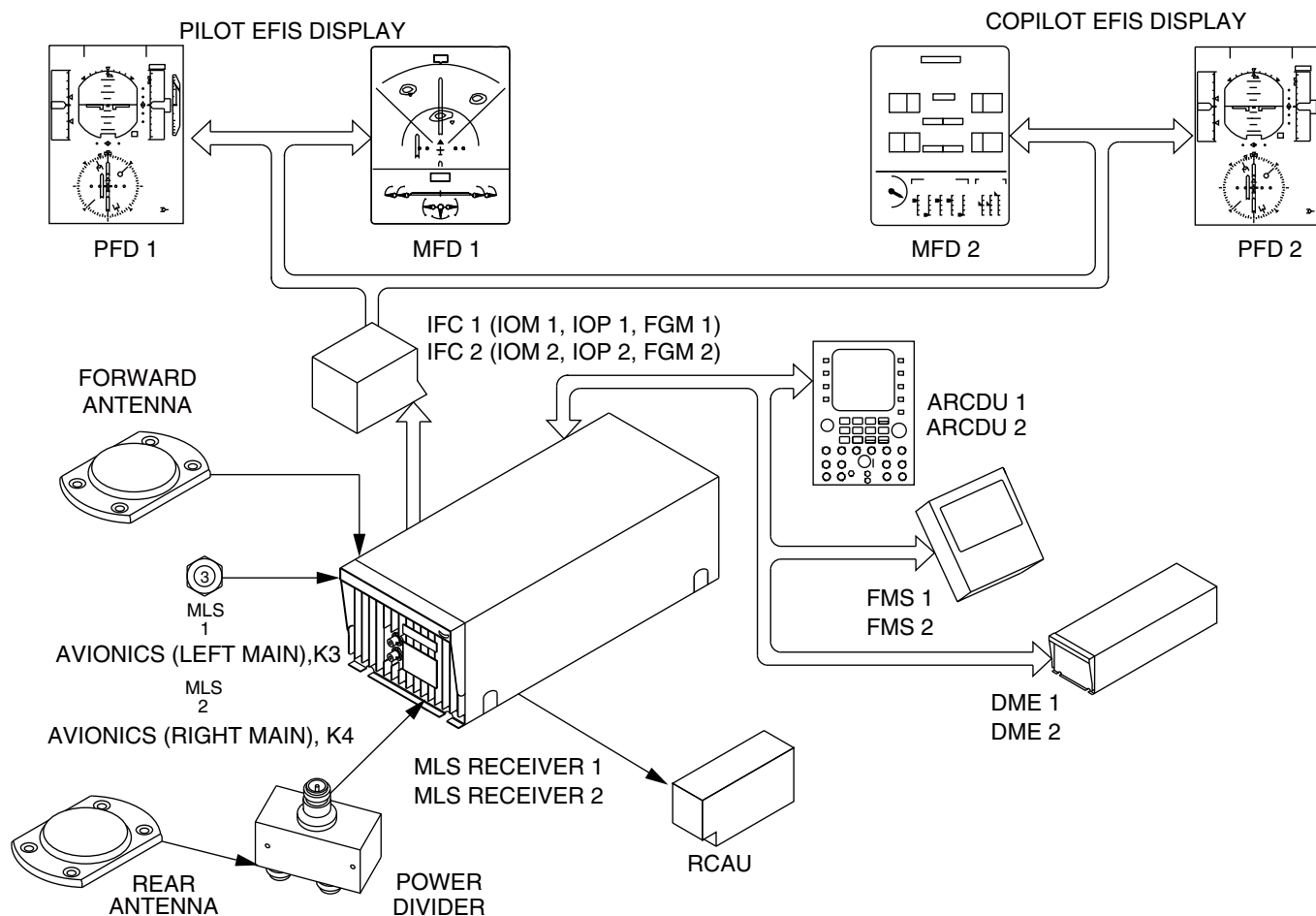
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MLS BLOCK DIAGRAM
Figure 1

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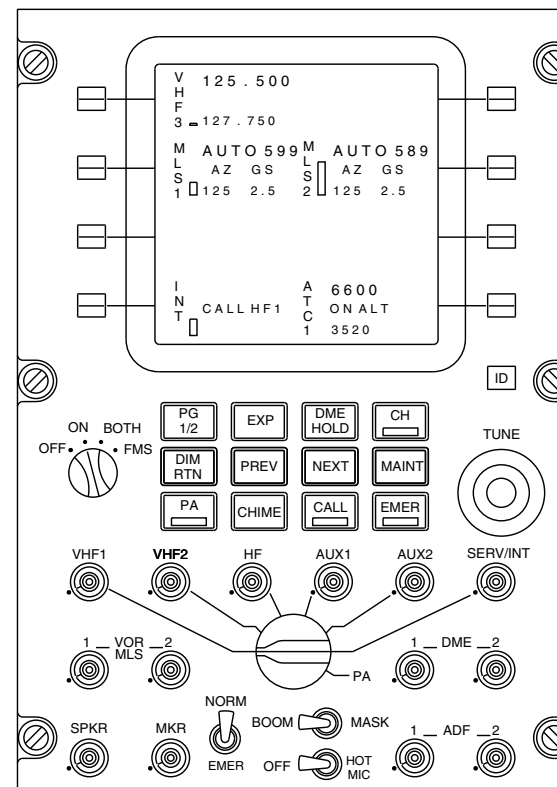
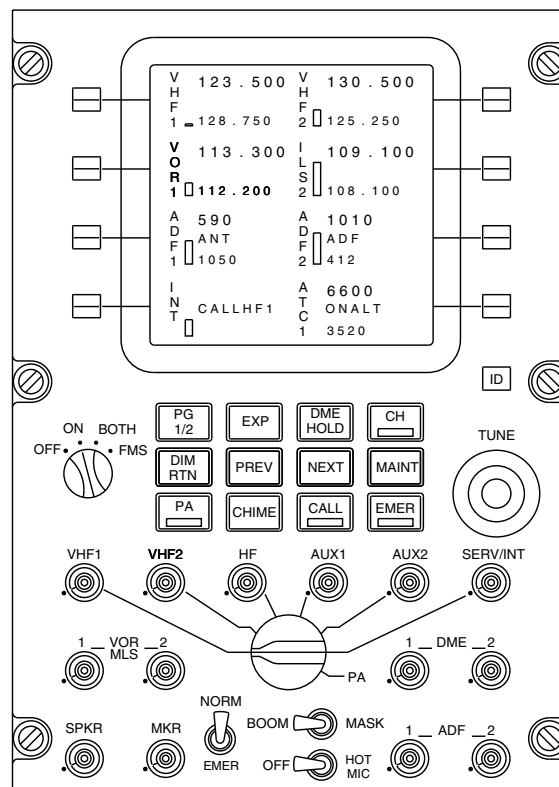
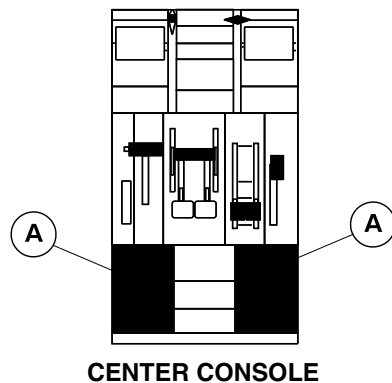
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ARCDU Main Pages
Figure 2

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Config 001

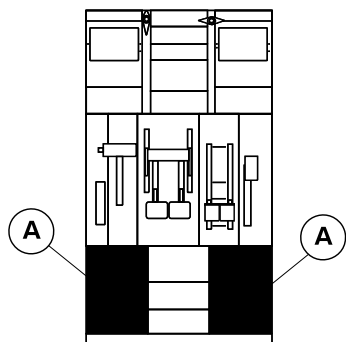
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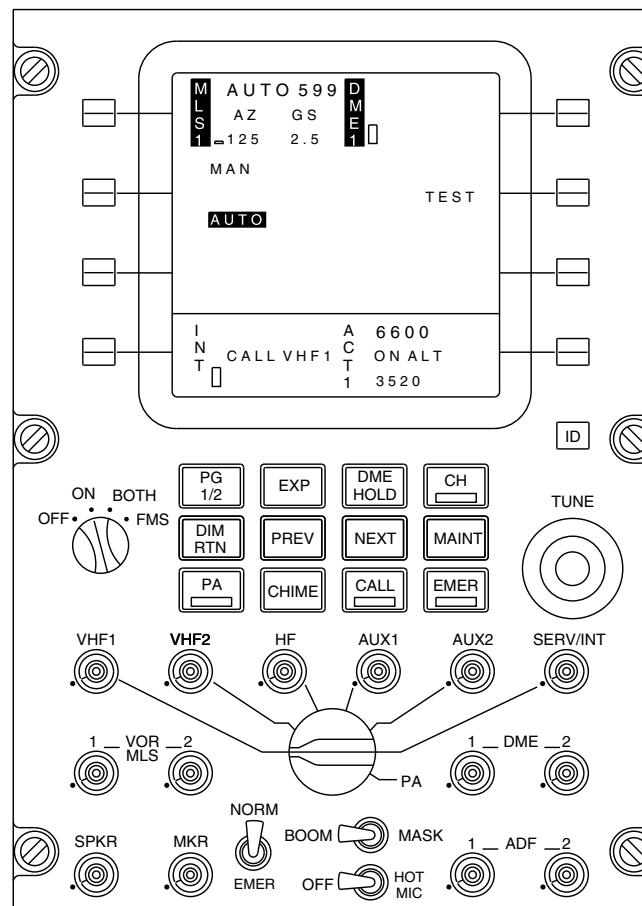


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CENTER CONSOLE



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ARCDU, MLS Main Page
Figure 3

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EFFECTIVITY:
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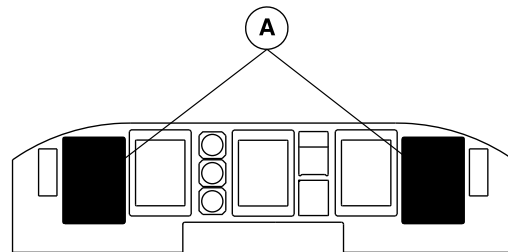
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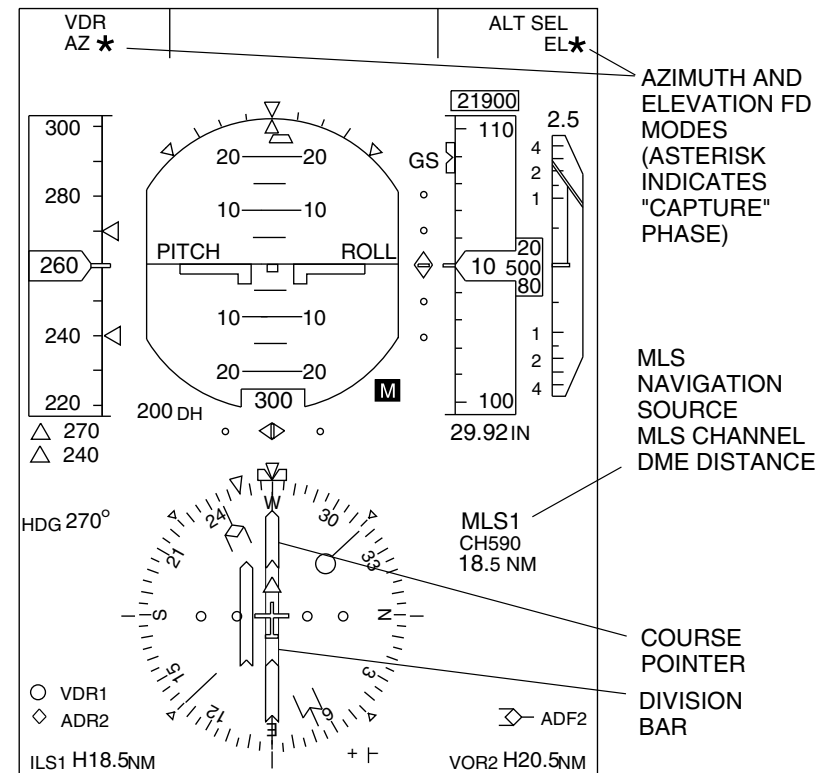


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MAIN INSTRUMENT PANEL



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MLS PFD INDICATION
Figure 4

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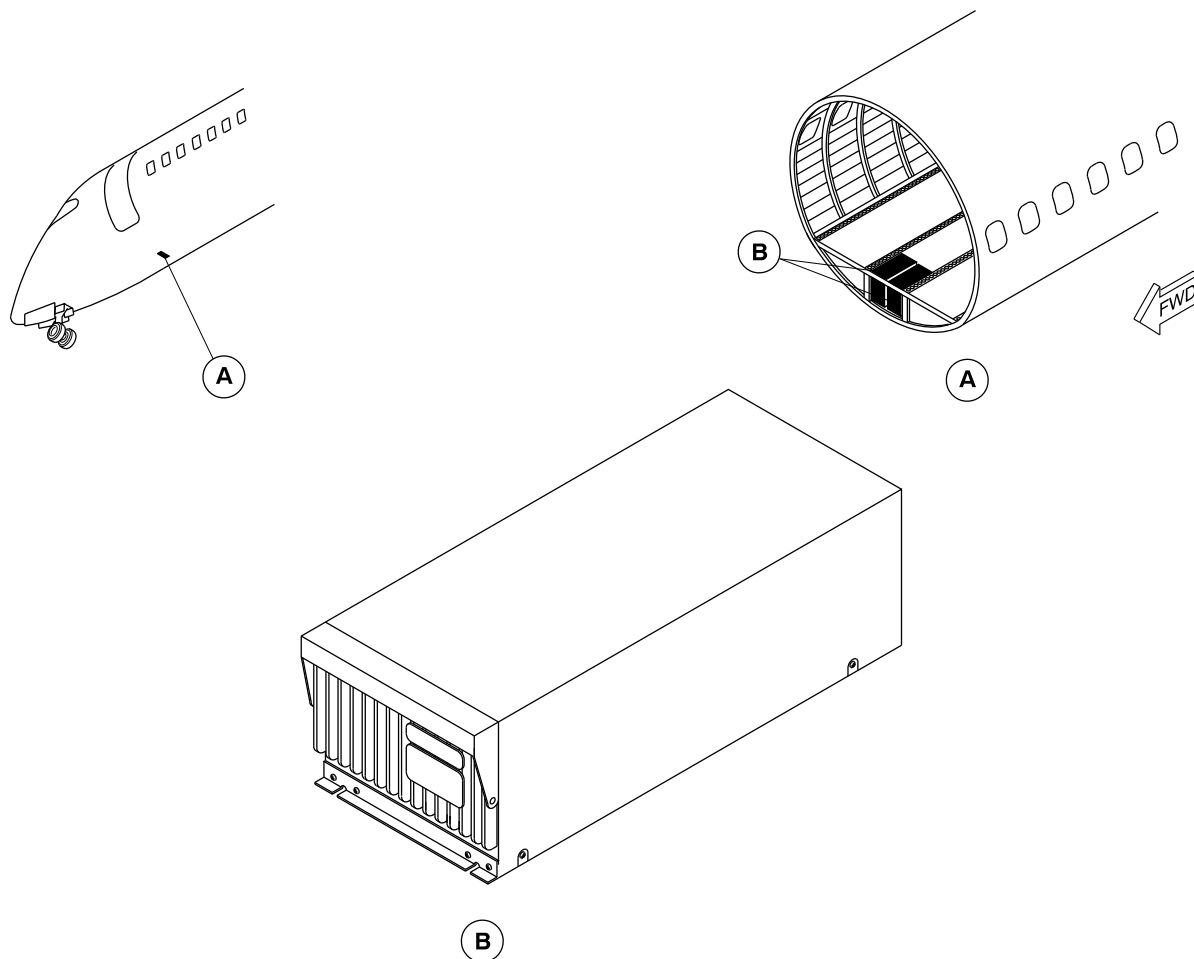
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MLS RECEIVER
Figure 5

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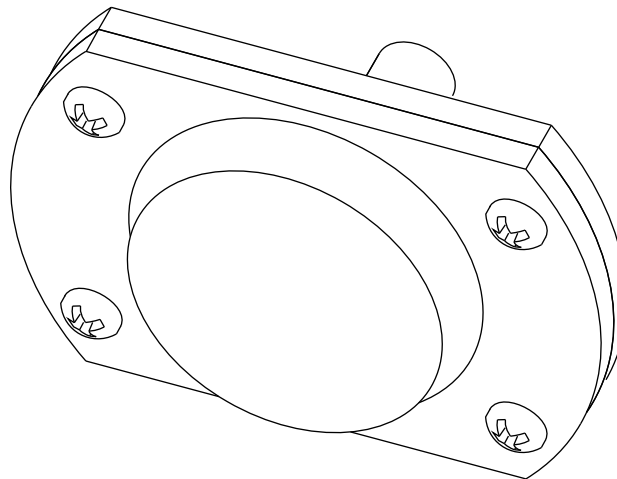
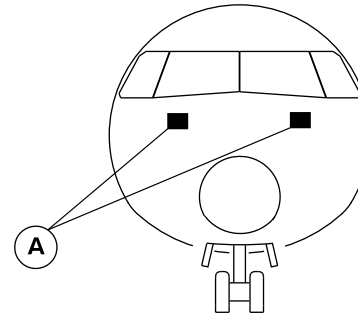
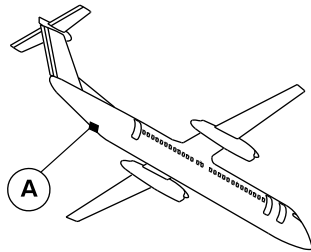
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MLS ANTENNAS
Figure 6

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Config 001

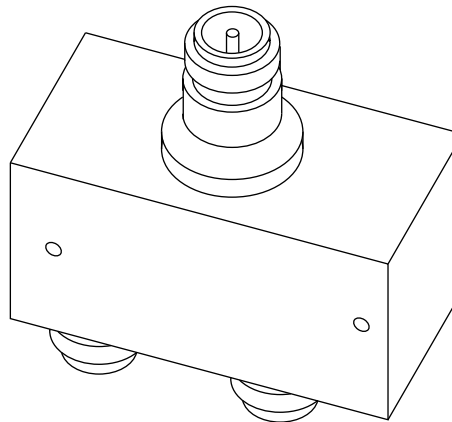
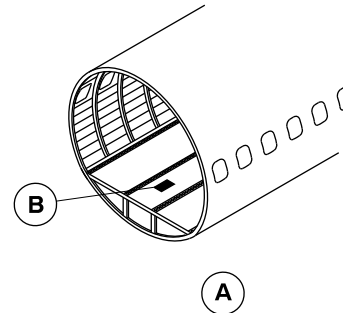
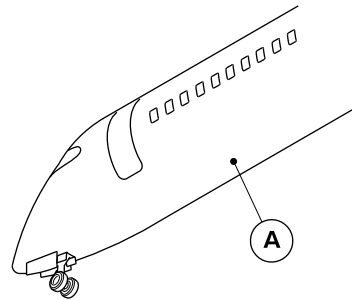
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POWER DIVIDER
Figure 7

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INERTIAL REFERENCE SYSTEM

Introduction

The Inertial Reference System (IRS) provides an additional independent source of aircraft position data to back-up the dual UNS-1Ew Flight Management System (FMS) in case of the Global Positioning System (GPS) signal loss.

The IRS is required as an independent inertial reference sensor of FMS during the Required Navigation Performance (RNP) operations.

General Description

[Refer to Figures 1 and 2.](#)

The IRS installation has the components that follow:

- Inertial Reference Unit (IRU) (34-32-30)
- Mounting tray (34-32-31)
- Aircraft personality module (APM) (31-41-21)

The inertial reference system consists of an IRU, a mounting tray and an Aircraft Personality Module (APM). The IRU is mounted on a mounting tray, which is installed on a platform assembly. The platform assembly is located in the center lower fuselage. The APM is installed on the platform assembly.

The IRS provides aircraft rates and accelerations, inertial velocities, wind speed and direction, ground speed and present position as well as attitude, heading and magnetic north reference data.

In this application, the IRU performs only the Inertial Reference function because the GPS Hybrid function is not supported by the current version of the FMS software. The IRS inertial outputs of attitude, heading and angular rates are not utilized in the current application.

The Inertial Reference (IR) function of the IRU contains three force rebalance accelerometers and three laser gyros, which are used to measure the inertial motion. The IR function requires system initialization (entry of latitude and longitude) that takes place from either the FMS 1 or the FMS 2. In addition, the IR function receives the air data information (altitude, altitude rate, and true air speed) from the Air Data Computer (ADC) 1 and the ADC 2. When the IRS is initialized, the IR component continuously calculates and transmits the inertial and position output.

The IRS provides the following outputs:

- Latitude and longitude
- N/S velocity, E/W velocity, and ground speed
- Inertial attitude
- True and magnetic heading
- Track angle true and track angle magnetic
- Track angle rate
- Wind speed and wind direction true
- Drift angle
- Along track and cross track accelerations

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- Along heading and cross heading accelerations.

Detailed Description

The Inertial Reference function of the IRU has the following modes of operation:

- Power-up
- Attitude (ATT)
- Stationary alignment
- Align In Motion (AIM)
- Navigation (NAV)
- Automatic Realignment (AR)
- End-of-flight.

Power-Up Mode: When the power is supplied, the IRU enters into the power-up mode. When it completes the power-up mode it enters the erect attitude mode. After completion of the erect attitude mode, the IRU transitions to the reversionary attitude mode and enters the stationary alignment mode on the stationary aircraft.

If IRU senses sufficient movement, it attempts to enter the align in motion mode, however the AIM entry conditions will not be satisfied because of the absence of the related GPS interface in the current application.

After it completes the stationary alignment mode, the IRU exits both the stationary alignment mode and the reversionary attitude mode and transitions to the navigation (NAV) mode. In the NAV mode, the IRU performs an automatic realignment when motionless on the ground as defined by the AR mode. When the IR senses no

movement after a flight, it enters an end-of-flight mode concurrent with the NAV mode or either of the two attitude modes.

In the power-up mode, the IRU does a configuration check to find some of its functionality and the characteristics of its installation. The IRU also checks the integrity of the Operations Per Second (OPS) and Operation Per Cycle (OPC) software in less than 20 seconds from off to power-up mode transition.

At the power-up, the IRU does the installation validation tests to make sure that the IRU and the APM are installed on the correct aircraft and installed in the correct location on the aircraft. At the power-up, it also finds if the internal and external APM data is blank, invalid, or valid. External APM data resides in the APM and internal APM data resides in the IRU.

Attitude Mode: After the IRU completes the power-up mode, it goes into the erect attitude mode where the platform leveling function quickly establishes the initial local level coordinate frame. The initial local leveling usually takes less than one second to complete but may take longer depending on the aircraft dynamics. After the IRU completes the platform leveling, it does the reversionary attitude function. The reversionary attitude function provides the reversionary pitch and the roll attitudes, magnetic heading referenced to a magnetic heading input, rotational rates and the linear accelerations. The device also operates in the stationary alignment mode when the reversionary attitude function is in progress.

Stationary Alignment Mode: The stationary alignment mode starts after the IRU completes the power-up and the erect attitude modes. In this mode, the IRU uses valid data from the FMS 1 or FMS 2 for the initial position entry. The IRU aligns its reference axes to the local vertical and estimates the horizontal earth rate components to calculate the latitude. It calculates the true north from the ratio of the horizontal earth rate components. The stationary alignment time



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changes as a function of latitude. The alignment time is calculated based on the last navigation latitude from the previous "End of Flight" event until latitude is entered in the alignment. If the last navigation latitude is not known, the IRU defaults the alignment time to 17 minutes

Automatic Realignment (AR) Mode: The IRU goes into the AR mode when the IRU is in the NAV mode and the aircraft is stationary for an extended period of time. The IRU ground speed must be less than 30 kts for the entire AR no motion period before it applies the AR estimate.

Air/Ground Function: The determine air/ground function provides the device verification of the "In Air" versus "On Ground" operational state.

The IRU defines it is "On Ground" when one of the conditions that follow exists:

- If the calculated ground speed is less than 60 kts, or
- If it does not calculate the ground speed, and the ADC true airspeed is valid and less than 110 kts.

The IRU defines it is "In Air" when one of the conditions that follow exists:

- If the calculated ground speed is more than 80 kts, or
- If it does not calculate the ground speed, and the ADC true airspeed is valid and more than 130 kts.

End of Flight Function: The End of Flight (EOF) event determination is necessary to provide an indication to system functions that an IRU transition to off mode is about to occur. This allows the IRU to store the operational and the performance data before the system is powered off.

The IRU senses an "End Of Flight" event after a transition from the "In Air" to the "On Ground" when one of the conditions that follow exists:

- The IRU is "motionless", GPS data is valid and the autonomous ground speed is less than 10 kts, or
- The IRU is "motionless" and the GPS data is invalid, or
- If it does not calculate the inertial ground speed.

Excessive Motion: Movement of the aircraft before completion of the alignment must be detected to ensure the accuracy of the alignment. Movement of the aircraft can be caused by push back from the gate, or taxiing. Such movement invalidates the current alignment and re-starts the alignment, 30 ±1 seconds after the excessive motion stops.

No Motion Period: After the AR no motion period while the IRU is in the navigation mode, it periodically applies the AR estimates.

No Motion Condition: A change in heading of less than 0.2 degrees in a 10 second interval or a change in velocity of less than 0.46 ft/sec (0.14 m/sec) in a 10 second interval.

AR activation conditions: After the AR no motion period, while the IRU is in the navigation mode, it periodically applies the AR estimates. The IRU continues to provide the valid navigation mode outputs during and after the AR estimates are applied. The three conditions in which the AR mode starts are as follow:

- Extended Alignment AR
- Pre-flight AR
- Post-flight AR.



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Extended Alignment AR: The extended alignment AR occurs if the alignment is extended after a stationary alignment to the NAV mode transition and if it does not detect too much movement.

Pre-Flight AR: The pre-flight AR condition occurs if the aircraft is stationary after exiting the extended alignment AR because of the movement. If the extended alignment AR has been exited and if it is prior to a first flight, the IRU goes into the pre-flight AR if movement is not sensed in the AR no motion time period. In the pre-flight AR period, the IRU accepts position updates from a general purpose bus. It will not accept the automatic position update from the GPS input bus to prevent the pilot entered position being overwritten.

Post-Flight AR: The post-flight AR condition occurs if no movement is sensed in the AR no motion time period after a flight and if the inertial ground speed is less than 30 kts.

Navigation Mode: In the navigation mode, the IRU uses the last correct FMS position data entered during the stationary alignment. In this mode, the IRU does unaided Schuler-tuned inertial navigation and provides outputs of the aircraft attitude, body rates and accelerations, true and magnetic heading, velocity vectors, and present position. The IRU provides unrestricted attitude and world-wide navigation. The IRU algebraically adds calculated magnetic variation from a magnetic variation topographical map to true heading and true track to give magnetic heading and track angle magnetic.

Inertial Reference System

The IRU is the main component of the IRS. The IRU has IR and Global Positioning (GP) sections. The IR section contains three force-rebalance accelerometers and three laser gyros which the IRU uses to measure the inertial motion. The IR section receives system

initialization data from the FMS. The IR section also receives Air Data System (ADS) altitude, altitude rate and true airspeed inputs on ARINC 429 input buses. From these measurements and inputs, the IRU continuously calculates the outputs of the body frame, local level frame, and the earth frame parameters. The GP function is not used in the current application.

Mounting Tray

The mounting tray is a low-profile, lightweight non-standard tray that consists of machined aluminum and stainless steel bushings and fittings. The two slotted round pins in the rear and one round pin in the front keep the IRU in alignment with the tray. Two captive screws on the IRU secure it to the tray.

Aircraft Personality Module

The APM is a self-contained, Non-Volatile Memory (NVM) device that is mounted externally from the IRU. It stores the program pin discrete data, mount misalignment Euler angles, aircraft type and serial number, and configuration-specific option selections. The IRU uses the APM data for computing mount misalignment and the aircraft-specific data.

Configuration Index File

The aircraft configuration file consists of a software profile that configures the IRU for operation in a unique aircraft installation. The software profile includes the following data:

- APM configuration index number
- Honeywell company identifier (ID)
- IRS equipment ID

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- Aircraft type
- Aircraft serial number
- Source destination identifier
- Program option selects
- Lever arm values
- Tray alignment Euler angles
- APM configuration Cyclic Redundancy Check (CRC)
- All APM installation file data CRC.

Power Supply

The IRU is powered from the left main bus, through a 3A circuit breaker (IRS). The IRS CB is located on the 28 V dc avionics circuit breaker panel at position L2.

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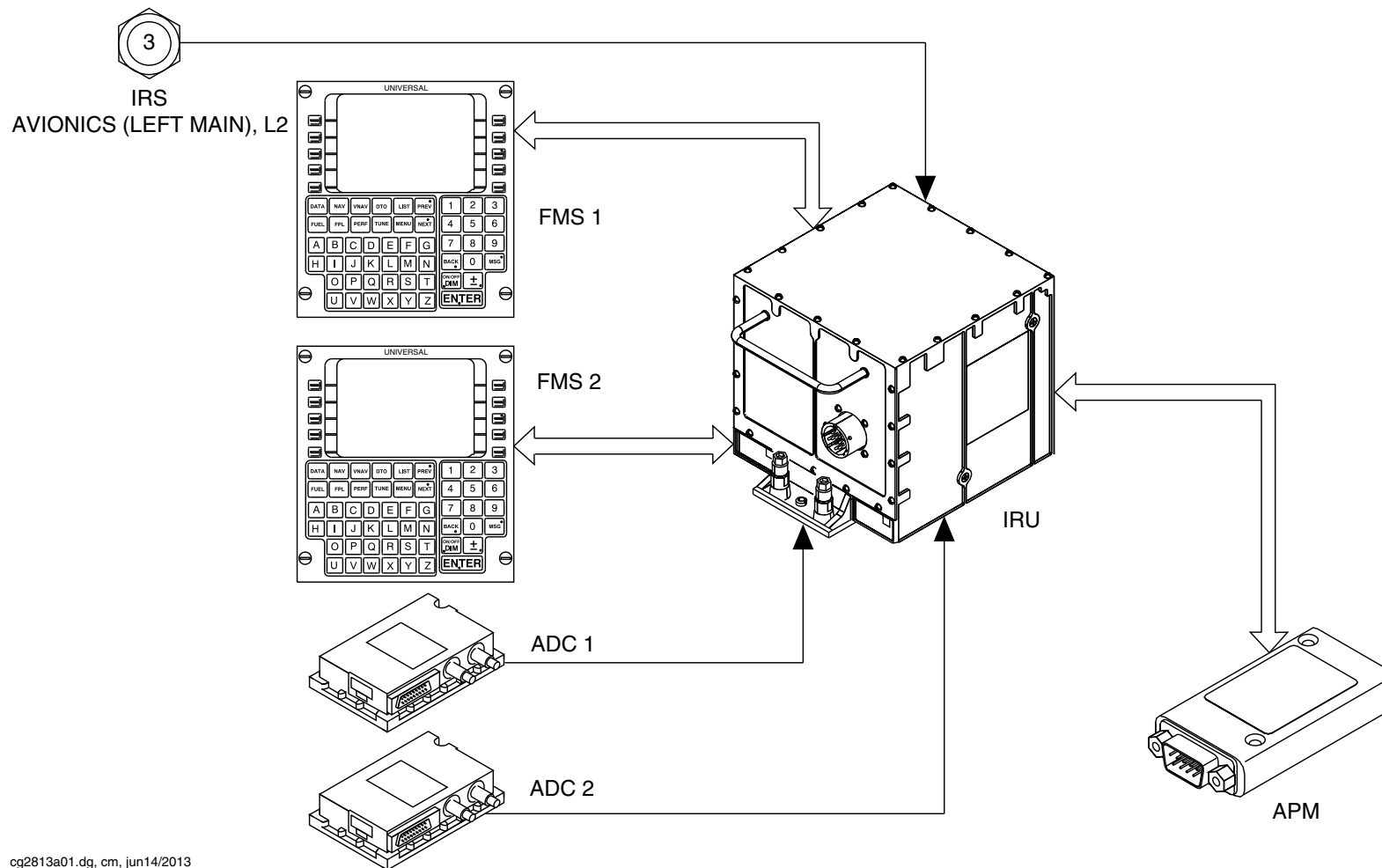
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Inertial Reference System – Block Diagram
Figure 1

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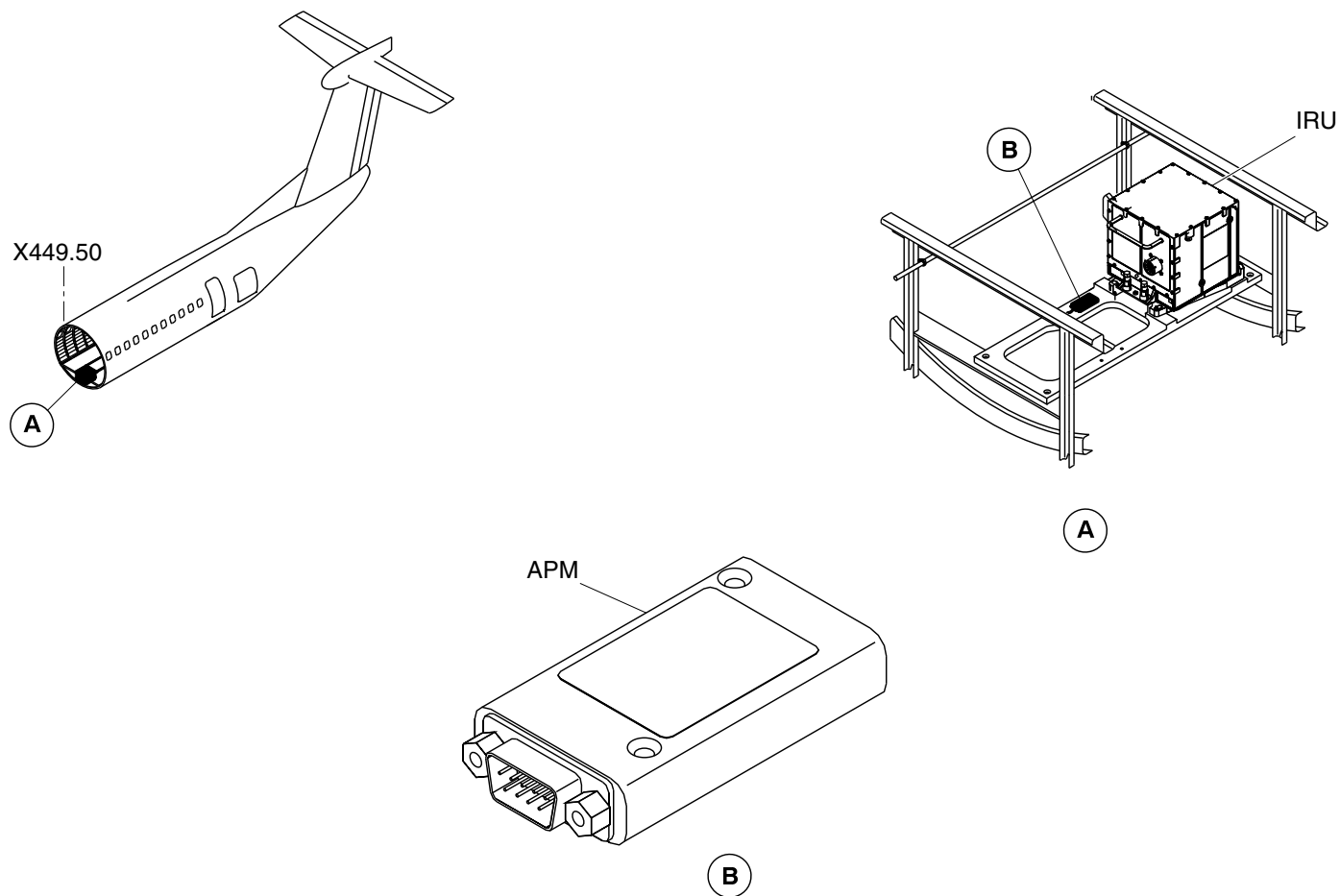
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Inertial Reference System – Component Location
Figure 2

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34-40-00-001

EXTERNAL PROTECTION SYSTEM

Introduction

The External Protection System (EPS) has different sub-systems to give visual and aural indications that relate to the conditions that follow:

- Weather
- Ground proximity
- Location and threat status of other aircraft
- Above Ground Level (AGL) altitude.

It also gives other aircraft and ground stations with data relating to the aircraft identification and location.

General Description

Refer to Figure 1.

Each sub-system calculates the operational data to give the necessary visual and aural indications.

The EPS does the functions that follow:

- Monitors the weather conditions and gives mapping capability
- Gives visual and aural indications that related to ground proximity.

- Gives the capability for other aircraft and ground stations to identify the aircraft and locate its position
- Gives visual and aural indications that relate to the location of other aircraft and coordinate and annunciate collision avoidance maneuvers with other aircraft
- Gives AGL altitude data that is used during approach and landing conditions.

The external protection system has the sub-systems that follow:

- 34-41-00 Weather Radar Systems
- 34-42-00 Ground Proximity Warning System (GPWS)
- 34-43-00 Traffic Alerting and Collision Avoidance System II (TCAS II)
- 34-44-00 Radio Altimeter.

Weather Radar Systems

The weather radar system helps the pilot to avoid thunderstorms and associated turbulence. When selected, it supplies a continuous visual colour display of the weather conditions to the Multi-Function Displays (MFD 1 and MFD 2).

Ground Proximity Warning System (GPWS)

The Ground Proximity Warning System (GPWS) gives aural and visual indications of possible Controlled Flight Into Terrain (CFIT).



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Traffic Alerting and Collision Avoidance System II (TCAS II)

The Traffic Collision Avoidance System II (TCAS II) gives aural and visual indications when the aircraft's position and flight path relative to an intruding aircraft may cause a potentially dangerous condition.

Radio Altimeter

The Radar Altimeter (RA) system gives Above Ground Level (AGL) altitude data for indication and to other systems.

It does Decision Height (DH) calculations during the approach phase of the flight.

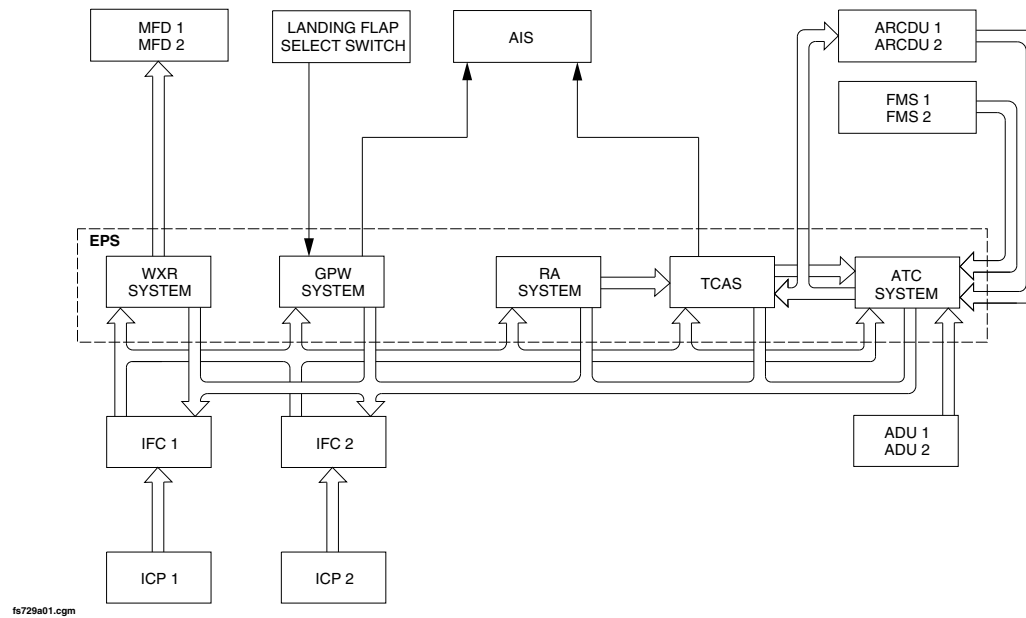
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EXTERNAL PROTECTION SYSTEM BLOCK DIAGRAM
Figure 1

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**ON A/C ALL

34-41-00-001

WEATHER RADAR SYSTEM

Introduction

The weather radar system helps the pilot to avoid thunderstorms and associated turbulence. When selected, it supplies a continuous visual colour display of the weather conditions to the Multi-Function Displays (MFD 1 and MFD 2).

General Description

Refer to Figure 1.

The Weather Radar (WXR) system is an advisory weather avoidance system. The radar system supplies the flight crew with operational features such as range selection, ground mapping, target alert, and fault annunciations.

The weather radar system is operated by a control panel on the center console. It has the components that follow:

- Transceiver (34-41-01)
- Antenna, Weather Radar (34-41-06)
- Panel, Radar Control (34-41-11)
- Couplers (34-41-16)
- Module, Configuration (34-41-21).

Detailed Description

Radar is a distance measuring system using the principle of radio echoing. The term RADAR is an acronym for RAdio Detecting And Ranging. The transmitter sends microwave energy in the form of pulses. These pulses are then transmitted to the antenna and focused into a radar beam. The radar beam is focused and transmitted by the antenna in such a way that it is most intense in the center of the beam and decreases in intensity near the edge. The same antenna is used to transmit and receive radar signals. When a pulse intercepts a target, the pulse is reflected as an echo, or return signal, back to the antenna. From the antenna, the returned signal is transmitted to the receiver transmitter unit. The echoes, or returned signals, are shown on the Multi-Function Display(s) (MFD)(s).

Refer to Figure 2.

Radio waves travel at the speed of 300 Mm/s and thus yield nearly instantaneous reply information. Radar ranging is a two-way process that requires 12.36 μ -sec for the radio wave to travel out and back for each nautical mile of target range.

Precipitation or objects more dense than water such as earth or solid structures are sensed by an X-band weather radar. The weather radar does not detect clouds, thunderstorms or turbulence directly. Instead, it detects precipitation which may be associated with thunderstorms and turbulence. The best radar reflectors are raindrops, wet snow or hail. The larger the raindrop the better it reflects. Large drops in a small concentrated area are characteristic of a severe thunderstorm, the MFD shows the storm as a strong echo. Drop size is the most important factor in high radar reflectivity. The larger the raindrop the better it reflects. Generally, ice, dry snow and dry hail have low reflective levels and often will not be shown on the MFD.

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Refer to Figure 3.

A cloud that has only small raindrops, such as fog or drizzle, will not produce a measurable radar echo.

Refer to Figure 4.

Calm water or water with swells do not reflect good radar signals. The pulse is reflected in a forward scatter angle without all portions being returned. The resulting display is no target.

Choppy water supplies better radar pulse reflections from the downwind sides of the waves. (The resulting display is a target whose intensity will change with the degree of chopiness.)

Illumination of terrain results in a “diffused” reflection of the beam. A portion of this reflected energy is scattered back toward the antenna and land features as well as lakes, large rivers, shore lines and ships come into view on the display.

Refer to Figure 5.

The incident angle at which the terrain is illuminated has a direct bearing on the detectable range and the area of illumination. A large incident angle gives the radar system a smaller detectable range of operation (due to a minimized reflection of direct radar energy). However, the illuminated area “A” is larger.

A smaller incident angle gives the radar a larger detectable range of operation because of an increase of direct radar energy reflected from the target to the antenna. The area of illumination “A” is smaller.

Concentration of the beam energy on a small area of terrain increases the magnitude of the echo intercepted by the antenna. The detectable range is therefore increased for mountainous terrain; the maximum distance at which this terrain can be monitored is greater because of the more direct reflection (or radar echo) produced.

Refer to Figure 6.

The antenna gain versus the angle characteristic is a continuous function at all angles. This means that there is a gain value associated with all forward angles relative to the selected tilt angle. Outside of the beamwidth, the antenna gain does not go to zero, but continues to reduce with increasing angles (continuous gain function).

Sidelobes are formed during transmission, but their gain value size makes them unimportant. A damaged radome can increase the gain value size of the sidelobes to a point where they cause a constant radar reflection from the ground. This is referred to as an “altitude ring” because the display will show a concentric ring at a distance equal to the slant range of the sidelobe to the ground.

The “cone” formed by the narrow beamwidth is where most of the radar energy is concentrated, so it is important to realize that at any given time whatever is within the cone (and sometimes other strong targets like clutter outside the cone) is what is being shown on the display.

When a radar pulse is transmitted into the atmosphere, it is gradually absorbed and scattered so that it loses its ability to return to the antenna. This attenuation or weakening of the radar pulse is caused by two primary sources, distance and precipitation. The Antenna/Receiver/Transmitter (ART) has several advanced features which significantly reduce the effects of attenuation (no airborne weather radar can eliminate them completely).

Attenuation because of distance is due to the fact that the energy of the radar pulse transmitted by the antenna is inversely proportional to the square of the distance. For example, the reflected radar energy from a target 60 mi (96.6 km) away will be one fourth, if the target is beam filling (object fills the cross-sectional area of the



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beam), of the reflected energy from an equivalent target 30 mi (48.3 km) away. Both the Sensitivity Timing Control (STC) and the Extended STC circuitry are used to compensate for distance attenuation. The ART has an STC range of 0 to approximately 40 nmi (0 to 74 km). The ART will also electronically compensate for the effects of distance attenuation with the net effect that targets do not appear to change colour as the distance decreases.

Outside the STC range the Extended STC circuitry increases the displayed intensity to more accurately represent storm intensity. The Extended STC will not, however, fully compensate for distance attenuation and, therefore, targets in this range can be expected to show more detail as the distance decreases until reaching the STC range.

Attenuation due to precipitation is far more intense and is less predictable than attenuation due to distance. As the radar pulses pass through moisture, some radar energy is reflected and absorbed. If the rain is very heavy or extends for many miles, the beam may not completely travel through the area of precipitation. If the beam is fully attenuated, the radar will display a “radar shadow” which is shown on the display as an end to the precipitation when, in fact, the heavy rain may extend for many more miles. Precipitation attenuation may cause the area of heaviest precipitation to be shown as the thinnest area of heavy precipitation. One cell that has heavy precipitation may fully block or shadow a second cell located behind it and prevent it from being shown on the radar.

The display may show 20 to 30 nmi (37 to 65 km) of precipitation ahead on the radar when, actually, the rain may extend a great distance. To reduce the effects of precipitation attenuation, the ART contains sophisticated weather attenuation compensation circuitry. The attenuation compensation feature is automatic in the Wx/WxA mode of operation. The compensation logic operates between 3 to

240 nmi (5.6 to 445 km), whenever a level 2 (yellow), 3 (red) or 4 (magenta) echo is displayed. The compensation circuits cause the software to measure each individual storm cell return and increase each individual cell return independently while the antenna scans the sector containing heavy rain. The compensation circuitry lets the radar beam effectively travel deeper into and through heavy rain to search for possible storm cells beyond. While attenuation compensation does not eliminate precipitation attenuation, it does let the radar travel through more rain at short ranges.

NOTE

Often the ground based ATC controller’s radar will have a better overall picture of a large rain area and the pilot can compare the controller’s information with his own radar picture to avoid the strongest cells in a general area of rain.

[Refer to Figure 7.](#)

The WXR system supplies the aircraft multi-function displays (MFDs) with a continuous visual colour display of the weather precipitation conditions. It also supplies the flight crew with additional operational features such as:

- Range selection
- Ground mapping
- Bad weather alert
- Target alert
- Fault annunciation.

The WXR system has a single ART computing radar transmitter and receiver, an ART configuration module for storing WXR system strapping data and a dedicated WXR mode selection control panel. Some of the WXR system related controls (i.e. WXR display



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brightness, range selection, arc/full navigation display mode selection) are controlled by the Electronic Flight Instrument System (EFIS) and are accessible to each flight crew member by their related EFIS Control Panel (EFCP).

The ART accepts mode commands and tilt, gain and range signals from the WX control panel over the ARINC 429 bus. Display data is sent from the ART to the Multi-Function Display (MFD) over an ARINC 453 data bus.

[Refer to Figure 16.](#)

The Weather Radar Control Panel (WRCP) has front-panel push buttons, rotary switches and controls. The WRCP has the WXR mode selectable parameters that follow:

- Tilt
- Gain
- Off, standby, test and on modes
- Weather, weather alert and ground map operational modes.

TILT: A rotary control that changes the antenna tilt angle and through a selectable range of $\pm 15^\circ$. This control also disables the automatic radar stabilization facility when the control knob is pulled. STAB OFF will be annunciated in yellow on each Multi-Function Display (MFD) if the radar image display is selected and the ARC option is selected by the related EFCP WX and ARC/FULL push-buttons.

GAIN: A rotary control that changes the gain value used by the Antenna/Receiver/Transmitter (ART) when operating in ground map (GND MAP) mode through a selectable range of 0 to -20 dB (CCW rotation reduces gain). The gain knob will only function when in the MAP mode.

OFF Mode: Selection of the WXR system OFF mode is made by the Control Panel (CP) rotary switch. When the OFF mode is selected, power is removed from the transmitter/receiver electronics within the Antenna/Receiver/Transmitter (ART) but the ART still has primary power.

[Refer to Figure 8.](#)

No radar transmissions will occur when the OFF mode is selected and WX OFF will be annunciated in white text on each Multi-Function Display (MFD) if the radar image display is selected by the related WX push-button switch. When the weather radar is set to the OFF mode, the ART remains active for up to a maximum time of 30 seconds. This time delay allows sufficient time for the ART to park the antenna at 0 degree azimuth and 30 degree tilt down position.

STANDBY (STBY) MODE: When the Stby mode is set, the transmitter/receiver electronics within the ART is powered and ART mode selection and status data is transmitted to each MFD by the ART data bus output. No radar transmissions will occur when the STBY mode is selected. WX STBY will be annunciated in white text on each MFD if the radar image display is selected by the related EFCP WX push-button switch. When the STBY mode is selected, the ART remains active and the antenna will be parked at 0 degree azimuth and 30 degree tilt down position.

TEST (TST) MODE: When the TST mode is selected the transmitter/receiver electronics within the ART are powered and the ART mode selection, the status data and display data are shown in

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the form of a four colour test pattern (i.e. Magenta/Red/Yellow/Green).

Refer to Figure 9.

The four colour test pattern is transmitted to each MFD by the ART data bus output approximately 7 seconds after the selection is made. The test pattern is sized to fit the 80 nmi (nautical mile) range setting and the test pattern will be scaled according to the range setting set by its related EFCP. The TST position function select switch inhibits the radar transmitter. WX TEST will be annunciated in yellow text on each MFD if the radar image display is selected by its related EFCP WX push-button switch. The ART test pattern is also shown on each MFD if the radar image display is selected and the ARC option is selected by the related EFCP WX and ARC/FULL push-buttons. When the TST mode is selected, the ART remains active and the antenna continuously scans ± 45 degree along the selected tilt angle axis at a rate of 25 degrees/second.

Selection of the WXR system TILT angle is made by the Control Panel (CP) rotary control (i.e. clockwise for up and counterclockwise for down) over a range of ± 15 degrees. The selected TILT angle setting will be annunciated in cyan text on each Multi-Function Display (MFD) if the radar image display is selected and the ARC option is selected by the ownside EFCP WX and ARC/FULL push-buttons.

ON MODE: When the ON mode is selected, the transmitter/receiver electronics and the radar transmitter within the ART are powered and ART mode selection, status data and the display data are transmitted

to each MFD by the ART data bus output approximately 60 seconds after the selection is made.

Refer to Figure 10.

Radar transmissions will occur continuously when the ON mode is selected and after a 60 second warm-up time. The applicable ON mode option selected (i.e. WX ON, WX ALERT or WX GMAP) will be annunciated in white text on each MFD if the radar image display is selected by its related EFCP WX push-button switch. The ART display data will also be displayed on each MFD if the radar image display is selected and the ARC option is selected by the ownside EFCP WX and ARC/FULL push-buttons. When the ON mode is selected, the ART will stay on and the antenna will continuously scan ± 45 degrees along the selected tilt angle axis at a rate of 25 degrees/second.

The ART uses a magnetron to transmit power levels of between 2.5 to 6 kW (i.e. 3.3 kW nominal) at an X-band frequency (i.e. 9375 ± 30 MHz) for a pulse width duration of 4 mseconds, at a pulse repetition rate of 106.5 ± 5 Hz by microwave coupling to its 12 in. (305 mm) vertically-polarised antenna. The calculated weather avoidance range capability of the WXR system is 227 nmi. The WX beam properties are as follows:

Table 1: WX Beam Illumination with 12 in. Antenna

Beam Coverage (Beam Angle $\pm 4^\circ$)	$\pm 10,600$ ft	$\pm 21,250$ ft	$\pm 31,900$ ft	$\pm 42,500$ ft
Range from ART	25 nmi	50 nmi	75 nmi	100 nmi



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When the ON mode is selected and the aircraft is on the ground for more than 30 seconds, then the selected option (i.e. WX ON, WX ALERT or WX GMAP) will be annunciated in yellow text and flash continuously at a frequency of 1 Hz.

WEATHER RADAR (WX) OPTION–The WX option is annunciated as WX ON on the MFDs and is the initial default option entered when the On mode is first selected. Thereafter, the WX option can be set when the applicable WX push–button switch is pushed. When selected, this option is used to supply 4 colour image display data that is representative of the four levels of weather precipitation sensed by the ART. The radar display is calibrated to show five levels of target intensity: Black (level 0), Green (level 1), Yellow (level 2), Red (level 3), and Magenta (level 4). The target intensity levels are defined as follows:

Table 2: WX Display Severity Indications

DISPLAY LEVEL (COLOUR)	STORM CATEGORY	VIP LEVEL	REMARKS
4 (Magenta)	Extreme	6	Severe turbulence, large hail, lightning, extensive wind gust, and turbulence. Rainfall of 50mm/hr +.
4 (Magenta)	Intense	5	Severe turbulence, lightning, organized wind gusts, hail likely.
3 (Red)	Very Strong	4	Severe turbulence likely, lightning. Rainfall from 12 to 50 mm/hr.
3 (Red)	Strong	3	Severe turbulence, possible lightning.



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DISPLAY LEVEL (COLOUR)	STORM CATEGORY	VIP LEVEL	REMARKS
2 (Yellow)	Moderate	2	Light to moderate turbulence is possible with lightning. Rainfall from 4 to 12mm/hr.
1 (Green)	Weak	1	Light to moderate turbulence is possible with lightning. Rainfall up to 4mm/hr.
0 (Black)	None	None	No radar return

These levels are applicable only when:

- The Wx and WxA modes are selected
- The displayed returns are within the STC range of the radar (approximately 40 miles)
- The returns are beam filling
- There are no intervening radar returns.

When the WX option is set, the receiver supplies a weather compensation capability over the range 0 to 240 nmi.

[Refer to Figure 11.](#)

The Target Alert feature alerts the pilot to the presence of a significant weather cell that are beyond the currently selected range. For this mode to be active, Wx or WxA mode must be selected. The criteria for a Target Alert is for the cell to be at least red intensity, within ± 10 degrees of aircraft heading, a minimum size of 2 nmi in range and 2 degrees in azimuth, and within the range of 80 to 240 nmi. When a Target Alert is issued, two red arcs separated by a black arc will be shown at the top of the display centred on the aircraft heading.

NOTE

Target Alert is not active in the ground map mode.

When a target alert condition has been identified by the ART, the status data is transmitted by the ART data bus output to the applicable MFDs. A TGT message will be annunciated in magenta text and flash continuously at 1 Hz on the right of the radar mode of the applicable Multi-Function Display (MFD)(s) if the radar image display is selected by its related EFCP WX push-button switch.

WEATHER RADAR ALERT (WXA) OPTION: The WXA option is annunciated as WX ALRT and can be set by depressing the appropriate WXA push-button switch that is located on the control panel. When the WXA option is set, the operation of the WXR system is identical to that of the WX option described above with the exception that when the Antenna/Receiver/Transmitter (ART) data bus display data transmitted to the Multi-Function Display (MFD) contains a target that has a display level of 4 (i.e. display colour is magenta). The displayed target area will flash continuously at a frequency of 1 Hz on the applicable Multi-Function Display (MFD)(s).



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GROUND MAP (GND MAP) OPTION: The GND MAP option is annunciated as WX GMAP and can be selected by depressing the appropriate GND MAP push-button switch that is located on the Control Panel (CP). When the GND MAP option is selected the target alert feature is not active and the WXR system is configured to enable a gain control capability to be operative. Selection of the required WXR system GAIN value is made by the Control Panel (CP) rotary control (i.e. clockwise for increases in gain and counterclockwise for gain reductions) over a range of 0 to -20 dB. GAIN MAN will be annunciated in white text on each Multi-Function Display (MFD) when the GND MAP option is selected if the radar image display is selected and the ARC option is selected by the ownside EFCP WX and ARC/FULL push-buttons. The GAIN MAN annunciation serves to remind the flight crew that the Antenna/Receiver/Transmitter (ART) gain is variable when the GND MAP option is selected.

The GND MAP option is a secondary objective of the Weather Radar (WXR) system and is used for gathering and presentation of terrain data. When selected, it supplies 3 colour image display data that equates to the levels of intensity of ground returns as measured by the Antenna/Receiver/Transmitter (ART). This display data is in the form of a topographical map that is representative of the ground terrain. The GND MAP display levels and colours are as follows:

- Display level 3 (Magenta) is used to represent intense levels of ground returns.
- Display level 2 (Yellow) is used to represent moderate levels of ground returns.
- Display level 1 (Green) is used to represent low levels of ground returns.

The EFCP selections are converted to serial ARINC 429 digital data and are transmitted to the applicable MFD. The WXR system selected range information is then transmitted to the applicable IOP.

The following switches on the EFCP are involved with the WXR system :

- FULL/ARC push-button: Sets the display mode to be either the forward 90 degree Arc or a full 360 degree presentation
- WX push-button: Adds the weather information to the MFD
- RANGE rotary switch: Selects the displayed range
- WXBRT rotary pot: Adjusts brightness of the weather radar symbology.

The WXR system selectable ranges are 10, 20, 40, 80, 160 or 240 nmi.

The Navigation Display (ND) page of the MFD shows the weather radar modes in the upper left corner as follows:

- WX OFF in white font. It shows that the radar is switched OFF on the WXR control panel. The information comes from the control panel output bus through the IOP. In this mode, the radar is not active (ie. the antenna is not transmitting high energy radiations)
- WX STBY in white font. It indicates that the radar is selected on the Standby mode through the WXR control panel. In this mode, the radar is not ON



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- WX TEST in white font for WX, in yellow font for TEST. It indicates that the radar is engaged in the TEST mode to perform self-tests and to send a multicolour arc display pattern to EFIS. [Refer to Figure 9.](#) In this mode, the radar is not active.
- [Refer to Figure 10.](#)
WX ON in white font. It indicates that the radar is active and is engaged in the weather basic mode to supply precipitation echoes on 4 intensity levels within the area scanned by the radar.
- [Refer to Figure 11.](#)
WX ALERT in white font. It indicates that the radar is active and is engaged in the weather alert mode. This mode is the same as the previous one, except that echoes with the greatest intensity are displayed flashing continuously.
- WX GMAP in white font. It indicates that the radar is active and is engaged in the ground map mode to supply ground targets on 3 intensity levels with the area scanned by the radar.

When the radar is OFF or in STBY, weather radar symbology is not shown on the MFDs except for the mode itself.

When the radar is active (WX ON, WX Alert or WX GMAP), the related mode is shown and flashes continuously in yellow when the aircraft is on the ground for more than 30 seconds.

If the Antenna/Receiver/Transmitter (ART) does not receive a valid ON discrete, the radar defaults to the WX OFF mode. The WX ON, WX ALERT, or WX GMAP mode annunciation flashes in amber when selected on the ground.

The ND page shows the radar image in the ARC mode when the image display is set to WX OFF. Two white dashed radial lines give the lateral boundaries of the sweep angle of ± 45 degrees.

The image erases after every radar mode, range change, when the image display is selected OFF, or in during confirmed radar failure.

An amber DSPLY OFF shows that the weather radar is on in the ARC mode but is not set on the EFIS Control Panel (EFCP).

The ART and Control Panel (CP) have an extensive initiated and continuous Built-In Test Equipment (BITE) capability that verifies their status through self-test. During all powered-up modes of operation, the ART and CP will be executing a comprehensive continuous self test routine.

When the ART BITE senses a failure condition then its status data will be transmitted to each MFD by the ASCAS 453 ART data bus output. [Refer to Figure 12.](#)

If an ART input stabilization data failure condition is identified and transmitted to the MFDs, then STAB FAIL will be annunciated in yellow text on each Multi-Function Display (MFD), if the radar image display is selected and the ARC option is selected by the ownside EFCP WX and ARC/FULL push-buttons. [Refer to Figure 13.](#) In this event the flight crew have the capability to disable the automatic radar stabilization facility by pulling the tilt control knob located on the Control Panel (CP) and as a consequence of this action, STAB OFF will be annunciated in yellow on each Multi-Function Display (MFD) if the radar image display is selected and the ARC option is selected by its related EFCP WX and ARC/FULL push-buttons.

The specific ART BITE failure conditions transmitted to the MFDs by the ART data bus data link and the subsequent action taken by the MFDs are summarized in the table that follows:



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MALFUNCTION	INDICATION
ART Receiver / Transmitter Processor Fault	WX FAIL shown in yellow text on MFD & WXR system failure status is reported to the CDS
ART Antenna Fault	WX FAIL shown in yellow text on MFD & WXR system failure status is reported to the CDS
ART Transmitter Fault	WX FAIL shown in yellow text on MFD & WXR system failure status is reported to the CDS
ARINC 429 Pitch & Roll Data Source Fault	STAB FAIL shown in yellow text on MFD.

If the radar is defective, none of the weather radar symbology is shown on the MFD(s) except for the failure message itself.

The ART LRU 28 Vdc is powered from the aircraft's Left Main Bus, through a 5 A circuit breaker. The ART CB is located on the 28 Vdc avionics circuit breaker panel at position H2 (WX RDR).

The CP LRU 28 Vdc is powered from the aircraft's Left Main Bus, through a 1 Amp circuit breaker. The CP circuit breaker is located on the 28 Vdc avionics circuit breaker panel at position H1 (WX CP).

The Weather Radar (WXR) System receives data from the sources that follow:

SOURCE	PARAMETER
WX CP	Ground Mapping
ART	Bad Weather Alert
ART	Fault Annunciations
EFCP 1, EFCP 2	Range Selection WX Display Brightness Arc/Full Navigation Display Antenna Tilt Antenna Gain Test

Transceiver

The Antenna/Receiver/Transmitter (ART) has a radar receiver and a radar transmitter with an installed antenna, and is constructed as a single unit. A configuration module is used to store system strapping information and it is an internal part of the ART, but it is not attached to it. The ART receives radar stabilization input for combined pitch, roll and tilt angles of up to ± 30 degrees.

The antenna is set up to operate at a 90 degree scan. The antenna array is attached to the microwave assembly on the ART and the two move together as the radar scans. The receiver and all processing and control circuits are located in the base of the ART.

The ART is processor controlled and supplies display and system fault data to the MFDs over an ARINC 453 data bus in order to



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supply the visual display and facilitate fault annunciation on the EFIS. The ART continuously monitors the data that follows:

- Pitch and Roll data received by an ARINC 429 interface with the validated AHRS source transmitted by IFC2
- Flight crew control data received by the two ARINC 429 control interfaces with IFC1 and IFC2
- Built in test parameters and system annunciators.

The Antenna/Receiver/Transmitter (ART) aircraft connector is a D-type 50 pin slide-lock connector type.

A label attached to the base plate of the Antenna/Receiver/Transmitter (ART) LRU states the unit serial number, TSO approval details, part number, applicable software standard & level (i.e. RTCA/DO-178A, Level 2 Essential), applicable qualification standard (i.e. RTCA/DO-160C) and its hardware and software modification status.

The ART is a digital computing radar Antenna/Receiver/Transmitter LRU that has the main modules that follow:

- Low voltage power supply unit
- Transmitter
- Receiver
- Automatic frequency control
- Weather data accumulator and 453 transmitter
- Logic and control
- Gyro and antenna stabilization.

- Low Voltage Power Supply Unit: The low voltage power supply unit receives unfiltered 28 Vdc from the aircraft bus and supplies the required operating voltages to the WXR. The low voltage power supply has the parts that follow: a hash filter to remove unwanted noise and spikes from the aircraft power bus, a low voltage power supply, and a timer circuit to hold the power supply on for a fixed amount of time after the unit is turned off. This time out is necessary so that the antenna is parked in the down position when the unit is turned off.
- Transmitter: The transmitter portion of the radar has of a high voltage power supply, pulse transformer, magnetron, magnetron isolator, circulator, harmonic filter and the antenna. The high voltage power supply and the pulse transformer supply the magnetron with a pulse of 3600 V at 3 A for 4 msec. The timing of the 4 msec pulse is generated by a pulse timing circuit on the logic board.
- Receiver: The transmitted rf pulse travels from the antenna at a rate approximately equal to the speed of light to a weather target and then is reflected back to the antenna. The antenna supplies a gain to reflected energy of 27.8 dBi. The received energy from the antenna is then passed through a harmonic filter into a three port circulator. The three port circulator prevents the received energy from passing towards the magnetron and allow the energy to pass to the receiver. From the three port circulator, the received energy passes through a diode limiter which is used to protect the receiver from the high amount of energy produced by the magnetron.



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- Automatic Frequency Control: The Automatic Frequency Control (AFC) is used to control the frequency of a local oscillator in the receiver module. The local oscillator is mixed with the receiver signal and a 59 MHz IF frequency is produced. In order for the 59 MHz IF frequency to remain at 59 MHz, the AFC circuit must adjust the local oscillator periodically to correct for magnetron frequency drift. The AFC circuitry has three basic parts; a frequency sample circuit, a microprocessor, and a local oscillator voltage drive circuit.
- Weather Data Accumulator and 453 Transmitter: The weather data accumulator circuit has the components that follow: video A/D, WX ROM, weather compensation and extended Sensitivity Timing Control (STC) circuit, WX RAM, radial sum circuit, 453 RAM, and a 453 serial data transmitter. The accumulator and transmitter converts video data from the ART into a 453 output data. It first converts the analog video into compensated digital data which is filtered by the weather accumulator. This filtered data is stored in the WX RAM where it is transferred to the 453 RAM. From the 453 RAM, the weather data is routed through the coupler to the displays over an ARINC 453 bus.
- Logic and Control: The logic and control section has a microprocessor with it's support circuits, and the ARINC 429 data receivers. The microprocessor is a Zilog Z8400 and is configured in a memory mapped I/O configuration.
- Gyro and Antenna Stabilization: The gyro and antenna stabilization circuit has three parts; the analogue gyro inputs, the azimuth and tilt motor drive circuits, and the Hall effect sensors. The azimuth motor drive circuit and the tilt motor drive circuit are located on the power supply/modulator/motor drive board. These drivers receive commands from the microprocessor to either step the motors up or down or not to step the motor at all. The azimuth and pitch motors are stepper motors and step the antenna in 0.05 degree steps. The two Hall effect sensors are used to determine the position of the antenna in both the tilt and azimuth axis. The input voltage from the Hall effect devices is then multiplexed to a 16 input A/D converter.

The complete Antenna/Receiver/Transmitter (ART) system (i.e. Receiver/Transmitter + Antenna + Configuration Module) has a weight not more than 11 lb (4.99 kg). The circular base of the ART has a diameter of 10.03 in. and the 12 in. (305 mm) antenna tilts and scans about a pivot axis that is located at 5.11 in. (129.79 mm) from the centre of the base. The aircraft connector interface is located at the bottom of the ART base in order to minimize the possibility of water ingress within the mating connector. In order to supply optimum Weather Radar (WXR) system performance, the radome will be compliant with Class A of RTCA/DO-213 and the radome installation will be sealed to prevent water ingress and the accumulation of condensation within the nose section.

The ART is installed to the forward bulkhead in the non-pressurised nose section at station X-162.2, Y+0.1 and Z+100.5. It is installed within 1degree of level in both the vertical and horizontal planes along the longitudinal (fore-aft) centreline of the aircraft. The location of each Weather Radar (WXR) system component has been chosen

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to allow free air circulation around the unit in order to limit its maximum operating temperature and increase its reliability.

Antenna, Weather Radar

Refer to Figure 14.

The vertically polarized antenna is a 12 in. (305 mm) aluminum alloy, slotted, flat plate antenna array that is attached to the microwave assembly on the ART and the two move together as the radar scans. The radar scan angle is set up to operate at a 90 degree scan.

Panel, Radar Control

Refer to Figure 15.

The Control Panel (CP) is processor based, and designed to facilitate mode selection of the WXR system within the IMS100 avionics suite.

The Control Panel (CP) has front push-buttons, rotary switches and knobs that accept mode selection from the flight crew and converts them to serial ARINC 429 digital data. The data is transmitted to IFC1 and IFC2, and a dedicated power on/off control discrete interface with the ART. The mode selections are converted by the radar control panel to serial ARINC 429 digital data that is sent to the symbol generator.

The Control Panel (CP) has a power supply/logic module, a front-panel switch/light module, and a wire harness. The front-panel switch/light module has front-panel push-buttons, back-lighting circuit, and the connections to the function switch, the TILT control

and GAIN control. The Control Panel (CP) module has the functional blocks that follow:

- Power supply
- Micro-computer
- Reset
- Analogue/digital converter
- Addressable latch
- ARINC 429 transmitter
- Front panel controls
- Front panel switches.
- Power Supply: A power supply/logic module that receives unfiltered 28 Vdc from the aircraft bus and 5 Vdc from the aircraft lighting bus. These voltages supply the power to the CPs circuits; the TTL logic circuits, the variable resistors bias voltage, and the control panel lighting.
- Micro-computer: The power supply/logic module has a micro-computer, an 8031 microprocessor and an EPROM chip, which monitors the front panel knobs, switches and push-buttons, and sends radar control status data and CPU status data to the IFCs by its ARINC 429 output data bus.
- Reset: The reset circuit resets the microprocessor to a known state in the event of dc power failure or if a loss of programming is detected.



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- Analogue/Digital Converter: The TILT and GAIN knobs are located on the front panel and are connected to variable resistors. When a knob is turned, the variable resistor voltage produces an analog signal. This analog signal is changed to digital data by a multiplexer A/D circuit. The multiplexing A/D circuit has an 8-channel multiplexer and an 8-bit A/D converter. The microprocessor controls an 8-channel multiplexer to select one of the analogue data sources. The output of the A/D converter is sent to the microprocessor data bus to be processed.
- Addressable Latch: Digital data from the micro-computer is latched to produce sensing signals, TTL data, and a watch-dog signal. Digital data transmitted by the micro-computer is latched by the addressable latch to produce a TTL data signal. The output of the latch is sent to a TTL-bipolar converter to change the data to an ARINC 429 format. The watch dog signal is transmitted by the micro-computer and latched by the addressable latch. The reset circuit monitors the addressable latch's output to determine if the micro-computer is sequencing through the programme properly. Data from the micro-computer is decoded and latched by the addressable latch to generate the sensing signals for the front panel switches. If a panel switch is closed the sensing signal is received back from the switch indicating the switch is in a closed state. Likewise if a panel switch is open, the sensing signal is not received back from the switch indicating the switch is in an open state.
- ARINC 429 Transmitter: Data from the micro-computer (i.e. switch states, TILT and GAIN knob states, and CPU status) is sent by the addressable latch circuit to a TTL to bipolar converter circuit which converts the digital signals into ARINC 429 format.
- Front Panel Controls: The TILT and GAIN knobs on the front panel are connected to variable resistors. When one of these knobs is turned, the variable resistor causes an analogue to be produced. This analogue voltage is converted to digital data on the power supply/logic module for processing. [Refer to Figure 16.](#)
- Front Panel Switches: The status of the front panel's push-button momentary switches, push-pull switch, and rotary switch is monitored by circuitry on the power supply/logic module. When the status of a switch is changed the power supply/logic module generates a 429 word indicating the new status. The OFF, STBY, TST, ON rotary switch status also supplies the On/Off control signal for the ART.

The Control Panel (CP) maximum weight is not more than 1.54 lbs and is packaged in a standard ARINC-306 control panel form factor, and is installed using four DZUS fasteners. The front panel has push-buttons, rotary switches and knobs.

The CP is installed in the flight compartment, in the forward center console at station X-65.3, Y0 and Z+121.9.

Couplers

The coupler is a transformer and is located in the wiring harness. It routes display data from the single WX receiver/transmitter to the two MFDs.



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Module, Configuration

[Refer to Figure 17.](#)

A configuration module is used to permanently store the aircraft radar configuration (scan angle, Weight On Wheels [WOW] override, gyro source, etc.) specific to the aircraft configuration. When the radar energizes, it reads the configuration module to set up the particular operating environment for this aircraft. The WOW switch input for the radar is supplied by the PSEU.

Prior to the installation of the configuration module, it is preprogrammed over the RS-232 bus with a configuration module programmer and a personal computer. The configuration module programmer and personal computer execute a dedicated software package to set a configuration outside the default parameters stored in the configuration module.

The data stored in the CM includes ART configurable parameters such as :

- Antenna scan angle (90 degree)
- WOW switch override (when configured, puts the system in standby mode when the aircraft is on the ground and will not let the ART transmit unless the override function is enabled by the WOW input from the PSEU)

- Target alert mode enable. When enabled, a target alert condition shall be communicated to the MFDs by the 453 data bus when Level #3 weather is present. Level #3 weather involves of heavy rainfall rate, possible severe turbulence and lightning within ± 10 degrees of aircraft heading with a minimum cell size of 2 nmi in range and 2 degree in azimuth, at a range greater than 80 nmi but less than 240 nmi with a selected range of 80 nmi or less; and
- Stabilization source data type (when configured to 429, enables the Antenna/Receiver/Transmitter (ART) to operate using attitude source data supplied by the ARINC 429 interface from IOP No. 2).

If the radar cannot read the configuration module the fault annunciation WX FAIL is shown on the MFD.

The configuration module is an internal part of the ART, but it is not attached to it.

Training Information Points

[Refer to Figure 18.](#)

Refer to Caution and Warning List # C34-1W-0 and #C34-3W-0.

The diagram illustrates the EFIS system architecture, showing the flow of data from various inputs to the displays. It is divided into two main sections: PILOT EFIS DISPLAY and COPILOT EFIS DISPLAY.

PILOT EFIS DISPLAY: This section includes the PFD 1 (Primary Flight Display) and MFD 1 (Multi-Function Display). The PFD 1 is connected to the MFD 1. The MFD 1 is connected to the EFCP 1 (EFIS Function Control Panel 1). The EFCP 1 is connected to the WX CP (Weather Control Panel) and the WXR ANTENNA (Weather Radar Antenna). The WX CP is connected to the WX RDR (Weather Radar Receiver) and the WXR ANTENNA. The WXR ANTENNA is connected to the WXR CP and the WXR CONFIG MODULE. The WXR CP is connected to the WXR ANTENNA and the WXR CONFIG MODULE. The WXR CONFIG MODULE is connected to the WXR ANTENNA and the WXR CP.

COPILOT EFIS DISPLAY: This section includes the PFD 2 (Primary Flight Display) and MFD 2 (Multi-Function Display). The PFD 2 is connected to the MFD 2. The MFD 2 is connected to the EFCP 2 (EFIS Function Control Panel 2). The EFCP 2 is connected to the WX CP and the WXR ANTENNA. The WX CP is connected to the WX RDR and the WXR ANTENNA. The WXR ANTENNA is connected to the WXR CP and the WXR CONFIG MODULE. The WXR CP is connected to the WXR ANTENNA and the WXR CONFIG MODULE. The WXR CONFIG MODULE is connected to the WXR ANTENNA and the WXR CP.

Other Components: The diagram also shows the IFC 1 (IOP 1, IOM 1) and IFC 2 (IOP 2, IOM 2) units, which are connected to the WXR CP and the WXR ANTENNA. The PSEU (Proximity Switch Electronic Unit) is connected to the WXR ANTENNA and the WXR CONFIG MODULE. The WXR CONFIG MODULE is connected to the WXR ANTENNA and the WXR CP.

WEATHER RADAR SYSTEM BLOCK DIAGRAM

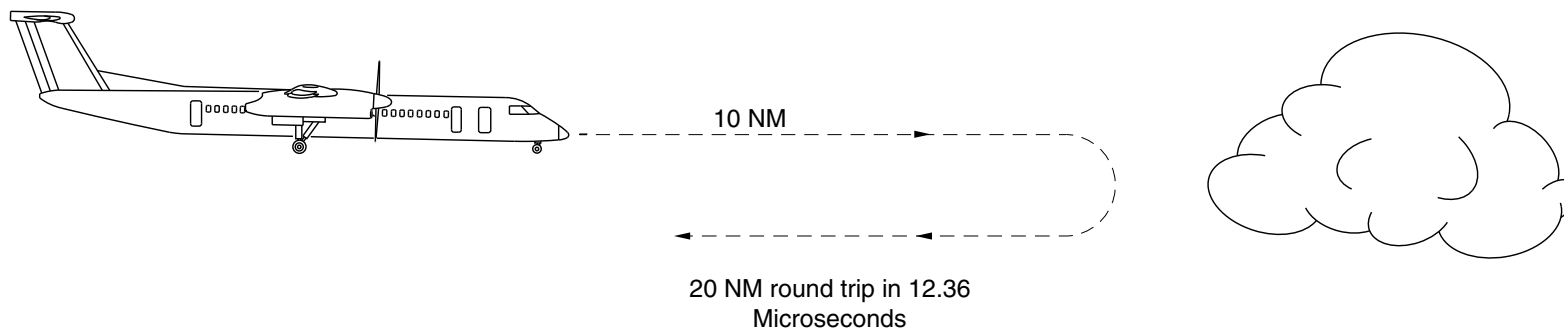
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fs529a01.dg, rc, aug16/2011

Radar Principle
Figure 2

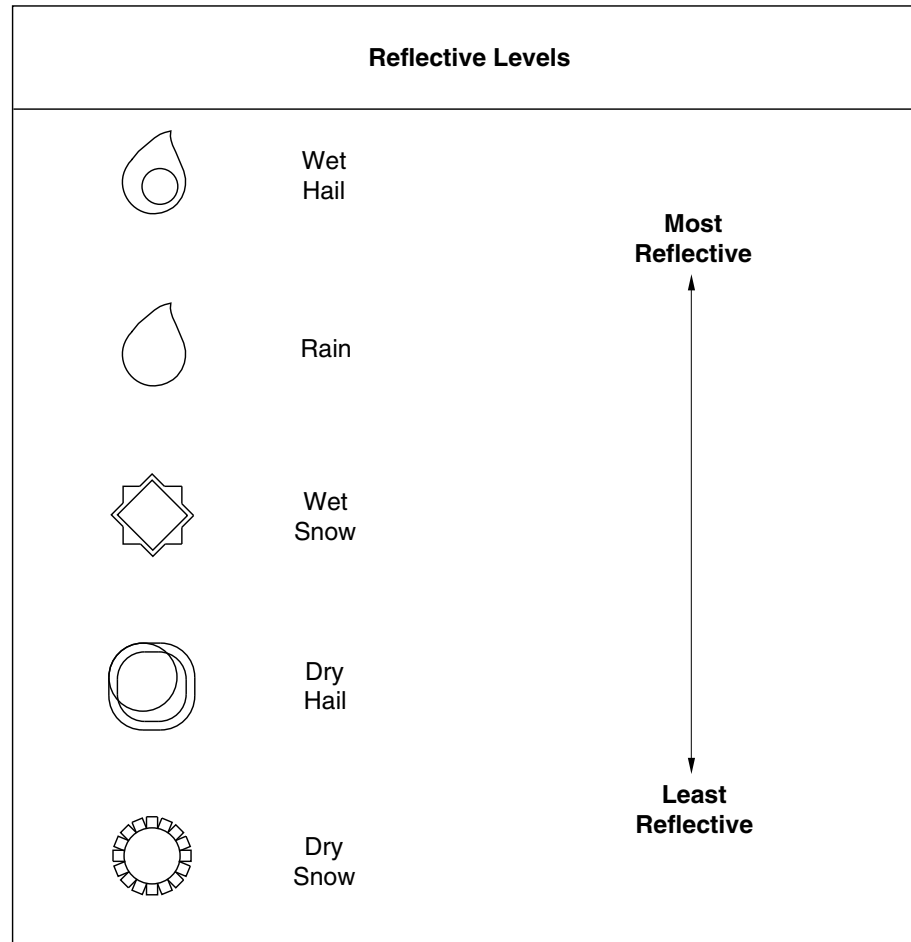
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Reflectivity Levels
Figure 3

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EFFECTIVITY:
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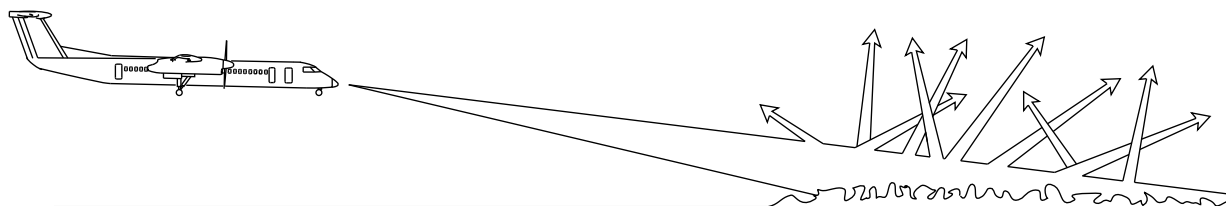
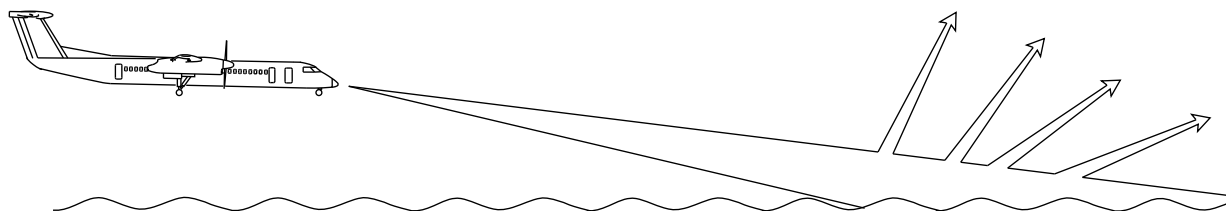
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GROUND MAPPING
Figure 4

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EFFECTIVITY:
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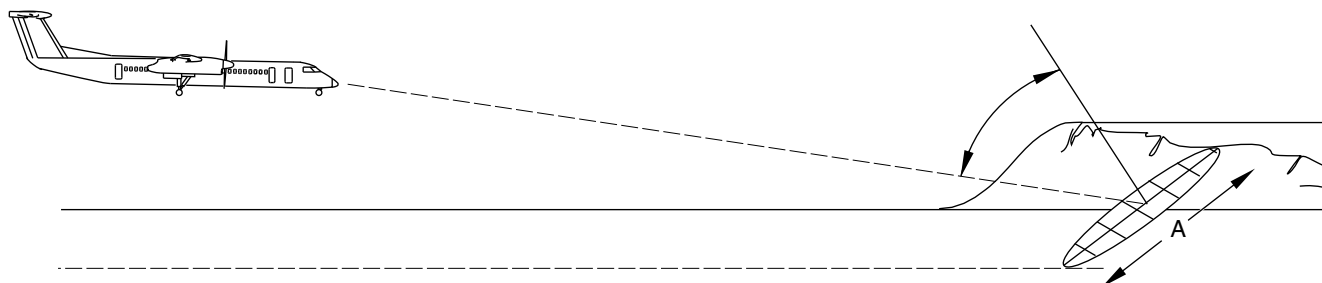
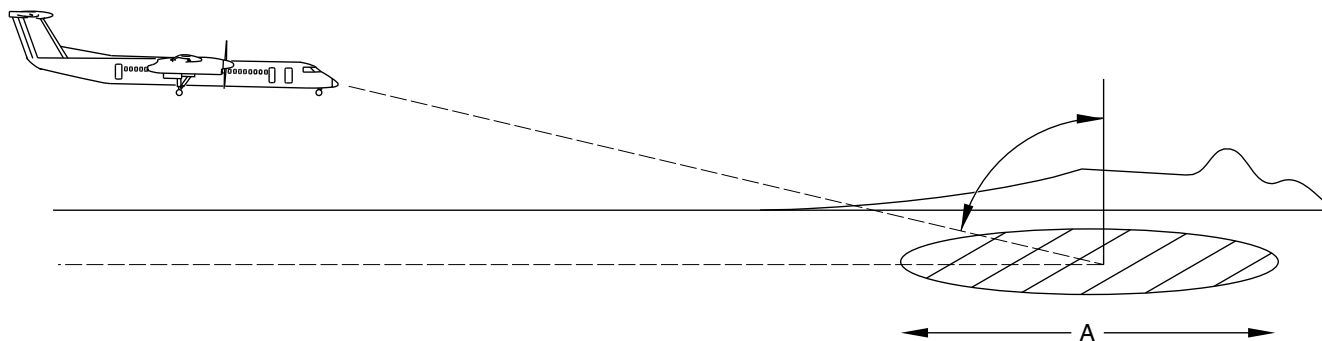
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LOOKING ANGLE
Figure 5

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EFFECTIVITY:
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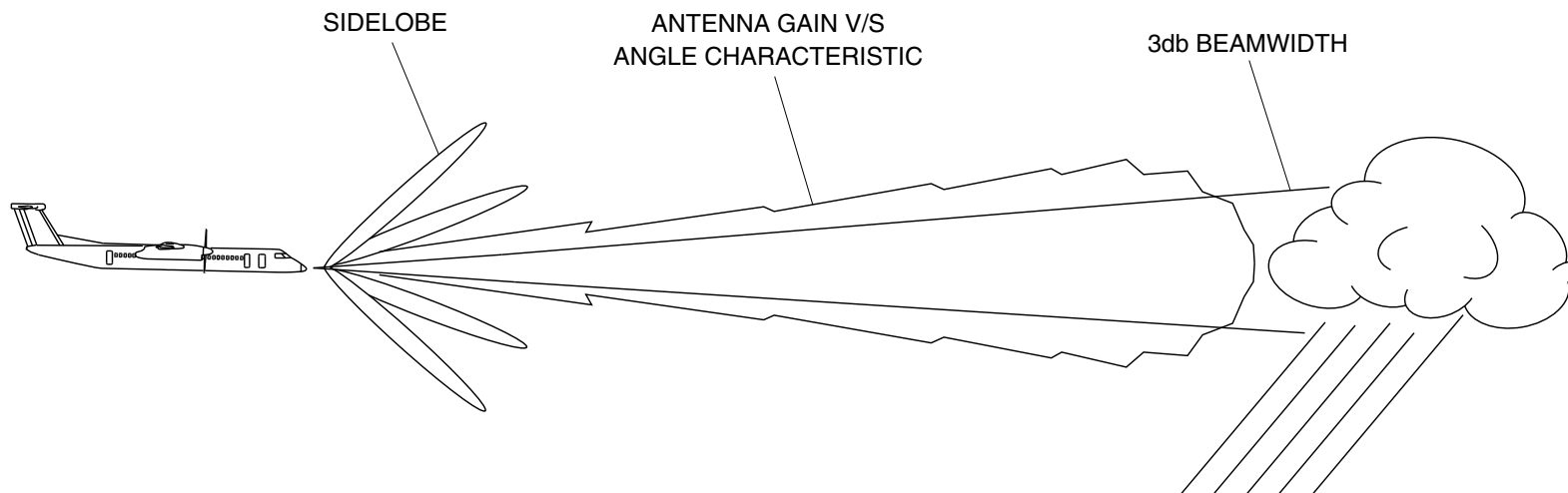
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fs599a01,dg, rc, aug16/2011

Radar Beam Illumination Characteristics
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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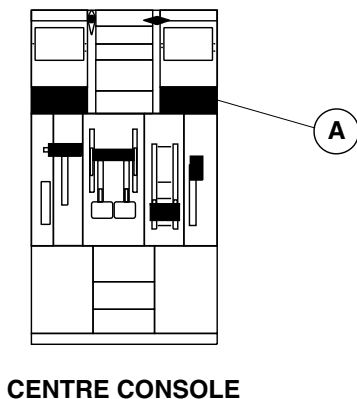
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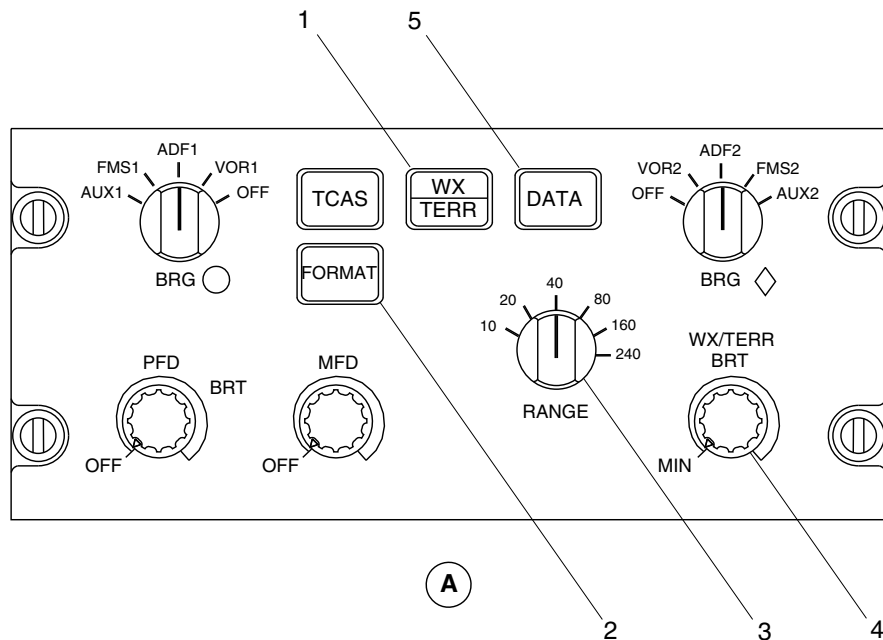
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CENTRE CONSOLE

LEGEND

1. Weather radar/terrain pushbutton switch.
2. FORMAT pushbutton switch.
3. Weather radar display range selector knob.
4. Weather radar display brightness knob.
5. DATA pushbutton switch.



fs634a01.dg, gv, 19/12/02

EFIS Control Panel – Weather Radar Controls
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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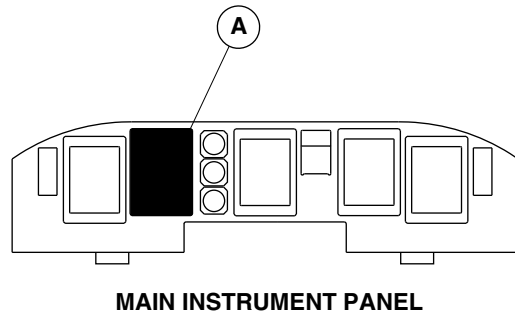
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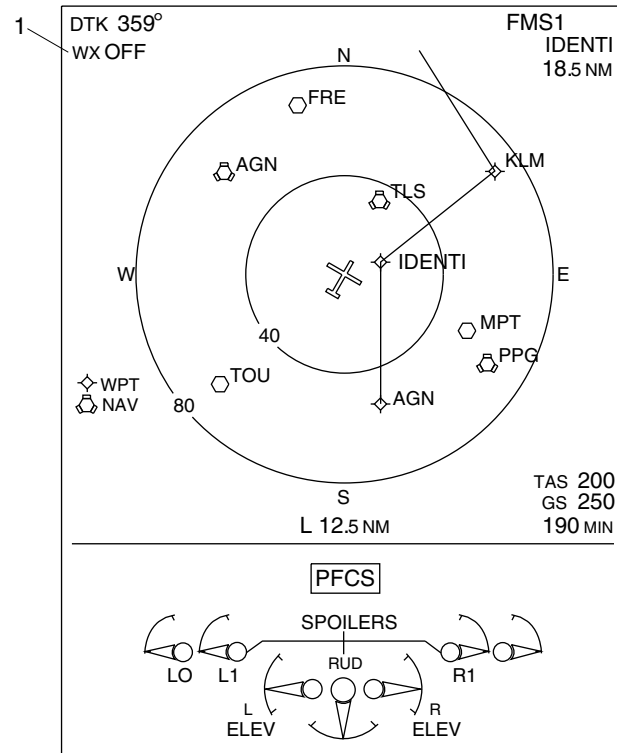
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LEGEND

1. Weather Radar Off Mode Indication.



A

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WEATHER RADAR OFF MODE INDICATION
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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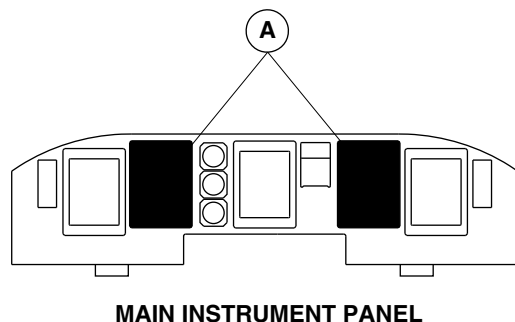
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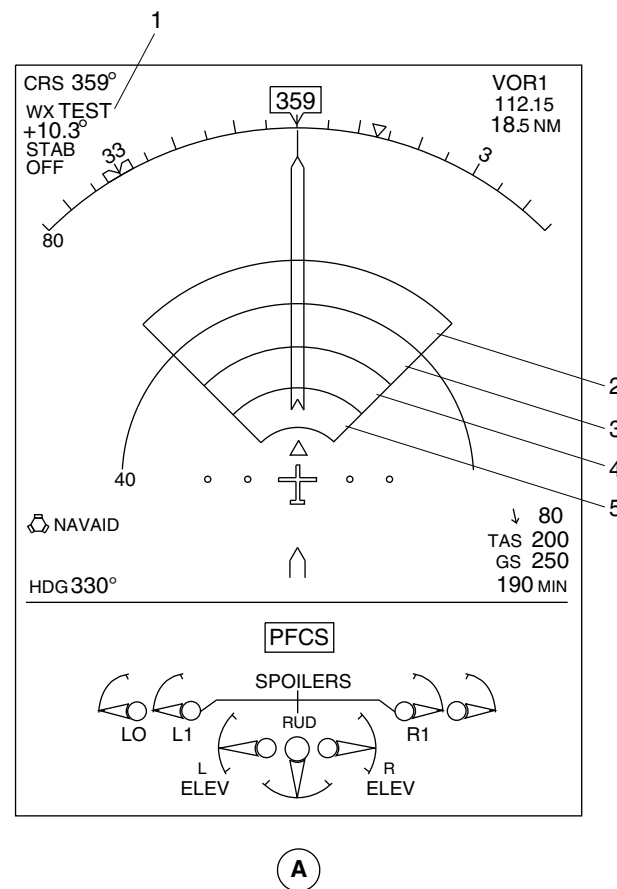
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LEGEND

1. Weather Radar Test Mode Indication.
2. Magenta.
3. Red.
4. Yellow.
5. Green.



fs631a01.cgm

WEATHER RADAR TEST MODE INDICATION
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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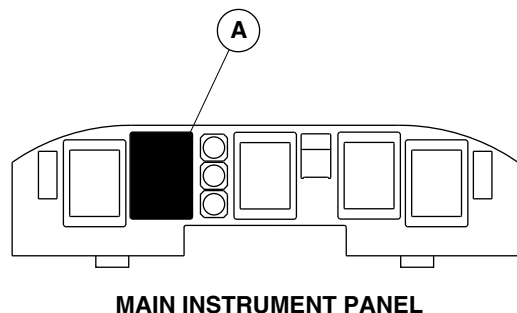
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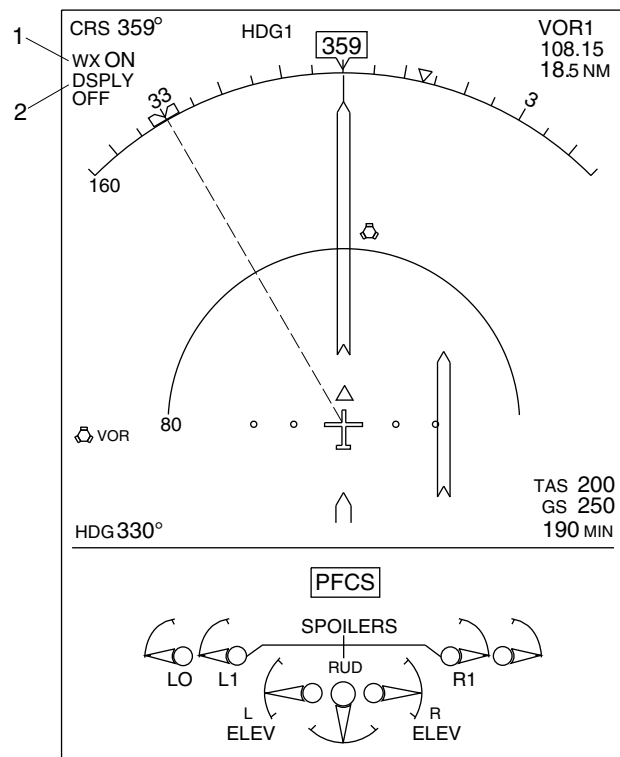
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LEGEND

1. Weather Radar On Mode Indication.
2. Weather Radar Image Display Off Status Message.



A

fs628a01.cgm

WEATHER RADAR ON MODE INDICATION
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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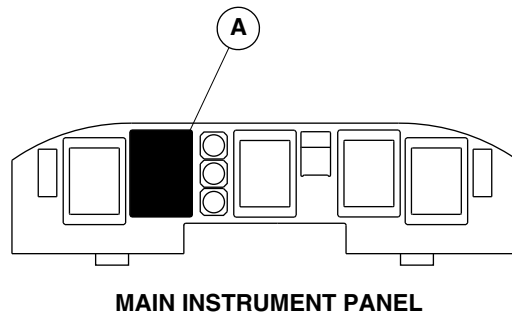
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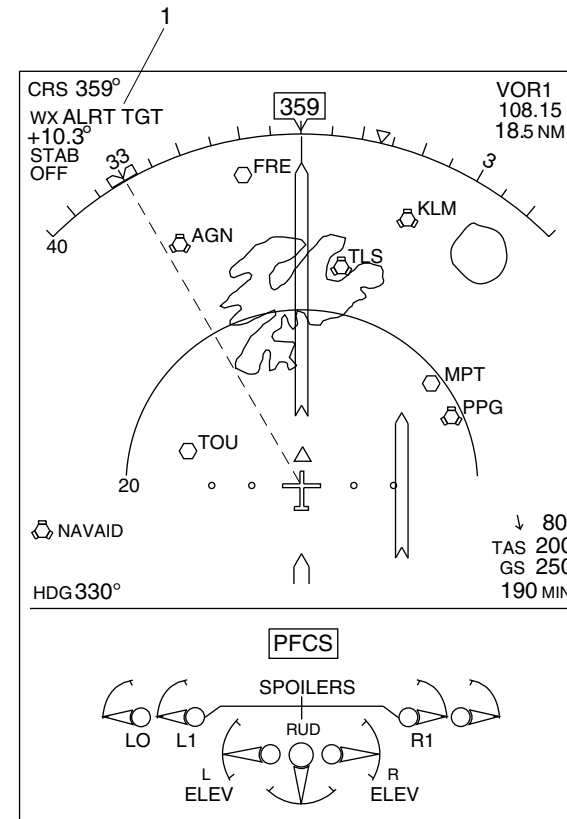
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LEGEND

1. Weather Target Alert Message.



A

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TARGET ALERT INDICATION
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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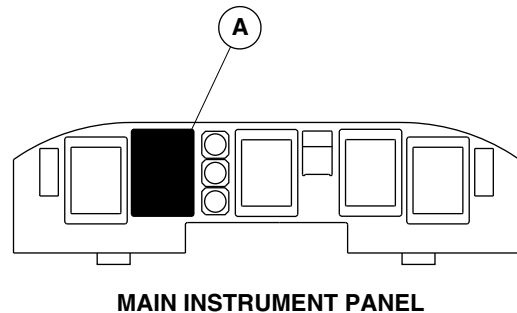
34-41-00

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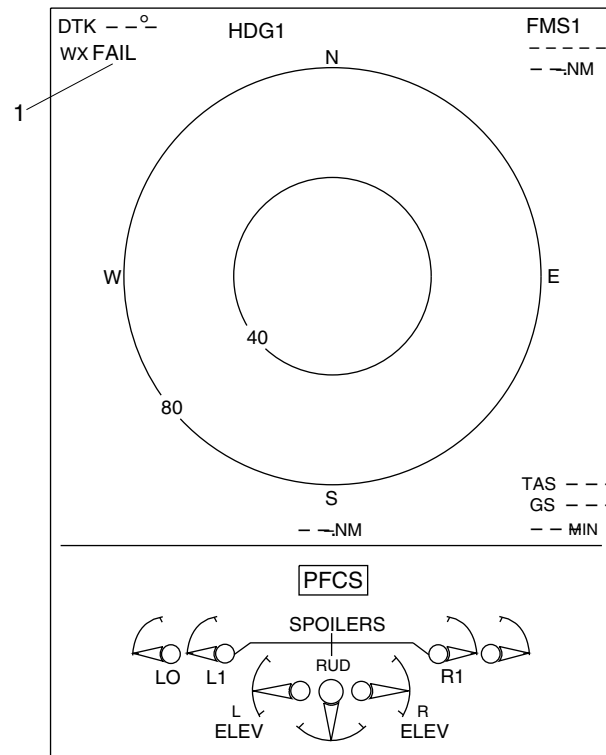
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LEGEND

1. Weather Radar Fail Annunciation.



A

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WEATHER RADAR FAIL ANNUNCIATION
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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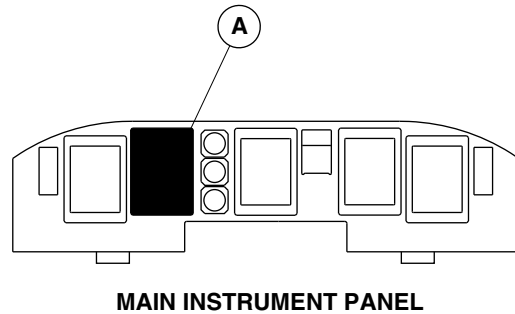
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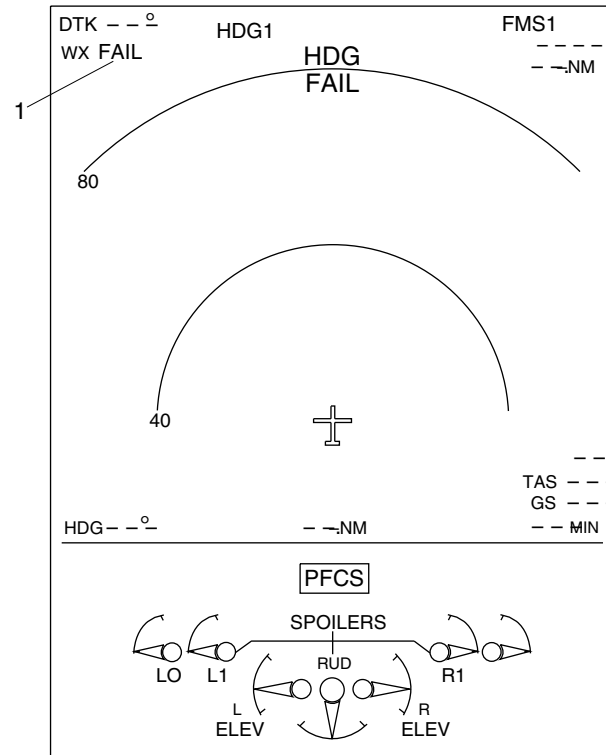
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LEGEND

1. Weather Radar Fail Annunciation.



fs630a01.cgm

WEATHER RADAR FAIL ANNUNCIATION, ARC MODE
Figure 13

PSM 1-84-2A
EFFECTIVITY:
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Config 001

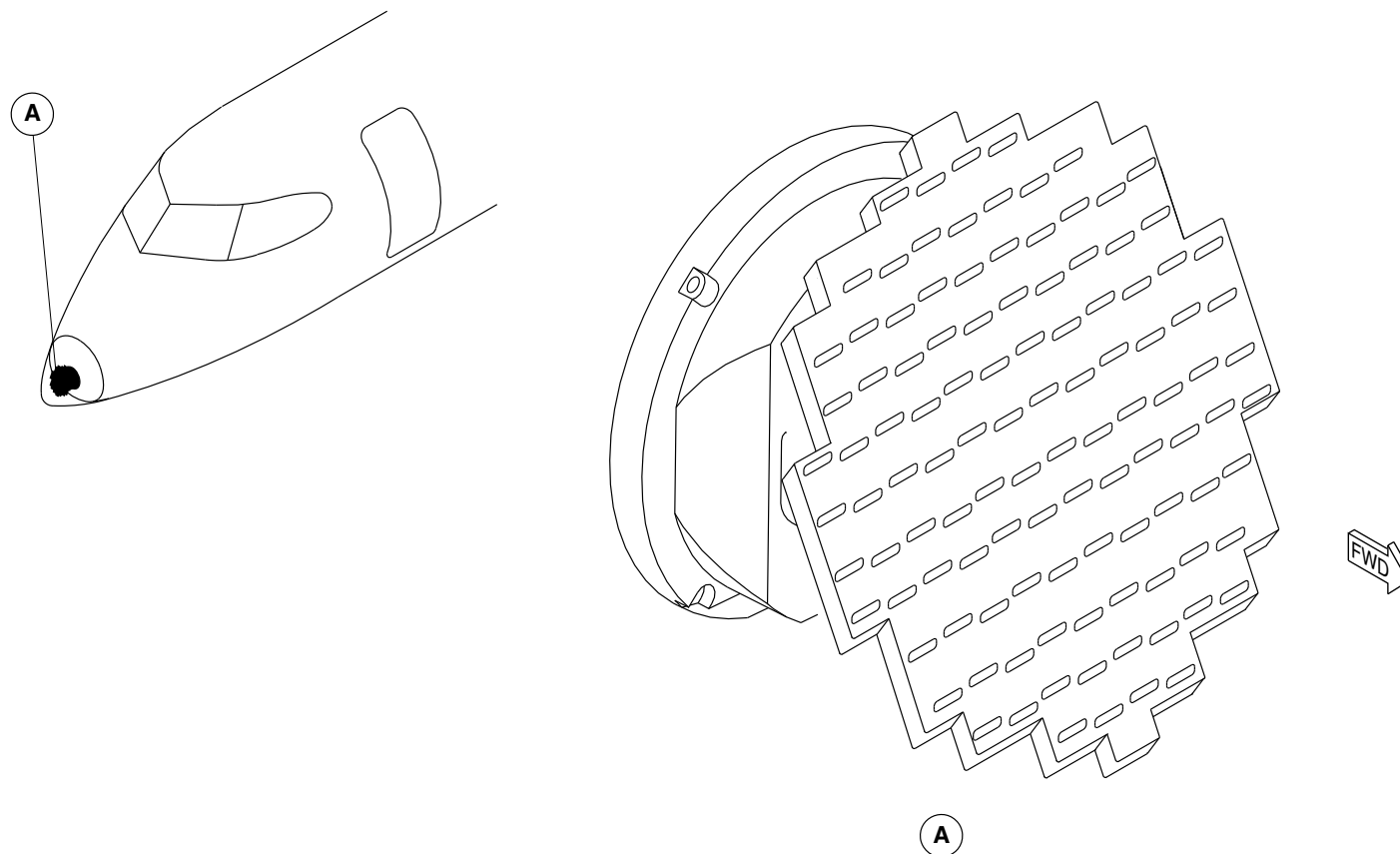
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fs600a01.cgm

WEATHER RADAR ANTENNA LOCATION
Figure 14

PSM 1-84-2A
EFFECTIVITY:
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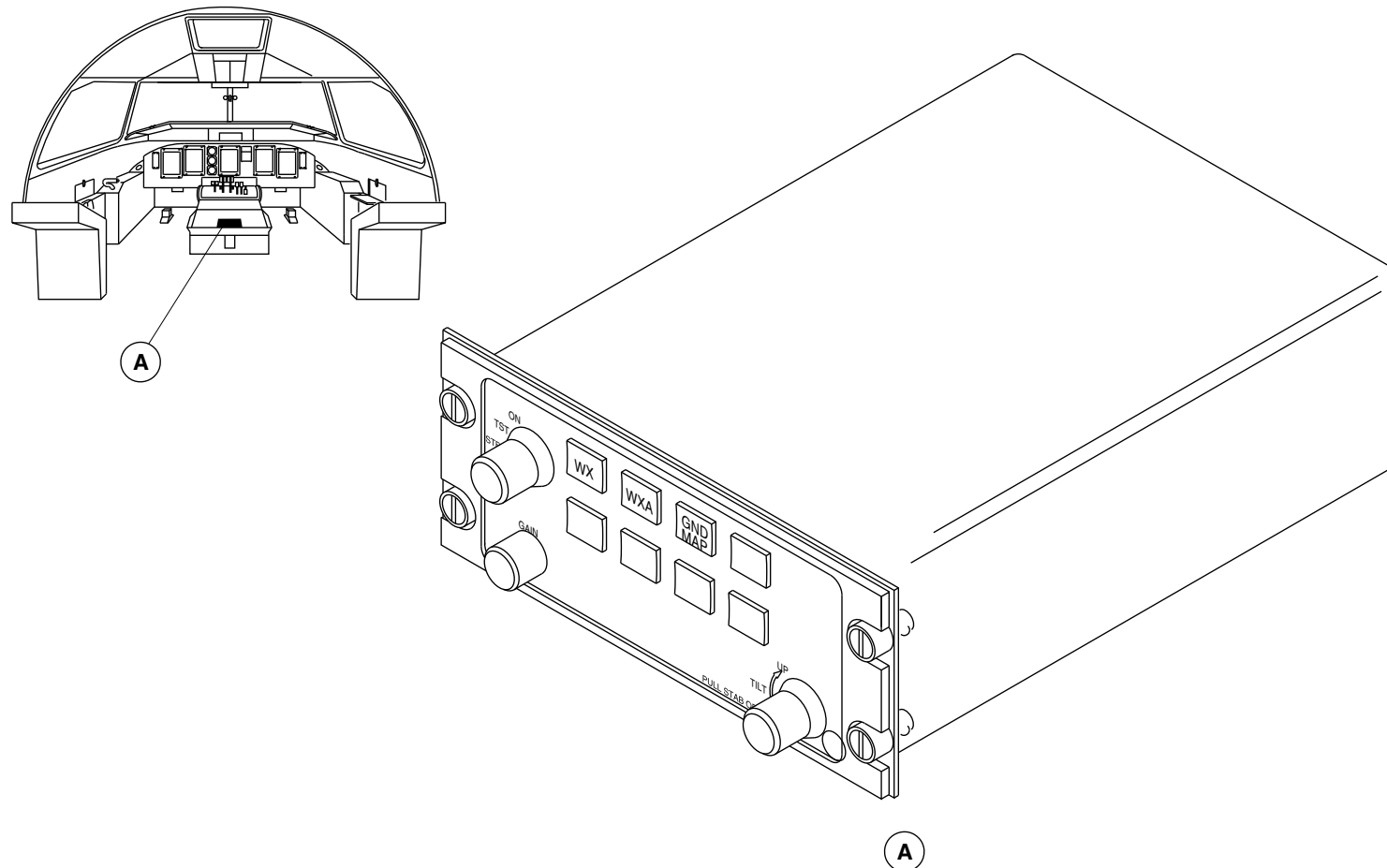
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WEATHER RADAR CONTROL PANEL LOCATION
Figure 15

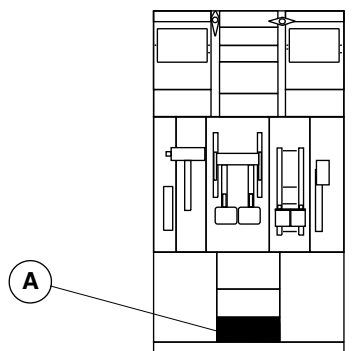
PSM 1-84-2A
EFFECTIVITY:
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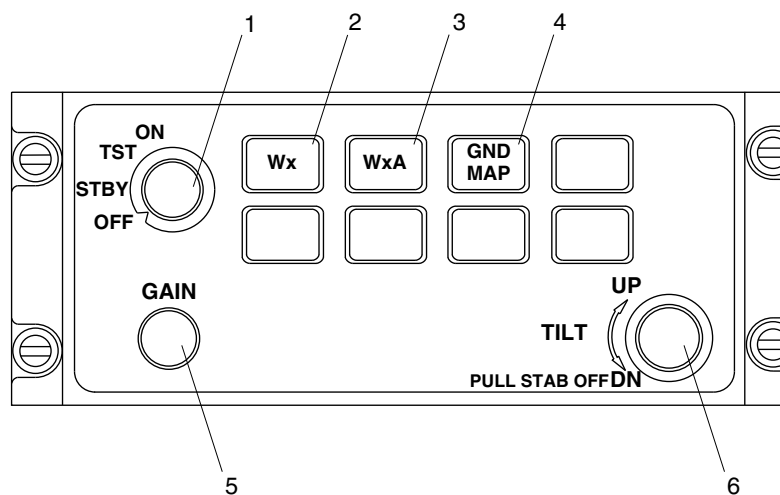
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

LEGEND

1. Weather Radar Control Knob
2. Weather Radar Mode Pushbutton Switch.
3. Weather Radar Alert Mode Pushbutton Switch.
4. Ground Mapping Mode Pushbutton Switch.
5. Weather Radar Gain Control Knob.
6. Weather Radar Antenna Tilt Knob.



A

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WEATHER RADAR CONTROL PANEL SWITCHES
Figure 16

PSM 1-84-2A
EFFECTIVITY:
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Config 001

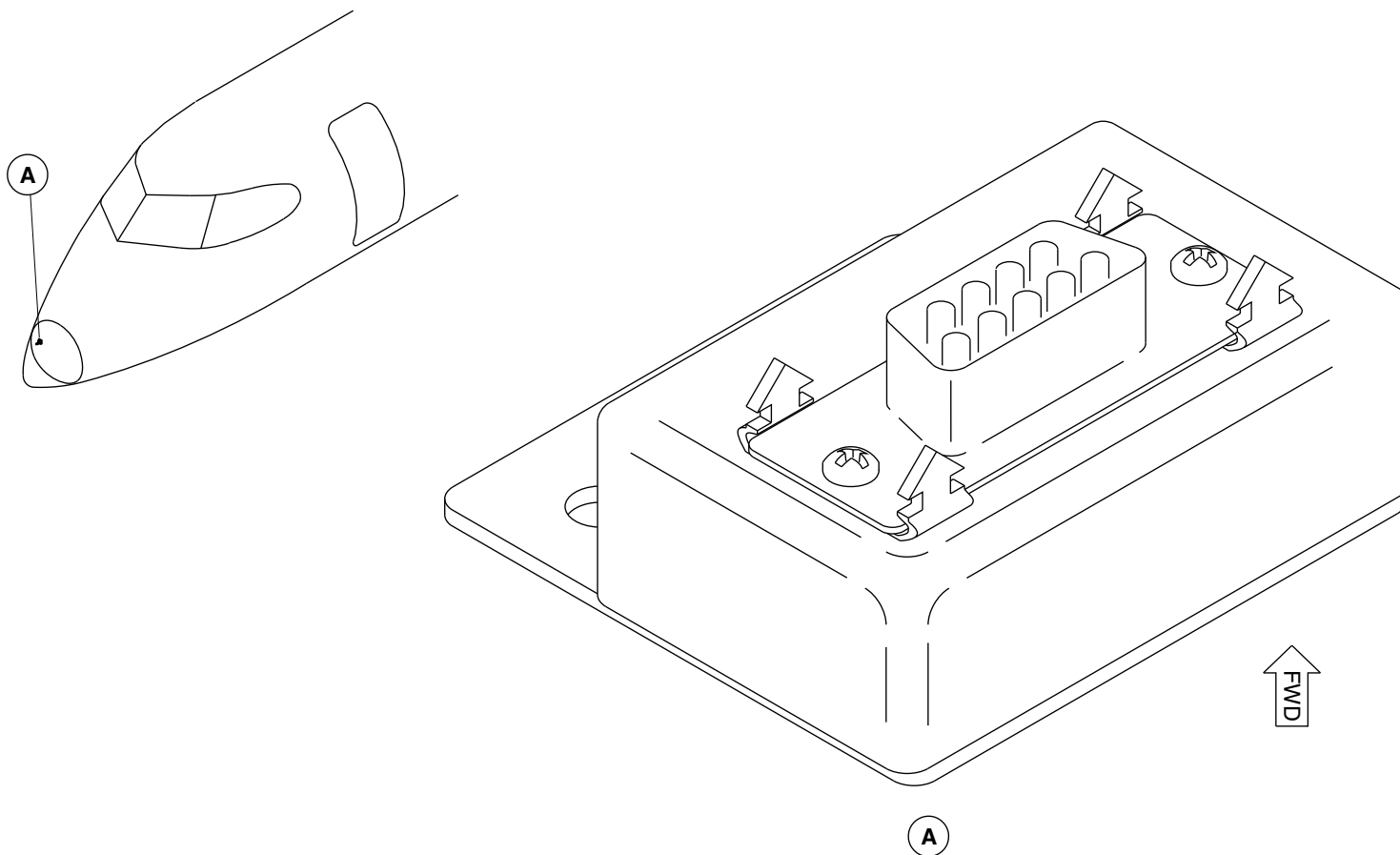
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CONFIGURATION MODULE LOCATION
Figure 17

PSM 1-84-2A
EFFECTIVITY:
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Config 001

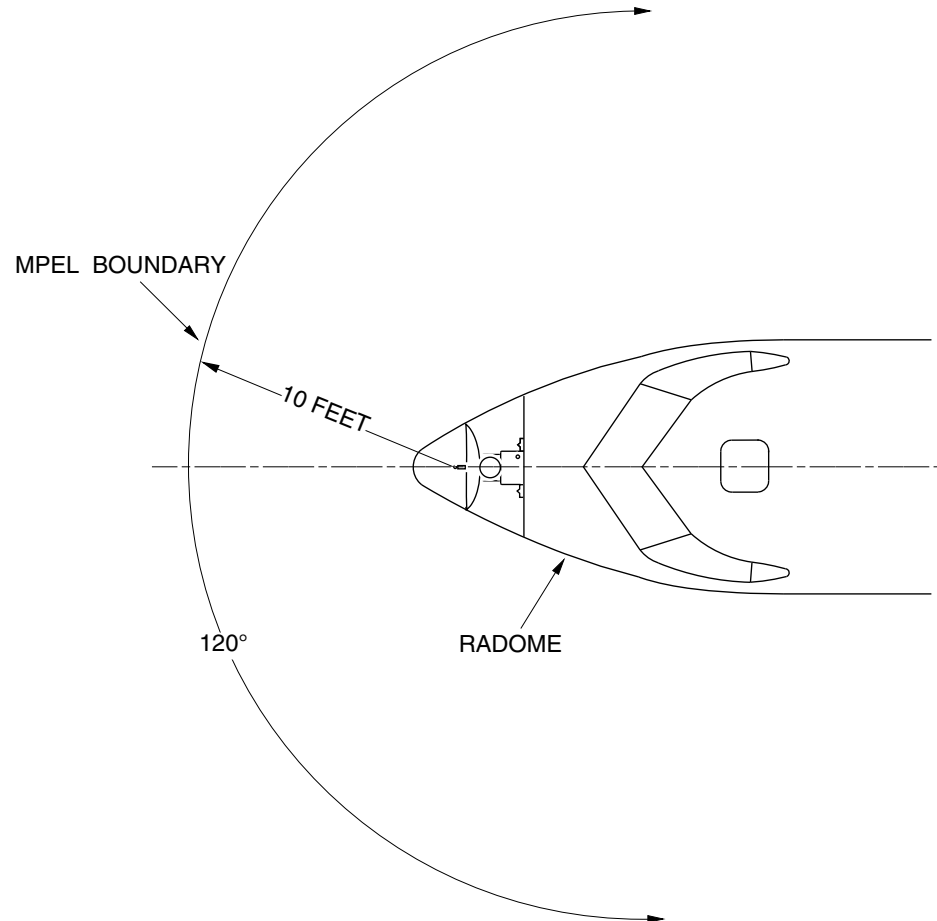
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MAXIMUM PERMISSIBLE EXPOSURE LEVEL (MPEL)
Figure 18

PSM 1-84-2A
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WEATHER RADAR SYSTEM (PRIMUS 660)

Introduction

The Primus 660 Weather Radar (WXR) system helps the pilot to avoid thunderstorms and associated turbulence. When selected, it supplies a continuous visual color display of the weather conditions to the Multi Function Displays (MFD 1 and MFD 2).

General Description

[Refer to Figure 19.](#)

The Primus 660 weather radar system is a light weight X-band digital radar with alpha numerics assigned for weather detection and ground mapping. The radar system also supplies the flight crew with operational features such as range selection, ground mapping, target alert, and fault annunciations.

The Primus 660 digital weather radar system detects and locates various types of storms within a 300 nm range ahead of the aircraft, and within an arc of approximately 120 degrees, and provides the pilot with a visual indication of their turbulence content based on the color of the display.

The weather radar system is operated by a control panel on the center console. The weather radar system has the components that follow:

- Receiver Transmitter (34-41-01)
- Antenna, Weather Radar (34-41-06)

- Panel, Radar Control (34-41-11)
- Coupler (34-41-26).

The Primus 660 weather radar system has a forced standby feature that inhibits the transmitter on the ground to eliminate the X-band microwave radiation hazard.

Detailed Description

Radar is a distance measuring system using the principle of radio echoing. The term RADAR is an acronym for RAdio Detecting And Ranging. The transmitter sends microwave energy in the form of pulses. These pulses are then transmitted to the antenna and focused into a radar beam. The radar beam is focused and transmitted by the antenna in such a way that it is most intense in the center of the beam and decreases in intensity near the edge. The same antenna is used to transmit and receive radar signals. When a pulse intercepts a target, the pulse is reflected as an echo, or return signal, back to the antenna. From the antenna, the returned signal is transmitted to the receiver transmitter unit. The echoes, or returned signals, are shown on the MFDs.

[Refer to Figure 20.](#)

Radio waves travel at the speed of 300 Mm/s and thus yield nearly instantaneous reply information. Radar ranging is a two way process that requires 12.36 μ s for the radio wave to travel out and back for each nautical mile of target range.

[Refer to Figure 21.](#)

Precipitation or objects more dense than water such as earth or solid structures are sensed by an X-band weather radar. The weather radar does not detect clouds, thunderstorms or turbulence directly. Instead, it detects precipitation which may be associated with



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thunderstorms and turbulence. The best radar reflectors are raindrops, wet snow or hail. The larger the raindrop the better it reflects. Large drops in a small concentrated area are characteristic of a severe thunderstorm, the MFD shows the storm as a strong echo. Drop size is the most important factor in high radar reflectivity. The larger the raindrop the better it reflects. Generally, ice, dry snow and dry hail have low reflective levels and often will not be shown on the MFD.

A cloud that has only small raindrops, such as fog or drizzle, will not produce a measurable radar echo.

[Refer to Figure 22.](#)

Calm water or water with swells do not reflect good radar signals. The pulse is reflected in a forward scatter angle without all portions being returned. The resulting display is no target.

Choppy water supplies better radar pulse reflections from the downwind sides of the waves. The resulting display is a target whose intensity will change with the degree of choppiness.

Illumination of terrain results in a "diffused" reflection of the beam. A portion of this reflected energy is scattered back toward the antenna and land features as well as lakes, large rivers, shore lines and ships come into view on the display.

[Refer to Figure 23.](#)

The incident angle at which the terrain is illuminated has a direct bearing on the detectable range and the area of illumination. A large incident angle gives the radar system a smaller detectable range of operation (due to a minimized reflection of direct radar energy). However, the illuminated "A" is larger.

A smaller incident angle gives the radar a larger detectable range of operation because of an increase of direct radar energy reflected from the target to the antenna. The area of illumination "A" is smaller.

Concentration of the beam energy on a small area of terrain increases the magnitude of the echo intercepted by the antenna. The detectable range is therefore increased for mountainous terrain, the maximum distance at which this terrain can be monitored is greater because of the more direct reflection (or radar echo) produced.

[Refer to Figure 24.](#)

The antenna gain versus the angle characteristic is a continuous function at all angles. This means that there is a gain value associated with all forward angles relative to the selected tilt angle. Outside of the beam width, the antenna gain does not go to zero, but continues to reduce with increasing angles (continuous gain function).

Sidelobes are formed during transmission, but their gain value size makes them unimportant. A damaged radome can increase the gain value size of the sidelobes to a point where they cause a constant radar reflection from the ground. This is referred to as an "altitude ring" because the display will show a concentric ring at a distance equal to the slant range of the sidelobe to the ground.

The "cone" formed by the narrow beam width is where most of the radar energy is concentrated, so it is important to realize that at any given time whatever is within the cone (and sometimes other strong targets like clutter outside the cone) is what is being shown on the display.

When a radar pulse is transmitted into the atmosphere, it is gradually absorbed and scattered so that it loses its ability to return to the antenna. This attenuation or weakening of the radar pulse is caused by two primary sources, distance and precipitation. The Receiver



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Transmitter Antenna (RTA) has several advanced features which significantly reduce the effects of attenuation (no airborne weather radar can eliminate them completely).

Attenuation due to precipitation is far more intense and is less predictable than attenuation due to distance. As the radar pulses pass through moisture, some radar energy is reflected and absorbed. If the rain is very heavy or extends for many miles, the beam may not completely travel through the area of precipitation. If the beam is fully attenuated, the radar will display a “radar shadow” which is shown on the display as an end to the precipitation when, in fact, the heavy rain may extend for many more miles. Precipitation attenuation may cause the area of heaviest precipitation to be shown as the thinnest area of heavy precipitation. One cell that has heavy precipitation may fully block or shadow a second cell located behind it and prevent it from being shown on the radar.

The WXR system supplies a continuous visual color display of the weather precipitation conditions to the MFDs. It also supplies the flight crew with additional operational features such as:

- Range selection
- Ground mapping
- Bad weather alert
- Target alert
- Fault annunciation.

Receiver Transmitter Antenna (RTA)

[Refer to Figure 32.](#)

The RTA is an integrated unit that incorporates a transmitter, receiver, and antenna into a single unit. The RTA is cantilever-mounted on the forward side of the bulkhead and protected by the nose cone radome. The bottom of the receiver transmitter antenna also contains maintenance related switches and an electrical connection is provided by a multi-pin connector.

The receiver transmitter antenna operates in the frequency range of 9350 MHz to 9400 MHz with a nominal magnetron power output of 10 kilowatts. The transmitter section delivers short, high-power pulses of radio-frequency energy to the antenna, which radiates the energy as a narrow beam in a direction determined by the antenna scan and tilt positions.

Pulses reflected back to the antenna by weather or ground targets are applied to the receiver section of the receiver transmitter antenna, where they are converted to an intermediate frequency, amplified, processed and applied to the indicator as video signals. Other interconnections provide sync pulses and display information to the indicator. The receiver transmitter antenna receives control information from the indicator through a Serial Control Interface (SCI) data bus.

Separate outputs from the receiver provide color video, x/y coordinates, Rho-theta and digital display information to each of the EFIS system, for conversion to the display format required by the EHSI instruments.

The antenna is a 18 in. (45.72 cm) diameter flat plate phased array combining receive and transmit signals of the radar system. The RTA is mounted on the pedestal assembly which provides antenna



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

movement in the vertical and horizontal planes. The receiver transmitter antenna contains the electromechanical drive components and feedback circuits for antenna rotation.

Logic drive signals from the receiver–transmitter operate the azimuth motor to sweep the antenna horizontally right and left for a total sweep of 60 or 120 degrees, as set by the control on the control panel. The radar scan rate is 24 sweeps per minute for the 60 degree sweep, and 12 sweeps per minute for the 120 degree sweep.

The receiver transmitter antenna circuits include antenna stabilization, data processing and timing, and regulated and unregulated dc voltage power supplies for operation of the radar system. In addition to the transmitting and receiving functions, the receiver transmitter antenna contains self–test circuits.

Vertical adjustment of the antenna is actuated by combined signals from the tilt control, signals from the vertical gyro output of the AHRS system and the airspeed and altitude signal from the air data computer. These signals permit the radar antenna to be tilted up or down 30 degrees. In normal flight, the antenna is automatically maintained in a vertical plane when the tilt control is selected to zero, or at a selected tilt angle, by servo and stabilization circuits, referenced to the AHRS pitch and roll outputs.

Multi Functional Display

The MFD provides the visual displays of weather and ground mapping targets.

Each digital display cell can be black or one of four colors, dependent on the intensity of the returned signal. In weather detection mode, target returns are displayed at one of five video levels (0, 1, 2, 3, and 4). Zero is represented by a black screen

because of weak or no returns, and levels 1, 2, 3, and 4 are represented by green, yellow, red, and magenta respectively to show progressively stronger returns. In ground mapping mode, video levels of increasing reflectivity are displayed as black, green, yellow, and magenta.

A 60 degree or 120 degree sector ahead of the aircraft can be shown, selectable by the SECT push–button on the Radar Control Panel (RCP). The distance of the displayed targets from the aircraft is indicated by five range marks. In addition to range, the relative position of the target in azimuth may be indicated by azimuth markers which, when selected by the AZ switch, are spaced at 30 degree intervals across the display. The antenna tilt angle, operating mode, range selection and other operating parameters are also displayed.

The radar has a forced standby function (FSTBY) a signal from the PSEU (weight on wheels) forces the radar in a standby mode on landing.

Weather Radar Coupler

[Refer to Figure 33.](#)

The coupler is a transformer and is located in the wiring harness. It routes display data from the single receiver/transmitter to the two MFDs. Thus, the coupler acts as an interface between the weather radar and the MFD over the ARINC 453/708 bus.

The A453/708 frame transmitted by the PRIMUS 660 radar contains an additional bit before the syncro start bit of the frame. This additional bit prevents the MFD from decoding the ARINC 453/708 frame. Thus, no radar data are displayed on the EFIS when the EFIS is directly connected to the radar.



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The coupler fetches the data frames, detects the synchro start bit and removes the additional data that was present before the synchro bit. The coupler does not alter the other portion of the input data frame transmitted by the radar.

The coupler has a continuous BIT capability that verifies its operation. If the coupler detects a failure or if no ARINC 453/708 frames are detected at the coupler input, a red LED located on the top of the unit will illuminate.

Power Supply

The RTA is powered from the left main bus, through a 7.5 A circuit breaker (WX RDR). The WX RDR CB is located on the 28 Vdc avionics circuit breaker panel at position H2. The RCP and the coupler is powered from the aircraft left main bus, through a 2 A circuit breaker (WX CP). The WX CP CB is located on the 28 Vdc avionics circuit breaker panel at position H1.

Operation and Operating Controls

[Refer to Figure 31.](#)

Operation of the radar system and the display is controlled by four push-buttons and three rotary potentiometer switches, located on the RCP. The display and various indications are directly related to these controls.

OFF

[Refer to Figure 25.](#)

- Selection of the WXR system in OFF mode is made by the control panel rotary switch. This position turns the radar system off.

- No radar transmissions will occur when the OFF mode is selected and WX OFF will be annunciated in white text on each MFD. When the weather radar is set to the OFF mode, the transmitter/receiver and the antenna scan are inhibited with the stowed in a tilt-up position.
- For power up, SBY or TST mode is selected. When power is applied, WAIT (in white) is annunciated in the upper left corner of the MFD for 40 to 100 seconds while the magnetron warms up.

Forced Standby (FSBY)

- The FSBY is an automatic non-selectable safety mode that inhibits the radar system when the aircraft is on the ground. When airborne, the system switches from the forced standby mode to the mode set on the function switch.
- In the forced standby mode, the transmitter and the antenna scan are inhibited, the display is erased from the indicator and the FSBY legend is displayed.
- To override the forced standby mode, the STAB push-button must be pushed four times, in less than three seconds. The system then switches from the forced standby mode and operates normally.



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SBY (Standby)

- This position puts the radar system in standby, a ready state with the antenna stopped, the transmitter inhibited, and the display memory erased. A white WX STBY message is displayed. If STBY is set before the RTA warm-up period is over (about 100 seconds), a white WAIT is displayed in the mode field. When the warm-up period is over, the system automatically displays STBY on the MFD.
- The transmitter/receiver electronics within the RTA is powered and RTA mode selection and status data is transmitted to each MFD by the RTA data bus output. No radar transmissions will occur, when the STBY mode is selected. WX STBY will be annunciated in white text on each MFD if the radar image display is selected by the related EFCP WX push-button switch. When the SBY mode is selected, the RTA remains active and the antenna will be stowed in a tilt-up position.

WX (Weather)

- This mode puts the radar system in the weather mode of operation. The weather mode is used during storm seeking. The radar displays a target storm based on the back-scatter intensity received from the rainfall in the storm.

- Since intensity varies in each storm cell, the digital map-form colors of the display also vary. Areas of heaviest rainfall will show in magenta, the next lower level in red, the next lower level in yellow, the next lower level in green, and the least or no rainfall in black. The radar colors, rainfall rates and the intensity levels (in dBZ) are shown in Table 1.

Table 1: Radar Colors, Rainfall Rates and dBZ Levels

DISPLAY LEVEL	RAINFALL RATE (MM/HR)	RAINFALL RATE (IN.)	DBZ	CALIBRATED RANGE (NM) FOR 18 IN. ANTENNA IN 200 NM RANGE
4 Magenta	>50	>1.92	>50	290
3 Red	12–50	0.45–1.92	40–50	170
2 Yellow	4–12	0.14–0.45	32–40	116
1 Green	1–4	0.04–0.14	23–32	75
0 Black	<1	<0.04	<23	–



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GMAP (Ground Mapping Mode)

- This mode puts the radar system in the ground mapping mode. The message WX GMAP is displayed in white on the MFD. When in this mode, the amount of reflected RF from ground surfaces is displayed in magenta, yellow, cyan or black from the most reflective to the least reflective surfaces. The pilot can therefore identify coastlines, hilly or mountainous regions, cities, or even large structures.
- In this Mode, RCT and TGT cannot be set.

TST (Test Mode)

[Refer to Figure 26.](#)

- This position puts the radar system in the test mode. Transmitter output power can be radiated. When TEST mode is selected, the message WX is displayed in white, TEST in yellow, and the test pattern is shown in multi colored arc display. When the TEXT FAULT feature is enabled, the status text messages for the crew use, fault codes, transmit status, maintenance messages, fault types and strap codes are displayed.
- In this mode, the radar is in a non-transmitting stage.

TILT

- This is a rotary control that is used to set the tilt angle of the antenna beam with relation to the aircraft horizontal attitude. Clockwise (cw) rotation tilts the antenna upward 0 to 15.9 degrees, counter-clockwise (ccw) rotation tilts the antenna downward 0 to –15.9 degrees. The most used range is between +5 and –5 degrees and is expanded for ease of operation. Tilt angle is displayed in cyan in 0.5 degrees increments.

GAIN

- This is a rotary push-pull control that is used to adjust the sensitivity of the radar receiver. When the knob is pulled out, variable gain is activated and the message GAIN MAIN is annunciated in white on the screen. When the knob is pushed in, the preset calibrated gain mode is activated and the gain of the receiver cannot be adjusted by rotation of the GAIN control. Calibrated gain is the normal mode for weather avoidance. In the GMAP mode the calibrated gain can also be used to resolve nearby strong target signals while ground mapping.
- Sensitivity increases with a clockwise rotation, the receiver gain is increased to the maximum when the gain control is at the full clockwise position. Setting RCT or TGT forces the system into calibrated gain.

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RCT (Rain Echo Attenuation Compensation Mode)

- This is a momentary–contact alternate action push–button that turns on or off the rain echo attenuation compensation mode. Pushing the RCT switch enables the Rain Echo Attenuation Compensation Technique (REACT) circuitry. RCT mode can only be set with the weather mode and a white WX RCT is displayed on the upper left corner of the MFD. RCT mode is active in all ranges. A subsequent push on the RCT switch disables the function.
- The RCT function adjust the calibration of the radar receiver automatically to compensate the attenuation losses. The adjustment is done by measuring of the signals and deducing from the attenuation through the target, then using this information to adjust the calibration appropriately. This adjustment is continuously done on each radar azimuth radial. When the maximum value to which the sensitivity can be set due to inherent noise generated in the receiver is reached, the REACT compensation ceases. At this point, a cyan field is displayed to indicate no further compensation is possible. As the cyan field annunciates at the end of the calibrated gain, any target that occurs in the cyan area is displayed in magenta (very strong).

TGT (Target Alert)

Refer to Figure 27.

- This is a momentary–contact alternate action push–button that turns on or off the target alert feature. When set, the message TGT is displayed in black box continuously flashing in magenta in the top centre of the radar map area. The target mode (WX TGT) is displayed in white in the upper left corner of the MFD. The TGT feature advises the pilot of any potentially hazardous targets directly in front of the aircraft outside the selected range.
- The target alert can be set in all ranges.
- In order to declare the target alert, the target must have specific depth and range characteristics, which are shown in Table 2.

Table 2: Target Alert Active Range

SELECTED RANGE (NM)	TARGET DEPTH (NM)	TARGET RANGE (NM)
10	5	10–60
20	5	20–65
40	5	40–85
80	5	80–125
160	5	160–204
240	5	240–270



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STAB Selector

- The radar is normally attitude stabilized. It automatically compensates for pitch and roll maneuvers. Attitude stabilization can be turned off by pushing the STAB control knob. The STAB OFF annunciator is displayed on the screen. Stabilization can be turned on when the STAB control knob is pushed in.
- The STAB control is also used to override the forced standby mode. When the STAB control knob is pushed four times in three seconds the system will switch from the forced standby mode and operates normally. The knob is also used to operate the hidden modes for in-flight analysis.

SECT (Scan Sector)

- This is a momentary-contact alternate action push-button that is used to set the antenna scan sector angle of either 120 or 60 degrees. There is a 12 scan per minute for the full 120 degrees scan sector and for a faster update a 24 scan per minute for the 60 degrees scan sector. Antenna sweep and display automatically adjusts to the set position.

EFIS (Electronic Flight Instrument System) Control Panel

Refer to Figure 31.

- Some of the WXR system related controls (i.e. WXR display brightness, range selection, arc/full navigation display mode selection) are controlled by the EFIS and are accessible to each flight crew member by their related EFIS Control Panel (EFCP).

- The EFCP has front-panel push buttons, rotary switches and controls. The EFCP provides radar range selection (RANGE selector) and the option of selecting the display of weather or terrain (or neither) (WX/TERR push-button). The WX/TERR push-button controls display of the radar image or the EGPWS terrain image in all formats except PLAN. The default function is WX (weather radar) in ON position with TERR (terrain) in the pop-up mode.

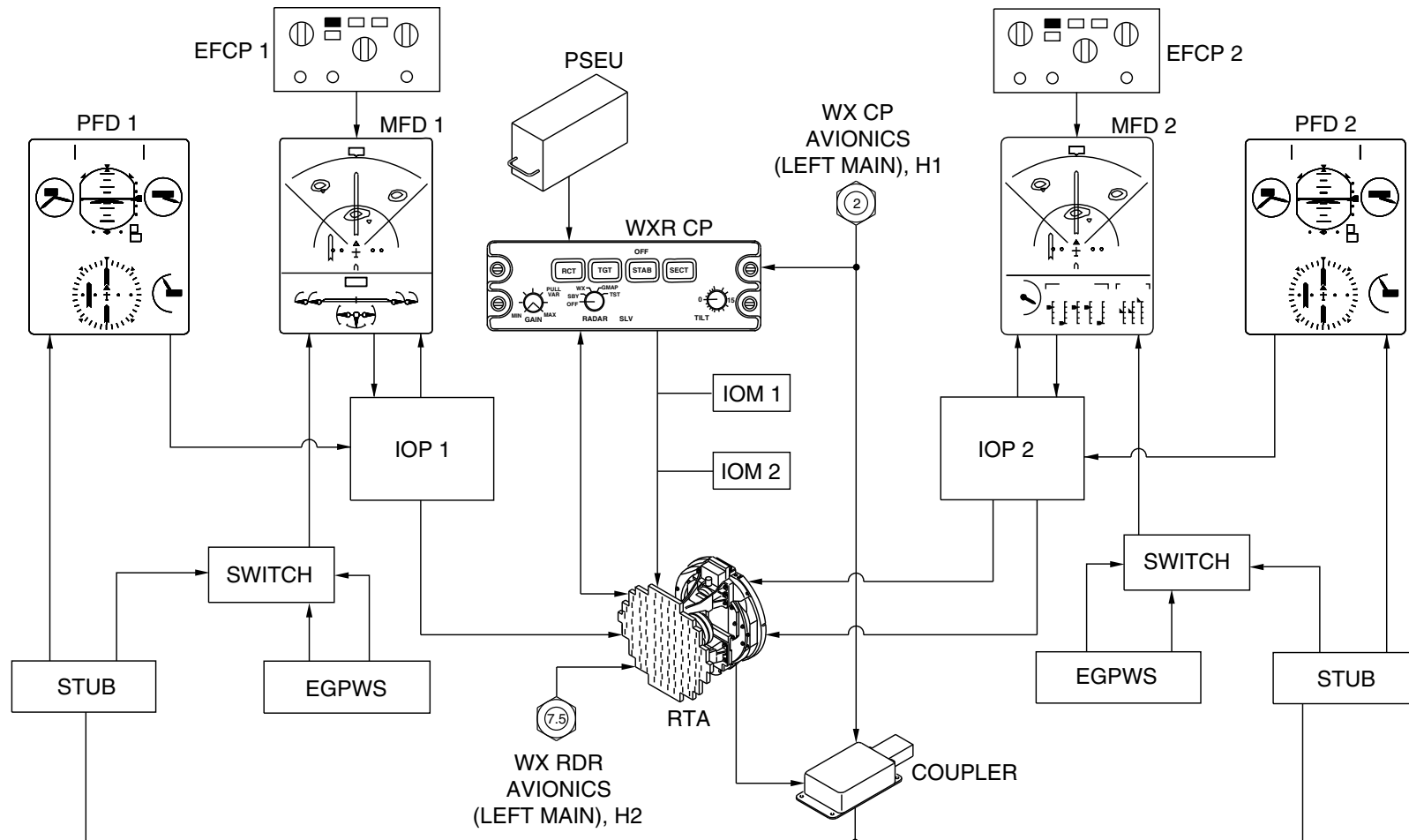
Fault Monitors

Refer to Figure 29.

- Critical functions in the receiver transmitter antenna is continuously monitored for faults. The detected faults and the brief text messages are displayed in the test pattern area on the MFD.
- All faults are stored in a nonvolatile memory. If the word FAIL shows on the display, in the ON mode, this means that a failure message is being presented on the maintenance page.
- The fault display format is shown on the MFDs and on the EFIS.



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Weather Radar System Block Diagram (Primus 660)
Figure 19

PSM 1-84-2A
EFFECTIVITY:
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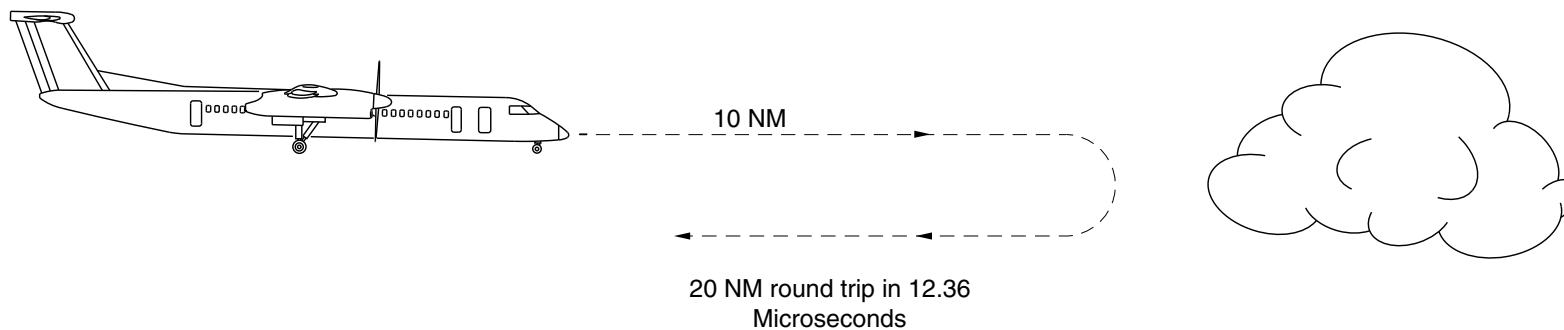
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Radar Principle
Figure 20

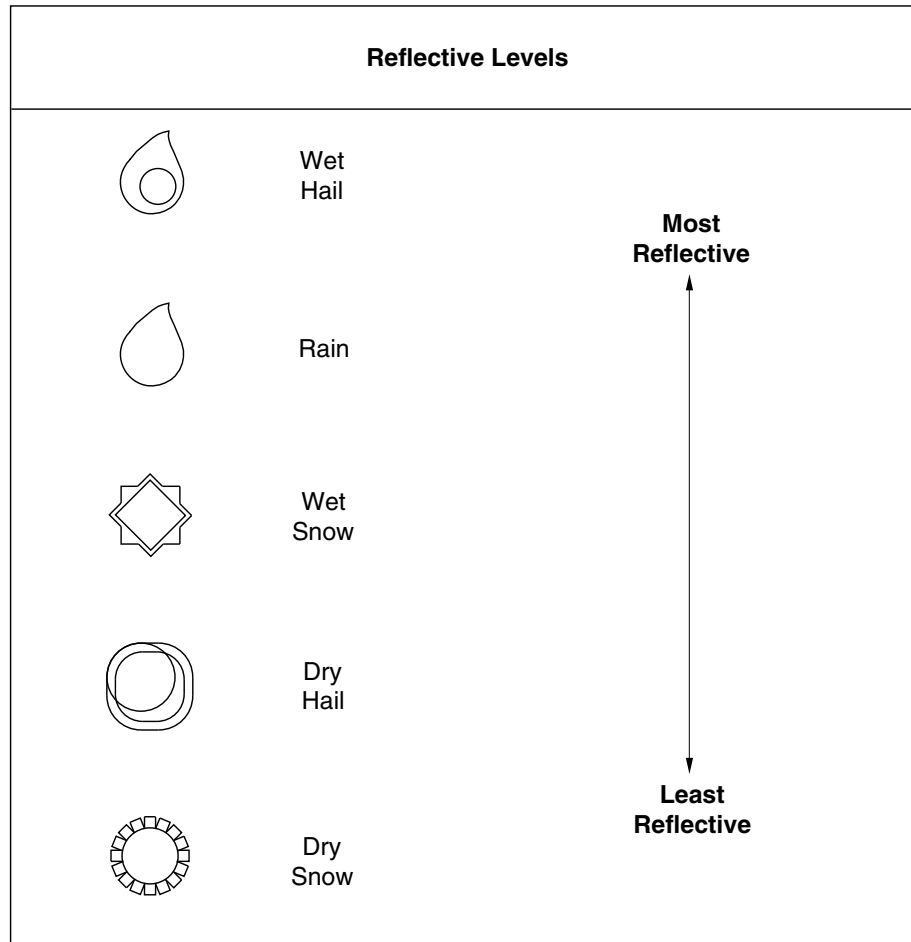
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EFFECTIVITY:
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Reflectivity Levels
Figure 21

PSM 1-84-2A
EFFECTIVITY:
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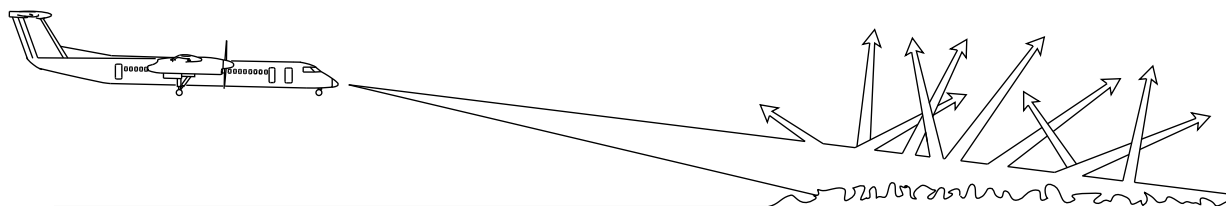
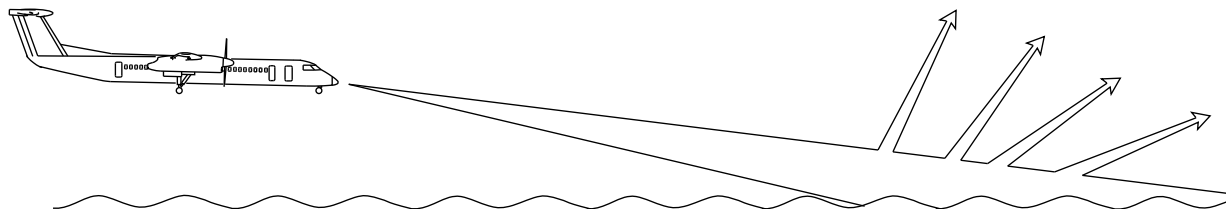
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GROUND MAPPING
Figure 22

PSM 1-84-2A
EFFECTIVITY:
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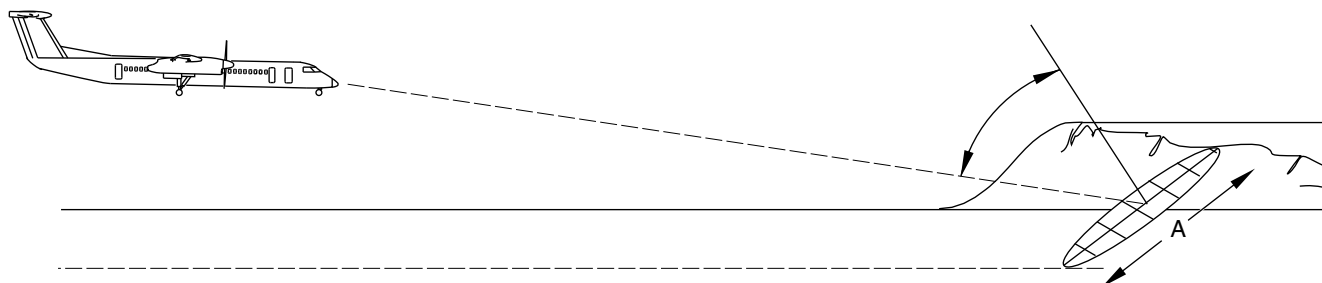
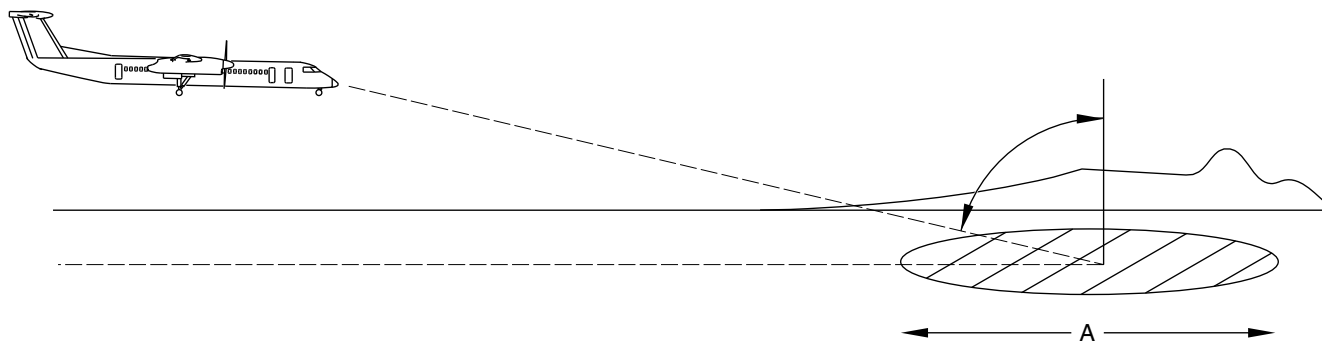
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LOOKING ANGLE
Figure 23

PSM 1-84-2A
EFFECTIVITY:
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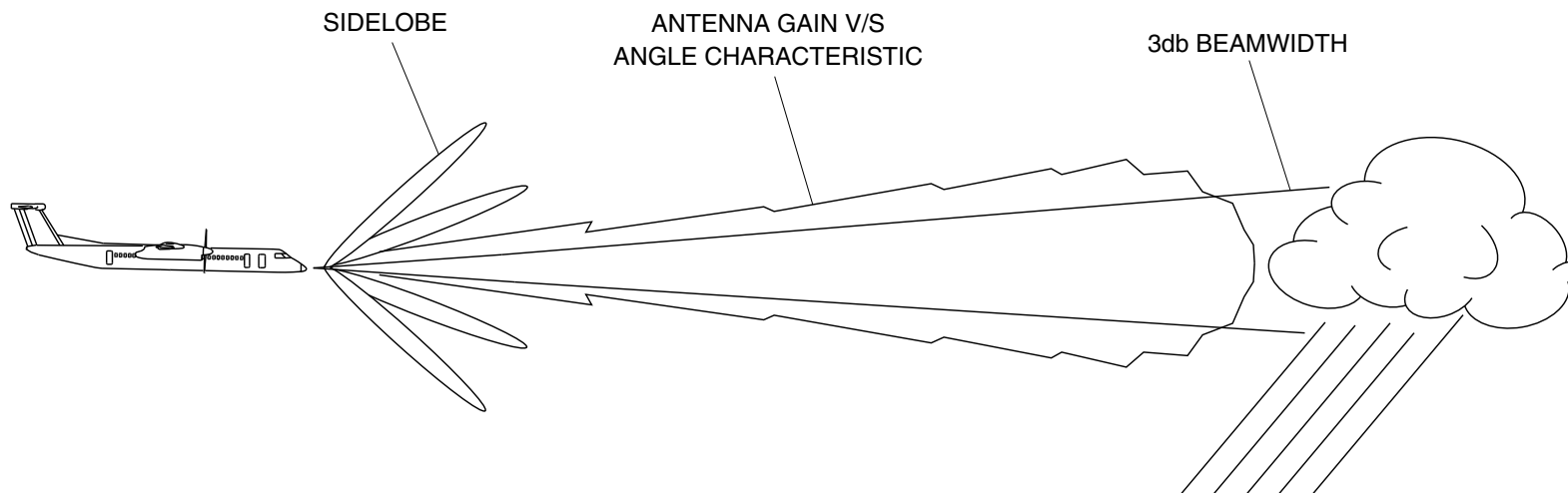
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Radar Beam Illumination Characteristics
Figure 24

PSM 1-84-2A
EFFECTIVITY:
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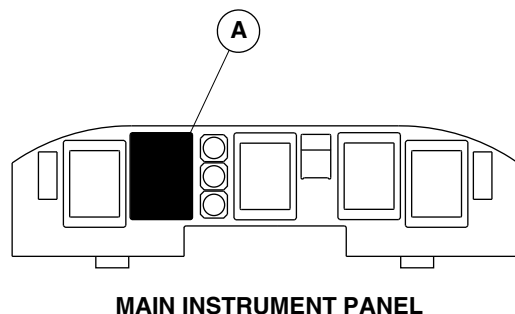
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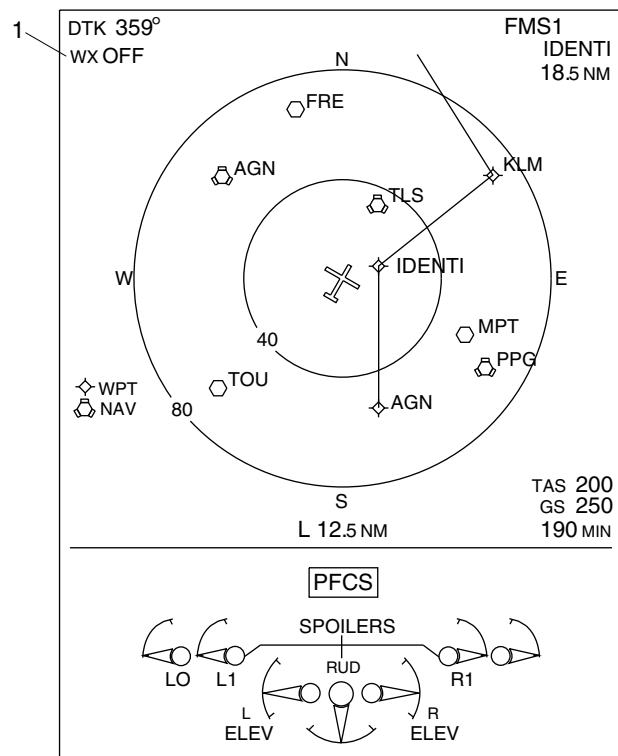
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LEGEND

1. Weather Radar Off Mode Indication.



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WEATHER RADAR OFF MODE INDICATION
Figure 25

PSM 1-84-2A
EFFECTIVITY:
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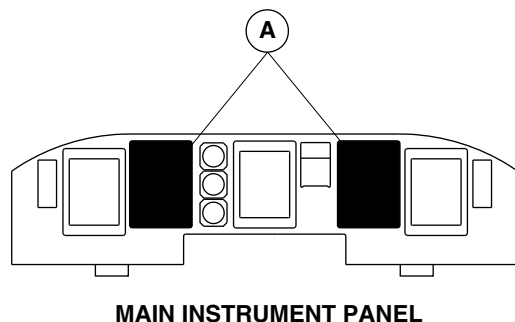
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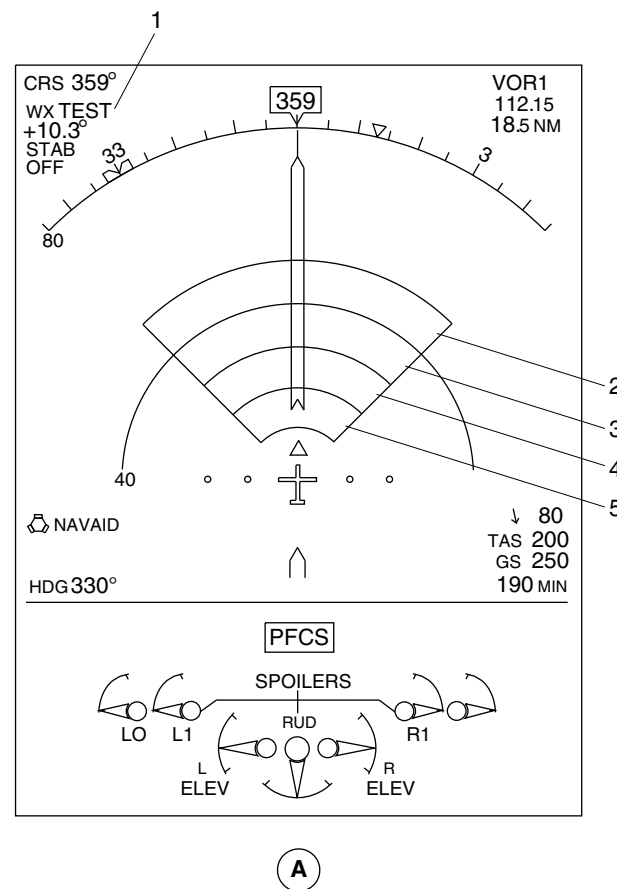
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LEGEND

1. Weather Radar Test Mode Indication.
2. Magenta.
3. Red.
4. Yellow.
5. Green.



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WEATHER RADAR TEST MODE INDICATION
Figure 26

PSM 1-84-2A
EFFECTIVITY:
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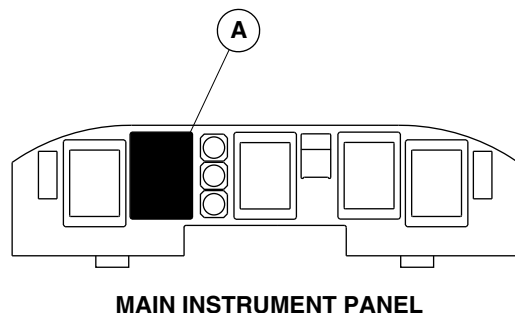
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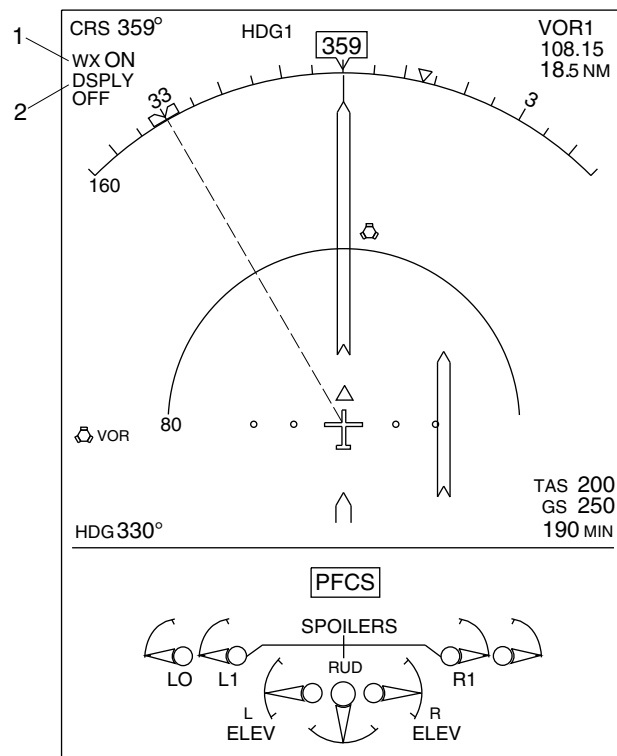
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LEGEND

1. Weather Radar On Mode Indication.
2. Weather Radar Image Display Off Status Message.



A

fs628a01.cgm

WEATHER RADAR ON MODE INDICATION
Figure 27

PSM 1-84-2A
EFFECTIVITY:
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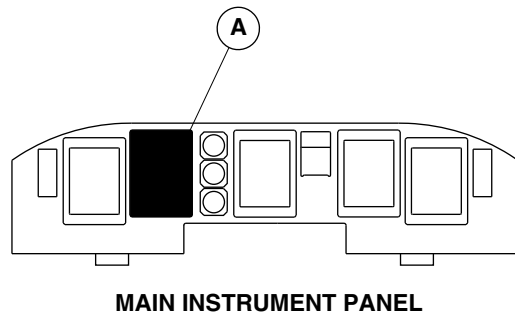
34-41-00

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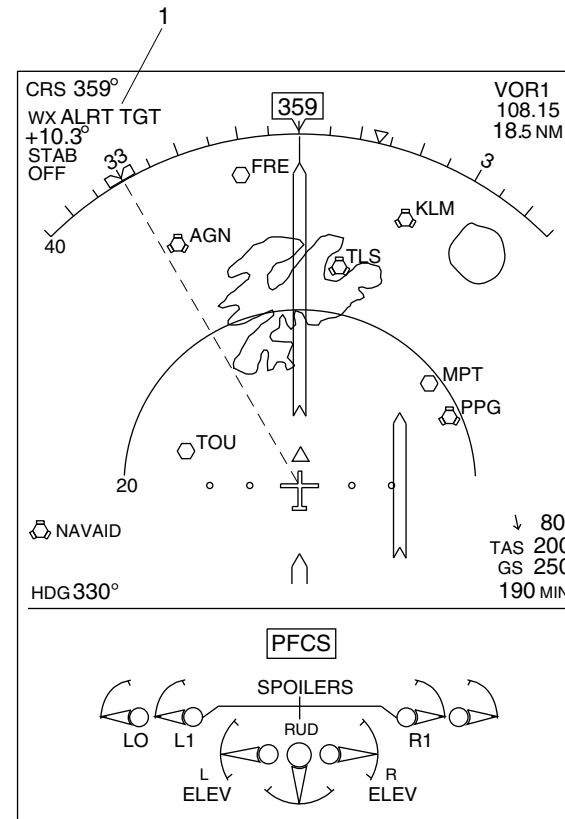
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LEGEND

1. Weather Target Alert Message.



fs626a01.cgm

TARGET ALERT INDICATION
Figure 28

PSM 1-84-2A
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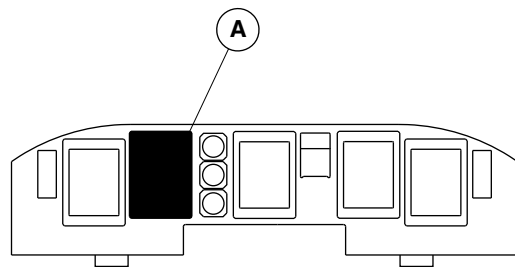
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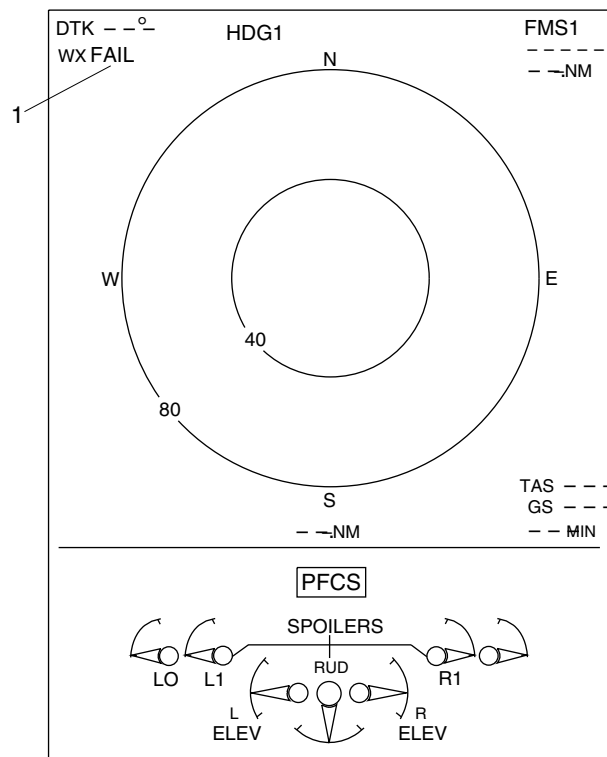
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Weather Radar Fail Annunciation.



fs629a01.cgm

WEATHER RADAR FAIL ANNUNCIATION
Figure 29

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 35 of 34-41-00
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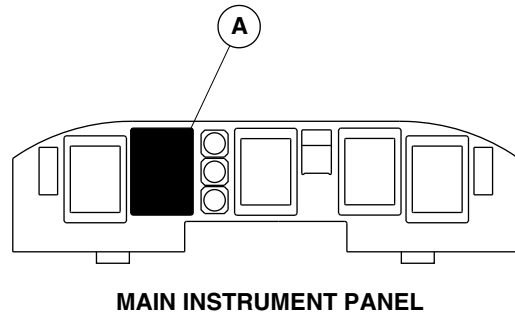
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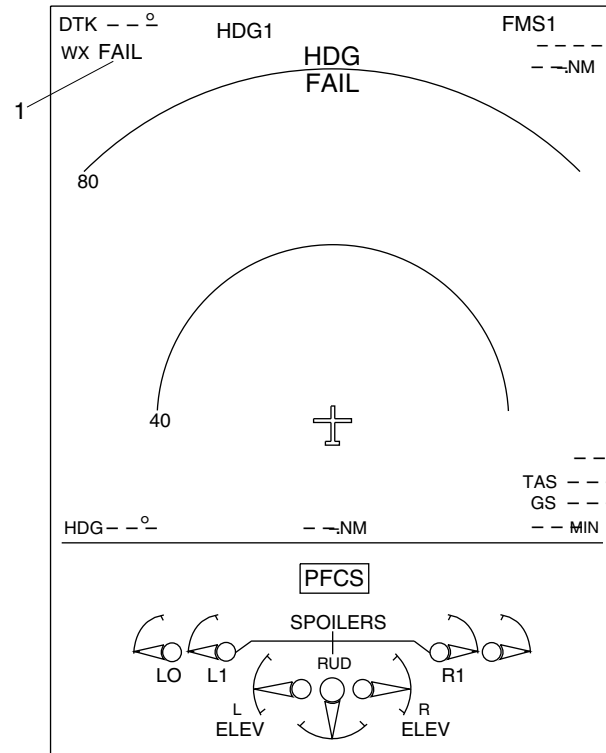
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LEGEND

1. Weather Radar Fail Annunciation.



fs630a01.cgm

WEATHER RADAR FAIL ANNUNCIATION, ARC MODE
Figure 30

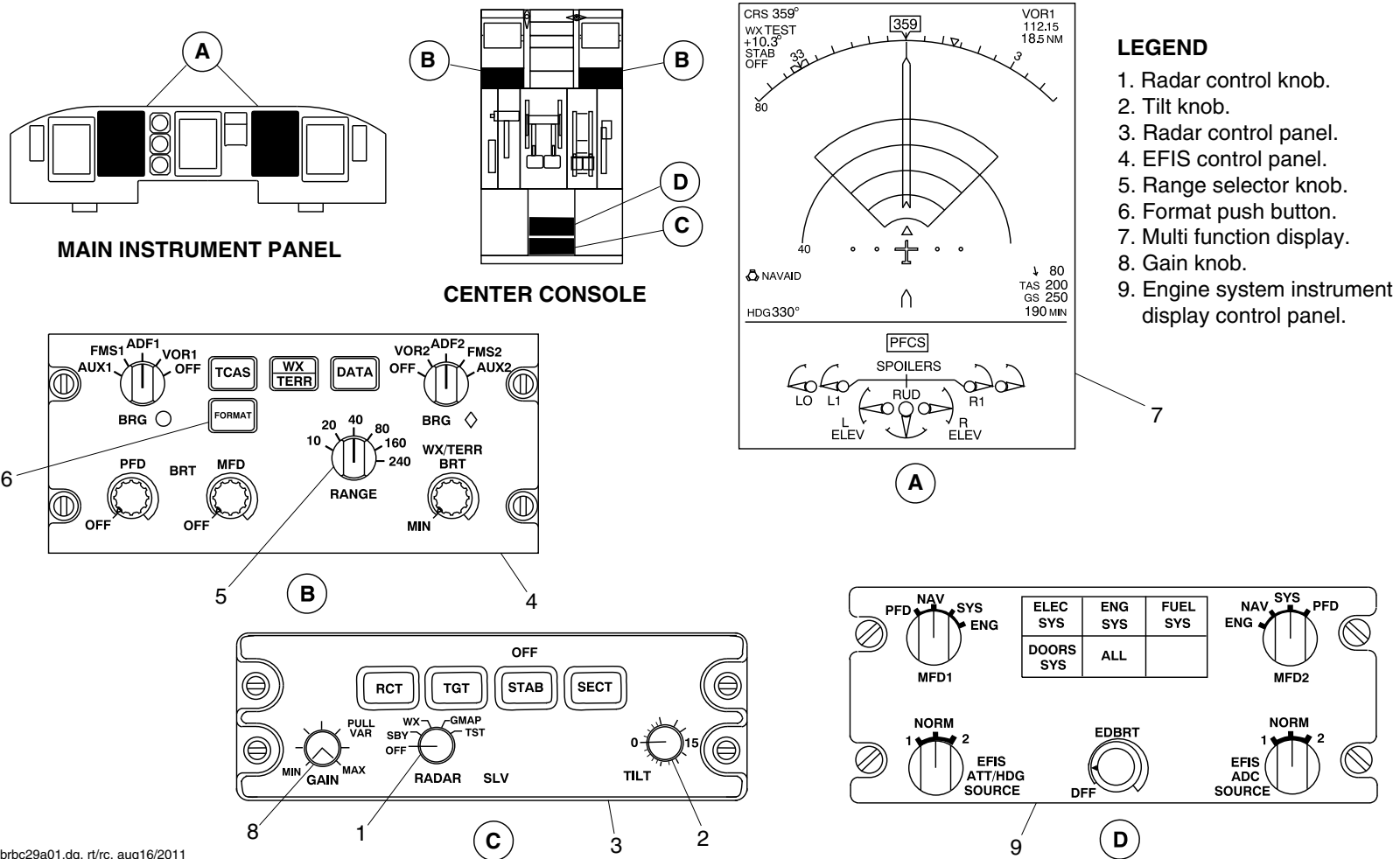
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brbc29a01.dg, rt/rc, aug16/2011

Weather Radar System (Primus 660)
Figure 31

PSM 1-84-2A
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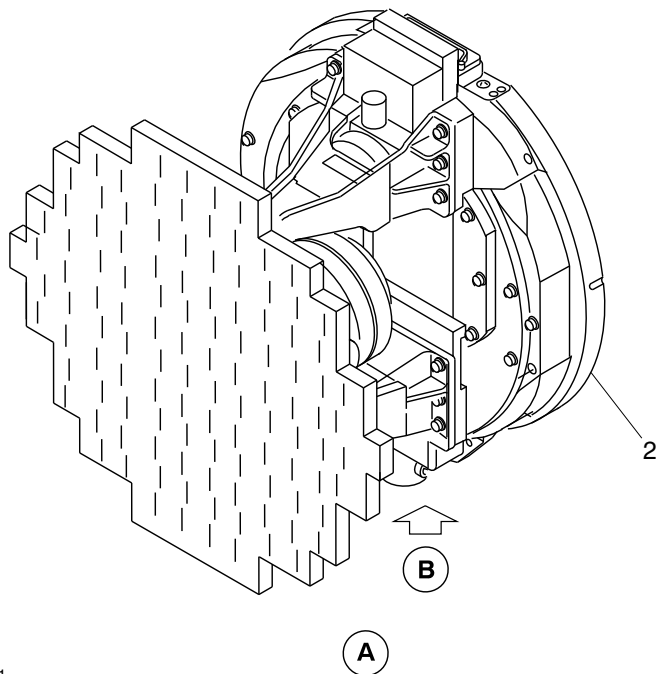
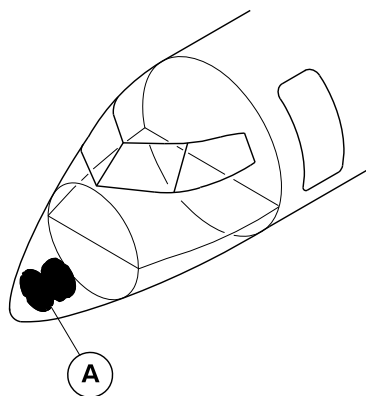
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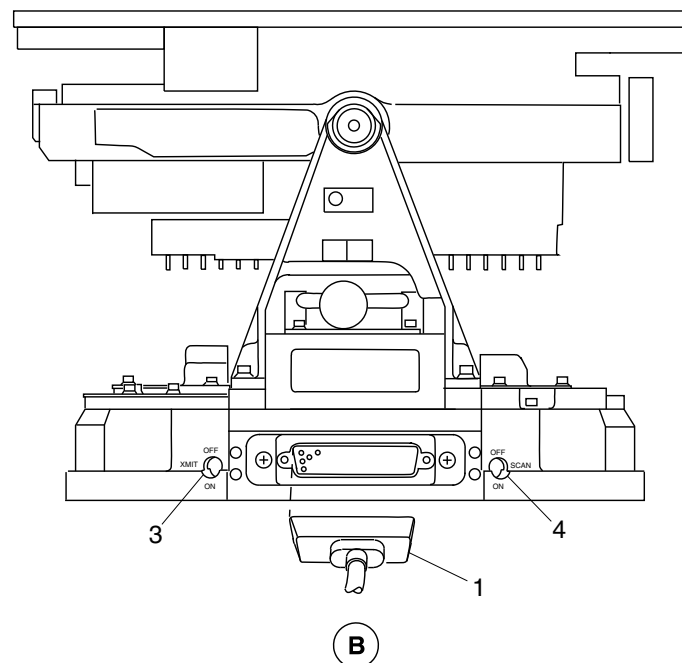
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LEGEND

1. Electrical connector.
2. Receiver transmitter.
3. Transmitter switch.
4. Scan switch.



brbc28a01.dg, rt/rc, aug16/2011

Receiver Transmitter (Primus 660)
Figure 32

PSM 1-84-2A
EFFECTIVITY:
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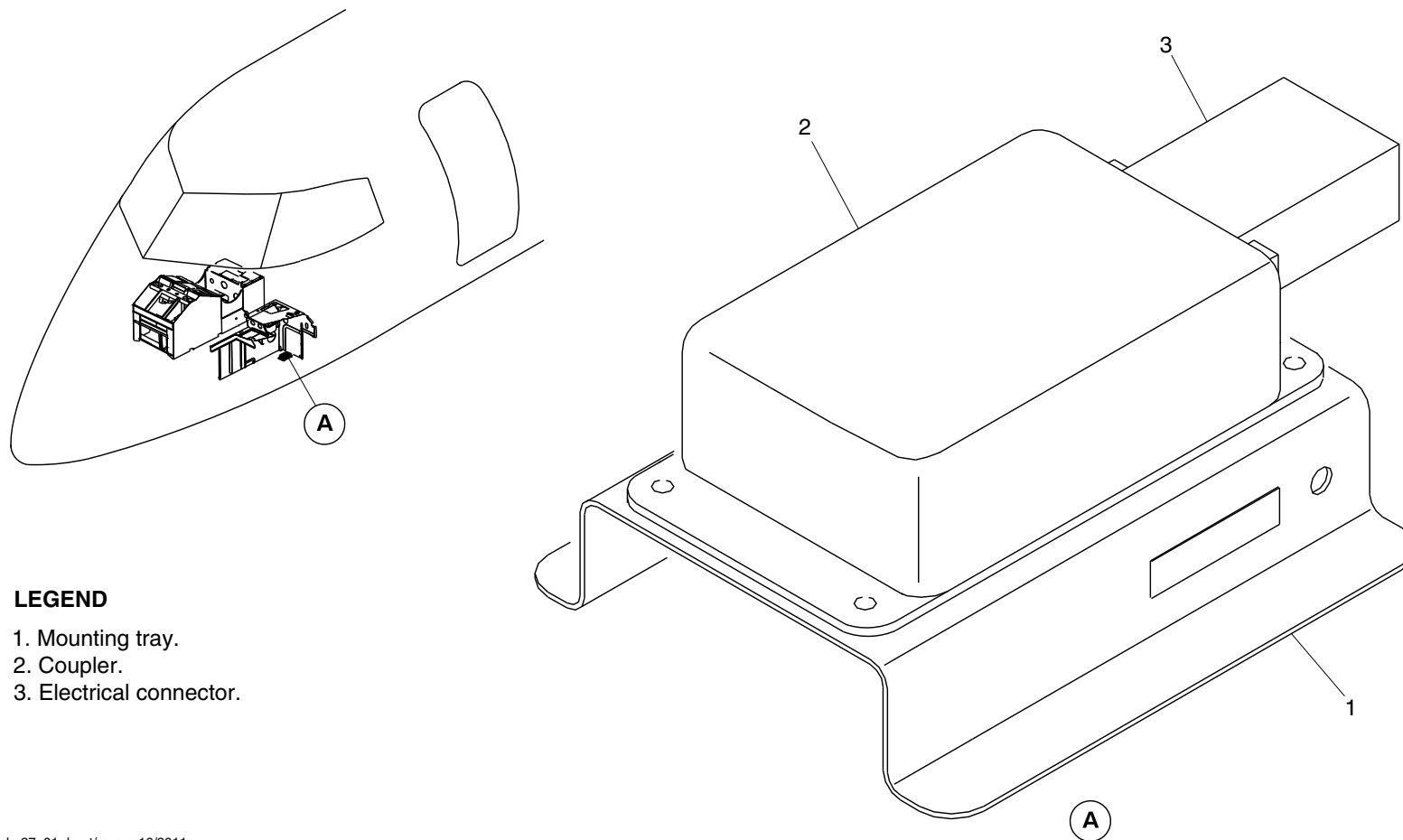
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LEGEND

- 1. Mounting tray.
- 2. Coupler.
- 3. Electrical connector.

brbc27a01.dg, rt/rc, aug16/2011

Weather Radar Coupler (Primus 660)
Figure 33

PSM 1-84-2A
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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34-42-00-001

GROUND PROXIMITY WARNING SYSTEM (GPWS)

Introduction

The Ground Proximity Warning System (GPWS) gives aural and visual indications of possible Controlled Flight Into Terrain (CFIT).

General Description

Refer to Figure 1.

The Ground Proximity Warning System (GPWS) gives visual and aural indications for the conditions that follow:

- Excessive descent rate
- Excessive closure rate to terrain
- Descent after takeoff
- Insufficient terrain clearance
- Excessive descent below the glideslope.

In addition, the Ground Proximity Warning System (GPWS) gives excessive bank angle and Decision Height (DH) callouts.

Ground Proximity Warning System (GPWS) lets the pilots override unwanted indications when landing with zero or partial flaps and cancel glideslope indications.

The Ground Proximity Warning System (GPWS) indications are automatically cancelled by the conditions that follow:

- Stall
- Ground Proximity Warning System (GPWS) malfunction sensed.

The Ground Proximity Warning System (GPWS) can be tested and diagnosed only when the aircraft is on the ground.

The Ground Proximity Warning System (GPWS) has the components that follow:

- Computer, GPWS (34-42-01)
- Switch, Annunciator-Flap Override (34-42-16)
- Switch, Annunciator-GPWS and GS Alert (34-42-21).

The Ground Proximity Warning System (GPWS) interfaces with other aircraft systems through the Ground Proximity Warning System Converter (GPWSC) located within the Integrated Flight Cabinet 1 (IFC1). It supplies data to the Ground Proximity Warning Computer (GPWC) from the aircraft data sources that follow:

- Proximity Sensor Electronics Unit (PSEU)
- Air Data Units (ADU1, ADU2)
- Attitude and Heading Reference System (AHRS1, AHRS2)
- Radio Altimeters (RA1, RA2)
- VHF Navigation Receivers (VHF NAV1, VHF NAV2)
- Index Control Panels (ICP1, ICP2).

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The Ground Proximity Warning System (GPWS) supplies data directly to the systems that follow:

- Caution and Warning Lights Panel
- Advisory Control Unit (ACU)
- Audio and Radio Management System (ARMS)
- Stall Protection System (SPS)
- GPWS Landing Flap Select Panel.

Detailed Description

The Ground Proximity Warning System (GPWS), uses inputs from other aircraft systems to continuously monitor the flight path of the aircraft when the Above Ground Level (AGL) altitude is between 50 and 2450 feet. If the sensed flight path would cause a possible impact into terrain, the system gives a visual and aural indication.

The Ground Proximity Warning System (GPWS) supplies ground proximity warning indications from calculations between the different input parameters that follow and pre-programming:

- Vertical speed, vertical speed validity
- Air speed
- Glideslope deviation, glideslope enable
- Bank angle
- Above ground level altitude radio altitude
- Radio altitude validity
- Flap setting

- Landing gear selection
- Decision Height (DH) selection
- Back course selection
- Stall indication.

The Ground Proximity Warning System (GPWS) operates in the modes that follows:

- Mode 1: Excessive descent rate
- Mode 2: Excessive closure rate to terrain
- Mode 3: Descent after takeoff
- Mode 4: Insufficient terrain clearance
- Mode 5: Descent below glideslope
- Mode 6: Minimum callout and excessive bank angle
- Ground self test
- Maintenance test

Mode 1, Excessive Descent Rate: The Mode 1 condition gives indications for excessive rates of descent with respect to the aircraft height above terrain. The indications are given well ahead of the possible impact with the ground. The Ground Proximity Warning Computer (GPWC) computes the possible flight into terrain from the



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Radio Altitude (RA) and Air Data Unit (ADU) Barometric Decent Rate (FPM) data.

[Refer to Figure 2.](#)

The Mode 1 condition gives "SINK RATE" and "PULL UP" aural alerts and turns and the PULL UP annunciator lights come on.

[Refer to Figure 3.](#)

The "SINK RATE" aural alert repeats every three seconds while the aircraft is in the initial alert area. As the aircraft goes into the second alert area in closer proximity to terrain, a "PULL UP" aural alert sounds in place of the "SINK RATE" aural alert.

The ILS glideslope selections change the alert area to minimize false warnings.

When the aircraft is above the glidepath, the alert area changes. The descent rate must increase for a GPWS alert output. This decreases the possibility of nuisance GPWS outputs.

When the aircraft is below the glidepath, a lower descent rate than when above the glidepath causes a GPWS alert output. The sensitivity of the GPWS is greater when the aircraft is below the glidepath.

When the aircraft is more than 200 feet Above Ground Level (AGL) altitude, 150 feet per minute is added for each dot of glideslope deviation, up to a maximum of 300 feet per minute.

The descent or climb rate is set to 300 feet per minute when the aircraft is above or below the glidepath with the GPWS Flap Override selected.

Mode 2, Excessive Closure Rate to Terrain: The Mode 2 condition gives indications when terrain below the aircraft is rising dangerously

fast. The Ground Proximity Warning Computer (GPWC) gives the indications well ahead of the projected collision with terrain.

The Ground Proximity Warning Computer (GPWC) monitors the conditions that follow to give the Mode 2 indication:

- Above Ground Level (AGL) altitude
- Air speed
- Terrain closure rate.

[Refer to Figure 4.](#)

The Mode 2 condition causes the "TERRAIN TERRAIN" and "PULL UP" aural alerts to sound and the PULL UP annunciator lights come on.

Mode 2 functions in two sub-modes:

- Mode 2A
- Mode 2B.

Mode 2A: Mode 2A functions when any one condition is met that follow:

- The flaps are not in the landing position, the GPWS flap override switch is not selected, and the aircraft is not on an ILS or MLS approach
- The glideslope mode has been manually cancelled
- The aircraft is more than 1.3 dots below the glidepath.

[Refer to Figure 5.](#)

When the Mode 2A warning condition is met, a "TERRAIN TERRAIN" aural alert sounds and the GPWS PULL UP annunciators come on. If the aircraft continues to go beyond the initial alert, the



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"PULL UP" aural alert sounds repeatedly and the PULL UP annunciators stay on.

The Mode 2A "PULL UP" aural alert will stop when the aircraft goes out of the warning area. The PULL UP annunciators will stay on until any condition occurs that follow:

- The barometric altitude increase is more than 300 feet
- An up acceleration that is a combination of barometric altitude gain and radio altitude, and time.

Mode 2B: Mode 2B functions when any one of the conditions is met that follow:

- The flaps are in the landing position
- The GPWS flap override switch is selected
- The aircraft is on an ILS or MLS approach and the aircraft is not more than 1.3 dots below the glidepath
- The glideslope cancel feature is not selected.

[Refer to Figure 6.](#)

When the landing gear is down, and the flaps are down or the GPWS flap override switch is set, a repetitive "TERRAIN TERRAIN" aural alert will sound. If the landing gear is not down, a repetitive normal "PULL UP" aural alert will sound when the "TERRAIN TERRAIN" aural alert is complete. When the aircraft goes out of the warning area, the aural alerts will stop and the PULL UP annunciators go out.

NOTE

Selecting the Flap Override annunciator switch cancels the Mode 2 warnings. This lets the aircraft manoeuvre near terrain when approaching an airport.

Mode 3, Descent After Takeoff: Mode 3, is functions during "Takeoff" or "Missed Approach" when the conditions that follow are met:

- Increase in airspeed
- 50 feet Above Ground Level (AGL) altitude
- Increase in barometric altitude
- Landing gear retraction.

The Mode 3 gives an indication to show too much altitude decrease after takeoff or after a missed approach. A barometric altitude decrease of approximately 10 percent of the Above Ground Level (AGL) altitude starts the Mode 3 alert.

[Refer to Figure 7.](#)

The Mode 3 condition gives a "DON'T SINK" aural alert and the GPWS "PULL UP" annunciator lights come on. An increased rate of climb cancels the "DON'T SINK" aural alerts and the PULL UP annunciator lights go out.

[Refer to Figure 8.](#)

When the Above Ground Level (AGL) altitude is more than 925 feet for 17 seconds, the Mode 3 condition is cancelled.

Aircraft speed, flap and landing gear position are used to calculate a warning area below the aircraft. Selecting the Flap Override annunciator switch cancels the Mode 3 alert conditions if the Above Ground Level (AGL) altitude is more than 50 feet. It lets the Above Ground Level (AGL) altitude decrease 20 percent before giving an indication.

When the aircraft is 700 feet Above Ground Level (AGL) altitude, it gives an additional altitude increase of 5 feet per second.



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Mode 4, Insufficient Terrain Clearance: The Mode 4 condition gives an indication for insufficient terrain clearance during the phases of flight that follow:

- Cruise
- Approach
- Descent
- Climbout.

The Mode 4 condition gives indications when the flight path is too low for Mode 2 excessive closure rates with terrain or Mode 1 excessive descent rate calculations. It uses the parameters that follow:

- Above Ground Level (AGL) altitude
- Airspeed
- Flight phase.

The Mode 4 condition functions in the modes that follow:

- Mode 4A
- Mode 4B
- Mode 4C.

[Refer to Figure 9.](#)

Mode 4A: When the aircraft is in a configuration with the gear and flaps selected up, the Ground Proximity Warning System (GPWS)

calculates a "floor" below the aircraft to show insufficient terrain clearance.

[Refer to Figure 10.](#)

During the initial approach, at speeds less than 178 knots and less than 500 feet Above Ground Level (AGL) altitude, the "TOO LOW GEAR" aural alert sounds and the PULL UP annunciator lights come on.

[Refer to Figure 10.](#)

When the air speed is more than 200 knots and the Above Ground Level (AGL) altitude is less than 750 feet, the "TOO LOW TERRAIN" aural alert sounds and the PULL UP annunciator lights come on. Lower Above Ground Level (AGL) altitudes are used to give the same alert at airspeeds 178 up to 200 knots.

[Refer to Figure 11.](#)

Mode 4B: Mode 4B operates during cruise and approach when the landing gear is down, the flaps not in the landing position and the GPWS flap override switch is not set.

[Refer to Figure 12.](#)

During the initial approach, at speeds less than 148 knots and the Above Ground Level (AGL) altitude is less than 170 feet, the "TOO LOW FLAPS" aural alert sounds and the PULL UP annunciator lights come on. When the flap override annunciator switch is set, the GPWS will not give any alerts for the flaps up landing.

The flight crew set the Landing Flap Selector Switch (LFSS) to supply the Ground Proximity Warning System (GPWS) with data relating to the specified landing flap position.



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Mode 4C: This is active during takeoff or go-around. The Mode 4C gives a minimum terrain clearance that increases with the Above Ground Level (AGL) altitude.

Refer to Figure 13.

When the Above Ground Level (AGL) altitude is more than 100 feet after takeoff or 200 feet when doing a go-around, a warning floor is calculated below the aircraft. It is 75 percent of the highest ACL altitude. The Mode 4C continues to operate until the approach mode is activated or the Above Ground Level (AGL) altitude goes below 30 feet.

The conditions that follow, cause the "TOO LOW TERRAIN" aural alert to sound and the PULL UP annunciator lights to come on:

- The aircraft descends after takeoff
- The terrain below the aircraft rises at a steeper slope than when the aircraft is climbing.

The warning will continue until the aircraft has sufficient clearance from the terrain.

NOTE

Warnings from Modes 4A, 4B, and 4C cannot occur at the same time.

Mode 5, Inadvertent Descent Below Glideslope: The Mode 5 operates when the conditions that follow are met:

- An ILS frequency is set
- The landing gear is down
- The Above Ground Level (AGL) altitude is less than 925 feet

- The aircraft is below the glidepath
- Glideslope cancel is not set.

Refer to Figure 14.

The Mode 5 has two aural alert areas depending on the deviation from the glidepath. Each uses a different aural alert volume level. The low volume level alert is specified as the soft alert and the louder alert is named the hard alert. The hard alert sounds two times as loud as a soft alert. The time between aural alerts change if on the glideslope deviation and Above Ground Level (AGL) altitude. The time between the aural indications decreases as the danger of the Mode 5 condition increases.

Refer to Figure 14.

When the aircraft is below the glideslope and into the soft alerting area, the "GLIDESLOPE" aural alert sounds and the amber BELOW GS annunciator switches come on. The volume of the initial "GLIDESLOPE" aural alert sounds is lesser than the other Ground Proximity Warning System's (GPWS) aural alerts. While the aircraft is below the glideslope, the aural repetition rate increases as the Above Ground Level (AGL) altitude decreases. If the aircraft goes into the hard alerting area, the audio volume level increases to that of the other aural alerts.

When the Above Ground Level (AGL) altitude is less than 150 feet, more glideslope deviation is necessary to cause the indications to come on.

The Mode 5 condition cancels when the Above Ground Level (AGL) altitude is less than 50 feet or by a Flap Override annunciator switch selection.



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Mode 6, Excessive Minimums or Bank Angle Callout: The mode 6 condition gives indications for excessive bank angles while manoeuvring close to the runway.

[Refer to Figure 15.](#)

A "MINIMUMS, MINIMUMS" aural alert sounds once for each approach as the aircraft descends through the decision height "DH" setting. It resets when the Above Ground Level (AGL) altitude is more than 925 feet.

[Refer to Figure 16.](#)

A "BANK ANGLE" aural alert gives indications for excessive bank angles depending on the Above Ground Level (AGL) altitude.

[Refer to Figure 17.](#)

The limit of the indication decreases from 50 degrees at 190 feet Above Ground Level (AGL) altitude to 15 degrees near ground level.

[Refer to Figure 18.](#)

The FLAP OVERRIDE annunciator switch located on the pilot's side console is pushed to prevent aural alerts when a landing is made with zero or partial flaps extended.

A black and yellow crosshatch indication in the switch comes on to show the manual selection of the GPWS FLAP OVERRIDE feature.

The switch is pushed again to manually cancel the GPWS FLAP OVERRIDE feature.

When the Above Ground Level (AGL) is less than 50 feet, the override feature automatically cancels.

[Refer to Figure 19.](#)

A red PULL UP annunciator switch located on the glareshield panel assembly comes on when the Ground Proximity Warning System (GPWS) calculates any of the warnings that follow:

- Mode 1, Excessive descent rate
- Mode 2, Excessive closure rate to terrain
- Mode 3, Descent after takeoff
- Mode 4, Insufficient terrain clearance.

The PULL UP lights stay on while the aircraft is in the warning area.

The amber BELOW G/S annunciator switches located on the glareshield panel assembly comes on to show a Mode 5 excessive descent below the Glideslope (GS).

One BELOW G/S annunciator switch or the other is pushed for 1 second or longer to cancel the aural indication. The amber light stays on to show the flight crew that the mode was intentionally cancelled.

The system cancels when any of these conditions are met:

- The Above Ground Level (AGL) is less than 50 feet
- The Above Ground Level (AGL) is more than 1900 feet
- The ILS frequency is de-selected.



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Refer to Figures 20 and 21.

The Landing Flap Selector Switch (LFSS) attached to the hydraulic control panel is turned to one of three landing flap settings for the insufficient terrain clearance mode as follows:

- 10 degrees
- 15 degrees
- 35 degrees.

A related indication on the hydraulic control panel comes on to show the landing flap position selection.

The aircraft flaps select lever is set at or more than the Landing Flap Selector Switch (LFSS) setting. A "TOO LOW FLAPS" aural alert will sound when the Above Ground Level (AGL) altitude becomes less than 170 feet.

Refer to Figure 22.

The alert indications have a priority as shown.

Ground Self Test: The GPWS can be self tested when the aircraft is on the ground. One PULL UP Annunciator Switch or the other is pushed to test the GPWS.

There are two possible test sequences as follows:

- Normal GPWS self test
- Fault GPWS self test.

The PULL UP, GPWS TEST annunciator switch is pushed and held to start a normal GPWS self test sequence.

The Ground Proximity Warning System (GPWS) test gives the indications that follow:

- The GPWS light located on the caution and warning panel comes on immediately
- The GPWS flap override annunciator switch on the pilot's side console comes on immediately
- Both BELOW GS annunciator switches come on
- The "GLIDESLOPE" aural alert sounds
- Both PULL UP annunciator switches turn on approximately 2 seconds after the annunciator switch is pushed
- The "PULL UP" aural alert sounds twice.

When the Pull Up annunciator switch is released, all lights go off.

NOTE

Allow thirty seconds before initiating another self test. If this time limit is not met, the Ground Proximity Warning System (GPWS) will not re-initialize.

The Ground Proximity Warning System (GPWS) can be tested while the GPWS caution light is on.

For the systems that do not supply correct data, the Ground Proximity Warning System (GPWS) gives the aural alerts that follow:

- "RADIO ALTITUDE FAULT"
- "GLIDESLOPE FAULT"
- "BARO RATE FAULT".

NOTE

A defective Internal Ground Proximity Warning Computer (GPWC) may not give an aural alert.



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Allow a minimum of one second before another self test sequence with the GPWS caution light on is started.

Maintenance Test: One Audio and Radio Control Display Unit (ARCDU1) or the other is used to do the maintenance test. It simulates Modes 1 through 6 indications and it makes sure that the interface with all the sensors and the Ground Proximity Warning System (GPWS) are correct. During the test, all the aural alerts sound and the related visual alerts turn on.

The Left Main Bus supplies the Ground Proximity Warning Computer (GPWC) with 28 VDC electrical power through a 3 Ampere circuit breaker. The circuit breaker is located on the 28 VDC avionics circuit breaker panel at position A1. The Ground Proximity Warning Computer (GPWC) uses less than 5 Watts of power in a non-alerting mode. It uses less than 20 Watts when an audio output is supplied to the flight compartment speakers. An internal heating circuit uses an additional 56 Watts when the unit is operating and the ambient temperature is below –20 degrees Celsius.

The Left Main Bus supplies 28 VDC electrical power to the GPWS and GS Alert annunciator switches through a 1 Ampere circuit breaker. The circuit breaker is located on the 28 VDC avionics circuit breaker panel in position B1.

The Ground Proximity Warning Computer (GPWC) receives data from other systems through a Ground Proximity Warning System Converter (GPWSC).

The Ground Proximity Warning System Converter (GPWSC) changes digital data into analog data. It is installed in the Stall Protection Module (SPM1) of Integrated Flight Cabinet (IFC1). The Ground Proximity Warning System Converter (GPWSC) operates independently of the stall protection system but uses the same Power Supply Module (PSM1).

Different systems supply digital data to the Ground Proximity Warning System Converter (GPWSC) through Input/Output Module (IOM1) and Input/Output Processor (IOP1). The Flight Data Processing System (FDPS) controls the data transmission to the Ground Proximity Warning System Converter (GPWSC).

The Ground Proximity Warning Computer (GPWC) receives data from the sources that follow through the Ground Proximity Warning System Converter (GPWSC):

Table 1: GPWSC Inputs

PARAMETER	SOURCE
Landing Gear Down	PSEU
Airspeed	ADU1, ADU2
Altitude Rate	AHRS1, AHRS2
Bank Angle	AHRS1, AHRS2
AGL altitude	RA1, RA2
GS Deviation	VHF NAV1, VHF NAV2
Backcourse Status	AHRS, VHF NAV
Higher DH setting	ICP1, ICP2

The Stall Protection Modules (SPM1, SPM2) send a signal directly to the Ground Proximity Warning Computer (GPWC) to prevent an aural indication during a stall condition.

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The GPWS Landing Flap Select Switch (LFSS) circuit receives a signal from the Flap Control Unit (FCU) when a flap selection is made. GPWS Landing Flap Select Switch (LFSS) supplies a signal to the Ground Proximity Warning Computer (GPWC) when there is a difference between the flap position and the landing flap selector switch.

The caution and warning panel monitors the Ground Proximity Warning System (GPWS). When Ground Proximity Warning System (GPWS) becomes unserviceable, an amber GPWS caution light located on the caution and warning panel comes on.

The Ground Proximity Warning Computer (GPWC) sends signals to the Advisory Control Unit (ACU) and the annunciator switches that follow come on:

- FLAP OVERRIDE
- PULL UP
- G/S

The Ground Proximity Warning Computer (GPWC) supplies the aural alerts through the Remote Control Audio Unit (RCAU) to the flight compartment speakers.

Computer, GPWS

[Refer to Figure 23.](#)

The GPWS Computer is an analog-computing device. Parameters from other aircraft systems are used to calculate aural and visual alerts.

The GPWC contains the modules that follow:

- Input/output

- Input receiver
- Logic processing
- Output driver
- Power supply.

Input/output module: The input/output module is used to filter and buffer all external interfaces.

Input receiver module: The input receiver module is used to calculate the input parameters that follow:

- Requisite discrete
- Analogue
- Pin programming.

Logic processing module: A logic processing module has the parameters that follow:

- Aural messages stored in ROM
- Validity logic
- Analogue comparators
- Mode logic and logic functions.

Output driver module: The output driver module has the annunciator lamp drivers and the audio signal power amplifier circuits.

Power supply module: A power supply module changes the aircraft supplied power for use by all the modules located in the GPCW. It does the functions that follow:

- Filtering
- Conversion



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- Stabilization.

The GPWS Computer (GPWC) weighs 4.2 pounds and is packaged in a rectangular aluminum chassis. The top, both sides and the most of the front panel has an adhesive backed blue semi-gloss polyester film material.

The GPWC is installed in the wardrobe on a shelf at station X-15.1, Y-24.3 and Z+154.8.

The GPWS Computer (GPWC) is cooled by free air circulation around the unit.

Switch, Annunciator–Flap Override

[Refer to Figure 24.](#)

The GPWS FLAP OVERRIDE switch is pushed to change the alert modes. A black and yellow crosshatch indication in the switch comes on to show the manual selection to the GPWS FLAP OVERRIDE feature.

The GPWS FLAP OVERRIDE switch is a guarded annunciator switch. A transparent switch cover prevents accidental selection of the switch.

The GPWS FLAP OVERRIDE annunciator switch is attached to pilot's left side console. DZUS fasteners and a related bonding wire directly connect to the console to make ground continuity between the side panel and console.

Switch, Annunciator–GPWS and GS Alert

[Refer to Figure 25.](#)

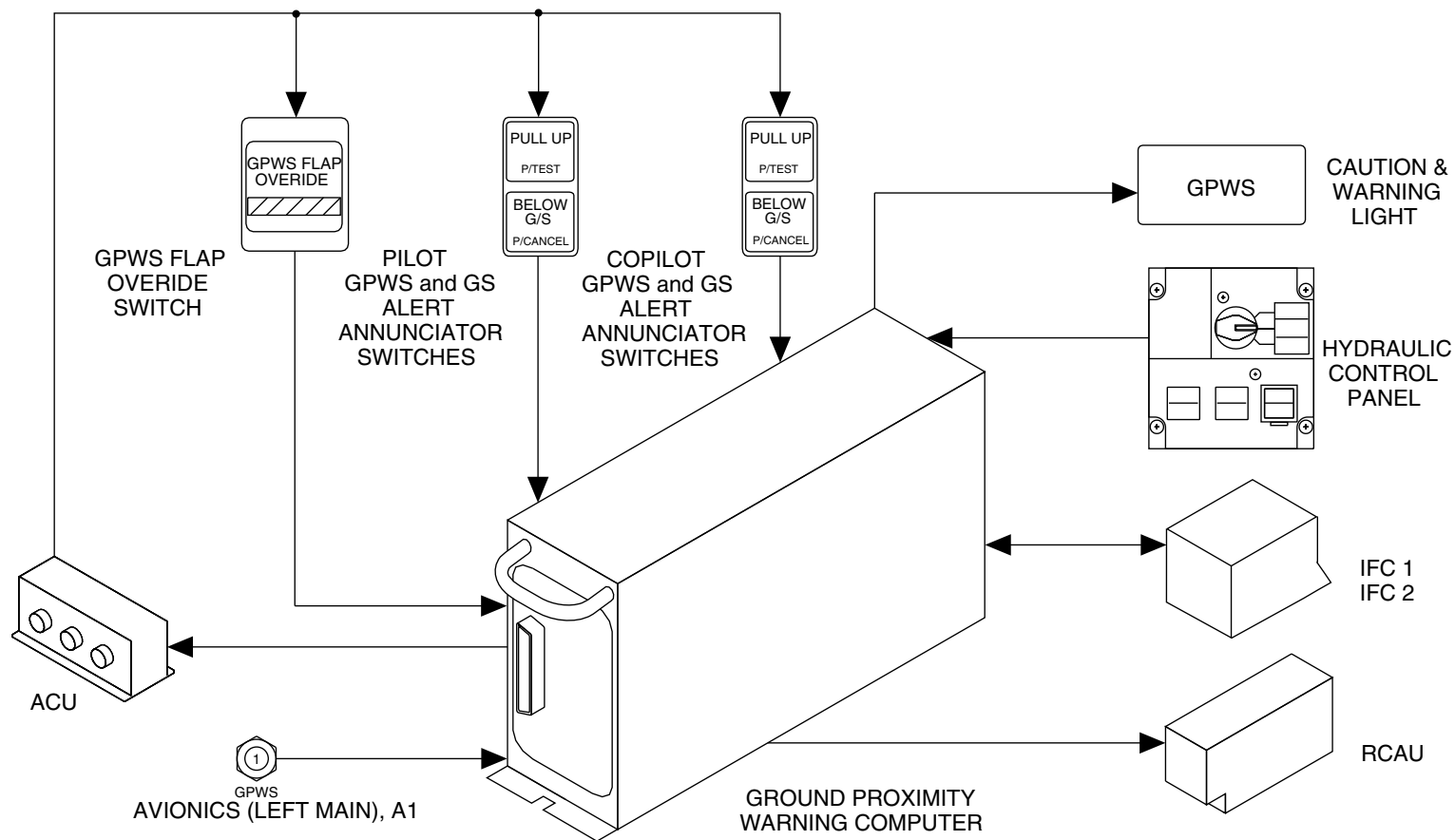
The GPWS and GS alert annunciator switches give the functions that follow:

- PULL UP visual indication
- GPWS TEST selection
- BELOW G/S visual indication
- BELOW G/S aural alert cancellation.

The GPWS and GS alert annunciator switches are vertically connected as a single component assembly. Two GPWS and GS alert annunciator switches are attached to the glareshield panel assembly.



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fs799a01.cgm

GPWS BLOCK DIAGRAM
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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Config 001

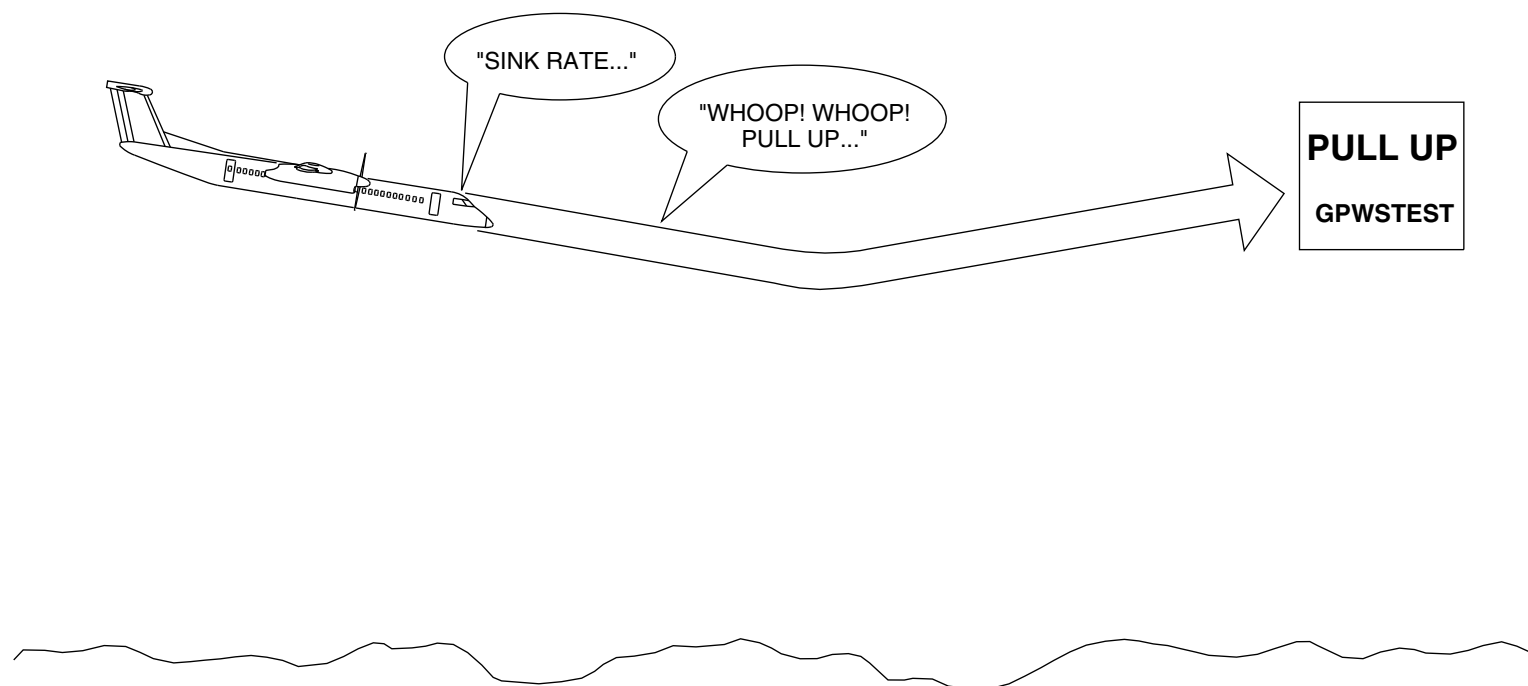
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fs802a01.cgm

GPWS MODE 1, EXCESSIVE DESCENT RATE INDICATIONS
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-42-00
Config 001

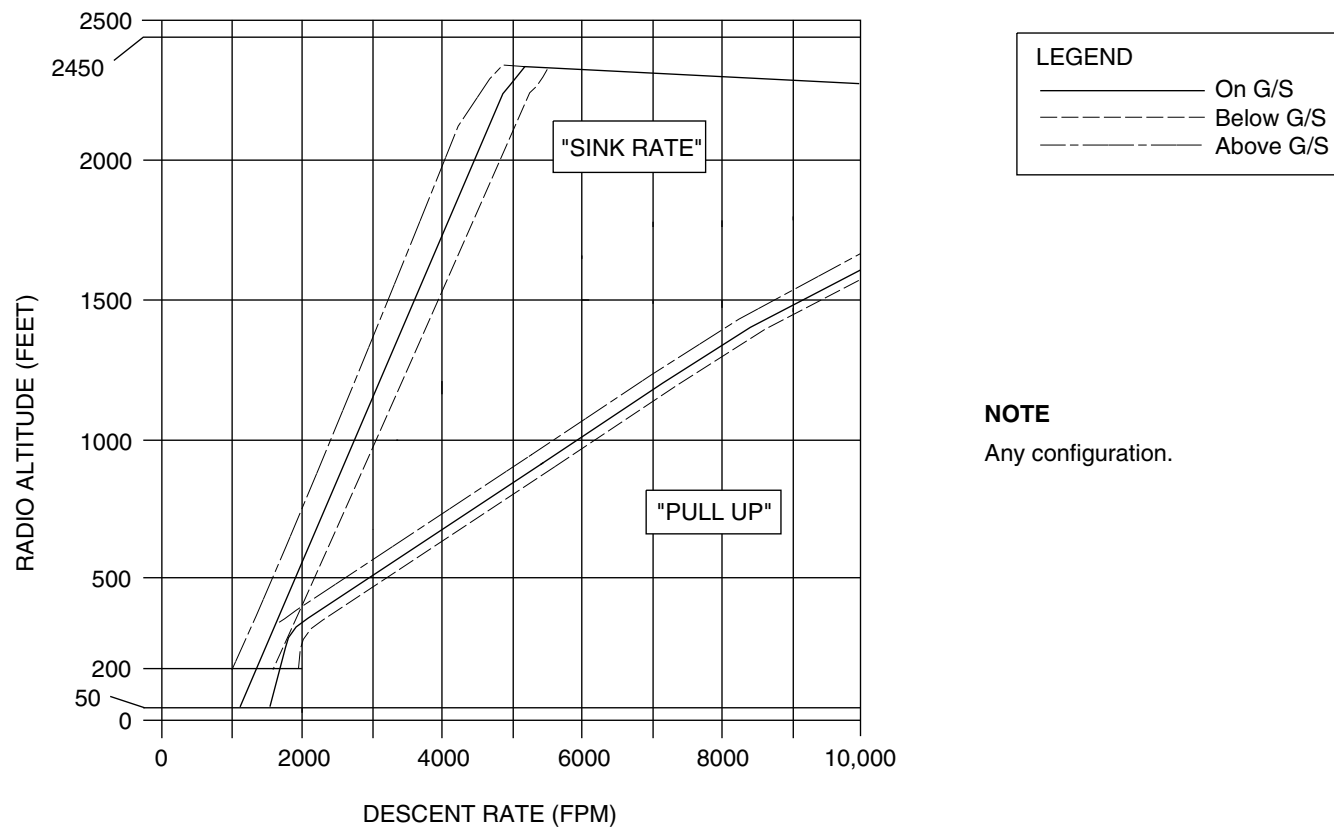
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fs801a01.cgm

GPWS MODE 1, EXCESSIVE DESCENT RATE
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-42-00
Config 001

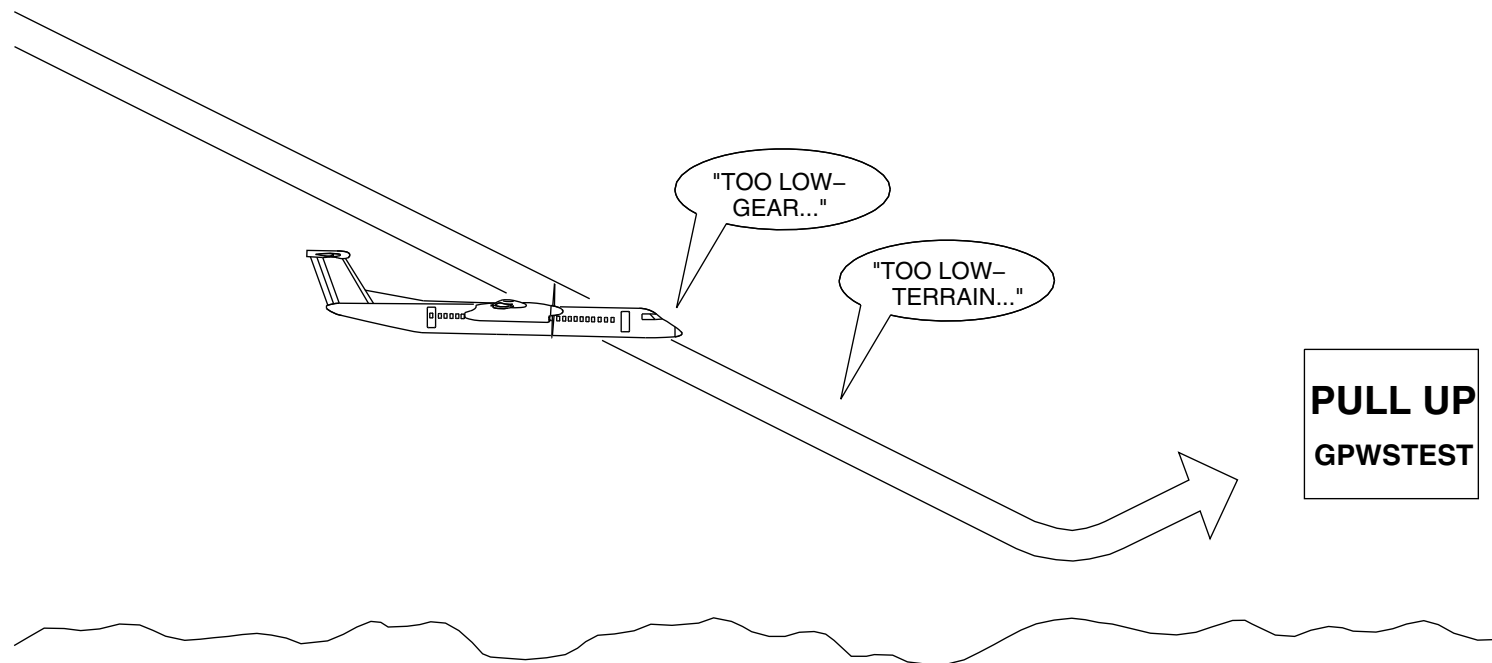
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fs803a01.cgm

GPWS MODE 2A, EXCESSIVE CLOSURE RATE INDICATIONS
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-42-00
Config 001

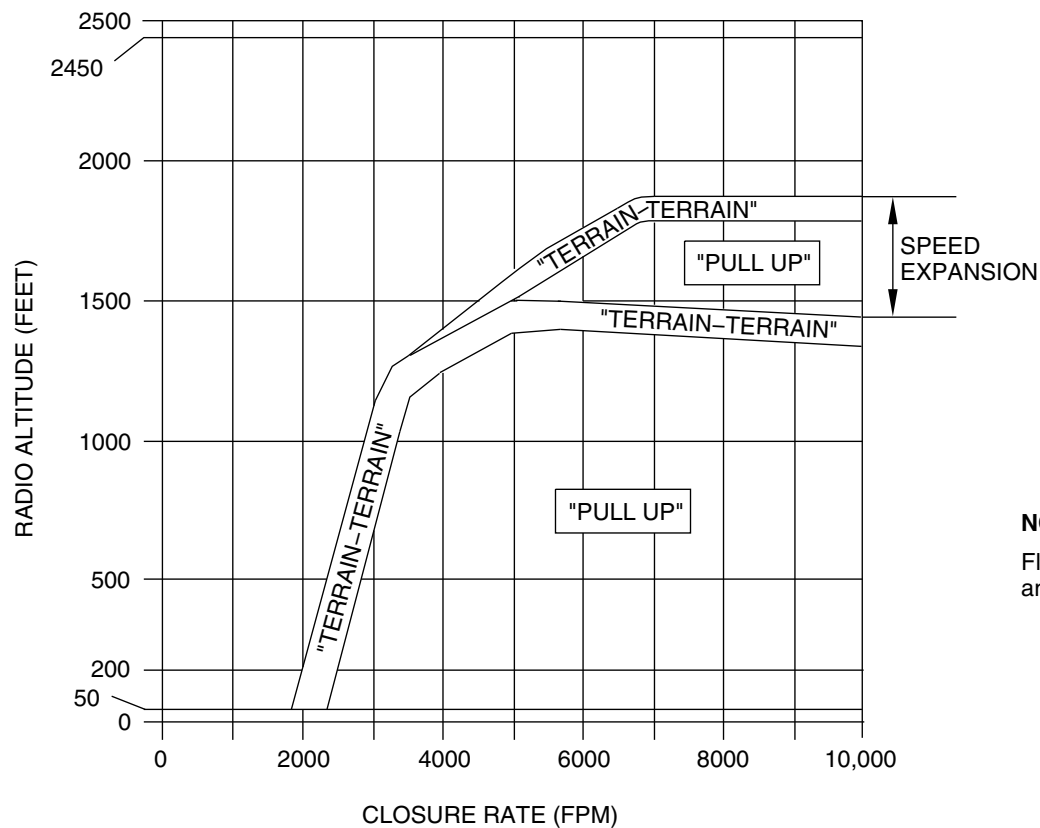
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NOTE

Flaps not in the landing range
and non ILS approach.

fs804a01.cgm

GPWS MODE 2A, EXCESSIVE CLOSURE RATE
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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Config 001

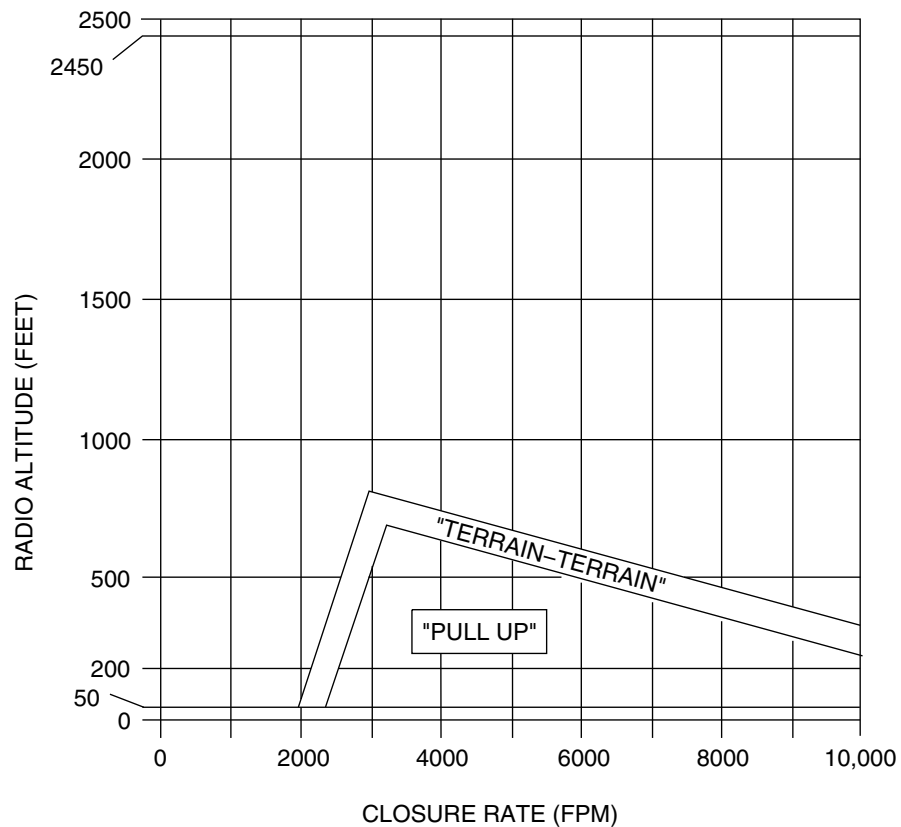
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NOTE

Flaps down or ILS approach.

fs817a01.cgm

GPWS MODE 2B, EXCESSIVE CLOSURE RATE
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

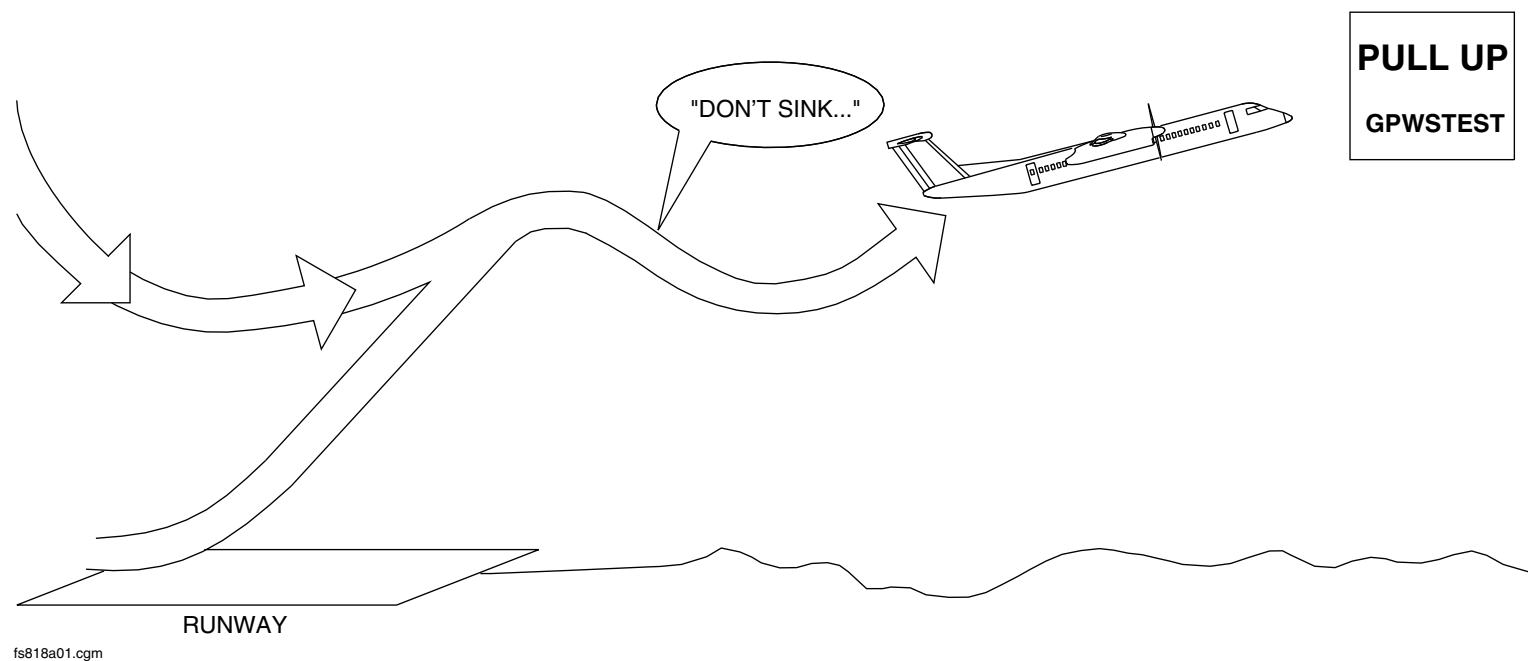
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GPWS MODE 3, DESCENT AFTER TAKEOFF INDICATIONS
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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Config 001

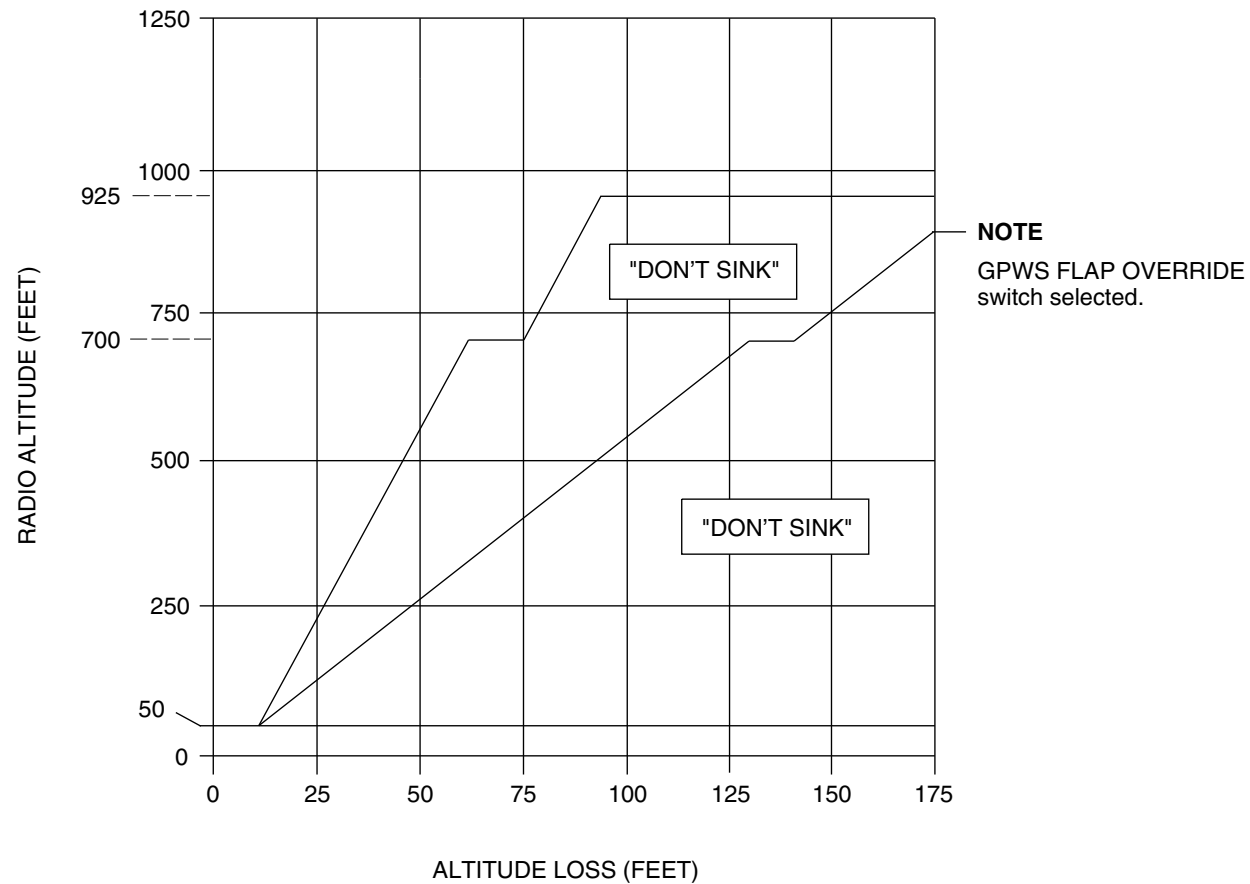
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fs819a01.cgm

GPWS MODE 3, DESCENT AFTER TAKEOFF
Figure 8

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-42-00
Config 001

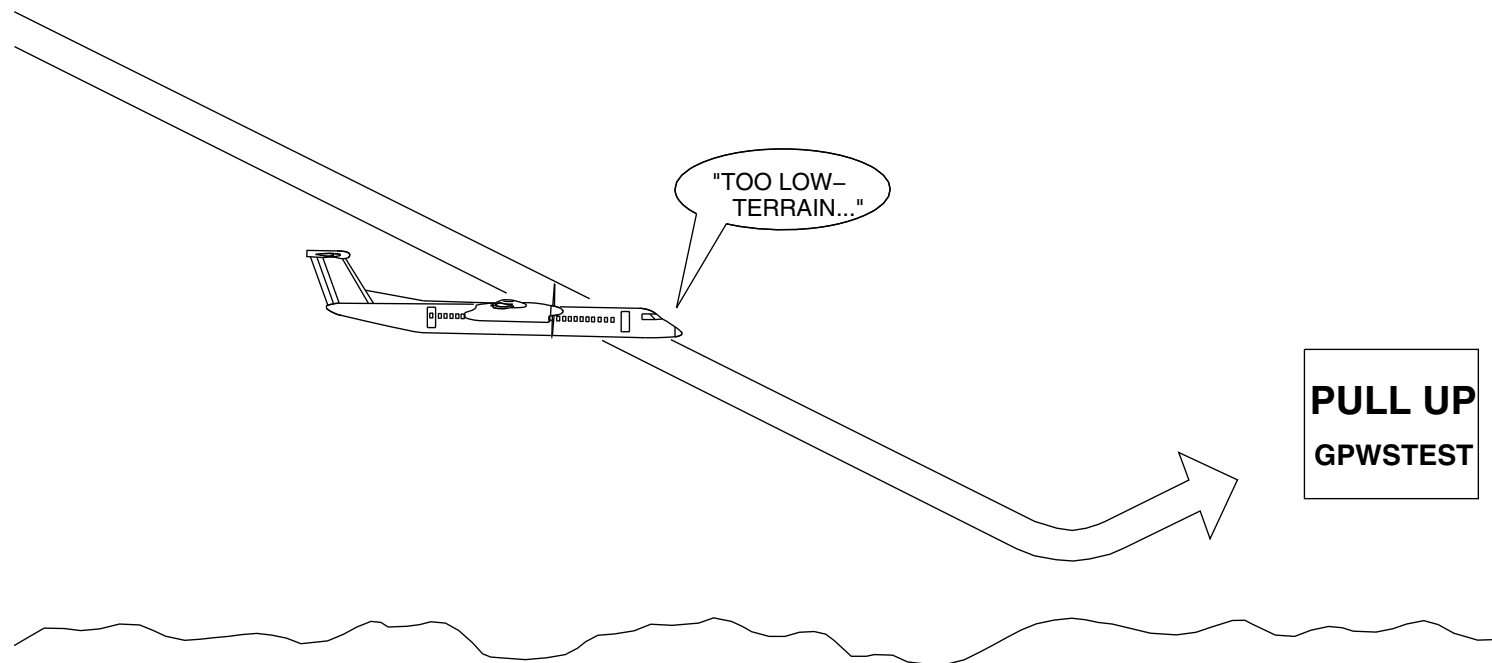
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fs820a01.cgm

GPWS MODE 4A, INSUFFICIENT TERRAIN CLEARANCE INDICATIONS
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-42-00
Config 001

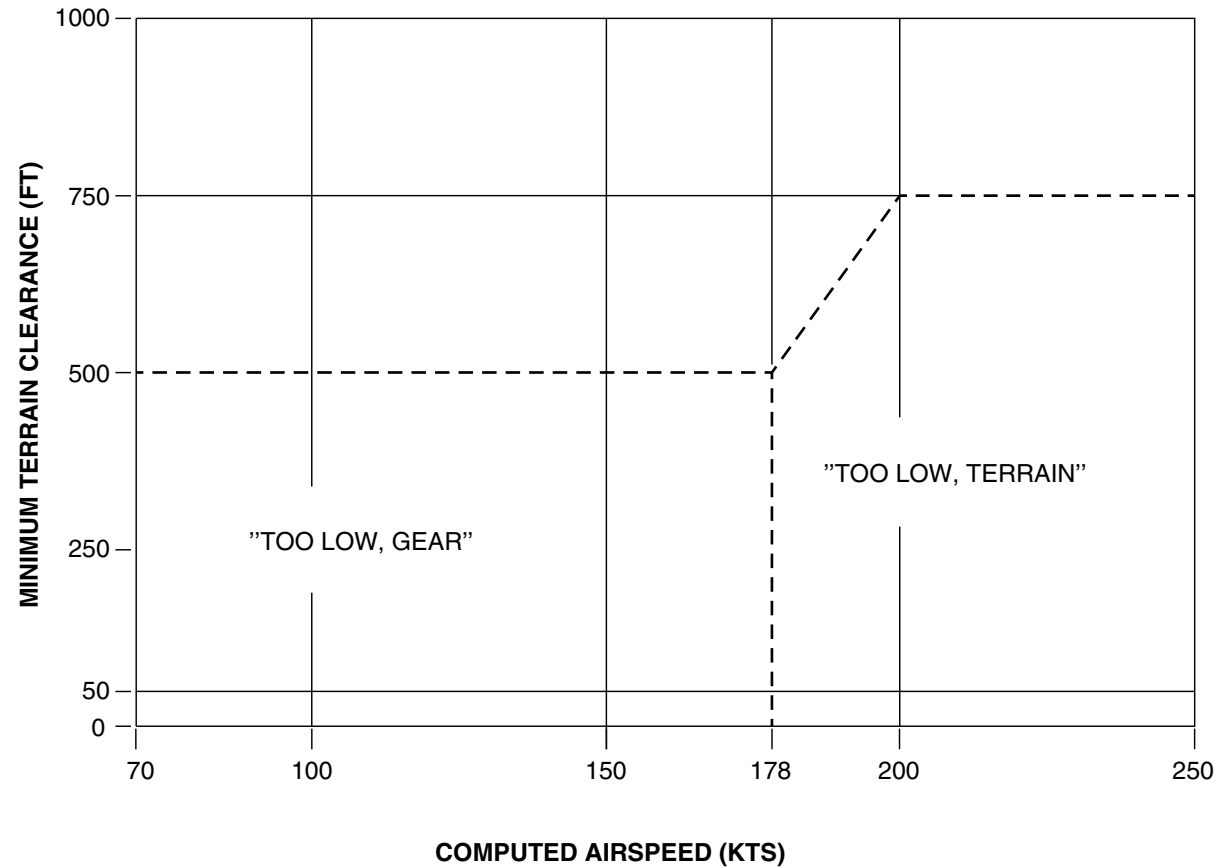
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NOTE

Gear up, Flaps up.

cg4457a01.dg, nn, apr01/2016

GPWS Mode 4A, Insufficient Terrain Clearance
Figure 10

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-42-00
Config 001

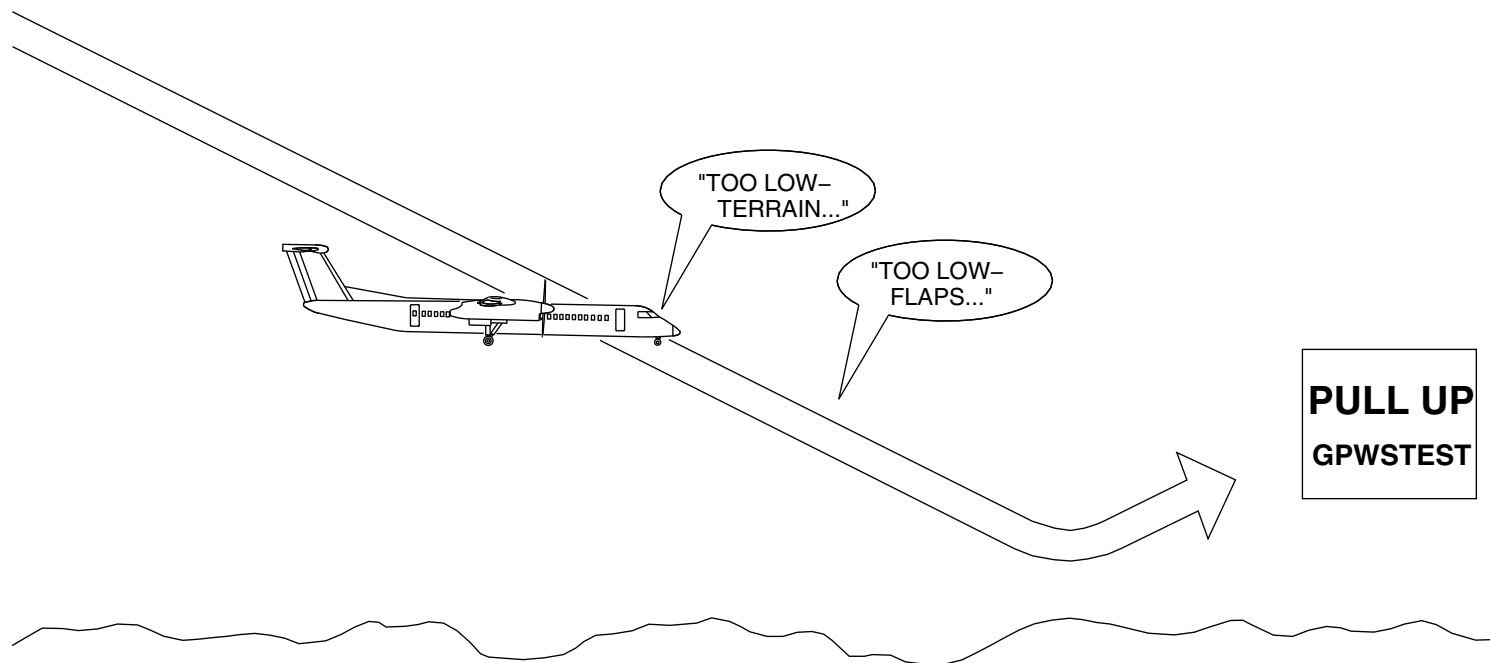
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fs823a01.cgm

GPWS MODE 4B, INSUFFICIENT TERRAIN CLEARANCE INDICATIONS
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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Config 001

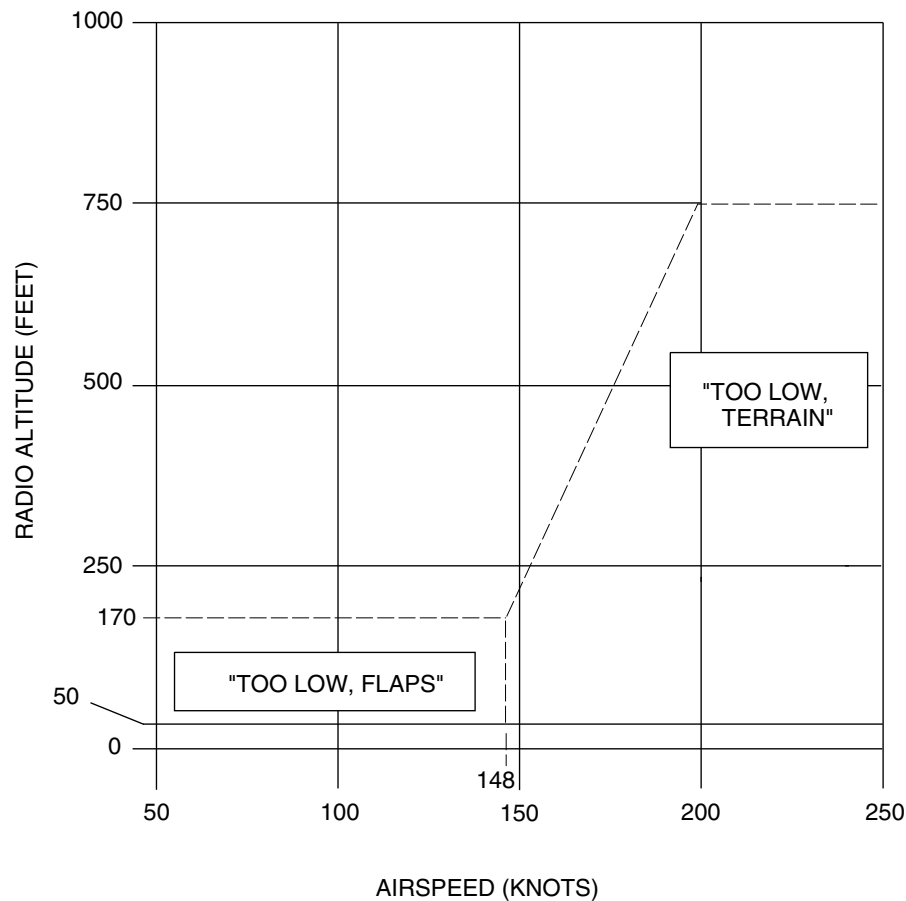
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NOTE

Gear down and flaps not in the landing range.

fs824a01.cgm

GPWS MODE 4B, INSUFFICIENT TERRAIN CLEARANCE
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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Config 001

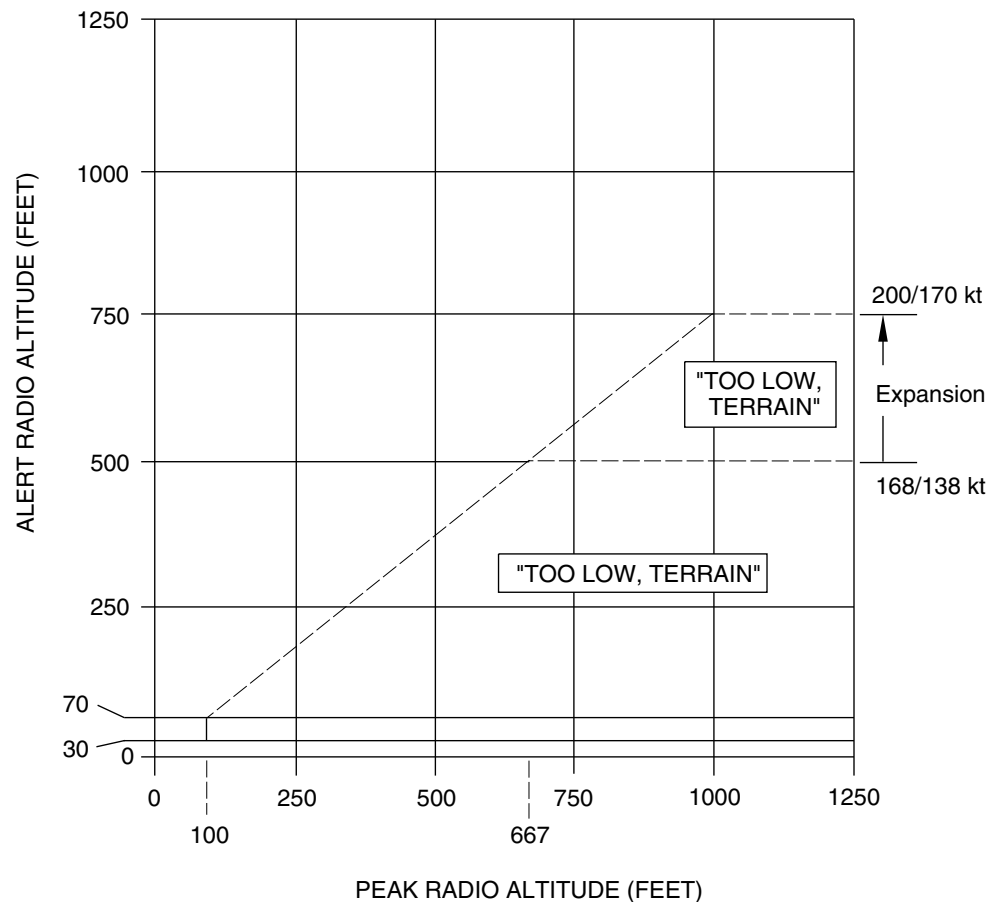
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NOTE

Takeoff with gear up or flaps not in landing range.

fs825a01.cgm

GPWS MODE 4C, INSUFFICIENT TERRAIN CLEARANCE
Figure 13

PSM 1-84-2A
EFFECTIVITY:
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Config 001

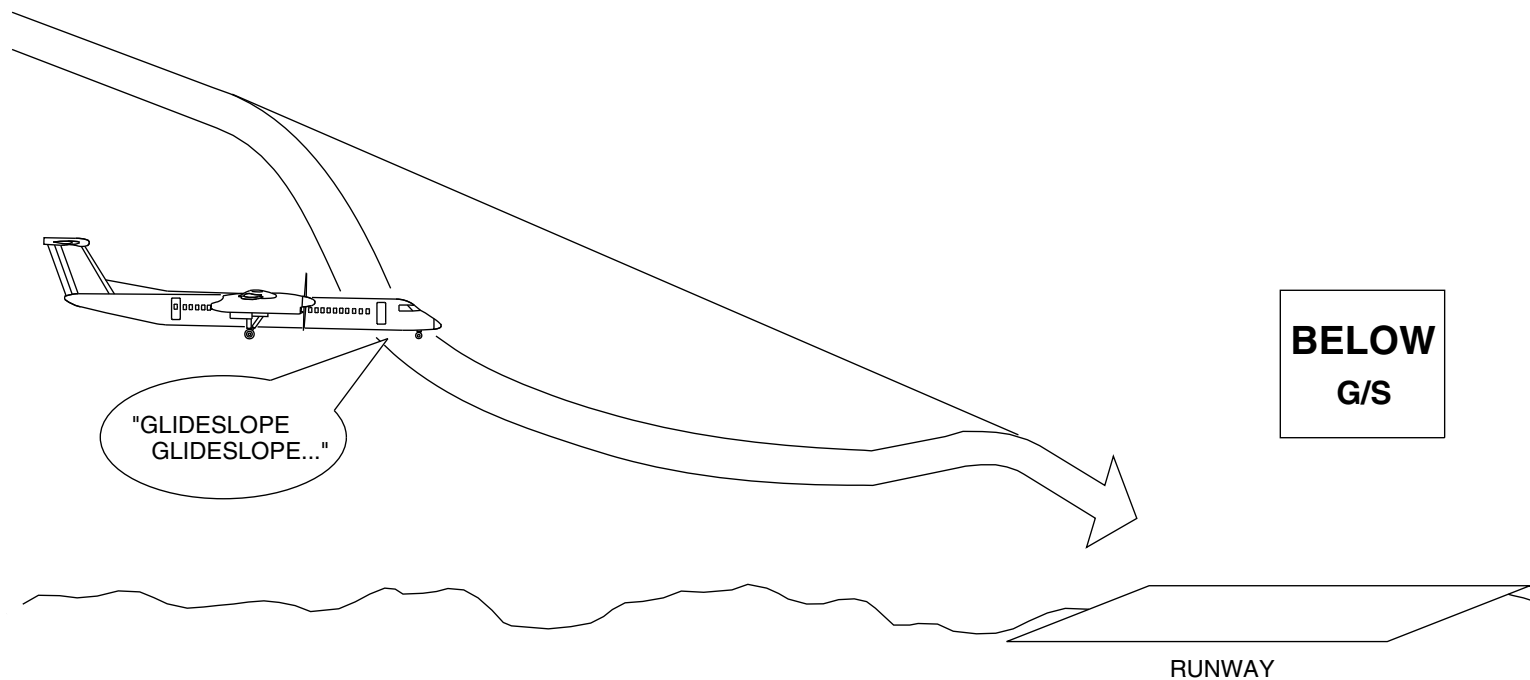
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fs826a01.cgm

GPWS MODE 5, DESCENT BELOW GLIDESLOPE INDICATIONS
Figure 14

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-42-00
Config 001

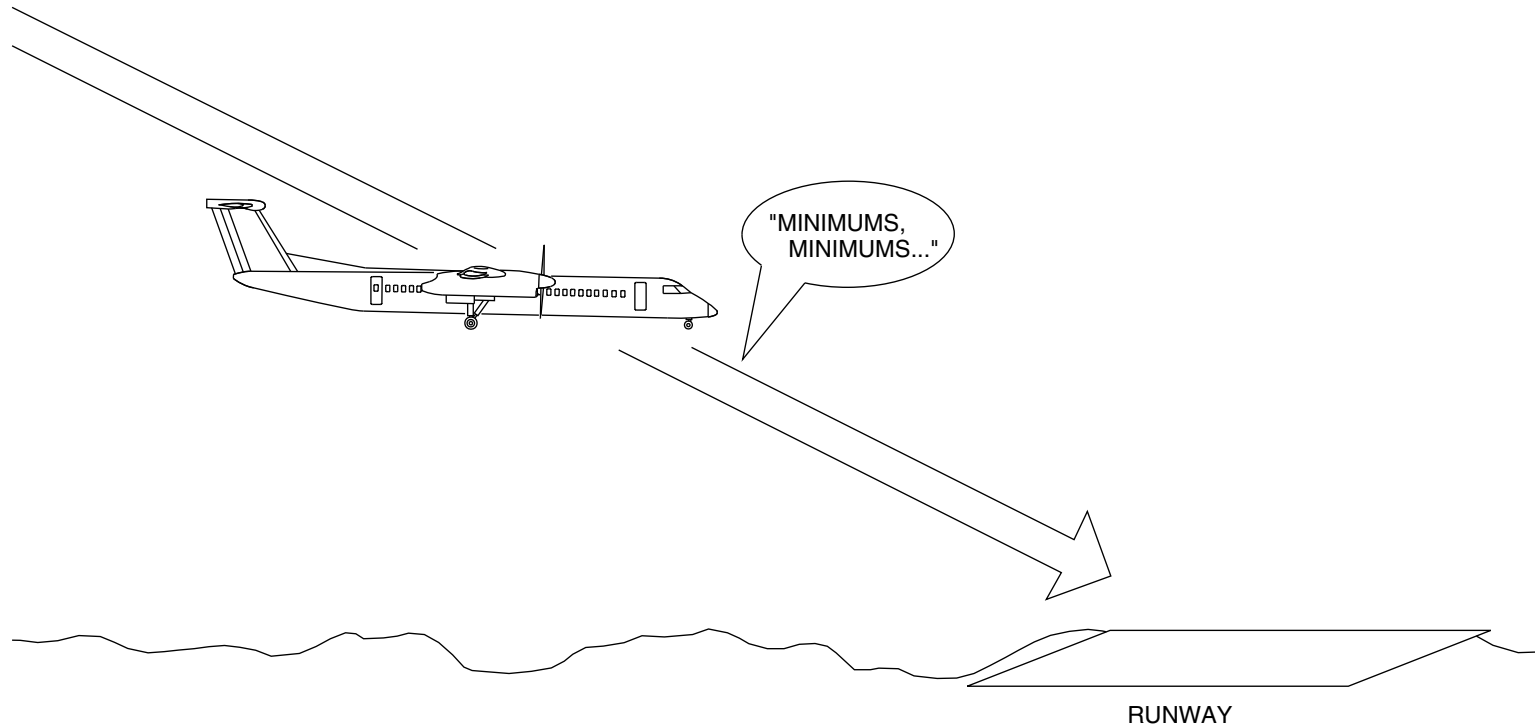
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fs830a01.cgm

GPWS MODE 6, DECISION HEIGHT CALLOUT
Figure 15

PSM 1-84-2A
EFFECTIVITY:
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Config 001

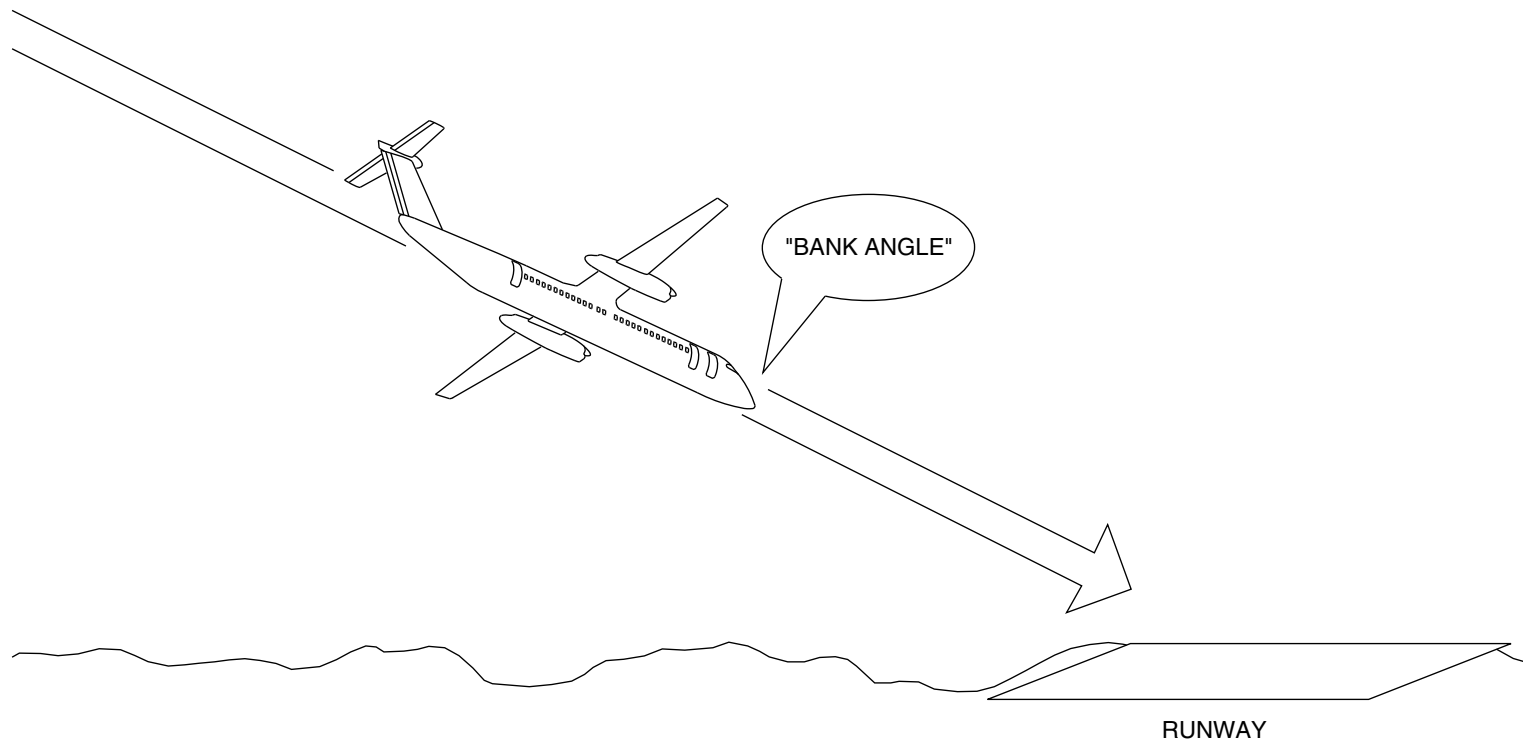
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fs828a01.cgm

GPWS MODE 6, EXCESSIVE BANK ANGLE CALLOUT
Figure 16

PSM 1-84-2A
EFFECTIVITY:
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Config 001

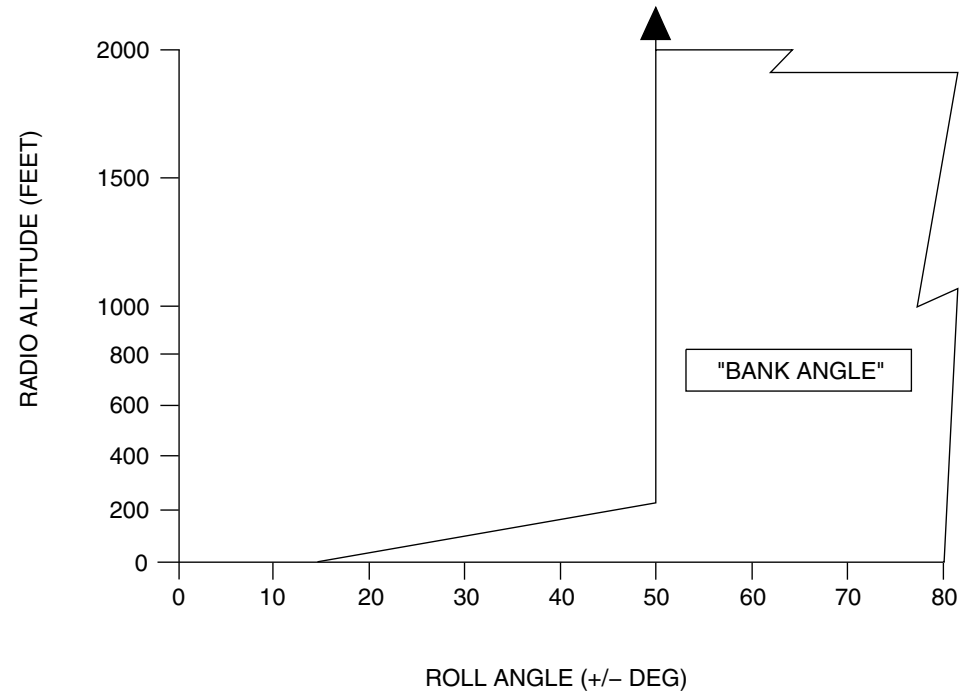
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fs829a01.cgm

GPWS MODE 6, EXCESSIVE BANK ANGLE
Figure 17

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-42-00
Config 001

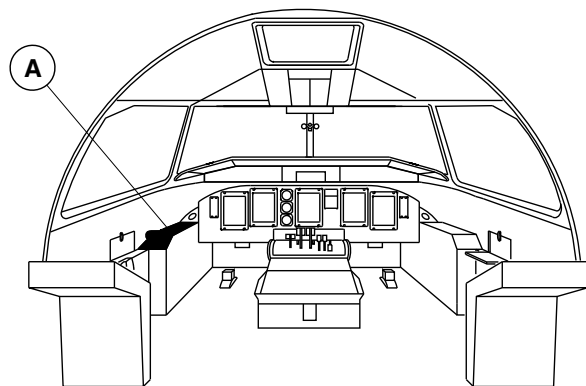
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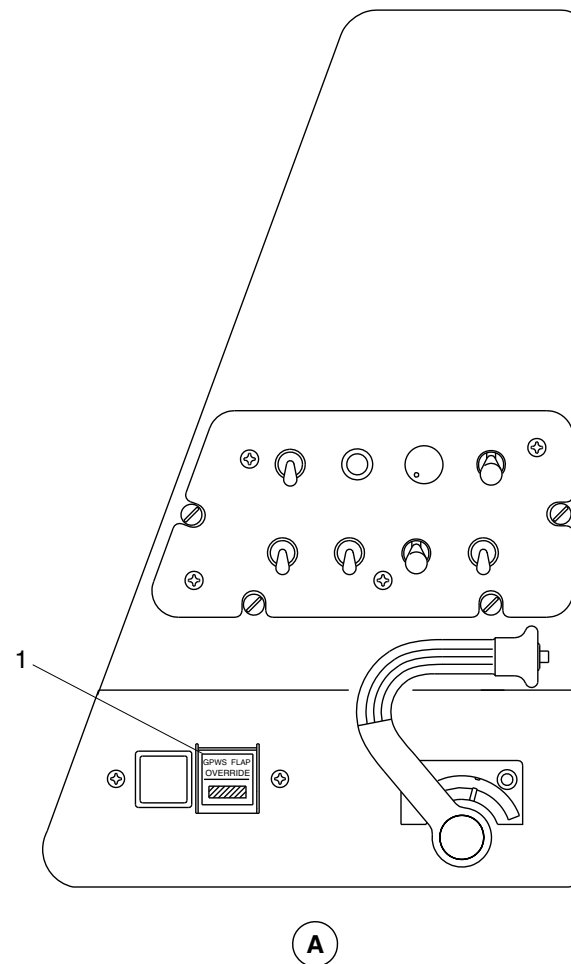
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LEGEND

1. GPWS Flap Override Annunciator Switch.



fs967a01.dg, gv, 19/12/02

GPWS Flap Override Annunciator Switch
Figure 18

PSM 1-84-2A
EFFECTIVITY:
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Config 001

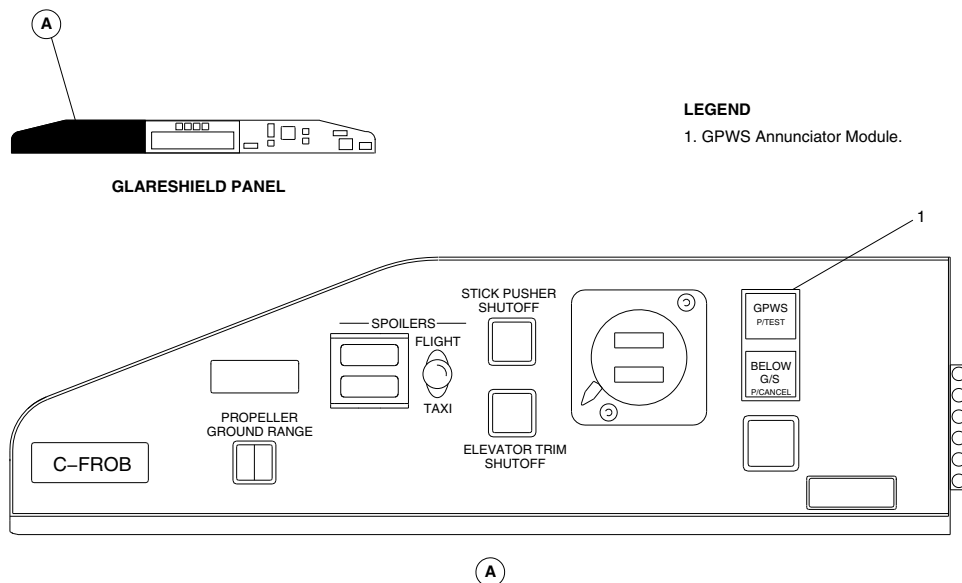
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fs965a01.cgm

GPWS AND GS ALERT ANNUNCIATOR SWITCHES
Figure 19

PSM 1-84-2A
EFFECTIVITY:
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Config 001

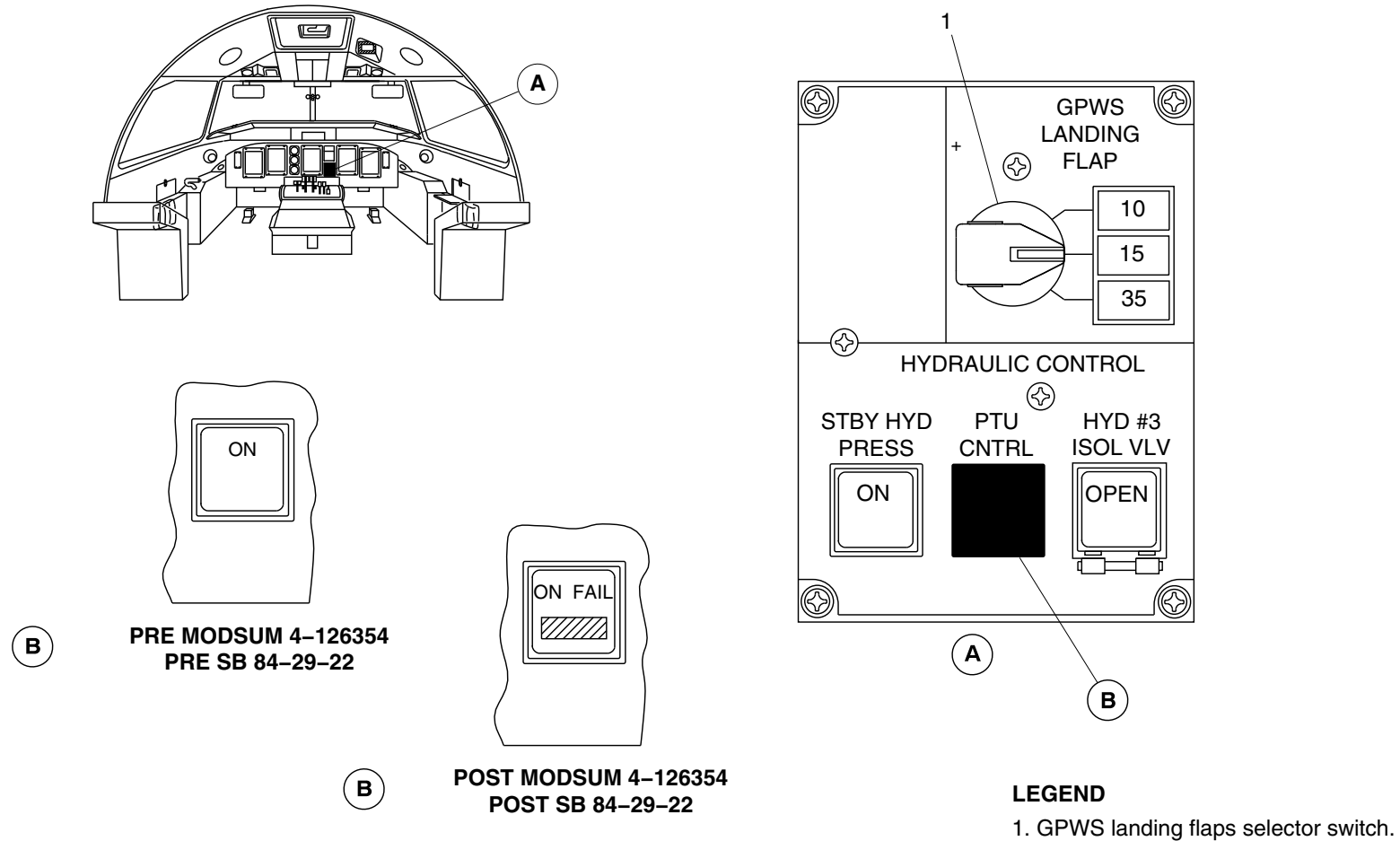
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cg0072a01.dg, dn/kg, nov21/2008

GPWS Landing Flaps Selector Switch
Figure 20

PSM 1-84-2A
EFFECTIVITY:
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Config 001

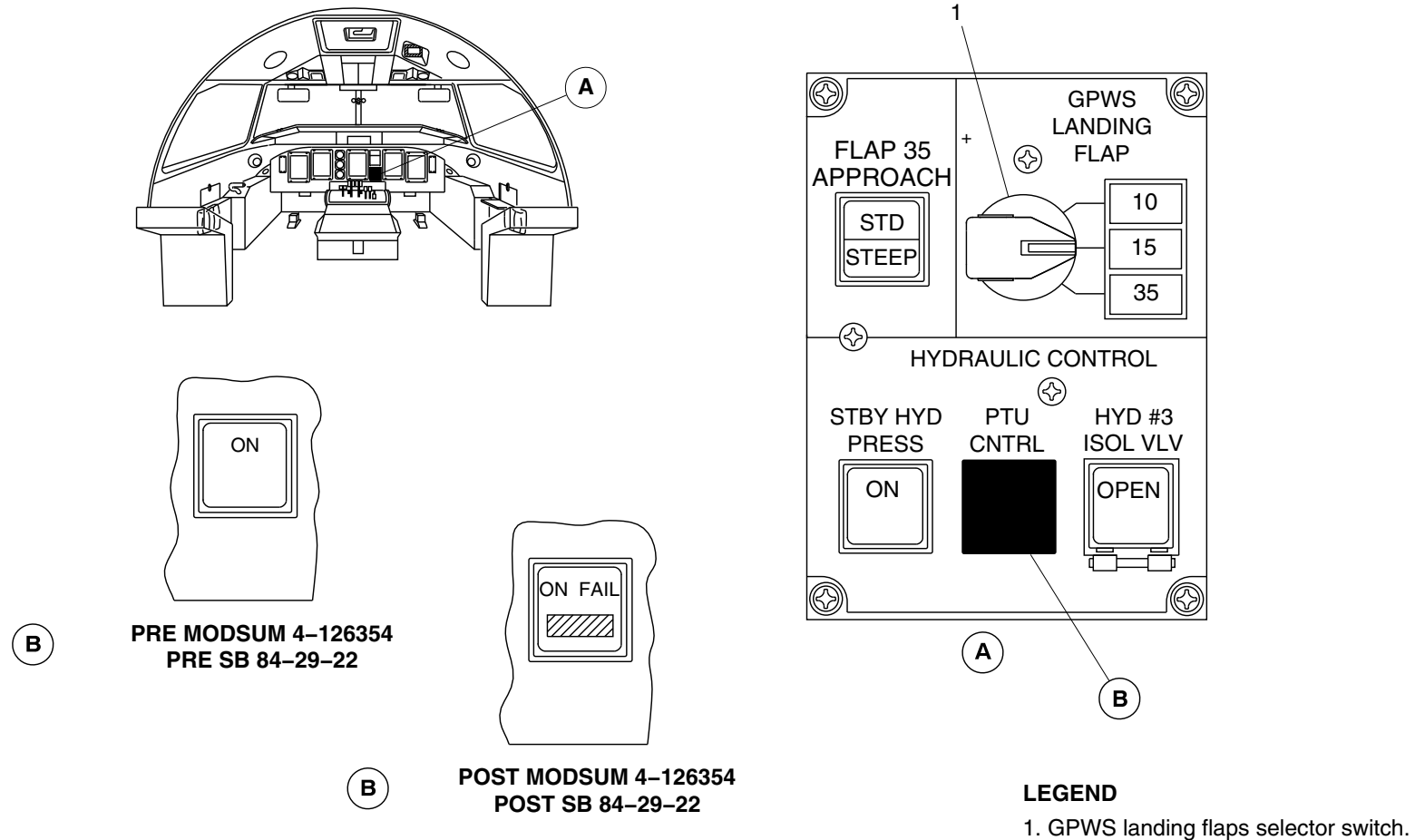
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fs966a01.dg, gv/ kg, nov21/2008

GPWS Landing Flaps Selector Switch (803SO90034)
Figure 21

PSM 1-84-2A
EFFECTIVITY:
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PRIORITY WARNING	ALERT CONDITION	MODE	AURAL WARNING	VISUAL
1	GPWC Invalid	- - -	- - - - -	GPWS Caution and Warning Panel
2	External Audio Suppression from Stall Protection Module	- - -	Aural Warnings Inhibited	Normal
3	Excessive Descent Rate	MODE 1	PULL UP (Immediate Repeat)	GPWS P/CANCEL
4	Excessive Closure Rate to Terrain	MODE 2	PULL UP (Immediate Repeat)	GPWS P/CANCEL
5	Excessive Closure Rate to Terrain	MODE 2	TERRAIN–TERRAIN (Once for each MODE 2 envelope penetration)	GPWS P/CANCEL
6	Excessive Closure Rate to Terrain	MODE 2	TERRAIN–TERRAIN (Pause...3 second repeat)	GPWS P/CANCEL
7	Insufficient Terrain Clearance	MODE 4	TOO LOW, TERRAIN (Pause...3 second repeat)	GPWS P/CANCEL
8	Decision Height Callout	MODE 6	MINIMUMS–MINIMUMS (Pause...3 seconds, 1 message for each approach)	- - - - -

fs840a01.cgm, apr02/2014

GPWS INDICATIONS PRIORITY
Figure 22 (Sheet 1 of 2)

PSM 1–84–2A
EFFECTIVITY:
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PRIORITY WARNING	ALERT CONDITION	MODE	AURAL WARNING	VISUAL
9	Insufficient Terrain Clearance	MODE 4	TOO LOW GEAR (Pause...3 second repeat)	GPWS P/CANCEL
10	Insufficient Terrain Clearance	MODE 4	TOO LOW FLAPS (Pause...3 second repeat)	GPWS P/CANCEL
11	Excessive Descent Rate and Descent Below Glideslope	MODE 1 & 5	GLIDESLOPE, SINK RATE (Pause...3 second repeat)	BELOW G/S P/CANCEL
12	Excessive Descent Rate	MODE 1	SINK RATE (Pause...3 second repeat)	GPWS P/CANCEL
13	Descent After Takeoff	MODE 3	DON'T SINK (Pause...3 second repeat)	GPWS P/CANCEL
14	Descent Below Glideslope	MODE 5	GLIDESLOPE (Variable pause)	BELOW G/S P/CANCEL
15	Altitude Callout	MODE 6	BANK ANGLE (Pause...3 second repeat)	-----

fs842a01.cgm

GPWS INDICATIONS PRIORITY
Figure 22 (Sheet 2 of 2)

PSM 1-84-2A
EFFECTIVITY:
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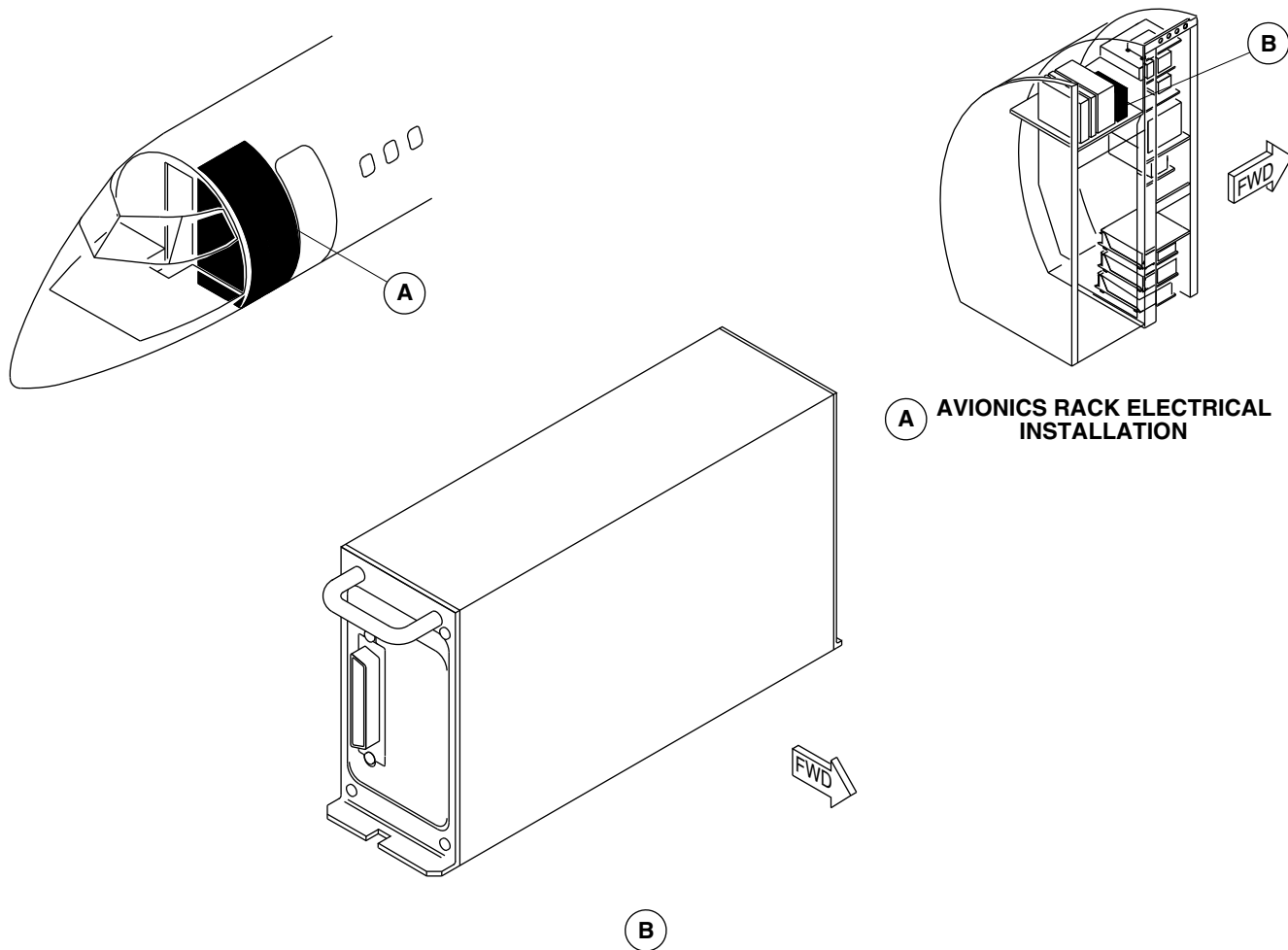
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fs910a01.cgm

COMPUTER, GPWS LOCATOR
Figure 23

PSM 1-84-2A
EFFECTIVITY:
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Config 001

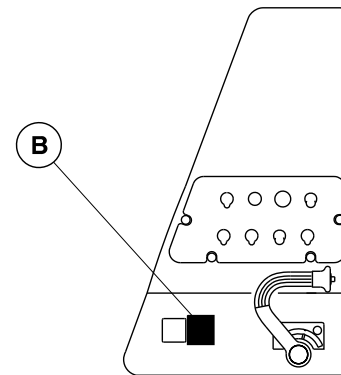
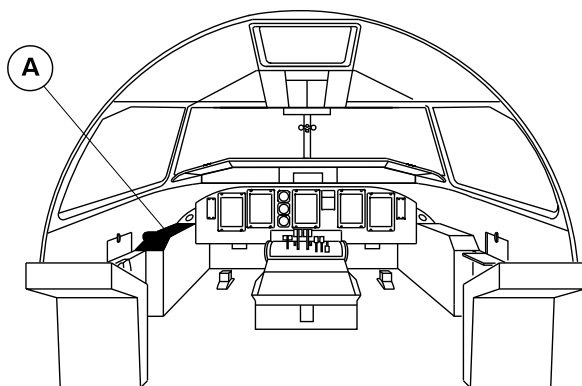
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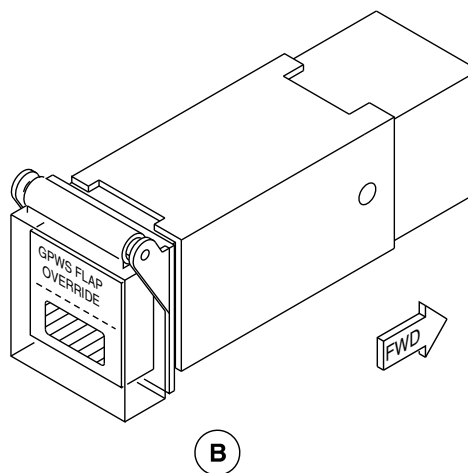


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A LEFT SIDE CONSOLE



B

fs963a01.dg, gv, 19/12/02

Switch, Annunciator–Flap Override Locator
Figure 24

PSM 1–84–2A
EFFECTIVITY:
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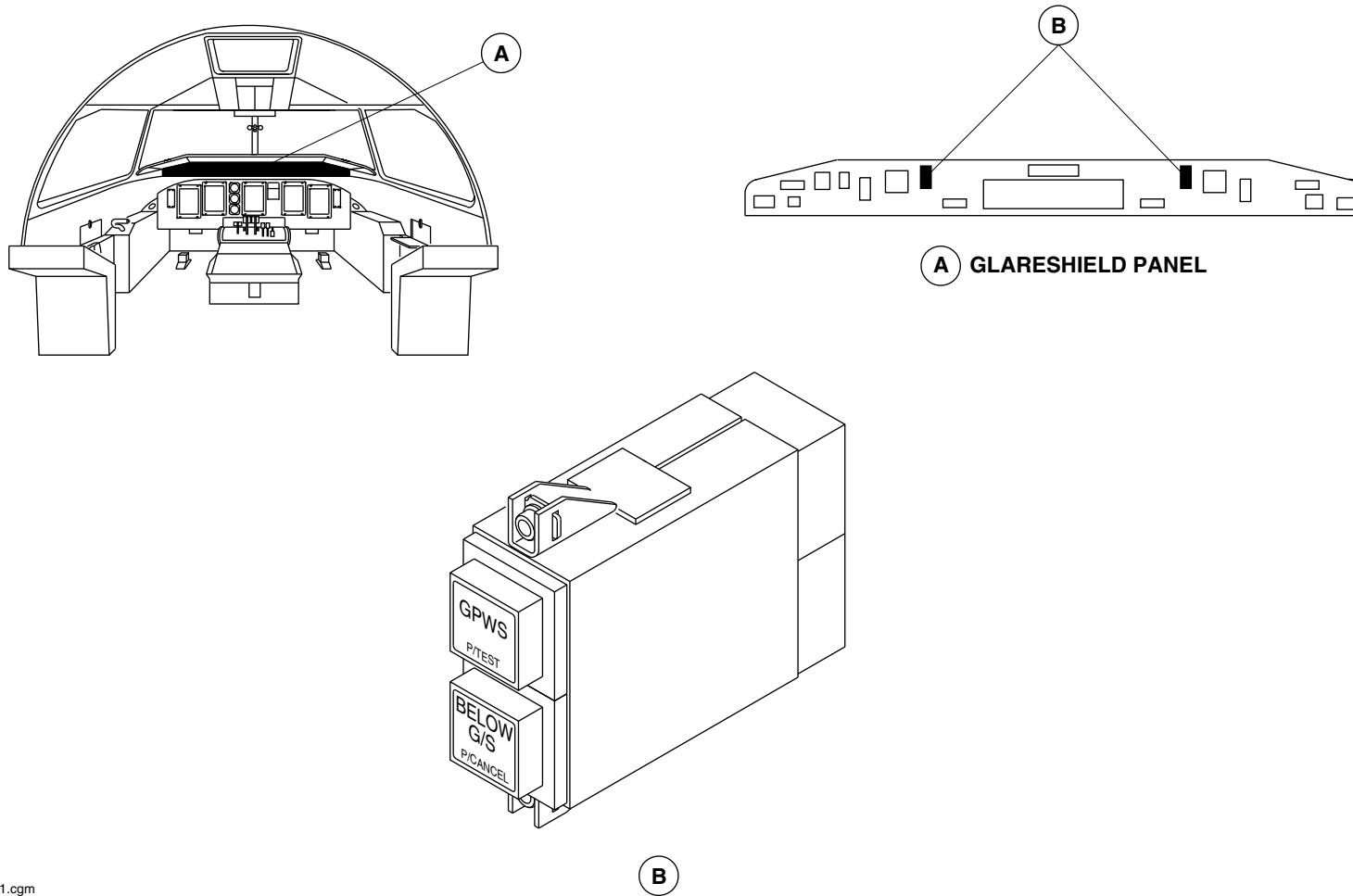
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fs962a01.cgm

SWITCH, ANNUNCIATOR-GPWS AND GS ALERT LOCATOR
Figure 25

PSM 1-84-2A
EFFECTIVITY:
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ENHANCED GROUND PROXIMITY WARNING SYSTEM (EGPWS)

Purpose

The Enhanced Ground Proximity Warning System (EGPWS) gives aural and visual indications of possible Controlled Flight Into Terrain (CFIT).

General Description

Refer to Figures 26 and 27.

The EGPWS has the units that follow:

- EGPWS computer
- FLAP OVERRIDE annunciator switch
- PULL UP and GS annunciator switch
- INHIBIT annunciator switch
- Switching relay.

The EGPWS gives different types of indications that follow:

- Basic Ground Proximity Warning
- Terrain Clearance Floor Alerting
- Terrain Alerting
- Terrain Awareness Display Mode.

Basic Ground Proximity Warning: The EGPWS makes the basic GPWS calculations that follows:

- Mode 1 – Excessive descent rate
- Mode 2 – Excessive closure rate to terrain
- Mode 3 – Descent after takeoff
- Mode 4 – Insufficient terrain clearance
- Mode 5 – Excessive descent below the glideslope
- Mode 6 – Advisory callouts.

Mode 1 – Excessive Descent Rate: The excessive descent rate mode gives audio and visual indications for excessive descent rates into terrain. A “SINK RATE” message sounds and the PULL UP annunciator light comes on. The sensitivity of the mode changes with glideslope position and when the GPWS FLAP OVERRIDE annunciator switch is set.

On aircraft with 803SO90034 installed, the “SINK RATE” message is inhibited when the steep approach mode is selected.

Mode 2 – Excessive Closure Rate To Terrain: The excessive closure rate to terrain mode gives audio and visual indications for dangerously high terrain closure rates. The excessive closure rate to terrain mode has two sub-modes that follow:

- Mode 2A – This mode is active when the flaps are not set to the landing position.
- Mode 2B – This mode is active when the flaps are set to the landing position or while on the glideslope.

A “TERRAIN, TERRAIN” message sounds and the PULL UP annunciator light comes on. If the high terrain closure rate continues,



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the "TERRAIN, TERRAIN" message changes to "PULL UP" and the PULL UP annunciator light stays on.

Mode 3 – Descent After Takeoff: The descent after takeoff mode gives audio and visual indications for excessive altitude loss after takeoff or go around while the aircraft is close to the ground and the flaps and landing gear is not set in the landing configuration. A "DON'T SINK, DON'T" message sounds and the PULL UP annunciator light comes on. The sensitivity of the mode changes when the GPWS FLAP OVERRIDE annunciator switch is set.

Mode 4 – Insufficient Terrain Clearance: The insufficient terrain clearance mode gives audio and visual indications for dangerous terrain clearance relative to flight condition, height above the ground, and speed. The insufficient terrain clearance mode has three sub-modes that follow:

- **Mode 4A** – This mode is active during cruise and approach with the landing gear up. A "TOO LOW TERRAIN" or "TOO LOW GEAR" message sounds depending on airspeed and the PULL UP annunciator light comes on.
- **Mode 4B** – This mode is active during and approach with the landing gear down and flaps up. A "TOO LOW TERRAIN" or "TOO LOW FLAPS" message sounds depending on airspeed and the PULL UP annunciator light comes on.
- **Mode 4C** – This mode is active during takeoff when the landing gear or flaps are up. A "TOO LOW TERRAIN" message sounds and the PULL UP annunciator light comes on.

Mode 5 – Excessive Descent Below the Glideslope: The excessive descent below the glideslope mode gives audio and visual

indications for descends below the glideslope on front-course Instrument Landing System (ILS) approaches. Two levels of audio indications (soft or hard) are given depending on the deviation from the glideslope. A "GLIDESLOPE" message sounds and the BELOW G/S annunciator light comes on.

On aircraft with ModSum 4-903538 installed, the EGPWS computer is equipped with the Localizer Performance with Vertical Guidance (LPV) glide path deviation monitoring capability. The Mode 5 of the EGPWS computer provides two levels of alerting when the aircraft flight path descends below the glideslope on the front-course ILS and LPV approaches.

Mode 6 – Advisory Callouts: The advisory callouts mode gives audio indications for the excessive bank angles, approaching minimums/minimums, and altitude awareness.

Terrain Clearance Floor (TCF) Alerting: The terrain clearance floor alerting gives more protection. It makes an increasing terrain clearance area around the airport runway to give CFIT protection when the mode 4 – insufficient terrain clearance mode give limited or no protection. The TCF mode is active during takeoff, cruise, and approach. It helps the basic mode 4 – insufficient terrain clearance mode by giving an insufficient terrain indication even if the aircraft is in a landing configuration. To make the TCF alerts calculations, the EGPWS uses the data that follows:

- Current aircraft location
- Designation runway center point
- Above Ground Level (AGL) altitude.

The TCF also has a Runway Field Clearance Floor (RFCF) alert to give short landing protection for runways that are significantly higher

PSM 1-84-2A

EFFECTIVITY:

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than the relative terrain. To make the RFCF alerts calculations, the EGPWS uses the data that follows:

- Current aircraft location
- Designation runway center point
- Above Ground Level (AGL) altitude
- Geometric or Above Sea Level (ASL) altitude.

NOTE

The EGPWS has a runway database that contains data records for all hard surface airport runways in the world 3500 ft. or more for coverage in the terrain database.

The TCF mode causes a “TOO LOW TERRAIN” message to sound and the PULL UP annunciator lights to come on.

Terrain Alerting: The terrain alerting mode gives audio and visual indications when the aircraft position relative to the local database–catalogued senses a terrain or obstacle (when and where available) threat.

The EGPWS terrain database has the earth’s surface divided into grid sets and cells relative to the geographic (latitude and longitude) coordinate system of World Geodetic System 1984 (WGS–84). The grid sets have the highest terrain altitude in each cell. The grid changes in resolution depending on geographic location and near airports.

The terrain alerting mode causes a “CAUTION TERRAIN” or “CAUTION OBSTACLE” message to sound and the PULL UP annunciator light to come on. If the condition continues to worsen, the “TERRAIN, TERRAIN, PULL UP” “OBSTACLE, OBSTACLE, PULL UP” message sounds and the PULL UP annunciator light stays on. The Multi–functional Display units (MFD1, MFD2) also

automatically show local terrain forward of the aircraft when a terrain alert condition is sensed.

Terrain Awareness Display Mode: In the terrain awareness display mode, the background display is calculated from the airplane altitude with respect to the terrain data in the digital elevation matrix overlays. The terrain awareness display modes are of two types:

- Standard display mode
- Peaks display mode.

Standard Display Mode – It provides a graphical plan–view image of the surrounding terrain as varying density patterns of green, yellow, and red.

Peaks Display Mode – It lets the terrain below the aircraft to be viewed during all flight conditions. At altitudes safely above the terrain for the set display range, the highest and lowest elevations are shown relative to the aircraft altitude. The peaks display mode has two elevation numbers to show the highest and lowest terrain currently being shown. Only one elevation number is shown when the aircraft is flying in the areas with no differences in terrain elevations (water or flat terrain).

The EGPWS has red PULL UP and amber BELOW G/S annunciator switches to show indications for the different conditions. The PULL UP indication comes on for the conditions that follow:

- Excessive descent rate
- Excessive closure rate to terrain
- Descent after takeoff
- Insufficient terrain clearance
- TCF alerting



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- Terrain alerting.

The light in the BELOW G/S annunciator switch comes on during an excessive descent below the glideslope condition.

The PULL UP annunciator switch is also used to test the GPWS and the BELOW G/S annunciator switch is used to mute the excessive descent below the glideslope condition.

The different EGPWS sounds are supplied through the Remote Control Audio Unit (RCAU) to the headphones and speakers as follows:

- “SINK RATE, SINK RATE”
- “PULL UP”
- “TERRAIN, TERRAIN”
- “OBSTACLE, OBSTACLE”
- “CAUTION TERRAIN”
- “CAUTION OBSTACLE”
- “TOO LOW TERRAIN”
- “TOO LOW GEAR”
- “TOO LOW FLAPS”
- “DON’T SINK”
- “GLIDESLOPE”.

The GPWS FLAP OVERRIDE annunciator switch is pushed to override unwanted indications when landing with zero or partial flaps and cancel glideslope indications.

The GPWS LANDING FLAPS switch on the HYDRAULIC control panel sets the landing flap value that is used for its calculation.

The EGPWS indications are automatically cancelled by the conditions that follow:

- Stall
- EGPWS malfunction is sensed.

The EGPWC supplies a signal to the Caution and Warning Panel (CWP) to make the GPWS caution light come on when the system has sensed a malfunction.

Basic Ground Proximity Warning System and Altitude Awareness Call-Out: The EGPWC makes the basic GPWS and Altitude Awareness Call-Out calculations. It uses data received from the units that follow through the two Input/Output Modules (IOM1, IOM2) and Input/Output Processors (IOP1, IOP2) that are located in their related Integrated Flight Cabinet (IFC1, IFC2):

- Proximity Sensor Electronics Unit (PSEU)
- Flap Position Indicator Unit (FPIU)
- Air Data Computers (ADC1, ADC2)
- Attitude and Heading Reference System (AHRS1, AHRS2)
- Radio Altimeters (RA1, RA2)
- VHF Navigation Receivers (VHF NAV1, VHF NAV2)
- Index Control Panels (ICP1, ICP2)
- Flight Guidance Control Panel.



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The Enhanced Ground Proximity Warning System (EGPWS) supplies data directly to the systems that follow:

- Caution and Warning Lights Panel
- Advisory Control Unit (ACU)
- Audio Integrating System (AIS)
- Stall Protection System (SPS).

The GPWS Landing Flap Select Panel supplies flaps position data to the EGPWS.

A related IOP1–EGPWS, IOP2–EGPWS bus supplies the data that follows:

- Vertical speed
- Air speed
- Glideslope deviation, glideslope enable
- Bank angle
- Above ground level altitude radio altitude
- Radio altitude validity
- Flap setting
- Landing gear selection
- Decision Height (DH) selection
- Back course selection
- Stall indication.

Detailed Description: Mode 1, Excessive Descent Rate:

Refer to Figures 28 and 29.

The Mode 1 condition gives indications for excessive rates of descent relative to the aircraft height above the terrain. The indications are given well ahead of the possible impact with the ground. The Ground Proximity Warning Computer (GPWC) calculates the possible flight into terrain from the Radio Altitude (RA) and Air Data Unit (ADU) Barometric Decent Rate (FPM) data.

The Mode 1 condition causes the "SINK RATE" and "PULL UP" aural indications to sound and the PULL UP annunciator lights come on.

The "SINK RATE" aural indication repeats every three seconds while the aircraft is in the initial indication area. As the aircraft goes into the second indication area that is closer to the terrain, a "PULL UP" aural indication sounds and not the "SINK RATE" aural indication.

An ILS glideslope selection changes the indication area to minimize incorrect warnings.

When the aircraft is above the glidepath, the indication area changes. The descent rate must increase for a EGPWS indication output. This decreases the possibility of nuisance EGPWS outputs.

When the aircraft is below the glidepath, a lower descent rate than when above the glidepath causes a EGPWS indication output. The EGPWS is more sensitive when the aircraft is below the glidepath.

When the aircraft is more than 200 feet AGL altitude, 150 feet per minute is added for each dot of glideslope deviation, up to a maximum of 300 feet per minute.

The descent or climb rate is set to 300 feet per minute when the aircraft is above or below the glidepath with the GPWS FLAP OVERRIDE annunciator switch selected.



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On aircraft with 803SO90034 installed the "SINK RATE" message is inhibited when the steep approach mode is selected. The steep approach mode is selected when the GPWS landing flap switch on the landing flap/hydraulic control panel is selected to 35 degree position and the flap 35 approach switchlight is pushed such that "STEEP" lens illuminates. The steep approach mode should be selected before the aircraft enters the steep approach phase of approach to landing. Once selected, the steep approach mode may be de-selected by one more push (depressing) of the flap 35 approach switchlight or by selection of the flap lever to any other position from the 35 degree detent. The de-selection of steep approach mode causes the EGPWS to return to standard mode automatically. The standard mode selection is also possible when the flap 35 approach switchlight is pushed such that the "STD" lens illuminates on the flap 35 approach switchlight. After this if required, the steep approach may be selected again.

Mode 2, Excessive Closure Rate to Terrain:

[Refer to Figure 30.](#)

The Mode 2 condition gives indications when terrain below the aircraft is rising dangerously fast. The Ground Proximity Warning Computer (GPWC) gives the indications well ahead of the projected collision with terrain.

The Ground Proximity Warning Computer (GPWC) monitors the conditions that follow to give the Mode 2 indication:

- AGL altitude
- Air speed
- Terrain closure rate.

The Mode 2 condition causes the "TERRAIN, TERRAIN" and "PULL UP" aural indications to sound and the PULL UP annunciator lights come on.

Mode 2 operates in the two sub-modes that follow:

- Mode 2A
- Mode 2B.

Mode 2A: [Refer to Figure 31.](#)

The Mode 2A "PULL UP" aural indication will stop when the aircraft goes out of the warning area. The PULL UP annunciators will stay on until any condition occurs that follow:

- The barometric altitude increase is more than 300 feet
- An up acceleration that is a combination of barometric altitude gain and radio altitude, and time.

Mode 2B: [Refer to Figure 32.](#)

Mode 2B functions when any one of the conditions is met that follows:

- The flaps are in the landing position
- The GPWS flap override switch is selected
- The aircraft is on an ILS or MLS approach and the aircraft is not more than 1.3 dots below the glidepath
- The glideslope cancel feature is not selected.

When the landing gear is down, and the flaps are down or the GPWS flap override switch is set, a repetitive "TERRAIN, TERRAIN" aural indication will sound. If the landing gear is not down, a repetitive normal "PULL UP" aural indication will sound when the "TERRAIN, TERRAIN" aural indication is complete. When the aircraft goes out of



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the warning area, the aural indications will stop and the PULL UP annunciators go out.

NOTE

Selecting the GPWS FLAP OVERRIDE annunciator switch cancels the Mode 2 warnings. This lets the aircraft maneuver near terrain when approaching an airport.

Mode 3, Descent After Takeoff:

[Refer to Figures 33 and 34.](#)

Mode 3 functions during a takeoff or missed Approach when the conditions that follow are met:

- Increase in airspeed
- 50 feet Above Ground Level (AGL) altitude
- Increase in barometric altitude
- Landing gear retraction.

The Mode 3 gives an indication to show too much altitude decrease after takeoff or after a missed approach. A barometric altitude decrease of approximately 10 percent of the Above Ground Level (AGL) altitude starts the Mode 3 indication.

The Mode 3 condition sounds a "DON'T SINK" aural indication and the PULL UP annunciator lights come on. An increased rate of climb cancels the "DON'T SINK" aural indications and the PULL UP annunciator lights go out.

When the Above Ground Level (AGL) altitude is more than 925 feet for 17 seconds, the Mode 3 condition is cancelled.

Aircraft speed, flap and landing gear position are used to calculate a warning area below the aircraft. Selecting the GPWS FLAP OVERRIDE annunciator switch cancels the Mode 3 indication conditions if the Above Ground Level (AGL) altitude is more than 50 feet. It lets the Above Ground Level (AGL) altitude decrease 20 percent before giving an indication.

When the aircraft is 700 feet Above Ground Level (AGL) altitude, it gives an additional altitude increase of 5 feet per second.

Mode 4, Insufficient Terrain Clearance:

The Mode 4 condition gives an indication for insufficient terrain clearance during the phases of flight that follow:

- Cruise
- Approach
- Descent
- Climbout.

The Mode 4 condition gives indications when the flight path is too low for Mode 2 excessive closure rates with terrain or Mode 1 excessive descent rate calculations. It uses the data that follow:

- AGL altitude
- Airspeed
- Flight phase.

The Mode 4 condition functions in the modes that follow:

- Mode 4A
- Mode 4B



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- Mode 4C.

Mode 4A: Refer to Figure 35 and 36

When the aircraft is in a configuration with the gear and flaps selected up, the EGPWS calculates a "floor" below the aircraft to show insufficient terrain clearance.

When the air speed is more than 250 knots and the AGL altitude is less than 1000 feet, the "TOO LOW TERRAIN" aural indication sounds and the PULL UP annunciator lights come on. Lower AGL altitudes are used to give the same indication at airspeeds 190 to 250 knots.

At speeds less than 190 knots and the AGL altitude is less than 500 feet, the "TOO LOW GEAR" aural indication sounds and the PULL UP annunciator lights come on.

Mode 4B: Refer to Figure 37 and 38

Mode 4B operates during cruise and approach when the landing gear is down, the flaps not in the landing position and the EGPWS flap override switch is not set.

NOTE

The Landing Flap Selector Switch (LFSS) supplies the EGPWS with data relating to the specified landing flap position.

When the air speed is less than 250 knots and the AGL altitude is less than 750 feet, the "TOO LOW TERRAIN" aural indication sounds and the PULL UP annunciator lights come on. Lower AGL altitudes are used to give the same indication at airspeeds 148 to 225 knots.

During the initial approach, at speeds less than 148 knots and the AGL altitude is less than 170 feet, the "TOO LOW FLAPS" aural indication sounds and the PULL UP annunciator lights come on.

When the flap override annunciator switch is set, the EGPWS will not give any indications for the flaps up landing.

Mode 4C: Refer to Figure 39 and 40

This sub-mode is active during a takeoff or go-around. The Mode 4C gives a minimum terrain clearance that increases with the Above Ground Level (AGL) altitude.

When the AGL altitude is more than 100 feet after takeoff or 200 feet when doing a go-around, a warning floor is calculated below the aircraft. It is 75 percent of the highest ACL altitude. The Mode 4C continues to operate until the approach mode is activated or the AGL altitude goes below 30 feet.

The conditions that follow, cause the "TOO LOW TERRAIN" aural indication to sound and the PULL UP annunciator lights to come on:

- The aircraft descends after takeoff
- The terrain below the aircraft rises at a steeper slope than when the aircraft is climbing.

The warning will continue until the aircraft has sufficient clearance from the terrain.

NOTE

Warnings from Modes 4A, 4B, and 4C cannot occur at the same time.

Mode 5, Excessive Descent Below Glideslope:

Refer to Figures 41 and 42.

The Mode 5 operates when the EGPWS senses the conditions that follow:

- An ILS frequency is set



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- The landing gear is down
- The AGL altitude is less than 925 feet
- The aircraft is below the glidepath
- Glideslope cancel is not set.

The Mode 5 has two aural indication areas that relate to the deviation from the glidepath. Each uses a different aural indication volume level. The low volume level indication is specified as the soft indication and the louder indication is named the hard indication. The hard indication sounds two times as loud as a soft indication. The time between aural indications change if on the glideslope deviation and AGL altitude. The time between the aural indications decreases as the danger of the Mode 5 condition increases.

When the aircraft is below the glideslope and into the soft indicating area, the "GLIDESLOPE" aural indication sounds and the amber BELOW GS annunciator switches come on. The volume of the initial "GLIDESLOPE" aural indication sounds is less than the other EGPWS aural indications. While the aircraft is below the glideslope, the aural repetition rate increases as the AGL altitude decreases. If the aircraft goes into the hard indication area, the audio volume level increases to that of the other aural indications.

When the AGL altitude is less than 150 feet, more glideslope deviation is necessary to cause the indications to come on.

The Mode 5 condition cancels when the AGL altitude is less than 50 feet or by a BELOW G/S annunciator switch selection.

Mode 6, Advisory Callouts:

The mode 6 condition gives indications for excessive bank angles while maneuvering close to the runway.

Refer to Figure 43.

On aircraft with ModSum 4–309203 or 4–410300 (without SB84–34–160) incorporated, the EGPWC gives the advisory callouts that follow:

Table 1: BASIC

CALLOUT	DESCRIPTION
"MINIMUMS, MINIMUMS"	At descent below decision height setting

On aircraft with ModSum 4–429519 or SB84–34–102 or SB84–34–47 incorporated, the EGPWC gives the advisory callouts that follow:

Table 2: MENU 85

CALLOUT	DESCRIPTION
"TWENTY FIVE HUNDRED"	At descent below 2500 ft.
"ONE THOUSAND"	At descent below 1000 ft.
"PLUS HUNDRED"	At descent below decision height setting plus 100 ft.
"MINIMUMS"	At descent below decision height setting
"FIFTY"	At descent below 50 ft.
"FORTY"	At descent below 40 ft.



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CALLOUT	DESCRIPTION
"THIRTY"	At descent below 30 ft.
"TWENTY"	At descent below 20 ft.
"TEN"	At descent below 10 ft.

NOTE

In a test scenario, when you push and hold the GPWS test switch, the 'PLUS HUNDRED' and 'MINIMUMS' callouts will be annunciated first. The actual sequence of these annunciations during the flight will depend on the Decision Height selection.

On aircraft with ModSum 4-457043 or SB84-34-115 or SB84-34-36 or SB84-34-83 or SB84-34-111 or SB84-34-137 or SB84-34-160 incorporated, the EGPWC gives the advisory callouts that follow:

Table 3: MENU 34

CALLOUT	DESCRIPTION
"TWENTY FIVE HUNDRED"	At descent below 2500 ft.
"FIVE HUNDRED"	At descent below 500 ft.
"MINIMUMS"	At descent below decision height setting
"ONE HUNDRED"	At descent below 100 ft.

CALLOUT	DESCRIPTION
"FIFTY"	At descent below 50 ft.
"FORTY"	At descent below 40 ft.
"THIRTY"	At descent below 30 ft.
"TWENTY"	At descent below 20 ft.
"TEN"	At descent below 10 ft.

NOTE

In a test scenario, when you push and hold the GPWS test switch, the 'MINIMUMS' callout will be annunciated first. The actual sequence of this annunciation during the flight will depend on the Decision Height selection.

On aircraft with ModSum 4-457062 incorporated, the EGPWC gives the advisory callouts that follow:

Table 4: MENU 42

CALLOUT	DESCRIPTION
"PLUS HUNDRED"	At descent below decision height setting plus 100 ft.
"MINIMUMS"	At descent below decision height setting
"FIFTY"	At descent below 50 ft.



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CALLOUT	DESCRIPTION
"FORTY"	At descent below 40 ft.
"THIRTY"	At descent below 30 ft.
"TWENTY"	At descent below 20 ft.
"TEN"	At descent below 10 ft.

Refer to Figure 44.

A "BANK ANGLE" aural indication gives indications for excessive bank angles relative to the AGL altitude.

Refer to Figure 45.

The limit of the indication decreases from 50 degrees at 190 feet AGL altitude to 15 degrees near ground level.

Terrain Clearance Floor (TCF) Alerting:

Refer to Figure 46.

A TCF is calculated to give more protection to the basic GPWS modes.

It gives an indication for insufficient terrain clearance during the phases of flight that follow:

- Take Off
- Cruise
- Approach

- Descent
- Climbout.

The EGPWS calculates an increasing terrain clearance area around an airport runway that is related to the distance from the runway.

The TCF is calculated with the data that follow:

- Current aircraft location
- Nearest runway center point position
- AGL altitude.

Refer to Figure 47.

If the aircraft goes below the TCF a "TOO LOW TERRAIN" aural indication sounds once and the PULL UP annunciator lights come on. The "TOO LOW TERRAIN" aural indication sounds again when AGL decreases 20 percent.

Terrain Alerting:

Refer to Figure 48.

The EGPWS calculates potential conflicts between the aircraft flight path and terrain or obstacles to give an Electronic Instruments System (EIS) indication.

The terrain is calculated with the data that follow:

- Aircraft position
- Aircraft altitude
- Ground track
- Ground Speed



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- Vertical Speed
- Roll attitude
- Terrain Clearance Floor
- Terrain and obstacle database.

Terrain Caution and Warning Alert Level:

Refer to Figure 49.

A terrain clearance area forward of the aircraft is continuously calculated. A terrain alert is calculated if the terrain clearance area does not agree with terrain elevation data in the terrain database.

Two alert areas are calculated as follows:

- Terrain caution alert level
- Terrain warning alert level.

If the aircraft goes into the terrain caution alert area, a "CAUTION TERRAIN, CAUTION TERRAIN" aural indication sounds and the EIS shows the conflicting terrain in yellow.

If the aircraft goes into the terrain warning alert area, a "TERRAIN, TERRAIN, PULL UP" aural indication sounds and the EIS shows the conflicting terrain in red.

NOTE

An obstacle alert has the same EIS indications but the aural indication is "CAUTION OBSTACLE, CAUTION OBSTACLE" or "OBSTACLE, OBSTACLE, PULL UP".

Terrain Awareness Display Modes

The terrain awareness alerting and display function maintains a background display of the local terrain forward of the airplane for the optional flight compartment display. In the event of terrain or obstacle caution or warning conditions, an aural alert and lamp outputs are triggered. The background image is then enhanced to highlight the related terrain or obstacle threats forward of the airplane. There are two types of terrain awareness display modes as follows:

- Standard Display Mode
- Peaks Display Mode.

Standard and Peaks Display Mode:

Refer to Figures 50 , 51 and 52.

The standard display mode lets the EIS show the terrain using different color patterns to indicate the vertical displacement between the terrain elevation and the aircraft altitude.

The red and yellow dot patterns indicate the terrain near or above the aircraft altitude. The solid yellow and red colors indicate alert and warning areas relative to the flight path of the aircraft. The medium and low density green display patterns indicate terrain that is below the aircraft and within 2000 feet of the aircraft altitude. The terrain more than 2000 feet below the aircraft is not displayed and the terrain display is typically blank during the en–route portion of the flight.

The peaks display mode lets the EIS show terrain around or below the aircraft. At altitudes that are safely above all terrain for the set indication range, the terrain is shown independently of aircraft



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altitude. The highest and lowest elevations are shown to increase situational awareness.

Terrain more than 2000 feet below the aircraft is not shown. The peaks display mode gives density patterns and level thresholds relative to the terrain elevations in the indication area.

The peaks display mode has two elevation numbers to show the highest and lowest terrain that is currently shown. The elevation numbers shows terrain in hundreds of feet above sea level, Mean Sea Level (MSL). The terrain elevation numbers are displayed with the highest terrain number on top, and the lowest terrain number below it. The terrain numbers are shown in the same color as their terrain color pattern. Only one elevation number is shown when flying above flat terrain, water, or the aircraft more than 2000 feet above the highest terrain.

The EGPWS supplies terrain indication data to the EIS Multi-Function Displays (MFD1, MFD2) for the standard and peaks display modes as follows:

COLOR	TERRAIN ELEVATION
Solid Red	Terrain warning
Solid Yellow	Terrain caution
High density red dots	More than 2000 ft. above aircraft
High density yellow dots	Between 1000 and 2000 ft. above aircraft

COLOR	TERRAIN ELEVATION
Medium density yellow dots	Between 500 below to 1000 ft. above the aircraft
Medium density yellow dots	Landing gear down – Between 250 below to 1000 ft. above the aircraft
Solid Green	Terrain/obstacle that is the highest elevation band within 500 below the aircraft. It is shown only when no red or yellow terrain are in the range of the indication.
Solid Green	Landing gear down – Terrain/obstacle that is the highest elevation band within 250 below the aircraft. It is shown only when no red or yellow terrain are in the range of the indication.
High density green dots	Between 500 and 1000 ft. below the aircraft
High density green dots	Landing gear down – Between 250 and 1000 ft. below the aircraft
High density green dots	Terrain/obstacle that is the middle elevation band when no red or yellow terrain are in the range of the indication (for peaks display mode only).
Light density green dots	Between 1000 and 2000 ft. below aircraft
Light density green dots	Terrain/obstacle that is the lower elevation band when no red or yellow terrain are in the range of the indication (for peaks display mode only).



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COLOR	TERRAIN ELEVATION
Black	No terrain/obstacles
Cyan	Water at sea level elevation (0 feet MSL) (for peaks display mode only)
Magenta	Unknown terrain. No terrain data in the database for the magenta area shown

The reference altitude is calculated down from the actual aircraft altitude to give a thirty second advance indication of terrain when descending more than 1000 fpm.

Enhanced Ground Proximity Warning Computer (EGPWC):

Refer to Figure 53.

The Enhanced Ground Proximity Warning Computer (EGPWC) receives different system inputs and makes calculations to give aural and visual indications when the boundaries of an alerting area is exceeded. Terrain display data is supplied to the Electronic Instrument System (EIS) for indication.

The Enhanced Ground Proximity Warning Computer (EGPWC) is installed in the wardrobe on a shelf at station X-15.1, Y-21.1 and Z+161.2. It weighs 7 pounds (3.18 Kg) and is packaged in an ARINC rectangular aluminum chassis.

The EGPWC is cooled by free air circulation around the unit.

On aircraft without the Modsum 4-903303 incorporated, the EGPWC has a front panel with the following:

- Status indicators

- Door
- Self-test switch
- Headphone jack
- PCMCIA interface
- Upload/download status indicators
- Test connector.

Status Indicators: The unit's front panel has three status Light Emitting Diodes (LEDs) to show EGPWS system and computer status. An amber LED, identifies EXTERNAL FAULT is on when a signal fault external to the EGPWC is sensed. A green LED, identified COMPUTER OK is on when the EGPWC is serviceable. A red LED, identified COMPUTER FAIL is on when the EGPWC has sensed an internal malfunction.

Door: A door gives access to the self-test switch, headphone jack, PCMCIA interface and status indicators, and test connector.

Self-Test Switch and Headphone Jack: A PRESS TO SELF-TEST switch gives same test as the flight compartment selection. A headphone jack lets the operator listen to the test.

PCMCIA Interface and Upload/Download Status Indicators: The PCMCIA interface is used to upload or download internal EGPWC data. Software and database is uploaded and history data is downloaded. The control of the upload or download process is done by inserting the PCMCIA card in the front panel access. LEDs give indication of PCMCIA interface operation.

Test connector: There is a RS-232 input/output port on the front and rear of each EGPWC for test control and data readout. These ports serve as access to signal monitoring, flight history, initiating BITE tests, updating databases and other functions. The interface is full



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duplex and is intended for human interaction, but can also be used for custom interface programs.

On aircraft with the Modsum 4–903303 incorporated, the EGPWC has a front panel with the following:

- Status indicators
- Door
- Self–test switch
- Headphone jack
- USB interface
- Two RJ45 Ethernet connectors
- Upload/download status indicators
- Test connector

Status Indicators: The unit's front panel has two status Light Emitting Diodes (LEDs) to show EGPWS system external fault and computer status. The External Fault LED will come on in yellow when the EGPWC has an external fault. The COMPUTER STATUS LED will come on in red when the EGPWC operates correctly and will turn to red when the EGPWC has an internal fault.

Door: A door gives access to the self–test switch, headphone jack, USB interface, status indicators, and ethernet connectors.

Self–Test Switch and Headphone Jack : A "PRESS TO SELF–TEST" switch gives same test as the flight compartment selection. A headphone jack lets the operator listen to the test.

USB Interface, Ethernet connectors and Data–load Status Indicators: The USB Interface and Ethernet connectors are used to upload or download internal EGPWC data. Software and database is uploaded

and history data is downloaded. The USB interface utilizes a standard Type A connector and is compatible with USB 2.0 and 3.0 mass storage devices. The USB interface supports only FAT32 formatted storage devices (default format provided from manufacturers), with a recommended storage capacity of 2 GB or greater. The EGPWC is not compatible with USB hubs or encrypted drives. LED above the USB connector gives indication of USB data load activity.

Test connector: There is one D–sub high–density 26–position connector on the front panel of EGPWC. The test connector can be connected to a portable PC to receive and control internal data. The front panel test connector is reserved for Honeywell use.

GPWS FLAP OVERRIDE Annunciator Switch:

[Refer to Figure 54.](#)

The GPWS FLAP OVERRIDE annunciator switch is attached to pilot's left side console. DZUS fasteners and a related bonding wire directly connect to the console to make ground continuity between the side panel and console.

The GPWS FLAP OVERRIDE switch is a guarded annunciator switch. A transparent switch cover prevents accidental selection of the switch.

The GPWS FLAP OVERRIDE switch is pushed to change the indication modes. A black and yellow crosshatch indication in the switch comes on to show the manual selection to the GPWS FLAP OVERRIDE feature.



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PULL UP and BELOW G/S Annunciator Switches:

Refer to Figure 55.

The PULL UP and BELOW G/S annunciator switches are vertically connected as a one assembly. Two PULL UP and BELOW G/S annunciator switches are attached to the glareshield panel assembly.

The PULL UP and BELOW G/S annunciator switches give the operation that follow:

- PULL UP indication
- GPWS TEST selection
- BELOW G/S indication
- BELOW G/S aural indication cancellation.

INHIBIT Annunciator Switch:

Refer to Figure 56.

Two INHIBIT annunciator switch are attached to the glareshield panel assembly. Each switchlight is located in the area of view of each pilot.

The INHIBIT annunciator switches come on when one TERRAIN INHIBIT annunciator switch or the other is pushed to prevent terrain indication on the Multi-Functional Displays (MFD1, MFD2) navigation pages.

Controls and Indications

Refer to Figure 57.

The FLAP OVERRIDE annunciator switch located on the pilot's side console is pushed to prevent aural indications when a landing is made with zero or partial flaps extended.

A black and yellow crosshatch indication in the switch comes on to show the manual selection of the GPWS FLAP OVERRIDE feature.

The switch is pushed again to manually cancel the GPWS FLAP OVERRIDE feature.

When the Above Ground Level (AGL) is less than 50 feet, the override feature automatically cancels.

Refer to Figure 58.

A red PULL UP annunciator switch located on the glareshield panel assembly comes on when the Enhanced Ground Proximity Warning System (EGPWS) calculates any of the warnings that follow:

PULL UP ANNUNCIATOR	VOICE
Mode 1, Excessive descent rate	"SINK RATE, SINK RATE", "PULL UP"
Mode 2, Excessive closure rate to terrain	"TERRAIN, TERRAIN", "PULL UP"
Mode 3, Descent after takeoff	"DON'T SINK, DON'T SINK"
Mode 4, Insufficient terrain clearance	"TOO LOW TERRAIN", "TOO LOW GEAR", "TOO LOW FLAPS"



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PULL UP ANNUNCIATOR	VOICE
Terrain Clearance Floor (TCF) alerting	“TOO LOW TERRAIN”
Terrain alerting	“CAUTION TERRAIN”, “CAUTION OBSTACLE”, “WARNING TERRAIN”, PULL UP”, “WARNING OBSTACLE, PULL UP”

The PULL UP lights stay on while the aircraft is in the warning area.

An amber PULL UP annunciator switch located on the glareshield panel assembly comes on when the Enhanced Ground Proximity Warning System (EGPWS) calculates the condition that follows:

BELOW G/S ANNUNCIATOR	VOICE
Excessive descent below the Glideslope (GS)	Soft “GLIDESLOPE”, Hard “GLIDESLOPE”

One BELOW G/S annunciator switch or the other is pushed for 1 second or longer to cancel the voice indication. The amber light stays on to show that the mode was manually cancelled.

The mode automatically stops when any of these conditions are set:

- The Above Ground Level (AGL) is less than 50 feet
- The Above Ground Level (AGL) is more than 1900 feet
- The ILS frequency is de-selected.

One TERRAIN INHIBIT annunciator switch or the other is pushed to prevent MFD terrain indications. The INHIBIT indication in the

annunciator switch comes on and the two MFDs show a TERRAIN INHIBIT message.

[Refer to Figures 59 and 60.](#)

The GPWS LANDING FLAP selector switch on the hydraulic control panel is set to one of three landing flap positions for the insufficient terrain clearance mode as follows:

- 10 degrees
- 15 degrees
- 35 degrees.

A related indication on the hydraulic control panel comes on to show the landing flap position selection.

The aircraft flaps select lever is set at the same GPWS LANDING FLAP selector switch selection.

[Refer to Figure 61.](#)

The EFIS Control Panel (EFCP) has pushbutton and rotary switches and knobs to control the WXR and EGPWS systems indications that follow:

- WX/TERR pushbutton switch
- FORMAT pushbutton switch
- RANGE selector knob
- WX/TERR BRT knob.

WX/TERR Pushbutton Switch: A short WX/TERR pushbutton switch selection changes the MFD navigation page weather radar/EGPWS terrain automatic pop-up indication to a continuous EGPWS terrain indication. One more short WX/TERR pushbutton switch selection



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changes the MFD navigation page indication back to a weather radar/EGPWS terrain automatic pop-up indication.

A long WX/TERR pushbutton switch selection causes the weather radar and EGPWS terrain indications to go out of view. One more long WX/TERR pushbutton switch selection causes the weather radar and EGPWS terrain indications to come into view again.

If the navigation page is set to show a weather radar indication (arc or map format) and an EGPWS terrain alert is sensed, the terrain indication automatically comes into view. If the navigation page is set to the plan format, an EGPWS terrain alert causes it to change the arc or map format. An EGPWS terrain alert always causes the range indication to change to 10 NM. An AUTO RANGE message below the outer range digital mark shows that the range indication does not relate to the range selection on the EFCP.

NOTE

An EGPWS terrain alert condition prevents the EFCP switch selections that follow:

- WX/TERR pushbutton
- RANGE rotary switch
- FORMAT pushbutton
- DATA pushbutton.

After the EGPWS terrain alert condition, the 10 nmi range indications stays in view. The WX/TERR, TCAS, FORMAT, DATA, or RANGE switch is set to reset the indication.

If the EGPWS terrain indication is set while the aircraft is on the ground for more than 30 seconds and the WX radar is transmitting (WX ON, WX ALERT or WX GMAP), the indication changes to WX.

The MFD defaults to the WX indication at power-up. If the EFCP malfunctions, the last selection stays set unless the MFD is set to a different indication or to off.

NOTE

A WX/TERR pushbutton switch selection has no effect when the navigation page is not in view.

WX/TER BRT knob: The WXBRT knob is turned to adjust the weather radar image brightness.

RANGE selector knob: The RANGE selector knob is turned to change the weather radar and EGPWS range indications on the navigation page. The WXR and EGPWS system selectable ranges are 10, 20, 40, 80, 160 or 240 nmi.

FORMAT Pushbutton Switch: A short FORMAT pushbutton switch selection changes the MFD navigation page indication from an FMS map format to a VOR/ILS arc format. One more short FORMAT pushbutton switch selection changes the MFD navigation page to show an MLS arc format, if the MLS is installed. One more short FORMAT pushbutton switch selection changes the MFD navigation page indication back to an FMS map format. A long FORMAT pushbutton switch selection changes the MFD navigation page indication from an FMS map or VOR/ILS arc format to a plan format.

NOTE

The MFD always shows the related navigation source except when only one source is installed in the aircraft.

DATA pushbutton: Alternate action on this push-button enables the optional FMS Data selection on MFD NAV page. A DATA pushbutton selection displays on the NAV page the 10 nearest navigation aids as derived from the FMS data base. One more DATA pushbutton selection changes the NAV page to show a presentation of optional navigation aids plus the 10 nearest airports as derived from the FMS



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data base. One more DATA pushbutton selection removes all options (default selection).

A Terrain Awareness Display (TAD) gives an indication of the surrounding terrain in different density of green, yellow and red dot patterns. The indication is shown on the navigation display page from the aircraft altitude relative to the terrain data in the EGPWS computer.

Refer to Figure 62.

The MFD navigation pages (ARC or MAP format) give the terrain indications that follow:

- Image
- Display mask
- Auto range message
- Mode messages
- Messages
- Peaks annunciation.

Terrain Image: The EGPWS supplies the terrain indication data to the MFD navigation pages that follows:

COLOR	TERRAIN/OBSTACLE AREA INDICATION
Solid Red	Terrain warning
Solid Yellow	Terrain caution
High density red dots	More than 2000 ft. above aircraft

COLOR	TERRAIN/OBSTACLE AREA INDICATION
High density yellow dots	Between 1000 and 2000 ft. above aircraft
Medium density yellow dots	Between 500 below to 1000 ft. above the aircraft
Medium density yellow dots	Landing gear down – Between 250 below to 1000 ft. above the aircraft
Solid Green	Terrain/obstacle that is the highest elevation band within 500 below the aircraft. It is shown only when no red or yellow terrain are in the range of the indication.
Solid Green	Landing gear down – Terrain/obstacle that is the highest elevation band within 250 below the aircraft. It is shown only when no red or yellow terrain are in the range of the indication.
High density green dots	Between 500 and 1000 ft. below the aircraft
High density green dots	Landing gear down – Between 250 and 1000 ft. below the aircraft
High density green dots	Terrain/obstacle that is the middle elevation band when no red or yellow terrain are in the range of the indication.
light density green dots	Between 1000 and 2000 ft. below aircraft



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COLOR	TERRAIN/OBSTACLE AREA INDICATION
light density green dots	Terrain/obstacle that is the lower elevation band when no red or yellow terrain are in the range of the indication.
Black	No terrain/obstacles
Cyan	Water at sea level elevation (0 feet MSL)
Magenta	Unknown terrain. No terrain data in the database for the magenta area shown

Terrain Display Mask: When the terrain display is set or during a terrain alert, two white dashed radials come into view to show the lateral limits of the terrain display mask. The heading arc scale is the top limit of the indication.

Auto Range Message: A terrain alert condition will cause the 10 nmi range and terrain images to come into view with a white AUTO RANGE message below the outer range digital mark.

Terrain Mode Messages: When the terrain display is set or during a terrain alert, the upper left part of MFD navigation pages show the terrain modes messages that follow:

- TERR ON
- TERR TEST.

TERR ON Terrain Mode Message: A white TERR ON terrain mode message is shown while the terrain image is in view.

TERR TEST Terrain Mode Message: A white and yellow TERR TEST terrain mode message is shown while the test indication is in view.

Terrain Messages: The EGPWS has terrain related messages that are shown in the upper part of the navigation display when the terrain mode is set that follow:

- TERRAIN INHIBIT
- TERR UNAVAILABLE.

TERRAIN INHIBIT Terrain Message: A white TERRAIN INHIBIT terrain message shows the TERRAIN INHIBIT annunciator switch selection.

TERR UNAVAILABLE Terrain Message: A white TERR UNAVAILABLE terrain message is shown when the EGPWS is not installed.

Peaks Annunciation: When the terrain image is shown, two three-number indications in the upper right part of the terrain mask show the altitude of the highest and lowest points of the terrain image. The peaks annunciation color is same as its related indication. The peaks annunciation goes out of view if the EGPWC malfunctions.

[Refer to Figure 63.](#)

If the EGPWS malfunctions, the MFD navigation page (arc or plan format) will show a yellow TERR FAIL message. The terrain mode message (TERR ON or TERR TEST) stays in view but the terrain images do not.

A TERR FAIL message is shown during a condition that follow:

- EGPWS senses an internal malfunction in the terrain circuitry or unreliable external sensor data
- IOP senses a malfunction from the EGPWS outputs



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- Terrain data bus has malfunctioned while the terrain display is not set off
- Range received from the terrain data bus output is not the same as the calculation made by the MFD while not in terrain alert mode
- MFD senses a malfunction of the data bus to the related IOP
- Terrain image is not supplied by the EGPWS to the MFD
- FMS is not energized.

Refer to Figure 64.

The GPWS caution light comes for the malfunction conditions that follow:

- EGPWS computer
- IOM1, IOP1 (FDPS1) and IOM2, IOP2 (FDPS2).

Interfaces

The left main bus supplies the Ground Proximity Warning Computer (GPWC) with 28 Vdc electrical power through a 5 Ampere circuit breaker. The circuit breaker is located on the 28 Vdc avionics circuit breaker panel at position A1. The Ground Proximity Warning Computer (GPWC) uses less than 15 Watts in a non alert or warning mode, and less than 22 Watts with an alert or warning. The unit is equipped with a heater blanket which turns on at –20°C or below and turns off at –10°C or above. Typical power consumption with the heater blanket on is 48 Watts.

The left main bus supplies 28 Vdc electrical power through a 1 Ampere circuit breaker to the lights in the annunciator switches that follow:

- PULL UP
- BELOW G/S
- FLAP OVERRIDE
- INHIBIT.

The circuit breaker is located on the 28 Vdc avionics circuit breakerpanel in position B1.

The Advisory Control Unit (ACU) supplies discrete inputs to make the lights come on.

The Enhance Ground Proximity Warning Computer (EGPWC) receives discrete inputs from the sources that follow:

DATA	SOURCE
Landing Flaps Switch 10, 15, 35 Degree Position	GPWS LANDING FLAPS SELECTOR SWITCH
Audio Suppression	SPM1, SPM2
EGPWC Flap Override	GPWS FLAP OVERRIDE Annunciator Switch
Self Test	The PULL UP, GPWS TEST Annunciator Switch
Glide Slope Cancel	BELOW G/S Annunciator Switch



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DATA	SOURCE
Terrain Inhibit	INHIBIT Annunciator Switch
Mode 6 Callout Low Volume	This discrete is not enabled in the DHC-8 Series 400
Steep Approach Enable	Rear connector top plug pin TP4C (Steep Approach Enabled) is permanently grounded.
Steep Approach Select	Steep Approach Switch
Mode 6 Altitude Callouts Enable	Pin 8C (Altitude Callouts Enable) of P1B-A1 is permanently grounded.

The Enhance Ground Proximity Warning Computer (EGPWC) receives digital inputs through the related IOP from sources that follow:

DATA	SOURCE
GPS Altitude	FMS
GPS Track Angle	FMS
GPS Latitude	FMS
GPS Longitude	FMS
GPS Ground Speed	FMS

DATA	SOURCE
GPS Vertical Speed	FMS
GPS Status	FMS
GPS Sensor Status	FMS
GPS Horizontal Integrity Limit	FMS
GPS Vertical Figure of Merit	FMS
GPS Horizontal Figure of Merit	FMS
FMS Destination Latitude	FMS
FMS Destination Longitude	FMS
FMS Track Angle	FMS
FMS Latitude	FMS
FMS Longitude	FMS
FMS Ground Speed	FMS
FMS True Heading	FMS
FMS Discrete Word	FMS
AGL Altitude	RA



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DATA	SOURCE
Decision Height	ICP1, ICP2
Localiser Deviation	VHF NAV1, VHF NAV2
Glideslope Deviation	VHF NAV1, VHF NAV2
MLS Azimuth Deviation	MLS1, MLS2
MLS Elevation Deviation	MLS1, MLS2
Standard Altitude	ADC1, ADC2
Baro Corrected Altitude	ADC1, ADC2
Calibrated Airspeed	ADC1, ADC2
Altitude Rate	ADC1, ADC2
Static Air Temperature	ADC1, ADC2
Navigation Source	FGCP
Backcourse Mode	FGCP
Magnetic Heading	AHRS1, AHRS2
Pitch Attitude	AHRS1, AHRS2
Roll Attitude	AHRS1, AHRS2

DATA	SOURCE
Flap Lever Position	FCU
Range Rings	EFCP1, EFCP2
Landing Gear Position	PSEU
Terrain Display Request	INHIBIT annunciator switch

The Enhance Ground Proximity Warning Computer (EGPWC) receives digital inputs from the FMS as follows:

DATA
FMS Track Angle
FMS Latitude
FMS Longitude
FMS Ground Speed
FMS True Heading
FMS Discrete Word

The Enhance Ground Proximity Warning Computer (EGPWC) receives digital inputs from the GPS as follows:



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DATA
GPS Altitude
GPS Track Angle
GPS Latitude
GPS Longitude
GPS Ground Speed
GPS Vertical Speed
GPS Sensor Status
GPS Hor Integ Limit
GPS Vert Figure of Merit
GPS Horizontal Figure of Merit
Latitude fine
Longitude fine
Horizontal Dilution of Precision (HDOP)
Vertical Dilution of Precision (VDOP)

The Enhance Ground Proximity Warning Computer (EGPWC) supplies the discrete outputs that follow:

DATA	TERMINATION
EGPWC Audio On	IOM1, IOM2
Glideslope Alert light	BELOW G/S Annunciator Switch
EGPWS Alert/Warning Light	The PULL UP, GPWS TEST Annunciator Switch
EGPWC System Status	Caution and Warning Panel
Pilot, Copilot Terrain Select	MFD1 Display relay, MFD2 Display relay

NOTE

The FLAP OVERRIDE annunciator switch output is supplied through the Advisory Control Unit (ACU) back to the annunciator switch to make the cross hatching lights come on. The INHIBIT annunciator switch output is supplied through the ACU back to the annunciator switch to make the INHIBIT lights come on.

The Enhance Ground Proximity Warning Computer (EGPWC) supplies the digital outputs that follow:



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DATA	TERMINATION
TERRAIN FAIL Message	Through FDPS1, FDPS2 MFDs
TERRAIN NOT AVAILABLE Message	Through FDPS1, FDPS2 to MFDs
Terrain Caution	Through FDPS1, FDPS2 to FDR
Terrain warning	Through FDPS1, FDPS2 to FDR
Pilot Terrain Pop up	Through FDPS1, FDPS2 to FDR
Copilot Terrain Pop up	Through FDPS1, FDPS2 to FDR
TERRAIN INHIBIT Message	Through FDPS1, FDPS2 to MFD1 MFD2
Peaks Values	MFD1, MFD2
Terrain Images (ARINC 453)	Through MFD Display relays to related MFD

Tests

[Refer to Figure 65.](#)

The EGPWC has a manual self-test sequence to show the status of the system. The self-test sounds the EGPWS status, makes the lights in the annunciator switches come on, and shows an MFD terrain test indication.

The EGPWS has six different levels of self-tests identify conditions that follow:

- Level 1 – Status of each of the primary functions of the system
- Level 2 – All current internal and external malfunctions in the system
- Level 3 – Configuration status of the EGPWC and the installation
- Level 4 – Faults sensed during last flight
- Level 5 – Alerts sensed during last flight
- Level 6 – Changes of the discrete inputs.

Before the EGPWS test is started, check the conditions that follow:

- Aircraft is on the ground
- On the EGPWC front panel, check that the green COMPUTER OK status indicator is on
- GPWS Landing Flap selector switch is set to the 35 degrees position
- FMS is initialized
- Course pointer is set less than 105 degrees of the aircraft heading
- The two TERRAIN INHIBIT annunciator switches are not set
- The two MFD are set to show navigation pages
- Terrain indication is shown.



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Level 1 Self Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed and held for more than three seconds to start a level 1 self test sequence. The EGPWS level 1 self test gives the indications that follow:

- GPWS caution light comes on
- The two BELOW G/S lights come on
- “GLIDESLOPE” voice is heard
- The two BELOW G/S lights go off
- The two PULL UP lights come on.
- “PULL UP” voice is heard
- The two PULL UP lights go off
- TERRAIN TEST mode continues for 12 seconds on the two MFDs
- “TERRAIN, TERRAIN, PULL UP” voice is heard.

On aircraft with ModSum 4–309203 or 4–410300 (without ModSum 4–429519 or SB84–34–102 or SB84–34–47 or ModSum 4–457043 or SB84–34–115 or SB84–34–36 or SB84–34–83 or ModSum 4–457062 or SB84–34–160) incorporated, if the PULL UP, GPWS TEST annunciator switch is pushed and held until the “GLIDESLOPE” voice is heard, the test continues to give the indications that follow:

- “GLIDESLOPE”
- “PULL UP”
- “TERRAIN, TERRAIN”
- “PULL UP”

- “SINK RATE”
- “PULL UP”
- “TERRAIN”
- “PULL UP”
- “DON’T SINK, DON’T SINK”
- “TOO LOW TERRAIN”
- “TOO LOW GEAR”
- “TOO LOW FLAPS”
- “TOO LOW TERRAIN”
- “GLIDESLOPE”
- “BANK ANGLE, BANK ANGLE”
- “MINIMUMS”
- “MINIMUMS”
- “TOO LOW TERRAIN”
- “CAUTION TERRAIN”
- “CAUTION TERRAIN”
- “TERRAIN”
- “TERRAIN”
- “PULL UP”
- “CAUTION OBSTACLE”
- “CAUTION OBSTACLE”
- “OBSTACLE”

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- “OBSTACLE”
- “PULL UP”.

On aircraft with ModSum 4–429519 or SB84–34–102 or SB84–34–47 incorporated, if the PULL UP, GPWS TEST annunciator switch is pushed and held until the “GLIDESLOPE” voice is heard, the test continues to give the indications that follow:

- “GLIDESLOPE”
- “PULL UP”
- “TERRAIN, TERRAIN”
- “PULL UP”
- “SINK RATE”
- “PULL UP”
- “TERRAIN”
- “PULL UP”
- “DON’T SINK, DON’T SINK”
- “TOO LOW TERRAIN”
- “TOO LOW GEAR”
- “TOO LOW FLAPS”
- “TOO LOW TERRAIN”
- “GLIDESLOPE”
- “BANK ANGLE, BANK ANGLE”
- “PLUS HUNDRED”
- “MINIMUMS”

- “2500”
- “1000”
- “50”
- “40”
- “30”
- “20”
- “10”
- “TOO LOW TERRAIN”
- “CAUTION TERRAIN”
- “CAUTION TERRAIN”
- “TERRAIN”
- “TERRAIN”
- “PULL UP”
- “CAUTION OBSTACLE”
- “CAUTION OBSTACLE”
- “OBSTACLE”
- “OBSTACLE”
- “PULL UP”.

On aircraft with ModSum 4–457043 or SB84–34–115 or SB84–34–36 or SB84–34–83 or SB84–34–111 or SB84–34–137 or SB84–34–160 incorporated, if the PULL UP, GPWS TEST



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annunciator switch is pushed and held until the “GLIDESLOPE” voice is heard, the test continues to give the indications that follow:

- “GLIDESLOPE”
- “PULL UP”
- “TERRAIN, TERRAIN”
- “PULL UP”
- “SINK RATE”
- “PULL UP”
- “TERRAIN”
- “PULL UP”
- “DON’T SINK, DON’T SINK”
- “TOO LOW TERRAIN”
- “TOO LOW GEAR”
- “TOO LOW FLAPS”
- “TOO LOW TERRAIN”
- “GLIDESLOPE”
- “BANK ANGLE, BANK ANGLE”
- “MINIMUMS”
- “2500”
- “500”
- “100”
- “50”

- “40”
- “30”
- “20”
- “10”
- “TOO LOW TERRAIN”
- “CAUTION TERRAIN”
- “CAUTION TERRAIN”
- “TERRAIN”
- “TERRAIN”
- “PULL UP”
- “CAUTION OBSTACLE”
- “CAUTION OBSTACLE”
- “OBSTACLE”
- “OBSTACLE”
- “PULL UP”.

On aircraft with ModSum 4–457062 incorporated, if the PULL UP, GPWS TEST annunciator switch is pushed and held until the “GLIDESLOPE” voice is heard, the test continues to give the indications that follow:

- “GLIDESLOPE”
- “PULL UP”
- “TERRAIN, TERRAIN”
- “PULL UP”



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- “SINK RATE”
- “PULL UP”
- “TERRAIN”
- “PULL UP”
- “DON’T SINK, DON’T SINK”
- “TOO LOW TERRAIN”
- “TOO LOW GEAR”
- “TOO LOW FLAPS”
- “TOO LOW TERRAIN”
- “GLIDESLOPE”
- “BANK ANGLE, BANK ANGLE”
- “PLUS HUNDRED”
- “MINIMUMS”
- “50”
- “40”
- “30”
- “20”
- “10”
- “TOO LOW TERRAIN”
- “CAUTION TERRAIN”
- “CAUTION TERRAIN”
- “TERRAIN”

- “TERRAIN”
- “PULL UP”
- “CAUTION OBSTACLE”
- “CAUTION OBSTACLE”
- “OBSTACLE”
- “OBSTACLE”
- “PULL UP”.

Level 2 Self Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed again less than 3 seconds after the completion of the level 1 self test to listen to currently sensed faults. The EGPWS level 2 self test gives the indications that follow:

- “Current Faults”
- “No Faults”.

Level 3 Self Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed again less than 3 seconds after the completion of the level 2 self test to listen to system configuration data. The EGPWS level 3 self test gives the indications that follow:

- Part Number
- Modification Status
- Serial Number
- Application Software Version
- Configuration Software Version
- Terrain Database Version
- Envelope Modulation Database Version



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- Boot Code Version
- Aircraft Type
- Audio Menu
- Altitude Call out Menu
- Configuration Option (831SO90180 only)
- Dual FMS (834SO00418–01 or 834SO90214 only)
- Dual RA (834SO00417 or 834SO90212 only)
- Dual MLS (834CH00398 only)
- Bank Angle
- Peaks
- GPS Altitude Reference (Aircraft with UNS FMS)
- Obstacle Awareness
- Terrain Display.

Level 4 Self Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed again less than 3 seconds after the completion of the level 3 self test to listen to faults that were sensed during the last 10 flight legs. The previous flight faults are not heard if the pilot PULL UP, GPWS TEST annunciator switch is pushed less than 3 seconds after the completion of the level 3 self test and held until the “PRESS TO CONTINUE” is heard to start a level 4 self test sequence.

Level 5 Self Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed again less than 3 seconds after the completion of the level 4 self test to listen to caution and warning alerts that were sensed during the last 10 flight legs. The previous flight alerts are not heard if the pilot PULL UP, GPWS TEST annunciator switch is

pushed less than 3 seconds after the completion of the level 4 self test and held until the “PRESS TO CONTINUE” is heard to start a level 5 self test sequence.

Level 6 Self Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed again less than 3 seconds after the completion of the level 5 self test to listen to changes in the discrete inputs. The EGPWS level 6 self test gives the voice indications that follow:

SELECTION	INDICATION
BELOW G/S pushed	“GLIDESLOPE CANCELLED”
BELOW G/S released	“GLIDESLOPE ENABLED”
GPWS LANDING FLAPS switch to 15	“FLAP SWITCH 35 SELECTED FALSE”, “FLAP SWITCH 15 SELECTED TRUE”
GPWS LANDING FLAPS switch to 10	“FLAP SWITCH 15 SELECTED FALSE”, “FLAP SWITCH 10 SELECTED TRUE”
FLAP OVERRIDE	“NOT LANDING FLAPS”
TERRAIN INHIBIT set	“TERRAIN OFF”
TERRAIN INHIBIT released	“TERRAIN ON”

A “DISCRETE INPUT TEST, PRESS TO CANCEL” sounds every 60 seconds during Level 6 Self Test.



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The pilot PULL UP, GPWS TEST annunciator switch is pushed again to stop the self test sequence.

NOTE

The copilot PULL UP, GPWS TEST annunciator switch selection causes a short level 1 self test only.

NOTE

Let thirty seconds pass before starting another self test.

Refer to Figure 66.

During a EGPWS self test the MFD navigation page shows TERR TEST mode message and a test indication.

On aircraft with SB84–34–123 or SB84–34–127 or ModSum 4–902388 or ModSum 4–902371 incorporated, the MFD's terrain display area shows the terrain database version number.

Refer to Figure 67.

On aircraft without the Modsum 4Q903303 incorporated, the EGPWC front panel LEDs gives the status indications that follow:

EXTERNAL FAULT-YELLOW	COMPUTER OK-GREEN	COMPUTER FAIL-RED	CORRECTIVE ACTION
Off	On	Off	Correct operation
On	On	Off	Do level 2 self test
Off	Off	Off	Make sure unit is energized

EXTERNAL FAULT-YELLOW	COMPUTER OK-GREEN	COMPUTER FAIL-RED	CORRECTIVE ACTION
Off	Off	On	Do level 2 self test
Off	On	On	Do level 2 self test
On	Off	Off	Do level 2 self test
On	Off	On	Do level 2 self test
On	On	On	Do level 2 self test

Refer to Figure 67.

On aircraft with the Modsum 4Q903303=YES, the EGPWC front panel LEDs gives the status indications that follow:

EXTERNAL FAULT	COMPUTER STATUS	CONDITION	CORRECTIVE ACTION
Off	Off	EGPWC Power off	Ensure that the EGPWC is installed in the rack correctly and power to the EGPWC is turned ON.
Off	On (GREEN)	Normal	None



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EXTERNAL FAULT	COMPUTER STATUS	CONDITION	CORRECTIVE ACTION
Off	On (RED)	EGPWC internal fault exists	Record the level 2 self test, if possible. Replace the EGPWC if required.
Off	Flashing GREEN	EGPWC is powering up	None
On (YELLOW)	Off	Invalid condition	Troubleshoot external faults using the level 2 self test. Do not replace the EGPWC.
On (YELLOW)	On (GREEN)	EGPWS external fault exists	Troubleshoot external faults using the level 2 self test. Do not replace the EGPWC.
On (YELLOW)	On (RED)	Both EGPWC internal and EGPWS external faults exist	Troubleshoot external faults using the level 2 self test. Replace the EGPWC, if required.

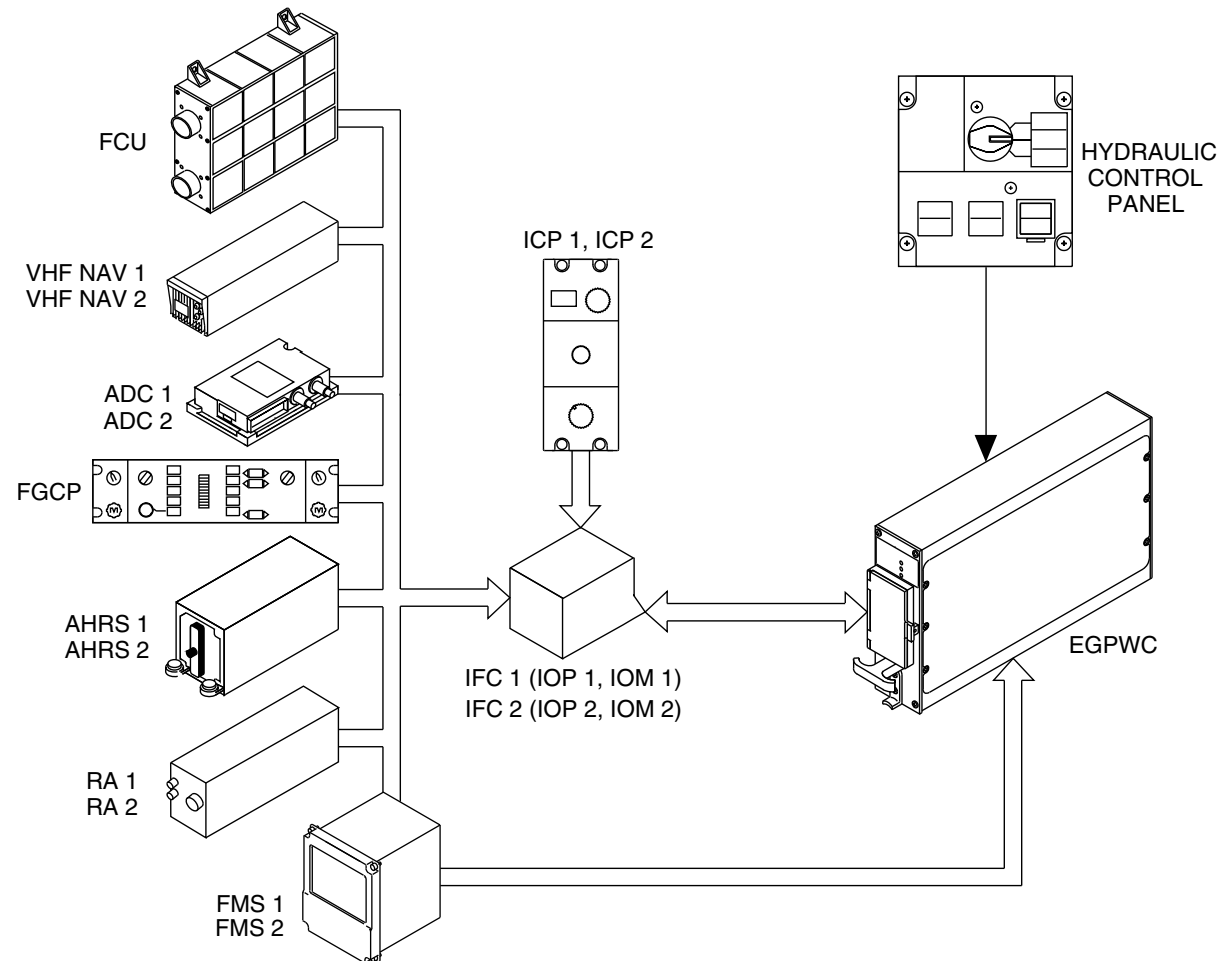
NOTE

The COMPUTER FAIL indication can come on during a self test.



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fs0a4a03.dg, gv, 16/12/02

EGPWS Block Diagram, Inputs
Figure 26

PSM 1-84-2A
EFFECTIVITY:
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[illegible]

EGPWS Block Diagram, Outputs
Figure 27

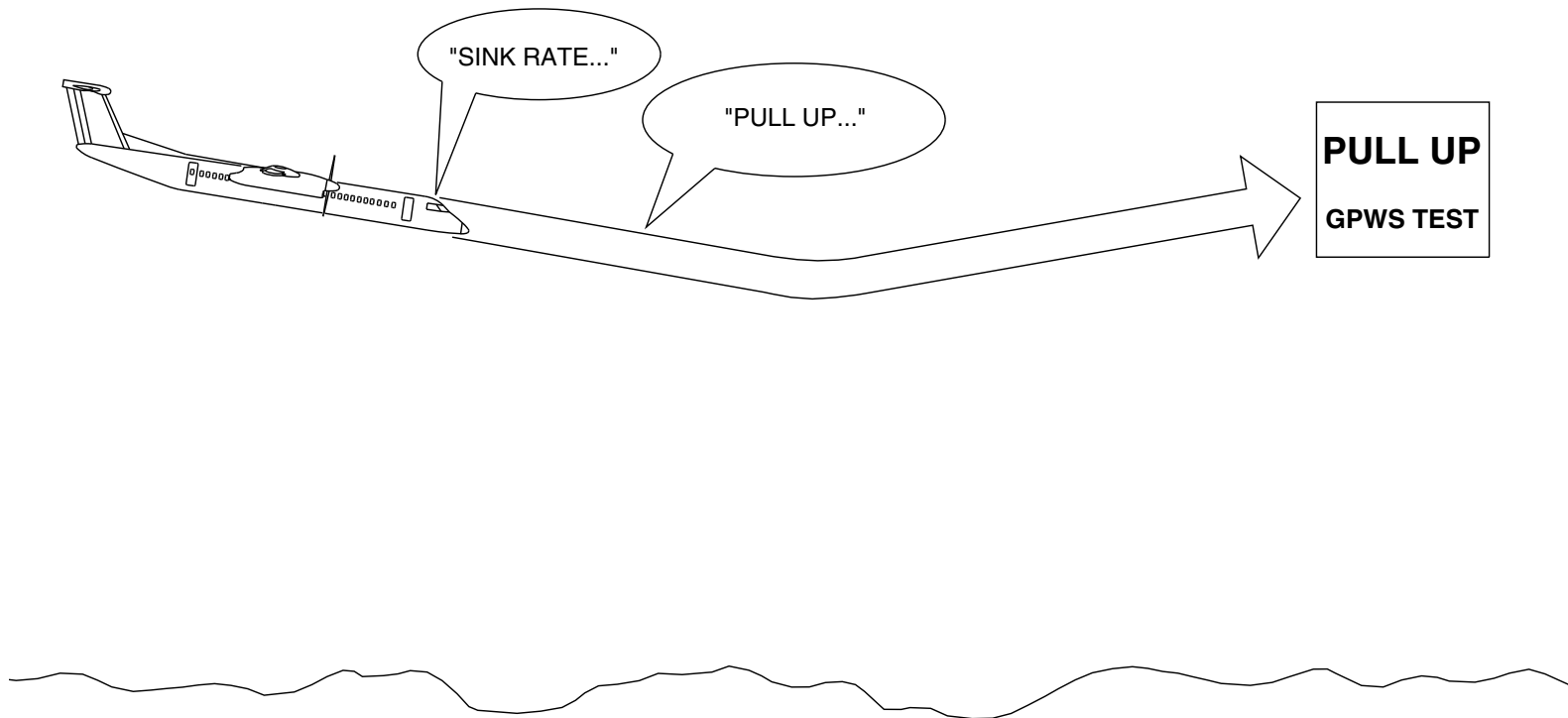
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fs802a01.dg, gv, 16/12/02

EGPWS Mode 1, Excessive Descent Rate Indications
Figure 28

PSM 1-84-2A
EFFECTIVITY:
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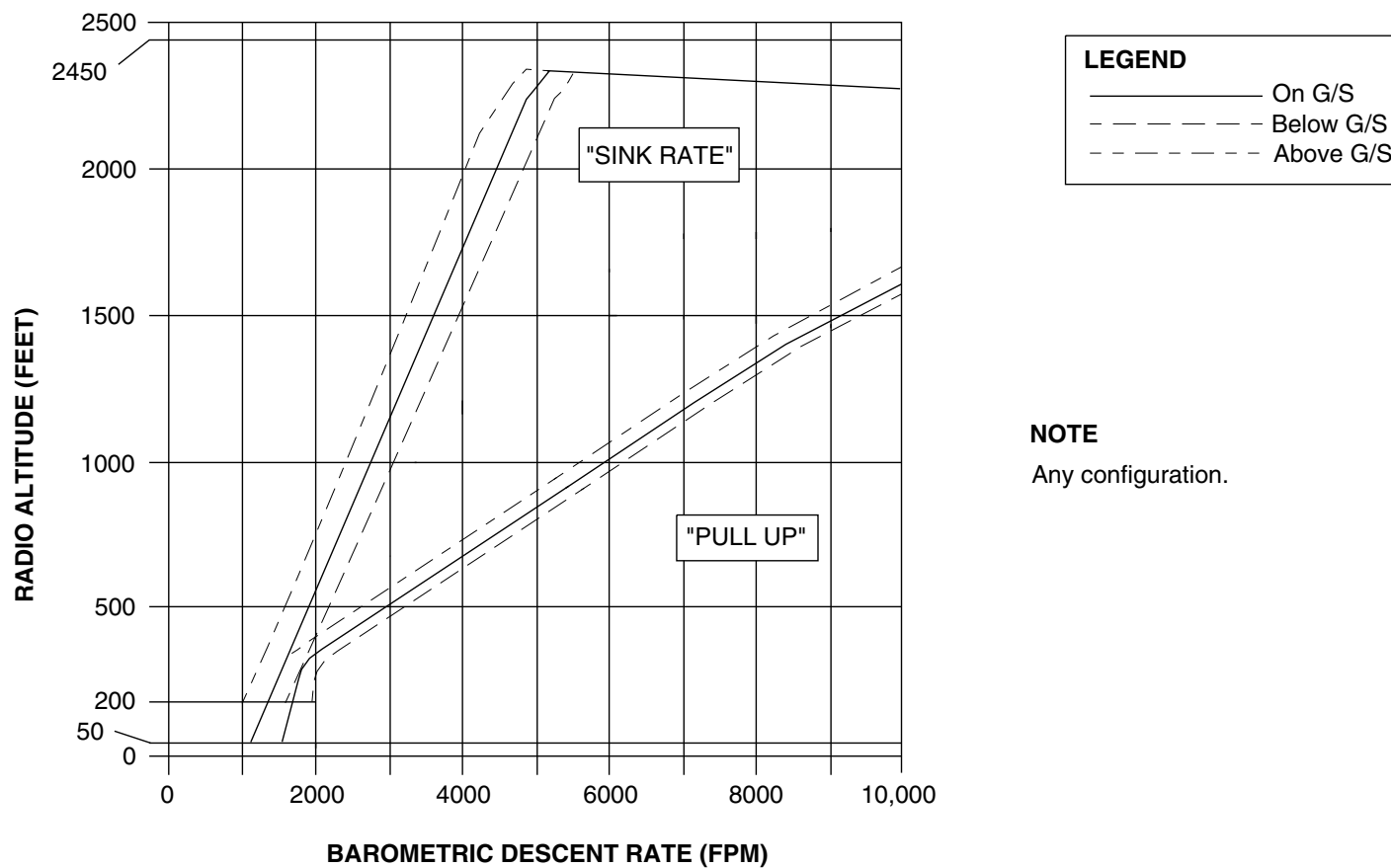
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fs801a01.dg, gv, 16/12/02

EGPWS Mode 1, Excessive Descent Rate
Figure 29

PSM 1-84-2A
EFFECTIVITY:
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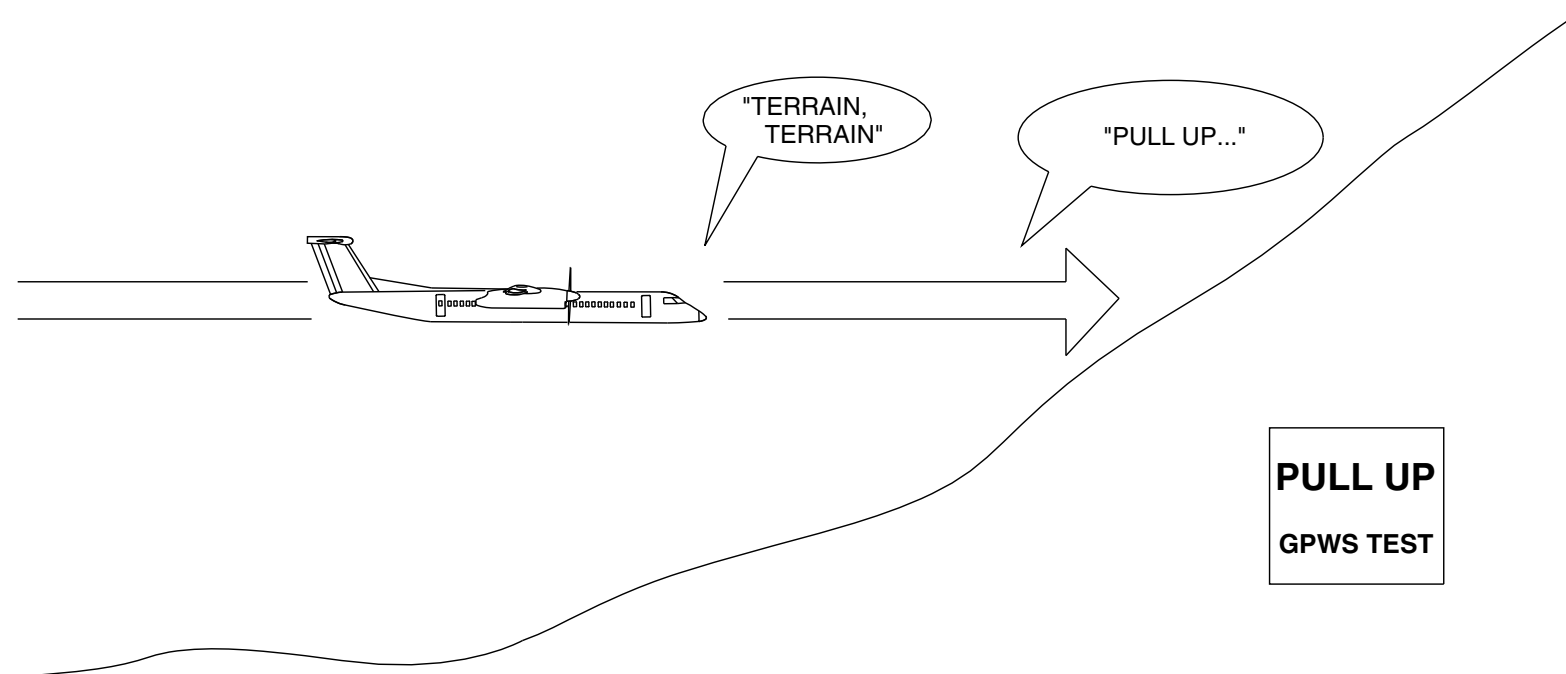
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fsy08a01.dg, gv, 16/12/02

EGPWS Mode 2, Excessive Closure Rate Indications
Figure 30

PSM 1-84-2A
EFFECTIVITY:
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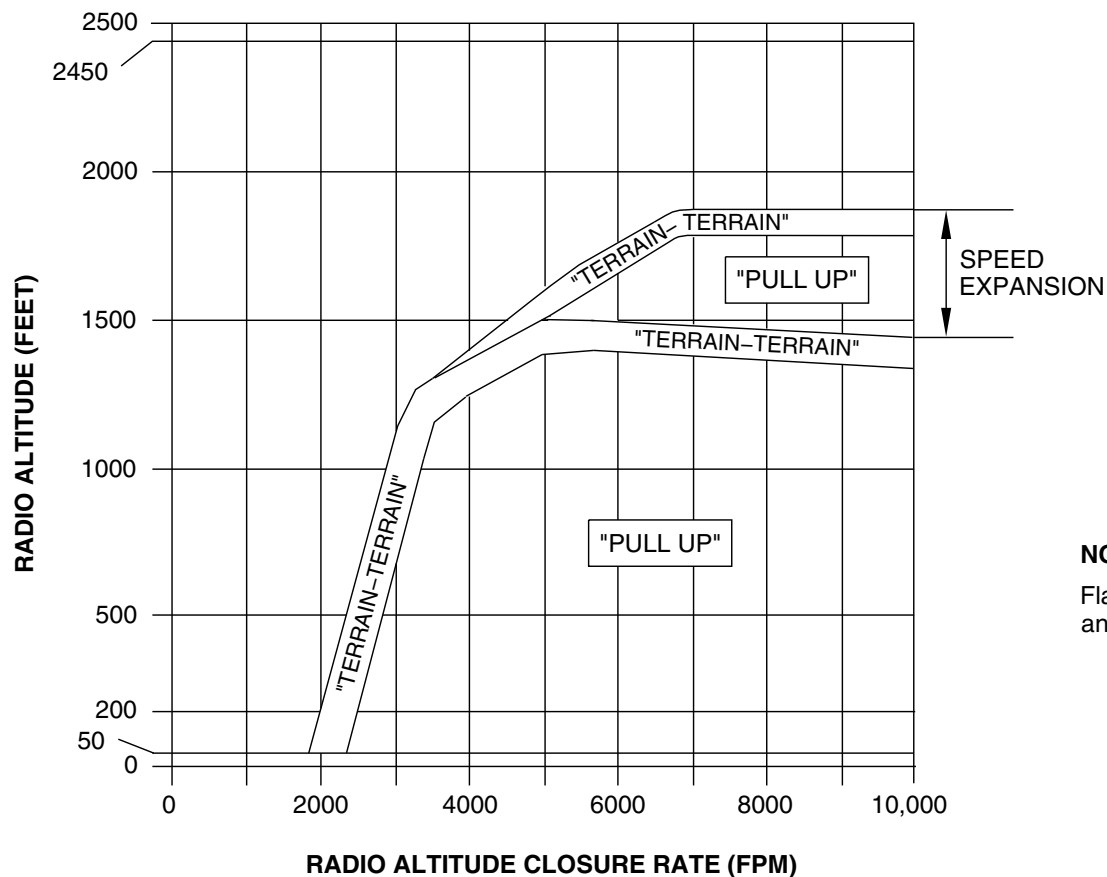
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NOTE

Flaps not in the landing range
and non ILS approach.

fs804a01.dg, gv, 16/12/02

EGPWS Mode 2a, Excessive Closure Rate
Figure 31

PSM 1-84-2A
EFFECTIVITY:
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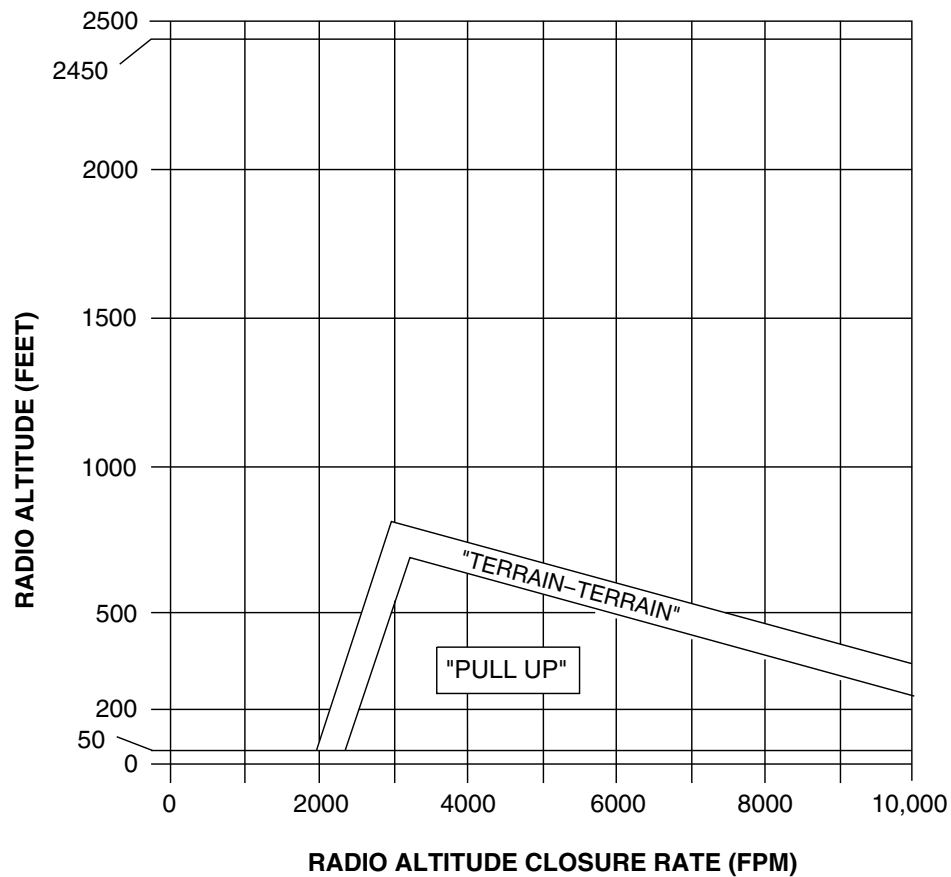
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NOTE

Flaps down or ILS approach.

fs817a01.dg, gv, 16/12/02

EGPWS Mode 2b, Excessive Closure Rate
Figure 32

PSM 1-84-2A
EFFECTIVITY:
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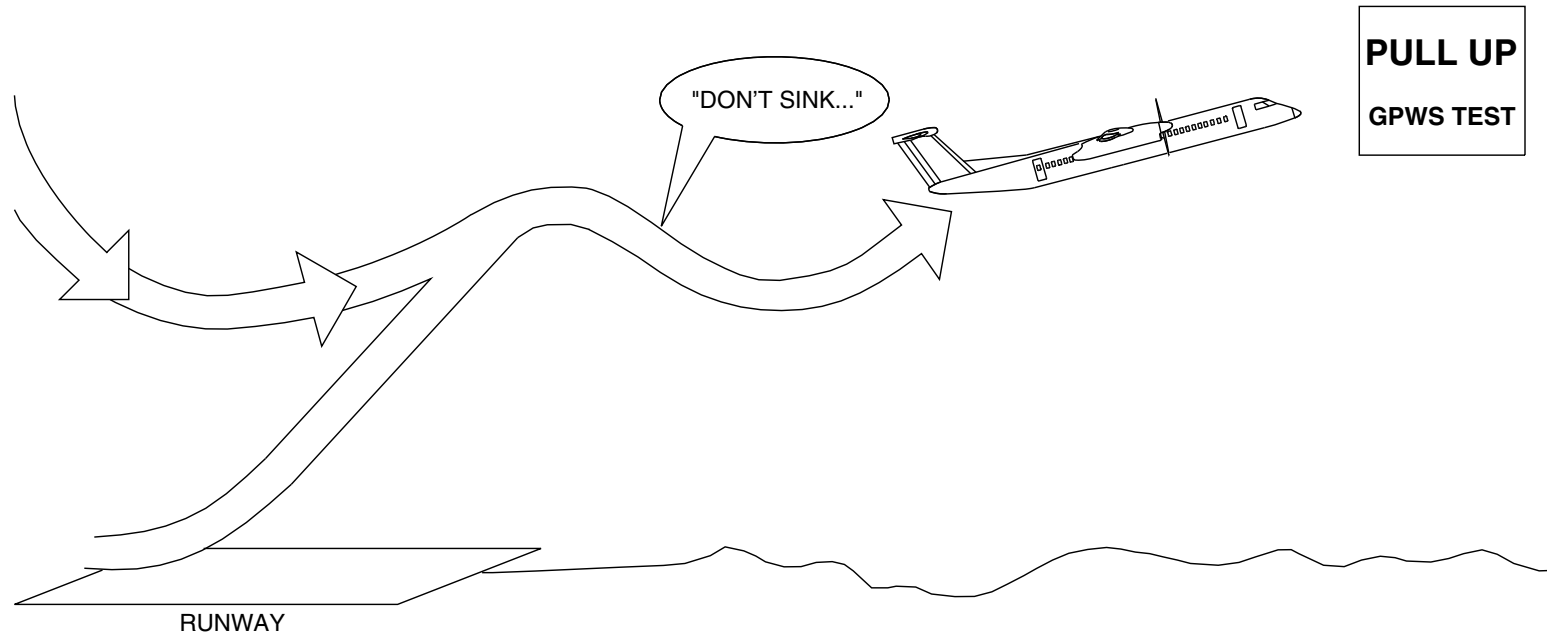
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fs818a01.dg, gv, 16/12/02

EGPWS Mode 3, Descent After Takeoff Indications
Figure 33

PSM 1-84-2A
EFFECTIVITY:
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Config 001

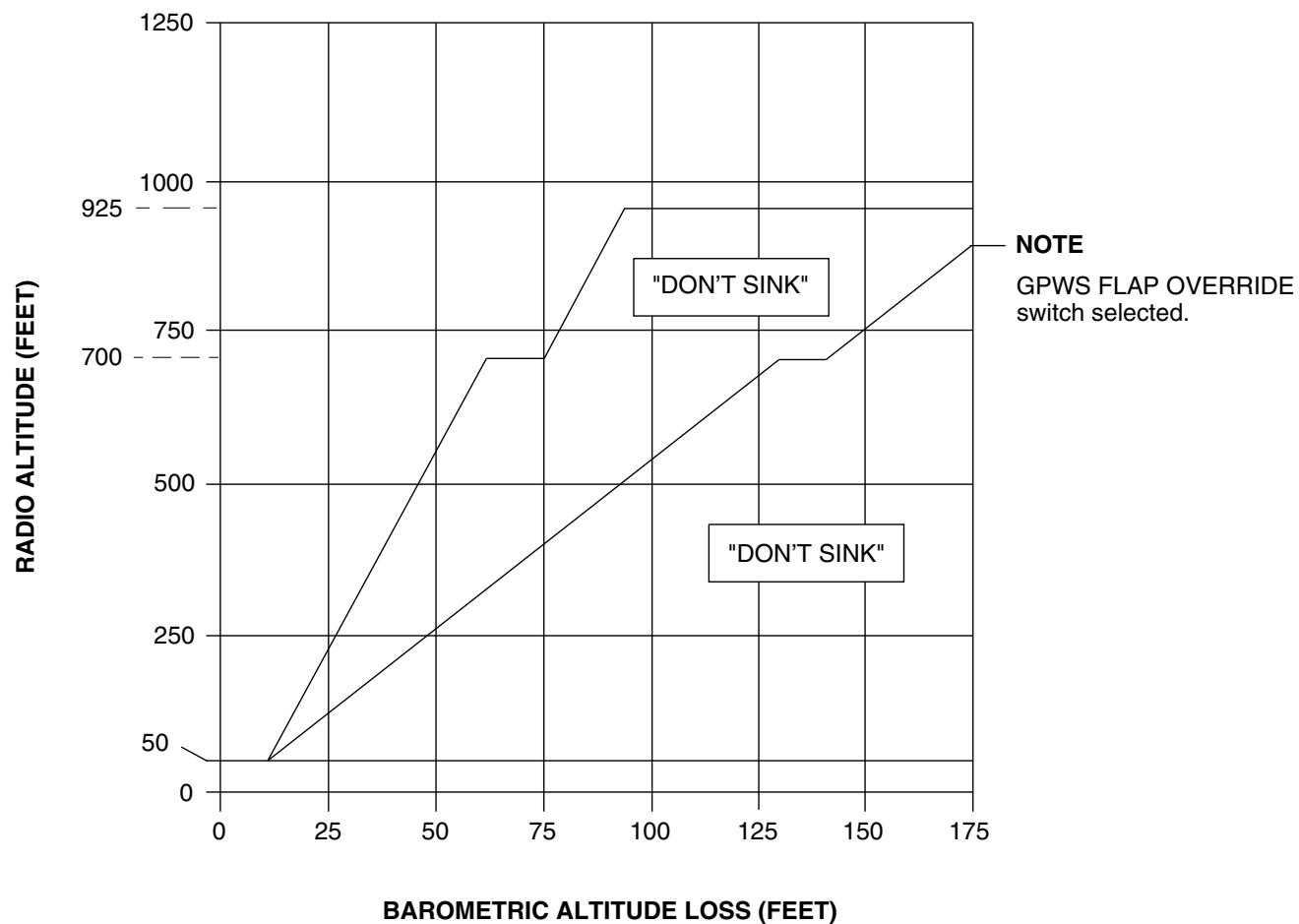
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fs819a01.dg, gv, 16/12/02

EGPWS Mode 3, Descent After Takeoff
Figure 34

PSM 1-84-2A
EFFECTIVITY:
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Config 001

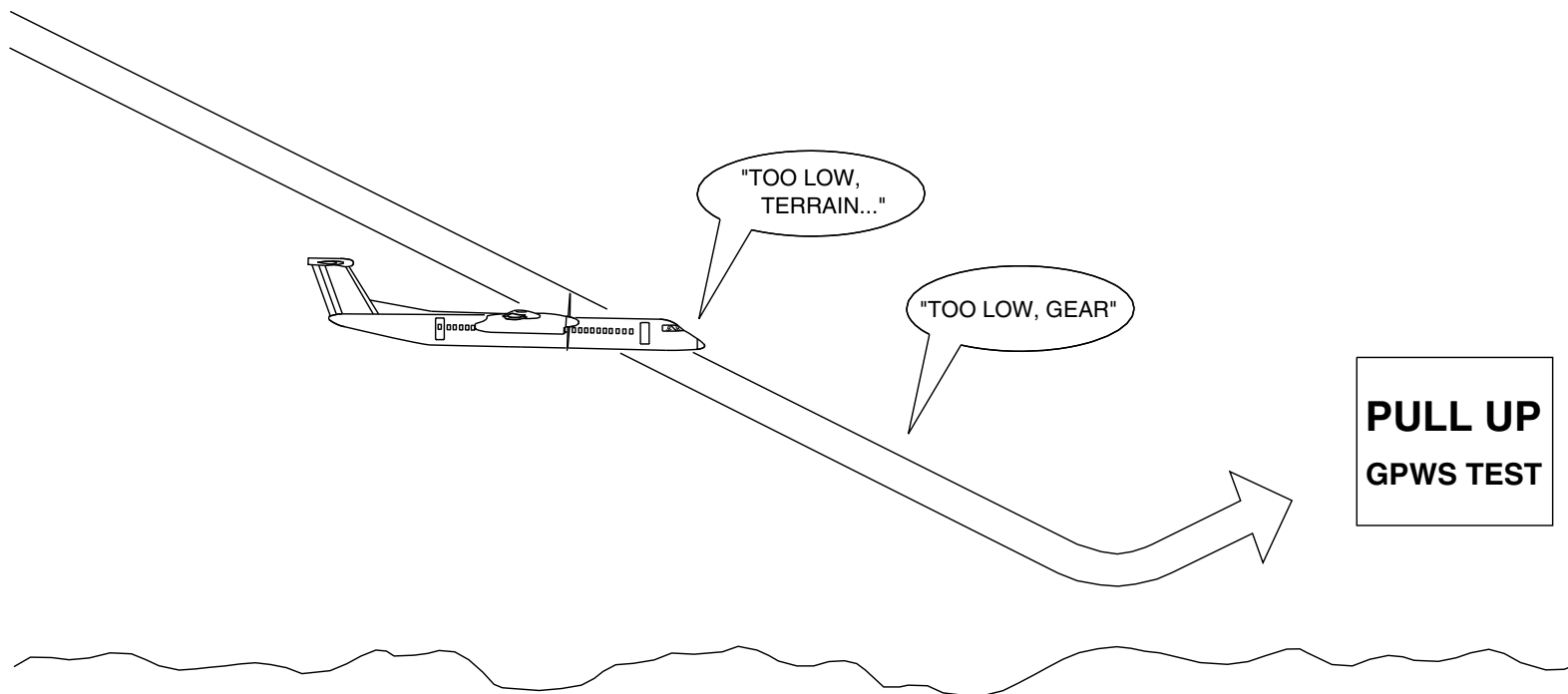
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fs820a01.dg, gv, 16/12/02

EGPWS Mode 4a, Insufficient Terrain Clearance Indications
Figure 35

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 39 of 34-42-00
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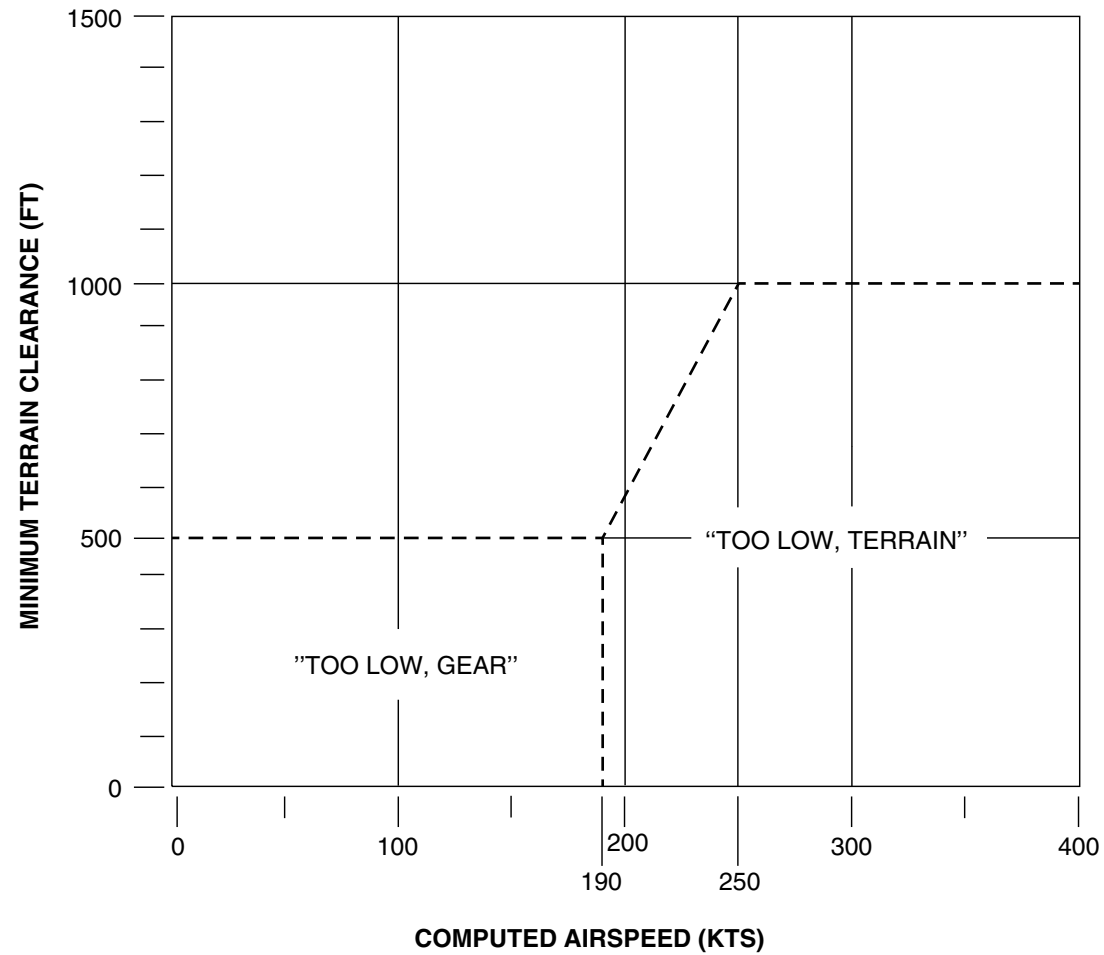
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NOTE

Gear up, Flaps up.

fs822a01.dg, nn/gw, mar29/2016

EGPWS Mode 4A, Insufficient Terrain Clearance
Figure 36

PSM 1-84-2A
EFFECTIVITY:
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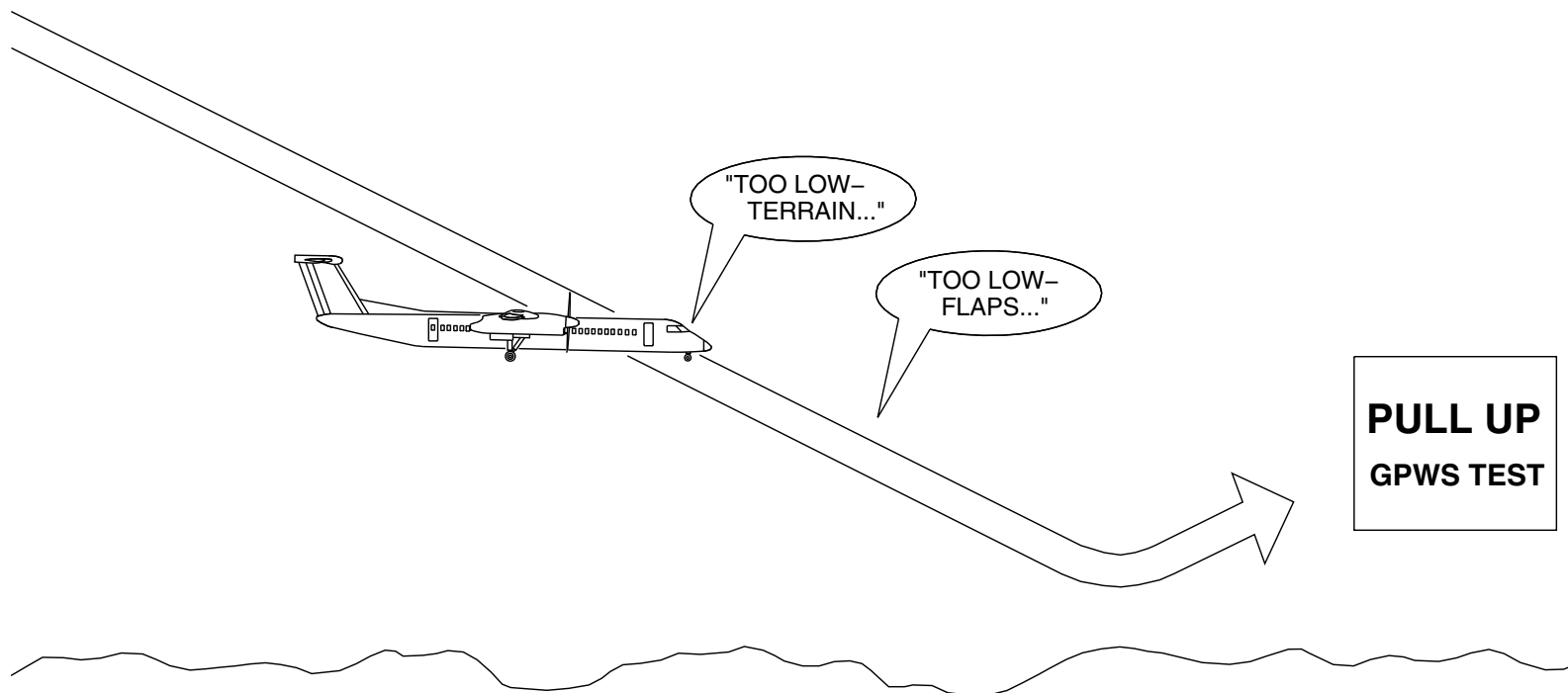
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fs823a01.dg, gv, 16/12/02

EGPWS Mode 4b, Insufficient Terrain Clearance Indications
Figure 37

PSM 1-84-2A
EFFECTIVITY:
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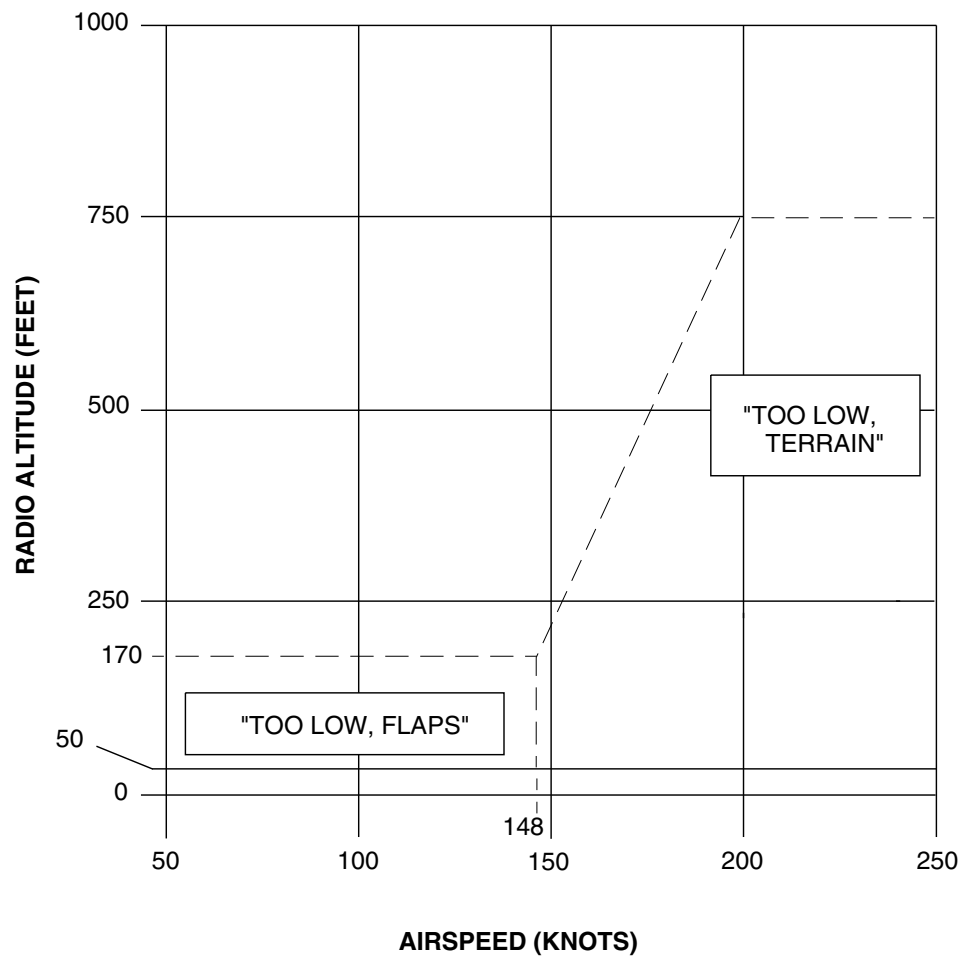
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NOTE

Gear down and flaps not in the landing range.

fs824a01.dg, gv, 16/12/02

EGPWS Mode 4b, Insufficient Terrain Clearance
Figure 38

PSM 1-84-2A
EFFECTIVITY:
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Config 001

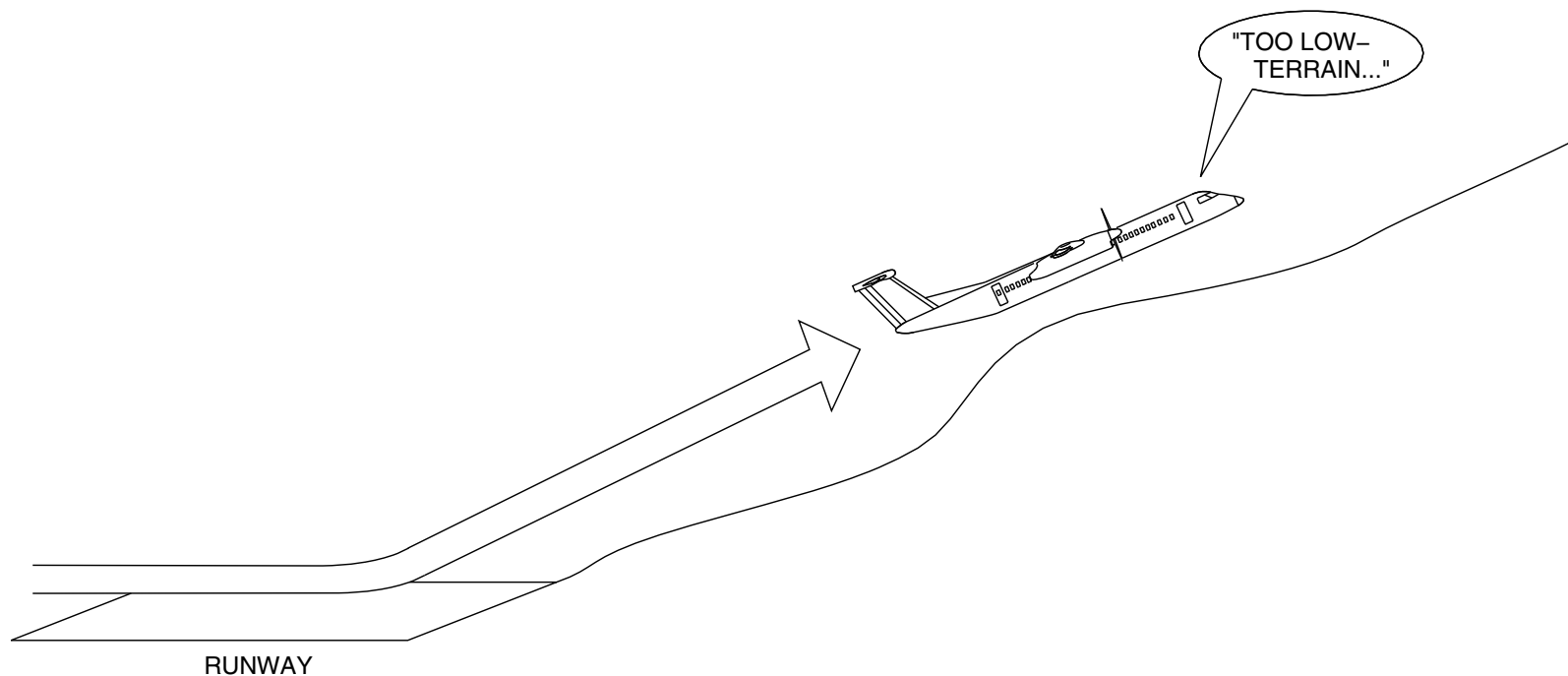
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fsy07a01.dg, gv, 16/12/02

EGPWS Mode 4c, Insufficient Terrain Clearance Indications
Figure 39

PSM 1-84-2A
EFFECTIVITY:
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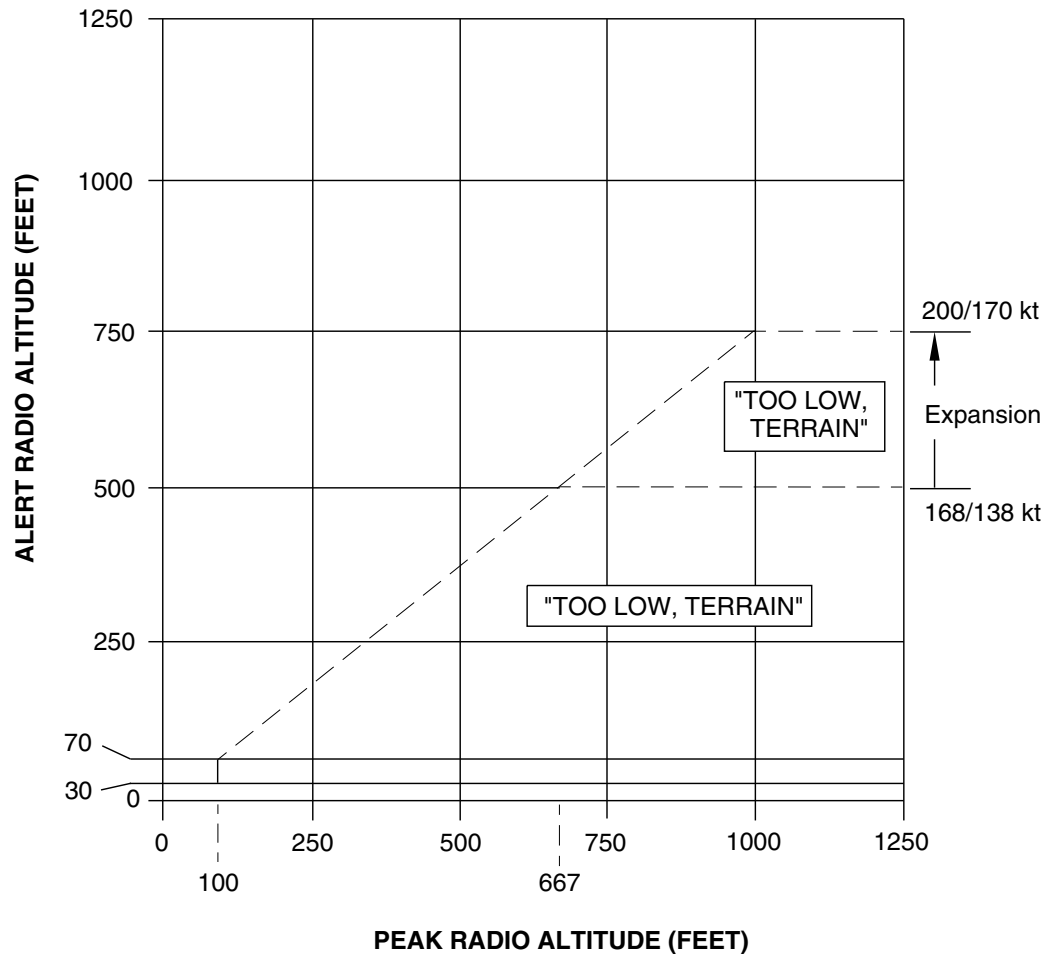
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NOTE

Takeoff with gear up or flaps not
in landing range.

fs825a01.dg, gv, 16/12/02

EGPWS Mode 4c, Insufficient Terrain Clearance
Figure 40

PSM 1-84-2A
EFFECTIVITY:
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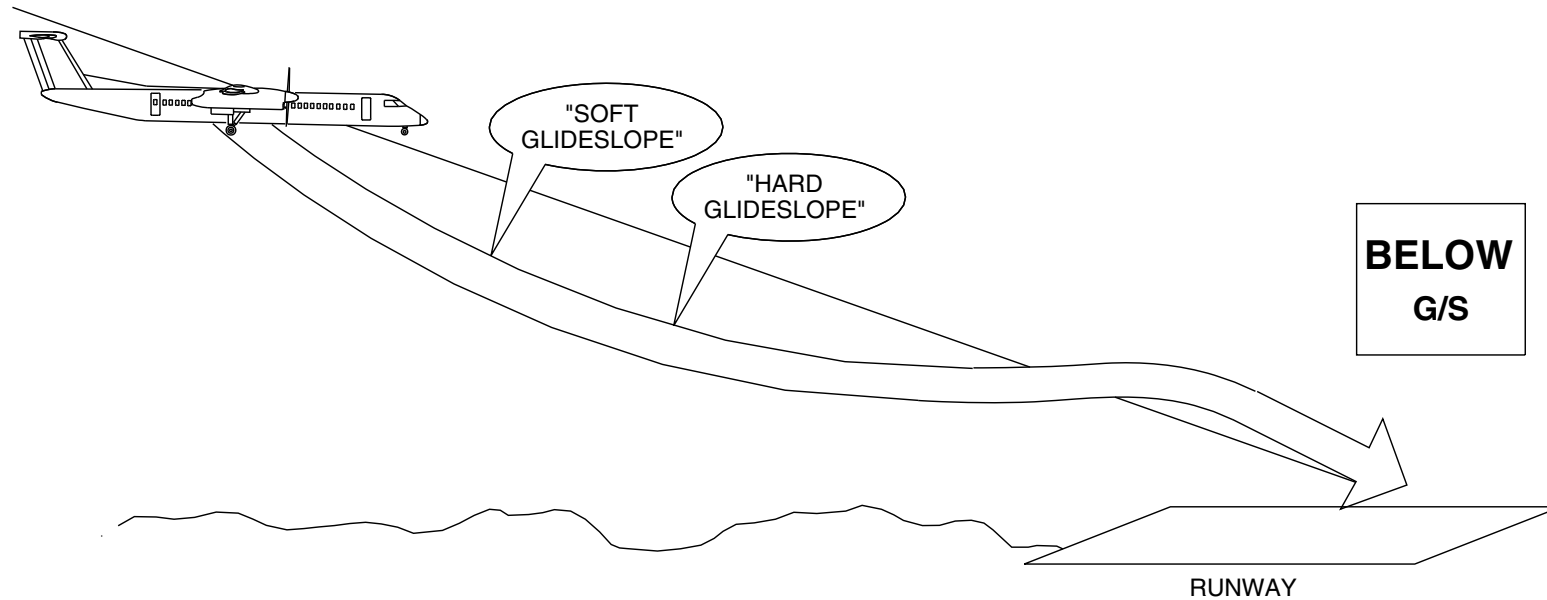
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fs826a01.dg, gv, 16/12/02

EGPWS Mode 5, Descent Below Glideslope Indications
Figure 41

PSM 1-84-2A
EFFECTIVITY:
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Config 001

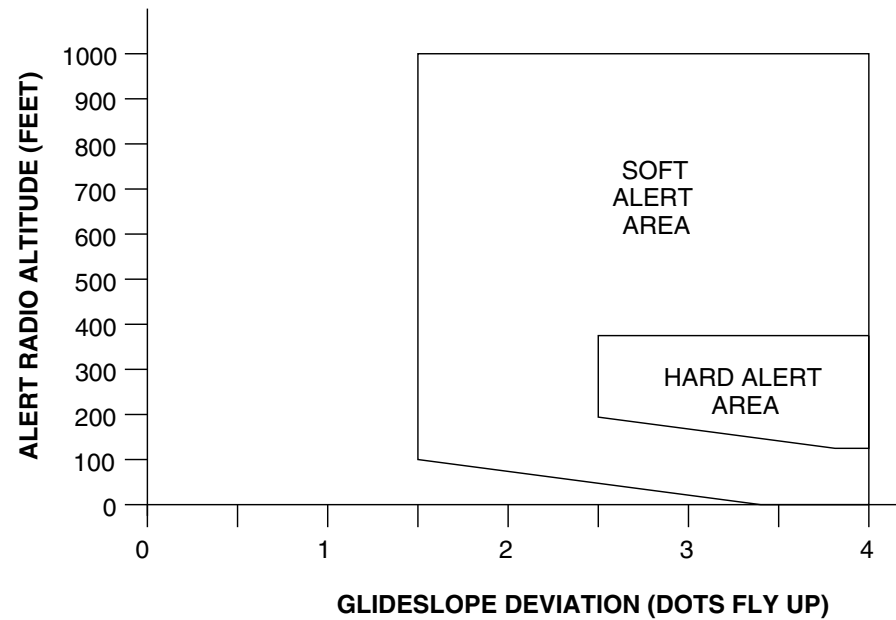
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fsz91a01.dg, gv, 16/12/02

EGPWS Mode 5, Descent Below Glideslope
Figure 42

PSM 1-84-2A
EFFECTIVITY:
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Config 001

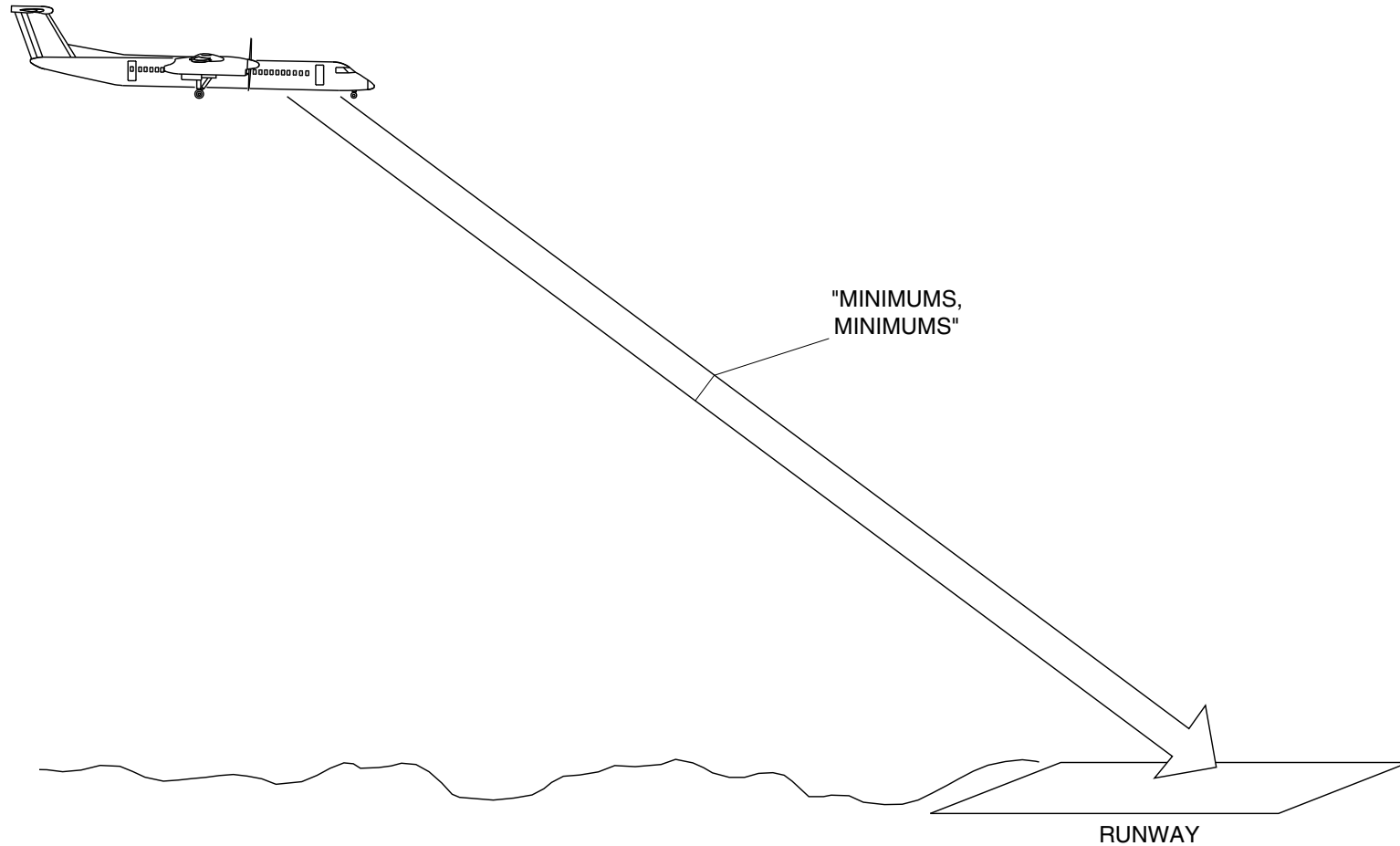
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MODSUM 4-309203 OR MODSUM 4-410300 AND PRE SB84-34-160

fsz90a01.dg, ck/nn, may09/2014

EGPWS Mode 6, Decision Height Callouts
Figure 43 (Sheet 1 of 4)

PSM 1-84-2A
EFFECTIVITY:
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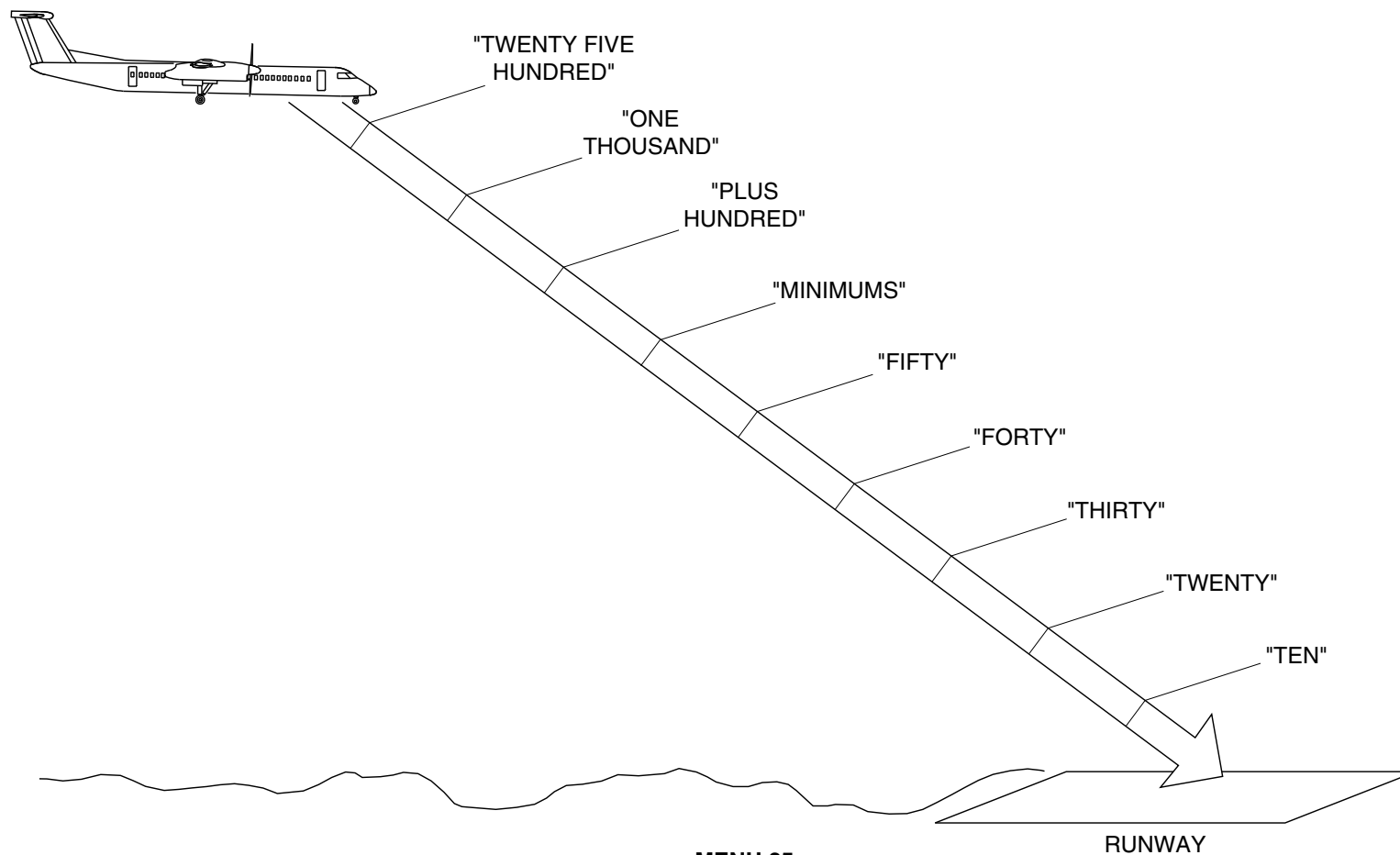
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fsz90a02.dg, gv/ck, aug08/2011

MENU 85
MODSUM 4-429519 OR SB84-34-102 OR SB84-34-47

EGPWS Mode 6, Decision Height Callouts
Figure 43 (Sheet 2 of 4)

PSM 1-84-2A
EFFECTIVITY:
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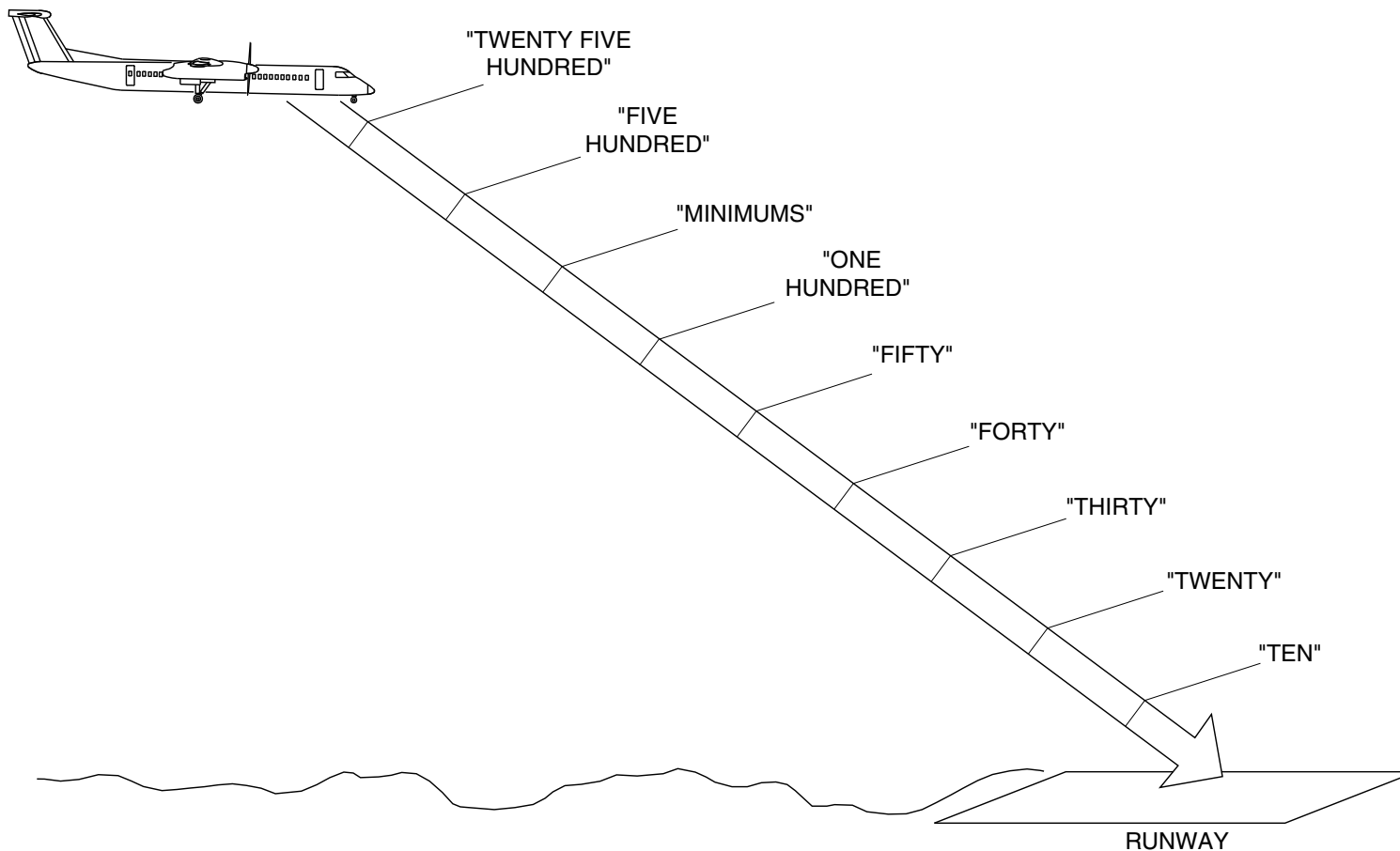
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MENU 34

MODSUM 4-457043 OR SB84-34-115 OR SB84-34-36 OR SB84-34-83 OR SB84-34-111 OR SB84-34-137 OR POST SB84-34-160

fsz90a03.dg, ck/nn, may09/2014

EGPWS Mode 6, Decision Height Callouts
Figure 43 (Sheet 3 of 4)

PSM 1-84-2A
EFFECTIVITY:
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Config 001

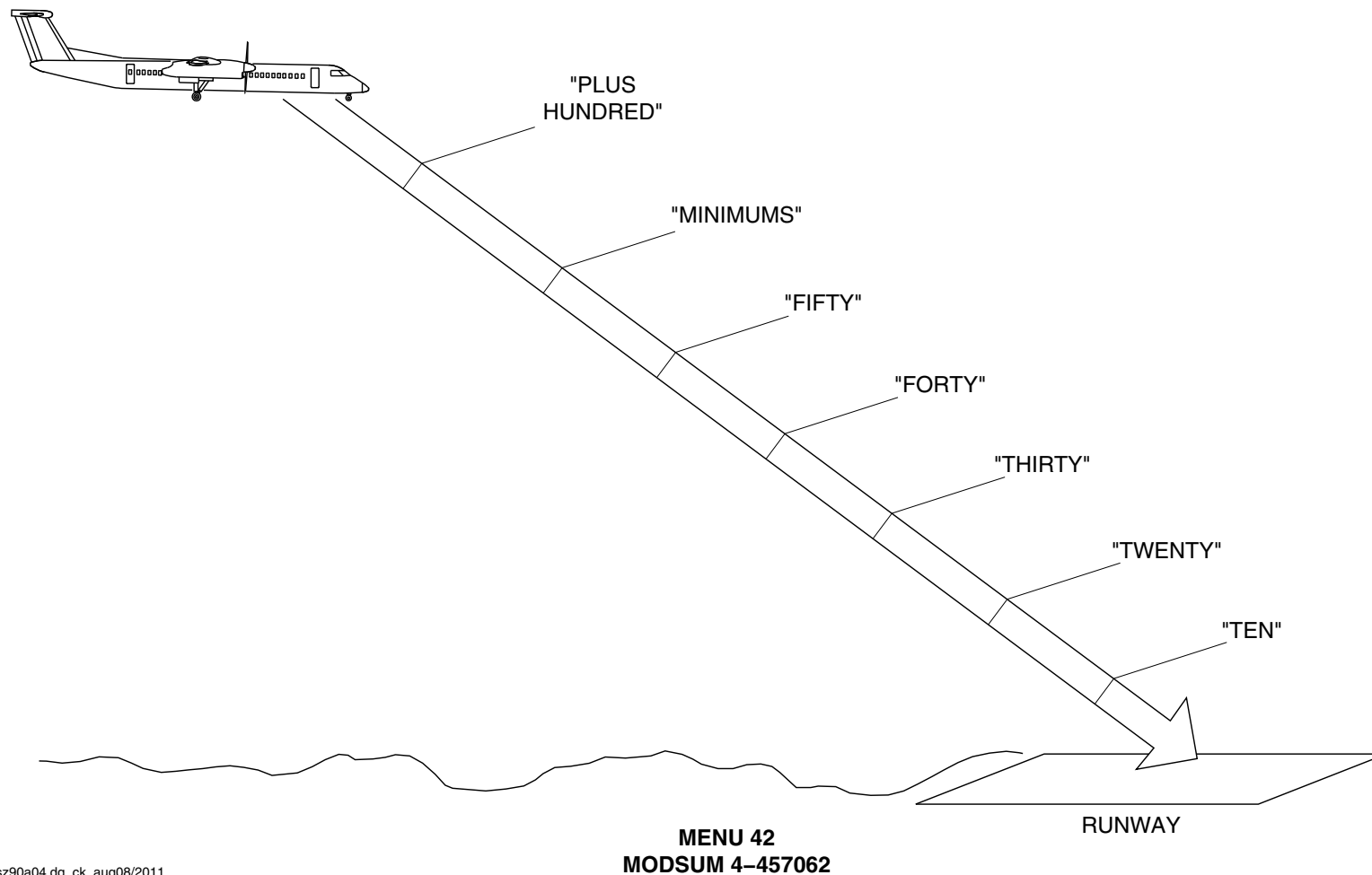
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fsz90a04.dg, ck, aug08/2011

EGPWS Mode 6, Decision Height Callouts
Figure 43 (Sheet 4 of 4)

PSM 1-84-2A
EFFECTIVITY:
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Config 001

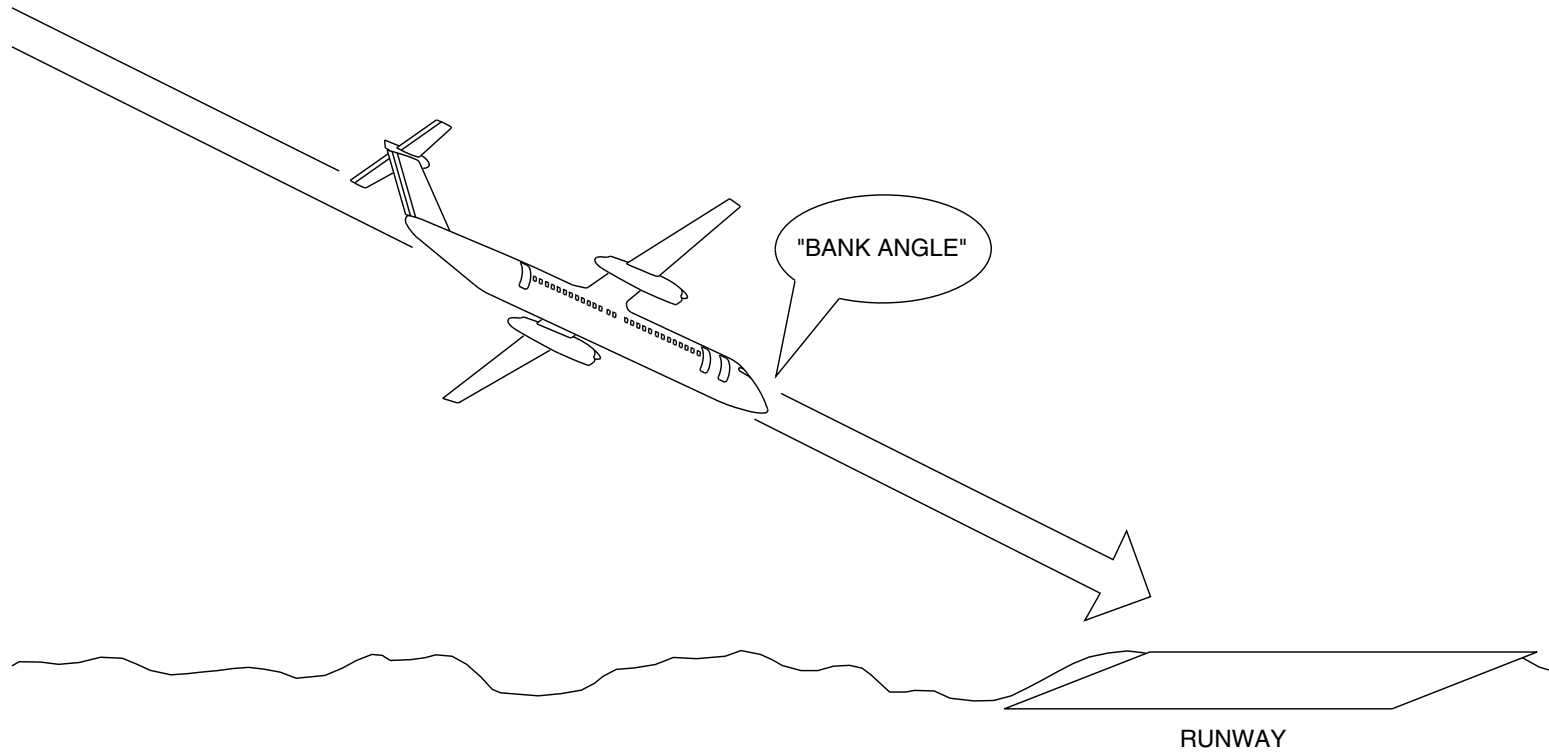
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fs828a01.dg, gv, 16/12/02

EGPWS Mode 6, Excessive Bank Angle Callout
Figure 44

PSM 1-84-2A
EFFECTIVITY:
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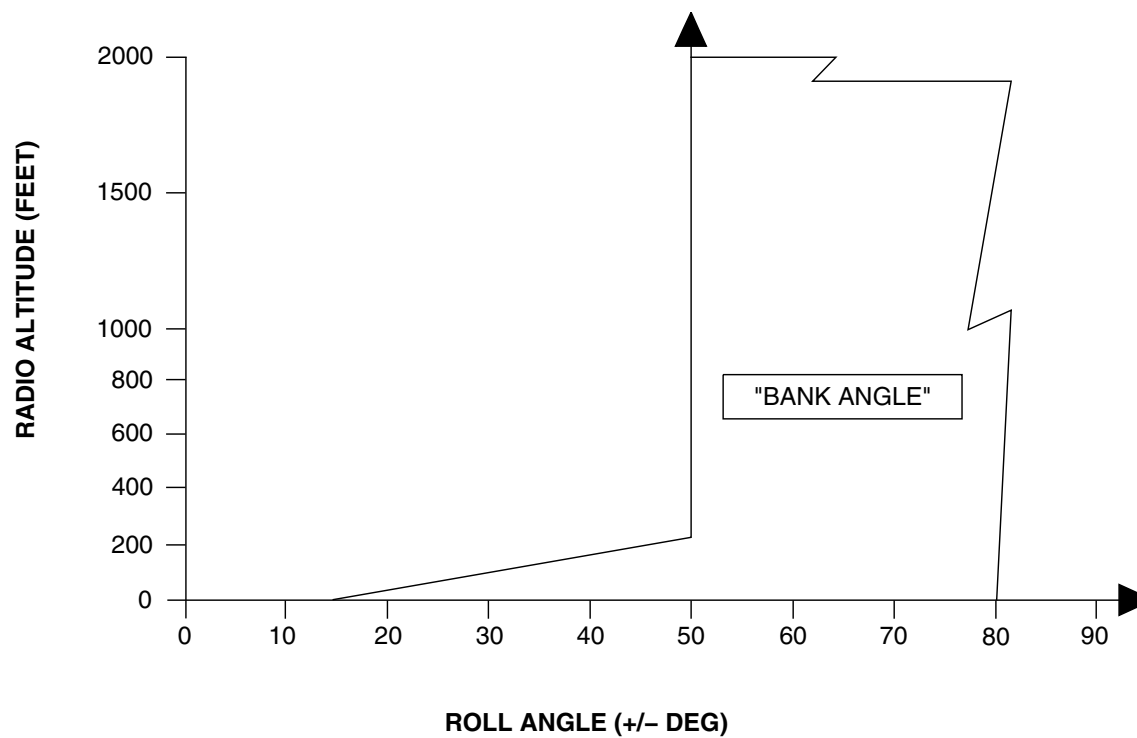
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fs829a01.dg, gv, 16/12/02

EGPWS Mode 6, Excessive Bank Angle
Figure 45

PSM 1-84-2A
EFFECTIVITY:
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Config 001

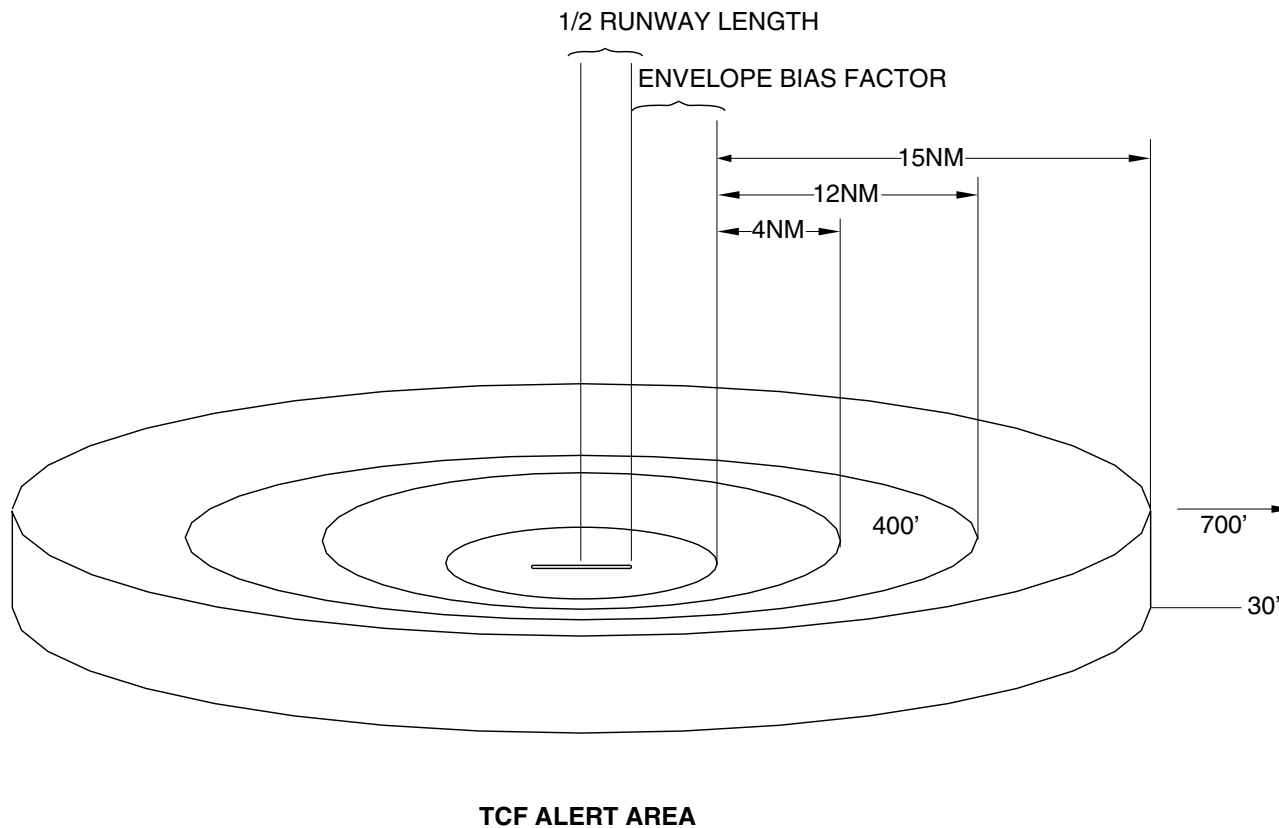
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fsz88a01.dg, gv, 16/12/02

Terrain Clearance Floor Indications
Figure 46

PSM 1-84-2A
EFFECTIVITY:
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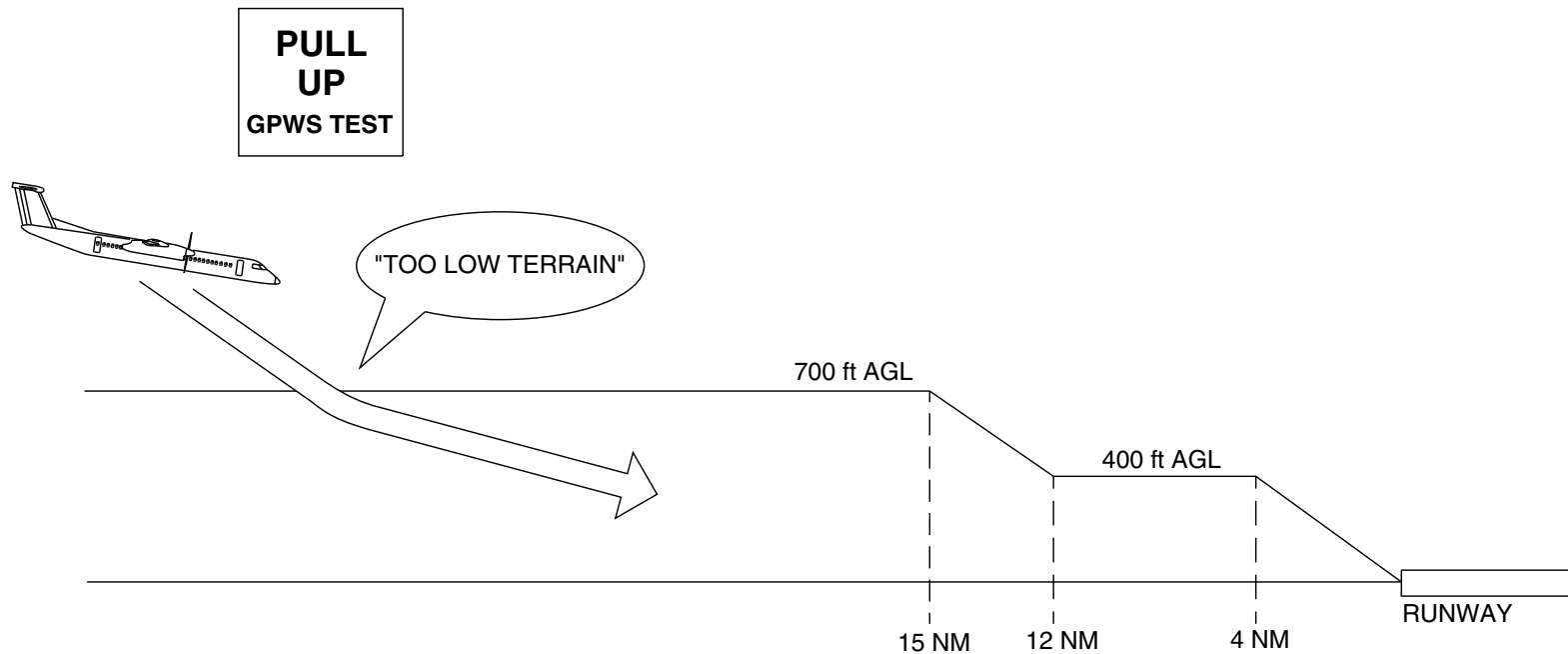
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fs0d8a01.dg, gv, 16/12/02

Terrain Clearance Floor
Figure 47

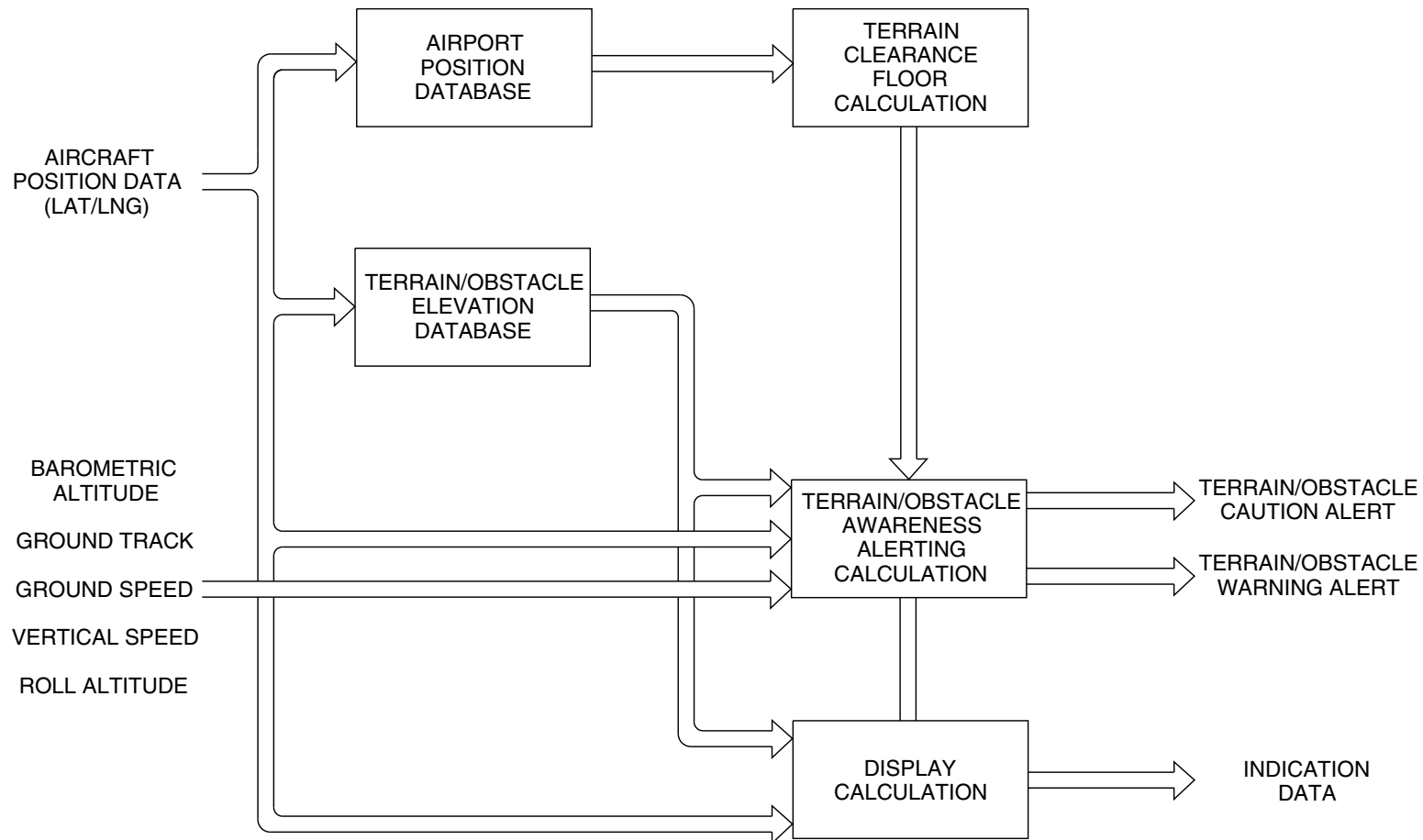
PSM 1-84-2A
EFFECTIVITY:
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fsz86a01.dg, gv, 16/12/02

Terrain Alerting Calculation
Figure 48

PSM 1-84-2A
EFFECTIVITY:
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Config 001

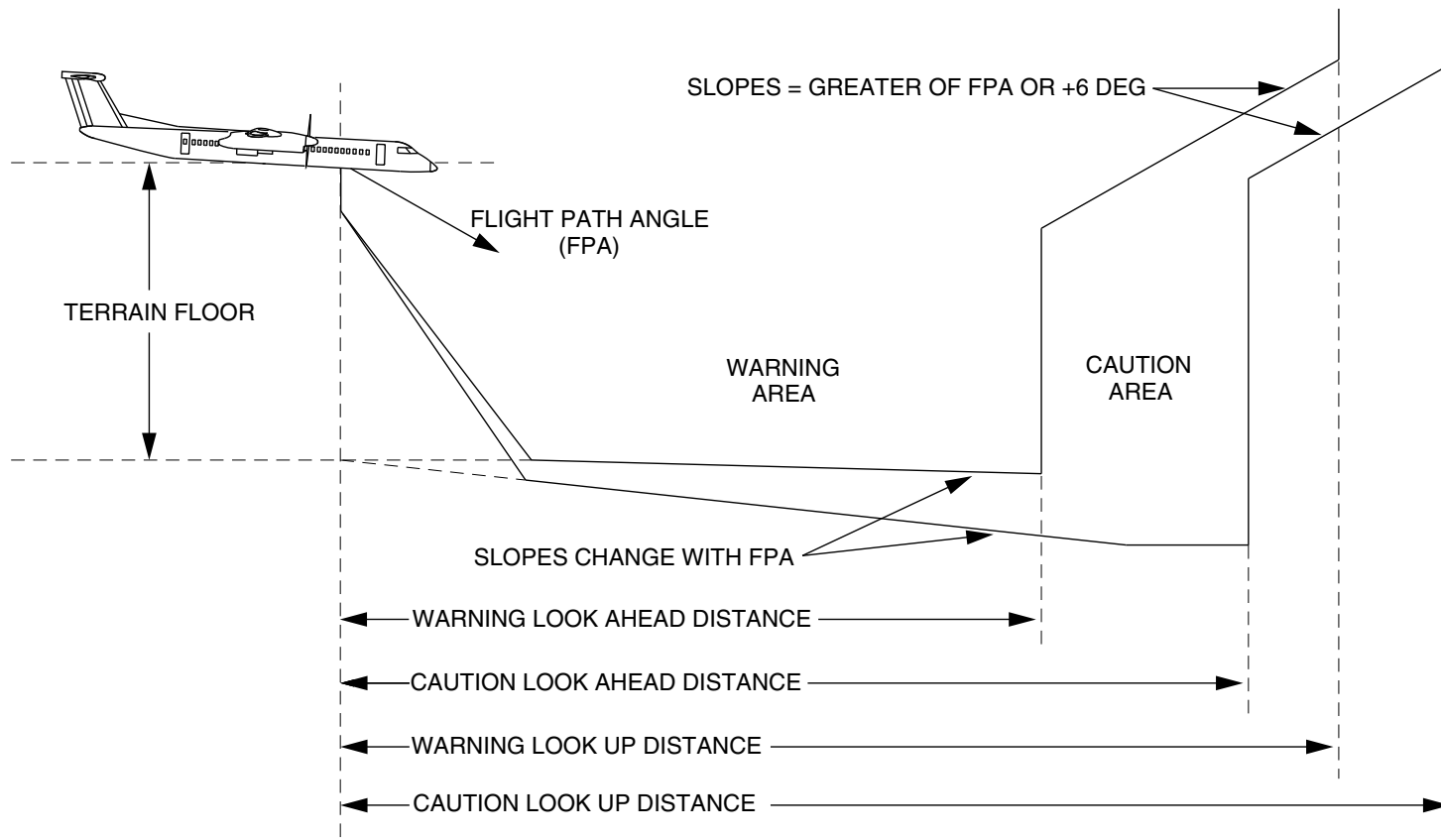
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LOOK AHEAD DISTANCES CHANGE WITH GROUND SPEED AND DISTANCE TO RUNWAY

TERRAIN FLOOR CHANGES WITH DISTANCES TO RUNWAY

fsz89a01.dg, gv, 16/12/02

Terrain Caution and Warning Areas
Figure 49

PSM 1-84-2A
EFFECTIVITY:
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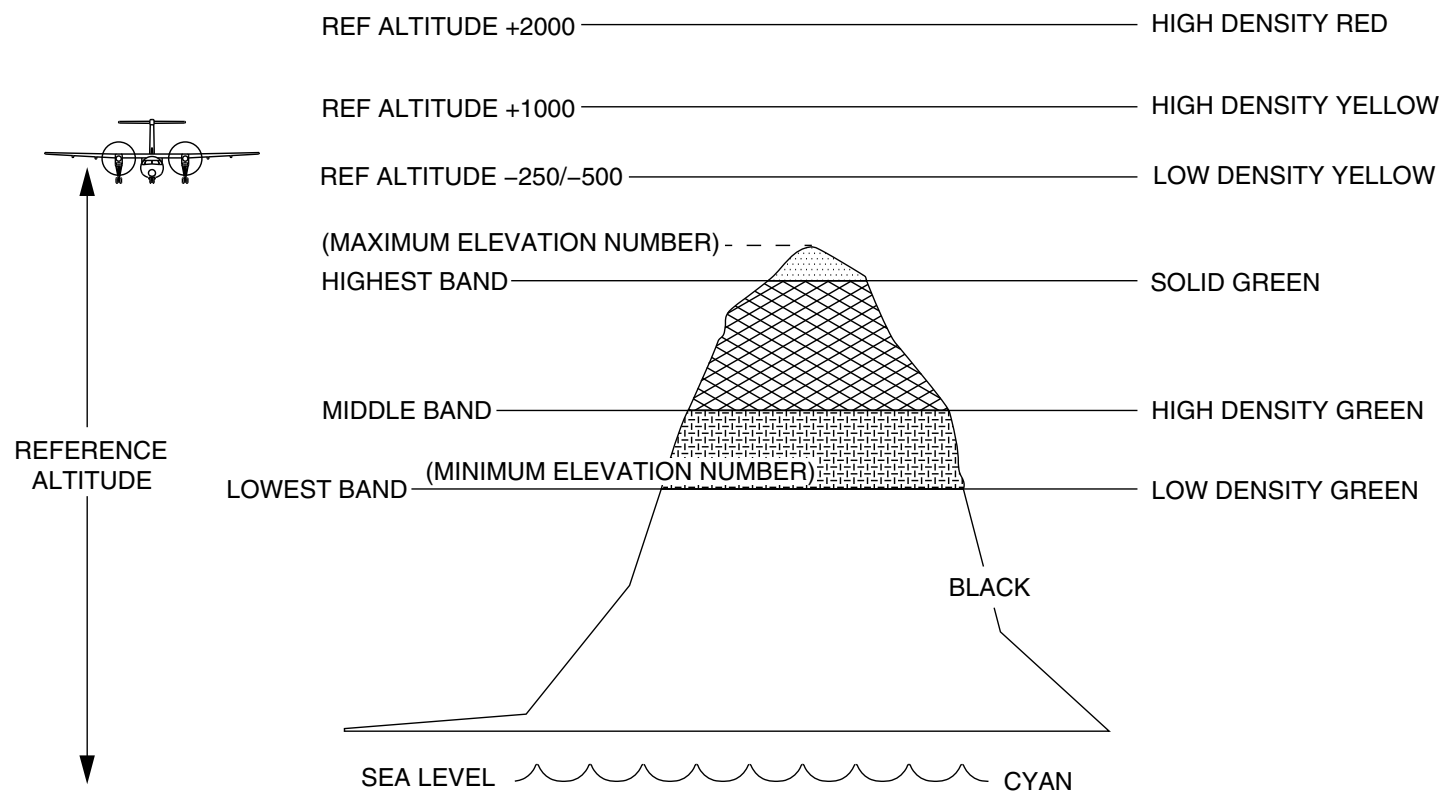
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fsz94a01.dg, gv/nn, oct16/2012

Peaks Display, Aircraft Above Terrain
Figure 50

PSM 1-84-2A
EFFECTIVITY:
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Config 001

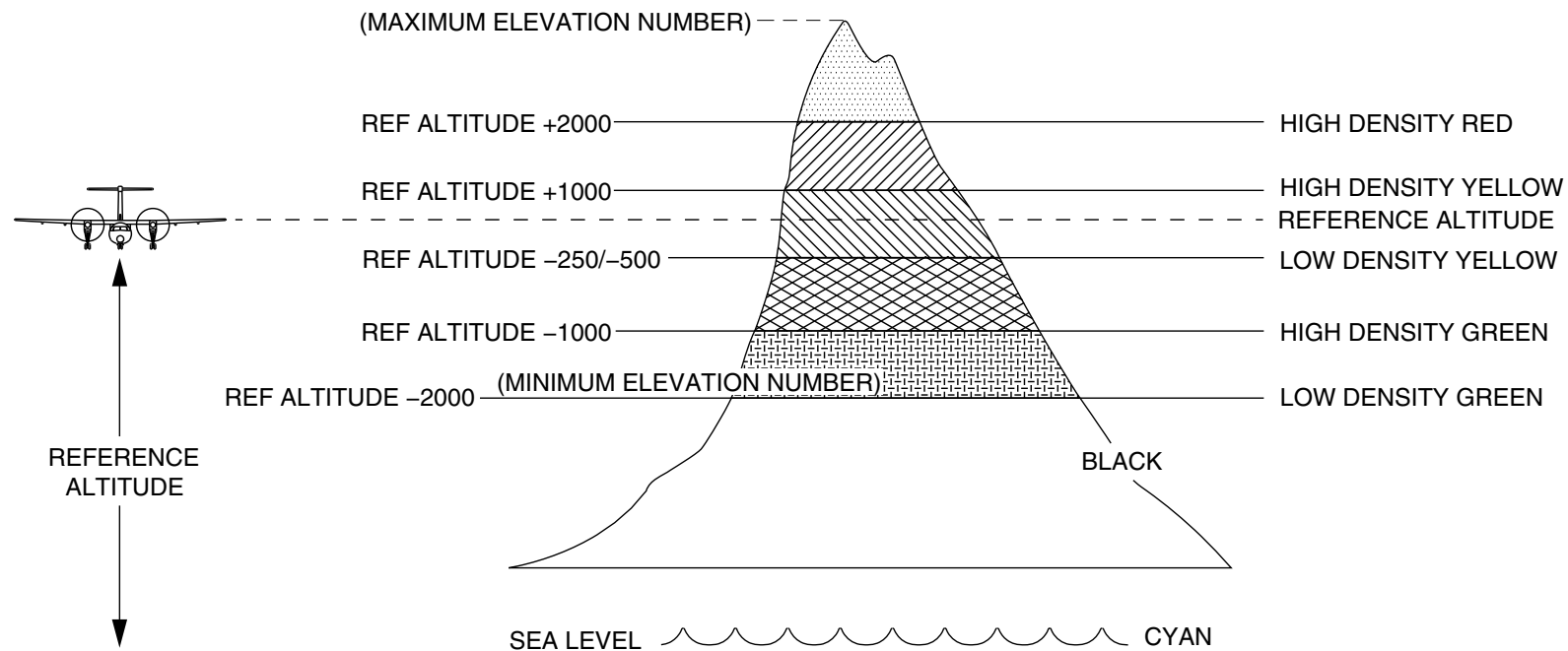
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fsz92a01.dg, gv/nn, oct16/2012

Peaks Display, Aircraft Adjacent to Terrain
Figure 51

PSM 1-84-2A
EFFECTIVITY:
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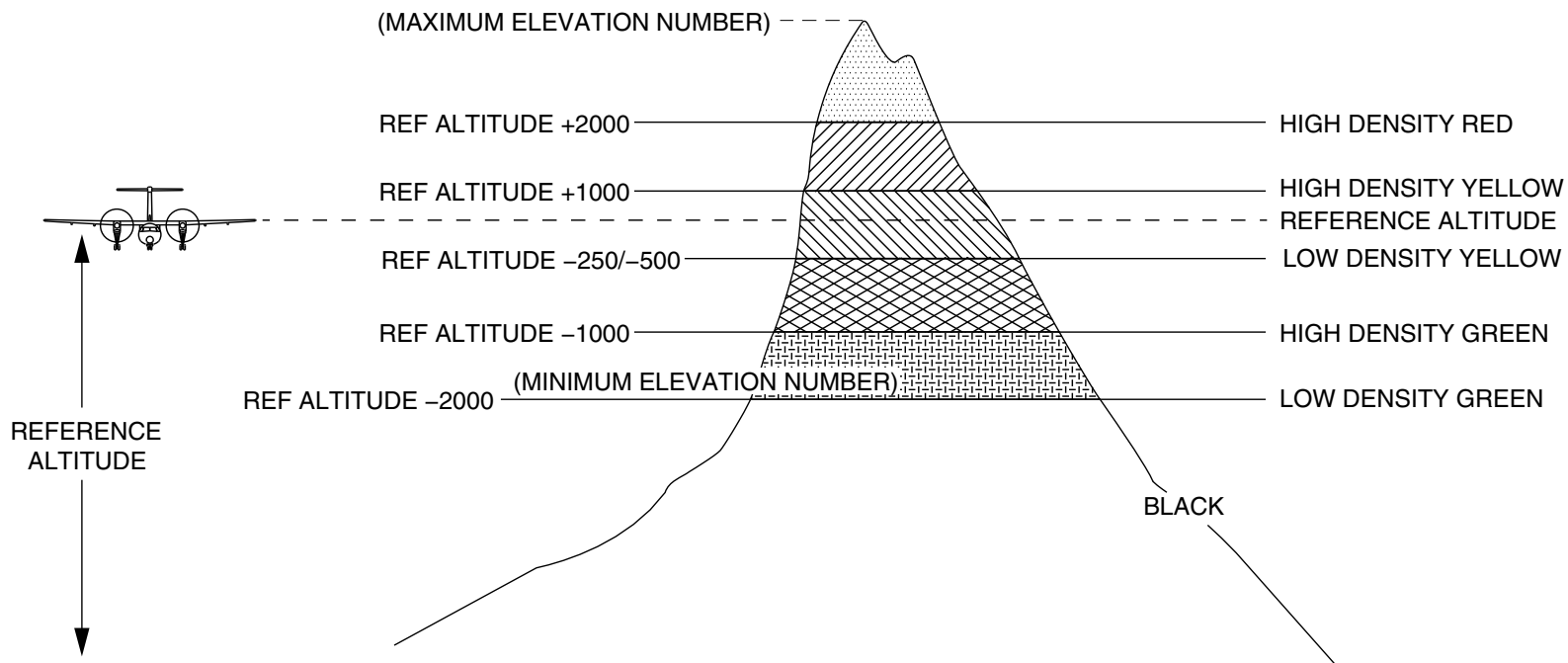
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cg2374a01.dg, nn, oct16/2012

Standard Display
Figure 52

PSM 1-84-2A
EFFECTIVITY:
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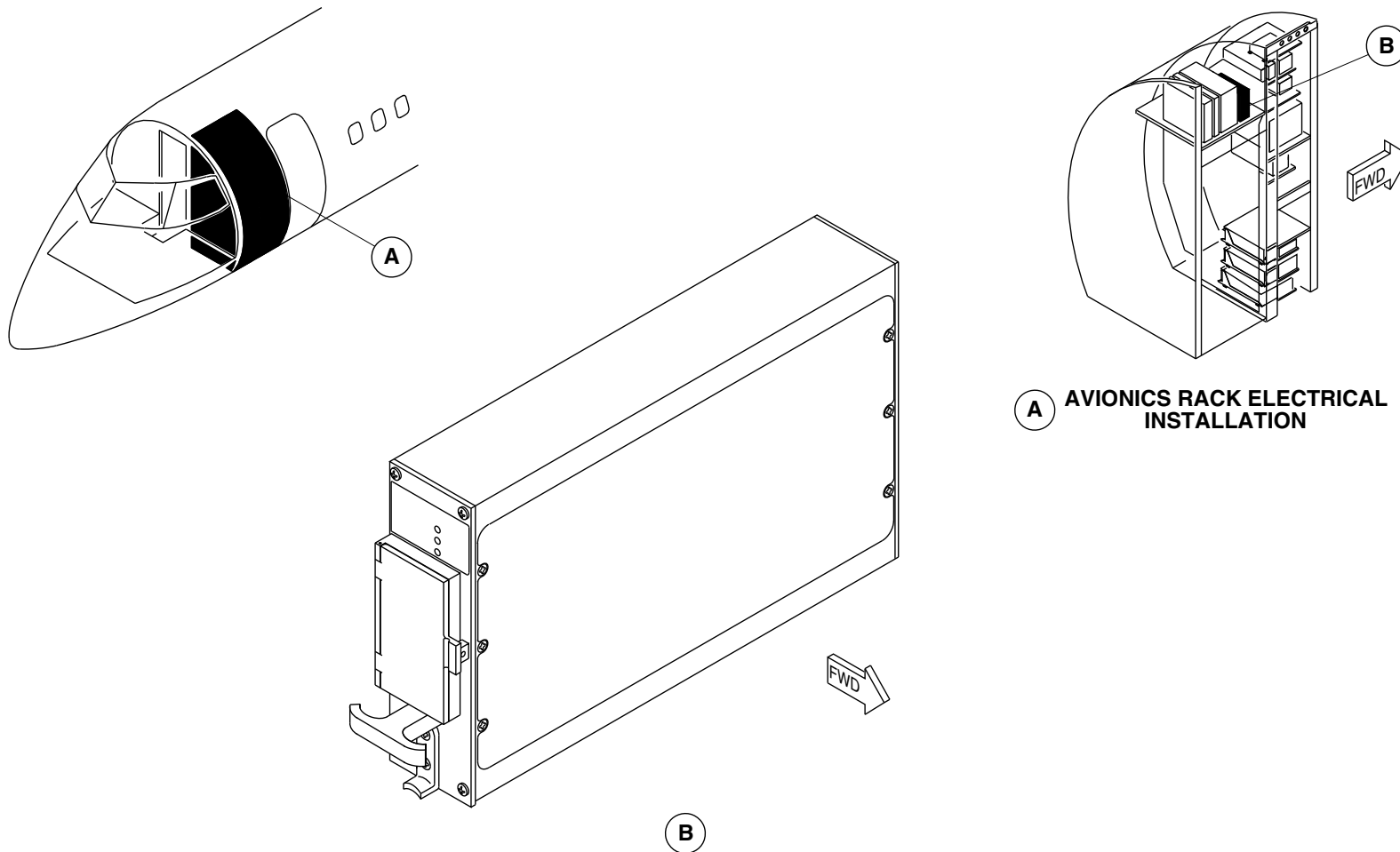
34-42-00

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fs0a8a01.dg, gv, 16/12/02

Computer, EGPWS Locator
Figure 53

PSM 1-84-2A
EFFECTIVITY:
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Config 001

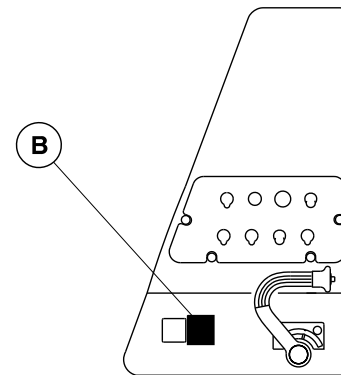
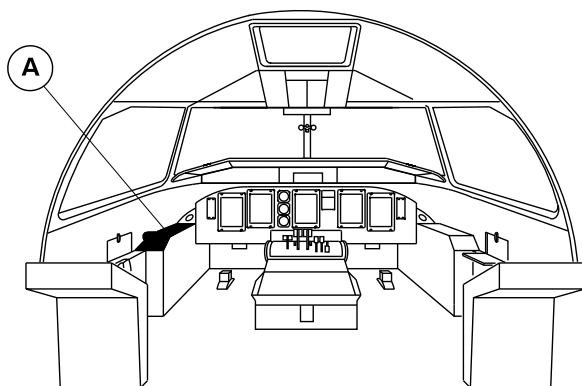
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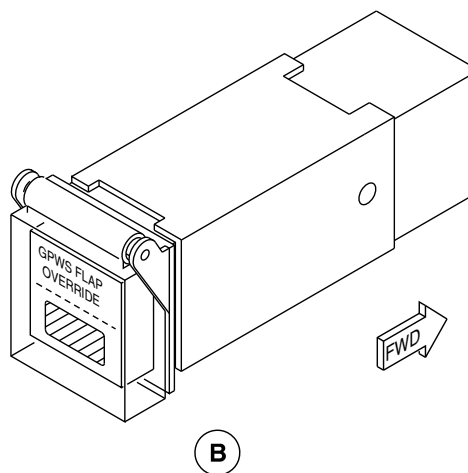


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A LEFT SIDE CONSOLE



fs963a01.dg, gv, 19/12/02

Switch, Annunciator–Flap Override Locator
Figure 54

PSM 1–84–2A
EFFECTIVITY:
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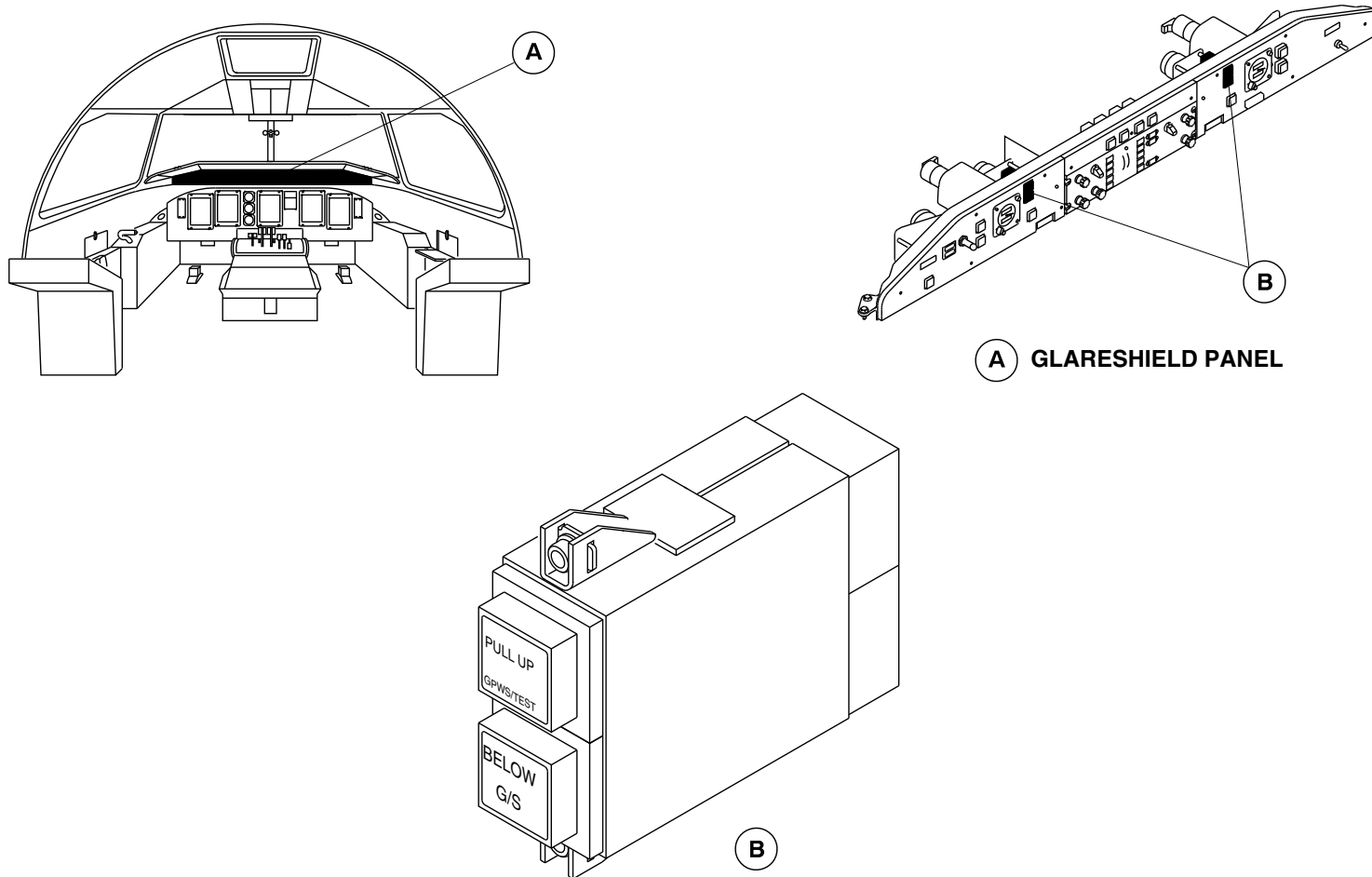
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fs962a01.dg, gv, 19/12/02

Pull Up and Below G/S Annunciator Switch Locator
Figure 55

PSM 1-84-2A
EFFECTIVITY:
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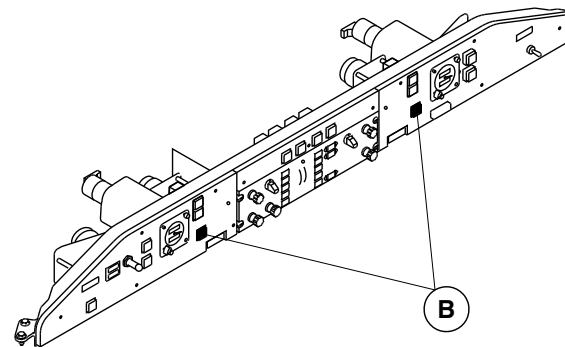
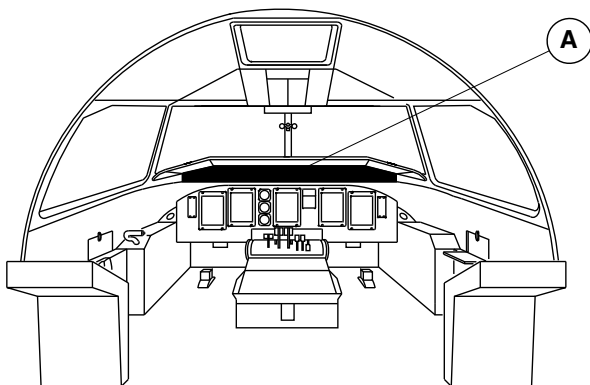
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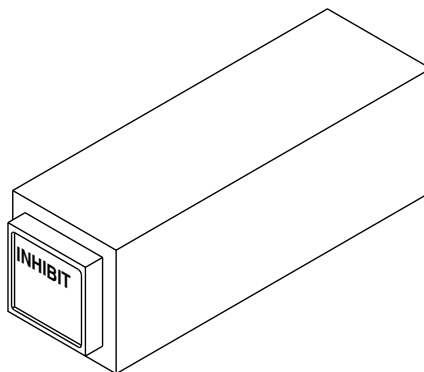


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A GLARESHIELD PANEL



B

fs0a6a01.dg, gv, 19/12/02

Inhibit Annunciator Switch Locator
Figure 56

PSM 1-84-2A
EFFECTIVITY:
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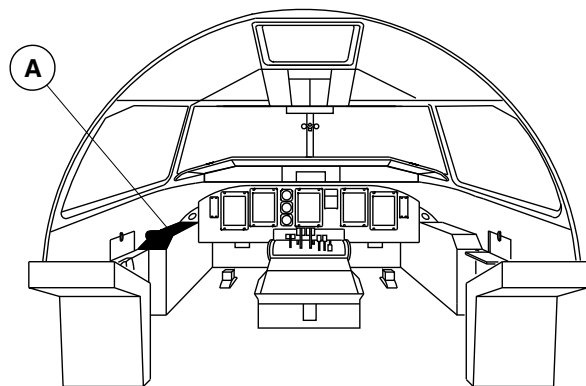
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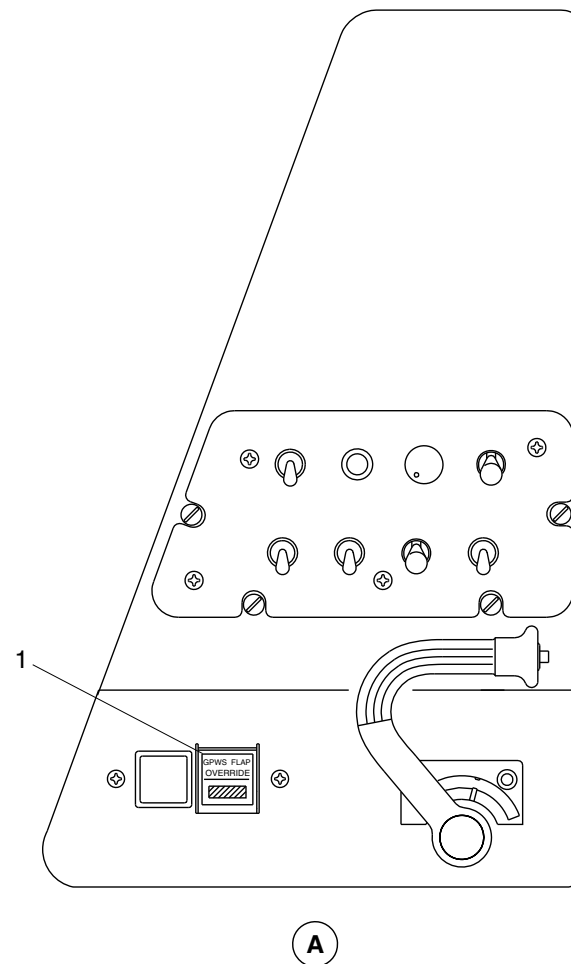
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LEGEND

1. GPWS Flap Override Annunciator Switch.



fs967a01.dg, gv, 19/12/02

GPWS Flap Override Annunciator Switch
Figure 57

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 39 of 34-42-00
Config 001

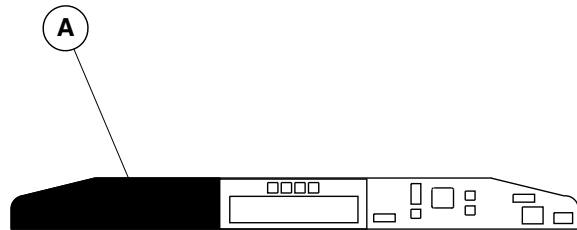
34-42-00

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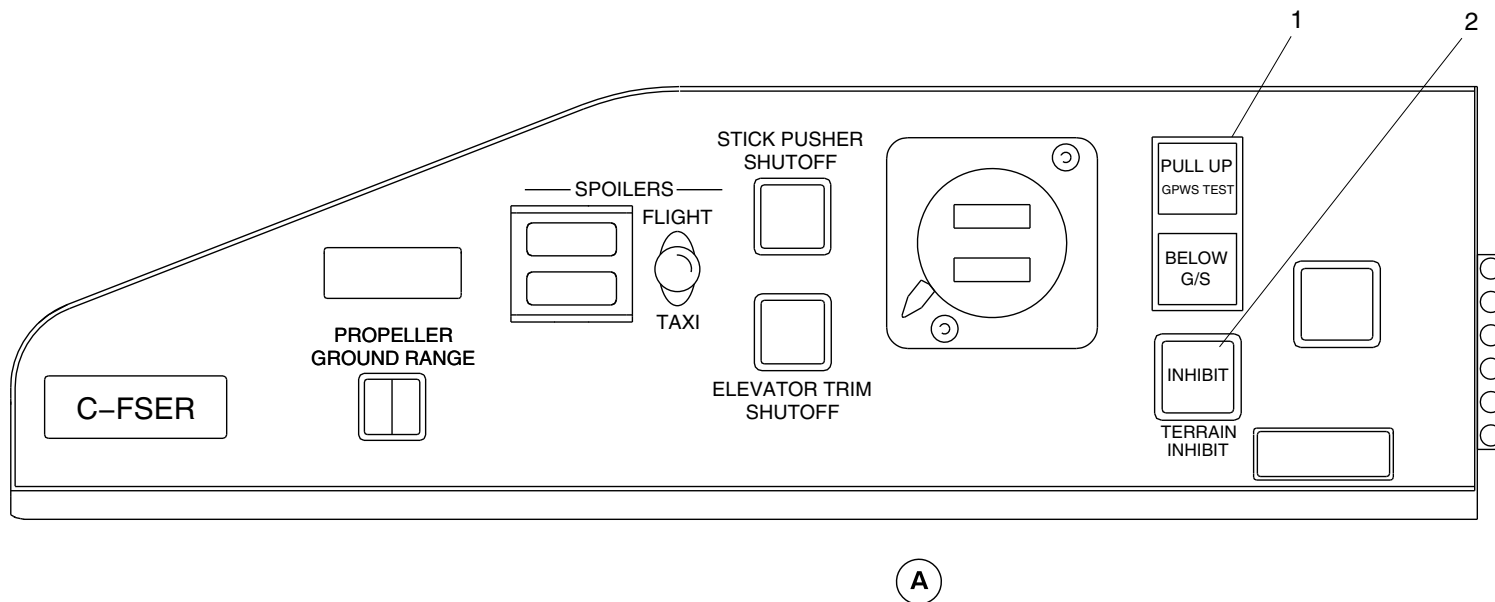
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. GPWS annunciator module.
2. TERRAIN INHIBIT annunciator switch.



fs0b0a01.dg, gv, 19/12/02

Pull Up, Below G/S and Inhibit Annunciator Switches
Figure 58

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 39 of 34-42-00
Config 001

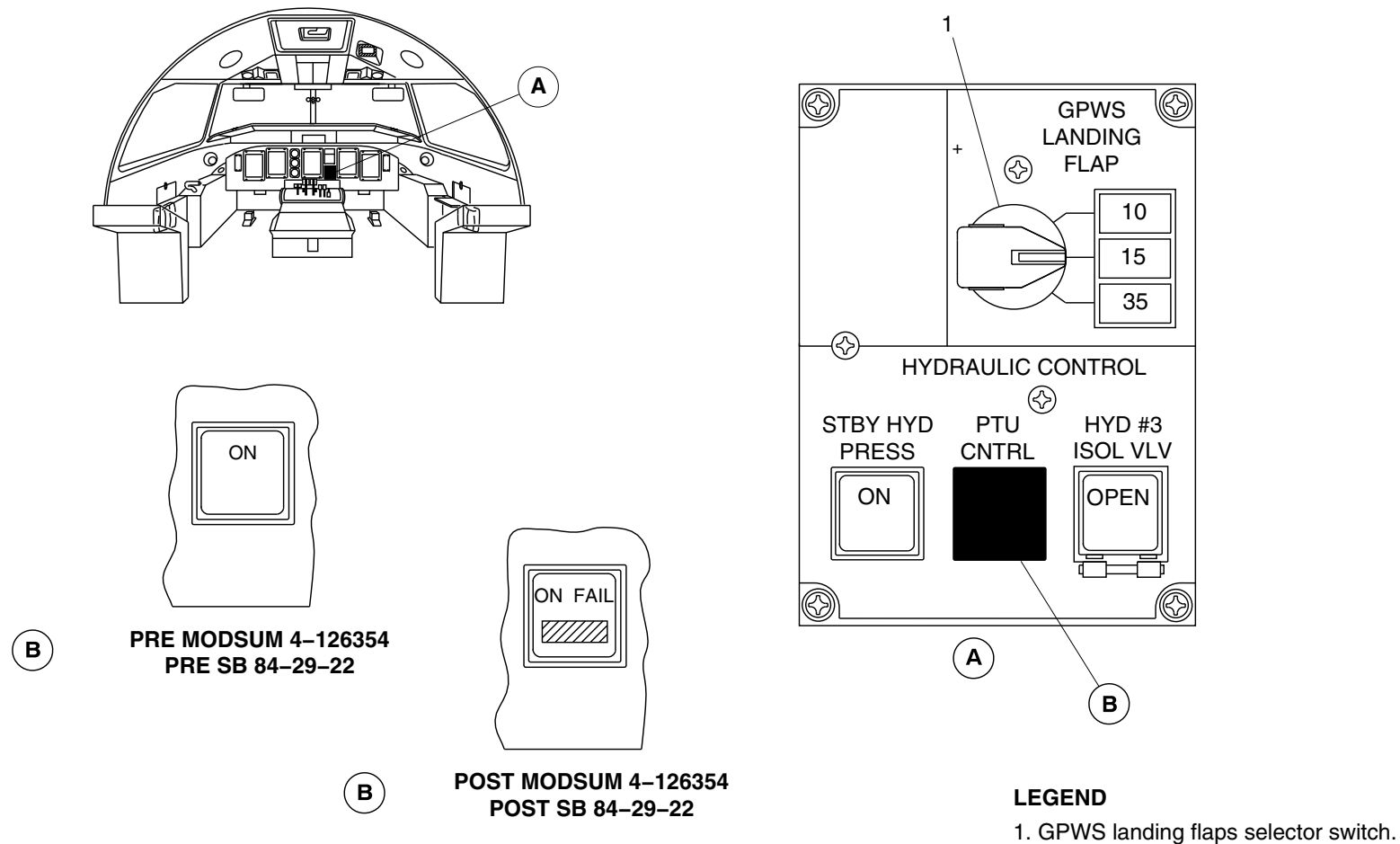
34-42-00

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



cg0072a01.dg, dn/kg, nov21/2008

GPWS Landing Flaps Selector Switch
Figure 59

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 39 of 34-42-00
Config 001

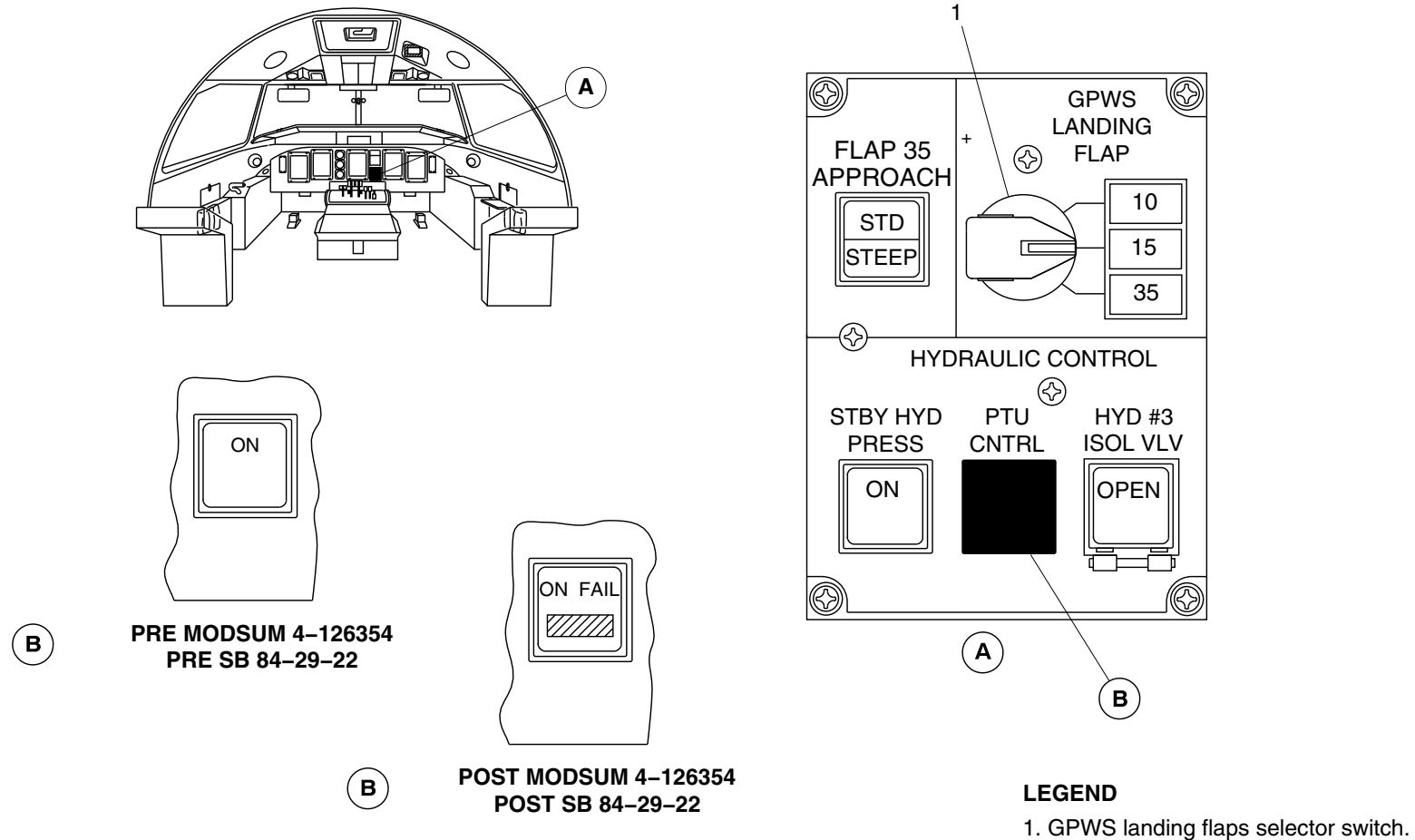
34-42-00

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fs966a01.dg, gv/ kg, nov21/2008

GPWS Landing Flaps Selector Switch (803SO90034)
Figure 60

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 39 of 34-42-00
Config 001

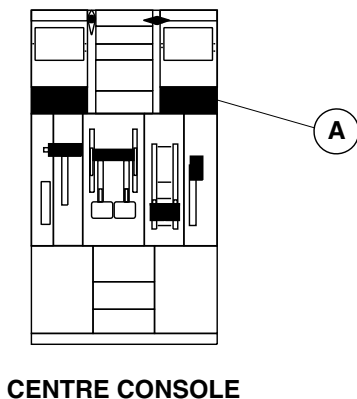
34-42-00

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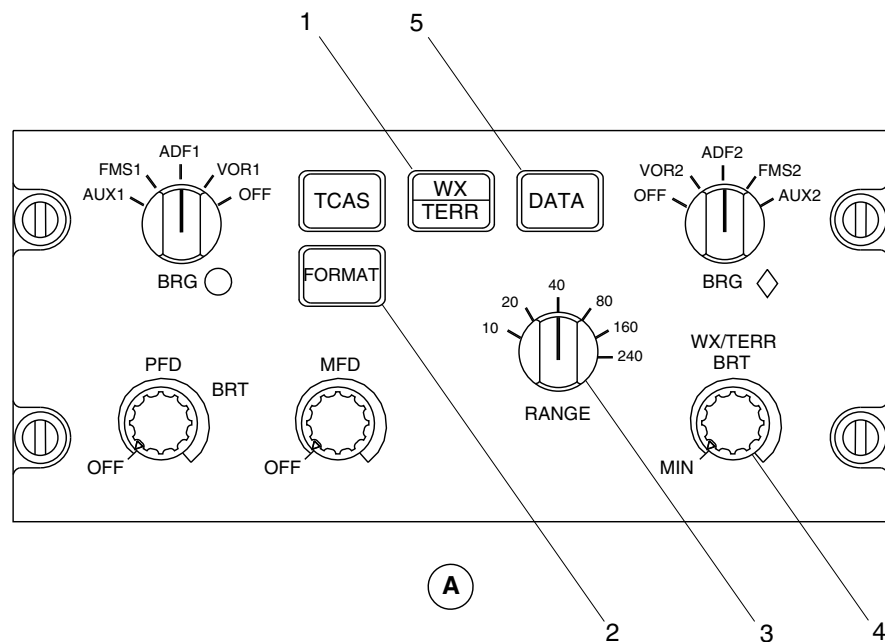
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

LEGEND

1. Weather radar/terrain pushbutton switch.
2. FORMAT pushbutton switch.
3. Weather radar display range selector knob.
4. Weather radar display brightness knob.
5. DATA pushbutton switch.



fs634a01.dg, gv, 19/12/02

EFIS Control Panel, Weather Radar Controls
Figure 61

PSM 1-84-2A
EFFECTIVITY:
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Config 001

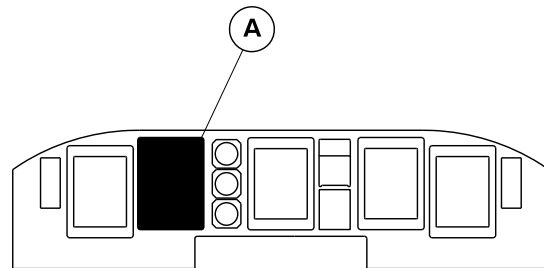
34-42-00

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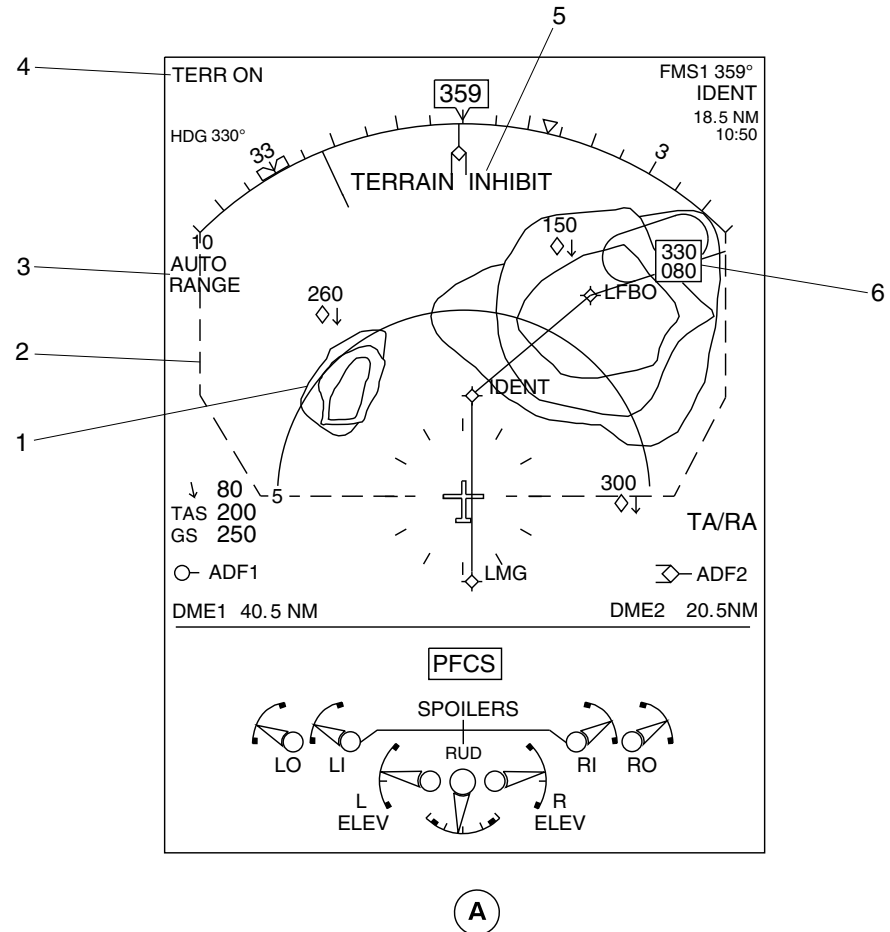
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Terrain Image.
2. Terrain Display Mask.
3. AUTO RANGE Message.
4. Terrain Mode Message.
5. Terrain Message.
6. Terrain Peaks Annunciation.



fs0a5a01.dg, gv, 19/12/02

Multi-Function Display (MFD1, MFD2), EGPWS Indications
Figure 62

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 39 of 34-42-00

Config 001

34-42-00

Config 001

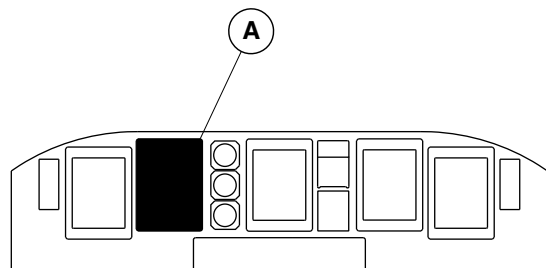
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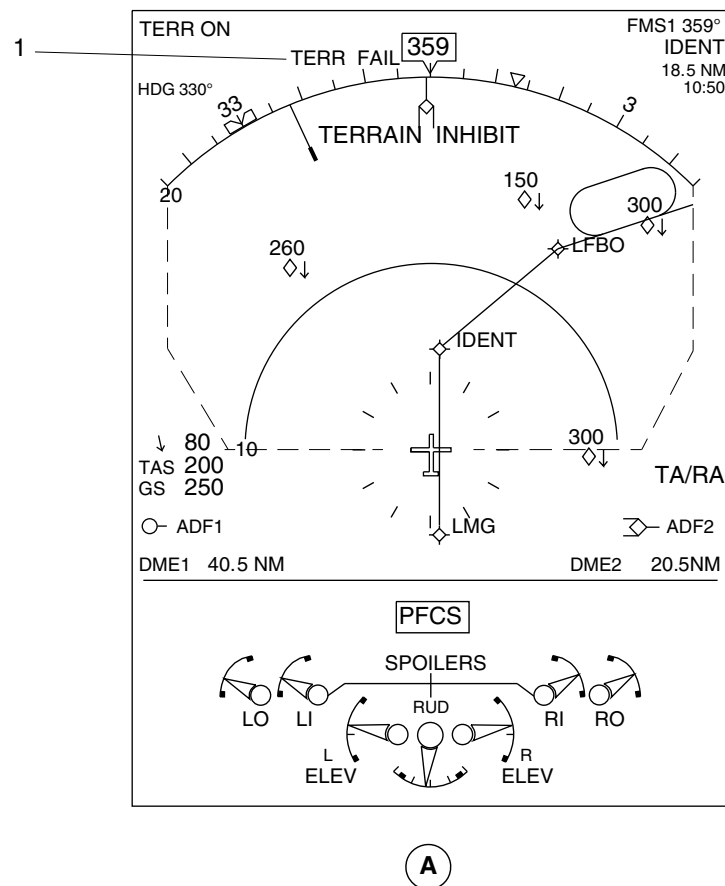
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. TERRAIN FAIL Message.



fs0a7a01.dg, gv, 19/12/02

Multi-Function Display (MFD1, MFD2), EGPWS Malfunction Indications
Figure 63

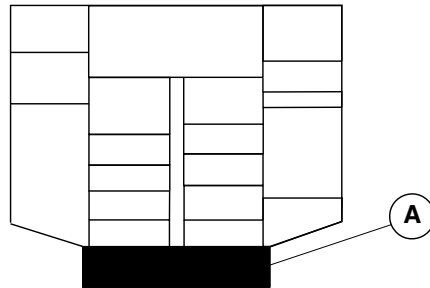
PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 39 of 34-42-00
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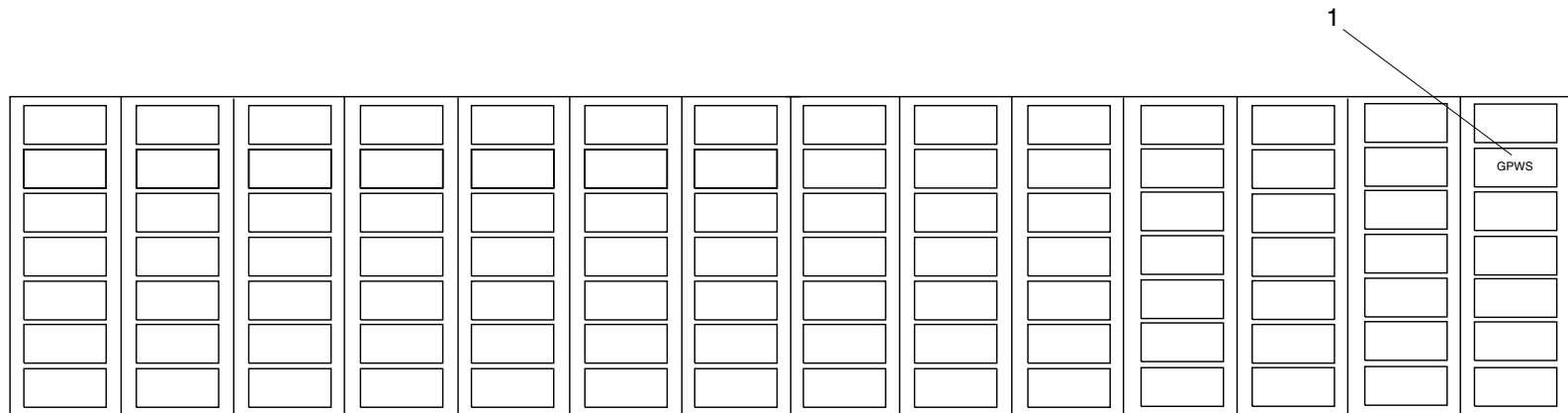
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

1. Ground Proximity Warning System (amber).



A

fsi49a01.dg, gv, 19/12/02

Caution and Warning Panel, GPWS Caution Indication
Figure 64

PSM 1-84-2A
EFFECTIVITY:
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Config 001

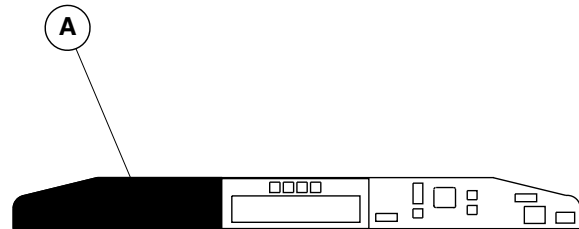
34-42-00

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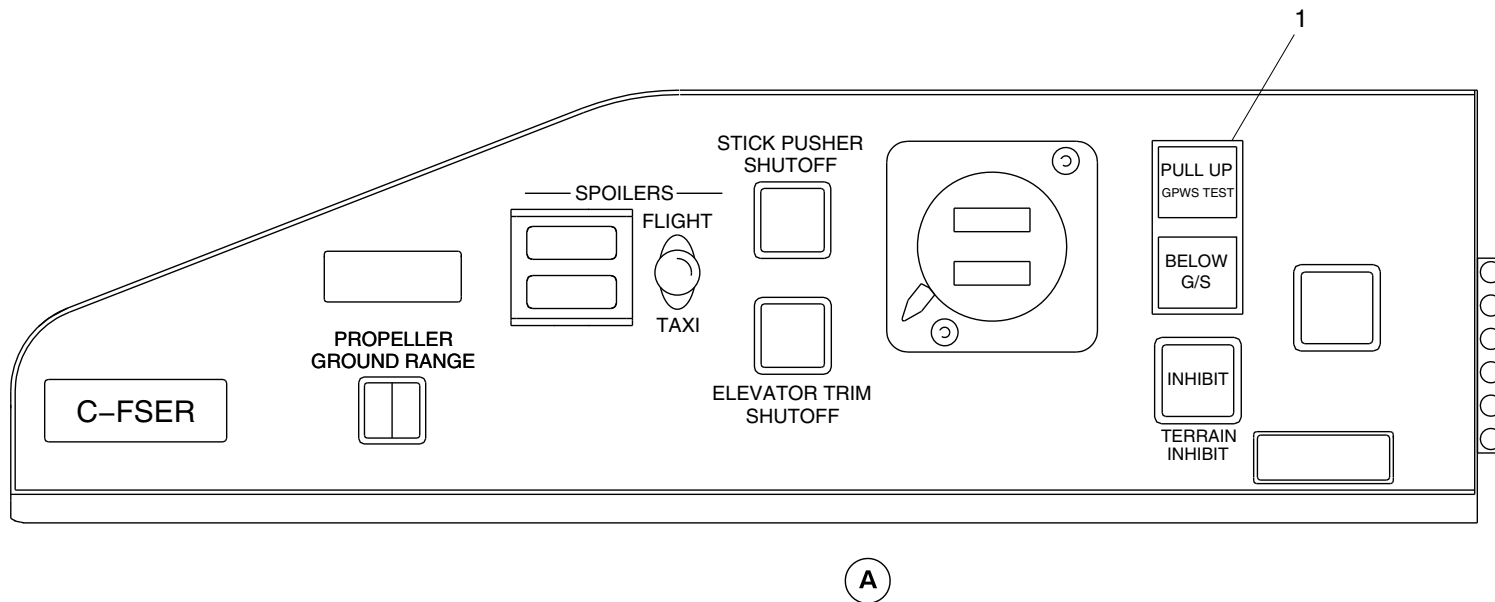
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. GPWS Annunciator Module.



fs0b0a02.dg, gv, 19/12/02

EGPWS Test Selection
Figure 65

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 39 of 34-42-00
Config 001

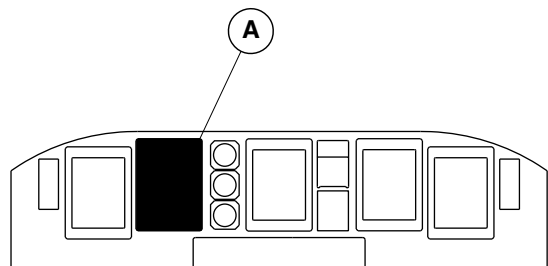
34-42-00

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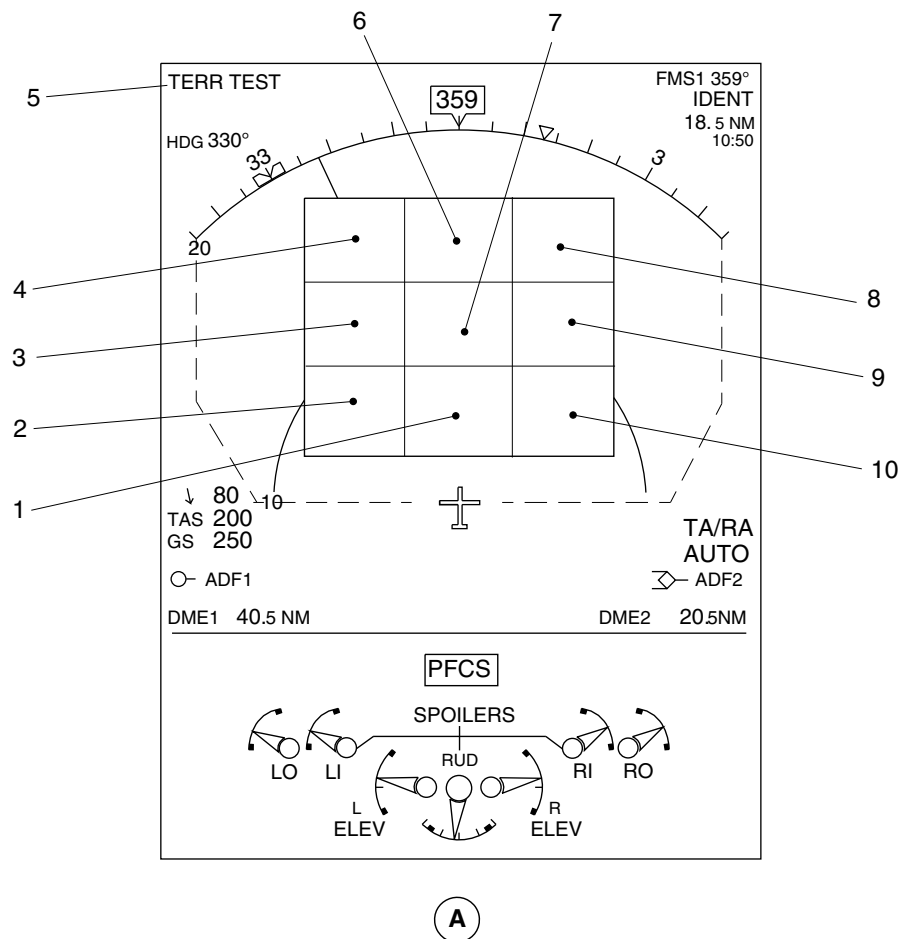
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Medium yellow.
2. Medium density green.
3. Solid red.
4. Magenta.
5. Terrain mode message.
6. High density red.
7. High density yellow.
8. Cyan.
9. Solid yellow.
10. Low density green.



fs0b1a01.dg, gv, 19/12/02

Multi-Functional Display, EGPWS Test Indication
Figure 66

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 39 of 34-42-00
Config 001

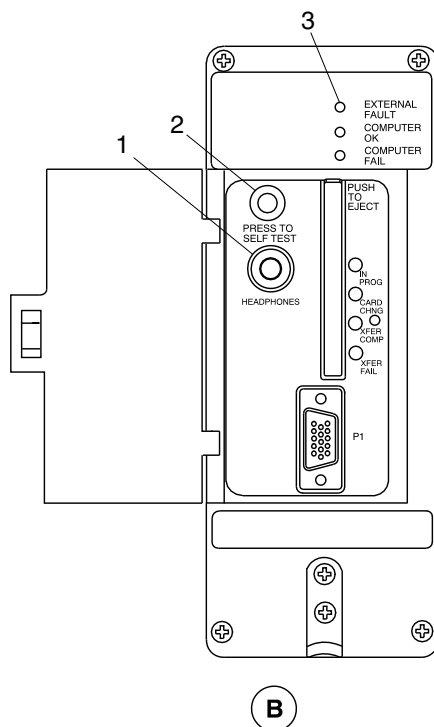
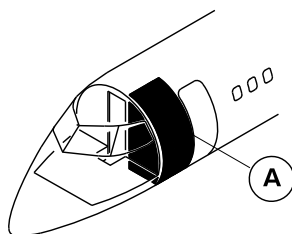
34-42-00

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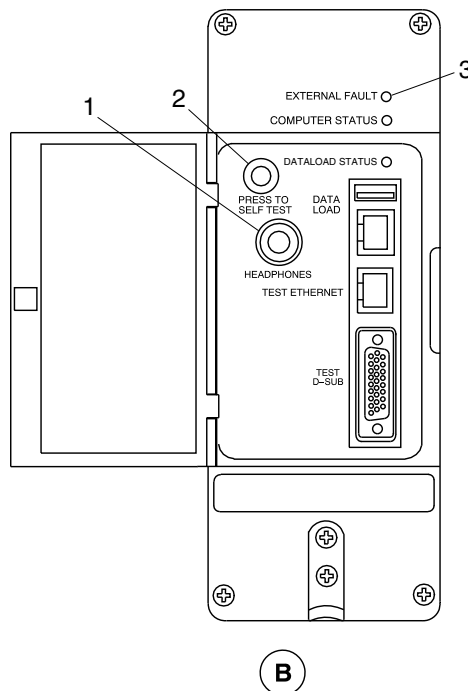


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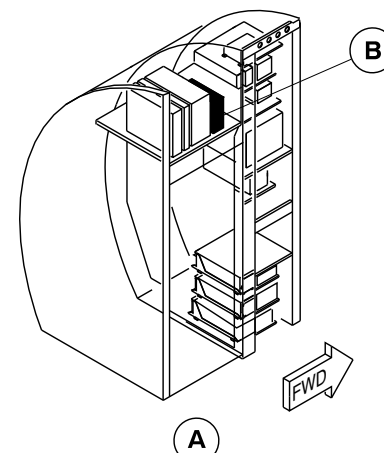
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



PRE MODSUM 4Q903303



POST MODSUM 4Q903303



**AVIONICS RACK ELECTRICAL
INSTALLATION**

LEGEND

1. Headphone jack.
2. Self-test switch.
3. Status indicator.

fs0a9a01.dg, pr, oct28/2015

EGPWC Test Selection
Figure 67

PSM 1-84-2A
EFFECTIVITY:
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**ON A/C ALL

34-43-00-001

TRAFFIC ALERTING AND COLLISION AVOIDANCE SYSTEM (TCAS)

Introduction

The Traffic Collision Avoidance System II (TCAS II) gives aural and visual indications when the aircraft's position and flight path relative to an intruder aircraft may cause a potentially dangerous condition.

General Description

Refer to Figures 1 and 2.

The TCAS II interrogates other aircraft's transponders and uses their replies to calculate the parameters of the intruder aircraft that follow:

- Range
- Relative bearing
- Altitude and vertical speed
- Closure rate.

The TCAS II compares this data with its own position to calculate a potential collision condition, and give an audio and visual indication for vertical avoidance maneuvers.

NOTE

Note: TCAS II is unable to detect other aircraft without an operating Air Traffic Control Radar Beacon System (ATCRBS) transponder (operating in Mode A and C) or a Mode S transponder.

The TCAS II calculates the time to, and the separation at the intruder aircraft's Closest Point of Approach (CPA).

The TCAS II will show four different symbols on the Electronic Instrument System (EIS). The type of symbol shown is based on the intruder aircraft's location and closure rate.

The symbols change shape and colour to show more important levels of urgency.

The traffic symbols can also have a related altitude indication and a trend arrow to show that the intruding aircraft is vertical speed is more than 500 ft/min.

The EIS shows the TCAS II symbols that follow:

- Non-threat traffic
- Proximity intruder traffic
- Traffic Advisory (TA)
- Resolution Advisory (RA).

Non-Threat Traffic: The hollow cyan diamond shows that an intruder aircraft's relative altitude is more than ± 1200 ft, or its range is more than 6 nm.



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On aircraft with ModSum 4Q309289 installed, the hollow cyan diamond shows that an intruder aircraft's relative altitude is between ± 1200 to 2700 ft, or its range is more than 6 nm.

It is not considered a threat.

Proximity Intruder Traffic: The filled cyan diamond shows that an intruder aircraft's relative altitude is less than ± 1200 ft, or its range is less than 6 nm.

It is not considered a threat.

TA: The traffic symbol changes to a filled yellow circle to show a possible dangerous condition. A TA is shown when the CPA is between 20 to 48 seconds.

RA: If the intruding aircraft continues to close, the TA indication changes to a filled red square. The CPA is between 15 to 35 seconds.

The RA traffic symbol is shown with a Inertial Vertical Speed Indicator (IVSI) RA to maintain a safe vertical separation. The IVSI RA indication has red and green arcs on the IVSI scale to show the required rate of climb or descent to prevent a possible collision.

The TCAS II calculates the RA indication on a five second crew reaction time to get sufficient separation. A two and one half seconds reaction time is given when there is a change in the RA indication.

NOTE

Note: Any aircraft that is the surveillance range of the TCAS II is an intruder. If the intruder aircraft causes an RA, it then referred to as a threat aircraft.

The TCAS II has different levels of protections as shown in the table that follow:

Table 1: Protection from the Intruder Aircraft Equipments

INTRUDER AIRCRAFT EQUIPMENT	PROTECTION
Mode A Transponder	TA
Mode C Transponder	TA, Vertical RA
Mode S Transponder	TA, Vertical RA
TCAS I	TA, Vertical RA
TCAS II	TA, Vertical RA, TCAS – TCAS Coordination
TCAS III	TA, Vertical RA, TCAS – TCAS Coordination

The TCAS II System has the components that follow:

- Processor, TCAS (34–43–01)
- Antenna, Directional (34–43–11).

The TCAS II interfaces with the systems that follow:

- Audio Integration System (AIS) (23–51–00)
- Unit, Audio and Radio Control Display (ARCDU) (23–81–01)
- Module, Input/Output Processor (31–41–06)
- Module, Input/Output (31–41–11)
- Module, Aircraft Configuration (31–41–21)



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- Unit, Proximity Sensor Electronic (32–61–01)
- Radio Altimeter (34–44–00)
- Air Traffic Control Transponder (34–54–00)
- Central Diagnostic System (45–00–00).

The ARCDU1 and ARCDU2 are used to make the TCAS II mode selections.

Detailed Description

The mode S transponder gives airborne to airborne altitude reporting data between TCAS equipped aircraft.

The TCAS Processor Unit (TPU) does the functions that follow:

- Surveillance
- Collision avoidance calculations
- Monitor intruder aircraft and own aircraft.

Surveillance: The TCAS II monitors a 14 nautical mile (26 kilometer) area around the aircraft independently of the Air Traffic Control (ATC) system. The TCAS II monitors for squitter transmissions from other mode S transponder equipped aircraft. The squitter transmission gives the aircraft's unique address. After the squitter is received, the TCAS II then interrogates that mode S address, and from the reply, it calculates the range, bearing, and if available, the altitude of the other aircraft.

To minimize interference with other aircraft and ATC, the interrogation rate changes with the range, closing speed, and when it calculates a TA.

The range and altitude of each mode S transponder equipped aircraft is monitored for collision avoidance. A relative bearing track is also calculated for a traffic advisory indication only.

The TCAS II uses a mode C all-call to interrogate the ATCRBS transponder equipped aircraft.

The mode A transponder equipped aircraft reply with no altitude data. The TCAS II uses their framing pulses to calculate their range and bearing. This data is shown on the EIS navigation pages as non-threat, proximity intruder, or TA indications.

The mode C transponder equipped aircraft reply with altitude data. The TCAS II uses their replies to calculate their range, bearing, and altitude. This data is shown on the EIS navigation pages as non-threat, proximity intruder, TA, or RA indications.

[Refer to Figure 3.](#)

When the two aircraft are less than 1.7 nmi (3.15 km) apart, one aircraft will start to reply to an interrogation before all of the data is received. This interference is called synchronous garble.

The TCAS II also receives unwanted ATCRBS transponders replies from interrogations from other TCAS equipped aircraft and ATC ground stations. This is called nonsynchronous garble or fruit replies.

The TCAS II decreases synchronous and nonsynchronous garble interference by the methods that follow:

- An ATCRBS P2 sidelobe suppression pulse is transmitted in an omnidirectional pattern on the directional antenna to control the effective beam width



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- A sequence of ATCRBS interrogation are transmitted at different power levels. The whisper–shout levels divide the surrounding area into range segments. The maximum number of whisper–shout levels is 24. They are used when interrogating in the forward direction.

The whisper–shout uses the differences between the ATCRBS transponder receiver sensitivities. A low power level (Whisper) is used for the first interrogation in the whisper–shout sequence. During the next sequence, a suppression pulse that is less than the previous interrogation is transmitted followed by an interrogation 2 µsec later that is more than the previous interrogation (Shout) transmission. This procedure suppresses most of the ATCRBS transponders that had replied to the previous interrogations and get replies from the ATCRBS transponders that did not reply to previous interrogations.

The whisper–shout procedure separates the ATCRBS replies into several groups to reduce the possibility of garbling.

- Each ATCRBS transponder is interrogated more than once to check the reply
- A P2 ATCRBS transponder suppression pulse is transmitted with a mode S interrogation transmission
- A P4 mode S suppression pulse is transmitted with a ATCRBS transponder interrogation transmission.

NOTE

Note: There is a extremely remote possibility that the TCAS II will track synchronous or nonsynchronous replies.

Collision Avoidance Calculations: The collision avoidance logic makes a comprise between necessary protection and not necessary indications. The sensitivity level is changed to change the safe area around the aircraft.

The TCAS II monitors the slant range and closure rate of each intruder aircraft to calculate the time in seconds to a CPA. If the intruder aircraft has a mode C ATCRBS transponder, the TCAS II also monitors its altitude to calculate the altitude of the intruder at the CPA. The altitude of the intruder is used to calculate its vertical speed.

The system can monitor forty–five aircraft and show the eight most dangerous intruder aircraft.

[Refer to Figure 4.](#)

When the aircraft is less than 1700 feet (518 meters) Above Ground Level (AGL), its TCAS II uses the difference between its aircraft pressure altitude and radio altitude to estimate the elevation of the ground. It then subtracts the estimated elevation of the ground value from the intruder aircraft's pressure altitude value to calculate its AGL altitude. If it is less than 180 feet (55 meters), the intruder aircraft is considered to be on the ground and an indication is not shown.

[Refer to Figure 5.](#)

If the a threat condition is calculated, a upward or downward RA is calculated based on a 1500 ft/sec (457 m/sec) vertical speed that gives more vertical separation at the CPA. The vertical separation



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changes between 400 and 740 feet (122 and 225.5 meters) depending the own aircraft's altitude.

[Refer to Figure 6.](#)

If the calculated vertical separation is not between 400 and 740 feet (122 and 225.5 meters) at the CPA, a crossover RA is then calculated.

The RA shows a safe vertical speed to climb or descent throughout the altitude of the threat aircraft.

[Refer to Figure 7.](#)

A change in the intruder aircraft's vertical speed has an effect on the calculated RA. If the threat aircraft does not have a TCAS installed, an increase vertical speed (climb or descent) or sense-reversal RA is calculated.

If the intruder aircraft has TCAS installed, an increase vertical rate (climb or descent) RA is calculated. It will not calculate a sense-reversal RA.

When the aircraft altitude is less than 1450 AGL, the increase descent RAs are not calculated and when the aircraft altitude is less than 1000 AGL, the descent RAs are not calculated.

NOTE

Note: A climb or increase climb RA is not shown during a high altitude or landing condition.

[Refer to Figure 8.](#)

If the intruder aircraft does not have a TCAS II or TCAS III system, an RA reversal is calculated when the intruder aircraft reverses its vertical speed to show an opposite RA indication.

[Refer to Figure 9.](#)

The TCAS II can monitor many aircraft situations. One RA is calculated that maintains a safe separation from each threat aircraft.

When the aircraft senses another aircraft that is equipped with TCAS, each aircraft interrogates the other's mode S transponder to make sure that complementary RA are calculated.

Coordination interrogations are transmitted once per second during the RA condition. The coordination interrogations have data that is related to its RA intention when the other aircraft causes the RA. If one aircraft has calculated a climb RA intention, it will transmit a message in the coordination interrogation to the intruder aircraft to make it calculate a descent RA intention. The amount of the descent RA depends on the geometry of the situation. The RA intentions are checked by each TCAS and opposite RA are calculated.

Each aircraft usually senses the other as a threat at different times. The first aircraft transmits its RA intention and, a small time later, the other aircraft transmits the opposite intention but if the two aircraft sense each other as threats and the same time, a RA is calculated based on geometry. If the two aircraft are flying level at the same



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altitude, the two systems will calculate the same RA. The aircraft with the higher mode S address will reverse its RA.

The TCAS system operates in the following modes:

- Active mode
- Standby Mode
- Test Mode
- System Mode.

Active Mode: When a mode S ATC Transponder (ATC1, ATC2) is on and the TCAS II system mode is not set to standby, the TCAS processor unit operates in an active mode, TA/RA mode or TA ONLY mode. The TA/RA active mode only operates when the aircraft is airborne. The TA ONLY active mode operates when the aircraft is on the ground and when its is airborne.

During the active mode, the TCAS Processor transmits mode S and ATCRBS ATC system interrogations to other aircraft at 1030 MHz through the top and bottom antennas. The TCAS processor unit decodes and calculates 1090 MHz replies received from ATC system equipped aircraft in ATCRBS or mode S formats.

The TCAS II system operates in collision avoidance environments that follow:

- ATCRBS
- Mode S.

ATCRBS Collision Avoidance Environment: The TCAS transmits an all call interrogation to other aircraft equipped with ATCRBS transponders. The intruder aircraft's pressure altitude (mode C data) is received, and the direction and range of the ATCRBS transponder equipped intruder aircraft in the surveillance area are calculated.

Mode S Collision Avoidance Environment: The intruder aircraft equipped with a mode S transponder continuously transmits squitter at one-second intervals. These signals alternate between the top and bottom antennas. The transmissions have data that specify its unique Mode-S, 24 bit address vertical status and equipment functions. The TCAS II equipped aircraft system monitors the squitter signals to indicate that a Mode-S ATC Transponder equipped aircraft has entered the surveillance area.

The TCAS II system then interrogates the intruder aircraft. The intruder aircraft answers to valid mode S interrogations with mode S surveillance signals. The TCAS II system uses this data to calculate the range and direction of each mode S transponder equipped intruder aircraft in its surveillance area. The TCAS II system individually interrogates and tracks each intruder aircraft using the unique Mode-S address of each aircraft.

If the intruder aircraft is equipped with TCAS, the Mode-S communications, that includes coordination maneuvers, between two TCAS equipped aircraft occur from the TCAS antenna of one aircraft to the mode S transponder antenna of the other aircraft. Aboard the TCAS equipped aircraft, data received by the Mode-S transponder is supplied to the TCAS Processor and the processor determines the content of the reply messages.

Standby Mode: During the standby mode, the TCAS processor unit stays in an energized state and its Built In Test (BIT) does continuous tests. The TCAS II does not transmit interrogation in the standby mode.

Test Mode: When the aircraft is on the ground and the TCAS II and ATC systems are set to the standby mode a test selection starts a self test to determine the system's operational status. The TCAS processor unit also sends a status signal directly to the Central Diagnostic System (CDS).



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Refer to Figure 10.

The ARCDU1 and ARCDU2 selections control the TCAS II modes to be used.

Table 2: ARCDU Selection

SELECTION	DESCRIPTION
OFF	The ARCDU is not energized
ON	The ARCDU is energized and it controls and tunes its related radio system
BOTH	The ARCDU is energized and it controls and tunes its related and opposite radio systems
FMS	The FMS controls and tunes its related and opposite radio systems

Refer to Figure 11.

The side key adjacent to the ATC display area is pushed and then the EXP key to show the TCAS/ATC particular page.

Refer to Figure 12.

A TCAS/ATC label is shown in black letters on a white background in the center of the top line. The TCAS/ATC particular page is used to control the TCAS II functions that follow:

- Intruder aircraft altitude, absolute or relative to own aircraft
- Surveillance area, normal, above, or below
- System mode, TA and RA, TA only, or standby

– Test.

Intruder Aircraft Altitude: The side key adjacent to the REL, ABS display area is pushed to change the highlighted selection. It moves in a wrap round manner from REL to ABS. The selection changes to black letters on a green background from the usually white on black background. The Multi Function Display (MFD) navigation page shows only a TCAS ABS message.

Surveillance Area: The side key adjacent to the ABOVE, NORM, BELOW display area is pushed to select the surveillance area. The selection highlights and moves in a wrap round manner from NORM, BELOW, to ABOVE. It changes to black letters on a green background from the usually white on black background. The MFD navigation pages show a TCAS ABV or TCAS BLW message.

The surveillance area is limited to the altitude that follows:

Table 3: Surveillance Area Selection

SELECTION	ALTITUDE ABOVE	ALTITUDE BELOW
ABOVE	9000 feet	2700 feet
NORM	2700 feet	2700 feet
BELOW	2700 feet	9000 feet

System Mode: The side key adjacent to the TA/RA, TA ONLY, SBY display area is pushed to set the system mode. The indication in the display area changes to reverse video to show the selection. The selection changes from SBY, TA/RA, to TA ONLY with each side key selection. The MFD navigation pages show a TCAS STBY or TCAS TA ONLY message.



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TA/RA: The TA/RA mode shows traffic and resolution advisory intruders. The TA ONLY mode shows traffic advisory intruders only. The standby mode sets the TCAS processor unit to the standby mode.

TA ONLY: The TCAS processor operates in the TA only mode when the radio altitude is less than 1000 feet (305 meters) AGL. When the AGL altitude is more than 1000 feet (305 meters), the TCAS II mode changes to TA/RA. When the AGL altitude is less than 100 feet (30.5 meters), the mode changes back to TA.

SBY: The TCAS processor unit and Mode S transponder do not transmit when the mode is set to STB.

NOTE

The ATC display area also shows a SBY indication.

Test Mode: The side key adjacent to the TEST display area is pushed to cause the TCAS processor unit to supply the test indications that follow:

- Intruder symbol test pattern and TCAS TEST are shown in white and yellow text on the MFDs and IVSI part of the PFD
- A climb/descent symbol test pattern is shown on the IVSI part of each PFD
- The "TCAS System Test OK" message sounds if a malfunction is not sensed and a "TCAS System Test Fail" if a malfunction is sensed.

The test mode shows the MFD indications that follow:

- RA symbol that is at the 3 o'clock position, 2 mile (3.22 kilometer) range, 200 feet (61 meters) above, and flying level

- TA symbol that is at a 9 o'clock position, 2 mile range (3.22 kilometer), 200 feet below, and climbing
- Proximity intruder traffic symbol that is at 1 o'clock position, 3.6 mile (5.79 kilometer) range, 1000 feet (305 meters) below, and descending
- Non-threat traffic that is at 11 o'clock position, 3.6 mile (5.79 kilometer) range, 1000 feet (305 meters) above, and flying level.

The test mode also shows the PFD IVSI indications that follow:

- Green arc from 0 to 250 ft/min
- Red arc from 0 to -6,000 ft/min
- Red arc from +2,000 to +6,000 ft/min.

[Refer to Figure 13.](#)

The MFD navigation displays shows TCAS automatic (pop-up) or continuous indications. When the navigation page is set to arc or map mode and the range is set to 40 nautical miles or less, the TCAS push-button switch on the Electronic Flight Control Panel (EFCP) is pushed to show TCAS traffic continuously. If the TCAS push-button switch is pushed again, the automatic mode starts again.

If the EFCP malfunctions, the automatic mode is set.

The automatic mode only shows TA and RA indications. When a TCAS TA or RA condition is sensed, the ARC or MAP format and the 10 nmi range is automatically shown. An AUTO RANGE message



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below the outer range digital mark shows that the range indication does not relate to the range selection on the EFCP.

NOTE

Note: A TA or RA condition prevents the EFCP switch selections that follow:

- TCAS push-button
- RANGE rotary switch
- FULL/ARC push-button.

After the TA or RA condition, the 10 nmi range indications stays in view. The TCAS push-button switch is pushed to reset the indication.

[Refer to Figure 14.](#)

The MFD navigation page shows the TCAS parameters that follow:

- Mode messages
- Range ring
- Intruder symbols
- No-bearing advisory.

Mode Messages: The TCAS mode selections are shown with the indication priority that follows:

- TCAS FAIL
- TCAS STBY
- TCAS TEST
- TCAS TA ONLY

- TCAS ABV or BLW, ABS.

NOTE

Note: When the operational mode is set to TA/RA, no message is shown. Also, no message is shown for the threat aircraft altitude relative (REL) and surveillance area normal (NORM) selection.

[Refer to Figure 15.](#)

TCAS FAIL Mode Message: The TCAS FAIL mode message is shown in yellow letters when the TCAS processor unit malfunctions or when the display unit does not receive valid data from the TCAS processor unit through the IOPs.

On aircraft with ARCDU P/N CDU3933AE04, the TCAS FAIL message is an inherent nuisance message and it must be ignored.

On aircraft with ARCDU P/N CDU3933AE05, the TCAS FAIL message is removed.

When TCAS FAIL mode message is shown, no other TCAS symbols are shown.

[Refer to Figure 16.](#)

TCAS STBY Mode Message: The TCAS STBY mode is shown in white letters when the TCAS processor units is in the standby mode.

NOTE

Note: When the mode S transponder is set to the standby mode, the TCAS will also automatically operate in the standby mode.



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When TCAS STBY mode message is shown, no other TCAS symbols are shown.

[Refer to Figure 17.](#)

TCAS TEST Mode Message: The TCAS TEST mode message is shown in white and yellow letters when the TCAS processor unit does a self test. TCAS is shown white and TEST is shown in yellow.

[Refer to Figure 18.](#)

TCAS TA ONLY Mode Message: The TA ONLY message shows the TA only operational mode selection. No RAs will not be shown.

[Refer to Figure 19.](#)

TCAC ABV or BLW and ABS Mode Message: The ABV or BLW message shows the above or below mode selection.

The ABS mode message shows absolute altitude mode selection. The absolute altitude is shown in hundreds of feet using two digits with the intruder symbol.

The relative altitude mode does not show a mode message.

Range Ring: The TCAS range ring shows a 2 nmi (1.85 km) circular area around the aircraft symbol. It is shown as white short line segments or dots when the RANGE rotary switch on the EFCP is set to the 10, 20, or 40 nmi range. When the RANGE rotary switch is set to a range that is more than 40 nmi it is out of view. A TA or RA condition will cause the 10 nmi range and TCAS range ring indications to come into view with a white AUTO RANGE message below the outer range digital mark.

In the automatic mode, the range ring comes into view only when a TA or RA condition occurs. During a TA or RA condition, another

range selection cannot be made. The auto range mode can be cancelled after the alert has ended.

The MFD shows the eight most dangerous intruder aircraft, as calculated by the TCAS. These intruder indications are shown on the MFD navigation pages to identify surrounding intruder aircraft that are equipped with mode S or ATCRB mode A or C transponders.

[Refer to Figure 20.](#)

The MFD shows the intruder symbols that follows:

- Non-threat traffic
- Proximity intruder traffic
- TA
- RA
- No-bearing advisory
- Off Scale traffic.

Non-Threat Traffic (Intruder): The MFD navigation page shows non-threat traffic as a open cyan diamond symbols with their relative altitude.

Proximity Intruder Traffic (Intruder): The MFD navigation page shows proximity intruder traffic as filled blue diamond symbols with their relative altitude.

Traffic Advisory (Intruder): The MFD navigation page shows traffic advisories as filled yellow circle symbols with their relative altitude.



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Resolution Advisory (Threat): The MFD navigation page shows resolution advisories as filled red circle symbols with their relative altitude.

NOTE

For better viewing, the full intruder symbol is shown on a rectangular black background.

No-Bearing Advisory: The no-bearing advisory; indication shows the two most dangerous TA or RA intruder aircraft that do not have bearing data. It is shown below the aircraft symbol when the range is set to 10, 20, or 40 nmi.

Each line is used to show data that follows:

- Threat level, RA or TA
- Range in nautical miles from 0.0 to 99.9 nmi with 0.1 nmi resolution
- Relative or absolute altitude in hundreds of feet
- Vertical speed trend in the same format as for the traffic symbols.

The indication is shown in red for an RA and yellow for a TA.

When an intruder aircraft is not reporting altitude, the altitude and trend arrow are not shown.

Off Scale Traffic: If the TA or RA aircraft range is more than the range selection, one half of the symbol is shown at the edge of the indication. Its position gives the bearing of the intruder.

The relative altitude mode shows the relative altitude between the own aircraft and the intruder is shown as follows:

- With + symbol if the intruder is at a higher altitude

- With – symbol if the intruder is at a lower altitude.

The relative altitude is shown in hundreds of feet using two digits. It has a range of –9900 feet to +9900 feet with 1000 feet accuracy. If the relative altitude is less than 1000 feet, a leading zero is used. A 00 indication shows the other aircraft at the same altitude. This indication is shown in the same color as the intruder symbol.

Their relative or absolute altitude is shown with the intruder indication.

A vertical arrow is also shown to the right of the symbol if the vertical speed of the intruder is more than 500 ft/min. An up-pointing arrow shows climb condition and a down-pointing arrow shows a descent condition. The arrow is shown in the same color as the symbol.

When an intruder aircraft is not reporting altitude, the altitude and trend arrow is not shown.

[Refer to Figure 21.](#)

The RAs are grouped as corrective advisories or preventive advisories. The vertical flight path of the aircraft must be changed when a corrective advisory is shown and maintained for a preventive advisory.

A pointer and vertical speed scale, which are part of the IVSI, shows the present vertical speed of the own aircraft. For TCAS II guidance, the IVSI scales show red and green command indications.

The red and green RA command indications give RA maneuvering guidance to the pilot.

If the pointer is in a red command area, the aircraft's vertical speed must be changed to put the pointer in the green command area. This is a corrective resolution advisory.



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If the pointer is not in a red command area, the aircraft's vertical speed must be maintained to keep the pointer out of the red command area. This is a preventive RA.

The TCAS RA indications also sound the messages that follow:

Table 4: RA Messages

RA Category	Corrective	Preventive
Climb	Climb, Climb, Climb	Monitor Vertical Speed, Monitor Vertical Speed
Descent	Descend, Descend, Descend	Monitor Vertical Speed, Monitor Vertical Speed
Crossover Climb	Climb, Crossing Climb, Climb, Crossing Climb	Monitor Vertical Speed, Monitor Vertical Speed
Crossover Descent	Descend, Crossing Descend, Descend, Crossing Descend	Monitor Vertical Speed, Monitor Vertical Speed

Vertical Speed Restricted (Climb)	Reduce Climb, Reduce Climb	Monitor Vertical Speed, Monitor Vertical Speed
Vertical Speed Restricted (Descend)	Reduce Descent, Reduce Descent	Monitor Vertical Speed, Monitor Vertical Speed

NOTE

The "Monitor Vertical Speed" message sounds once if downgraded from a previous corrective advisory.

The RA increases and reversal from the data shown in Table 4 are shown in the table that follows:

Table 5: RA Messages

RA Category	Corrective	Preventive
Change from Climb to Descent	Descend, Descend NOW, Descend, Descend NOW	Not Applicable
Change from Descent to Climb	Climb, Climb NOW, Climb, Climb NOW	Not Applicable



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Increase Climb Rate	Increase Climb, Increase Climb	Not Applicable
Increase Descent Rate	Increase Descent, Increase Descent	Not Applicable

TCAS II indications are shown in the table that follows:

Table 6: TCAS II Indications

Aural	Visual
CLIMB, CLIMB, CLIMB	VSI RED from –6000 to +1500 ft/min, and GREEN from +1500 to +2000 ft/min.
DESCEND, DESCEND, DESCEND	VSI RED from +6000 to –1500 ft/min, and GREEN from –1500 to –2000 ft/min.
MONITOR VERTICAL SPEED, MONITOR VERTICAL SPEED	Current vertical speed is outside the RED arc as shown on the VSI
REDUCE CLIMB, REDUCE CLIMB	VSI shows prohibited vertical speed in RED. Goal is vertical speed in GREEN.
CLIMB, CROSSING CLIMB, CLIMB, CROSSING CLIMB	Same as 'CLIMB", and further shows that own flight path will cross that of intruder.

DESCEND, CROSSING DESCEND, DESCEND, CROSSING DESCEND	Same as "DESCEND", and further shows that own flight path will cross that of intruder.
CLEAR OF CONFLICT	VSI RED and GREEN arcs removed. Range is increasing and sufficient.
INCREASE CLIMB, INCREASE CLIMB	Follows a "CLIMB" advisory. VSI RED from –6000 to +2500 ft/min, and GREEN from +2500 to +3500 ft/min.
INCREASE DESCENT, INCREASE DESCENT	Follows a "DESCEND " advisory. VSI RED from +6000 to –2500 ft/min, and GREEN from –2500 to –3500 ft/min.
CLIMB, CLIMB NOW, CLIMB, CLIMB NOW	Follows a "DESCEND" advisory when it has been found that a reversal of vertical speed (direction) is needed to give sufficient separation.
DESCEND, DESCEND NOW, DESCEND, DESCEND NOW	Follows a "CLIMB" advisory when it has been found that a reversal of vertical speed (direction) is needed to give sufficient separation.



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Examples of the conditions shown in Table 4, 5 and 6 are shown in figures 23 through 30.

[Refer to Figure 22.](#)

If the intruder is more than 10 nmi (18.5 km) ahead, at 12:00 o'clock, and 200 feet (61 meters) below, an off-scale indication is shown on the MFD. A "TRAFFIC, TRAFFIC" message also sounds.

[Refer to Figure 23.](#)

If the intruder is 4 nautical miles (7.4 kilometers) ahead, at 12:00 o'clock, and 200 feet (61 meters) below, an RA indication is shown on the MFD. A 1500 ft/min (457 m/min) rate of climb is shown on the IVSI and a "CLIMB, CLIMB, CLIMB" message also sounds.

[Refer to Figure 24.](#)

If the intruder is 4 nautical miles (7.4 kilometers) ahead, at 12:00 o'clock, and 200 feet (61 meters) above, an RA indication is shown on the MFD. A 1500 ft/min (457 m/min) rate of descent is shown on the IVSI and a "DESCEND, DESCEND, DESCEND" message also sounds.

[Refer to Figure 25.](#)

If the intruder is 4 nautical miles (7.4 kilometers) ahead, at 12:00 o'clock, and 600 feet (183 meters) below, an RA indication is shown on the MFD. A do not descent indication is shown on the IVSI and a "MONITOR VERTICAL SPEED" message also sounds.

[Refer to Figure 25.](#)

If an intruder is at 12:00 o'clock and 500 feet (152 meters) above, and another is at 12:00 o'clock and 500 feet (152 meters) below, two RA indications are shown on the MFD. A do not descent indication is

shown on the IVSI and a "MONITOR VERTICAL SPEED" message also sounds.

[Refer to Figure 25.](#)

If the intruder is 4 nautical miles ahead, at 12:00 o'clock, and 500 feet (152 meters) above, and you are climbing at 2000 ft/min (610 m/min), an RA indication is shown on the MFD. A do not descent indication is shown on the IVSI and a "MONITOR VERTICAL SPEED" message also sounds.

[Refer to Figure 26.](#)

If the intruder is 4 nautical miles ahead, at 12:00 o'clock, 100 ft (30.5 m) below and descending, and you are descending at 1500 ft/min (457 m/min), a RA indication is shown on the MFD. A 1500 ft/min (457 m/min) rate of climb is shown on the IVSI and a "CLIMB, CLIMB NOW" message also sounds if the aircraft descending.

[Refer to Figure 27.](#)

If the intruder has passed and is not a threat, a "CLEAR OF CONFLICT" message sounds.

NOTE

The "Monitor Vertical Speed" message sounds once if downgraded from a previous corrective advisory.

The aircraft electrical system left main bus supplies 28 Vdc electrical power through a 5 A circuit breakers to the TCAS processor unit. The circuit breaker is located in position F1 on the 28 Vdc avionics circuit breaker panel.

The Distance Measuring Equipment (DME) and the ATC Mode S Transponders (ATC1, ATC2) operate in the same frequency range. Each sends a suppression signal to the other systems before transmitting.



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The TCAS parameters that follow are controlled using discrete signals and ARINC 429 low speed busses (12.0 to 14.5 kilobits/sec):

The TCAS system interfaces with the following:

- Input Output Module (IOM)
- Input–Output Processor (IOP)
- Proximity Sensing Electronics Unit (PSEU)
- Radio Altimeter (RA)
- Mode–S ATC Transponders.

IOM: The IFCs have the TCAS interfaces that follow:

- The TCAS TA and RA event discrete signals are sent to the IOMs and are monitored by the Flight Data Recorder (FDR) system
- A TCAS valid discrete is sent to the IOMs and is monitored by the CDS to know the status of the TCAS Processor
- The climb inhibit and increase climb inhibit discrete signals are sent from the IOMs and are used by the TCAS processor unit
- A TA aural traffic discrete is sent to the IOMs and is monitored by the Warning Tone Generator (WTG) to calculate the prioritizing of all the aircraft warnings
- The TA/RA indications are supplied to the PFDs and MFDs through ARINC 429 data busses and the IOPs
- The heading and attitude data is supplied to the TCAS processor unit from IOP2 through an ARINC 429 data bus.

PSEU: The PSEU supplies the TCAS processor unit with the discrete signals that follow:

- A Landing gear status discrete output is supplied to indicate that the landing gears of the aircraft are extended
- An air/ground status discrete output is supplied to indicate that the aircraft is on the ground or airborne.

RA: The RA supply AGL data to the TCAS processor unit through an ARINC 429 data bus.

Mode S ATC Transponders: The phase shift modulated signal received from the TCAS processor unit is changed into a Transistor–Transistor Logic (TTL) level signal. The TTL signal represents the 56 or 112 bit word from either TCAS II or the ATC ground station transmitter. The transponder decodes the last 24 bits of the signal, and then the remaining received message. The last 24 bits of the signal are the strapped address of the transmitting aircraft.

Audio Integrating System: The TCAS processor unit supplies an analog audio announcement output to the Remote Control Audio Unit (RCAU) part of the AIS with audio, when enabled by the WTG.

Processor, TCAS

[Refer to Figure 28.](#)

The TCAS Processor supplies intruder position and status information to the PFDs and MFDs through the IFCs and directly to the AIS.

The TPU has the following main assemblies:

- RF front assembly
- Receiver assembly



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- Transmitter/modulator assembly
- Digital computer I/O assembly
- Power supply assembly.

RF front end assembly: The RF front end assembly interfaces between each TCAS antenna and the TPU transmitter and receiver modules. This assembly helps to do multifunctional tasks, including bearing processing. Other multifunctional tasks include forming transmitter beam characteristics, antenna cable calibration, and switching between top and bottom antennas. The front end assembly is used in both transmit and receive modes.

Receiver assembly: The receiver assembly has a 1090 MHz receiver and a 1030 MHz receiver. The 1090 MHz receiver has three subassemblies; an RF board, an Intermediate Frequency (IF) board, and a video processor board. The video processor board does frequency conversion, amplification, detection, decoding, and intruder bearing estimates. The 1030 MHz receiver uses the TCAS broadcast from other aircraft to count TCAS equipped aircraft. These data are sent to the digital computer I/O assembly's computer board.

Transmitter/modulator assembly: The transmitter/modulator assembly supplies the transmit function for the TPU. It has five major subassemblies; a 1030 MHz reference, a transmitter, a modulator, a whisper-shout attenuator, and a whisper-shout driver.

Digital computer I/O assembly: The digital computer I/O assembly has two boards; a computer board and an input/output board. The computer board is the main processing unit of the TPU. The board uses 32-bit bus architecture with an Intel 80960 KB microprocessor that runs at a 16 MHz clock rate. The computer board controls the video processor board and the I/O board. It does all the serial data transmission through ARINC 429, RS-232, and RS-422. The I/O

board is the interface between the computer board and the various LRUs of the TCAS II system.

Power Supply Module: The power supply module changes the aircraft 28 to ± 12 , +5, +30, and -250 Vdc. The power supply also supplies +15 Vdc for the audio amplifier (located on the I/O board). During power interruptions, the power supply holds a +5 Vdc sustaining power for 1 second.

The TCAS processor weighs 16.5 lb (7.48 kg). It is contained in a 1/2 ATR short form rack. The TPU is installed in an ARINC 404A compatible rack that weighs 3.0 lb (1.36 kg). The processor mounting tray is located with the two ATC transponder mounting trays.

The TPU mounting trays are installed below the floor in the forward portion of the center fuselage. The TCAS processor unit mounting tray is located at station X+64.6, Y-3.5, and Z+89.6.

The TCAS CM contains the EEPROM memory and is used to store the TCAS II system strapping data and bearing calibration information that is input to the TCAS processor. The CM weighs 0.1 lb (0.045 kg) and it is mounted under the floor at station X+71.20.

On aircraft with ModSum 4Q309289 installed, the TCAS processor weighs (nominal) 13.47 lb (6.11 kg). The TPU is installed in an ARINC 600 compatible rack.

On aircraft with ModSum 4Q309289 installed, the TPU mounting trays are installed below the floor in the forward portion of the center fuselage. The TCAS processor unit mounting tray is located at station X+71.2, Y0.0, and Z+86.35.

On aircraft with ModSum 4Q309289 installed, the compact flash card is installed in the slot at the front face of the TCAS processor to upload the new operating software and record the internal and external data.



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On aircraft with ModSum 4Q309289 installed, the CM is not installed.

On aircraft with SB84–34–135 incorporated, the TCAS processor can perform the system self test with the ATC in a standby mode. It can operate with a new control panel/transponder interface and ensures a correct TCAS status display when the system is first powered up.

Antenna, Directional

Refer to Figure 29.

The directional antenna lets the bearing of intruder aircraft to be calculated by giving a means for the TCAS processor unit to transmit interrogations and receive reply receptions on one of four antenna beams. By selecting the beam, the TCAS processor electronically points the antenna in a surveillance direction during TCAS transmit and receive operations.

The directional antenna is an electronically steerable phased array consisting of four top loaded monopole elements.

The antenna pattern is pointed in one of four directions without physically moving the antenna. This is done by independently changing the phase to each of the four antenna elements contained in the directional antenna. The phasing of these signals directly affect the beam, therefore, the phase match of the coaxial cables is very important.

The four beam forming elements are located on the perimeter of the antenna, and the whole assembly is contained in a aerodynamic plastic enclosure.

During TCAS receptions, each of the four directional antenna elements receives any 1090 MHz signal that passes by the element.

The phasing of these received signals calculated by the direction from which the rf energy is received. These signals are directed onto the same four cables that connect transmit signals between the TCAS processor unit and the directional antenna.

Each part of the four ports of the directional antenna has a resistor located across the antenna element to ground. Each of the four resistors is the same value. The TCAS processor unit periodically does a continuity check on the antenna ports and should sense the correct resistance value through an analogue to digital converter if the port is not shorted or not open. The directional antenna is a passive device and does not require aircraft electrical power to function.

The antenna is a low profile (maximum overall height of 1.36 in. (34.54 mm)) teardrop that is made from advanced thermoplastic material. All radiated elements are preformed and encapsulated within the antenna as a single void-free assembly. An aluminum base plate is pre-located within the antenna housing together with the four relevant connector types. Each antenna has a maximum weight of 2.0 lb (0.94 kg).

An aluminum foil gasket with an elastomeric sealant makes sure consistent distribution of conducting contacts of the gasket to the airframe. The pressure applied during installation gives a contouring feature. It lets the gasket conform to the two mating surfaces.

The top antenna is located at station X+21.3, Y+11.0, and Z+182.6. The bottom antenna is located at station X–45.0, Y0.0 and Z+89.0. Each antenna is attached to the aircraft with four mounting screws.

On aircraft with ModSum 4Q309289 installed, the antenna has a low profile height of 0.806 in. (20.47 mm). The antenna consists of a molded radome with radiating/receiving elements and is completely filled with a rigid foam. The top and the bottom antennas are

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directional antennas. Each antenna is preinstalled with an adapter plate and weighs 3.0 lb (1.36 kg).

On aircraft with ModSum 4Q309289 installed, the top antenna is located at station X+11.0 and Y+4.0. The bottom antenna is located at station X-45.0 and Y0.0. Each antenna is attached to the aircraft with four mounting screws and washers.

Light, Annunciator

[Refer to Figure 30.](#)

On aircraft with ModSum 4-309289 installed, the annunciator lights installed on the front face of the TCAS processor give the indications of the failure as follows:

TCAS PASS: This Light Emitting Diode (LED) comes on if the unit passes its own internal Built In Test Equipment (BITE) test.

TCAS FAIL: This LED comes on if the unit has not passed its own internal BITE test.

TOP ANT: This LED comes on when there is a failure in the DC resistance test of the top antenna.

BOT ANT: This LED comes on when there is a failure in the DC resistance test of the bottom antenna.

C/F LOAD STATUS: This LED comes on red if there is an error in software upload through the compact flash drive. The light C/F LOAD STATUS comes on green, if the software upload is successful.

TA DISP: This LED comes on if the traffic advisory display discrete signals from the No. 1 or the No. 2 TCAS system has failed.

RA DISP: This LED comes on if the resolution advisory display discrete signals from the No. 1 or the No. 2 TCAS system has failed.

RAD ALT: This LED comes on if the radio altitude source from the No. 1 or No. 2 radio altimeter system is invalid or has failed.

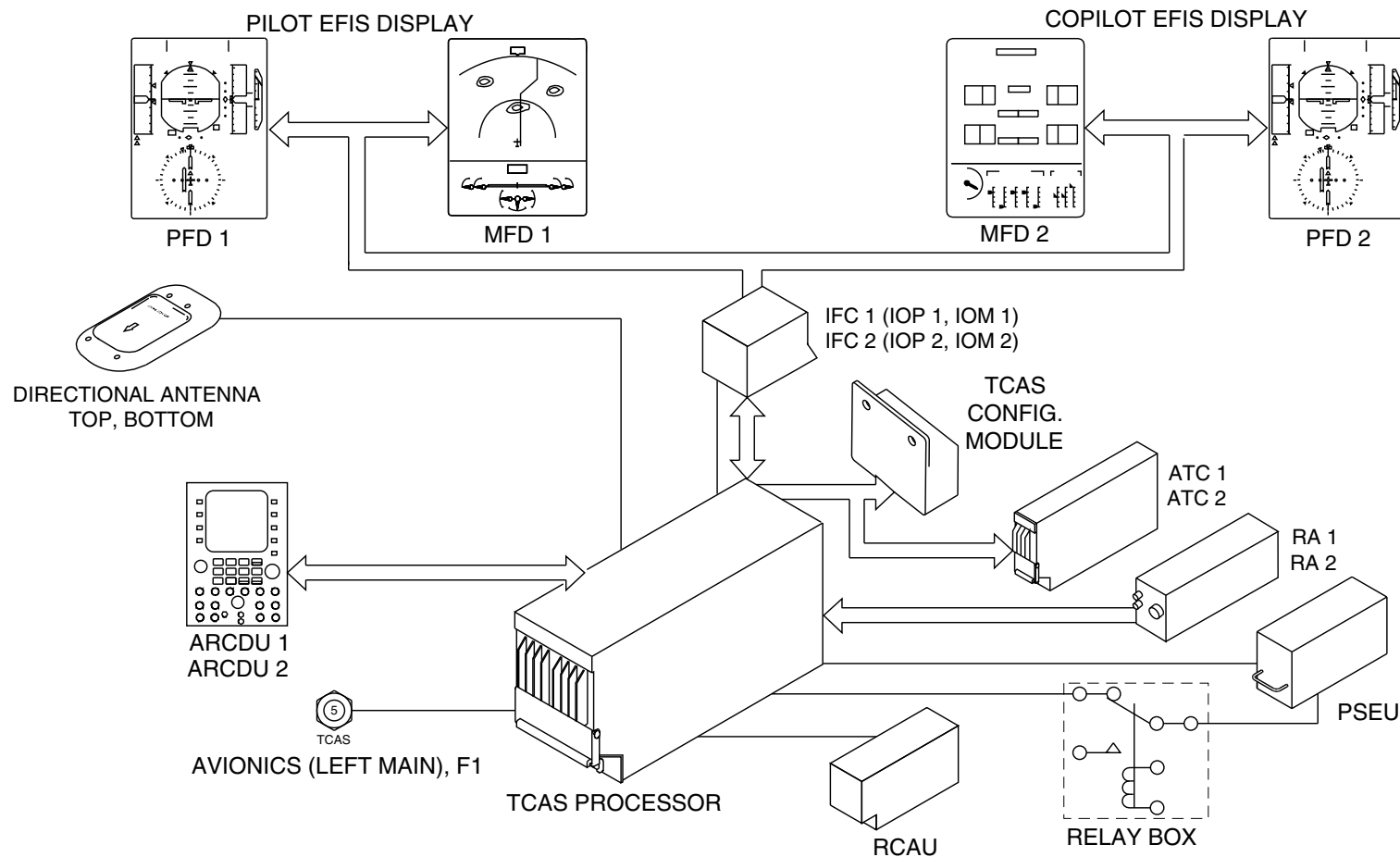
XPDR BUS: This LED comes on if the mode S transponder signals from the Air Traffic Control (ATC) No. 1 or the No. 2 system is invalid or has failed.

XFER IN PROCESS: This LED comes on to show the software upload status through the compact flash drive. When the TCAS processor reads the compact flash drive, the LED blinks. If there is an error in the software upload through the compact flash drive, the LED turns to red. The LED turns to green, if the software upload is successful.



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PRE MODSUM 4-309289

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TCAS II Block Diagram
Figure 1 (Sheet 1 of 2)

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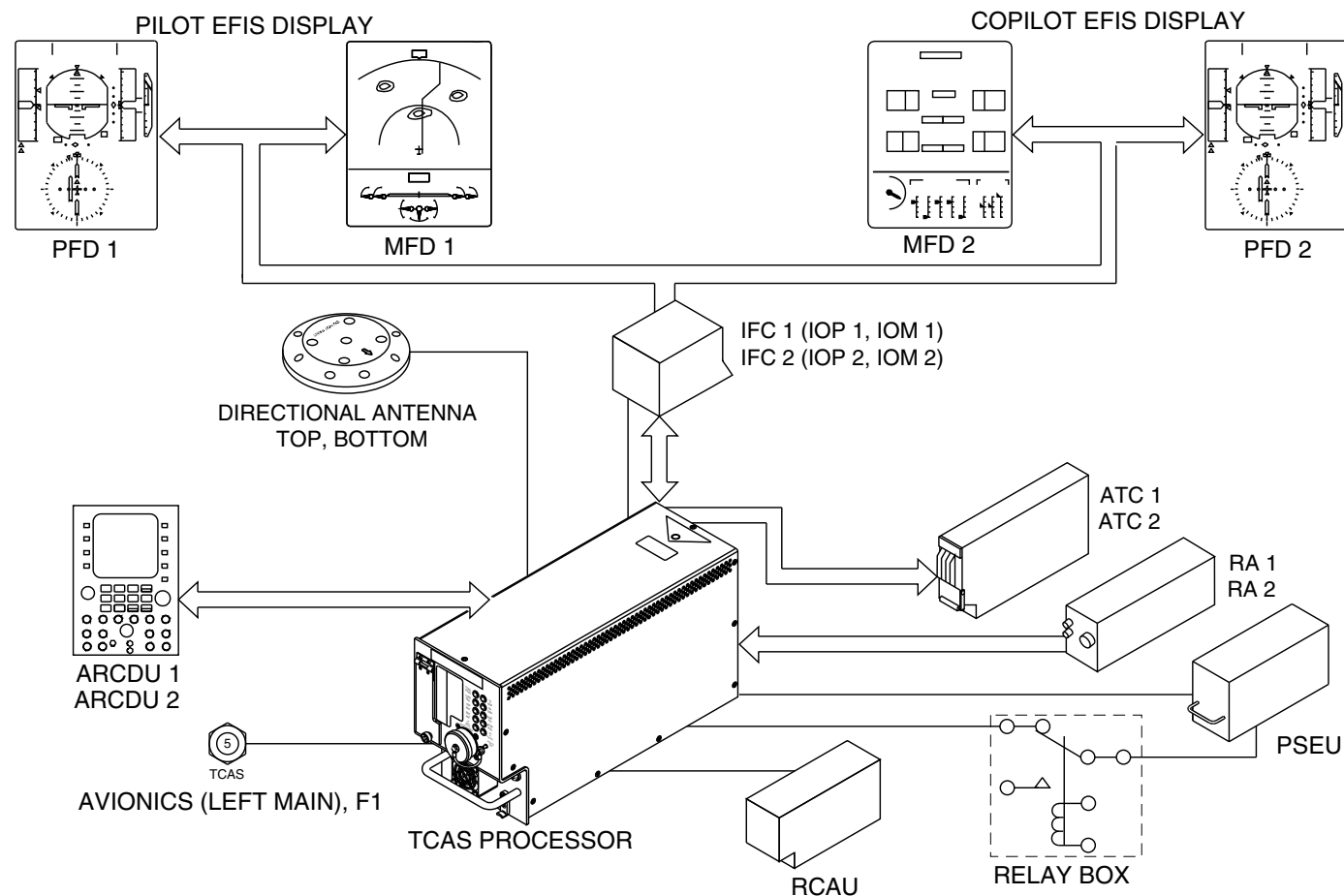
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POST MODSUM 4-309289

SHEET 2

TCAS II Block Diagram
Figure 1 (Sheet 2 of 2)

PSM 1-84-2A
EFFECTIVITY:
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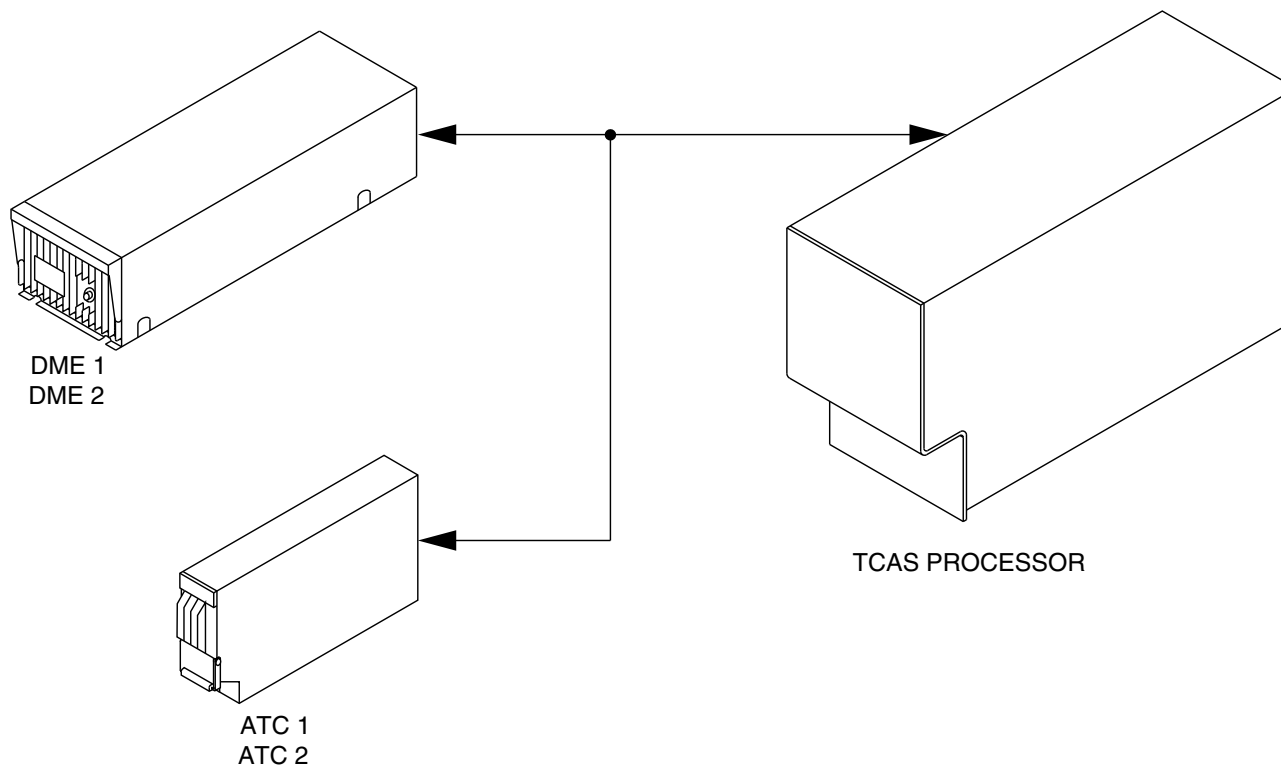
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SHEET 2

TCAS II Block Diagram, Suppression
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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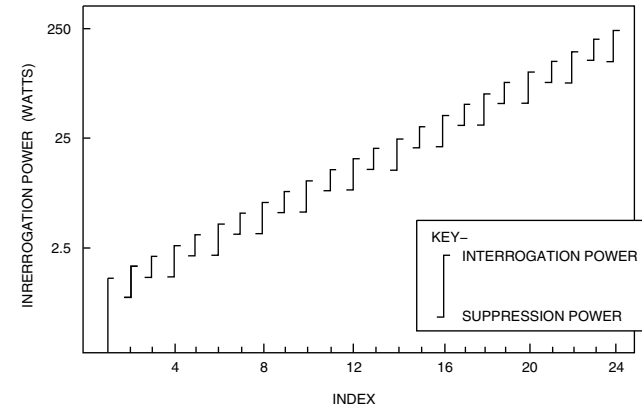
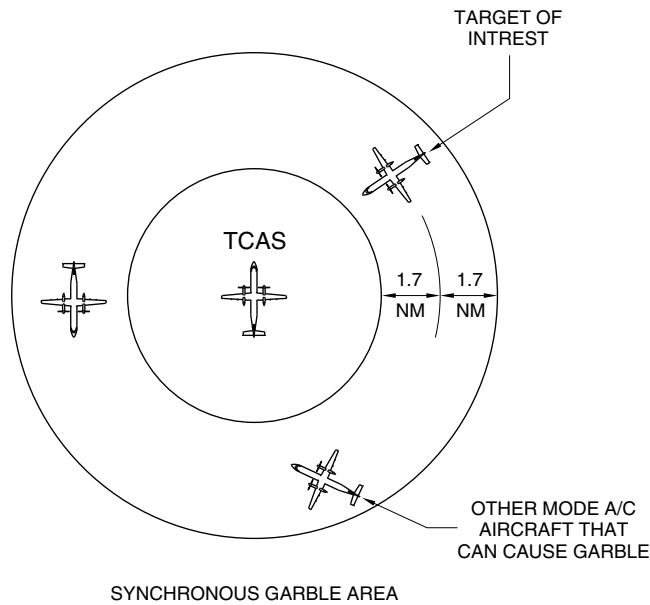
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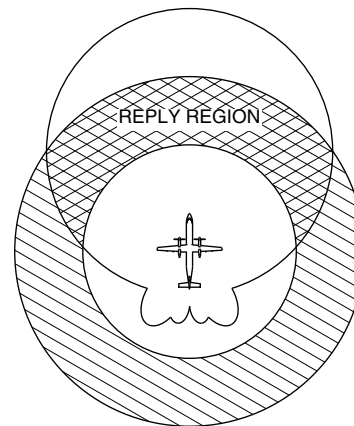


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WHISPER-SHOUT INTERROGATION SEQUENCE



DIRECTIONAL TRANSMISSION

fsn40a01.cgm

TCAS SYNCHRONOUS GARBLE
Figure 3

PSM 1-84-2A
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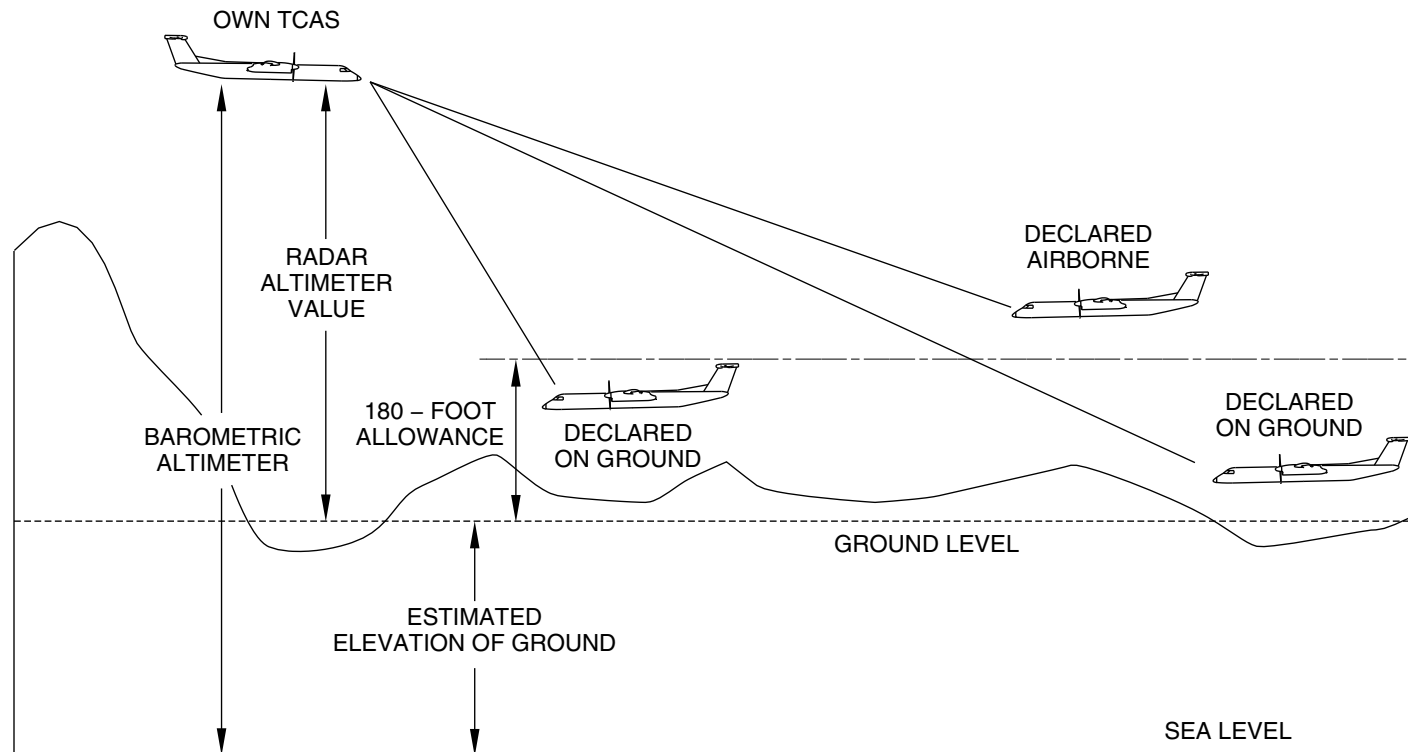
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1700 FEET ABOVE GROUND LEVEL
(TRESHOLD BELOW WHICH TCAS CHECKS FOR TARGETS ON THE GROUND)



fsn33a01.cgm

TCAS ON GROUND CALCULATION
Figure 4

PSM 1-84-2A
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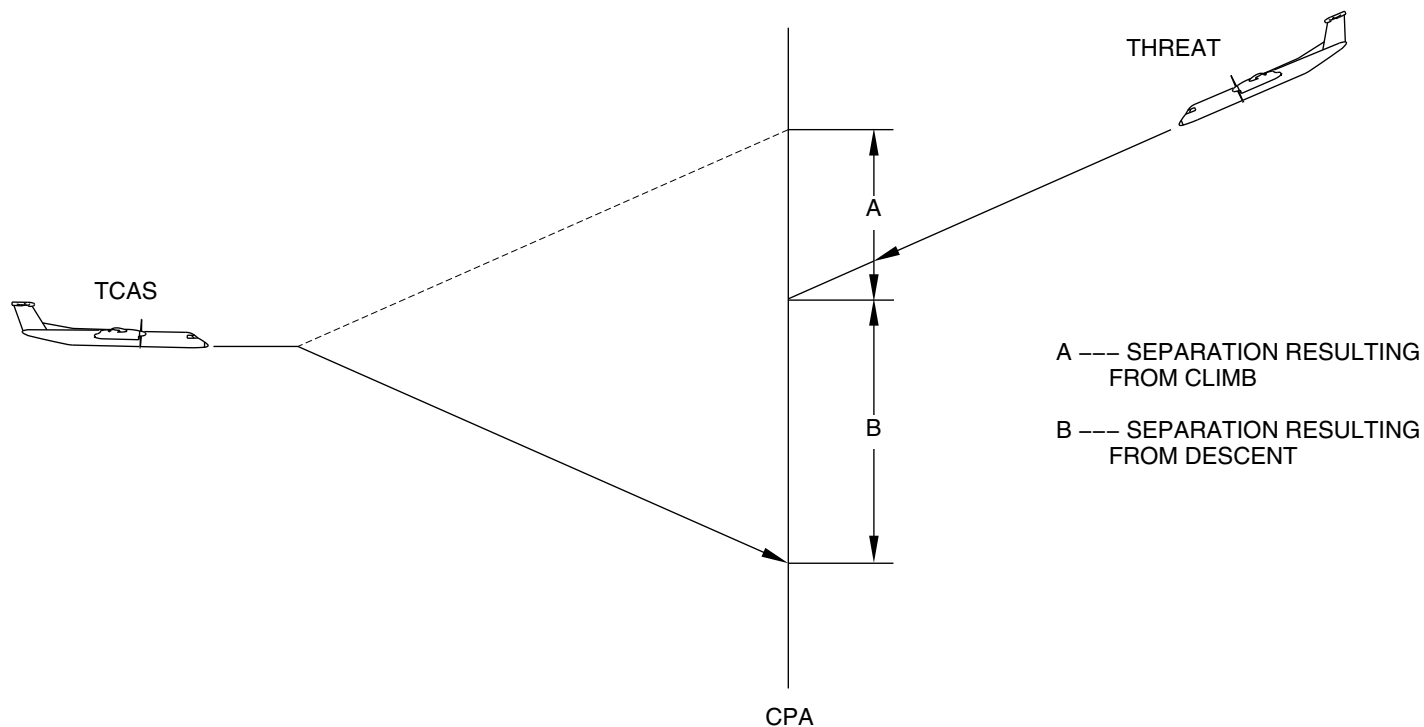
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TCAS RESOLUTION ADVISORY (RA) CALCULATION
Figure 5

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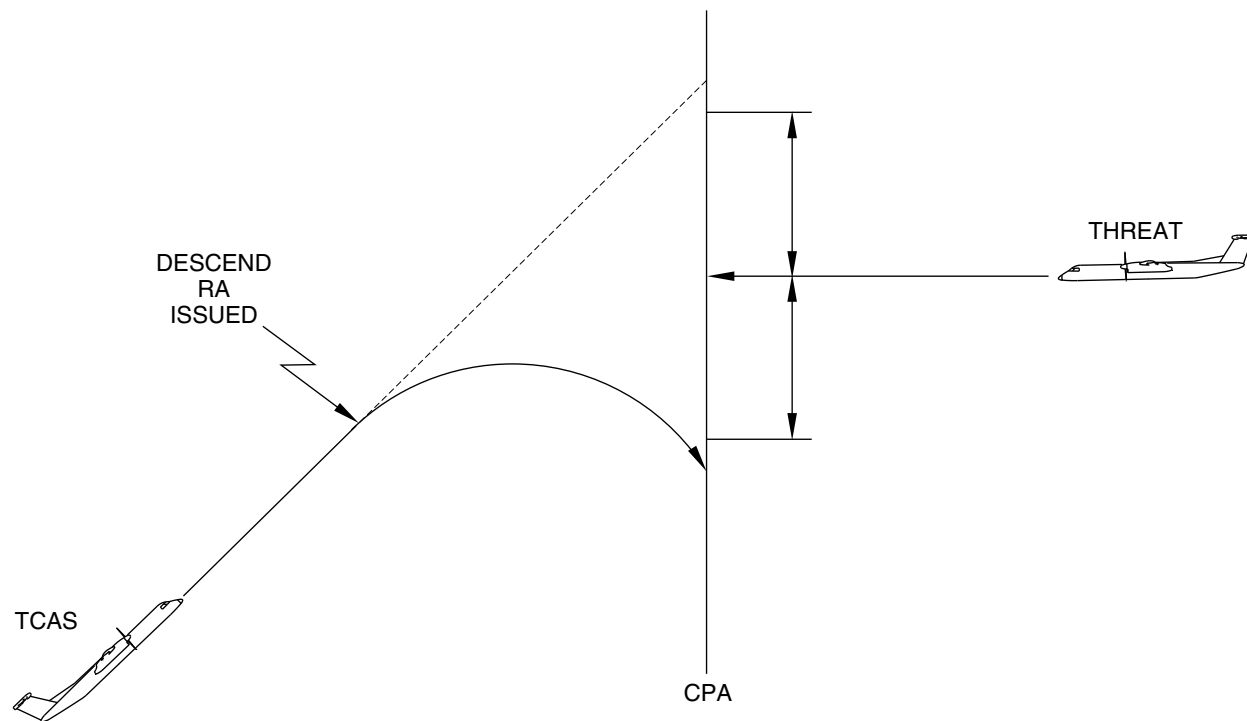
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f3n35a01.cgm

TCAS NON-CROSSING RA CALCULATION
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-43-00
Config 001

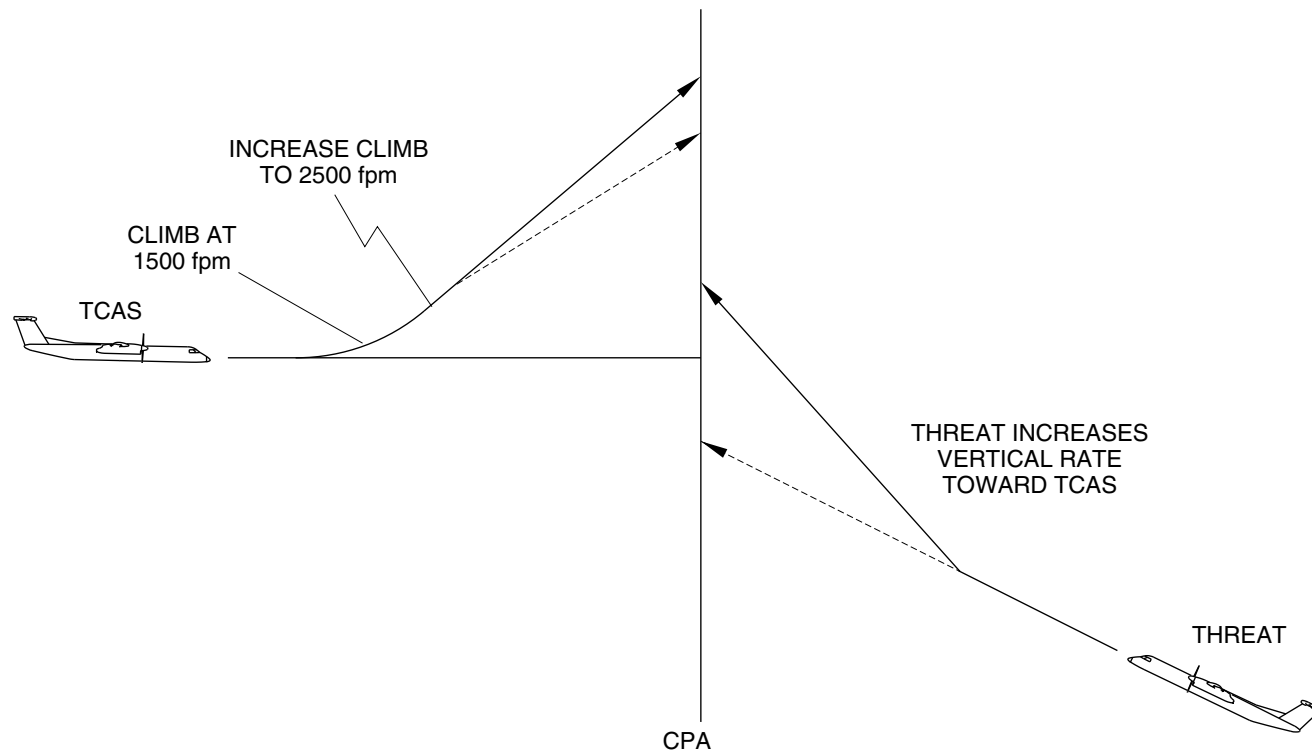
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fsn38a01.cgm

TCAS INCREASE VERTICAL RATE CALCULATION
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-43-00
Config 001

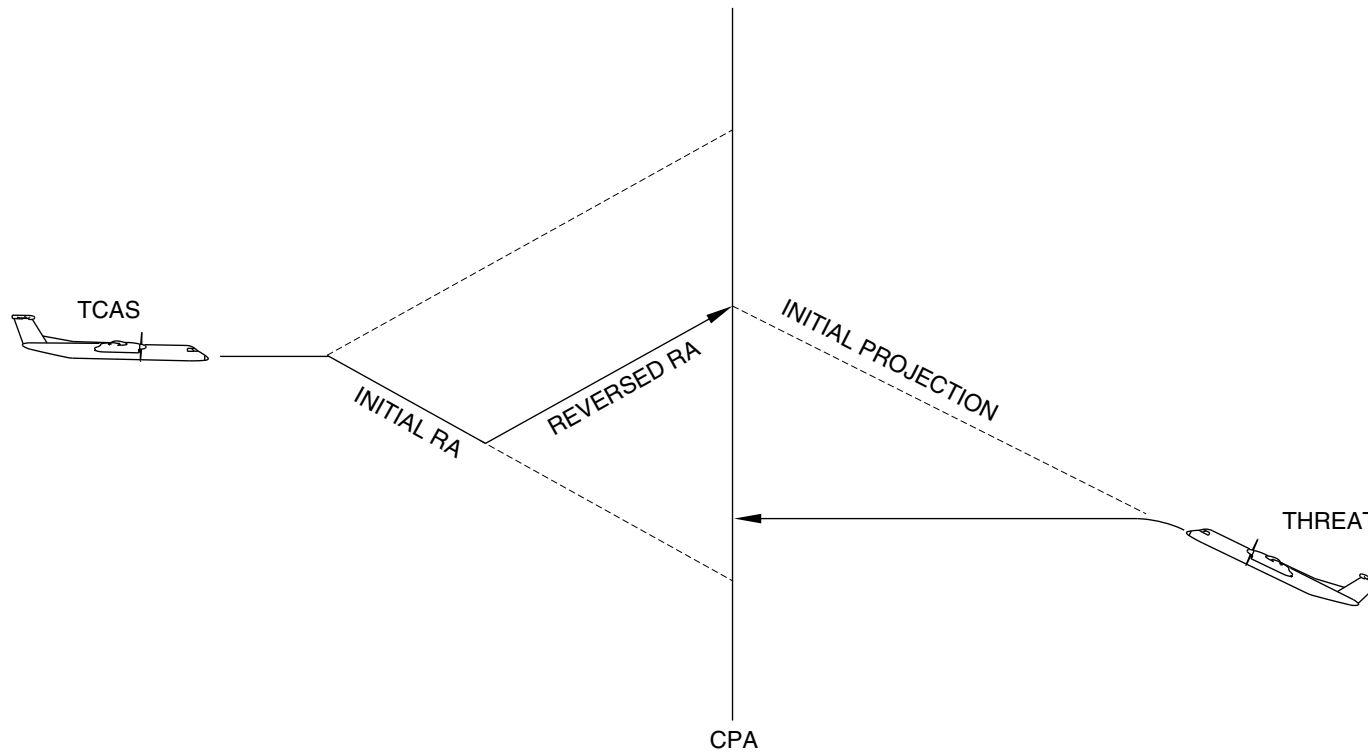
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fsn45a01.cgm

TCAS RA REVERSAL CALCULATION
Figure 8

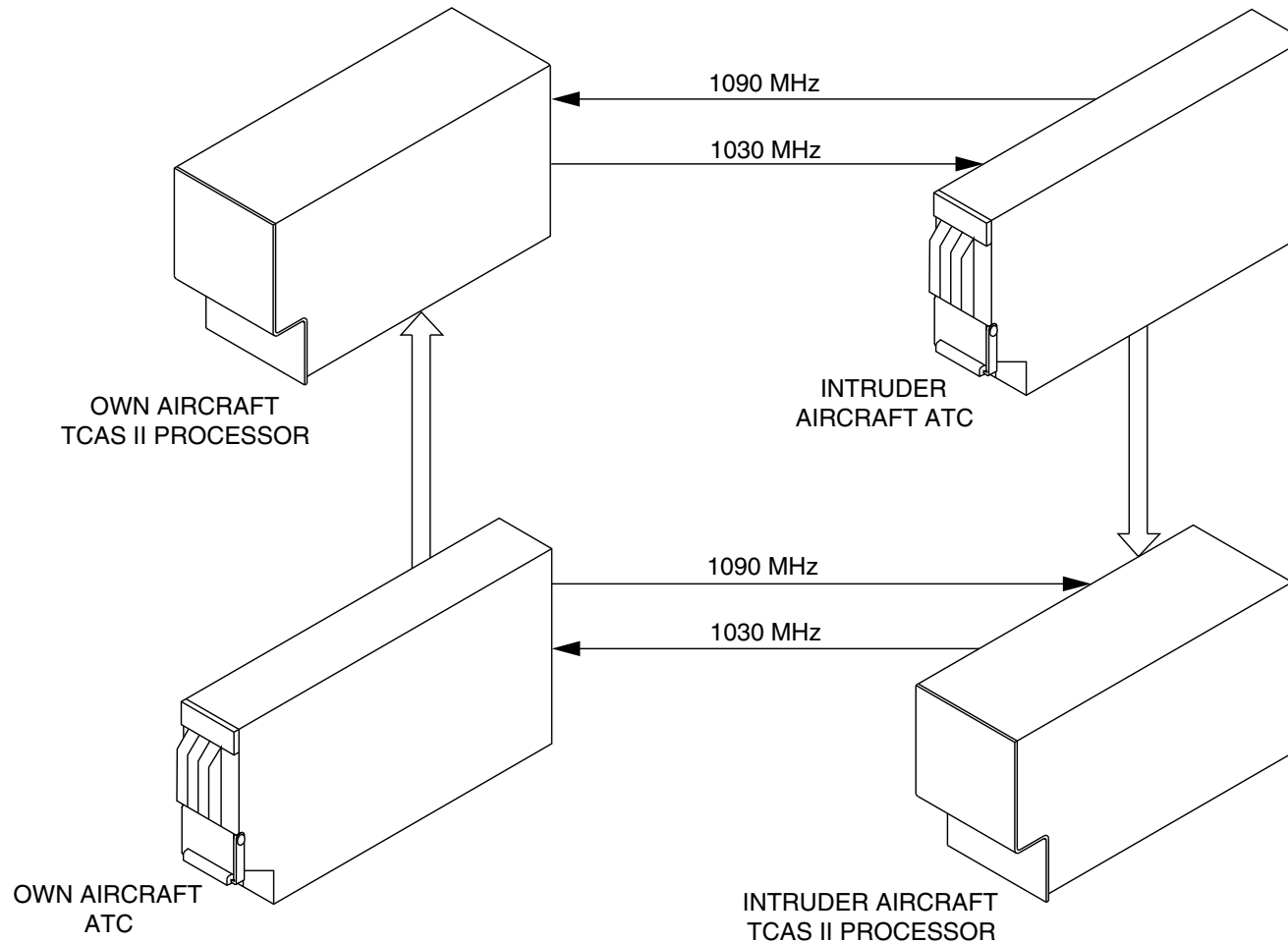
PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-43-00
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fsn32a01.dg, ms, mar12/2012

TCAS to TCAS Communication
Figure 9

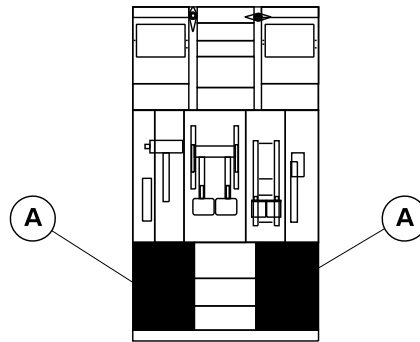
PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-43-00
Config 001

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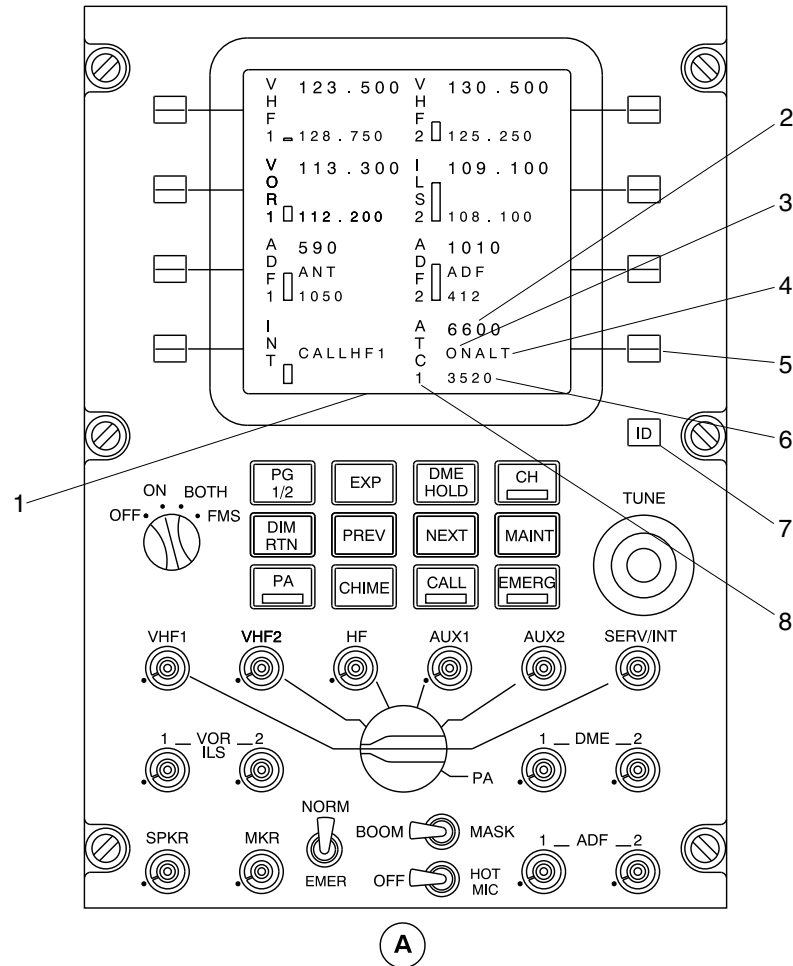
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTER CONSOLE

LEGEND

1. Display area.
2. Active code.
3. Transponder mode.
4. Altitude reporting.
5. Side key.
6. Preset code.
7. Specific Purpose Identifier (SPI) key.
8. Label.



fs837a01.dg, ms, mar12/2012

ARCDU, TCAS/ATC Controls and Indications
Figure 10

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-43-00
Config 001

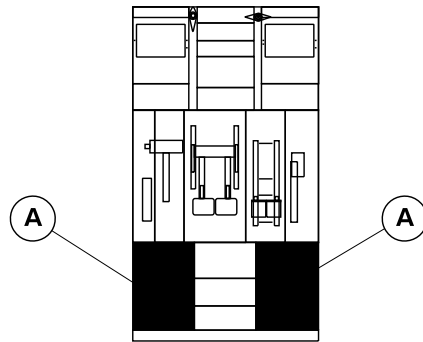
34-43-00

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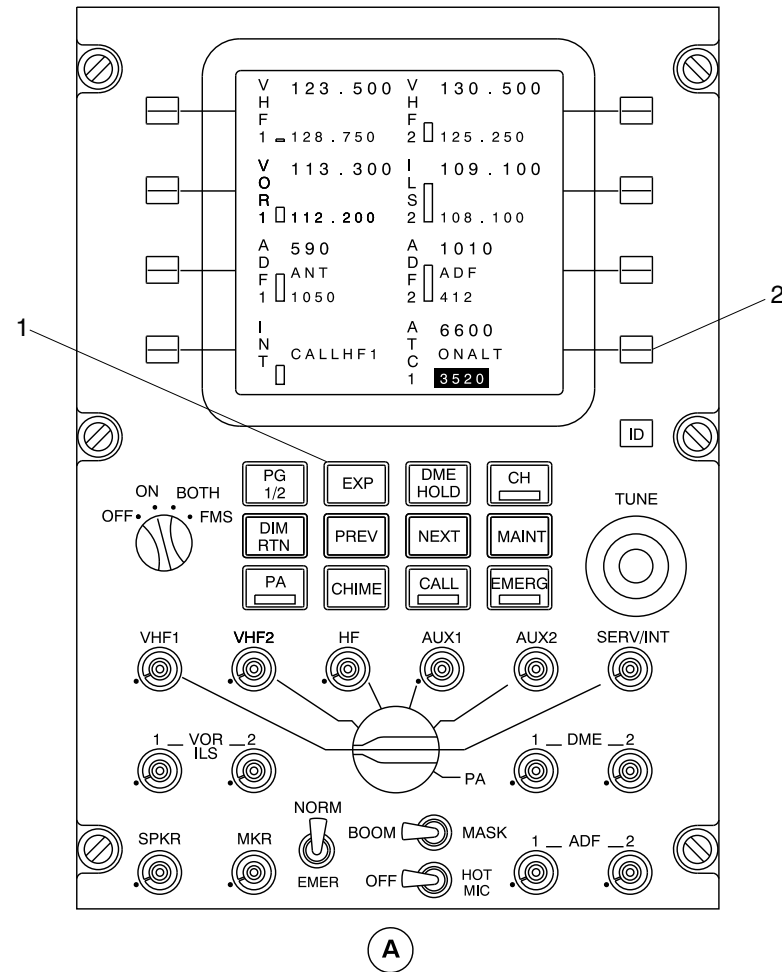
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTER CONSOLE

LEGEND

1. Expand key.
2. Side key.



fs837a04.dg, ms, mar12/2012

ARCDU, TCAS/ATC Particular Page Selection
Figure 11

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-43-00
Config 001

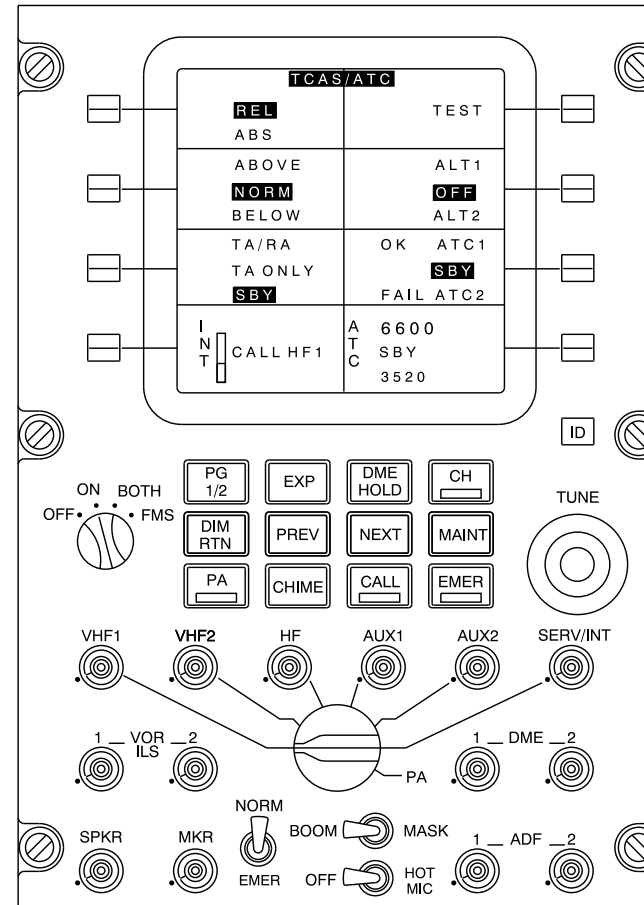
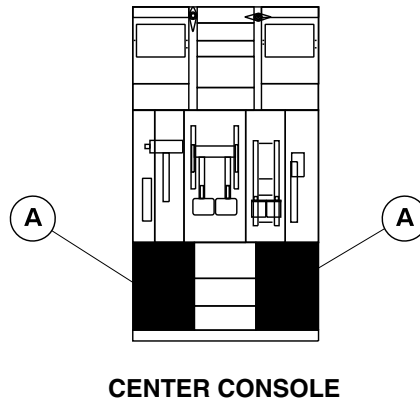
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fsc34a14.dg, ms, mar12/2012

ARCDU, TCAS/ATC Particular Page
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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Config 001

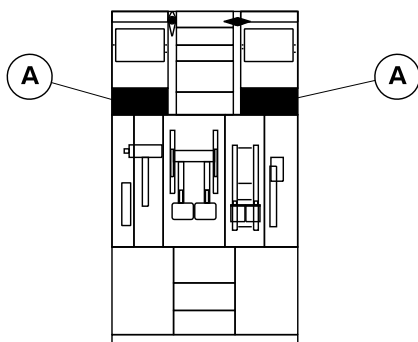
34-43-00

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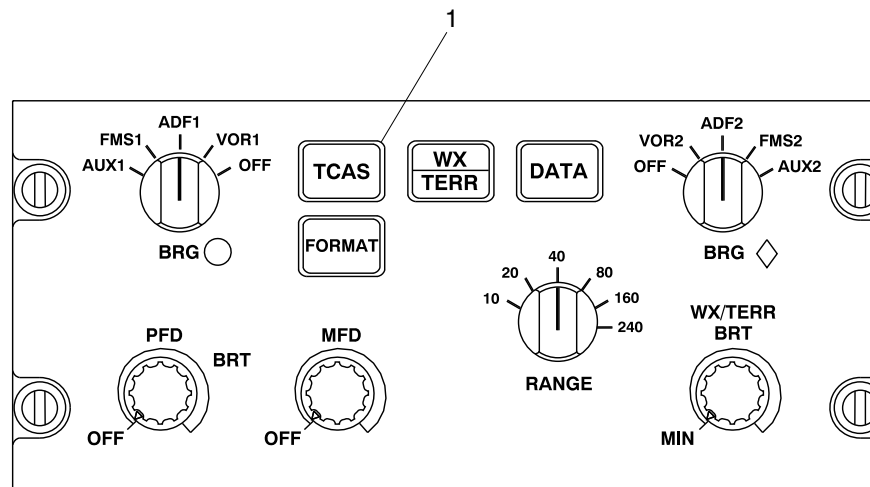
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTER CONSOLE

LEGEND

1. TCAS pushbutton switch.



fag30a03.dg, ms, mar12/2012

EFCP, TCAS POP-UP or Continuous Indication Selection
Figure 13

PSM 1-84-2A
EFFECTIVITY:
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A diagram of the front panel of the device. It features a central black rectangular area, which is the location of the front panel lock. A callout line points from a circle containing the letter 'A' to this black area. The front panel also includes several other components: a small rectangular display on the left, a large rectangular display in the center, and a large rectangular display on the right. There are also several small rectangular buttons or indicators arranged around the central display.

1. AUTO RANGE message.
2. TCAS traffic indication.
3. No-bearing advisory.
4. TCAS range ring.
5. Mode messages.

MFD, TCAS Indications
Figure 14

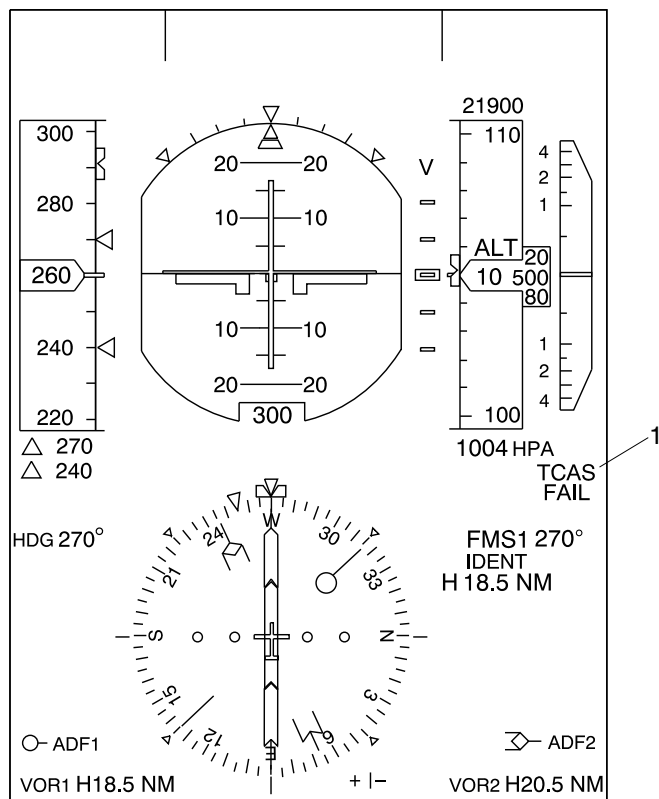
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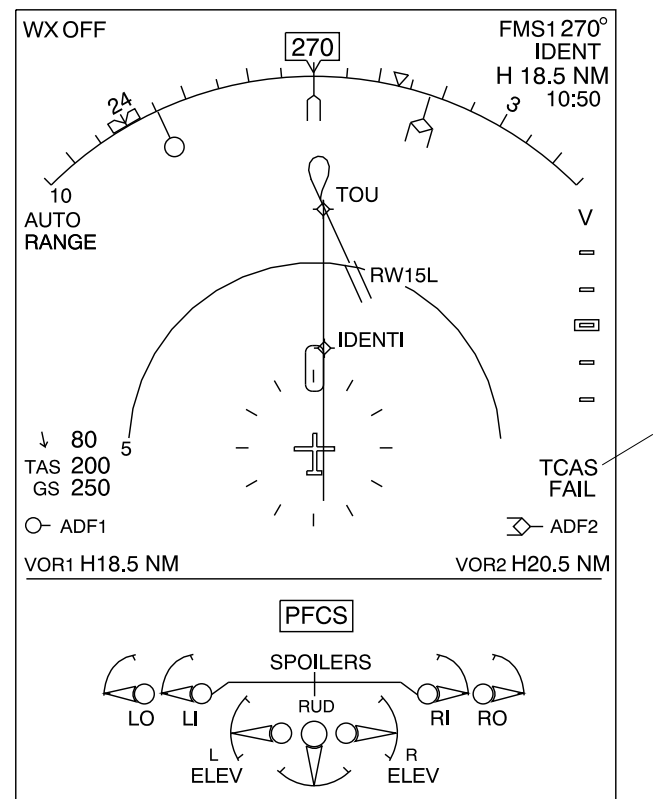


PFD

LEGEND

1. TCAS FAIL message.

fae54a15.dg, ms, mar12/2012



MFD

MFD, TCAS FAIL Message
Figure 15

PSM 1-84-2A
EFFECTIVITY:
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Config 001

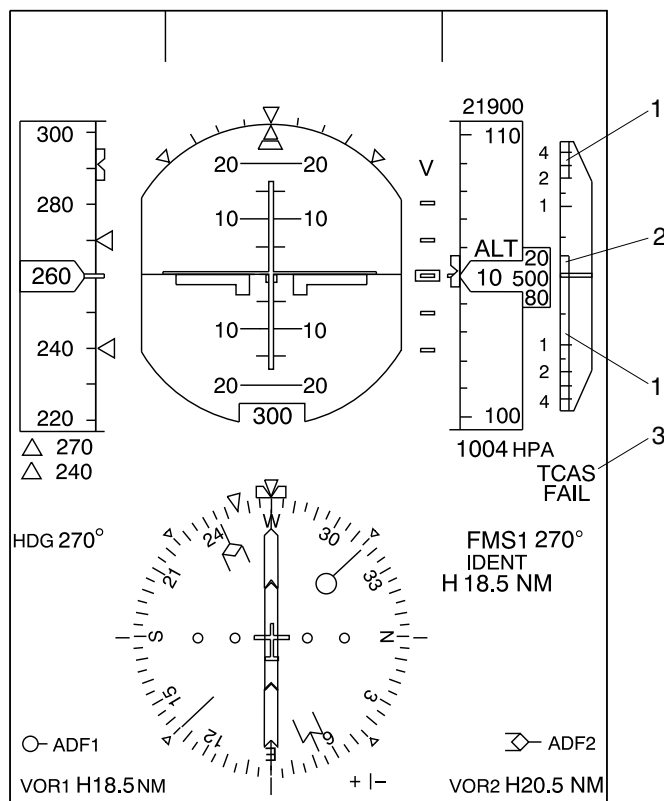
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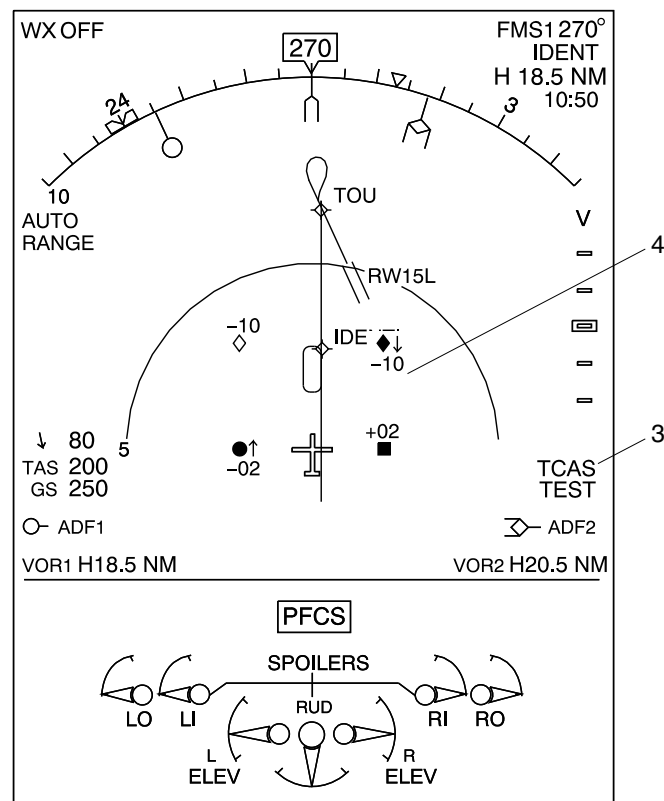
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



PFD

LEGEND

1. Red RA band.
2. Green fly-to band.
3. TCAS TEST message.
4. TCAS test indication.



MFD

MFD, TCAS TEST Message
Figure 17

fae54a16.dg, ms, mar12/2012

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-43-00
Config 001

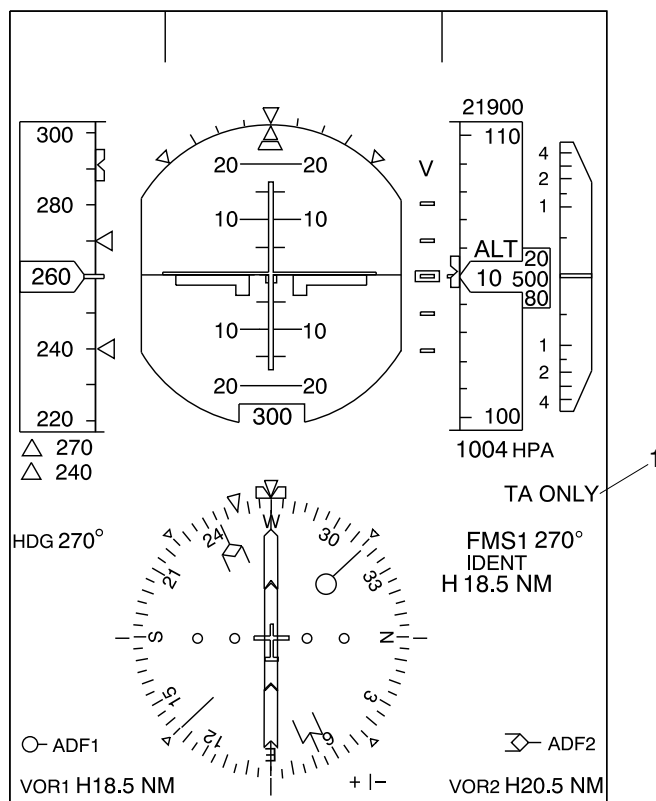
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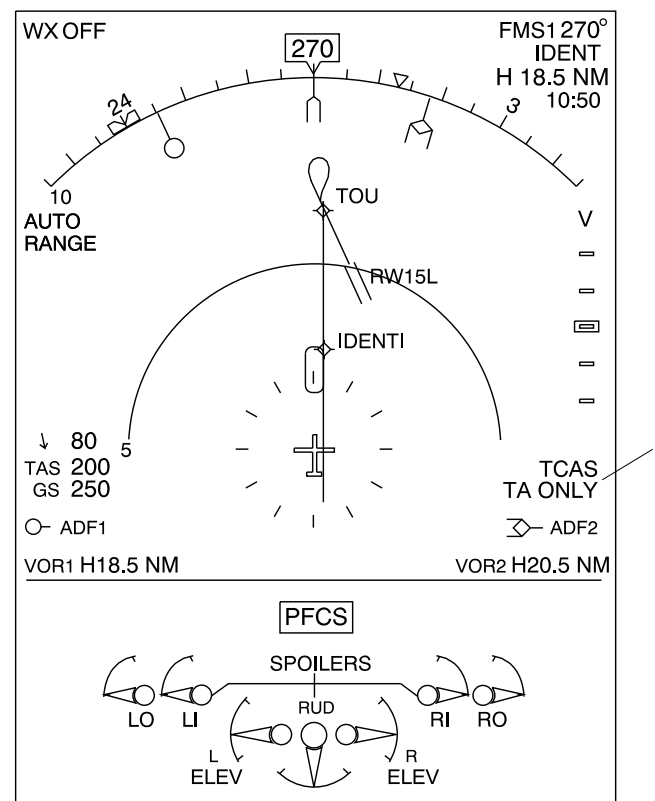


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



PFD



MFD

LEGEND

1. TA ONLY message.

fae54a13.dg, ms, mar12/2012

MFD, TA ONLY Message
Figure 18

PSM 1-84-2A
EFFECTIVITY:
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Config 001

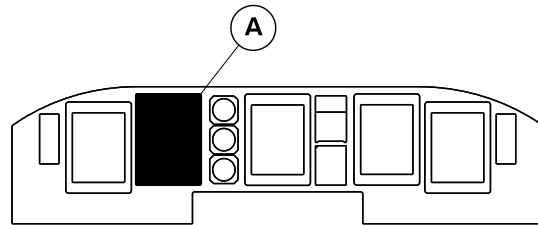
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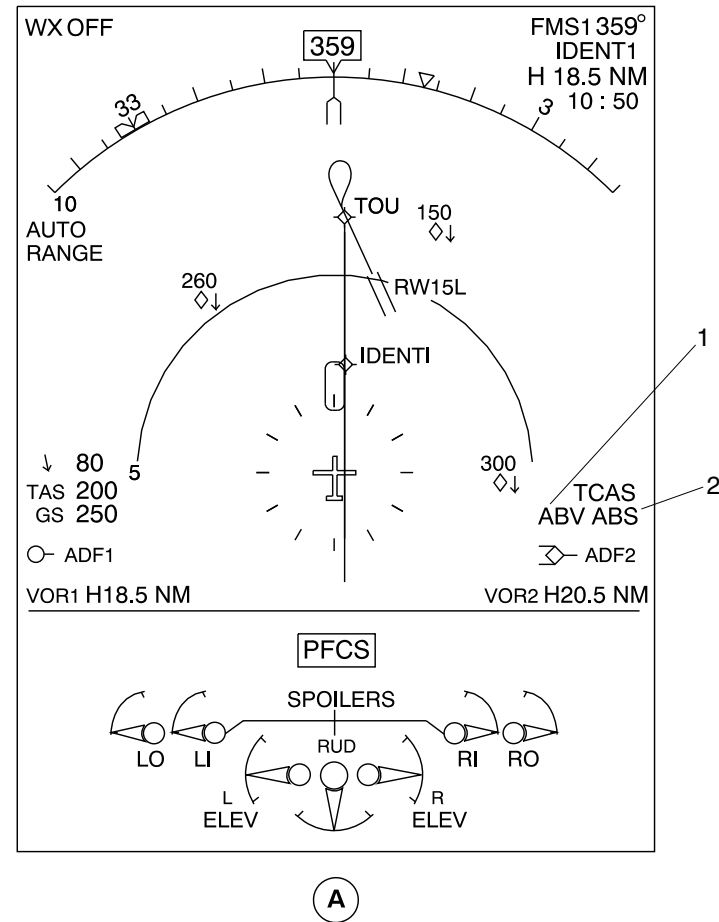
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Traffic above message.
2. Traffic absolute message.



fae54a18.dg, ms, mar12/2012

MFD, TCAS ABV or BLW and ABS Message
Figure 19

PSM 1-84-2A
EFFECTIVITY:
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Config 001

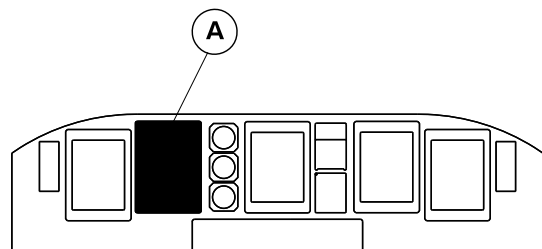
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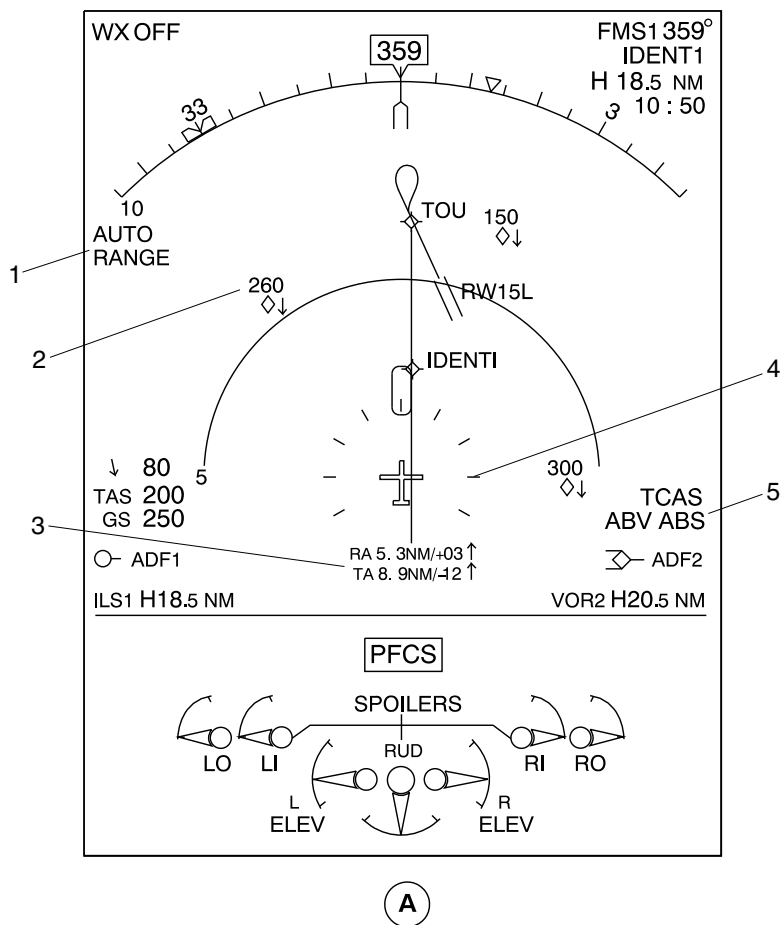
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. AUTO RANGE message.
2. TCAS traffic indication.
3. No-bearing advisory.
4. TCAS range ring.
5. Mode messages.



fae54a02.dg, ms, mar12/2012

MFD, TCAS Traffic Indications
Figure 20

PSM 1-84-2A
EFFECTIVITY:
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Config 001

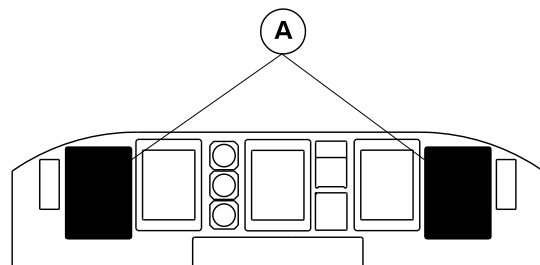
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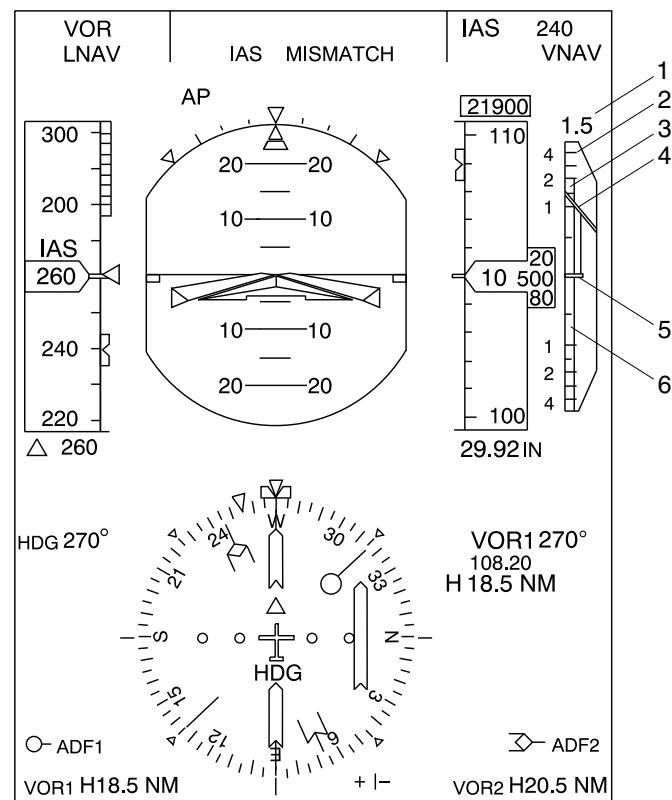
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Digital readout.
2. Scale.
3. Green fly-to band.
4. Pointer needle.
5. Zero reference line.
6. Red RA band.



A

fac30a11.dg, ms, mar12/2012

TCAS, PFD Indications
Figure 21

PSM 1-84-2A
EFFECTIVITY:
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Config 001

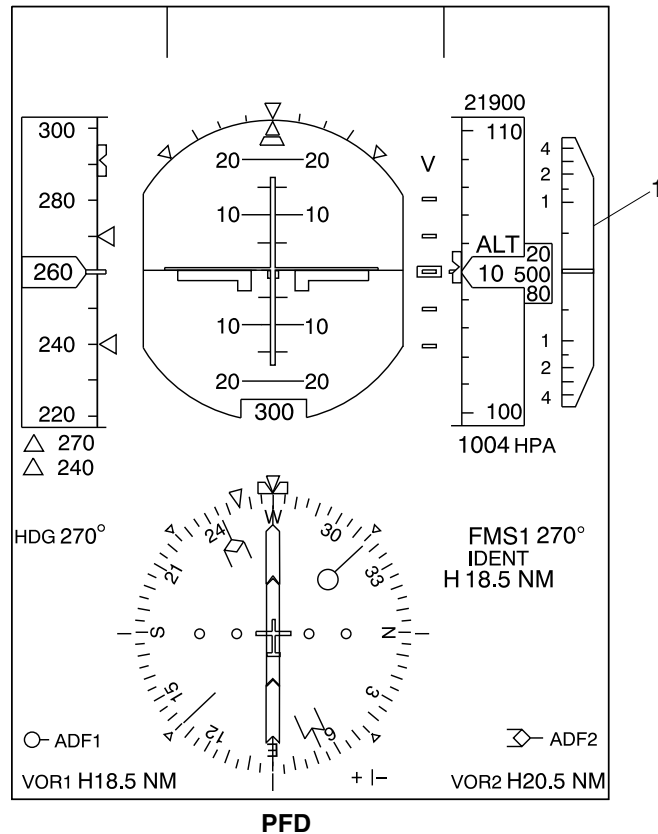
34-43-00

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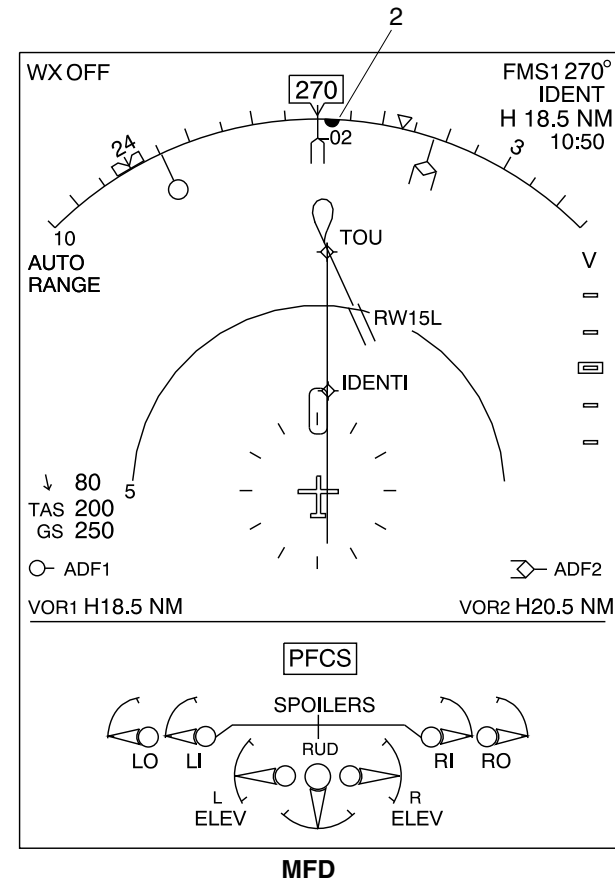
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



PFD

LEGEND

1. IVSI.
2. Traffic advisory symbol.



MFD

fae54a07.dg, ms, mar12/2012

TCAS, Traffic, Traffic Indications
Figure 22

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-43-00
Config 001

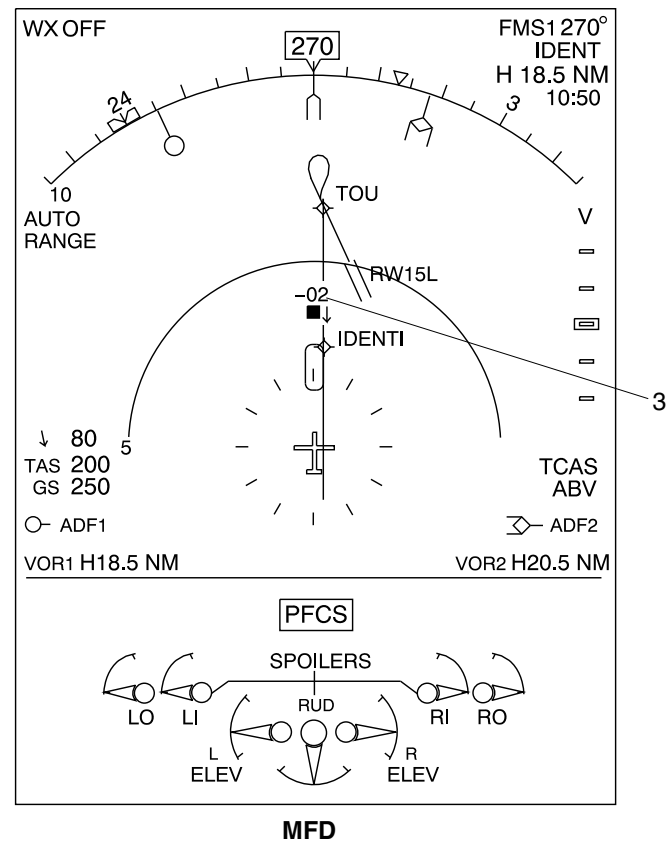
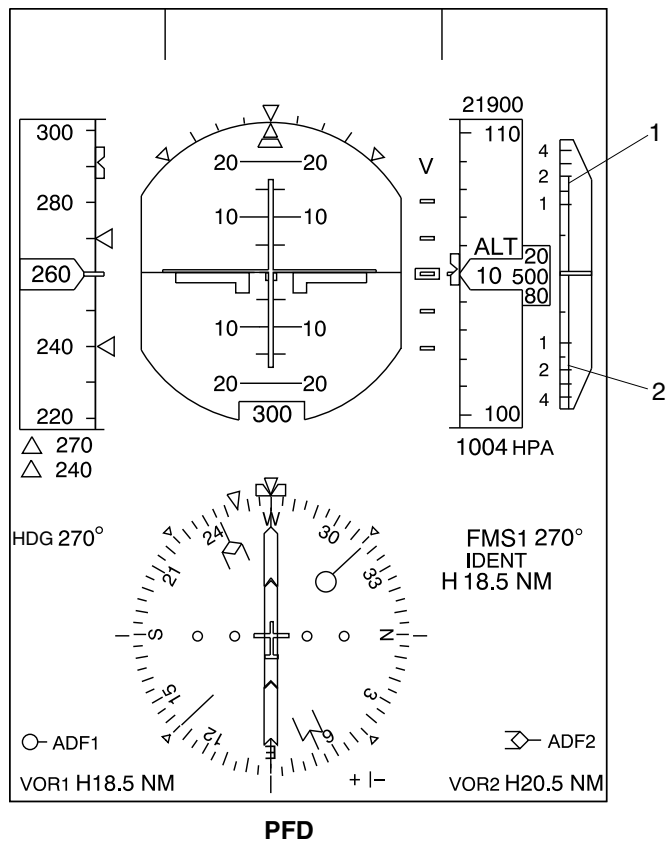
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LEGEND

1. Green fly-to band.
2. Red RA band.
3. Resolution advisory symbol.

fae54a10.dg, ms, mar12/2012

TCAS, Climb, Climb, Climb Indications
Figure 23

PSM 1-84-2A
EFFECTIVITY:
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Config 001

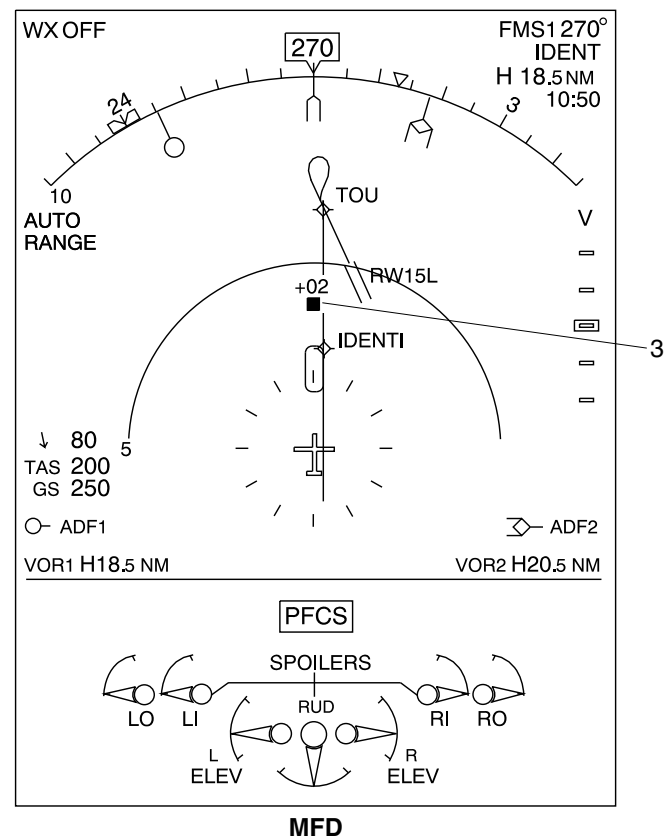
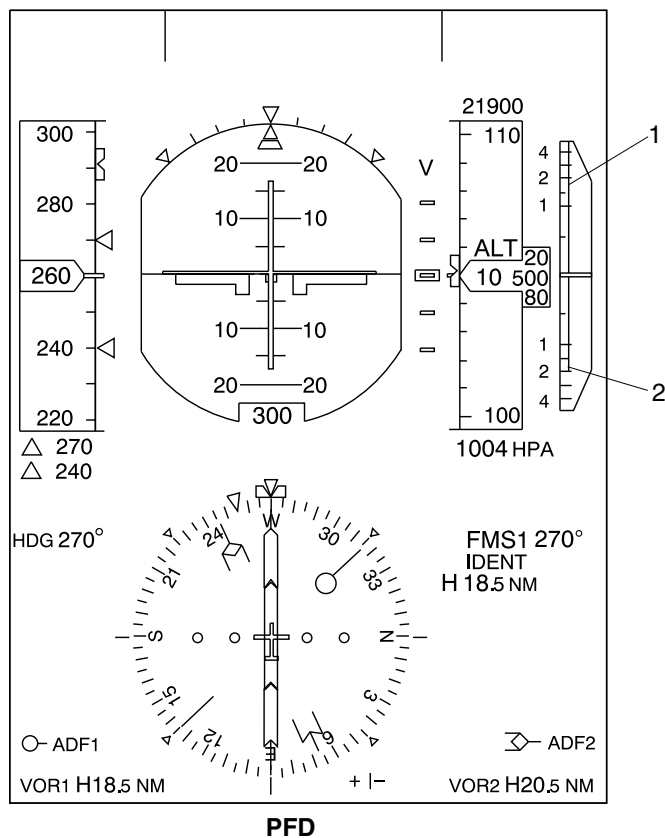
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LEGEND

1. Red RA band.
2. Green fly-to band.
3. Resolution advisory symbol.

fae54a05.dg, ms, mar12/2012

TCAS, Descend, Descend, Descend Indications
Figure 24

PSM 1-84-2A
EFFECTIVITY:
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Config 001

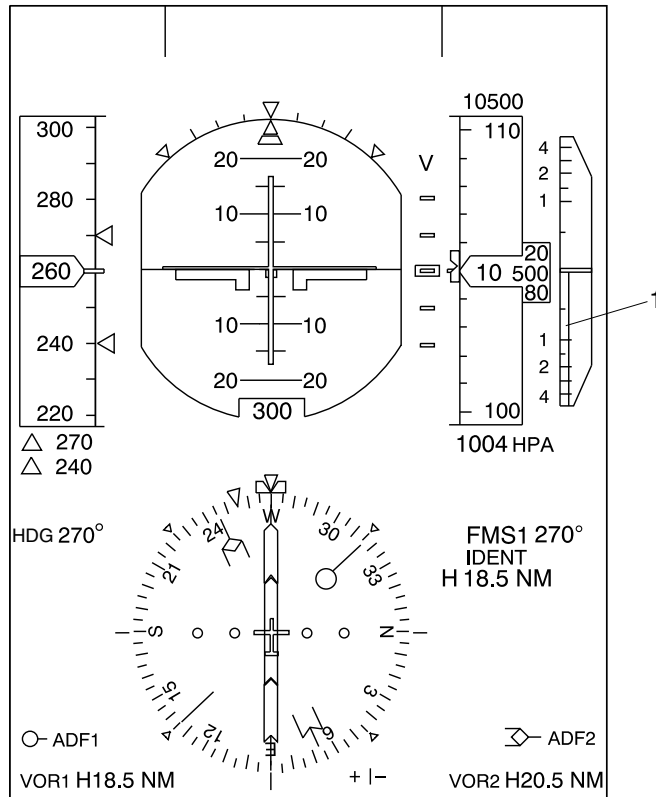
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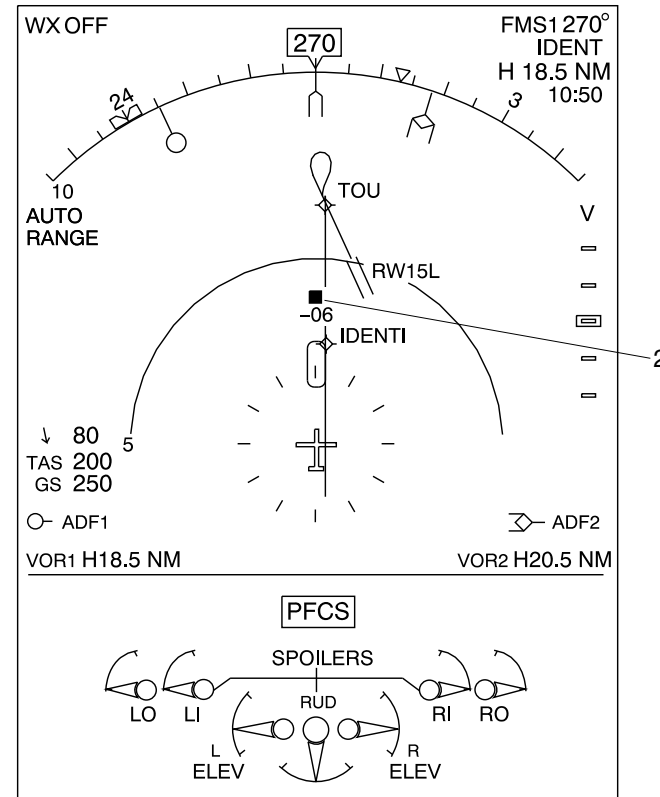


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PFD



MFD

LEGEND

1. Red RA band.
2. Resolution advisory symbol.

fae54a08.dg, ms, mar08/2012

TCAS, Monitor Vertical Speed Indications (Do Not Descent)
Figure 25 (Sheet 1 of 3)

PSM 1-84-2A
EFFECTIVITY:
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Config 001

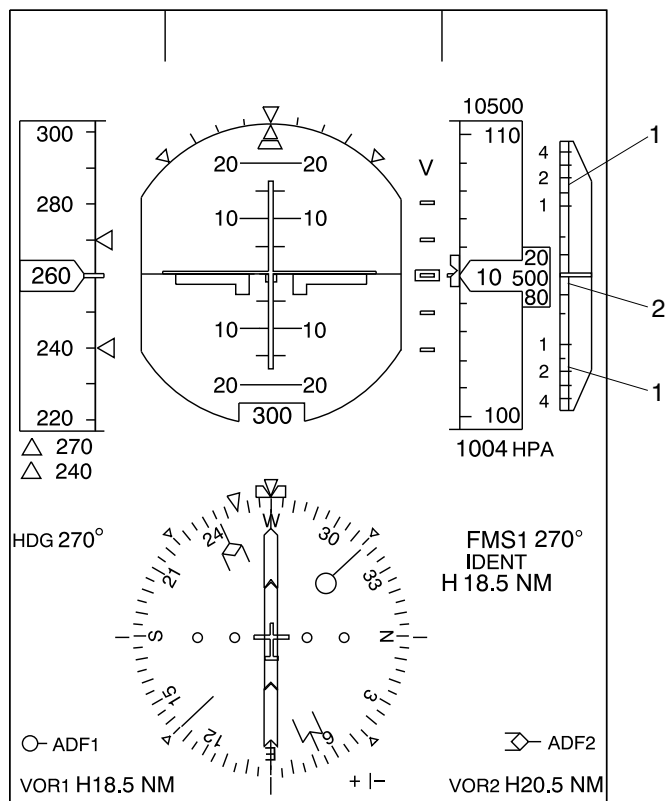
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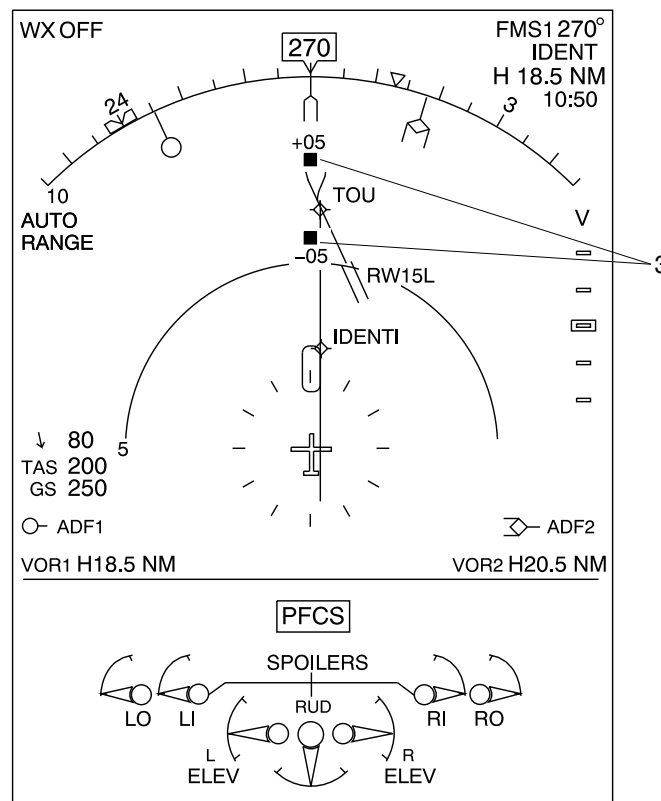
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PFD

LEGEND

1. Red RA band.
2. Green fly-to band.
3. Resolution advisory symbol.



MFD

fae54a09.dg, ms, mar12/2012

TCAS, Monitor Vertical Speed Indications (Do Not Descent)
Figure 25 (Sheet 2 of 3)

PSM 1-84-2A
EFFECTIVITY:
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Config 001

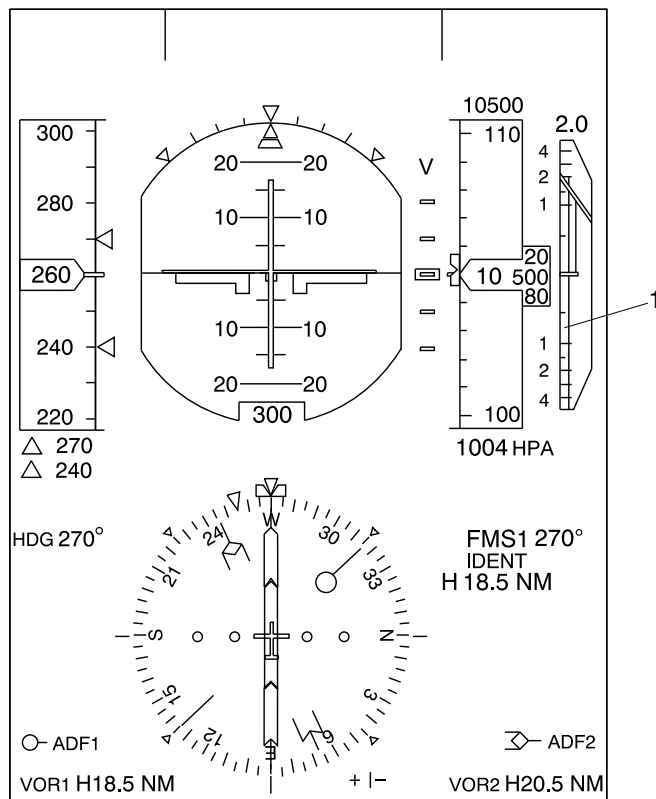
34-43-00

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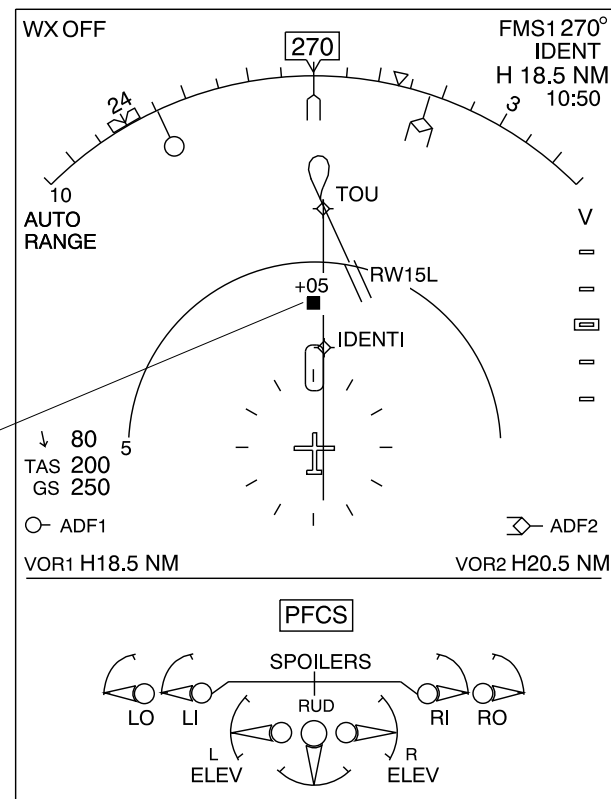


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PFD



MFD

LEGEND

1. Red RA band.
2. Resolution advisory symbol.

fae54a06.dg, ms, mar12/2012

TCAS, Monitor Vertical Speed Indications (Do Not Descent)
Figure 25 (Sheet 3 of 3)

PSM 1-84-2A
EFFECTIVITY:
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Config 001

34-43-00

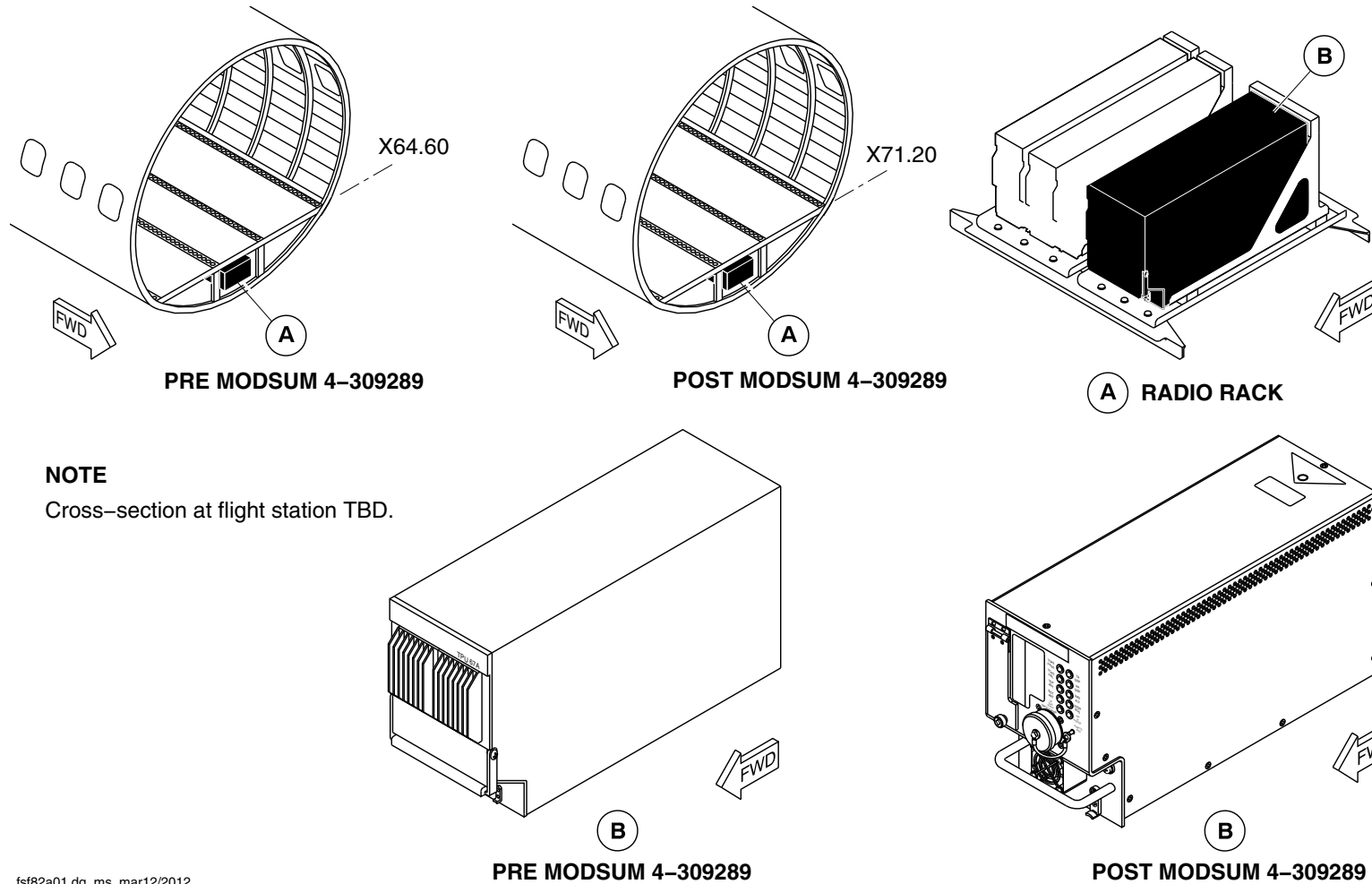
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fsf82a01.dg, ms, mar12/2012

TCAS Processor Unit Locator
Figure 28

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-43-00
Config 001

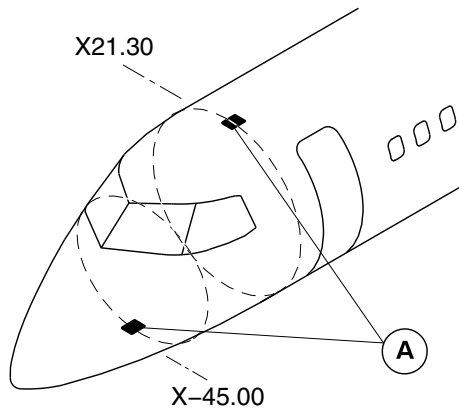
34-43-00

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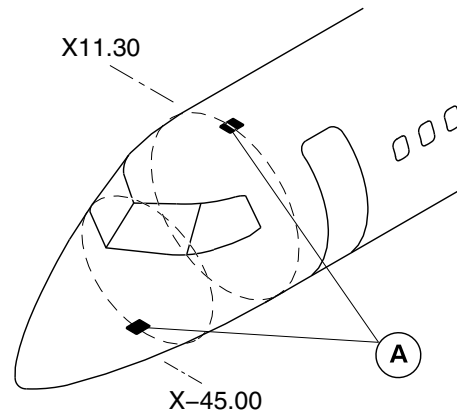


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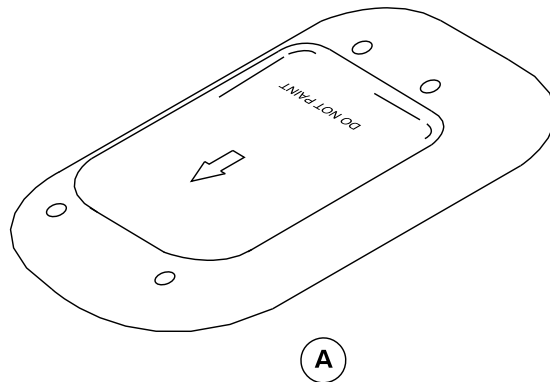
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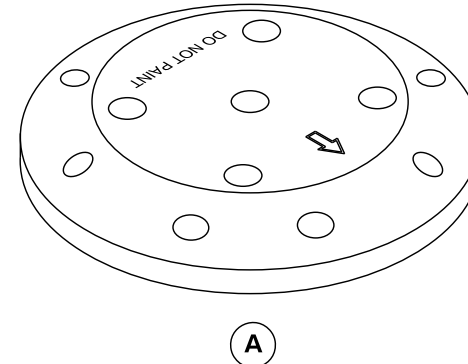
PRE MODSUM 4-309289



POST MODSUM 4-309289



PRE MODSUM 4-309289



POST MODSUM 4-309289

fs376a01.dg, ms, mar12/2012

TCAS Antenna Locator
Figure 29

PSM 1-84-2A
EFFECTIVITY:
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Config 001

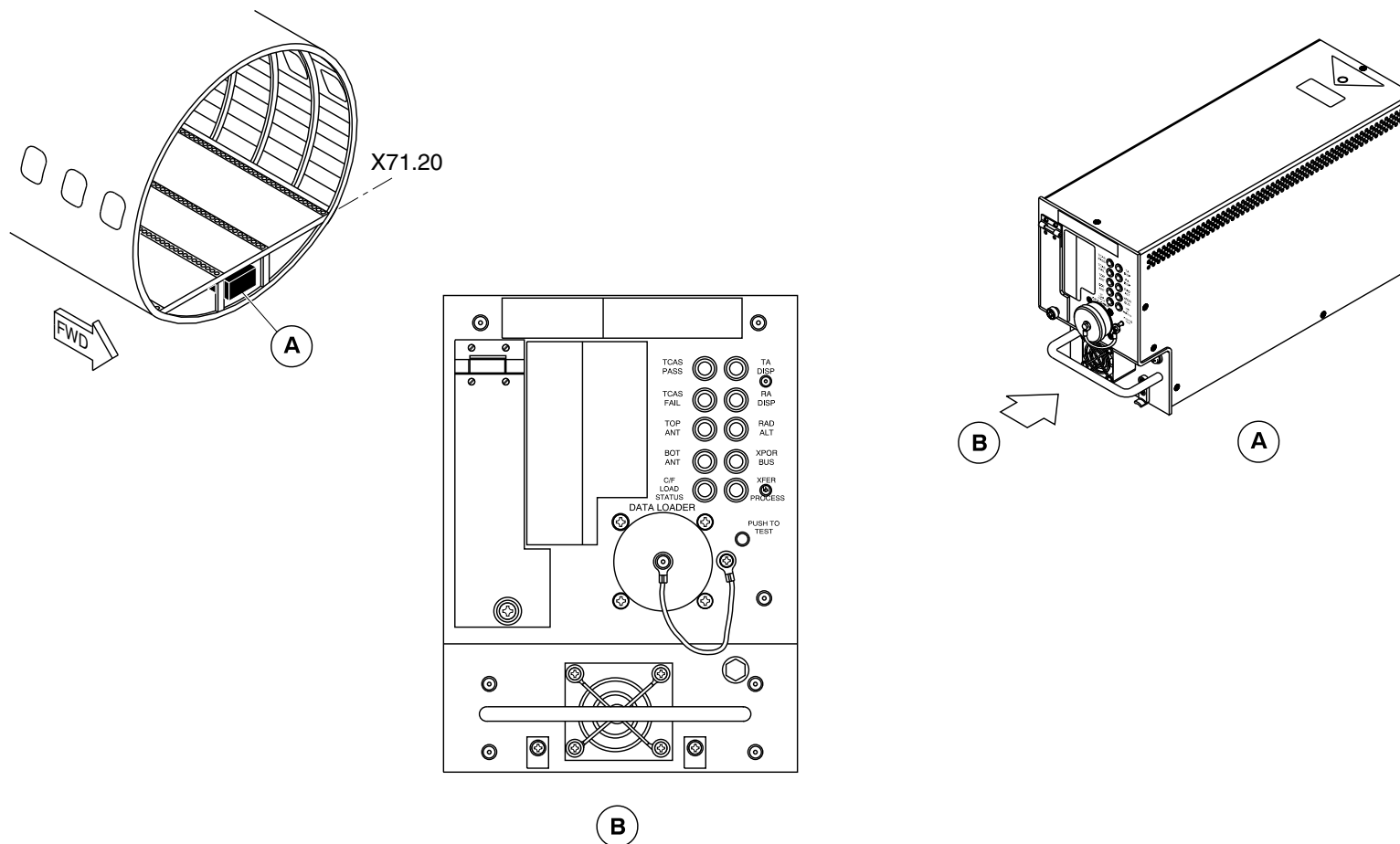
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cg2884a01.dg, ak, sep10/2013

TCAS Annunciator light
Figure 30

PSM 1-84-2A
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**ON A/C ALL

34-44-00-001

RADIO ALTIMETER (HONEYWELL)

Introduction

The Radio Altimeter (RA) System calculates the low level Above Ground Level (AGL) altitude for display and for use by other aircraft systems.

The System allows the Decision Height (DH) altitude to be set for use during the approach phase of a flight.

General Description

[Refer to Figure 1.](#)

The RA system transmits a signal to the ground and then it receives the reflected signal to calculate the AGL altitude.

The aircraft Primary Flight Displays (PFD1, PFD2) show the RA system parameters that follow:

- Above Ground Level (AGL) altitude
- RA mismatch
- Rising runway symbol
- DH status.

The RA System is used to make DH calculations. The AGL altitude data is compared to the DH value set by the flight crew. The Primary Flight Displays (PFD1, PFD2) shows the selected value. The

Electronic Instrument System (EFIS) and the Audio and Radio Management System (ARMS) are used to alert the flight crew both visually and aurally when the aircraft reaches the set value.

In addition, the RA system supplies a signal to the Stall Protection System to prevent it from operating when the altitude is less than 200 ft (60.96 m) AGL.

The RA system has the components that follow:

- Transmitter/receiver (34-44-01)
- Transmitter/receiver Antenna (34-44-11 and 34-44-16).

The RA system supplies data directly to the Traffic Collision Avoidance System (TCAS) and through both Integrated Flight Cabinets (IFC1, IFC2) to the systems that follow:

- Auto Flight Control System (AFCS)
- Stall Protection Module (SPM1, SPM2)
- Primary Flight Displays (PFD1, PFD2)
- Solid State Flight Data Recorder (SSFDR)
- Ground Proximity Warning System (GPWS)
- Central Diagnostic System (CDS).

Detailed Description

A Frequency Modulated Carrier Wave (FMCW) is transmitted by the RA transmitter/receivers (RA1, RA2). The Frequency Modulated Carrier Wave (FMCW) continuously changes its frequency and compares the time it takes for the transmitted signal to go to the ground and back to the new frequency.



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The received signal is mixed with the transmitted signal to make an Intermediate Frequency (IF). The IF is directly proportional to the time that it takes the signal to go to the ground and back. The Intermediate Frequency (IF) is changed into ARINC 429 and analogue outputs.

The RA transmitter/receiver gives continuous AGL altitude data from 2500 ft (762 m) until landing. The center operating frequency of the transmitter that is routed through the transmit antenna is 4300 ± 15 MHz. It is continuously modulated at a 100 Hz rate, with a FM deviation of 100 ± 5 MHz. The nominal transmitted power of the signal is 160 mW.

When two RA systems are installed, both transmitter/receivers will continuously function at the same time. An external hardware strap, which is part of the aircraft wiring, is connected to the second RA transmitter/receiver interface connector. When the strap connection is sensed, the RA transmitter/receiver will use a different modulation frequency, 105 Hz.

During power-up, each RA transmitter/receiver does a comprehensive continuous self test routine. The status of the transmitter/receiver is sent to the Central Diagnostic System (CDS).

If a single RA system is installed and the transmitter/receiver fails, then its failure status will be shown.

If a dual RA system is installed and one transmitter/receiver fails, then its failure status will not be shown to the flight crew or to the TCAS. Each Integrated Flight Cabinet IFC1, IFC2, receives the same

AGL data. If both transmitter/receivers fail, then its failure status will be shown on the Primary Flight Displays (PFD1, PFD2).

[Refer to Figure 2.](#)

The PFD1 and PFD2 display altitude AGL as four white numbers followed by a RA label. It shows the altitude from zero to 2500 ft (762 m) AGL.

The altitude increases and decreases in 5 ft (1.52 m) increments below 200 ft (61 m) AGL and increases and decreases in 10 ft increments above 200 ft (61 m) AGL.

A rising runway symbol is used to give an analogue indication of AGL altitude. It moves vertically, linearly and correspondingly to the AGL altitude.

It comes into view at the bottom of the attitude sphere when the aircraft is at 200 ft (61 m) AGL altitude. It moves towards the airplane symbol as the altitude AGL decreases and touches the bottom of the airplane symbol at touchdown.

The rising runway symbol goes out of view when radio altitude is not available or the aircraft is above 200 ft (61 m) AGL altitude.

[Refer to Figure 3.](#)

The Flight Data Processing System (FDPS) senses AGL differences between the RA transmitter/receivers (RA1, RA2), or between an output and shown parameter. If there is a mismatch condition, the FDPS causes the RA label to change from white to yellow and flash initially and then stay on steady.



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In addition, the advisory display areas of the Primary Flight Displays (PFD1, PFD2) show a RAD ALT MISMATCH message. It comes into view and flashes in yellow text and then stays on steady.

Refer to Figure 4.

A failure annunciation shows four red dashes in place of the AGL altitude on the PFD1 and PFD2. If a TCAS is installed, a TCAS FAIL message will also be shown in yellow text on the PFD1 and PFD2.

Refer to Figure 5.

For DH calculations, the DH knob located on the Index Control Panel (ICP1, ICP2) is turned to set a DH altitude.

Refer to Figure 6.

The related PFD1 and PFD2 shows the selection as three digits in cyan followed by a white DH label. It shows the altitude selection in one-foot increments from zero through 999 ft (304.49).

Three white dashes replace the digits when PFD1 and PFD2 do not receive valid data from the Integrated Flight Cabinets (IFC1, IFC2).

The DH label shows that the aircraft is at the selected height.

When aircraft is at or less than 100 ft (30.5 m) AGL above the selected DH, the Primary Flight Displays (PFD1, PFD2) will show a white box. When the aircraft goes below the DH, a yellow DH label comes into view inside the white box.

The DH indication is out of view when the altitude AGL is greater than the DH setting.

The 28 Vdc left and right main buses supply power to the RA transmitter/receivers (RA1, RA2) through 3 ampere circuit breakers. The RA transmitter/receivers (RA1, RA2) system circuit breakers are

located in positions L3 and L4 on the 28 Vdc avionics circuit breaker panel.

When two RA transmitter/receivers are installed, each unit has a related transmit and receive antenna.

In this configuration both RA transmitter/receivers (RA1, RA2) supply data to both Integrated Flight Cabinets (IFC1, IFC2) and the TCAS. When one RA transmitter/receiver (RA1, RA2) fails, the Integrated Flight Cabinets (IFC 1, IFC2) use data from the other RA transmitter/receiver (RA1, RA2). No indication of the status of the failed RA transmitter/receiver is shown.

NOTE

The status of the failed RA transmitter/receiver (RA1, RA2) is transmitted and stored in the Central Diagnostic System (CDS).

Both RA transmitter/receivers (RA1, RA2) supply a discrete health status output to each IOM1 and IOM2 for CDS monitoring purposes.

Both RA transmitter/receivers (RA1, RA2) supply radio height data to each Input Output Processors (IOP1, IOP2). A FDPS within each IOP controls the supply of data to the systems that follow:

- Ground Proximity Warning System (GPWS)
- Auto Flight Control System (AFCS)
- Stall Protection System (SPS)
- Flight Data Recorder (FDR)
- Primary Flight Displays (PFD1, PFD2).

Each FDPS does a discrepancy check between the RA AGL altitude data that is received by the opposite IOP and the RA AGL altitude data that is shown on the PFD. If a difference is sensed between



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these inputs, then a RAD ALT MISMATCH message comes into view on the opposite PFD.

The FDPS monitors the status of its related DH setting. The IOP1 sends the DH status to the GPWS. When the higher DH setting is reached the IOP1 FDPS sends the DH data to the GPWS. The "MINIMUMS, MINIMUMS" aural warning comes on if the highest DH setting is more than 50 ft (15.2 m).

Each Stall Protection Module (SPM1, SPM2) supplies a discrete RA Test #1 Enabled discrete input that is used to test the SPS to RA system interface. The SPS test switch selection supplies a discrete RA Test #2 Enabled discrete input that is also used to test the SPS to RA system interface. Each Stall Protection Module (SPM1, SPM2) supplies an analogue test voltage input that is used to test the SPS to RA system interface.

Each Stall Protection Module (SPM1, SPM2) receives a discrete 200 ft (60.96 m) altitude trip output that is used to prevent SPS stick pusher operation when the AGL altitude is below 200 ft (60.96 m) .

Each RA transmitter/receiver sends data to the TCAS through a single data bus.

Transmitter/Receiver

[Refer to Figure 7.](#)

The RA transmitter/receiver is a digital computing unit that has the modules that follow:

- An RF Board
- Power Supply and Logic Board.

Digital controlled circuitry is used to meet the functional requirements as controlled by the embedded software.

All of the transmitter/receiver parts are attached to a one-piece chassis.

The transmitter/receiver weights 2.8 lb (1.27 kg). It is 11 in. (279 mm) long, 3 in. (76 mm) wide, and 3.5 in. (89 mm) high. The aircraft interface connectors and the zero ft calibration switch are located on the front face of the unit.

Each transmitter/receiver is installed in a mounting rack that is located in the forward area of the center fuselage of the aircraft, below the floor. The RA 1 mounting rack is located at station X 208.6, Y-3.5 and Z 87.5, and the RA 2 mounting rack is located at station X 208.6, Y+3.5 and Z 87.5.

Antenna

[Refer to Figure 8.](#)

The RA microstrip antenna operates in the RA frequency band of 4200 through 4400 MHz. The antennas are linear polarized and have a beam width coverage more than 45 degrees in the roll plane and more than 40 degrees in the pitch plane.

The antenna is made from lightweight, durable Teflon and glass material with an aluminum base and TNC coaxial connector.

Each antenna weighs 3 oz (90 g). It has a 0.15 in. (3.81 mm) high profile and is 3.65 (92.7 mm) by 3.5 in. (89 mm) wide. Each antenna can send or receive transmissions.

Each antenna is installed to the aircraft structure with four mounting screws located at each corner. A "dry fit" sealing gasket electrically bonds each antenna to the aircraft structure.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The gasket is an aluminum foil gasket with an elastomeric sealant. The pressure applied during installation gives a contouring feature that lets the gasket form itself to the two surfaces. The moisture free, contoured RF seal requires no maintenance.

The RA 1 and RA 2 transmit and receive antennas are located inline and spaced 20 in. (508 mm) apart on the bottom of the centre fuselage section.

On aircraft with ModSum 4–201612 and without ModSum 4–902453 or SB 84–34–151 incorporated, the arrangements of the antenna are as follows:

TX1 (X126.0, Y2.6 and Z81.0), RX1 (X146.0, Y2.6 and Z81.0), TX2 (X166.0, Y2.6 and Z81.0) and RX2 (X186.0, Y2.6 and Z81.0)

On aircraft with ModSum 4–300800 or 4–309215 or 4–902453 or SB 84–34–151 incorporated, the arrangements of the antenna are as follows:

TX1 (X126.0, Y2.6 and Z81.0), RX1 (X146.0, Y2.6 and Z81.0), RX2 (X166.0, Y2.6 and Z81.0) and TX2 (X186.0, Y2.6 and Z81.0)

Antenna Cables

The RA system uses coaxial cables to connect the antennas to the transmitter/receiver. The coaxial cable is a transmission line with two conductors. A dielectric insulator separates the center conductor and the braided outer conductor.

The RA transmitter/receiver can use different lengths of coaxial cables. It is calibrated to 0 ft during installation when a recessed calibration switch located on the front faceplate of the Transmitter/Receiver is pushed. The 0 ft calibration of the RA system

is done when the aircraft is static and in the normal Weight On Wheels (WOW) position.

The specified installation length is between 20 and 55 ft (6.1 to 16.8 m).

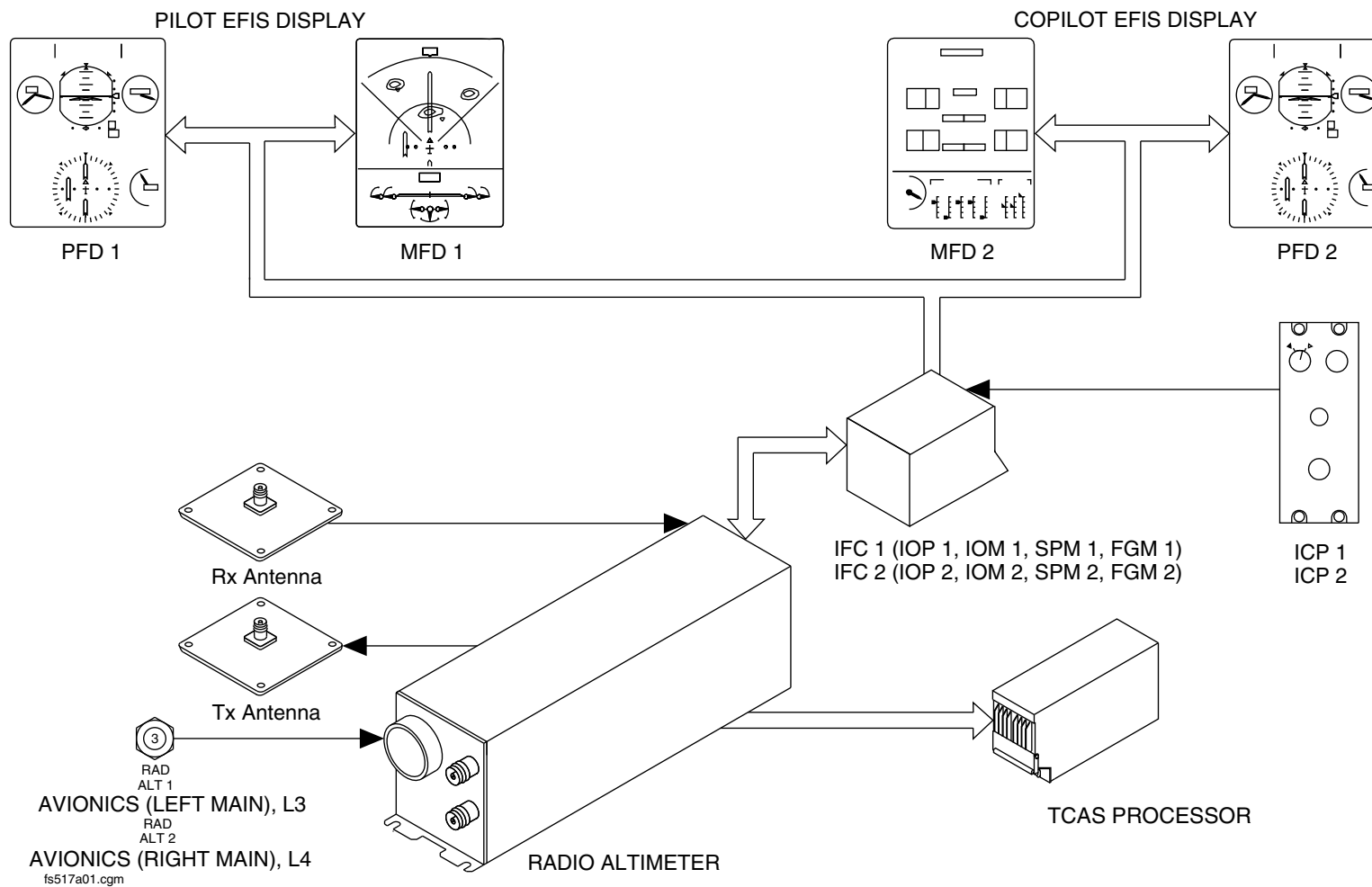
Beta Lockout

The delayed WOW signal during landing causes a delay in the release of the propeller beta lockout function, which prevents the crew to select the propeller ground beta range and increases the rollout and stopping distance.

The occurrences of the delayed WOW signal is estimated by the addition of the two radio altimeter discrete signals with the WOW signals to the Propeller Electronic Controls (PECs). This is done by the addition of a configuration module to each radio altimeter transceiver, which provides the 25 ft. (7.62 m) trip point discrete signal input to the PECs. This allows the beta lockout system to be disabled before the main landing gear touchdown at 25 ft. (7.62 m) AGL.



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RADIO ALTIMETER BLOCK DIAGRAM
Figure 1

PSM 1-84-2A
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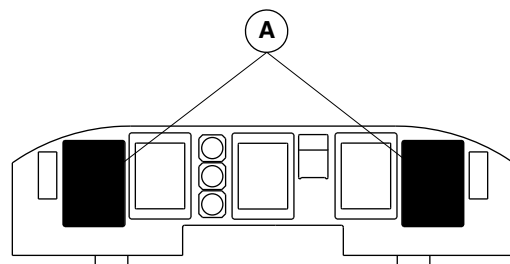
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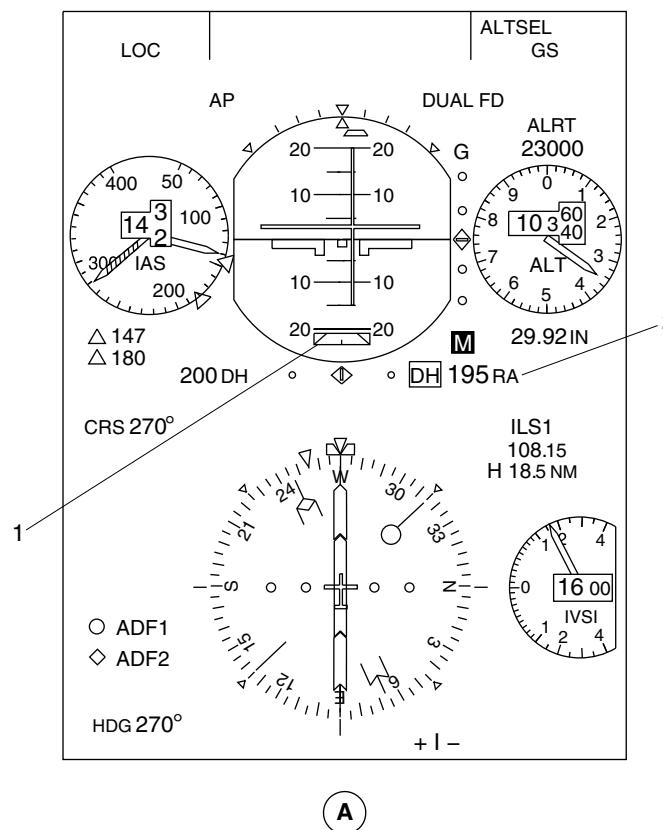
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Rising Runway Indication.
2. Radio Altitude Display.



fs639a01.cgm

RADIO ALTIMETER INDICATIONS
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

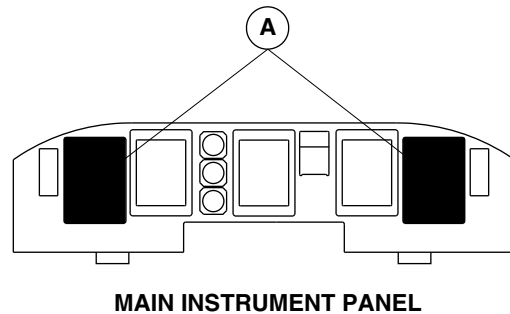
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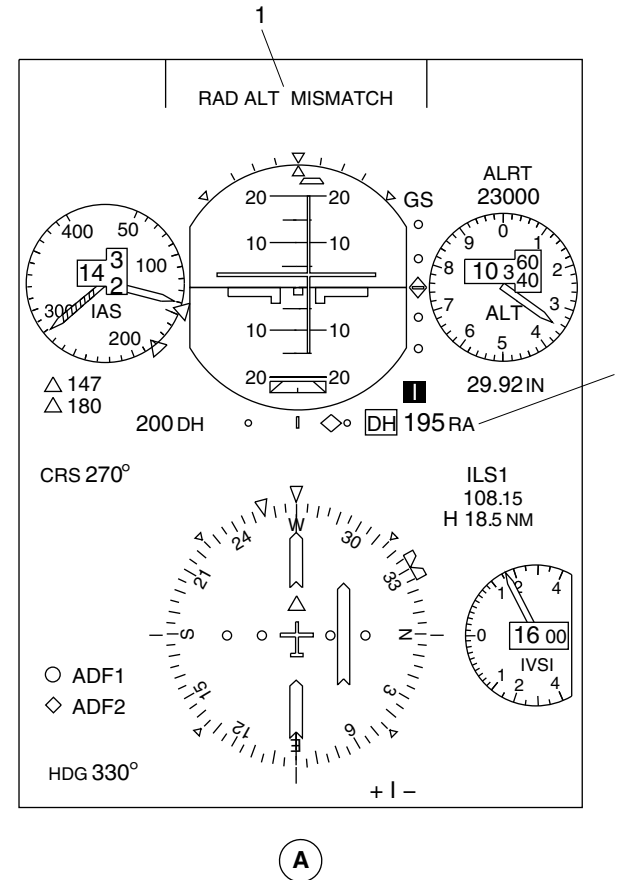
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Radio Altimeter Mismatch Message.
2. Radio Altitude Display.



fs641a01.cgm

RADIO ALTIMETER MISMATCH INDICATION
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-44-00
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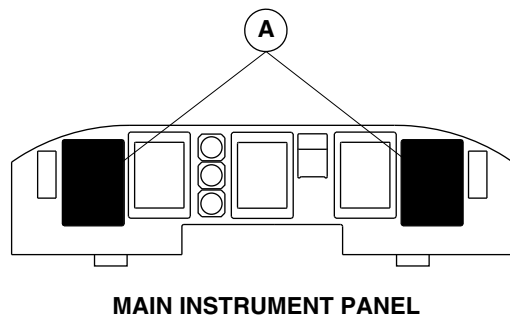
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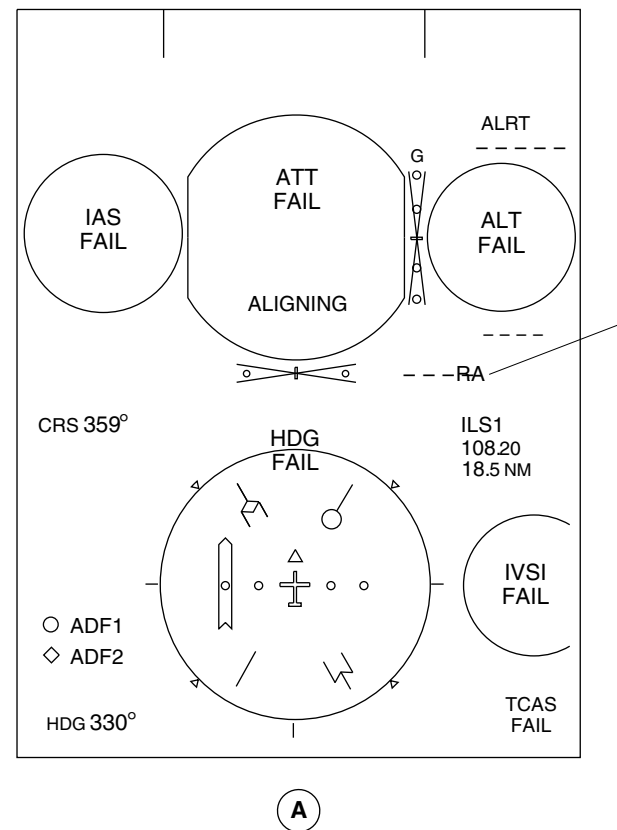
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Radio Altimeter Fail Annunciation.



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RADIO ALTIMETER FAIL ANNUNCIATION
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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Config 001

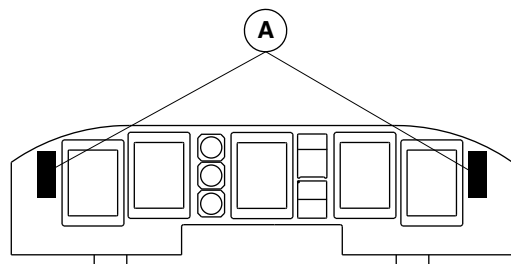
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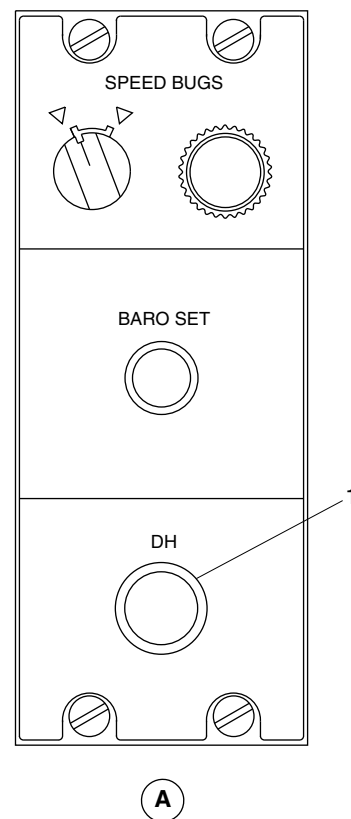
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Decision Heading Selector Knob.



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INDEX CONTROL PANEL, DH KNOB
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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Config 001

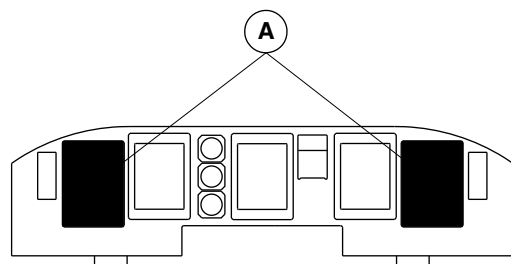
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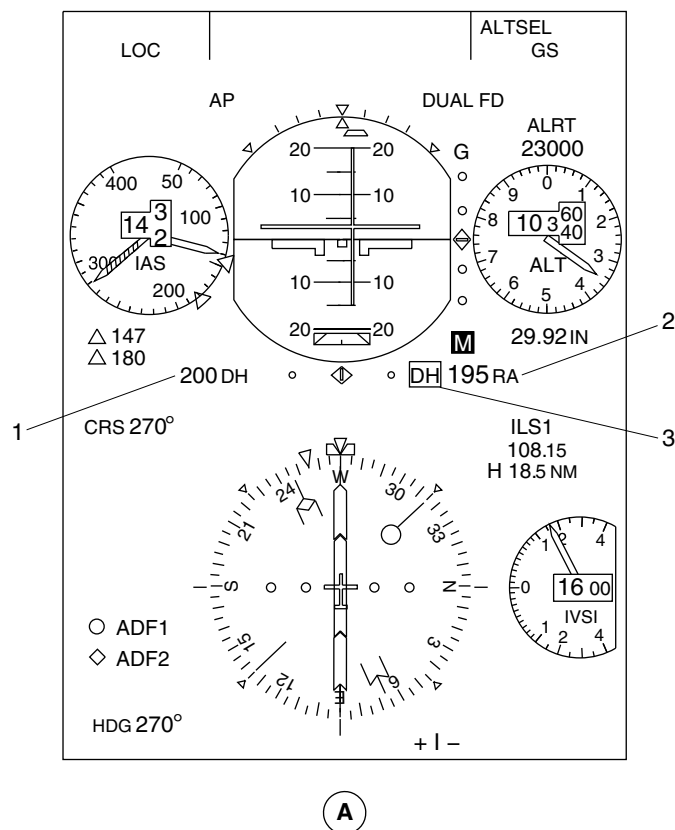
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Decision Height Indication.
2. Radio Altitude Display.
3. Decision Height Annunciation.



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RADIO ALTIMETER DECISION HEIGHT PROCESSING INDICATIONS
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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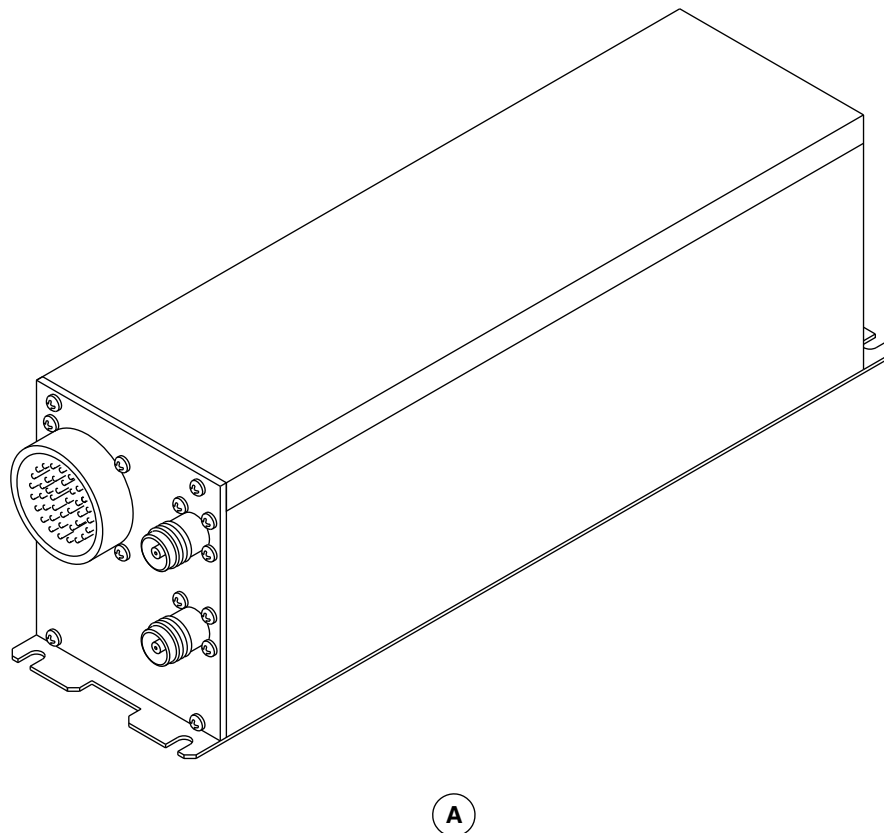
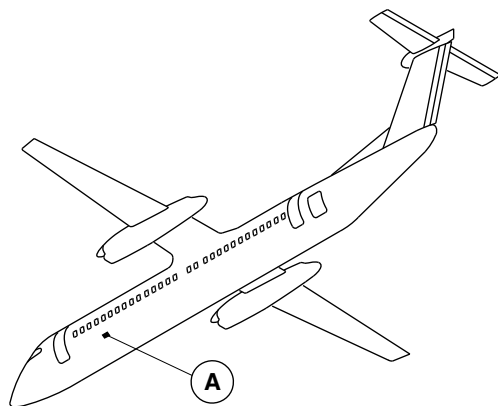
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RADIO ALTIMETER TRANSMITTER/RECEIVER LOCATOR
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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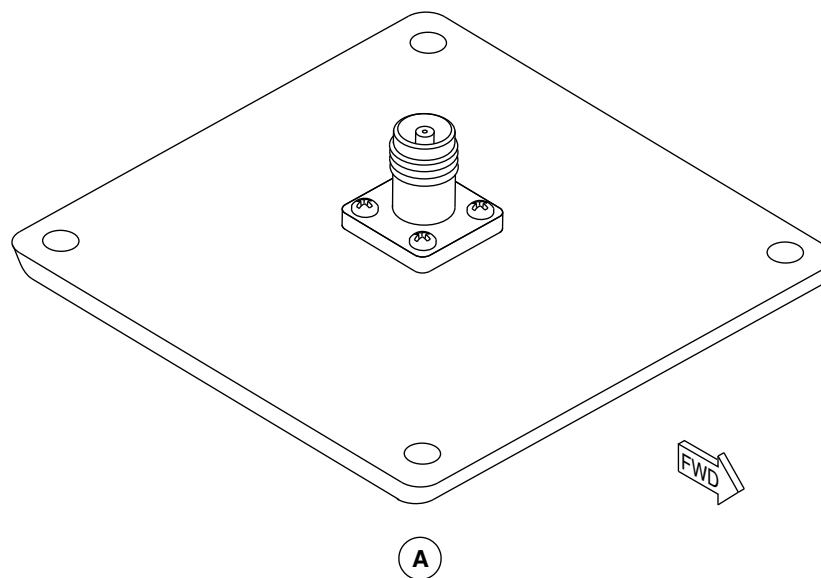
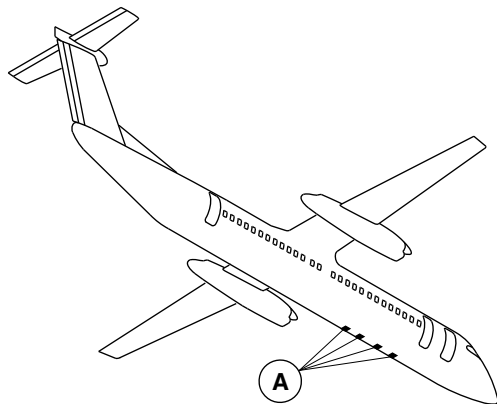
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RADIO ALTIMETER ANTENNAS LOCATOR
Figure 8

PSM 1-84-2A
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RADIO ALTIMETER (COLLINS)

Introduction

The Radio Altimeter (RA) system calculates the low level Above Ground Level (AGL) altitude for display and for use by other aircraft systems.

The system allows the Decision Height (DH) altitude to be set for use during the approach phase of a flight.

General Description

[Refer to Figure 1.](#)

The RA system transmits a signal to the ground and then it receives the reflected signal to calculate the AGL altitude.

The aircraft Primary Flight Displays (PFD1, PFD2) show the RA system parameters that follow:

- Above Ground Level (AGL) altitude
- RA mismatch
- Rising runway symbol
- DH status.

The RA System is used to make DH calculations. The AGL altitude data is compared to the DH value set by the flight crew. The Primary Flight Displays (PFD1, PFD2) show the selected value. The

Electronic Instrument System (EFIS) and Audio and Radio Management System (ARMS) are used to alert the flight crew both visually and aurally when the aircraft reaches the set value.

In addition, the Radio Altimeter system supplies a signal to the Stall Protection System to prevent it from operating when the altitude is less than 200 ft (60.96 m) AGL.

The Radio Altimeter (RA) system has the components that follow:

- Transmitter/receiver (34-44-01)
- Transmitter/receiver Antenna (34-44-11 and 34-11-16).

Digital and analog outputs are available from RA. ARINC 429 digital outputs are routed directly to No. 1 and No. 2 Input Output Processors (IOPs), Traffic Alert and Collision Avoidance System (TCAS) and the Head-Up Guidance System (HGS) (if installed). From the IOPs, radio altitude is further distributed to the systems that follow:

- Auto Flight Control System (AFCS)
- Stall Protection Module (SPM1, SPM2)
- Primary Flight Displays (PFD1, PFD2)
- Flight Data Recorder (FDR)
- Enhanced Ground Proximity Warning System (EGPWS)
- Central Diagnostic System (CDS).

Analog output from the No. 1 RA is directly given to the HGS in the ALT-55 format. The format of ALT-55 is as follows:

- 20mV/foot from -20 to 500 feet
- 10.4 Vdc + 3mV/foot from 500 to 2500 feet

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- 18 Vdc above 2500 feet

Detailed Description

A Frequency Modulated Carrier Wave (FMCW) is transmitted by the RA transmitter/receivers (RA1, RA2). The Frequency Modulated Carrier Wave (FMCW) continuously changes its frequency and compares the time it takes for the transmitted signal to go to the ground and back to the new frequency.

The received signal is mixed with the transmitted signal to make an Intermediate Frequency (IF). The IF is directly proportional to the time that it takes the signal to go to the ground and back. The Intermediate Frequency (IF) is changed into ARINC 429 and analog outputs.

The RA transmitter/receiver gives continuous AGL altitude data from 2500 ft (762 m) until landing. The center operating frequency of the transmitter that is routed through the transmit antenna is 4300 ± 15 MHz. It is continuously modulated at any selectable frequency (50 or 51 Hz) rate, with a linear FM deviation of 100 MHz (nominal). The nominal transmitted power of the signal is 350 mW.

When two RA systems are installed, both transmitter/receivers will continuously function at the same time. An external hardware strap, which is part of the aircraft wiring, is connected to the second RA transmitter/receiver interface connector.

Pin-programming at the rear LRU connector provides source identification and distinguishes the modulating frequencies. Electrical grounds at pins 38 and 40 respectively, identify the No. 1 and No. 2 units and modulate the RF carrier at 50 Hz and 51 Hz respectively.

In normal powered-up mode of operation, the RA transmitter/receiver LRU does a comprehensive self test routine. The

status of the transmitter/receiver is sent to the Central Diagnostic System (CDS).

[Refer to Figure 2.](#)

All AGL altitude data and trip point status information from the RA transmitter/receiver is communicated directly to the relevant Integrated Flight Cabinets (IFCs) and the optional Traffic Alert and Collision Avoidance System II (TCAS II). The AGL altitude data together with the associated DH status information, and rising runway symbology annunciation are displayed on the PFDs, as controlled by the ownside IFC and PFD.

The PFDs show the altitude from zero to 2500 ft (762 m) AGL. When the aircraft RA AGL is greater than 2500 ft (762 m), the RA display symbols are removed from the PFDs.

The altitude increases and decreases in 5 ft (1.52 m) increments below 200 ft (61 m) AGL and increases and decreases in 10 ft (3.05 m) increments above 200 ft (61 m) AGL.

A rising runway symbol gives an analog indication of AGL altitude. It moves vertically, linearly and correspondingly to the AGL altitude.

It comes into view at the bottom of the attitude sphere when the aircraft is at 200 ft (61 m) AGL altitude. It moves towards the airplane symbol as the altitude AGL decreases and touches the bottom of the airplane symbol at touchdown.

The rising runway symbol goes out of view when radio altitude is not available or the aircraft is above 200 ft (61 m) AGL altitude.

Failure of one RA will cause the affected side to display immediately radio altitude from the valid system. No flagged altitude readout will result. However, the advisory area of the engine display will show the message RA1 FAIL or RA2 FAIL in white characters.

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DH annunciation is determined from the higher of the two possible pilot/co-pilot PFD DH settings.

Failure of both RA will result in the display of red dashes adjacent to a red “RA” legend in place of displayed height information.

[Refer to Figure 3.](#)

The Flight Data Processing System (FDPS) senses AGL differences between the RA transmitter/receivers (RA1, RA2), or between an output and shown parameter. If there is a mismatch condition, the FDPS causes the RA label to change from white to yellow and flash initially and then stay on steady.

In addition, the advisory display areas of the Primary Flight Displays (PFD1, PFD2) show a RAD ALT MISMATCH message. It comes into view and flashes in yellow text and then stays on steady.

[Refer to Figure 4.](#)

A failure annunciation shows four red dashes in place of the RA altitude on the PFD1 and PFD2 and “RA” will be annunciated in red. If a Traffic Collision Avoidance System (TCAS) is installed, a TCAS FAIL message will also be shown in yellow text on the PFD1 and PFD2.

The rising runway symbol will also be removed from the PFDs.

[Refer to Figure 5.](#)

For DH calculations, the DH knob located on the Index Control Panel (ICP1, ICP2) is turned to set a DH altitude.

[Refer to Figure 6.](#)

The related PFD1 and PFD2 shows the selection as three digits in cyan followed by a white DH label. It shows the altitude selection in one-foot increments from zero through 999 ft (304.49 m).

Three white dashes replace the digits when PFD1 and PFD2 do not receive valid data from the Integrated Flight Cabinets (IFC1, IFC2).

The DH label shows that the aircraft is at the selected height.

When aircraft is at or less than 100 ft (30.5 m) AGL above the selected DH, the Primary Flight Displays (PFD1, PFD2) will show a white box. When the aircraft goes below the DH, a yellow DH label comes into view inside the white box.

The DH indication is out of view when the altitude AGL is greater than the DH setting.

The 28 Vdc left and right main buses supply power to the RA transmitter/receivers (RA1, RA2) through 3 ampere circuit breakers. The RA transmitter/receivers (RA1, RA2) system circuit breakers are located in positions L3 and L4 on the 28 Vdc avionics circuit breaker panel.

When two RA transmitter/receivers are installed, each unit has a related transmit and receive antenna.

In this configuration, both RA transmitter/receivers (RA1, RA2) supplies data to both Integrated Flight Cabinets (IFC1, IFC2) and the Traffic Collision Avoidance System (TCAS). When one RA transmitter/receiver (RA1, RA2) fails, the Integrated Flight Cabinets (IFC 1, IFC2) use data from the other RA transmitter/receiver (RA1, RA2). No indication of the status of the failed RA transmitter/receiver is shown.

NOTE

The status of the failed RA transmitter/receiver (RA1, RA2) is transmitted and stored in the Central Diagnostic System (CDS).

Both RA transmitter/receivers (RA1, RA2) supply a discrete health status output to each IOM1 and IOM2 for CDS monitoring purposes.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Both RA transmitter/receivers (RA1, RA2) supply radio height data to each Input Output Processors (IOP1, IOP2). A Flight Data Processing System (FDPS) within each IOP controls the supply of data to the systems that follow:

- Enhanced Ground Proximity Warning System (EGPWS)
- Auto Flight Control System (AFCS)
- Stall Protection System (SPS)
- Flight Data Recorder (FDR)
- Primary Flight Displays (PFD1, PFD2).

Each FDPS does a discrepancy check between the RA AGL altitude data that is received by the opposite IOP and the RA AGL altitude data that is shown on the PFD. If a difference is sensed between these inputs, then a RAD ALT MISMATCH message comes into view on the opposite PFD.

The FDPS monitors the status of its related DH setting. The IOP1 sends the DH status to the EGPWS. When the higher DH setting is reached, the IOP1 FDPS sends the DH data to the EGPWS. The "MINIMUMS, MINIMUMS" aural warning comes on if the highest DH setting is more than 50 ft (15.2 m).

Each Stall Protection Module (SPM1, SPM2) supplies a discrete RA Test #1 Enabled discrete input that is used to test the SPS to RA system interface. The SPS test switch selection supplies a discrete RA Test #2 Enabled discrete input that is also used to test the SPS to RA system interface. Each Stall Protection Module (SPM1, SPM2) supplies an analog test voltage input that is used to test the SPS to RA system interface.

Each Stall Protection Module (SPM1, SPM2) receives a discrete 200 ft (60.96 m) altitude trip output that prevents the SPS stick pusher operation when the AGL altitude is below 200 ft (60.96 m).

Each RA transmitter/receiver sends data to the Traffic Alert and Collision Avoidance System (TCAS) through a single data bus.

The RA No. 1 valid discrete is also input to the HGS when applicable. When interfaced to the optional HGS installation, each RA transmits radio altitude to the HGS via an ARINC 429 LS interface. The No. 1 RA also provides an analog DC radio altitude and discrete valid signal.

Transmitter/Receiver

[Refer to Figure 7.](#)

The RA is a digital computing LRU that contains four functional modules that follow:

- Transmitter
- Receiver
- Power Supply
- Digital

Connection to other on-board systems is made through a lighting protection assembly that is a part of the rear connector.

There is also an interconnection card that provides only "pass-through" connections between the power supply and the digital card.

All of the transmitter/receiver parts are attached to a one-piece chassis.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The transmitter/receiver weighs 4.5 lb (2.04 kg). It is 13.92 in. (353.69 mm) long, 3.56 in. (90.42 mm) wide, and 3.33 in. (84.58 mm) high. The aircraft interface connectors are located on the front face of the unit.

The two RA transmitter/receiver LRUs are located under the floor in the forward part of the centre fuse of the aircraft at the station points that follow:

- No.1 RA: between X 196 and X 216
- No.2 RA: between X 196 and X 216

Antenna

Refer to Figure 7.

The RA microstrip antenna operates in the RA frequency band of 4200 through 4400 MHz. The antennas are linear polarized and have a beam width coverage of more than 45 degrees in the roll plane and more than 40 degrees in the pitch plane.

The antenna is made from lightweight, durable Teflon and glass material with an aluminum base and TNC coaxial connector.

Each antenna weighs 0.2 lb (0.09 kg). It is 0.15 in. (3.81 mm) high, 3.65 (92.7 mm) long and 3.5 in. (89 mm) wide. Each antenna can send or receive transmissions.

Each antenna is installed on the aircraft with four mounting screws located at each corner. A "dry-fit" sealing gasket electrically bonds each antenna to the aircraft structure.

The RA 1 and RA 2 transmit and receive antennas are located inline and spaced 20 in. (508 mm) apart on the bottom of the centre fuselage section.

On aircraft with ModSum 4–457079 and without ModSum 4–902454 or SB 84–34–155 incorporated, the arrangements of the antenna are as follows:

TX1 (X126.0, Y2.6 and Z81.0), RX1 (X146.0, Y2.6 and Z81.0), TX2 (X166.0, Y2.6 and Z81.0) and RX2 (X186.0, Y2.6 and Z81.0)

On aircraft with ModSum 4–902454 or SB 84–34–155 incorporated, the arrangements of the antenna are as follows:

TX1 (X126.0, Y2.6 and Z81.0), RX1 (X146.0, Y2.6 and Z81.0), RX2 (X166.0, Y2.6 and Z81.0) and TX2 (X186.0, Y2.6 and Z81.0)

Antenna Cable

The RA system uses coaxial cables to connect the antennas to the transmitter/receiver. The coaxial cable is a transmission line with two conductors. A dielectric insulator separates the center conductor and the braided outer conductor.

The RA transmitter/receiver can use different lengths of coaxial cables. It is calibrated to zero ft during installation when a recessed calibration switch located on the front faceplate of the transmitter/receiver is pushed. The zero ft calibration of the RA system is done when the aircraft is static and in the normal Weight On Wheels (WOW) position.

Total length of the coaxial cables is as follows:

- No. 1 RA: 21.25 ft (6.48 m)
- No. 2 RA: 21.80 ft (6.64 m)

[illegible]

Radio Altimeter Block Diagram

Figure 1

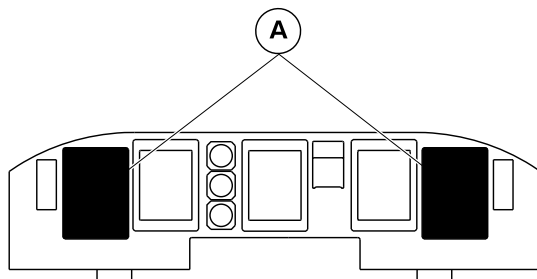
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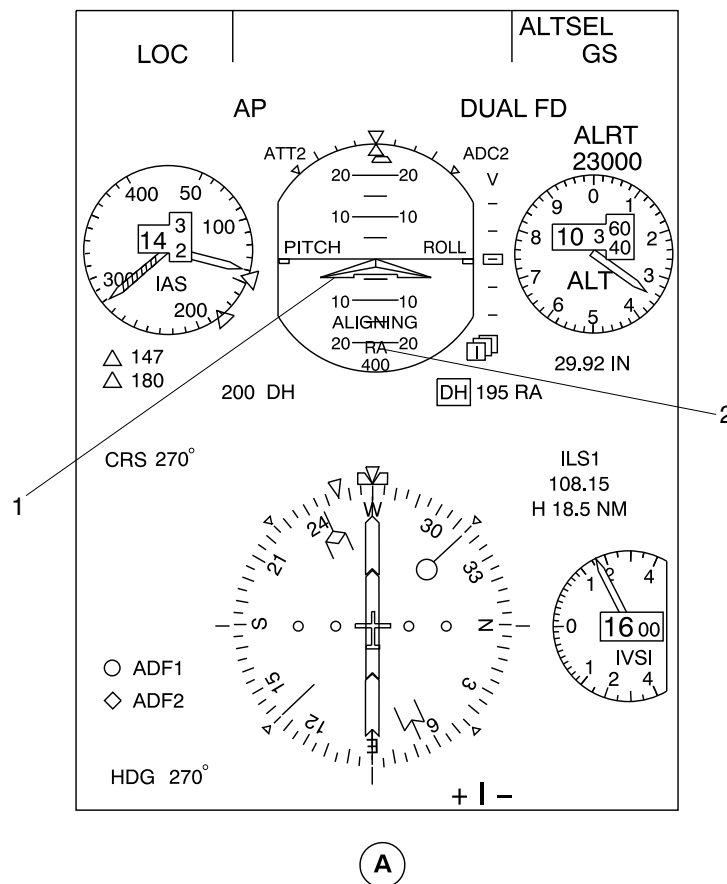
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

- 1. Rising runway indication.
- 2. Radio altitude display.



Radio Altimeter Indication
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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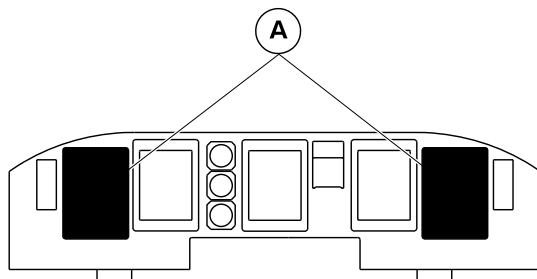
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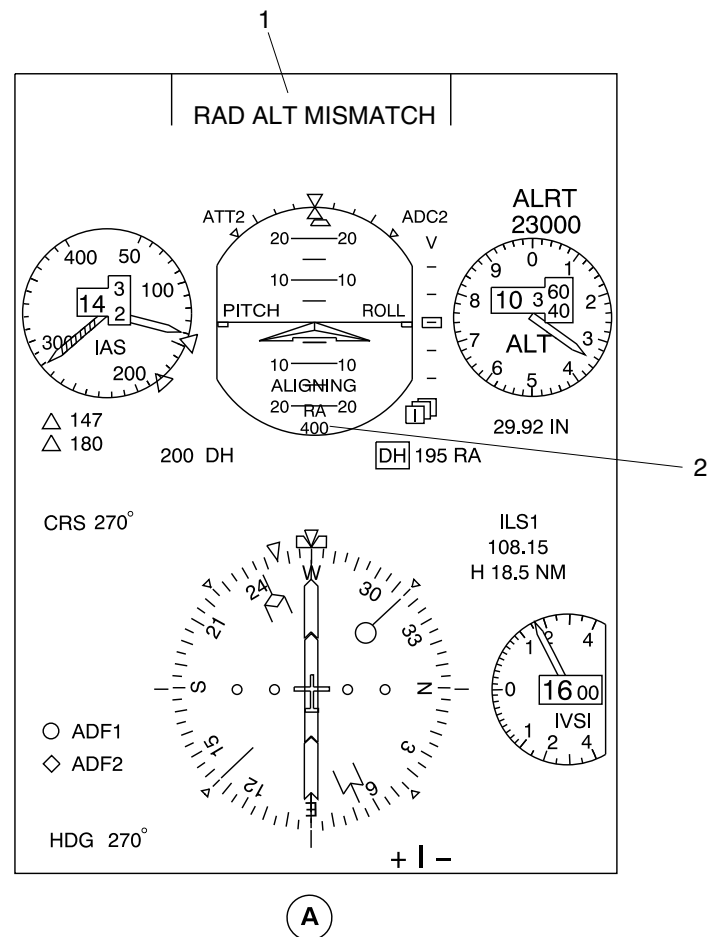
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

- 1. Radio altimeter mismatch message.
- 2. Radio altitude display.



cg1201a01.dg, av, oct12/2010

Radio Altimeter Mismatch Indication
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

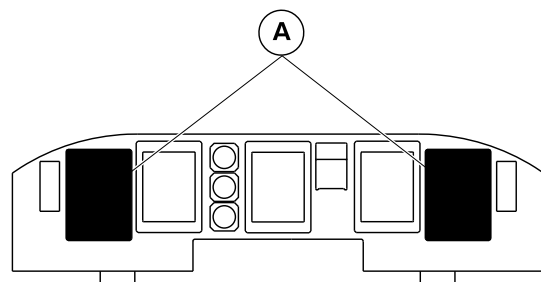
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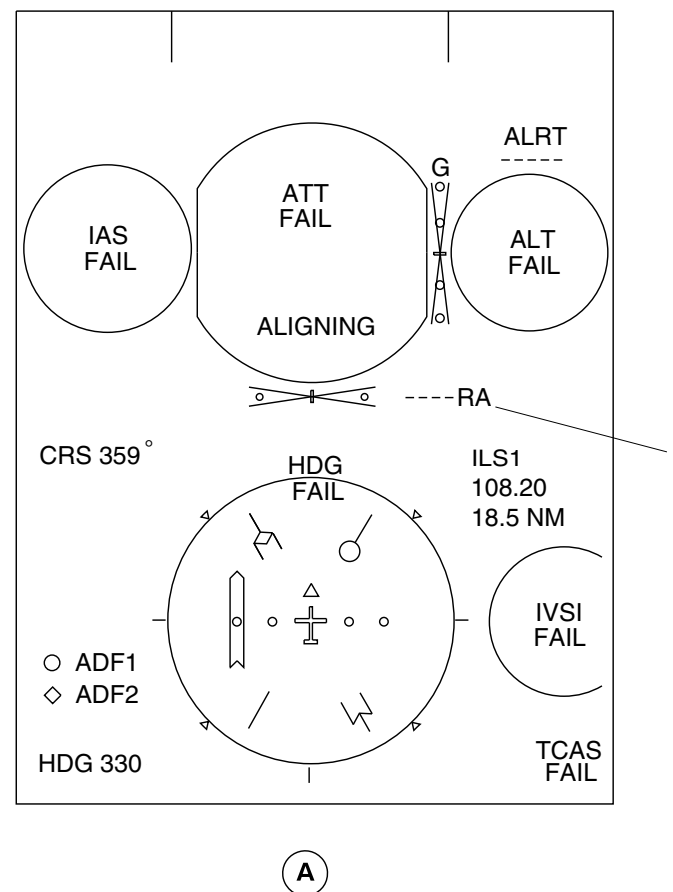
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Radio altimeter fail annunciation.



cg1202a01.dg, av/gv, oct18/2010

Radio Altimeter Fail Annunciation
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-44-01
Config 001

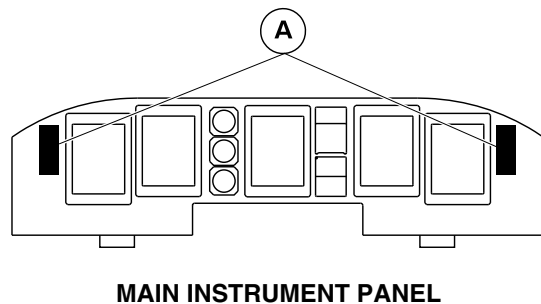
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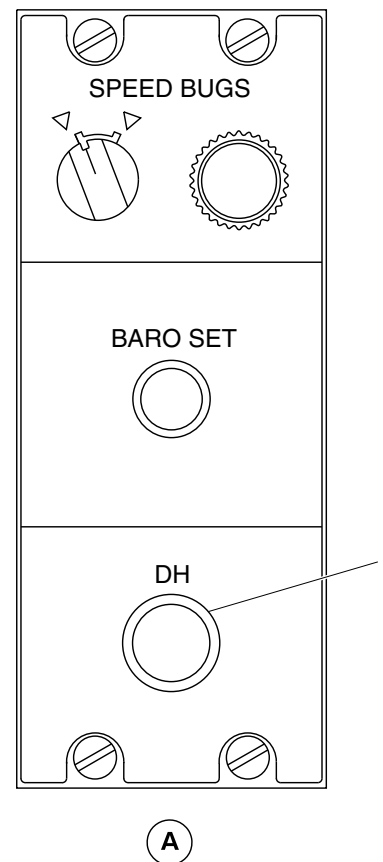
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Decision heading selector knob.

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Index Control Panel
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-44-01
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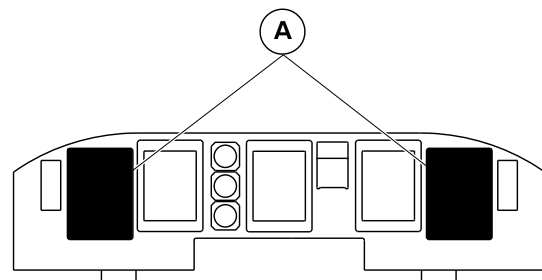
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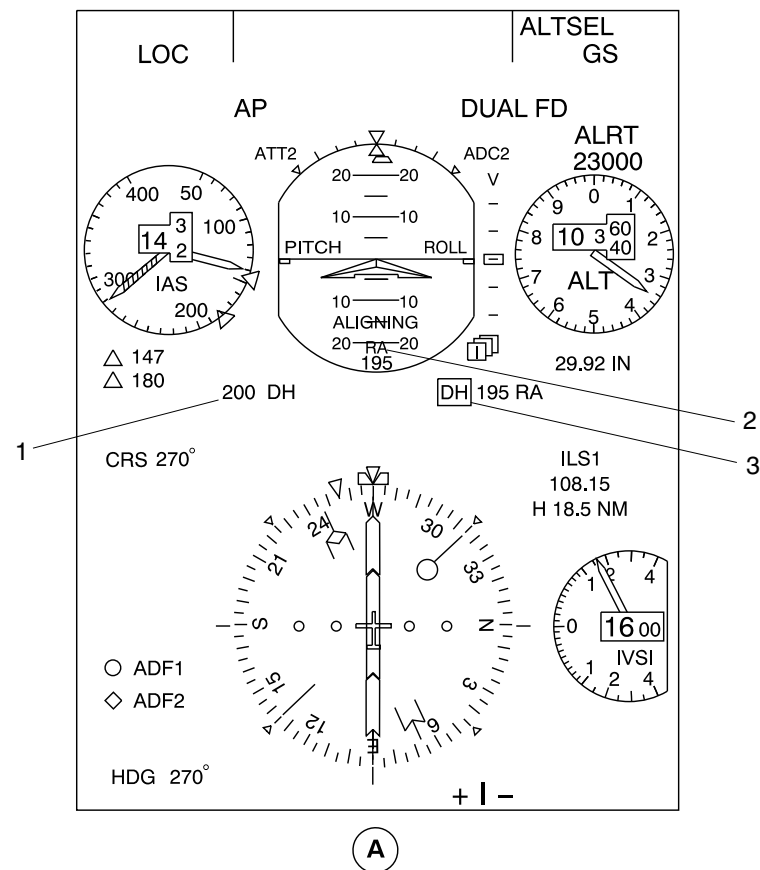
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Decision height indication.
2. Radio altitude display.
3. Decision height annunciation.



cg1204a01.dg, av, oct12/2010

Radio Altimeter Decision Height Processing Indication
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

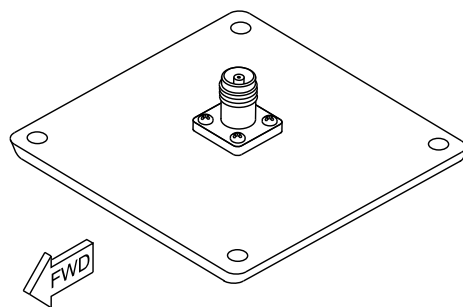
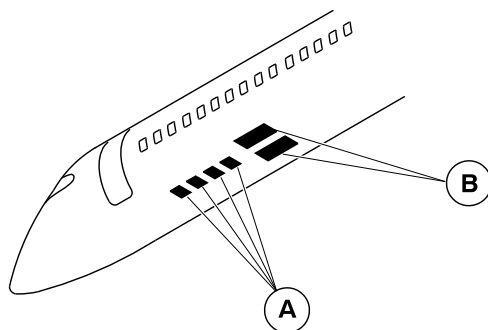
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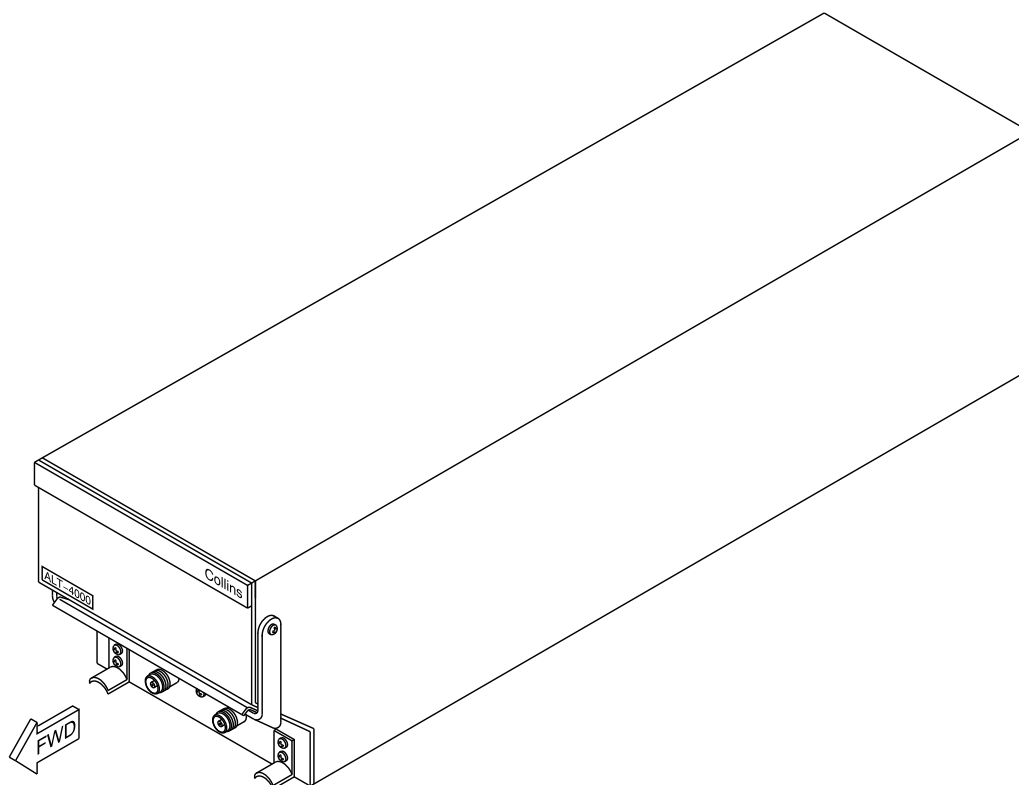


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A RX/TX ANTENNA



B RADIO ALTIMETER

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Radio Altimeter System
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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**ON A/C ALL

34-50-00-001

DEPENDENT POSITION DETERMINING

Introduction

The dependent position determining system has different sub-systems to give visual and aural indications that relate to the conditions that follow:

- VHF Omni-range (VOR)
- Instrument Landing System (ILS)
- Magnetic bearing to a fixed Non-Directional Beacon (NDB)
- Slant range from the aircraft to the DME ground station
- Aircraft identity and its altitude parameters to Air Traffic Control (ATC) ground stations and to other aircraft with Traffic Alert and Collision Avoidance System (TCAS).

General Description

[Refer to Figure 1.](#)

Each sub-system calculates the operational data to give the necessary visual and aural indications.

The dependent position determining system has the sub-systems that follow:

- 34-51-00 VHF Navigation Systems

- 34-52-00 Automatic Direction Finding (ADF)
- 34-53-00 Distance Measuring Equipment (DME)
- 34-54-00 Air Traffic Control Transponder.

VHF Navigation Systems

The Very High Frequency Navigation (VHF NAV) system receives radio signals to give the navigation data that follows:

- VHF Omni-range (VOR)
- Instrument Landing System (ILS).

VHF Omni-range (VOR): The Very High Frequency Omni-range (VOR) ground stations form part of the Victor airways navigation system. The VOR gives guidance on any track to or from the station.

Instrument Landing System (ILS): The Instrument Landing System (ILS) is an integrated system that gives the approach flight path to landing on a runway.

Automatic Direction Finding (ADF)

The Automatic Direction Finding (ADF) system shows the magnetic bearing to a fixed Non-Directional Beacon (NDB).

Distance Measuring Equipment (DME)

The airborne Distance Measuring Equipment (DME) measures the line of sight slant range from the aircraft to the DME ground station. It uses the slant distance data to calculate the ground speed and the time-to-go parameters.



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Air Traffic Control Transponder

The Mode S Transponder system transmits a signal to identify the aircraft and its altitude parameters to Air Traffic Control (ATC) ground stations and to other aircraft equipped with Traffic Alert and Collision Avoidance System (TCAS).

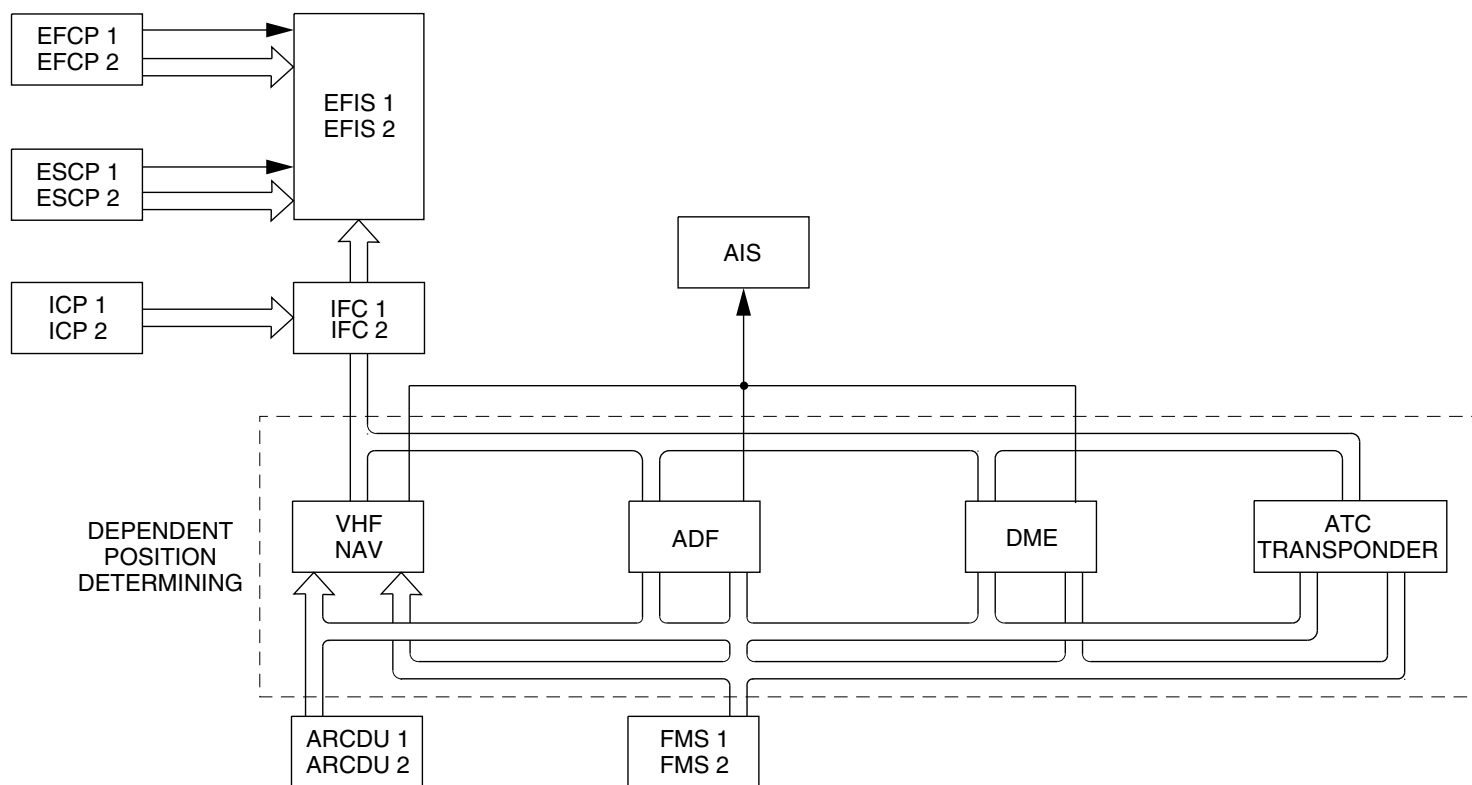
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EFFECTIVITY:
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DEPENDENT POSITION DETERMINING BLOCK DIAGRAM
Figure 1

PSM 1-84-2A
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

34-51-00-001

VHF NAVIGATION SYSTEMS

Introduction

The Very High Frequency Navigation (VHF NAV) system receives radio signals to give the navigation data that follows:

- VHF Omni-range (VOR)
- Instrument Landing System (ILS).

VHF Omni-range (VOR): The Very High Frequency Omni-range (VOR) ground stations form part of the Victor airways navigation system. The VOR gives guidance on any track to or from the station.

Instrument Landing System (ILS): The Instrument Landing System (ILS) is an integrated system that gives the approach flight path to landing on a runway.

General Description

[Refer to Figures 1 and 2.](#)

The two VHF Navigation System operates in VOR or ILS mode depending on frequency selection.

In the VOR mode, the navigation receiver measures the deviation from the selected radial of the Very High Frequency Omni-range (VOR) ground transmitting station.

In the Instrument Landing System (ILS) mode, the Navigation Receiver uses the following landing aids:

- Localizer (LOC)
- Glideslope (GS)
- Marker beacon (MB).

The Localizer (LOC) system helps to guide the aircraft to the centreline approach path to the runway. The Glideslope (GS) system gives the approach angle of an aircraft. It helps to guide the aircraft down an approximate 3 degree glideslope. The Marker Beacons (MB) designate and mark an area above themselves. Up to three MBs are located near the centreline of the approach to the runway. They use the following designations:

- Outer
- Middle
- Inner.

The Very High Frequency Omni-Range (VOR) and Localizer (LOC) ground stations periodically transmit an identification signal. The navigation receiver supplies the audio to the Audio and Radio Management System (ARMS). The Marker Beacon (MB) continuously transmits its identification signal. The navigation receiver, receives the Marker Beacon (MB) audio when directly above the transmitter. The audio is supplied to the Audio and Radio Management System (ARMS).

The VHF receivers are controlled by the components that follow:

- Audio and Radio Control Display Units (ARCDU1, ARCDU2)
- Flight Management System (FMS).



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The ARCDU manually tunes the VOR and ILS (localizer and paired glideslope) frequencies. The FMS provides automatic tuning.

The ARCDU or FMS is used to self-test the marker beacons.

The Electronic Instrument System (EIS) shows the information that follows:

- VOR bearing course
- VOR lateral deviation
- Localizer lateral deviation
- Glideslope vertical deviation
- Marker beacon passage.

The Very High Frequency (VHF) Navigation System has the components that follow:

- Receiver, VHF Navigation (34–51–01)
- Antenna, Glideslope (34–51–11)
- Antenna, VOR/LOC (34–51–16)
- Coupler, Antenna VOR/LOC (34–51–21)
- Splitter, Antenna VOR/LOC (34–51–26)
- Antenna, Marker (34–51–31)
- Splitter, Marker (34–51–36).

The VHF Navigation Receivers (VHF NAV1, VHF NAV2) supply navigation data to the systems that follow:

- Flight Guidance Modules (FGM1, FGM2)
- Audio and Radio System (ARMS)

- Input/Output Processors (IOP1, IOP2)
- Distance Measuring Equipment (DME)
- Flight Management System (FMS).

Detailed Description

Refer to Figure 3.

The VHF Navigation Receivers (VHF NAV1, VHF NAV2) receive radio signals and data from the navigation ground stations that follow:

- VOR/DME
- VORTAC
- ILS.

VHF Omni-range (VOR): The VHF Omni-range (VOR) calculates the data that follows:

- The aircraft's position with reference to a VOR ground station radial
- A flight path toward or away from the station.

The VOR ground station uses two antennas. The VOR ground station uses the same carrier frequency to transmit the signals to the aircraft that follow:

- Reference Phase, Frequency Modulated (FM)
- Variable phase, Amplitude Modulated (AM).

Reference Phase: A non-directional (omni-directional) antenna transmits the Reference Phase as a Frequency Modulated (FM)

PSM 1–84–2A

EFFECTIVITY:

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signal. It has a 9960 Hz sub-carrier that deviates 480 Hz 30 times a second. Its phase is the same through 360 degrees of heading.

Variable Phase: A directional antenna that turns at 30 revolutions per second and transmits a horizontally polarized signal. The phase of the received signal varies at a rate that is the same as the antenna rotation. The aircraft receives a Radio Frequency (RF) signal that appears as 30 Hz Amplitude Modulation (AM). Its phase varies through 360 degrees of heading.

The two signals align exactly in phase on magnetic north. There is phase differential at any other point of azimuth around the station. The variable phase signal lags the reference signal. The VHF navigation receiver separates the 30 Hz frequency reference and variable phase signals from the RF carrier. It measures the difference in phase and shows the result in terms of a selected omni-bearing line and the amount of deviation from it. It does not calculate direction.

A VOR deviation pointer will centre on either of two course selector settings that are 180 degrees apart to show the magnetic bearing to or from the VOR ground station. Bearing pointers are used to show the magnetic bearing to the VOR ground station.

The VOR ground stations transmits the audio identifiers that follow:

- Three-letter code group
- Voice.

The VOR ground stations continue to identify the code groups during weather and voice broadcasts. The relative audio levels of the

broadcasts are set so that the code or voice transmission is easy to hear without interference from the other.

[Refer to Figure 4.](#)

Instrument Landing System (ILS): The Instrument Landing System (ILS) is a precision instrument approach system. It guides the aircraft to a position from which the flight crew can make a landing.

The Instrument Landing System (ILS) supplies the information that follows to the VHF navigation system:

- Localizer
- Glideslope
- Marker Beacon.

[Refer to Figure 5.](#)

Localizer: The localizer system is used to give the position of the aircraft relative to a straight approach line to the runway. The ground based localizer antenna system sends two transmission patterns at the same frequency. One modulates at 90 Hz and the other at 150 Hz. The radio beams overlap to give a 5 degree angle to the centreline of the runway.

There are two approaches to a runway centreline:

- Front course (Localizer or ILS)
- Back course.

Front Course: A front course approach occurs when the aircraft is flown towards the departure end of the runway. The 90 Hz area is to the left and the 150 Hz is to the right.



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Back Course: A back course approach occurs when the aircraft is flown away from the departure end of the runway. The areas reverse. The 150 Hz area is to the left and the 90 Hz is to the right.

The back courses are classified into one of the three categories as follows:

- Usable
- Restricted
- Unusable.

Usable: The usable category is suitable for back course approaches.

Restricted: It is not safe for back course approaches. The flight crew may use the back course for transitions, intersections, holding patterns, or missed approaches.

Unusable: The back course is not suitable or usable for any purpose.

The localizer projects at least 25 mi (40.2 km) from the airport and at an altitude of at least 2000 ft (610 m). A 5 degree angular course width makes an actual width that varies directly with the distance to the transmitter. At touchdown, the course width is usually from 50 to 100 ft (15.2 to 30.5 m) wide. It expands out to about 4600 ft. (1402 m) at a distance of 20 mi (32.2 km) from the transmitter.

When directly on the localizer course, the aircraft receives the 90 and 150 Hz signals equal in strength. If the aircraft is flying to the left of the centreline, the 90 Hz signal is relatively stronger than the 150 Hz signal. Deviation bars move to right to show that the position of the aircraft is to the left of the runway centreline. If the aircraft is flying to the right of the centreline, the 150 Hz signal is stronger than the 90 Hz signal. The deviation bar moves to the left. The sensing reverses when flying a back course. On the back course, the aircraft is navigated away from the deviation bar to get back on course.

Aural identification of the localizer is the same as that of its navigational facility designations and with the letter I or X as prefixes. For example, the Toronto City Centre airport identifier is YTZ. Therefore, the runway 08 designator localizer is identified as ITZ. If the localizer is offset from the runway centreline by more than 3 degrees, the prefix is X. Therefore, the localizer (LOC DME–B) approach to the Toronto City Centre airport, is identified as XTC, because it is offset more than 3 degrees from runway 26.

[Refer to Figure 6.](#)

Glideslope: The glideslope system uses a transmitter and aircraft receiver to give the aircraft approach altitude. It gives a 3 degree glideslope angle from the ground horizontal. The glideslope extends out from the runway for 7 mi (11.3 km).

A directional antenna array located near the approach end of the runway transmits the glideslope signal. The signal has two transmission patterns of Radio Frequency (RF) energy that overlap. The upper transmission pattern contains 90 Hz modulation and the bottom transmitting pattern contains 150 Hz modulation. The equal amplitude intersection of the two transmission patterns form the glideslope. A typical glideslope angle is between 2.25 and 3.0 degrees. When the aircraft is on the glideslope, it receives the 90 and 150 Hz signals equal in strength. If the aircraft is flying above the glideslope, the 90 Hz signal is relatively stronger than the 150 Hz signal. Deviation bars move down to show that the position of the aircraft is above the glideslope. If the aircraft is flying below the glideslope, the 150 Hz signal is relatively stronger than the 90 Hz signal. The deviation bar moves up. Typically, full-scale deflection may vary between 0.4 to 0.75 degrees depending on the installation.

The glideslope altitude at the outer marker is usually within 50 ft (15.2 m) vertically of the procedure turn or minimum holding altitude. The distance between the outer marker and the runway is 4 to 7 mi



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(6.4 to 11.3 km). The glideslope altitude at the middle marker is usually the lowest published general aviation minimum ceiling for the approach. The distance between the middle marker and the runway changes between 3250 to 3750 ft (991 to 1143 m). The glideslope intersects the runway directly opposite to the transmitter. It is between 750 and 1250 ft (229 and 381 m) beyond the threshold.

The Back course may have one or more false glideslope. They are unusable and the flight crew must ignore their indications.

Refer to Figure 7.

Marker Beacons: The Marker Beacons are navigation aids that transmit a low power signal to mark an area above themselves. They are located along the centreline of the runway. The markers are designated from the runway as:

- Outer, 4 to 7 mi (6.4 to 11.3 km)
- Middle, 3250 to 3750 ft (991 to 1143 m)
- Inner, 300 to 2650 ft (91.4 to 807.7 m).

A large number of Instrument Landing Systems (ILS) have middle and outer marker beacons. The inner marker is rarely used.

Each marker has the following identifying signal and audio tone:

- Outer marker, two dashes modulated by 400 Hz
- Middle marker, alternating dot and dashes modulated by 1300 Hz
- Inner marker, six dots per second modulated by 3000 Hz.

As the aircraft approaches the runway, each successive marker has a higher tone. The length of time that the aircraft receives the marker beacon information varies with the speed and altitude of the aircraft.

The reception time of the middle marker is about half that of the outer marker. The reception time of the inner marker is about half that of the middle marker.

NOTE

It is short enough to be missed if the flight crew is not alert.

The Marker Beacon outputs the information that follows:

Table 1: Table 1.

MARKER	AUDIO TONE (HZ)	INDICATION
Outer	400	Blue O
Middle	1300	Amber M
Inner or Airway	3000	White I

The Electronic Instrument System (EIS) shows a positive visual identification when a marker beacon signal is received and the Audio and Radio Management System (ARMS) gives an audio indication.

The VHF Navigation Receivers (VHF NAV1, VHF NAV2) receive VOR and localizer signals in the same frequency band. Their range is from 108.000 to 117.950 MHz in 50 kHz increments. This gives 160 VOR channels and 40 localizer channels. As Internationally accepted, localizer channels are allocated the odd-tenth frequencies between 108.00 and 111.950 MHz. For example, 108.100 MHz is a localizer frequency. These channels automatically tune a paired UHF glideslope frequency. The glideslope portion of the VHF Navigation Receivers (VHF NAV1, VHF NAV2) receives glideslope signals in the frequency range 329.15 to 335.00 MHz spaced 150 kHz apart. There are 40 localizer and glideslope pairs. The VOR channels in the 108



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to 111.95 MHz band are assigned "even tenth" frequencies. For example, 108.000 MHz is a VOR frequency.

When a localizer frequency selection is made, the navigation receiver automatically tunes its paired glideslope frequency. The table that follows shows the localizer and glideslope pairing:

Table 2: Table 2.

LOCALIZER (MHZ)	GLIDESLOPE (MHZ)
108.10	334.70
108.15	334.55
108.30	334.10
108.35	333.95
108.50	329.90
108.55	329.75
108.70	330.50
108.75	330.35
108.90	329.30
108.95	329.15
109.10	331.40

LOCALIZER (MHZ)	GLIDESLOPE (MHZ)
109.15	331.25
109.30	332.00
109.50	332.60
109.55	332.35
109.70	333.20
109.75	333.05
109.90	333.80
109.95	333.65
110.10	334.40
110.15	334.25
110.30	335.00
110.35	334.85
110.50	329.60
110.55	329.45
110.70	330.20



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LOCALIZER (MHZ)	GLIDESLOPE (MHZ)
110.75	330.05
110.90	330.80
110.95	330.65
111.10	331.70
111.15	331.55
111.30	332.30
111.35	332.15
111.50	332.90
111.55	332.75
111.70	333.50
111.75	333.35
111.90	331.10
111.95	330.95

The marker beacon receiver demodulates the amplitude modulated, 75 MHz marker beacon signal.

[Refer to Figure 8.](#)

The Audio and Radio Control Display Units (ARCDU1, ARCDU2) are used to set the VHF navigation modes and to manually tune the VHF Navigation Receivers (VHF NAV1, VHF NAV2) frequencies. The Flight Management System (FMS) will also tune the navigation receivers.

The ARCDU selections control the VHF Navigation frequencies and the VOR or ILS mode to be used.

Table 3: Table 3.

SELECTION	DESCRIPTION
OFF	The ARCDU is not energized
ON	The ARCDU is energized and it controls and tunes its related radio system
BOTH	The ARCDU is energized and it controls and tunes its related and opposite radio systems
FMS	The FMS controls and tunes its related and opposite radio systems

[Refer to Figure 9.](#)

The side key adjacent to the VHF navigation display area is pushed to highlight the preset indication. It is pushed again to make the preset frequency active and the frequency that was active, the new preset frequency. If a VOR frequency is set, the display area will



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

show a VOR1 or VOR2 label. If an ILS frequency is set, the display area will show an ILS1 or ILS2 label.

When the side key adjacent to the VHF navigation display area is pushed, the preset indication in the display area changes to reverse video to show the selection. The TUNE double rotary knobs located at the middle right part of the ARCDU are turned to change the preset indication.

The side key adjacent to the VHF navigation display area is pushed and then the CH key to set the channel memory mode for VHF navigation. The VHF navigation channel memory mode lets the preset frequency area show one of eight different programmed channel.

If the preset frequency shown relates to a channel memory, a channel memory annunciation is shown on the second data line that relates to the channel number.

The TUNE double rotary knobs are turned to change the preset frequency indication from eight programmed channels. The channel memory annunciation changes with the preset (channel memory) frequency indication.

NOTE

Note: If the preset frequency does not relate to a channel memory when the mode is set, no channel memory annunciation is shown. A TUNE double rotary knobs selection causes the display area to start with CH1 when turned clockwise or CH8 when turned counterclockwise.

The CH key is pushed again to change from the channel memory mode back to the frequency selection mode. The channel memory annunciator will go out of view.

NOTE

Note: The TUNE double rotary knobs selection can roll over.

[Refer to Figure 10.](#)

The side key adjacent to the VHF navigation display area is pushed and then the EXP key to show the VHF navigation particular page.

[Refer to Figure 11.](#)

The particular page is used to change the VHF Navigation Receiver (VHF NAV1, VHF NAV2) modes and functions that follow:

- FMS automatic tuning
- Channel programming
- Marker Beacon (MKR) test
- Marker Beacon (MKR) sensitivity.

FMS Automatic Tuning: The side key adjacent to the FMS AUTO TUNE legend is pushed to set the automatic tuning mode. When the Flight Management System (FMS) is automatically tuning, the indication in the display area changes to reverse video to show the automatic tuning mode selection. The FMS AUTO TUNE legend changes from a white letters to black letters on a green background. The VOR active frequency indication changes to show a green AUTO label legend. A new navigation frequency selection causes the FMS automatic tune mode to automatically stop.

Channel Programming: Eight more preset channels are set on a different page for VHF navigation tuning.



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Marker Beacon (MKR) Test: The side key adjacent to the MKR TEST legend is pushed to start a three-second test of the marker receiver. The two Primary Flight Displays (PFDs) show the Inner (I), Outer (O) and Middle (M) marker indications the marker receiver sounds the three marker audio tones through the Audio Integration System (AIS).

The white MKR TEST legend changes to reverse video during the test.

Marker Beacon (MKR) Sensitivity: The side key adjacent the MKR SENS legend is pushed to toggle between LO and HI sensitivities. The indication changes to reverse video to show the selection.

NOTE

Note: When the HI sensitivity is set, the VHF navigation display area will show a green MKR HI indication on the second data line.

[Refer to Figure 12.](#)

The side key adjacent to the CHANNELS legend is pushed to show the channel programming page.

[Refer to Figure 13.](#)

The channel programming page has the indications that follow:

- A index of all radios with channel programming
- Channel memory annunciators (CH1 through CH)
- Programmed frequency values for the radio selection.

The two display areas at the top of the screen show the radios that can be channel programmed. The left display area shows the radios that are related to the pilot's side and the right display area shows the radios that are related to the copilot's side.

One radio is always set and is shown in reverse video.

NOTE

Note: The radio for the related particular page is automatically set. The side key adjacent to display area is pushed to show the channel memory frequencies for a different radio.

Four other display areas each show two channel memory annunciators with their channel memory frequency values that relate to the radio selection.

NOTE

Note: When the ARCDU is set to the ON position, only the related radios can be channel programmed.

[Refer to Figure 14.](#)

The side key adjacent to the channel memory display area is pushed to highlight the top memory frequency indication. It is pushed two times to highlight the bottom memory frequency indication.

When the side key adjacent to the channel memory display area is pushed, the indication in the display area changes to reverse video to show the selection. The TUNE double rotary knobs located at the middle right part of the ARCDU are turned to change the preset indication.

The new memory frequency is set by any condition that follows:

- A side key is pushed
- Five seconds after a TUNE double rotary knobs selection



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- The EXP or PG 1/2 key is pushed to show a different page.

Refer to Figure 15.

The VOR/ILS Potentiometer push–button switches control the audio parameters that follow:

- On
- Off
- Level.

The top part of the potentiometer is pushed to turn the related audio signal to the headphone on or off. The VOR ILS1, VOR ILS2, or MKR push–button switch is pushed on the ARCDU to make the audio available. When the SPKR push–button potentiometer switch is pushed, the VHF Navigation audio is routed to its related flight compartment speaker.

The VHF navigation display area shows the volume level as a vertical bar graph. The height of the bar graph shows the volume selection level. When the audio is set to the off position, the bar graph indication changes to white. It changes to green when the audio is set to the on position.

The VHF navigation audio is available at the observer's headphone jack when the VOR/ILS1, VOR/ILS2 is selected on the observer's audio control panel. An MLS/VOR toggle switch on the headset

panel switches the audio to the headset between the VOR/LOC and MLS.

Refer to Figure 16.

When a VOR1 or VOR2 bearing is selected on the EFIS Control Panels (EFCP1, EFCP2), a distinctive pointer comes into view on the Primary Flight Display.

Refer to Figure 17.

The bearing pointers in the Electronic Horizontal Situation Indicators (EHSI1, EHSI2) show the lateral direction to the navigation aid as follows:

- A white, single bar pointer with a white circle to identify VOR1
- A green, wide two–bar pointer with a green diamond to identify VOR2.

The bottom left part of the Primary Flight Displays (PFD1, PFD2) shows a reminder label for VOR1 and the bottom right part shows a reminder label for VOR2. Each reminder label is related to the EFIS Control Panel selection. Each indication has a symbol that is similar to its bearing pointer with a VOR1 or VOR2 legend.

NOTE

Note: The reminder bearing symbol and legend are the same color, white or green.

Refer to Figure 18.

When a VOR signal has malfunctioned, its bearing pointer goes out of view, but its reminder label stays in view to show the bearing pointer selection.



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If the pilot's EFIS Control Panel (EFCP1) malfunctions, the bearing pointers will automatically show the information that follows:

- The single bar indicates Automatic Direction Finder (ADF1)
- The double bar indicates Very High Frequency Omni-range (VOR2).

If the copilot's EFIS Control Panel (EFCP2) fails, the bearing pointers will automatically show the information that follows:

- The single bar indicates Very High Frequency Omni-range (VOR1)
- The double bar indicates Automatic Direction Finder (ADF2).

The Audio and Radio Control Display Units (ARCDU1, ARCDU2) monitor the ARINC 429 data bus through the Flight Data Processing System (FDPS) for a malfunction of the VHF Navigation system. The ARCDUs show a red FAIL message in the frequency preset area of the VHF navigation display area when a system malfunctions.

[Refer to Figure 19.](#)

The NAV SOURCE rotary switch located on the Flight Guidance Control Panel (FGCP) is used to set the VHF navigation source.

[Refer to Figure 20.](#)

The Primary Flight Display (PFD) shows the navigation source selection in the upper right part of the Electronic Horizontal Situation Indicators (EHSI1, EHSI2). The PFDs show the navigation source annunciations that follow:

- VOR1, VOR2
- ILS1, ILS2

- FMS1, FMS2
- MLS1, MLS2.

The VOR or ILS indication depends on the VHF Navigation frequency selection. A VOR frequency selection will cause the EHSI to show VOR deviation.

The Primary Flight Display will show the VOR parameters that follow:

- Navigation Source Annunciation
- VHF Navigation Frequency
- Selected Course Digital Value
- Selected Course Pointer
- Course Deviation Scale and Bar
- TO/FROM pointer
- Course Deviation Failure Flag.

Navigation Source Annunciation: The navigation source annunciation shows which navigation source is selected by the related NAV SOURCE rotary switch on the Flight Guidance Control Panel (FGCP). When the related source is selected, the annunciation is shown in cyan characters. If an opposite navigation source is selected, the annunciation is shown in yellow fonts.

If the section data has malfunctioned, four white dashes replace the source annunciation.

VHF Navigation Frequency: The frequency of the navigation station is shown in cyan numbers from 108.00 to 117.95.

Selected Course Digital Value: The selected course is controlled by its related COURSE knob on the Flight Guidance Control Panel (FGCP). The selected course digital value is shown by 3 cyan



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numbers from 1 through 360. A degree symbol is shown in white adjacent to the digital value.

If the selection data has malfunctioned, three white dashes replace the source annunciation.

Selected Course Pointer: The COURSE knob is turned to set the VOR course. It turns a cyan pointer around the 360 degree heading scale to show the selected course and supply the navigation receiver with the desired course. Each COURSE knob controls its related course pointer.

When no heading data is received or not correct, the selected course pointer goes out of view.

Course Deviation Scale and Bar: The course deviation scale and bar is part of the course pointer. It shows lateral deviations from the selected VOR course. The course deviation scale and bar moves with the course pointer and the bar moves independently left or right of the course pointer.

The bar is shown to the right of the aircraft symbol when the aircraft is flying to the left of the desired course and is shown to the left of the aircraft symbol when the aircraft is flying to the right of the desired course. The lateral deviation bar centers in the course pointer when the aircraft is on course. Two dots of deviation shows that the aircraft is 10 degrees from the desired course.

TO/FROM pointer: The TO/FROM pointer symbol is part of the course pointer indication. It turns with the course pointer to show the direction of navigation relative to a ground VOR transmitter. The TO/FROM pointer symbol is also shown in cyan.

The conditions that follows cause the TO/FROM pointer symbol to go out of view:

- The VOR ground station is out of range
- Navigation receiver malfunctions
- No VOR course data is available
- No VOR navigation source selection data is available.

[Refer to Figure 21.](#)

Course Deviation Failure Flag: The course deviation failure flag is a red cross shown with the course deviation scale. The conditions that cause the course deviation failure flag to come in view are as follows:

- The VOR ground station is out of range
- Navigation Receiver malfunctions
- No VOR course data available
- No VOR navigation source selection data available.

[Refer to Figure 22.](#)

The MFD1 and MFD2 reversion switches located on the ESID Control Panel (ESCP) are used to set the navigation page. The navigation page comes into view in the map mode format.

[Refer to Figure 23.](#)

The FORMAT push-button switch on EFCP is pushed to change to the navigation page indication from a map (FMS) to arc (VOR/ILS)



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indication independently of the PFD. It is pushed again to show the map indication.

Refer to Figure 24.

The navigation page is shown on the upper part of MFD1 and permanent data is shown on the bottom. The arc mode shows the same VOR navigation parameters as the Primary Flight Display (PFD1, PFD2) indication on a 90 degree heading arc.

Refer to Figure 25.

The navigation page, arc format mode shows a course deviation failure flag for the same conditions as for the Primary Flight Displays (PFD1, PFD2). It shows a red cross on top of the course deviation scale.

Refer to Figure 26.

An ILS frequency selection will cause the related PFD to show the ILS parameters that follow:

- Localizer deviation
- Glideslope deviation
- Marker beacon passage.

Localizer Deviation: The Primary Flight Display will show the localizer parameters that follow:

- Navigation Source Annunciation
- VHF Navigation Frequency
- Selected Course Digital Value
- Selected Course Pointer
- Course Deviation Scale and Bar

- Course Deviation Failure Flag
- Expanded Localizer.

Navigation Source Annunciation: The navigation source annunciation shows which navigation source is selected by the related NAV SOURCE rotary switch on the Flight Guidance Control Panel (FGCP). When the related source is selected, the annunciation is shown in cyan characters. If an opposite navigation source is selected, the annunciation is shown in yellow fonts.

If the section data is not correct, four white dashes replace the source annunciation.

VHF Navigation Frequency: The frequency of the navigation station is shown in cyan numbers from 108.00 to 117.95.

Selected Course Digital Value: The selected course is controlled by its related COURSE knob on the Flight Guidance Control Panel (FGCP). The selected course digital value is shown by 3 numbers from 1 through 360. A degree symbol is shown in white after the digital value.

If the section data has malfunctioned, three white dashes replace the source annunciation.

Selected Course Pointer: The COURSE knob is turned to set the inbound or back course. It turns a cyan pointer around the 360 degree heading scale to show the selected course. Each COURSE knob controls its related course pointer.

NOTE

Note: When no heading data is received or if it is not correct, the localizer selected course pointer does not go out of view.

Course Deviation Scale and Bar: The course deviation scale and bar is part of the course pointer. It shows lateral deviations from the



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localizer. The course deviation scale and bar moves with the course pointer and the bar moves independently left or right of the course pointer.

The bar is shown to the right of the aircraft symbol when the aircraft is flying to the left of the localizer. The bar is shown to the left of the aircraft symbol when the aircraft is flying to the right of the localizer. The lateral deviation bar centers in the course pointer when the aircraft is on the localizer. Two dots of deviation show the aircraft 2 1/2 degrees from the desired course.

Expanded Localizer: An expanded localizer scale is shown below the Electronic Attitude Direction Indicator (EADI). It is used during a Category II precision approach. The expanded localizer scale is 6 times as sensitive as the localizer display. If the AGL altitude is less than 1200 ft. (366 m) and the localizer pointer goes off the scale, the pointer and scale will flash for 5 seconds and then go steady. The Flight Mode Annunciators (FMA) will show an LOC EXCESS DEV message.

Glideslope Deviation: The PFD will show the glideslope parameters that follow:

- Vertical deviation pointer and scale
- Vertical Deviation Source.

Vertical Deviation Pointer and Scale: The Primary Flight Displays (PFD1, PFD2) have a filled cyan pointer and white scale to show deviation from the glideslope. The vertical deviation pointer and scale is shown to the right of the attitude sphere.

If the AGL altitude is less than 1200 ft (366 m) and the glideslope pointer goes off the scale, the pointer and scale will flash for 5 seconds and then go steady. The Flight Mode Annunciators (FMA) will show an GS EXCESS DEV message.

Vertical Deviation Source: The ILS vertical deviation source indication shows a cyan G above the scale to identify the scale.

Marker Beacon: The marker beacon reception is automatic and its indication is always available. It is not a function of an ILS frequency selection.

The PFDs show the Marker Beacon (MB) passage near the attitude sphere. An O, M, or I indication is shown in as a black letter on a blue, yellow or white rectangle background.

NOTE

When the marker beacon receiver is tested, the PFD shows all marker beacon indications at the same time.

[Refer to Figure 27.](#)

The Flight Data Processing System (FDPS) senses signal differences between the ILS1 and the ILS2. If the difference is excessive, a yellow LOC MISMATCH message in the centre row of the centre column. The indications flash for five seconds when they come into view and then go steady.

A GS MISMATCH message is shown at the same location as the LOC MISMATCH message. If a glideslope and localizer mismatch condition occur at the same time, the glideslope mismatch message is shown. It is a more important mismatch message.

[Refer to Figure 28.](#)

Course Deviation Failure Flag: The course deviation failure flag is a red cross shown with the course deviation scale. The conditions that cause the course deviation failure flag to come in view are as follows:

- The navigation ground station is out of range
- Navigation receiver malfunctions



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- No data is available
- No navigation source selection data is available.

The PFDs show the Failure Flags that follow:

- Course Deviation
- Expanded Localizer
- Glideslope.

[Refer to Figure 29.](#)

The MFD1 and MFD2 Reversion switches located on the ESID Control Panel (ESCP) are used to set the navigation page. The navigation page comes into view in the map mode format. The FORMAT push-button switch on EFCP is pushed to change to the navigation page indication from a map (FMS) to arc (VOR/ILS) indication independently of the PFD. The arc mode shows localizer and glideslope parameters as its related PFD indication on a 90 degree heading arc.

[Refer to Figure 30.](#)

The navigation page, arc format shows a Course Deviation Failure Flag for the same conditions as for the PFDs. The course deviation failure flag is a red cross shown with the course deviation scale.

The Audio and Radio Control Display Units (ARCDU 1, ARCDU 2) monitor the ARINC 429 data bus through the Flight Data Processing System (FDPS) formal functions. The ARCDUs show a red FAIL message in the frequency preset area of the VHF navigation display area for system malfunctions.

The left essential and right main busses supply 28 Vdc electrical power through 2 A VHF Navigation Receivers (VHF NAV1, VHF NAV2) circuit breakers. The circuit breakers are located in position

G1 on the Left Essential and D5 on the 28 Vdc avionics circuit breaker panel.

The VHF Navigation (VHF NAV1, VHF NAV2) system has two navigation receivers with common VOR/LOC, glideslope, and marker antennas.

The VOR/Localizer antenna has two low profile blades attached to each side of the vertical stabilizer. It has phase-matched coaxial cables to connect the two antenna blades to a coupler located in the empennage and adjacent to the blades. From the coupler, a single coaxial cable is routed to a VOR/LOC antenna splitter and then to each VHF navigation radio.

Each navigation receiver connects to a single glideslope antenna with two RF connectors. Two separate coaxial cables connect the VHF navigation receivers to the antenna.

Each VHF navigation receiver connects to a single marker beacon antenna. A single coaxial cable connects the antenna to a splitter attached to the avionics rack. The splitter connects the two coaxial cables to the VHF navigation receivers.

The VHF Navigation Receivers (VHF NAV1, VHF NAV2) supply navigation information for display through the Input/Output Processors (IOP1, IOP2). In addition, it supplies VOR and ILS information to the Flight Guidance Modules (FGM1, FGM2) for autopilot and flight director processing.

The Input/Output Processor (IOP1, IOP2) and Flight Guidance Modules (FGM1, FGM2) are located inside its related Integrated Flight Cabinet (IFC1, IFC2).

The VHF Navigation parameters that follow are controlled using ARINC429 Low speed busses (12.0 to 14.5 kilobits per second):



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Table 4: Table 3.

Parameter	Source
Frequency	ARCDU
VOR bearing	IOP 1, IOP 2
Left/Right Deviation	IOP 1, IOP 2
Glideslope Deviation	IOP 1, IOP 2
Outer Marker	IOP 1, IOP 2
Middle Marker	IOP 1, IOP 2
Inner Marker	IOP 1, IOP 2

The Distance Measuring Equipment (DME1, DME2) is tuned by the VHF Navigation Receivers (VHF NAV1, VHF NAV2) or Microwave Landing System Receivers (MLS1, MLS2).

Part of the Flight Guidance Control Panel (FGCP) navigation source selection sends ARINC 429 tuning information to the DME. The navigation source selection causes an Input/Output Module (IOM1, IOM2) discrete to control a relay that routes the tuning information to the DME. The Input/Output Module (IOM1, IOM2) is part of the Flight Data Processing System (FDPS).

The Flight Management System (FMS) uses up to two sources of VOR information for the VORTAC Position Unit (VPU) mode. The Flight Management System (FMS) uses VOR information to calculate position-fixing algorithms.

Receiver, VHF Navigation

[Refer to Figure 31.](#)

The VHF Navigation Receiver has three separate receivers. They receive and process VHF omnirange, localizer, glideslope, and marker beacon data. It supplies left/right and up/down data for indication and Auto Flight Control System (AFCS) guidance.

The VHF Navigation Receiver has the sub-assemblies that follow:

- VOR/Localizer Receiver
- Glideslope Receiver
- Marker Beacon Receiver
- Navigation Converter
- Module Navigation Processor Module
- Power Supply.

The interconnections between assemblies are made using ribbon cables. Maintenance is simplified through modular construction. Modules are hinged and attached to a rigid main frame to allow easy access. The circuit boards plug into connectors to allow troubleshooting by substitution. The case has a front panel handle for easy removal and installation.

The VHF Navigation Receiver weighs 6.12 lb (2.78 kg). It has subassemblies mechanically packaged into a 4 in. (102 mm) high, 4.1 in. (104 mm) wide by 13.33 in. (338.6 mm) long case. A single recessed DPX rack-mounted rear-panel connector and three front-panel coaxial RF connectors interconnect the aircraft wire harness and antenna installation.



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The VHF Navigation Receivers are installed one above the other on the first and second shelves of the avionics rack, at station point X-34.00.

The VHF Navigation Receivers are located in racks that have forced-air cooling.

Antenna, Glideslope

[Refer to Figure 32.](#)

The Glideslope Antenna receives radio frequencies between 328 and 336 MHz.

The antenna encases a preformed radiating element in thermoplastic within a single assembly. Two Radio Frequency (RF) TNC female connectors are attached to an aluminum base.

An aluminum foil gasket with an elastomeric sealant makes sure consistent distribution of conducting contacts of the gasket to the airframe structure. The pressure applied during installation gives a contouring feature and lets the gasket conform to the two mating surfaces.

The glideslope antenna is located on the forward pressure bulkhead at station points X -158.00, Y 0.00, Z 115.04. The antenna is attached to the airframe structure with six mounting bolts.

Antenna, VOR/LOC

[Refer to Figure 33.](#)

The VOR/Localizer Antenna receives radio frequencies between 108 and 118 MHz. The antennas have the features that follow:

- Directional

- Horizontally polarized
- Loop
- End fed.

The VOR/Localizer antenna has two low profile blades installed to each side of the vertical stabilizer. Each antenna blade has a preformed radiating element in a thermoplastic assembly. A Radio Frequency (RF) BNC female connector is attached to the aluminum base. Phase-matched coaxial cables connect the two antenna blades to a coupler housed within the empennage and adjacent to the blades.

An aluminum foil gasket with an elastomeric sealant makes electrical contact between the gasket and airframe structure. The pressure applied during installation gives a contouring feature to let the gasket contact the two mating surfaces.

The antenna has two low profile blades installed on each side of the vertical stabilizer at station points X.994.00, Z.284.00. The antenna is attached to the airframe structure with four mounting bolts.

Coupler, Antenna VOR/LOC

[Refer to Figure 34.](#)

The VOR/Localizer antenna couple is used to connect the two VOR/Localizer antenna blades.

Splitter, Antenna VOR/LOC

[Refer to Figure 35.](#)

The VOR/LOC antenna splitter connects the antenna to each VHF navigation radio.



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Antenna, Marker

[Refer to Figure 36.](#)

The Marker Antenna receives radio frequencies at 75 MHz.

The antenna blade has a preformed radiating element in a thermoplastic assembly. A Radio Frequency (RF) BNC female connector is attached to the aluminum base.

An aluminum foil gasket with an elastomeric sealant makes electrical contact between the gasket and airframe structure. The pressure applied during installation gives a contouring feature to let the gasket contact the two mating surfaces.

The antenna is attached to the airframe structure with six mounting bolts at station point X 513.7.

Splitter, Marker

[Refer to Figure 37.](#)

The Marker Splitter connects the Marker Antenna to each VHF Navigation radio.

Antenna Cable

Coaxial cables connect the navigation receivers to the antennas that follow:

- VOR/Localizer
- Glideslope
- Marker Beacon.

The coaxial cables are transmission lines with one conductor surrounding another. A dielectric insulator separates the two conductors. The centre conductor is seven strand silver-plated copper wire and the outer conductor is a tinned copper wire braid.

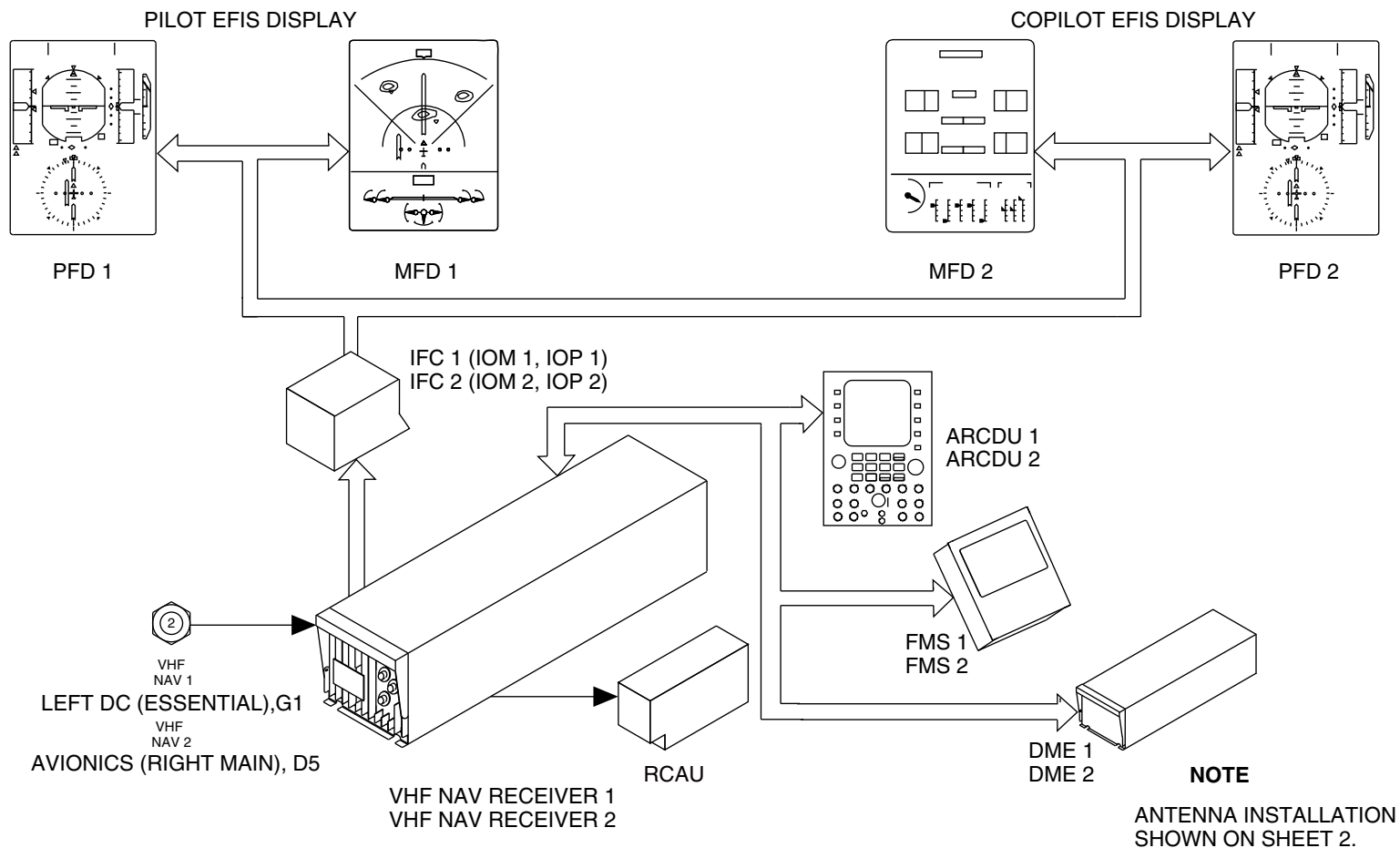
NOTE

The coaxial cables that connect the two blades of the VOR/Localizer antenna to the coupler are phase-matched within 2 degrees.



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fa607a01.cgm

VHF NAVIGATION SYSTEM BLOCK DIAGRAM
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

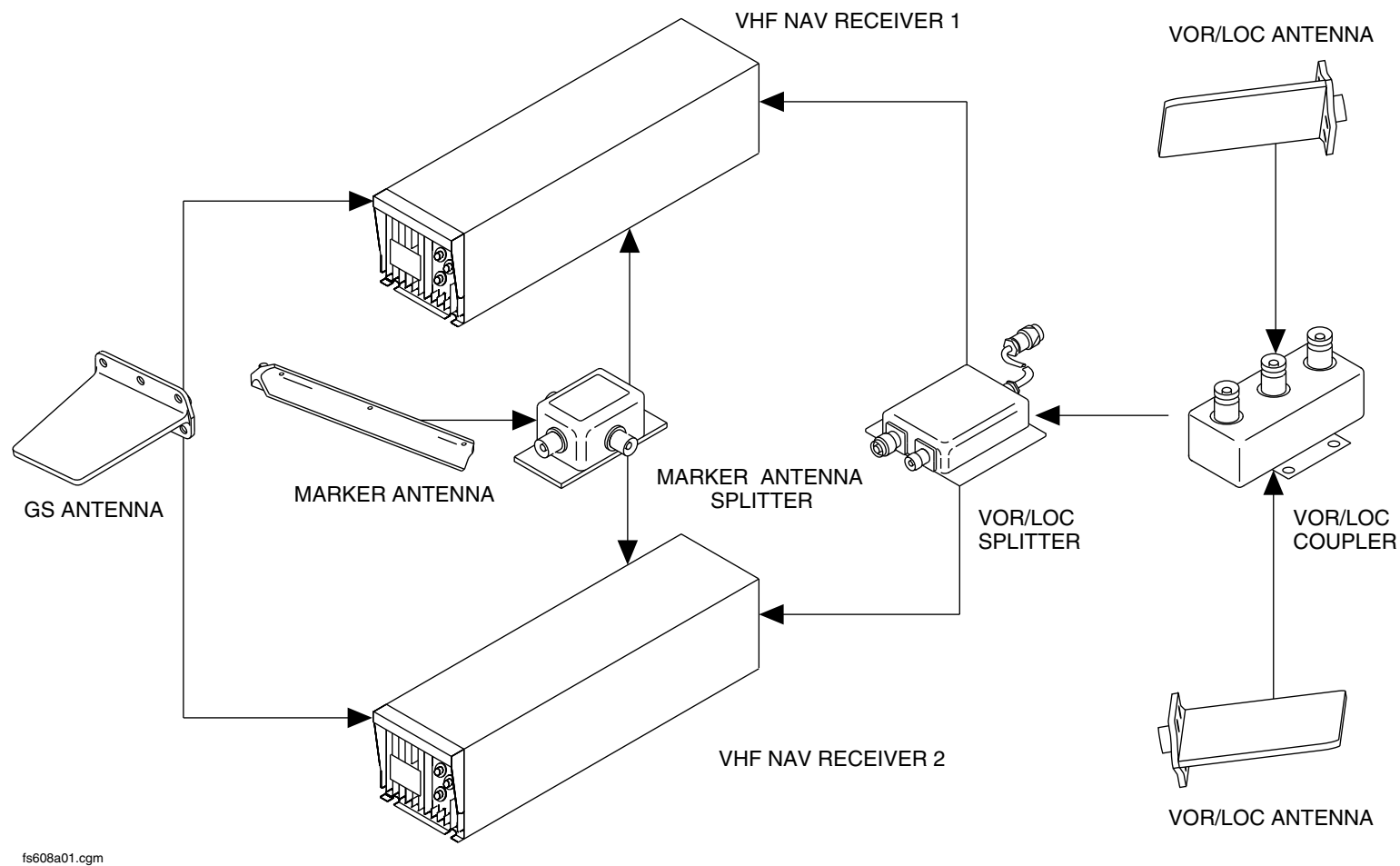
34-51-00

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VHF NAVIGATION ANTENNAS BLOCK DIAGRAM
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

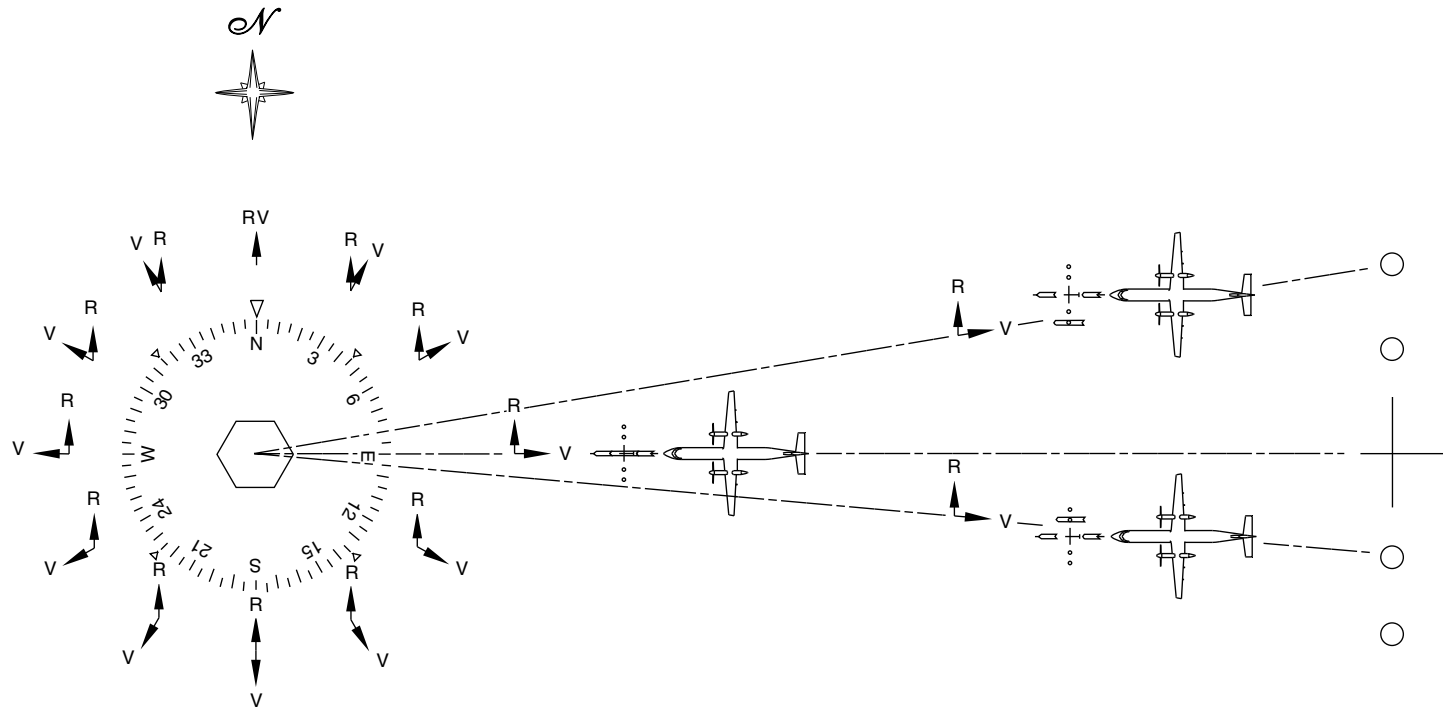
34-51-00

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fs524a01.cgm

VOR NAVIGATION PRINCIPLES
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

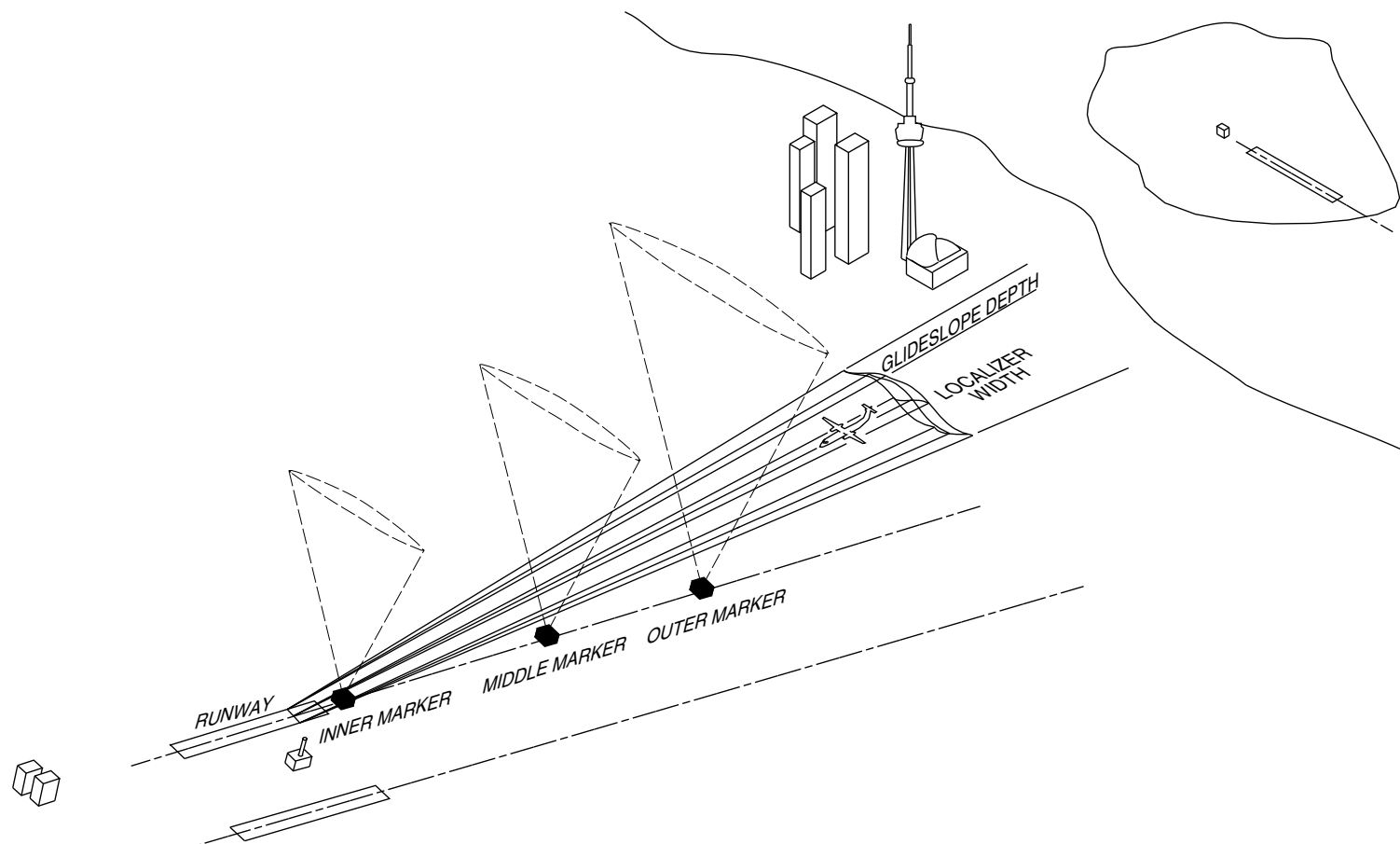
34-51-00

Config 001
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fs476a01.cgm

INSTRUMENT LANDING SYSTEM (ILS) PRINCIPLES
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

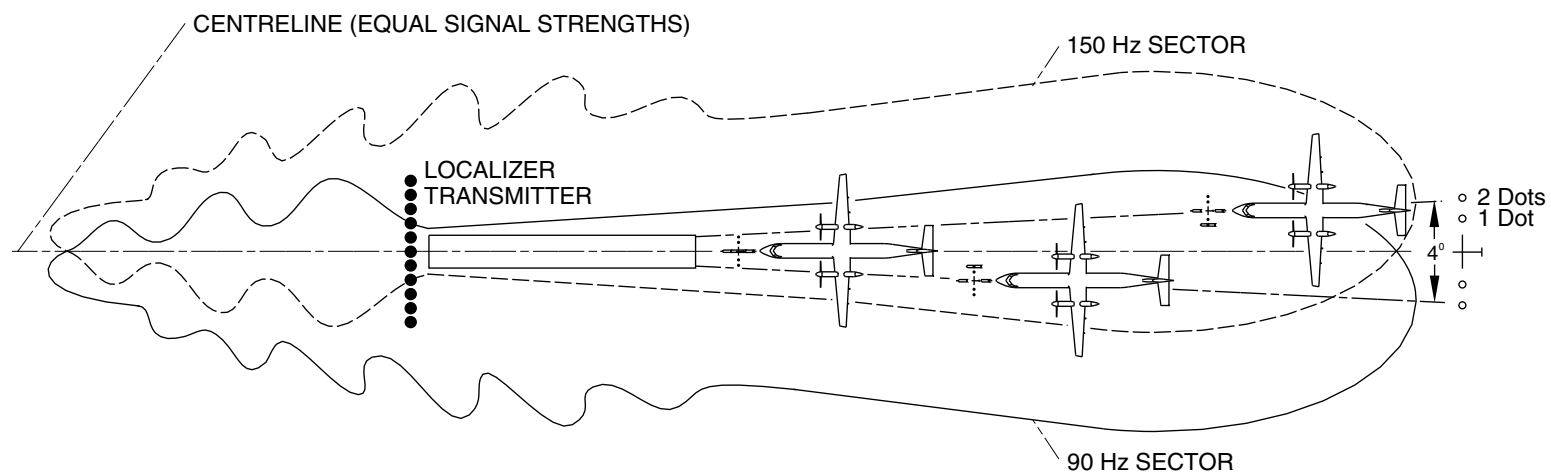
34-51-00

Config 001
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LOCALIZER INDICATION PRINCIPLES
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

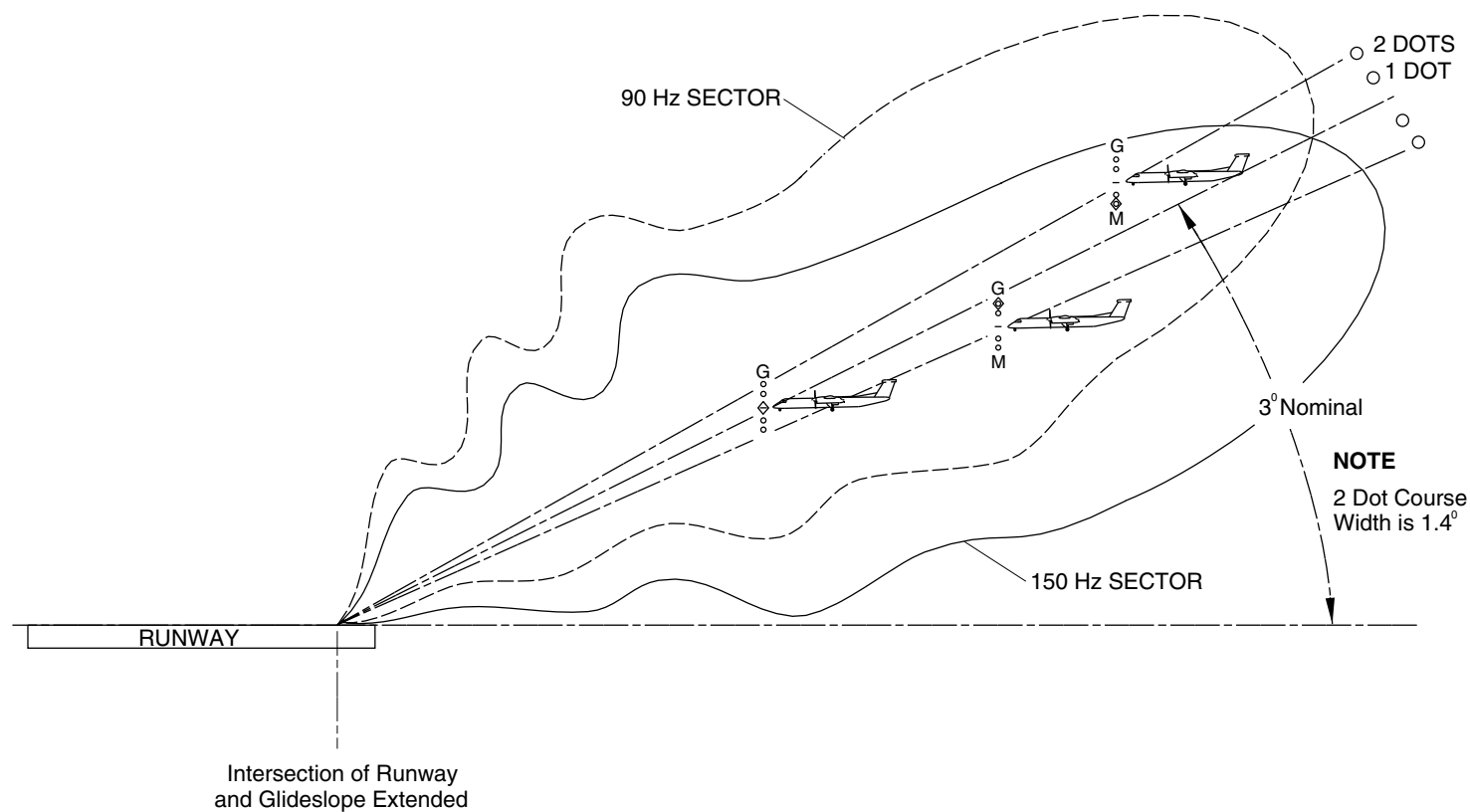
34-51-00

Config 001
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fs500a01.cgm

GLIDESLOPE INDICATION PRINCIPLES
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

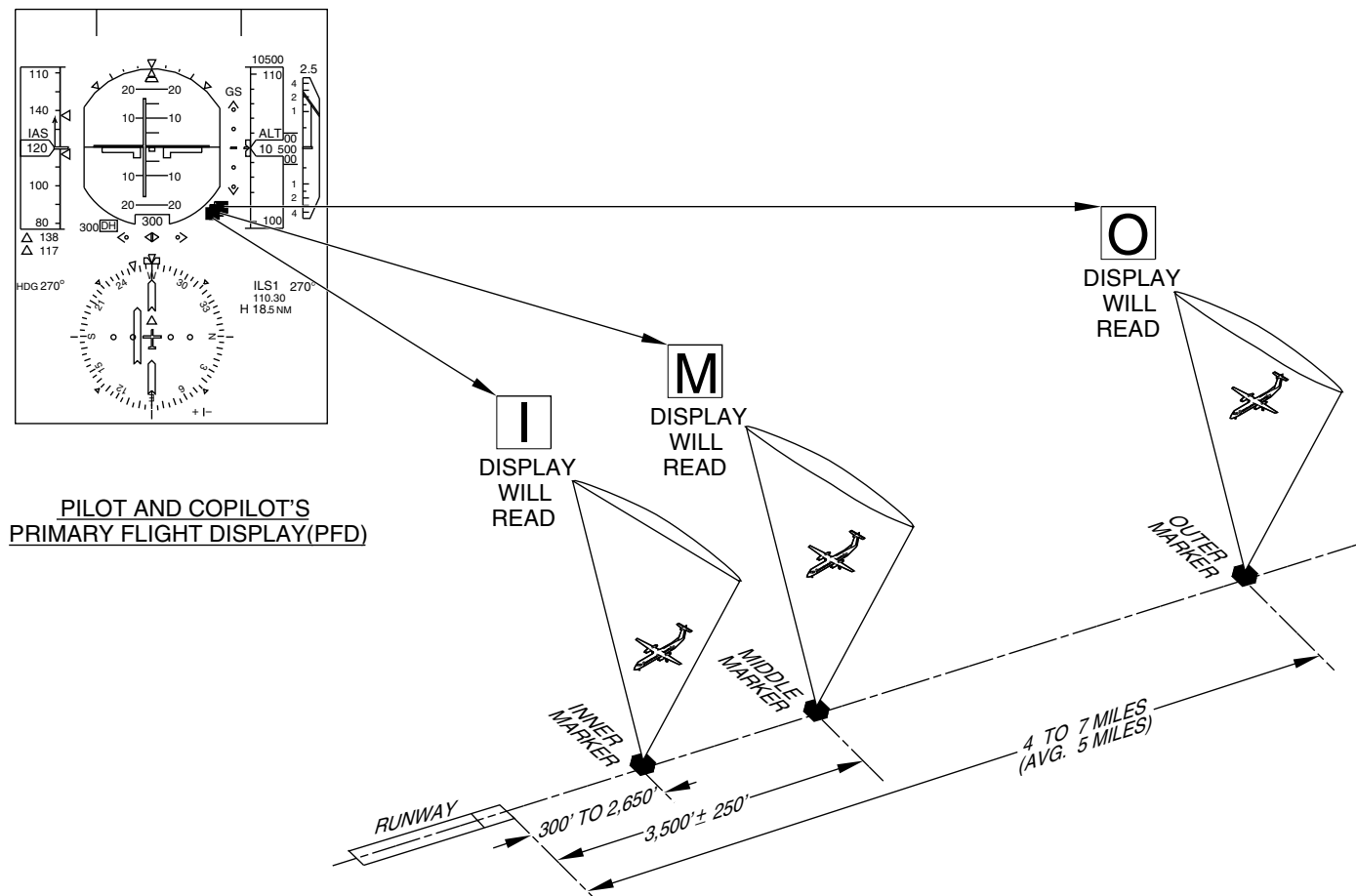
34-51-00

Config 001
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MARKER BEACON INDICATION PRINCIPLES
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

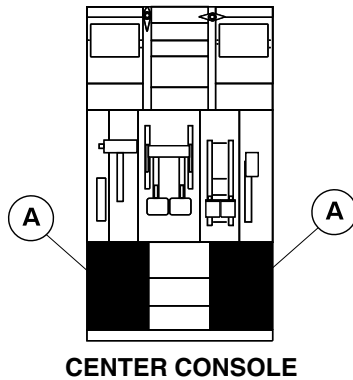
34-51-00

Config 001
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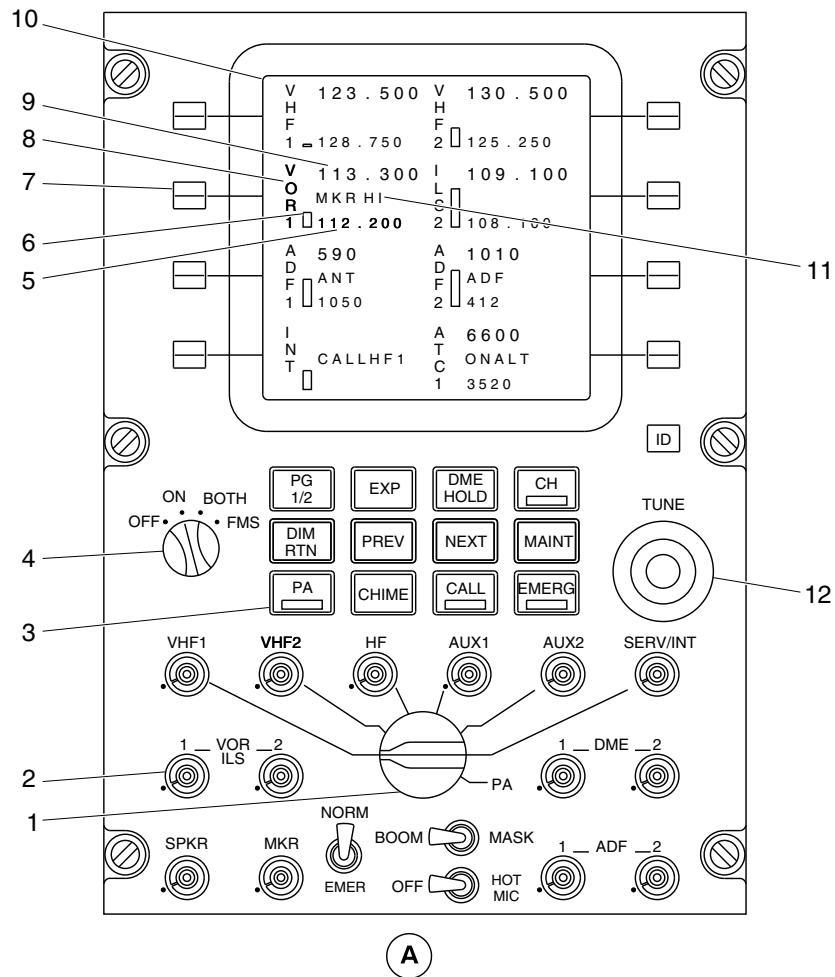
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LEGEND

1. Microphone/Interphone selector switch.
2. VOR potentiometer/push-button switches.
3. Keys.
4. Rotary switch.
5. Preset frequency.
6. Level adjust bar graph.
7. Side key.
8. Label.
9. Active frequency.
10. Display area.
11. VHF navigation mode.
12. Tuning knob.



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ARCDU, VHF Navigation Controls and Indications
Figure 8

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

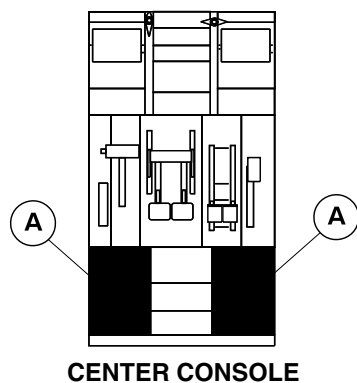
34-51-00

Config 001
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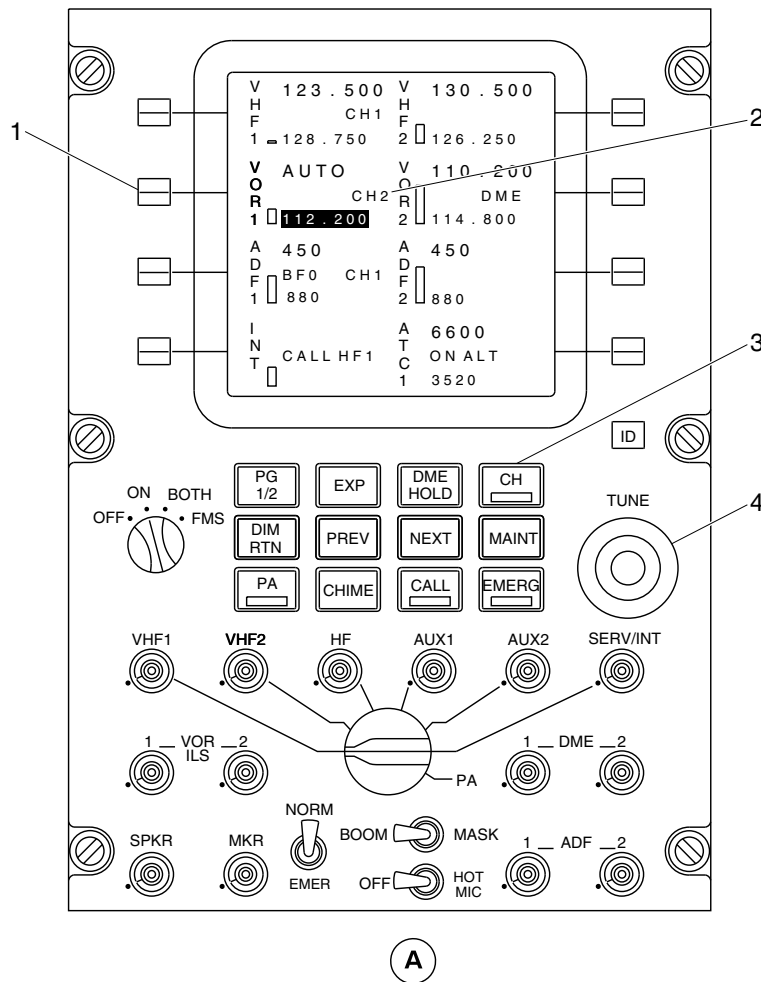
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Side key.
2. Channel memory annunciation.
3. Channel key.
4. Tuning knob.



cg5718a01.dg, sk, jul08/2019

ARCDU, VHF Navigation Frequency Control
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

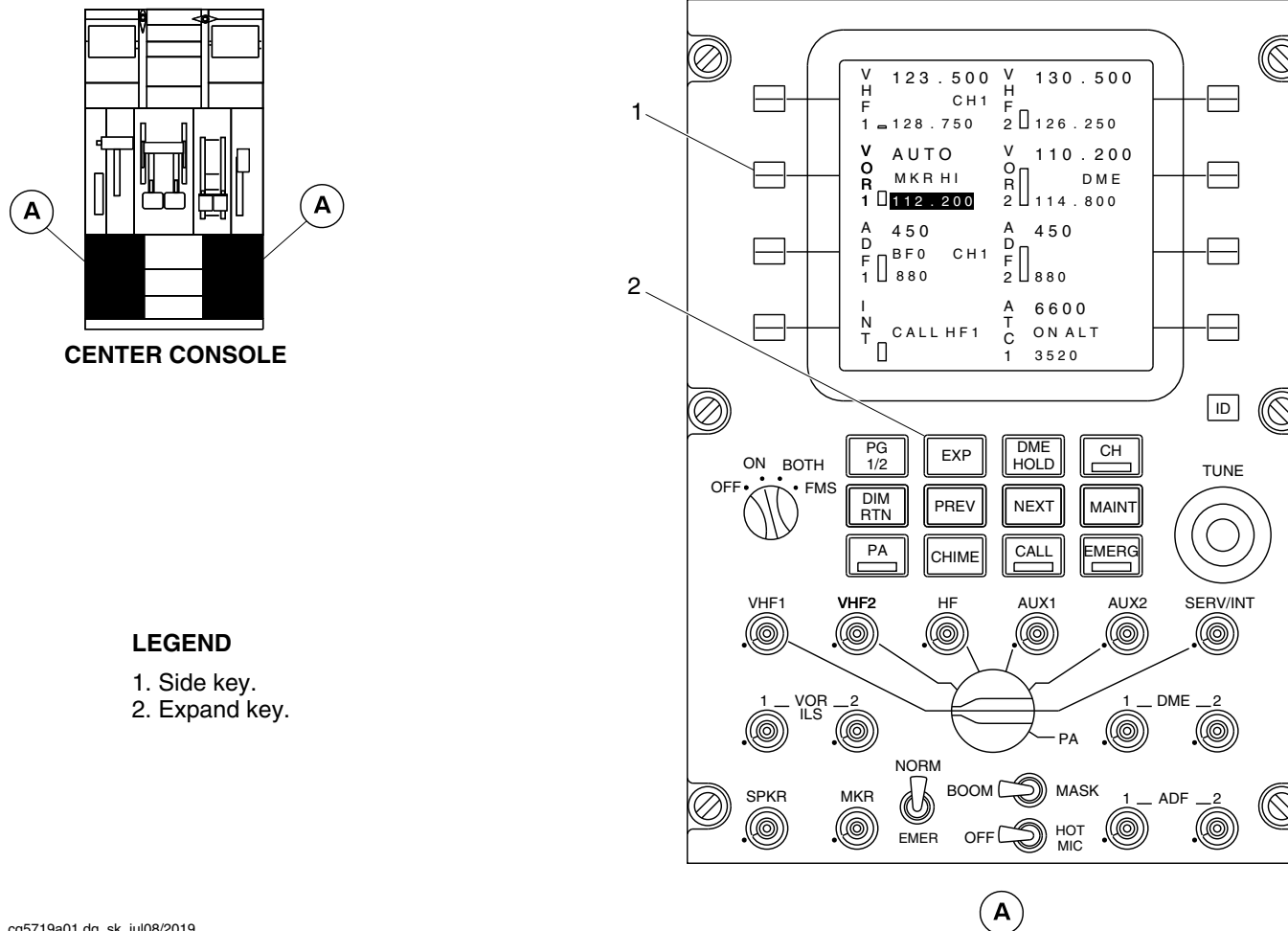
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



cg5719a01.dg, sk, jul08/2019

ARCDU, VHF Navigation Particular Page Selection
Figure 10

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

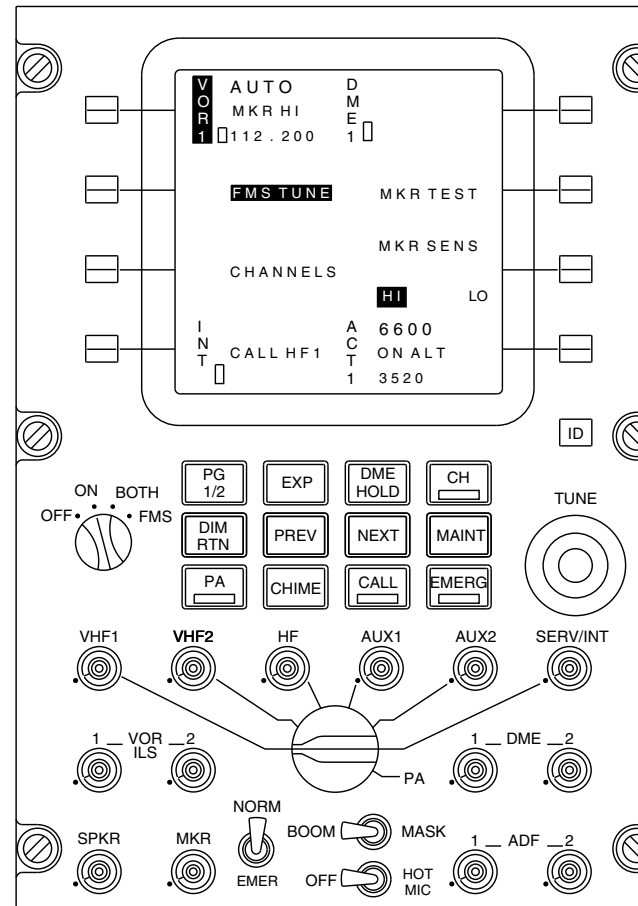
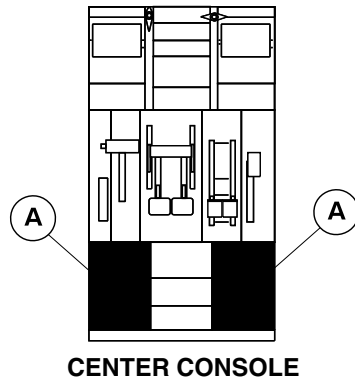
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A

cg5720a01.dg, sk, jul08/2019

ARCDU, VHF Navigation Particular Page
Figure 11

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

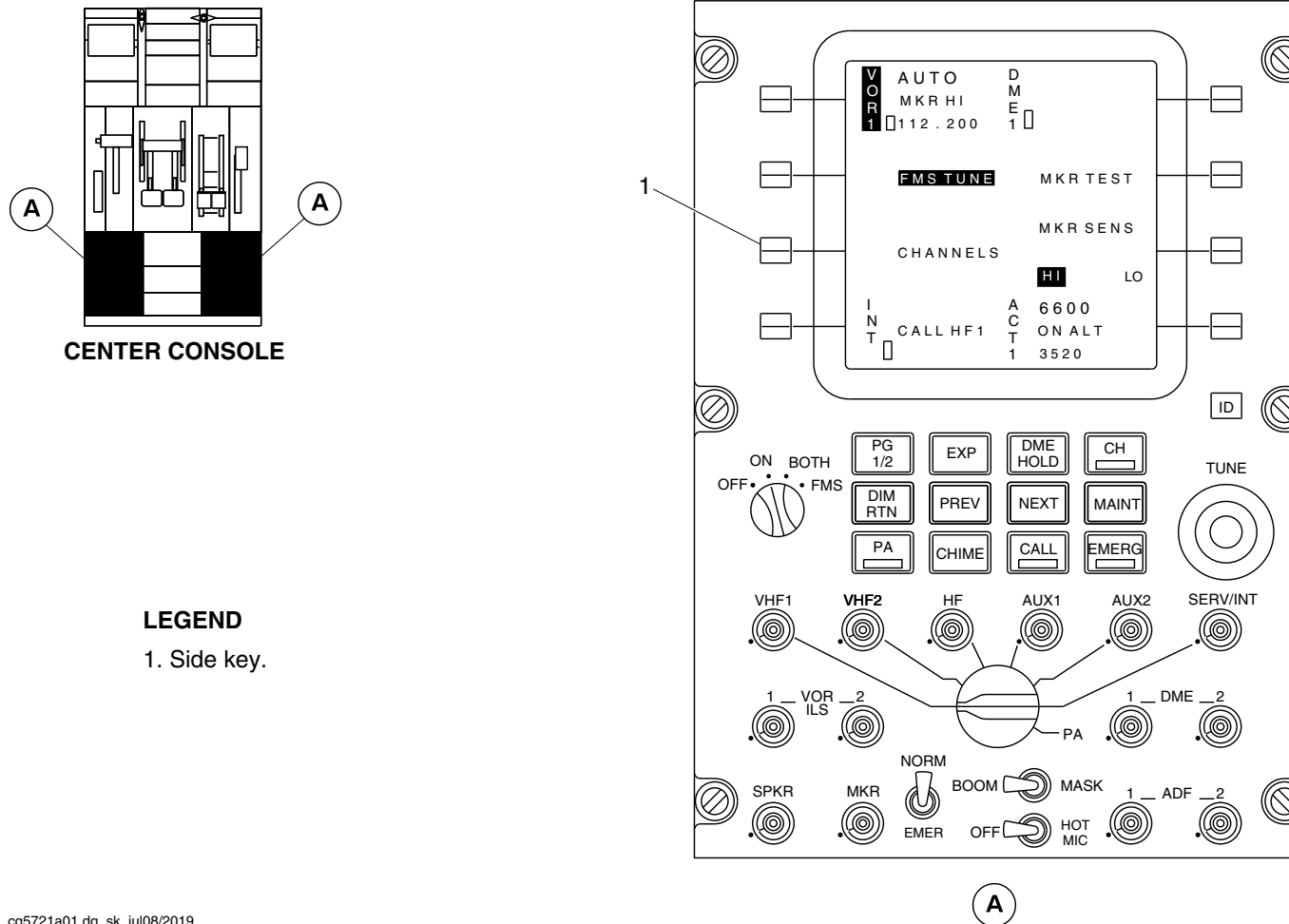
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



cg5721a01.dg, sk, jul08/2019

ARCDU, VHF Navigation Channels Selection
Figure 12

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

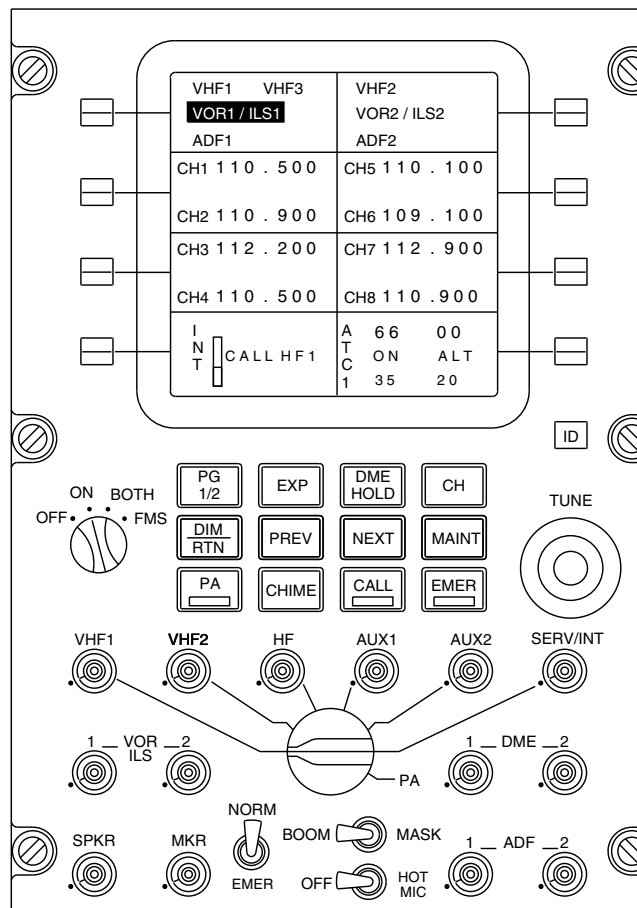
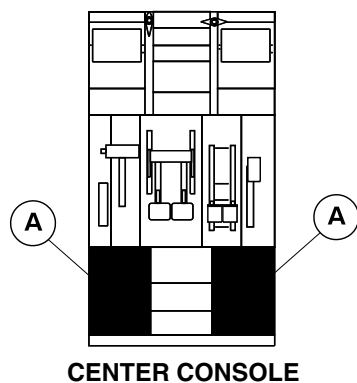
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



cg5722a01.dg, sk, jul08/2019

ARCDU, VHF Navigation Channels Page
Figure 13

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

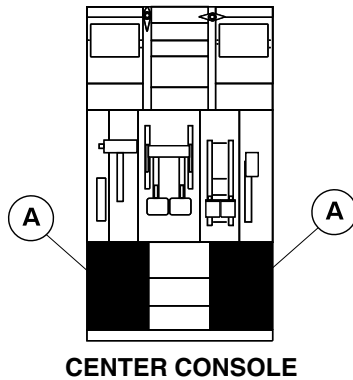
34-51-00

Config 001
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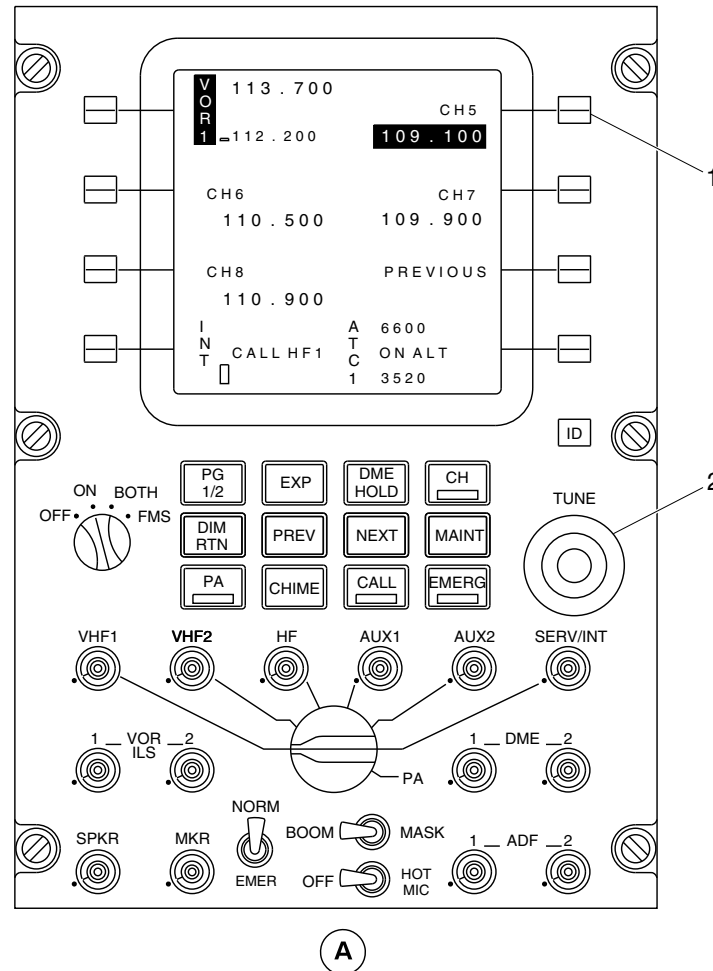
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Side key.
2. Tuning knob.



cg5723a01.dg, sk, jul08/2019

ARCDU, VHF Navigation Channel Memory Programming
Figure 14

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

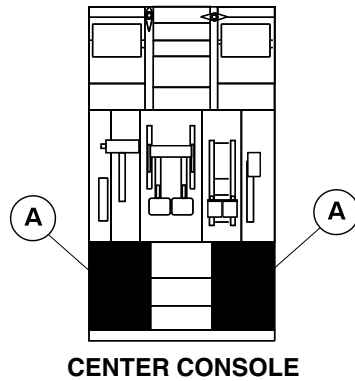
34-51-00

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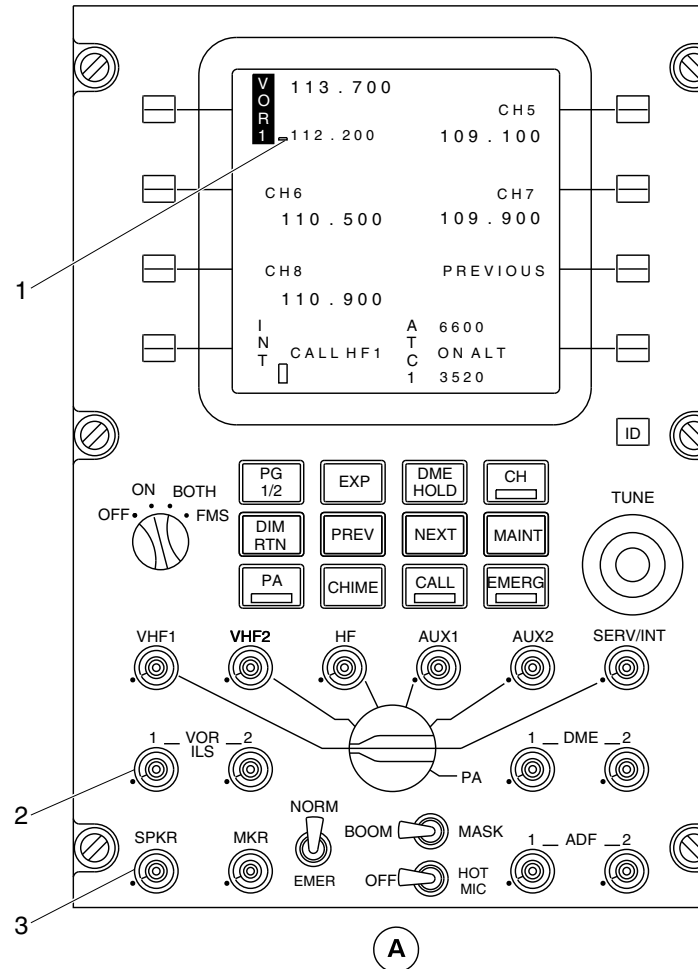
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Level adjust bar graph.
2. VOR potentiometer/Push-button switches.
3. Speaker potentiometer/Push-button switch.



cg5724a01.dg, sk, jul08/2019

ARCDU, VHF Navigation Audio Control
Figure 15

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

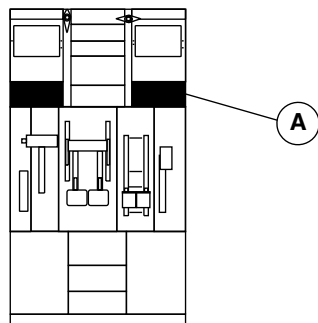
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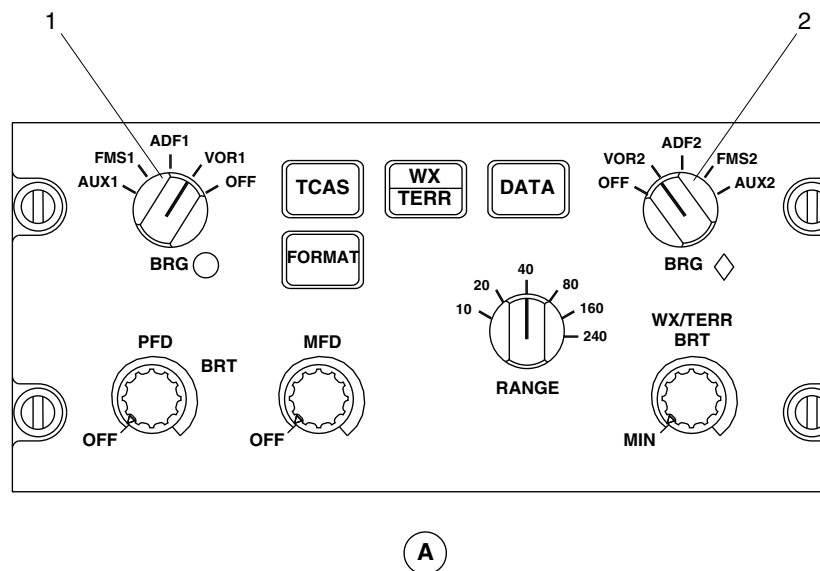
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

LEGEND

1. Bearing 1 Selector Knob.
2. Bearing 2 Selector Knob.



fa728a01.cgm

EFCP, VOR BEARING SELECTION
Figure 16

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

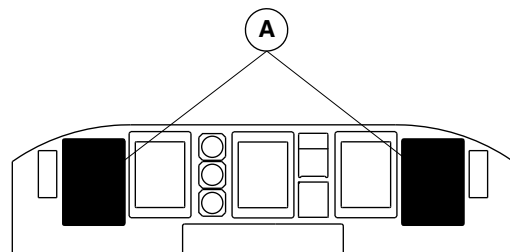
34-51-00

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OF CANADA LIMITED

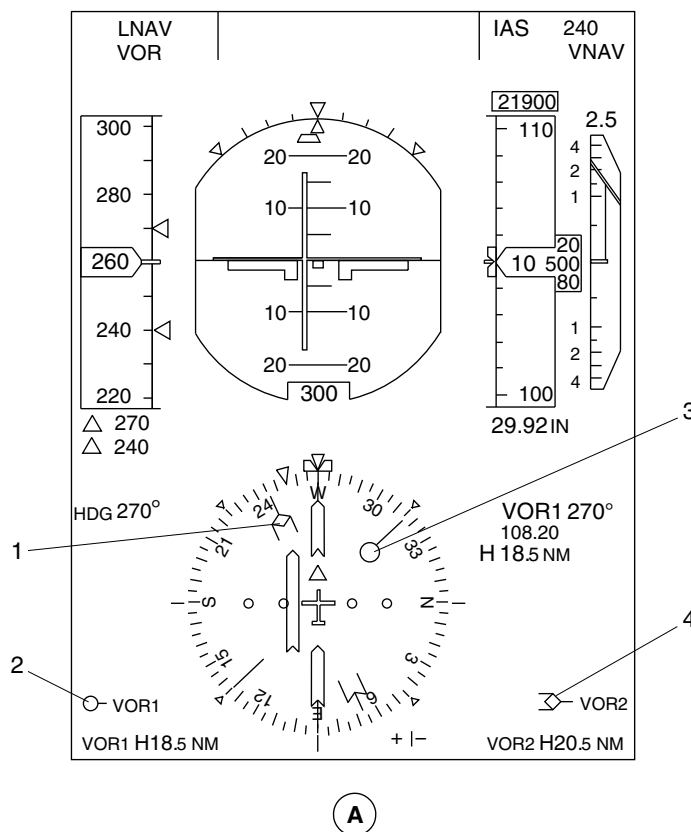
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. No. 2 Bearing Pointer.
2. No. 1 Bearing Reminder Label.
3. No. 1 Bearing Pointer.
4. No. 2 Bearing Reminder Label.



fa730a01.cgm

EIS, VOR BEARING INDICATION
Figure 17

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

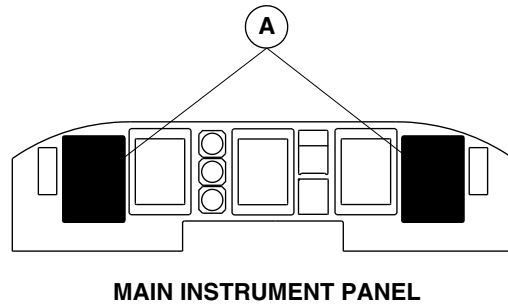
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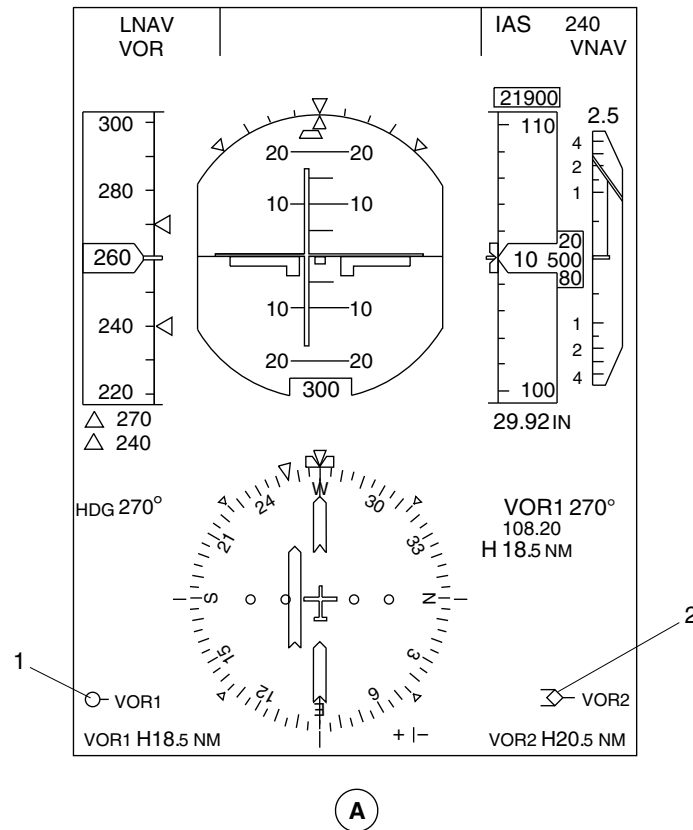
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. No. 1 Bearing Reminder Label.
2. No. 2 Bearing Reminder Label.



fa730a02.cgm

EIS, VOR BEARING MALFUNCTION INDICATION
Figure 18

PSM 1-84-2A
EFFECTIVITY:
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Config 001

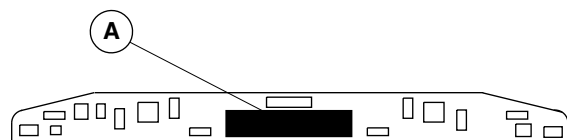
34-51-00

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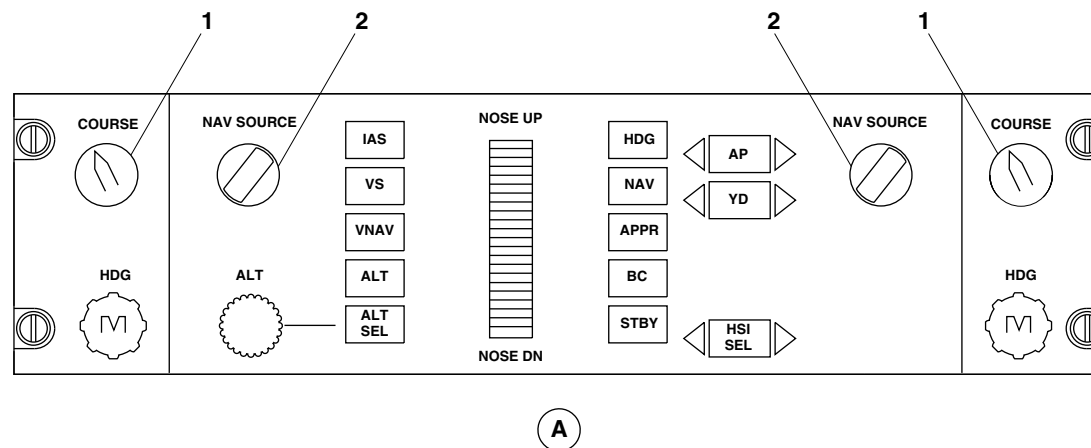
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. Course Selection Knob.
2. Navigation Source Selection Knob.



fs754a01.cgm

AFCP, NAVIGATION SOURCE SELECTION
Figure 19

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

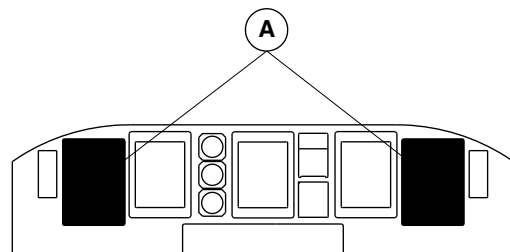
34-51-00

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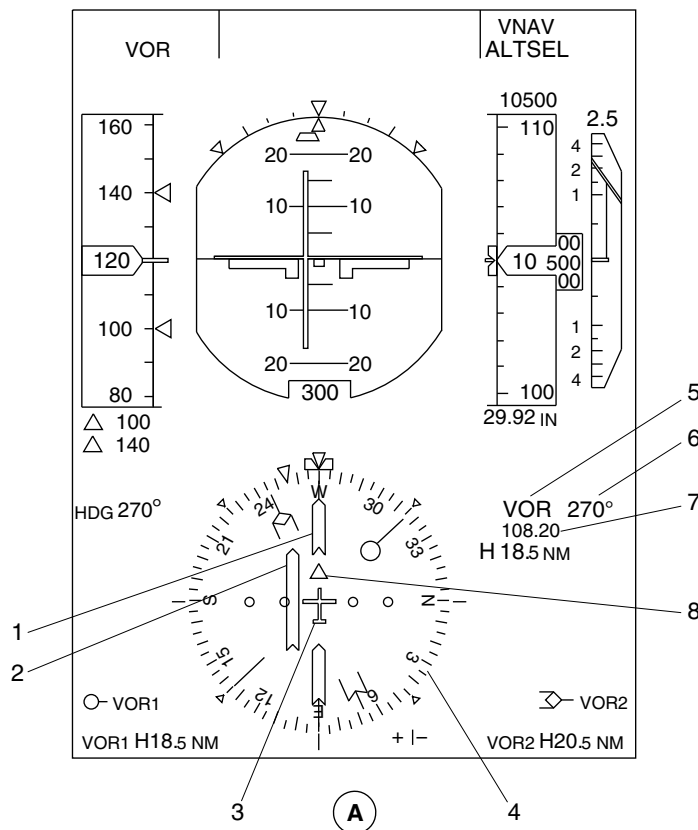
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Selected Course Pointer.
2. Course Deviation Scale & Bar.
3. Aircraft Symbol.
4. Heading Scale.
5. Navigation Source Annunciation.
6. Selected Course Digital Value.
7. VHF Navigation Frequency.
8. TO/FROM Pointer.



fa757a01.cgm

PFD, VOR COURSE INDICATION
Figure 20

PSM 1-84-2A
EFFECTIVITY:
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Config 001

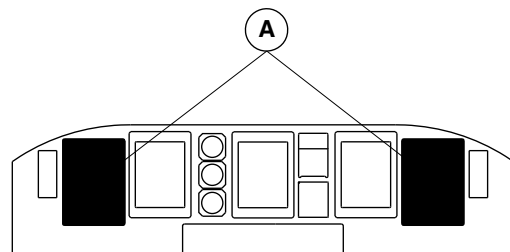
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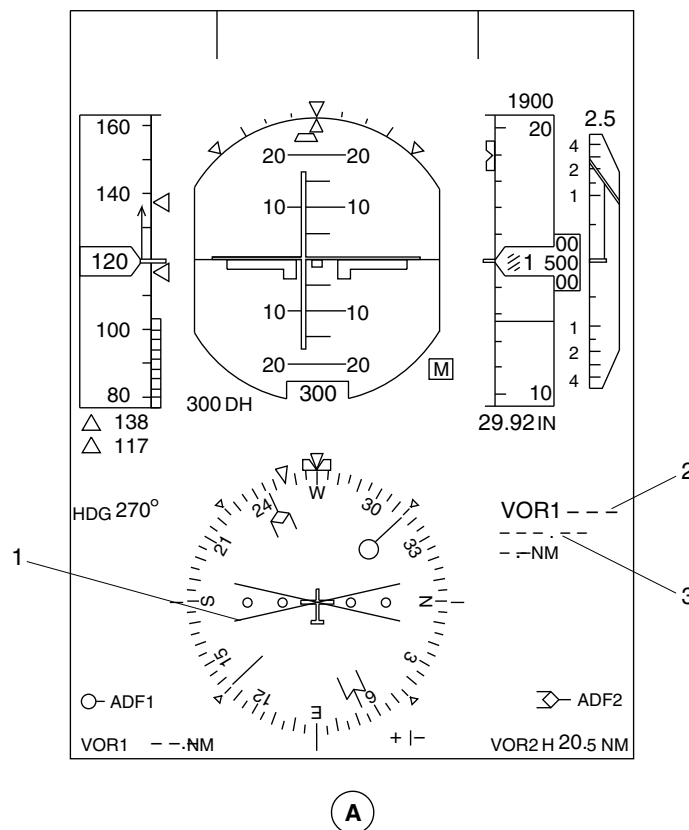
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Course Deviation Failure Flag.
2. Selected Course Fail Annunciation.
3. VOR Navigation Frequency Fail Annunciation.



fa758a01.cgm

PFD, VOR MALFUNCTION INDICATION
Figure 21

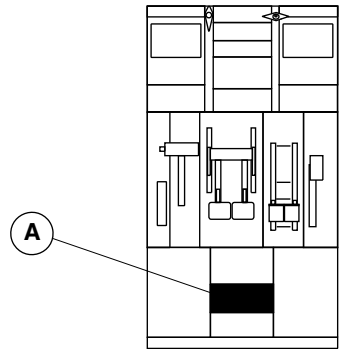
PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

34-51-00

Config 001
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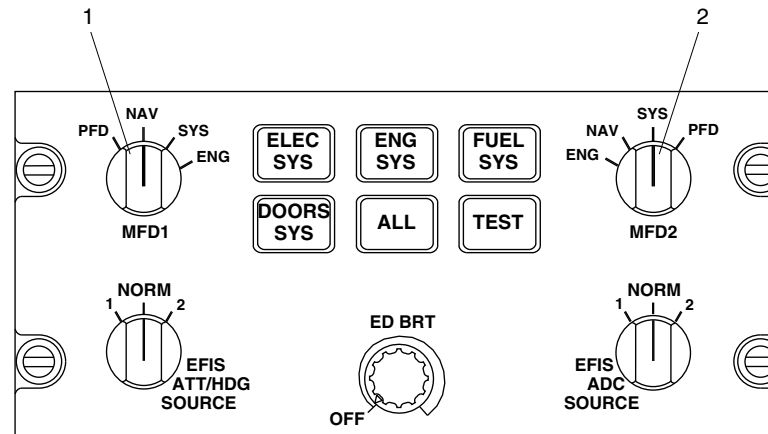
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

LEGEND

1. MFD1 Reversion Selector.
2. MFD2 Reversion Selector.



A

fs750a01.cgm

ESCP, NAVIGATION PAGE SELECTION
Figure 22

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

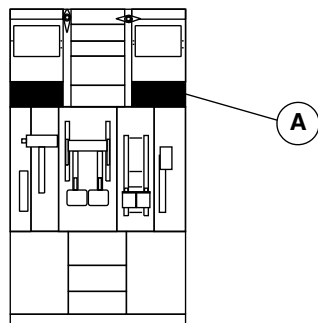
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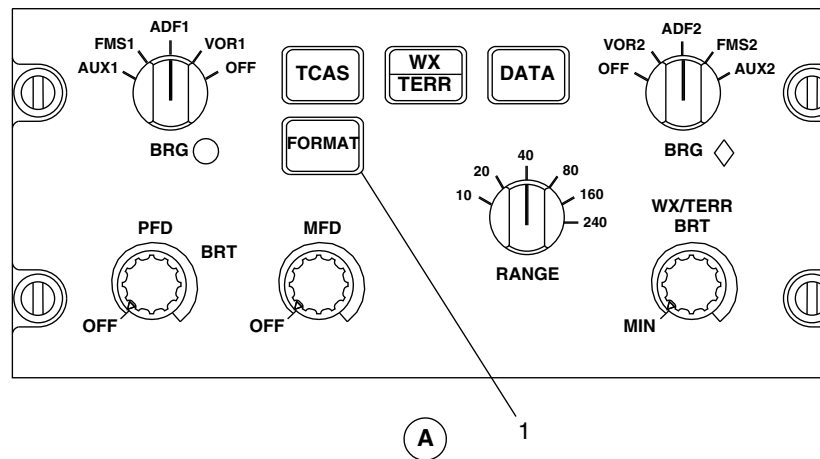
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

LEGEND

1. FORMAT Pushbutton Switch.



fae51a01.cgm

EFCP, ARC FORMAT SELECTION
Figure 23

PSM 1-84-2A
EFFECTIVITY:
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Config 001

34-51-00

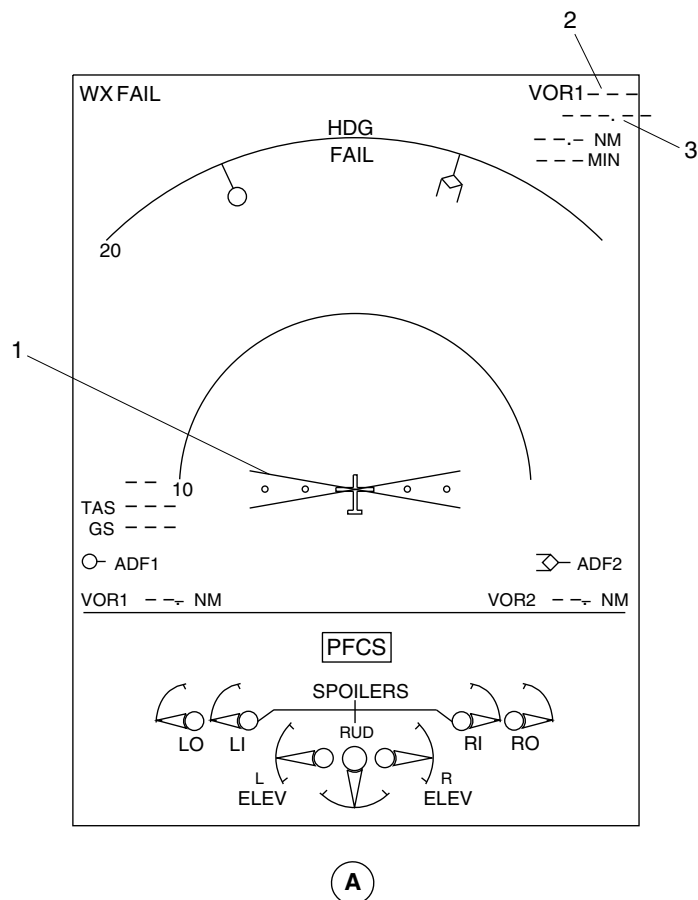
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A line drawing of the main instrument panel. It features a central rectangular display area, flanked by various control buttons and switches. A callout line points from a circle labeled 'A' to a specific control panel on the left side of the panel.

MAIN INSTRUMENT PANEL

1. Course Deviation Failure Flag.
2. Selected Course Fail Annunciation.
3. VHF Navigation Frequency Fail Annunciation.



NAVIGATION PAGE, ARC FORMAT VOR MALFUNCTION INDICATION
Figure 25

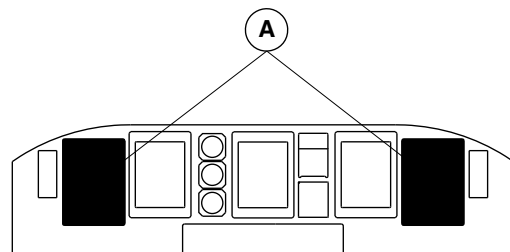
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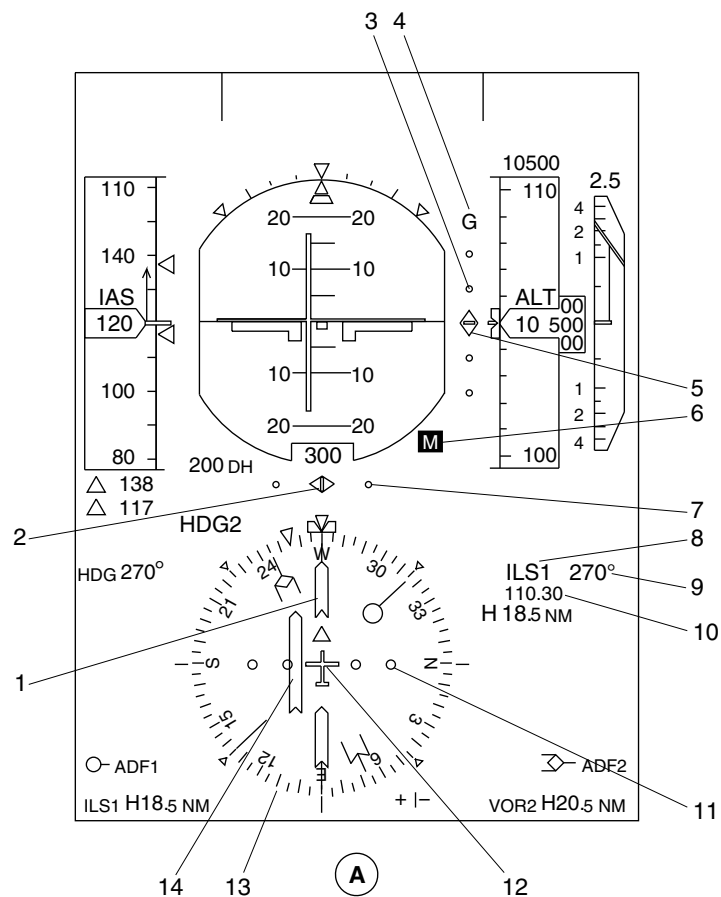
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Selected Course Pointer.
2. Expanded Localizer Pointer.
3. Glide Slope Scale.
4. Vertical Deviation Source.
5. Vertical Deviation Pointer.
6. Marker Beacon Annunciator.
7. Expanded Localizer Scale.
8. Navigation Source Annunciation.
9. Selected Course Digital Value.
10. VHF Navigation Frequency Identifier.
11. Course Deviation Scale.
12. Aircraft Symbol.
13. Heading Scale.
14. Course Deviation Bar.



fa755a01.cgm

PFD, ILS COURSE INDICATION
Figure 26

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

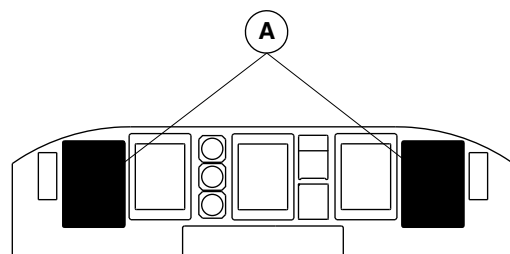
34-51-00

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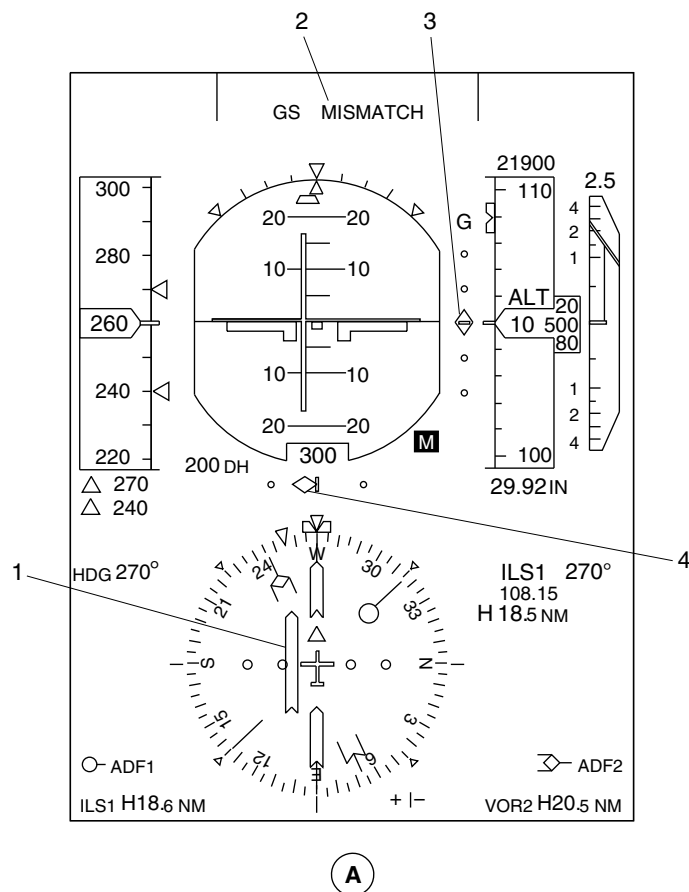
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Course Deviation Bar.
2. Mismatch Message.
3. Vertical Deviation Pointer.
4. Expanded Localizer Pointer.



fa751a01.cgm

PFD, ILS MISMATCH INDICATIONS
Figure 27

PSM 1-84-2A

EFFECTIVITY:

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Config 001

34-51-00

Config 001

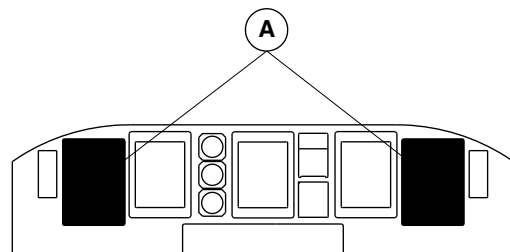
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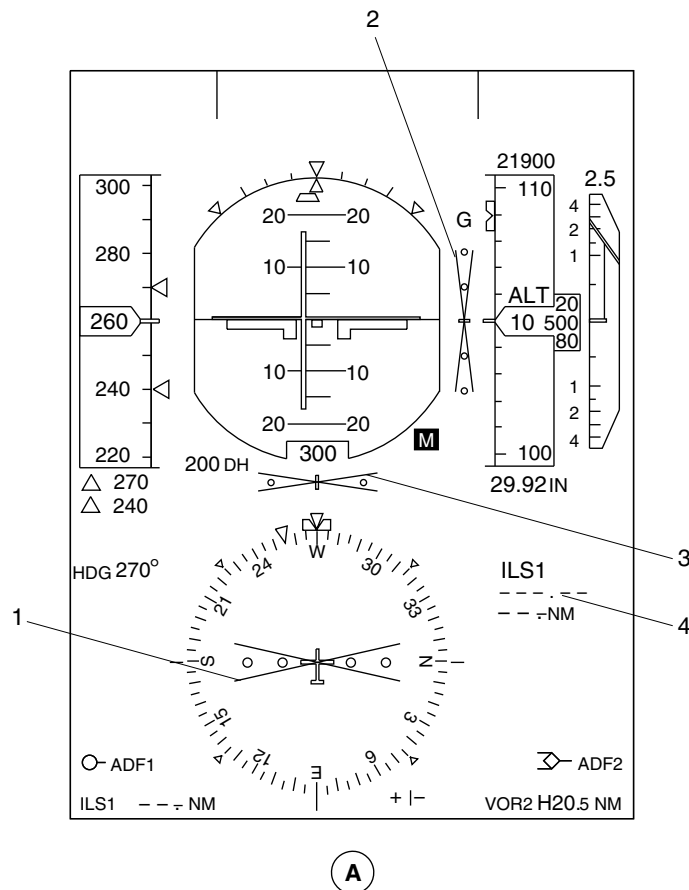
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Course Deviation Failure Flag.
2. Vertical Deviation Failure Flag.
3. Expanded Localizer Failure Flag.
4. VHF Navigation Frequency Fail Annunciation.



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PFD, ILS MALFUNCTION INDICATIONS
Figure 28

PSM 1-84-2A
EFFECTIVITY:
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Config 001

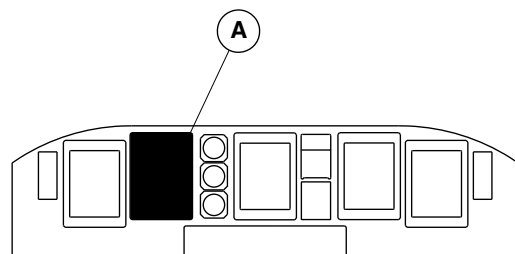
34-51-00

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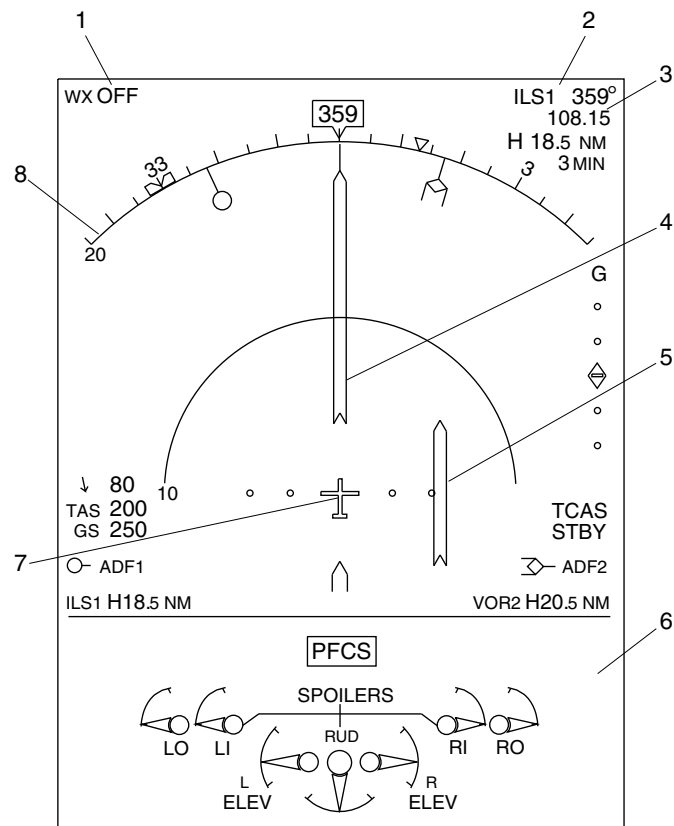
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Selected Course Digital Value.
2. Navigation Source Annunciation.
3. Radio Navigation Frequency Identifier.
4. Selected Course Pointer.
5. Course Deviation Scale and Bar.
6. Permanent System Data Area.
7. Aircraft Symbol.
8. Heading Scale in Arc Format.



fa752a01.cgm

NAVIGATION PAGE, ARC FORMAT ILS INDICATION
Figure 29

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

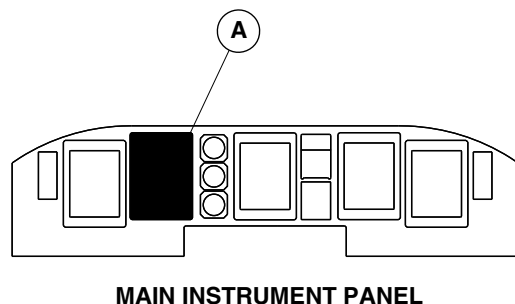
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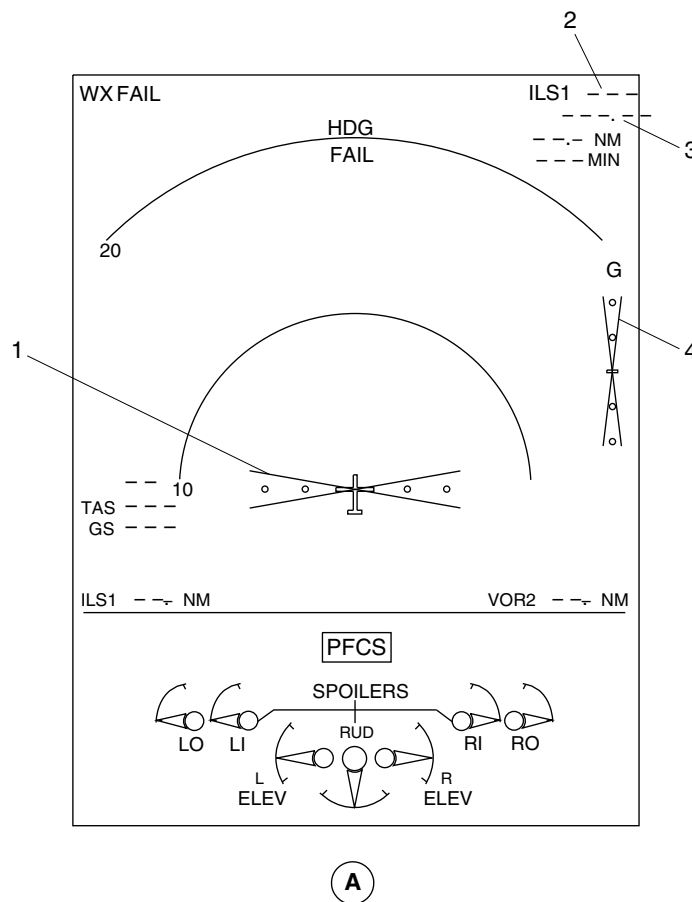
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Course Deviation Failure Flag.
2. Selected Course Fail Annunciation.
3. Radio Navigation Frequency Fail Annunciation.
4. Vertical Deviation Failure Flag.



fa753a01.cgm

NAVIGATION PAGE, ARC FORMAT ILS MALFUNCTION INDICATION
Figure 30

PSM 1-84-2A
EFFECTIVITY:
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Config 001

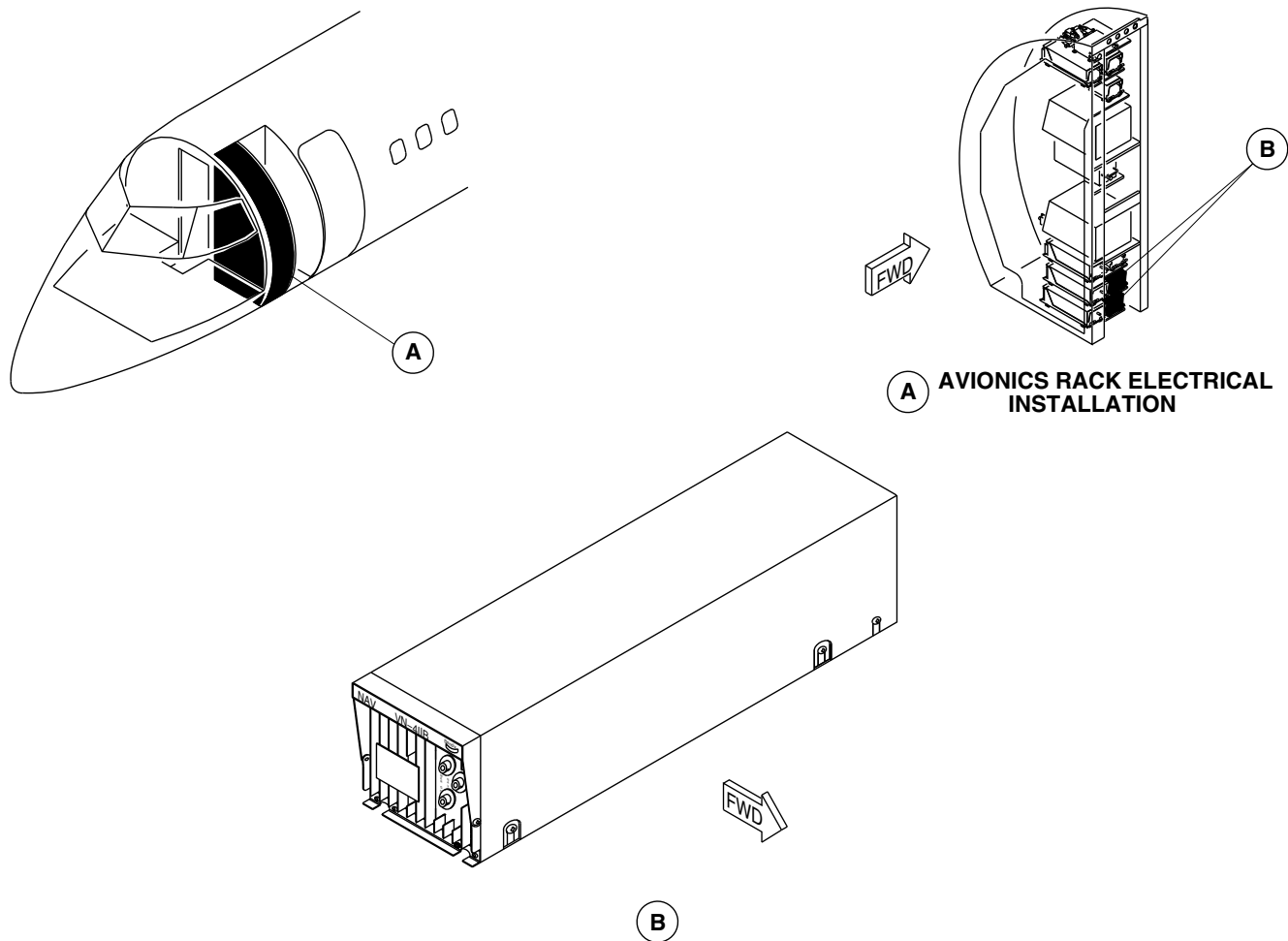
34-51-00

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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NAVIGATION RECEIVER LOCATOR
Figure 31

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

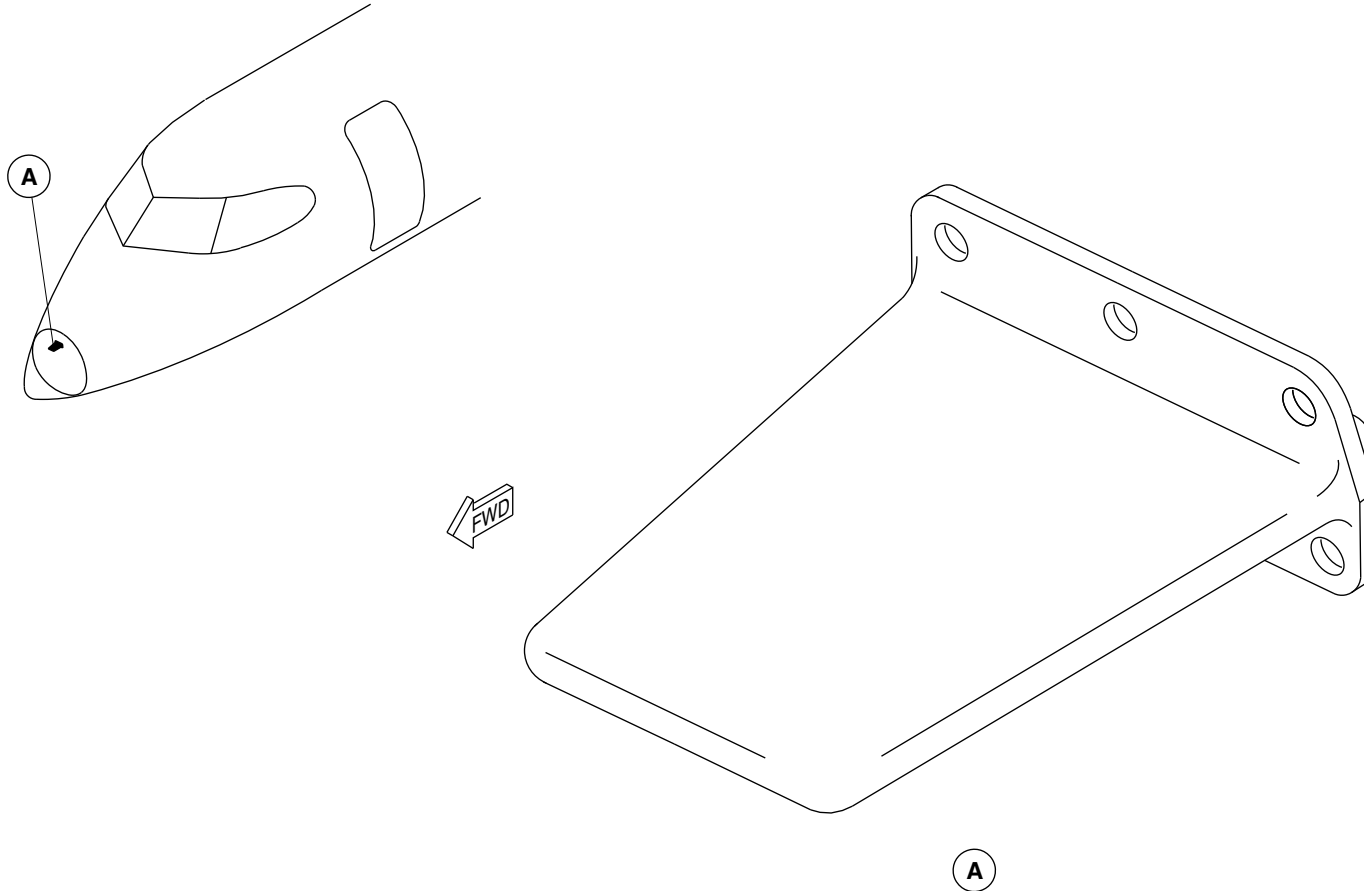
34-51-00

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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GLIDESLOPE ANTENNA LOCATOR
Figure 32

PSM 1-84-2A
EFFECTIVITY:
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Config 001

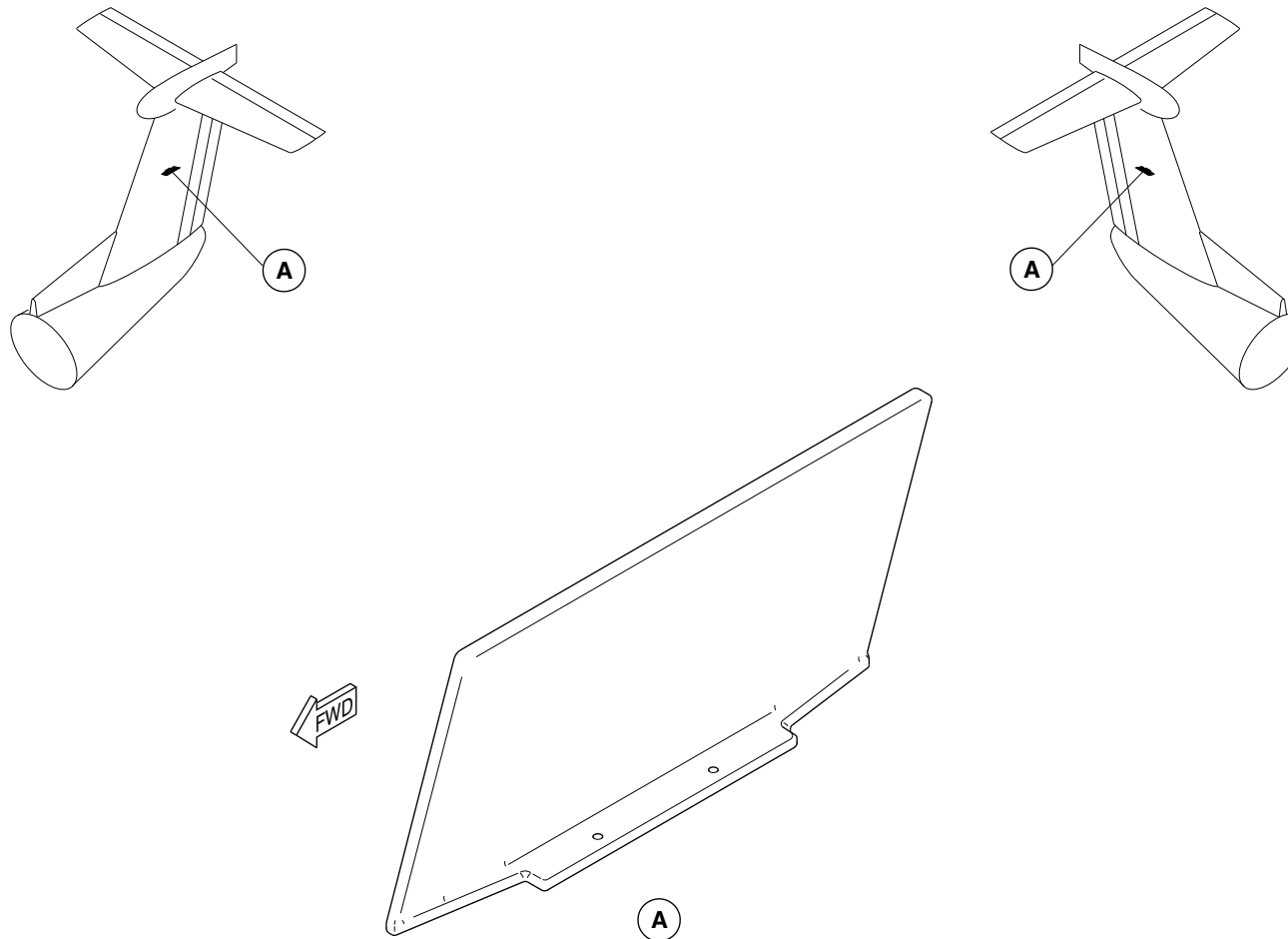
34-51-00

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VOR/LOC ANTENNA LOCATOR
Figure 33

PSM 1-84-2A
EFFECTIVITY:
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Config 001

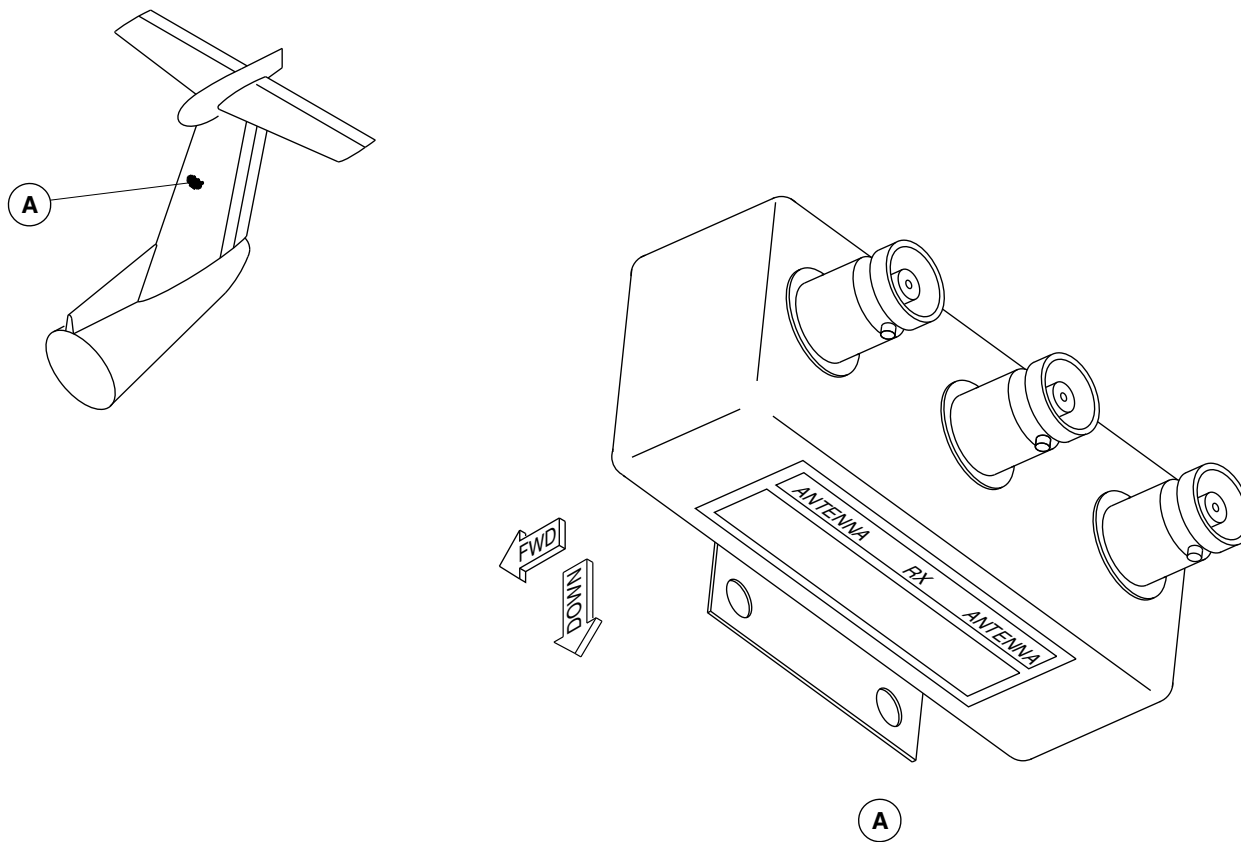
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fsa00a01.cgm

VOR/LOC ANTENNA COUPLER LOCATOR
Figure 34

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

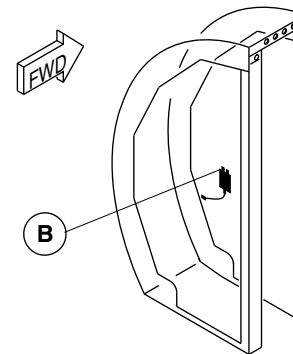
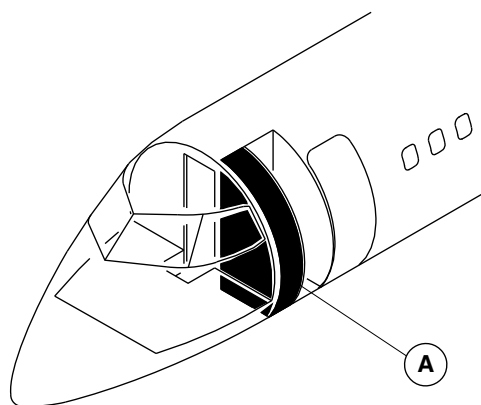
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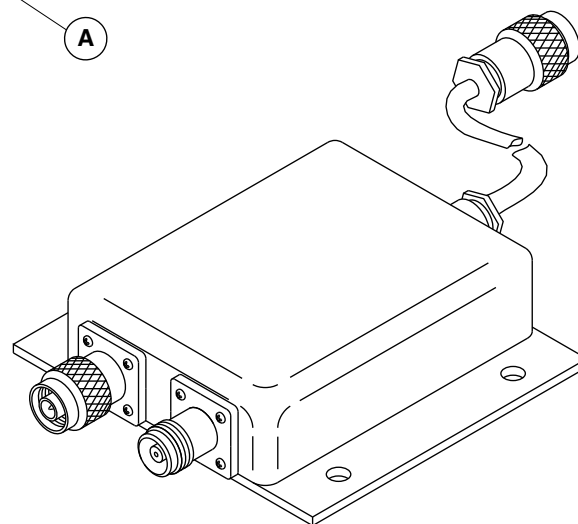


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A AVIONICS RACK ELECTRICAL
INSTALLATION



B

fs478a01.cgm

VOR/LOC ANTENNA SPLITTER LOCATOR
Figure 35

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

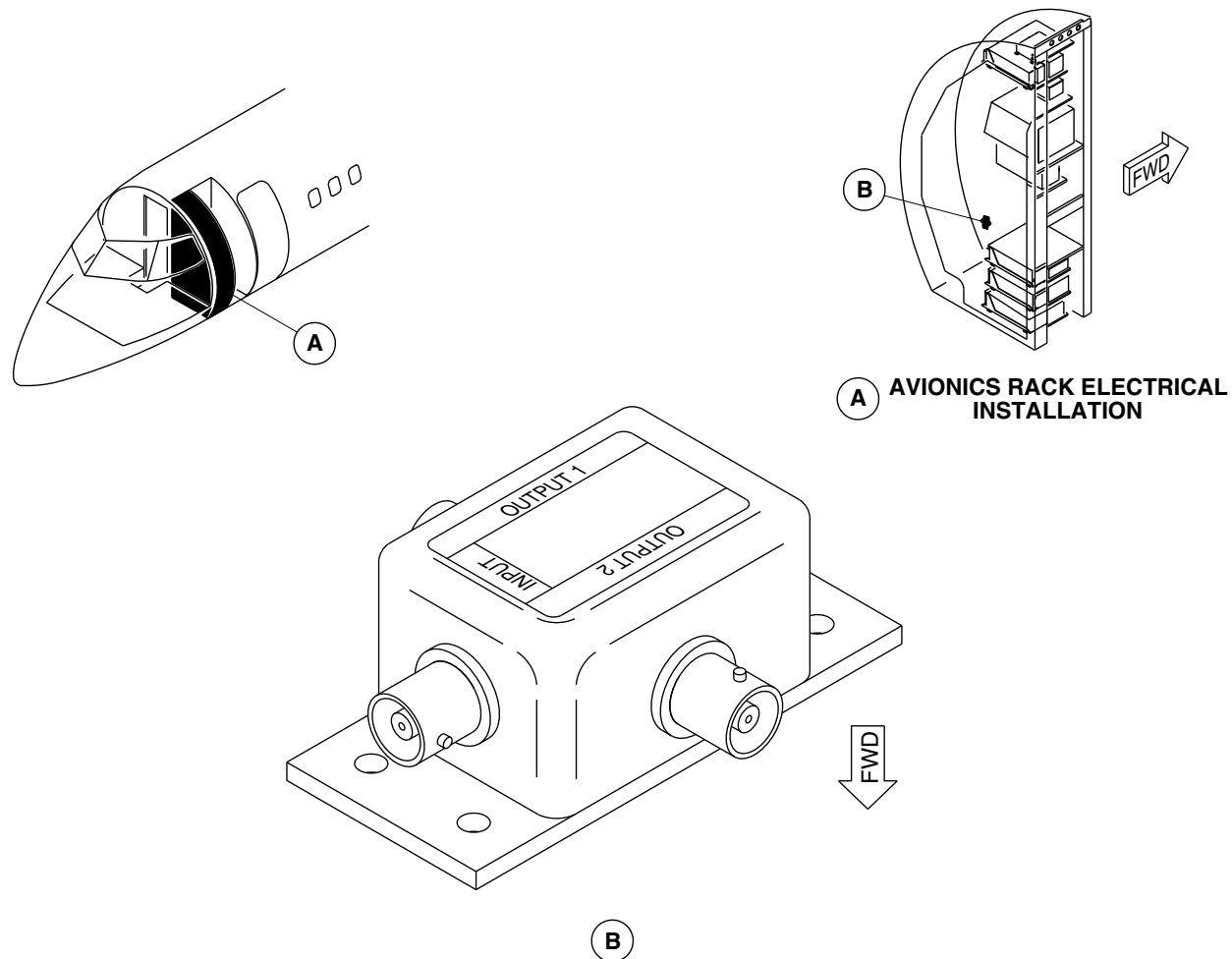
34-51-00

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs137a01.cgm

MARKER ANTENNA LOCATOR
Figure 36

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
Config 001

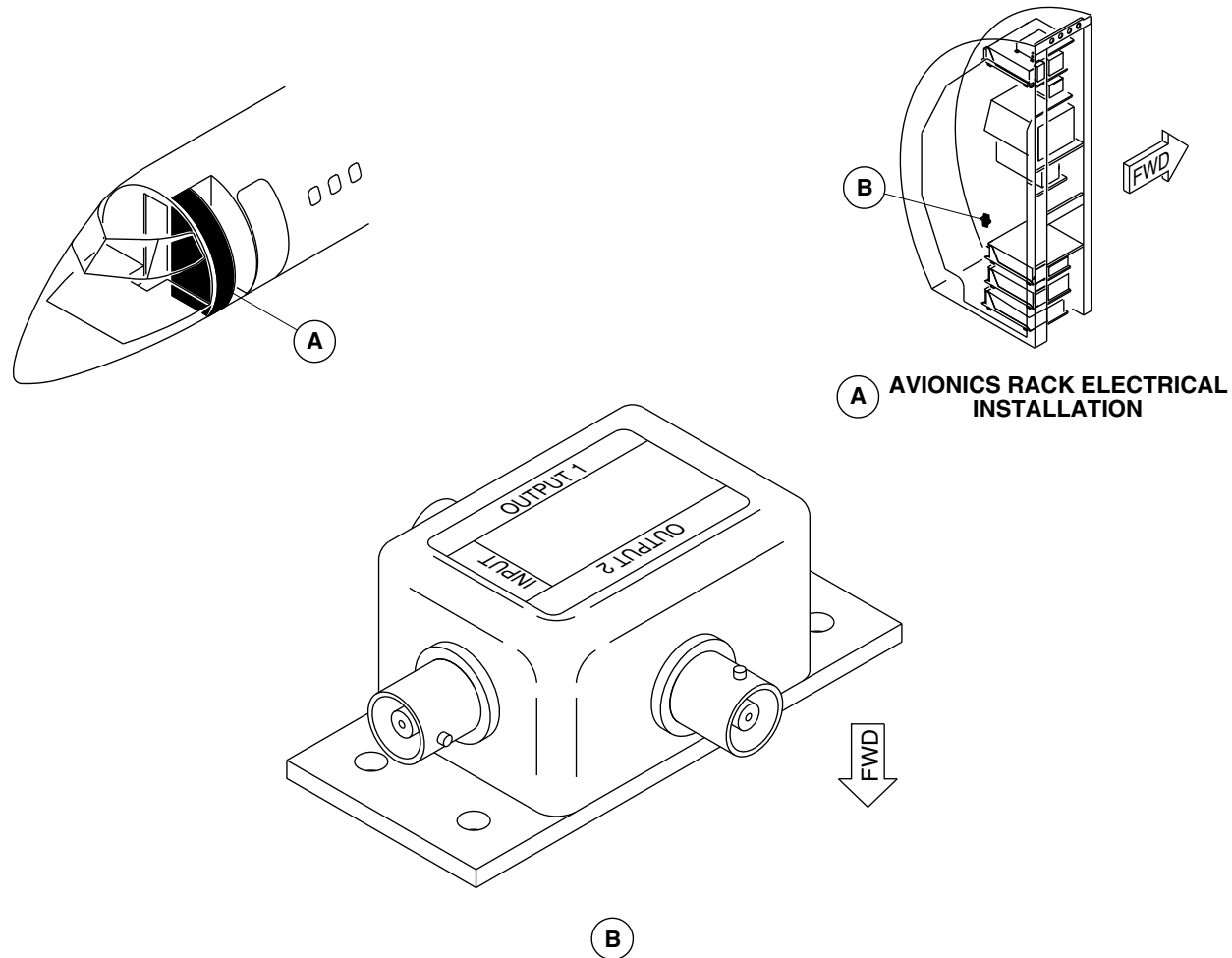
34-51-00

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs137a01.cgm

MARKER ANTENNA SPLITTER LOCATOR
Figure 37

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-51-00
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

34-52-00-001

AUTOMATIC DIRECTION FINDING (ADF)

Introduction

The Automatic Direction Finding (ADF) system shows the magnetic bearing to a fixed Non-Directional Beacon (NDB).

General Description

[Refer to Figure 1.](#)

Two Automatic Direction Finding (ADF 1, ADF 2) systems function independently to continuously show the magnetic bearing to the selected ground stations.

On aircraft with SB84-34-150 OR ModSum 4-458734 OR 4-113793 incorporated, the No. 2 ADF system is disabled and its related components are removed from the aircraft.

On aircraft with SB84-34-181 incorporated, the No. 2 ADF system is disabled and its related components are removed from the aircraft except No. 2 ADF antenna.

The Automatic Direction Finding (ADF 1, ADF 2) system removes the audio identification from the transmitting signal and sends it to the Audio and Radio Management System (ARMS).

The ADF system is controlled by the components that follow:

- Audio and Radio Control Display Units (ARCDU1, ARCDU 2)

- Flight Management System (FMS 1, FMS 2).

The Audio and Radio Control Display Units (ARCDU1, ARCDU 2) are used to manually tune the ADF system frequencies and to control the audio level. The Flight Management System (FMS) is used to tune the ADF frequencies when the Audio and Radio Control Display Unit (ARCDU1, ARCDU 2) malfunctions. The observer uses an audio control panel to control audio level only.

The Audio and Radio Control Display Unit (ARCDU 1, ARCDU 2) or the Flight Management System (FMS 1, FMS 2) are used to manually start the Automatic Direction Finding (ADF) system self-tests.

The Electronic Instrument System (EIS) shows the Automatic Direction Finding (ADF) system bearing information.

The Automatic Direction Finding (ADF) system has the components that follow:

- Receiver, ADF (34-52-01)
- Antenna, ADF (34-52-11).

The ADF Receivers (ADF 1, ADF 2) supply data to the systems that follow:

- Audio and Radio Management System (ARMS)
- Input/Output Processors (IOP 1, IOP 2)
- Flight Management System (FMS).



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Detailed Description

Refer to Figure 2.

The Automatic Direction Finding (ADF) system gives a means to point to the relative bearing of the selected ground station.

Refer to Figure 3.

The ground stations transmit radio waves that induce voltages in the loop windings of the ADF antenna. The loop antenna has two mutually perpendicular windings. One axis of one winding is aligned with the longitudinal axis of the aircraft and the other winding is aligned with the lateral axis. The voltages induced are proportional to the angle between the heading of the aircraft and the track to the station. The loop antenna has two positions 180 degrees apart where the voltages induced are the same. A sense antenna signal gives a reference phase to which the loop signals are combined to prevent bearing ambiguity.

The Automatic Direction Finding (ADF) system receives Low Frequency (LF) Amplitude Modulated (AM) Radio Frequency (RF) signals from the stations that follow:

- Non-directional beacons
- Broadcast stations
- Maritime distress.

The Non-directional beacons and the AM broadcast stations function between 190 and 1860 kHz. The maritime distress frequency range is between the 2181 and 2183 kHz.

The Automatic Direction Finding (ADF 1, ADF 2) systems function in the modes that follow:

- ADF, Automatic Direction Finding
- ANT, Antenna
- BFO, Beat Frequency Oscillator
- Test.

ADF Mode: In the ADF mode, the Electronic Instrument System (EIS) shows the bearing of the aircraft relative to the selected ground station.

ANT Mode: In the ANT mode, the receiver functions as an audio receiver to identify stations. The loop antenna is disabled to give a higher audio sensitivity than in ADF mode. The ADF bearing pointer parks at 90 degrees relative bearing.

Beat Frequency Oscillator (BFO) Mode: Some ground stations operate with an interrupted continuous wave (CW) carrier. They do not use a modulated RF carrier for audio identification. The ADF receiver supplies an audio frequency that beats with the incoming intermittent carrier wave to give an audio output that is the same as the ground station identifying transmissions. In the BFO mode, an intermittent 1000 Hz audio tone is heard when the ADF receives a valid transmission that identifies the station.

The ADF system continues to show the bearing of the aircraft relative to the selected ground station in the BFO mode when ADF mode is selected.



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Test mode: The TEST mode is a confidence test. It causes the bearing pointer of the ADF receiver under test to park at a relative bearing of 90 degrees.

Refer to Figure 4.

The Radio Control Display Units (ARCDU 1, ARCDU 2) selections control the Automatic Direction Finding (ADF) frequencies and modes to be used.

Table 1: Table 1.

Selection	Description
OFF	The ARCDU is not powered. The related FMS controls and tunes as a backup
ON	The ARCDU controls and tunes its related radio system
BOTH	The ARCDU controls and tunes its related and opposite radio systems
FMS	The FMS controls and tunes its related and opposite radio systems

Refer to Figure 5.

A minimum of two steps are required to change the ADF active frequency.

The side key adjacent to the ADF display area is pushed to highlight the preset frequency. It is pushed again to make the preset frequency active. The frequency that was active now becomes the new preset frequency.

When the side key adjacent to the ADF display area is pushed, the preset frequency in the display area highlights. The TUNE double rotary knobs located at the lower right side of the ARCDU are turned to change the preset frequency. The Side key is pressed again to set the new active frequency. The frequency that was active now becomes the new preset frequency.

The CH key is pushed to set the channel memory mode from eight preset frequencies using a channel memory selection. A green bar in the CH key comes on and a channel memory annunciator comes into view in green characters below the active frequency indication. In addition, the channel memory number is shown adjacent to the preset frequency.

The side key adjacent to the ADF display area is pushed to change the preset frequency. When the TUNE double rotary knobs are turned, the channel memory number shown adjacent to the preset frequency changes. If a channel memory number is not shown in the display area, it starts with CH1. The programmed frequency for the channel memory is shown in place of the preset value as black characters on a cyan background. The double rotary knobs are turned to select the different preset channel memories. The rotary knob has a rollover capability. It shows the first preset channel again after it shows the last channel when the rotary knob is continuously turned in one direction or the other. The ADF side key is pushed again to validate the channel memory selection. Two successive pushes of the ADF side key causes the preset frequency to replace the active frequency. The active frequency now becomes the preset frequency.

The channel memory number appears in green characters below the active frequency. This shows that the active frequency is associated with a channel memory number. When the CH key is pushed again, the channel memory mode changes to the normal frequency



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

selection. All channel memory annunciators are removed and the green light in the CH key goes out.

If the selections that follow are not made within five seconds, the previous preset frequency is restored:

- Turn one or the other TUNE double rotary knob
- Push the same side key again.

[Refer to Figure 6.](#)

The ADF Particular Page comes into view when the ADF side key is pushed followed by the EXP key on the keyboard.

[Refer to Figure 7.](#)

The particular page can be used to change the ADF Receiver (ADF 1, ADF 2) modes and functions that follow:

- 0.5 or 1 kHz tuning
- CHANNELS, Channel programming
- ADF, Automatic Direction Finding or ANT, Antenna mode
- TEST, Test
- BFO, Beat Frequency Oscillator on or off.

Tuning Features: The side key adjacent the 0.5, 1.0 kHz legend is pushed to alternately switch between the 0.5 and 1.0 kHz tuning feature. The selected text is black letters on a green background. The non–selected text is in white text on the black background. If 0.5 kHz is selected, the ADF receiver is tuned in 0.5 kHz increments. If 1 kHz is selected, the ADF receiver is tuned in 1 kHz steps.

Channel Programming: Eight preset channels are shown on two pages for ADF tuning. The preset and active frequencies can be tuned from both channel pages.

[Refer to Figure 8.](#)

The side key adjacent to the CHANNEL legend is pushed to access the Channel Programming mode.

[Refer to Figure 9.](#)

Channel presets are labeled as CH1 through CH8 in white characters on two pages. Each channel window shows the current preset channel number in cyan fonts below its preset labels.

[Refer to Figure 10.](#)

The second channel memory page comes into view when the side key adjacent to the NEXT legend is pushed or the NEXT key is pushed. The side key adjacent to the PREVIOUS legend is pushed or the PREV key is pushed to see the first channel memory page. The PG 1/2 key is pushed to show the first main page.

[Refer to Figure 11.](#)

When the side key related to the channel being changed is pushed, the current frequency value changes to black characters on a cyan background. The two TUNE double rotary knobs located at the lower right side of the ARCDU is turned to change the frequency. The side key is pushed again to validate the new frequency. The channel window shows the new frequency in cyan characters on the black display. If the new selected channel is associated to a preset radio frequency, both values change together. If the modified channel is the active radio frequency, the label will flash for five seconds at a 1 hertz rate. This shows that it is not possible to directly change an active frequency.



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If the frequency selections are not made within five seconds, the previous preset frequency restores.

Refer to Figure 6.

ADF and ANT mode: The side key adjacent to the ADF, ANT legend is pushed to switch between the ADF and ANT modes. The selected text is black letters on a green background. The non-selected text is in white text on the black background.

Test Mode: Push the side key adjacent to the TEST legend to test the ADF receiver. The TEST legend changes to black letters on a green background from the usually white on black background. The bearing pointer of the ADF receiver under test is parked at a relative bearing of 90 degrees. The display area shows OK in green letters for five seconds, if the test result is correct. It shows FAIL in red letters when the test result is not correct.

BFO ON or BFO OFF: The side key adjacent to the BFO ON, BFO OFF legend is pushed to alternately switch the BFO on or off. The selected text changes to black letters on a green background and the non-selected text is in white text on the black background.

Audio Control: [Refer to Figure 12.](#) The ADF Potentiometer push-button switches control the audio parameters that follow:

- On
- Off
- Level.

The top part of the potentiometer is pushed to set the specific audio signal to the headphone on or off. The ADF 1, or ADF 2 switch is pushed on the Audio and Radio Control Display Unit (ARCDU 1, ARCDU 2) to make its audio available to that flight crew member. When the SPKR push-button potentiometer switch is pushed, the

selected ADF audio is routed to the related flight compartment speaker.

The ADF receiver display areas shows the volume level as a vertical bar graph. The height of the bar graph shows the volume selection level. When the audio is selected off, the bar graph displays in white. The bar graph changes to green when the audio is selected on.

The ADF audio is available at the observer's headphone jack when the ADF 1 or ADF 2 is selected on the observer audio control panel.

[Refer to Figure 13.](#)

When an ADF 1 or ADF 2 bearing is selected at the EFIS Control Panels (EFCP 1, EFCP 2), a distinctive pointer comes into view on the Primary Flight Display.

[Refer to Figure 14.](#)

The bearing pointers in the Electronic Horizontal Situation Indicators (EHSI 1, EHSI 2) show the lateral direction to the navigation aid as follows:

- A white, single bar pointer with a white circle to identify the ADF 1
- A green, wide double-bar pointer with a green diamond to identify the ADF 2.

The lower left portion of the Primary Flight Displays (PFD 1, PFD 2) shows a reminder label. The reminder label is related to the EFIS



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Control Panel selection. It has the same two symbols in white along with ADF 1 and ADF 2 legends.

[Refer to Figure 15.](#)

When the ADF signal is invalid or failed, the pointers are removed from the Primary Flight Displays (PFD 1, PFD 2). It continues to display the reminder as the bearing selection.

If the pilot's EFIS Control Panel (EFCP 1) fails, the bearing pointers default to show the information that follows:

- The single bar indicates Automatic Direction Finder (ADF 1)
- The double bar indicates Very High Frequency Omni-range (VOR 2).

If the copilot's EFIS Control Panel (EFCP 1) fails, the bearing pointers default to indicate the following:

- The single bar indicates Very High Frequency Omni-range (VOR 1)
- The double bar indicates Automatic Direction Finder (ADF 2).

The Audio and Radio Control Display Units (ARCDU 1, ARCDU 2) monitor the ARINC 429 data bus through the Flight Data Processing System (FDPS) for a malfunction of the Automatic Direction Finding (ADF) system. The Audio and Radio Control Display Units (ARCDU 1, ARCDU 2) show a red FAIL message. It comes into view in the frequency-preset area of the ADF display area for system failures.

The aircraft Left Essential and Right Main buses supply 28 Vdc electrical power through 2 ampere circuit breakers to the ADF

Receivers (ADF 1, ADF 2). The circuit breakers are located in position H3 and H4 on the 28 Vdc Avionics circuit breaker panel.

The Automatic Direction Finding (ADF1, ADF2) systems have two ADF receivers with related ADF antennas.

The ADF Receivers (ADF 1, ADF 2) supply bearing information for display through both Input/Output Processor (IOP 1, IOP 2). The Input/Output Processor (IOP 1, IOP 2) is located inside its related Integrated Flight Cabinet (IFC 1, IFC 2).

The ADF parameters that follows are controlled using ARINC 429 Low speed buses (12.0 to 14.5 kilobits/sec):

Table 2: Table 2

Parameter	Source
Frequency	ARCDU
Failure Warning	IOP 1, IOP 2
No Computed data	IOP 1, IOP 2
Normal Operation	IOP 1, IOP 2

Receiver, ADF

[Refer to Figure 16.](#)

The ADF Receiver receives a combined sense and loop relative bearing signal from the ADF Antenna to calculate the bearing to the station. It uses a self calibration mode that gives better better bearing



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accuracy and digital filtering to give better bearing stability in severe environmental noise conditions.

The ADF receiver has the sub-assemblies that follow:

- Receiver Module
- Analog Module
- Input/Output (I/O) Module
- Processor Module
- Power Supply.

The interconnections between assemblies are made using ribbon cables. Maintenance is simplified through modular construction. Modules are hinged and attached to a rigid main frame to allow easy access. The circuit boards plug into connector to allow troubleshooting by substitution. The case has a front panel handle for easy removal and installation.

The ADF receiver weighs 5.82 lb (2.64 kg). It has subassemblies mechanically packaged into a 4 in. (102 mm) high, 4.1 in. (104 mm) wide, by 13.33 in. (338.6 mm) long case. A single recessed DPX rack-mounted rear-panel connector and two front-panel coaxial RF connectors are installed for connection to the aircraft wire harness and antenna installation.

The ADF receivers are installed in mounting racks side by side on the third shelf of the avionics rack at station points X -28.00 and X -34.00.

Antenna, ADF

Refer to Figure 17.

The ADF antenna receives radio frequencies between 190 and 1860 kHz and between 2181 and 2183 kHz. The antenna has electrical circuits that sum the loop and sense signals to make an Intermediate Frequency signal. The Intermediate Frequency signal is sent to the ADF receiver.

The ADF antenna weighs five pounds. It has a combined loop and sense element with a solid-state amplifier. The antenna enclosure has a low profile to reduce aerodynamic drag. It is 2.85 in. (72.4 mm) high, 6.70 in. (170.2 mm) wide, by 13.80 in. (350.5 mm) long.

The ADF antennas are attached with six mounting bolts to the bottom of the aircraft fuselage along the centerline at station points X 246.00 and X 291.00.

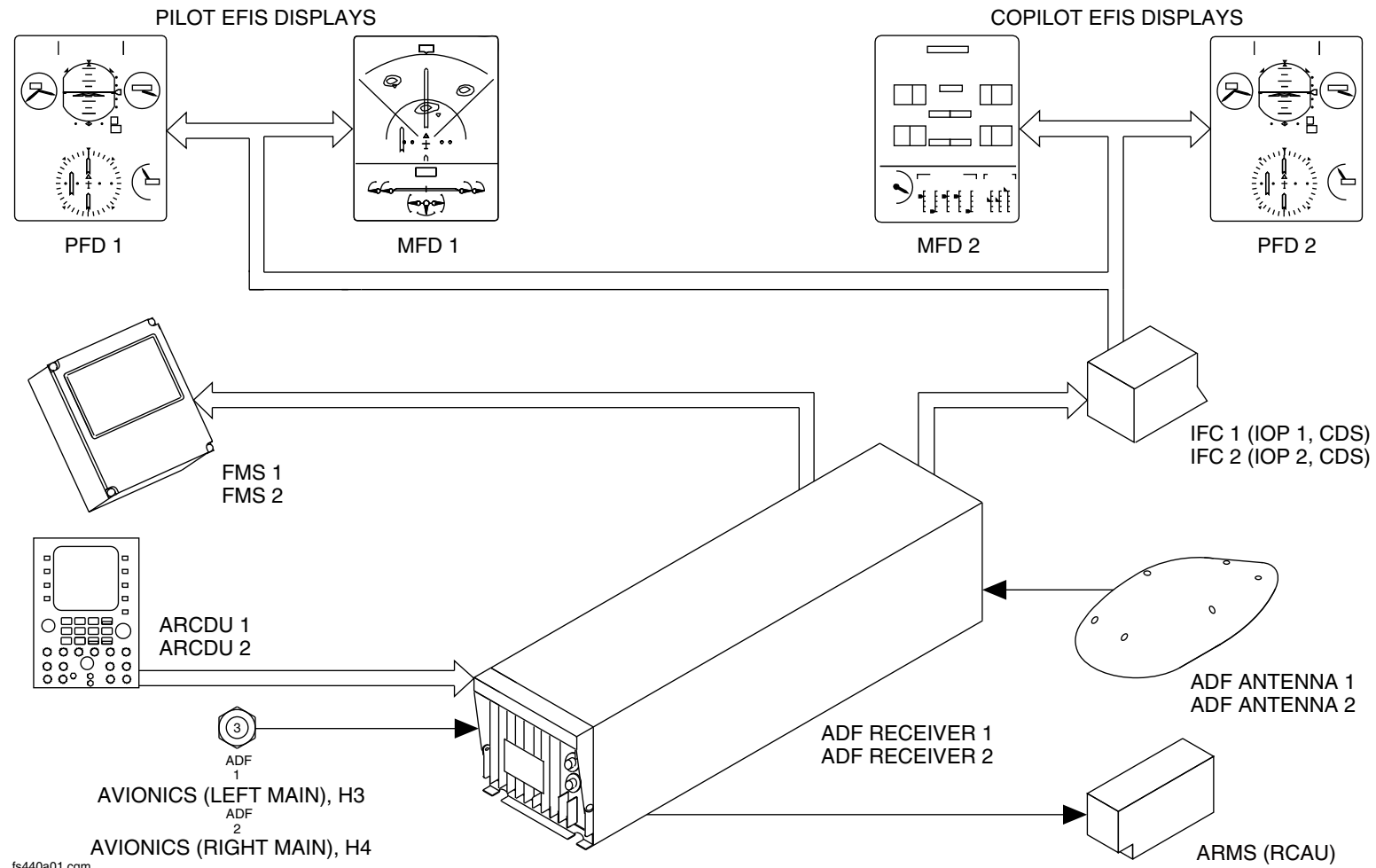
Antenna Cable

The Automatic Direction Finding (ADF) systems use twinax antenna cables to connect the ADF receiver to its respective antenna. The twinax cable is a transmission line with two conductors twisted together as a balanced wire line with a copper shielding braid around both wires. Two insulated braids are connected to the system ground and to the airframe ground. The two balanced signal-carrying wires are twisted to cancel any random induced voltage pickup. The antenna cable length is not critical.



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ADF BLOCK DIAGRAM
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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Config 001

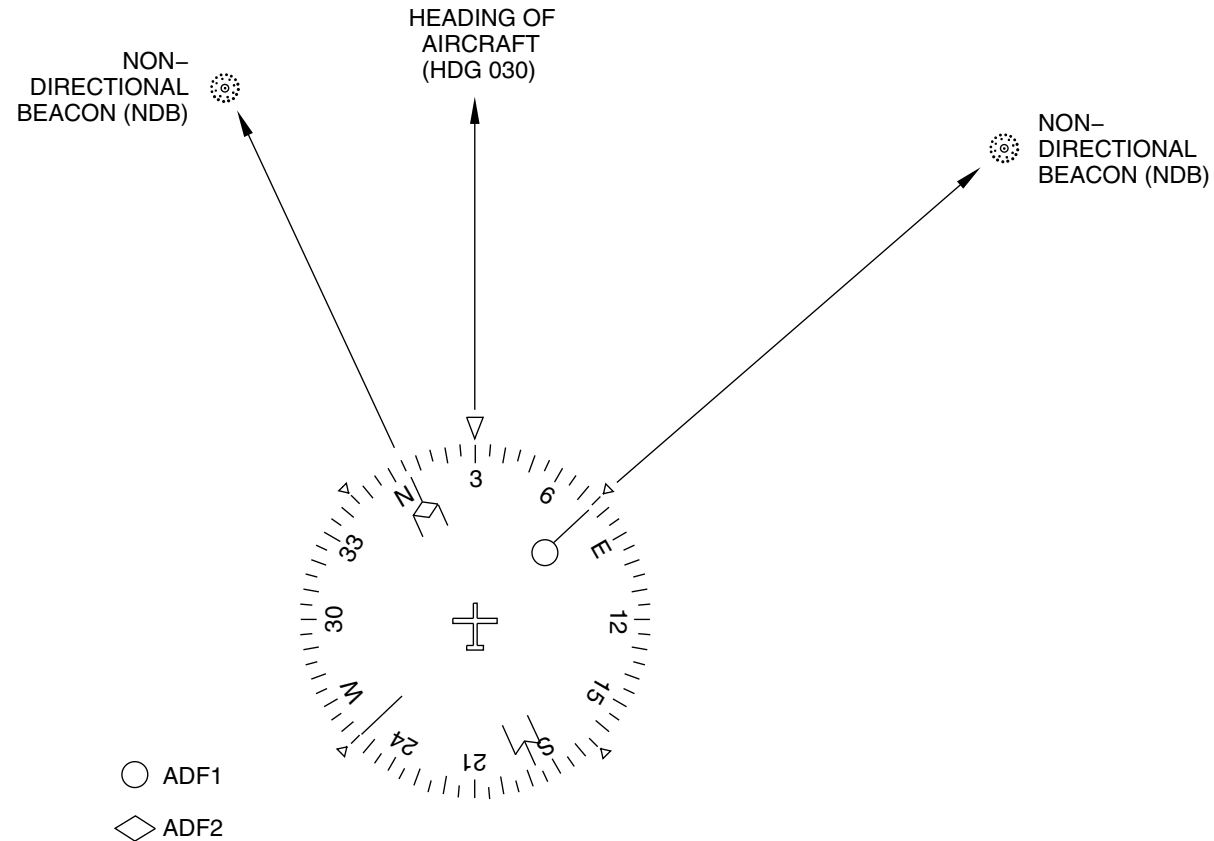
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MAGNETIC BEARING INDICATION
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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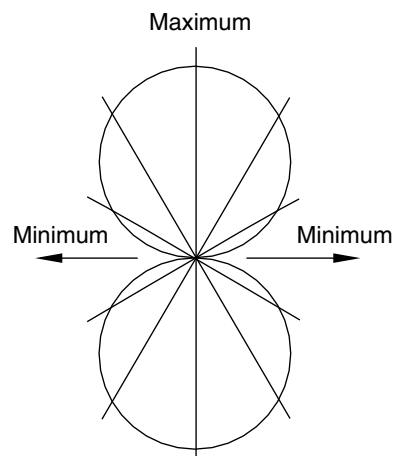


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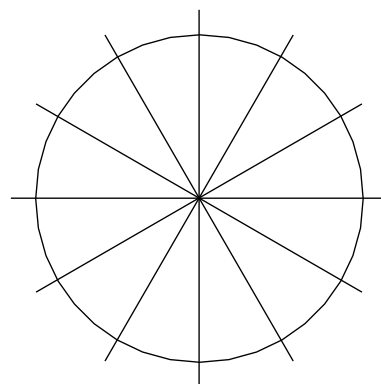
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ANTENNA SIGNALS

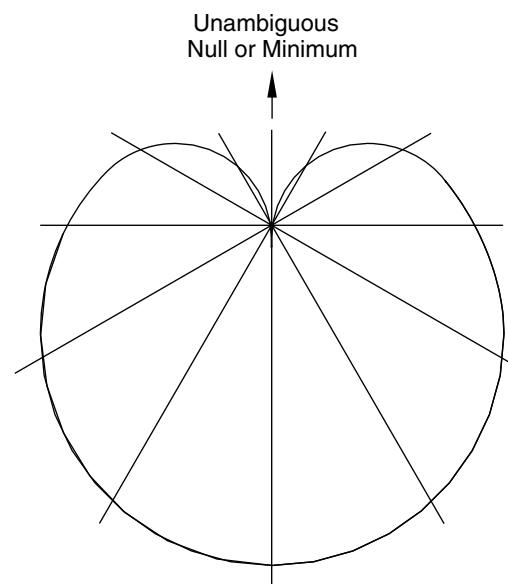
Direction of Incoming Signal



LOOP



SENSE



COMBINED

fs477a01.cgm

NDB ANTENNA SIGNALS
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-52-00
Config 001

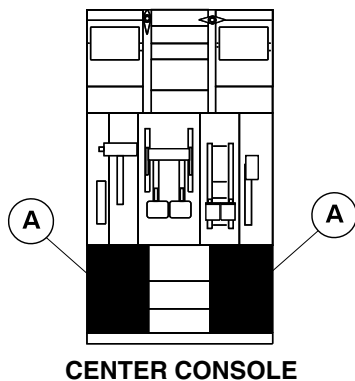
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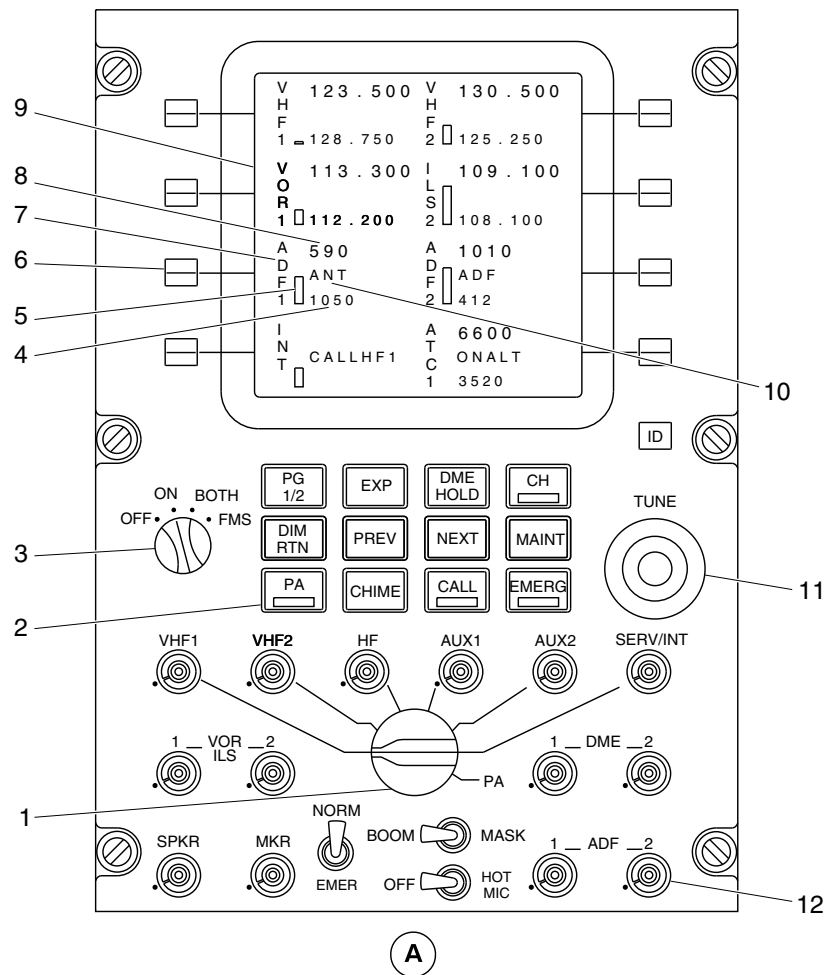
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Microphone/Interphone selector switch.
2. Keys.
3. Rotary switch.
4. Preset frequency.
5. Level adjust bar graph.
6. Side key.
7. Label.
8. Active frequency.
9. Display area.
10. Automatic direction finding mode.
11. Tuning knob.
12. Potentiometer/Push-button switches.



ADF, ARCDU
Figure 4

fs620a01.dg, sk, jul08/2019

PSM 1-84-2A
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Config 001

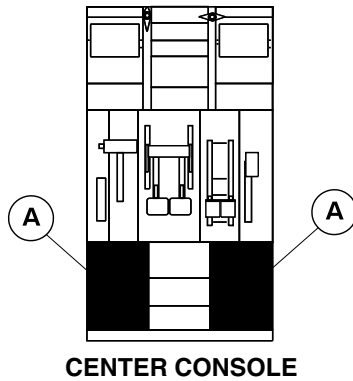
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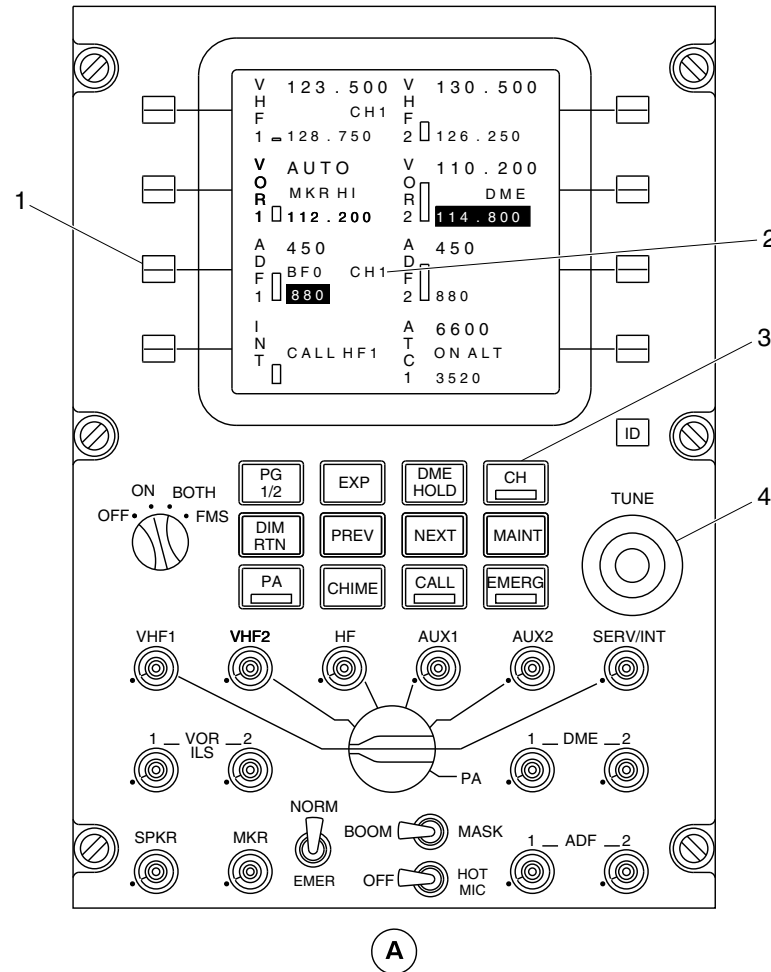
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LEGEND

1. Side key.
2. Channel memory annunciation.
3. Channel key.
4. Tuning knob.



cg5725a01.dg, sk, jul08/2019

ARCDU ADF Frequency Control
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-52-00
Config 001

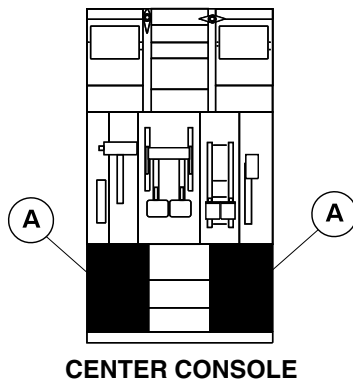
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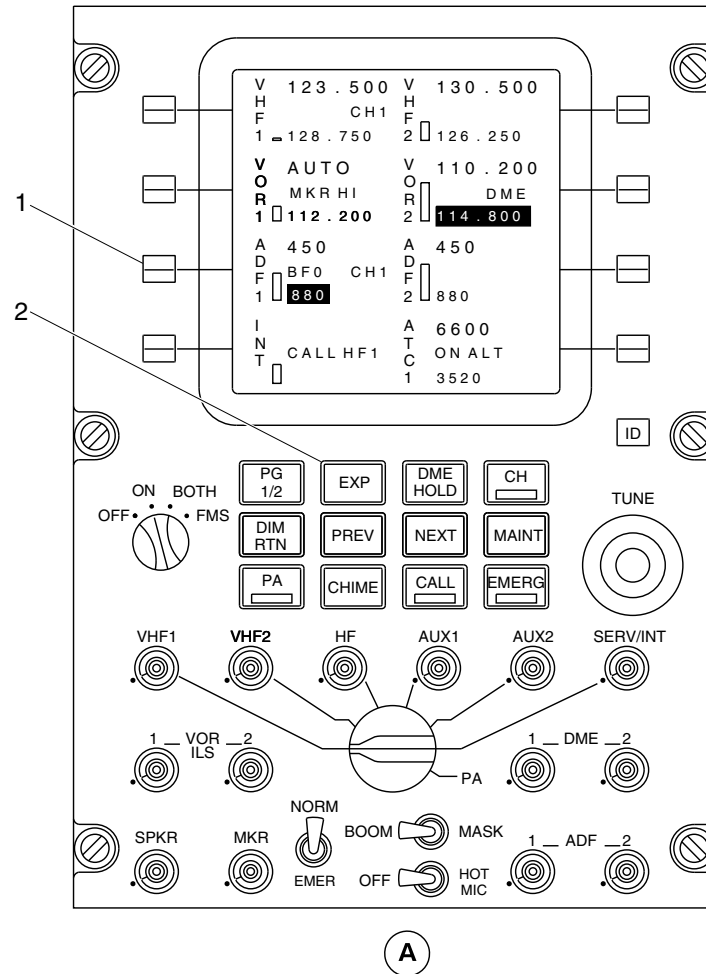
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LEGEND

1. Side key.
2. Expand key.



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ARCDU ADF Particular Page Access
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-52-00
Config 001

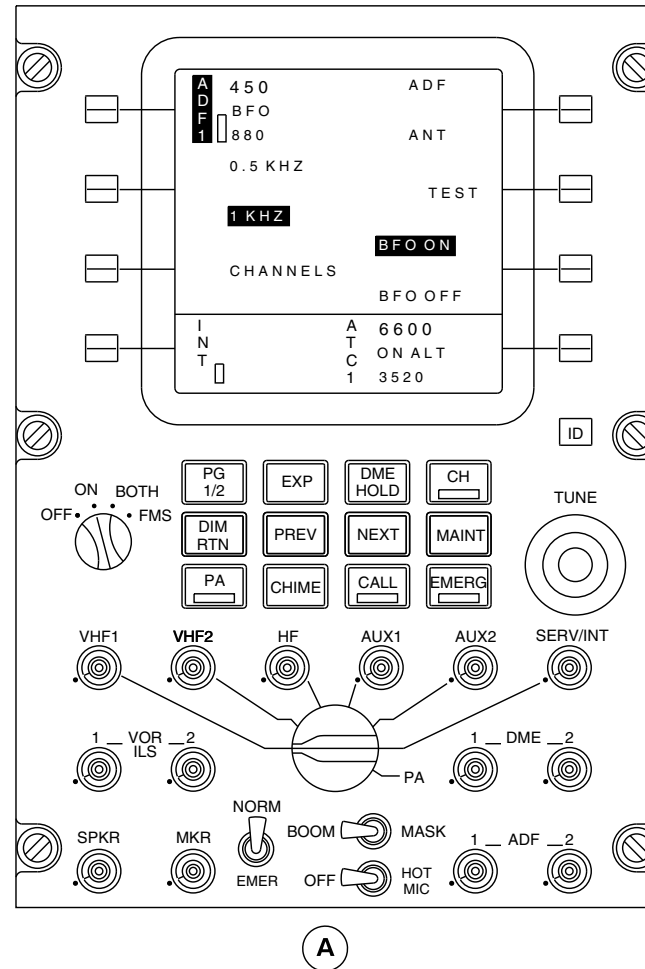
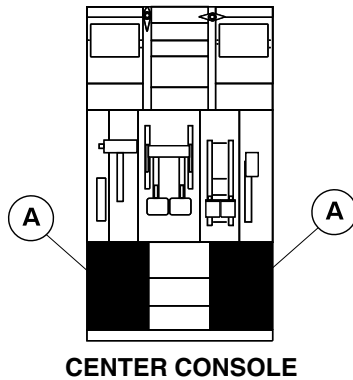
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ARCDU ADF Particular Page
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-52-00
Config 001

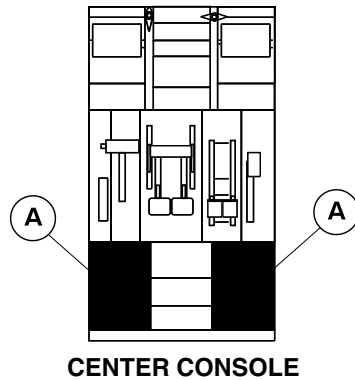
34-52-00

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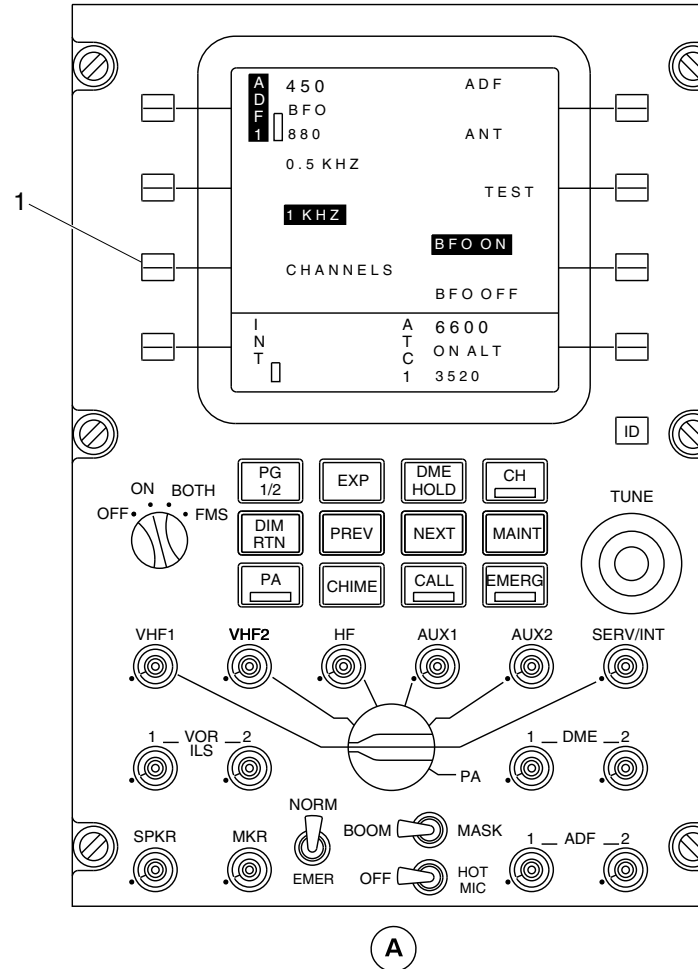
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LEGEND

1. Side key.



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ARCDU ADF Channels Access Page
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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Config 001

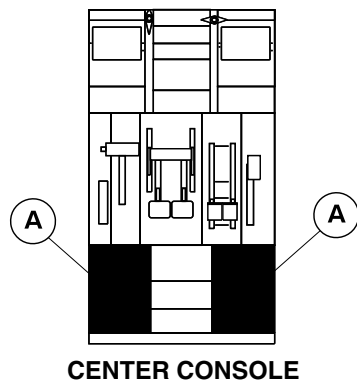
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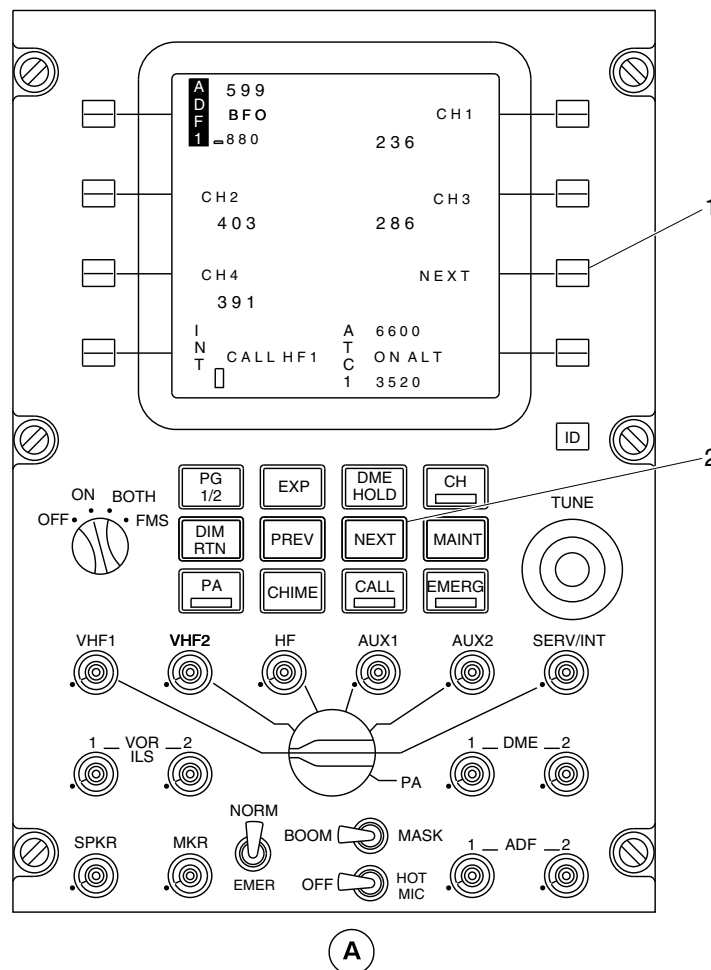
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LEGEND

1. Side key.
2. Next page key.



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ARCDU ADF First Channels Page
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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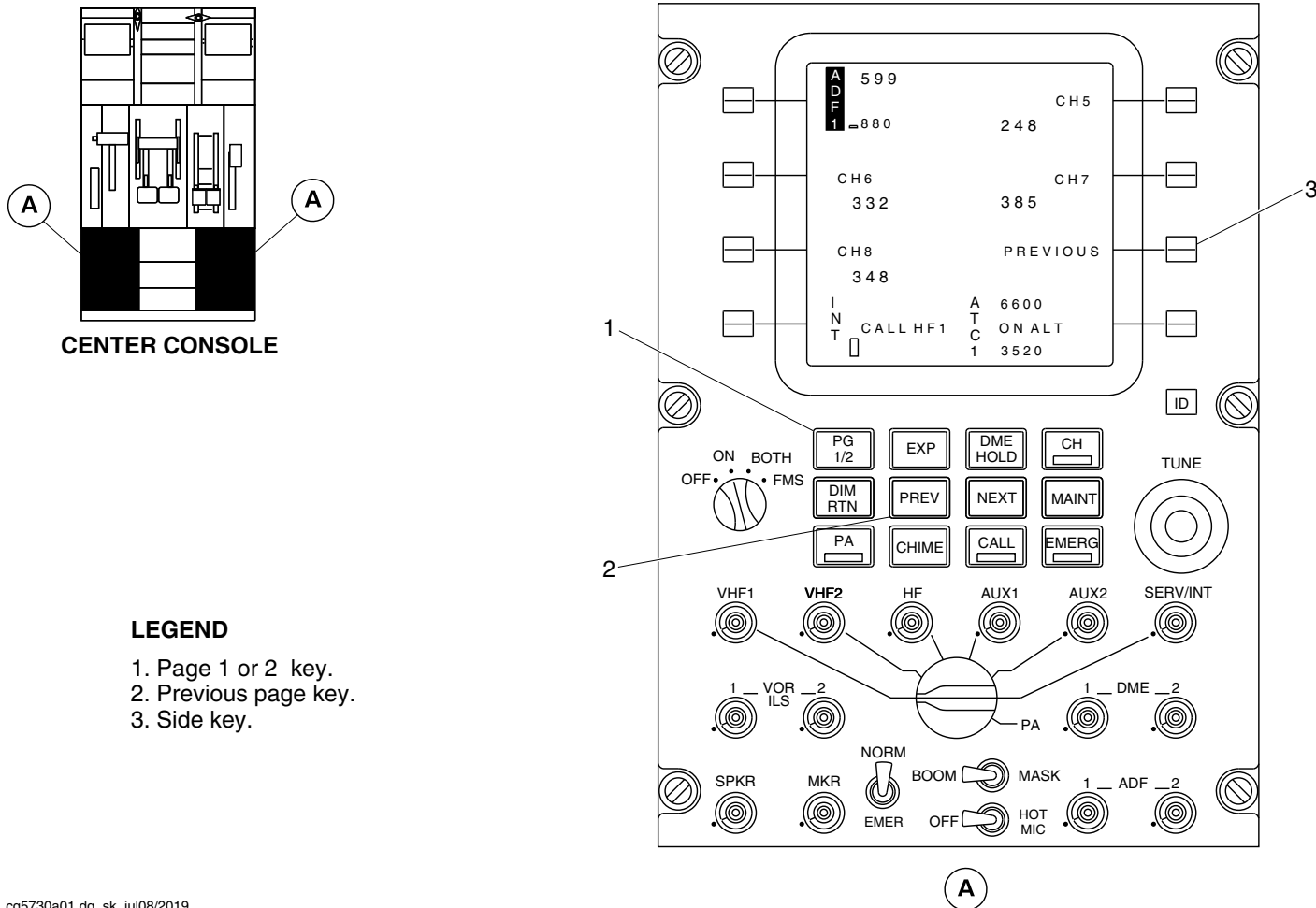
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ARCDU ADF Second Channels Page
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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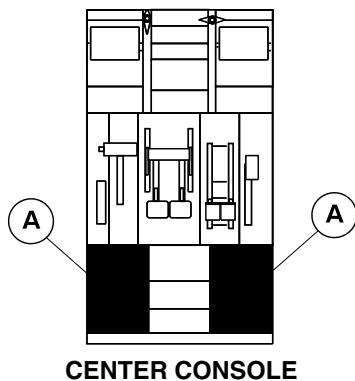
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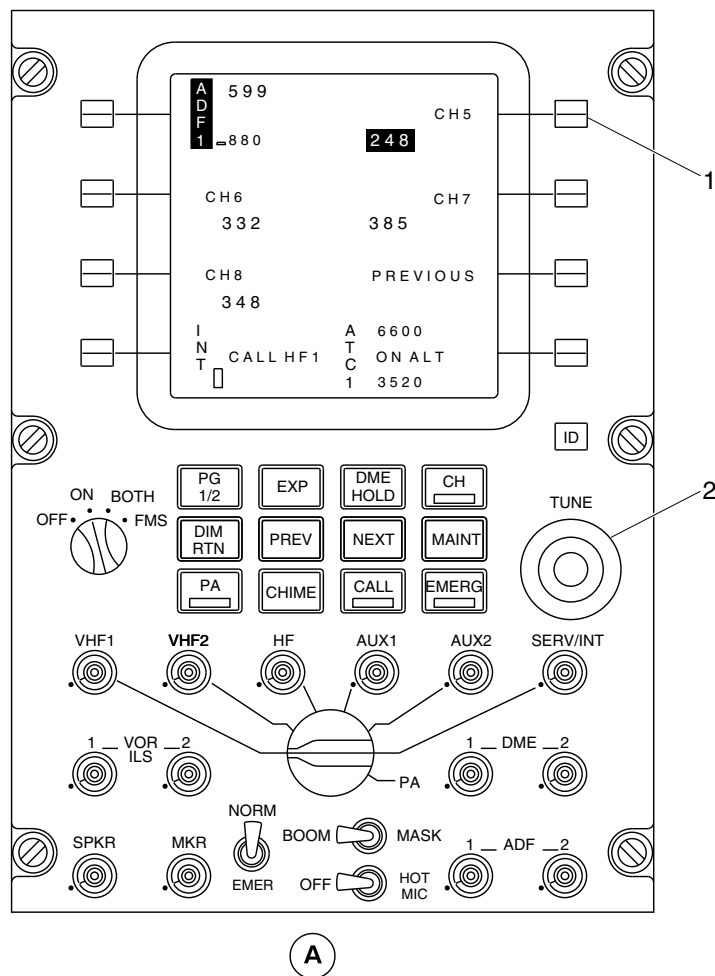
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LEGEND

- 1. Side key.
- 2. Tuning knob.



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ARC DU ADF Preset Channel Programming
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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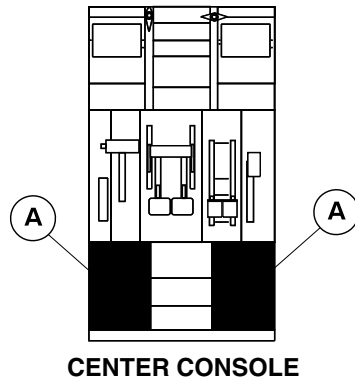
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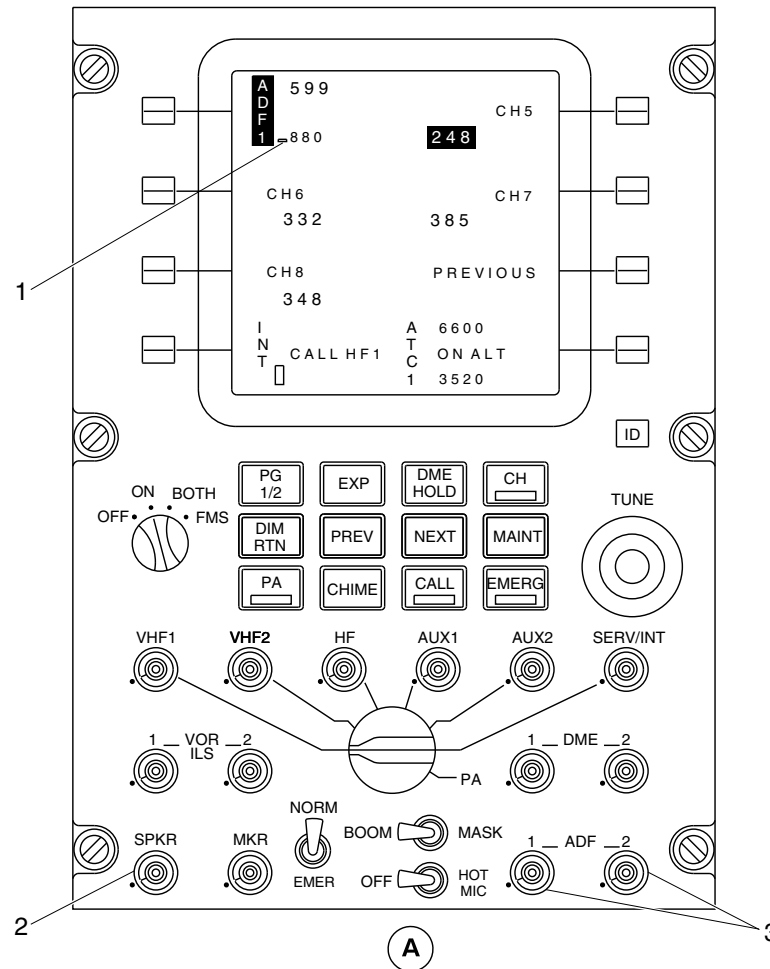
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Level adjust bar graph.
2. Speaker potentiometer/Push button switch.
3. ADF potentiometer/Push button switches.



cg5732a01.dg, sk, jul08/2019

ARCDU ADF Audio Control
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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Config 001

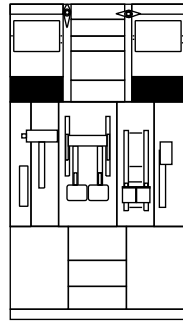
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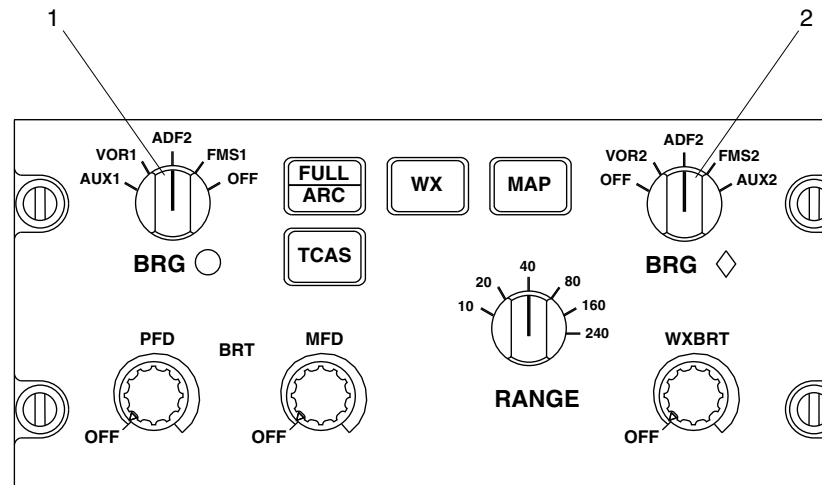
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

LEGEND

1. Bearing 1 Selector Knob.
2. Bearing 2 Selector Knob.



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EFCP ADF BEARING SELECTION
Figure 13

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-52-00
Config 001

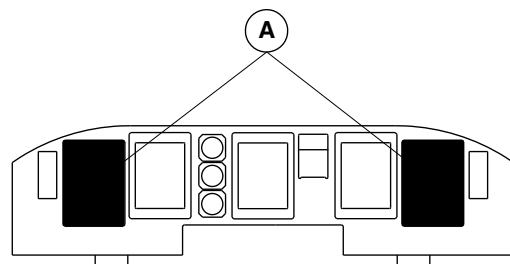
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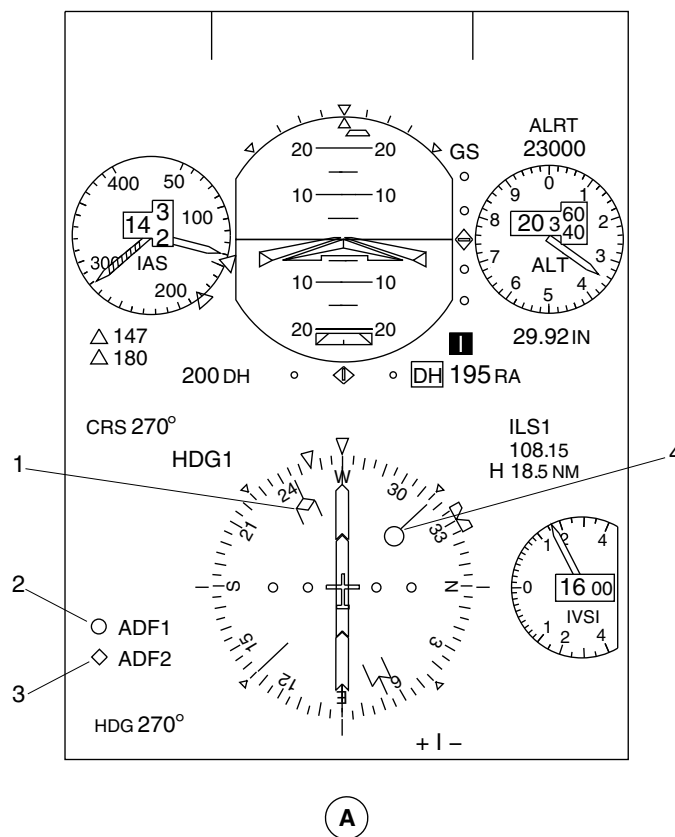
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. #2 Bearing Pointer.
2. #1 Bearing Reminder Label.
3. #2 BearingReminder Label.
4. #1 Bearing Pointer.



fs618a01.cgm

EIS ADF NORMAL BEARING INDICATION
Figure 14

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-52-00
Config 001

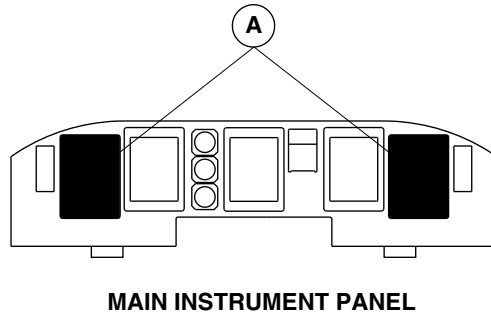
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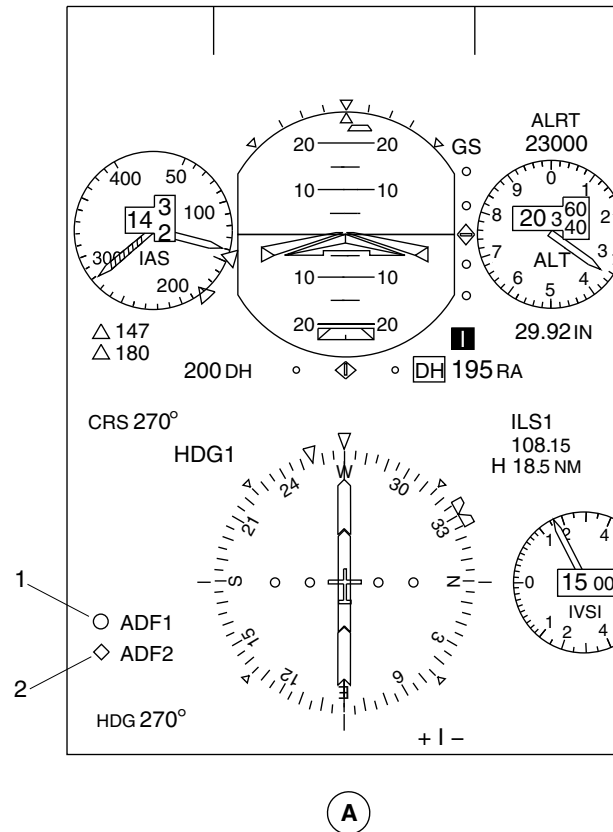
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. #1 Bearing Reminder Label.
- 2. #2 Bearing Reminder Label.



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EIS ADF MALFUNCTION INDICATION
Figure 15

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-52-00
Config 001

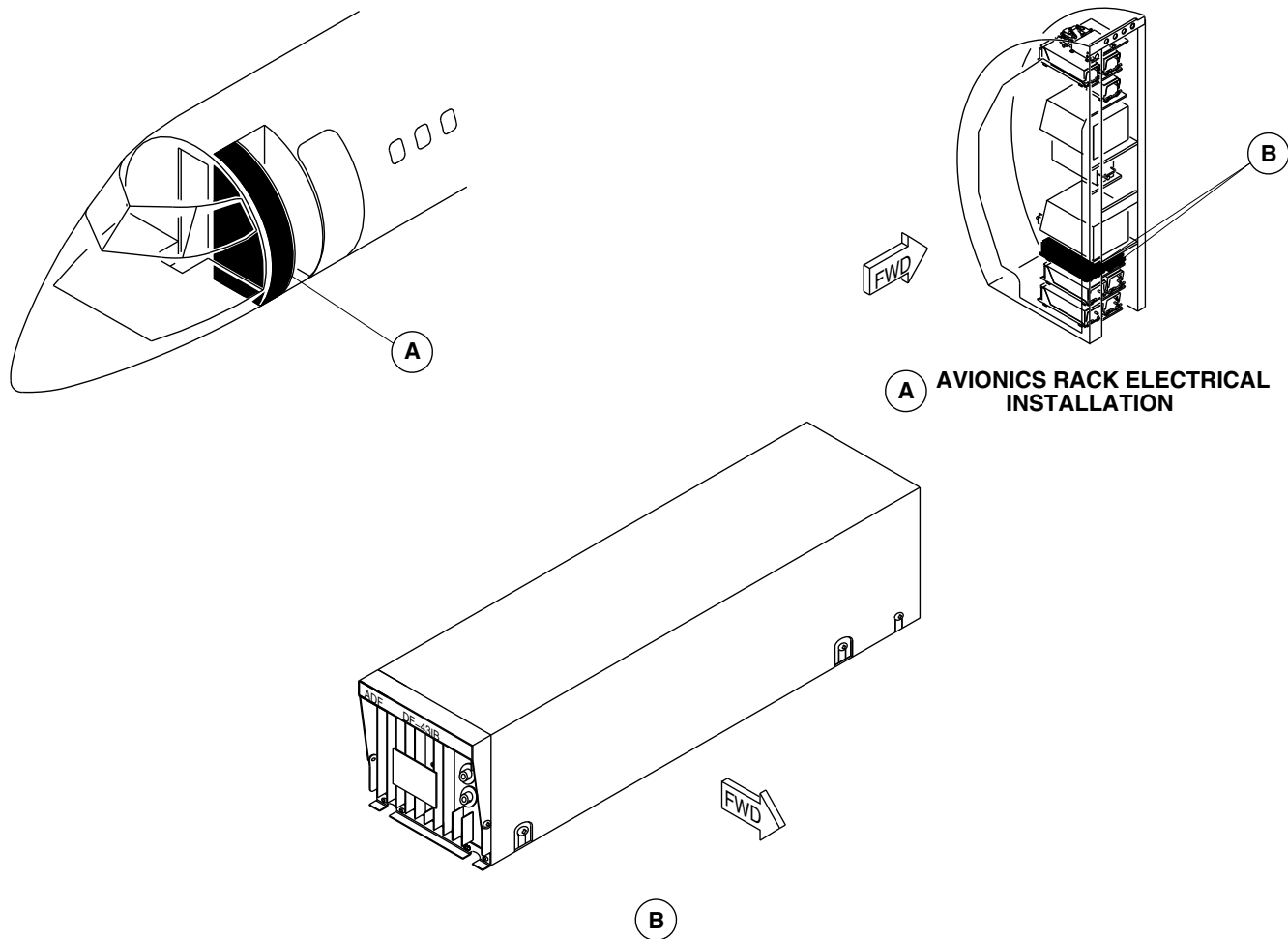
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ADF RECEIVER LOCATOR
Figure 16

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-52-00
Config 001

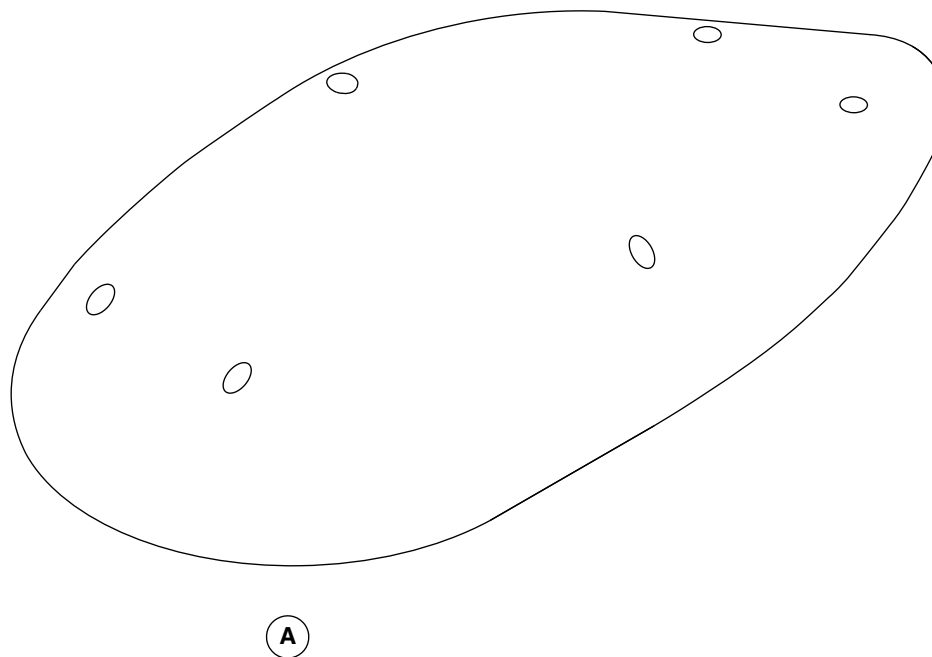
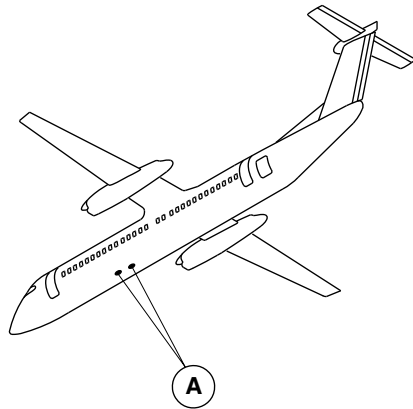
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ADF ANTENNA LOCATOR
Figure 17

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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DISTANCE MEASURING EQUIPMENT (DME)

Introduction

The airborne Distance Measuring Equipment (DME) measures the line of sight slant range from the aircraft to the DME ground station. It uses the slant distance data to calculate the ground speed and the time-to-go parameters.

General Description

Refer to Figure 1.

The Distance Measuring Equipment (DME) system transmits Radio Frequency (RF) pulse-pairs to a ground station.

It measures the time it takes for the signal to travel to the station and back to make the calculations that follow:

- Distance to station
- Ground Speed
- Time-to-go.

The VHF Navigation Receivers (VHF NAV1, VHF NAV2) tune the Distance Measuring Equipment (DME1, DME2) from an Audio and Radio Control Display Units (ARCDU1, ARCDU2) selection. A DME HOLD selection is used to keep the current DME station active when a new frequency is selected. The Microwave Landing System Receivers (MLS1, MLS2) tune the Distance Measuring Equipment

(DME1, DME2) to its paired channel selection when the Microwave Landing System is used.

The Audio and Radio Control Display Units (ARCDU1, ARCDU2) are used to make the DME audio selections. The observer uses a Audio Control Panel (ACP) to set the DME audio.

The Electronic Flight Instrument System (EFIS1, EFIS2) shows the Distance Measuring Equipment (DME) parameters that follow:

- Slant Range
- Time To Station
- Ground Speed
- DME Hold.

The Distance Measuring Equipment (DME) system supplies data from multiple ground stations to the Flight Management System (FMS) to calculate its position-fixing algorithm.

The Distance Measuring Equipment (DME1, DME2) has the components that follow.

- Transceiver, DME (34-53-01)
- Antenna, DME (34-53-11).

The Distance Measuring Equipment (DME1, DME2) supplies data to the systems that follow:

- Audio and Radio System (ARMS)
- Input/Output Processors (IOP1, IOP2)
- Traffic Collision Avoidance System (TCAS)
- ATC Transponder (ATC1, ATC2)
- Flight Management System (FMS).

PSM 1-84-2A

EFFECTIVITY:

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Detailed Description

The airborne Distance Measuring Equipment (DME) interrogates the ground station with pulse pair transmissions spaced either 12 or 36 microseconds apart. The ground station replies with pulse pairs transmissions spaced either 12 or 30 microseconds apart. The ground station has a built in 50 microsecond delay between reception of an interrogation and transmission of the reply to allow operation at close range.

The airborne DME senses the time interval between transmission of an interrogation and reception of the reply to calculate the slant range.

The Distance Measuring Equipment (DME) ground station transmits at a constant Pulse Repetition Frequency (PRF) of 2700 pulse pairs per second. It consists of replies to interrogations and random pulses called squitter. If the DME ground station receives more than 2700 interrogations per second, it will reply to the stronger interrogation rather than increase its Pulse Repetition Frequency (PRF) rate. The airborne DME contains circuitry that lets it identify replies to its own interrogations from squitter or replies to other Distance Measuring Equipment (DME).

The airborne Distance Measuring Equipment (DME) operates in the search mode when its transmissions are sent and replies are not received. In the search mode, the DME transmits about 100 pulse pairs per second.

Once the Distance Measuring Equipment (DME) receives replies from the ground station, it operates in the tracking mode and reduces the interrogations to about 30 pulse pairs per second.

The Distance Measuring Equipment (DME) must receive replies from the selected station within 40 seconds. It will not transmit again until it receives squitter from the selected ground station.

The DME transceivers (DME1, DME2) operate within the frequency range of 962 to 1213 MHz. Each transceiver uses two hundred DME channels that are paired with VHF Navigation frequencies between 108.00 and 117.95 MHz. The channels are divided into 100 X and 100 Y channels. The X channel spacing of the transmitted pulse-pairs is 12 microseconds for both the interrogation and reply signals. The Y channel spacing is 30 microseconds for the interrogation and 36 microseconds for the reply.

The DME ground stations transmits a three-letter code group as an identifier.

When a VHF Omni-range (VOR) or Instrument Landing System (ILS) frequency selection is made, the navigation receiver automatically tunes a paired DME frequency.

The Radio Control Display Units (ARCDU1, ARCDU2) are used to control the VHF Navigation and paired DME frequencies.

[Refer to Figure 2.](#)

The DME HOLD key is pushed on the Audio and Radio Control Display Unit (ARCDU1, ARCDU2) to hold the DME channel active while a new VHF Navigation frequency is selected.

The ARCDU shows the held frequency adjacent to a DME hold legend. The held frequency is shown in a green font on a white background below the active VHF NAV frequency.

The Primary Flight Display (PFD1, PFD2) and related Navigation Page show a yellow H to the left of the displayed distance. It shows that the Distance Measuring Equipment (DME) is not tuned to the station frequency that is shown directly above it.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

When the DME Hold key is pushed again it cancels the DME hold function.

NOTE

The conditions that follow also cancel the DME hold function:

- The Microwave Landing System (MLS) is selected as the navigation source
- The Flight Management System (FMS) is autotuning

[Refer to Figure 3.](#)

The DME Potentiometer push–button switches give the audio control functions that follow:

- On
- Off
- Level.

The top part of the potentiometer is pushed to set the specific audio signal to the headphone on or off. The DME1 or DME2 push–button switch is pushed on the Audio and Radio Control Display Unit (ARCDU1, ARCDU2) to make its audio available to that flight crew member. When the SPKR push–button potentiometer switch is pushed, the DME audio is routed to its related flight compartment speaker.

The DME display areas shows the volume level as a vertical bar graph. The height of the bar graph shows the volume selection level. When the audio is selected off, the bar graph displays in white. The bar graph changes to green when the audio is selected on.

The DME audio is available at the observer's headphone jack when the DME 1 or DME 2 is selected on the observer's Audio Control Panel (ACP).

[Refer to Figure 4.](#)

The upper right part of the Electronic Horizontal Situation Indicator (EHSI) area of the Primary Flight Display (PFD1, PFD2) shows the DME distance in a blue font. It shows the distance to station information from 0 to 300 nmi. The range resolution is 0.1 nmi for distances to 99.9 nmi and 1 nmi for distances greater than 99.9 nmi.

[Refer to Figure 5.](#)

The Navigation page of the Multi–Function Displays (MFD1, MFD2) shows the DME functions that follow:

- Distance
- GS, Ground Speed
- Time to station.

The distance is shown in the same location on the MFD as on the PFD. The ground speed and time to station are shown in a blue font in the lower right area of the Navigation Page.

[Refer to Figures 6 and 7.](#)

The numerical values for all indications change to three white dashes when there is a source failure or the distance is out of range.

If the DME fails, the Audio and Radio Control Display Unit (ARCDU1, ARCDU2) shows a FAIL legend in red characters below the active frequency.

The Left and Right Main DC buses supply 28 Vdc electrical power to the Distance Measuring Equipment (DME1, DME2) through 2 A circuit breakers. The circuit breakers are located in positions G3 and G4 on the 28 Vdc avionics circuit breaker panel.

The DME, ATC transponder and the Traffic Collision Avoidance System (TCAS) operate in the same frequency range. Suppression



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

pulses are output to momentarily suppress the transmissions of the other L-band equipment.

Each Distance Measuring Equipment (DME1, DME2) transceiver interfaces with the systems that follow so that only one system transmits at a time:

- The other DME transceiver
- Traffic Collision Avoidance System (TCAS)
- Transponders (ATC1, ATC2).

Each Distance Measuring Equipment (DME1, DME2) systems has a transceiver with a related antenna.

The Distance Measuring Equipment (DME1, DME2) supplies navigation information for display through the Input/Output Processors (IOP1, IOP2). The Input/Output Processors (IOP1, IOP2) and are located inside its related Integrated Flight Cabinet (IFC1, IFC2).

The DME sends the parameters that follow on an ARINC 429 Low speed data bus:

Table 1: Table 1

Parameter	Source
DME Frequency	ARCDU
DME Distance (Range)	IOP1, IOP2

DME Time-to-Station	IOP1, IOP2
DME Ground Speed	IOP1, IOP2

The Distance Measuring Equipment (DME1, DME2) are tuned by the navigation sources that follow:

- VHF Navigation Receivers (VHF NAV1, VHF NAV 2)
- Microwave Landing System (MLS1, MLS2) receivers.

Part of the Flight Guidance Control Panel (FGCP) navigation source selection routes ARINC 429 tuning information to the DME. The navigation source selection causes an Input/Output Module (IOM1, IOM2) discrete to control a relay to route tuning information to the DME. The Input/Output Modules (IOM1, IOM2) are located inside of its related Integrated Flight Cabinet.

The Distance Measuring Equipment (DME1, DME2) supplies data to the Flight Management System (FMS). It uses Distance Measuring Equipment (DME) data for the VORTAC Position Unit (VPU) mode to calculate its position-fixing algorithms.

Transceiver, DME

[Refer to Figure 8.](#)

The DME Transceiver gives slant range distance data from DME ground stations. The transceiver transmits pulse pair interrogation signals to a ground station which in turn sends back coded reply signals for each interrogation signal received. The DME Transceiver measures the two way transmission delay from the aircraft to the ground station and changes it to a distance value in nautical miles. Microprocessor circuits use the distance data to calculate ground speed and time-to-go data.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Each DME ground station periodically transmits an audio identification signal which is received, decoded, and then sent to the audio system.

The DME Transmitter is tuned by external sources using the ARINC 429 format. The system is capable of tracking three DME ground stations for the Flight Management System (FMS) VORTAC Position Unit (VPU) mode to calculate its position-fixing algorithms.

The DME transceiver has these sub-assemblies that follow:

- Transmitter
- Modulator
- Synthesizer
- VCO
- Receiver
- Master Processor
- Range Processor
- Power Supply.

The DME Transceiver has a solid-state transmitter which gives 325 W of output power. It uses two separate microprocessors. One microprocessor makes all the calculations and the other controls the digital input/output and frequency tuning interfaces.

A range processor is a unique distance convertor that can range on data in less than 200 msec and has a resolution of less than 0.01 nmi. This allows fast acquisition of ground speed and time-to-go, even when tracking 3 three stations.

The power supply module is activated by a hard-wired low-level enable signal at the rear connector.

The DM-441B DME transceiver weighs 6.28 lb (2.85 kg). It has subassemblies mechanically packaged into a 4 in. (102 mm) high, 5 in. (127 mm) wide by 13 in. (330 mm) long case. A single recessed DPX rack-mounted rear-panel connector and a front-panel TNC coaxial connector is used for connection to the aircraft wire harness and the antenna cable.

On aircraft with ModSum 4-113507 incorporated, the CDM-451 DME transceiver weighs 3.60 lb (1.63 kg). It has subassemblies mechanically packaged into a 3.87 in. (98.17 mm) high, 3.27 in. (83.01 mm) wide and 12.62 in. (320.55 mm) long case. A single recessed DPX rack-mounted rear-panel connector and a front-panel TNC coaxial connector is used for connection to the aircraft wire harness and the antenna cable.

The DME transceivers are installed in mounting trays one above the other on the two lower shelves of the avionics rack, at approximately station point X -28.0.

The DME transceivers are installed in the avionics rack that is supplied with forced air cooling.

Antenna, DME

[Refer to Figure 9.](#)

The L-band DME antenna transmits and receives radio frequencies in the 950 to 1250 MHz frequency bands. The antenna has the features that follow:

- Omni-directional
- Quarter wavelength
- Vertically polarized
- Monopole



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- End fed.

The L-band antenna is an advanced lightweight durable thermoplastic blade antenna. The antenna encases the pre-formed radiating element within a single assembly. The antenna's Radio Frequency (RF) coaxial type HN female connector is attached to an aluminum base plate.

The L-band antenna weighs less than 0.25 pounds (0.11 kilograms). The DME ANT 1 is installed in the forward, left-side area below the forward fuselage section at station points X -94.41, Y -15.41. DME No.2 antenna is installed on the aircraft centerline at X 479.5. Each antenna is installed to the aircraft with five mounting screws. The antennas are bonded to the aircraft structure using a "wet fit" sealing gasket.

The gasket consists of an aluminum foil gasket with an elastomeric sealant. The pressure applied during installation gives a contour feature. It lets the gasket form itself precisely to the two metallic surfaces to give good conduction between the antenna and airframe. The moisture free, contoured seal requires no maintenance.

Antenna Cable

Coaxial cables connect the antennas to the DME transceivers. The coaxial cable is a transmission line with one conductor surrounding another. A dielectric insulator separates the two conductors. The center conductor is seven strand silver-plated copper wire and the outer conductor is a tinned copper wire braid.

Two types of dc resistance coaxial cables are installed because of different cable lengths in the aircraft. The DME2 has a lower loss cable because of the longer run between the transceiver and the antenna.

The table that follows show the difference between the two co-axial cables.

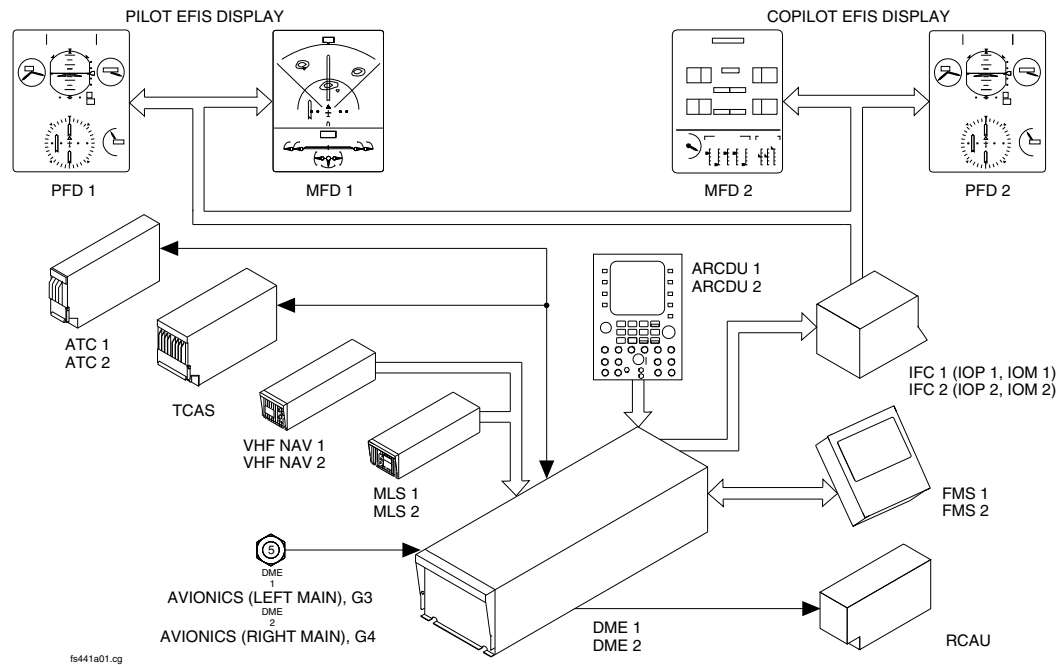
Table 2: Table 2

Cable	Attenuation	DC Resistance
DME 1	7dB / 100 ft	1.8 Ohms / 100 ft
DME 2	4dB / 100 ft	0.95 Ohms / 100 ft



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DME BLOCK DIAGRAM
Figure 1

PSM 1-84-2A
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Config 001

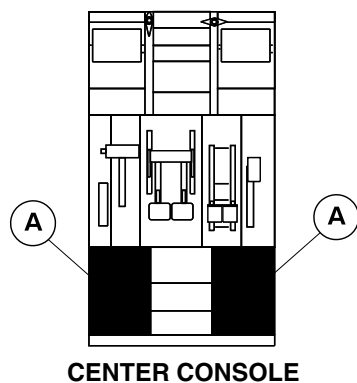
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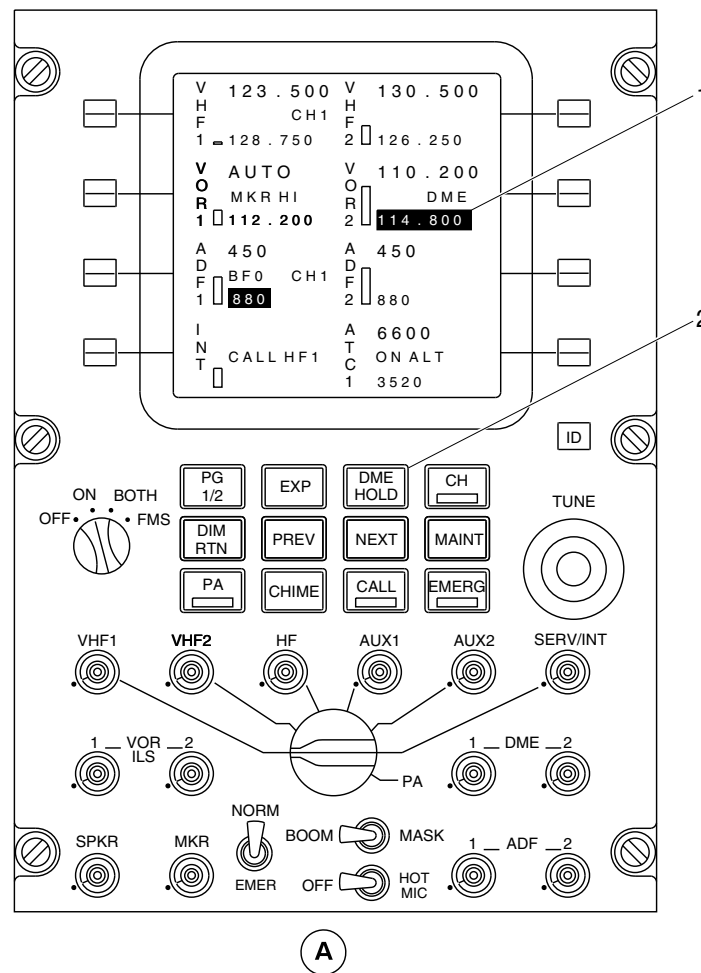
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LEGEND

1. DME hold indication.
2. DME hold key.



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ARCDU DME Hold Function
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-53-00
Config 001

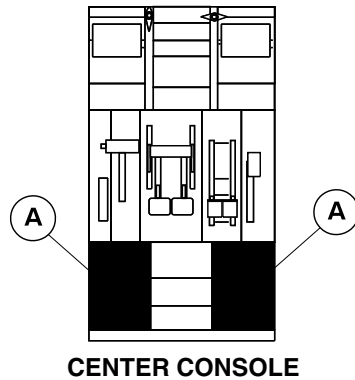
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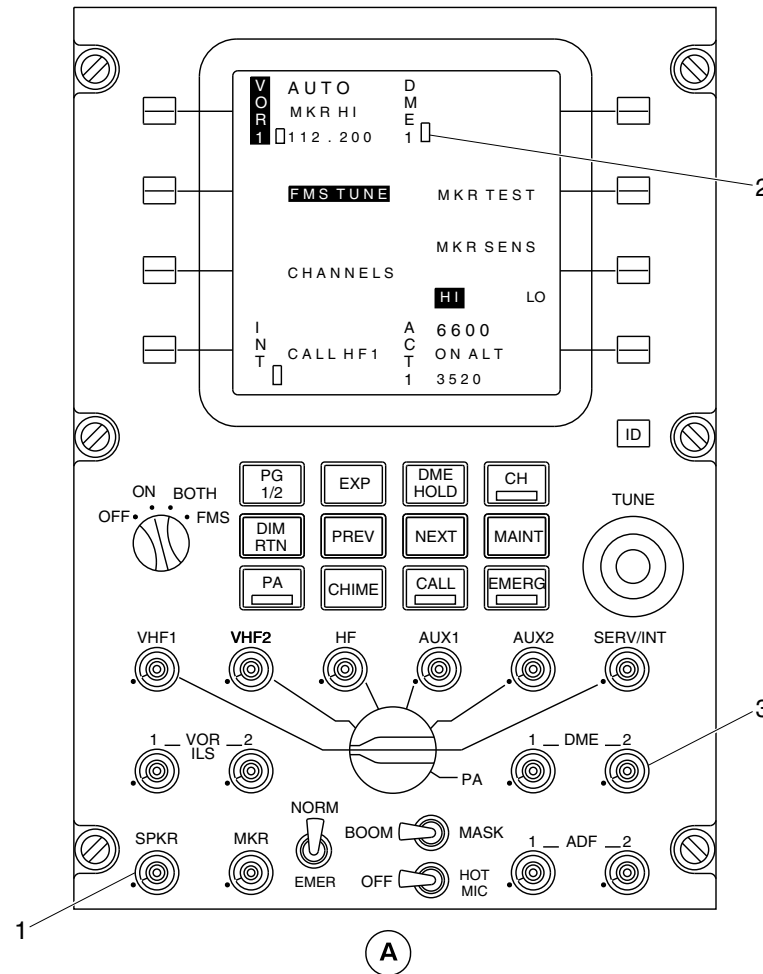
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LEGEND

1. Speaker potentiometer/Push-button switch.
2. Level adjust bar graph.
3. DME potentiometer/Push-button switches.



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ARCDU DME Audio Control
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

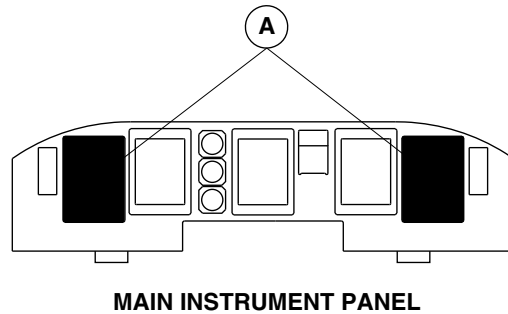
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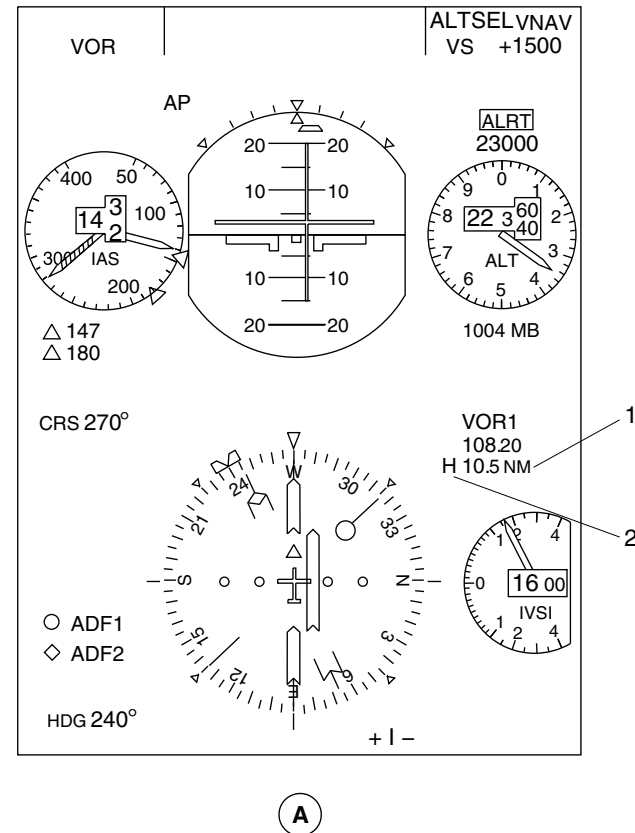
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LEGEND

- 1. DME Distance.
- 2. DME Hold Annunciation.



PFD DME NORMAL INDICATION
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-53-00
Config 001

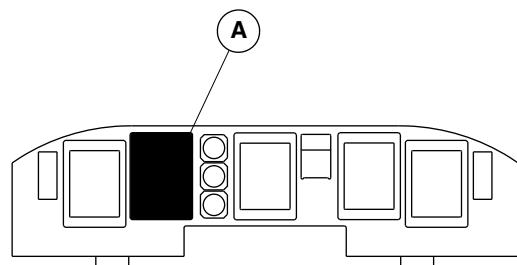
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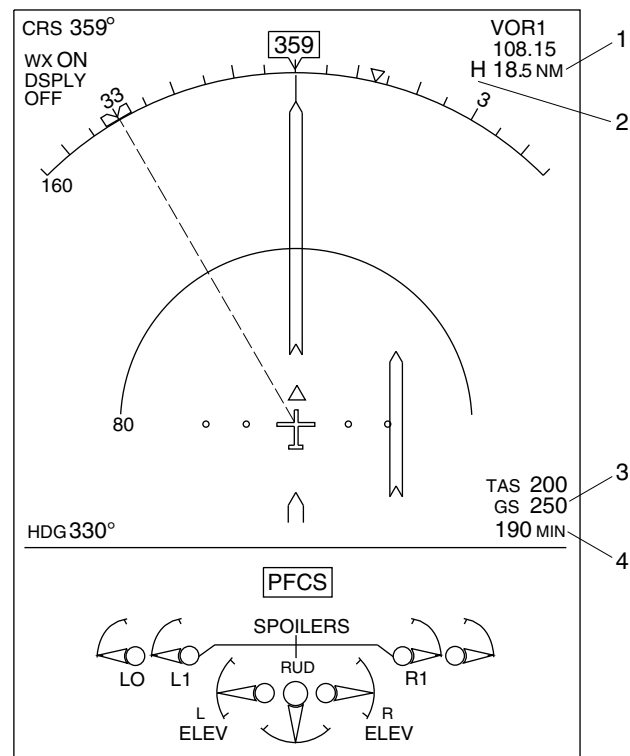
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MAIN INSTRUMENT PANEL

LEGEND

1. DME Distance.
2. DME Hold Annunciation.
3. Ground Speed Digital Value.
4. Time To Go Digital Value.



A

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MFD DME NORMAL INDICATION
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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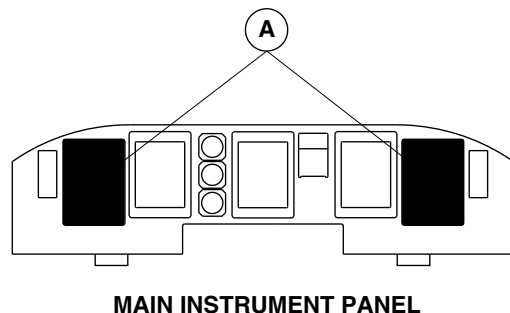
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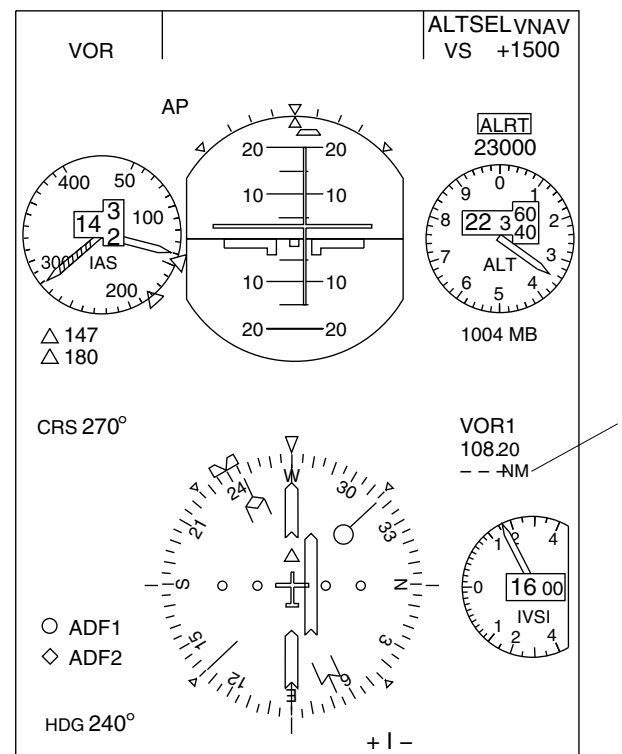
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LEGEND

1. DME Fail Annunciation.



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PFD DME FAIL INDICATION
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

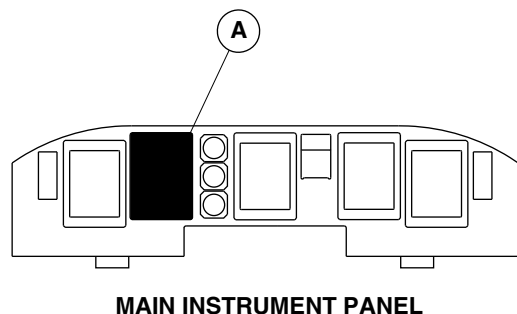
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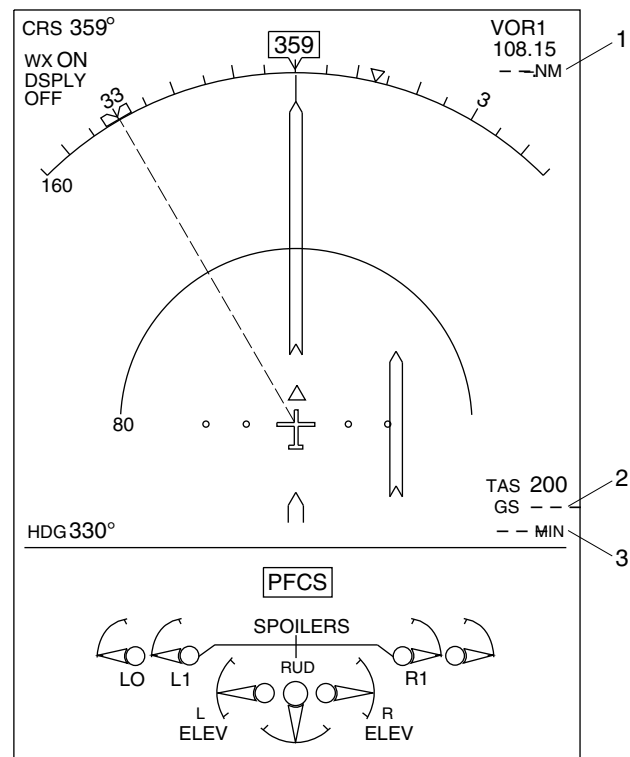
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LEGEND

1. DME Fail Annunciation.
2. Ground Speed Fail Annunciation.
3. Time To Go Fail Annunciation.



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MFD DME FAIL INDICATION
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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Config 001

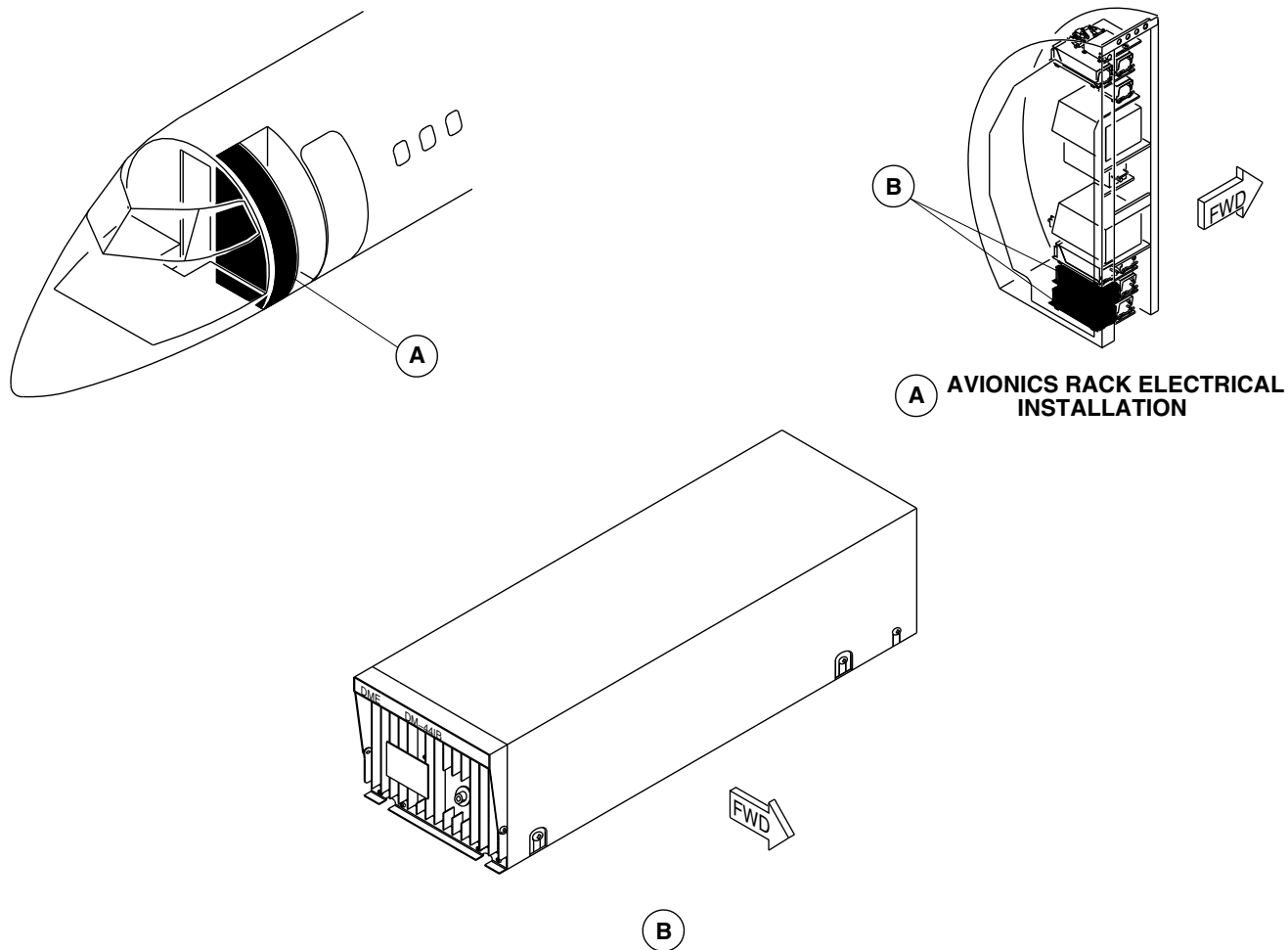
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DME TRANSCEIVER LOCATOR
Figure 8

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-53-00
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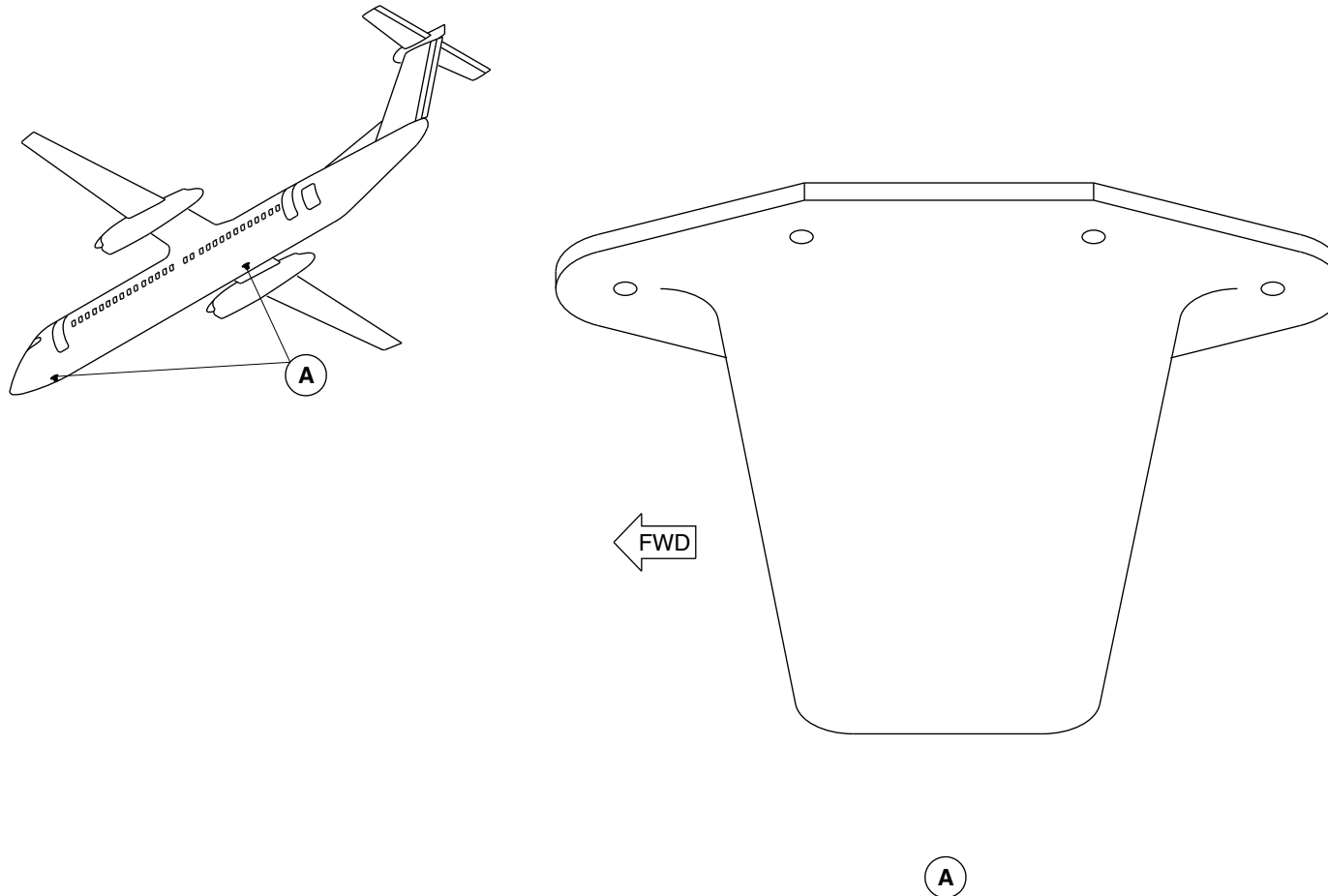
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DME ANTENNA LOCATOR
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

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AIR TRAFFIC CONTROL TRANSPONDER

Introduction

The Mode S transponder system transmits a signal to identify the aircraft and its altitude parameters to Air Traffic Control (ATC) ground stations and to other aircraft equipped with Traffic Alert and Collision Avoidance System (TCAS).

On aircraft with SB84-34-158 or SB84-34-159 or SB84-34-193 or SB84-34-212 or 4-113827 or 4-113826 incorporated, the ATC NXT-600 transponder is installed, which provides the aircraft with Mode A, C, S and Automatic Dependent Surveillance-Broadcast (ADS-B) Out transponder air-to-air and air-to-ground data transfer capabilities.

General Description

Refer to Figures 1 and 2.

The Mode S transponder replies to ATC ground station interrogations and other TCAS equipped aircraft. It transmits the information that follows:

- Altitude
- Maximum airspeed
- TCAS
- Mode S Identifier of the aircraft.

In addition, the interrogating ATC ground station or other aircraft fitted with TCAS determines aircraft's range and azimuth.

The Mode S transponder system has two transponders (ATC1, ATC2). One transponder operates in the active mode and the other will automatically operate in the standby mode.

The active Mode S transponder is selected to operate in one of the sub-modes that follow:

- Air Traffic Control Radar Beacon System (ATCRBS), Mode A and Mode C
- Mode S Surveillance
- Test.

In addition, the pilots can select the altitude reporting data source.

The Mode S transponder system has the components that follow:

- Receiver, Mode S Transponder (34-54-01)
- L-Band Antenna, Transponder (34-54-11)
- Coaxial Cable Switch, Antenna (34-54-16)
- Switch, ATC ID.

A coaxial cable switch connects each transponder to an upper and lower fitted L-band omni-directional antennae.

The Mode S transponder system interfaces with both of the Audio and Radio Control Display Units (ARCDU1, ARCDU2) and the Flight Management System (FMS). Pilot inputs to the ARCDU and the FMS control the operating modes of the transponder. In addition to mode selections, the ARCDU1 and the FMS sends a pilot selected four-digit octal code to the transponders.



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The Mode S transponder system interfaces with the Air Data System (ADS). It converts altitude information from the ADS into formats required for Mode C and Mode S replies.

On aircraft with SB84-34-158 or SB84-34-159 or SB84-34-193 or SB84-34-212 or 4-113827 or 4-113826 incorporated, the ATC NXT-600 transponder supports the ADS-B functionality, which includes Mode A and C ATCRBS, Mode Select (S), TCAS II and antenna diversity operation. The ADS-B system is enabled during all phases of the flight operation including the airport surface movement operations.

Detailed Description

The Mode S transponder functions in air traffic alert and collision avoidance environments.

The Mode S transponder receives 1030 MHz interrogations on the aircraft top and bottom antennas from ground ATC ground stations and from TCAS equipped aircraft. The transponder decodes and processes the interrogations and if required, transmits its replies at 1090 MHz in one of the modes that follow:

- ATCRBS
- Mode S Surveillance
- TCAS Environment
- Squitter Transmission (Unsolicited transmissions).

The replies have data that relates to the altitude, maximum airspeed, TCAS, and Mode S Identifier of the aircraft. The replies will be transmitted from the antenna that had the stronger interrogation signal strength to more accurately determine the aircraft position.

[Refer to Figure 3.](#)

ATCRBS Environment: The airborne Mode S transponder system is part of the ATCRBS. It transmits a coded response to a coded interrogation from an ATC ground station.

There are three types of radar at each the ATC ground stations:

- Primary Surveillance Radar (PSR)
- Secondary Surveillance Radar (SSR)
- Omni-directional.

The PSR determines the relative bearing and distance of the aircraft. It operates on the normal radar principle to process energy reflected from the aircraft under surveillance.

The SSR determines the aircraft parameters that follows:

- Identity
- Distance
- Altitude.

The SSR transmits two pulses on a 1030 MHz carrier to interrogate the airborne transponder. The airborne transponder replies with a code consisting of up to 16 pulses. The spacing between the two interrogation pulses determines the modes as follows:

- Mode A, Code
- Mode C, Altitude.

The Mode A interrogation has an 8 microsecond time interval between P1 and P3 pulses and Mode B interrogation has a 21 microsecond time interval between P1 and P3 pulses.



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The transponder must reply to ATCRBS interrogations from the transmitting antenna main beam and not from the side lobes. Each interrogation contains a third pulse at the same frequency transmitted by an omni-directional antenna. This pulse is transmitted two microseconds after the first SSR pulse.

When the aircraft receives an interrogation from the main beam, the P2 pulse has a lower amplitude than the P1 and P2 pulses, the transponder will reply.

When the aircraft receives an interrogation from a side lobe where the P2 pulse is equal or greater than the P1 and P2 pulses, the transponder will not reply.

[Refer to Figure 4.](#)

The transponder replies to Mode A interrogations with a pilot selected four-digit octal code. After the Mode A reply, an 18 second Special Purpose Identifier (SPI) can be transmitted when one of the ATC ID switches is pushed.

The transponder replies to Mode C interrogations. It transmits aircraft barometric altitude information.

The transponder uses 13 serial pulses to give the aircraft identification when replying to a Mode A interrogation or altitude data when replying to Mode C interrogation. The coded pulses are transmitted between two framing pulses located at the beginning and end of data.

The framing pulses, F1 and F2 are spaced 20.3 microseconds apart for both Mode A and Mode C replies. The transponder receive the aircraft code from the ARCDU1, ARCDU2 and altitude from the Air Data Units (ADU1, ADU2).

When no altitude data is available during Mode C interrogations, the transponder replies with only F1 and F2 framing pulses.

The SPI pulse is transmitted when the ID key is pushed on the ARCDU to give a positive aircraft identification to ATC. The SPI pulse is transmitted 4.35 microseconds after the F2 framing pulse. The SPI pulse is transmitted for 20 seconds after the ID switch is pushed.

[Refer to Figure 5.](#)

Mode S Air Traffic Control Environment: When an aircraft equipped with a transponder enters the airspace that is monitored by ground ATC facilities or TCAS equipped aircraft, it is interrogated by an ATCRBS/Mode S All-Call interrogation. The ATCBRS transponders reply with a Mode A or Mode C format and the Mode S transponders as installed on this aircraft reply with a Mode S format that includes an unique address. The address with the aircraft location is entered into a roll call file. On the next interrogation, that Mode S transponder is interrogated. The azimuth and range are known for all aircraft on roll-call. Each aircraft is interrogated according to a precomputed schedule. The schedule has ATCRBS/Mode S All-Calls that provide tracking of known ATCRBS aircraft and the acquisition of more ATCBRS and Mode S aircraft that enter the airspace.

The ATCBRS/Mode S All-Call interrogation has the same pulses as the ATCRBS format but with an additional P4 pulse. The P4 pulse width is 1.6 microseconds. There is also an ATCBRS Only All-Call interrogation that is the same as the ATCRBS/Mode S All-Call interrogation except that the P4 pulse width is 0.8 microseconds. The Mode S transponders will reply to standard ATCRBS interrogations but will not reply to the ATCBRS Only All-Call interrogations. The Mode S transponders reply to interrogations with a Mode S reply if the amplitude of the P4 pulse is more than or equal to the P1 and P3



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pulses or with an ATCRBS reply if the amplitude of the P4 pulse is less than or equal to the amplitude of the P1 and P2 pulses.

The Mode S interrogation makes use of Differential Phase–Shift Keying (DPSK) modulation to transmit a 4 Million bit per second (Mbps) data burst. The signal has P1, P2, and P6 pulses.

The P1 and P2 pulses before the P6 pulse is used to suppress replies from ATCRBS transponders to prevent interference caused by random triggering of ATCRBS transponder by the Mode S Interrogation. A series of chips that has data within the P6 pulse starts 0.5 microseconds after the sync phase reversal. A chip is an unmodulated interval that continues for 0.25 microsecond. When the chip is preceded by a phase reversal, it is a hi and when it is not preceded by a phase reversal, it is a lo. The last chip is followed by a 0.5 microsecond guard interval that prevents the trailing edge of the P6 pulse from interfering with the demodulation process. The Mode S interrogation uses one of two Uplink Formats. The short format uses 56 bits of DPSK data or long format with 112 bits of DPSK data.

The Uplink Format has the data fields that follow:

- Uplink Format Number
- Data Field
- Address.

Uplink Format Number: The first part of all uplink formats is the Format Number. It is a 5 bit coded binary data field.

Data Field: The Uplink Format (UF) has 25 Mode S interrogation formats. Presently, only seven of the UFs are used in the data field as follows:

UPLINK FORMAT NUMBER	FUNCTIONS
0	Short Air–Air Surveillance
4	Surveillance, Altitude Request
5	Surveillance, identity Request
11	Mode S–Only, All–Call
16	Long Air–Air Surveillance
20	Comm–A, Altitude Request
21	Comm–A, Identity Request

Address: The last 24 bits of each uplink format represents the unique 24 bit Mode S transponder address plus parity bit on top of the address.

[Refer to Figure 6.](#)

The Mode S reply has a 4 pulse preamble followed by the reply data pulses. The reply data pulses are 56 pulses if the reply is a short format or 112 pulses if the reply is a long format. Pulses that are 0.5 microseconds in duration are placed within 1.0 microsecond windows of the data block. If the pulse is in the first half of the window, it is hi. If the pulse is in the second half of the window, it is lo. The downlink reply has 25 formats. The short 56 bits reply format is used for surveillance, squitter, and TCAS acquisition. The long 112 bits reply gives the surveillance data as the short format but it has 56 bits of non–surveillance type messages.



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Only seven downlink formats are used. The first data field in all downlink formats is the format number. It is coded in binary form. The last 24 bits of each downlink format except All-call replies is the unique 24 bit Mode S transponder address plus parity bit on top of the address.

DOWNLINK FORMAT NUMBER	FUNCTIONS
0	Short Air–Air Surveillance
4	Surveillance, Altitude Request
5	Surveillance, identity Request
11	Mode S–Only, All–Call
16	Long Air–Air Surveillance
20	Comm–B, Altitude Request
21	Comm–B, Identity Request

TCAS Environment: The Mode S transponder is part of the TCAS. It supplies an air–to–air data link for altitude reporting information for the TCAS.

The Mode S transponder transmits a squitter signal to identify itself. It is the initial indication that a Mode S equipped aircraft has entered a surveillance area. ATC ground stations and TCAS equipped aircraft monitors the 1090 MHz frequency for squitter. After a valid identification squitter is received, the aircraft's unique Mode S

address is added to a list of aircraft the other TCAS equipped aircraft will interrogate.

The aircraft also receives squitter signals from other Mode S equipped aircraft. Its TCAS compiles a list of Mode S equipped aircraft in the area to interrogate. In addition, the TCAS processor transmits interrogations to any transponder Mode A or Mode C equipped aircraft in the area. These aircraft respond to the ATCRBS type interrogation.

The aircraft interrogates each of the aircraft from the compiled list. The TCAS receives replies through antennas. The TCAS directional antenna receives phase information related to the actual bearing of the other aircraft. The TCAS processor uses altitude reporting from the replying transponder. The TCAS processor computes the range of the other aircraft by using the round trip time difference between message interrogation and reply. It calculates the range of the replying aircraft using the speed of the returning signal and elapsed time.

The TCAS processor calculates the parameters that follow:

- Relative altitude
- Altitude rate
- Range
- Bearing.

The TCAS processor calculates which aircraft is on a possible collision course. The TCAS processor alerts the pilots with an aural and visual indication. The TCAS processor issues an alert if it determines that the flight path of the other aircraft passes within 1200 ft (366 m). It process each aircraft within the area individually. It determines a safe corrective or preventive action and coordinates it



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with other TCAS equipped aircraft through an air to air Mode S data link.

Squitter: The Mode S transponder transmits its unique address at a periodic rate, once every second known as "squitter". The transmissions alternate between the upper and lower antennae. Other aircraft equipped with TCAS receive the squitter signal and then interrogate the Mode S transponder equipped aircraft based on the address received from the squitter. These interrogations are used for Mode S target tracking and collision threat assessment.

The squitter transmissions have the data that follow:

- A unique Mode S address
- Altitude
- Equipment capabilities of the sending aircraft

The ARCDU1, ARCDU2 or the FMS control the Mode S transponders (ATC1, ATC2).

If the ARCDU1 fails and switched to OFF, then the ARCDU2 or the FMS maintains and gives full transponder control from pilot commands.

The transponder system defaults to ATC 1 the conditions when both ARCDU malfunction, and both ARCDUs transfer switches are set to the OFF position.

The transponder uses default ARCDU control logic and pilot commands through the FMS. The opposite Mode S transponder, ATC2 operates in standby.

The Mode S transponders (ATC1, ATC2) functions in the modes that follow:

- Active (ATC1 or ATC2)

- Standby (SBY)

- Initiated Test (TEST).

Active: The selected Mode S transponder receives interrogations through the upper and lower antennae from ATC ground stations and from TCAS equipped aircraft. The transponder decodes and processes the interrogations and if required, replies in Mode A, C, or S format. The antenna that receives the stronger interrogation signal strength is used to transmit replies. The transponder operates in the L radio Frequency band. It receives interrogations at 1030 MHz and transmits replies at 1090 MHz.

Standby Mode: The Mode S transponder functions in the standby mode when it is not selected. The Mode S transponder stays in a powered state in the standby mode. Its Built In Test (BIT) does a Radio Frequency (RF) loop-around checkout in addition to the normal continuous test routine. The Mode S transponder will not respond to any ATCRBS or Mode S interrogations when in the standby operation mode.

Initiated Test Mode: The selected Mode S transponder will not respond to any ATCRBS or Mode S interrogations when in the test operation mode.

The Mode S transponder continuously carries out system fault isolation. It finds the source of a failure at the component or Aircraft wiring level.

The transponder sends a failure message to the ARCDUs through a data bus. A Light Emitting Diode (LED) code inside the transponder indicates a control data failure. When a failure that does not affect the ATC surveillance function is detected, the transponder will not send a failure message to ARCDU.



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Each Mode S transponder has an initiated and continuous Built In Test Equipment (BITE) capability. It includes an RF loop-around test that is only executed when a transponder is in the standby operation mode. Each transponder reports its overall status and its interfaces with other systems to each ARCDU through a discrete health status interface. Each ARCDU shows the status of each Mode S transponder.

The Central Diagnosis System (CDS) continuously monitors the discrete health status output from each Mode S transponder.

On aircraft with SB84-34-158 or SB84-34-159 or SB84-34-193 or SB84-34-212 or 4-113827 or 4-113826 incorporated, the self-test monitor is executed on ground only. The self-test monitor function is active when the test switch on the front panel of the NXT-600 transponder is pushed or after the power-on cold start. When the NXT-600 transponder is powered on, all the LEDs on the front panel come on in red color. After the completion of power-on test, all the LEDs go off except of the XPDR LED that will turn from red to green and then goes off. If there is a failure detected during the power-on test, the applicable LED shown in the table that follows will turn to red:

LED	DESCRIPTION
XPDR P/F	Will turn to red, if an internal failure occurs. Will turn to green, if the test passes and the LED goes off.
ADS-B	ADS-B fail
TOP ANT	Top antenna fault

LED	DESCRIPTION
BOT ANT	Bottom antenna fault
CTRL PNL	ARCDU control signal fail
ALT	ADU signal fail

On aircraft without SB84-34-158 or SB84-34-159 or SB84-34-193 or SB84-34-212 or 4-113827 or 4-113826 incorporated, a seven segment LED installed to the Input/Output processor board of each transponder provides system fault code diagnostic information.

On aircraft with SB84-34-158 or SB84-34-159 or SB84-34-193 or SB84-34-212 or 4-113827 or 4-113826 incorporated, a seven segment LED installed to the front panel of each NXT-600 transponder provides the self-test monitor fault code diagnostic information.

These hexadecimal fault codes represent a combination of transponder subassemblies and external interface fault conditions. The single seven-segment display indicates either a two-digit fault code for the most recently detected fault or 00H for no faults. The code can be seen on the input/output processor board through cooling holes on the side panel of the transponder. The seven-segment display is continuously active. The table that follows shows the detailed definition of BITE fault codes.



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FAULT CODE	FAILED ASSEMBLY	FAILED SYSTEM
00H	None	No Faults
10H	Power Supply	+5 Vdc
17H	IOP/DPL Board	Power Fail Stuck Hi
18H	None	Illegal Mode S Address
16H	IOP/DPL Board	Internal Suppression Stuck Hi
20H	IOP/DPL Board	RAM Read/Write Failure
21H	IOP/DPL Board	ROM Read/Write Failure
23H	IOP/DPL Board	RLSI Failure
24H	IOP/DPL Board	EEPROM Read/Write Failure
27H	IOP/DPL Board	Squitter Tx Data Failure
28H	IOP/DPL Board	Squitter Rate Out of Spec
29H	IOP/DPL Board	TCAS 429XT Loopback Failure
31H	TX Assembly	Transmitter Failure
D0H	IOP/DPL Board	Comm A/B Bus Failure

FAULT CODE	FAILED ASSEMBLY	FAILED SYSTEM
D1H	IOP/DPL Board	Comm C Bus Failure
E3H	Control Panel	No Control Data
F1H	TCAS	TCAS Interface Failure
F4H	Altitude 706	706 Air Data Failure

If the Mode S transponder determines a failure that causes the system not to function for ATC purposes, the ARCDUs show "FAIL" in the code area of the display.

[Refer to Figure 7.](#)

The ARCDUs selections control the ATC transponder mode and codes to be used.

SELECTION	DESCRIPTION
OFF	The ARCDU is not powered. The related FMS controls and tunes as a backup
ON	The ARCDU controls and tunes its related radio system
BOTH	The ARCDU controls and tunes its related and opposite radio systems
FMS	The FMS controls and tunes its related and opposite radio systems



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The ATC display area is shown on both main pages and all particular pages. A double horizontal line separates the ATC display area from the particular pages. The main page shows the information that follows in the ATC display area.

- Active ATC label
- Active Code
- Mode, On or Standby
- Altitude, On or Off.
- Specific Purpose Identifier (ID).

[Refer to Figure 8.](#)

A minimum of two steps are required to change the active code.

The side key adjacent to the ATC display area is pushed to highlight the preset code. It is pushed again to make the preset code active. The code that was active now becomes the new preset code.

When the side key adjacent to the ATC display area is pushed, the preset code in the ATC transponder display area highlights. The TUNE double rotary knobs located at the lower right side of the ARCDU are turned to change the preset code. The Side key is pressed again to set the new active code. The code that was active now becomes the new preset code.

When a side key is selected, the flight crew must follow up that selection with one of the following actions within five seconds:

- Rotate either TUNE double rotary switch
- Push the same side key again

If not, the previous preset data to restores.

[Refer to Figure 9.](#)

One ARCDU or handwheel located ATC IDENT push-button switch is pushed to cause the transponder to transmit an identifier for 17 seconds. The ATC display area shows the Specific Purpose Identifier (SPI) as an ID legend in green fonts for the duration of the identifier.

[Refer to Figure 10.](#)

The ATC Particular Page comes into view when the ATC side key is pushed followed by the EXP key on the keyboard.

[Refer to Figure 11.](#)

An ATC label is shown in black letters on a white background in the center of the first line. The ATC particular page is used to change the following Mode S transponder (ATC 1, ATC 2) modes and functions:

- Standby, Active
- Test
- Altitude source selection.

Standby and Active Modes: The side key adjacent to the ATC1, SBY, ATC2 display area is pushed to change the highlighted selection. It moves in a wrap round manner from ATC1, SBY, to ATC2. The selection changes to black letters on a green background from the usually white on black background. The first main page shows a SBY or ON legend in green letters in the ATC display area.

If ATC1 is selected the ATC display area will show ATC1 and ATC2 when ATC2 is selected.

Test Mode: The side key adjacent to the TEST legend is pushed to test the transponder that is operating in the standby mode. It is possible to test both transponders at the same time by setting both transponders to the standby operating mode. The TEST legend



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changes to black letters on a green background from the usually white on black background. The ATC1, SBY, ATC2 display area shows the test results adjacent to the ATC1 and ATC2 legends. The display area shows OK in green fonts for five seconds if the test result is successfully completed. It shows FAIL in red fonts when the test result shows a malfunction. The test feature is not available when the aircraft is in the air. An attempt to test the transponders while the aircraft is in the air will causes the ATC label to flash for five seconds at a one hertz rate.

The side key adjacent to the ALT1, OFF, ALT2 display area is pushed to select the encoding altitude source. The selection highlights and moves in a wrap round manner from ALT1, OFF, to ALT2. It changes to black letters on a green background from the usually white on black background. The main pages show an ALT legend in the ATC display area. It does not come into view if both transponders are operating in the standby mode. If the selected ADU malfunctions, a FAIL label comes into view next the ALT1 or ALT2 legend.

The aircraft electrical system left main and right main buses supply 28 Vdc power through 5 ampere circuit breakers to the ATC Mode S transponders (ATC1, ATC2) and to the two coaxial cable switches. The circuit breakers are located in positions M3 and M4 on the 28 Vdc Avionics circuit breaker panel. Each Mode S transponder consumes less than 45 W and the coaxial cable switches use less than 7.5 W.

The onboard Distance Measuring Equipment (DME) and the TCAS operate in the same frequency range. Each sends a suppression signal to the other systems before transmitting. The transponder receives the suppression signal and stops transponder operations while the DME is transmitting. Prior to transmitting, the transponder

sends a suppression signal to momentarily inhibit operation of the DME.

ADS-B Out Function

The ATC NXT-600 transponder also provides the ADS-B Out function using the Mode S DF17 extended squitter. This is a function for airborne and surface aircraft, which transmits the horizontal and vertical position, velocity and other pertinent surveillance data. The NXT-600 transponder will automatically transmit DF17 data based on the onboard navigation sensors with or without an interrogation from a ground station or an aircraft.

The NXT-600 transponder has two dedicated ARINC 429 data inputs for receiving the necessary GPS parameters to support the ADS-B out functionality. The ADS-B functionality supports a better use of airspace, better surface surveillance and enhanced safety.

With the ADS-B function, the aircraft broadcasts its identification, position, altitude, velocity and other data (e.g. accuracy of its position) automatically without the need of interrogations from another aircraft or ground station. The ground portion is comprised of the ADS-B ground stations, which receive these broadcasts and direct them to the ATC automation systems for the presentation on a controller's display.

The NXT-600 transponder replies to the ATC SSR ground based interrogations with mode 3/A (aircraft identification or 4096ID code) and mode C (barometric altitude reporting).

The ADS-B out system provides the data parameters that follow:

- Aircraft Identification
- Mode A Code

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 34-54-00

Config 001

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- ICAO 24-bit Aircraft Address
- Airborne Horizontal Position – Latitude and Longitude
- Airborne Horizontal Position Quality: NIC
- Horizontal Position Quality: NACp
- Horizontal Position Quality: SIL
- Horizontal Position Quality: SDA
- Pressure Altitude (including NICbaro)
- SPI
- Emergency Status
- Emergency Indication
- 1090 ES Version Number
- Airborne Velocity Over Ground – (East/West and North/South)
- Horizontal Velocity Quality: NACv
- Emitter Category
- Vertical Rate
- Surface Horizontal Position – Latitude and Longitude
- Surface Horizontal Position Quality: NIC
- Surface Ground Track
- Movement (Surface Ground Speed)
- Length/Width of Aircraft
- GPS Antenna Offset

- Geometric Altitude
- Geometric Altitude Quality: GVA (also includes Selected Altitude, Barometric Pressure Setting, ACAS Operational, ACAS Resolution Advisory, Selected Heading and the ADS-B IN Capability).

Receiver, Mode S Transponder

[Refer to Figures 12 and 13.](#)

The Mode S transponder decodes the received phase shift modulated signal and changes it into a TTL level signal. The TTL level signal shows the 56 or 112 bit word from the TCAS or ATC ground station transmitter. The transponder decodes the last 24 bits of the signal and the remaining received message is decoded software. The last 24 bits of the signal represents the strapped address of the transmitting aircraft.

The Mode S transponder is a Line Replaceable Unit (LRU). It has main modules to receive and to calculate data.

- Power supply board
- Filter board
- Antenna switchboard
- RF/IF board
- I/O processor board
- Modulator board
- Transmit assembly
- Pin diode driver board
- Test modulator board



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- Test oscillator board.

On aircraft without SB84-34-158 or SB84-34-159 or SB84-34-193 or SB84-34-212 or 4-113827 or 4-113826 incorporated, the Mode S transponder weighs less than 8.7 lb (3.95 kg) and is packaged in ARINC 1/4 ATR short form. Each transponder is installed in an ARINC 404A compatible mounting tray. The mounting tray weighs less than 1.2 lb (0.54 kg).

The transponder mounting trays are installed below the floor in the forward portion of the center fuselage. The ATC 1 and ATC 2 mode S transponder mounting trays are located between station X+49.50 and X+71.20, between Y+0.0 and Y+9.25 and Z+89.6.

On aircraft with SB84-34-158 or SB84-34-159 or SB84-34-193 or SB84-34-212 or 4-113827 or 4-113826 incorporated, the NXT-600 transponder weighs 5.2 lb (2.36 kg) and is packaged in ARINC 1/2 ATR short form. Each transponder is installed in an ARINC 404A compatible mounting tray. The NXT-600 transponder mounting trays are installed below the floor in the forward portion of the centre-fuselage. The ATC 1 NXT-600 transponder mounting tray is located between station X+49.50 and X+71.20 and between Y+0.0 and Y+9.25. The ATC 2 NXT-600 transponder mounting tray is located between station X+71.20 and X+94.80 and Y+0.0 and Y+9.25.

Refer to Figure 14.

The unique Mode S strapping is part of the data in the Mode S squitter and reply messages. Specific rear mating pin connections to ground potential determines the system's wire strapping. The transponder uses software to read and identify the configuration of

the wire strap. Unique Mode S strapping determines the following aircraft parameters:

- Mode S address
- Maximum airspeed
- Antenna configuration
- Altitude source type.

The Mode S strap gives the aircraft a 24 bit unique Mode S address.

Some Mode S replies from the transponder contain information related to the designated maximum cruising speed of the aircraft. The airspeed program strap sets the maximum airspeed between 150 and 300 knots (278 and 556 km/h).

The ATC transponder system provides a weight-on-wheels signal input (ground to open) to the transponder on connection J1A, pin 79 described as the Air/Ground ATCRBS Inhibit discrete input (not). The discrete is in the active state when a logic 0 (aircraft electrical ground) is sensed. Ground is applied when the aircraft is on ground. As per Mode S requirements, the transponder will only transmit on the ground when interrogated by its unique 24 bit address and if the airspeed is less than 100 kt (185.2 km/h).

For the minimum operation of the ATCRBS/Mode S system, the airspeed of 100 kt (185.2 km/h) will allow the Mode S transponder to report an airborne state and takes precedence over pin 79 status (the Air/Ground Inhibit discrete input status of the transponder). If the discrete input at pin 79 remains stuck in the on-ground condition and airspeed is more than 100 kt (185.2 km/h), this will cause the transponder to transmit the airborne status.



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L-Band Antenna, Transponder

Refer to Figure 15.

The L-band antenna transmits and receives radio frequencies in the 950 to 1250 MHz frequency bands. The L-band antenna has the features that follow:

- Omni-directional
- Quarter wavelength
- Vertically polarized
- Monopole
- End fed.

The L-band blade antennae are compatible with the Mode S transponder.

The L-band antenna encases a preformed radiating element in thermoplastic within a single assembly. An Radio Frequency (RF) connector is attached to an aluminum base.

An aluminum foil gasket with an elastomeric sealant makes sure consistent distribution of conducting contacts of the gasket to the airframe. The pressure applied during installation gives a contouring feature. It lets the gasket conform to the two mating surfaces.

The Mode S transponder L-band antenna weighs less than 0.25 pounds (0.11 kilograms). The upper antenna is attached to the fuselage at station X+146, Y-2.0, and Z+182.7. The lower antenna is attached to the fuselage at X+11.9, Y+2.6, and Z+80.3. Each antenna is installed to the aircraft with five mounting screws.

Coaxial Cable Switch, Antenna

Refer to Figure 16.

Two antenna coaxial switches connect the shared upper and lower L-band antennae to the active Mode S transponder. The antenna switch defaults to connect upper and lower L-band antennas to the ATC1 transponder.

The antenna coaxial switches are attached to the fuselage at station X+71.20, Y+9.35. Each antenna coaxial switch is installed to the aircraft using mounting screws. It electrically bonds to the aircraft structure.

Switch, ATC ID

Refer to Figure 17.

Two ATC identifier switches gives a remote selection of the SPI. The switch selection supplies a discrete input to its related ARCDU.

The ATC identifier switch is a device that opens or closes the electrical circuit. It is a usually open and pushed momentarily for close. Electrical wires are soldered to the switches.

The ATC identifier switches attach to the housing covers. In turn, the housing covers are attached to the pilot and copilot hand-wheel assemblies with set screws.

Training Information Points

Do not operate the Mode S transponder when the ARCDUs show the codes that follow:

- 0000, reserved
- 7500, hijack



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- 7600, radio failure
- 7700, emergency.

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-54-00
Config 001

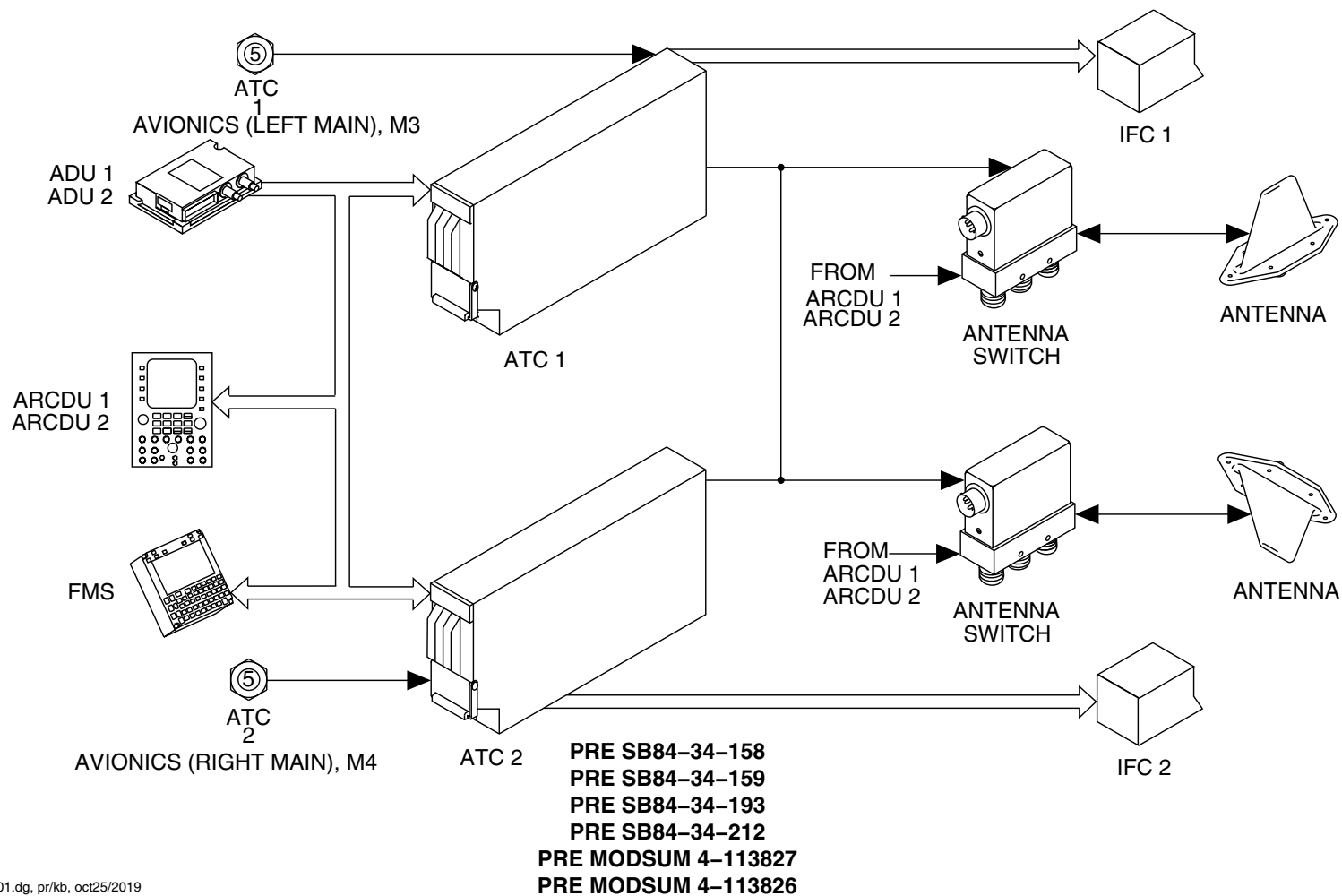
34-54-00

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fs079a01.dg, pr/kb, oct25/2019

ATC Block Diagram
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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Config 001

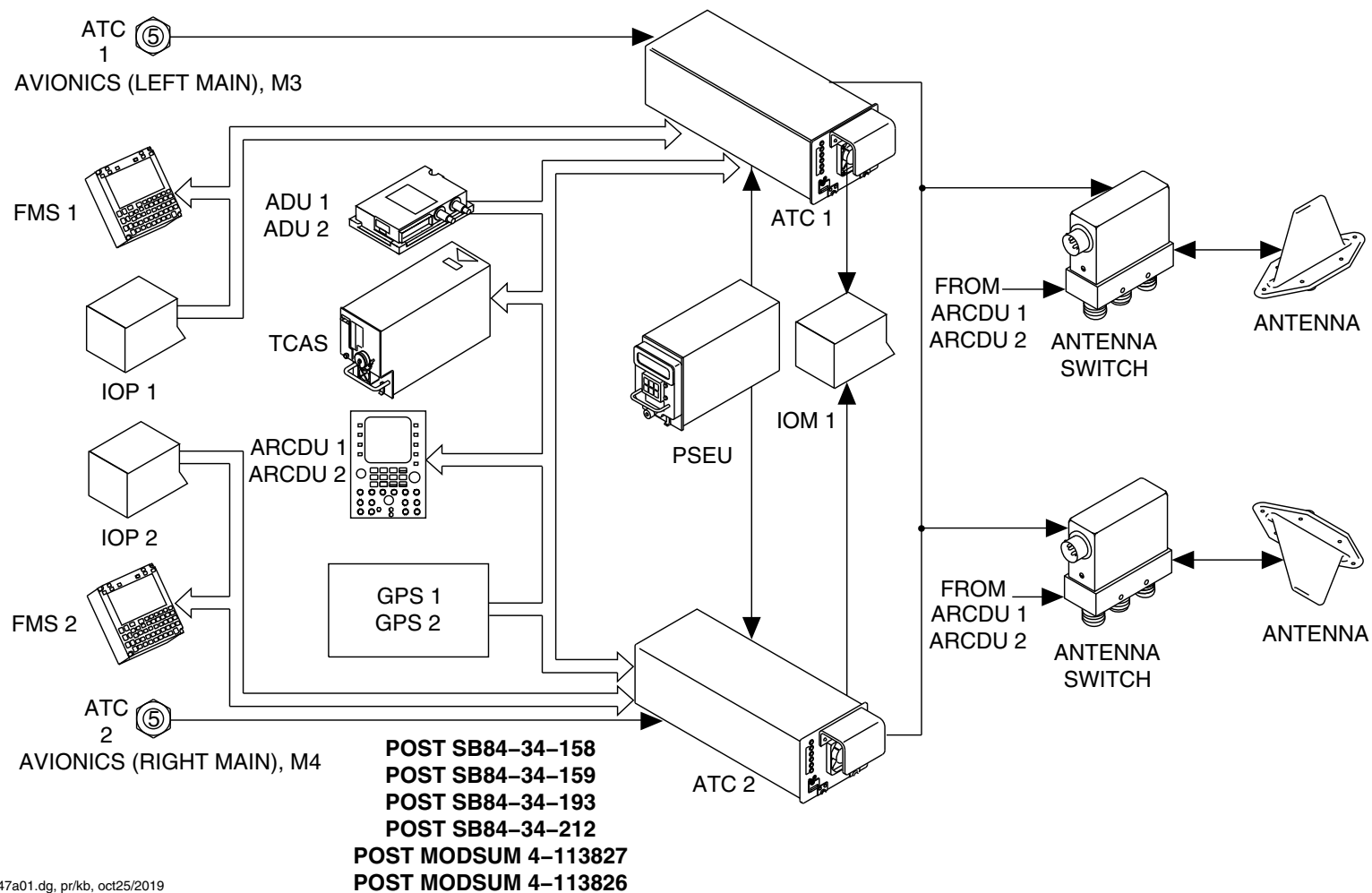
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NXT-600 ATC Block Diagram
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

34-54-00

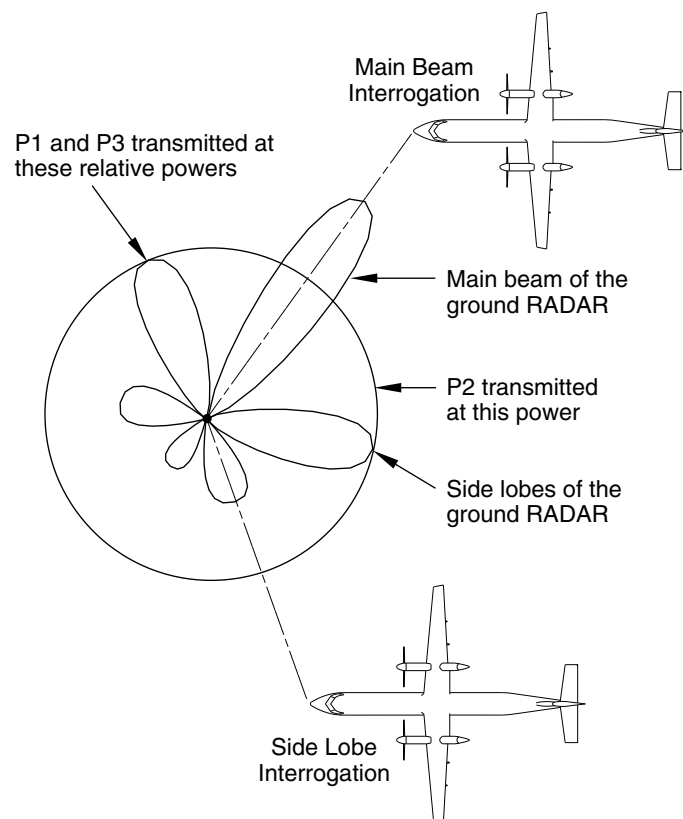
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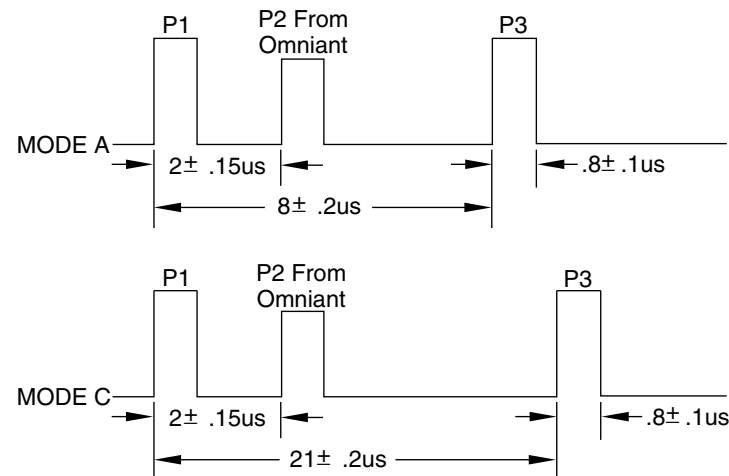
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

COMPOSITE ANTENNA PATTERN OF THE GROUND STATION

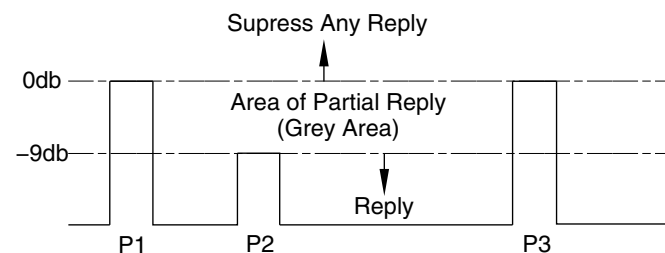


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INTERROGATION CODING CHARACTERISTICS



CONDITION OF REPLY AND SIDE LOBE SUPPRESSION



ATCRBS INTERROGATION
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

34-54-00

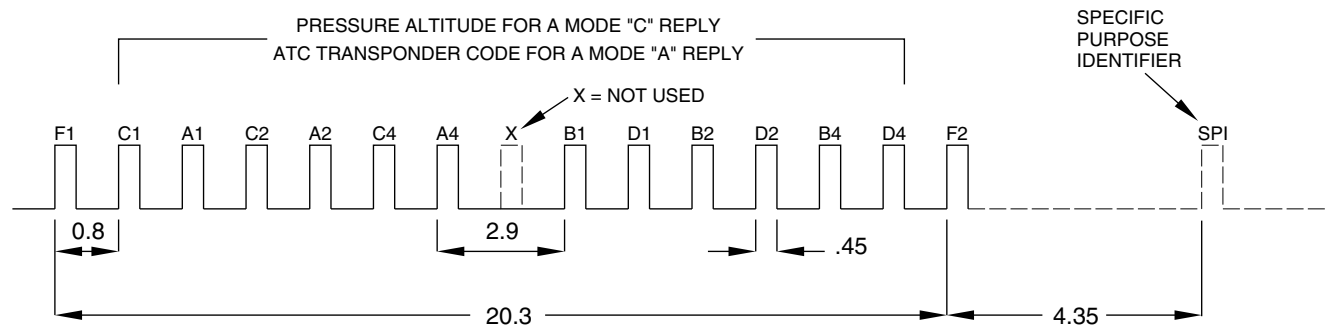
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MODE C AND MODE A REPLY MESSAGE FORMATS



NOTE: TIMES NOTED IN μ SEC.

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ATCRBS REPLIES
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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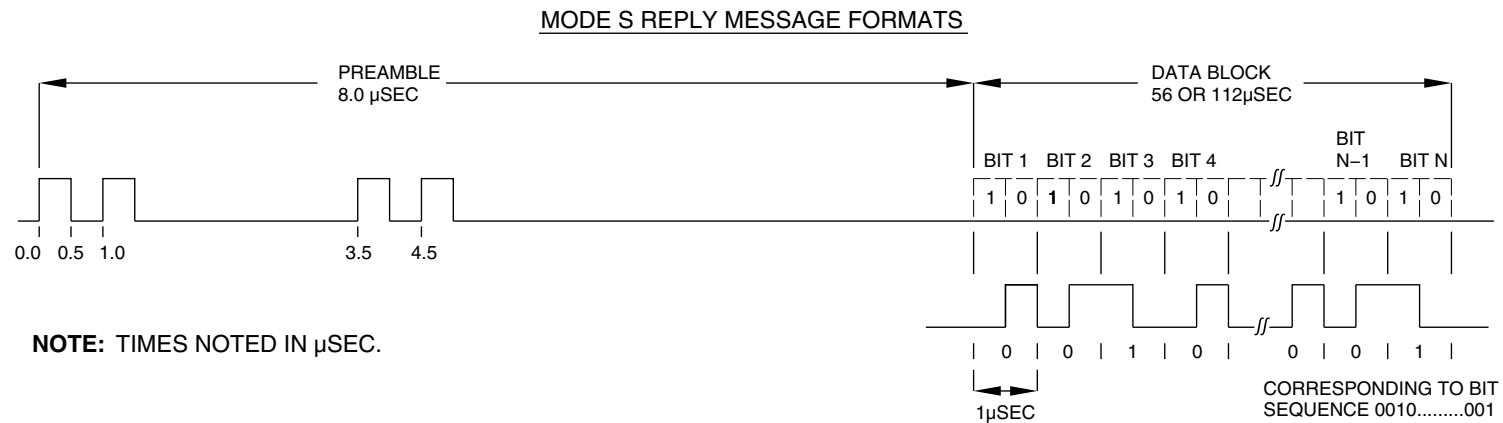


MODE S INTERROGATION, UPLINK FORMAT



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MODE S REPLIES
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

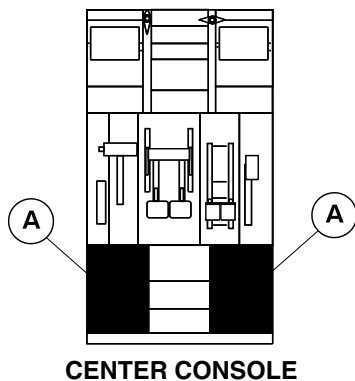
34-54-00

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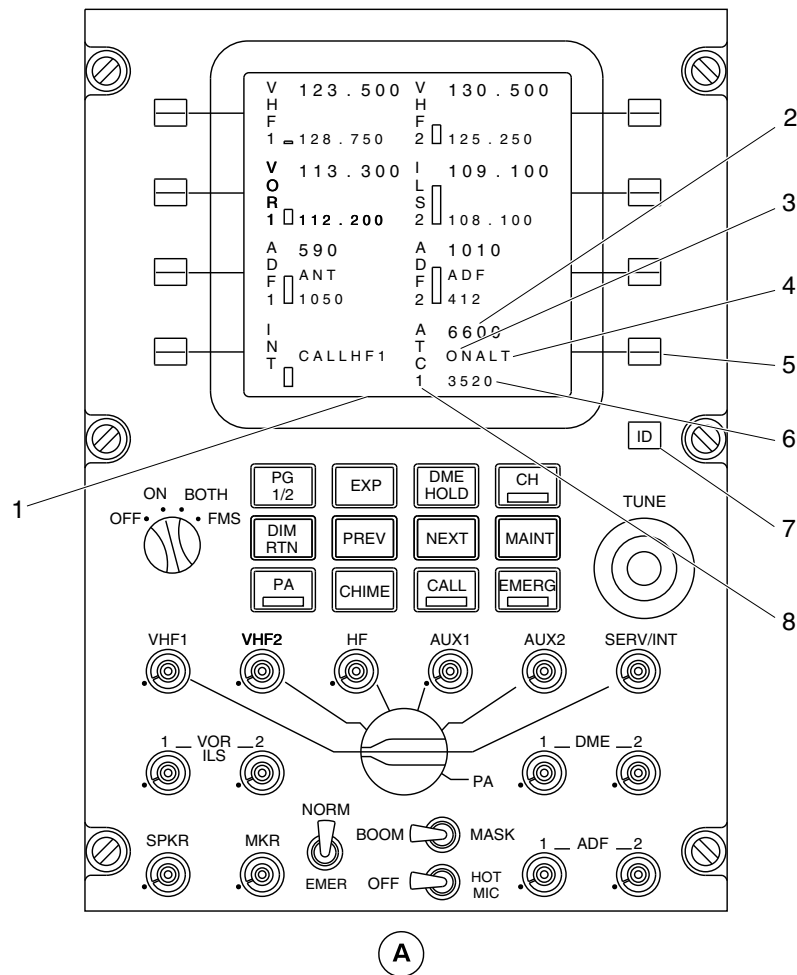
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Display area.
2. Active code.
3. Transponder mode.
4. Altitude reporting.
5. Side key.
6. Preset code.
7. Specific purpose identifier (SPI) key.
8. Label.



cg5742a01.dg, sk, jul08/2019

Air Traffic Control Transponder, ARCDU
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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Config 001

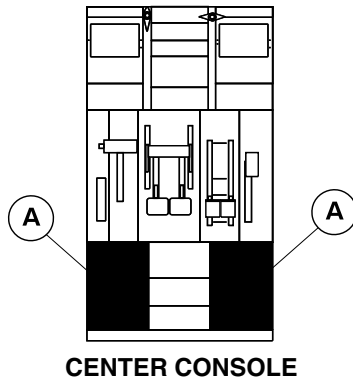
34-54-00

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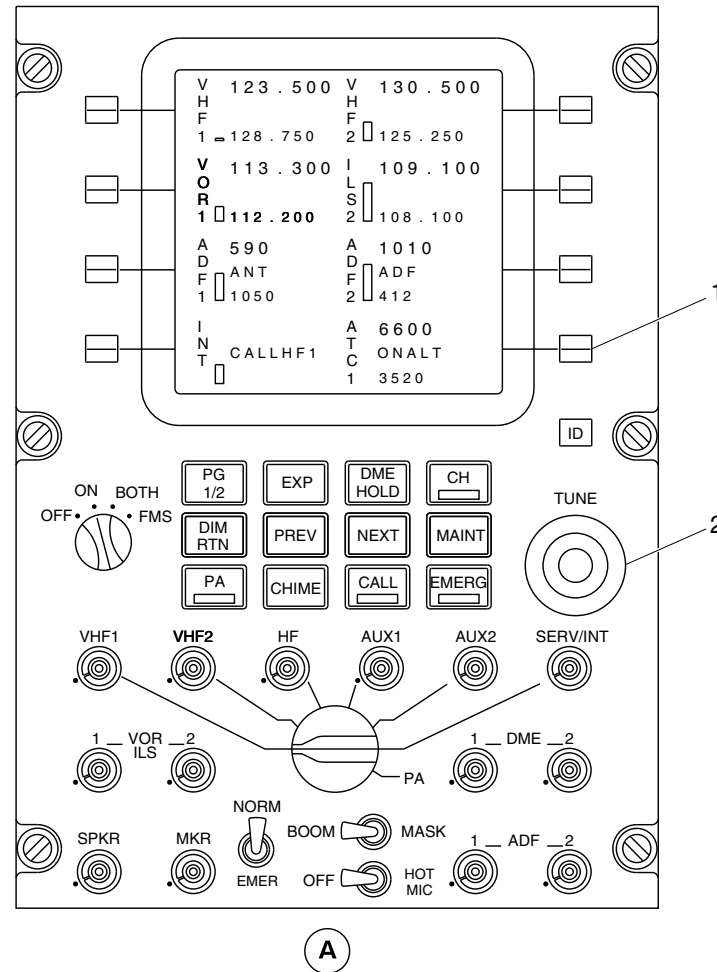
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Side Key.
2. Tuning Knob.



cg5733a01.dg, sk, jul08/2019

ARCDU ATC Mode a Control
Figure 8

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-54-00
Config 001

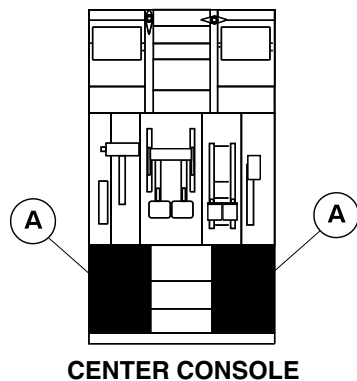
34-54-00

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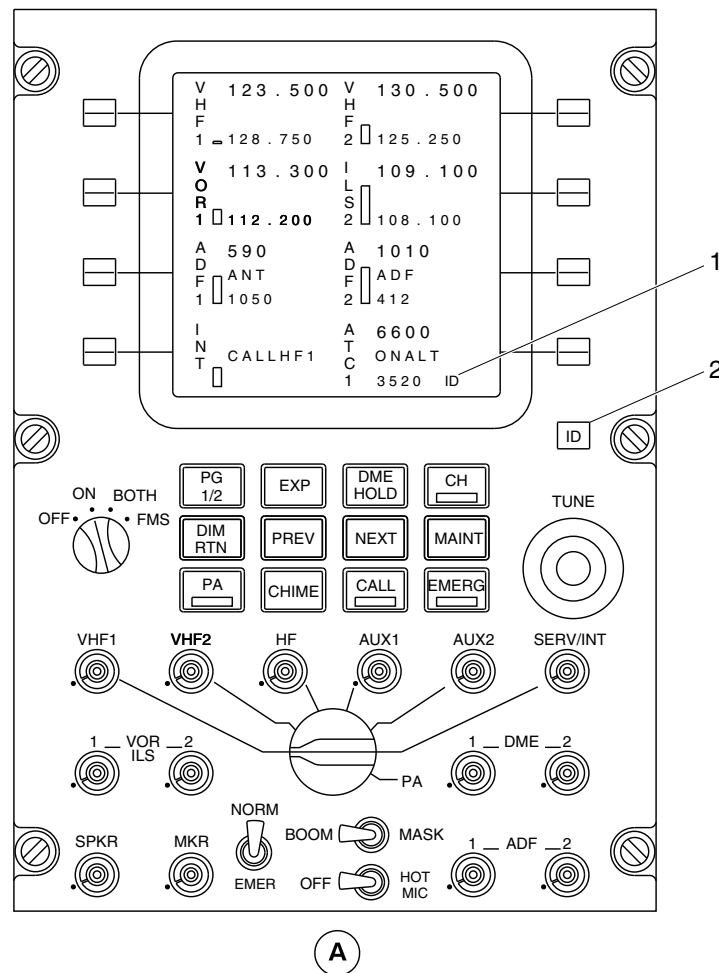
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LEGEND

1. Specific purpose identifier.
2. Specific purpose identifier (SPI) key.



cg5734a01.dg, sk, jul08/2019

ARCDU ATC Identifier Control
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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Config 001

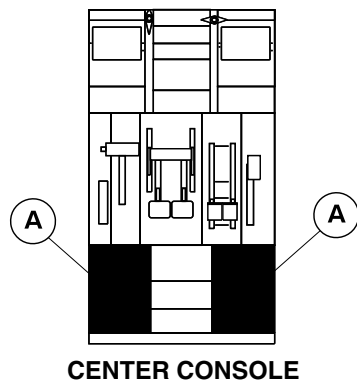
34-54-00

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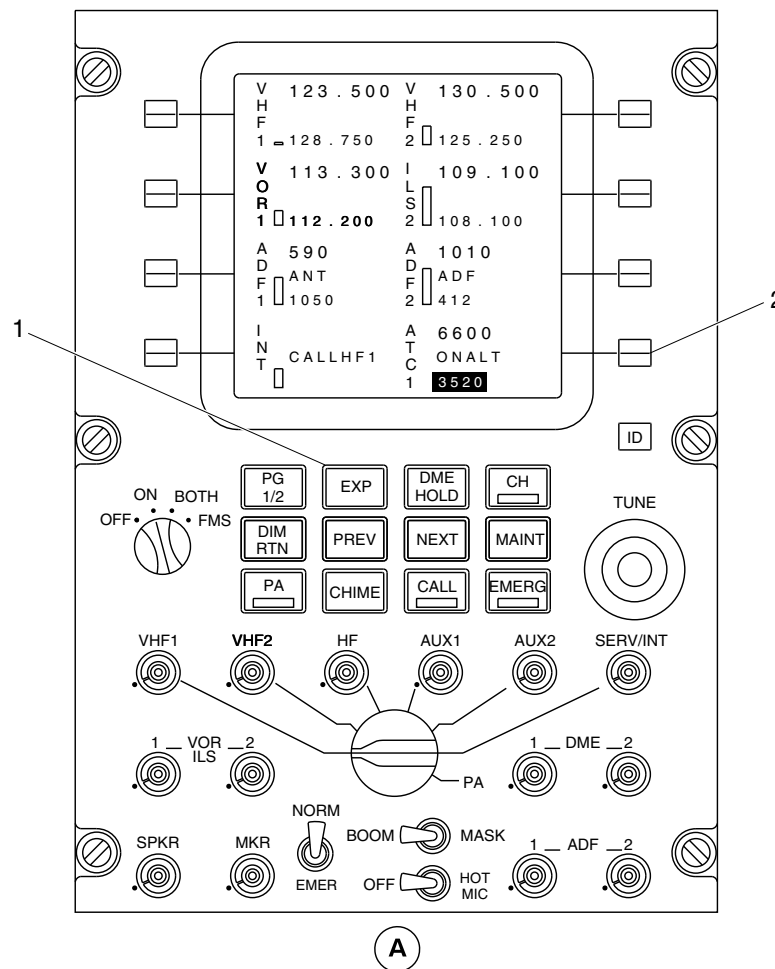
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LEGEND

- 1. Expand key.
- 2. Side key.



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ARCDU ATC Particular Page Access
Figure 10

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-54-00
Config 001

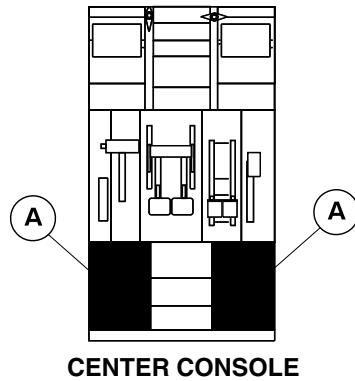
34-54-00

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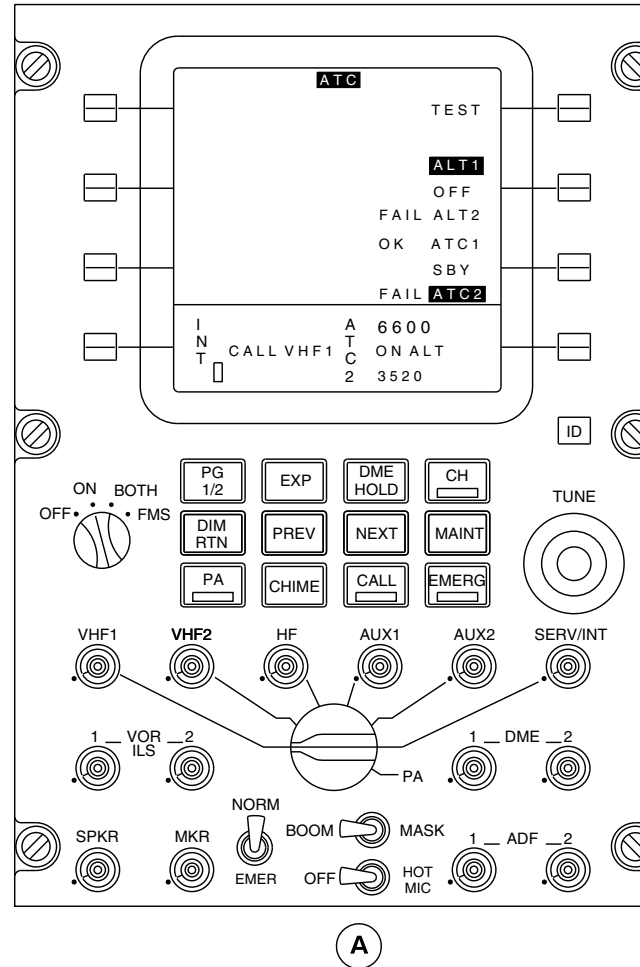


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CENTER CONSOLE



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ARCDU ATC Particular Page
Figure 11

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-54-00
Config 001

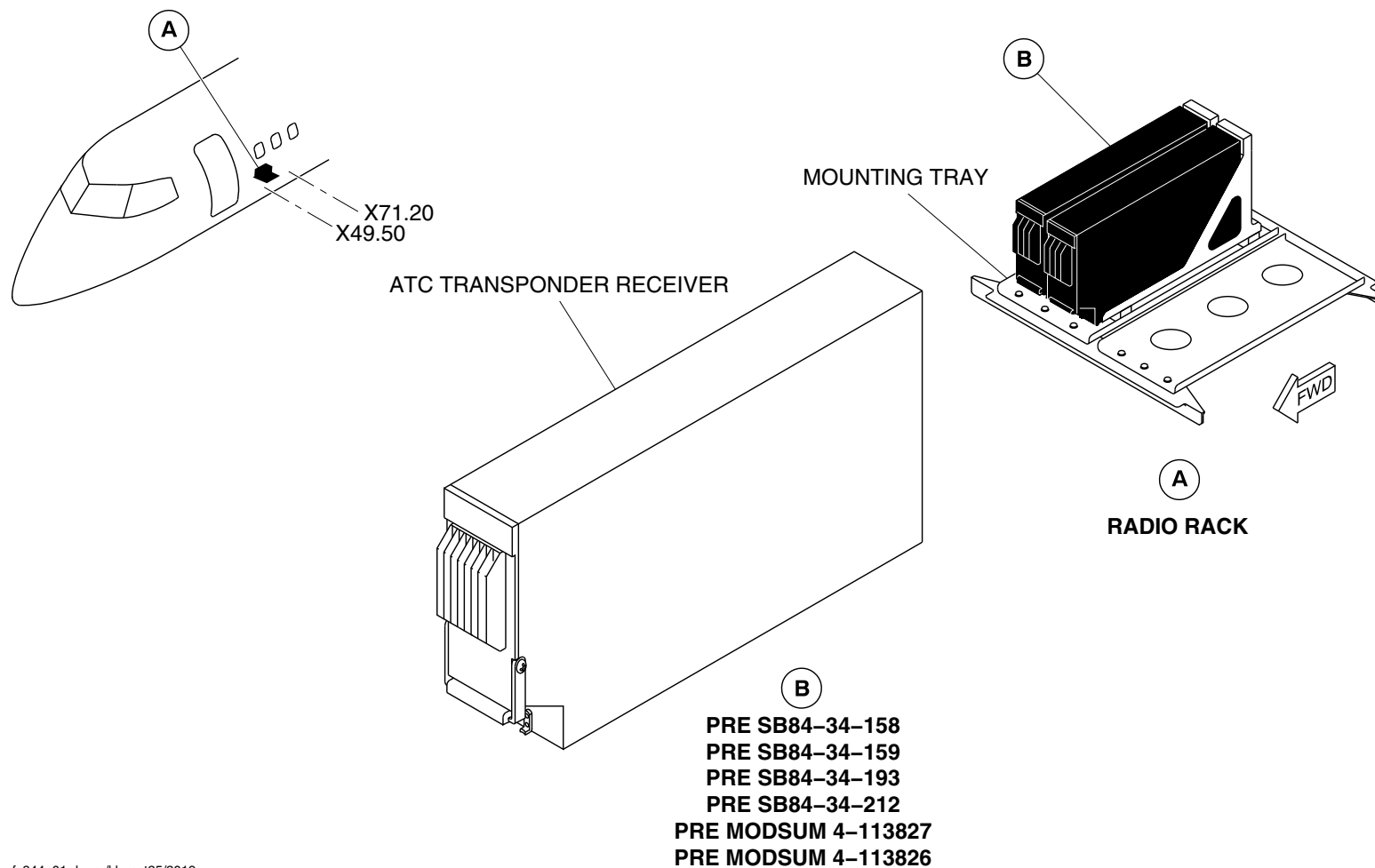
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Mode S Transponder Receiver Locator
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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Config 001

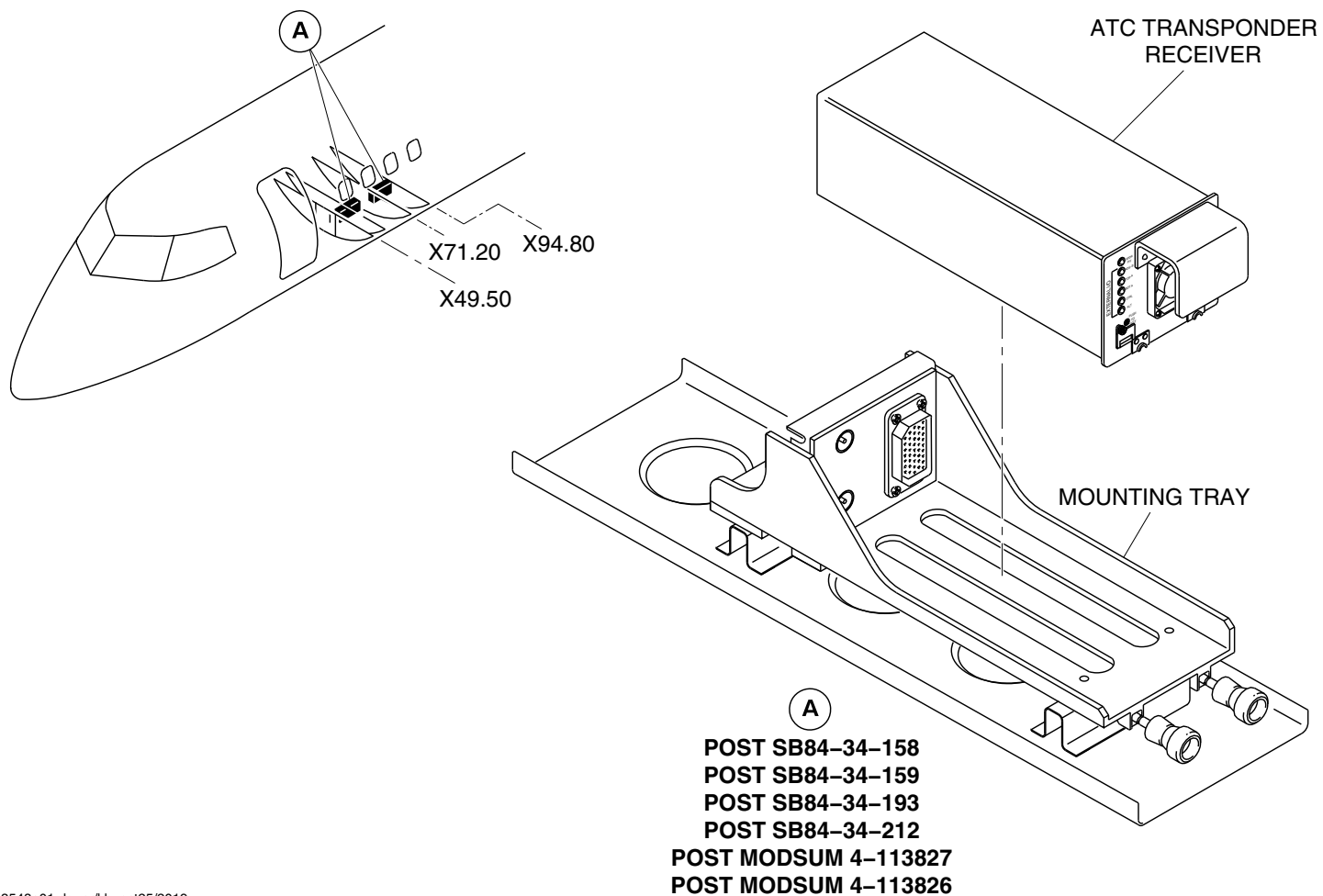
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cg3548a01.dg, pr/kb, oct25/2019

NXT-600 ATC Transponder Receiver
Figure 13

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-54-00
Config 001

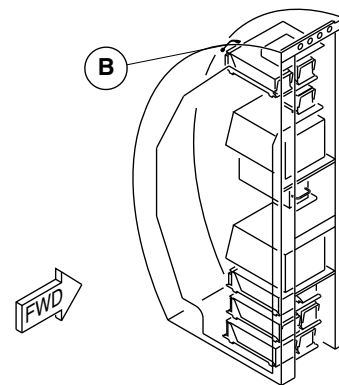
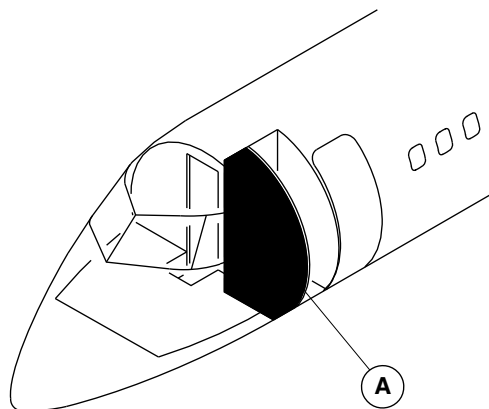
34-54-00

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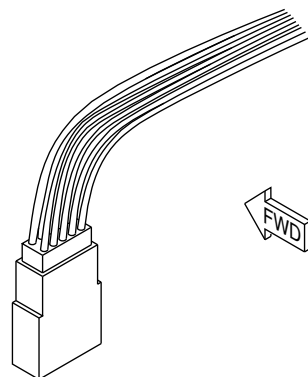


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A AVIONICS RACK ELECTRICAL
INSTALLATION



B

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MODE S TRANSPONDER ADDRESS CONNECTOR LOCATOR
Figure 14

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-54-00
Config 001

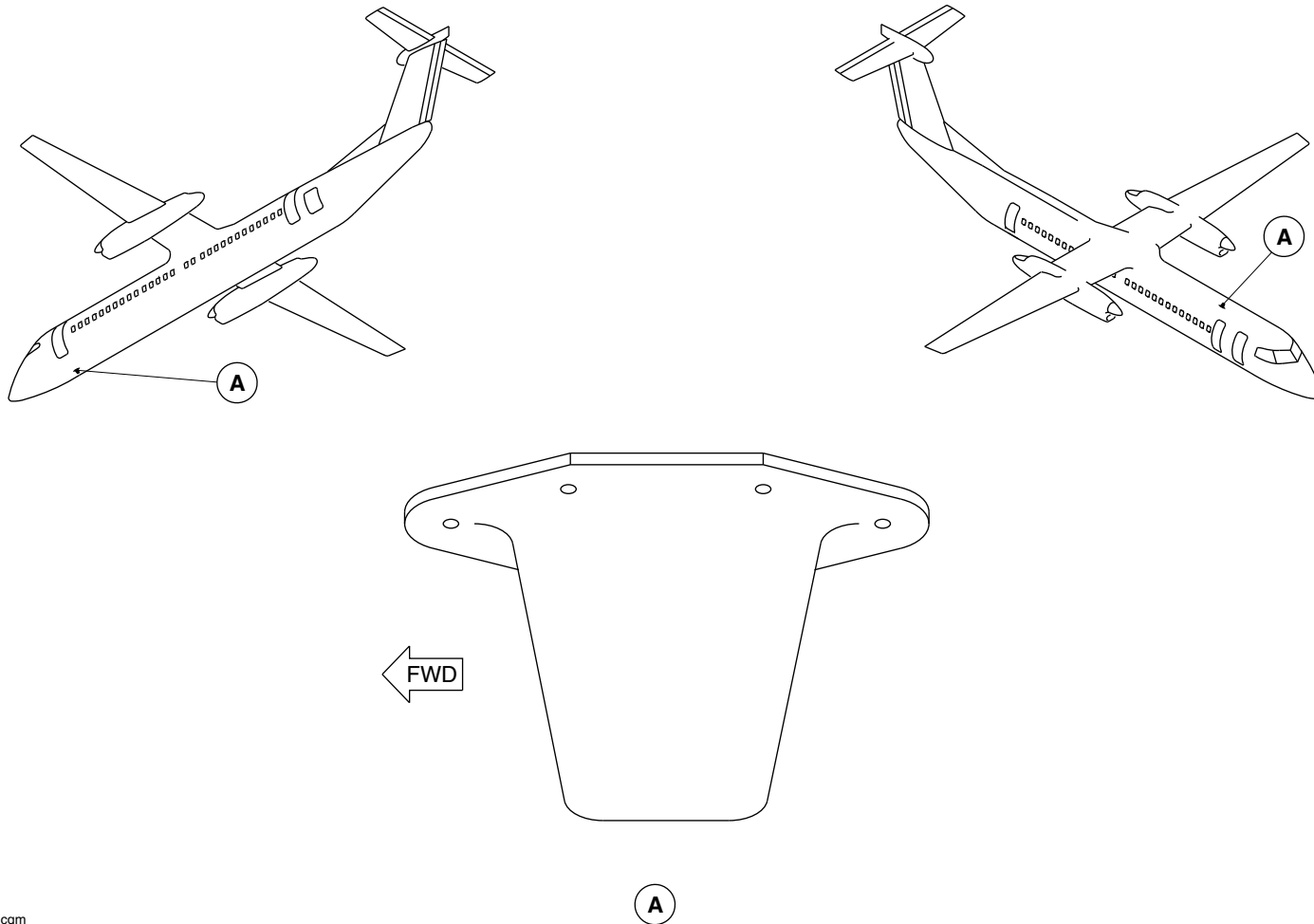
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TRANSPONDER L-BAND ANTENNA LOCATOR
Figure 15

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-54-00
Config 001

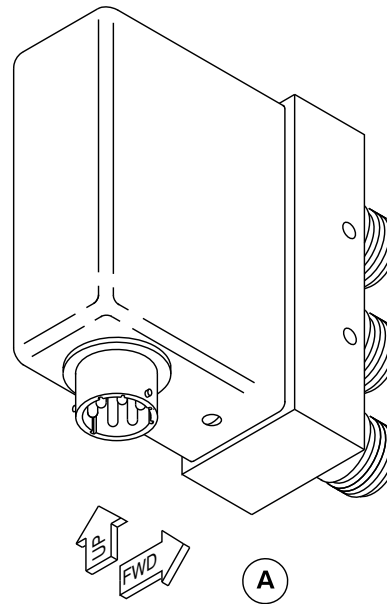
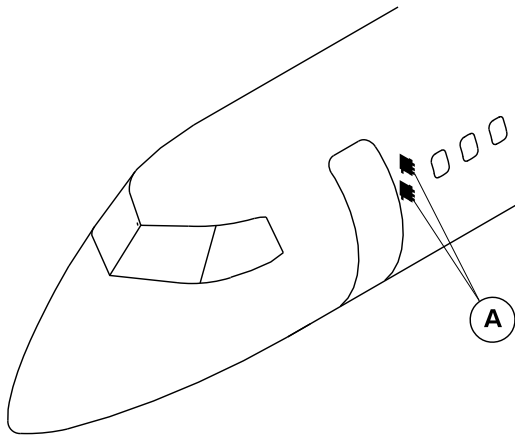
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fs846a01.dg, pr, aug11/2014

TRANSPONDER COAXIAL CABLE ANTENNA SWITCH LOCATOR
Figure 16

PSM 1-84-2A
EFFECTIVITY:
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Config 001

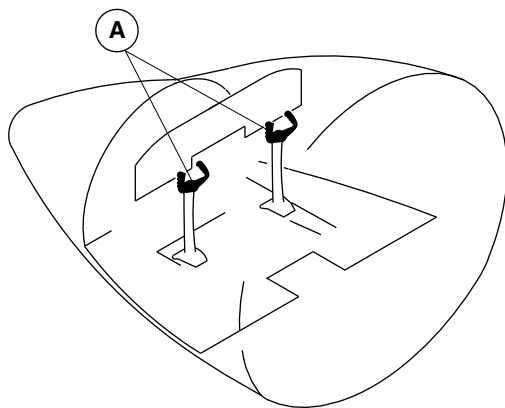
34-54-00

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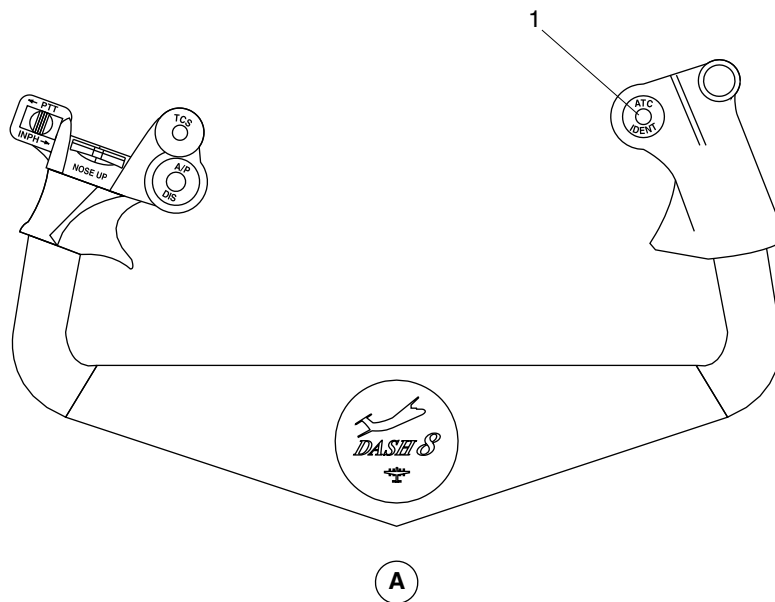
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LEGEND

1. ATC Ident Button.



NOTE

Pilot's equipment shown.
Copilot's equipment similar.

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TRANSPONDER IDENTIFIER SWITCH LOCATOR
Figure 17

PSM 1-84-2A
EFFECTIVITY:
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34-60-01-001

FLIGHT MANAGEMENT COMPUTING

Introduction

The Multi-Functional Control Display Unit (MCDU) automates many routine pilot tasks and manual calculations.

It contains a Flight Management System (FMS), Global Positioning System (GPS) and VORTAC Position Unit (VPU) in a single system.

The MCDU with its other interfacing equipment in the aircraft does the functions that follow:

- Calculates the aircraft position and velocity
- Guidance for the Automatic Flight Control System (AFCS)
- Manages the aircraft flight path from origin to destination
- Tunes communication and navigation radios
- Makes fuel management calculations.

GENERAL DESCRIPTION

The FMS calculates lateral and vertical guidance cues for presentation on the Electronic Flight Instrument System (EFIS) Display Units (DU). The guidance cues show comparisons between the current aircraft position and the selected flight plan. By following these cues, the pilots guide the aircraft along a path defined in the active flight plan. The pilots may couple the FMS lateral and vertical

guidance cues to the AFCS. In doing so, the aircraft follows the FMS cues automatically.

The pilots create flight plans for navigating from a navigation database using:

- Standard Instrument Departures
- Standard Terminal Arrival Routes
- Approaches
- Airways
- Company Routes.

The EFIS displays the FMS flight plan in a graphical map format.

The FMS provides an alternative means of tuning radio frequencies or setting Air Traffic Control (ATC) codes. It interfaces with the following navigation and communication systems:

- Very High Frequency Communication Transceivers (VHF COM1, VHF COM2)
- Very High Frequency Navigation Receivers (VHF NAV1, VHF NAV2)
- Automatic Direction Finder Receivers (ADF1, ADF2)
- ATC Transponders (ATC1, ATC2).

The FMS provides the following features for fuel management:

- Fuel flow indication
- Total fuel remaining indication
- Trip fuel planning.

PSM 1-84-2A

EFFECTIVITY:

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Using signals received from the NAVSTAR/GPS satellite network, the GPS computes a four-dimensional:

- Latitude
- Longitude
- Altitude
- Current Time.

The MCDU interfaces with the following systems to provide accurate determination of the aircraft's position information management:

- Very High Frequency Omni Range (VOR) 34-51-00
- Distance Measuring Equipment (DME) 34-53-00
- Distance GPS Antenna 34-55-01
- Configuration Module 34-61-06.

In addition, the MCDU interfaces with the following systems through the Input/Output Processors of the Integrated Flight Cabinets (IFC1, IFC2):

- Flight Guidance Modules (FGM1, FGM2) 22-00-00
- Air Data Units (ADU1, ADU2) 34-11-00
- Attitude and Heading Reference System (AHRS1, AHRS2) 34-21-00
- Fuel Flow Sensors 73-11-16.

DETAILED DESCRIPTION

The MCDU is the primary pilots interface with the FMS. The MCDU portion of the FMS consists of a keyboard and display. It provides

readout and verification of pilot programmed data. In addition, MCDU displays advisory data from the FMS in a scratchpad at the bottom of the display or on a dedicated message page.

The Left Main 28 Vdc Bus powers the MCDU.

The FMS completes a self-test sequence to ensure correct operation before entering an operational mode. The self-test includes the following tests:

- CPU
- Operational Program CRC
- RAM
- Navigation Database CRC
- NVM
- Configuration Module CRC
- GPS Sensor RAM
- GPS Sensor ROM
- GPS Sensor ASIC
- GPS Sensor RF.

CPU Test: The CPU Test tests the arithmetic unit and internal registers of the FMS host microprocessor.

Operational Program CRC Test: The Operational Program CRC Test verifies that the FMS operational software program loaded in the FMS EEPROM memory is valid.

RAM Test: The RAM Test verifies that the FMS host processor RAM memory is operating correctly. Each RAM memory location is tested



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

with various data patterns and proper addressing of RAM is confirmed.

These tests must all pass for FMS use. The FMS System Valid discrete output is available only if these tests pass:

- CPU
- Operational Program CRC
- RAM.

Navigation Database CRC Test: The Navigation Database CRC Test verifies that the FMS navigation database loaded in the FMS EEPROM memory is valid. A failure of this test inhibits the use of the navigation database. It causes the DATABASE FAIL message to appear on the MCDU.

NVM Test: The NVM Test confirms that stored data in the FMS non-volatile memory is valid. A failure of this test deletes the invalid data item and causes the MEMORY FAIL message on the MCDU.

Configuration Module CRC Test: The Configuration Module CRC Test confirms that the FMS has correctly loaded the aircraft specific configuration options. A failure of this test causes the CONFIG MODULE FAIL message to appear on the MCDU. In this condition, the FMS operates in a default configuration.

The FMS configuration module is installed in the harness adjacent to the MCDU.

GPS Sensor RAM Test: The GPS Sensor RAM Test performs a read and write test of system RAM. The test consists of writing alternating patterns of ones and zeros to every byte of system RAM and confirming that the value read back matches the value written.

GPS Sensor ROM Test: The GPS Sensor ROM Test performs a checksum test of system ROM. It computes a long 32-bit checksum and compares it against a checksum stored at the end of system ROM.

GPS Sensor ASIC Test: The GPS Sensor ASIC Test performs a self-test of the signal processing ASIC module. It utilizes the built in test functions of the ASIC to test channel control registers and integrators.

GPS Sensor RF Test: The GPS Sensor RF Test tests the GPS Radio Frequency section. The test consists of averaging 100 samples and comparing the average magnitude with a predetermined threshold.

A failure of any of the GPS Sensor Tests will cause the GPS FAIL message on the MCDU.

The MCDU displays a self-test page during the power on tests. If any part of the power on test fails, the self-test page remains in view and prompts the user to:

- Restart the self-test
- Manually continue to the initialization pages.

The ability to continue to the initialization pages depends on which test failed.

The FMS performs two types of start up. It performs a warm or cold start depending on the length of the power interruption.

A warm start occurs when power is removed from the FMS for less than:

- 150 seconds when the aircraft is on the ground
- 7 seconds when the aircraft is in the air.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The FMS returns to the active page, retaining all data, after a warm start.

A cold start occurs anytime when the conditions for a warm start is not met. A cold start initializes the FMS. It detects inoperative hardware that may jeopardize FMS operation. All data, including existing flight plans clear during a cold start initialization.

Navigation Management: The FMS uses an internal GPS or VPU to determine the following:

- Latitude and longitude position
- Ground speed
- Track angle
- The current wind vector (direction and magnitude).

The FMS bases position fixing in the following order of preference:

1) GPS

2) VPU.

The MCDU indicates which sensor it is using to fix the system position. The pilot can override the FMS sensor priority by deselecting one or more sensors through the MCDU.

The MCDU displays each sensor's position in the following formats:

- Latitude and longitude.
- Difference of latitude and longitude between the sensor and FMS position.
- Radial distance in nautical miles between the sensor and FMS position.

The FMS provides dedicated MCDU pages for each sensor. The MCDU pages include a separate section to display messages that relate to the status or operation of each configured sensor.

GPS Sensor: The MCDU presents the following GPS information:

- Latitude and longitude position
- A radial difference in nautical miles between the FMS and the GPS position
- PS horizontal position error
- GPS time in Universal Time Coordinated (UTC)
- Using altitude annunciation
- GPS operating in Dead Reckoning (DR) mode annunciation
- RAIM availability at the current position
- FDE availability at the current position
- RAIM will not be available at the Active Flight Plan Destination annunciation
- GPS integrity mode
- GPS accuracy or integrity is not adequate annunciation
- Satellite status for up to eight satellites.

The FMS contains an embedded GPS sensor. The GPS section receives signals transmitted by the NAVSTAR/GPS satellite network to compute aircraft position and velocity.

The GPS sensor receives the C/A code signals transmitted by the satellite network on the L1 frequency of 1575.42 MHz. The sensor uses eight signal-processing channels to track eight satellites



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

simultaneously. At least four satellites must be in view of the antenna for the sensor to operate. The GPS antenna is small "patch" antenna. It fits on the top of the aircraft fuselage.

The GPS sensor uses data in the satellites' navigation messages to calculate:

- Latitude
- Longitude
- Altitude
- Horizontal velocity
- Vertical velocity
- Current UTC.

The GPS sensor provides a figure of merit associated with its position fix. The FMS host processing estimates the Total System Error (TSE).

The estimate of the TSE consists of the following elements:

- Flight Technical Error (FTE)
- Navigation Sensor Error (NSE).

The FTE measures the deviation of the indicated aircraft position from the defined flight path.

NSE is the error in indicating aircraft position relative to the true aircraft position. NSE indicates accuracy of the differential corrections computed at the ground stations and transmitted to the aircraft.

The GPS uses RAIM with FDE. The RAIM function provides a capability to detect faults in the set of satellites used for navigation. It

alerts the crew to the possibility of an error in the GPS computed position. When FDE is available, the fault can be isolated to a particular satellite. The GPS receiver then automatically deselects this satellite. Detection limits for the RAIM and FDE functions are set according to the flight phase.

- Approach
- Terminal
- En-route
- Oceanic or remote.

The Primary Flight Display (PFD) presents a yellow "GPS INTEG" message to the pilots if:

- The satellite geometry is not adequate to support at least Receiver Autonomous Integrity Monitoring (RAIM) error detection
- If an uncorrectable satellite error occurs.

The GPS sensor updates the FMS time and date when operating. A hardware calendar and clock integrated chip keeps date and time current even while the FMS is turned off. The FMS uses a saved shut down position, calendar and time to initialize the system on a cold start. This speeds up GPS acquisition time, the time-to-first-fix, to about one-minute.

VPU Sensor: The VPU sensor function provides the following information for display on the MCDU:

- Latitude and longitude position
- A radial difference in nautical miles between the FMS position and the VPU position
- VPU Estimated Position Uncertainty (EPU)



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- VPU mode
- Identifier for each navigation aid being tuned
- Frequency for each navigation aid being tuned
- VOR bearing for each available VOR channel
- DME range for each available DME channel
- Dead reckoning annunciation.

The VPU uses information from the following sensors to compute its position-fixing algorithm:

- Up to two channels of VOR
- Up to six channels of DME.

Barometric corrected altitude data from the Air Data Unit (ADU) provides slant range correction of the DME data.

The following are definitions of VPU modes:

- No Measurements: No acceptable VOR or DME data available for navigation.
- Bad Geometry: The available VOR or DME data does not have an acceptable geometry for navigation.
- VOR/DME: The VPU is determining a position based primarily on a single VOR/DME navigation aid.
- DME/DME: The VPU is determining a position based on two or more DME navigation aids.

When no sensors are available for navigation, the FMS provides a dead reckoning mode of navigation. Dead reckoning updates the position and velocities using True Airspeed (TAS) from the ADU and heading from the Attitude and Heading Reference System (AHRS).

The PFD displays a yellow "DR" message to alert the pilots of the dead reckoning mode.

The FMS Navigation function determines an appropriate Required Navigation Performance (RNP) value. Normally, the RNP value is determined automatically based on a value either associated with the active route or based on the current flight phase. The pilots can override the automatic value with a manual entry when necessary.

The FMS separately computes and displays the TSE. The TSE is a figure of merit associated with the capability to accurately guide the aircraft along the defined flight plan path. The FMS monitors the TSE by comparing it to the RNP. The Primary Flight Display presents a yellow "RNP" message when the TSE becomes larger than the RNP. This alerts the crew that the FMS is unable to guarantee the RNP.

In the GPS mode of operation, the FMS RNP capability is set to 0.3 Nautical Miles.

In the VPU mode of operation, the FMS RNP capability is 1.0. It is capable of this performance only if adequate DME is available. In the dead reckoning mode of navigation, the FMS does not use the RNP capability.

The FMS monitors the position difference between its position estimate and the positions reported by each position sensor. Deselected and inoperative sensors are not included in the monitor. All other sensors are included even when not being used to fix the FMS system position. Each sensor used in the monitor selects a comparison threshold, which is equal to 1.5 times its reported EPU, or 0.3 nautical miles, whichever is greater. The MCDU message page shows a SENSOR MIS COMPARE message if the radial difference between the FMS position estimate and the sensor position is greater than the threshold for any monitored sensor.



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The MCDU displays magnetic variation associated with the aircraft's present position. In normal operation, the FMS automatically determines the magnetic variation based on a table contained inside the navigation database. In addition, the FMS accepts an operator entered value for magnetic variation. The operator inputted value takes priority over the automatic value. The FMS starts in the automatic variation mode.

The FMS magnetic variation table conforms to the International Geomagnetic Reference Field model. The National Geophysical Data Center publishes this table every 5 years. The table is valid for a 5 year period. The magnetic variation table remains current through FMS database loading. The FMS reverts to a default table in case of a database failure. The default table reverts to the year in which the software was developed.

The standard navigation database magnetic variation table is between South 60 and North 70 latitudes. The range expands for special application by using a custom database. It displays an action message when the aircraft flies outside the range of the magnetic variation table in the automatic magnetic variation mode.

Polar Navigation

The FMS uses an internal coordinate system for position and velocity estimating which is compatible with navigating in Polar Regions. The FMS uses the GPS at all degrees of latitude. The VPU sensor is not used at latitudes of 85 degrees or higher. The limited availability of VOR/DME navigation aids confines the use of the VPU mode.

In addition, the Dead Reckoning mode is available at any latitude under the following conditions:

- TAS data is available from the ADU
- Heading data is available from the AHRS.

Navigation modes requiring heading inputs are retained at high latitudes (out of range of the magnetic variation table) providing one of the following conditions is met:

- The FMS is receiving true heading data from the AHRS.
- The FMS is receiving magnetic heading data from the AHRS and a pilot enter and updates the magnetic variation manually.

Flight Plan Creation: A flight planning function lets the pilots program a three-dimensional flight plan by using:

- Standard Instrument Departures (SID)
- Airways
- Standard Arrivals (STAR)
- Approaches
- Individual waypoints.

In most cases, the flight plans elements come from a navigational database stored in the FMS. In addition, the pilots can create pilot defined waypoints by specifying any of the following parameters from a facility:

- Latitude and longitude
- Fix
- Bearing
- Distance.

The flight planning function also provides calculated data for the flight plan such as distances and Estimated Time En-route (ETE). The operator can program vertical profile information such as speed constraints and altitude constraints into the flight plan.



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The operator manually enters the flight planning data or retrieves it from the database. Flight planning data consists of the following elements:

Table 1: Flight Planning Data

ELEMENT	PILOT ENTRY	DATABASE
Origin Airport	—	X
Destination Airport	—	X
Waypoints	X	X
Runways	—	X
Airways	—	X
Standard Instrument Departure (SID)	—	X
Standard Terminal Arrival Route (STAR)	—	X
Instrument Approach Procedure (IAP)	—	X
Flight Plan Breaks	X	X
Holding Patterns	X	—
Direct To a Waypoint	—	X

ELEMENT	PILOT ENTRY	DATABASE
Selected Heading	X	—
Heading to Intercept a Flight Plan Leg	X	—

The pilots load a company route from the navigational database into the active or secondary flight plan. This function is available when the navigation database actually contains company routes. The pilots select the company routes by specifying a unique route identifier or by entering an origin–destination pair of airport identifiers and selecting from a list of available routes that go between the specified airports. Company routes may contain any of the flight planning elements defined in the navigation database.

The navigation database receives an update every 28 days.

The flight planning function computes an initial waypoint for those lateral leg types not defined by fixed waypoint termination. It computes an initial waypoint based on the leg geometry when inserting the leg into the flight plan.

The FMS determines the computed waypoint after allowing for capture of the desired course or heading.

The computed waypoint is determined as the following geodetic positions:

CR or VR leg: [Refer to Figure 1.](#)

CD or VD leg: [Refer to Figure 2.](#)

CI or VI leg: [Refer to Figure 3.](#)

PI: [Refer to Figure 4.](#)



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CA or VA leg: [Refer to Figure 5.](#)

FC leg: [Refer to Figure 6.](#)

FD leg: [Refer to Figure 7.](#)

FA leg: [Refer to Figure 8, 9, 10, 11, and 12](#)

Active Flight Plan: The FMS utilizes one active and one secondary flight plan. Each flight plan consists of up to 100 lateral legs.

A flight plan may be strung by selecting any of the following:

- Planning elements from various pages of the MCDU
- Stored flight plan
- Company route.

The FMS accepts only one data entry for each of the following elements:

- Origin Airport
- Destination Airport
- SID
- STAR
- IAP.

The operator accesses the modification pages on the MCDU to edit the active or secondary flight plan. Modifying any of the flight plan elements is allowed but it is subject to the following conditions:

- The operator must confirm modifying entries prior to affecting the active flight plan
- Illegal legs or leg sequences cannot be created

- The IAP from the Final Approach Fix (FAF) through the Missed Approach Point (MAP) cannot be modified
- The active lateral guidance leg may be changed but not deleted

The pilot can perform the following flight plan operations on either the active or secondary flight plans through the MCDU:

- Erase the flight plan but not the active lateral guidance leg
- Review the flight plan to confirm the correct origin, destination, waypoint/leg sequence, and coordinates for each waypoint
- Resolve duplicate waypoint selection by reviewing coordinates and country name for each waypoint
- Save the flight plan as a stored flight plan
- Copy the active flight plan to the secondary flight plan
- Swap the active and secondary flight plans
- Load a stored flight plan or company route.

The operator uses the MCDU to enter the following data at each waypoint:

- Altitude constraint
- Speed constraint
- Flight path angle.

The FMS computes and displays the following information for each leg the active and secondary flight plan:

- Leg distance in nautical miles
- Magnetic or true course



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- Distance to the destination in nautical miles
- Estimated Time of Arrival (ETA) to the waypoint.

The active flight plan erases on power up cold start or at transition to the done flight phase.

Direct To: The FMS provides the ability to manually enter a Direct To leg (DF). The operator specifies the desired fix waypoint from the following:

- Active flight plan
- Navigation database
- Manual pilot entry.

Manual Holding Pattern: The FMS allows the operator to manually enter a holding pattern at any flight plan waypoint or present position. The operator specifies the following holding pattern definition parameters through the hold page of the MCDU:

- Leg length specified in minutes or nautical miles
- Holding inbound course
- Turn direction, right hand or left hand holding pattern
- Termination criteria at the waypoint, automatic or manual.

The operator can modify the holding pattern definition parameters while in the holding pattern.

If the operator modifies the inbound course or turn direction while holding, a Direct To is inserted to the hold waypoint with a modified holding pattern.

The holding pattern entry is not available at latitudes greater than 85 degrees north or south.

Each flight plan leg contains a Magnetic Variation (Mag Var) that associates with the waypoint termination. The leg reference Mag Var is determined in the following order of priority:

1. The magnetic declination of the VOR or TACAN if either is the termination waypoint
2. The magnetic declination of a recommended navigation aid associated with the leg if it exists
3. The Mag Var of the destination airport provided the leg associates with a STAR or IAP, or the origin airport if the leg associates with a SID
4. The Mag Var computed from the Navigation Database Mag Var model using the waypoint termination position

If the Mag Var is invalid, the course associating with the leg references to true north.

Company Routes: The operator can load a company route from the navigation database into the active or secondary flight plan. The operator enters the company route through the MCDU FLT PLAN INIT page. This function is only present when the navigation database contains company routes. The operator selects the company routes by specifying a unique route identifier or by entering an origin and destination pair of airport identifiers. The operator selects a route from a list of available routes that go between the specified airports.

Pilot Defined Waypoints: The pilots can define their own waypoints for use by the FMS. The pilot defined waypoints is either a latitude and longitude coordinate pair or a bearing and distance from another waypoint. The reference waypoint is called the parent waypoint and the resultant computed waypoint is called the offset waypoint. If the pilots do not supply an identifier with the defined waypoint entry, the



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FMS assigns an appropriate default identifier. The default identifier is unique relative to identifiers contained in the navigation database.

The entry of a pilot defined waypoint by bearing and distance is subject to the following conditions:

- The bearing should be entered relative to magnetic or true north depending on the selected mode.
- When using a magnetic bearing, the system determines the magnetic variation according to the following priority:

1. Station declination from the navigation database if the parent waypoint is navigation aid
2. Airport Mag Var from the navigation database if the parent waypoint is an airport reference point, runway, or terminal waypoint.
3. Navigation database Mag Var model at the parent waypoint coordinates
4. If a magnetic variation is invalid, the bearing entry is interpreted as a true entry even though the mode is magnetic

The pilot defined waypoint erase on power up cold start. They erase unless they exist in stored flight plans.

Stored Flight Plans: The pilots use the MCDU to store up to 50 flight plans, each containing up to 100 legs in a non-volatile section of memory. The pilot may create a stored flight plan by selecting with restrictions. The FMS accepts only one data entry for each of the following elements:

- Origin Airport
- Destination Airport
- SID

- STAR
- IAP.

The MCDU presents a list of stored flight plans to the pilots. It lists the identifier associated with each stored flight plan in alphabetical order. In addition, the MCDU provides access to the stored flight plans.

The pilots are able to perform the following stored flight plan functions through the MCDU:

- Select any stored flight plan and make it the active flight plan
- Erase the stored flight plan
- Review the stored flight plan to confirm the correct origin, destination, waypoint/leg sequence, and coordinates for each waypoint.

The FMS automatically reviews the stored flight plan against the navigation database in order to verify coordinates and flight planning elements. The FMS identifies any invalid element from the non-volatile memory.

Trip Planning: The FMS provides the capability to calculate flight planning information for the active and stored flight plans. The trip planning function computes the following information:

- Desired track for each leg
- Leg distance
- Distance to destination, the sum of leg distances
- Estimated Time en route for each leg
- ETA at the destination



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- Required fuel for each leg
- Remaining fuel at the termination waypoint
- Fuel required to destination
- Remaining fuel at destination
- Fault Detection and Exclusion (FDE) along the planned route if applicable
- GPS Receiver Autonomous Integrity Monitor (RAIM) at the destination.

Lateral Navigation: The Lateral Navigation function compares the active flight plan with the aircraft's present position and velocity in the horizontal plane. The FMS calculates guidance cues for display on the EFIS. This provides the pilots with maneuvering information in order to fly the aircraft on the programmed horizontal path.

The FMS produces a roll steering command for the AFCS. It couples to the Lateral Navigation (LNAV) modes to control the aircraft along its roll axis.

The FMS provides the ability to enter and fly parallel tracks offset by up to 20 nautical miles from the active flight plan. The offset distance entry, in increments of 1 nautical mile, can be left or right of the flight plan. When the offset is programmed, the Lateral Navigation function provides guidance to a parallel track offset from the active track in the flight plan. The PFD displays a XTRK message when the parallel track offset mode is active.

Lateral Guidance Leg Construction: The Lateral Guidance function provides control to each of legs as assembled by the flight planning function. The navigation database defines all path parameters for the following lateral leg types:

Table 2: Lateral Leg Types

ACRONYM	DEFINITION
TF	Track to a Fix Refer to Figure 9.
CF	Course to a Fix Refer to Figure 10.
FM	Course from a Fix to Manual termination Refer to Figure 11.
VM	Heading to a Manual termination Refer to Figure 12.
IF	Initial Fix

Other leg types require the lateral guidance function to compute additional parameters to define the path for guidance.

A termination waypoint position is undefined for Course and Heading Legs excluding a Course to a Fix (CF). The lateral guidance function calculates initial position of the termination waypoint. It continuously updates initial position of the termination waypoint based on the current aircraft state and capture state of the desired course or heading.



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Table 3: Course and Heading Legs

LEG	DESCRIPTION
CR	Geodetic position of the specified course and a radial out of the reference navigation aid intercept.
VR	Geodetic position of the specified heading and a radial out of the reference navigation aid intercept.
CD	Geodetic position of the specified course and a DME distance from the reference navigation aid intercept.
VD	Geodetic position of the specified heading and a DME distance from the reference navigation aid intercept.
CI	Geodetic position of the specified course and the great circle course of the next leg in the flight plan intercept.
VI	Geodetic position of the specified heading and the great circle course of the next leg in the flight plan intercept.

LEG	DESCRIPTION
CA	Geodetic position at which time the aircraft is predicted to reach termination altitude based on the specified course, current aircraft altitude, and current flight path angle. It assumes a flight path angle of 10 for level flight.
VA	Geodetic position at which time the aircraft is predicted to reach termination altitude based on the specified heading, current aircraft altitude, and current flight path angle. It assumes a flight path angle of 10 for level flight.

If the lateral guidance function cannot determine the intercept point, the geodetic position becomes the waypoint at the time of capturing the specified course.

A termination waypoint position is calculated by the flight planning function for course from a fix legs.

Table 4: Course from a Fix Leg

LEG	DESCRIPTION
FC	Geodetic position of the course from the waypoint out a specified distance in nautical miles.
FD	Intercept point between the specified course from a waypoint and a DME distance from the reference navigation aid.
FA	Geodetic position at which time the aircraft is predicted to reach the termination altitude based on the specified course from a fix, current aircraft altitude, and current flight path angle. It assumes a flight path angle of 10 for level flight.



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For FA legs, a termination waypoint position is undefined. The later guidance function calculates the initial position of the termination waypoint. It is continuously updated based on the current aircraft state.

[Refer to Figure 13.](#)

DME Arc to a Fix Legs (AF): The termination of the previous leg lies on the DME arc between the defined boundary radial and the TO fix in the specified turn direction for a DME Arc to a Fix Leg (AF).

[Refer to Figure 14.](#)

Constant Radius to a Fix (RF): The termination of the previous leg lies on the arc and the course is the same as the initial course of the RF. The course at the termination of the RF leg is the same as the initial course of the next leg.

[Refer to Figure 15.](#)

Direct to Fix Legs (DF): The operator constructs a DF Leg with a circular arc segment and a computed great circle course to the specified fix. The circular arc allows capture of the great circle course without overshoot. An operator entry of the current aircraft position or previous leg when pre-programmed in the flight plan initiates the DF Leg.

Holding Patterns: The operator constructs holding patterns (HA, HF, HM) around the specified holding fix. While holding, the track updates the following parameters when required:

- Wind
- Airspeed
- Pilot entries.

The FMS computes the turn radius based on the following:

- Estimate average ground speed in the highest speed turn. It is a function of the current TAS and wind
- Nominal 25 degree bank angle.

When the operator specifies the inbound leg length in terms of time, the FMS computes a distance based on the ground speed from the inbound leg. The ground speed is a function of the following:

- Current TAS
- Winds
- Inbound course.

The MCDU displays an action message if the current Ground Speed (GS) will cause an overshoot of the planned turn radius. The message appears on the message page three minutes prior to entering a planned holding pattern and during a hold. This prevents the aircraft from flying outside of the protected airspace.

[Refer to Figure 16.](#)

Holding Pattern Entry Procedures: The Lateral Guidance Function transitions to a holding pattern by using one of the appropriate entry procedures:

- Direct
- Offset/Teardrop
- Parallel.

The boundary between the offset/teardrop and parallel entry zones is five degrees. The offset/teardrop entry takes precedence over a parallel entry when flying inbound on the reciprocal course to the holding course.

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Refer to Figure 17.

The Procedure Turn to Course to a Fix (PI – CF) combination is constructed like a standard procedure turn. The Procedure Turn (PI) has the following segments:

- The first segment (SEG1) is the outbound segment. It begins at the PI waypoint, along the reciprocal CF leg, for a specified distance.
- The second segment (SEG2) is the turn out segment. It begins at the end of the SEG1, along the specified PI course, to a computed distance.
- The third segment (SEG3) is the course reversal turn. It begins at the end of the SEG2 and arcs in the specified direction to overfly the intercept fix on the CF leg without an overshoot.

The procedure turn does not extend beyond the boundary distance less 0.5 nautical miles.

Refer to Figure 18.

Table 5: Boundary Distance

FIRST SEGMENT DISTANCE	BOUNDARY DISTANCE
0 Nautical Miles	Less than 10 Nautical Miles
3.5 Nautical Miles	Greater than 10 Nautical Miles and less than 15
5.3 Nautical Miles	Greater than 15 Nautical Miles

When the PI leg is active, the FMS produces an action message if the current ground speed will cause an overshoot of the procedure turn. It displays on the MCDU message page.

Parallel Offset: The FMS provides the ability to enter and fly, parallel tracks offset by up to 20 nautical miles from the active flight plan. The offset distance entry can be left or right of the flight plan. It is in increments of 1 nautical mile.

The FMS only allows entry from any of the following active leg types:

- Track to a fix (TF)
- Direct to a fix (DF)
- Course to a fix (CF)
- Course from a fix to manual termination (FM).

The parallel offset function manually cancels by any of the following pilot actions:

- 0 nautical mile offset entry
- Enters CLEARS into the data field
- Modifies the active lateral flight plan.

The Lateral Guidance function provides guidance to a parallel track offset from the active track in the flight plan. The following items references to the offset path and waypoint while the Parallel Offset function is active:

- Roll steering command
- Leg transitions
- Leg sequencing
- Crosstrack cue



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- Track angle error cue
- Desired track cue.

The FMS computes offset waypoint positions so that they intersect a track parallel to the active primary track. The offset waypoint positions lay at the desired offset distance and a line bisecting the course change between leg segments of the active primary track.

If the route terminates at a flight plan break, the offset waypoint is a beam to the original flight plan waypoint at the desired offset distance.

Any of the following conditions automatically cancels the parallel offset function:

- A course change between the active and next leg in the active primary flight plan that exceeds 90 degrees
- Encountering a flight plan break
- FMS enters the approach mode.

The next leg in the active primary flight plan being:

- Heading
- Course (excluding CF)
- Holding
- Procedure Turn
- Initial Fix.

The MCDU displays an Attention Message for 30 seconds prior to an automatic cancellation of a parallel offset function.

Both capture and return of the parallel offset path is done at an intercept angle of 45 degrees or less. Capture of the parallel path

initiates immediately after activation of the offset entry on the LNAV page. A return to the parent path initiates immediately from any of the following functions:

- Pilots manually cancel the parallel offset function
- Changing the active primary track.

Otherwise, the system re-captures the path prior to the condition causing the offset cancellation.

Heading and Selected Course Modes: The Lateral Guidance function provides a heading mode where the following guidance cues are provided to a selected heading:

- Crosstrack (XTK)
- Bearing (BRG)
- Desired Track (DTK).

A pilot activates the heading mode through the following selections on the MCDU:

- Heading page
- Specifying the desired heading
- Executing the change.

When active, the FMS allows the pilot to specify the following parameters:

- Selected heading
- Intercept waypoint and track (INTCPT/TRK)
- To or from the waypoint (TO/FROM).



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The pilots enter an intercept waypoint and select TO the waypoint to fly a course into the waypoint. A new CF leg is inserted in the flight plan and then a heading to intercept (VI) leg.

The pilots select FROM the waypoint to fly the course from the waypoint. A new fix to manual (FM) is inserted in the flight plan and then a heading to VI leg. The heading mode provides an alert to the pilot for the following conditions:

- The programmed intercept causes the aircraft to intercept the leg on the FROM side of the fix
- No intercept is found.

The heading mode cancels itself when an intercept termination exits. The heading to intercept leg is sequenced and the capture of the next leg is initiated.

The heading and selected course mode is not available at latitudes greater than 85 degrees north or south.

Lateral Leg Transitions

The FMS constructs curved path transitions to provide a smooth geometry for guidance from one leg to the next.

In order to minimize overshoot, roll anticipation is used to start the roll maneuver towards the desired bank angle prior to sequencing onto or off of all circular arc segments. The roll anticipation distance is based on the following parameters:

- Current ground speed
- Current roll angle
- Maximum roll rate of three degrees per second.

Lateral Leg Sequencing: The FMS sequences the lateral legs as they are passed during the course of a flight. Sequencing is independent of LNAV engagement. At the sequence, the NEXT leg becomes the ACTIVE leg. The point of sequencing is a function of the transition type between the two legs.

Manual termination and initial fix legs require pilot action to sequence the leg. In addition, sequencing the Missed Approach Point is inhibited until the pilot activates the missed approach procedure.

Vertical Navigation: Vertical Navigation (VNAV) is a FMS function that allows the pilots to define a vertical flight plan. This provides the pilots with guidance cues in order to fly the aircraft on the programmed vertical path. VNAV is available in all flight phases except Go-Around (GA).

The FMS produces a pitch steering command for the AFCS. It couples to the AFCS VNAV mode to control the aircraft along its pitch axis.

The pilots engage the VNAV mode through the Flight Guidance Control Panel (FGCP). The FMS outputs the active VNAV sub mode for annunciation on the Electronic Instrument System (EIS).

VNAV manages the climb profile according to a FMS recommended climb speed schedule. During cruise, the FMS manages crew-initiated changes in the cruise flight level. The pilots manage descents with a path mode, utilizing a series of glide paths, or through a FMS recommended descent speed schedule. The pilots can modify the speed schedules.

The FMS manages the following vertical definitions of the flight path:

- Speed
- Angle and altitude constraints at waypoints



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- Speed limits associated with specified altitudes or leg types
- Cruise Altitude.

The pilot or co-pilot sets the pre-selected altitude to the cleared altitude. The FMS does not change the pre-selected altitude or fly through the selected altitude.

The PFD annunciates four FMS VNAV sub-modes to the pilots:

- Vertical Flight Level Change (VFLC)
- Vertical Path (VPTH)
- Altitude Hold (VALT)
- Altitude Capture (VASL).

Each sub-mode displays the unique effects induced by power lever movement.

Vertical Flight Level Change: The VFLC sub-mode controls the aircraft ascent or descent to a constant speed target. The elevator controls the aircraft's speed by changing the pitch of the aircraft. Moving the power levers during this sub-mode changes the vertical speed and pitch of the aircraft. The VFLC sub-mode controls:

- Climb or cruise when changing the flight level
- Descent for non-path referenced guidance.

VPTH: The Vertical Path sub-mode controls the aircraft to a constant flight path angle. Power lever movements during this sub-mode result in speed changes as the vertical path is fixed. This sub-mode is the normal sub-mode used in descent. It provides a series of flight path angles for each flight plan altitude constraint. The

pilots select this sub-mode following a vertical direct to path in descent.

The Vertical Path sub-mode does not provide control to speed constraints. The FMS provides advisory data to help control aircraft speed. The speed target displayed on the VNAV page changes in anticipation of the constrained speed.

VALT: The FMS, Vertical VALT sub-mode controls the following parameters in all flight phases:

- Flight plan imposed level segments
- Cruise flight level
- Pre-selected altitude.

A cruise flight level must be defined for VNAV operations. If one is not entered, the value defaults to a pre-selected altitude after takeoff.

In the Vertical VALT sub-mode, power lever movements cause a change in aircraft speed.

Altitude capture (VASL): All flight phases use VASL sub-mode when VNAV is engaged to capture the:

- Flight plan imposed level segments
- Cruise flight level
- Pre-selected altitude.

In the VASL sub-mode, power lever movements cause a change in aircraft speed.

The Vertical Navigation VNAV mode observes the flight plan altitude constraints for climb and descent. The pilots use the Vertical Direct



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function in order to fly VNAV directly to the pre-selected or pilot selected altitude.

The airplane never flies through the pre-selected altitude.

Vertical Track Alert (VTA): The FMS provides a VTA for each vertical transition to or from a level segment. The VTA indicates to the pilots that the FMS is preparing to initiate a vertical track change. The pilots may need to adjust the power levers to support the maneuver. The FMS issues a VTA 60 seconds prior to a change from level flight to climb or descent. If climbing or descending, the VTA is issued 1000 feet prior to the altitude of the impending level off. A FMS message associates with the VTA.

The VNAV mode must be engaged to issue a VTA.

Initiate Climb or Descent: The pilots can initiate a climb or descent to a pre-selected altitude level flight by selecting the INITIATE CLIMB or INITIATE DESCENT mode on the VNAV page. The VNAV mode transitions from VALT to VFLC.

Elements of the Vertical Profile: The vertical profile for climb and descent is defined by the following:

- VNAV mode
- Managed speed target when selected to VFLC
- Vertical elements of the flight plan.

The flight plan includes the following vertical elements:

- Speed constraints
- Altitude constraints
- Flight path angle constraints and default flight path angle for descent profile only

- Climb and descent speed restrictions
- ICAO procedure speed restrictions.

The vertical profile is specified in the flight plan from the following:

- Database as part of the associated procedure
- Pilot entered or modified.

For VPTH descents, the angle of the flight path angle is determined by the following constraints at the waypoints:

- Altitude
- Flight path angle.

The pilot defined default flight path angle is used in the absence of a flight path angle constraint. All flight paths angles are limited to be within minus 1 and minus 6 degrees except when using vertical direct to function. The default flight path angle is displayed on the initialization pages. The default flight path angle can be modified by the pilots through the initialization pages.

The FMS Computer calculates the Top of Descent (TOD) by locating the first altitude constraint at a descent waypoint and then projecting backwards from that point. It uses the default flight path angle to the cruise altitude. If there are no descent altitude constraints in the flight plan, then the TOD is located by projecting the default flight path angle backwards from the destination waypoint.

The EFIS Map identifies the TOD as #TOD with the constrained altitude. The Top of Climb (TOC) is identified by #TOC with the constrained altitude.

Vertical Navigation VNAV Climb: All VNAV climbs are flown using VFLC.



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The FMS holds the climb speed using the following parameters:

- Pre-programmed, IAS/Mach speed profiles
- Crew entered speed target change
- Upper limit for Vmo or Mmo
- Lower limit
- Applicable speed limits and constraints.

Moving the power levers during VNAV Climb changes the vertical speed and pitch of the aircraft.

The FMS will not allow the aircraft to climb above the selected altitude.

[Refer to Figure 19.](#)

1. A VFLC selection engages the VNAV mode to the altitude constraint defined in the flight plan. The VNAV page identifies the altitude constraint as the TARGET ALT and provides a waypoint name for it. In addition, the VNAV page annunciates a VTA 1000 feet prior to reaching the target altitude. The FMS controls the following transitions automatically:

- VFLC to VASL
- VASL to VALT.

2. VALT is active until the constrained waypoint is reached. The MCDU annunciates a VTA 60 seconds prior to reaching the waypoint. When the constraint is sequenced, the mode switches to VFLC and the FMS commands a climb. Moving the power levers changes the vertical speed and pitch of the aircraft.

3. The VNAV page the pre-selected altitude as the TARGET ALT. The FMS issues a VTA 1000 feet prior to reaching the target altitude.

4. The pilots initiates the transition from VALT to VFLC. The pilots select INITIATE CLIMB on the VNAV page. The selection becomes available when a value for the pre-selected altitude is defined.

5. The MCDU annunciates a VTA 1000 feet prior to reaching the target (cruise) altitude.

6. The FMS maintains control at the target (cruise) altitude. Redefining the pre-selected altitude allows the pilots to engage the VFLC. The FMS manages the climb or descent to the newly selected altitude.

[Refer to Figure 20.](#)

Airmass Descent:

1. The VNAV mode transitions from VALT to VFLC when a new altitude is selected and the pilots initiates a descent.

2. To continue the descent, a new altitude is selected and the pilots re-initiates a descent. A transition from VALT to VFLC occurs.

3. The FMS maintains control in VALT mode.

[Refer to Figure 21.](#)

Path Descent: VNAV descents are flow using a defined descent path (VPTH). The FMS provides the following VPTH parameters:

- Guidance to the VNAV flight path
- Vertical deviation for display on EFIS and VNAV page
- Prevents descents through the selected altitude by automatically transitioning to VALT
- Over speed protection through a transition to VFLC mode with a Vmo or Vmo as a speed target.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

1. The first descent VTA occurs sixty seconds prior to the TOD. The FMS computes the TOD.
2. The VNAV transitions from VALT to the defined VPTH with over-speed protection.
3. There is a flight path angle change at the first waypoint.
4. If no further path is defined at the waypoint, the VNAV mode transitions from VPTH to Flight Level Change (VFLC).
5. The MCDU annunciates a VTA 1000 feet prior to reaching the pre-selected altitude. The VNAV mode transitions from Flight Level Change (VFLC) to VASL to VALT automatically.

[Refer to Figure 22.](#)

Late Path Descent: A Late Path Descent scenario can occur if the pre-selected altitude is lowered after reaching the TOD.

1. A Late Path Descent causes the MCDU to announce a VTA. It announces the a VTA before reaching the TOD.
2. The pilots select a lower altitude and initiates a managed descent using the MCDU.
3. The VNAV transitions from VALT to VFLC after the crew initiates the descent.
4. The VNAV mode transitions from VFLC to defined VPTH. The VNAV mode continues operating in the defined VPTH until capturing

the pre-selected altitude. At altitude capture, the VNAV mode transitions from the defined VPTH to VASL to VALT automatically.

[Refer to Figure 23.](#)

Early Descent to Capture Path: The Early Descent to Capture Path scenario occurs when the crew is directed to descent prior to reaching the TOD.

1. The pilots select a lower altitude and initiates a managed descent using the MCDU before reaching the TOD.
2. The VNAV transitions from VFLC to VALT after the path is captured.
3. The VNAV transitions from the defined VPTH to VASL to VALT automatically.

[Refer to Figure 24.](#)

Late Path Descent using the Vertical Direct To Function: The pilots use the Vertical Direct To Function for a late path descent when the pre-selected altitude and cruise altitude are the same at the TOD.

1. The FMS issues a VTA sixty seconds prior to the TOD.
2. At the TOD, the pilots enter a lower pre-selected altitude and initiate the Vertical Direct To function.
3. The VNAV mode transitions from VALT to defined VPTH. The VNAV mode automatically transitions from defined VPTH to VASL to VALT at the pre-selected altitude.

[Refer to Figure 25.](#)

Early Path Descent using the Vertical Direct To Function: The pilots use the Vertical Direct To Function for an early path descent by lowering the pre-selected altitude before reaching the TOD.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

1. The FMS issues a VTA sixty seconds prior to the TOD.
2. The pilots use the Vertical Direct To function by selecting a lower altitude and initiating a waypoint and altitude combination. The VNAV mode transitions from VALT to the defined VPTH after executing the modified flight plan.
3. The VNAV mode transitions from VASL to VALT at the pre-selected altitude.

Fuel Management: The FMS provides the following fuel managing features:

- Fuel Flow
- Total Fuel Remaining
- Gross Weight
- Trip Fuel Planning.

The FMS computes Fuel on Board (FUEL) from the following parameters:

- Initial pilot entered fuel quantity
- Integrated or pilot manually entered fuel flows.

The integrated fuel flow is the automatic mode. If the fuel flow inputs fail, the system accepts manually entered values.

The pilots enter a reserve fuel value to restrict the amount of fuel that is used in the range and endurance calculations.

In addition, the FMS computes the Gross Weight depending on the following parameters:

- Fuel on board
- Basic operating weight

- Passenger weight
- Cargo weight.

The fuel management function provides the pilots with other fuel related data:

- Hours of flight
- Nautical miles per unit of fuel
- Range.

FMS Radio Tuning: The pilots can manually tune the following radio frequencies through the FMS.

- Very High Frequency Navigation Receivers
- (VHF NAV1, VHF NAV2)
- Very High Frequency Communication
- Transceivers (VHF COM1, VHF COM2)
- Automatic Direction Finder Receivers (ADF1, ADF2).

In addition, the pilots can also set the ATC transponder (ATC1, ATC2) codes.

A navigation auto-tuning function provides automatic selection of up to six navigation aids in order to support automatic VPU navigation. (The VPU uses inputs from VOR Receivers and DME sensors). The FMS uses an algorithm that analyzes the position of all available navigation aids within the radio line of sight. It determines a DME set that best supports multiple position fixing accuracy.

The FMS always auto tunes the remote channels of the active DME receivers. The pilots enable the auto tune mode through an Audio and Radio Control Display Unit (ARCDU) selection or TUNE key selection on the MCDU.



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To prevent misleading operation, the FMS automatically cancels the navigation auto tune mode whenever a pilot selects a:

- VOR or Localizer as the navigation source
- New radio frequency through the ARCDU

The ARCDU continuously indicates the auto tune status of each navigation source.

Test and Maintenance

Power On Self Test: The FMS completes a self-test sequence to ensure correct operation before entering an operational mode. The self-test includes the following tests:

- CPU
- Operational Program CRC
- RAM
- Navigation Database CRC
- NVM
- Configuration Module CRC
- GPS Sensor RAM
- GPS Sensor ROM
- GPS Sensor ASIC
- GPS Sensor RF.

CPU Test: The CPU Test tests the arithmetic unit and internal registers of the FMS host microprocessor.

Operational Program CRC

The Operational Program CRC Test verifies that the FMS operational software program loaded in the FMS EEPROM memory is valid.

RAM Test: The RAM Test verifies that the FMS host processor RAM memory is operating correctly. Each RAM memory location is tested with various data patterns and proper addressing of RAM is confirmed.

These tests must all pass for FMS use. The FMS System Valid discrete output is available only if these tests pass:

- CPU
- Operational Program CRC
- RAM.

Navigation Database CRC Test: The Navigation Database CRC Test verifies that the FMS navigation database loaded in the FMS EEPROM memory is valid. A failure of this test inhibits the use of the navigation database. It causes the DATABASE FAIL message to appear on the MCDU.

NVM Test: The NVM Test confirms that stored data in the FMS non-volatile memory is valid. A failure of this test deletes the invalid data item and causes the MEMORY FAIL message on the MCDU.

Configuration Module CRC Test: The Configuration Module CRC Test confirms that the FMS has correctly loaded the aircraft specific configuration options. A failure of this test causes the CONFIG MODULE FAIL message to appear on the MCDU. In this condition, the FMS operates in a default configuration.

GPS Sensor RAM Test: The GPS Sensor RAM Test performs a read and write test of system RAM. The test consists of writing alternating patterns of ones and zeros to every byte of system RAM and confirming that the value read back matches the value written.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

GPS Sensor ROM Test: The GPS Sensor ROM Test performs a checksum test of system ROM. A long 32 bit checksum is computed and compared against a checksum stored at the end of system ROM.

GPS Sensor ASIC Test: The GPS Sensor ASIC Test performs a self-test of the signal processing ASIC module. It utilizes the built in test functions of the ASIC to test channel control registers and integrators.

GPS Sensor RF Test: The GPS Sensor RF Test tests the GPS Radio Frequency section. The test consists of averaging 100 samples and comparing the average magnitude with a predetermined threshold.

A failure of any of the GPS Sensor Tests will cause the GPS FAIL message on the FMS CDU.

Fault Monitoring: The FMS monitors the following functions during operation:

MESSAGE	DESCRIPTION
AIR DATA INPUT FAIL	message indicates that the FMS has lost the Air Data source
HEADING INPUT FAIL	message indicates that the FMS has lost the heading source
GPS FAIL	message indicates that the GPS sensor within the FMS has failed

The PFD indicates a MSG message when the FMS is selected as the navigation source. Otherwise, the MCDU annunciates failures message on the messages page.

Ground Maintenance: The FMS provides the following maintenance functions:

- Configuration Module Programming
- CDU Parallax Adjustment
- Navigation Database Loading
- GPS Almanac Acquisition
- GPS Receiver Test
- Annunciator Test.

Configuration Module Programming: The Configuration Module Programming function uses selected installation options to program the FMS.

CDU Parallax Adjustment: The CDU Parallax Adjustment function adjusts the MCDU display to eliminate parallax misalignment between the text and the line select keys.

Navigation Database Loading: The Navigation Database Loading function loads new navigation database information into the FMS. An updated database is distributed every 28 days on a Personal Computer Memory Card International Association (PCMCIA) memory card. This card inserts into a slot located at the top of the MCDU. The worldwide database contains and stores information relating to:

- Waypoints
- Navigation aids
- Airports
- Altitudes at appropriate waypoints
- Standard Instrument Departure (SID)



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- STAR
- Airway and Approach procedures
- RNP
- Magnetic variation table
- Operator generated waypoints
- Flight Plans
- Company Routes.

GPS Almanac Acquisition Function: The GPS Almanac Acquisition Function commands the embedded GPS sensor to go to search-the-sky mode. In this mode, the GPS acquires the current almanac data from the GPS satellite network. This function returns the GPS to service after taking the system out of storage.

GPS Receiver Test: The GPS Receiver Test function allows the GPS sensor testing with a single channel test set.

Annunciator Test: An External discrete signal from the Integrated Flight Cabinet (IFC) commands the FMS Annunciator Test function.

It turns on all of the annunciator lights on its MCDU. It activates the auto tune indications on the ARCDU.

Airborne Flight Information System (AFIS): The AFIS provides the integrated flight planning and performance management interfaces to the FMS. Its interface consists of the following capabilities:

The FMS accesses ground based Global Data Center computers from a Personal Computer (PC) to generate or retrieve the following:

- Performance optimized flight plans
- En-route flight plan

- Pre departure clearance requests
- Sending or receiving messages
- Current aviation weather.

The FMS communicates with the Global Data Center using following communications networks:

- ARINC
- SITA
- AVICOM
- Air Canada VHF
- Inmarsat satellite.

The AFIS interfaces with the FMS using an ARINC 739 data bus. Navigation and flight plan information transfers between the FMS and the AFIS. The FMS and the AFIS shares the MCDU. This makes AFIS features available to the pilot.

The AFIS annunciator turns on when the AFIS is requesting control of the MCDU. The operator accesses the AFIS Menu Page by pressing the AFIS Key.

[Refer to Figure 26.](#)

MCDU keyboard selections provide FMS power on and off control, brightness adjustment, and access to specific pages and their associated data using the following types of keys:

- On and Off
- Brightness adjustment
- Alpha and numeric
- Mode



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- Function
- Line Select Keys

The MCDU provides displays colour characters and symbols.

The first line of the display is the title line. It displays the title and any of the following:

- Current flight phase
- Associated flight phase
- Operational mode.

When required, the bottom line of the display serves as a scratchpad. It displays operator data entries and messages from subsystems up to 24 characters in length. Data is first entered into the scratchpad and then into a desired data field.

FMS pages divide into data fields. A data field consists of a label line and a data line below it. For data entry, each field associates with a dedicated LSK. Some pages divide into left and right fields. In this case, a dedicated left LSK associates with each left field and a right LSK associates with each right field. If the data field's label uses the entire width of the display, the MCDU uses a dedicated left LSK for data entry. Some pages have more than two data fields per line. In these cases, data entry is only allowed in the left and right most fields.

A white dot character at the extreme left or right of the data line indicates which LSK associates with the data field. If the field contains data, pushing the LSK next to the white dot character copies the field's data into the scratchpad. In addition, the operator can enter data into the scratchpad using the alphanumeric keys. Then, pushing the LSK next to the white dot character moves the scratchpad's data into the data field.

The FMS checks all data entries into the scratchpad for correct format and range for the corresponding data field. The scratchpad displays a message to indicate any incorrect format or range entries. The message page does not record these types of messages.

The original data remains in the data field where the incorrect entry was attempted. The operator entered data remains in the scratchpad with the error message inhibiting its display.

The operator performs one of the following actions to correct an incorrect entry:

- Clear the last character from the scratchpad by selecting the CLR function key once and then correcting the data. The operator clears the scratchpad completely by repeatedly selecting the CLR function key or by holding it down.
- Re-entering the data into the scratchpad. The FMS clears the scratchpad, both incorrect entry and error message, upon selection of the first alpha or numeric key for the entry.
- Selecting the correct LSK if a wrong LSK selection is the cause of the entry error. This action allows data entry without the need for clearing the scratchpad message and re-entering the data.

The Operator uses the CLR function key to clear data or messages from the scratchpad or data from a field.

A single push of the CLR function key clears messages from scratchpad. Pushing the CLR function key for less than two seconds clears the last character of the data entry in the scratchpad. Pushing and holding the CLR function key for more than two seconds clears the entire data entry from the scratchpad.



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The crew clears data fields by selecting the CLR key while the scratchpad is empty. A CLEAR prompt appears in the scratchpad. Then selecting a LSK clears the data field corresponding to that key. The CLEAR prompt deletes from the scratchpad. It remains in view if the FMS does not allow the clear action. In this case, the crew deletes the CLEAR prompt by selecting the CLR function key again or by making any alpha or numeric entry into the scratchpad.

The MCDU uses two character sizes. Generally, information displayed in large font is database or pilot entries. Information displayed in small characters represents default, predicted, or system calculated values.

- White
- Green
- Yellow
- Cyan
- Magenta.

White Characters: The MCDU uses white characters to display the following:

- Page titles
- Field Labels
- Page selection or reversion prompts
- Information Only Messages
- Scratchpad Data
- Scratchpad Messages.

Green Characters: The MCDU uses green characters to display data associated with the active waypoint.

Yellow Characters

The MCDU uses yellow characters to display the following:

- Data which is selected but not active
- Caution messages on the Message Page
- Important prompts
- Data
- Annunciations.

Cyan Characters: The MCDU uses cyan characters to display data that is selectable, but is not active.

Magenta Characters: The MCDU generally uses magenta characters to display the following:

- FMS commanded parameters
- TO waypoint and associated data
- Current time
- Speed
- Altitude targets
- Altitude constraints or limits.

The FMS generates the following symbols:

- >Pushing the associated LSK results in the display of a different MCDU.



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- * Pushing the associated LSK changes the field's parameters and the active state of FMS. The page displayed does not change.
- ← → Prompts the operator to data can be selected from a list by pressing an associated LSK. Once pressed, the arrows disappear.
- ← → This character appears next to a waypoint in the flight plan to indicate a specified turn direction.
- ← → This character appears in conjunction with wind data. These symbols indicate crosswind direction of the associated wind vector component.
- ↑↓ This character appears in conjunction with wind data. These symbols indicate head or tailwind component of the associated wind vector.
- • This character appears on the far left or right side of the display in the data line. This symbol indicates data contained in the field adjacent to the Line Select Key is not fixed.
- (/) This symbol separates data that appears horizontally in a single line.

ON KEY

[Refer to Figure 26.](#)

Depress and release the ON key to apply power to the system. There is a warm-up period of approximately 30 seconds. The display illumination will initially be set at 80% of full bright.

Depressing the ON Key for approximately three seconds will initiate the system power off sequence. During the sequence the display will

show SYSTEM TURNING OFF. This is to prevent inadvertent system shutdown.

NOTE

The system is also capable of being turned on and off by cycling aircraft power.

BRT KEY

[Refer to Figure 26.](#)

The Brightness Key is used to change the illumination of the display and the annunciator lights.

NOTE

The illumination of all the keys is controlled through the aircraft panel lighting control.

PREV KEY

[Refer to Figure 26.](#)

The Previous Key is used to display the previous page of a section or subsection. This key also allows the operator to remain in a section or subsection by looping from the first to the last and back to the first page of that section or subsection.

NEXT KEY

[Refer to Figure 26.](#)

The Next Key is used to display the next page of a section or subsection. This key also allows the operator to remain in a section or subsection by looping from the first to the last and back to the first page of that section or subsection.



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CLR KEY

[Refer to Figure 26.](#)

The Clear Key is used to clear a single keystroke from the scratchpad (by pressing the key once) or to clear the scratchpad completely (by repeatedly pressing the key or holding it down).

SP KEY

[Refer to Figure 26.](#)

The Space Key is used to enter a blank space in the data entry.

— KEY

[Refer to Figure 26.](#)

The Line Select Keys (LSK) are used to place data entered in the scratchpad in the appropriate data field. A white “a” is displayed on either side of the display to indicate Line Select Keys that are active for data uploading on each individual page.

There are six Line Select Keys arranged along the left side of the display and six along the right. These keys are aligned horizontally with the associated data line.

The Line Select Keys are referenced by the naming convention LSK 1L through LSK 6R with LSK 1L located in the upper left of the display and LSK 6R located at the lower right of the display.

MODE KEYS

ACARS /ANNUNCIATOR KEY

[Refer to Figure 26.](#)

The ACARS Annunciator is lit when the ACARS system is requesting control of the MCDU. Access the ACARS Menu Page by pressing the ACARS Key.

MSG/ANNUNCIATOR

[Refer to Figure 26.](#)

The Message Annunciator is lit when an FMS message requiring pilot attention or action is displayed on the Message Page. The Message Page is accessed by pressing the MSG Key.

PERF

[Refer to Figure 26.](#)

Press the Perf Key to access the Performance Initialization pages. These pages are used to initialize the performance function. The pages include:

Performance Initialization Pages

- Fuel Flow Page
- Climb Speed Page
- Cruise Speed Page
- Descent Speed Page

TUNE

[Refer to Figure 26.](#)

Press the Tune Key to access the radio tuning pages. These pages display left and right side tuning of NAV, ADF, and transponder and the active and preset frequencies for COMM 1, 2, and 3.



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EXEC/ANNUNCIATOR

[Refer to Figure 26.](#)

The Execute Annunciator is lit when a modified flight plan exists. In this case the modification must be executed (by selection of the EXEC Key) in order to confirm the modifications and to make them part of the active flight plan.

D ARROW KEY

[Refer to Figure 26.](#)

Press the Direct-To Key to access the Direct-To and subsequent Closest Airport Pages. Allows both Lateral and Vertical Direct-To entries to be made.

HDG KEY

[Refer to Figure 26.](#)

Pressing the Heading Key accesses the Heading Page. This page allows direct entry of a heading to be flown and provides capability to define a path to be intercepted by the specified heading.

FPL KEY

[Refer to Figure 26.](#)

The Flight Plan Key allows access to the Active Flight Plan. From the Flight Plan pages the following pages are accessed:

- Lateral Modification Page
- Next Waypoint List Page
- Airways Pages
- Hold Page

- Vertical Modification Page

DEP ARR KEY

[Refer to Figure 26.](#)

Pressing the Depart/Arrive Key displays the Depart/Arrive Page. This page allows initialization of an origin and/or destination and provides access to the Departure, Arrival, Approach and Flight Plan pages.

LNAV KEY

[Refer to Figure 26.](#)

Pressing the LNAV Key provides access to the Lateral Navigation Pages and the following subsequent page:

- GPS Sensor Pages
- VPU Sensor Pages
- Navaid Inhibit Page

VNAV KEY

[Refer to Figure 26.](#)

Pressing the VNAV Key gives access to the Vertical Navigation Page.

FMS MENU

[Refer to Figure 26.](#)

Pressing the FMS Menu Key displays the FMS Menu Page and the following subsequent pages:

- Initialization Page
- Waypoint Information Page
- Flight Plan Initialization Pages



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- Route List Pages
- Select Route Pages
- Performance Init Pages
- Stored Routes Pages
- Secondary Flight Plan Index Page
- Pilot Waypoint List Page
- Performance Initialization Pages
- Fuel Page
- Destination RAIM Status Page
- Flight Log Page
- Cold Temperature Compensation Page
- Maintenance Page
- Database Status Page
- Keyboard Compensation Page
- Parallax Adjustment Page
- GPS Test Page
- Database Update Page
- Configuration Options Page
- Unit Information Page

ALPHA KEYS

[Refer to Figure 26.](#)

The alpha keys are used to enter the 26 letters of the alphabet, and a period.

NUMERIC KEYS

[Refer to Figure 26.](#)

The numeric keys are used to enter numbers 0 to 9, $\pm$ and $\div$.

NOTE

Repeated presses of the + key toggles between + and –.

SCRATCHPAD LINE

[Refer to Figure 26.](#)

The bottom line (line 14) of the display serves as a scratchpad line used by the pilot for data entry and by the connected subsystem (e.g. FMS) for the display of certain messages.

SECTION 2

PAGE DESCRIPTIONS AND ORGANIZATION

The following discussion is to acquaint the user with the organization of the pages and the type of data contained on these pages. This discussion will describe the page displays associated with a specific Mode Key and show the organization of these page displays.

POWER-UP

ON KEY



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SELF TEST PAGE

[Refer to Figure 26.](#)

Displayed during the first 30 seconds after the system is turned on while the computer performs extensive internal tests. If a test detects a fault, the SELF TEST Page will remain displayed and will indicate the fault conditions, otherwise, the system will proceed to the FMS IDENT Page.

FMS IDENT PAGE

NAV DATABASE

[Refer to Figure 27.](#)

This is the navigation database identification.

DATABASE EXPIRES

[Refer to Figure 27.](#)

The date displayed in this field is the date the current navigation database expires.

FMS PART

[Refer to Figure 27.](#)

This is the part number of the Global Star 2100 installed in the aircraft.

DATE

[Refer to Figure 27.](#)

The current Greenwich Date is displayed (and entered if needed) as day, month and year. Values last known to the FMS are pre-filled. If the date is not available, yellow boxes will be displayed.

UTC

[Refer to Figure 27.](#)

Time of day is displayed in Coordinated Universal Time. Values last known to the FMS are pre-filled. If the time is not available, yellow boxes will be displayed. The DATE and UTC fields can be modified, in which case the entered data would be displayed in large font. A value must be present in these fields before proceeding.

INITIALIZATION>

[Refer to Figure 27.](#)

This prompt is displayed after a date and time are entered. Selecting this prompt will initialize the system date and time, however, GPS date and time takes priority if already acquired. LSK 6R must be pressed to exit this page at initial startup, otherwise the user may select other mode keys to exit this page.

INITIALIZATION 1/1 PAGE

GPS POSITION

[Refer to Figure 28.](#)

This field displays the GPS position in latitude and longitude.

ORIGIN

The origin identifier and position last known to the system or entered from the database is displayed here.

RWY, WPT, OR GATE

[Refer to Figure 28.](#)

A single field is provided for entry of a runway, waypoint or gate (gate is displayed only if there are gates in the database). Once an ident is entered, a corresponding position is displayed under the appropriate



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field label. This label is displayed as either RWY POSITION, WPT POSITION or GATE POSITION. If nothing is entered in the identifier field, no position will be displayed.

INIT POSITION

[Refer to Figure 28.](#)

This field displays the position with which the FMS is initialized. This position may be selected by pressing LSK 1R, 2R or 3R, whichever corresponds to the desired selection.

Pressing LSK 5R initializes the FMS position, displays the initialization position in green and displays the FLT PLAN INIT> prompt. Selection of LSK 6R on the initialization page will display the FLT PLAN INIT page. Attempting to access any other pages before the position is confirmed results in CONFIRM INIT POSITION being displayed in the scratchpad field.

FLT PLAN INIT 1/2

[Refer to Figure 29.](#)

This is the first of two Flight Plan Initialization Pages. It provides a window to select routes, display or modify performance data, select origin/destination pairs, modify flight plans and initialize a selected flight plan. If an attribute of a selected route is changed, MODIFY will be displayed at the top left.

COMPANY ROUTE

This field displays the identifier for a selected company route.

ORIGIN

This field allows selection of the four character identifier of the origin airport of the flight.

DEST

This field allows selection of the four character identifier of the destination airport.

DEPART TIME

This field allows entry of the four character estimated time of departure in UTC format.

FLT NO

This field displays the assigned flight number. This field is not filled from another source and must be entered manually. Up to seven characters maybe entered.

<PERF INIT

Pressing LSK 5L will access the PERF INIT 1/2 Page to initialize performance data.

<SECOND

Pressing LSK 6L will access the SECOND FLIGHT PLAN INIT page for selection of the secondary flight plan.

ROUTE LIST

Pressing LSK 5R will access the COMPANY ROUTE LIST 1/X Page. This page allows selection of a company route from the database. Routes between the selected origin/destination pair are listed. Routes are displayed based on the ORIGIN or ORIGIN/DEST pair. All route are displayed if neither ORIGIN or DEST are specified.

FLT PLAN>

Pressing LSK 6R will access the Flight Plan Pages. These pages allow the building of new flight plans and the modification of existing flight plans.



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FLT PLAN INIT 2/2

[Refer to Figure 30.](#)

This is the second of two Flight Plan Initialization Pages. It provides a window to select routes, display or modify performance data, select origin/destination pairs, modify flight plans and initialize a selected flight plan. See Figure 2–6. If an attribute of a selected route is changed, MODIFY will be displayed at the top left.

STORED ROUTE

This field displays the identifier for a selected stored route.

All other fields on this page are the same as those on the FLT PLAN INIT 1/1 Page. The only difference is they pertain to a stored route instead of a company route.

DEP/ARR KEY

DEP ARR KEY

DEP/ARR 1/2 PAGE

[Refer to Figure 31.](#)

This is the first of two Departure/Arrival Pages. It contains the ORIGIN and DEST fields for the active flight plan. It also contains links to the departure, arrival and approach selection pages.

Selection of the NEXT key or the DEP/ARR key will access the DEP/ARR 2/2 page, which contains the same information for the secondary flight plan.

ORIGIN

[Refer to Figure 32.](#)

This field allows selection of the four character identifier of the origin airport of the flight. If the ORIG has been entered during the initialization process, then it is shown here. See Figure 2–8. If any data is not yet defined, the field will contain dashes. This field is modifiable on this page, and will result in a modified flight plan.

DEST

This field allows selection of the four character identifier of the destination airport. If the DEST has been entered during the initialization process, then it is shown here. If any data is not yet defined, the field will contain dashes. This field is modifiable on this page, and will result in a modified flight plan. Entry of DEST will delete any procedures associated with the prior DEST, but will not affect the rest of the flight plan.

<DEPART

Pressing LSK 4L will access the DEPART 1/1 Page to allow stringing of the origin runway, SID and SID/enroute transition for the active flight plan. The <DEPART prompt is displayed only if an origin is defined and the aircraft is not airborne. Selection of this prompt when there are no procedures associated with the defined origin will result in the scratchpad message NO PROCEDURE AVAILABLE.

ARRIVE>

Pressing LSK 4R will access the ARRIVE 1/1 Page to allow stringing of a runway, STAR or STAR/en–route transition for the active flight plan. The ARRIVE> prompt is displayed only if a destination is defined. Selection of this prompt when there are no procedures associated with the defined destination will result in the scratchpad message NO PROCEDURE AVAILABLE.



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APPROACH>

Pressing LSK 5R will access the APP 1/1 Page to allow stringing of the destination runway, approach and approach transition for the active flight plan. The APPROACH> prompt is displayed only if a destination is defined. Selection of this prompt when there are no procedures associated with the defined destination will result in the scratchpad message NO PROCEDURE AVAILABLE.

<RETURN

Pressing LSK 6R will access the Active Flight Plan Page.

FLT PLAN

Pressing LSK 6R will access the Active Flight Plan Page.

DEP/ARR 2/2 PAGE

[Refer to Figure 33.](#)

This is the second of two Departure/Arrival Pages. It contains the ORIGIN and DEST fields for the secondary flight plan. It also contains links to the departure, arrival and approach selection pages.

Selection of the PREV key or the DEP/ARR key will access the DEP/ARR 1/2 page, which contains the information for the active flight plan.

DEPART 1/N PAGE

[Refer to Figure 34.](#)

This is the first DEPARTURE/RUNWAY Page. It displays a list of runways defined at the ORIGIN airport. If the list is longer than four entries, then the NEXT key will provide the next page of runways (DEPART 2/N). The runways are initially cyan, indicating that they are available, but not selected.

RUNWAY — LENGTH

These fields display the identifier and length of runways defined at the ORIGIN airport. Selection of a runway modifies the flight plan and automatically goes to the DEPART/SID page.

RWY

This field displays the selected runway. Dashes are displayed if no runway is selected.

SID>

This field displays the selected SID. Dashes are displayed if no SID is selected. Pressing LSK 5R will access the DEPART/SID Page. If no SIDs are defined for the selected runway, this prompt does not appear.

FLT PLAN>

Pressing LSK 6R will access the Active Flight Plan Page.

DEPART/SID PAGE

[Refer to Figure 35.](#)

This is the first DEPARTURE/SID Page. It displays a list of SIDs defined for the selected runway. If the list is longer than eight entries, then the NEXT key will provide the next page of SIDs. The SIDs are initially cyan, indicating that they are available, but not selected.

SIDS

This field displays the identifier of SIDs defined for the selected runway. Selection of a SID modifies the flight plan and automatically goes to the DEPART/TRANSITION page.

<RWY



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

This field displays the selected runway. The selected runway is not modifiable on this page. Modification must be done on the DEPART/RUNWAY Page. Dashes are displayed if no SID is selected. Pressing LSK 5L will access the DEPART/RUNWAY Page.

SID

This field displays the selected SID. Dashes are displayed if no SIDs selected.

FLT PLAN>

Pressing LSK 6R will access the Modify Flight Plan Page for the active flight plan.

DEPART/TRANSITION PAGE

[Refer to Figure 36.](#)

This is the first DEPARTURE/TRANSITION Page. It displays a list of transitions defined for the selected runway/SID. If the list is longer than eight entries, then the NEXT key will provide the next page of transitions. The transitions are initially cyan, indicating that they are available, but not selected.

TRANSITIONS

This field displays the identifier of transitions defined for the selected runway/SID. Selection of a transition adds the selection to the departure procedure and returns to the DEPART/RUNWAY page.

<RWY

This field displays the selected runway. The selected runway is not modifiable on this page. Modification must be done on the DEPART/RUNWAY Page. Dashes are displayed if no runway is selected. Pressing LSK 5L will access the DEPART/SID Page. If no

SIDs are defined for the selected runway, this prompt does not appear.

SID>

This field displays the selected SID. The selected SID is not modifiable on this page. Modification must be done on the DEPART/SID Page. Pressing LSK 5R will access the DEPART/SID Page. If no SIDs are defined for the selected runway, this prompt does not appear.

TRAN

This field displays the selected transition. Dashes are displayed if no transition is selected.

FLT PLAN

Pressing LSK 6R will access the Modify Flight Plan Page for the active flight plan.

ARRIVE 1/N PAGE

[Refer to Figure 37.](#)

This is the first ARRIVAL/RUNWAY Page. It displays a list of runways defined at the DEST airport. If the list is longer than four entries, then the NEXT key will provide the next page of runways (ARRIVE 2/N). The runways are initially cyan, indicating that they are available, but not selected.

RUNWAY – LENGTH

These fields display the identifier and length of runways defined at the DEST airport. Selection of a runway modifies the flight plan and automatically goes to the ARRIVE/STAR page.

RWY



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This field displays the selected runway. Dashes are displayed if no runway is selected.

STAR>

This field displays the selected STAR. Dashes are displayed if no STAR is selected. Pressing LSK 5R will access the ARRIVE/STAR Page. If no STARs are defined for the selected runway, this prompt does not appear.

FLT PLAN>

Pressing LSK 6R will access the Active Flight Plan Page.

ARRIVE/STARS PAGE

[Refer to Figure 38.](#)

This is the first ARRIVE/STAR Page. It displays a list of STARs defined for the selected runway. If the list is longer than eight entries, then the NEXT key will provide the next page of STARs. The STARs are initially cyan, indicating that they are available, but not selected.

STAR

This field displays the identifier of STARs defined for the selected runway. Selection of a STAR modifies the flight plan and automatically goes to the ARRIVE/TRANSITION page.

<RWY

This field displays the selected runway. The selected runway is not modifiable on this page. Modification must be done on the ARRIVE/RUNWAY Page. Dashes are displayed if no runway is selected. Pressing LSK 5L will access the ARRIVE/RUNWAY Page.

STAR

This field displays the selected STAR. Dashes are displayed if no STAR is selected.

FLT PLAN

Pressing LSK 6R will access the Modify Flight Plan Page for the active flight plan.

ARRIVE/TRANSITION PAGE

[Refer to Figure 39.](#)

This is the first ARRIVE/TRANSITION Page. It displays a list of transitions defined for the selected runway/STAR. If the list is longer than eight entries, then the NEXT key will provide the next page of transitions. The transitions are initially cyan, indicating that they are available, but not selected.

TRANSITIONS

This field displays the identifier of transitions defined for the selected runway/SID. Selection of a transition adds the selection to the departure procedure and returns to the ARRIVE/RUNWAY page.

<RWY

This field displays the selected runway. The selected runway is not modifiable on this page. Modification must be done on the ARRIVE/RUNWAY Page. Dashes are displayed if no runway is selected. Pressing LSK 5L will access the ARRIVE/RUNWAY Page.

STAR>

This field displays the selected STAR. The selected STAR is not modifiable on this page. Modification must be done on the ARRIVE/STAR Page. Pressing LSK 5R will access the ARRIVE/STAR Page. If no STARs are defined for the selected runway, this prompt does not appear.



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TRAN

This field displays the selected transition. Dashes are displayed if no transition is selected.

FLT PLAN>

Pressing LSK 6R will access the Modify Flight Plan Page for the active flight plan.

APPROACH 1/N PAGE

This is the first APPROACH Page. It displays a list of approaches defined at the DEST airport. If the list is longer than four entries, then the NEXT key will provide the next page of approaches (APPROACH 2/N). The approaches are initially cyan, indicating that they are available, but not selected.

APPROACHES

This field displays the identifier of approaches defined at the DEST airport. Selection of an approach modifies the flight plan and automatically goes to the APPROACH/TRANSITION Page.

<RWY

This field displays the selected runway. The selected runway is not modifiable on this page. Modification must be done on the APPROACH/RUNWAY Page. Dashes are displayed if no runway is selected. Pressing LSK 5L will access the APPROACH/RUNWAY Page.

APP

This field displays the selected APP. Dashes are displayed if no APP is selected.

FLT PLAN

Pressing LSK 6R will access the Active Flight Plan Page.

APPROACH/TRANSITION PAGE

This is the first APPROACH/TRANSITION Page. It displays a list of transitions defined for the selected approach. If the list is longer than eight entries, then the NEXT key will provide the next page of transitions. The transitions are initially cyan, indicating that they are available, but not selected.

TRANSITIONS

This field displays the identifier of transitions defined for the selected approach. Selection of a transition adds the selection to the departure procedure and returns to the APPROACH/RUNWAY page.

<RWY

This field displays the selected runway. The selected runway is not modifiable on this page. Modification must be done on the APPROACH/RUNWAY Page. Dashes are displayed if no runway is selected. Pressing LSK 5L will access the APPROACH/RUNWAY Page.

APP>

This field displays the selected APP. The selected APP is not modifiable on this page. Modification must be done on the APPROACHES Page. Pressing LSK 5R will access the APPROACHES Page. If no APPs are defined for the selected runway, this prompt does not appear.

TRAN

This field displays the selected transition. Dashes are displayed if no transition is selected.

FLT PLAN>



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Pressing LSK 6R will access the Modify Flight Plan Page for the active flight plan.

FLIGHT PLAN KEY

FPL KEY

ACTIVE FLT PLAN 1/N PAGE

[Refer to Figure 40.](#)

The ACTIVE FLT PLAN 1/N page is accessed by pressing the FPL Key. The waypoints, airports, etc. defined in the active flight plan are displayed on these pages.

Data associated with the active flight plan is shown in green, with the exception that the current leg is shown in magenta.

FROM

The identifier for the flight plan from waypoint is shown in this field.

ETA

This field displays the estimated time of arrival at the next waypoint.

SPD/ALT

This field displays the planned speed and altitude for the associated leg of the flight plan. The speed can be in either knots or MACH, and the altitude can be either feet or flight level.

Associated with each waypoint, on the line above the Identifier, ETA and speed/altitude are the desired track and leg distance.

<MORE DATA

[Refer to Figure 41.](#)

Selection of LSK 6L results in the replacement of the ETA, SPD/ALT and leg distance with FUEL, TRUE WIND and distance to destination. Re-selection of LSK 6L switches back to the original data.

FUEL

This field displays the estimated fuel remaining at the waypoint.

TRUE WIND

This field displays the wind velocity and wind true bearing for the associated leg of the flight plan.

SECONDARY>

[Refer to Figure 42.](#)

Selection of LSK 6R results in display of the secondary flight plan. Data associated with the secondary flight plan is shown in white.

LATERAL ACTIVE MODIFY AT 1/1 PAGE

[Refer to Figure 43.](#)

Selection of LSK xL on ACTIVE FLT PLAN 1/N page will bring up the lateral MODIFY AT page for the associated waypoint.

Additions and modifications to a flight plan are made using the MODIFY page. The chosen modification point and the latitude/longitude are displayed below the title line.

They are displayed in green to indicate that the active flight plan is being modified.

NEXT WPT



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Selection of LSK 2L allows insertion of a waypoint into the modified flight plan following the modification point and returns to the MODIFY FLT PLAN N/N page.

ATO DIST

Selection of LSK 3L allows entry of an Along Track Offset distance and creates a waypoint based upon the ATO DIST from the modification point, then returns to the MODIFY FLT PLAN N/N page.

<HOLD

Selection of LSK 4L results in the display of the ACTIVE HOLDING AT 1/1 page for the modification point to activate a holding pattern at the modification point.

<WPT INFO

Selection of LSK 6L results in display of the WAYPOINT INFO 1/1 page for the modification point.

NEW DEST

Selection of LSK 2R allows entry of a new destination and deletes all waypoints between the modification point and the new destination, then returns to the MODIFY FLT PLAN N/N page.

DELETE UP TO WPT

Selection of LSK 3R allows entry of a waypoint and deletes all waypoints between the modification point and the selected waypoint, the returns to MODIFY FLT PLAN N/N page.

DELETE AAAAA*

Selection of LSK 5R deletes modification point, the returns to the MODIFY FLT PLAN N/N page.

FLT PLAN

Selection of LSK 6R displays the ACTIVE FLT PLAN 1/N page.

[Refer to Figure 44.](#)

When the modification point is the origin airport or runway, the ACTIVE MODIFY AT 1/1 page changes as follows.

The <HOLD, <AIRWAYS and DELETE NNNNN* prompts/functions are not available.

[Refer to Figure 45.](#)

When the modification point is the destination airport or runway, the ACTIVE MODIFY AT 1/1 page changes as follows.

No lateral modification functions are available.

[Refer to Figure 46.](#)

When the modification point is the position selected by pressing the left LSK next to the FROM waypoint on the flight page), the ACTIVE MODIFY AT 1/1 page changes as follows.

The NEST WPT, ATO DIST, NEW DEST, DEL UP TO WPT, <AIRWPTS and DELETENNNNN* prompts/functions are not available.

The <AIRWAYS prompt is replaced by the OFFSET DIS prompt, used to enter or modify (if an offset is currently active) an offset to the current position.

ACTIVE HOLDING AT 1/1 PAGE

[Refer to Figure 47.](#)

The ACTIVE HOLDING AT 1/1 page is selected by LSK 4L on the ACTIVE MODIFY AT 1/1 page. If there is not a holding pattern at the selected waypoint, all data will be in cyan. Data will also be in cyan if a modification is pending (INSERT* has not been pressed).



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INBOUND CRS

Selection of LSK 2L allows insertion of an inbound course for the holding pattern. The default value will be the desired track.

INBOUND TIME

Selection of LSK 3L allows insertion of a value for the length of time for the inbound leg. This field is linked with the LEG DIST field so that when data is entered in one field, the other field is calculated. The default for INBOUND TIME is one minute.

LEG DIST

Selection of LSK 4L allows insertion of a value for the length of the inbound leg. This field is linked with the INBOUND TIME field so that when data is entered in one field, the other field is calculated. The default for LEG DIST is calculated based on the default value of one minute inbound time.

TURN DIR

Selection of LSK 5L allows toggling of direction of the first turn between RIGHT/LEFT.

<RETURN

Selection LSK 6L goes back to the lateral ACTIVE MODIFY AT 1/1 page.

ENTRY TYPE

This field displays the type of entry into the holding pattern (DIRECT, TEARDROP OR PARALLEL).

EXIT TYPE

Selection of LSK 5R allows toggling between AUTO/MANUAL exit from the holding pattern.

INSERT*

Selection of LSK 6R applies the changes to the holding pattern to the flight plan. The prompt changes to FLT PLAN> and the data fields change from cyan to green. If the page is exited before INSERT* is selected, the data changes are lost.

ACTIVE AIRWAYS AT 1/1 PAGE

[Refer to Figure 48.](#)

The ACTIVE AIRWAYS AT 1/1 page is selected by LSK 5L on the ACTIVE MODIFY AT 1/1 page.

AIRWAY

Selection of LSK 1L, 2L or 3L allows selection of an airway and go to the AIRWAY WPTS N/N page for the selected airway.

<RETURN

Selection LSK 6L goes back to the previous page.

AIRWAY WPT N/N PAGE

[Refer to Figure 49.](#)

The AIRWAY WPT N/N page is selected by a left LSK on the ACTIVE AIRWAYS AT 1/1 page. The airway identifier and the starting waypoint are shown in yellow to indicate modification in progress. The display defaults to the page which contains the starting waypoint.

AIRWAY

[Refer to Figure 50.](#)

Selection of an LSK associated with a waypoint selects allows selection of a terminating waypoint for the airway.



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<RETURN

Selection LSK 6L goes back to the previous page.

FLT PLAN

Selection of LSK 6R goes to the MODIFY FLT PLAN page.

VERTICAL ACTIVE MODIFY AT 1/1 PAGE

[Refer to Figure 51.](#)

Selection of LSK xR on ACTIVE FLT PLAN 1/N page will bring up the vertical ACTIVE MODIFY AT page for the associated waypoint. Selection of a climb or cruise waypoint brings up a screen as shown.

Additions and modifications to a flight plan are made using the MODIFY page. The waypoint and data are displayed in green to indicate that the active flight plan is being modified. If the selected waypoint is the “TO” waypoint, the data associated with the waypoint is shown in magenta.

ALT CNSTR

Selection of LSK 2R allows entry of an altitude constraint at the selected waypoint. The altitude constraint can be AT, AT or ABOVE, or AT or BELOW. The entry can also be an altitude window.

SPD CNSTR

Selection of LSK 3R allows entry of a speed constraint at the selected waypoint.

[Refer to Figure 52.](#)

When the modification point is a descent waypoint, the ACTIVE MODIFY AT 1/1 page changes as follows.

This item is added to the screen.

FPA AT

Selection of LSK 4R allows entry of the flight path angle to the destination airport.

For approach, approach transition, missed approach waypoints, or if the waypoint is within 30 NM of the destination, APT TEMP and APT ELEV are added to the screen.

APT TEMP

Selection of LSK 2L allows entry of the temperature at the destination airport.

APT ELEV

This field displays the elevation of the destination airport. When a valid temperature is inserted, the cold temperature compensation can be computed. If the correction is valid, the following items are displayed.

HAA

This field displays the height above the aerodrome.

MSL ALT

This field displays the Mean Sea Level altitude of the destination runway.

CORR

This field displays the cold temperature compensation value for the airport, if valid.

COMP ALT

This field displays the compensated Mean Sea Level altitude of the destination runway.



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Refer to Figure 53.

When the modification point is the origin waypoint, the ACTIVE MODIFY AT 1/1 page appears like this.

Refer to Figure 54.

When the modification point is the destination waypoint, the ACTIVE MODIFY AT 1/1 page appears like this (Figure 2–32). The flight path angle value defaults to –3.0 degrees.

LNAV KEY

LNAV 1/3 PAGE

Refer to Figure 55.

The LNAV 1/3 page is accessed by pressing the LNAV key. The LNAV 1/3 page provides lateral guidance information relative to the active and next waypoint.

Data associated with the active waypoint is shown in magenta.

TITLE LINE

Displays an annunciator when certain lateral modes are active (e.g. HOLD–R).

TO

The identifier (up to seven characters) for the active waypoint. If a turn direction is specified, a right or left arrow will be post fixed to the identifier. If a flyover is specified, a triangle symbol is post fixed. If both a turn direction and flyover are specified, the turn direction symbol is shown.

DTK

The desired track relative to the active waypoint.

DIST

The distance to go relative to the active waypoint.

ETE

The estimated time en–route relative to the active waypoint.

NEXT

The identifier (up to seven characters) for the next waypoint. If the next leg is a holding pattern, then HOLD–R or HOLD–L is displayed next to the NEXT title. The holding pattern entry type is displayed in parentheses (DIRECT, TEARDROP or PARALLEL). If the next leg is a procedure turn, then PROC TURN is displayed next to the NEXT title. If the next leg is an arc turn, then ARC–R or ARC–L is displayed next to the NEXT title.

DTK

The desired track for the next waypoint. The desired track when the next leg is a holding pattern is the hold course. The desired track when the next leg is a procedure turn is the outbound course.

DIST

The distance to go to the next waypoint.

ETE

The estimated time en–route for the next waypoint.

DEST

The identifier (up to four characters) for the current destination.

FUEL

The estimated fuel on board at the current destination. Displayed in yellow if less than Fuel Reserves.



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DIST

The distance to the current destination.

ETA

The estimated time of arrival at the current destination.

BRG

The bearing to the active waypoint.

XTK

The current value of cross-track error. The color changes to yellow when a XTK DEV WARNING is active on the message page.

HDG

The current value of aircraft heading.

TRS

The current value of aircraft airspeed.

GS

The current value of aircraft ground speed.

CDI SCL

The maximum CDI scale deflection value. The default value is 5.0 NM for en-route, 1.0 NM for terminal and 0.3 NM for approach. The value is also pilot enterable. Display field limit: 0.3–5.0 NM.

H/WIND

The current value of the head/tail wind component. A down arrow indicates a head wind.

TRUE WIND

The current value of the wind direction/magnitude.

X/WIND

The current value of the crosswind component. A right or left arrow indicates direction of the crosswind component.

[Refer to Figure 56.](#)

When a parallel offset is active, the LNAV 1/3 page changes as follows.

TITLE LINE

OFST R4 – – Shows the offset direction and value (e.g. R4).

[Refer to Figure 57.](#)

When a holding pattern is active, the LNAV 1/3 page changes as follows.

TITLE LINE

HOLD–R or HOLD–L – – Displayed when a right turning hold or left turning hold is active.

EXIT HOLD – – Displayed when a hold is active and the pilot has selected the EXIT HOLD* prompt.

HOLD ENTRY OR HOLD SEG

Displayed on lines associated with LSK 4R/L when a holding pattern is active. Choices for entry are DIRECT, TEARDROP or PARALLEL. Choices for segment are TEARDROP, PARALLEL, OUTBOUND or INBOUND.

EXIT HOLD*

RESUME HOLD*



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These prompts are displayed for all holds. Selection of EXIT HOLD* (LSK 6R) causes the holding pattern to exit upon crossing the fix. Selection of RESUME HOLD* (LSK 6R) continues the hold.

The desired track is relative to the active segment of the hold. The distance is the along track distance to the hold fix.

NOTE

For an HF type leg, EXIT HOLD is displayed at the entry of the hold and the holding pattern is automatically exited. For an HA type leg, the hold automatically changes to an exit state after reaching the specified altitude.

[Refer to Figure 58.](#)

When a procedure turn is active, the LNAV 1/3 page changes as follows.

TITLE LINE

PROC TURN – – Displayed when a procedure turn is active.

PROC SEG

Displayed on lines associated with LSK 4R/L when a procedure turn is active. Choices for the active segment are OUTBOUND, TURNOUT or INBOUND.

The desired track is relative to the active segment of the procedure turn. The distance is the along track distance to the intercept waypoint.

[Refer to Figure 59.](#)

When an arc turn is active, the LNAV 1/3 page changes as follows.

TITLE LINE

ARC–R or ARC–L – – Displayed when a right arc turn or left arc turn is active.

The desired track is relative to the current position along the arc. The distance is the along track distance to the bisector at the arc fix.

[Refer to Figure 60.](#)

When a manual heading is active, the LNAV 1/3 page changes as follows.

TITLE LINE

HDG MODE – – Displayed when a manual heading is active.

The desired track, bearing, estimated time en–route, distance and crosstrack error are referenced to the next leg.

[Refer to Figure 61.](#)

When the FMS flight mode is APPROACH, the LNAV 1/3 page changes as follows.

GO MISSED*

This prompt is displayed for APPROACH mode. Selection of GO MISSED* (LSK 6R) enables the missed approach, enabling MAP sequencing and changes the missed approach waypoint data from cyan to green.

LNAV 2/3 PAGE

[Refer to Figure 62.](#)

The LNAV 2/3 page is accessed by pressing the NEXT key or the LNAV key from the LNAV 1/3 page, or pressing the PREV key from the LNAV 3/3 page. The LNAV 2/3 page provides FMS position, navigation data and system error data.

FMS POSITION



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The current aircraft position.

HDG

The current value of aircraft heading.

TK

The current value of aircraft track angle.

DRIFT

The current value of aircraft drift angle.

MAG VAR

The current value of magnetic variation.

EPE

The current estimated position error of the FMS.

TSE

The current total system error of the FMS. TSE will be displayed in yellow if the value is greater than RNP.

RNP

The current value for required navigation performance of the FMS.

[Refer to Figure 63.](#)

LNAV 3/3 PAGE

The LNAV 3/3 page is accessed by pressing the NEXT key or the LNAV key from the LNAV 2/3 page, or pressing the PREV key from the LNAV 1/3 page. The LNAV 3/3 page provides FMS position, the difference between FMS position and each available sensor position, and status of each position sensor. If there are more than five

sensors, an additional LNAV 4/4 page can be accessed. Similar sensors will be grouped together on the same page.

FMW POSITION

The current aircraft position.

<(SENSOR NAME)

The identifiers of the configured sensors (e.g. GPS1, VPU).

DIST

The radial position difference between this sensor and the FMS position.

STATUS

Indicates which sensors are being used for FMS position determination. The sensor(s) being used are shown in green, with a status of ACTIVE. Sensors that are available, but not being used are cyan, with a status of AVAILABLE. Sensors that are not navigating show the status NOT IN NAV.

[Refer to Figure 64.](#)

GPS1 SENSOR 1/2 PAGE

The GPS1 SENSOR 1/2 page is accessed by pressing the LSK next to the identifier for GPS1 (e.g. LSK 2L). This page displays FMS position, GPS1 position and the difference between them. Also displayed are details about the GPS1 status.

FMS POSITION

The current aircraft position calculated by the FMS.

GPS POSITION

The current aircraft position calculated by GPS1.



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DIFFERENCE

The difference between FMS and GPS1 aircraft position.

HPE

The current horizontal position error of the GPS.

MODE

The current navigation mode. Choices are ENROUTE, TERM, APPRCH, OCEANIC, or UNKNOWN.

UTC

The current value of universal coordinated time.

STATE

The current GPS operating state. Choices are INIT, SRCH, ACQ, TRANS, NAV, DR, or UNKNOWN.

ALT

Indication of altitude availability. Choices are YES or NO.

POS UNCERT

The amount of uncertainty in the current position of the GPS.

* SATS

The number of satellites being tracked (up to 99).

RAIM

The availability of RAIM at the destination. Choices are AVAIL, NOFDE or UNAVAIL.

<RETURN

Selection of LSK 6L results in display LNAV 3/3 page.

[Refer to Figure 65.](#)

GPS1 SENSOR 2/2 PAGE

The GPS SENSOR 2/2 page is accessed by pressing the NEXT key from the GPS1 SENSOR 1/2 page. This page displays the status of each GPS channel.

PRN

The Psuedo Range Number.

AZM

The current azimuth of the satellite from the aircraft position.

EL

The current elevation of the satellite from the aircraft position.

SNR

The current value of signal-to-noise ratio for this satellite.

HLTH

The health status of the satellite. Choices are GOOD, WEAK, BAD or DESEL.

T

The current status of the satellite tracking. Y — being tracked, N — not being tracked.

<RETURN

Selection of LSK 6L results in display of LNAV 3/3 page.

[Refer to Figure 66.](#)

VPU SENSOR 1/2 PAGE



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The VPU SENSOR 1/2 page is accessed by pressing the LSK next to the identifier for VPU (e.g. LSK 4L). This page displays FMS position, VPU position and the difference between them. Also displayed are mode status and estimated position error.

FMS POSITION

The current aircraft position calculated by the FMS.

VPU POSITION

The current aircraft position calculated by the VPU.

DIFFERENCE

The difference between FMS and VPU aircraft position.

MODE

The current VPU navigation mode. Choices are NO MEASUREMENTS, BAD GEOMETRY, VOR/DME or DME/DME.

EPE

The estimated position error of the VPU.

<RETURN

Selection of LSK 6L results in display of LNAV 3/3 page.

NAVAID INHIBIT>

Selection of LSK 6R results in display of NAVAID INHIBIT 1/1 page.

[Refer to Figure 67.](#)

NAVAID INHIBIT 1/1 PAGE

The NAVAID INHIBIT 1/1 page is accessed by pressing LSK 6R on the VPU SENSOR 1/2 page. This page displays up to ten nav aids excluded from use for navigation.

IDENT/FREQ

FREQ/IDENT

The identifier and frequency for an excluded nav aid.

<RETURN

Selection of LSK 6L results in display of VPU SENSOR 1/2 page.

NAV TUNE LIST>

Selection of LSK 6R results in display of NAV TUNE LIST 1/1 page.

[Refer to Figure 68.](#)

VPU SENSOR 2/2 PAGE

The VPU SENSOR 2/2 page is accessed by pressing the NEXT key from the VPU SENSOR 1/2 page. This page displays the identifier, frequency, range and bearing for the currently tuned nav aids.

IDENT

The identifier for a tuned nav aid. Up to four characters.

FREQ

The frequency for a tuned nav aid.

RNG

The distance to the nav aid.

BRG

The bearing to the nav aid.

<RETURN



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Selection of LSK 6L results in display of LNAV 3/3 page.

[Refer to Figure 69.](#)

VNAV KEY

VNAV 1/1 PAGE

The VNAV 1/1 page is accessed by pressing the VNAV key. The VNAV 1/1 page provides vertical guidance information relative to the active and next altitude target. The information provided is dependent on the active phase of flight.

Data associated with the active target is shown in magenta.

ACTIVE:

The active vertical mode.

XX

The active target altitude. XX will be TO (climb or descent) or AT (cruise).

TARGET SPD

The climb target speed.

TO XXX

The active vertical destination identifier or altitude. XXX indicates the type of destination (e.g. WPT, TOC, TOD, MAP, FAF, etc).

DIST

The distance to the altitude target.

ETE

The estimated time en-route to the altitude target.

CRUISE ALT

The cruise altitude for the next cruise segment.

CLIMB GRAD

The climb gradient for the next segment.

NEXT:

The vertical mode for the next segment.

XX

The active target altitude. XX will be TO (climb or descent) or AT (cruise).

BARO ALT

The current barometric altitude.

[Refer to Figure 70.](#)

VNAV cruise example. The fields are the same as for climb, except CLIMB GRAD is removed.

[Refer to Figure 71.](#)

VNAV descent example. The fields are the same as for climb, except for REQ VSPD and VERT DEV.

REQ VSPD

The required vertical speed for a descent segment. LO (negative) / HI (positive).

VERT DEV



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The vertical deviation for a descent segment. LO (negative) / HI (positive).

[Refer to Figure 72.](#)

TUNE KEY

TUNE 1/1 PAGE

The TUNE 1/1 page is accessed by pressing the TUNE Key. The COMM radios that have been configured for tuning through the MCDU are displayed on these pages. This page also displays the NAV radios, transponders and ADFs that can be tuned.

The COMM radios can be tuned manually or by selecting frequencies from the COMM TUNE LIST page.

The NAV radios can also be manually tuned by entering the desired frequency or selecting a navaid identifier from the NAV TUNE LIST page.

ACTIVE

The active frequency for each COMM is displayed in this field.

PRESET

This field displays the standby frequency for each COMM. The active and preset frequencies can be toggled by first pressing the LSK adjacent to the preset frequency which will become the active frequency. This enters the frequency into the scratchpad field. Then, press the LSK adjacent to that COMM's active field. The preset frequency is now placed in the field and the former active frequency pushes over the preset field.

NAV1 AUTO

The number one NAV radio's tuned frequency and associated identifier are displayed in this field. AUTO is displayed showing that auto-tuning is selected. The frequency and identifier are displayed in small font when auto tune is selected. If auto tune is not selected, AUTO is not displayed and the frequency and identifier are displayed in large font.

NOTE

The FMS cannot auto tune a NAV channel if that channel's raw data is currently displayed on the EFIS. If the pilot tries to select auto-tuning in this context, NOT ALLOWED is displayed in the scratchpad field. Manually setting the channel to a particular frequency is still allowed.

NAV2

The number two NAV radio's frequency and identifier are displayed in this field.

XPDR1 AND XPDR2

Transponders and their associated squawk is displayed here.

ADF1 AND ADF2

These fields display the Automatic Direction Finder systems installed and the frequencies to which they are presently tuned.

COMM LIST>

Pressing LSK 6R adjacent to this prompt will display the COMM TUNE LIST 1/1 Page. This page displays a list of the radio frequencies available at any selected airport.

<NAV LIST

Pressing LSK 6L adjacent to this prompt will display NAV TUNE LIST 1/1 page.



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If the list tunable items is too large to fit on a single page, then a second page will be used to display the data. The second page is accessed by pressing the NEXT KEY or a second press of the TUNE KEY.

[Refer to Figure 73.](#)

COMM TUNE LIST 1/2 PAGE

This page is accessed by pressing LSK 6R on the first TUNE page.

AIRPORT

This page is accessed by pressing the selected airport name and identifier. The system defaults to the IRIFIN airport when departing any changes to the DESTINATION airport when arriving. The default identifier is displayed in small font.

The pilot can select any airport in the database by entering the identifier in the scratchpad field and pressing LSK 1L. A pilot entered airport is displayed in large font.

All frequencies displayed in the various fields are those available at the selected airport. Any of these frequencies can be tuned by pressing the LSK adjacent to the desired frequency. This places the frequency in the scratchpad field on the first TUNE page. Next, press the LSK adjacent to the desired field of entry.

If the list of frequencies is too large to fit on a single page, then additional pages will be used to display the data. The additional pages are accessed by pressing the PREV or NEXT KEY.

[Refer to Figure 74.](#)

MORE>

When the MORE> prompt is selected, additional data is provided for each frequency. The COMM TUNE LIST page is expanded to show

only one frequency per page. Pressing <RETURN (LSK 6L) returns to the condensed version of the page.

SECTOR

The SECTOR description is a range of radials. If the radial is referenced to any point other than the airport reference point, the identifier will be displayed to the right of the frequency.

ALTITUDE

The ALTITUDE description is used to indicate altitude restrictions. The restriction can be a BELOW restriction, and ABOVE restriction or two altitudes indicating an altitude band. The altitude is provided in terms of feet or Flight Level (e.g. BELOW FL230, ABOVE 1000 or FL240 — 1000).

DESCRIPTION

The DESCRIPTION field is used for any additional information about the restriction on the frequency.

The SECTOR, ALTITUDE and DESCRIPTION fields are optional, and will be blank when no data is provided.

[Refer to Figure 75.](#)

NAV TUNE LIST 1/1 PAGE

This page is accessed by pressing LSK 6L on the first TUNE page and on the NAVAID INHIBIT page.

The displayed navaid identifiers are those closest to the current FMS position within 250 NM. This page will not show TERMINAL class navaids when the difference between aircraft altitude and navaid site elevation is more than 12,000 feet. LOW altitude class navaids are not shown when the difference is greater than 18,000 feet. Navaids that have been inhibited are shown in yellow.



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Any of these frequencies can be tuned by pressing the LSK adjacent to the desired frequency. This places the frequency in the scratchpad field on the first TUNE page. Next, press the LSK adjacent to the desired NAV field.

[Refer to Figure 76.](#)

DIRECT TO KEY

DIRECT TO N/N PAGE

The DIRECT TO N/N page is accessed via the DIRECT-TO key on the MCDU, allowing both lateral and vertical Direct To selection.

When Direct-To is selected, the current TO waypoint in the active flight plan is displayed in LSK 2L. All waypoints and altitudes that are selectable for going direct are displayed in cyan. The remaining waypoints are displayed as they would be displayed in the active flight plan.

FROM

The identifier for the flight plan from waypoint is shown in this field.

[Refer to Figure 77.](#)

Waypoint

This field displays the identifier of waypoints in the active flight plan. Selection of LSK 2L, 3L, 4L or 5L initiates a lateral Direct-To the associated waypoint. Selection of LSK 2R, 3R, 4R or 5R initiates a vertical Direct-To the associated waypoint. Either results in display of the MODIFY FLT PLAN 1/3 page with the associated waypoint as the direct to waypoint, ready for execution.

ETA

This field displays the estimated time of arrival at the next waypoint.

SPD/ALT

This field displays the planned speed/altitude for the associated leg. The speed is in knots or MACH, and altitude is in feet or flight level.

Associated with each waypoint, on the line above the Identifier, ETA and speed/altitude are the desired track and leg distance. The FPA value is displayed in yellow if the magnitude is greater than 6.0.

<RETURN

Selection of LSK 6L results in display of the previous page.

[Refer to Figure 78.](#)

CLOSEST AIRPORT>

Selection of LSK 6R results in display of DIRECT CLOSEST APT 1/2 page.

This page displays up to eight closest airports within 250 NM of the current FMS position, in order of closest to furthest. If an airport listed is in the active flight plan, it will be displayed in green, all others are displayed in cyan.

AIRPORT NAME

This field displays the name of the airport.

AIRPORT IDENTIFIER

This field displays the identifier of the airport.

MAXLEN (FT)

This field displays the length of the longest runway at the airport.

BRG

This field displays the bearing from the current position to the airport.



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DIST (NM)

This field displays the distance from the current aircraft position to the airport.

<RETURN

Selection of LSK 6L results in display of the DIRECT TO N/N page.

[Refer to Figure 79.](#)

HDG KEY

HEADING 1/N PAGE

The HEADING 1/N page is accessed via the HDG key on the MCDU, allowing definition of a heading to be flown and a path to be intercepted by the specified heading.

HDG

This field displays the selected heading.

NOTE

[Refer to Figure 80](#) and [81](#) For systems that have a heading pre-selector input, the heading field defaults to the pre-selector value and the field is not modifiable. For systems that do not have a pre-selector input, the heading field defaults to the current heading and is modifiable.

NOTE

[Refer to Figure 82.](#), If the heading page is accessed while heading mode is active, all active parameters will be displayed in green.

[Refer to Figure 83.](#)

INTERCEPT LEG

These fields display the flight plan waypoints and flight plan leg track.

[Refer to Figure 84.](#)

*TO/FROM WPT

This prompt is displayed only when an intercept leg has been defined. It is used to specify whether to fly the specified course inbound to the waypoint or outbound from the waypoint. The default is TO, but is selectable.

INSERT*

Pressing the INSERT* prompt makes the change to the modified flight plan.

[Refer to Figure 85.](#)

PERF KEY

PERF INIT PAGE

PERF INIT 1/2

The PERF INIT 1/2 Page contains performance initialization data for the flight plan displayed in the title line. System defaults are shown in small font. If modifications are made, the operator entered values will be displayed in large font. All entries will become the system default values as long as they are not cleared by the operator. Clearing an entered value returns the value to the system default. Changes made on this page are intended to be long term changes which apply for general use on the aircraft in which the system is installed. Temporary changes to these values can be made on other pages, but entries here remain the system defaults even on power down.

CRUISE ALT

This field displays and allows modification of the system default aircraft cruise altitude. Changes to the active CRUISE ALT results in



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a modified flight plan. The CRUISE ALT is displayed as boxes until the pilot enters the cruise altitude or sets the altitude pre-selector. The pre-selector input to this page does not cause a modified flight plan and the pre-selector values are displayed in small font.

FUEL FLOW

This field displays and allows modification of the system default aircraft fuel flow. Selection of LSK 2L results in the display of the FUEL FLOW page. The FUEL FLOW page allows modification and selection of the predicted fuel flows and the prediction model for the phases of flight. This page will be explained in later paragraphs.

CLIMB SPD

This field displays and allows modification of the system default aircraft climb speed. Selection of LSK 3L results in the display of the CLIMB SPD page. The CLIMB SPD page allows modification and selection of the climb speeds for the flight. This page will be explained in later paragraphs.

CRUISE SPD

This field displays and allows modification of the system default aircraft cruise speed. Selection of LSK 4L results in the display of the CRUISE SPD page. The CRUISE SPD page allows modification and selection of the cruise speed for the flight. This page will be explained in later paragraphs.

DESCENT SPD

This field displays and allows modification of the system default aircraft descent speed. Selection of LSK 5L results in the display of the DESCENT SPD page. The DESCENT SPD page allows modification and selection of the descent speed for the flight. This page will be explained in later paragraphs.

TRANS ALT

This field displays and allows modification of the system default aircraft departure transition altitude. This value is available from the flight plan for airports and terminal procedures. If the transition altitude is not available, a system default is displayed.

TRAN LVL

This field displays and allows modification of the system default aircraft arrival transition altitude. This value is available from the flight plan for airports and terminal procedures. If the transition altitude is not available, a system default is displayed. Altitudes at or above this value are displayed as flight levels (FLXXX).

DEP SPD LTM

This field displays and allows modification of the system default speed limit for departure from that airspace. The default value is 250/10000.

ARR SPD LIM

This field displays and allows modification of the system default speed limit for arrival into that airspace. The default value is 250/10000.

DES FPA

This field displays and allows modification of the system default aircraft flight path angle for descents. The default value is –3.0 degrees, with a range of –1.0 to –6.0 degrees. A high angle descent option is configurable for greater than –6.0 degree descents.

<FLT PLAN INIT

Pressing LSK 6L will access the FLT PLAN INIT page.

FLT PLAN>



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Pressing LSK 6R will access the FLT PLAN page.

[Refer to Figure 86.](#)

PERF INIT 2/2

The PERF INIT 2/2 Page contains the entries for weight and the calculated gross weight for the flight plan displayed in the title line. Weights are displayed in pounds or kilograms as configured on the maintenance pages.

BASIC OPERATING WEIGHT

This field displays the basic aircraft operating weight and is retained in non-volatile memory. When the ZERO FUEL WEIGHT is cleared, the BASIC OPERATING WEIGHT reverts to the value in NVM. If the BOW NVM value is cleared or invalid, the ZERO FUEL WEIGHT is boxed, otherwise the ZERO FUEL WEIGHT is displayed with dashes.

+CARGO WEIGHT

This field displays the weight of the cargo. CARGO WEIGHT is added to BASIC OPERATING WEIGHT, which results in ZERO FUEL WEIGHT. If CARGO WEIGHT is not entered, it is set to dashes when the ZERO FUEL WEIGHT is crew entered.

=ZERO FUEL WEIGHT

This field displays the ZERO FUEL WEIGHT. ZERO FUEL WEIGHT is the sum of BASIC OPERATING WEIGHT and CARGO WEIGHT. A zero fuel weight value can be entered at LSK 3L, in which case CARGO WEIGHT and BASIC OPERATING WEIGHT are displayed with dashes.

+FUEL WEIGHT

This field displays the weight of the fuel on board the aircraft. After FUEL WEIGHT is entered, the GROSS WEIGHT is computed.

=GROSS WEIGHT

This field displays the gross weight of the aircraft, including the basic operating weight, the cargo weight and the fuel weight. The value is displayed in small font and is not enterable.

<FLT PLAN INIT

Pressing LSK 6L will access the FLT PLAN INIT page.

FLT PLAN>

Pressing LSK 6R will access the FLT PLAN page.

[Refer to Figure 87.](#)

FUEL FLOW 1/1

The FUEL FLOW 1/1 Page allows entry, display, and selection of the predicted fuel flows and prediction model for the flight average, climb, cruise, and descent phases of the trip. The fuel flow used in performance calculations is set on this page.

The FLIGHT AVERAGE, CLIMB PHASE, CRUISE PHASE, DESCENT PHASE fuel burn rates, and the current selection are retained in non-volatile memory.

Selecting <RETURN or a list selection switches to the PERF INIT 1/2 Page.

PREDICTION MODEL — PERF TABLE>

This selects the performance table as the model for predicting fuel flow. This is the default selection.

PREDICTION MODEL — AVERAGE>



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This selects the flight average value as the model for predicting fuel flow.

PREDICTION MODEL — FLT PHASE>

This selects the flight phase values as the model for predicting fuel flow.

FLIGHT AVERAGE

This field displays and allows modification of the aircraft average fuel burn rate.

CLIMB PHASE

This field displays and allows modification of the aircraft climb phase fuel burn rate.

CRUISE PHASE

This field displays and allows modification of the aircraft cruise phase fuel burn rate.

DESCENT PHASE

This field displays and allows modification of the aircraft descent phase fuel burn rate.

[Refer to Figure 88.](#)

CLIMB SPD 1/1

The CLIMB SPD 1/1 Page allows entry, display, and selection of the climb speeds for the trip. Entries are entered and displayed in CAS only, Mach only or CAS/Mach, depending upon the system configuration.

The NORMAL CLIMB and CRUISE CLIMB speeds are retained in non-volatile memory. The NORMAL CLIMB selection is the default.

The MANUAL CLIMB speed is retained only until power down, and must be entered again upon power up.

Selecting <RETURN or a list selection switched to the PERF INIT 1/2 page.

< MANUAL CLIMB

This line allows selection of the MANUAL CLIMB speed for the performance calculations (LSK 2L) and modification of the MANUAL CLIMB speed valve (LSK 2R).

<NORMAL CLIMB

This line allows selection of the NORMAL CLIMB speed for the performance calculations (LSK 4L) and modification of the NORMAL CLIMB speed valve (LSK 4R).

<CRUISE CLIMB

This line allows selection of the CRUISE CLIMB speed for the performance calculations (LSK 4L) and modification of the CRUISE CLIMB speed valve (LSK 4R).

[Refer to Figure 89.](#)

CRUISE SPD 1/1

The CRUISE SPD 1/1 Page allows entry, display, and selection of the cruise speeds for the trip. Entries are entered and displayed in CAS only, Mach only or CAS/Mach, depending upon the system configuration.

The LONG RNG CRUISE and MAX END CRUISE speeds are retained in non-volatile memory. The MAX END CRUISE selection is the default. The MANUAL CRUISE speed is retained only until power down, and must be entered again upon power up.



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Selecting <RETURN or a list selection switched to the PERF INIT 1/2 page.

<MANUAL CRUISE

This line allows selection of the MANUAL CRUISE speed for the performance calculations (LSK 1L) and modification of the MANUAL CRUISE speed valve (LSK 1R).

<NORMAL CRUISE

This line allows selection of the NORMAL CRUISE speed for the performance calculations (LSK 2L) and modification of the NORMAL CRUISE speed valve (LSK 2R).

<LONG RNG CRUISE

This line allows selection of the LONG RNG CRUISE speed for the performance calculations (LSK 3L) and modification of the LONG RNG CRUISE speed valve (LSK 3R).

<MAX SPD CRUISE

This line allows selection of the MAX SPD CRUISE speed for the performance calculations (LSK 4L) and modification of the MAX SPD CRUISE speed valve (LSK 4R).

<MAX END CRUISE

This line allows selection of the MAX END CRUISE speed for the performance calculations (LSK 5L) and modification of the MAX END CRUISE speed valve (LSK 5R).

[Refer to Figure 90.](#)

DESCENT SPD 1/1

The DESCENT SPD 1/1 Page allows entry, display, and selection of the cruise speeds for the trip. Entries are entered and displayed in

CAS only, Mach only or CAS/Mach, depending upon the system configuration.

The Mmo/Vmo DESCENT speed is retained in non-volatile memory and is the default. The MANUAL DESCENT speed is retained only until power down, and must be entered again upon power up.

Selecting <RETURN or a list selection switched to the PERF INIT 1/2 page.

<MANUAL DESCENT

This line allows selection of the MANUAL DESCENT speed for the performance calculations (LSK 2L) and modification of the MANUAL DESCENT speed valve (LSK 2R).

<NORMAL DESCENT

This line allows selection of the NORMAL DESCENT speed for the performance calculations (LSK 3L) and modification of the NORMAL DESCENT speed valve (LSK 3R).

<Mmo/Vmo DESCENT

This line allows selection of the Mmo/Vmo DESCENT speed for the performance calculations (LSK 4L) and modification of the Mmo/Vmo DESCENT speed valve (LSK 4R).

[Refer to Figure 91.](#)

FMS MENU KEY

FMS MENU 1/3 PAGE

The FMS MENU 1/3 page allows access to other FMS functions not available through a dedicated key, and allows erasure of the active flight plan.

<INITIALIZATION

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 34-60-01

Config 001

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Config 001
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This LSK allows access to the INITIALIZATION page (LSK 1L). See the initialization section for more details on this page.

<WAYPOINT INFO

This LSK allows access to the WAYPOINT INFO page (LSK 2L).

<FLIGHT PLAN INIT

This LSK allows access to the FLIGHT PLAN INIT 1/2 page (LSK 3L). See the initialization section for more details on this page.

<STORE DELETE ROUTES

This LSK allows access to the STORE ACTIVE ROUTE 1/3 page (LSK 4L).

<SECOND FLIGHT PLAN INDEX

This LSK allows access to the SECOND INDEX 1/2 page (LSK 5L).

ERASE FLIGHT PLAN*

This LSK allows access to the MODIFIED FLIGHT PLAN page (LSK 6L).

[Refer to Figure 92.](#)

FMS MENU 2/3 PAGE

The FMS MENU 2/3 page allows access to more FMS functions not available through a dedicated key.

This LSK allows access to the PILOT WPT LIST 1/N page (LSK 1L). See the flight plan section for more details on this page.

PERF INIT

This LSK allows access to the PERF INIT 1/2 page (LSK 2L). See the performance initialization section for more details on this page.

FUEL

This LSK allows access to the FUEL LB 1/1 page (LSK 3L).

<DEST RAIM

This LSK allows access to the DEST RAIM STATUS 1/1 page (LSK 4L).

<FLT LOG

This LSK allows access to the FLIGHT LOG 1/1 page (LSK 5L).

<COLD TEMP COMP

This LSK allows access to the COLD TEMP COMP °C 1/1 page (LSK 6L). This page is only available if configured in the configuration options.

[Refer to Figure 93.](#)

FMS MENU 3/3 PAGE

The FMS MENU 3/3 page allows access to FMS maintenance functions. Some functions are not available while airborne, and some are only available when in maintenance mode.

<MAINTENANCE

This LSK allows access to the MAINTENANCE 1/1 page (LSK 1L).

[Refer to Figure 94.](#)

WAYPOINT INFO 1/1 PAGE

The WAYPOINT INFO 1/1 page provides information about a selected waypoint/navaid.

IDENT



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This field allows selection of the seven character identifier of the waypoint. When first accessed, the field will contain dashes. Selection of LSK 2L will either down select the contents of the IDENT field to the scratchpad, or if the scratchpad contains an identifier, selection will up select the identifier to the IDENT field.

<RETURN

This LSK returns to the previous screen (LSK 6L).

[Refer to Figure 95.](#)

When an identifier is selected, additional data is displayed on the screen, depending on the type of waypoint/navaid selected. If the ident is a navaid, the following data from the navigation database appears.

NAME

The line associated with LSK 1 shows the long name of the navaid.

IDENT

The line associated with LSK 2 shows the identifier of the navaid.

COUNTRY

The line associated with LSK 2 shows the country name of the navaid.

LATITUDE/LONGITUDE

The position of the navaid.

MEDGE

The magnetic declination of the navaid.

ELEV

The elevation of the navaid.

BRG/DIS

The bearing and distance to the navaid.

FRQ KHZ

The assigned frequency of the navaid.

TYPE

The type of navaid.

CLASS

The class of navaid.

If the ident is a pilot defined waypoint, the following data appears.

[Refer to Figure 96.](#)

IDENT

The line associated with LSK 2 shows the identifier of the waypoint.

PILOT DEFINED

The line associated with LSK 2 shows that the waypoint has been defined by the pilot.

LATITUDE/LONGITUDE

The position of the waypoint.

MAGVAR

The magnetic variation of the waypoint.

BRG/DIS

The bearing and distance to the waypoint.



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RESERVED BY STORED ROUTE

A message indicating that the selected waypoint is used by a stored route.

[Refer to Figure 97.](#)

WPT RESERVED BY FLT PLAN

A message indicating that the selected waypoint is used by the active flight plan. See Figure 2–73. A prompt is displayed at LSK 6R to delete the waypoint, resulting in a modified flight plan.

[Refer to Figure 98.](#)

If the ident is a runway, the following data appears.

NAME

The line associated with LSK 1 shows the long name of the airport. IDENT The line associated with LSK 2 shows the identifier of the airport.

COUNTRY

The line associated with LSK 2 shows the country name of the airport.

LATITUDE/LONGITUDE

The position of the airport.

MAGDEC

The magnetic variation of the airport.

ELEV

The elevation of the airport.

BRG/DIS

The bearing and distance to the airport.

<RETURN

Pressing LSK 6L will access the previous screen.

RUNWAY INFO>

Pressing LSK 6R will access the RUNWAY AT NNNN 1/N page.

[Refer to Figure 99.](#)

RUNWAYS AT AAAA 1/2 PAGE

The RUNWAYS AT AAAA 1/2 page provides information about the runways at the selected airport.

IDENT

This field displays the identifier of the runway.

LEN–FT

This field displays the length of the runway. The units are configurable between feet and meters on the configuration options page.

BRG

This field displays the bearing of the runway.

ELEV

This field displays the elevation of the runway.

<RETURN

Pressing LSK 6L will return to the WAYPOINT INFO 1/1 page.

[Refer to Figure 100.](#)

DUPLICATES 1/2 PAGE



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If the identifier entered matches more than one waypoint in the navigation database and/or the pilot defined list, the DUPLICATES 1/2 page will appear to resolve the selection.

TITLE LINE

The title line displays the entered identifier (e.g. CC).

SECOND LINE

The second line displays the reference position (e.g. EED). The reference position will be either current aircraft position (PPOS) if doing a tactical leg change, or will be the insertion point (EED) for a strategic insert.

LAT/LONG

The latitude and longitude of the associated waypoint.

DISTANCE

The distance from the reference point to the associated waypoint. The list of duplicate waypoints is sorted by this distance, from closest to farthest.

BEARING

The bearing from the reference point to the associated waypoint.

←AAAAA

The country code for the waypoint. Pressing the left LSK will select the waypoint for completion of the waypoint selection operation that is in progress.

WAYPOINT TYPE

The associated waypoint type.

INFO>

Pressing the right LSK will display the WAYPOINT INFO page for the associated waypoint.

<RETURN

Pressing LSK 6L will return to the WAYPOINT INFO 1/1 page.

[Refer to Figure 101.](#)

STORE ACTIVE ROUTE 1/3 PAGE

The STORE ACTIVE ROUTE 1/3 page provides capability to store the active flight plan as a route.

[Refer to Figure 102.](#)

ACTIVE ROUTE IDENT

This field displays the identifier of the stored route. If an active flight plan exists, the system defaults the route ident to ten characters made up of four for current ORIGIN, followed by four for the current DEST, followed by a number which starts at one and defaults to the next available number (up to 99). If a default can not be determined, the ident field contains dashes and the *STORE prompt is replaced by a <RETURN prompt.

ROUTES AVAIL

The total number of stored routes available.

*STORE

Pressing LSK 6L will return to the WAYPOINT INFO 1/1 page.

CANCEL*

Pressing LSK 6R will return to the previous page without storing the route.

<RETURN



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Pressing LSK 6L will return to the WAYPOINT INFO 1/1 page.

[Refer to Figure 103.](#)

STORE SECOND ROUTE 2/3 PAGE

The STORE SECOND ROUTE 2/3 page provides capability to store the secondary flight plan as a route. The functions are the same as for STORE ACTIVE ROUTE 1/3 page, except applying to the secondary flight plan.

[Refer to Figure 104.](#)

DELETE STORED ROUTE 3/3 PAGE

The DELETE STORED ROUTE 3/3 page provides capability to delete a single route or to delete all stored routes. The ident field is dashed upon entry to the page. The ident can be up selected from the scratchpad or selected from the route list page.

ROUTE LIST>

Pressing LSK 5R will display the STORED ROUTE LIST 1/N page.

<RETURN

Pressing LSK 6L will return to the FMS MENU 1/3 page.

DELETE

Pressing LSK 6L will delete the stored route and cause the ident field to go back to dashes and the *DELETE prompt to change back to <RETURN, and the CANCEL* prompt shall be removed.

CANCEL*

Pressing LSK 6R will not delete the route and will cause the ident field to go back to dashes and the *DELETE prompt to change back to <RETURN, and the CANCEL* prompt shall be removed.

[Refer to Figure 105.](#)

DELETE ALL>

Pressing LSK 6R will display the DELETE STORED ROUTES 1/1 page.

[Refer to Figure 106.](#)

STORED ROUTE LIST 1/N PAGE

The STORED ROUTE LIST 1/N page provides a list of stored routes with the ORIGIN–DEST pair.

IDENT

Pressing LSK 1L, 2L, 3L, 4L or 5L will select the identifier and return to the DELETE STORED ROUTE 3/3 page with the ROUTE IDENT filled with the selected identifier.

ORIGIN–DEST

This field contains the origin–destination pair associated with the stored route.

<RETURN

Pressing LSK 6L will return to the DELETE STORED ROUTE 3/3 page without filling in the ROUTE IDENT.

[Refer to Figure 107.](#)

DELETE STORED ROUTES 1/1 PAGE

ROUTES 1/1 page provides the capability to delete all stored routes at once.

* DELETE

Pressing LSK 6R will delete all of the stored routes.



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<RETURN

Pressing LSK 6L will return to the DELETE STORED ROUTE 3/3 page.

[Refer to Figure 108.](#)

SECOND INDEX 1/2 PAGE

The SECOND INDEX 1/2 page provides access to other pages and actions related to the secondary flight plan.

<FLIGHT PLAN INIT

Pressing LSK 1L results in display of the SECOND FLIGHT PLAN INIT 1/2 page.

<PERF INIT

Pressing LSK 2L results in display of the SECOND PERF INIT 1/2 page.

*COPY FROM ACTIVE

Pressing LSK 3L will copy the active flight plan into the secondary flight plan.

*SWAP WITH ACTIVE

Pressing LSK 4L will switch the active flight plan and secondary flight plan.

*MOVE TO ACTIVE

Pressing LSK 5L will copy the secondary flight plan into the active flight plan.

ERASE FLT PLAN

Pressing LSK 6R will erase the secondary flight plan.

[Refer to Figure 109.](#)

SECOND INDEX 2/2 PAGE

The SECOND INDEX 2/2 page provides access to other pages and actions related to the secondary flight plan.

<DEST RAIM

Pressing LSK 1L results in display of the DEST RAIM STATUS 1/1 page.

[Refer to Figure 110.](#)

PILOT WPT LIST 1/N PAGE

The PILOT WPT LIST 1/N page provides a list of pilot defines waypoints. A reserved (R) indication is also shown if the waypoint is used in a stored route.

Selection of any LSK with an associated identifier will down select the identifier (e.g. ANIDENT) to the scratchpad for up selection on a subsequent page (e.g. WPT INFO).

IDENT

This field displays the identifier of the pilot defined waypoint.

<RETURN

Pressing LSK 6L will return to the previous page.

WPT INFO>

Pressing LSK 6R results in display of the WAYPOINT INFO 1/1 page.

[Refer to Figure 111.](#)

FUEL LB 1/1 PAGE



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The FUEL LB 1/1 page provides the status of the fuel usage. All quantities are displayed in pounds, reflecting the selection of pounds option on the maintenance page. If kilograms had been selected, all occurrences of LB would change to KG, and quantities would be shown in kilograms.

[Refer to Figure 112.](#)

Upon power-up, the remaining fuel and reserve fuel quantities need to be verified. The values displayed shall be the values from the end of the last flight retrieved from non-volatile memory. After verifying the values, pressing the VERIFY POWERUP INPUTS* prompt will change the values to green.

REMAINING

This field displays and allows modification of the remaining fuel onboard the aircraft.

RESERVE

This field displays and allows modification of the reserve fuel desired.

HOURS

This field displays the flight time remaining in hours and minutes (HHMM) until the fuel reserve limit is reached, based on current fuel usage data from the sensors or a manually entered flow value.

RANGE

This field displays the aircraft range in NM until the fuel reserve limit is reached, based on current fuel usage data from the sensors or a manually entered flow value.

NM/LB or NM/KG

This field displays the nautical miles flown per pound (or kilogram) of fuel used, based on current fuel usage data from the sensors or a manually entered flow value.

[Refer to Figure 113.](#)

MAN FLOW

This field displays and allows entry of a manual fuel flow value as an alternative to sensors, if sensors are not available. When a manual value is entered, the ELAPSED TIME field starts counting up from zero. The per engine fuel flow and fuel used values change to cyan because sensor usage has been deselected. The total fuel flow is set to the manual flow value.

[Refer to Figure 114.](#)

ELAPSED TIME

This field displays the elapsed time since the manual flow value has been verified. When the elapsed time reaches 15 minutes, the VERIFY FUEL INPUTS* prompt appears, and manual fuel flow, elapsed time and remaining fuel change to yellow. Entering a new value of manual flow or pressing LSK 6R to verify current value of manual flow will turn these fields back to green.

ENGX FLOW

This field displays the rate of fuel flow to engine X. The number of engines displayed reflects aircraft configuration.

ENGX USED

This field displays the amount of fuel used by engine X. The number of engines displayed reflects aircraft configuration.

TOTAL FLOW

This field displays the total rate of fuel flow to all engines.



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TOTAL USED

This field displays the total amount of fuel used by all engines.

[Refer to Figure 115.](#)

If a fuel sensor becomes invalid, the corresponding engine flow and used fields turn yellow. The total flow, total used and remaining fuel fields turn yellow also.

[Refer to Figure 116.](#)

If the sensor becomes reliable again, the associated engine fields and the total fields return to green, but the remaining fuel field remains yellow and must be verified manually by selecting LSK 6R or by entering a manual value using LSK 1L.

[Refer to Figure 117.](#)

DEST RAIM STATUS 1/1 PAGE

The DEST RAIM STATUS 1/1 page displays the predicted GPS RAIM status at a specified point and time. The destination coordinates and ETA are automatically loaded from the flight plan (active or secondary). COMPUTING is displayed while the GPS sensor is calculating the availability of RAIM at the destination for the time period 15 minutes before the ETA to 15 minutes after the ETA. Once the calculation is complete, the availability is displayed in five minute increments before and after ETA.

[Refer to Figure 118.](#)

Once the calculation is complete, the availability is displayed in five minute increments before and after ETA.

DEST

This field displays and allows modification of the destination identifier from the active or secondary flight plan, depending on which FPL was being viewed previously. If the destination is modified here, the flight plan is not affected.

RAIM@ETA

This field displays whether RAIM is AVAILABLE or NOT AVAIL at the estimated time of arrival at the destination.

ETA

This field displays and allows modification of the estimated time of arrival at the destination. If the ETA is modified here, the flight plan is not affected.

[Refer to Figure 119.](#)

EXCLUDED SATELLITES

This field displays and allows modification of satellites to be excluded from the RAIM calculation. Up to eight satellites can be excluded. Satellites can be cleared one at a time using the *CLEAR* token.

@ETA-NN or @ETA+NN

This field displays whether RAIM is AVAILABLE or NOT AVAIL at the times before and after the estimated time of arrival at the destination.

<RETURN

Pressing LSK 6L will return to the prior page.

COMPUTE*



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Pressing LSK 6R will cause the RAIM calculation to be recomputed. This prompt only appears when data has been changed on this page.

[Refer to Figure 120.](#)

FLIGHT LOG 1/1 PAGE

The FLIGHT LOG 1/1 page displays information about the flight and is normally accessed after the flight is complete.

DATE

This field displays the date of the flight.

UTC

This field displays the current time.

TAKE-OFF

This field displays takeoff time.

FLT TIME

This field displays the actual flight time.

LAND

This field displays landing time.

DIS GND

This field displays the ground distance covered by the flight. This is computed by multiplying the integral of the ground speed by the actual flight time.

DIS AIR

This field displays the distance in the air covered by the flight. This is computed by multiplying the integral of the airspeed by the actual flight time.

AVG GS

This field displays the average ground speed during the flight.

AVG TAS

This field displays the average true airspeed during the flight.

FUEL USED

This field displays the total amount of fuel used during the flight.

AVG NM/LB

This field displays the average nautical miles flown per pound of fuel during the flight.

[Refer to Figure 121.](#)

<RETURN

Pressing LSK 6L will return to the prior page.

If the page is accessed during flight, the landing time is dashed, and computations are based upon data up to the current time.

[Refer to Figure 122.](#)

COLD TEMP COMP °C 1/1 PAGE

The COLD TEMP COMP °C 1/1 page displays and allows modification of the destination airport temperature. This page is only available if configured in the CONFIG OPTIONS. The temperature scale is configured on the maintenance page for °F/°C.

DEST AIRPORT TEMP



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This field displays and allows modification of the destination airport temperature.

DEST AIRPORT ELEVATION

This field displays the destination airport elevation. This data is retrieved from the navigation database and is not modifiable on this page.

[Refer to Figure 123.](#)

MAINTENANCE 1/1 PAGE

The MAINTENANCE 1/1 page allows access to other pages and actions related to FMS status and installation. The DATABASE LOAD prompt will not be displayed, nor will the page be available when airborne. The GPS TEST, CONFIG OPTIONS and KEYBOARD COMPENSATION prompts will not be displayed nor the pages available when not in maintenance mode.

<DATABASE STATUS

Pressing LSK 1L results in display of the DATABASE STATUS 1/1 page.

<DATABASE LOAD

Pressing LSK 2L results in display of the DATABASE UPDATE 1/1 page.

<KEYBOARD COMPENSATION

Pressing LSK 4L results in display of the KEYBOARD BACKLIGHT COMPENSATION 1/1 page.

*LBS/KGS

Pressing LSK 5L results in the toggling of the units for weight between pounds and kilograms.

*°F/°C

Pressing LSK 6L results in the toggling of the units for temperature between degrees Fahrenheit and degrees Celsius.

UNIT INFORMATION>

Pressing LSK 1R results in display of the UNIT INFORMATION 1/1 page.

PARALLAX>

Pressing LSK 2R results in display of the PARALLAX ADJUSTMENT 1/1 page.

GPS TEST>

Pressing LSK 3R results in display of the GPS TEST 1/1 page.

CONFIG OPTIONS>

Pressing LSK 4R results in display of the A/C CONFIG MODULE 1/32 page.

RUNWAY LENGTH METERS/FT

Pressing LSK 6R results in the toggling of the units for runway length between feet and meters.

[Refer to Figure 124.](#)

DATABASE STATUS 1/1 PAGE

The DATABASE STATUS 1/1 page allows the operator to view information about the current database. If the current database is invalid, the statistics fields will be replaced with yellow dashes. If the part number is longer than fourteen characters, it will be split and the second half will be right justified on the line below.

TYPE:



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This field displays the current database type.

REVISION:

This field displays the revision date of the current database.

EXPIRES:

This field displays the expiration date of the current database.

PART NO:

This field displays the part number of the current database.

<RETURN

Pressing LSK 6L will go to the DELETE STORED ROUTE 3/3 page.

[Refer to Figure 125.](#)

DATABASE UPDATE 1/1 PAGE

The DATABASE UPDATE 1/1 page allows the operator to view information about the current database. If the new database is invalid, the statistics fields will be replaced with yellow dashes. The data shown is the same as on the DATABASE STATUS 1/1 page, but both the new and current database are shown.

*UPDATE

Pressing LSK 6L will begin the update process.

CANCEL*

Pressing LSK 6R will cancel the update process and either return to the MAINTENANCE 1/1 page (manual prompting) or COLD START the system (automatic prompting).

[Refer to Figure 126.](#)

KEYBOARD BACKLIGHT COMPENSATION 1/1 PAGE

The KEYBOARD BACKLIGHT COMPENSATION 1/1 page allows the operator to increase (LSK 2R) or decrease (LSK 2L) the keyboard backlight brightness by one step per LSK press.

<RETURN

Pressing LSK 6L will return to the MAINTENANCE 1/1 page.

[Refer to Figure 127.](#)

UNIT INFORMATION 1/1 PAGE

The UNIT INFORMATION 1/1 page displays the specific hardware and software configuration of the FMS.

FMS P/N

This field displays the part number of the FMS.

FMS S/N

This field displays the serial number of the installed MCDU.

CPU S/W

This field displays the part number of the CPU software in the MCDU.

GPS S/W

This field displays the part number of the GPS software in the MCDU.

VIDEO S/W

This field displays the part number of the video driver software in the MCDU.

422 S/W



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This field displays the part number of the RS-422 software in the MCDU.

CPU S/W

This field displays the part number of the 429 data bus software in the MCDU.

<RETURN

Pressing LSK 6L will return to the MAINTENANCE 1/1 page.

[Refer to Figure 128.](#)

PARALLAX ADJUST 1/1 PAGE

The PARALLAX ADJUST 1/1 page allows the operator to adjust the CDU display to eliminate parallax between the LCD display and the line select keys from the pilot's viewing position. Each press of the PREV key shifts the display upward one pixel. Each press of the NEXT key shifts the display downward one pixel. When the limits are reached, a message SHIFT LIMIT REACHED is shown in the scratchpad. Parallax settings are saved to non-volatile memory and are restored at power-up.

<RETURN

Pressing LSK 6L will return to the MAINTENANCE 1/1 page.

[Refer to Figure 129.](#)

GPS TEST 1/1 PAGE

The GPS TEST 1/1 page allows the operator to put the GPS sensor into one of two test modes.

ACQUIRE ALMANAC*

Pressing LSK 4R selects a mode that causes the GPS sensor to update its almanac data.

TRACK ONLY SAT 11*

Pressing LSK 5R selects a mode that causes all channels of the GPS sensor to track satellite number 11.

<RETURN

Pressing LSK 6L will return to the MAINTENANCE 1/1 page.

[Refer to Figure 130.](#)

A/C CONFIG MODULE 1/32 PAGE

The A/C CONFIG MODULE 1/32 page is the first of 32 pages used to setup the configuration and operation of the FMS. Pages 2–31 are similar pages.

SELECTION

Pressing LSK 1L sequences through the possible choices for the aircraft data source.

<EXIT

Pressing LSK 6L will go to the A/C CONFIG MODULE 1/32 page.

[Refer to Figure 131.](#)

A/C CONFIG MODULE 32/32 PAGE

The A/C CONFIG MODULE 32/32 page is the last of 32 pages used to setup the configuration and operation of the FMS.

<SAVE AND RESTART

Pressing LSK 2L saves the configuration module settings and restarts the FMS.



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<QUIT WITHOUT SAVE

Pressing LSK 3L returns to the MAINTENANCE 1/1 page without saving the configuration data.

<EXIT

Pressing LSK 6L will go to the previous configuration page.

[Refer to Figure 132.](#)

MESSAGE KEY

FMS MESSAGES 1/1 PAGE

System and Sensor Messages are displayed on this page. System messages are displayed first followed by sensor messages. See Section 4 for a detailed description of the messages.

The Navigation Database (NDB) type and expiration date is displayed here as a system message. As many pages as necessary are used to display all of the current messages. The message annunciator is lit when there are unread messages to be read. If a message requiring action to be taken is displayed, the annunciator will remain lit until the action is taken.

System messages are displayed in order of priority. Sensor messages are displayed by sensor. The sensor name is displayed in front of the first message for that particular sensor. If there are no abnormal conditions related to sensors, then no sensors will be displayed.

Yellow is used if the message refers to an abnormal condition. If the message requires action to be taken, it will be continually displayed in yellow. If the message is one requiring only attention, the color is changes to cyan after being viewed by the operator.

The message is displayed in green if it refers to a normal condition.

In some cases, a message is used to communicate the mode or status of the system (APPROACH ARMED) and an operation is provided (CANCEL APPROACH) to change the mode. These messages are the highest priority messages.

APPROACHED ARMED is green because it is the normal mode of operation. Pressing LSK 6R will cancel the approach mode and change the message page to reflect the new mode.

APPROACH CANCELED is shown in yellow because it is the abnormal mode of operation. Pressing LSK 6R would change the mode back to approach armed.

EXEC KEY

The Execute Key is used to confirm modifications to the active flight plan. When pressed, the modifications are confirmed and made part of the active flight plan. The annunciator is lit when modifications have been made to the active flight plan and need to be confirmed.

[Refer to Figure 133.](#)

NORMAL PROCEDURES

INITIALIZATION OF A FLIGHT PLAN

The purpose of this section is to guide the user through Global Star 2100 power on and flight plan initialization. The steps include initializing the FMS position, defining a departure and arrival airport as well as terminal area procedures (e.g. SID, STAR and Approach), stringing an airway, and preparing a secondary flight plan to an alternate destination. The flight plan is from Kansas City, MO to Houston, TX with the alternate to San Antonio, TX. The items to be setup are as follows:

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 34-60-01

Config 001

34-60-01

Config 001
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DEPARTURE: SID:	KMCI RACER2 WITH BUM (BUTLER) EN-ROUTE TRANSITION
Airway:	J87.TNV (Navasota)
Arrival: STAR: Approach:	KHOU STRUK8 with no enroute transition VOR DME Rwy 35 with HUB (Hobby) procedure turn transition
Alternate: STAR:	KSAT MARCS3 with WINDU enroute transition

[Refer to Figure 134.](#)

FMS Power On/Initialization

1. Select the ON button. The FMS powers up and the SELF TEST page is displayed until the self test is complete.

[Refer to Figure 135.](#)

Upon successful completion of the self test, the database will be verified.

[Refer to Figure 136.](#)

When the database verification is complete, the FMS IDENT page will be displayed.

2. Verify the DATE and UTC . Modify any incorrect values by entering the correct value into the scratchpad and up selecting it to the correct field using LSK 2R or 3R.

NOTE

[Refer to Figure 137.](#)

If the FMS should happen to power up with no values for the date and time, the fields will be filled with yellow boxes. Values for both fields must be entered before the initialization can continue. Attempts to select any other pages (with the exception of the FMS messages page) will result in an ENTER DATE AND UTC message in the scratchpad.

[Refer to Figure 138.](#)

3. Select the INITIALIZATION> prompt (LSK 6R). The INITIALIZATION 1/1 page will be displayed, showing a list of positions that may be selected as the FMS position.

4. If the ORIGIN field is already contains KMCI and the associated position, skip to step 5. If the ORIGIN field is does not already contain KMCI and the associated position, type KMCI into the scratchpad line and up select KMCI to the ORIGIN field by pressing LSK 2L.

NOTE

The ORIGIN field prefills with the airport from the database closest to the last known position as shown in the CURRENT POSITION field.

[Refer to Figure 139.](#)

5. Select the ORIGIN position as the INIT POSITION by pressing LSK 2R. The ORIGIN position is copied to the INIT POSITION and changes to green. The FMS position is initialized. The FLT PLAN INIT prompt is displayed and can be used to initialize company or



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pilot defined flight plans. For this example, there are no company or pilot defined flight plans from KMCI to KHOU.

[Refer to Figure 140.](#)

1. Select the DEP/ARR button. The DEP/ARR 1/2 page will be displayed.

NOTE

[Refer to Figure 141.](#)

The ORIGIN field is prefilled with the ORIGIN data from the initialization. If no origin data had been entered during initialization, the field would contain dashes. The origin data (KMCI) would be entered on the scratchpad and up selected to the ORIGIN field with LSK 1L, and a modified flight plan would be created.

[Refer to Figure 142.](#)

2. Select the <DEPART prompt (LSK 4L). The departure runways page will be displayed.

[Refer to Figure 143.](#)

3. Select runway RW01R (LSK 2L). The departure SIDS page will be displayed.

A modified flight plan is created and the selected runway is inserted into the modified flight plan, as shown by the

[Refer to Figure 144.](#)

4. Select the RACER2 departure (LSK 2L). The departure transitions page will be displayed.

The selected SID is inserted into the modified flight plan, as shown by the SID> prompt. This page displays a list of transitions available for the selected departure.

[Refer to Figure 145.](#)

5. Select the BUM transition (LSK 1L). The Flight Plan page will be displayed. The selected transition is inserted into the modified flight plan.

A DISCONTINUITY is displayed in the flight plan if no waypoint link exists. Discontinuities can be cleared from the flight plan by pressing the CLR button and selecting the appropriate left LSK. When a discontinuity is cleared, a track is defined between the waypoints on each side of the discontinuity and the waypoint are linked together. The ERASE* prompt provides a means to cancel all flight plan modifications.

Airways

Airways or holding patterns can be inserted at a specific waypoint in the flight plan. The Lateral Modification (MODIFY AT) page is used to make additions or changes to the lateral flight plan.



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The J87 airway will be inserted into the flight plan starting at the Butler (BUM) VOR.

[Refer to Figure 146.](#)

1. Select the waypoint BUM (LSK 4L). The lateral MODIFY AT BUM page is displayed. This page allows lateral modifications at BUM, such as stringing airways or defining holding patterns.

[Refer to Figure 147.](#)

2. Select the <AIRWAYS prompt (LSK 5L). The AIRWAYS AT BUM page is displayed. This page displays a list of air-ways that run through BUM.

[Refer to Figure 148.](#)

3. Select the route J87 (LSK 4L). The lateral AIRWAY WPTS page is displayed. This page displays all waypoints associated with J87.

BUM is displayed in yellow as the starting or from waypoint using (FR).

[Refer to Figure 149.](#)

4. Select the waypoint TNV (LSK 2L). TNV is designated as the (TO) waypoint for the airway, and all waypoints between BUM and TNV are colored yellow and inserted into the modified flight plan.

[Refer to Figure 150.](#)

5. Select the FLT PLAN> prompt (LSK 6R). The FLT PLAN page is displayed, with the airway waypoints inserted at the end of the flight plan.

[Refer to Figure 151.](#)

Arrivals

1. Select the DEP/ARR button. The DEP/ARR 1/2 page will be displayed.

[Refer to Figure 152.](#)

2. Type the destination identifier (KHOU) into the scratchpad. Select LSK 1R to up select KHOU to the DEST field.

[Refer to Figure 153.](#)

3. Select LSK 1R to up select KHOU to the DEST field. Now that a destination is identified, the ARRIVE> and APPROACH> prompts are displayed.

[Refer to Figure 154.](#)

4. Select the ARRIVE> prompt (LSK 4R). The ARRIVE RUNWAYS page is displayed. This page displays a list of runways and associated lengths at KHOU.

The runway will be specified later the choosing the approach procedure into KHOU.

NOTE

As in runway specific departures, runway specific arrivals are not displayed until the appropriate runway is selected.

[Refer to Figure 155.](#)

5. Select the STAR> prompt (LSK 5R). The ARRIVE STARS page is displayed. This page displays a list of arrivals into KHOU.

[Refer to Figure 156.](#)

6. Select the STRUK8 arrival (LSK 1R). The ARRIVE TRANSITIONS page is displayed. This page displays a list of en-route transitions for the STRUK8 arrival. The STAR is inserted into the modified flight plan.



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The STRUK8 arrival begins at TNV, therefore no en–route transition is chosen. If desired, select the flight plan prompt to review the STAR in the modified flight plan.

[Refer to Figure 157.](#)

Approaches

1. Select the ARR/DEP button. The DEP/ARR 1/2 page will be displayed.

[Refer to Figure 158.](#)

2. Select the APPROACH> prompt (LSK 5R). The APP APPROACHES page is displayed. These pages displays a list of all non precision approaches for KHOU.

The desired approach is the VOR/DME to RWY 35, found on the second page of available approaches.

[Refer to Figure 159.](#)

3. Select the NEXT button or the DEP/ARR button. The DEP/ARR 2/2 page will be displayed.

[Refer to Figure 160.](#)

4. Select the VOR/DME–35 approach (LSK 1L). The APP TRANSITIONS page is displayed. This page displays a list of available approach transitions for the VOR/DME–35 approach.

[Refer to Figure 161.](#)

5. Select the HUB transition (the procedure turn transition) (LSK 1L). The MODIFY FLT PLAN 5/7 page is displayed with the HUB waypoint.

The modified flight plan can be reviewed and selected as the active flight plan.

[Refer to Figure 162.](#)

6. Select the button. The active flight plan waypoints are displayed in green and the missed approach procedure is displayed in cyan.

[Refer to Figure 163.](#)

Secondary Flight Plan

The secondary flight plan provides a way to operate on a flight plan without affecting the active flight plan. The active flight plan can be copied to the secondary flight plan, modified and then moved back to the active flight plan. For this scenario, the secondary flight plan will contain the alternate destination.

[Refer to Figure 164.](#)

1. Select the FMS MENU button. The FMS MENU 1/3 page will be displayed.

[Refer to Figure 165.](#)

2. Select the <SECOND FLT PLAN INDEX prompt (LSK 5L). The SECOND INDEX 1/2 page is displayed.

The active flight plan will be copied into the secondary flight plan and then modified with an arrival to KSAT.

[Refer to Figure 166.](#)

3. Select the <COPY FROM ACTIVE prompt (LSK 3L). The SECOND FLT PLAN 1/4 page is displayed, containing an exact copy of the active flight plan.

[Refer to Figure 167.](#)

4. Select the DEP/ARR button. The DEP/ARR 2/2 page will be displayed.



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The SECOND DEP/ARR 2/2 page is used to load the secondary flight plan's departure and arrival just like we did for the active flight plan on the ACTIVE DEP/ARR 1/2.

[Refer to Figure 168.](#)

5. Type KSAT into the scratchpad and up select it to the DEST field (LSK 1R).

[Refer to Figure 169.](#)

6. Select the ARRIVE> prompt (LSK 4R). The ARRIVE RUNWAYS page is displayed. This page displays a list of runways and associated lengths at KSAT.

[Refer to Figure 170.](#)

7. Select the STAR> prompt (LSK 5R). The ARRIVE STARS page is displayed. This page displays a list of arrivals into KSAT.

[Refer to Figure 171.](#)

8. Select the MARCS3 arrival (LSK 3L). The ARRIVE TRANSITIONS page is displayed. This page displays a list of en-route transitions for the MARCS3 arrival. The STAR is inserted into the secondary flight plan.

[Refer to Figure 172.](#)

9. Select the WINDU transition (LSK 1R). The SECONDARY FLT PLAN 3/5 page is displayed with the WINDU waypoint.

The endpoint of the J87 airway can be changed to TORN to better join the airway with the MARCS3 arrival. This can be done by either clearing BILEE, JUDDY and TNV from the SECOND FLT PLAN page or by selecting any airway waypoint and redefining the endpoint of the J87 airway.

[Refer to Figure 173.](#)

10. Select the CLR button, then press the LSKs adjacent to waypoints BILEE, JUDDY, TNV and any DISCONTINUITY.

[Refer to Figures 174 and 175.](#)

The SECOND FLT PLAN pages now define an alternate flight plan to KSAT. The secondary flight plan sequences along with the active flight plan when the waypoints in both are identical. For this scenario the secondary flight plan will sequence along with the active flight plan up to TORN.

The secondary flight plan can be activate at any time by using the SECOND INDEX page and selecting the MOVE TO ACTIVE prompt.

FMS MESSAGES

GENERAL

The FMS has three types of messages:

- Scratchpad Messages
- System Messages
- Sensor Messages

Scratchpad messages will always occur on the scratchpad line of the CDU. The message will always be related to a CDU entry made by the operator.

System messages occur on the FMS MESSAGES Page. These messages all refer to the status of the FMS system operation.

Sensor messages occur on the FMS MESSAGES Page. These messages all refer to the status of individual position sensors that the FMS is configured to use.



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The FMS can also drive external annunciators to indicate various conditions. The external annunciation is driven by dedicated discrete outputs from the FMS or digitally via the ARINC 429 link.

SCRATCHPAD MESSAGES

A scratchpad message is always in the large font colored white. Scratchpad messages are as follows:

MESSAGE TEXT	DESCRIPTION
ACT FLT PLN WILL BE LOST	Displayed on the DATABASE UPDATE page to warn the operator that loading a new database will cause the active flight plan to be erased.
CARD NOT PRESENT	Displayed on the MAINTENANCE page if the operator attempts to select the DATABASE LOAD prompt when no PCMCIA card has been inserted.
CONFIRM INIT POSITION	Displayed on the IDENT or INITIALIZATION page if the operator attempts to select another page prior to confirming the initial position.
FLIGHT PLAN FULL	Proposed modification to the flight plan would exceed the maximum number of legs allowed for the flight plan.

MESSAGE TEXT	DESCRIPTION
FLIGHT PLAN STORAGE FULL	Attempting to save a new stored flight plan when all available stored flight plans have been defined.
FORMAT ERROR	The FMS has rejected the entry because it is not in the correct format expected by the system.
FPLN OPER IN PROGRESS	A new flight plan entry is made before the prior entry has been processed by the system or when a flight plan entry is attempted while dual FMS system are synchronizing flight plans.
INTERCEPT BEYOND FIX	Displayed when the selected heading entered on the HEADING page will cause the airplane to intercept the selected course beyond the active TO waypoint.
MODIFIED FLT PLN EXISTS	Displayed when storing the active flight plan and a modified flight plan exists.
NO ARC INTERCEPT	Displayed when the selected heading entered on the HEADING page will not cause the airplane to intercept the selected next arc leg.
NO AUTO LEG CHANGE	Displayed when a leg is created which will not sequence automatically.



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MESSAGE TEXT	DESCRIPTION
NO COURSE INTERCEPT	Displayed when the selected heading entered on the HEADING page will cause the airplane to diverge (or fly parallel to) the selected course to the active TO waypoint.
NO PROCEDURES AVAILABLE	Attempt to select the ARRIVE, DEPART or APPROACH page when no procedures of the corresponding type are available for the selected airport.
NO ROUTES AVAILABLE	Attempt to access the company routes list page when no routes are available for the selected origin–destination pair.
NON-UNIQUE IDENTIFIER	Attempt to store a new pilot–defined waypoint or stored route with an identifier that is already in use.
NOT ALLOWED	Action is not allowed for the specified field (e.g. CLR or down select attempt).
NOT AN AIRPORT	Displayed when a non–airport identifier is up–selected to a field then it accepts only an airport identifier.
NOT IN DATABASE	Entry was not found in the navigation database.

MESSAGE TEXT	DESCRIPTION
NVM WRITE FAILED	Displayed when an operation that writes data to non–volatile memory fails.
OUT OF RANGE	Entry is not within acceptable range.
PLEASE WAIT	The FMS has not yet updated the CDU in response to the last entry (occurs only if the response can not be displayed within one second).
PLT DEFINED WPTS FULL	Proposed pilot defined waypoint would exceed the maximum number of pilot defined waypoints allowed.

In most cases, scratchpad messages are cleared by the CLR key or by pressing the next key which will go into the scratchpad line. The PLEASE WAIT and FPLN OPER IN PROGRESS messages clear themselves once the system has updated the CDU in response to the entry.

SYSTEM MESSAGES

System messages provide information about the status of the FMS system. Three classes of messages are used:

Attention Message – These messages require pilot attention. When added to the message page, the MSG annunciator will be lighted. Viewing the message causes the MSG annunciator to extinguish.

Action Message – These messages require the pilot to take action. When added to the message page, the MSG annunciator will be



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lighted. Viewing the message will not cause the MSG annunciator to extinguish. The annunciator will only extinguish when the cause of the message is cleared.

Status Message – These messages do not require pilot attention and do not light the MSG annunciator. Status messages can be checked at any time by reviewing the message page.

System messages are displayed as a list on the system message page in order of importance according to priority. Messages with a mode change prompt are the highest priority in order to assure they are displayed on the first message page. Messages are in large font, with color coding as follows:

- Green – A message that refers to a normal condition or status.
- Yellow – A message that refers to an abnormal condition or status.
- White – Mode change prompts on the system message page.
- Cyan – A yellow “attention” message is changed to cyan after viewing. The color change is reset when the message is removed.

In the following table, the T column shows the type of message (C – crew “action” required, A – “attention”, S – “status” only), the C column shows the message color (G – green, Y – yellow, W – white, C – cyan), and the SYS column tells which subsystem triggered the message.

The messages are listed in order of priority, beginning with the highest priority messages.

MESSAGE TEXT	T	C	SYS	DESCRIPTION
APPROACH ARMED	S	G	FLM	Indicates that the approach mode is armed and provides a white prompt (CANCEL APPROACH*) that allows it to be canceled. Appears when within 30 nm of destination and an approach is programmed on the flight plan.
APPROACH NEXT	S	G	FLM	Indicates that the airplane is about to go to approach mode and provides a white prompt (CANCEL APPROACH*) that allows it to be canceled. Appears when within 3 nm of the FAF.
APPROACH ACTIVE	S	G	FLM	Indicates that the approach sequence is being executed and provides a white prompt (CANCEL APPROACH*) that allows it to be canceled. Appears while in the approach mode.

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MESSAGE TEXT	T	C	SYS	DESCRIPTION
APPROACH CANCELED	S	G	FLM	Indicates that the approach has been canceled by the pilot and provides a yellow prompt (ARM APPROACH*) that allows it to be rearmed. Remains while within 30 nm of the destination and an approach is programmed on the flight plan.
APPROACH CANCELED	A	Y	FLM	Same as previous message, except aircraft has come within 3 nm of the FAF. This is a reminder the approach mode is not enabled.
APPROACH INVALID	A	Y	FLM	Indicates that the approach is invalid because of GPS integrity problems. Crosstrack and Vertical Deviation indications will also be flagged. Provides a yellow prompt (CANCEL APPROACH TO CLR FLAG*) that allows the pilot to clear the approach warn annunciation.

MESSAGE TEXT	T	C	SYS	DESCRIPTION
NOT IN DUAL FPLN MODE – SELECT FMS FOR SYNC	C	Y	FPLN	In a dual FMS installation this message indicates that the flight plans are out of sync. The operator must select the flight plans from FMS1 or FMS2. After this selection is made, the FMS will re-synchronize their flight plans.
MAG VAR WARNING – VAR ENTRY REQUIRED	C	Y	NAV	Indicates that the FMS is operating in the “MAG” display mode and the aircraft position is out of range of the magnetic variation model. The pilot must enter a manual magnetic variation.
NO NAV	S	Y	NAV	Indicates that the FMS is not navigating. Typically occurs on the ground prior to GPS sensor availability. The external DR annunciator will also be annunciated.
DR TRK/GS	S	Y	NAV	Indicates that the FMS is dead reckoning on last known track and groundspeed. The external DR annunciator will also be annunciated.



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MESSAGE TEXT	T	C	SYS	DESCRIPTION
DR HDG/TAS	S	Y	NAV	Indicates that the FMS is dead reckoning on heading, true airspeed and last known winds. The external DR annunciator will also be annunciated.
RNP EXCEEDED	A	Y	NAV	Indicates that the FMS Total System Error (TSE) is larger than the Required Navigation Performance (RNP).
SENSOR MIS-COMPARE	A	Y	NAV	Indicates that there is a discrepancy between the multiple sensors providing data to the FMS. Note: The tolerances are based upon the individual sensor EPEs.
FMS MIS-COMPARE	A	Y	NAV	Indicates that there is a discrepancy between the two FMS positions in a dual system. Note: The tolerance is based upon the two FMS EPEs.
RNP OVERRIDDEN	S	Y	NAV	Indicates that the pilot has overridden the system RNP value. This message initially occurs as status message. See next message.

MESSAGE TEXT	T	C	SYS	DESCRIPTION
RNP OVERRIDDEN	A	Y	NAV	Indicates that the pilot has overridden the system RNP value, and subsequent changes to the automatic RNP are lower than the pilot entered value.
XTK DEV WARNING	A	Y	LG	Indicates that the FMS crosstrack deviation is more than the Required Navigation Performance (RNP) value.
EXCEEDING BOUNDARY LIMIT – REDUCE SPEED	A	Y	LG	While holding or flying a procedure turn and within 3 minutes of entry, indicates that the TAS is high enough to cause the aircraft to stray outside the boundaries of the protected airspace.
NO AUTO LEG CHANGE	A	Y	LG	A discontinuity exists after the TO waypoint such that the FMS will not automatically sequence legs.
UNABLE ALT AT XXXXX	A	Y	PERF	Occurs when the FMS determines that the aircraft can not attain the altitude that is required at the current waypoint. The text XXXXX is the waypoint name.



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MESSAGE TEXT	T	C	SYS	DESCRIPTION
CHECK ALT CNST AT XXXXX	A	Y	PERF	Occurs when the FMS finds a crew entered constraint at a waypoint that will cause a descent in the climb flight phase or a climb in the descent flight phase. The text XXXXX is the waypoint name.
CHECK SPD CNST AT XXXXX	A	Y	PERF	Occurs for crew entered constraints that cause decelerations in climb or accelerations in descent. The text XXXXX is the waypoint name.
DISCON IN VERT PATH	A	Y	PERF	Occurs when the pilot enters a flight path angle constraint that does not connect with a flight path angle constraint programmed at a prior waypoint.
MMO EXCEEDED	A	Y	VG	Occurs when the current aircraft Mach exceeds maximum operating Mach.
VMO EXCEEDED	A	Y	VG	Occurs when the current indicated airspeed exceeds maximum operating airspeed.

MESSAGE TEXT	T	C	SYS	DESCRIPTION
1 MIN TO CLIMB	A	Y	VG	Displayed when the FMS is within one minute of initiating a climb.
1 MIN TO DESCEND	A	Y	VG	Displayed when the FMS is within one minute of initiating a descent.
1000 FT TO LEVEL OFF	A	Y	VG	Displayed when the FMS is within 1000 feet of starting a transition from climb or descent to level flight.
MAX ALT IS XXXXX	S	Y	PERF	Displayed when the FMS determines that the aircraft will cross the descent path before reaching cruise altitude.
VNAV NOT AVAILABLE	A	Y	VG	Displayed when the VNAV function is not available due to a failure of the baro altitude input.
INSUFFICIENT FUEL	A	Y	PERF	Displayed when the FMS has determined that fuel remaining at the destination is less than reserve fuel.



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MESSAGE TEXT	T	C	SYS	DESCRIPTION
CONFIG MODULE FAIL	A	Y	IO	Indicates that the FMS could not read the aircraft configuration data from the FMS configuration module. The FMS will be operating in a default configuration.
AIR DATA INPUT FAIL	A	Y	BIT	Displayed when the input source for altitude, TAS, VSPD, etc has failed.
HEADING INPUT FAIL	A	Y	BIT	Displayed when the input source for heading has failed.
FUEL FLOW N INVALID	A	Y	PERF	Displayed when the fuel flow input for engine number N has failed.
MEMORY FAIL	A	Y	PU	Indicates all or part of the data in the FMS nonvolatile memory was lost on power up.
FMS LINK FAIL	A	Y	IO	Indicates that the cross fill connection between the two FMS in a dual installation has failed.

MESSAGE TEXT	T	C	SYS	DESCRIPTION
FMS LINK FAIL – SW VERSIONS DIFFERENT – DUAL FPLNING NOT AVAIL	A	Y	IO	Indicates that the cross fill connection between the two FMS is invalid because the two FMS have different software revisions. Flight plans will not be synchronized.
FMS LINK FAIL – CONFIG DATA WRONG – DUAL FPLNING NOT AVAIL	A	Y	IO	Indicates that the cross fill connection between the two FMS is invalid because the two FMS are configured incorrectly. Flight plans will not be synchronized.
FMS DATABASES DIFFERENT – DUAL FPLNING NOT AVAIL	A	Y	IO	Indicates that the two FMS in a dual installation have different navigation databases. Flight plans will not be synchronized.



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MESSAGE TEXT	T	C	SYS	DESCRIPTION
SYNCHRONIZING FPLNS	S	Y	IO	Indicates that the two FMS in a dual installation are synchronizing flight plans.
FMS OVERHEATING – SWITCH FMS POWER OFF	C	Y	BIT	Indicates that the internal FMS temperature has exceeded 85 °C. NOTE The FMS will shutdown automatically is the internal temperature goes above 90 °C.

SENSOR MESSAGES

Sensor messages provide information about the status of position sensors connected to the FMS.

Sensor messages operate like system messages except as follows:

- Sensor messages are grouped by sensor instead of priority. The sensor name is provided as a label on the sensor message page prior to the list of messages from that sensor.
- Messages from deselected sensors will not light the message annunciator.

GPS Sensor Messages

The label on the sensor message page is GPS, GPS1 or GPS2.

Messages are in large font, with color coding as follows:

- Green – A message that refers to a normal condition or status.
- Yellow – A message that refers to an abnormal condition or status.
- White – Mode change prompts on the system message page.

In the following table, the T column shows the type of message (C – crew “action” required, A – “attention”, S – “status” only), the C column shows the message color (G – green, Y – yellow, W – white), and the SYS column tells which subsystem triggered the message.

MESSAGE TEXT	T	C	SYS	DESCRIPTION
DESELECTED	S	Y	NAV	Indicates that the GPS sensor has been deselected.
NO NAV	S	Y	NA	Indicates that the GPS sensor has not provided a valid position since the last cold start.
DR	A	Y	NAV	Indicates that the GPS sensor is not currently providing a valid position.
FAIL	A	Y	NAV	Indicates that the GPS sensor has failed.



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MESSAGE TEXT	T	C	SYS	DESCRIPTION
RAIM NOT AVAILABLE	A	Y	NAV	Indicates that RAIM is currently not available.
ACCURACY WARNING	A	Y	NAV	Indicates a possible GPS position error (poor satellite geometry or a RAIM error detected).
NO RAIM AT NNNN –NO GPS APR MODE	A	Y	NAV	Displayed when an approach is programmed on the flight plan and the GPS has determined that RAIM will not be available when the aircraft arrives at the destination. The text NNNN is the destination airport identifier.

VPU Sensor Messages

The label on the sensor message page is VPU.

Messages are in large font, with color coding as follows:

- Green – A message that refers to a normal condition or status.
- Yellow – A message that refers to an abnormal condition or status.
- White – Mode change prompts on the system message page.

In the following table, the T column shows the type of message (C – crew “action” required, A – “attention”, S – “status” only), the C

column shows the message color (G – green, Y – yellow, W – white), and the SYS column tells which subsystem triggered the message.

MESSAGE TEXT	T	C	SYS	DESCRIPTION
DESELECTED	S	Y	NAV	Indicates that the VPU sensor has been deselected.
NO NAV	S	Y	NAV	Indicates that the VPU sensor has not provided a valid position since the last cold start.
DR	A	Y	NAV	Indicates that the VPU sensor is not currently providing a valid position.

EXTERNAL ANNUNCIATORS

External annunciators are used to indicate various conditions. The external annunciation is driven either by dedicated discrete outputs from the FMS or digitally via the ARINC 429 link.

FMS Discrete Outputs

The FMS provides the following discrete outputs to be used to drive external annunciator lamps.

OUTPUT NAME	DESCRIPTION
APPROACH MODE	Lighted when the FMS approach mode is active.
DR	Lighted when the FMS is dead reckoning.



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OUTPUT NAME	DESCRIPTION
REMOTE MSG	Lighted when the MSG annunciator on the CDU is lighted. (new message on the CDU message page).
SEL XTK WARN	Lighted when the FMS is operating along a parallel offset lateral track that has been programmed (selected crosstrack).
WPT	Lighted when the FMS is about to start sequencing the TO waypoint (30 seconds prior to starting the lateral maneuver).
NAV 1 AUTO TUNE	Lighted when the FMS is auto-tuning the NAV 1 (VOR/DME) frequency.
NAV 2 AUTO TUNE	Lighted when the FMS is auto-tuning the NAV 2 (VOR/DME) frequency.
HEADING MODE	Lighted when the FMS heading mode is active.

FMS ARINC 429 Data Outputs

The FMS provides the following ARINC 429 data outputs to be used to drive annunciations on an interfaced display system. The actual usage of this data depends on the display system and the availability of data bits in the defined interface. The following list contains all the possible annunciations, some of which will not be used in certain installations.

OUTPUT NAME	DESCRIPTION
Waypoint Alert	Indicates that the FMS is within 30 seconds of starting the lateral maneuver to turn to the next leg.
Dead Reckon	Indicates when the FMS is in the dead reckoning mode (NO NAV, DR HDG/TAS, DR TRK/GS).
FMS Hdg Mode	Indicates the FMS heading sub-modes active. Redundant with LNAV sub-mode.
Integrity Warn	Indicates when TSE is larger than RNP. Same condition as the RNP exceeded message.
Parallel XTK Offset	Indicates when the FMS is operating on a parallel offset track (selected crosstrack).
Message Alert	Remote message alert. Indicates whenever the MSG annunciator on the FMS CDU is lighted.
Vertical Track Change Alert	Visual vertical track change alert. Indicates within one minute of an FMS initiated climb or descent, or within 1000 feet of a transition to level flight.



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OUTPUT NAME	DESCRIPTION
Vertical Track Change Horn	Audible VTA alert. Indicates initial departure from the cruise altitude in coupled VNAV mode.
Max Speed Alert	Indicates that the FMS is limiting the speed to VMO during a coupled VNAV path, possibly resulting in a deviation from the programmed vertical path.
LNAV Sub-mode	Indicates when the FMS is in the heading select mode or the heading intercept mode.
VNAV Sub-mode	Indicates the FMS VNAV sub-mode when the FMS is VNAV coupled.
Lat. Dev. Scaling in Transition	Indicates the FMS lateral deviation scaling is in transition under FMS control.
GPS APPR Integrity	Indicates the GPS is in the approach integrity mode.
GPS Integrity	Set when the GPS sensor indicates the RAIM is not available.
GPS Integrity Mode Ann.	Indicates the GPS integrity mode.

LIST OF ABBREVIATIONS AND DEFINITIONS

ABBREVIATION	DEFINITION
ACARS	Airborne Communications Addressing and Reporting System
ACQ	Acquisition
ADF	Automatic Direction Finder
ALT	Altitude
APP(R)	Approach
APR	Approach
APRCH	Approach
ARC	DME Arc
ARINC	Aeronautical Radio, Inc.
ARR	Arrive, Arrival
ATO	Along Track Offset
AVG	Average
AZM	Azimuth
BARO	Barometric Altitude
BIT	Built In Test



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ABBREVIATION	DEFINITION
BRG	Bearing
CAS	Calibrated Airspeed
CDI	Course Deviation Indicator
CDU	Control Display Unit
CLR	Clear
CNST(R)	Constraint
COMM	Communications
COMP	Compensated, Compensation
CORR	Correction
CPU	Central Processing Unit
CRS	Course
DEL	Delete
DEP	Depart(ure)
DES	Descent
DESEL	Deselect

ABBREVIATION	DEFINITION
DEST	Destination
DEV	Deviation
DIR	Direct
DIS(T)	Distance
DISCON	Disconnect
DME	Distance Measuring Equipment
DR	Dead Reckoning
DTK	Desired Track
EFIS	Electronic Flight Instrument System
EL	Satellite Elevation Above the Horizon
ELEV	Elevation
EPE	Estimated Position Error
ETA	Estimated Time of Arrival
ETE	Estimated Time Enroute
EXEC	Execute



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ABBREVIATION	DEFINITION
FAF	Final Approach Fix
FDE	Fault Detection and Exclusion
FLM	
FMS	Flight Management System
FPA	Flight Path Angle
FPL/FPLN	Flight Plan
FPM	Feet Per Minute
FR	From
FREQ/FRQ	Frequency
FT	Feet
GND	Ground
GPS	Global Positioning System
GRAD	Gradient
GS	Groundspeed
HAA	Height Above Aerodrome

ABBREVIATION	DEFINITION
HDG	Heading
HI	High
HLTH	Satellite Health
HPE	Horizontal Position Error
HR	Hour
IAF	Initial Approach Fix
ICAO	International Civil Aviation Organization
IDENT	Identifier
INIT	Initialize, Initialization
IO	Input–Output
Jeppesen	Name of Navigation Database Source
KG	Kilogram/Kilograms
kHz	Kilohertz
LAT/LONG	Latitude/Longitude
LB	Pound/Pounds

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ABBREVIATION	DEFINITION
LCD	Liquid Crystal Display
LG	Leg
LIM	Limit
LNAV	Long–Range Navigation
LO	Low
LSK	Line Select Key
LVL	Level
MAG	Magnetic
MAG DEC	Magnetic Declination
MAG VAR	Magnetic Variation
MAP	Missed Approach Point
MAX	Maximum
MAXLEN	Maximum Length (Runway)
MCDU	Multi–Functional Control Display Unit
MHz	Megahertz

ABBREVIATION	DEFINITION
MIN	Minute
Mmo	Maximum Operating
MACH	Speed
MSG	Message
MSL	Mean Sea Level
NAV	Navigation
NAVAID	Navigation Aid (Typically a Radio Station Providing VOR and/or DME Position Data)
NDB	NAV Database or Non–Directional Beacon
NM	Nautical Miles
NVM	Non–Volatile Memory
OFST	Offset
OPER	Operation
ORIG	Origin
PCMCIA	Personal Computer Memory Card Interface Association

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ABBREVIATION	DEFINITION
PERF	Performance
PLT	Pilot
POS	Position
PPOS	Present Position
PREV	Previous
PRN	Pseudo Random Noise (GPS Satellite Number)
PROC	Procedure
PU	Power Up
RAIM	Receiver Autonomous Integrity Monitoring
REF	Reference
REQ	Required
RNG	Range
RNP	Required Navigation Performance
RWY	Satellite

ABBREVIATION	DEFINITION
SAT	Runway
SCL	Scale
SEG	Segment
SEL	Select
SID	Standard Instrument Departure
SNR	Signal to Noise Ratio
SPD	Speed
SRCH	Search
STAR	Standard Terminal Arrival SYNC Synchronize(d)
SYS	System
T	True (Heading or Track) or whether a Satellite is being tracked or not
TAS	True Airspeed
TERM	Terminal
TK	Track Angle



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ABBREVIATION	DEFINITION
TOC	Top of Climb
TOD	Top of Descent
TRAN(S)	Transition(s)
TRK	Track
TSE	Total System Error
TX	Transmit
UTC	Universal Coordinated Time
VAR	Magnetic Variation
VERT DEV	Vertical Deviation
VG	Vertical Guidance Subsystem
VHF	Very High Frequency
Vmo	Maximum Operating Airspeed
VNAV	Vertical Navigation
VOR	VHF Omni Range
VPU	VORTAC Position Unit

ABBREVIATION	DEFINITION
VSPD	Vertical Speed
VTA	Vertical Track Change Alert
WPT(S)	Waypoint(s)
XPDR	Transponder
XTK	Crosstrack

[Refer to Figure 176.](#)

When a FMS1 or FMS2 bearing is selected on the EFIS Control Panels (EFCP 1, EFCP 2), a distinctive pointer comes into view on the Primary Flight Display.

[Refer to Figure 177.](#)

The bearing pointers in the Electronic Horizontal Situation Indicators (EHSI 1, EHSI 2) show the lateral direction to the navigation aid as follows:

- A white, single bar pointer with a white circle to identify FMS1
- A green, wide two-bar pointer with a green diamond to identify FMS2.

The bottom left part of the Primary Flight Displays (PFD1, PFD2) shows a reminder label for FMS1 and the bottom right part shows a reminder label for FMS2. Each reminder label is related to the EFIS



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Control Panel selection. Each indication has a symbol that is similar to its bearing pointer with a FMS1 or FMS2 legend.

NOTE

Note: The reminder bearing symbol and legend are the same colour, white or green.

[Refer to Figure 178.](#)

When a FMS has malfunctioned, its bearing pointer goes out of view, but its reminder label stays in view to show the bearing pointer selection.

[Refer to Figure 179.](#)

The NAV SOURCE rotary switch located on the Flight Guidance Control Panel (FGCP) is used to select the FMS navigation source.

[Refer to Figure 180.](#)

The Primary Flight Display (PFD) shows the selection in the upper right part of the Electronic Horizontal Situation Indicators (EHSI1, EHSI2). The Primary Flight Displays show the Navigation Source Annunciations that follow:

- VOR1, VOR2
- ILS1, ILS2
- FMS1, FMS2
- MLS1, MLS2.

The Primary Flight Display will show the FMS parameters that follow:

- Vertical deviation
- Lateral deviation (crosstrack).

Vertical Deviation: The PFD shows the vertical parameters that follow:

- Vertical deviation pointer and scale
- Vertical deviation source indication.

Vertical Deviation Pointer and Scale: The Primary Flight Displays (PFD1, PFD2) use a pointer and scale to show deviation from the FMS vertical path. The vertical scale and its magenta pointer are shown to the right of the attitude sphere.

The sensitivity of the FMS vertical path deviation scale changes between en-route and approach flight. During en-route, 2 dots shows 500 ft vertical deviation and during approach 2 dots deviation shows 150 ft vertical deviation.

When the pointer goes beyond the scale (the pointer is deviated 120% or more), the diamond indication changes to a fixed arrow that is half of the original symbol.

Vertical Deviation Source: The Vertical Deviation Source Indication shows a magenta V symbol above the scale for FMS vertical path deviation.

Lateral Deviation: The Primary Flight Display will show the FMS lateral parameters that follow:

- Navigation source annunciation
- FMS waypoint identifier
- Selected Desired Track (DTK) digital value
- Selected DTK Pointer
- Selected DTK Scale and Bar
- To/From pointer



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- Track angle
- FMS general messages
- FMS lateral scale sensitivity messages
- FMS Navigation performance messages
- FMS Waypoint and Vertical track alert messages
- FMS distance
- FMS Failure Flags.

Navigation Source Annunciation: The Navigation Source Annunciation shows which navigation source is selected using the related NAV SOURCE rotary switch on the Flight Guidance Control Panel (FGCP). When the related FMS source is selected, the annunciation is shown in magenta letters. If an opposite navigation source is selected, the annunciation is shown in yellow fonts.

On aircraft with V800 or V811 Avionics Suite installed, the NAV Source Annunciation is always shown in magenta for the single FMS configuration.

If the section data has malfunctioned, six white dashes replace the source annunciation.

FMS Identifier: A maximum of 6 magenta letters are used to show the active waypoint identifier. The indication is replaced by 5 white dashes when the data in the FMS malfunctions.

Selected Desired Track (DTK) Digital Value: The selected desired track indication is controlled by its related COURSE knob on the Flight Guidance Control Panel (FGCP). The selected desired track digital value is shown by 3 magenta numbers from 1 through 360. A degree symbol is shown in white adjacent to the digital value.

If the selection data has malfunctioned, three white dashes replace the source annunciation.

Selected DTK Pointer: The COURSE knob is turned to set the desired track. It turns a cyan pointer around the 360 degree heading scale to show the selected desired track. Each COURSE knob controls its related DTK pointer.

When no heading data is received or not correct, the selected DTK pointer goes out of view.

Selected DTK Scale and Bar: The DTK scale and bar is part of the DTK pointer. It shows lateral deviations from the desired track. The DTK scale and bar with the DTK pointer and the bar moves independently left or right of the DTK pointer.

TO/FROM pointer: The TO/FROM pointer symbol is part of the DTK pointer indication. It turns with the DTK pointer to show the direction of navigation in relation to the active waypoint. The TO/FROM pointer symbol is also shown in magenta.

The bar is shown to the right of the aircraft symbol when the aircraft is flying to the left of the desired course and is shown to the left of the aircraft symbol when the aircraft is flying to the right of the desired course. The lateral deviation bar centers in the course pointer when the aircraft is on course.

The TO/FROM pointer symbol goes out of view if the FMS malfunctions.

Track Angle: The aircraft track is shown by a magenta triangle that moves on the heading scale.

When heading source malfunction or the data is not received valid from the FMS, the indication goes out of view.



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FMS General Messages: The PFDs show the FMS general messages that follow:

MESSAGE	COLOUR	DESCRIPTION
MSG	Yellow	Alerts the pilots to a MCDU message
XTRK	Green	Aircraft is along the flight plan, but with a parallel offset, as set by the MCDU

FMS lateral scale sensitivity messages: A digital indication of the scale sensitivity is not required. When the FMS is the selected navigation source, the PFD shows the lateral scale sensitivity messages from FMS as follows:

MESSAGE	COLOUR	DESCRIPTION
APPR	Green	FMS is operating in the approach mode
TERM (TBC)	Green	FMS is operating in terminal mode

FMS Navigation performance messages: When FMS is the selected navigation source, the PFDs show the navigation performance messages in order of importance from the FMS as follows:

PRIORITY	MESSAGE	COLOUR	DESCRIPTION
1	NAV DR	Yellow	FMS is in dead reckoning mode
2	NAV INTEG	Yellow	Required Navigation Performances (RNP) are not guaranteed by FMS
3	GPS INTEG	Yellow	GPS integrity (RAIM) is decreased

FMS Waypoint and Vertical track alert messages: When FMS is the selected navigation source, the PFDs show the waypoint and vertical track alert messages from the FMS as follows:

MESSAGE	COLOUR	DESCRIPTION
WPT ALERT	Magenta	FMS is about to start sequencing the TO Waypoint. (It occurs 30 seconds before the lateral maneuver is started).
VERT ALERT	Magenta	FMS is about to start a vertical track change, and that throttle adjustment to do the maneuver may be required. It occurs 60 seconds before a change from flight level to climb or descent, or 1000 feet prior to level off if climbing or descending.



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FMS distance: The FMS distance indication shows the FMS distance in magenta letters and numbers at the top right part of the navigation page when the navigation source is set to FMS.

The top right part of the Electronic Horizontal Situation Indicator (EHSI) area of the PFD shows the FMS distance as magenta numbers from 0 through 300 nmi. The resolution of the indication is 0.1 nmi for distances from 0 through 99.9 nmi and 1 nmi for distances more than 99.9 nmi.

On aircraft with V800 or V811 Avionics Suite installed, the FMS crosstrack resolution is 0.01 nmi in the approach phase.

The numerical values for the FMS distance to go indications change to three white dashes when the FMS distance data is not valid.

FMS Failure Flags: The course deviation failure flag is a red cross shown with the course deviation scale. The conditions that follow will cause it to come into view:

- FMS malfunctions
- No FMS desired track data available
- No FMS navigation source selection data available.

[Refer to Figure 181.](#)

The PFDs show the failure flags that follow:

- Vertical Deviation, FMS vertical path
- Crosstrack deviation.

[Refer to Figure 182.](#)

The MFD1 and MFD2 reversion switches located on the ESID Control Panel (ESCP) are used to select the navigation page. The navigation page comes into view in the map mode format.

[Refer to Figure 183.](#)

The navigation page comes into view in the map mode format. The map mode shows the same basic FMS parameters as its related PFD indication on a 90 degree heading arc as follows:

- FMS bearing pointers
- Navigation source annunciation
- FMS waypoint identifier
- Selected Desired Track (DTK) digital value
- Track angle
- FMS distance.

[Refer to Figure 184.](#)

It also shows the additional FMS parameters that follows:

- Desired track pointer
- Actual track (Drift angle)
- Ground speed digital value
- Wind speed indication
- Wind direction indication
- FMS abnormal status messages
- Map symbology.

Desired Track Pointer: The desired track pointer is a magenta indication that is shown in the ARC page format only. It turns around the aircraft symbol relative to the DTK digital value to show the difference between the current flight path calculated by the FMS and the flight plan map.



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If the aircraft heading or FMS data has malfunctioned, the desired track pointer goes out of view.

FMS Actual Track (Drift Angle): The actual track pointer is a magenta indication that is shown in the ARC page format only. It turns around the heading scale to show the aircraft actual track.

When two FMS's are installed, the navigation page shows the actual track indication from the selected FMS. If it malfunctions, the actual track indication is supplied by the opposite FMS.

If the aircraft heading is invalid or no valid data is received from the selected FMS, the actual track pointer is out of view.

FMS Abnormal Status Messages: When FMS is the selected navigation source, the navigation page shows the abnormal status messages in order of importance from the FMS as follows:

PRIORITY	MESSAGE	COLOUR	DESCRIPTION
1	NAV DR	Yellow	FMS is in dead reckoning mode
2	NAV INTEG	Yellow	Required Navigation Performances (RNP) are not guaranteed by FMS
3	GPS INTEG	Yellow	GPS integrity (RAIM) is decreased

DME/FMS Ground Speed: The ground speed indication shows the FMS groundspeed indication in magenta numbers and letters at the bottom left part of the MFD navigation page when the navigation source is set to FMS.

The indication has magenta numbers from 0 through 999 kts with a GS label. The resolution of the indication is 1 kt.

Wind Speed Indication: The wind speed indication gives a digital indication of the wind speed from FMS calculations. It is shown in white characters from 0 to 999 knots with 1 kt increments and a GS label also in white characters.

The digital value comes into view when the magnitude of the wind is more than 6 kts. When the wind speed is less than 3 kts, the indication changes to a green LGT/VAR message to show that the wind is light and variable.

When two FMS's are installed, the navigation page shows the wind speed indication from the selected FMS. If it malfunctions, the wind speed indication is supplied by the opposite FMS.

When the wind speed data is not valid from the FMS or when the navigation source selection data is not valid, the 3 digits change to 3 white dashes.

Wind Direction Indication: A green arrow that represents the wind direction turns relative to the magnetic heading in the arc mode format only.

The wind speed indication goes out of view when wind magnitude is less 3 kts and comes into view when the wind speed is more than 6 kts.

When two FMS's are installed, the navigation page shows the wind direction indication from the selected FMS. If it malfunctions, the wind speed indication is supplied by the opposite FMS.



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Refer to Figure 184.

Map Symbology: The FMS calculates the map symbols that follows:

- WPT
- VOR
- VOR/DME
- DME/TACAN
- PSEUDO-WPT
- AIRPORT
- ORIGN/DEST RUNWAY
- HOLDING PATTERN
- PROCEDURE TURN (LEFT)
- PROCEDURE TURN (RIGHT).

A map is shown in the editing area of the navigation page in map or plan formats. The map has continuous lines, curves and straight lines to connect the waypoints. The data shown depends on the range selection and current position of the aircraft relative to the flight plan.

The entire map is oriented relative to magnetic heading in the map format and magnetic north in the plan format. The Display Units (DUs) continuously update the position of the map on the screen with the current aircraft heading and position.

When the map data from the selected FMS is not valid it goes out of view.

Refer to Figure 185.

An offset path selection causes the FMS to calculate a new map so that the active and next legs are shown parallel to the original legs. It is laterally moved by the selected offset distance. The original FROM and TO waypoints and the remaining part of the flight plan is not changed.

When the TO waypoint is sequenced, it changes to the FROM waypoint, and the NEXT leg changes to the active leg. If the new active leg cannot be flown with an offset, the FMS will not shift it to give an advance indication of the aircraft programmed path.

The active leg and active waypoint are shown in magenta. The remaining part of the flight plan is in white color. All the identifiers or constraints labels shown are the same color as the related symbol. A white waypoint symbol (star) with a WPT label is shown in place of the VOR/DME station symbol when a valid flight plan is shown.

WPT: Waypoints along the flight plan are always shown as stars even when they relate to a nav aids or an airport but not a runway. Each waypoint symbol has an identifier that is shown in the same color as the waypoint to the upper right side of the waypoint symbol. It has a maximum of 6 characters with a second line to show other data if it exists.

VOR, VOR/DME, DME/TACAN (Optional): The MAP pushbutton switch on the EFIS Control Panel (EFCP) is pushed to show FMS nav aids in the map editing area of the navigation page in ARC or FULL format. The 10 nearest VOR, VOR/DME, and DME/TACAN navigation aids that are not part of the map is shown.



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The DATA pushbutton switch is pushed to show FMS navigation aids and airports when the navigation page is set to MAP or PLAN format. The first selection causes the navigation page to show the nearest navigation aids. It is pushed again to shown the nearest airports. The third selection causes the navigation page to show the nearest navigation aids and airports. The fourth selection causes the data indications to go out of view.

The navigation aids are shown as different blue symbols that relates to the navigation aid type. The latitude and longitude coordinates of the navigation aid is calculated by the selected FMS. When available, the navigation aid symbol has an identifier that is shown in the same color as the waypoint to the upper right side of the navigation aid symbol.

When two FMS's are installed, the navigation page shows the navigation aid symbols from the selected FMS. If it malfunctions, the navigation aid symbols are supplied by the opposite FMS.

PSEUDO-WPT: An PSEUDO-WPT symbol shows the FMS calculated VNAV altitude profile point along the active flight plan. It is a small circle with a character number and label to identify the type of altitude profile point as follows:

- TOC, Top Of Climb
- TOD, Top Of Descent
- PRESL, preselect altitude.

Pseudo-waypoint is shown as a white symbol. If it part of the active leg, it changes to magenta (TBC).

A maximum of 10 pseudo-waypoints are supplied by the FMS.

AIRPORTS: The DATA pushbutton switch on the EFIS Control Panel (EFCP) is pushed again to show airports in the map editing area of

the navigation page in arc or plan format. The 10 nearest airports that are not part of the map is shown.

NOTE

Note: A DATA pushbutton switch selection has no effect when the navigation page is set to show an ARC VOR/ILS or ARC MLS format.

The airport symbols are shown as blue symbols. The latitude and longitude coordinates of the airport is calculated by the selected FMS. When available, the airport symbol has an identifier that is shown in the same color as the waypoint to the upper left side of the airport symbol.

When two FMS's are installed, the navigation page shows the airport symbols from the selected FMS. If it malfunctions, the airport symbols are supplied by the opposite FMS.

The DATA pushbutton switch on the EFIS Control Panel (EFCP) is pushed again to show navigation aid and airports in the map editing area of the navigation page in ARC or FULL format.

Reminder labels at bottom left part of the navigation page shows the DATA pushbutton switch selections as follows:

- A blue navigation aid symbol with a NAV label
- A blue airport symbol with an ARPT label.

An FMS Map drawing status message FPLN IN PROG is shown in white letters when the FMS flight plan calculation is in progress. This indication can come into view after a range selection.

RUNWAY: Two different symbols are used to show the origin or destination airport or runway along the flight plan:

- Runway



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- Airport.

Runway: When the heading of the runway is programmed by the pilots, a runway symbol is shown as two parallel lines relative to the magnetic heading.

Airport: If the heading of the runway is not programmed, the airport is shown as a waypoint symbol.

HOLDING PATTERN: The holding pattern symbol is shown when an holding pattern is programmed along the flight plan. Two different holding pattern symbols are shown that relate to the display range selection or the location of the holding pattern on the flight plan as follows:

- Variable size
- Fixed size.

Variable Size: The symbol is shown to scale relative to the FMS graphic pattern when the holding pattern symbol is part of the active or the next leg and display range is set to 80 NM or less.

Fixed Size: The symbol is oriented relative to the holding pattern azimuth calculated by the FMS.

PROCEDURE TURN: The procedure turn symbol is shown when a procedure turn is programmed along the flight plan. Two different holding pattern symbols are shown that relate to the display range selection or the location of the procedure turn symbol on the flight plan as follows:

- Variable size
- Fixed size.

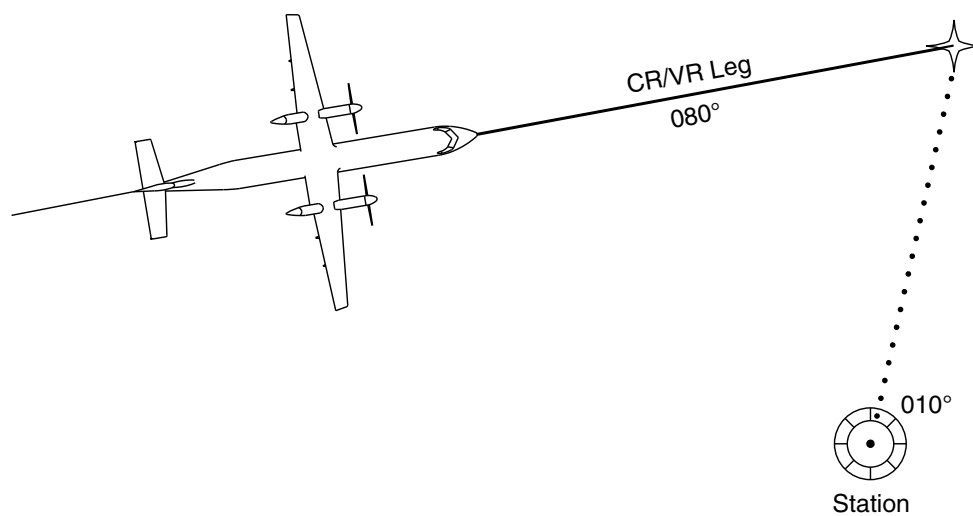
Variable Size: The symbol is shown to scale relative to the FMS graphic pattern when the holding pattern symbol is part of the active or the next leg and display range is set to 80 NM or less.

Fixed Size: The symbol is oriented relative to the holding pattern azimuth calculated by the FMS.



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fsp80a01.cgm

CR/VR LEG
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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Config 001

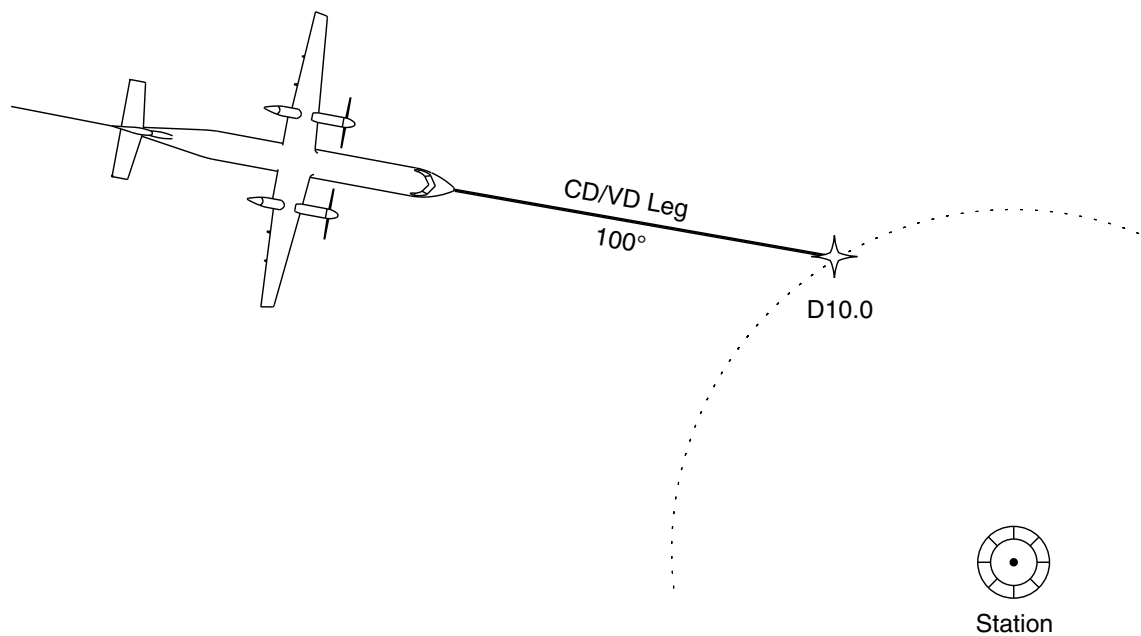
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fsp77a01.cgm

CD/VD LEG
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

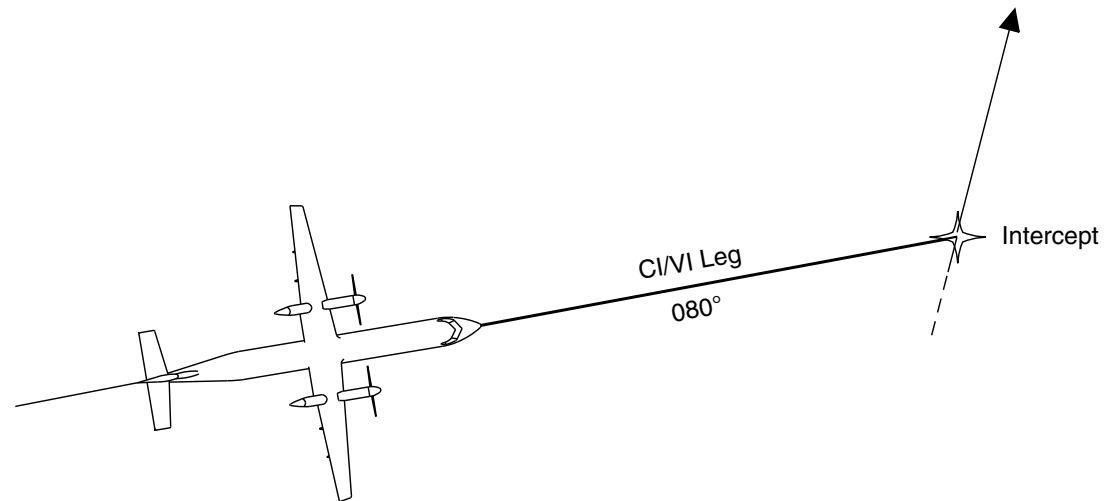
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fsp79a01.cgm

CI/VI LEG
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

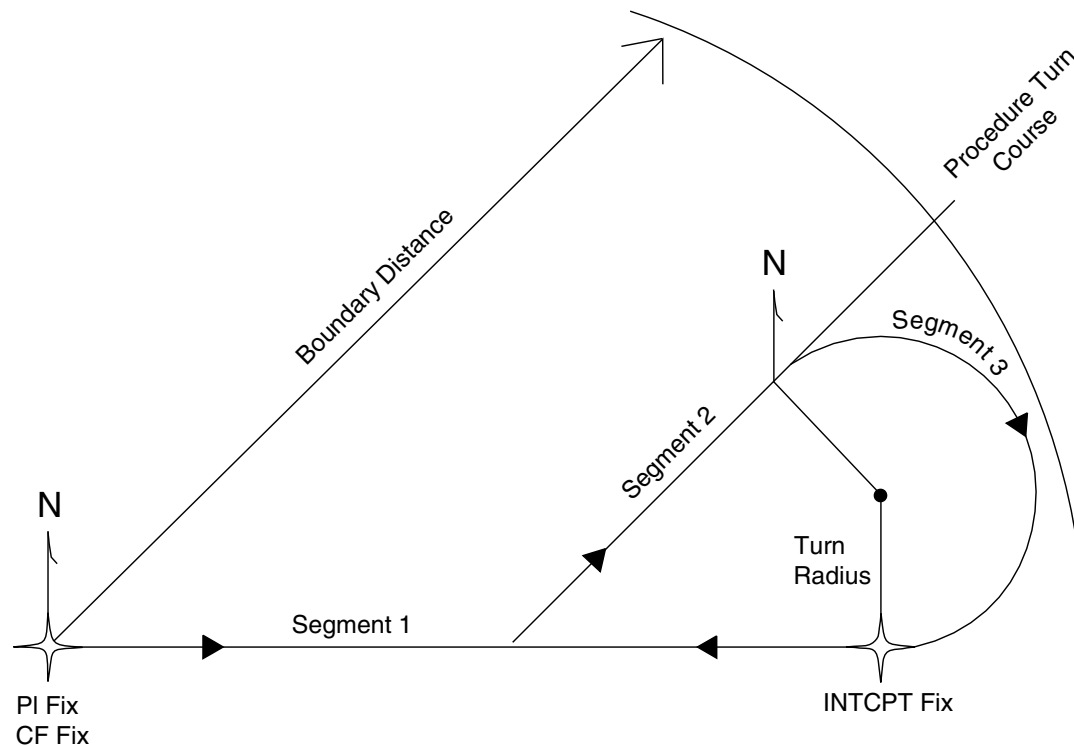
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fsp90a01.cgm

PROCEDURE TURN (PI)
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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Config 001

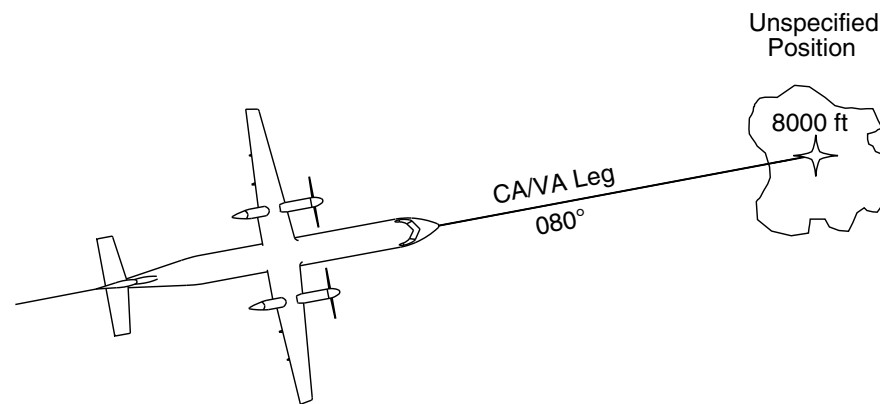
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fsp76a01.cgm

CA/VA LEG
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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Config 001

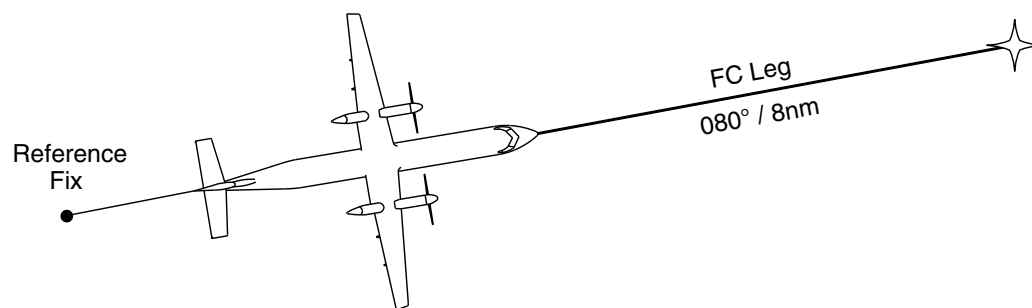
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fsp82a01.cgm

FC LEG
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

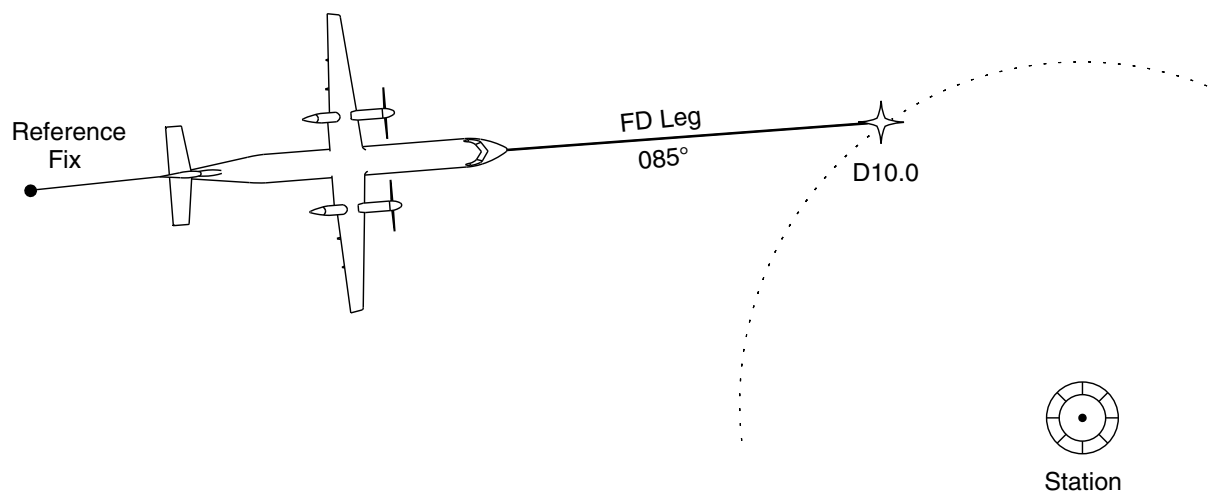
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fsp83a01.cgm

FD LEG
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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Config 001

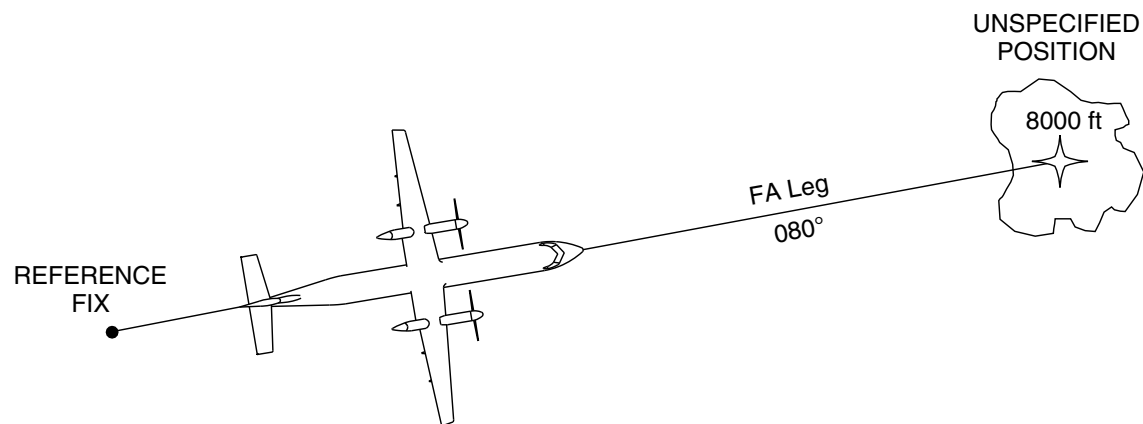
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fsp81a01.cgm

FA LEG
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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Config 001

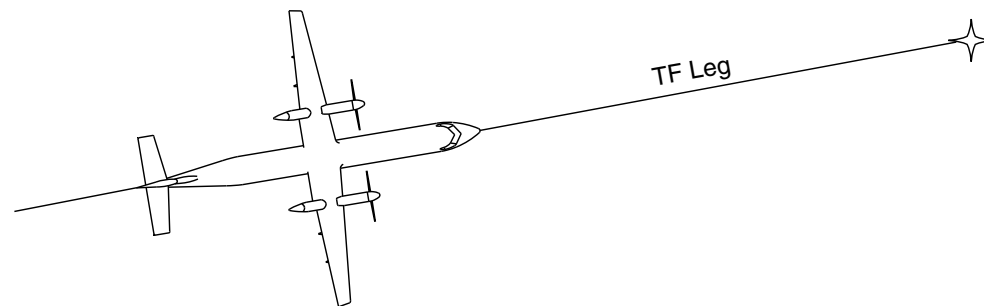
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fsp85a01.cgm

TRACK TO A FIX (TF) LEG
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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Config 001

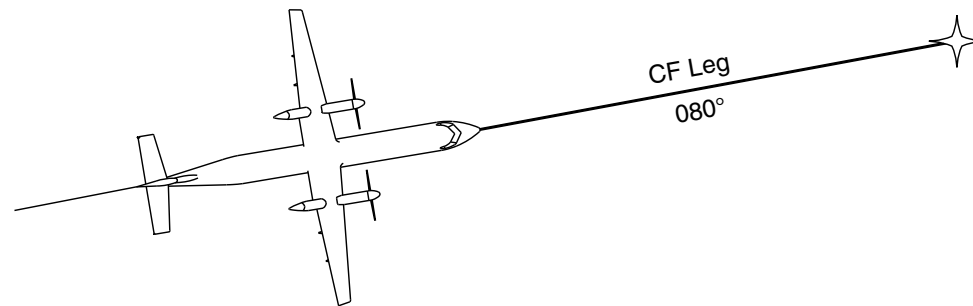
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fsp78a01.cgm

COURSE TO A FIX (CF) LEG
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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Config 001

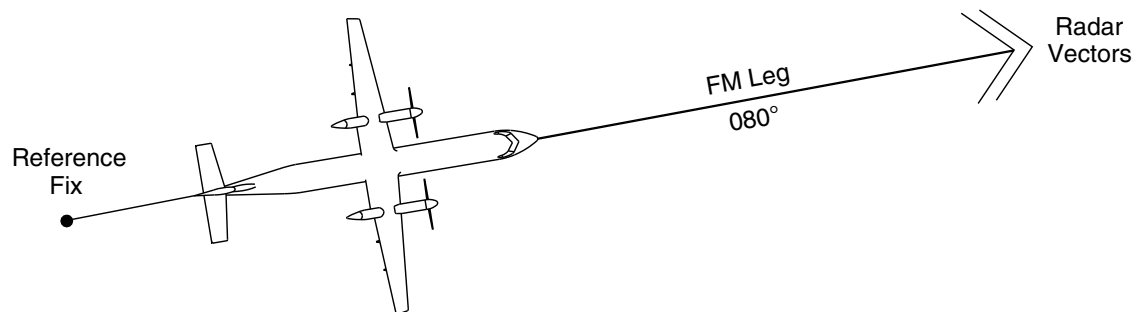
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fsp84a01.cgm

FIX TO A MANUAL TERMINATION (FM) LEG
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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Config 001

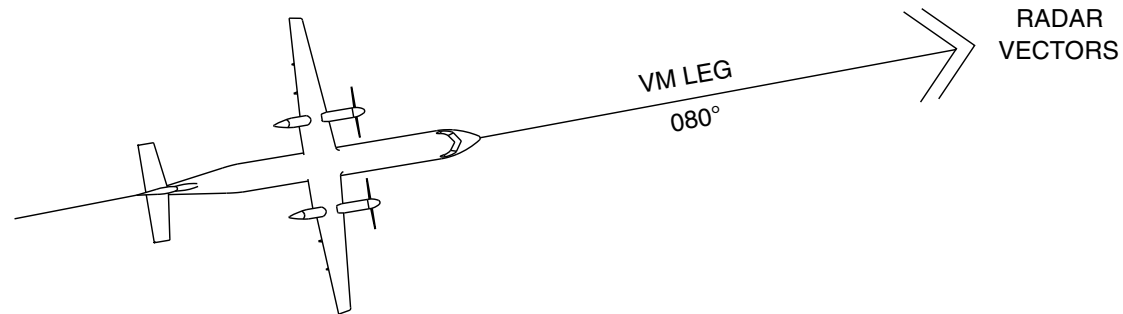
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fsp86a01.cgm

HEADING TO A MANUAL TERMINATION (VM) LEG
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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Config 001

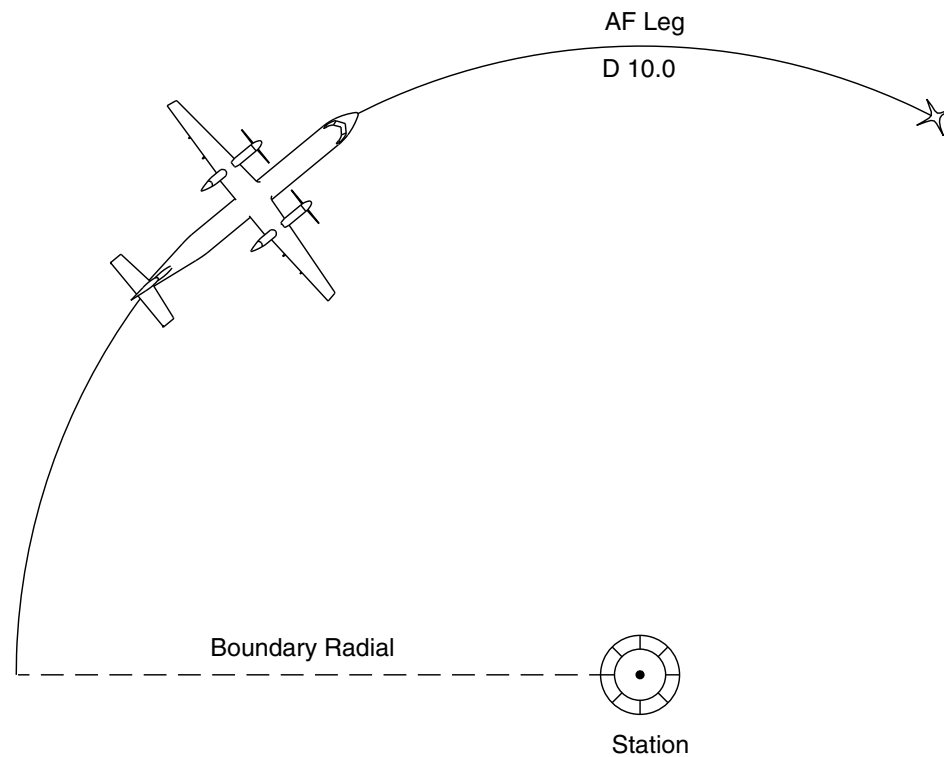
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fsp92a01.cgm

AF LEG
Figure 13

PSM 1-84-2A
EFFECTIVITY:
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Config 001

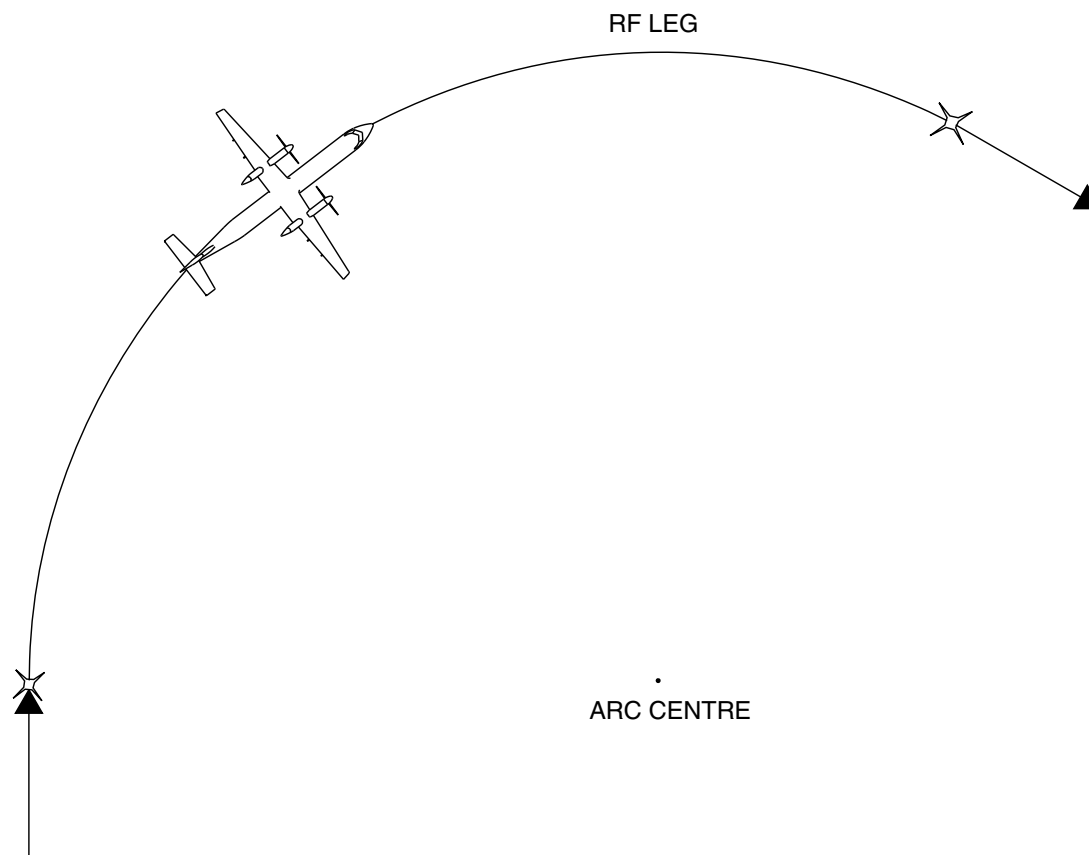
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fsp89a01.cgm

RF LEG
Figure 14

PSM 1-84-2A
EFFECTIVITY:
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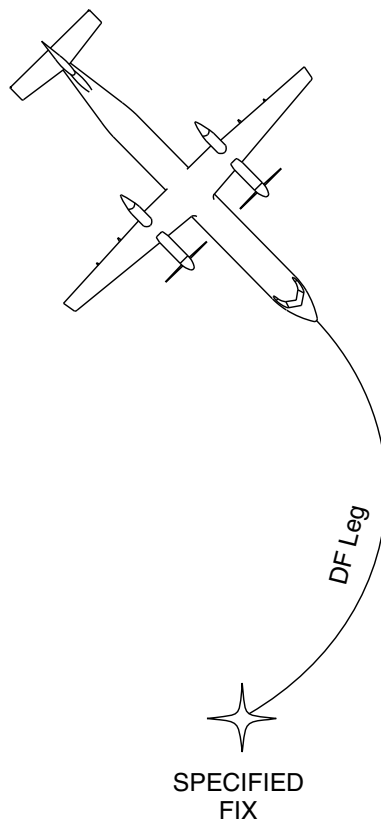
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fsp91a01.cgm

DF LEG
Figure 15

PSM 1-84-2A
EFFECTIVITY:
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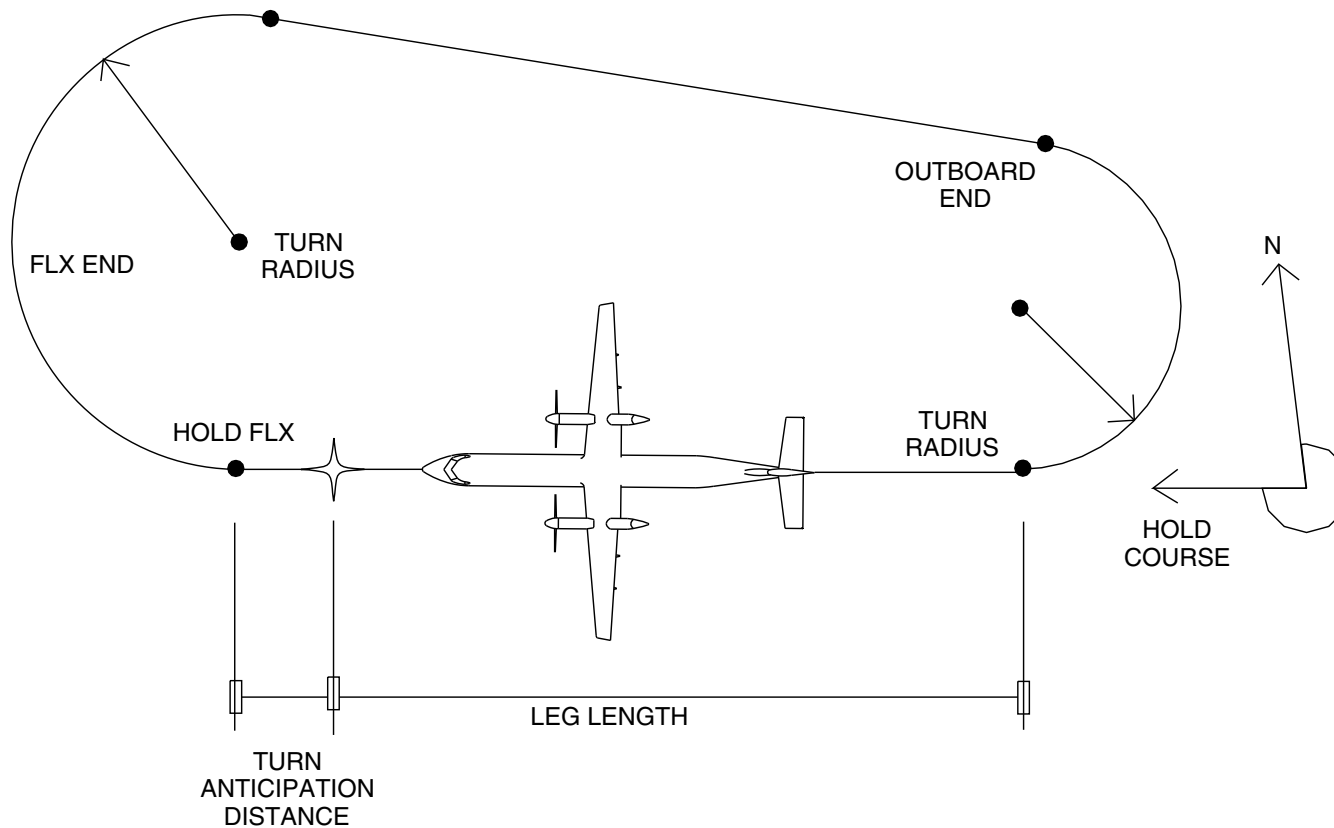
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fsp87a01.cgm

HOLDING PATTERN
Figure 16

PSM 1-84-2A
EFFECTIVITY:
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Config 001

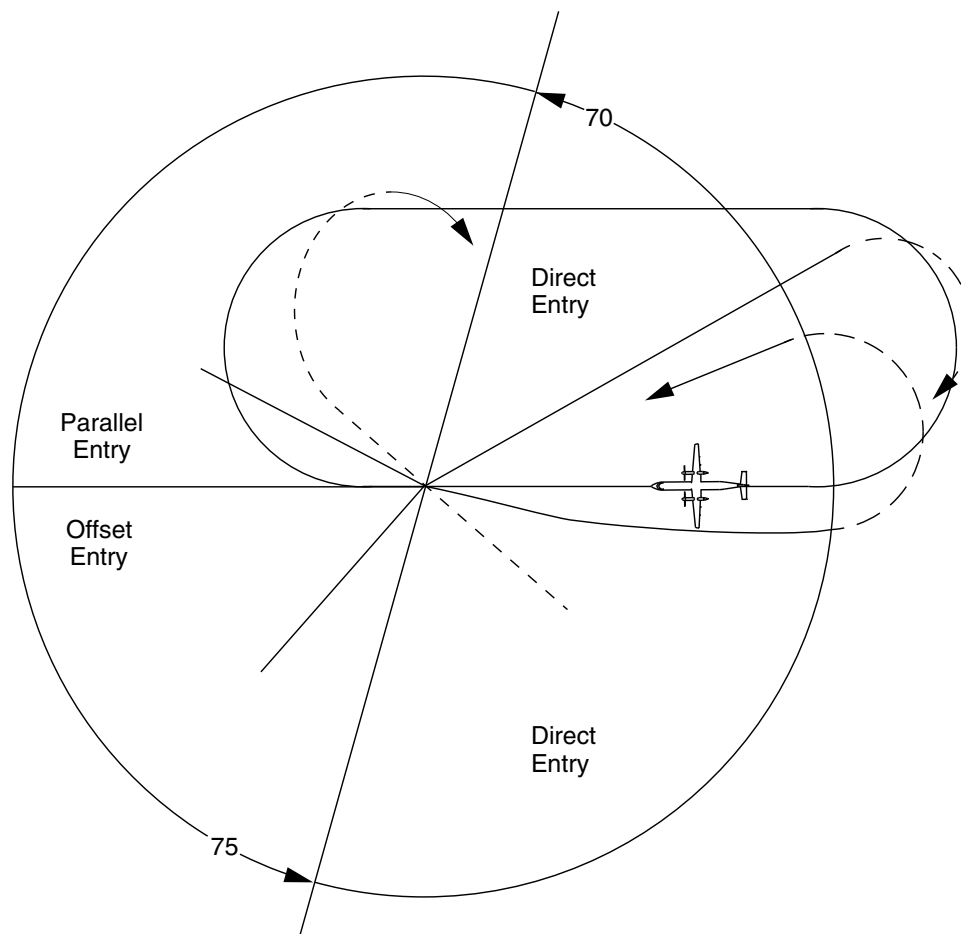
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fsp88a01.cgm

HOLDING PATTERN ENTRY PROCEDURE ZONES
Figure 17

PSM 1-84-2A
EFFECTIVITY:
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Config 001

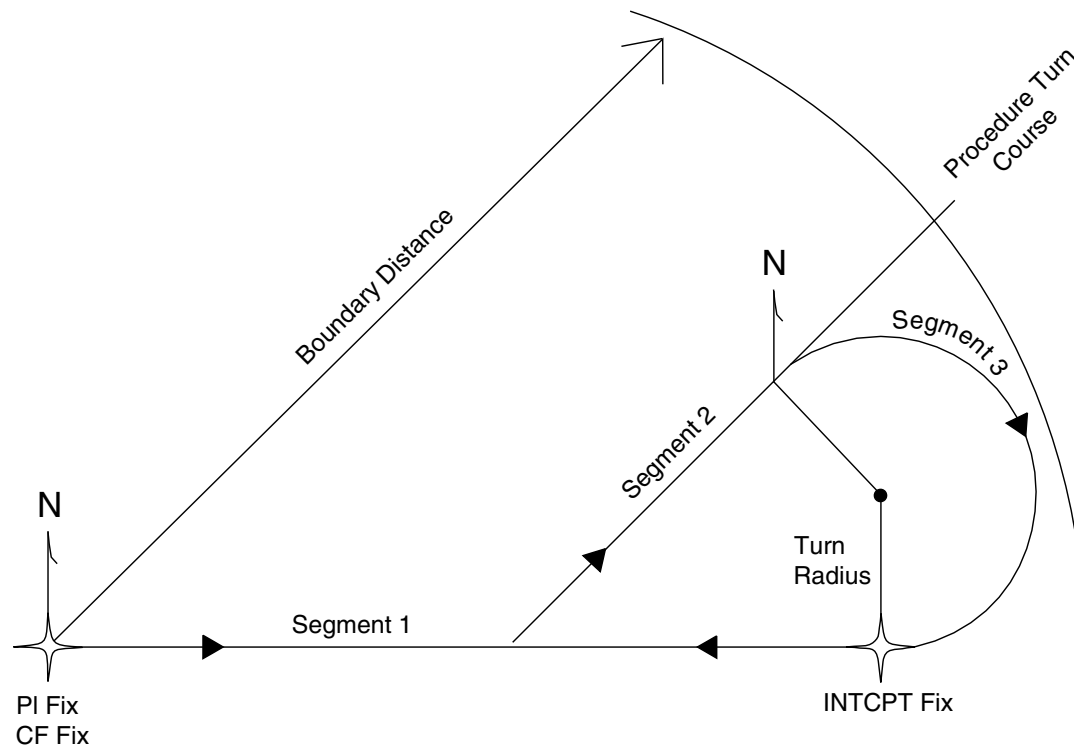
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fsp90a01.cgm

PROCEDURE TURN, STANDARD ENTRY
Figure 18

PSM 1-84-2A
EFFECTIVITY:
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Config 001

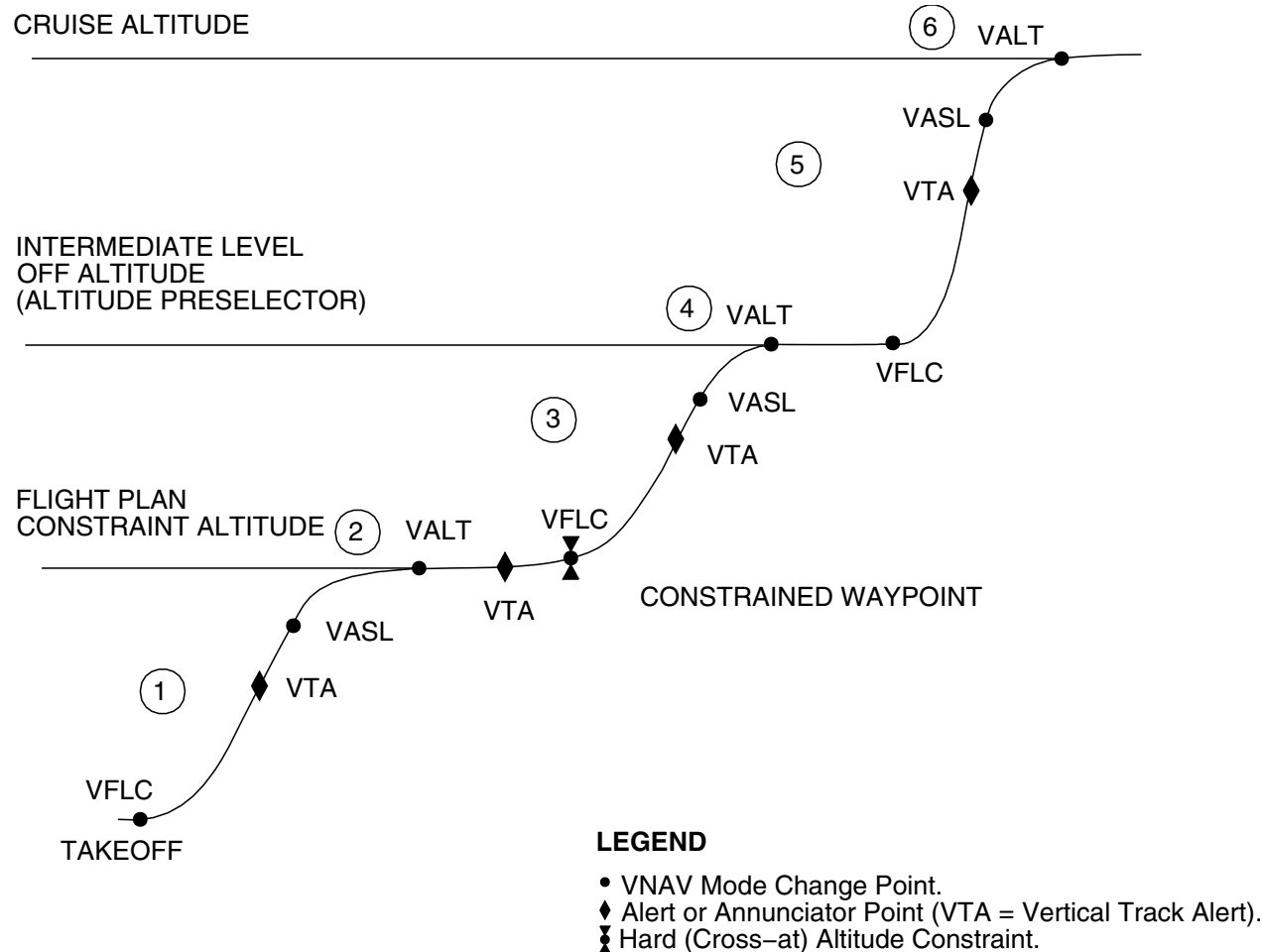
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fsp69a01.com

VNAV CLIMB
Figure 19

PSM 1-84-2A
EFFECTIVITY:
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Config 001

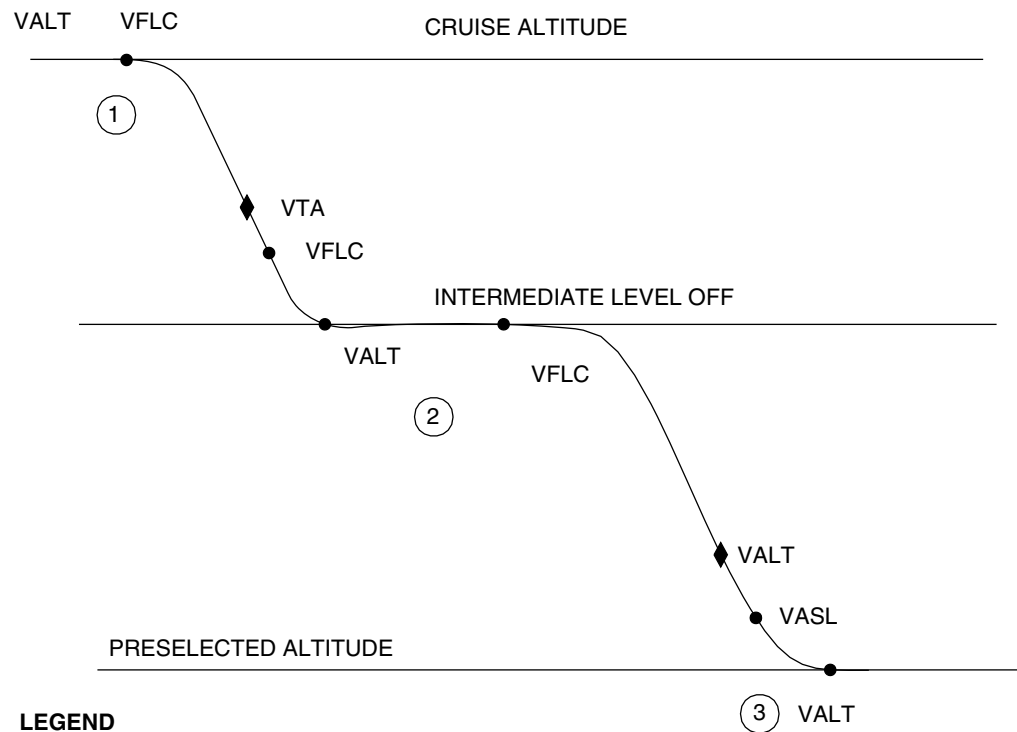
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- VNAV Mode Change Point.
- ◆ Alert or Annunciator Point (VTA = Vertical Track Alert).
- ⏏ Hard (Cross-at) Altitude Constraint.

fsp70a01.cgm

AIRMASS DESCENT
Figure 20

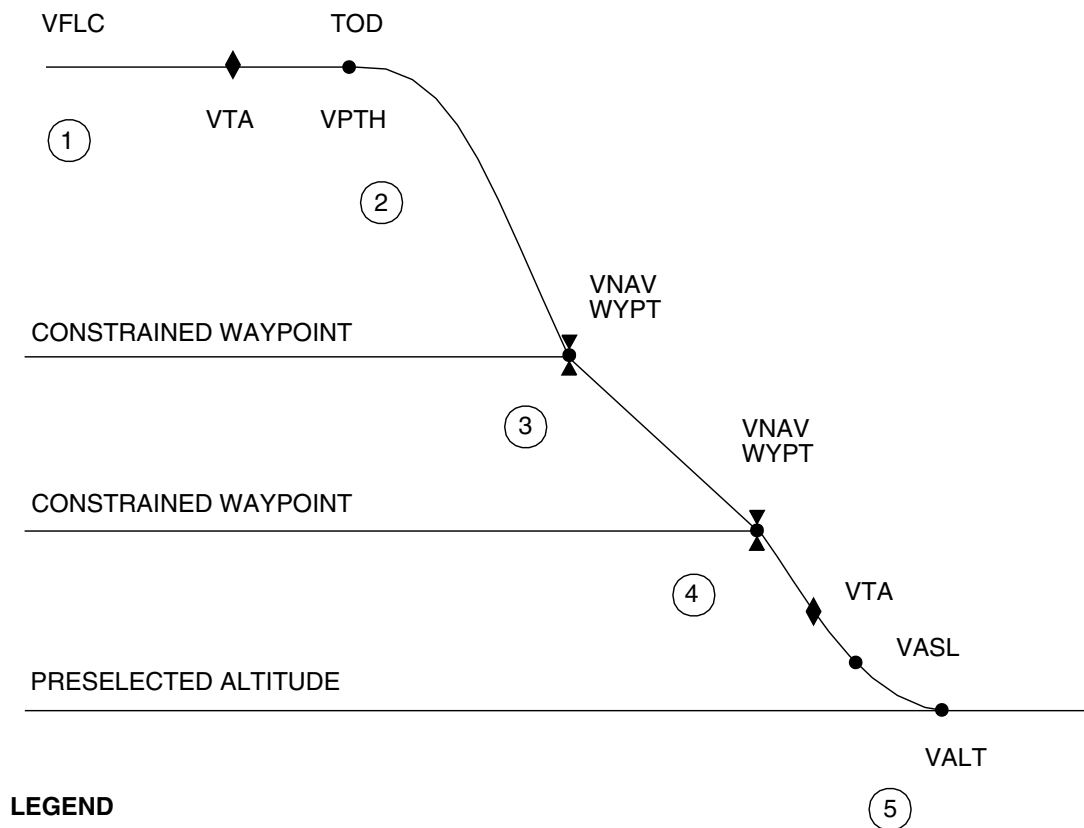
PSM 1-84-2A
EFFECTIVITY:
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Config 001

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- VNAV Mode Change Point.
- ◆ Alert or Annunciator Point (VTA = Vertical Track Alert).
- ⌵ Hard (Cross-at) Altitude Constraint.

fsp71a01.cgm

PATH DESCENT
Figure 21

PSM 1-84-2A
EFFECTIVITY:
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Config 001

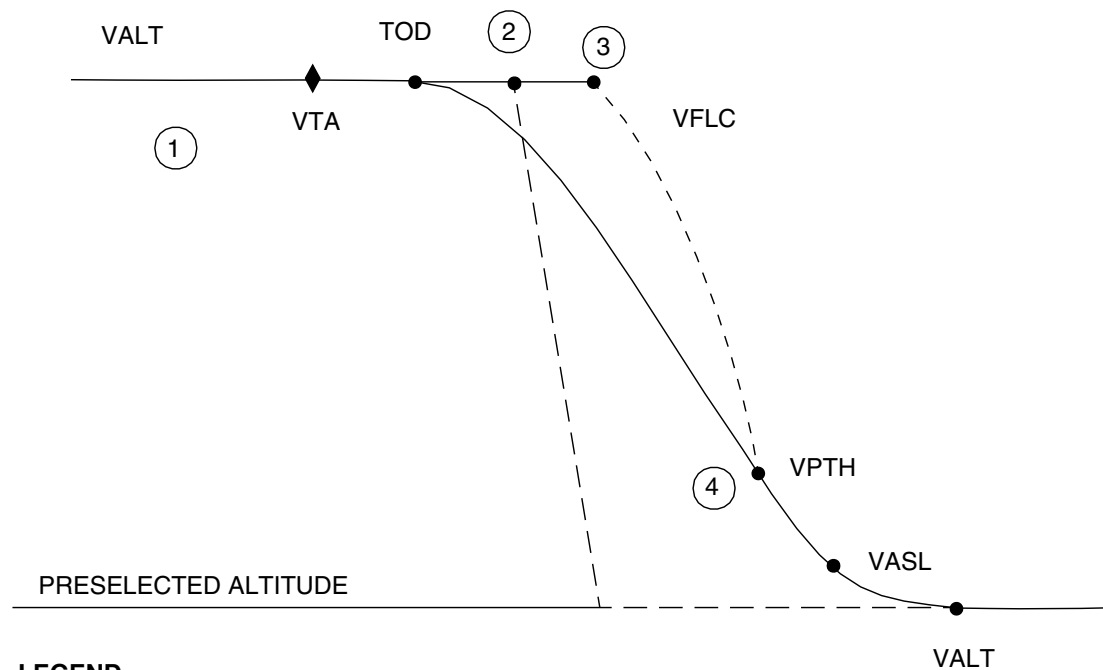
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LEGEND

- VNAV Mode Change Point.
- ◆ Alert or Annunciator Point (VTA = Vertical Track Alert).
- ⌵ Hard (Cross-at) Altitude Constraint.

fsp72a01.cgm

LATE PATH DESCENT
Figure 22

PSM 1-84-2A
EFFECTIVITY:
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Config 001

34-60-01

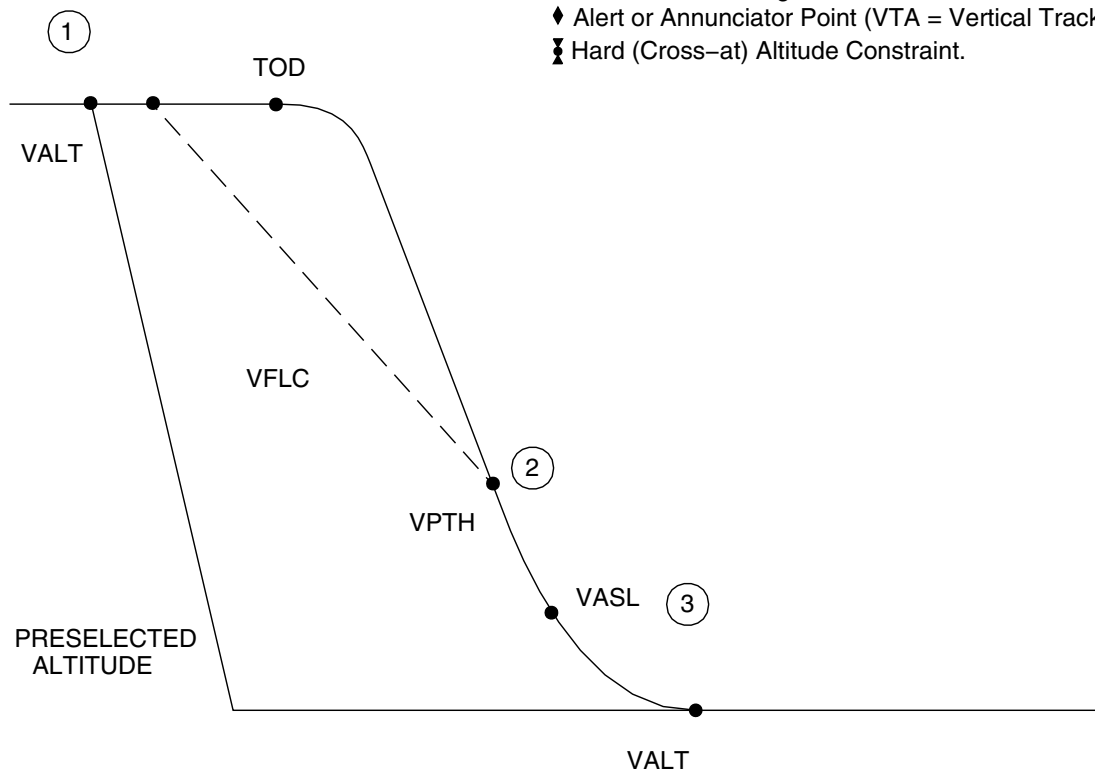
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

- VNAV Mode Change Point.
- ◆ Alert or Annunciator Point (VTA = Vertical Track Alert).
- ⏏ Hard (Cross-at) Altitude Constraint.



fsp73a01.cgm

EARLY DESCENT TO CAPTURE PATH
Figure 23

PSM 1-84-2A
EFFECTIVITY:
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Config 001

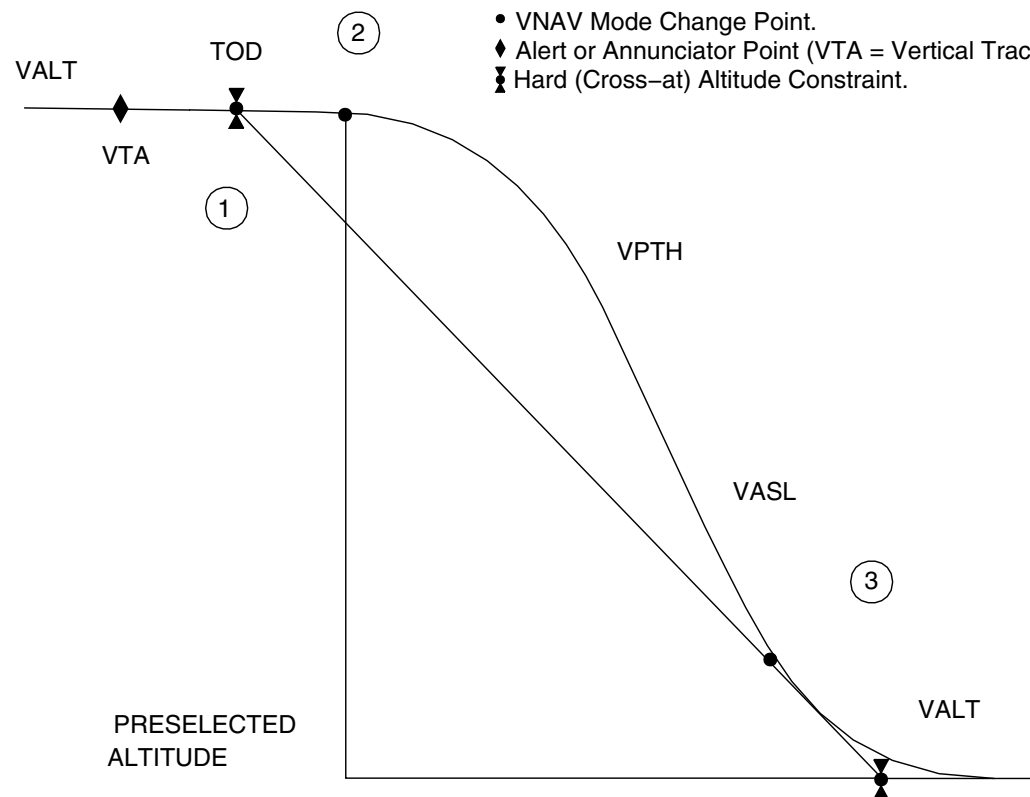
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LEGEND

- VNAV Mode Change Point.
- ◆ Alert or Annunciator Point (VTA = Vertical Track Alert).
- ✚ Hard (Cross-at) Altitude Constraint.



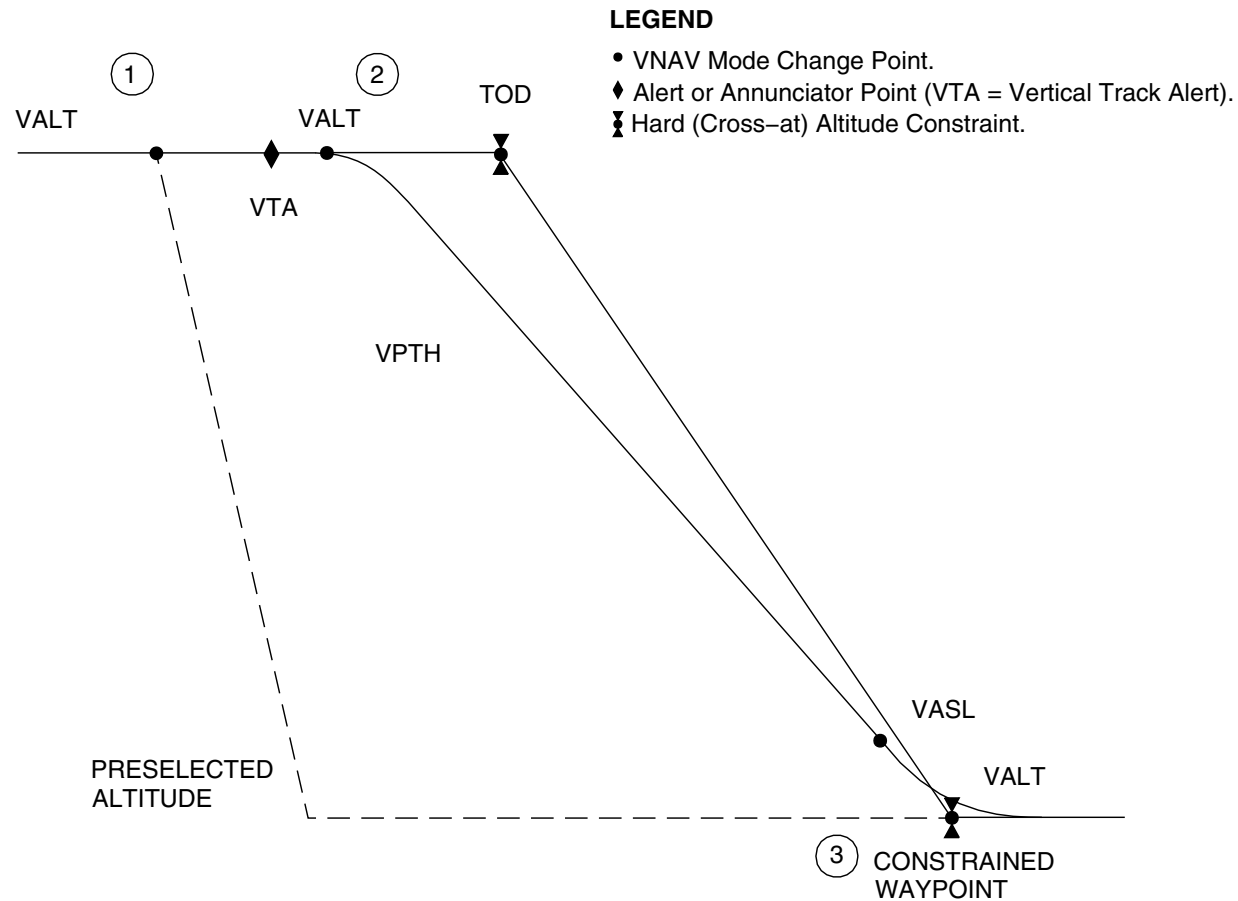
LATE PATH DESCENT USING VERTICAL DIRECT TO FUNCTION
Figure 24

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fsp74a01.cgm

EARLY DESCENT USING VERTICAL DIRECT TO FUNCTION
Figure 25

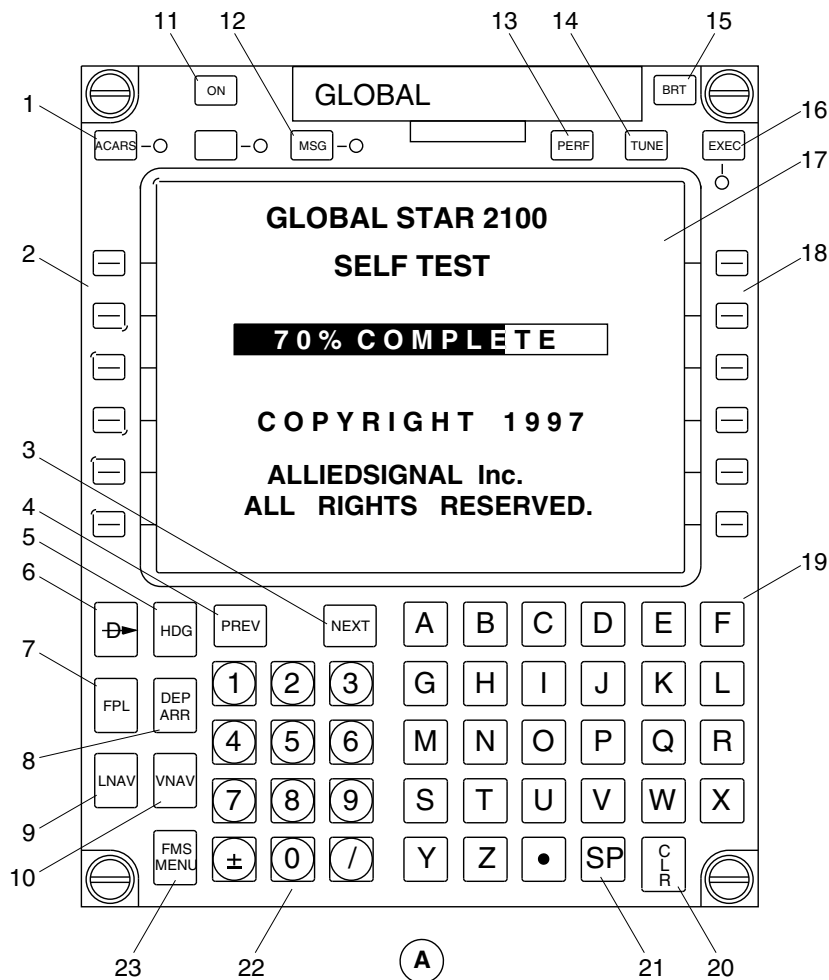
PSM 1-84-2A
EFFECTIVITY:
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1. ACARS Key/Annunciator.
2. Left Line Select Keys.
3. NEXT Key.
4. Previous Key.
5. Heading Key.
6. Direct-To Key.
7. Flight Plan Key.
8. Depart/Arrive Key.
9. LNAV Key.
10. VNAV Key.
11. ON Key.
12. Message Key/Annunciator.
13. PERF Key.
14. TUNE Key.
15. Brightness Key.
16. EXEC Key/Annunciator.
17. Display.
18. Right Line Select Keys.
19. Alpha Keys.
20. Clear Key.
21. Space Key.
22. Numeric Keys.
23. FMS MENU Key.



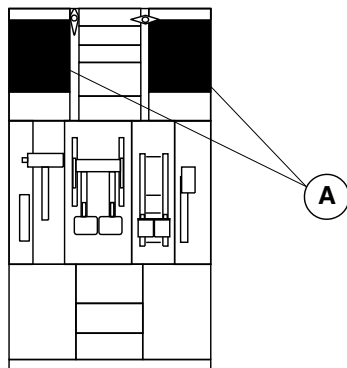
PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

Config 001
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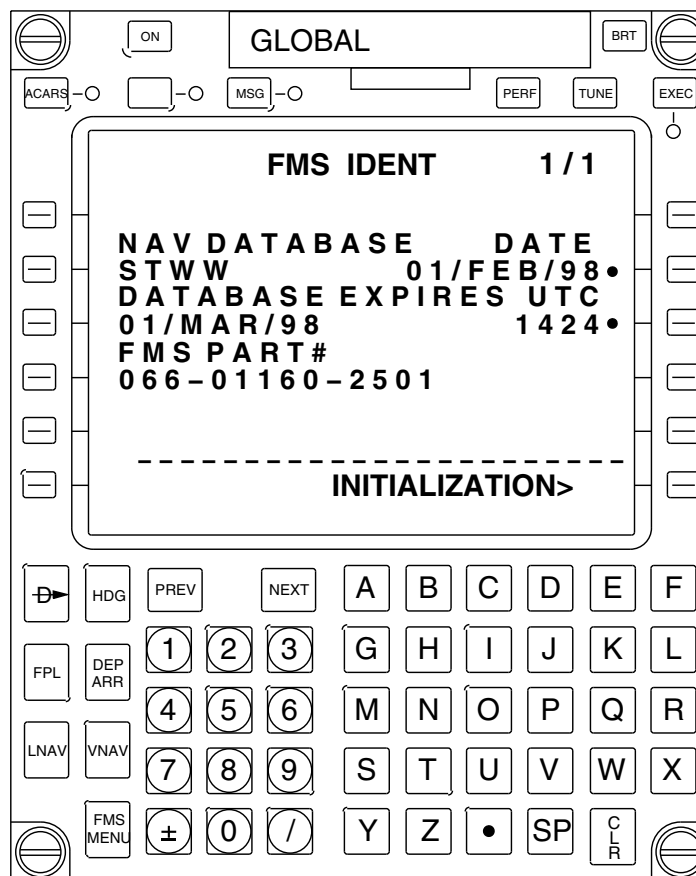


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a02.cgm

FMS IDENT PAGE
Figure 27

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

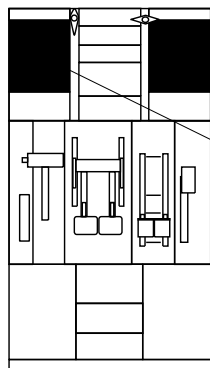
34-60-01

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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

ON GLOBAL BRT

ACARS MSG PERF TUNE EXEC

INITIALIZATION 1 / 1

GPS POSITION

N4726 . 91 / W12218 . 22 →

ORIGIN

• KSEA N4726 . 33 / W12218 . 77 →

RWY OR WPT

• 06L

RWY POSITION

N4726 . 33 / W12228 . 66 →

INIT POSITION

00000 . 00 / 000000 . 00 •

FMS - - - - -

<IDENT

HDG PREV NEXT A B C D E F

FPL DEP ARR 1 2 3 G H I J K L

LNAV VNAV 4 5 6 M N O P Q R

FMS MENU 7 8 9 S T U V W X

± 0 / Y Z • SP CLR

A

fsq03a03.cgm

GPS POSITION PAGE
Figure 28

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

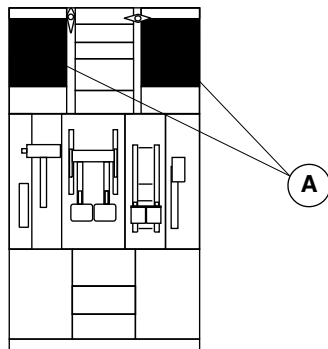
34-60-01

Config 001
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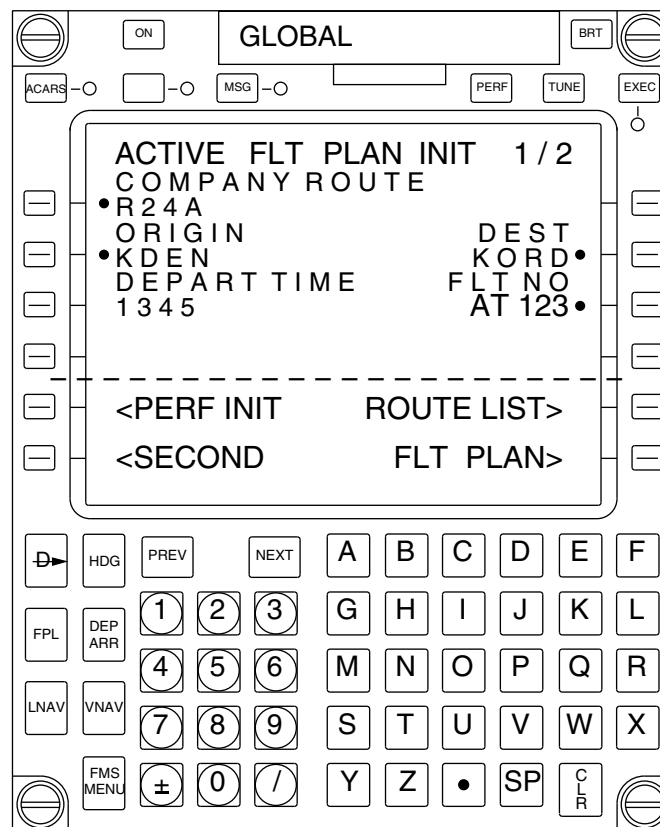


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a05.cgm

FLIGHT PLAN INIT 1/2 PAGE
Figure 29

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

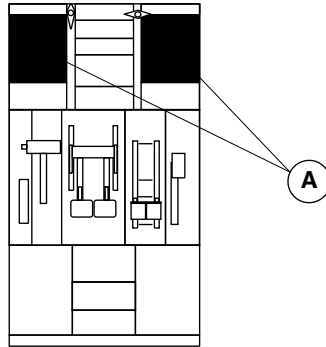
34-60-01

Config 001
Page 129
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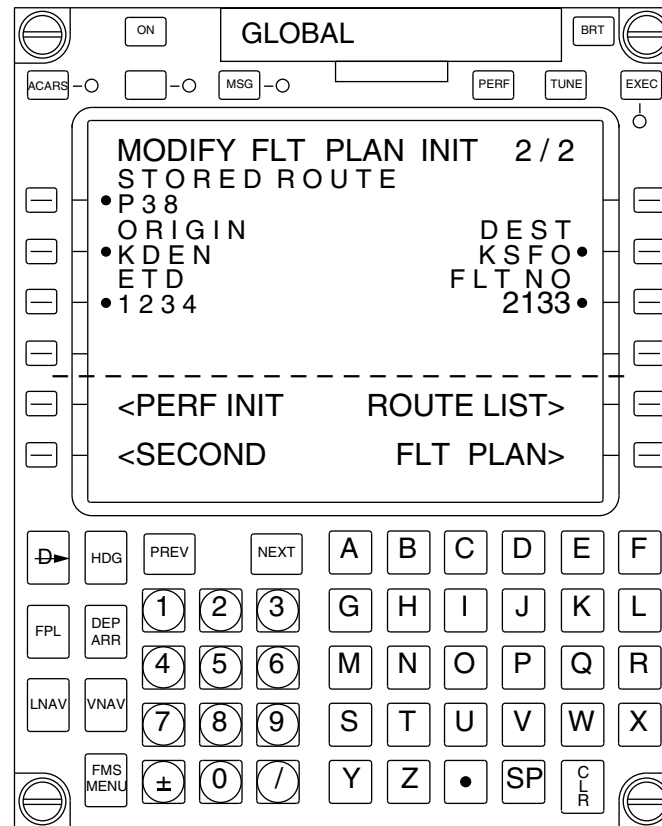


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a06.cgm

FLIGHT PLAN INIT 2/2 PAGE
Figure 30

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

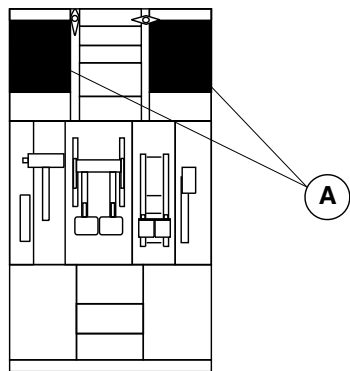
34-60-01

Config 001
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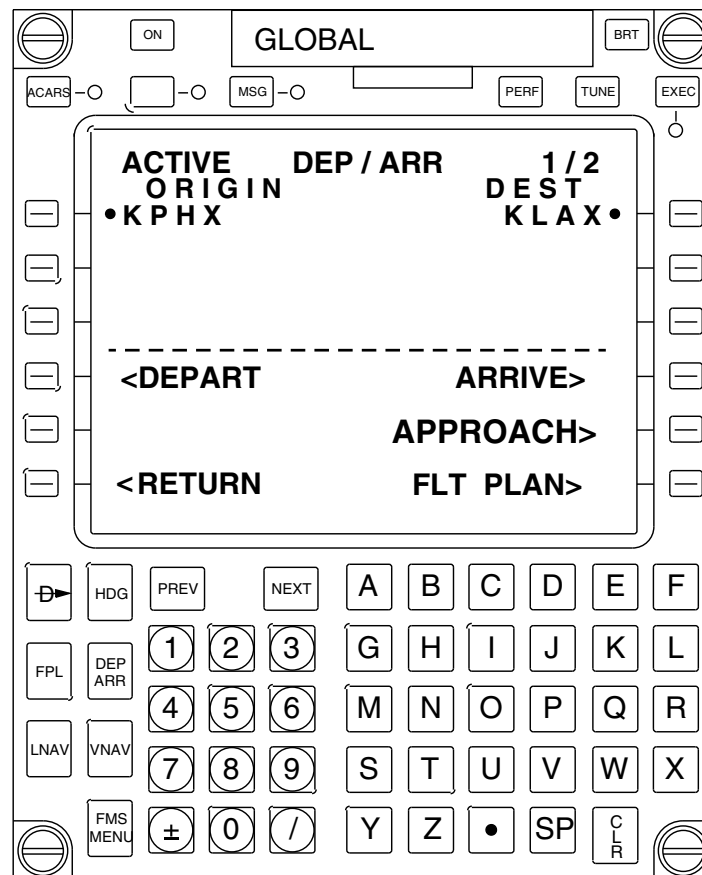


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a07.cgm

DEPART/ARRIVAL 1/2 PAGE
Figure 31

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

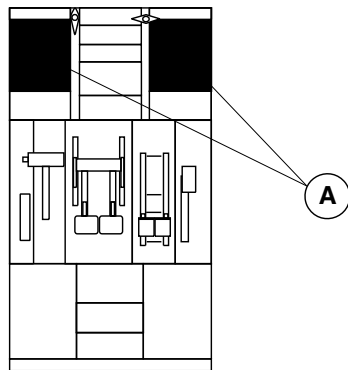
34-60-01

Config 001
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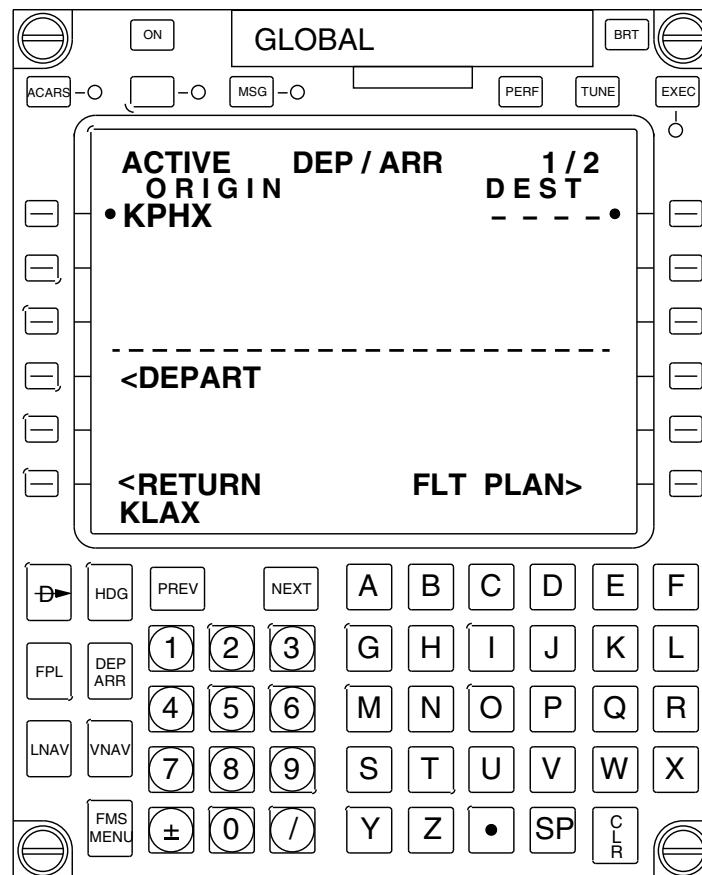


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a08.cgm

DEPART/ARRIVAL 1/2 PAGE ORIGIN
Figure 32

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

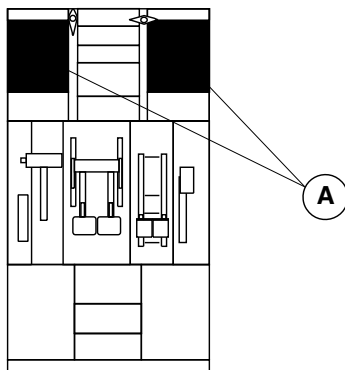
34-60-01

Config 001
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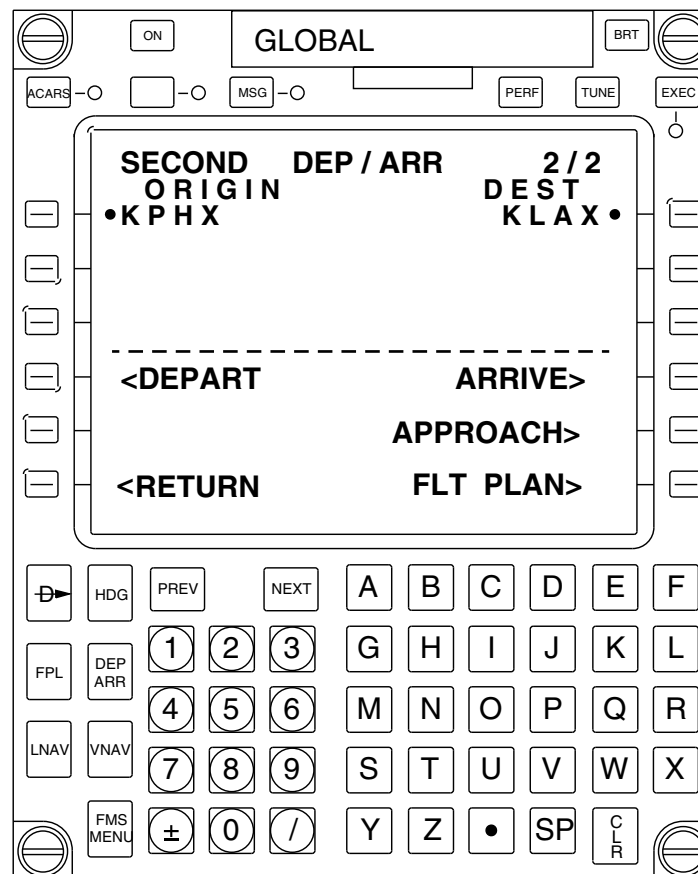


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



fsq03a09.cgm

DEPART/ARRIVAL 2/2 PAGE ORIGIN
Figure 33

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

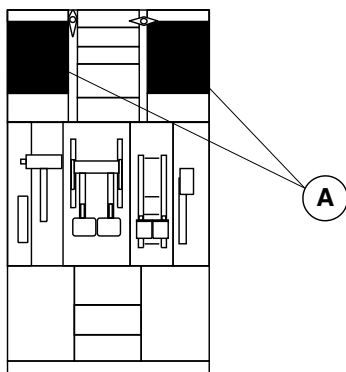
34-60-01

Config 001
Page 133
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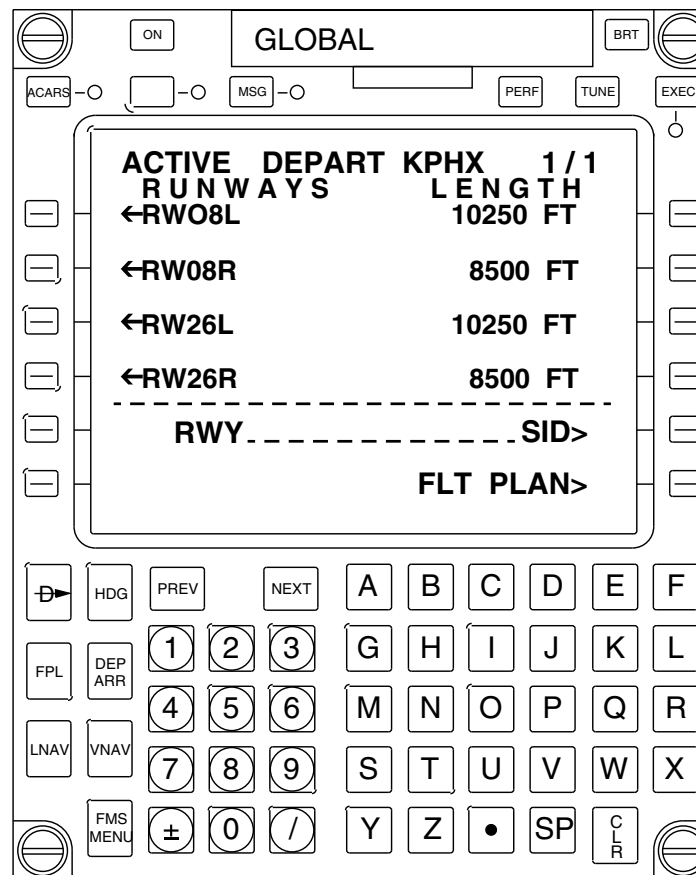


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a10.cgm

DEPART 1/N PAGE
Figure 34

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

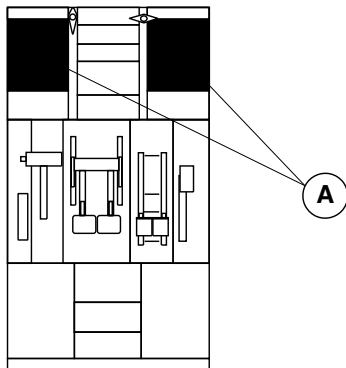
34-60-01

Config 001
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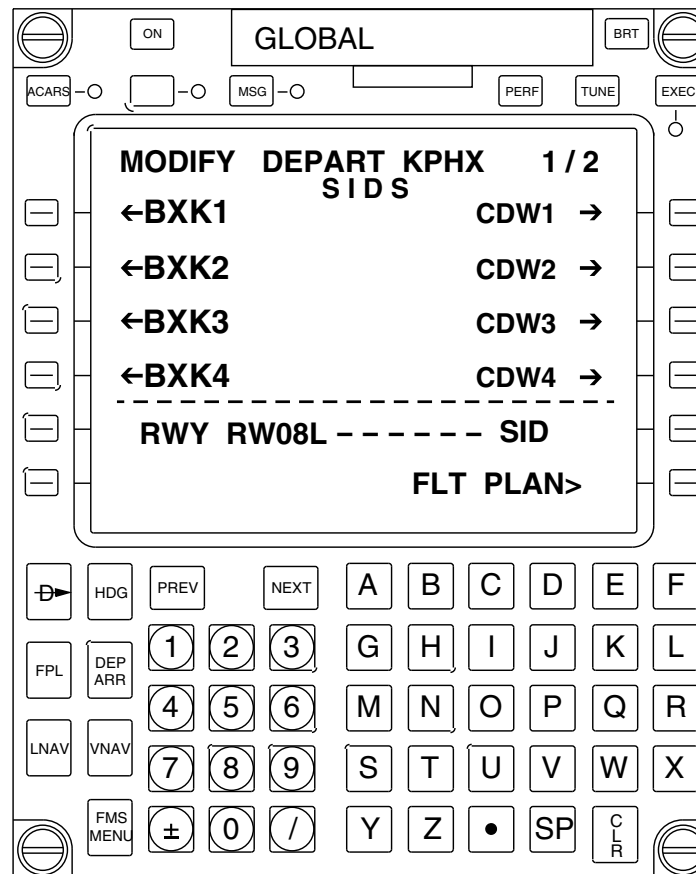


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a11.cgm

DEPART/SID PAGE
Figure 35

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

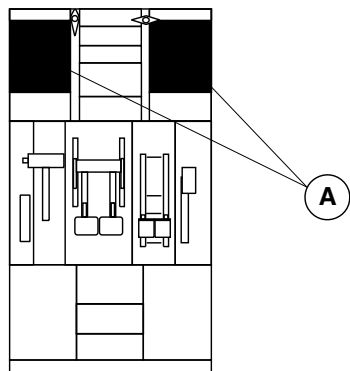
34-60-01

Config 001
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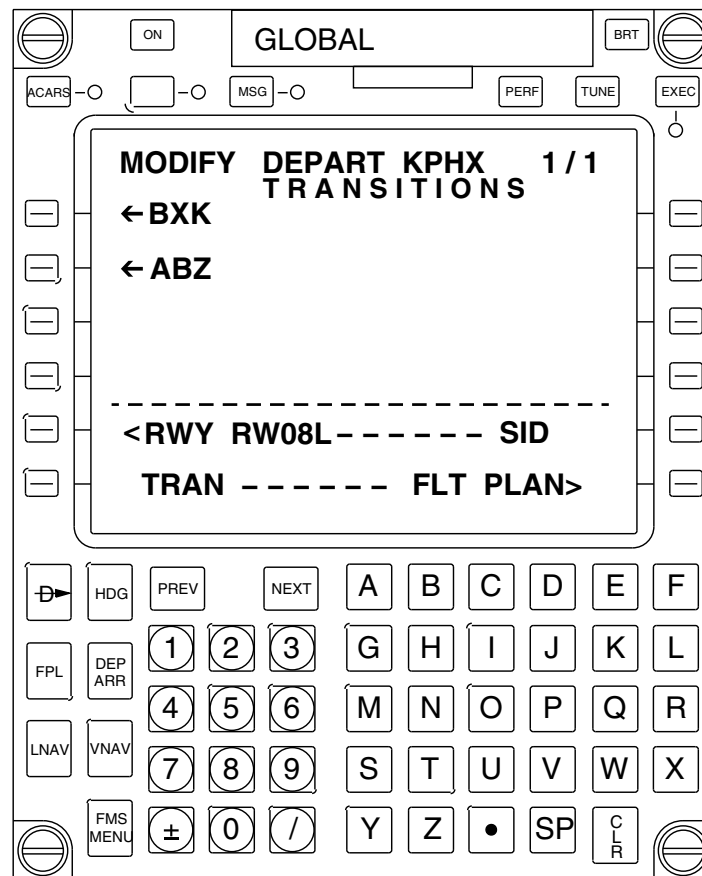


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a12.cgm

DEPART/TRANSITION PAGE
Figure 36

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

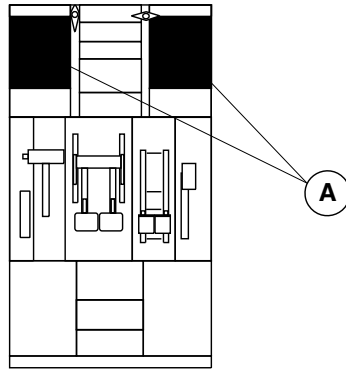
34-60-01

Config 001
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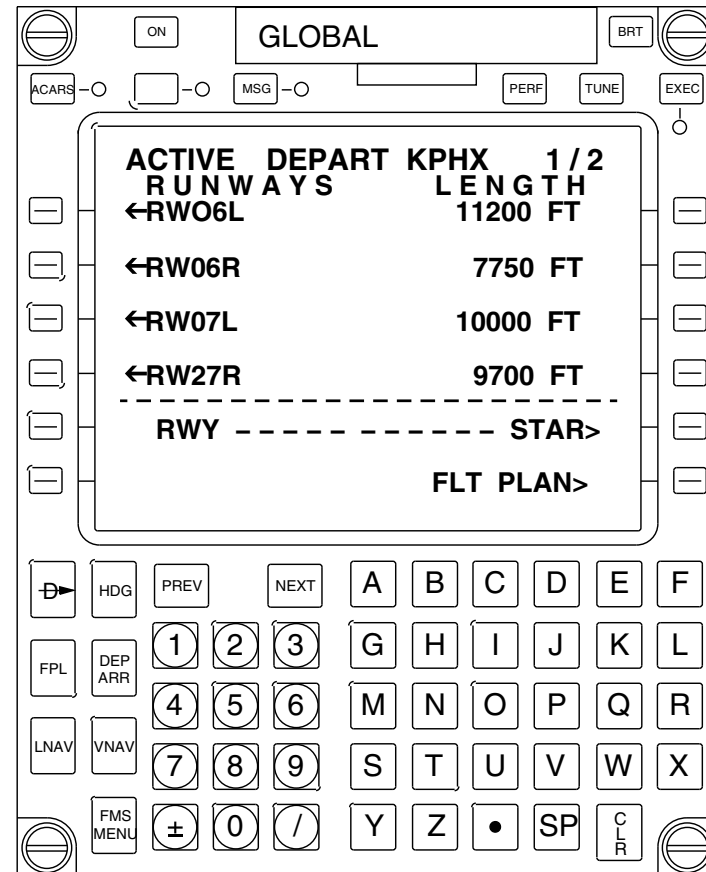


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a13.cgm

ARRIVE 1/N PAGE
Figure 37

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

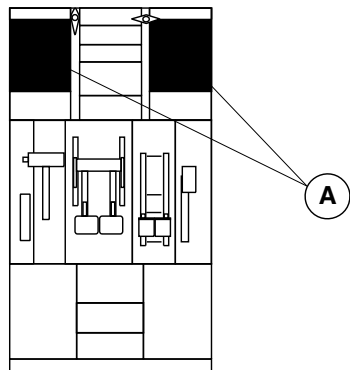
34-60-01

Config 001
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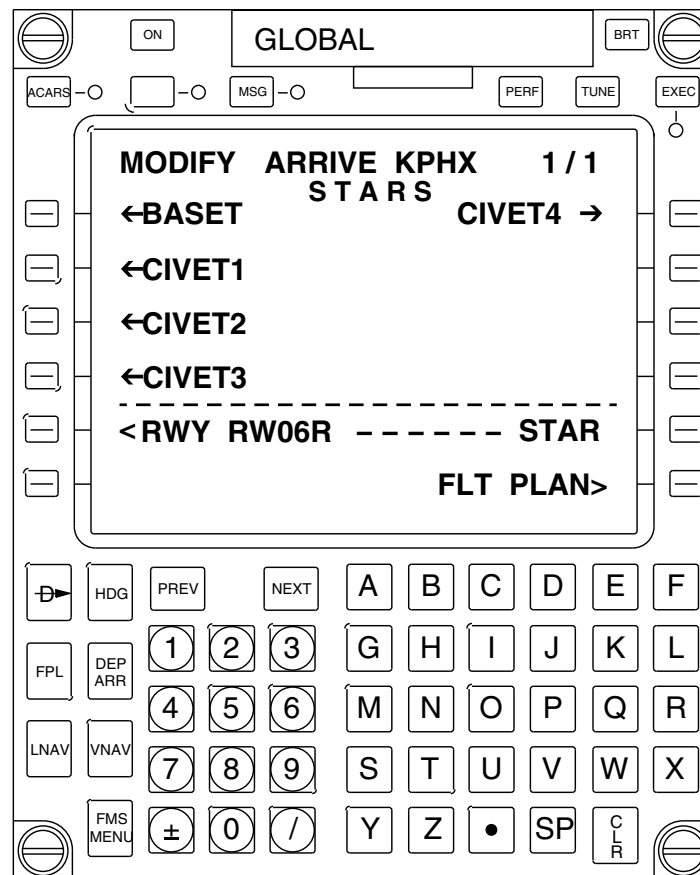


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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a14cgm

ARRIVE/STARS PAGE
Figure 38

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

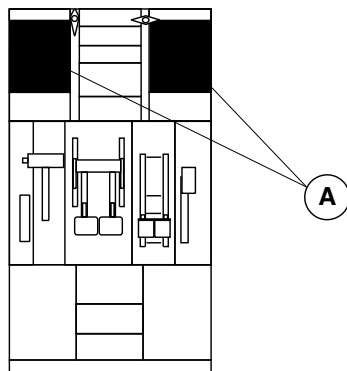
34-60-01

Config 001
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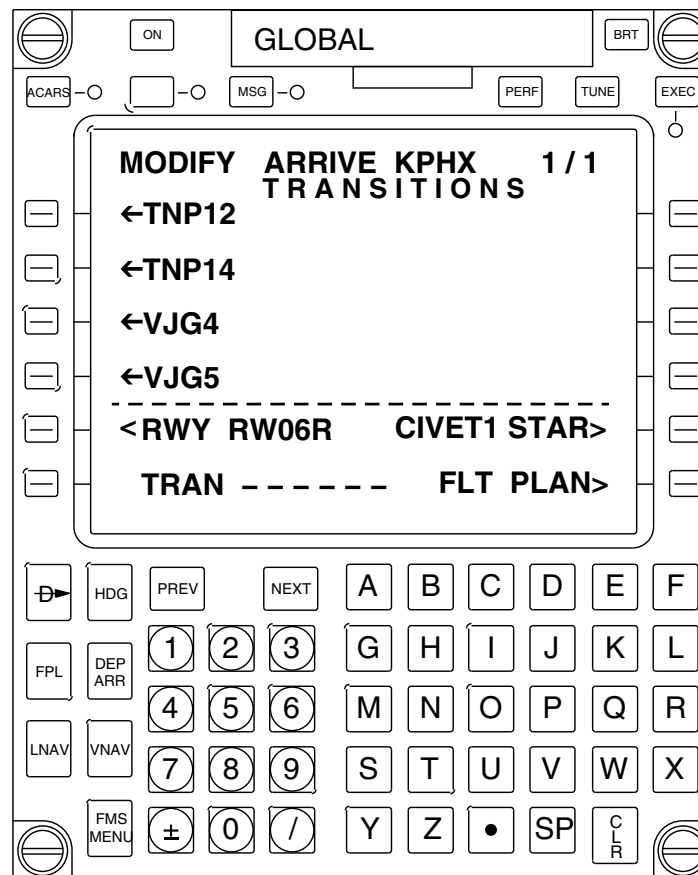


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a15.cgm

ARRIVE/TRANSITION PAGE
Figure 39

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

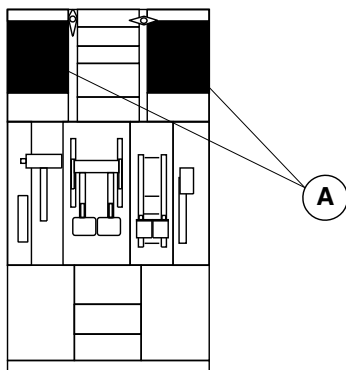
34-60-01

Config 001
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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

ON GLOBAL BRT

ACARS MSG PERF TUNE EXEC

ACTIVE	FLT PLAN	1 / 5
FROM	ETA	SPD / ALT
•RW08L	1NM	
•(R-350)	0800	-- / 2000 •
075°	4NM	
•D075D	0801	220 / 3000 B •
H190°	13NM	
•(INTC)	0803	-- / 4500 •
273°	3NM	
•BXK	0808	-- / 5800 •

<MORE DATA SECONDARY>

HDG PREV NEXT A B C D E F

FPL DEP ARR 1 2 3 G H I J K L

LNAV VNAV 4 5 6 M N O P Q R

FMS MENU 7 8 9 S T U V W X

± 0 / Y Z • SP CLR

A

fsq03a16.cgm

ACTIVE FLIGHT PLAN 1/N PAGE
Figure 40

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

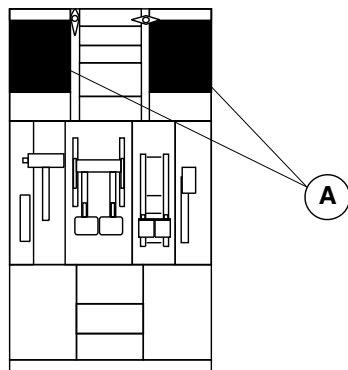
34-60-01

Config 001
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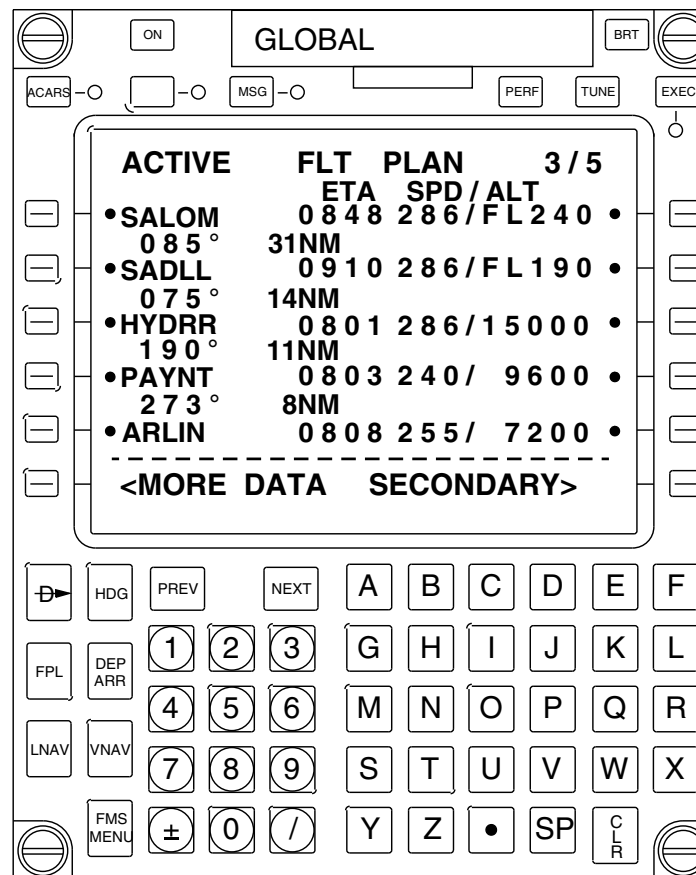


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a17.cgm

ACTIVE FLIGHT PLAN 1/N PAGE MORE DATA
Figure 41

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

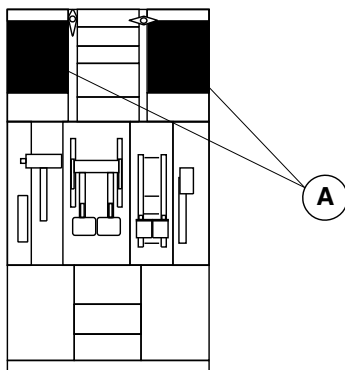
34-60-01

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

ON GLOBAL BRT

ACARS MSG PERF TUNE EXEC

SECOND FLT PLAN 1 / 5

FROM ETA SPD / ALT

• RW26R
H2 5 8 ° 1NM
• (1533) 0 8 0 0 - - - / 1533 B •
H2 4 0 ° 3NM
• (D-4) 0 8 0 1 - - - / 3000 B •
H2 4 0 ° 5NM
• (D-9) 0 8 0 3 - - / - - - •
H1 9 0 ° 12NM
• (INTC) 0 8 0 8 - - / - - - •

<MORE DATA ACTIVE>

HDG PREV NEXT A B C D E F

FPL DEP ARR 1 2 3 G H I J K L

LNAV VNAV 4 5 6 M N O P Q R

FMS MENU 7 8 9 S T U V W X

± 0 / Y Z • SP CLR

A

fsq03a18.cgm

ACTIVE FLIGHT PLAN 1/N PAGE SECONDARY
Figure 42

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

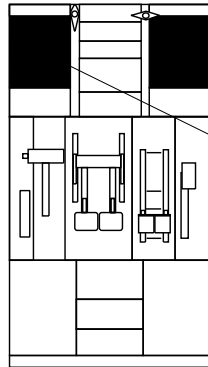
34-60-01

Config 001
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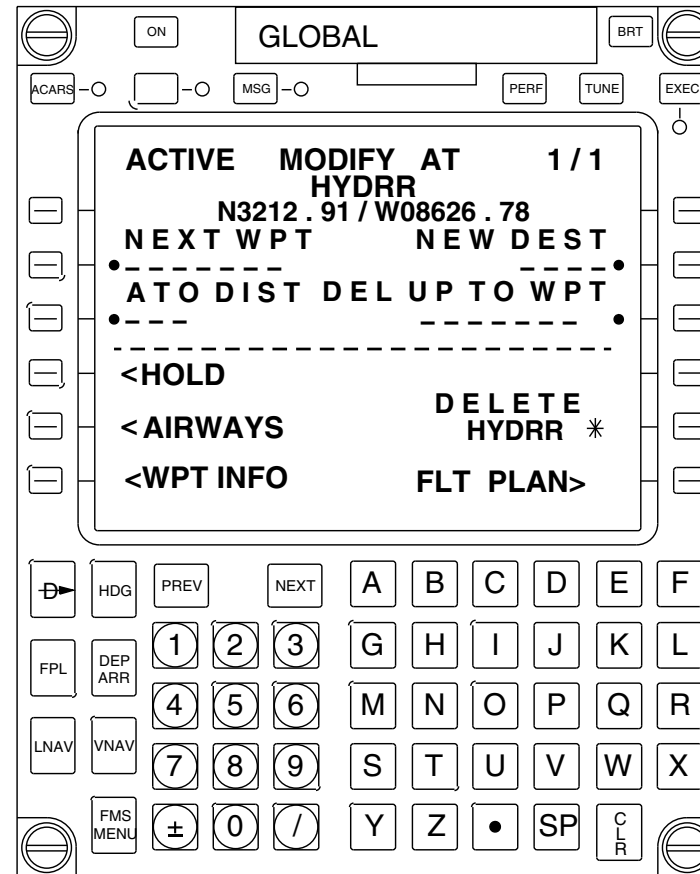


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a19.cgm

LATERAL ACTIVE MODIFY AT 1/1 PAGE
Figure 43

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

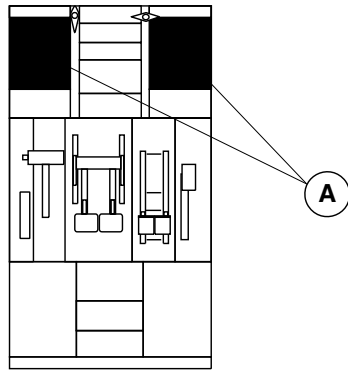
34-60-01

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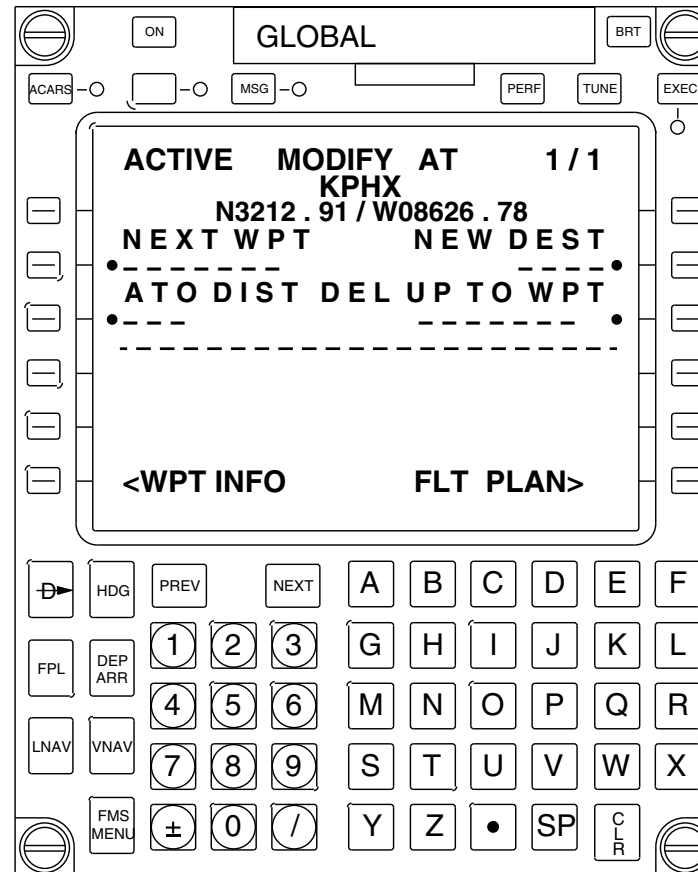


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a20.cgm

LATERAL ACTIVE MODIFY AT 1/1 PAGE ORIGIN AIRPORT
Figure 44

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

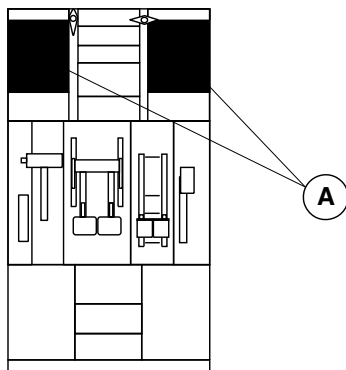
34-60-01

Config 001
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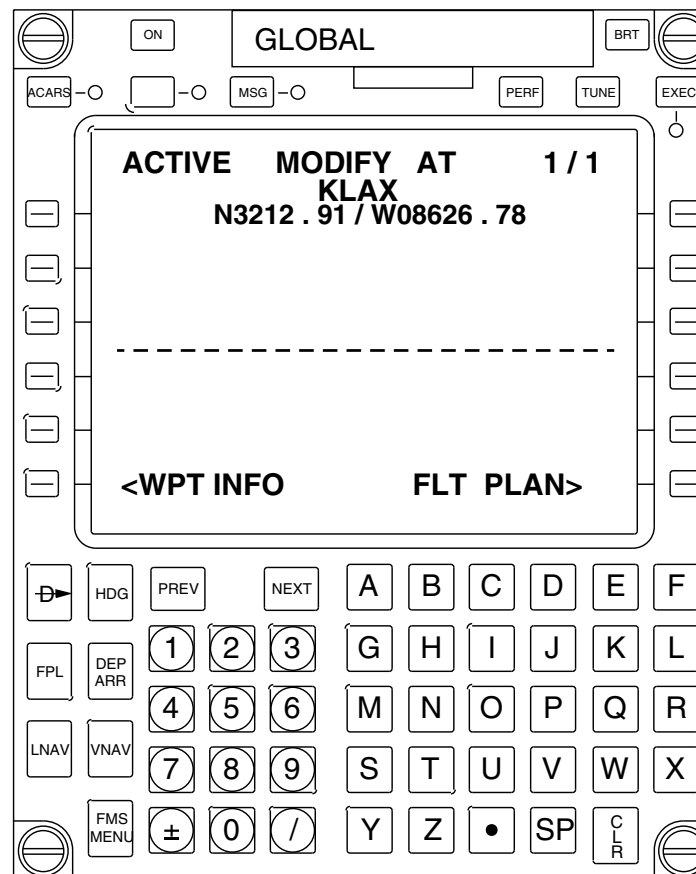


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a21.cgm

ACTIVE MODIFY AT 1/1 PAGE
Figure 45

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

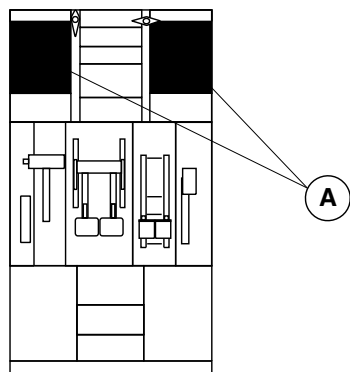
34-60-01

Config 001
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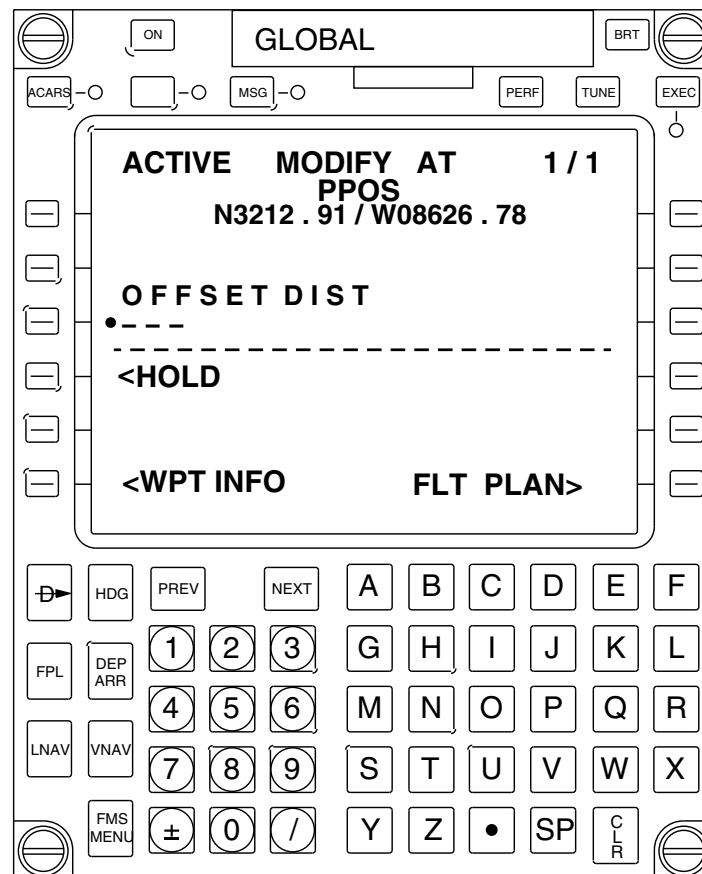


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a22.cgm

LATERAL ACTIVE MODIFY AT 1/1 PAGE FROM WAYPOINT
Figure 46

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

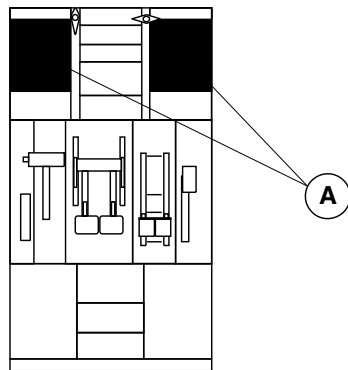
34-60-01

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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

ON GLOBAL BRT

ACARS MSG PERF TUNE EXEC

ACTIVE MODIFY AT 1/1
TANDY

INBOUND CRS ENTRY TYPE
• 253° DIRECT •

INBOUND TIME
• 1.0

LEG DIST
• 3.7

TURN DIR-----EXIT TYPE
*RIGHT/ LEFT AUTO /MANUAL *

<RETURN INSERT *

HDG PREV NEXT A B C D E F

FPL DEP ARR 1 2 3 G H I J K L

LNAV VNAV 4 5 6 M N O P Q R

FMS MENU 7 8 9 S T U V W X

± 0 / Y Z • SP CLR

A

fsq03a23.cgm

ACTIVE HOLDING AT 1/1 PAGE
Figure 47

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

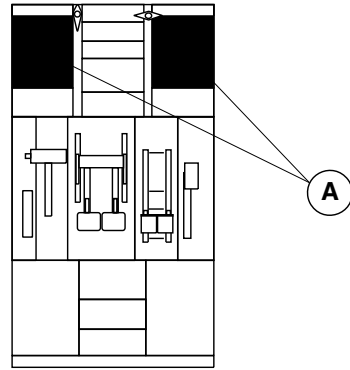
34-60-01

Config 001
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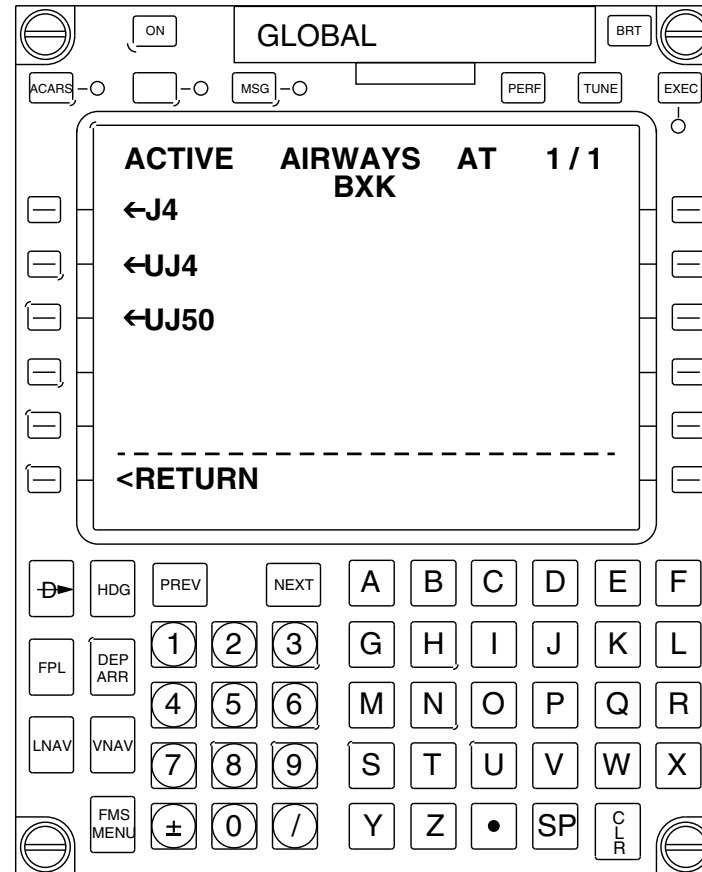


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a24.cgm

ACTIVE AIRWAYS AT 1/1 PAGE
Figure 48

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

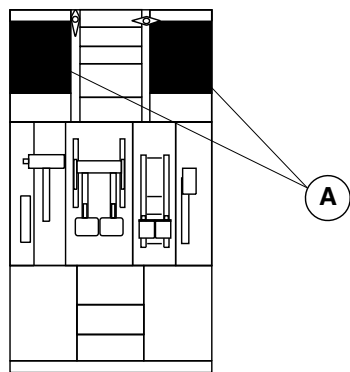
34-60-01

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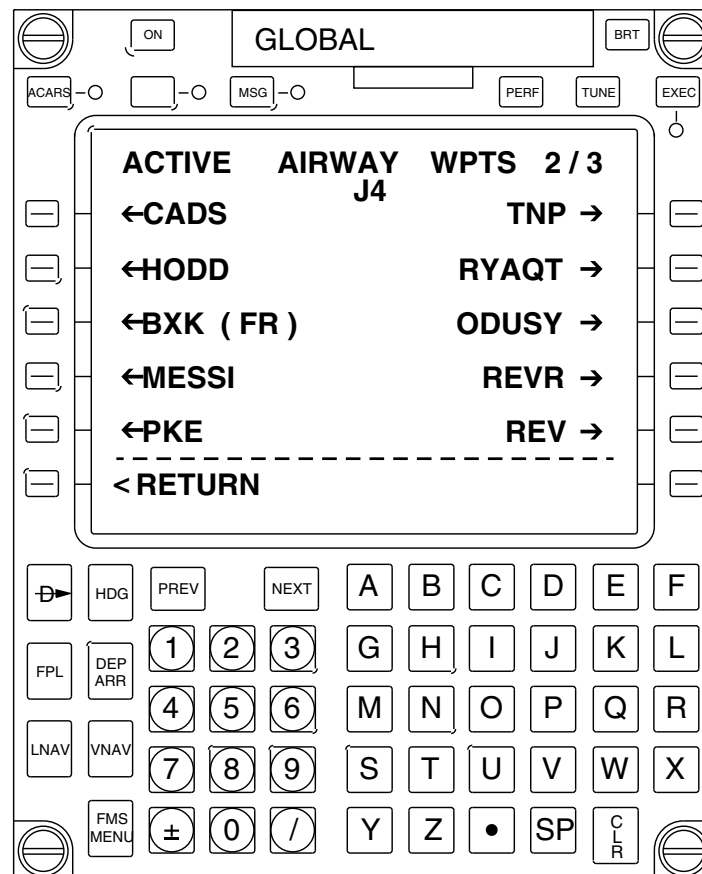


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a25.cgm

AIRWAY WPT AT N/N PAGE
Figure 49

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

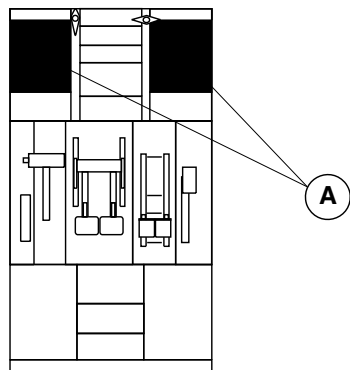
34-60-01

Config 001
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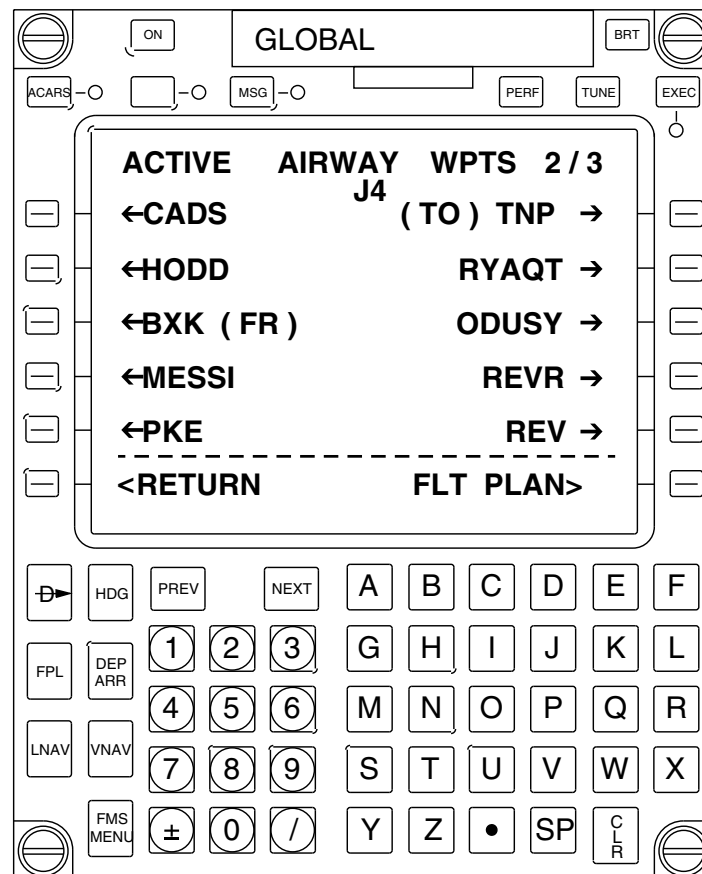


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a26cgm

AIRWAY WPTS
Figure 50

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

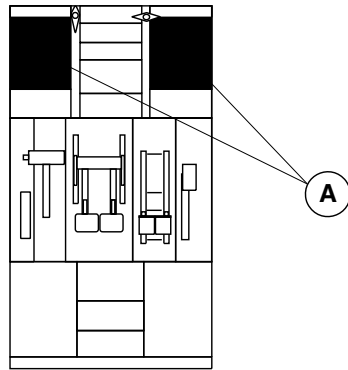
34-60-01

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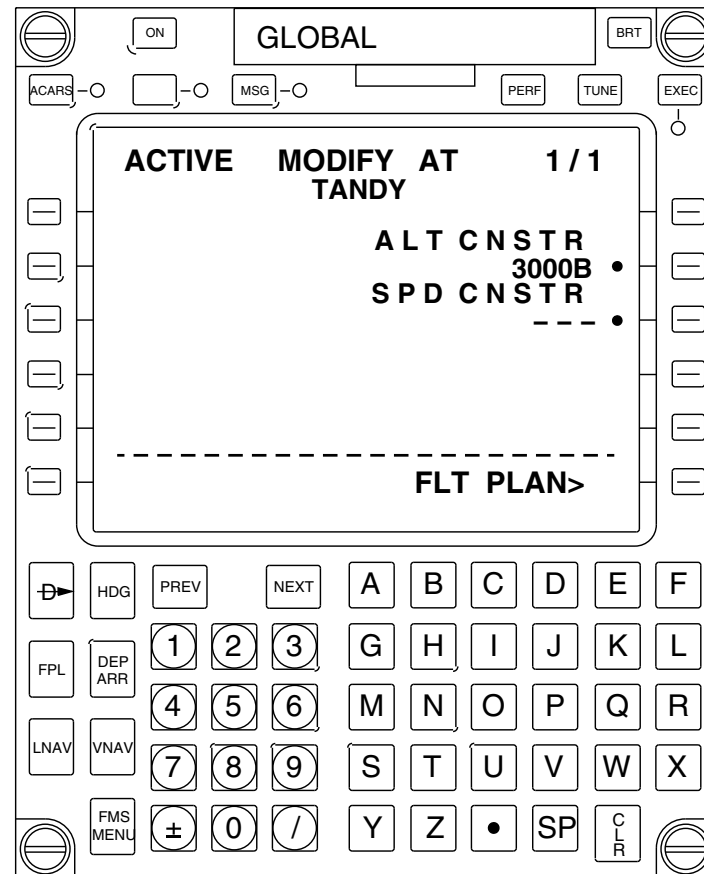


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a27.cgm

VERTICAL ACTIVE MODIFY AT 1/1 PAGE
Figure 51

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

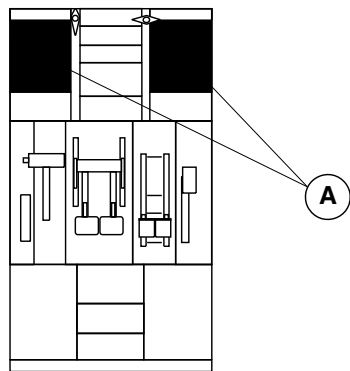
34-60-01

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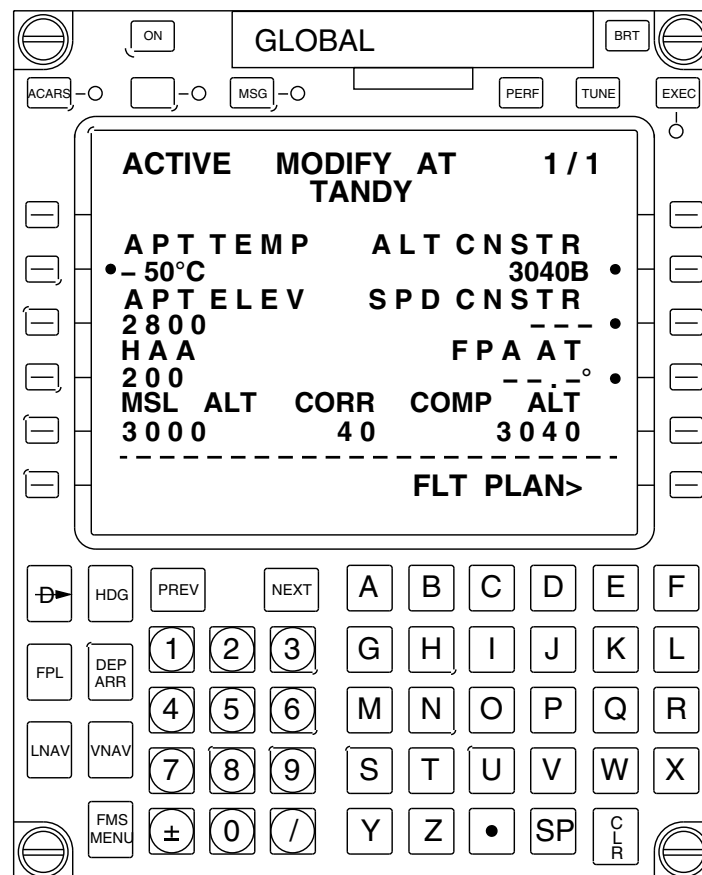


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq03a28.cgm

VERTICAL ACTIVE MODIFY AT 1/1 PAGE DESCENT WAYPOINT
Figure 52

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

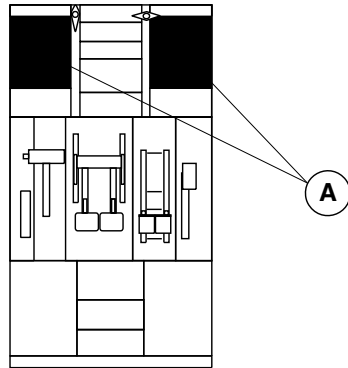
34-60-01

Config 001
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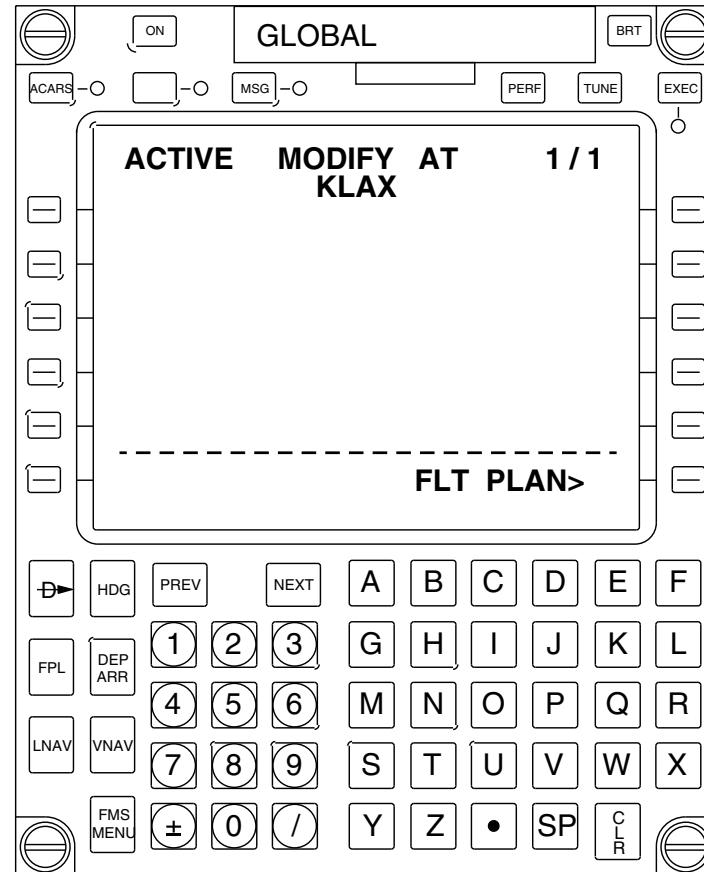


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CENTRE CONSOLE



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fsq03a29.cgm

VERTICAL ACTIVE MODIFY AT 1/1 PAGE ORIGIN WAYPOINT
Figure 53

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

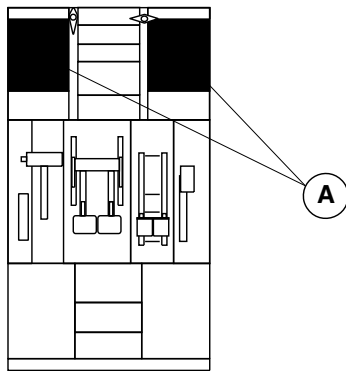
34-60-01

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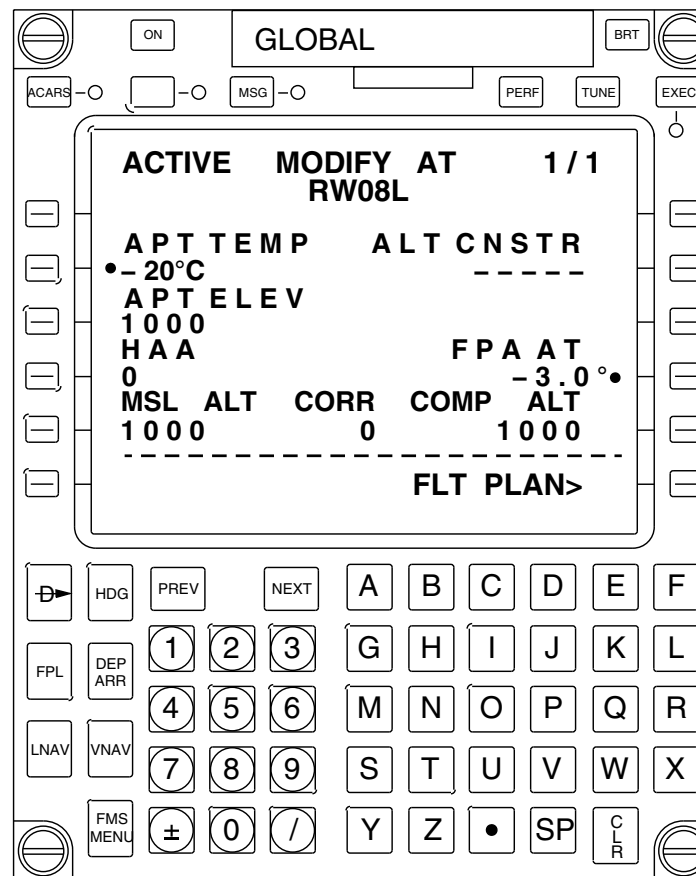


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VERTICAL ACTIVE MODIFY AT 1/1 PAGE DESTINATION WAYPOINT
Figure 54

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

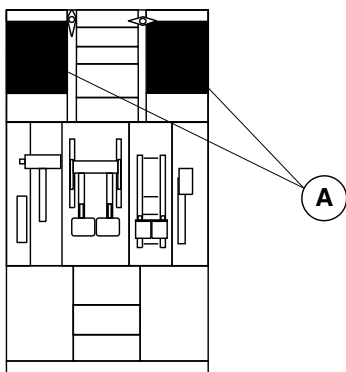
34-60-01

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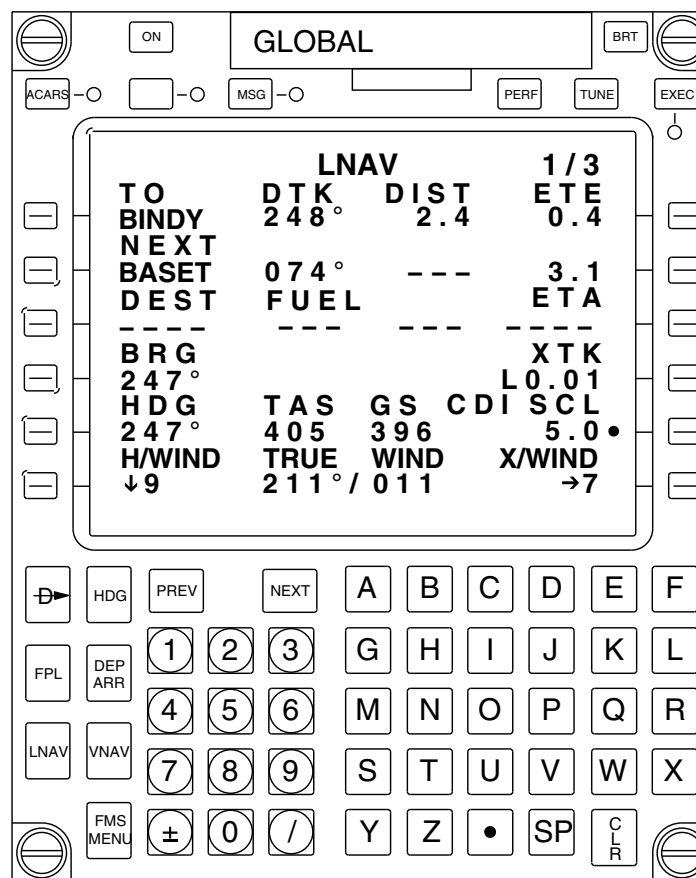


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LNAV 1/3 PAGE
Figure 55

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

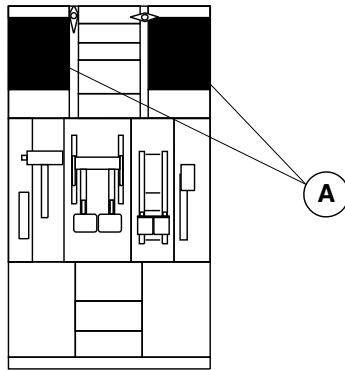
34-60-01

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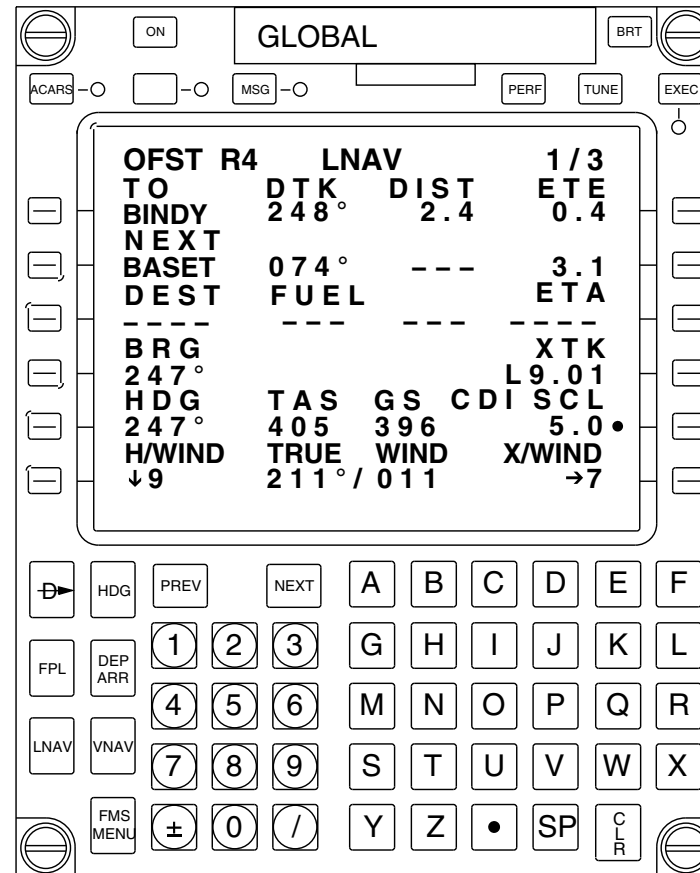


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LNAV 1/3 PAGE PARALLEL OFFSET
Figure 56

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

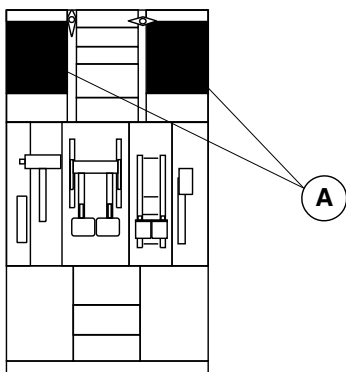
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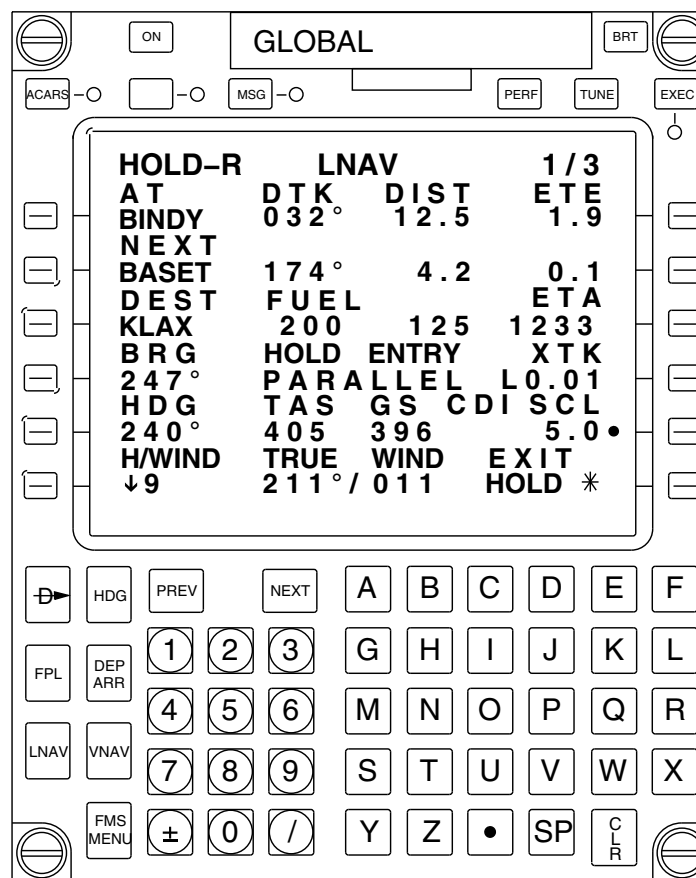


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LNAV 1/3
Figure 57

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

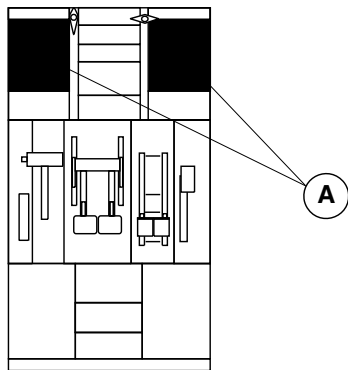
34-60-01

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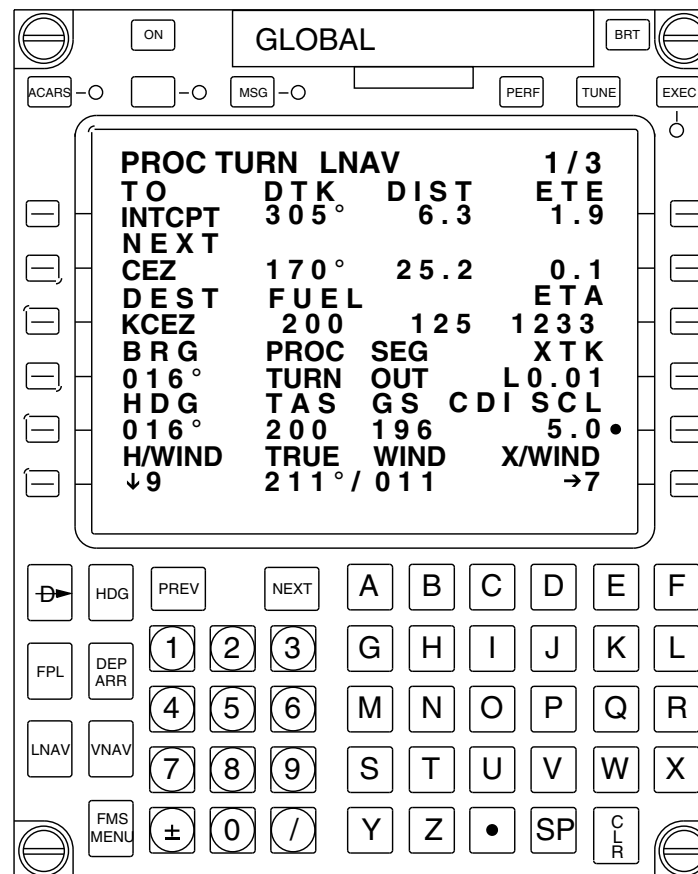


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fsq03a34.cgm

LNAV 1/3 PAGE PROCEDURE TURN
Figure 58

PSM 1-84-2A
EFFECTIVITY:
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Config 001

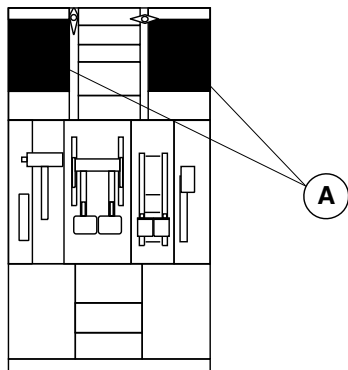
34-60-01

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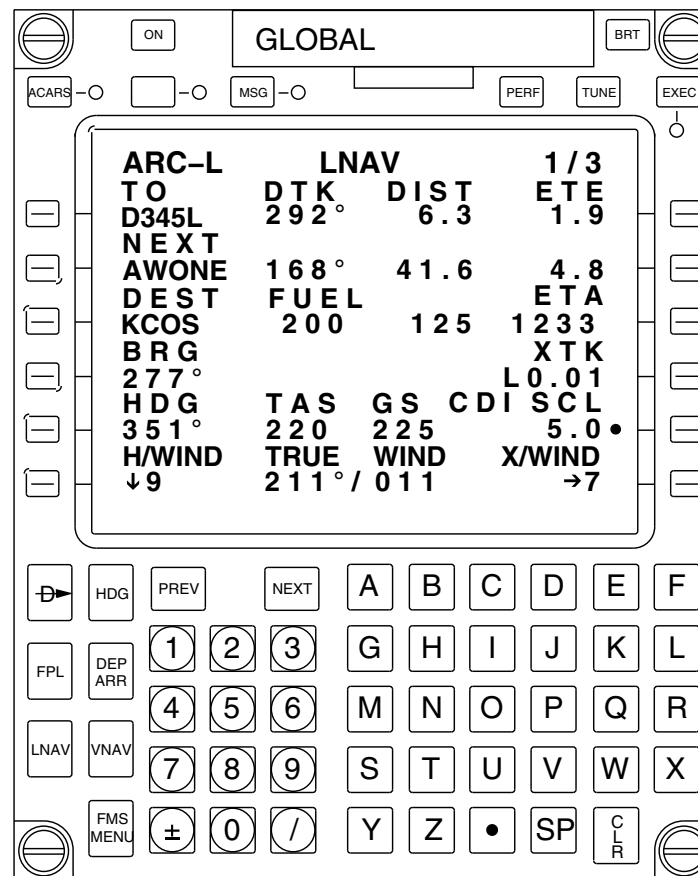


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LNAV 1/3 PAGE ARC TURN
Figure 59

PSM 1-84-2A
EFFECTIVITY:
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Config 001

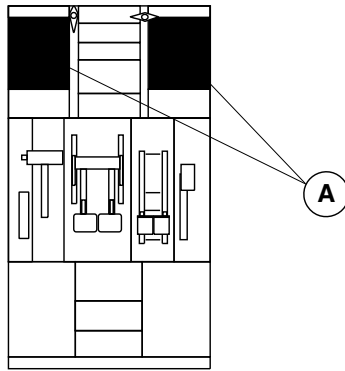
34-60-01

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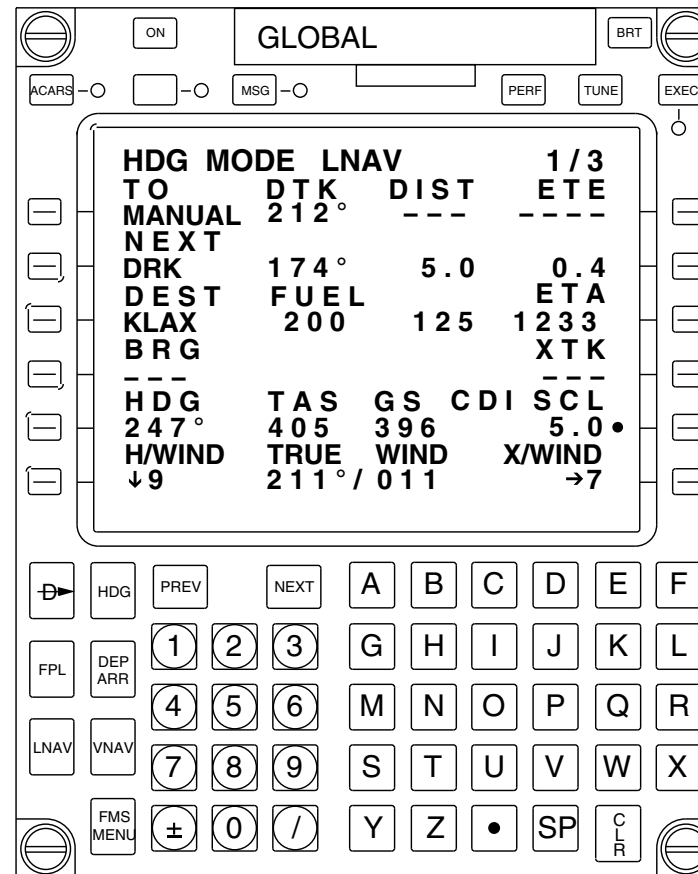


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LNAV 1/3 PAGE MANUAL HEADING
Figure 60

PSM 1-84-2A
EFFECTIVITY:
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Config 001

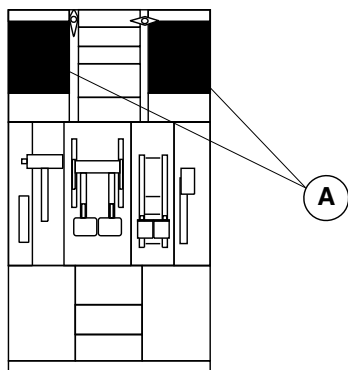
34-60-01

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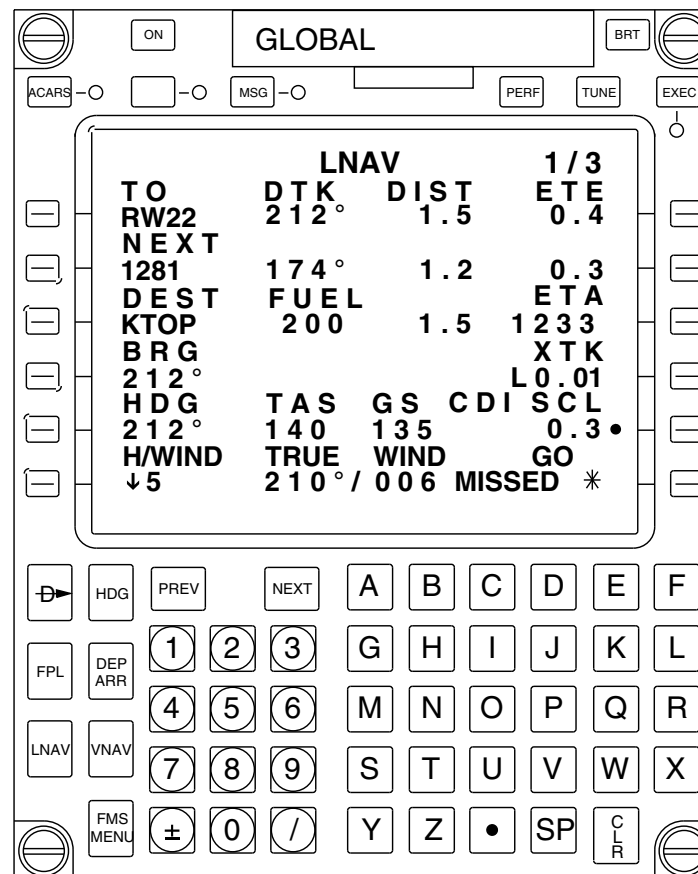


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LNAV 1/3 PAGE FMS FLIGHT MODE IS APPROACH
Figure 61

PSM 1-84-2A
EFFECTIVITY:
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Config 001

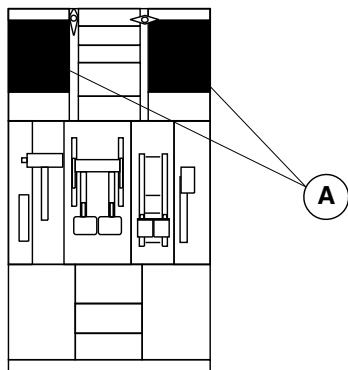
34-60-01

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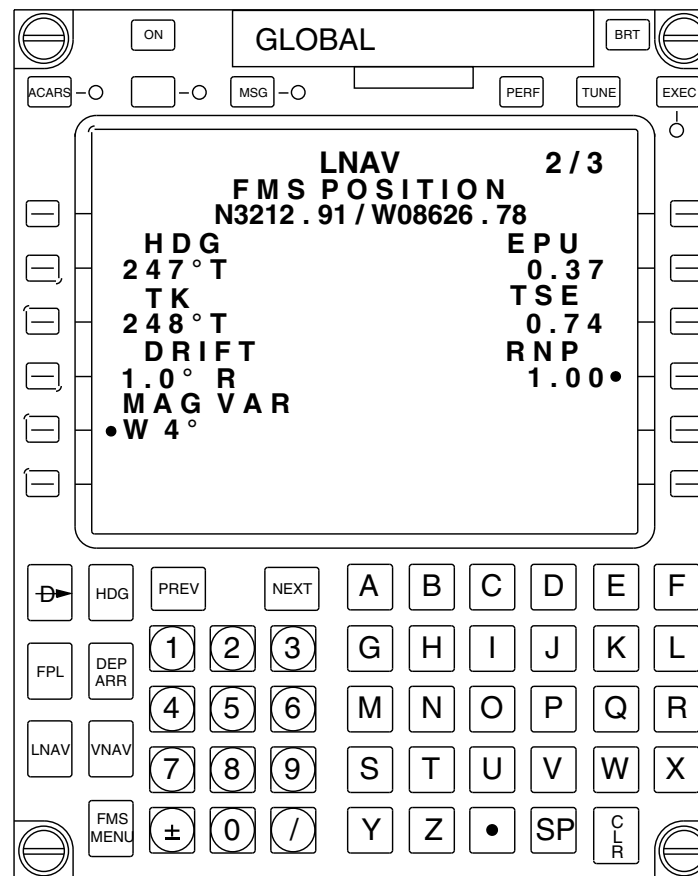


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LNAV 2/3 PAGE
Figure 62

PSM 1-84-2A
EFFECTIVITY:
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Config 001

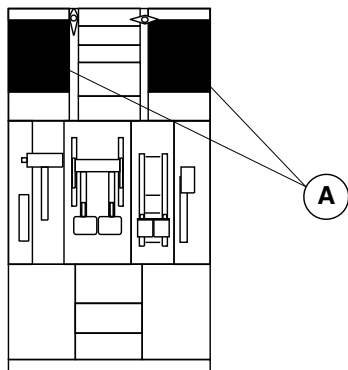
34-60-01

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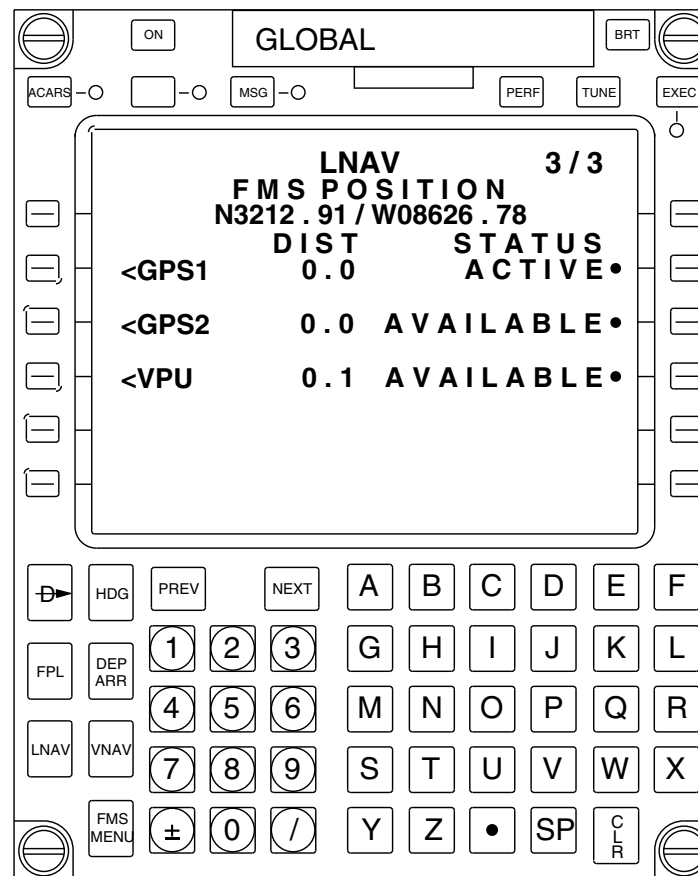


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LNAV 3/3 PAGE
Figure 63

PSM 1-84-2A
EFFECTIVITY:
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Config 001

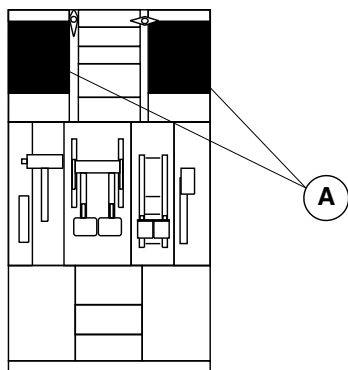
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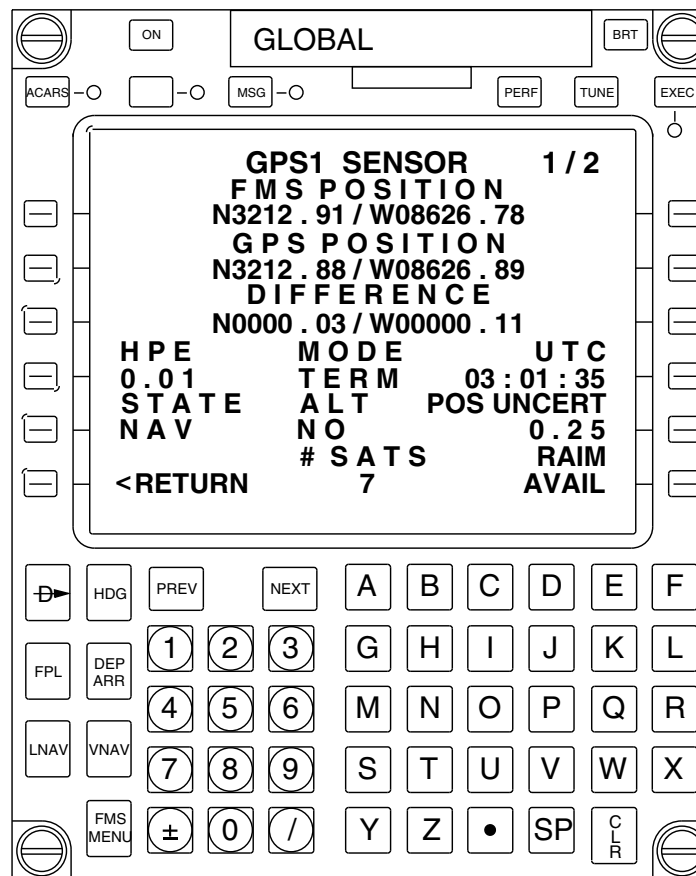


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GPS SENSOR 1/2 PAGE
Figure 64

PSM 1-84-2A
EFFECTIVITY:
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Config 001

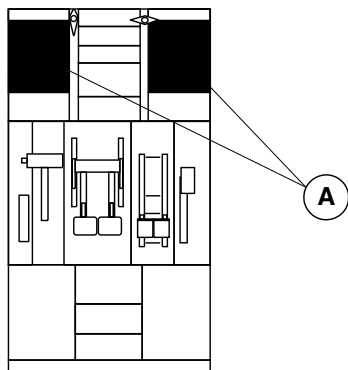
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☐ BRT

☐ ACARS

☐ MSG

☐ PERF

☐ TUNE

☐ EXEC

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PRN	AZM	EL	SNR	HLTH	T
	165	47	45	GOOD	Y
13	315	49	45	GOOD	Y
14	021	68	44	GOOD	Y
20	247	32	44	GOOD	Y
18	093	82	45	WEAK	Y
3	122	03	44	BAD	N
11	202	16	44	DESEL	Y
19	193	75	45	GOOD	Y

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GPS SENSOR 2/2 PAGE
Figure 65

PSM 1-84-2A
EFFECTIVITY:
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Config 001

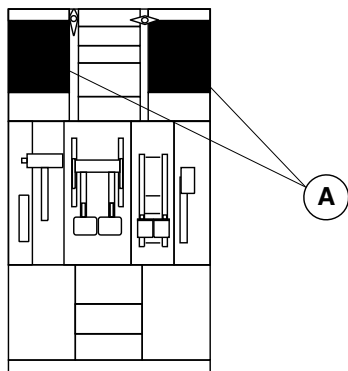
34-60-01

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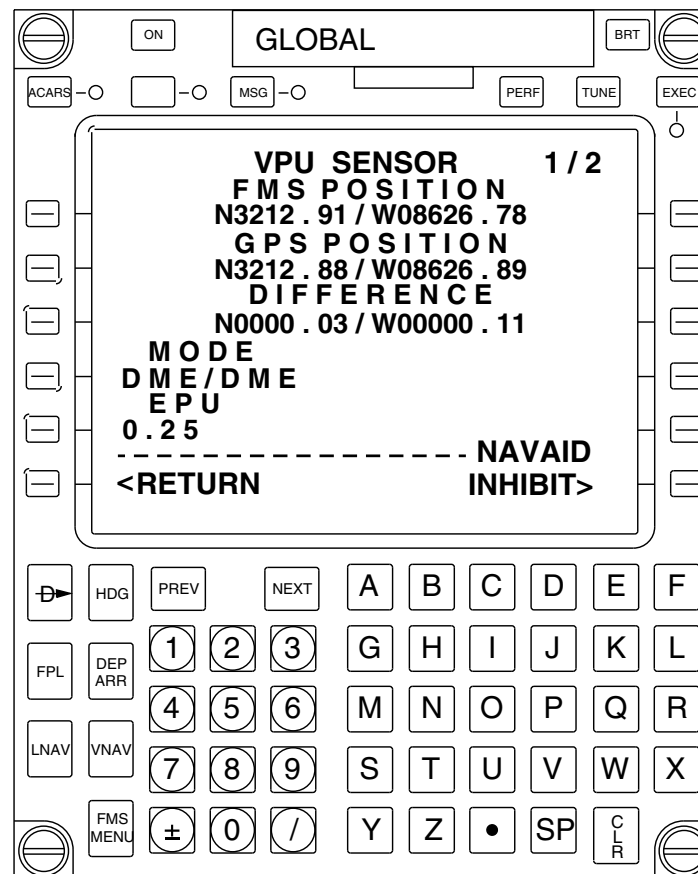


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VPU SENSOR 1/2 PAGE
Figure 66

PSM 1-84-2A
EFFECTIVITY:
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Config 001

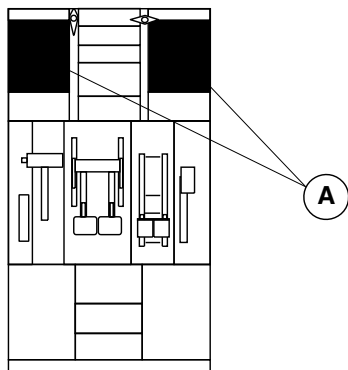
34-60-01

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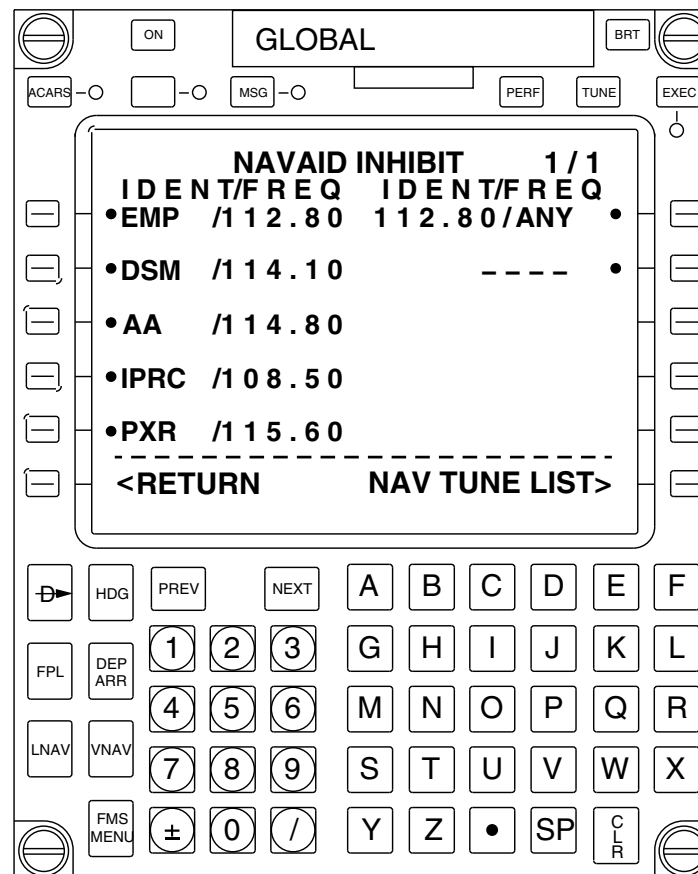


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NAVAID INHIBIT 1/1 PAGE
Figure 67

PSM 1-84-2A
EFFECTIVITY:
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Config 001

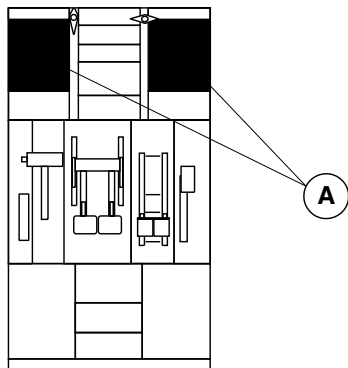
34-60-01

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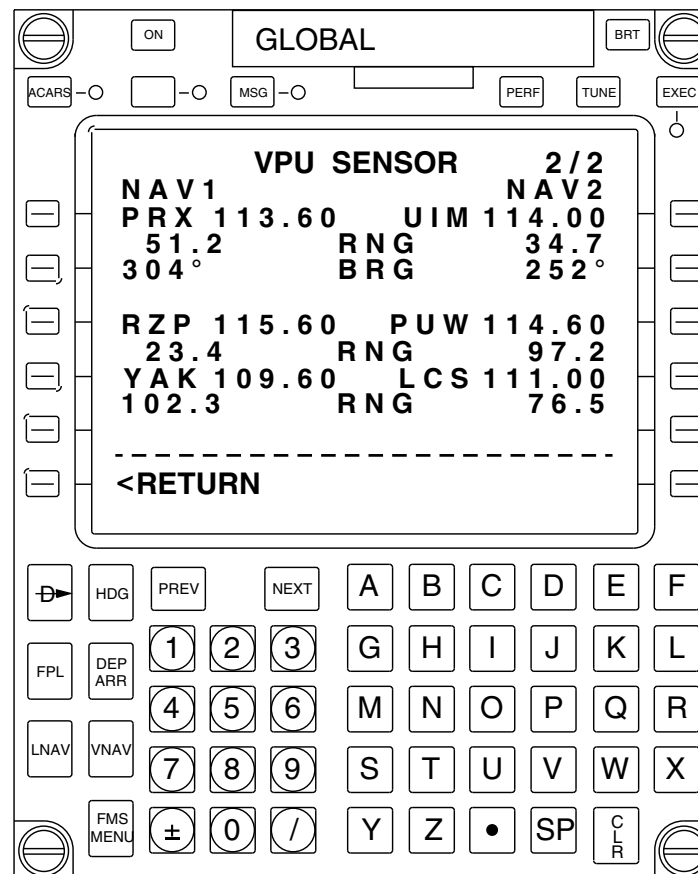


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VPU SENSOR 2/2 PAGE
Figure 68

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

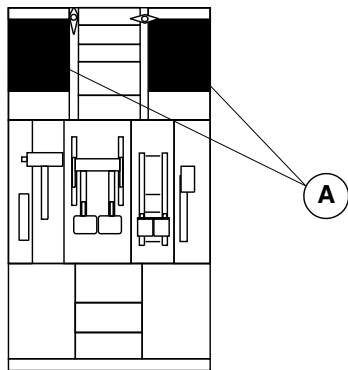
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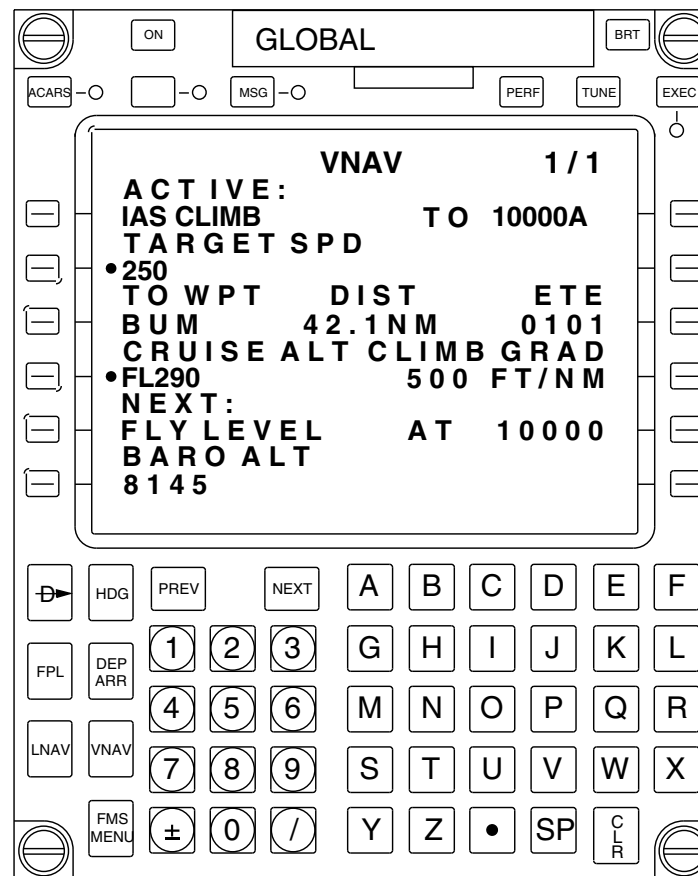


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VNAV 1/1 PAGE
Figure 69

PSM 1-84-2A
EFFECTIVITY:
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Config 001

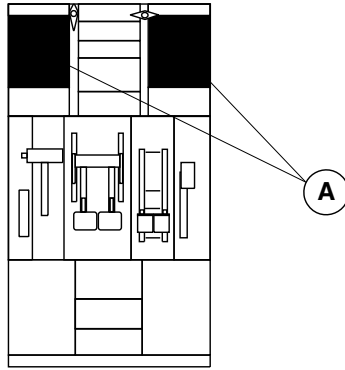
34-60-01

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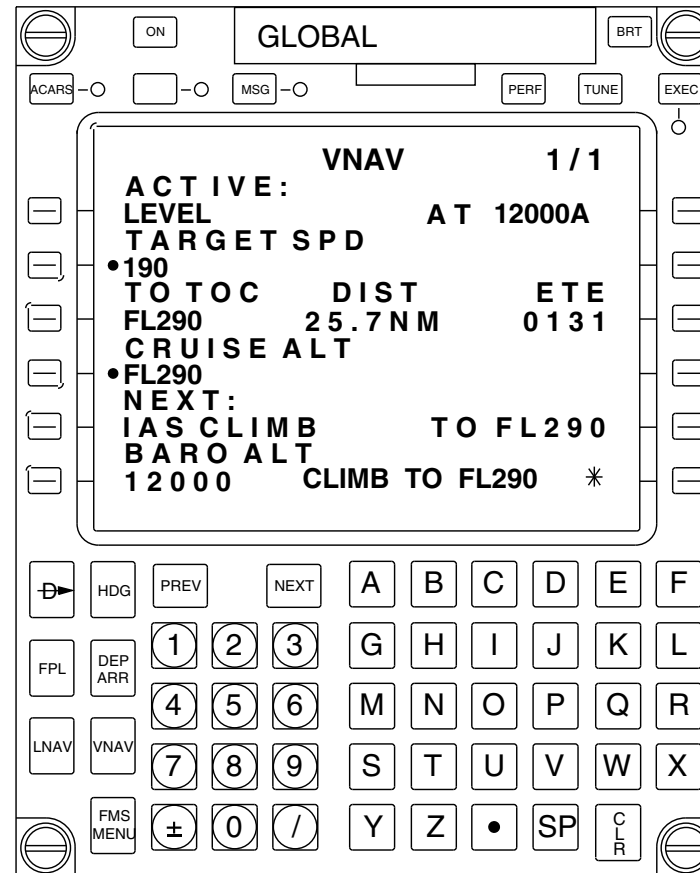


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VNAV 1/1 PAGE CRUISE
Figure 70

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

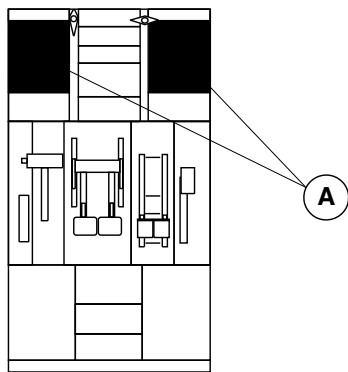
34-60-01

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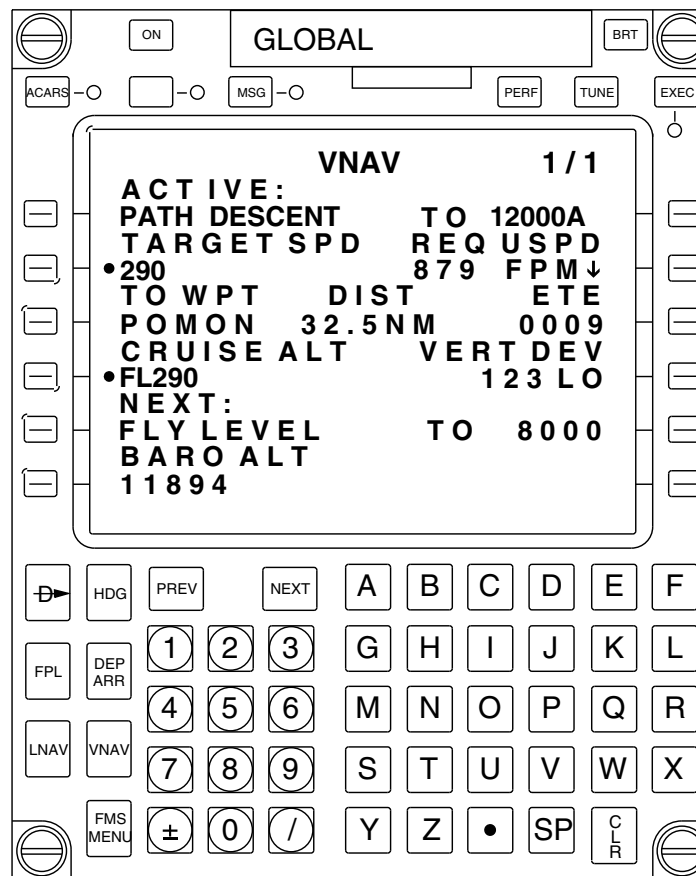


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VNAV 1/1 PAGE DESCENT
Figure 71

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

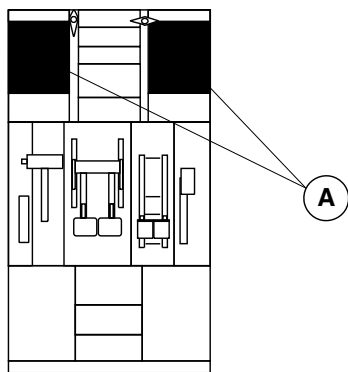
34-60-01

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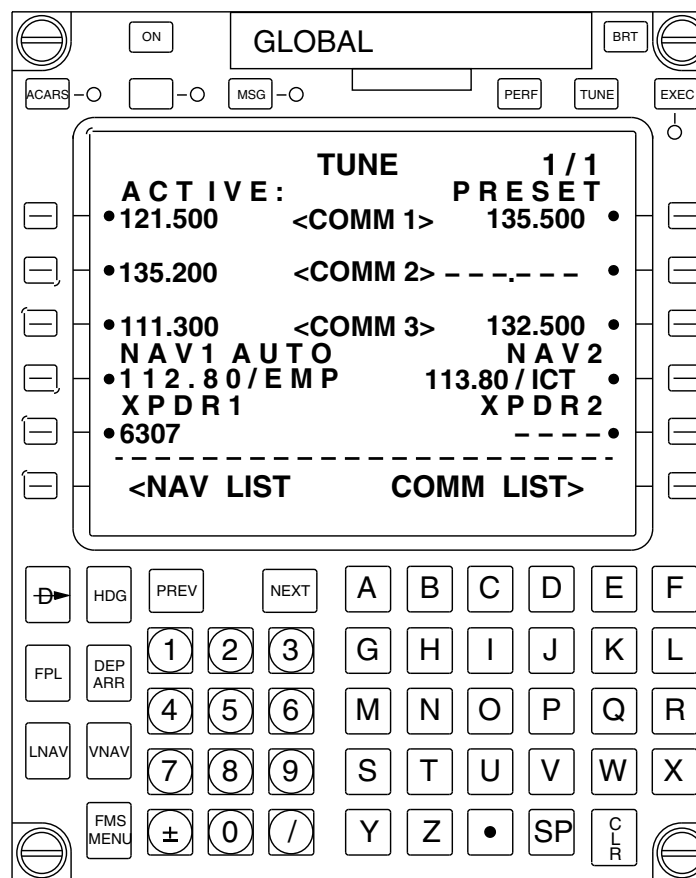


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TUNE 1/1 PAGE
Figure 72

PSM 1-84-2A
EFFECTIVITY:
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Config 001

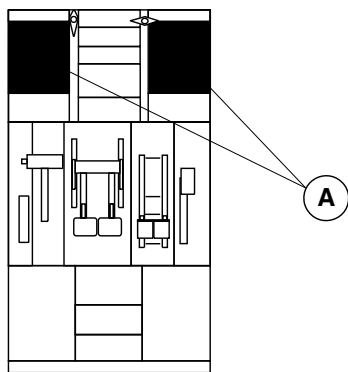
34-60-01

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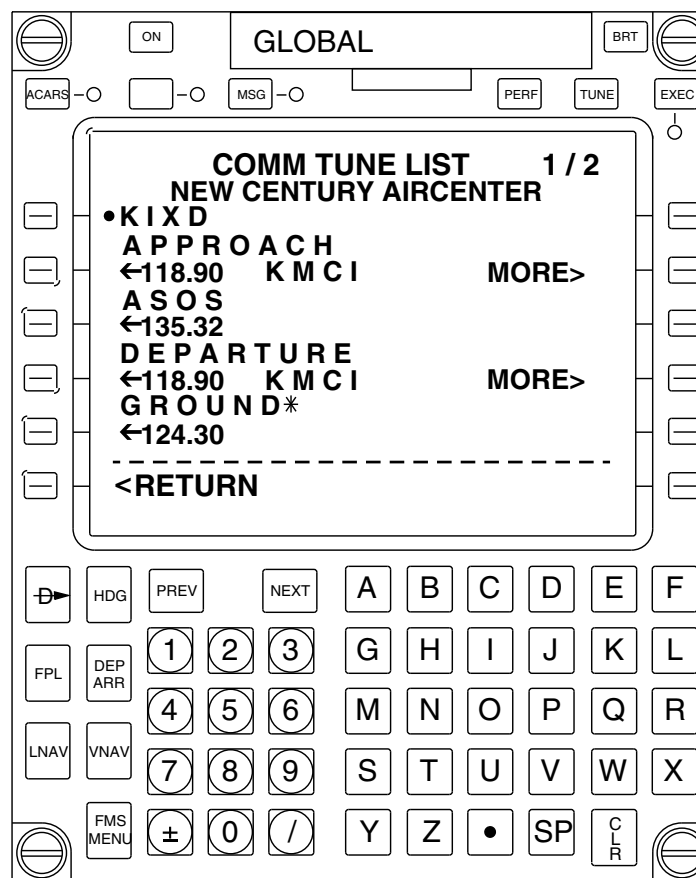


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COMM TUNE LIST 1/2 PAGE
Figure 73

PSM 1-84-2A
EFFECTIVITY:
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Config 001

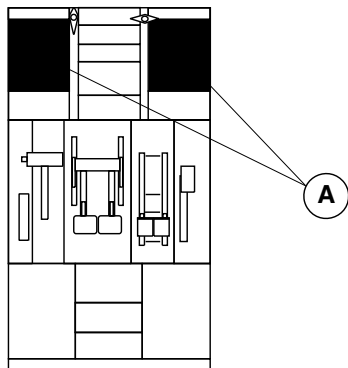
34-60-01

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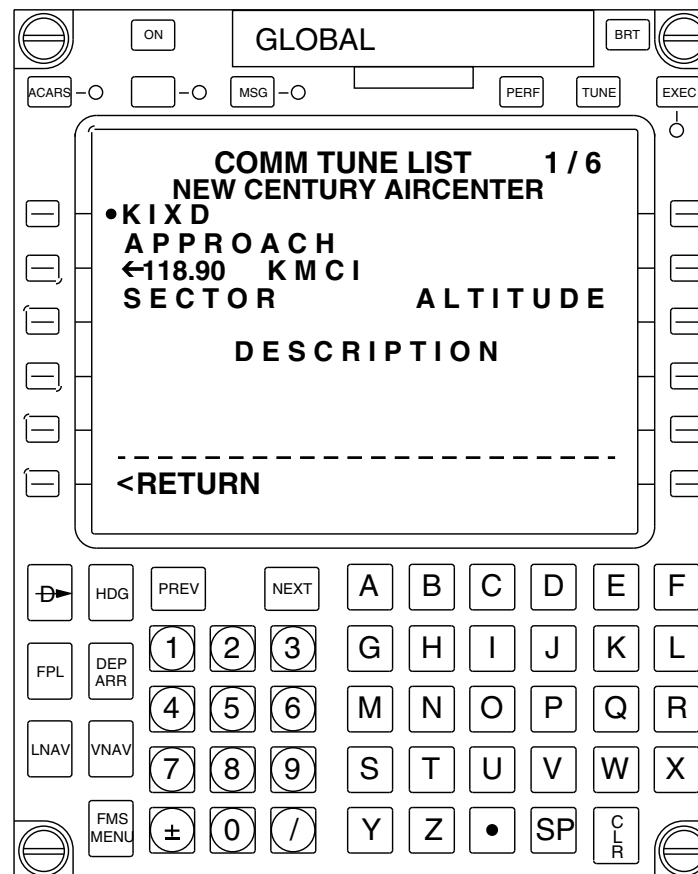


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COMM TUNE LIST 1/2 PAGE MORE
Figure 74

PSM 1-84-2A
EFFECTIVITY:
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Config 001

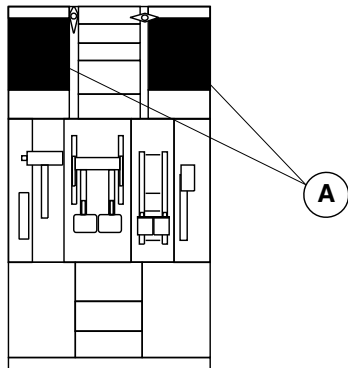
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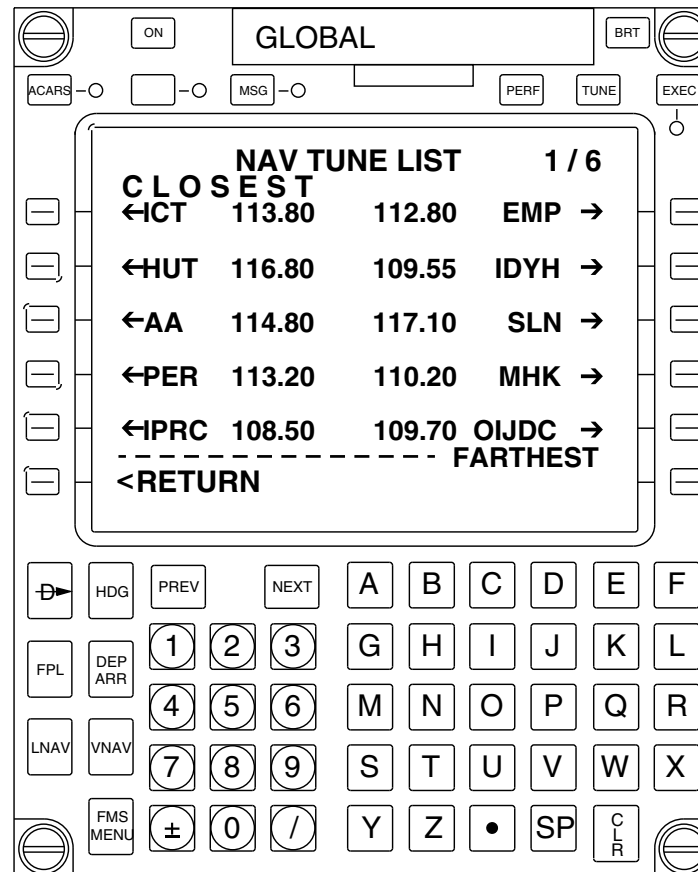


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NAV TUNE LIST 1/1 PAGE
Figure 75

PSM 1-84-2A
EFFECTIVITY:
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Config 001

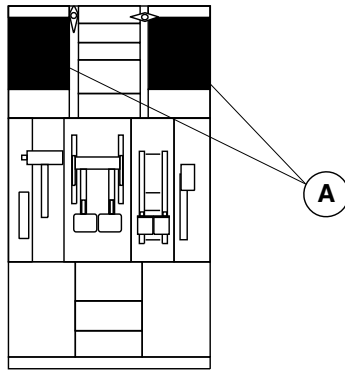
34-60-01

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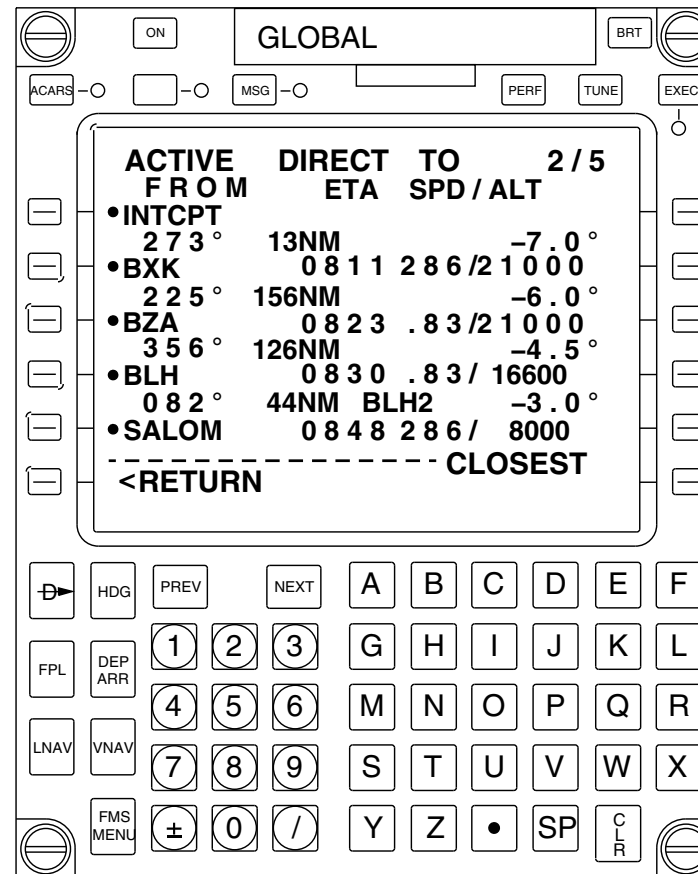


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DIRECT TO N/N PAGE
Figure 76

PSM 1-84-2A
EFFECTIVITY:
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Config 001

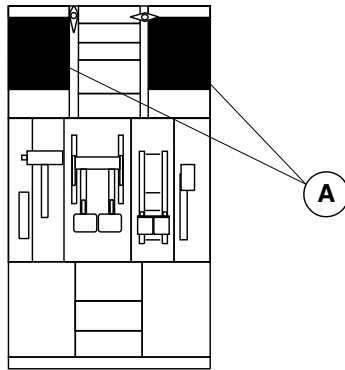
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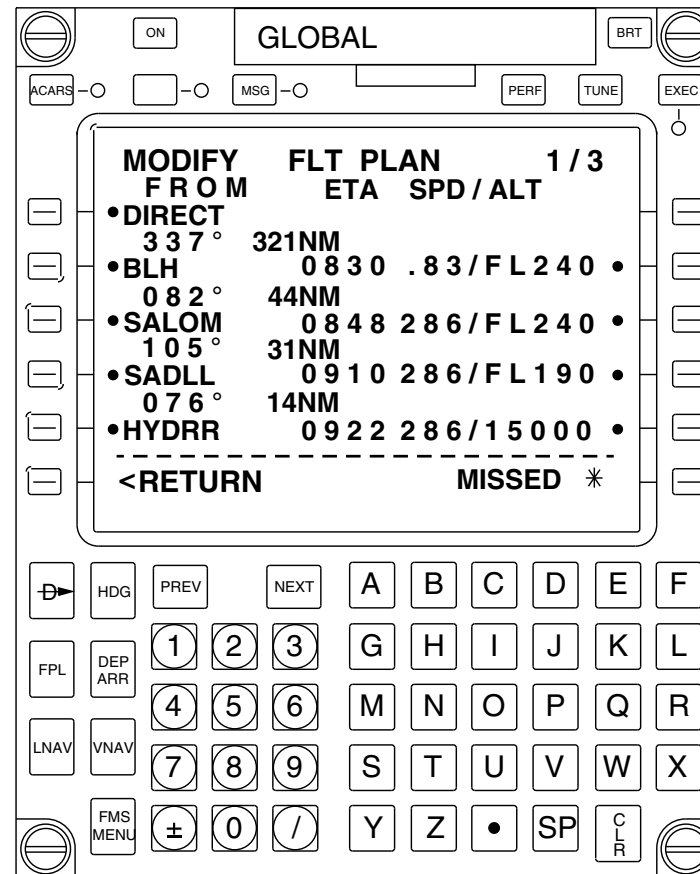


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MODIFY FLT PLAN 1/3 PAGE
Figure 77

PSM 1-84-2A
EFFECTIVITY:
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Config 001

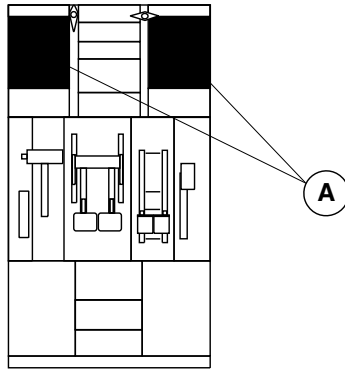
34-60-01

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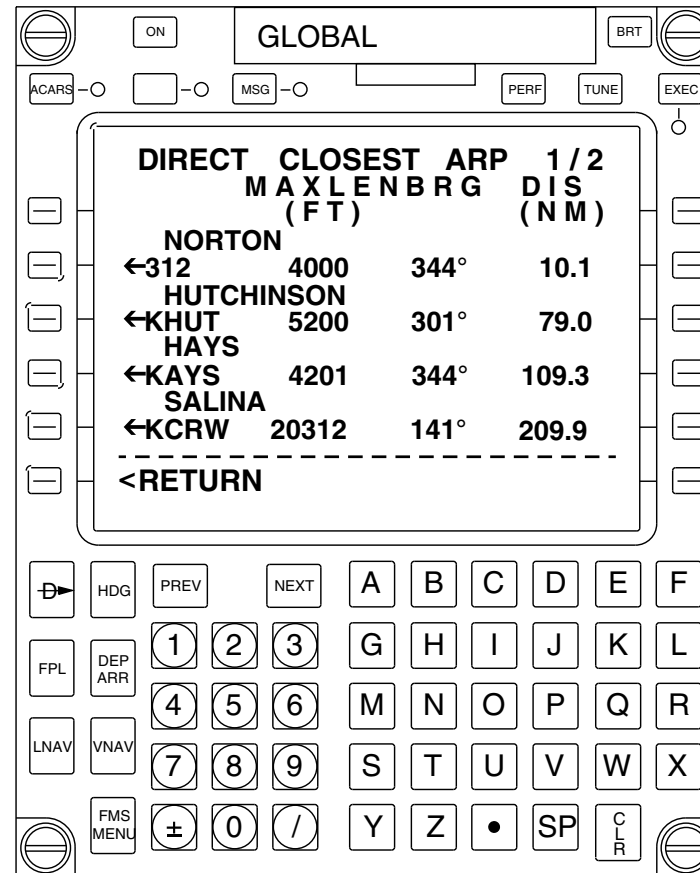


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a54.cgm

CLOSEST AIRPORT 1/2 PAGE
Figure 78

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

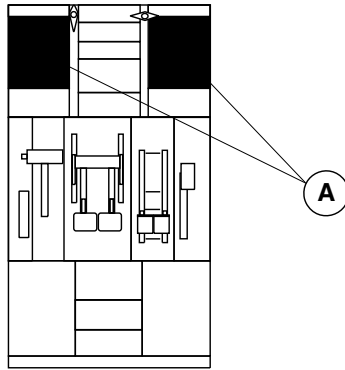
34-60-01

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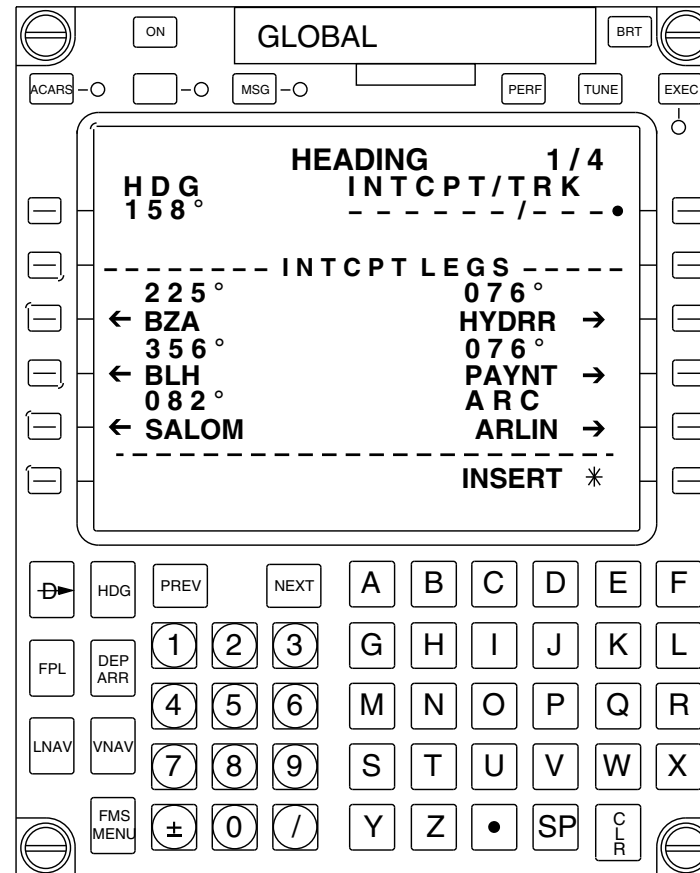


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a55.cgm

HEADING 1/N PAGE
Figure 79

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

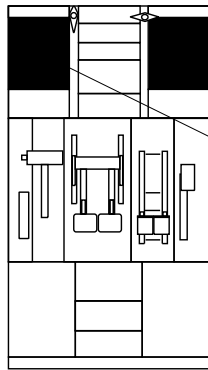
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CENTRE CONSOLE

ON GLOBAL BRT

ACARS MSG PERF TUNE EXEC

HEADING 1 / 4

HDG • 158°

INTCPT/TRK

INTCPT LEGS

225° 076°

←BZA HYDRR →

356° 076°

←BLH PAYNT →

082° ARC

←SALOM ARLIN →

INSERT *

HDG PREV NEXT A B C D E F

FPL DEP ARR 1 2 3 G H I J K L

LNAV VNAV 4 5 6 M N O P Q R

FMS MENU 7 8 9 S T U V W X

± 0 / Y Z • SP CLR

A

fsq03a56.cgm

HEADING 1/N PAGE HEADING FIELD DEFAULTS
Figure 80

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

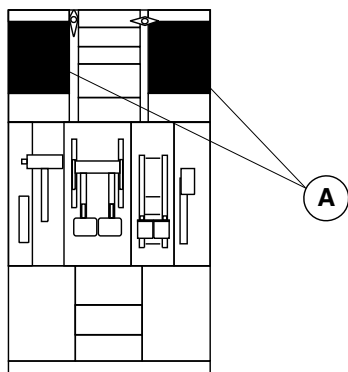
34-60-01

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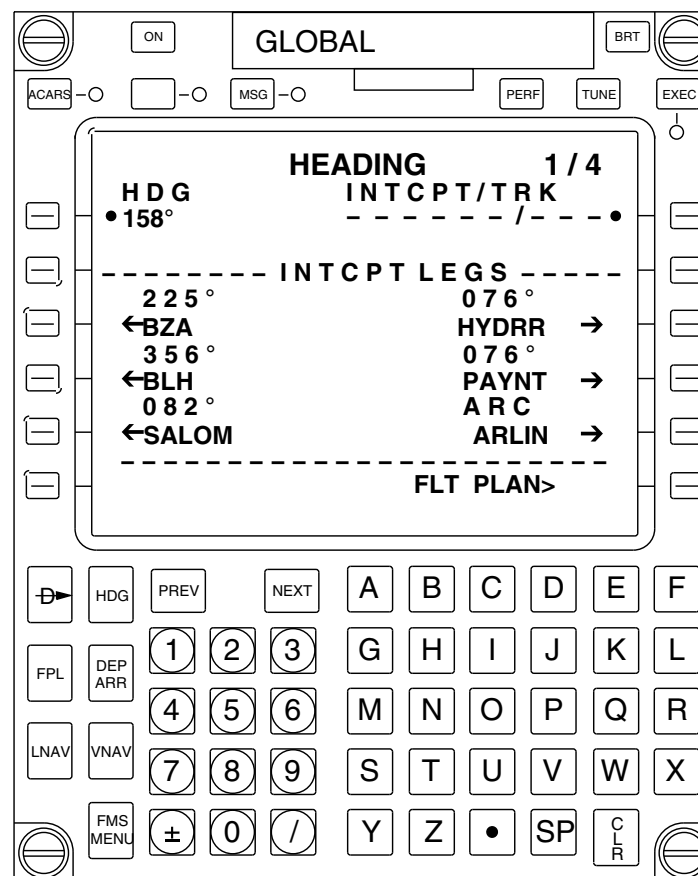


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq03a57.cgm

HEADING 1/N PAGE FIELD PLAN WAYPOINTS
Figure 81

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

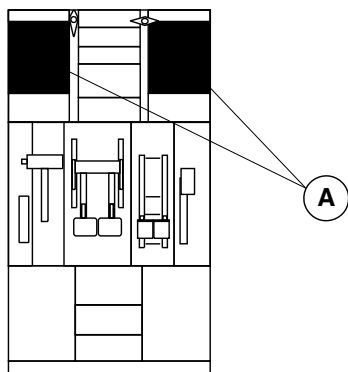
34-60-01

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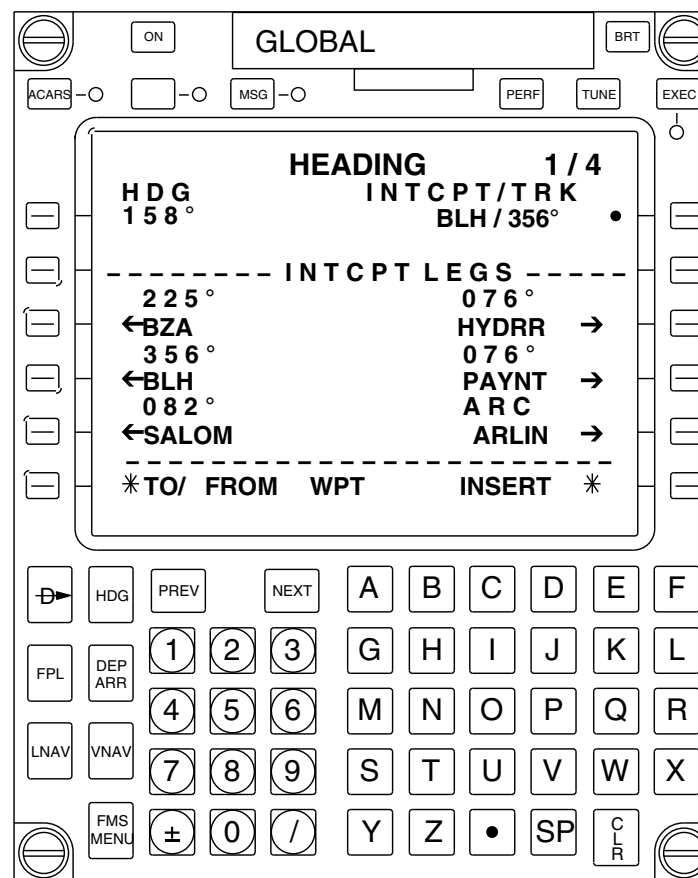


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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HEADING 1/N PAGE TO/FROM WAYPOINT
Figure 82

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

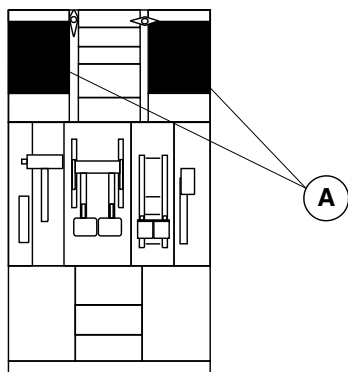
34-60-01

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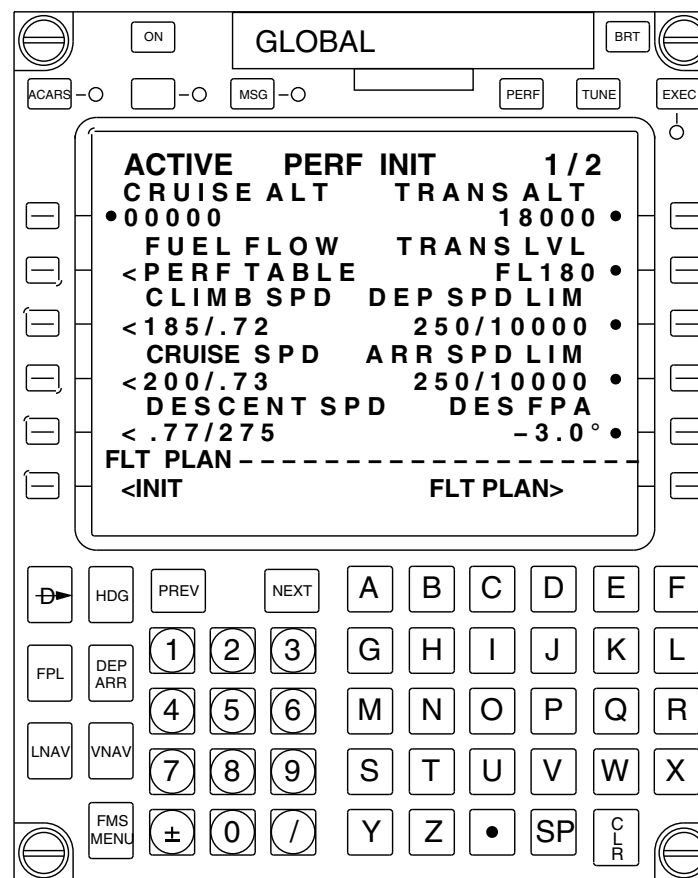


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CENTRE CONSOLE



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fsq03a59.cgm

PERF INIT 1/2 PAGE
Figure 83

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

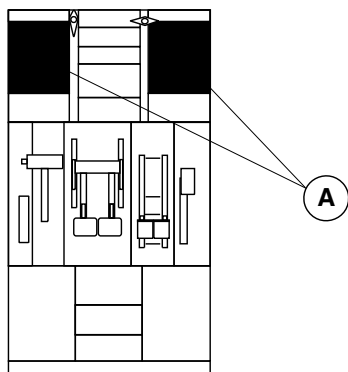
34-60-01

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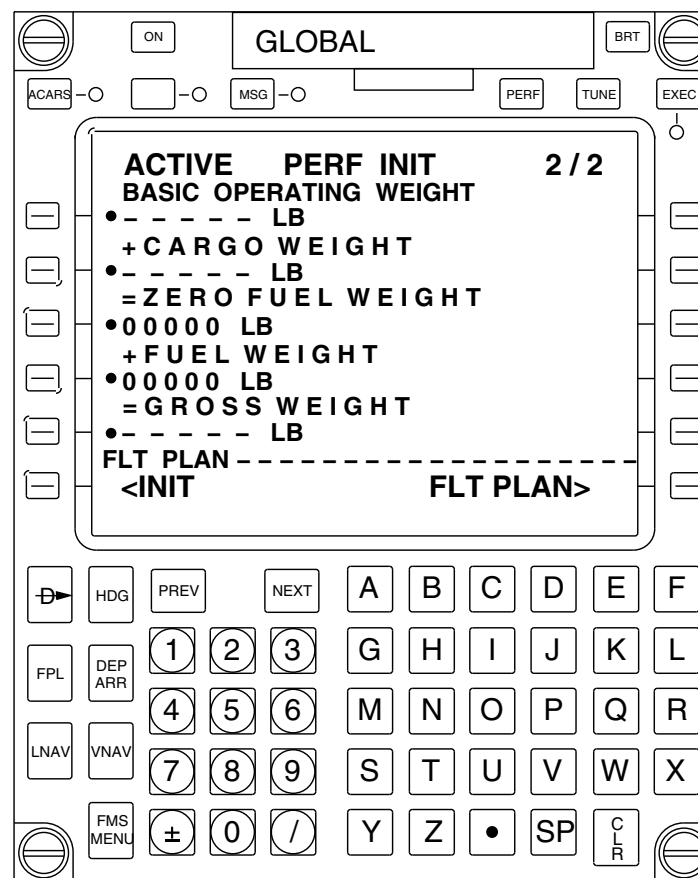


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CENTRE CONSOLE



A

fsq03a60.cgm

PERF INIT 2/2 PAGE
Figure 84

PSM 1-84-2A
EFFECTIVITY:
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Config 001

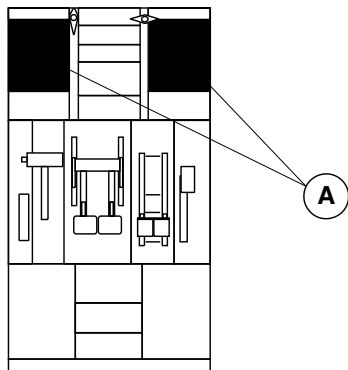
34-60-01

Config 001
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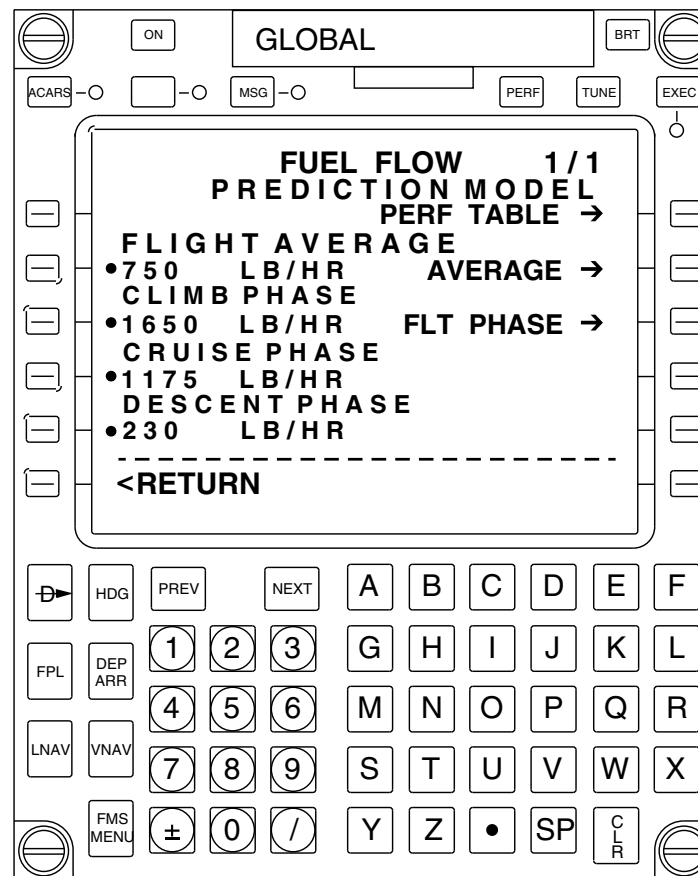


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq03a61.cgm

FUEL FLOW 1/1 PAGE
Figure 85

PSM 1-84-2A
EFFECTIVITY:
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Config 001

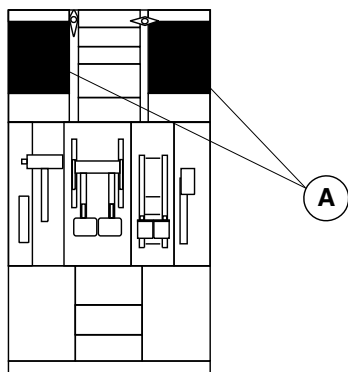
34-60-01

Config 001
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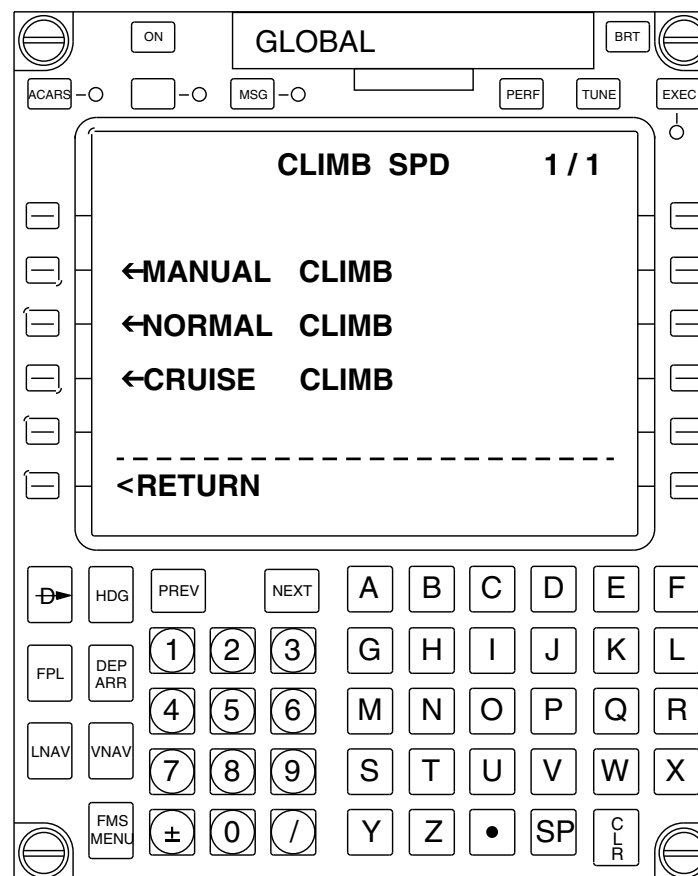


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CENTRE CONSOLE



A

fsq03a62.cgm

CLIMB SPEED 1/1 PAGE
Figure 86

PSM 1-84-2A
EFFECTIVITY:
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Config 001

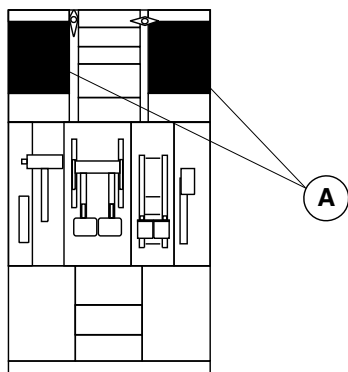
34-60-01

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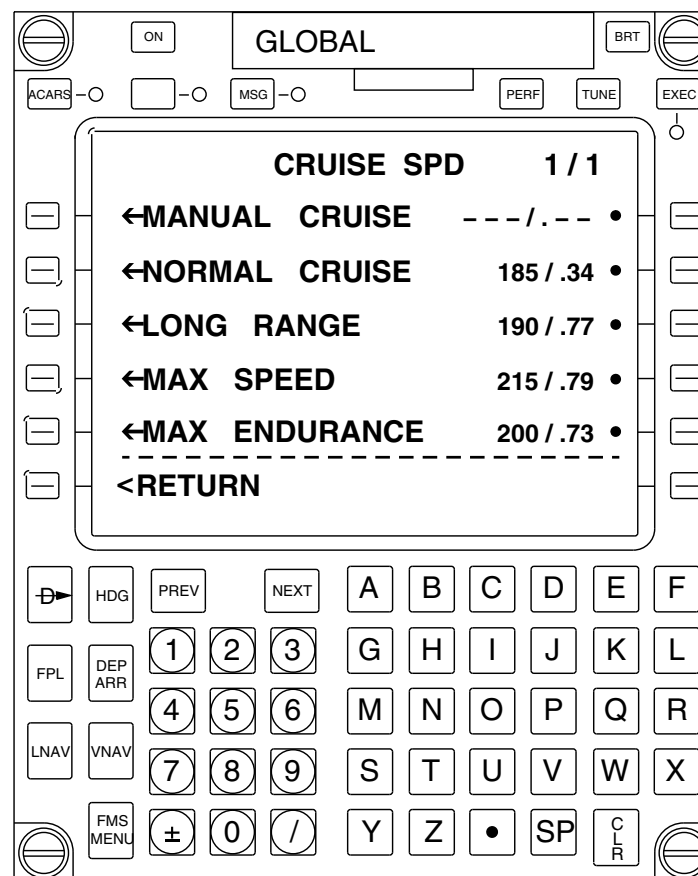


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CENTRE CONSOLE



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CRUISE SPEED 1/1 PAGE
Figure 87

PSM 1-84-2A
EFFECTIVITY:
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Config 001

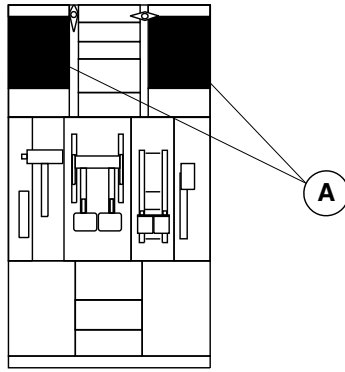
34-60-01

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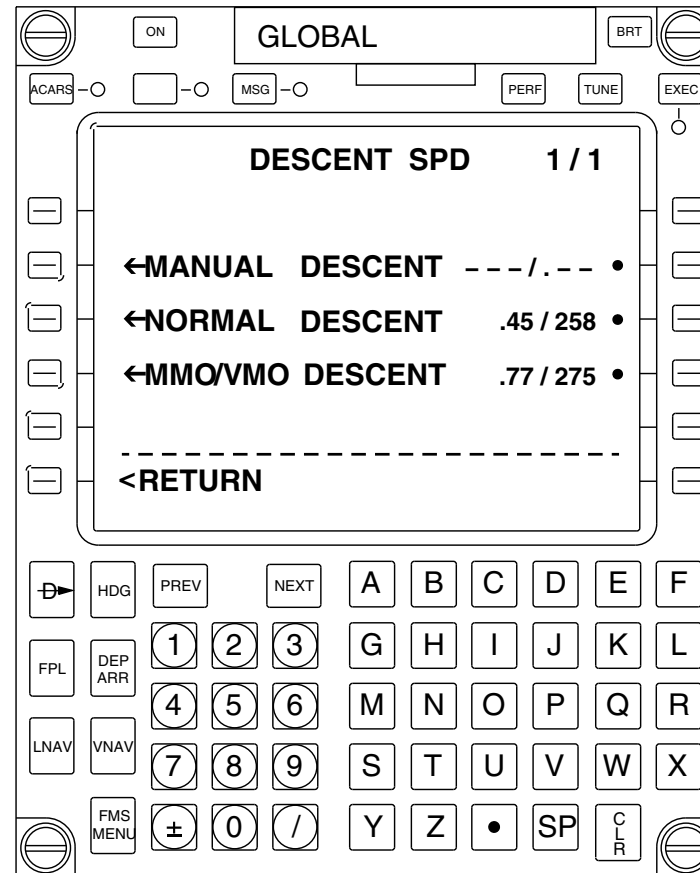


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a64.cgm

DESCENT SPEED 1/1 PAGE
Figure 88

PSM 1-84-2A
EFFECTIVITY:
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Config 001

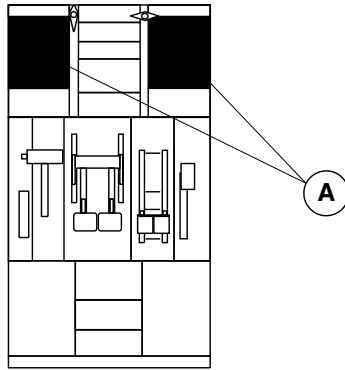
34-60-01

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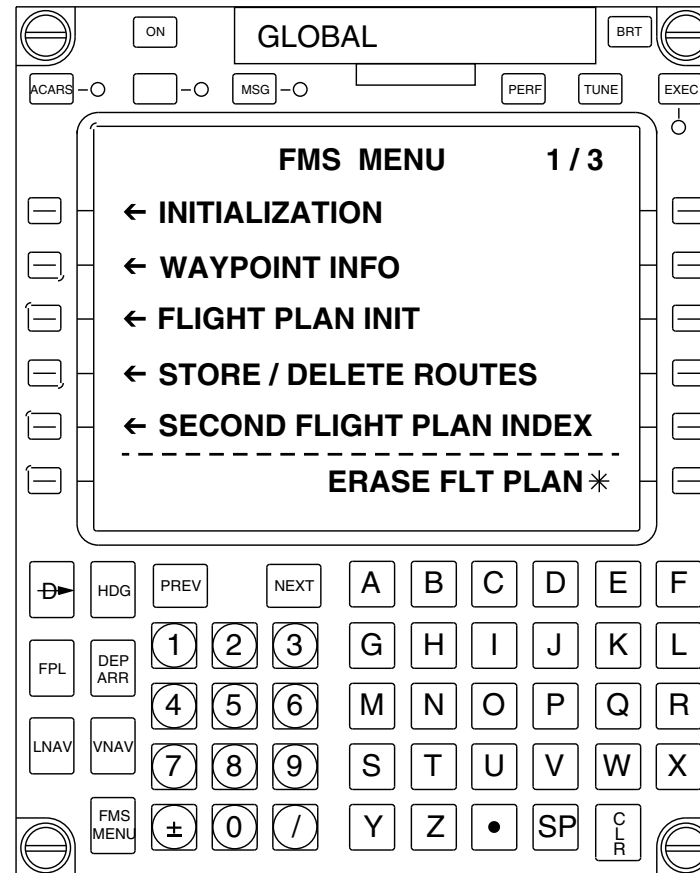


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CENTRE CONSOLE



A

fsq03a65.cgm

FMS MENU 1/3 PAGE
Figure 89

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

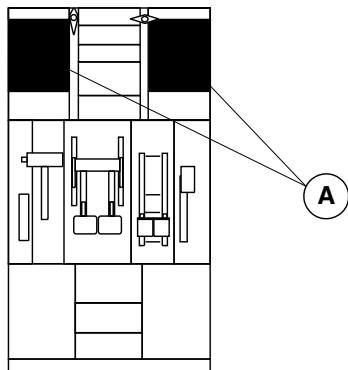
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CENTRE CONSOLE



A

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FMS MENU 2/3 PAGE
Figure 90

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

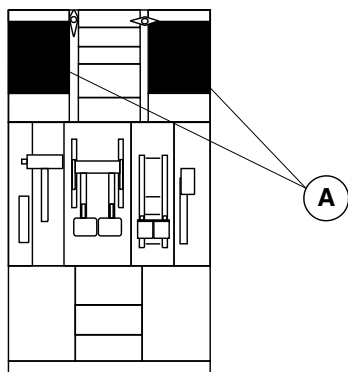
34-60-01

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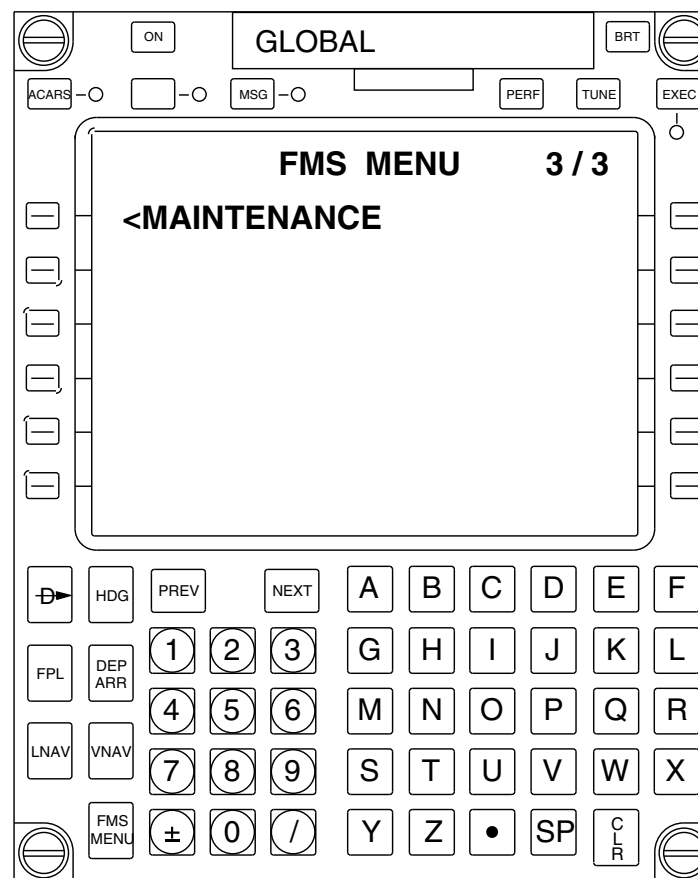


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CENTRE CONSOLE



A

fsq03a67.cgm

FMS MENU 3/3 PAGE
Figure 91

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

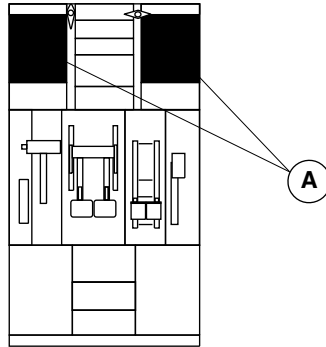
34-60-01

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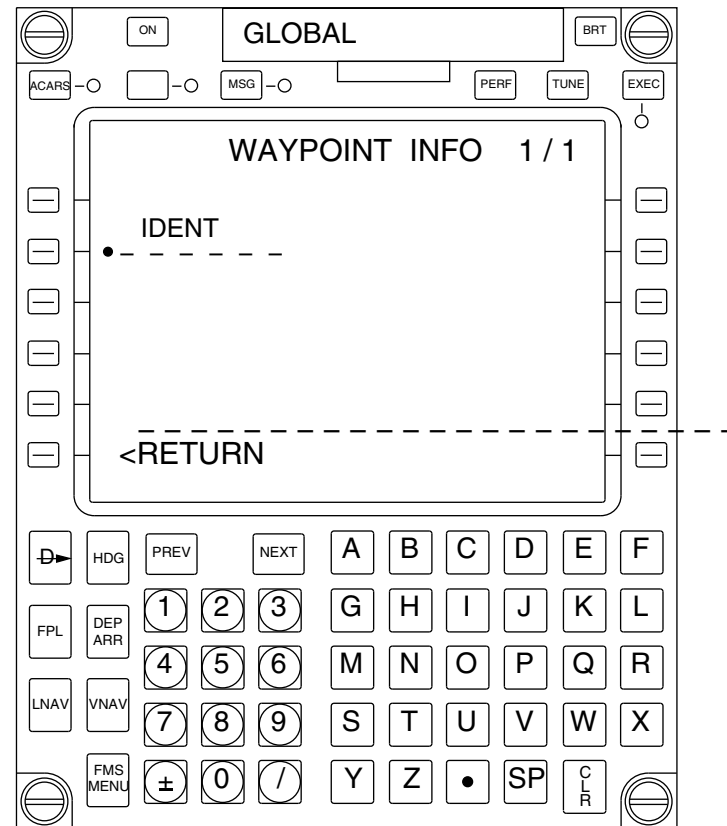


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a68.cgm

WAYPOINT INFORMATION 1/1 PAGE
Figure 92

PSM 1-84-2A
EFFECTIVITY:
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Config 001

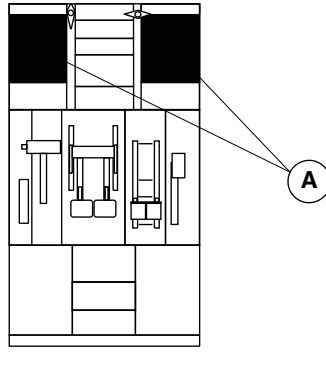
34-60-01

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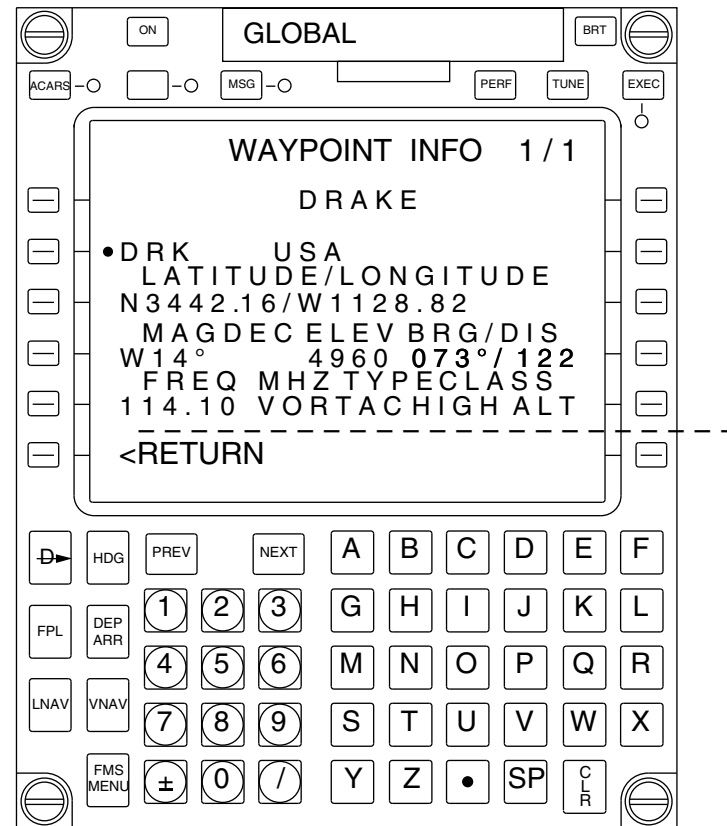


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a69.cgm

WAYPOINT INFORMATION 1/1 PAGE NAVAID
Figure 93

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

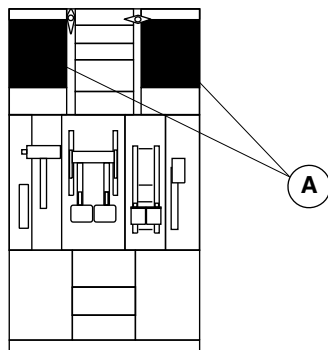
34-60-01

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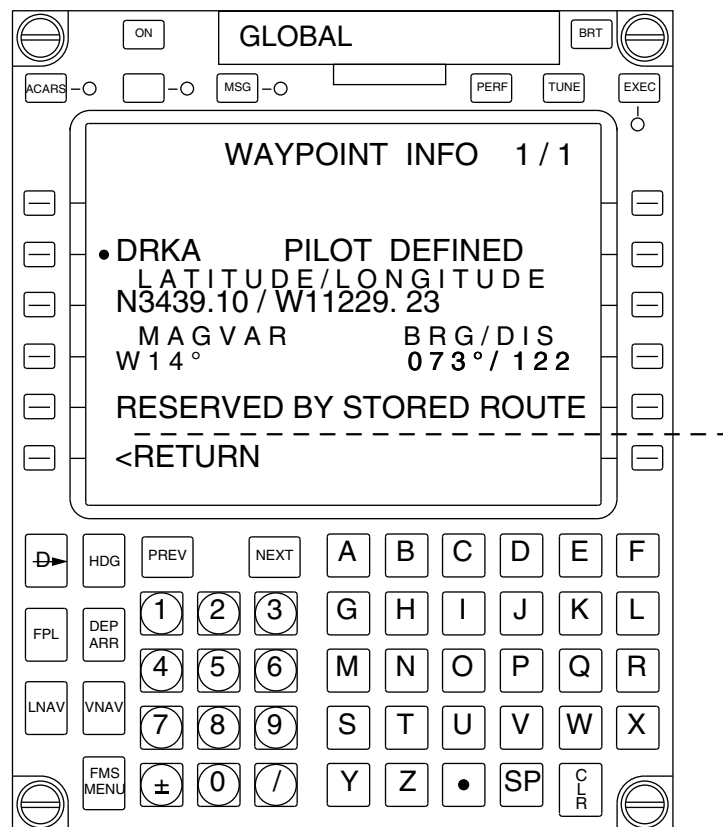


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a70.cgm

WAYPOINT INFORMATION 1/1 PAGE PILOT DEFINED
Figure 94

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

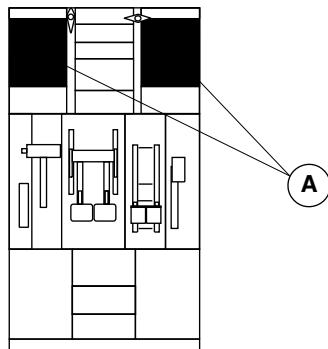
34-60-01

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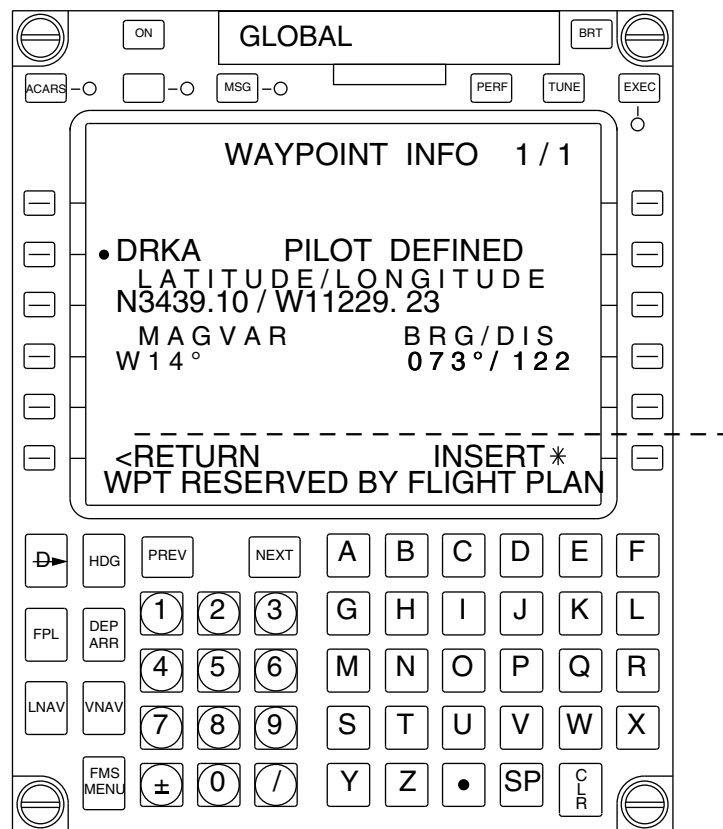


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a71.cgm

WAYPOINT INFORMATION 1/1 PAGE ACTIVE FLIGHT PLAN
Figure 95

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

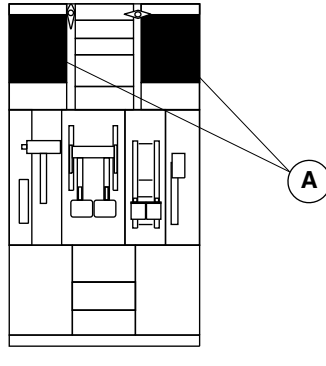
34-60-01

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CENTRE CONSOLE

ON GLOBAL BRT

ACARS MSG PERF TUNE EXEC

WAYPOINT INFO 1 / 1

LOVE

• KPRC USA
LATITUDE/LONGITUDE
N3439.10/W1128.23
MAGVAR ELEV BRG/DIS
W14° 5010 073°/122
CLASS
LARGE

<RETURN RUNWAY INFO>

HDG PREV NEXT A B C D E F

FPL DEP ARR 1 2 3 G H I J K L

LNAV VNAV 4 5 6 M N O P Q R

FMS MENU 7 8 9 S T U V W X

± 0 / Y Z • SP CLR

A

fsq03a72.cgm

WAYPOINT INFORMATION 1/1 PAGE RUNWAY
Figure 96

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

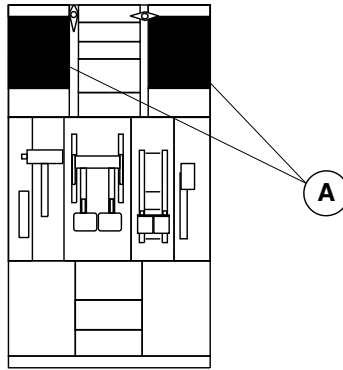
34-60-01

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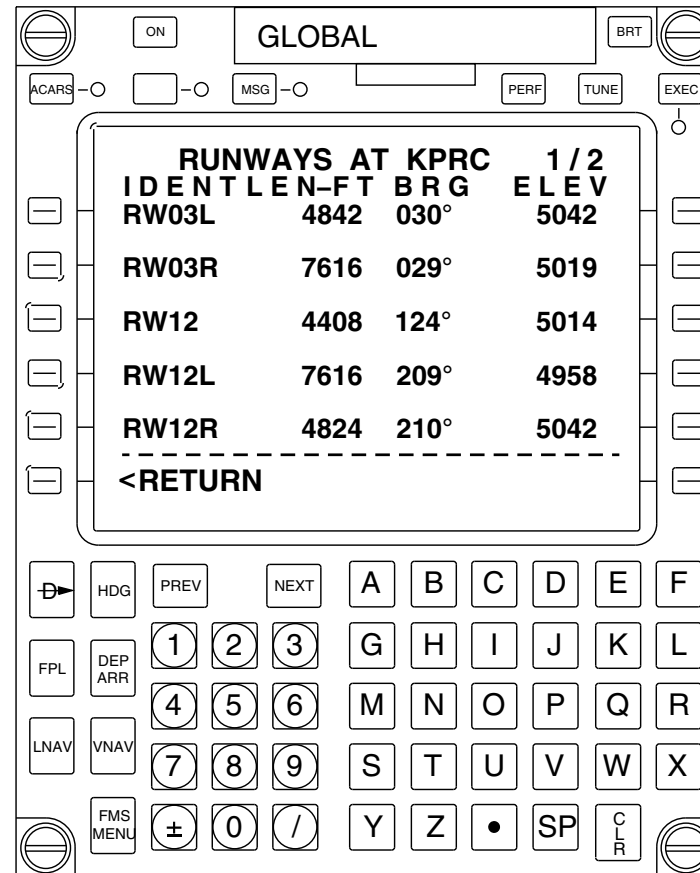


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CENTRE CONSOLE



A

fsq03a73.cgm

RUNWAYS AT AAAA 1/2 PAGE
Figure 97

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

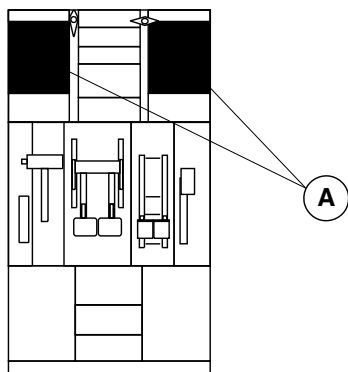
34-60-01

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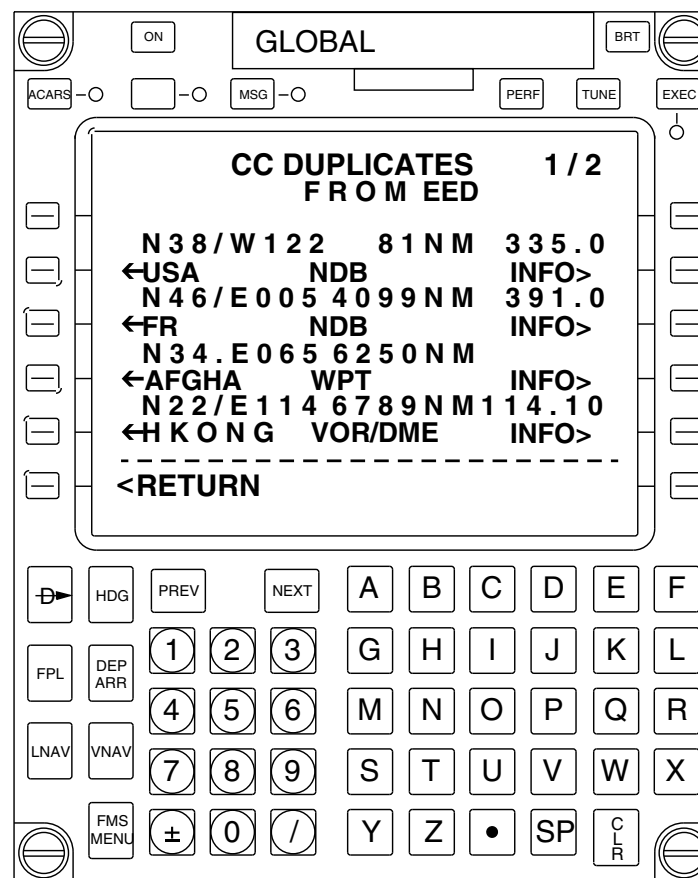


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A

fsq03a74.cgm

DUPLICATES 1/2 PAGE
Figure 98

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

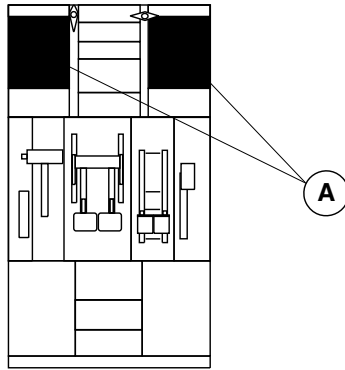
34-60-01

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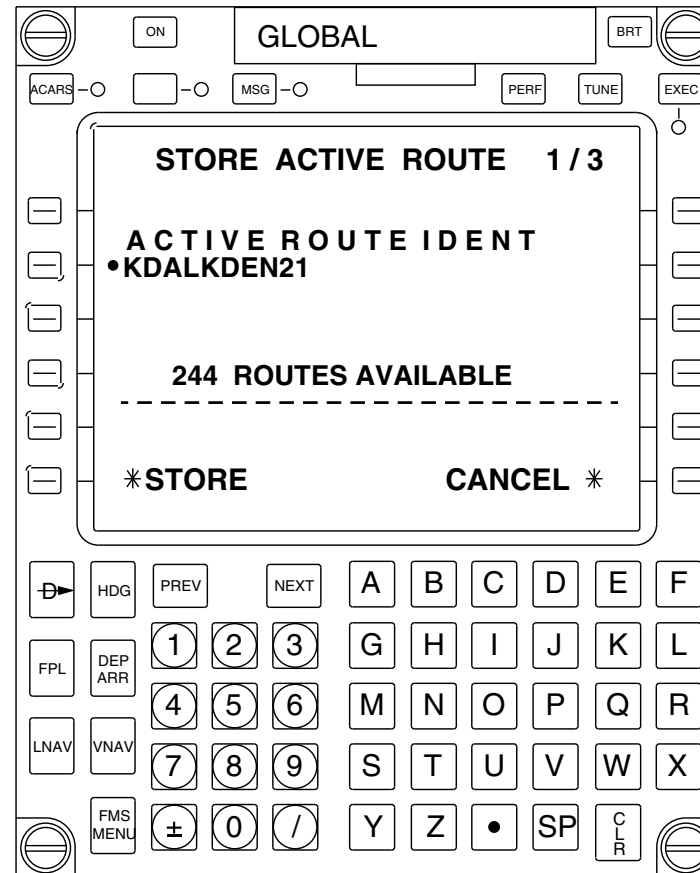


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A

fsq03a75.cgm

STORE ACTIVE ROUTE 1/3 PAGE
Figure 99

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

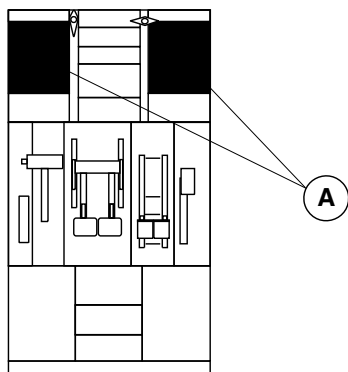
34-60-01

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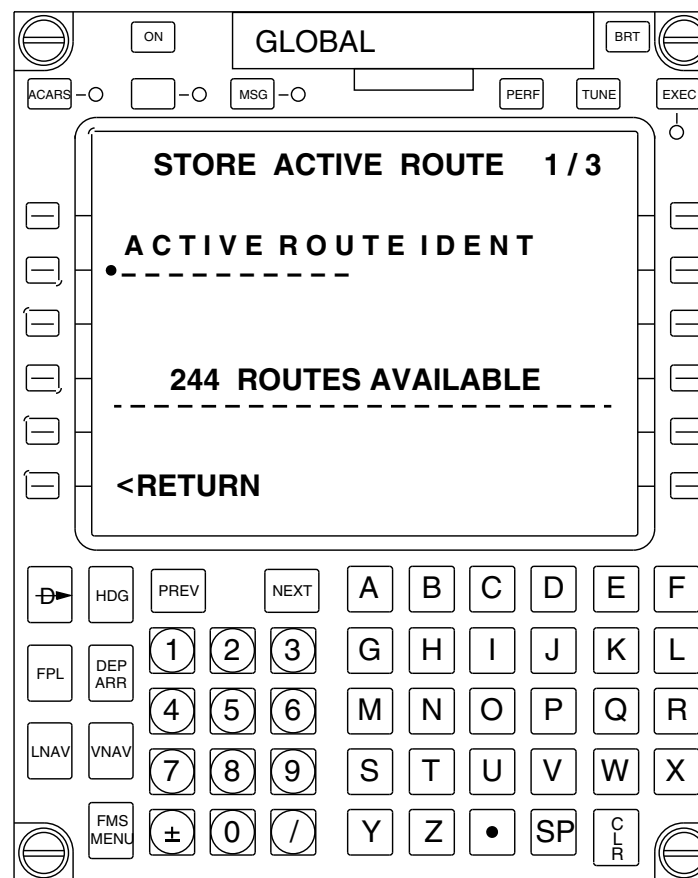


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CENTRE CONSOLE



A

fsq03a76.cgm

STORE ACTIVE ROUTE 1/3 PAGE RETURN PROMPT
Figure 100

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

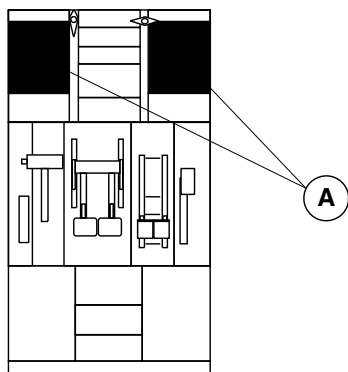
34-60-01

Config 001
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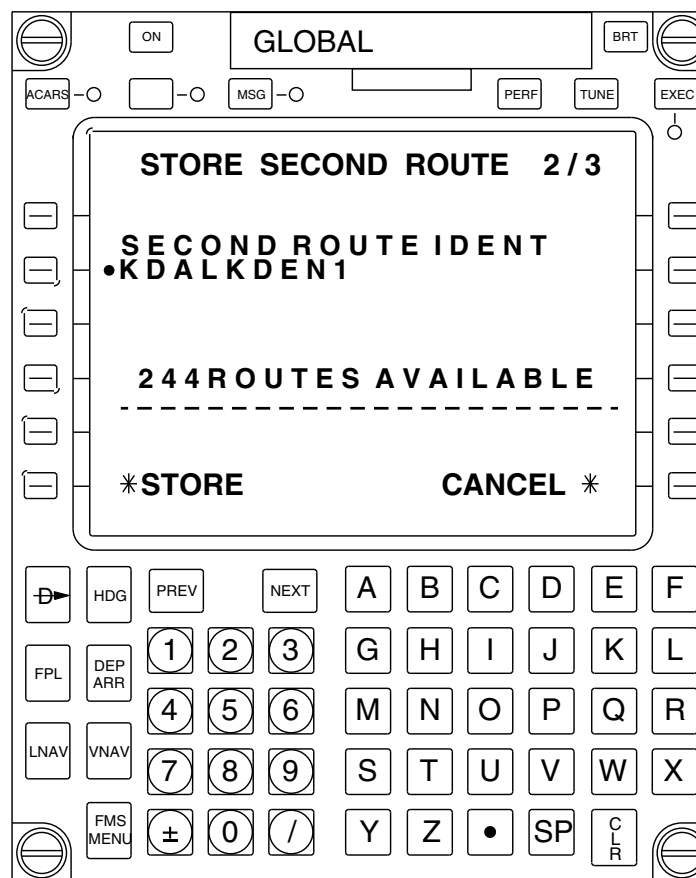


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A

fsq03a77.cgm

STORE SECOND ROUTE 2/3 PAGE
Figure 101

PSM 1-84-2A
EFFECTIVITY:
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Config 001

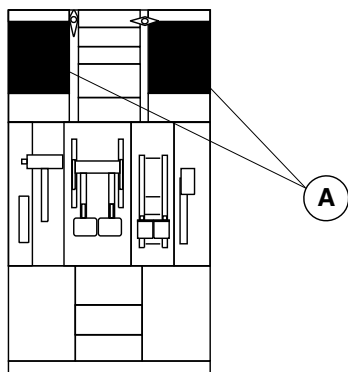
34-60-01

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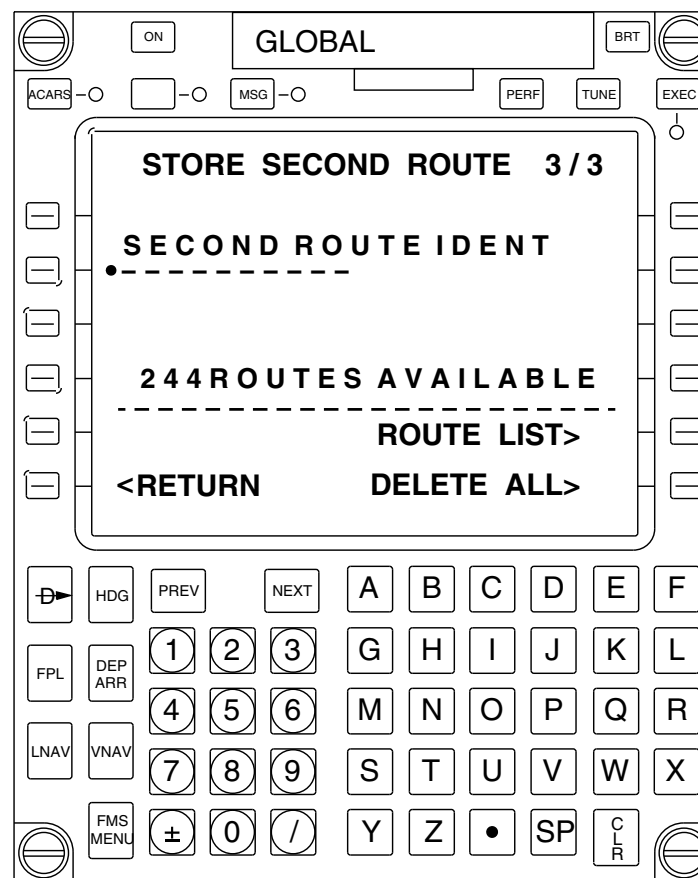


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a78.cgm

DELETE STORED ROUTE 3/3 PAGE
Figure 102

PSM 1-84-2A
EFFECTIVITY:
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Config 001

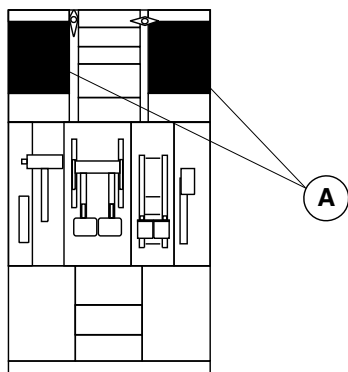
34-60-01

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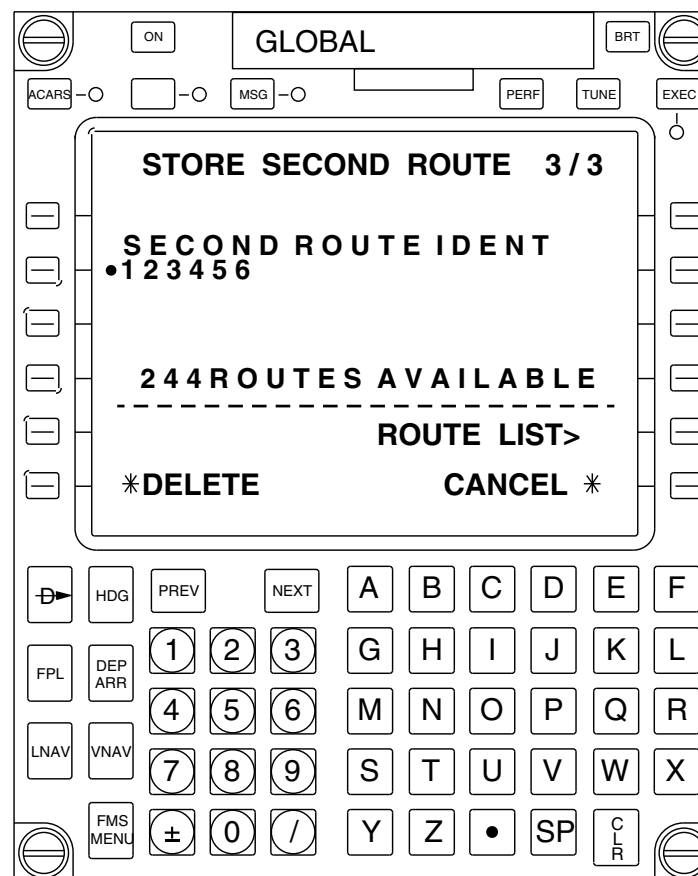


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fsq03a79.cgm

DELETE STORED ROUTE 3/3 PAGE IDENT
Figure 103

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

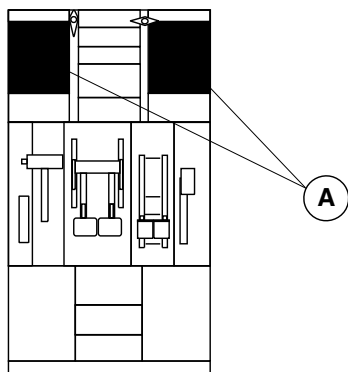
34-60-01

Config 001
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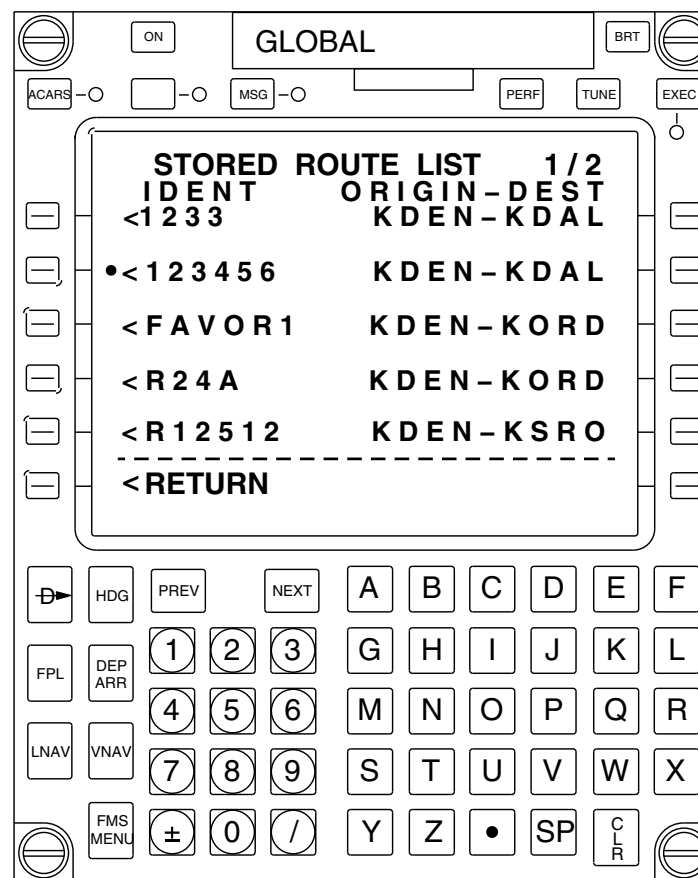


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a80.cgm

STORED ROUTE LIST 1/N PAGE
Figure 104

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

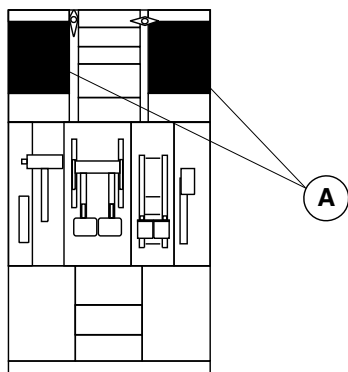
34-60-01

Config 001
Page 204
Sep 05/2021

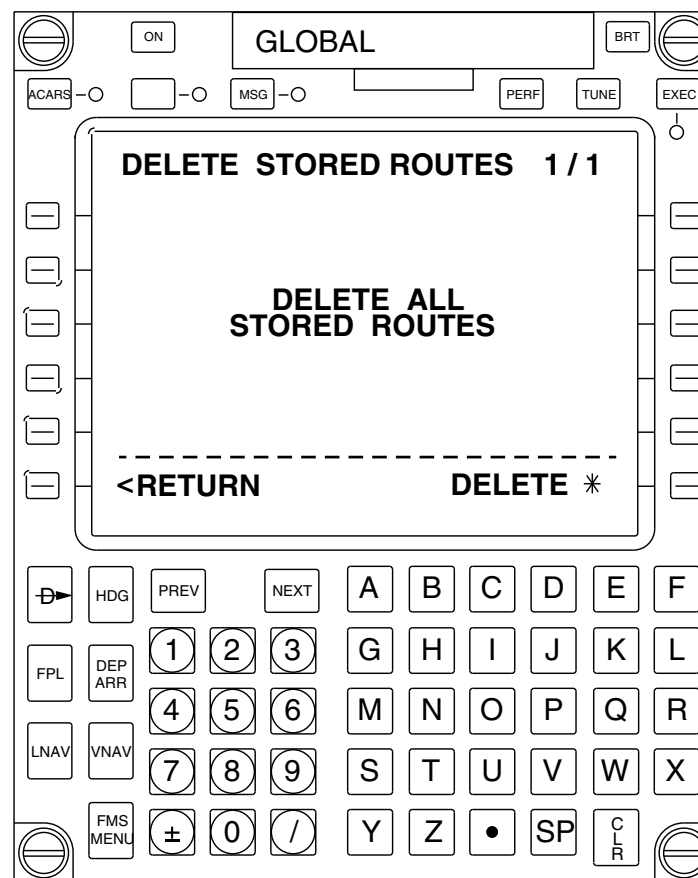


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a81.cgm

DELETE STORED ROUTES 1/1 PAGE
Figure 105

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

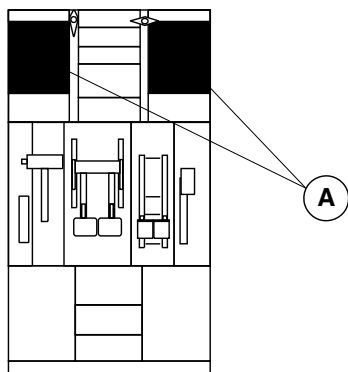
34-60-01

Config 001
Page 205
Sep 05/2021

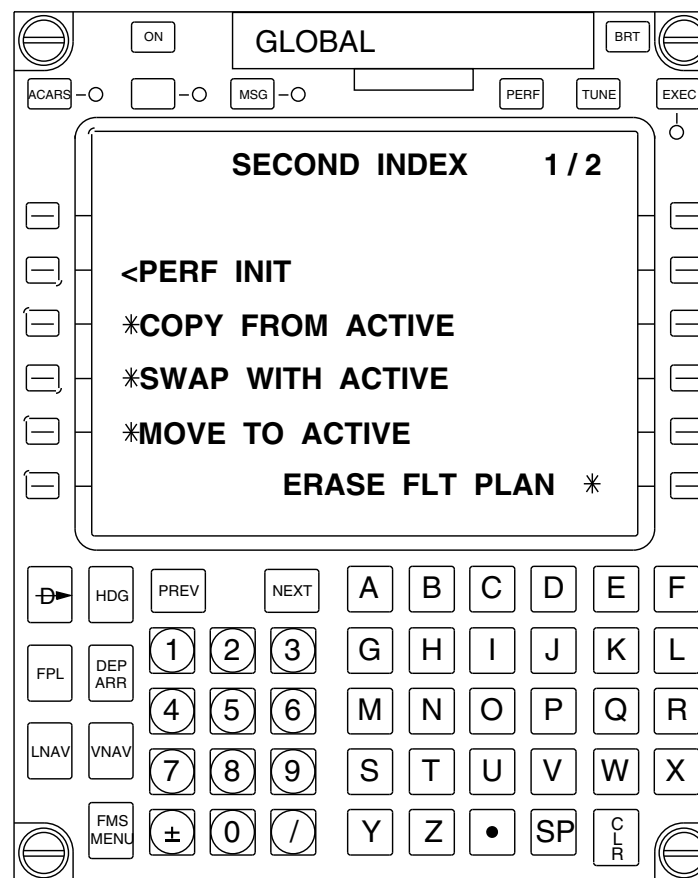


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a82.cgm

SECOND INDEX 1/2 PAGE
Figure 106

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

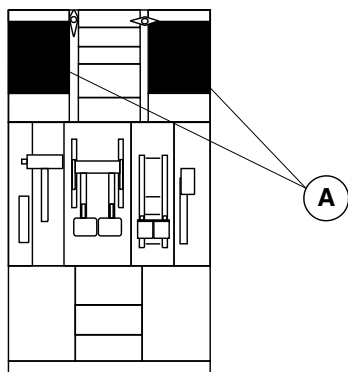
34-60-01

Config 001
Page 206
Sep 05/2021

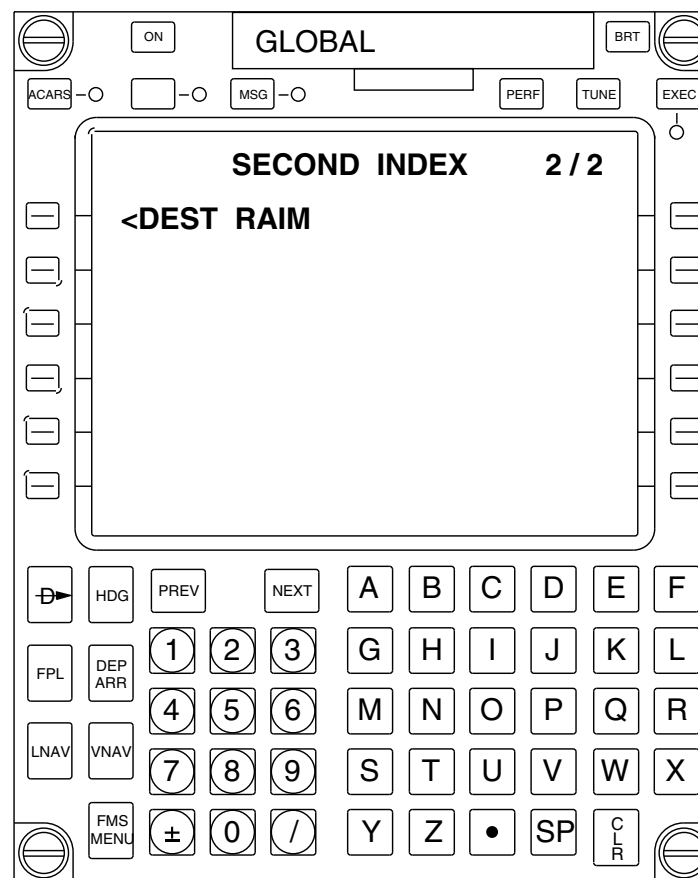


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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a83.cgm

SECOND INDEX 2/2 PAGE
Figure 107

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

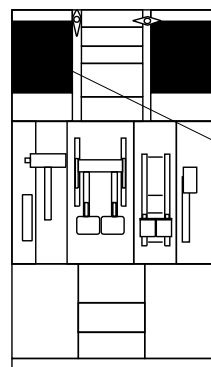
34-60-01

Config 001
Page 207
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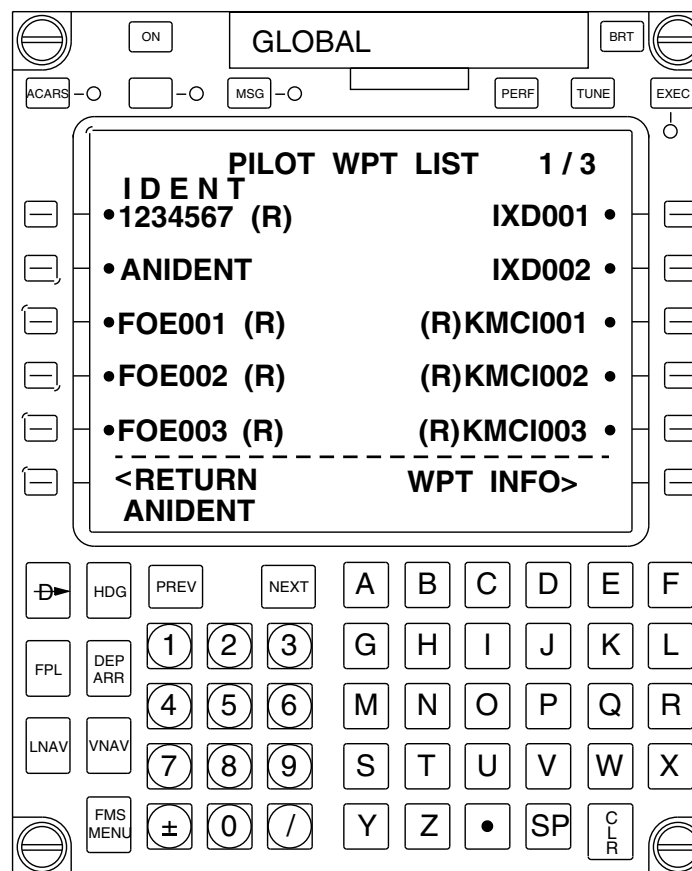


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a84.cgm

PILOT WAYPOINT LIST 1/N PAGE
Figure 108

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

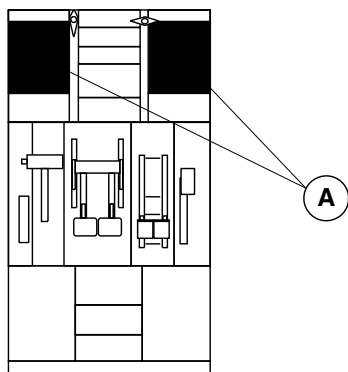
34-60-01

Config 001
Page 208
Sep 05/2021

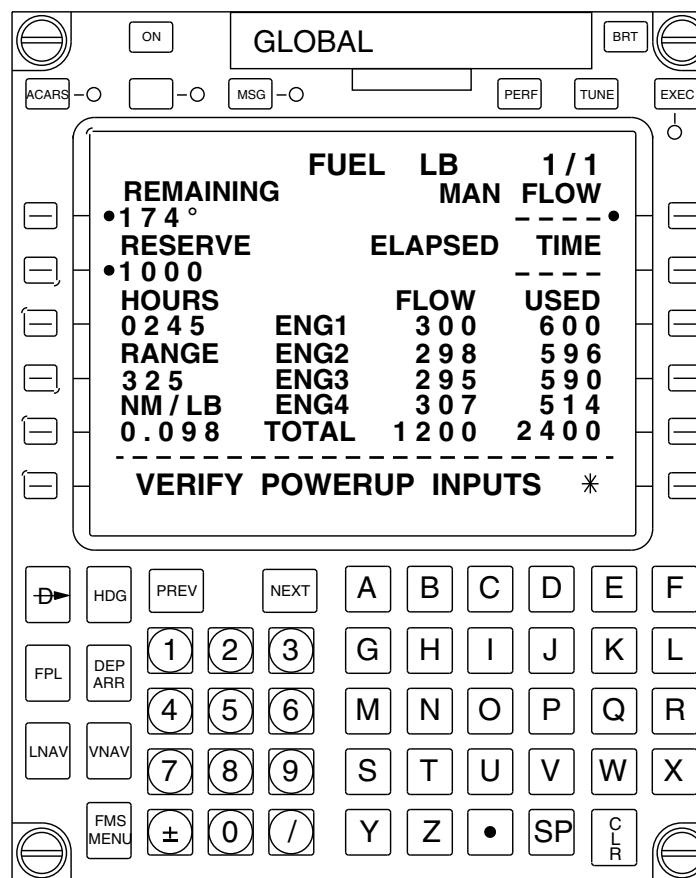


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a85.cgm

FUEL LB 1/1 PAGE
Figure 109

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

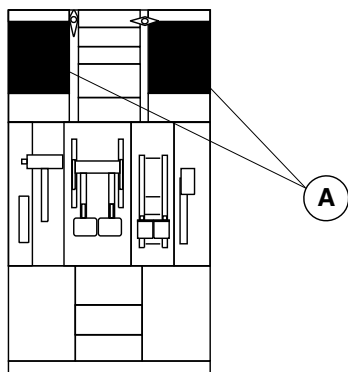
34-60-01

Config 001
Page 209
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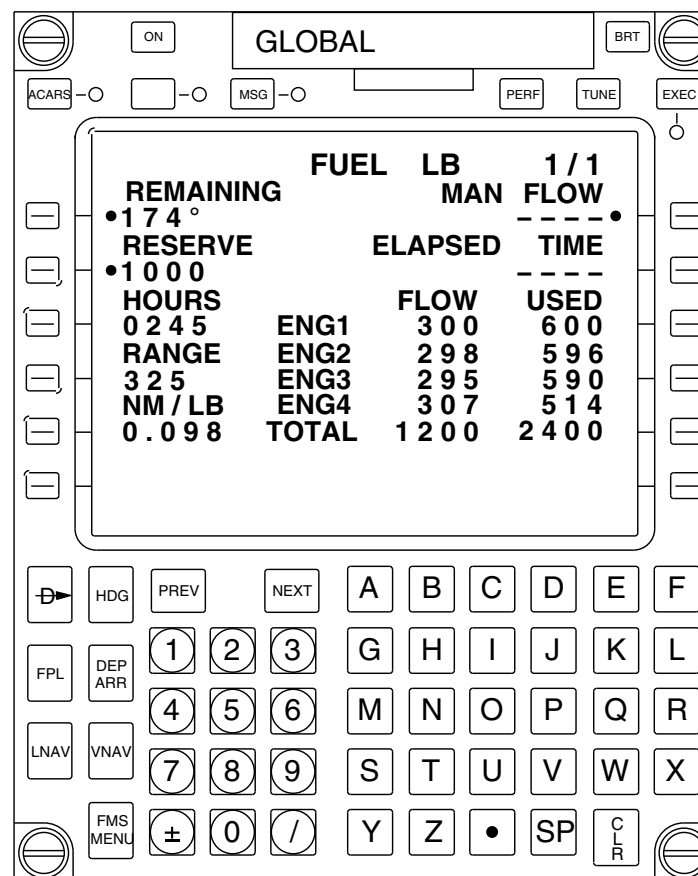


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a86.cgm

FUEL LB 1/1 PAGE VERIFY POWER INPUTS
Figure 110

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

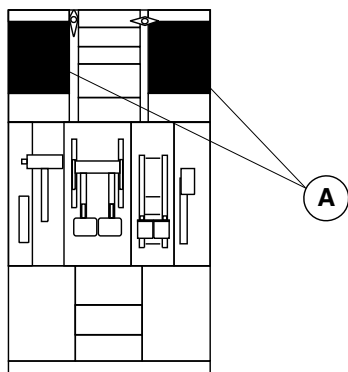
34-60-01

Config 001
Page 210
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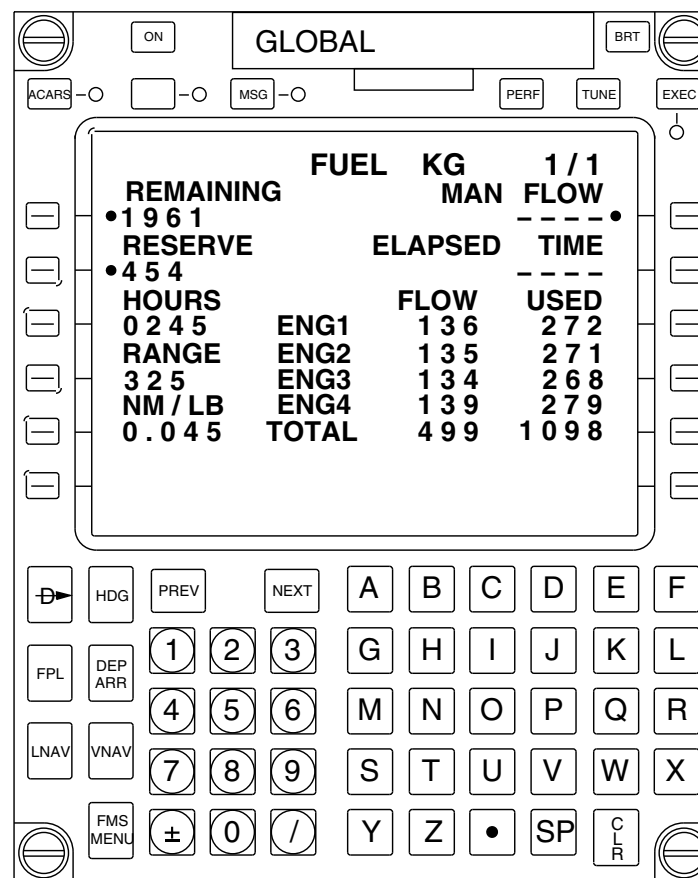


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a87.cgm

FUEL LB 1/1 PAGE MAIN FLOW
Figure 111

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

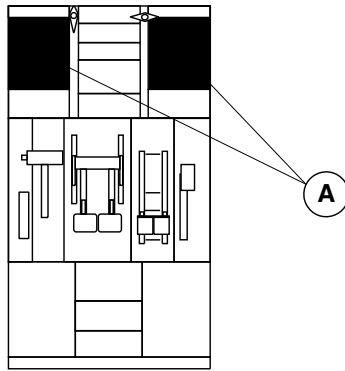
34-60-01

Config 001
Page 211
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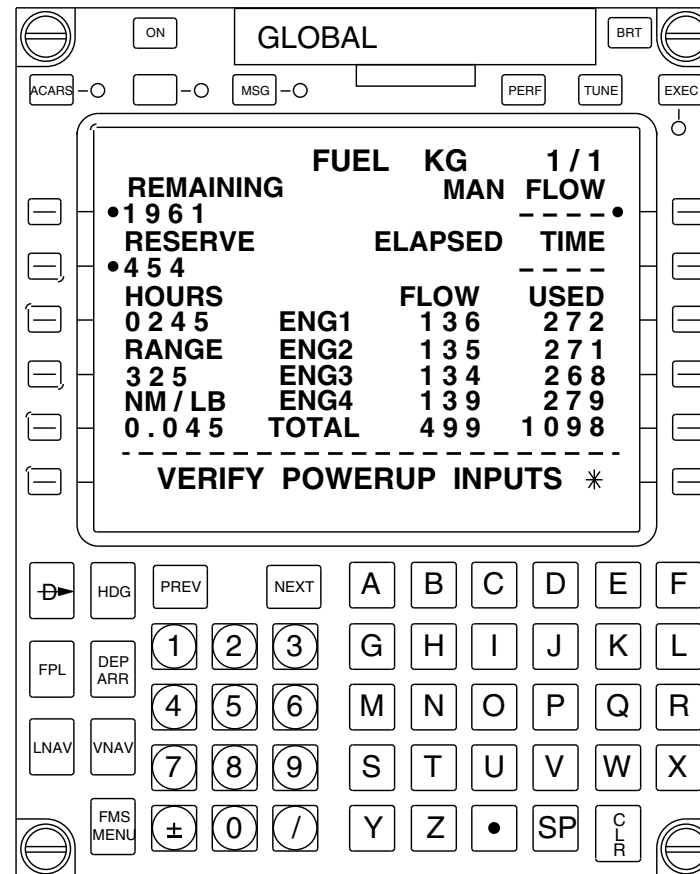


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a88.cgm

FUEL LB 1/1 PAGE ELAPSED TIME
Figure 112

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

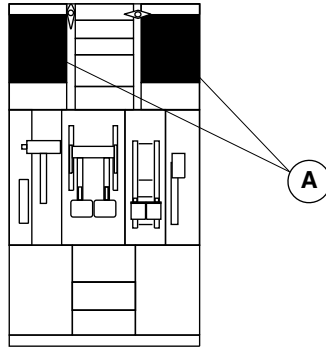
34-60-01

Config 001
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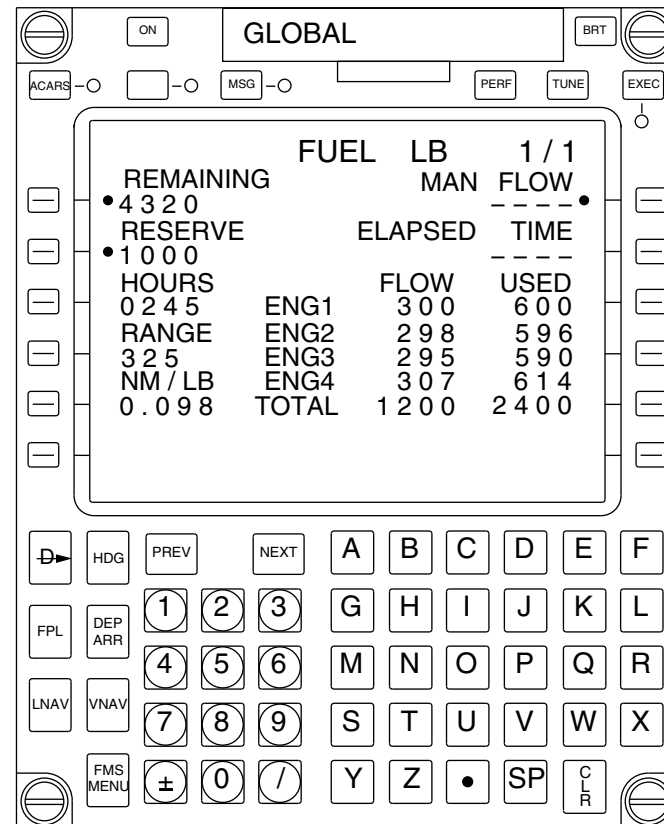


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a89.cgm

FUEL LB 1/1 PAGE TOTAL USED INVAILD SENSOR – YELLOW
Figure 113

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

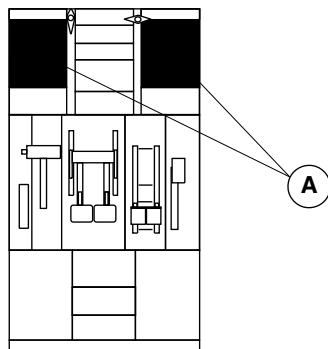
34-60-01

Config 001
Page 213
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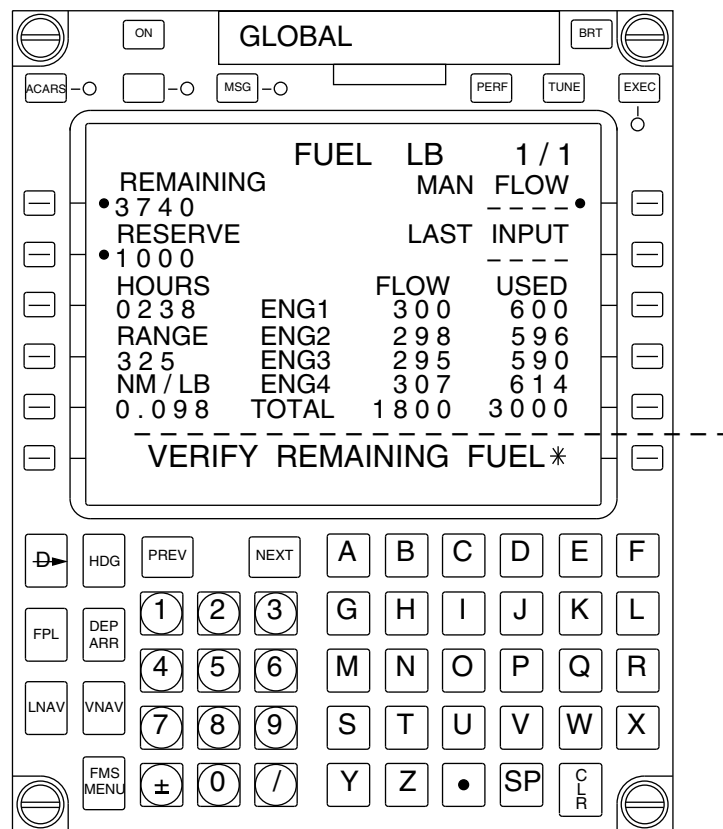


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a90.cgm

FUEL LB 1/1 PAGE TOTAL USED VALID SENSOR – GREEN
Figure 114

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

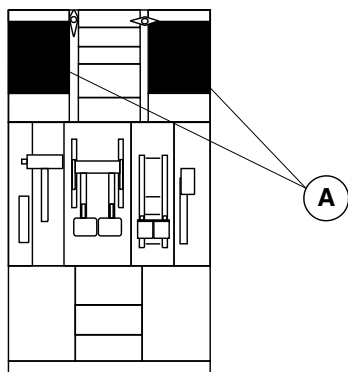
34-60-01

Config 001
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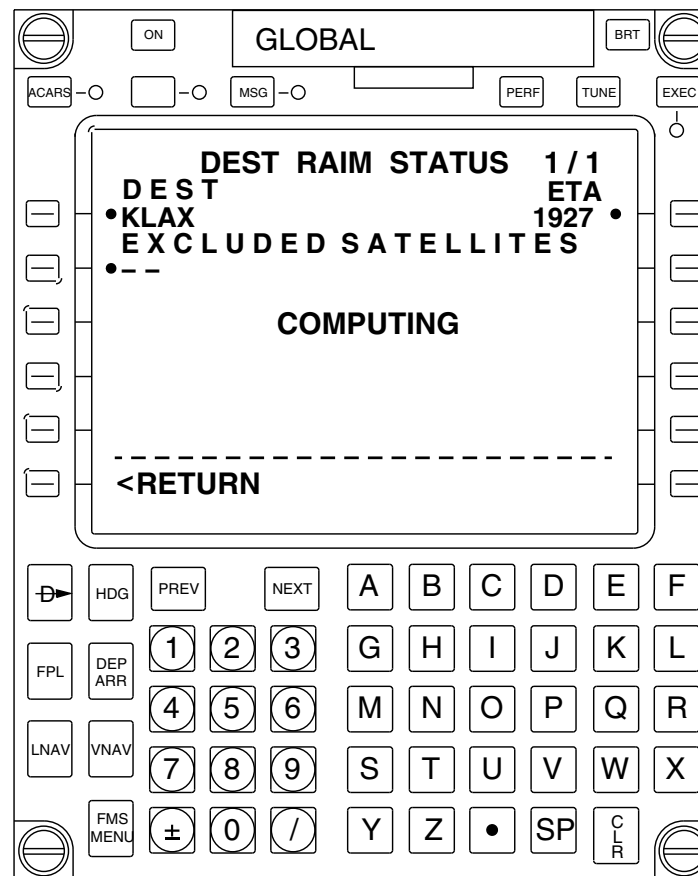


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a91.cgm

DEST RAIM STATUS 1/1 PAGE COMPUTING
Figure 115

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

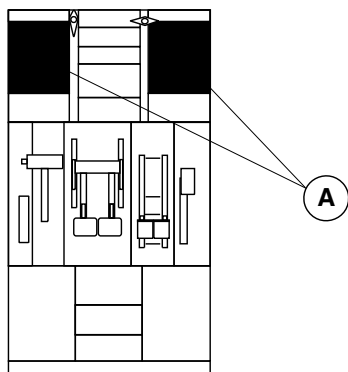
34-60-01

Config 001
Page 215
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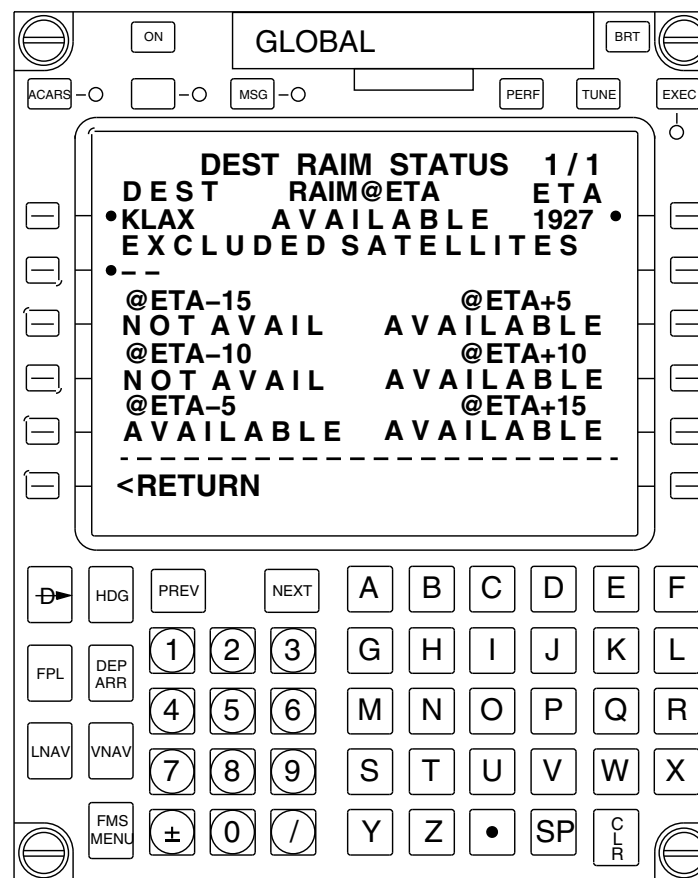


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a92.cgm

DEST RAIM STATUS 1/1 PAGE AVAILABLE
Figure 116

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

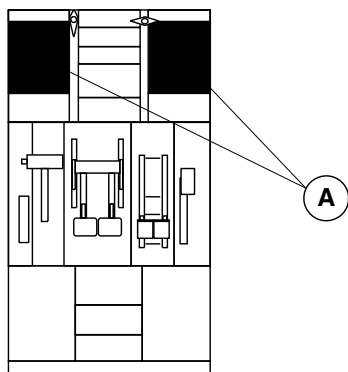
34-60-01

Config 001
Page 216
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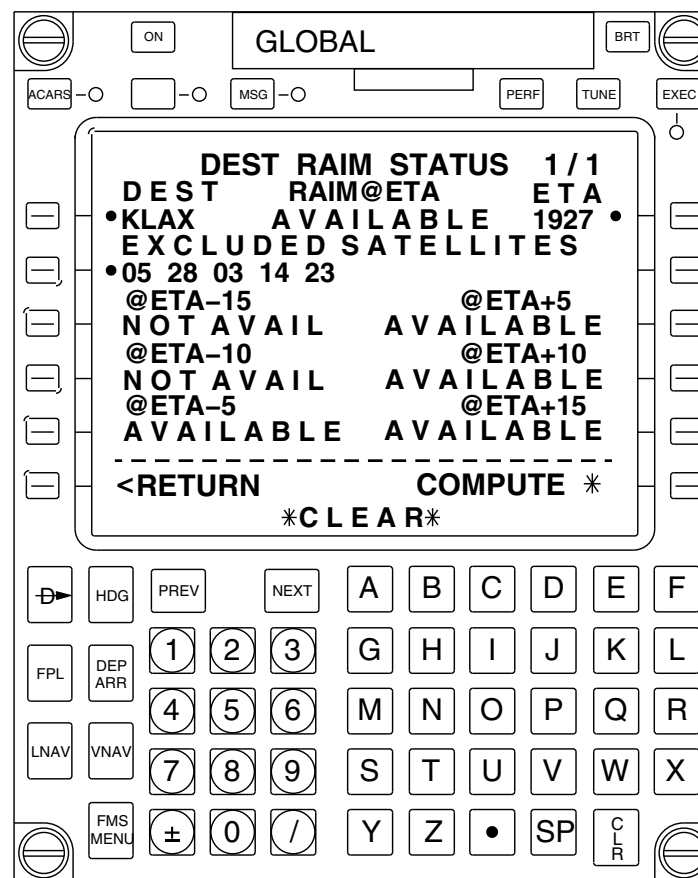


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a93.cgm

DEST RAIM STATUS 1/1 PAGE EXCLUDED SATELLITES
Figure 117

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

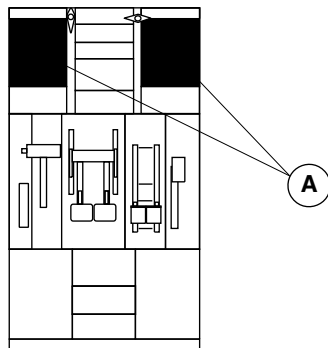
34-60-01

Config 001
Page 217
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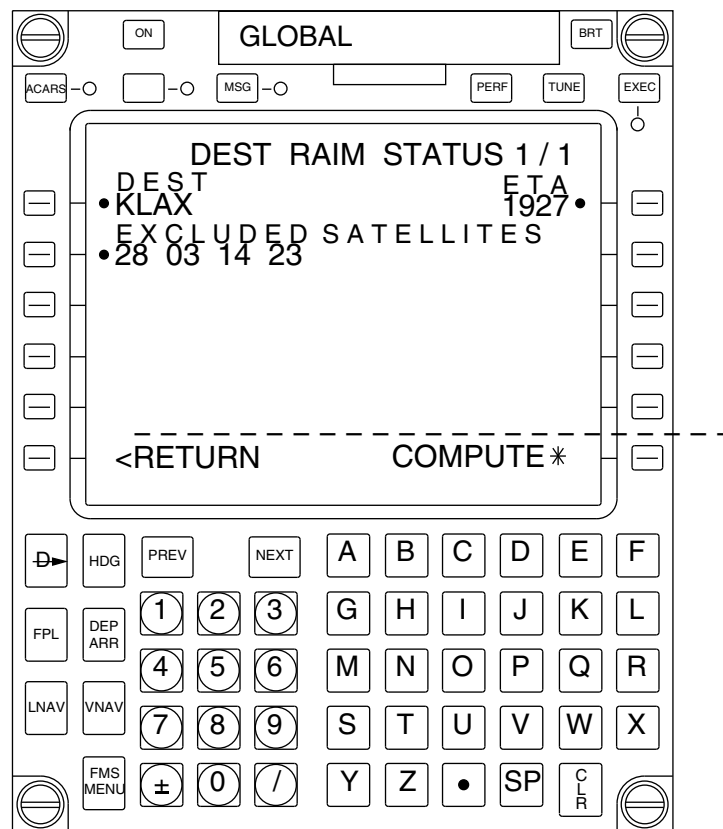


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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a94.cgm

FLIGHT LOG 1/1 PAGE
Figure 118

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

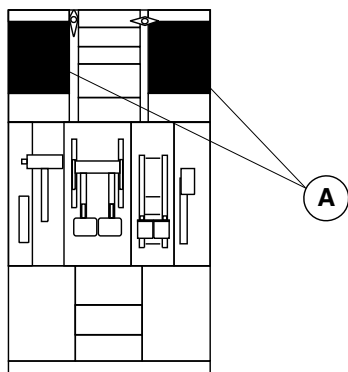
34-60-01

Config 001
Page 218
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DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

ON GLOBAL BRT

ACARS MSG PERF TUNE EXEC

FLIGHT LOG 1/1

DATE	10/JUN/95	UTC	2255
TAKEOFF	2043	FLT TIME	01+35
DIS GND	991	DIS AIR	927
AVG GS	426	AVG TAS	446
FUEL USED	7104 LB	AVG NM/LB	0.097

<RETURN

HDG PREV NEXT A B C D E F

FPL DEP ARR 1 2 3 G H I J K L

LNAV VNAV 4 5 6 M N O P Q R

FMS MENU 7 8 9 S T U V W X

± 0 / Y Z • SP CLR

A

fsq03a96.cgm

FLIGHT LOG 1/1 PAGE FUEL USED
Figure 119

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

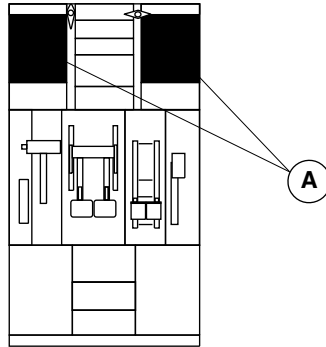
34-60-01

Config 001
Page 219
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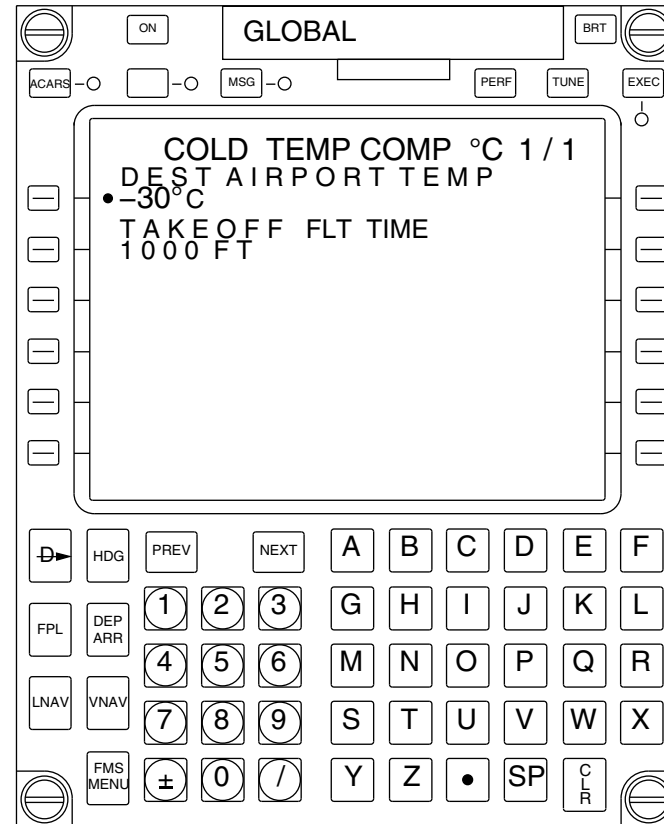


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a97.cgm

COLD TEMP COMP DEGREE CELSIUS 1/1 PAGE
Figure 120

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

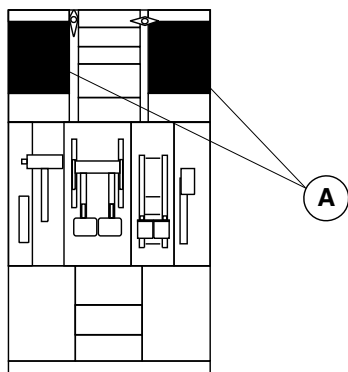
34-60-01

Config 001
Page 220
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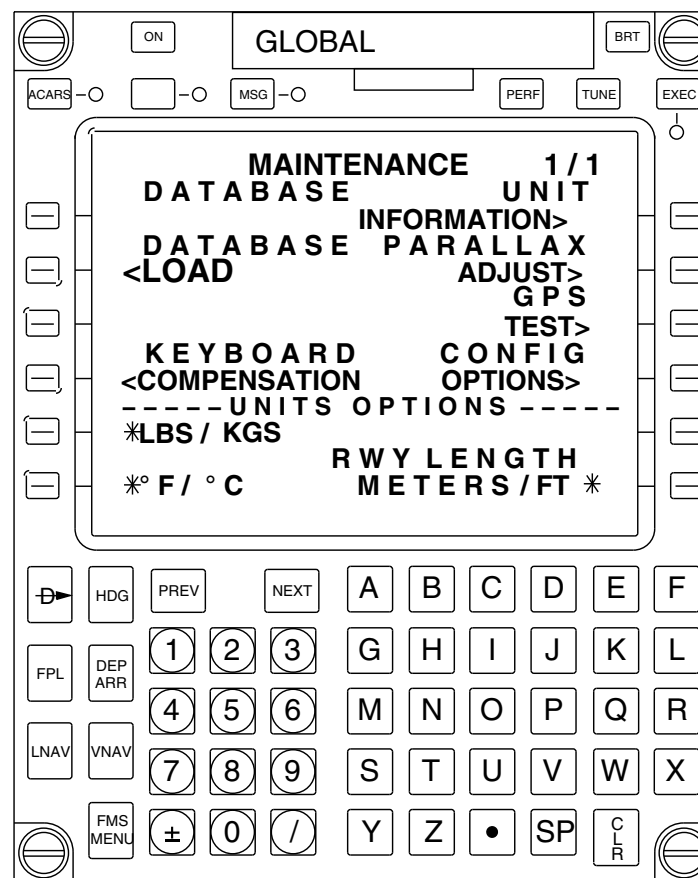


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a98.cgm

MAINTENANCE 1/1 PAGE
Figure 121

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

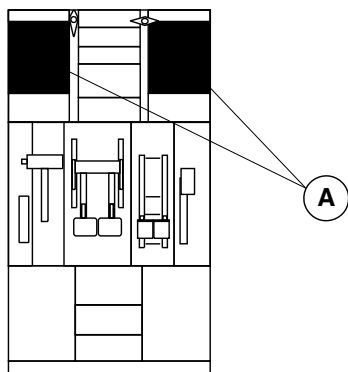
34-60-01

Config 001
Page 221
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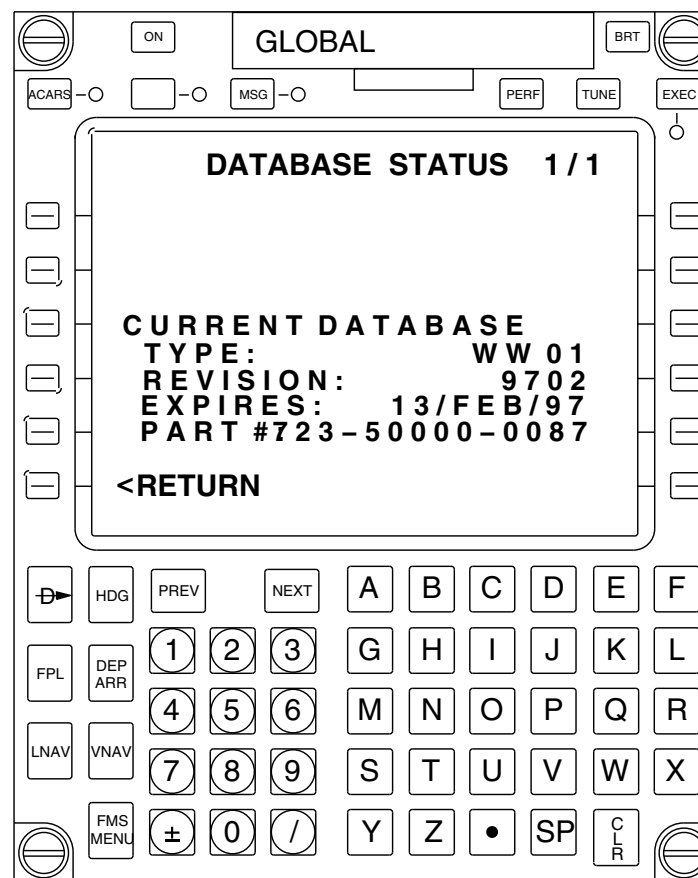


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq03a99.cgm

DATABASE STATUS 1/1 PAGE
Figure 122

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

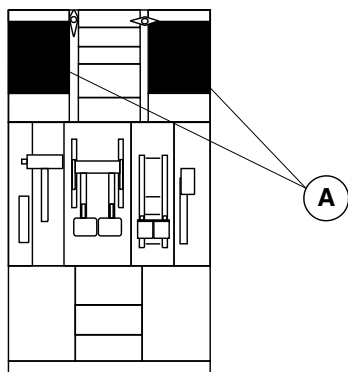
34-60-01

Config 001
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DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

ON GLOBAL BRT

ACARS MSG PERF TUNE EXEC

DATABASE UPDATE 1 / 1

NEW DATABASE
TYPE: WW 01
REVISION: 9703
EXPIRES: 13/MAR/97

CURRENT DATABASE
TYPE: WW 01
REVISION: 9702
EXPIRES: 13/FEB/97

*UPDATE CANCEL *

HDG PREV NEXT A B C D E F

FPL DEP ARR 1 2 3 G H I J K L

LNAV VNAV 4 5 6 M N O P Q R

FMS MENU 7 8 9 S T U V W X

± 0 / Y Z • SP CLR

A

fsq68a01.cgm

DATABASE UPDATE 1/1 PAGE
Figure 123

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

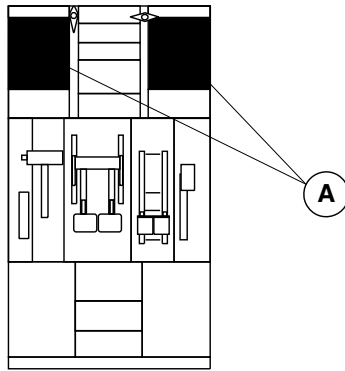
34-60-01

Config 001
Page 223
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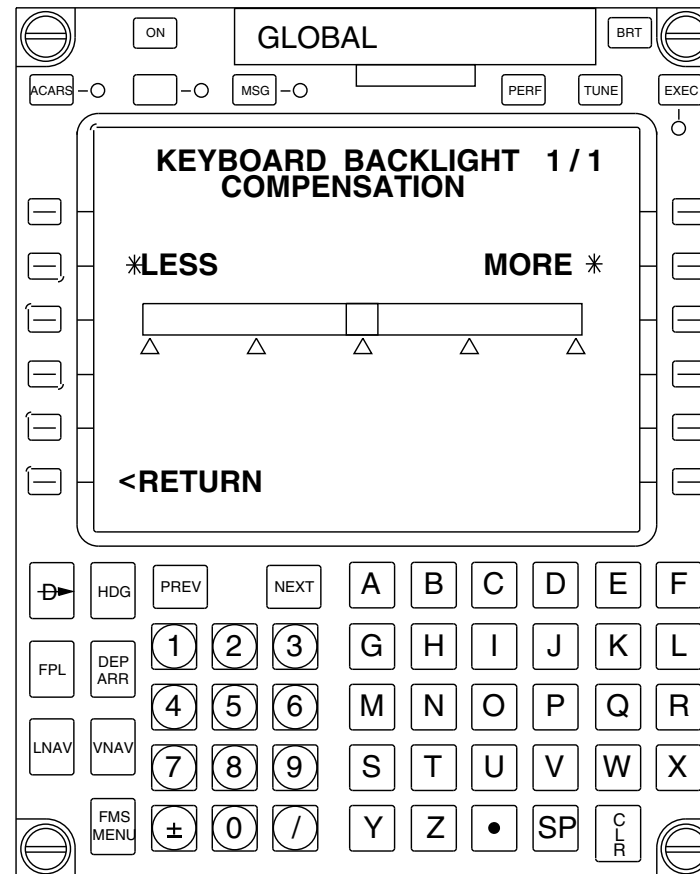


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a02.cgm

KEYBOARD BACKLIGHT COMPENSATION 1/1 PAGE
Figure 124

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

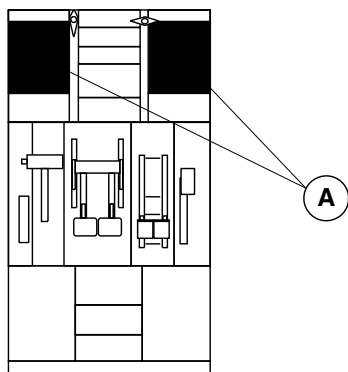
34-60-01

Config 001
Page 224
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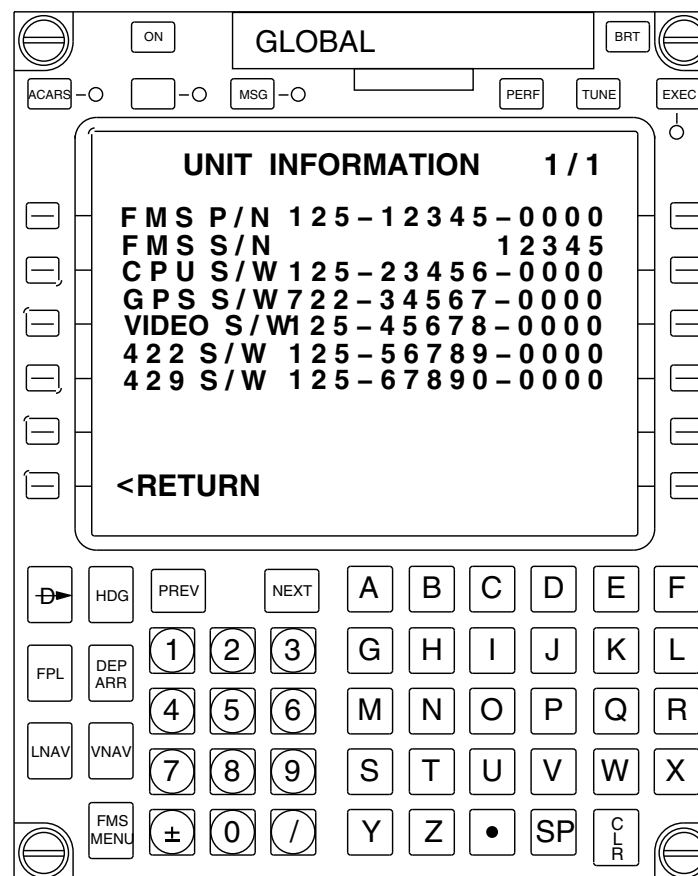


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a03.cgm

UNIT INFORMATION 1/1 PAGE
Figure 125

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

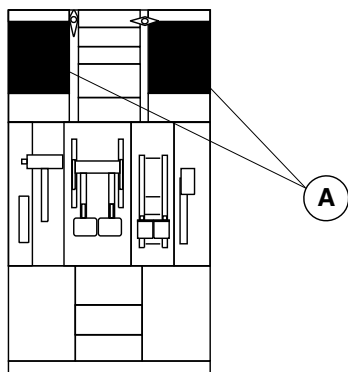
34-60-01

Config 001
Page 225
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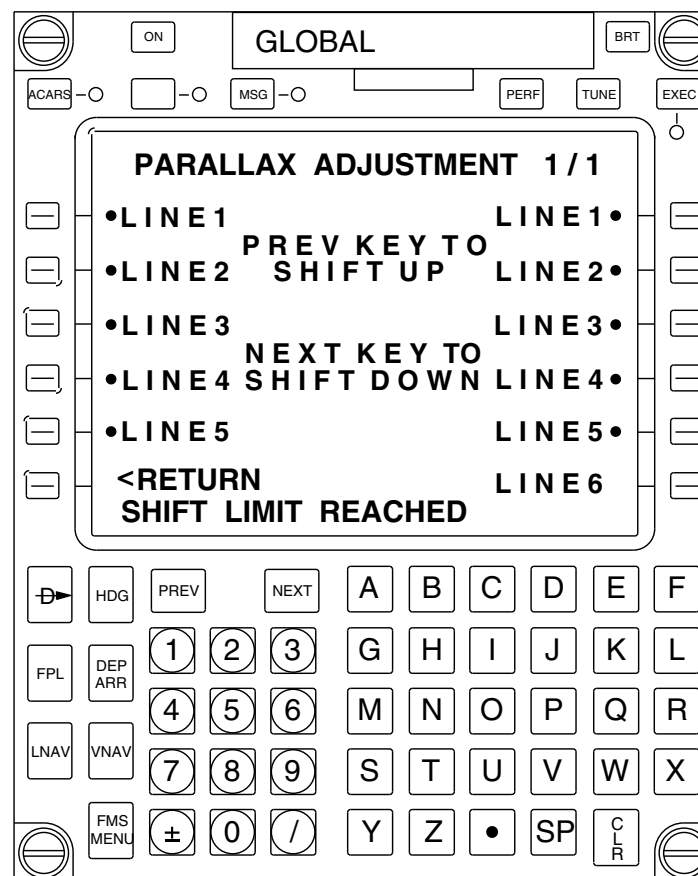


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

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CENTRE CONSOLE



A

fsq68a04.cgm

PARALLAX ADJUST 1/1 PAGE
Figure 126

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

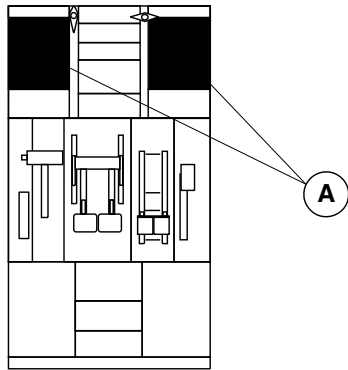
34-60-01

Config 001
Page 226
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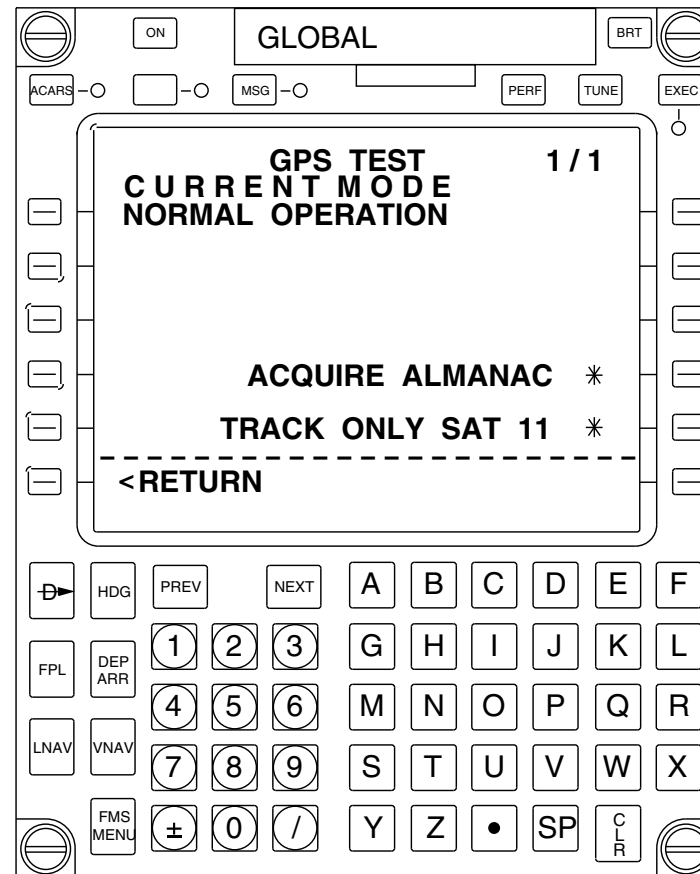


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a05.cgm

GPS TEST 1/1 PAGE
Figure 127

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

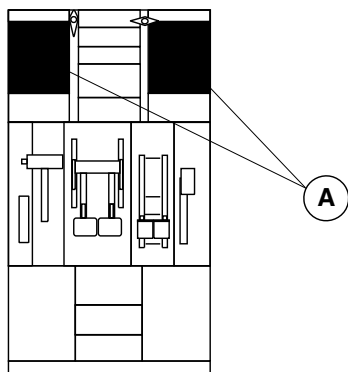
34-60-01

Config 001
Page 227
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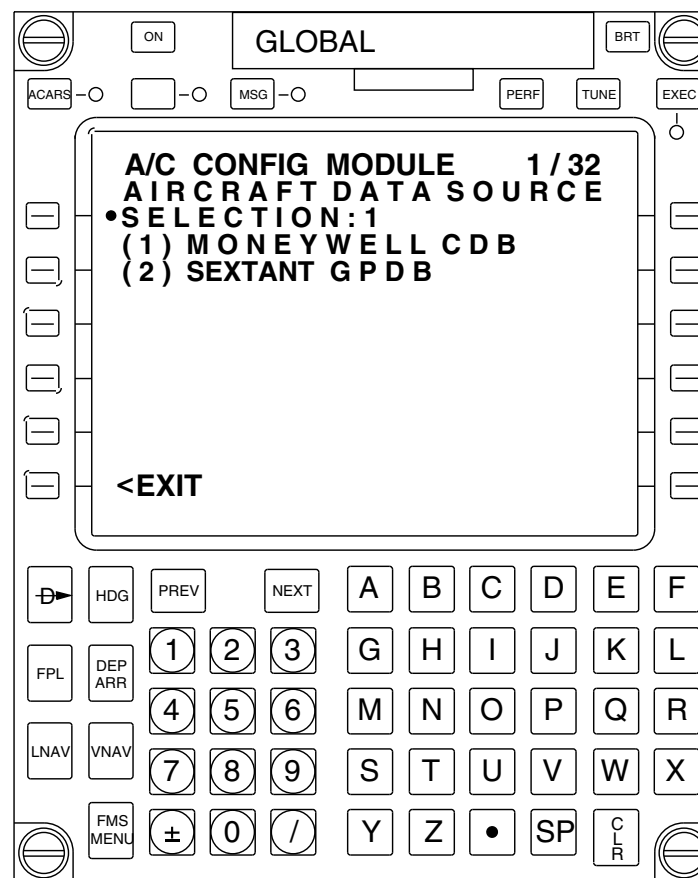


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a06.cgm

A/C CONFIG MODULE 1/32 PAGE
Figure 128

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

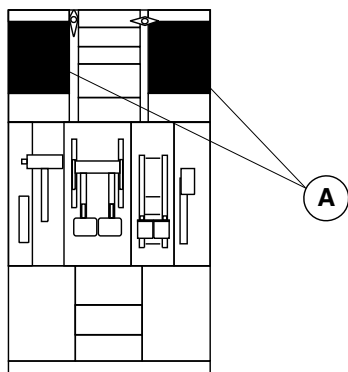
34-60-01

Config 001
Page 228
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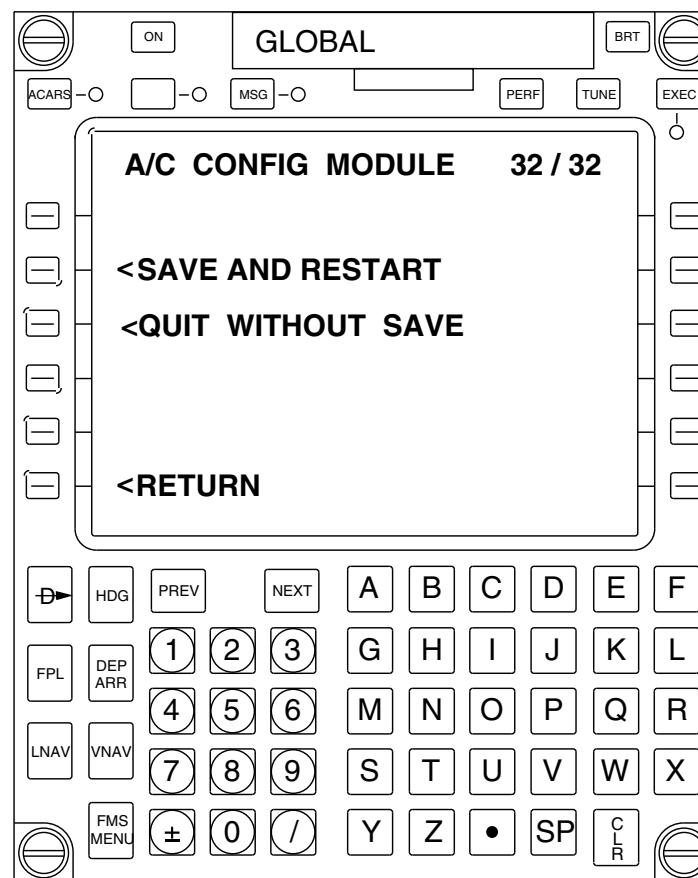


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a07.cgm

A/C CONFIG MODULE 32/32 PAGE
Figure 129

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

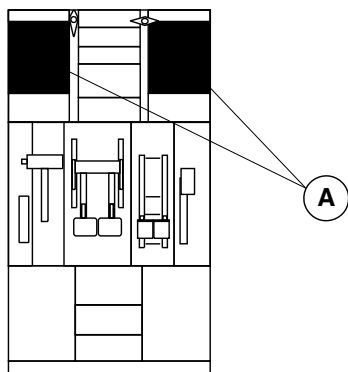
34-60-01

Config 001
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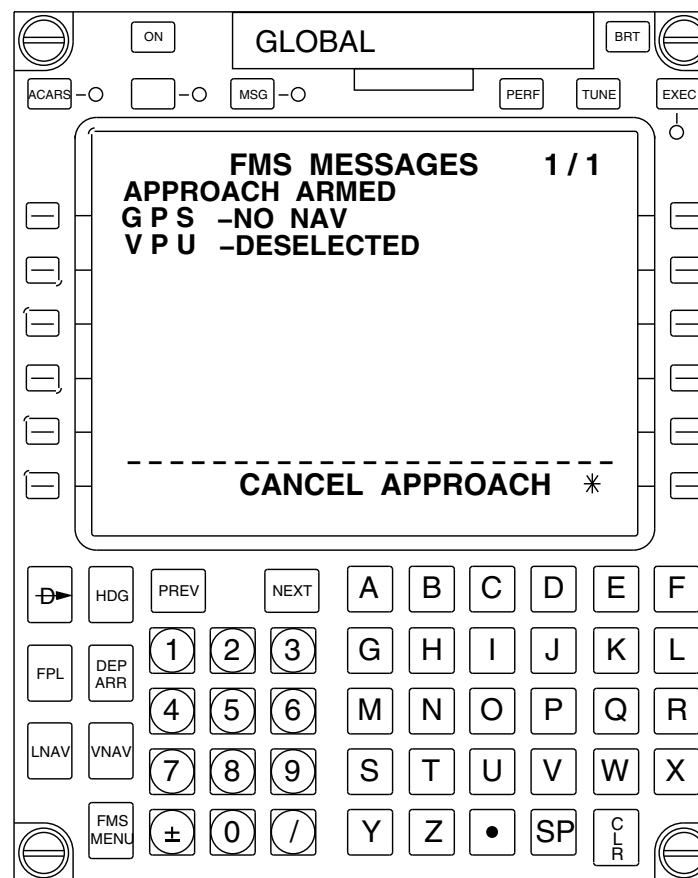


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a08.cgm

FMS MESSAGES 1/1 PAGE
Figure 130

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

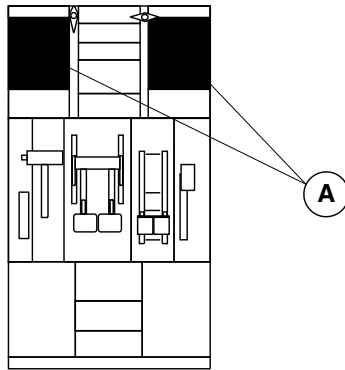
34-60-01

Config 001
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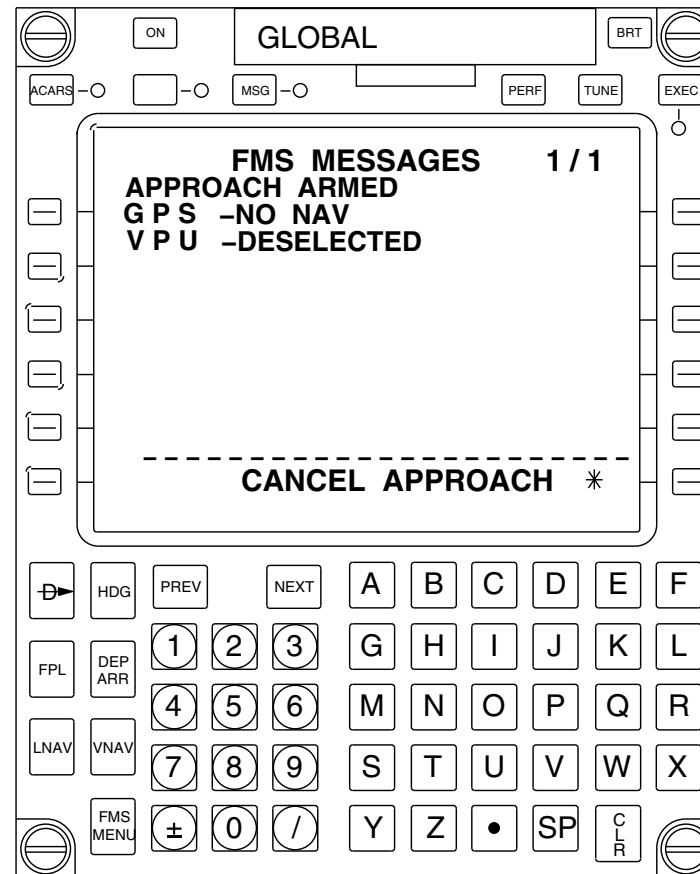


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a09.cgm

FMS MESSAGES 1/1 PAGE APPROACH ARMED
Figure 131

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

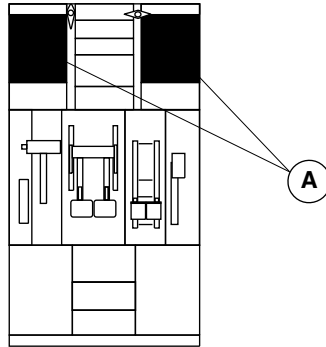
34-60-01

Config 001
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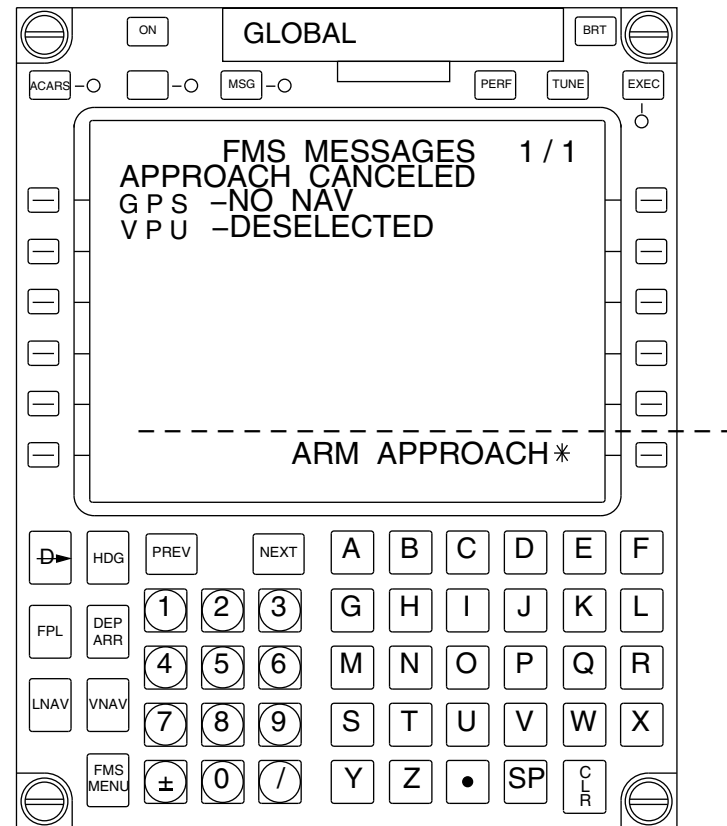


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a10.cgm

FMS MESSAGES 1/1 PAGE APPROACH CANCELLED
Figure 132

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

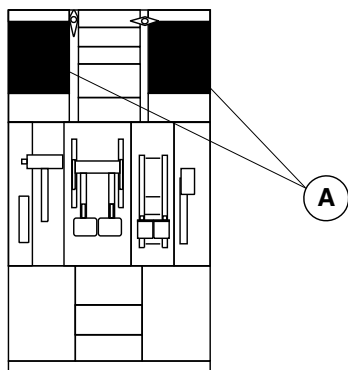
34-60-01

Config 001
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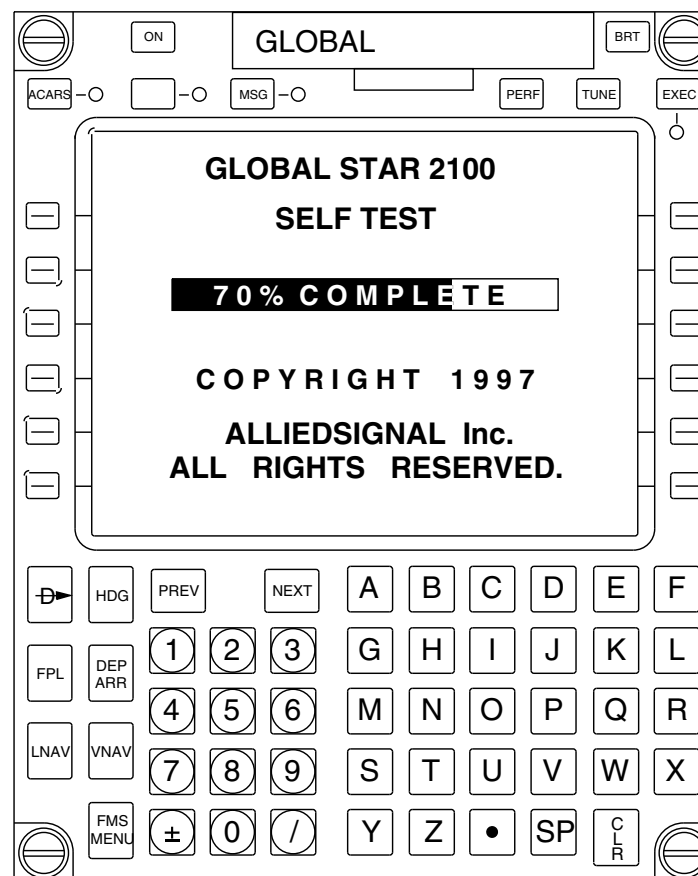


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a11.cgm

FMS POWER ON/INITIALIZATION
Figure 133

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

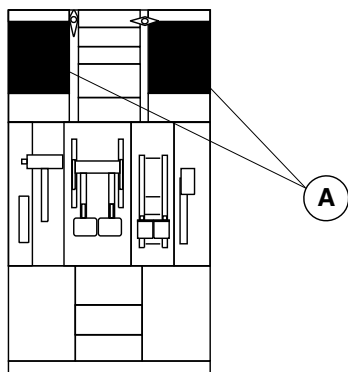
34-60-01

Config 001
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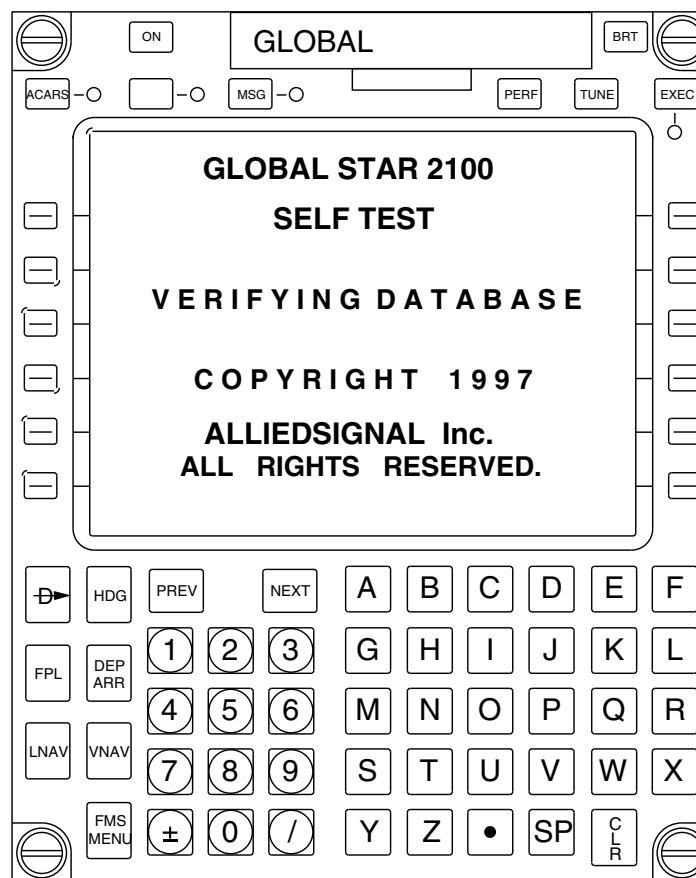


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a12.cgm

SELF TEST
Figure 134

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

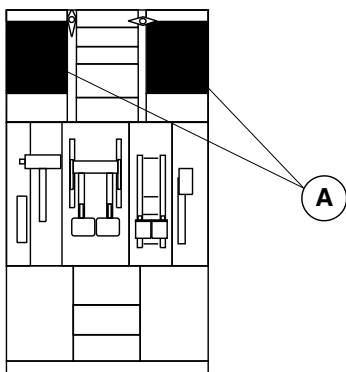
34-60-01

Config 001
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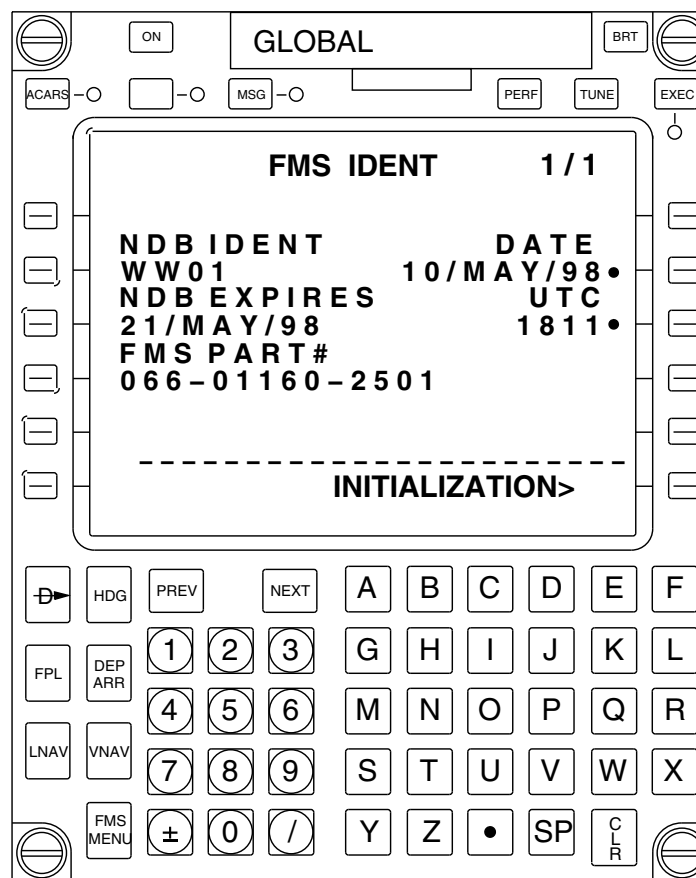


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a13.cgm

FMS IDENT
Figure 135

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

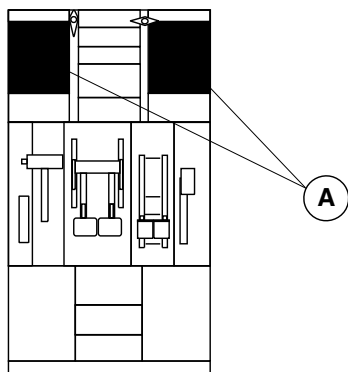
34-60-01

Config 001
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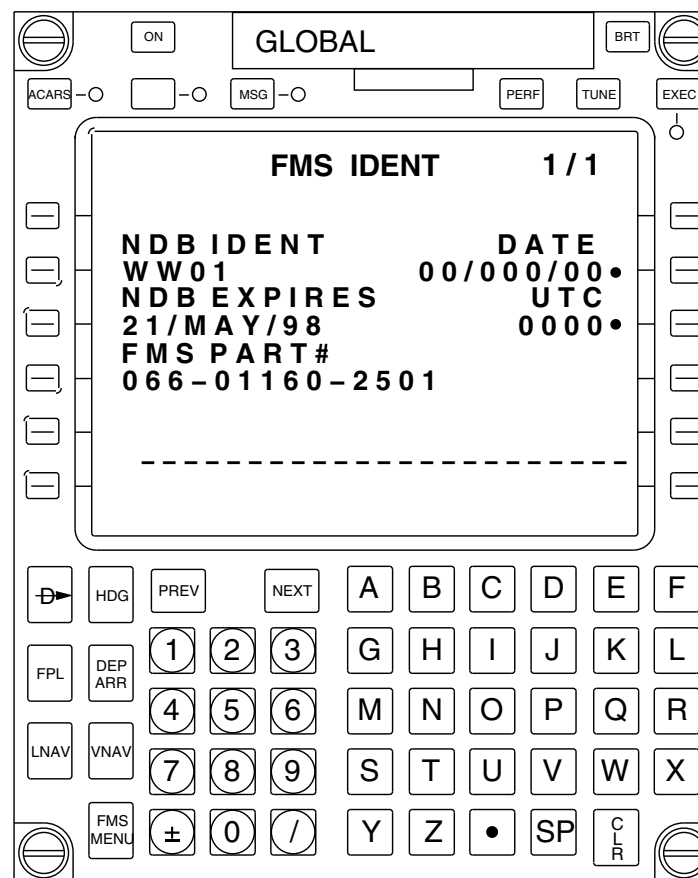


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a14.cgm

FMS IDENT NO DATA
Figure 136

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

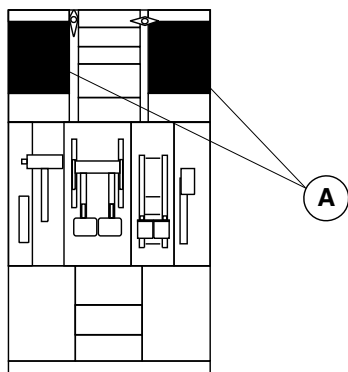
34-60-01

Config 001
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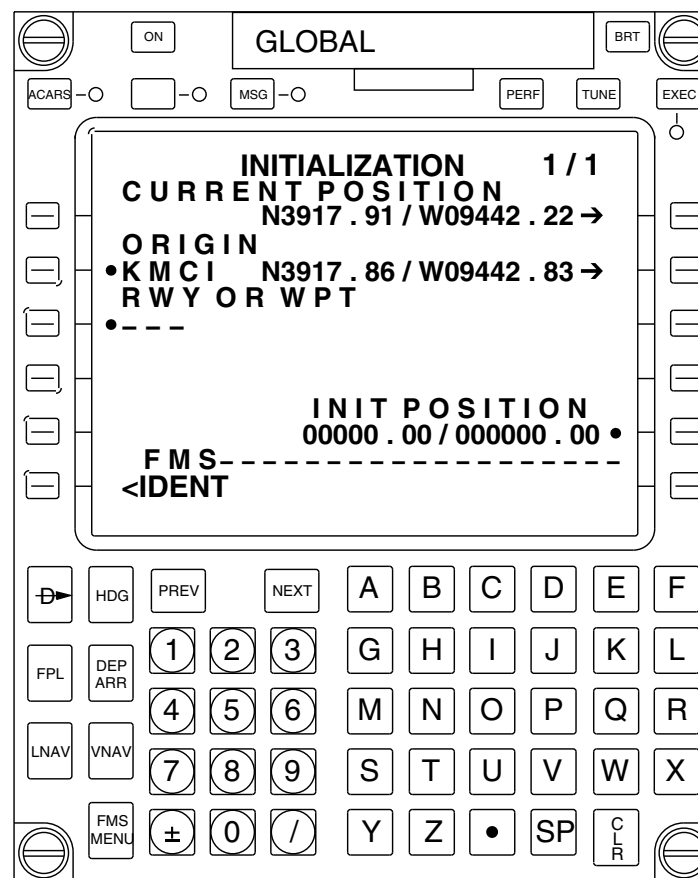


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a15.cgm

INITIALIZATION 1/1 PAGE
Figure 137

PSM 1-84-2A
EFFECTIVITY:
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Config 001

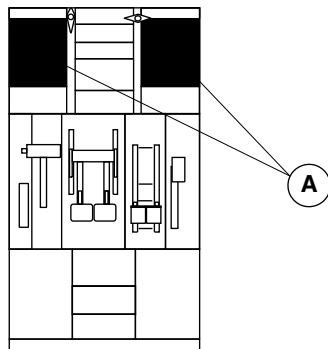
34-60-01

Config 001
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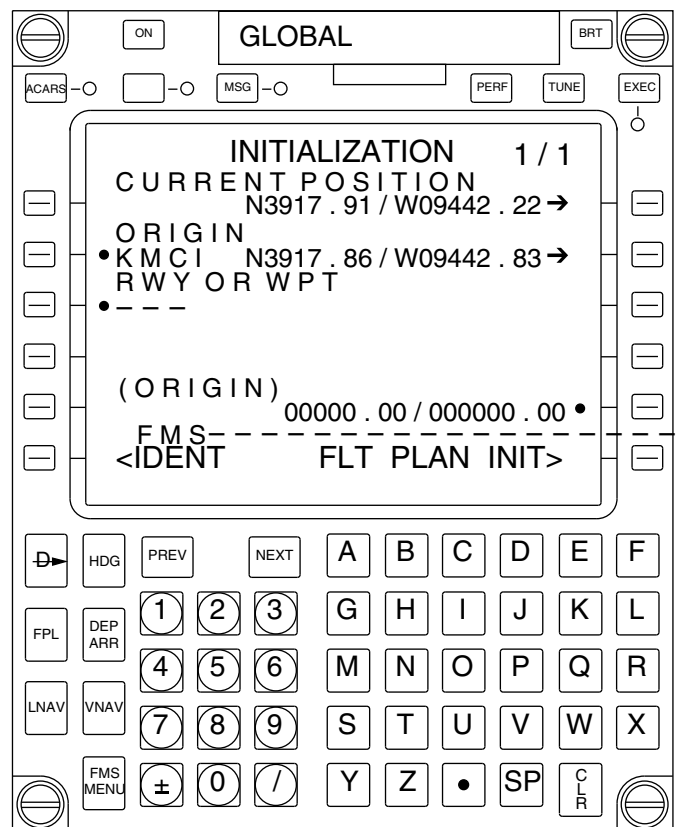


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a16.cgm

INITIALIZATION 1/1 PAGE PILOT-DEFINED FLIGHT PLANS
Figure 138

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

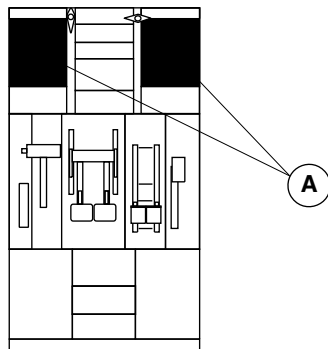
34-60-01

Config 001
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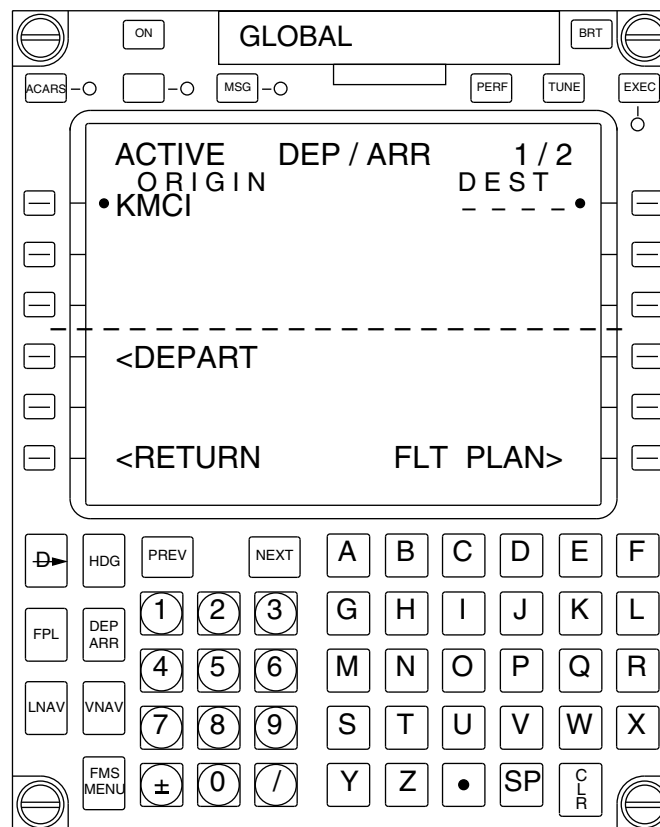


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a17.cgm

DEPART/ARRIVAL – 1/2 PAGE
Figure 139

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

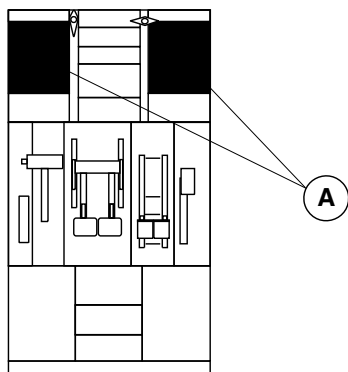
34-60-01

Config 001
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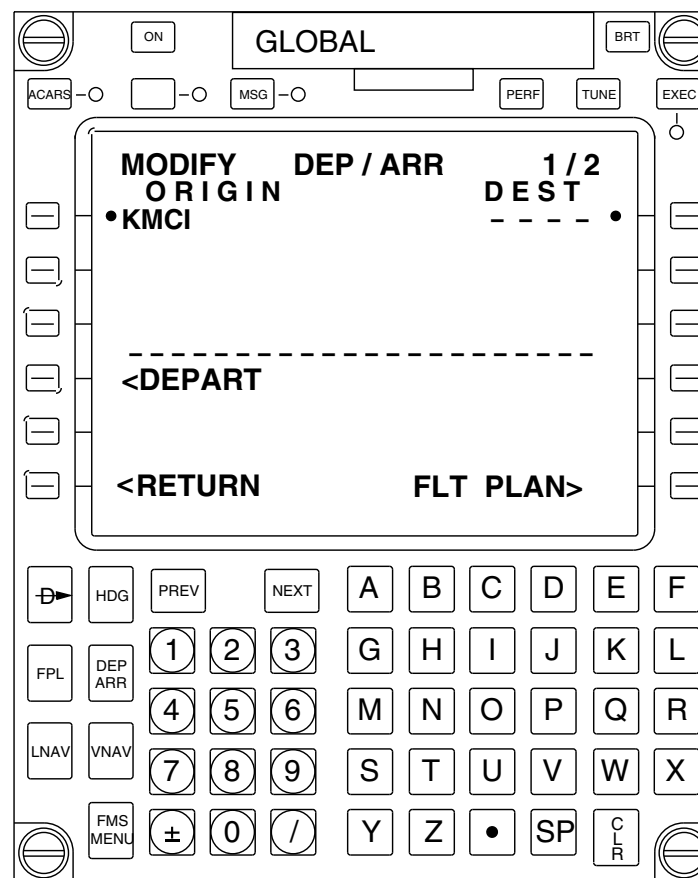


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a18.cgm

DEPART/ARRIVAL 1/2 PAGE MODIFIED FLIGHT PLAN
Figure 140

PSM 1-84-2A
EFFECTIVITY:
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Config 001

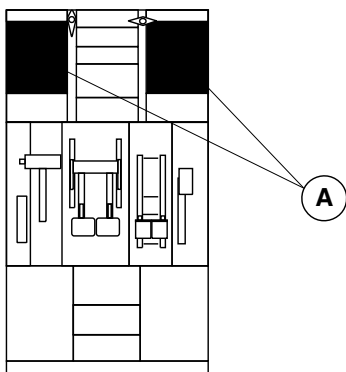
34-60-01

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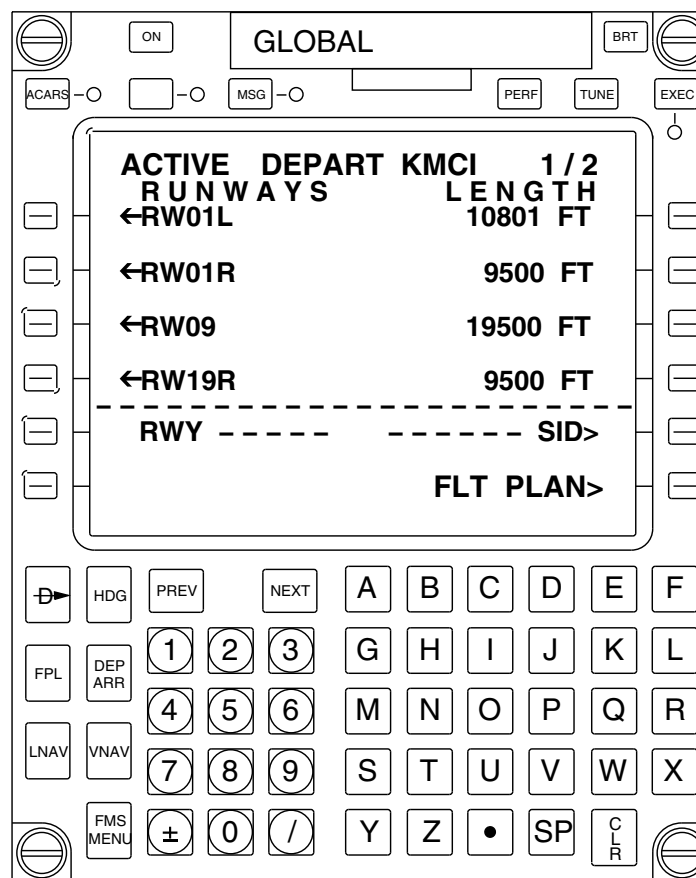


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a19.cgm

DEPART 1/2 PAGE DEPARTURE RUNWAYS
Figure 141

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

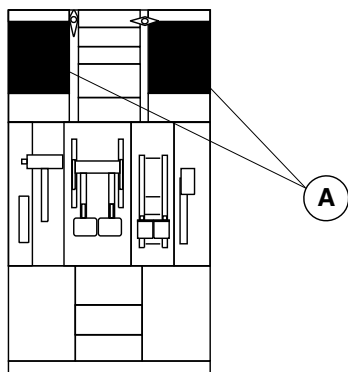
34-60-01

Config 001
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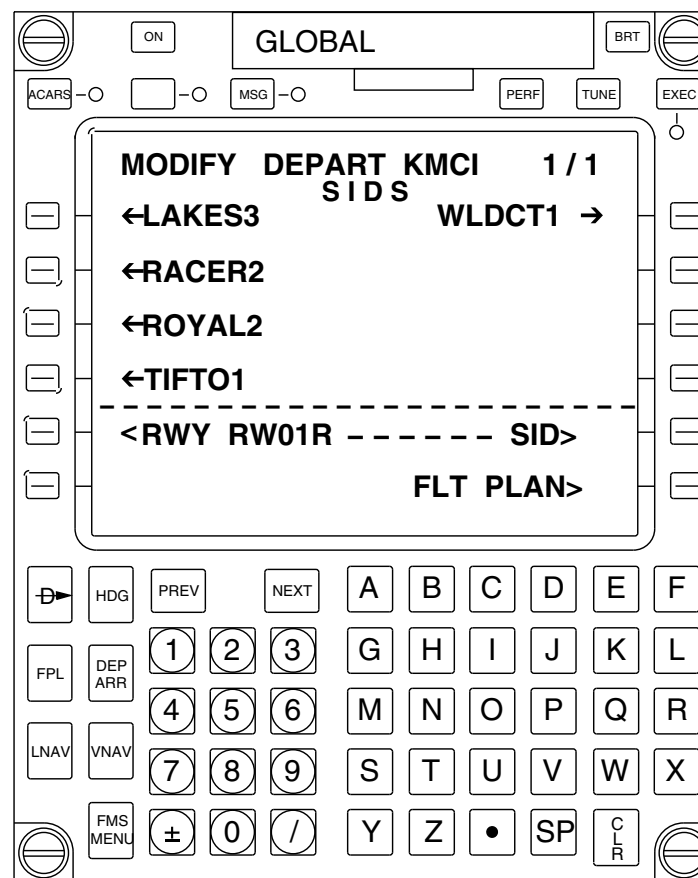


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a20.cgm

DEPART 1/1 PAGE DEPARTURE SIDS
Figure 142

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

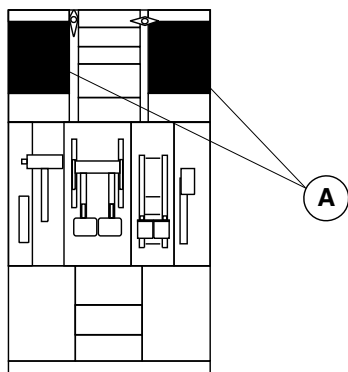
34-60-01

Config 001
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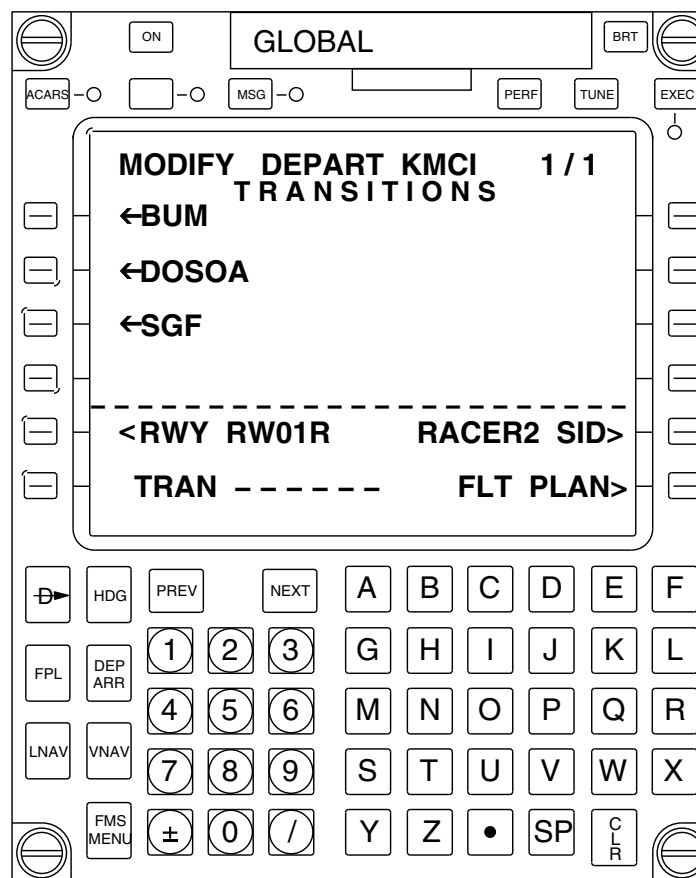


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a21.cgm

DEPART 1/1 PAGE DEPARTURE TRANSITION
Figure 143

PSM 1-84-2A
EFFECTIVITY:
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Config 001

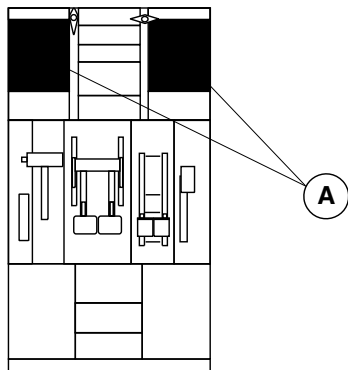
34-60-01

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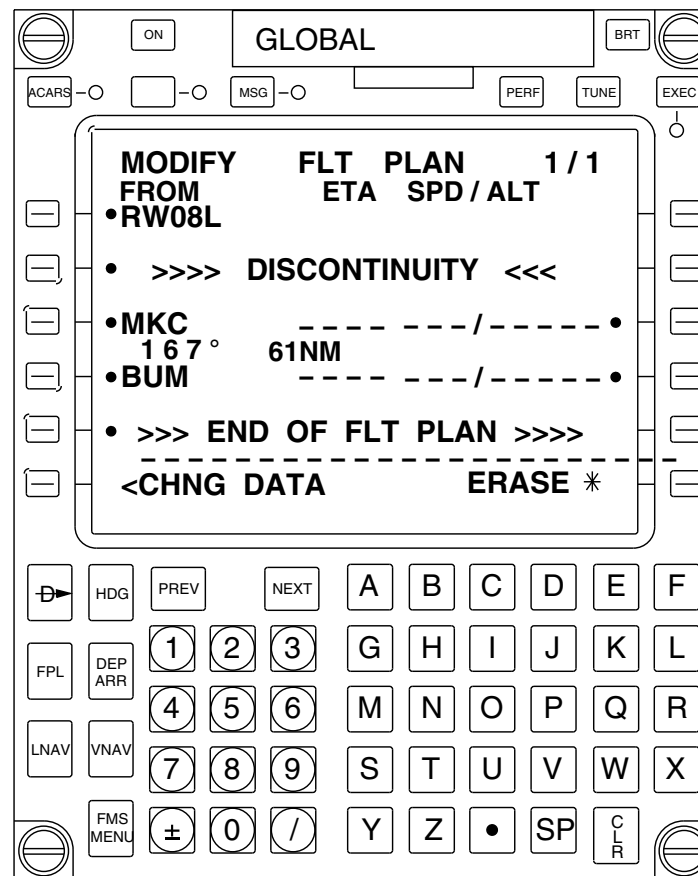


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a22.cgm

FLIGHT PLAN 1/1 PAGE
Figure 144

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

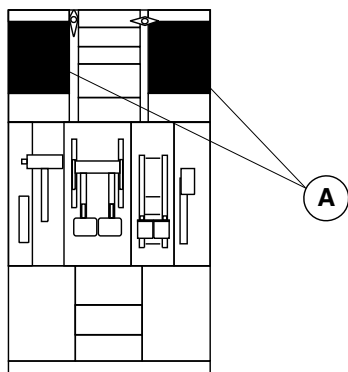
34-60-01

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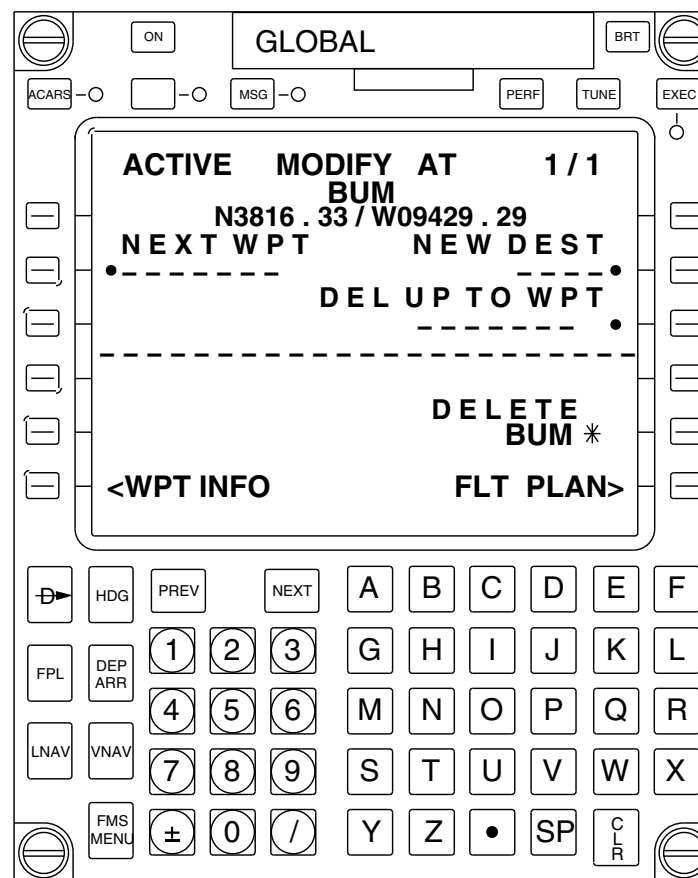


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a23.cgm

LATERAL MODIFY AT BUM 1/1 PAGE
Figure 145

PSM 1-84-2A
EFFECTIVITY:
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Config 001

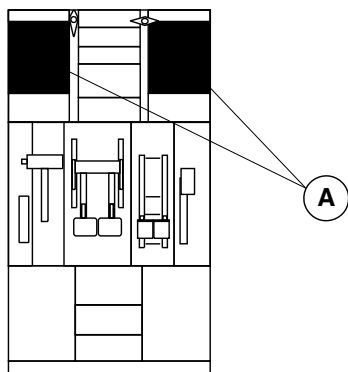
34-60-01

Config 001
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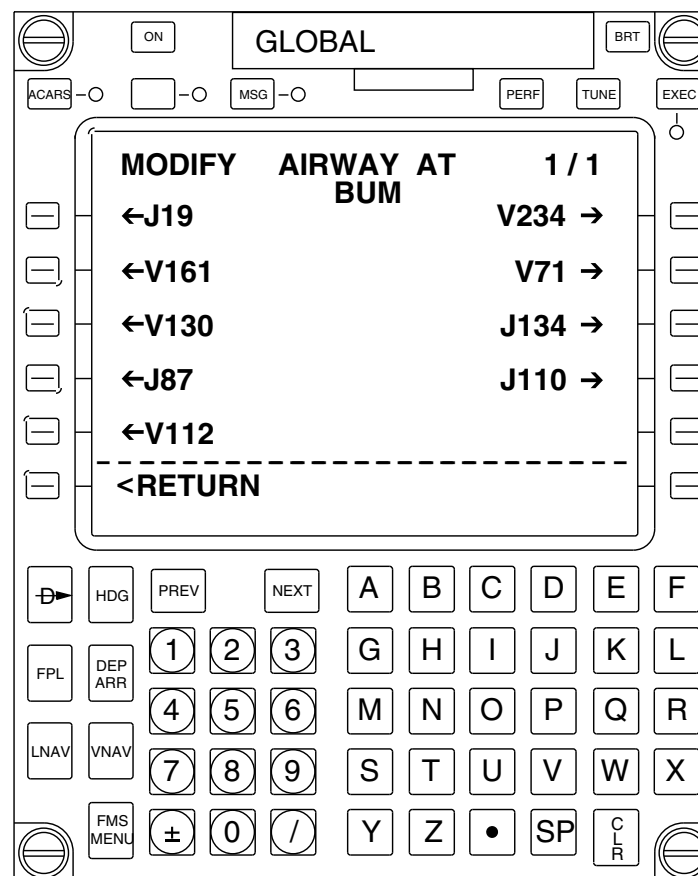


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a24.cgm

AIRWAYS AT BUM 1/1 PAGE
Figure 146

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

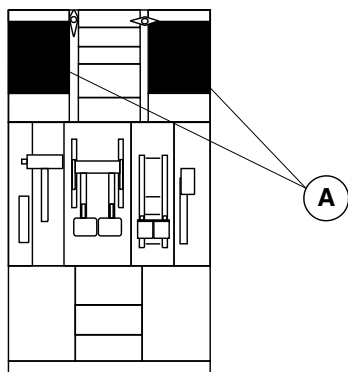
34-60-01

Config 001
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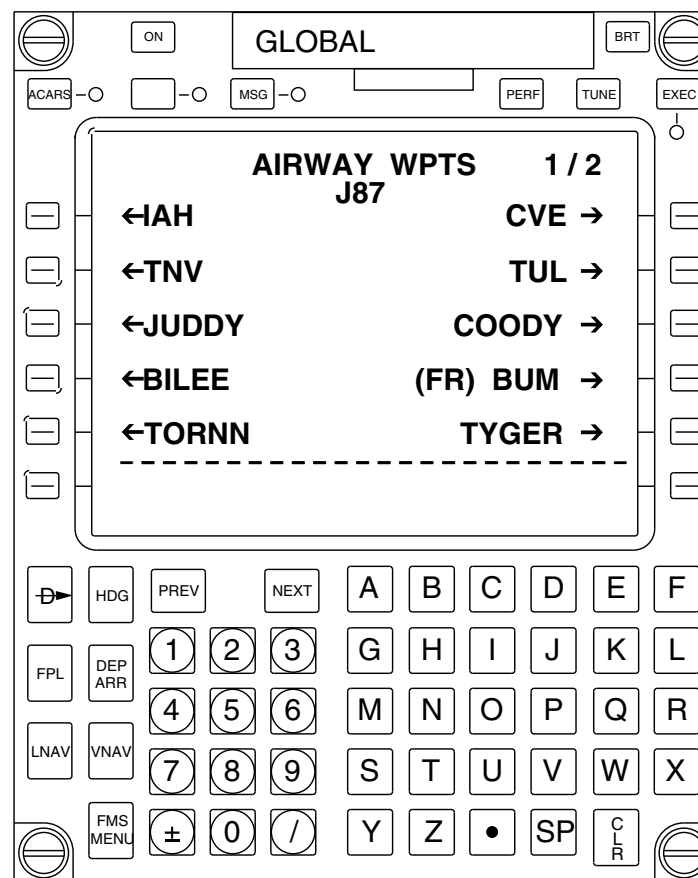


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a25.cgm

LATERAL AIRWAY WPTS 1/1 PAGE
Figure 147

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

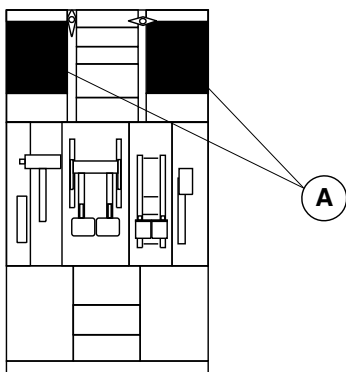
34-60-01

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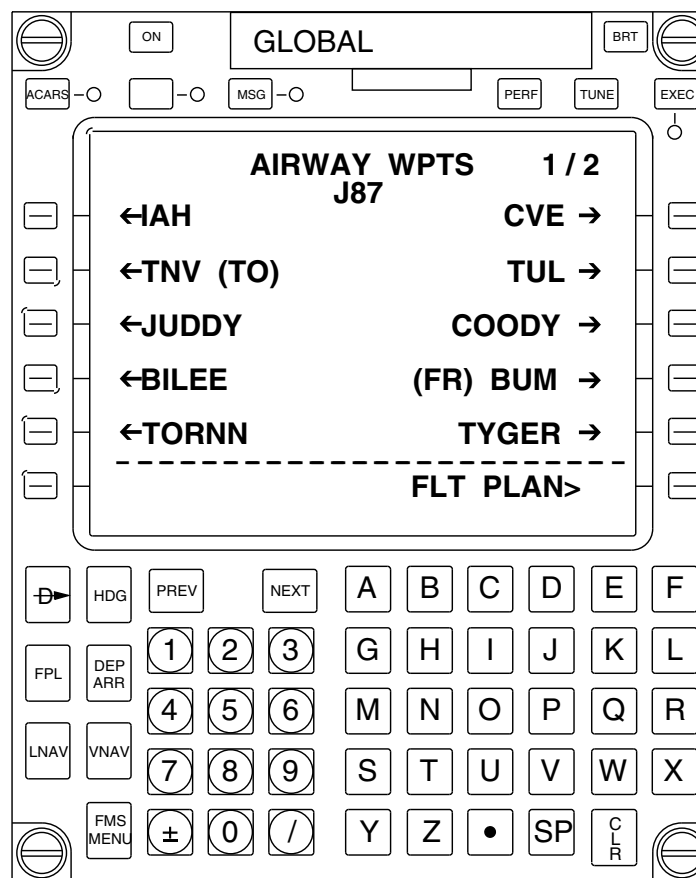


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a26.cgm

AIRWAY WPTS 1/1 PAGE MODIFIED FLIGHT PLAN
Figure 148

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

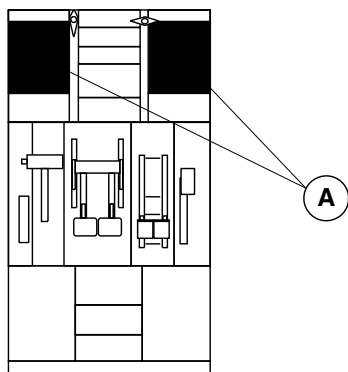
34-60-01

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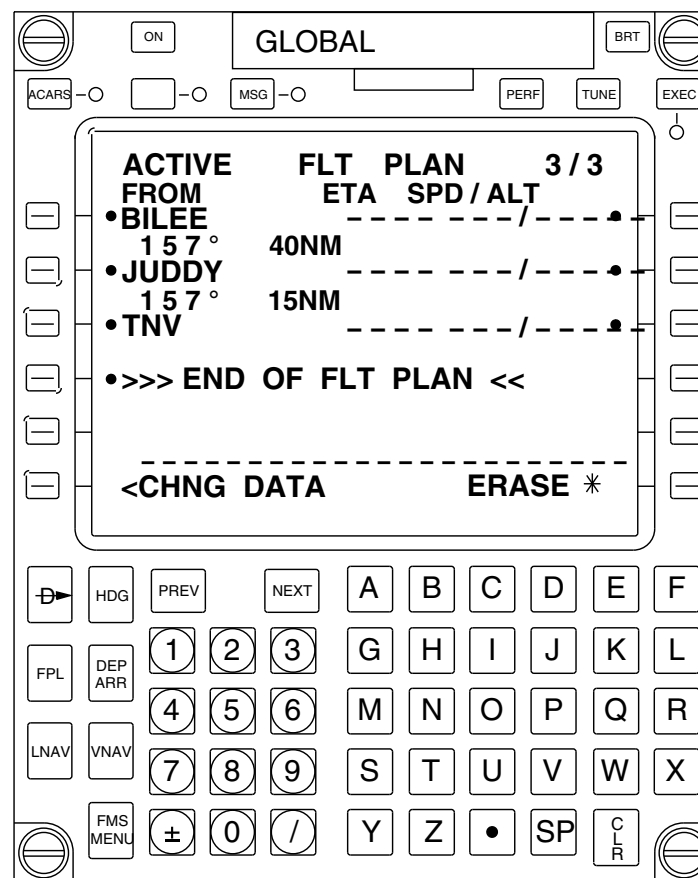


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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a27.cgm

FLIGHT PLAN 3/3 PAGE
Figure 149

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

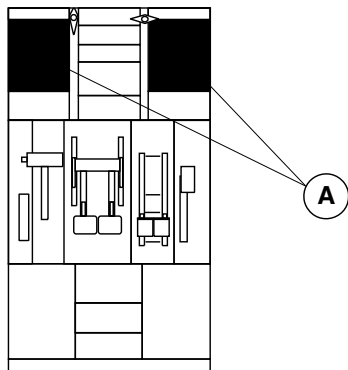
34-60-01

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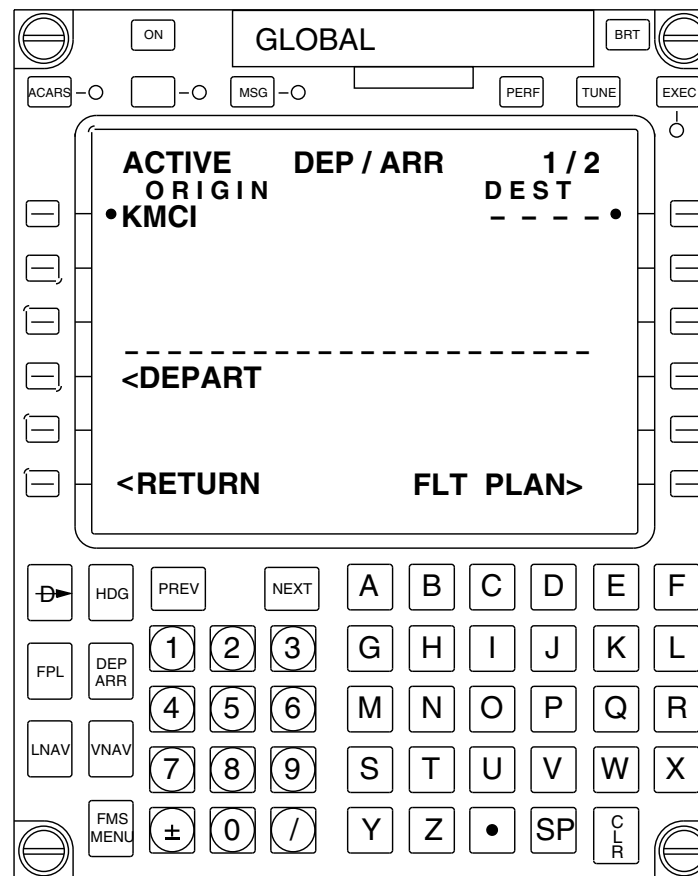


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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ARRIVAL/DEPART 1/2 PAGE
Figure 150

PSM 1-84-2A
EFFECTIVITY:
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Config 001

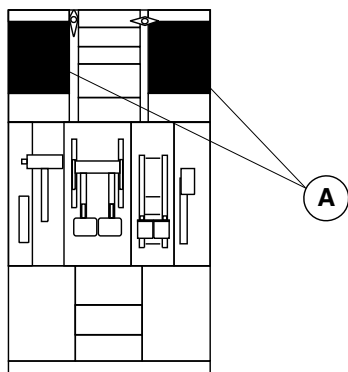
34-60-01

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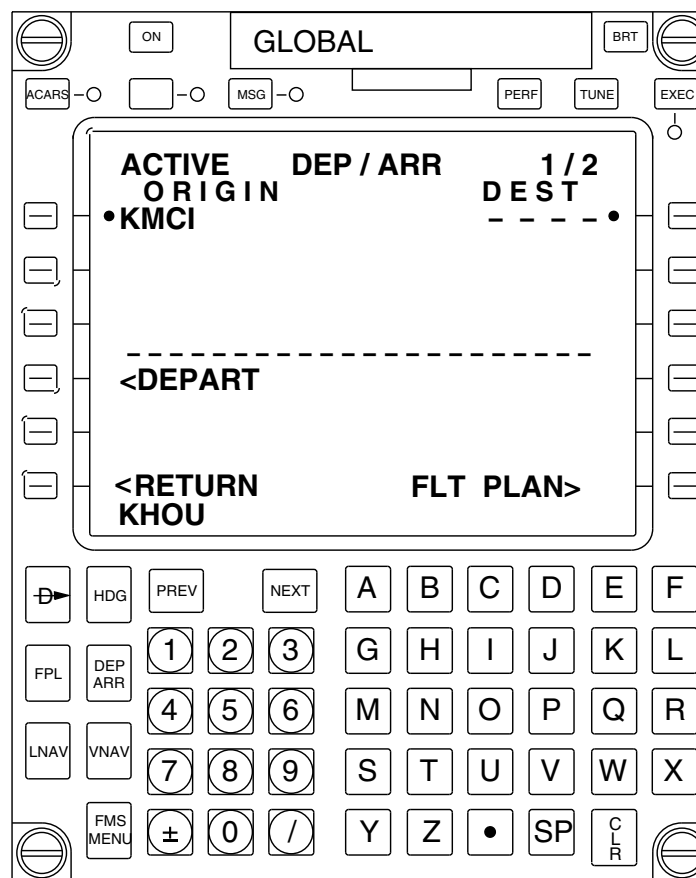


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a29.cgm

ARRIVAL/DEPART 1/2 PAGE (KHO)
Figure 151

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

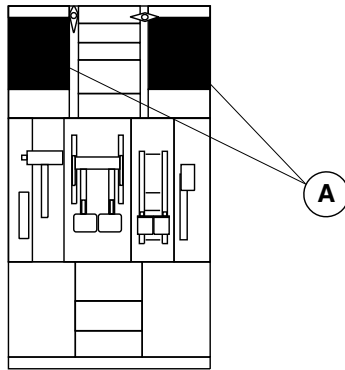
34-60-01

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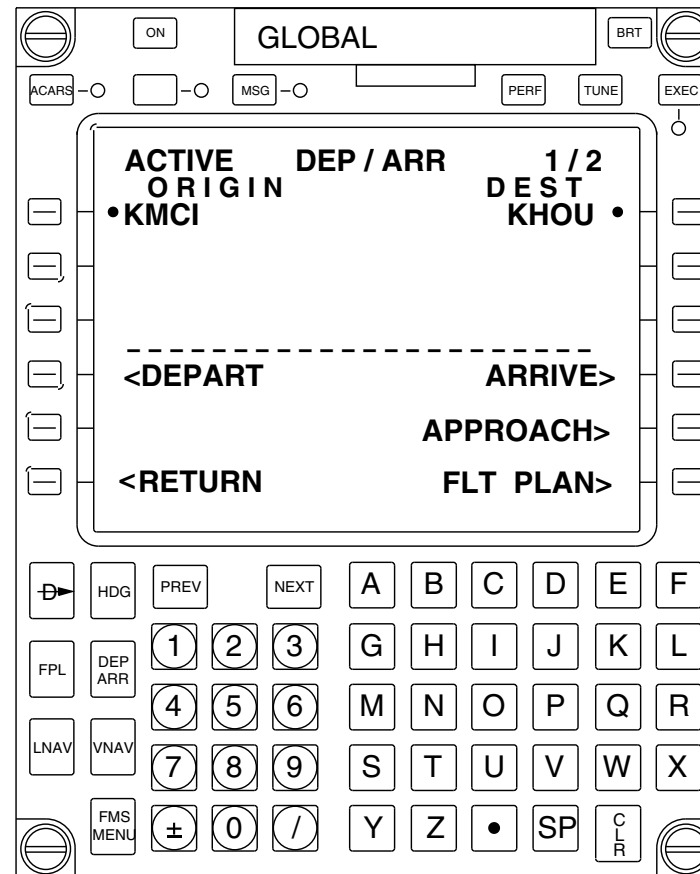


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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fsq68a30.cgm

ARRIVAL/DEPART 1/2 PAGE DESTINATION
Figure 152

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

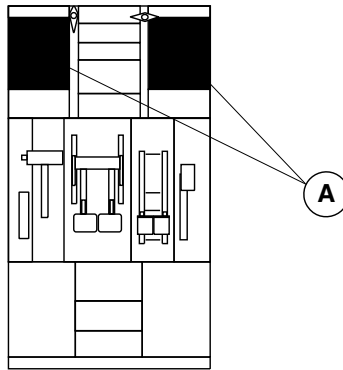
34-60-01

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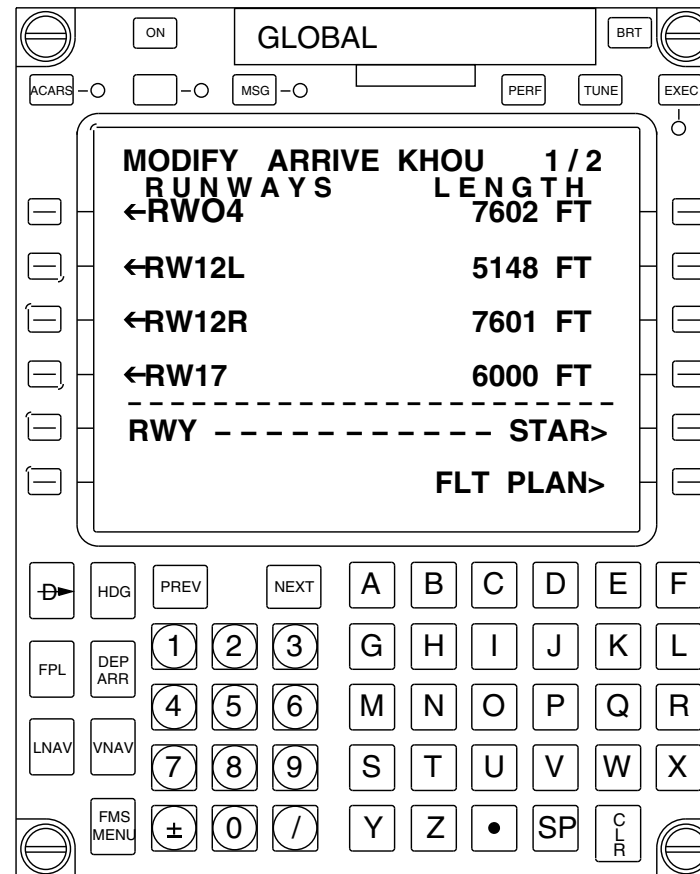


DE HAVILLAND AIRCRAFT
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fsq68a31.cgm

ARRIVE RUNWAYS 1/2 PAGE
Figure 153

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

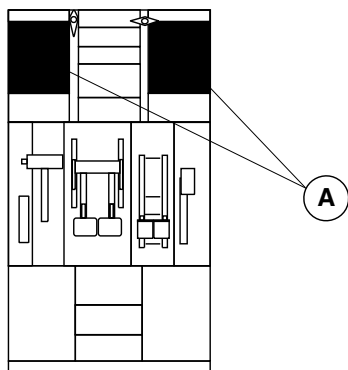
34-60-01

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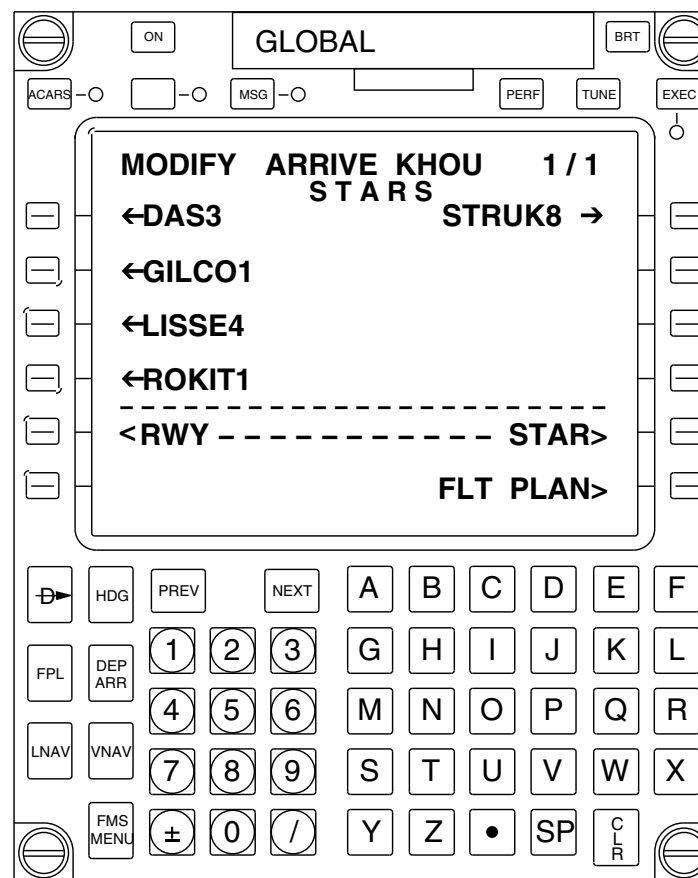


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OF CANADA LIMITED

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CENTRE CONSOLE



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fsq68a32.cgm

ARRIVE STARS 1/1 PAGE
Figure 154

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

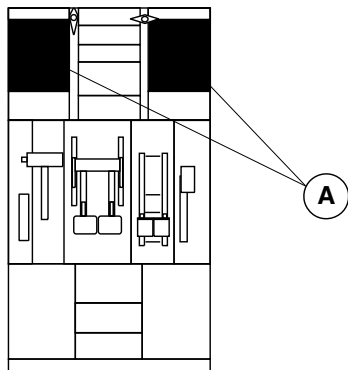
34-60-01

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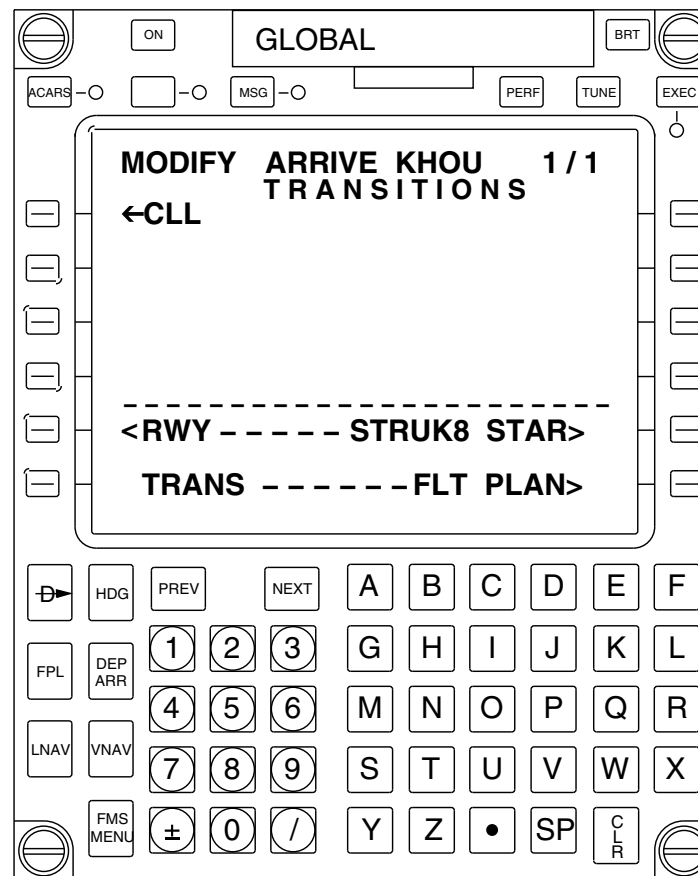


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CENTRE CONSOLE



A

fsq68a33.cgm

ARRIVE TRANSITIONS 1/1 PAGE
Figure 155

PSM 1-84-2A
EFFECTIVITY:
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Config 001

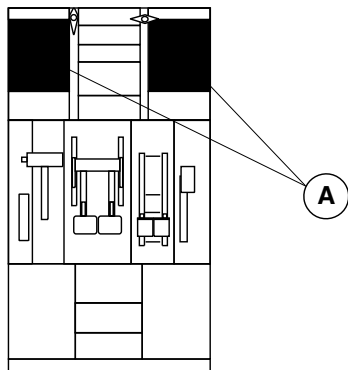
34-60-01

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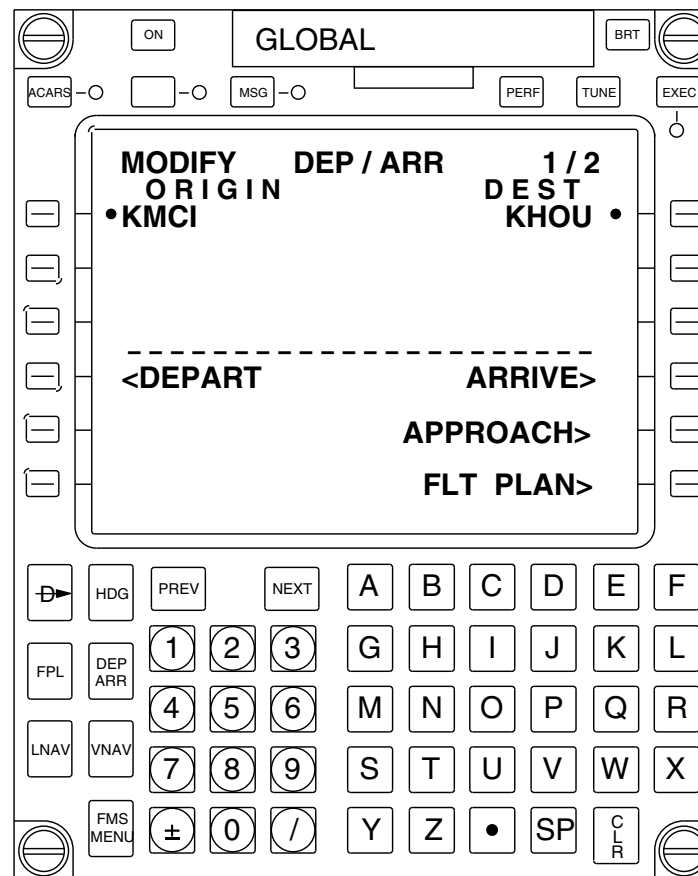


DE HAVILLAND AIRCRAFT
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A

fsq68a34.cgm

ARRIVE/DEPART 1/2 PAGE
Figure 156

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

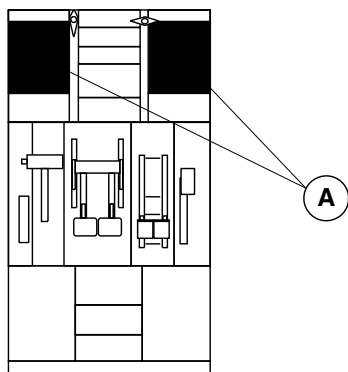
34-60-01

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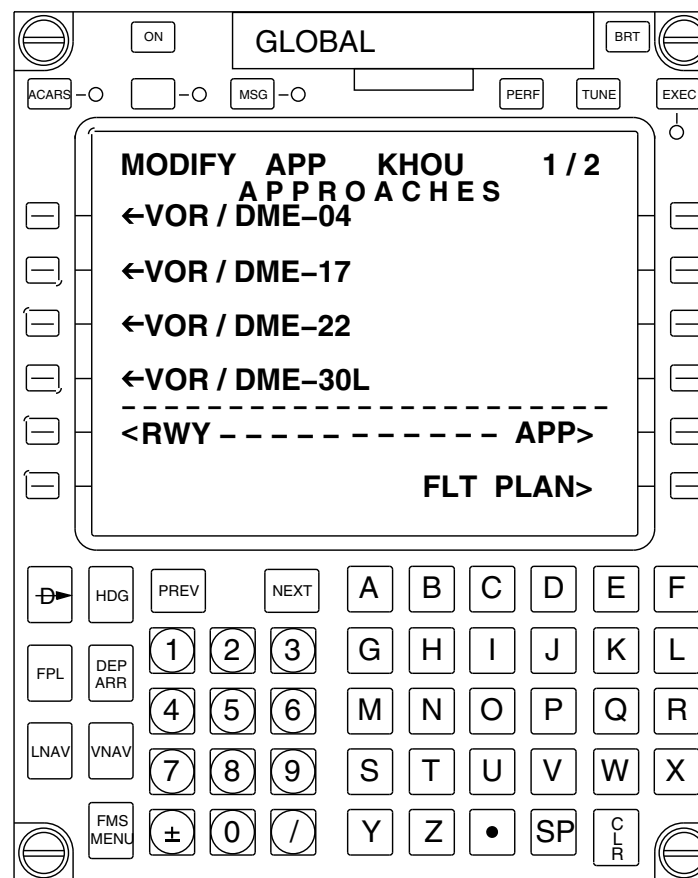


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A

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APPROACHES PAGE
Figure 157

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

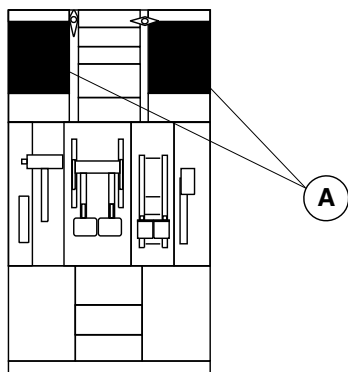
34-60-01

Config 001
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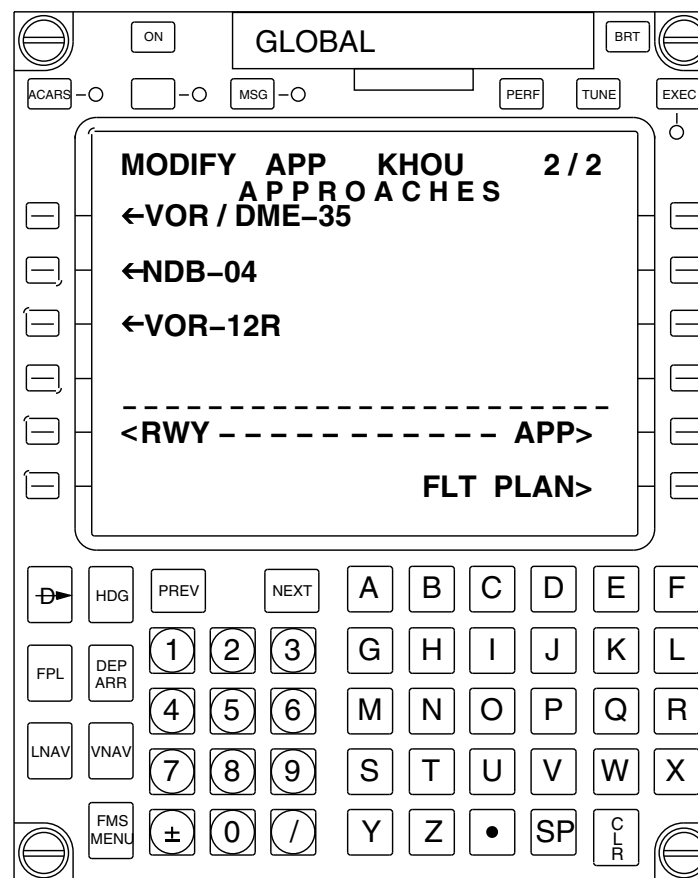


DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a36.cgm

DEPART/ARRIVE 2/2 PAGE
Figure 158

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

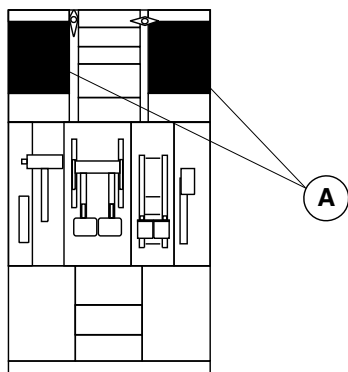
34-60-01

Config 001
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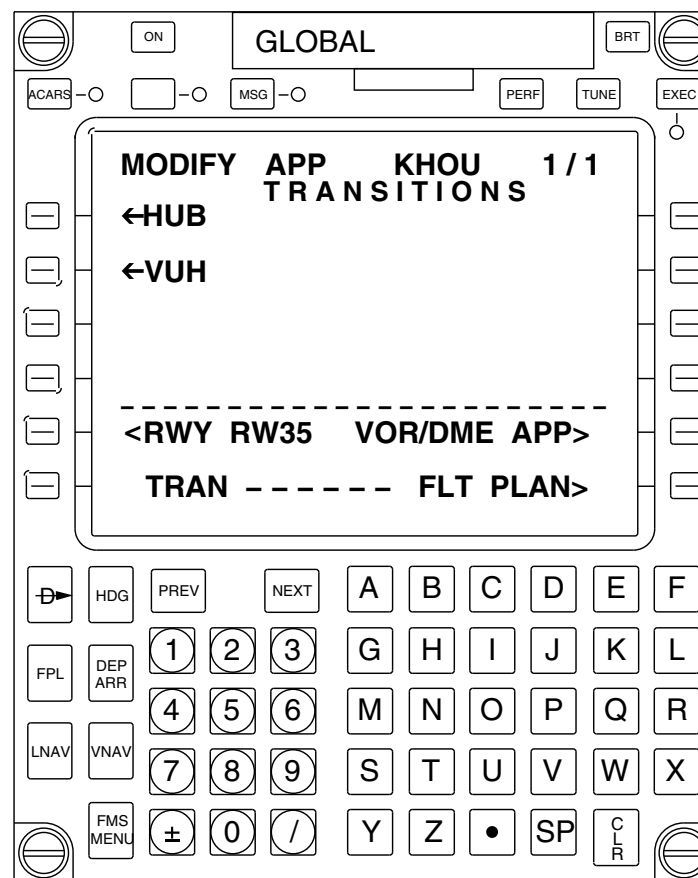


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a37.cgm

APP TRANSITIONS PAGE
Figure 159

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

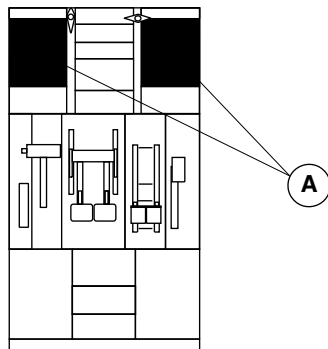
34-60-01

Config 001
Page 259
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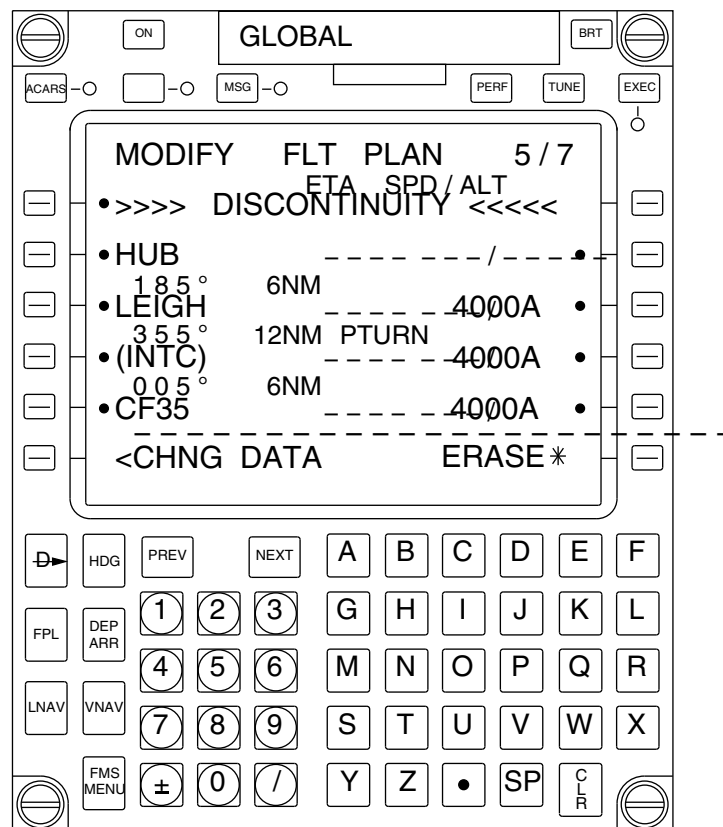


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a38.cgm

MODIFY FLIGHT PLAN 5/7 PAGE
Figure 160

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

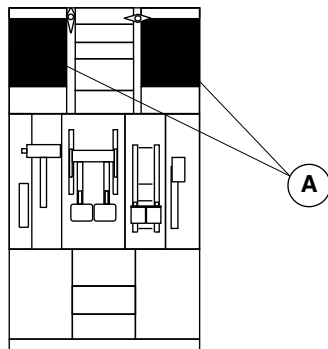
34-60-01

Config 001
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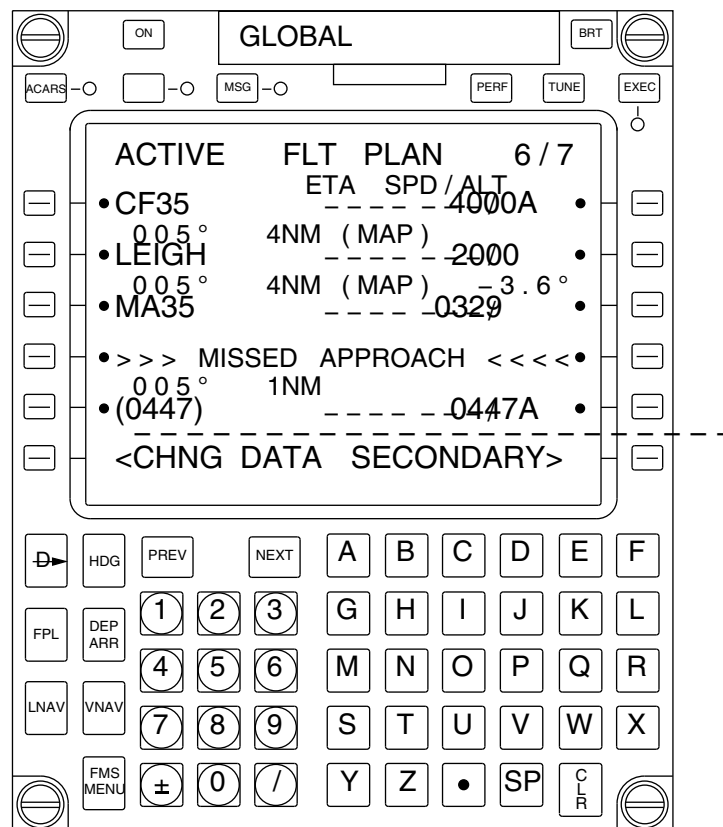


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



fsq68a39.cgm

ACTIVE FLIGHT PLAN WAYPOINT – GREEN
Figure 161

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

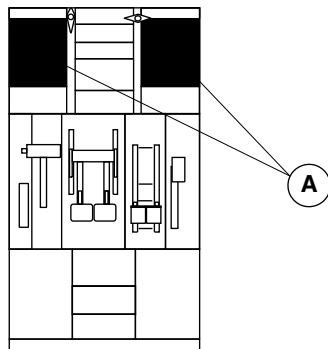
34-60-01

Config 001
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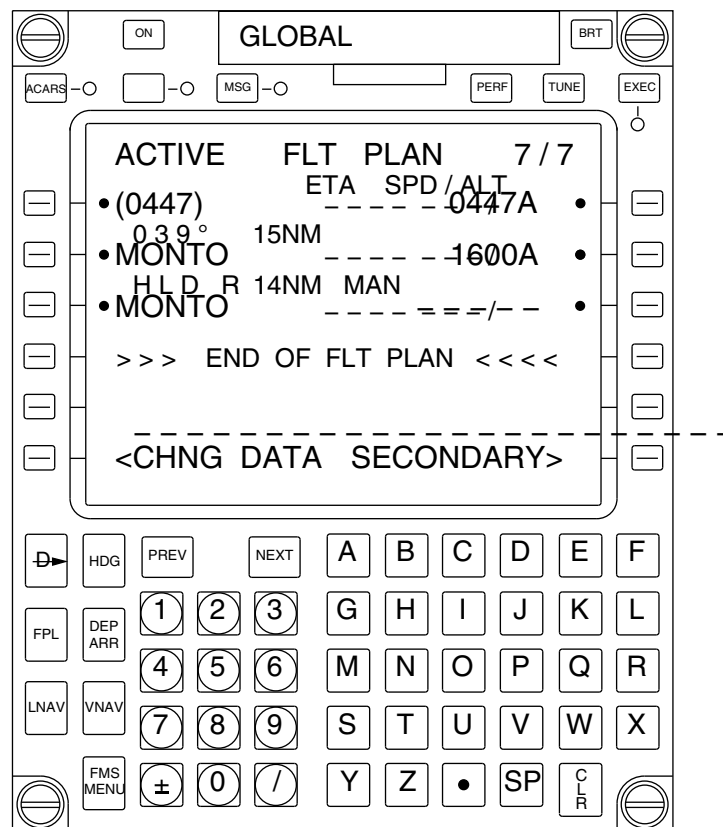


DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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ACTIVE FLIGHT PLAN MISSED PROCEDURE – CYAN
Figure 162

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

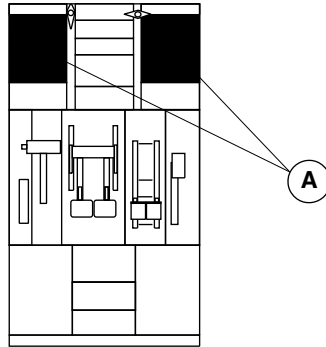
34-60-01

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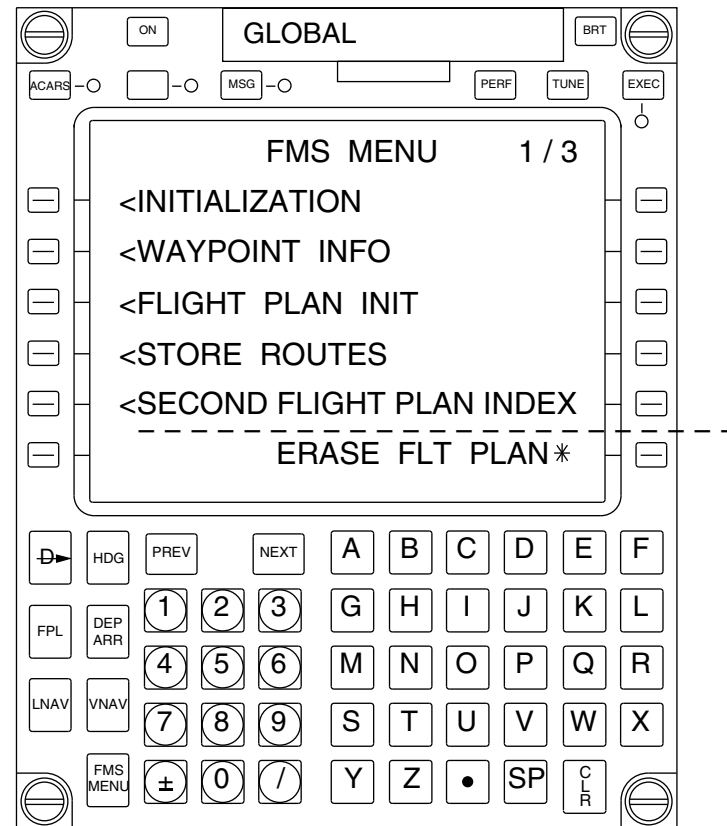


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a41.cgm

FMS-MENU 1/3 PAGE
Figure 163

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

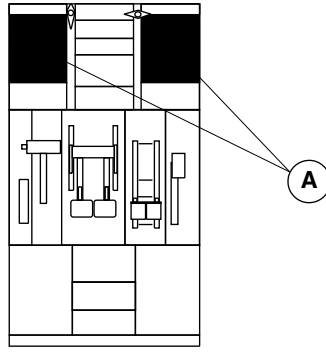
34-60-01

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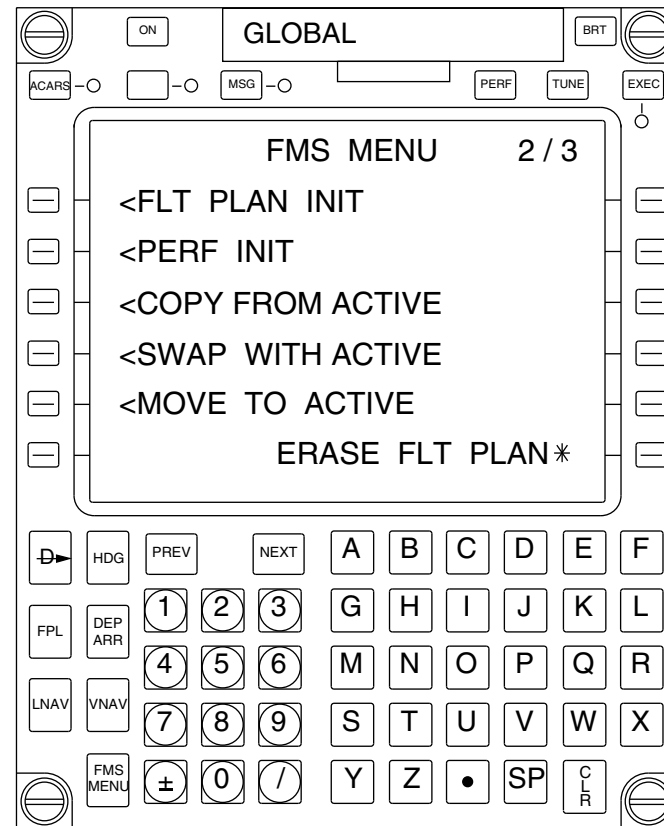


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a42.cgm

SECOND-INDEX 1/2 PAGE
Figure 164

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

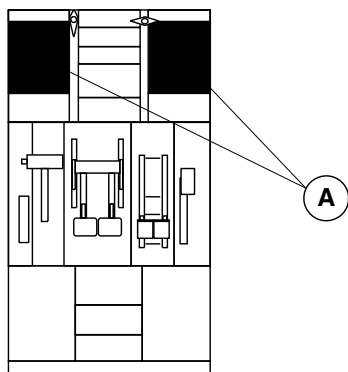
34-60-01

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

ON GLOBAL BRT

ACARS MSG PERF TUNE EXEC

SECOND FLT PLAN 1 / 4

ETA SPD / ALT

•RW01R	0 9 7 °	6NM	
•MKC	1 6 7 °	61NM	
•BUM	2 0 0 °	108NM	
•COODY	2 0 2 °	31NM	
•TUL			

<CHNG DATA ACTIVE>

HDG PREV NEXT A B C D E F

FPL DEP ARR 1 2 3 G H I J K L

LNAV VNAV 4 5 6 M N O P Q R

FMS MENU 7 8 9 S T U V W X

± 0 / Y Z • SP CLR

A

fsq68a43.cgm

SECOND FLIGHT PLAN 1/4 PAGE
Figure 165

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

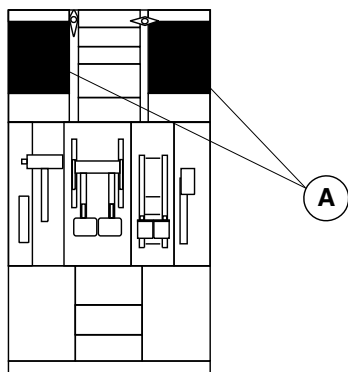
34-60-01

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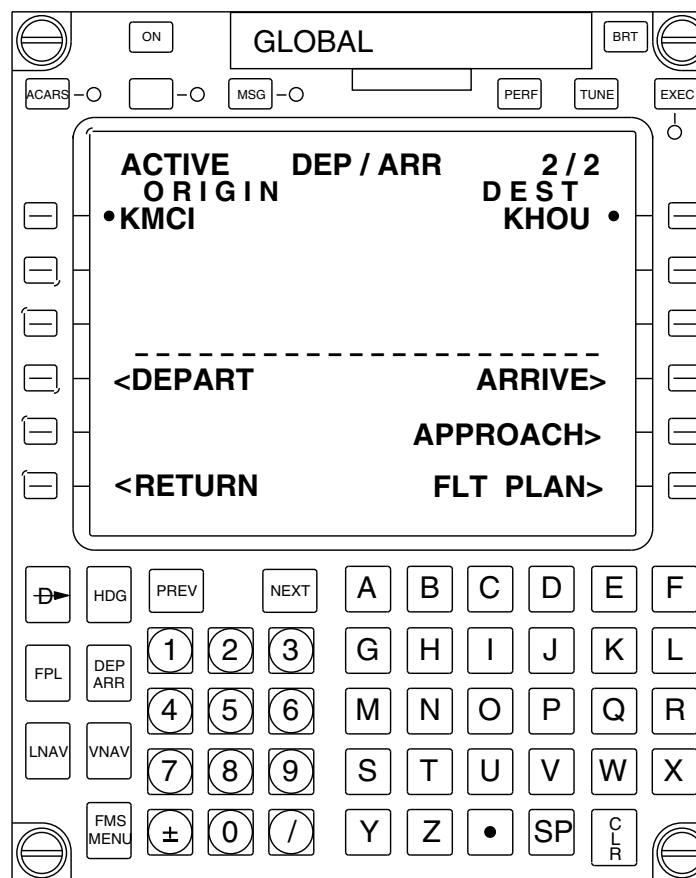


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a44.cgm

DEPART/ARRIVAL 2/2 PAGE
Figure 166

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

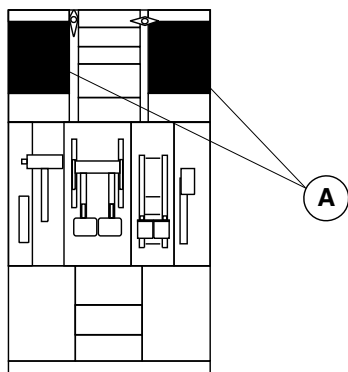
34-60-01

Config 001
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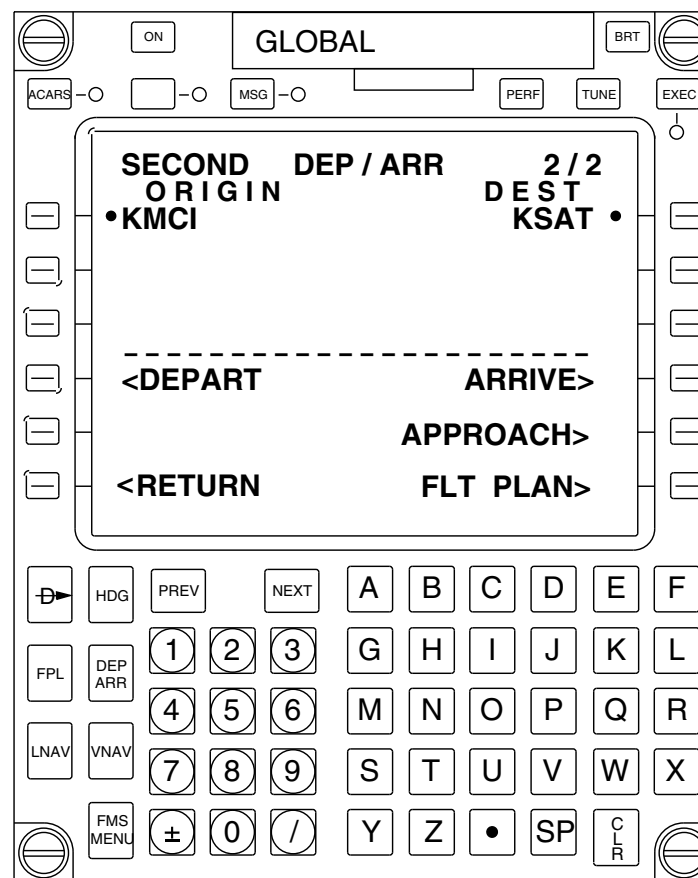


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a45.cgm

DEPART/ARRIVAL 2/2 PAGE DESTINATION FIELD
Figure 167

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

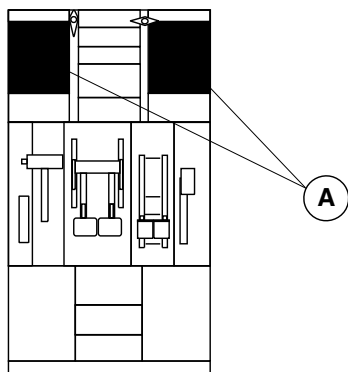
34-60-01

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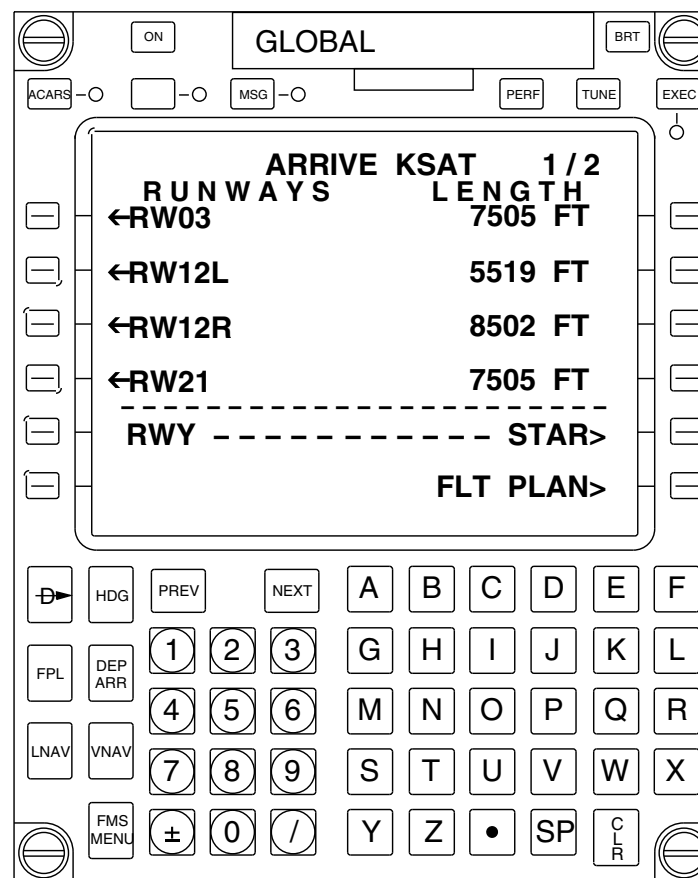


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a46.cgm

ARRIVE 1/2 PAGE ARRIVE RUNWAYS
Figure 168

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

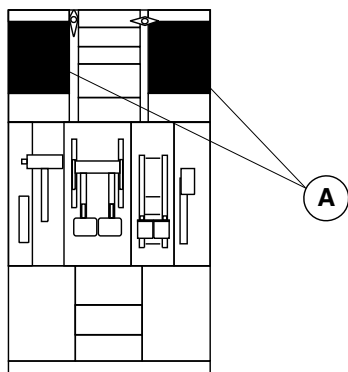
34-60-01

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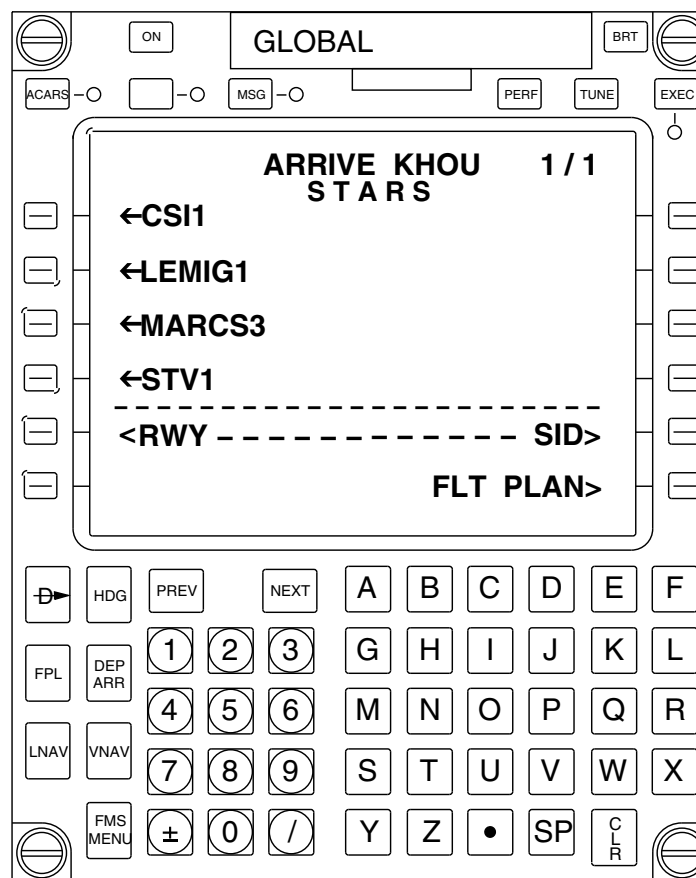


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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a47.cgm

ARRIVE STARS PAGE
Figure 169

PSM 1-84-2A
EFFECTIVITY:
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Config 001

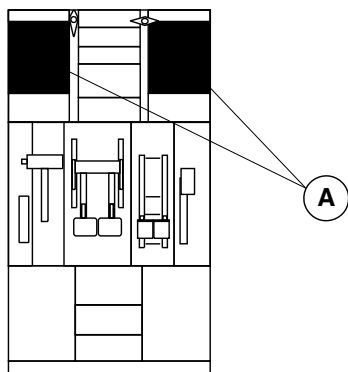
34-60-01

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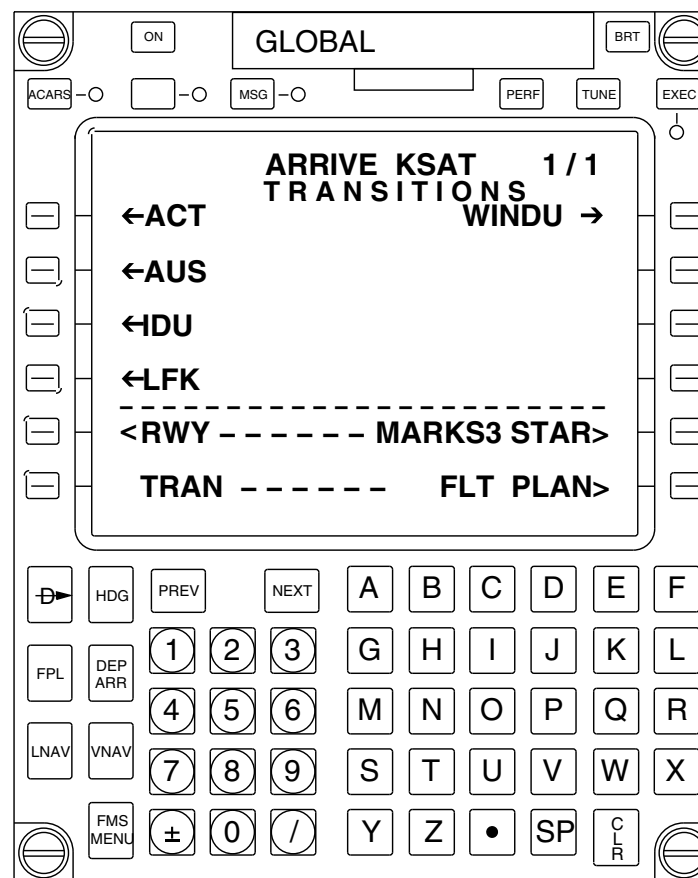


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a48.cgm

ARRIVE TRANSITIONS PAGE
Figure 170

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

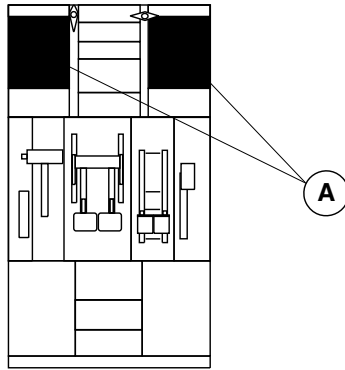
34-60-01

Config 001
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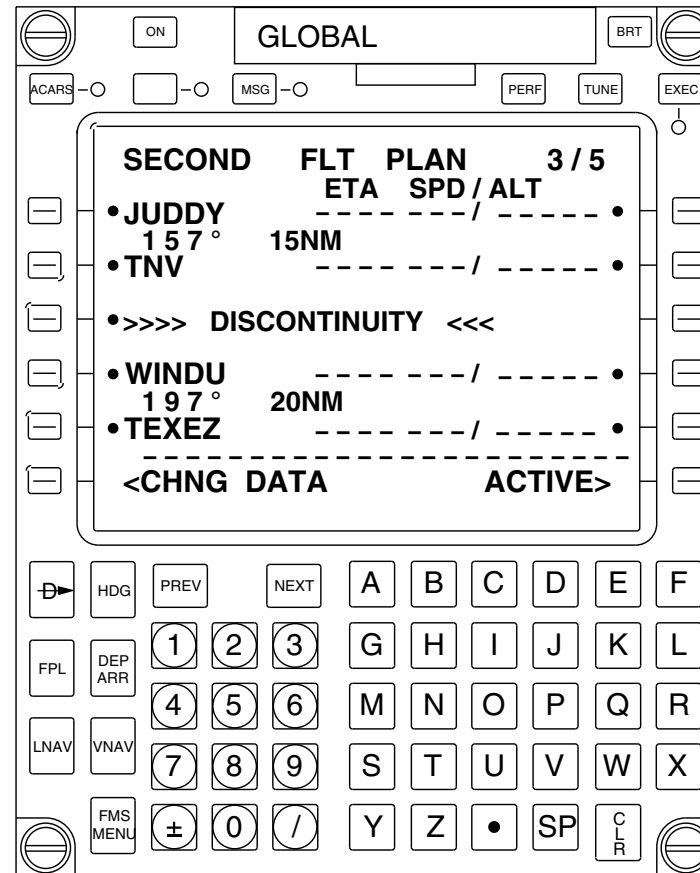


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a49.cgm

SECONDARY FLT PLAN 3/5 PAGE
Figure 171

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

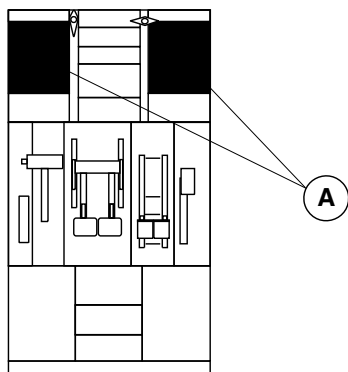
34-60-01

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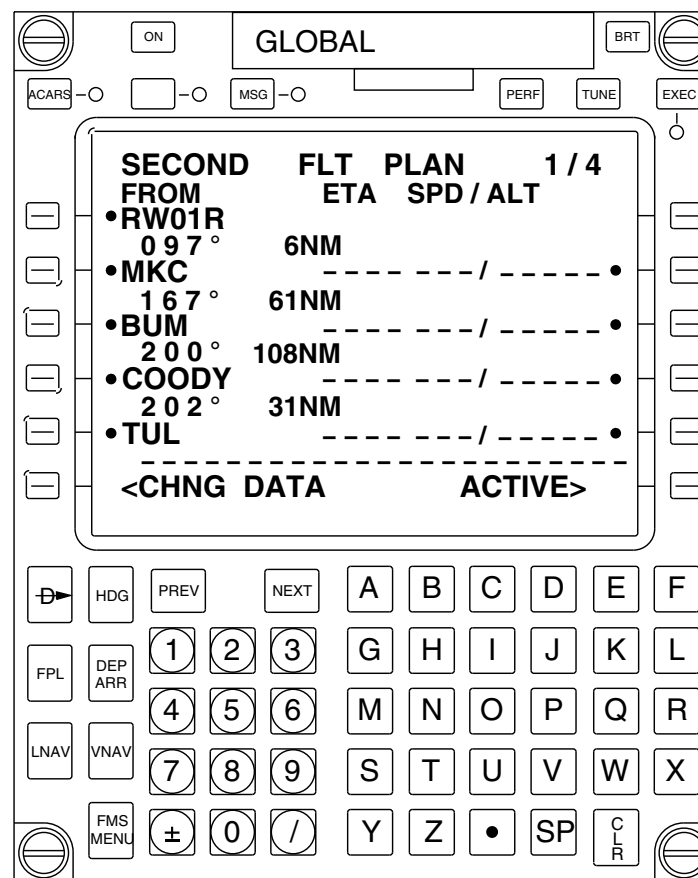


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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SECONDARY FLT PLAN 1/4 PAGE
Figure 172

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

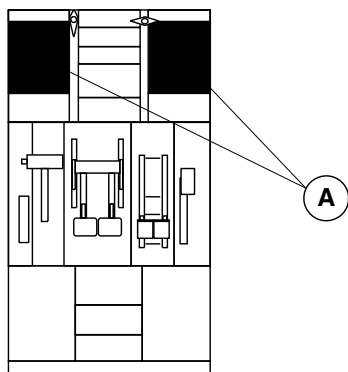
34-60-01

Config 001
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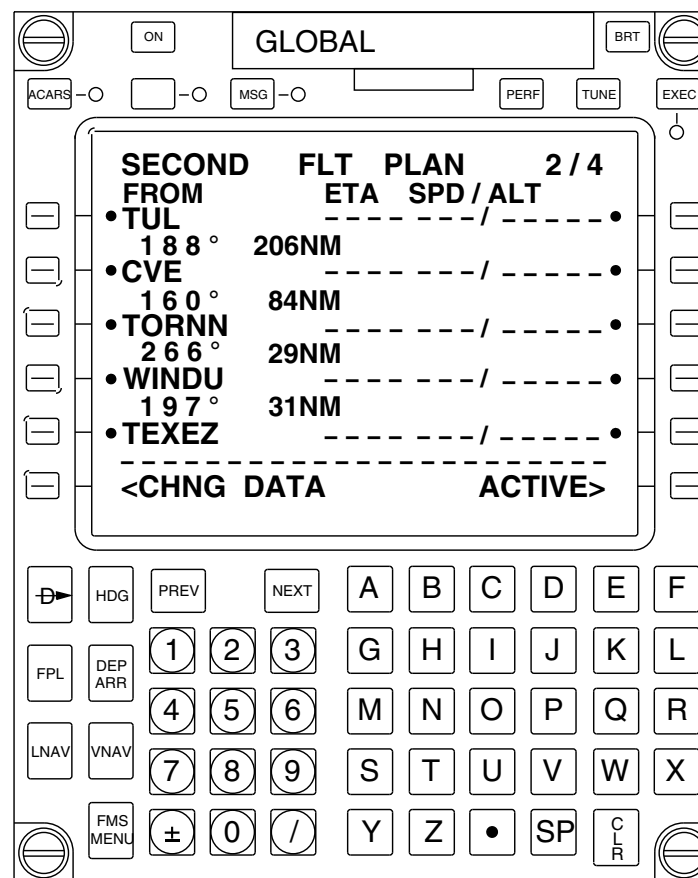


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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



A

fsq68a51.cgm

SECONDARY FLT PLAN 2/4 PAGE
Figure 173

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

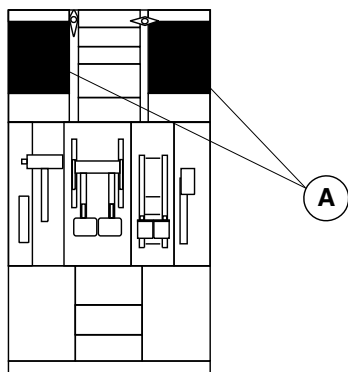
34-60-01

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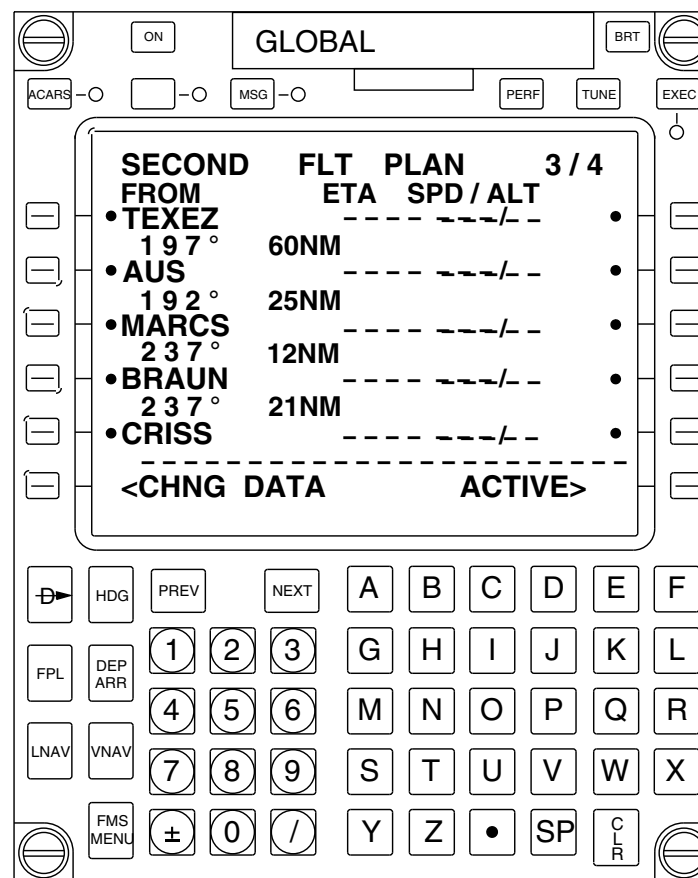


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a52.cgm

SECONDARY FLT PLAN 3/4 PAGE
Figure 174

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

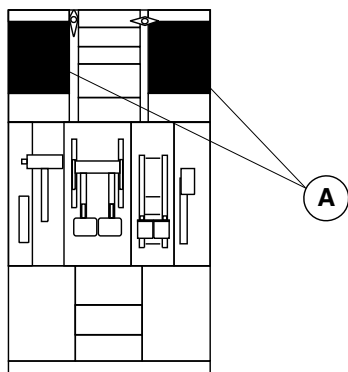
34-60-01

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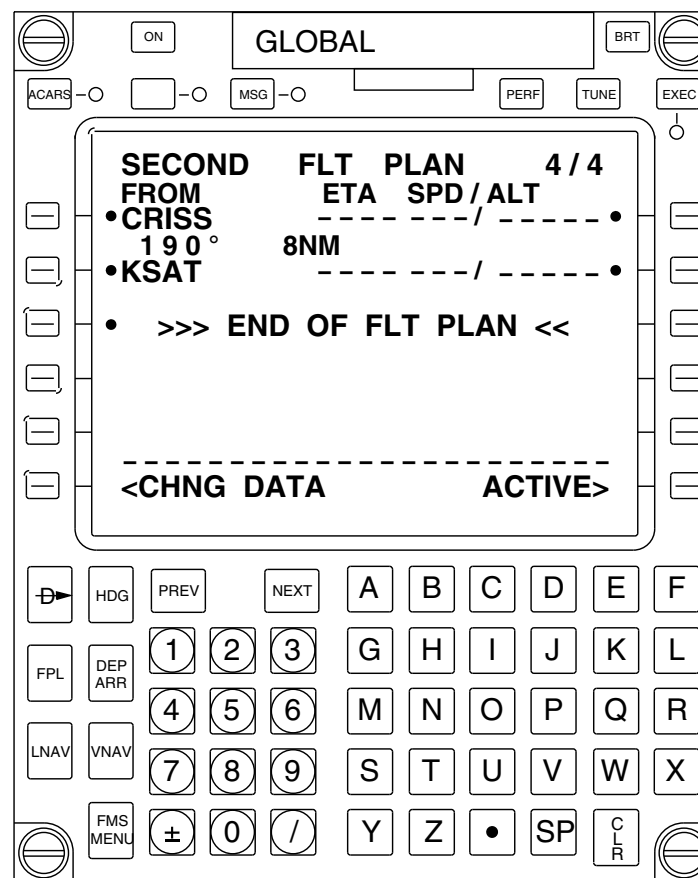


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE



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fsq68a53.cgm

SECONDARY FLT PLAN 4/4 PAGE
Figure 175

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

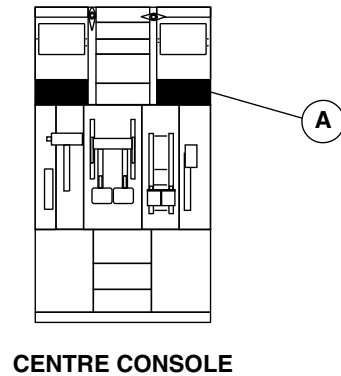
34-60-01

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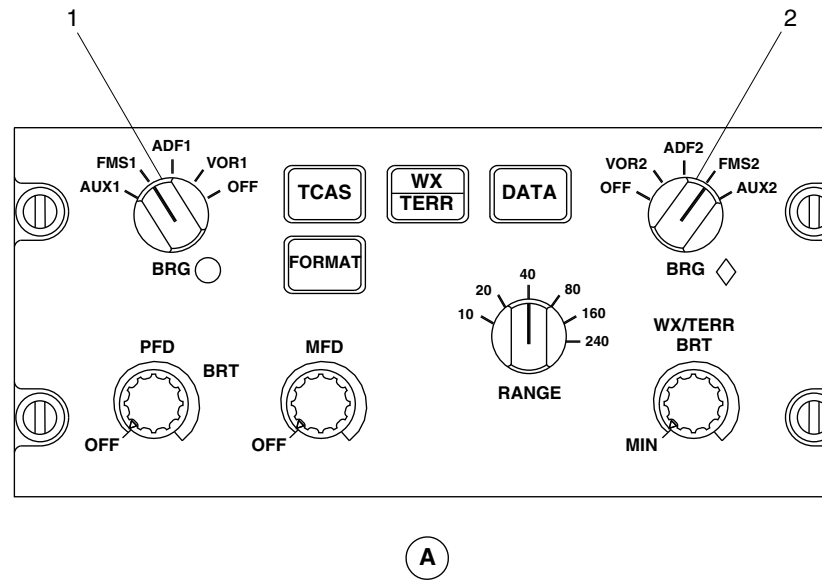
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. BRG 1 Selector Knob.
2. BRG 2 Selector Knob.



far13a01.cgm

EFCP FMS BEARING SELECTION
Figure 176

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

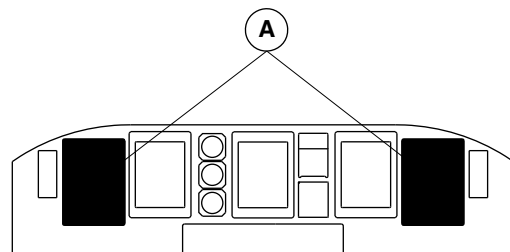
34-60-01

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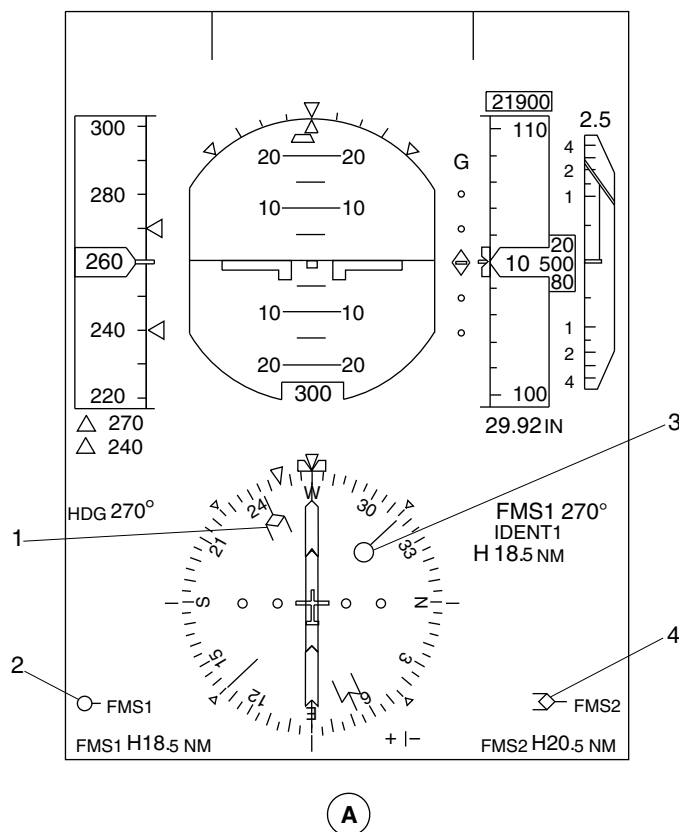
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. No. 2 Bearing Pointer.
2. No. 1 Bearing Reminder Label.
3. No. 1 Bearing Pointer.
4. No. 2 Bearing Reminder Label.



far08a01.cgm

EIS FMS BEARING INDICATION
Figure 177

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

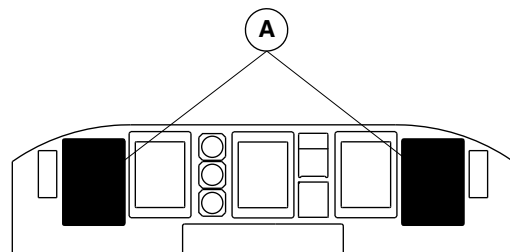
34-60-01

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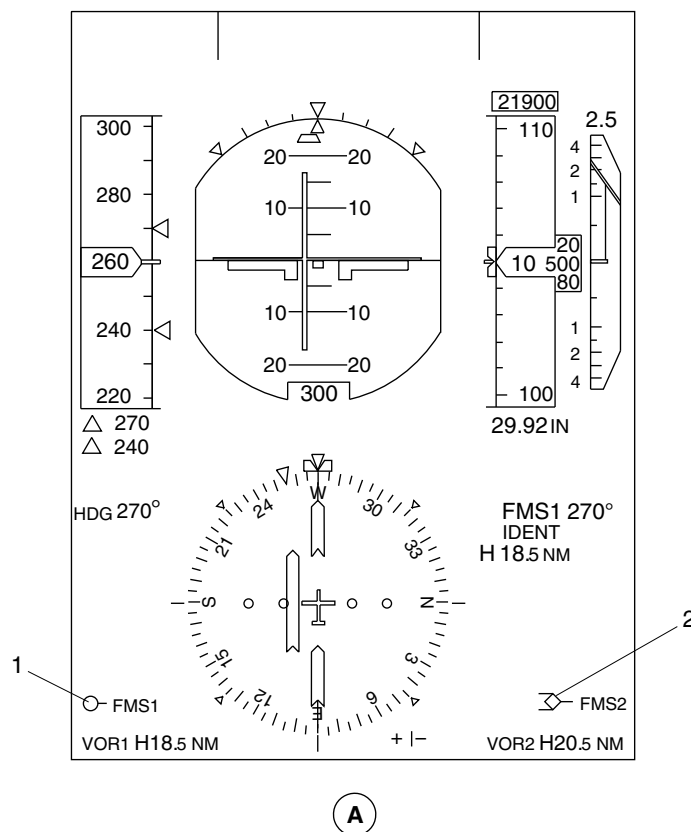
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. No. 1 Bearing Reminder Label.
2. No. 2 Bearing Reminder Label.



fa730a03.cgm

EIS FMS BEARING MALFUNCTION INDICATION
Figure 178

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

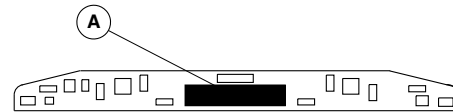
34-60-01

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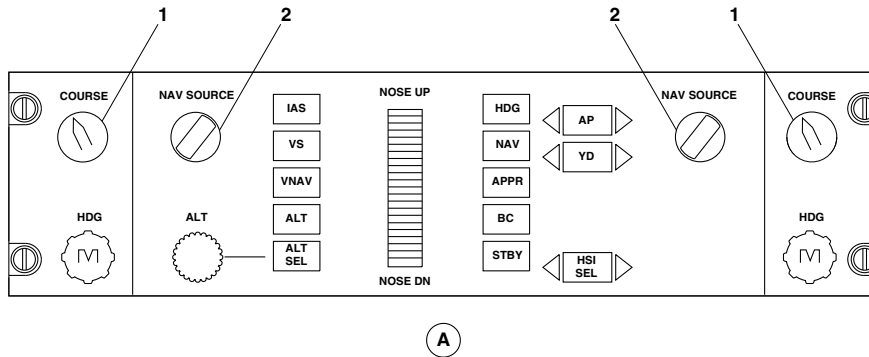
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. Course Selection Knob.
2. Navigation Source Selection Knob.



fs754a01.cgm

AFCP NAVIGATION SOURCE SELECTION
Figure 179

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 34-60-01
Config 001

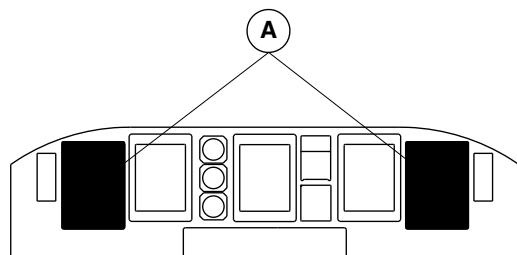
34-60-01

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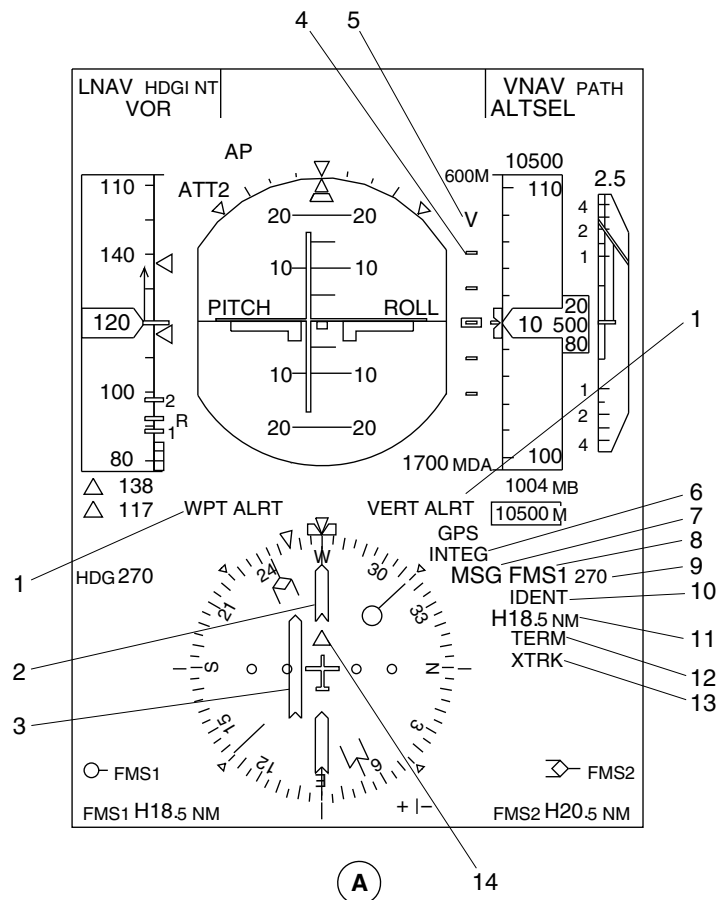
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. FMS Waypoint and Vertical Track Alert Messages.
2. Selected DTK Pointer.
3. Selected DTK Scale and Bar.
4. Vertical Deviation Scale and Pointer.
5. Vertical Deviation Source Identifier.
6. Track Angle (FMS Data).
7. FMS Navigation Performance Messages.
8. Vertical Deviation Source Identification.
9. Selected Desired Track (DTK) Digital Value.
10. FMS Waypoint Identifier.
11. FMS Distance.
12. FMS Lateral Scale Sensitivity Messages.
13. FMS General Messages.
14. To/From Pointer.



PFD FMS NORMAL DTK INDICATION
Figure 180

PSM 1-84-2A

EFFECTIVITY:

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Config 001

34-60-01

Config 001

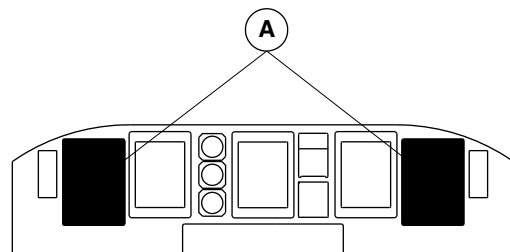
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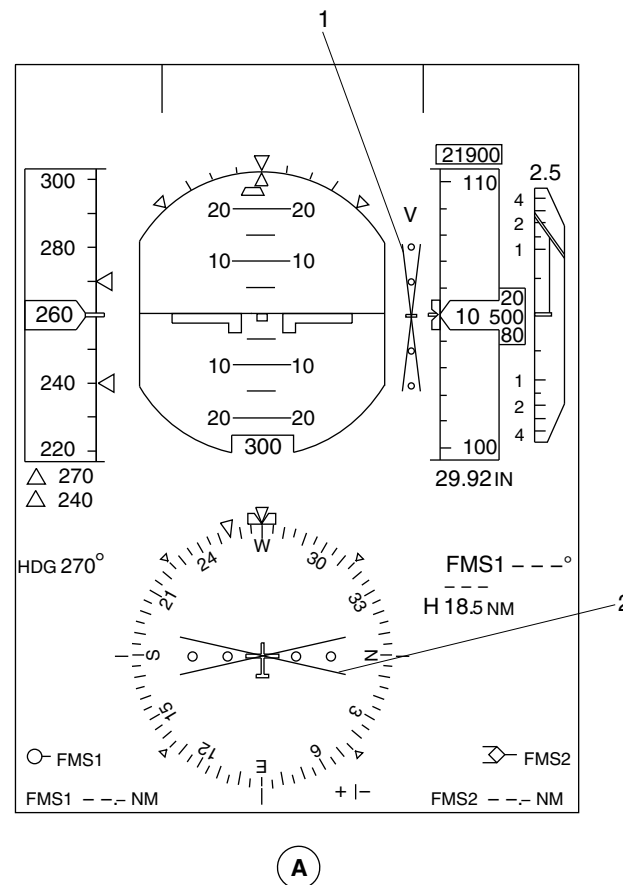
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Vertical Deviation Failure Flag.
2. Cross Track Deviation Failure Flag.



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PFD FMS MALFUNCTION INDICATION
Figure 181

PSM 1-84-2A
EFFECTIVITY:
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Config 001

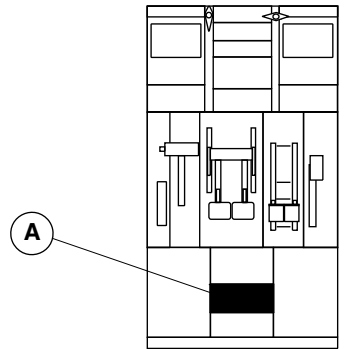
34-60-01

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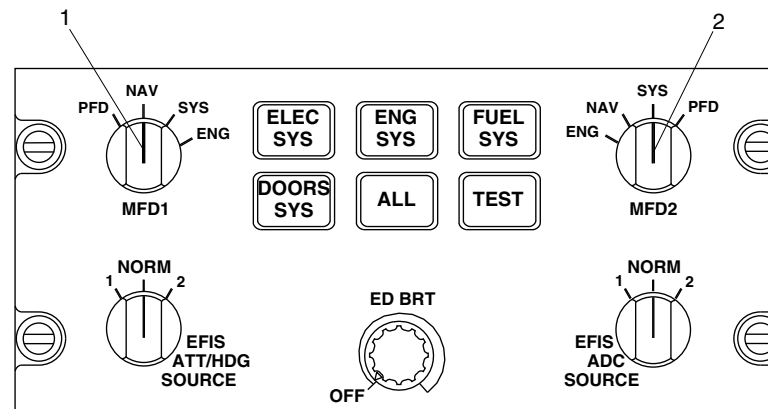
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

LEGEND

1. MFD 1 Reversion Selector.
2. MFD 2 Reversion Selector.



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ESCP NAVIGATION PAGE SELECTION
Figure 182

PSM 1-84-2A
EFFECTIVITY:
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Config 001

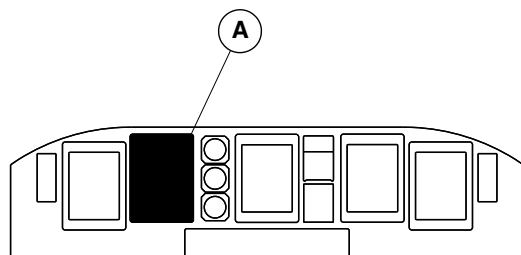
34-60-01

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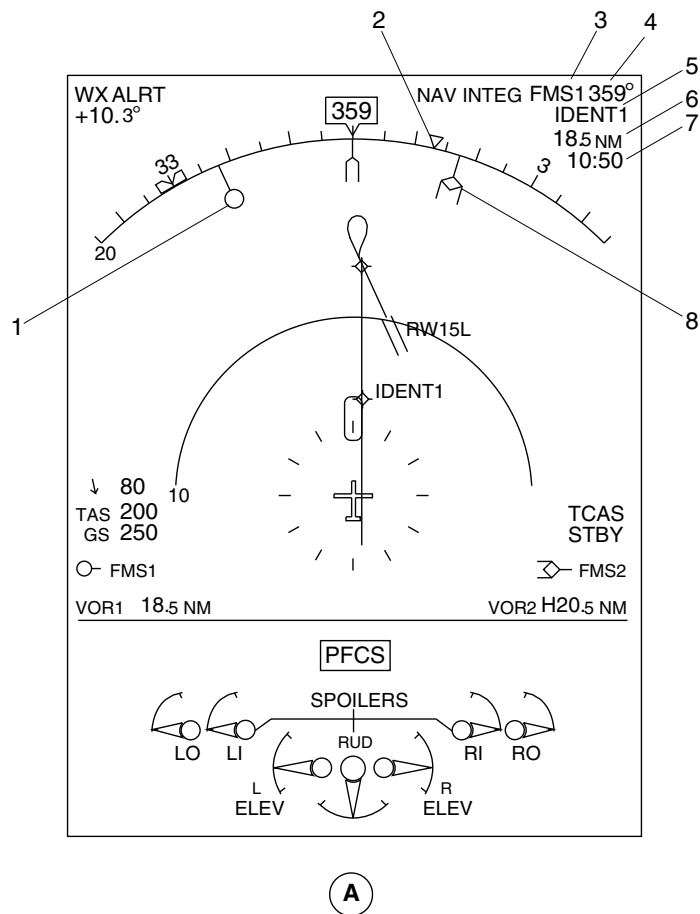
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. No. 1 Bearing Pointer.
2. Actual Track (Drift Angle).
3. Navigation Source Annunciation.
4. Selected Desired Track (DTK) Digital Value.
5. FMS Waypoint Identifier.
6. FMS Distance.
7. Estimated Time To Waypoint.
8. No. 2 Bearing Pointer.



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NAVIGATION PAGE ARC FORMAT FMS NORMAL INDICATION (BASIC)
Figure 183

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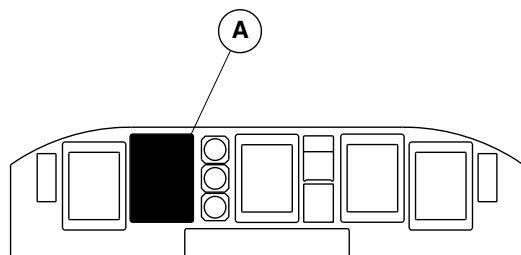
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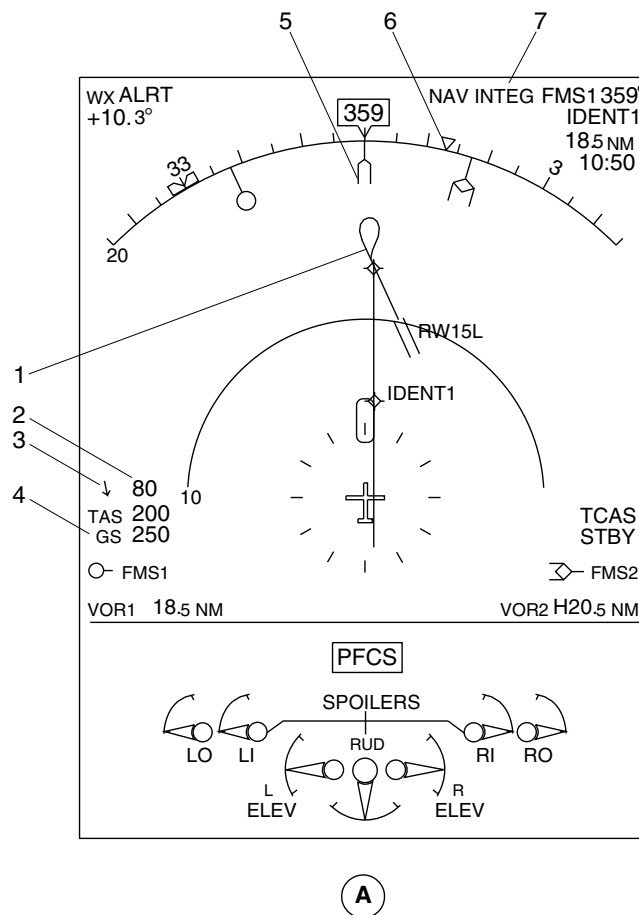
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MAIN INSTRUMENT PANEL

LEGEND

1. Map Symbology.
2. Wind Speed Indication.
3. Wind Direction Indication.
4. Ground Speed Digital Value.
5. FMS Desired Track Pointer.
6. Actual Track (Drift Angle).
7. FMS Abnormal Status Messages.



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NAVIGATION PAGE ARC FORMAT FMS NORMAL INDICATION
Figure 184

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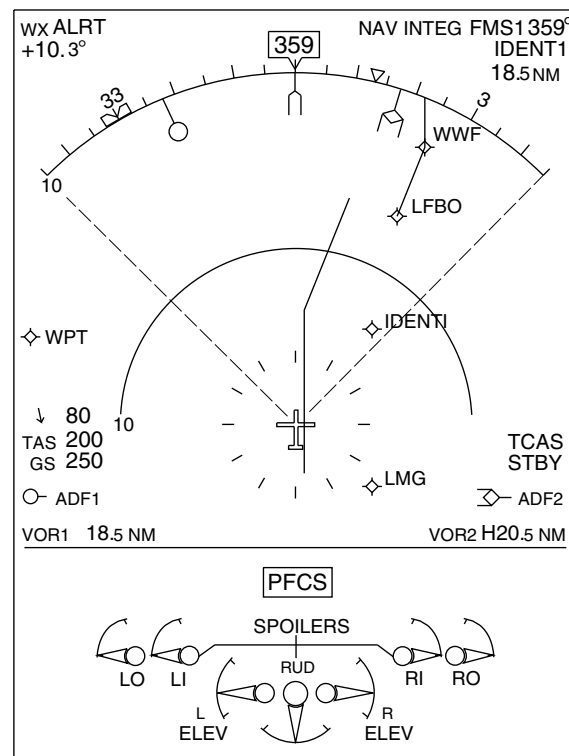
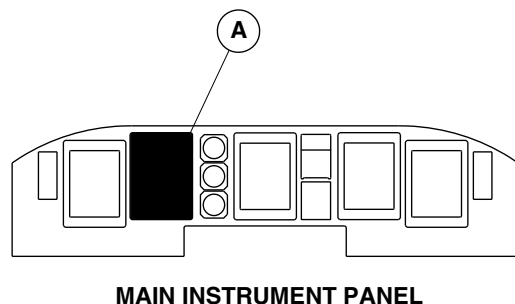
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NAVIGATION DISPLAY ARC FORMAT OFFSET INDICATION
Figure 185

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FLIGHT MANAGEMENT SYSTEM – UNS-1C/1E/1EW

INTRODUCTION

The Flight Management Systems (FMS) are integrated navigation management systems providing the flight crew with centralized control for:

- Navigation
- Flight planning
- Fuel management data
- Communication and navigation radio management.

Although similar in size, appearance and function, the UNS-1E is an upgraded version of the UNS-1C with the following improvements:

- The CPU board, motherboard, and filter board have been replaced to expand capacity and provide Ethernet capability.
- The LCD in the CDU has been replaced with a new LCD to provide better contrast, readability and viewing angle. The CDU software requires only a minor change to enable it to handle faster communications with the upgraded CPU board.
- The GPS receiver has been replaced with a new GPS/GLONASS receiver.

- The Ethernet replaces the serial link to the DTU. This speeds up loading of the navigation databases from about 10 minutes to less than one minute.

The FMS UNS-1Ew is an upgraded version of the FMS UNS-1E with the following improvements:

- The FMS UNS-1Ew contains a Wide Area Augmentation System (WAAS) that has cross channel monitoring to meet the integrity requirements.
- The FMS UNS-1Ew also contains two remote Level of Service (LOS) annunciators. One LOS annunciator is adjacent to each PFD.
- The existing GPS antennas are replaced with antennas having the same footprint but are designed and qualified to include the WAAS operation.
- The DTU for the UNS-1Ew is an enhanced version of the basic DTU that provides enhanced data transfer capability.

Single or Dual UNS-1C or UNS-1E or UNS-1Ew FMS installations are available depending on customer configurations.

NOTE

UNS-1C, UNS-1E and UNS-1Ew condensed operational procedures can be found in the Q400 Aeroplane Operating Manual (AOM). For complete operating instructions, refer to:

- UNS-1C FMS Operator's Manual (OM), Report No. 2423sv601 (dated July 11, 1996 or later)
- UNS-1E FMS Operator's Manual (OM), Report No. 2423sv810/901 (dated August 22, 2001 or later)
- UNS-1Ew FMS Operator's Manual (OM), Report No. 2423sv1000/1100.

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COMPONENT LOCATION

Refer to Figure 1.

The flight management system consists of a Flight Management Computer (FMC) located in the centre console and an FMS configuration module that is attached to the back of the FMC. The Data Transfer Unit (DTU) or Solid State Data Transfer Unit (SSDTU) is installed under the lower shelf in the copilot's side panel. The system antennas are mounted on top of the fuselage.

GENERAL DESCRIPTION

Refer to Figure 2.

The UNS FMS is a cockpit-mounted single Line Replaceable Unit (LRU) consisting of:

- Multi-functional Control Display Unit (MCDU)
- Flight Management Computer (FMC)
- Global Positioning System (GPS).

Position information is derived from VOR#1, DME#1 and GPS#1 for the pilot's side FMS, and VOR#2, DME#2 and GPS#2 for the optional co-pilot's side FMS. True airspeed, altitude, static air temperature data, fuel flow data, etc., are routed to each FMS from the Flight Data Processing System (FDPS).

The Q400 Series UNS-1C FMS configuration requires the following minimum equipment to be functional and operating:

Description	Part Number	Model Number	FMS #1 Quantity	FMS #2 Quantity
FMS Hardware P/N	1017-41-221	UNS-1C	1	1
FMS Software P/N	SCN 604.1	–	1	1
Antenna	10706	Universal Avionics	1	1
Data Transfer Unit	1405-01-1 or 1406-01-1	Universal Avionics	1	(part of FMS #1)

System capabilities for UNS-1C include:

- Full EFIS interface including GAMA 429 Map display
- Full compatibility with all ARINC 424 leg types
- Remote navigation tuning of VOR/DME for approach navigation as well as third DME channel
- GPS navigation
- Full IFR/IMC, supplementary means navigation capability
- Ability to provide complete coupled lateral guidance for all types of air traffic control routes (SIDs, Airways, STARTs, Approaches)
- TSO C129A Class A1/B1/C1 GPS navigation capability.

The Q400 Series UNS-1E FMS configuration requires the following minimum equipment to be functional and operating:

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Description	Part Number	Model Number	FMS #1 Quantity	FMS #2 Quantity
FMS Hardware P/N	2017-41-221	UNS-1E	1	1
FMS Software P/N	SCN 801.5 or SCN 802 or SCN 802.2 or SCN 802.8 or SCN 803.0	–	1	1
	SCN 802.8 or SCN 801.5 with MOD 23 World Magnetic Model (WMM) (SB84-34-183 or SB84-34-184)			
Antenna	10705	Universal Avionics	1	1
Data Transfer Unit	1406-01-1 or 1408-00-1	Universal Avionics	1	(part of FMS #1)

System capabilities for UNS-1E include:

- Full EFIS interface including GAMA 429 Map display

- Full compatibility with all ARINC 424 leg types
- Remote navigation tuning of VOR/DME for approach navigation as well as third DME channel
- GPS navigation
- Full IFR/IMC, supplementary means navigation capability
- Enroute, Terminal and Approach LNAV operations with AFCS coupling
- Enroute, Terminal and Approach VNAV operations with AFCS coupling
- TSO C129A Class A1/B1/C1 GPS navigation capability.

The Q400 Series UNS-1Ew FMS configuration requires the following minimum equipment to be functional and operating:

Description	Part Number	Model Number	FMS #1 Quantity	FMS #2 Quantity
FMS Hardware P/N	3017-41-221	UNS-1Ew	1	1
FMS Software P/N	SCN 1000.1 or SCN 1000.5 or SCN 1100.1 or SCN 1000.6 or SCN 1000.8	–	1	1

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GPS/ WAAS Antenna	10708	Universal Avionics	1	1
Data Transfer Unit	1406-01-1 or 1408-00-1	Universal Avionics	1	(part of FMS #1)

System capabilities for UNS-1Ew include:

- Full EFIS interface including GAMA 429 Map display
- Full compatibility with all ARINC 424 leg types
- Remote navigation tuning of VOR/DME for approach navigation as well as third DME channel
- GPS navigation including WAAS satellites
- Full IFR/IMC navigation capability
- Enroute, Terminal and Approach LNAV operations with AFCS coupling
- Enroute, Terminal and Approach VNAV operations with AFCS coupling
- TSO C129A Class A1/B1/C1 GPS navigation capability
- Approach operations using lateral and vertical guidance from the WAAS satellite system
- Enhanced VNAV modes of operation (Vpath, Vapp, Vflc).

The FMS UNS-1Ew provides GPS navigation data augmented by the Satellite Based Augmentation System (SBAS) with one of the following subsystems:

- Wide Area Augmentation System (WAAS) developed by the Federal Aviation Administration (FAA) in U.S.
- Multi functional Satellite Augmentation System (MSAS) developed by the Japan Civil Aviation Bureau (JCAB) in Japan.
- European Geostationary Navigation Overlay Service (EGNOS) developed by the European Space Agency (ESA) in Europe.
- GPS and Geo Augmented Navigation system (GAGAN) developed by the Raytheon in India.

For all phases of the flight and non-SBAS approaches, the FMS navigates and provides guidance using a multi sensor filter with GPS/WAAS sensor position or velocity as one of the inputs to the filter. During a SBAS approach the FMS uses GPS/WAAS sensor position or velocity directly for navigation and guidance.

The FMS with WAAS is categorized as a Gamma 3 class that covers navigation and guidance functions for all the flight phases to Localizer Performance with the Vertical guidance (LPV) approach level of service. The WAAS subsystem provides three dimensional position and velocity, time and WAAS protection levels in addition to providing ARINC inputs and outputs, critical analog output, and discrete inputs and outputs.

The FMS provides Lateral Navigation (LNAV) guidance and Vertical Navigation (VNAV) (approved for UNS-1E and UNS-1Ew on Q400) guidance based on data from the navigation receivers and air data sensors. An FMS approach mode allows the operator to fly flight



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director or autopilot non-precision approaches. The FMS provides the information for:

- Course
- Course Deviation
- Bearing-to-waypoint
- Distance-to-waypoint.

This information is displayed on the pilot's and co-pilot's PFD and MFD and is coupled to the AFCS for flight director and autopilot functions.

During flight the FMS automatically updates the fuel on board and gross weight as well as predicted fuel requirements and estimated time of arrival based on:

- Fuel flow
- Ground speed
- Enroute flight plan winds (if entered).

A self-contained database stored in non-volatile memory provides the FMS with information on over 100,000 waypoints, nav aids and airports. Data base updates are provided by Jeppesen.

DETAILED DESCRIPTION

SENSORS

The FMS accepts primary position information from short and long range navigation sensor(s) which include:

- VOR
- DME

- GPS.

The FMS also receives true airspeed and altitude information from the air data computer and fuel flow data from the aircraft's fuel flow sensors.

BEST COMPUTED POSITION

Primary position data received from each sensor is combined in a Kalman Filter contained within the FMS to derive a Best Computed Position (BCP). Utilizing this BCP, the FMS navigates the aircraft along the pilot-programmed flight path, which can include:

- Departure procedures (SID's)
- Airways
- In-flight maneuvers
- Holding patterns
- Arrival procedures (STAR's)
- Approaches
- Runways.

The FMS best computed position is determined by using position inputs from all available navigation sensors and DME distance information from the scanning DME. This DME derived position is then integrated with position information from the GPS, radial from the VOR, TAS from the air data computer and heading information to derive the best computed position as a weighted average of the various sensor inputs. Once the best computed position is obtained, secondary navigational functions are computed for display and include:

- Waypoint



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- ETA
- Distance to waypoint
- Wind
- Ground speed.

Estimated winds may be defined at various waypoints in the flight plan to give a more accurate ETA.

FLIGHT PLANNING

Flight planning is accomplished by accessing the internal Navigation Data Base, which includes navigation and pilot-defined data, and company defined routes created on Universal's Off-line Flight Planning Program. The pilot selects from the following parameters to create the desired flight plan:

- Waypoints
- Routes
- Airways
- Arrivals
- Departures
- Approaches
- Runways.

The user may also create tactical waypoints, which are temporary pilot defined waypoints limited to use with the flight plan. Accurate and quick flight planning is facilitated by features such as SIDs, STARS and approaches for use in terminal areas, and High and Low Altitude airways for the enroute phase.

MANEUVERS

In-flight maneuver capabilities under FMS control include:

- Controlled Departures
- Procedural turns
- Turn anticipation
- Direct-To functions
- VNAV (approved for UNS-1E and UNS-1Ew on Q400)
- Heading Mode
- Holding patterns
- Selected crosstrack
- Pseudo-VOR
- Arrivals
- Approaches.

WAYPOINT SEQUENCING

The FMS utilizes automatic turn anticipation for leg changes along the flight plan. Turns based on current altitudes are commenced at a distance based upon:

- Groundspeed
- Leg change magnitude
- Roll steering bank limits.

DIRECT-TO

The FMS provides maximum flexibility when defining direct-to leg changes. Direct-to (DTO) maneuvers can be to waypoints or airports on or off the flight plan. If a TO waypoint off the flight plan is chosen, the system will prod the pilot to redefine the NX waypoint in order to



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link the newly defined leg into the flight plan. The pilot may specify the turn direction when performing a DTO.

HEADING MODE

The HDG function of the NAV mode allows the pilot to select and fly a desired heading using keyboard entries from the FMS. The Heading Mode may be armed to automatically intercept the current NAV leg.

SELECTED CROSSTRACK

A course may be established to provide guidance with respect to an offset course, parallel to the leg defined by the FR/TO waypoints. When Selected Crosstrack (SXTK) is entered, the SXTK annunciator on the instrument panel will illuminate. XTK will indicate crosstrack from the selected crosstrack, the system will provide steering to fly the selected crosstrack and the HSI will be referenced to the parallel (offset) course only until the next waypoint sequences. Then SXTK automatically cancels.

VERTICAL NAVIGATION (approved for UNS-1E and UNS-1Ew on Q400)

The Vertical Navigation (VNAV) function allows the flight crew to define a desired vertical flight profile along the flight plan. It also computes aircraft deviation from that profile. Crossing altitudes provide a nearly seamless transition from cruise, enroute descent and approach phases of flight.

Two VNAV pages displaying six VNAV waypoints are available under the VNAV function key. Up to either VNAV waypoints are automatically pre-filled in reference to flight plan waypoints. Waypoint not on the flight plan cannot be accepted.

Additional VNAV features include a computed Top-of-Descent and Target Vertical Speed, support for holding patterns and vertical direct-to. For the initial VNAV leg, flight path angle in degrees is displayed under the Target Vertical Speed data field.

HOLDING PATTERNS

Holding patterns may be programmed at any waypoint on or off the flight plan or stored in the Navigation Data Base as part of SID, STAR and approach procedures. If defined at a fix on the flight plan, the pattern may be either armed (ARM HOLDING) or directly activated (DTO HOLD). If the fix is not on the flight plan, the hold may only be activated directly. When armed, the aircraft will continue along the current flight plan and when the hold fix is reached, holding will automatically commence. If the hold is directly activated, the aircraft will proceed from present position direct to the holding fix and automatically perform the appropriate entry to the pattern (parallel, direct or teardrop). Once established in the pattern, the FMS will fly the flight plan legs, compensating for existing wind conditions. Estimated Time of Arrival (ETA) over the holding fix is constantly displayed.

TERMINAL AREA PROCEDURES

The FMS can fly published SIDs and STARs with procedural legs such as:

- Heading-to-radial
- Heading-to-altitude
- Course-from-fix-to-DME-distance
- DME-arcs
- Fifteen additional leg types.

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SIDs and STARs can be easily linked into a flight plan by selecting from menus of procedures available at the departure and destination airports.

APPROACH MODE

The FMS approach mode provides the operator with ILS-like guidance while flying FMS approaches. During an FMS approach, the system provides signals like those of an ILS for flight director or autopilot coupled approaches. Altitude restrictions may be entered if vertical guidance is desired.

The following approaches may be accessed directly from the navigation data:

- VOR
- RNAV
- TACAN
- GPS
- GLS
- NDB
- ILS
- LOC
- BC.

The pilot can also define approaches by entering the approach waypoints and the reference navaid, however there are no provisions for pilot-defined GPS or NDB approaches.

DATA BASES

FMS data bases are categorized into the Navigation Data Base and the Pilot Defined Data Base, which includes a Company Routes Data Base.

DUAL CYCLE DATA BASE LOADING

The UNS-1E has two separate data base memory banks to hold the electronic data base. This feature allows a future (not yet effective) data base to be loaded and stored until its effective date. At the first power-on cycle after the effective start date of the new data base, it will automatically be used for navigation.

NAVIGATION DATA BASE

The self-contained worldwide Navigation Data Base, stored in non-volatile memory, provides the FMS with information for over 100,000 waypoints, navaids, airports and over 12,000 SIDs, STARs and approaches. The data base includes:

- VHF navaids
- NDB navaids
- Enroute waypoints
- Terminal waypoints
- High Altitude airways
- Low Altitude airways.

PILOT DEFINED DATABASE

The pilot defined database stores all data created by the pilot and has the capability to store up to approximately 200 pilot defined routes utilizing up to 2,500 legs. It also has the capacity to store:

- 200 waypoints
- 100 runways



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- 100 approaches
- Alignment points
- Airport reference points
- SIDs
- STARs
- Routes.

COMPANY DATA BASE

A company routes database can be created using Universal's Off-line Flight Planning Program and loaded via disk. A company route can be copied into the active flight plan where it can be edited, but the pilot cannot alter the basic route data base.

PERFORMANCE DATA BASE

The PERF page provides a synopsis of pertinent in-flight performance information. Some aircraft will utilize advanced performance features, with pages for Take-off, Climb/Cruise and Landing data. This page is read-only and no entries can be made.

SENSOR WATCHDOG

A sensor "watchdog" automatically detects sensor information that appears erroneous which could cause an error in the best computed position. This is accomplished by comparing each sensor, the unweighted average of all sensors and deselecting faulty sensors. If the difference exceeds a preset value, the "watchdog" will activate the appropriate messages to alert the pilot and in extreme cases automatically deselect the sensor.

NAVIGATION AND STEERING OUTPUTS

The FMS provides desired track, bearing, crosstrack, lateral deviation and related data to the flight guidance system for the HSI displays, and roll steering command for the autopilot/flight director system. The roll steering output limits for both "Direct To" and normal leg-to-leg changes are configuration specific and are set during initial configuration of the system. These specified limits are restricted to a maximum roll angle of 30°. The roll rate is limited to 3° per second in all installations.

FUEL

The aircraft's engine fuel flow system provides the inputs necessary to integrate real time fuel management information with navigational functions. During the pre-departure phase, the pilot inputs the fuel on board the aircraft to initialize the fuel management functions. During flight, the FMS automatically updates the fuel on board and gross weight as well as provides continuous estimates of fuel requirements for the programmed flight plan based upon fuel flow and groundspeed.

The FMS also provides specific range and endurance data to aid the pilot in optimizing fuel consumption to obtain maximum range or endurance.

NOTE

Fuel display parameters are advisory only and do not replace primary fuel quantity or fuel flow gauges for fuel load range planning.

MULTIPLE FMS'S

[Refer to Figures 5 and 6.](#)

In dual installations, each FMS provides system initialization and flight plan information through an interconnecting bus to other systems. While all long range navigation sensors are interfaced with



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both units, each FMS acts as the “master” to the associated sensor. Each FMS computes an independent position using all available navigation sensors and provides selected information to the associated flight instruments. Each FMS also shares its best computed position with the other system and continuously monitors the other system’s position output.

RADIO TUNING

Incorporated into the FMS, an optional TUNE mode provides the operator with a centralized means to tune the aircraft’s radios, select and store preset frequencies for each radio and view the selected (preset or active) frequencies for each radio.

ANNUNCIATIONS

MESSAGES

The FMS contains an array of messages that alert the operator to system status and flight plan sequencing. When a message is active, “MSG” will appear on the display and the message can be accessed by pressing the MSG key.

The Message function can also be utilized for AFIS and UniLink communications in applicable installations.

REMOTE ANNUNCIATORS

Outputs are provided to cause the six external annunciators to alert the pilot of system status or flight plan sequencing. These annunciations may be incorporated into the EFIS display:

MSG

A new message has been generated.

WPT

(Steady) Lateral waypoint alert.
(Flashing) Vertical waypoint
Flight Path Angle alert.

SXTK

FMS is in selected crosstrack mode.

FMS HDG

FMS is in Heading Mode.

FMS APPR

FMS is in Approach Mode.

GPS INTEG

RAIM is not available or a fault is detected. DME may be still in use.

SYSTEM COMPONENTS

The basic FMS installation consists of the FMS (containing the display, keyboard, navigation computer and a GPS sensor), a configuration module, a DTU or a SSDTU, a GPS antenna/WAAS antenna and a Radio Tuning Unit.

FMS DISPLAY

The FMS utilizes an active matrix flat panel color liquid crystal display (LCD). The high resolution display provides alphanumeric characters, icons, and graphics capabilities to facilitate on screen data recognition. The display allows for data display on eleven lines with 24 characters each in two different character sizes. Graphic displays for special photo, search and surveillance flight patterns are provided. The display also supports NTSC RGB format video.

FMS KEY BOARD

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The full alphanumeric keyboard contains dedicated functions keys that when used in conjunction with the ten line elect keys, provides the operator with all the controls necessary to communicate with the internal navigation computer and all associated sensors. The keyboard allows manual data entry, system mode selection and control (selection and de-selection) of navigation sensors.

FMS NAVIGATION COMPUTER

The FMS has a powerful central processing unit (navigation computer), navigation sensor interface circuits, flight guidance system interface and the Navigation Data Base.

The unit processes data from DME, VOR, Air Data, Fuel Flow and up to five long range navigation sensors. A radar joystick input can also be accommodated. At installation, the system is configured with the number and type of long range navigation sensors, which will optimize the FMS system to the aircraft's specific requirements. The long range navigation sensors act in addition to the multi-channel scanning DME and VOR sensing capability integrated into the basic circuit.

SHORT RANGE SENSORS

DME

The FMS unique auto-scanning DME-DME-DME position function uses the internal Navigation Data Base to continually map the stations surrounding the aircraft (VORTACs, DMEs, ILS-DMSs, LOC-DMEs, TACANs). The FMS then automatically tunes and reads these, one every four seconds, utilizing the blind channel of an external digital DME or Universal's Radio Reference Sensor (VOR, DME, TACAN). Range information from all responding DMEs within approximately 300 nautical miles of the FMS position are corrected for slant range error then compared with those computed from the

Navigation Data Base to verify reasonableness and to ensure DME position integrity is maintained.

Flights that never leave areas of multiple DME coverage can expect exceptional position accuracy. Over-water flights will retain DME accuracy long into the flight, reestablishing this same accuracy well before landfall as DMEs are again received.

VOR

The aircraft's digital VOR receiver or Universal's RRS provides the FMS with azimuth information from the referenced VOR navaid. This azimuth data combined with the scanning DME distances provides accurate navigation information for the FMS system throughout the enroute, terminal and approach modes of operation in accordance with the criteria set forth in AC 20-130A when within range of the required ground based navaid.

RADIO REFERENCE SENSOR (RRS)

Universal's Radio Reference Sensor (RRS) functions as a multi-channel DME-TACAN receiver/transmitter and VOR receiver, able to provide DME, VOR and TACAN data simultaneously to two FMS systems. The RRS provides the required navigation information via an ARINC 429 data bus without interfering with the aircraft's existing radios. RRS operation is fully automatic required no pilot intervention.

LONG RANGE SENSORS

GPS SENSOR

The FMS CDU presents the following GPS information:

- Latitude and longitude position
- A radial difference in nautical miles between the FMS and the GPS position

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- GPS horizontal position error
- GPS position uncertainty
- GPS time in Universal Time Coordinated (UTC)
- Using altitude annunciation
- GPS operating in Dead Reckoning (DR) mode annunciation
- Receiver Autonomous Integrity Monitoring (RAIM) availability at the current position
- FDE availability at the current position
- When RAIM will not be available at the Active Flight Plan Destination annunciation
- GPS integrity mode
- GPS accuracy or integrity is not adequate annunciation
- Satellite status for up to eight satellites.

For the UNS-1C and UNS-1E, the GPS sensor is the source of three-dimensional position and velocity, time and status information. The unit comprises an embedded 12-channel GPS receiver which receives signals transmitted by the U.S. GPS satellite network to process measurements from all satellites in view simultaneously.

For the UNS-1C and UNS-1E, the associated GPS active antenna is powered by 5 VDC from the FMS and it is installed on the top of the aircraft forward fuselage.

For the UNS-1Ew, the GPS sensor is the source of three-dimensional position and velocity, time and status information. The unit comprises an embedded 12-channel GPS receiver which receives signals transmitted by the U.S. GPS and WAAS satellite

network and ground support to process measurements from all satellites in view simultaneously.

For the UNS-1Ew, the related GPS active antenna operates on 4.5 to 15.0 VDC that is supplied from the FMS. The GPS/WAAS antenna is installed on the top of the aircraft forward fuselage.

The GPS sensor uses data in satellite navigation messages to calculate:

- Latitude
- Longitude
- Altitude
- Horizontal velocity
- Vertical velocity
- Current Universal Time Coordinated (UTC).

The Primary Flight Display (PFD) presents a yellow GPS INTEG message to the pilots if:

- The satellite geometry is not adequate to support at least RAIM error detection
- An uncorrectable satellite error occurs.

CONFIGURATION MODULE

A configuration module is installed on the FMS rear connector. It is programmed at the time of installation, and defines all the aircraft interfaces unique to the particular installation.

AIR DATA COMPUTER (ADC)

The FMS accepts digital or analog air data from a remote air data computer. True Airspeed, barometrically corrected or non-corrected

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altitude, and Static Air Temperature are utilized. Baro-corrected altitude is required for vertical guidance display during approaches and for VNAV (Approved for UNS-1E and UNS-1Ew on Q400) below 18,000 feet.

DATA TRANSFER UNIT (DTU)

The Data Transfer Unit (DTU) is designed to be mounted in the aircraft and used to update the FMS navigation data base within minutes. A portable unit is also available for multi-aircraft operations.

The DTU is connected directly to the FMS such that the data is transmitted directly to the FMS during the update process. The update data will refresh the entire navigation data base with each update.

A subscription service is available through Universal, which provides navigation data base updates on a 28 day cycle.

The DTU is also used to load aircraft specific performance data into the FMS for computation and display of aircraft performance data. The DTU may also load pilot-defined data created with off-line flight planning software loaded into an IBM compatible computer. FMS in-flight data parameters can be written to a disk utilizing the DTU. Maintenance log diagnostic history is also downloaded to a disk using the DTU. FMS software updates can also be loaded via the DTU.

Solid State Data Transfer Unit (SSDTU)

The fixed Solid State Data Transfer Unit (SSDTU) or portable SSDTU replaces the ZIP disk, floppy disk and the CD/DVD-ROM technology with the flash memory technology. The SSDTU is capable of supporting all the products previously supported by the DTU-100 (e.g. UNS-1C or UNS-1C+ or UNS-1E or UNS-1Ew).

Two data storage device ports are available on the faceplate of SSDTU. These ports support the Universal Serial Bus (USB), Secure Digital (SD) data storage devices and USB CD/DVD ROM disk drives. The unit has a high speed USB and a SD media port on the faceplate. These media ports function as disk drives.

RADIO TUNING UNIT (RTU)

An optional Radio Tuning Unit (RTU) allows the operator to use the FMS control display unit to tune a selected radio and store up to four preset frequencies for later use by each selected radio. COMMs, NAVs, ATC, ADFs and TACAN are selectable through the RTU, depending upon the specific aircraft configuration. Both 25 kHz and 8.33 kHz spacing is supported.

CONFIGURATION

Single or Dual UNS FMS UNS-1C or UNS-1E or UNS-1Ew installation can be used on the Q400.

SINGLE SYSTEMS

[Refer to Figures 3 and 4.](#)

In a single FMS UNS-1C or UNS-1E or UNS-1Ew installation, the FMS simultaneously accepts inputs from either digital DME and VOR receivers or DME/VOR/TACAN from an RRS, multiple long-range navigational sensors, the Air data Computer and engine fuel flow sensors.

MULTIPLE SYSTEMS

[Refer to Figures 5 and 6.](#)

In multiple system installations, communication between FMSs is accomplished through an interconnection bus. There is no "Master/Slave" type relationship between the FMS's. Each system makes data available (Initialization, Flight Plan, Fuel) to other



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interfaced system(s). Crossfill of this data can be initiated by the sending or receiving system. Each system acts as “MASTER” of its associated sensors, even though that sensor may also be providing data to another system. Typically, the on-side compass and air data computer will supply heading and air data, while fuel flow data will be shared by the systems.

Each FMS computes its own independent position using all available navigation sensors and provides selected information to the associated flight instruments. Each FMS also shares its best computed position with the other system and continuously monitors the other system's position output.

GPS/WAAS Approach Guidance

The WAAS navigation function is applicable for the dual FMS UNS-1Ew only. WAAS is a precise navigation system that provides more accuracy, availability, continuity and integrity necessary to enable reliance on the GPS for all phases of flight. The WAAS subsystem provides a three dimensional position and velocity, time and WAAS protection levels in addition to the ARINC inputs and outputs, critical analog output, discrete inputs and outputs.

WAAS collects the GPS satellite data from the ground stations. This data is transmitted through ground communications lines to the master stations which calculate and implement data corrections and provide additional integrity to the resultant signal. The corrected data is transmitted to the aircraft through two or more geostationary satellite communications links. The aircraft uses the WAAS signal in addition to the GPS service to conduct conventional area navigation procedures and also to fly LPV instrument approaches, which is equivalent to approaches conducted with the Instrument Landing System (ILS).

LPV AND LNAV/VNAV APPROACHES AND SBAS LOS ANNUNCIATION

The lateral and vertical guidance provided the FMS for SBAS approaches is referred as the Level of Service (LOS). The LOS is provided by a WAAS subsystem for SBAS approaches. When an approach is approved for the SBAS and has the LPV or LNAV/VNAV LOS or both, then the FMS will provide annunciation for the LOS.

The LOS annunciations of LPV or LNAV/VNAV indicate that the GPS satellite system provides the lateral and vertical guidance during the SBAS approach. If the LNAV only is shown on the SBAS LOS indicator, then LNAV uses vertical guidance based on the barometric altitude from the aircraft air data system. LOS is selected either automatically or manually.

LPV and LNAV/VNAV approaches use lateral and vertical guidance from the FMS (GPS) for a controlled descent to the runway. Baro-altitude from the air data system is not used for the LPV and LNAV/VNAV approaches. An LPV approach is similar to an LNAV/VNAV approach except that it has lower approach minima and this requires a dual installation.

When an approach is approved for LPV operations, the navigation database will include a final approach segment data block for the approach and a 'W' designator is displayed on the approach selection lists (on the FMS Arrival page) to indicate LPV availability. The FMS will automatically select the LOS, (example: LNAV only, LNAV/VNAV or LPV). The remote annunciators on the left and right pilots' panels will illuminate accordingly. When an approach is approved for LNAV/VNAV LOS and not LPV LOS, WGS-84 EOA altitude is included in the navigation database to allow the FMS to use GPS/WAAS altitude for the approach.



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FMS MCDU CONTROL KEYS

Refer to Figure 7.

1. ON/OFF – DIM key (momentary action)
PUSH – provides power, display dimming and unit shutdown functions.
2. ENTER key (momentary action)
PUSH – stores input data to memory. The cursor marks data that can be changed using the normal input processes (dark letters on a light background). When variable data is marked by the cursor, pressing the ENTER key stores data to memory.
3. $\pm$ key (momentary action)
PUSH – used in conjunction with the alphanumeric keys to enter data. It reverses the selected entry data field from + to –, N to S, and L to R. In the alpha field, it is used as a dash or period.
4. BACK key (momentary action)
PUSH – When the cursor is over a data entry field, the BACK key functions as a delete or backspace key.
5. MSG key (momentary action)
PUSH – displays an active message when the 'MSG' annunciation appears on the screen. The message alerts the operator to system status and flight plan sequencing. Selecting the UniLink option will access the UniLink menu page.
6. NEXT key (momentary action)
PUSH – used to cycle forward, one page at a time, through multiple pages of the same mode.

7. PREV key (momentary action)
PUSH – used to cycle backwards, one page at a time, through multiple pages of the same mode.
8. LSK key (momentary action)
PUSH – Line Select Keys (LSK's) are used to enter data utilizing the alphanumeric keys. Pressing the ENTER key completes the entry. 10 LSK's (5 down left side and five down right side of display) align with associated data lines.

FMS MCDU FUNCTION KEYS

Refer to Figure 8.

1. DATA key (momentary action)
PUSH – to obtain information and status about the FMS, Navigation Data Base and sensors which operate with the FMS. The function is also used to select/deselect individual sensors and to add/delete or change pilot defined locations.
2. NAV key (momentary action)
PUSH – to display all navigation data normally required by the flight crew. There are normally two Navigation pages. Additional pages are added when APPROACH or HEADING is selected. Pressing the PREV or NEXT function keys cycles through the associated pages.
3. VNAV key (momentary action)
PUSH – to define a desired vertical flight profiles along the flight plan route. It computes the aircraft deviation from that profiles for display.



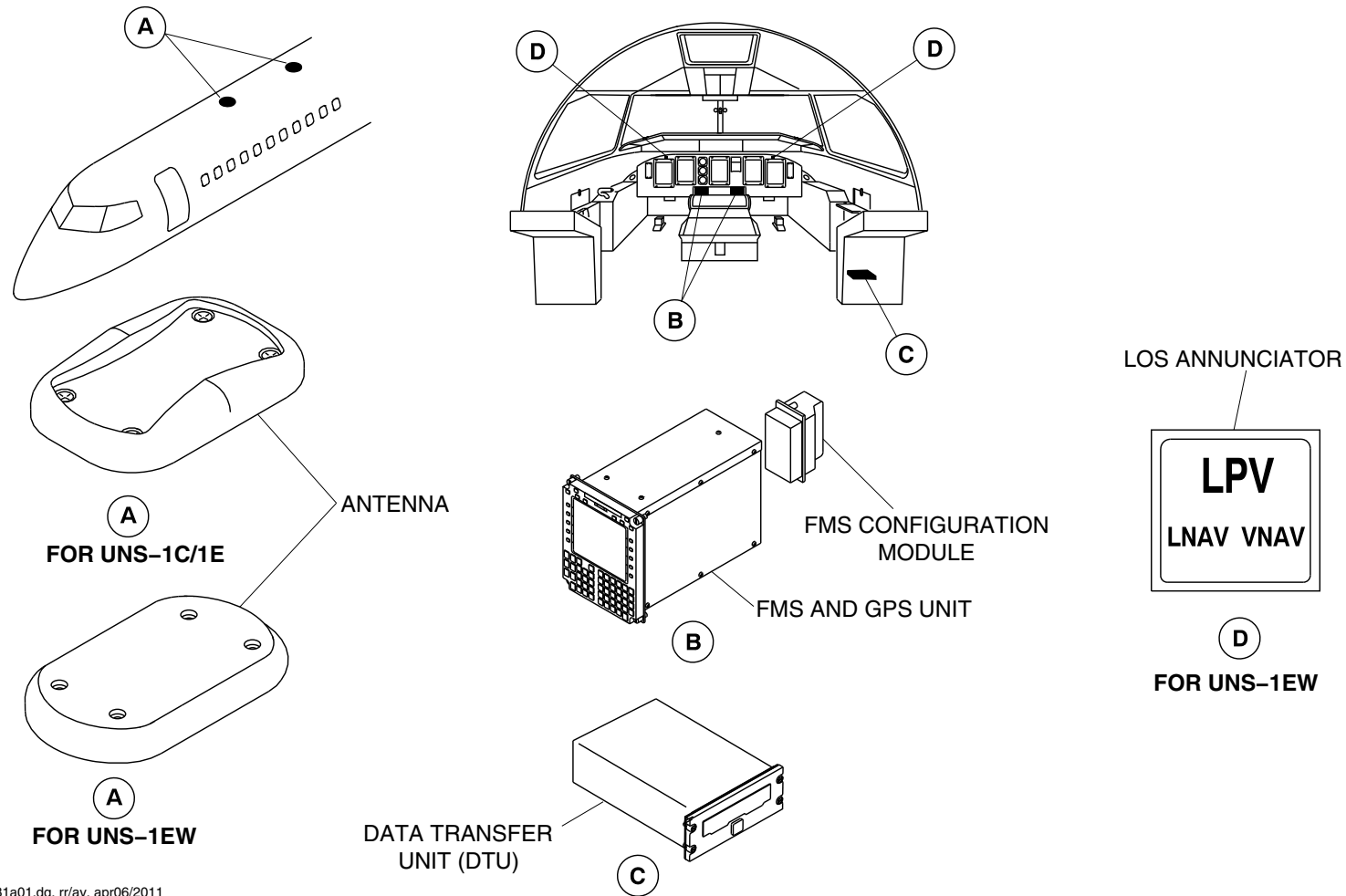
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4. DTO key (momentary action)
PUSH – accesses the 'DIRECT To' page and is specifically dedicated to changing the flight plan in response to the 'direct to' clearances. If the 'direct to' location is off the flight plan, provisions are made to link the location into the flight plan.
5. LIST key (momentary action)
PUSH – to display a list of options during data entry appropriate to the entry being made.
6. FUEL key (momentary action)
PUSH – to provide access to all fuel management functions. These include fuel and weight entry, range and endurance estimates, fuel requirement summary, projected landing weight, fuel flow and fuel consumption.
7. FPL key (momentary action)
PUSH – to access the Flight Plan pages or to access stored arrivals and routes. The FPL function is also used to create a new flight plan, alter current flight plan, insert a SID, STAR and approach into the flight plan.
8. PERF key (momentary action)
PUSH – to view a synopsis of in-flight performance information.
9. TUNE key (momentary action)
PUSH – to tune, select and store pre-selected frequencies and view selected frequencies (active and preset) for each radio.
10. MENU key (momentary action)
PUSH – to display a list of alternate formats or options for the FUEL, FPL, NAV, VNAV or TUNE modes. The letter 'M' will appear in a box on the title line of any page in which the MENU key is active.



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FMS System – Component Locations
Figure 1

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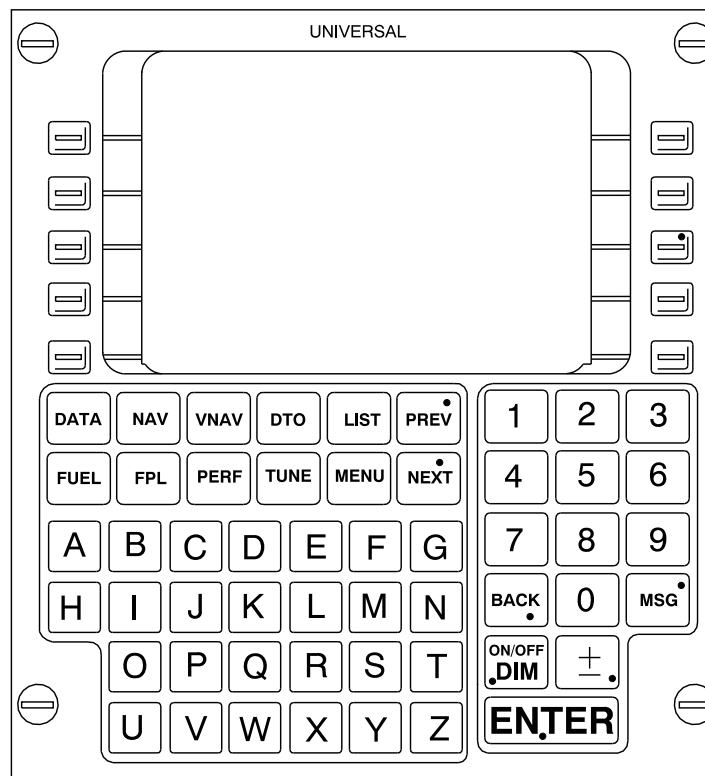
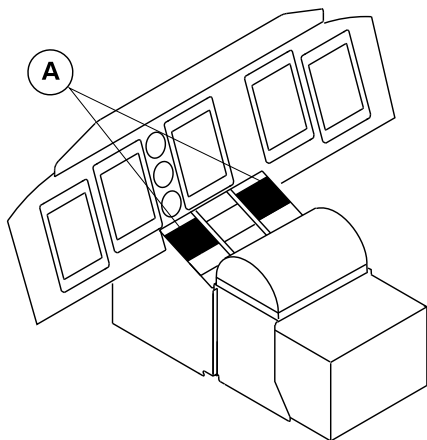
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UNS-1C/1E/1Ew FMS
Figure 2

PSM 1-84-2A
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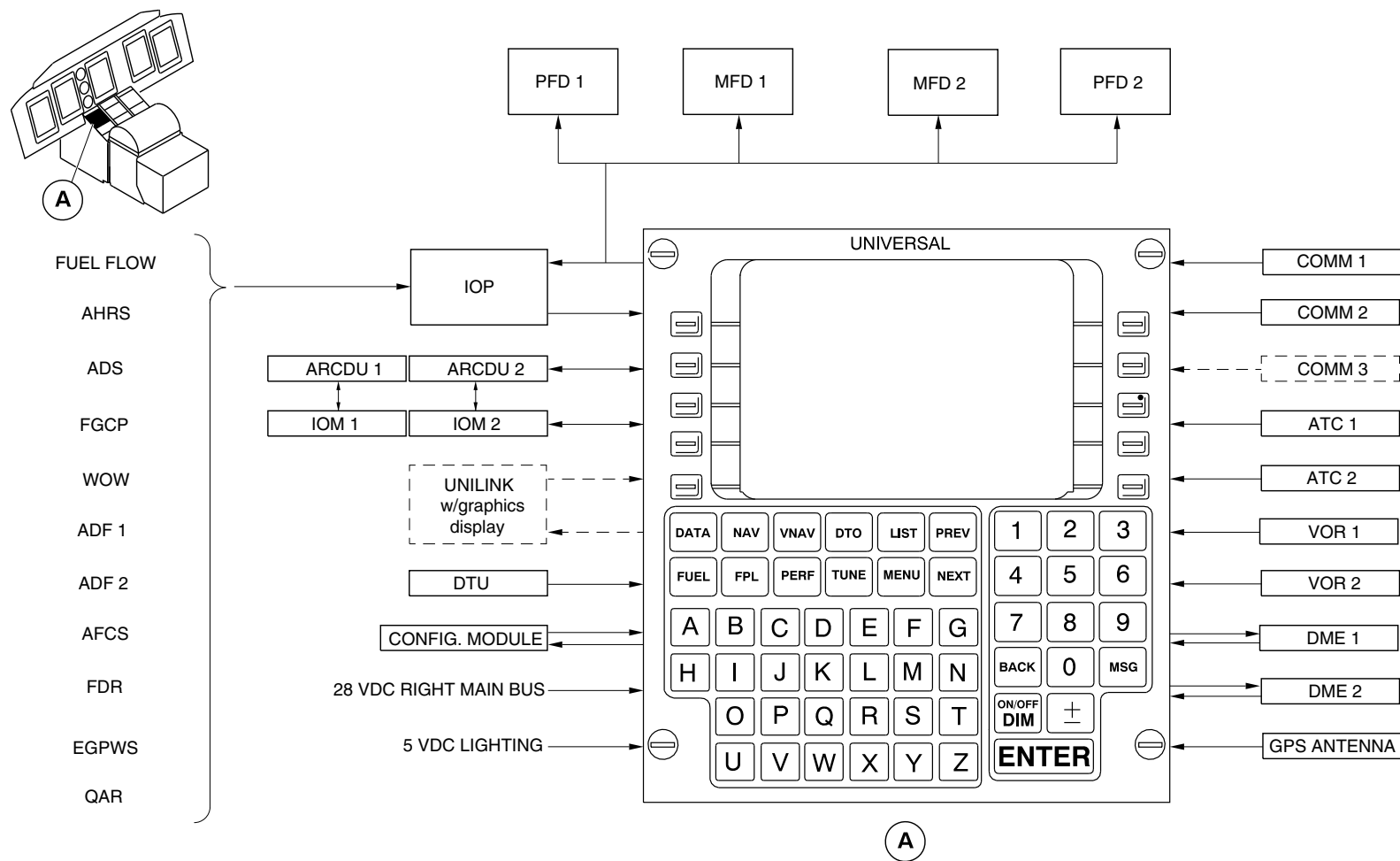
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Single UNS-1C/1E FMS Configuration
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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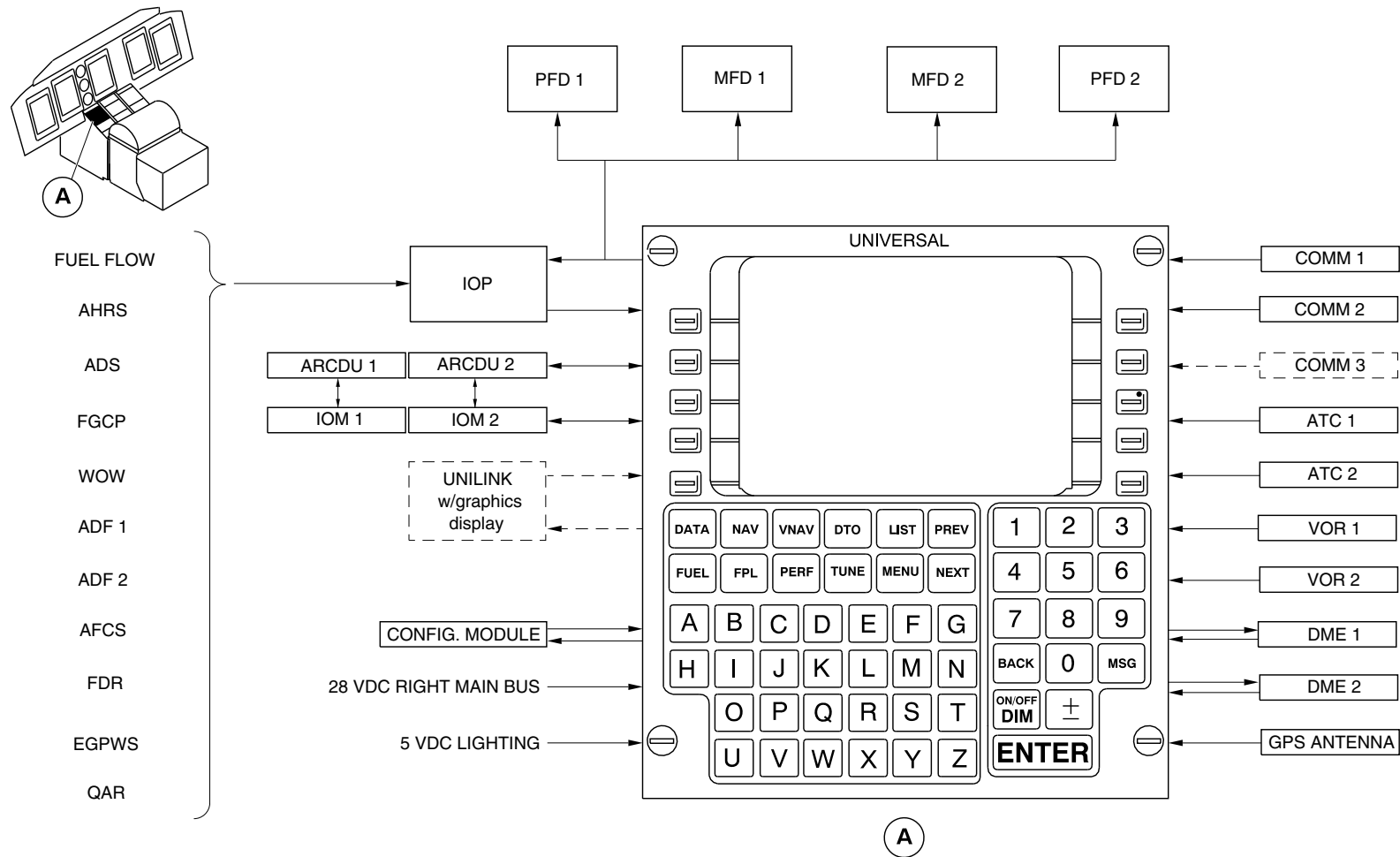
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Single UNS-1Ew FMS Configuration
Figure 4

PSM 1-84-2A
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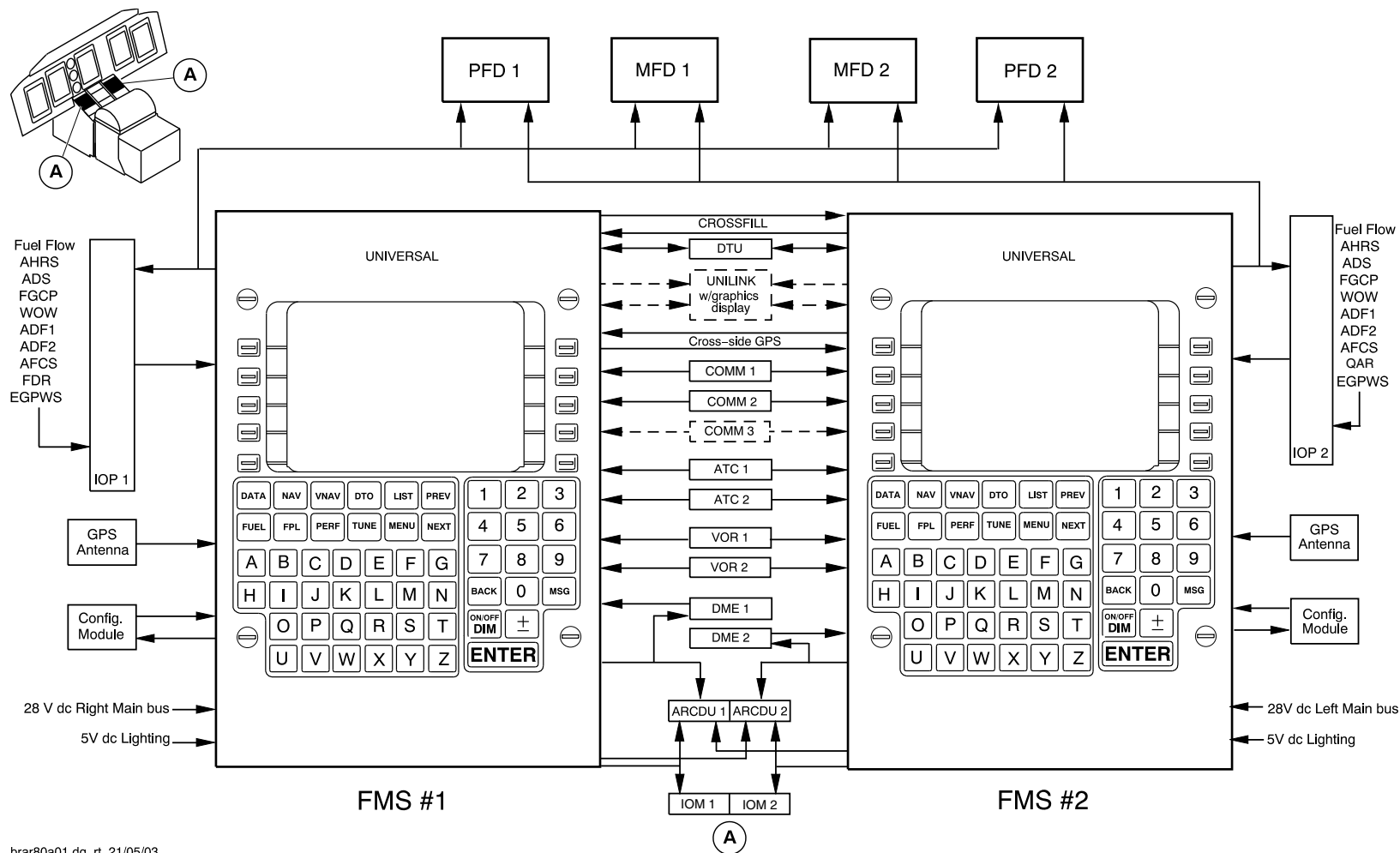
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Dual UNS-1C/1E FMS Configuration
Figure 5

PSM 1-84-2A
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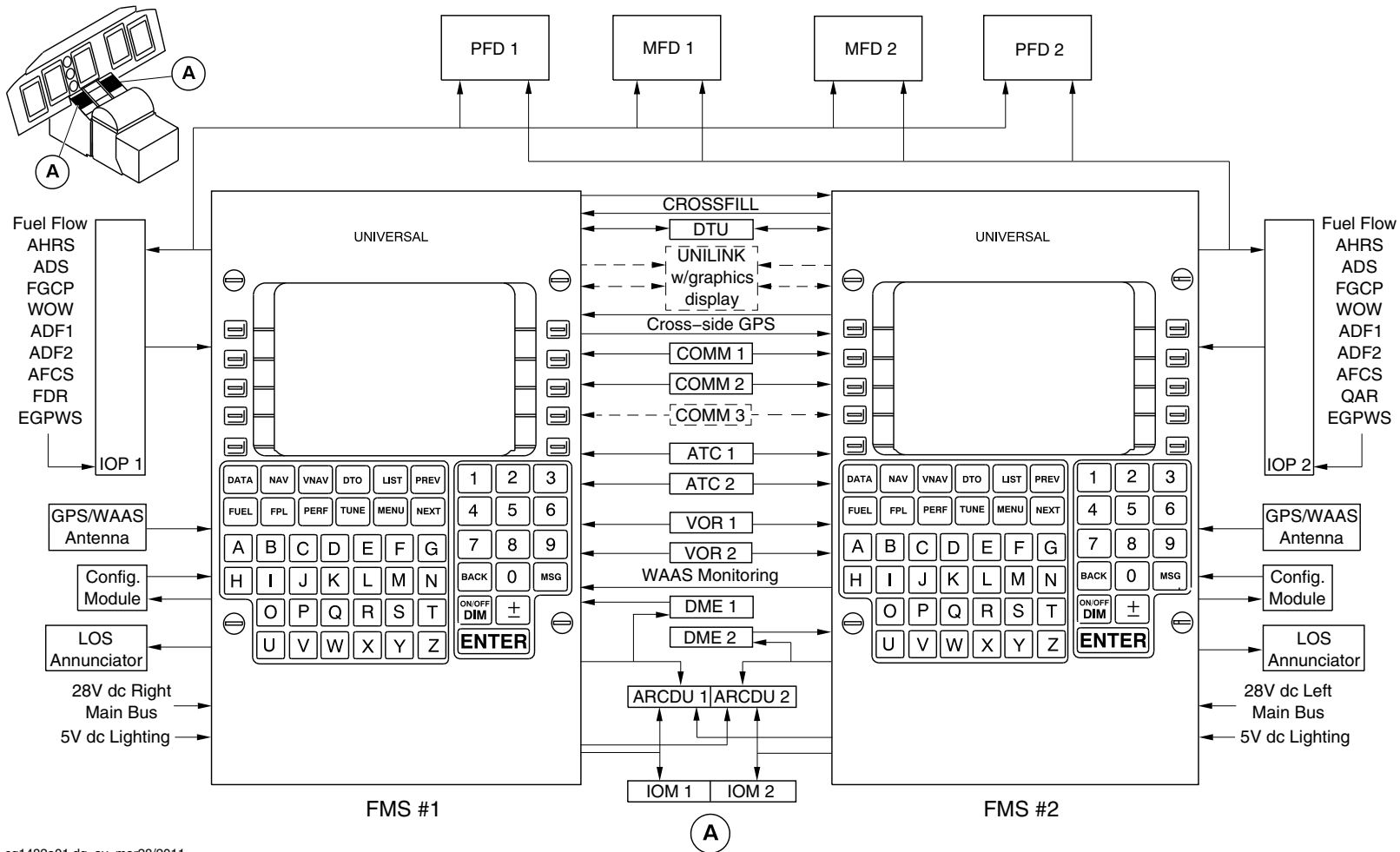
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Dual UNS-1Ew FMS Configuration
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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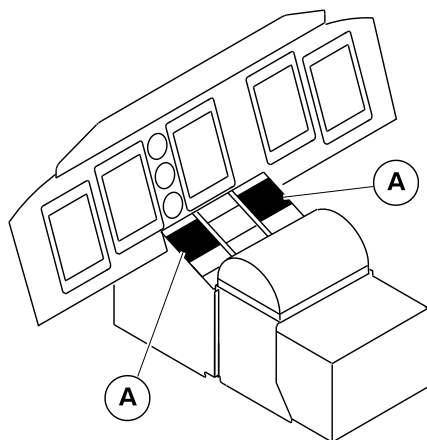
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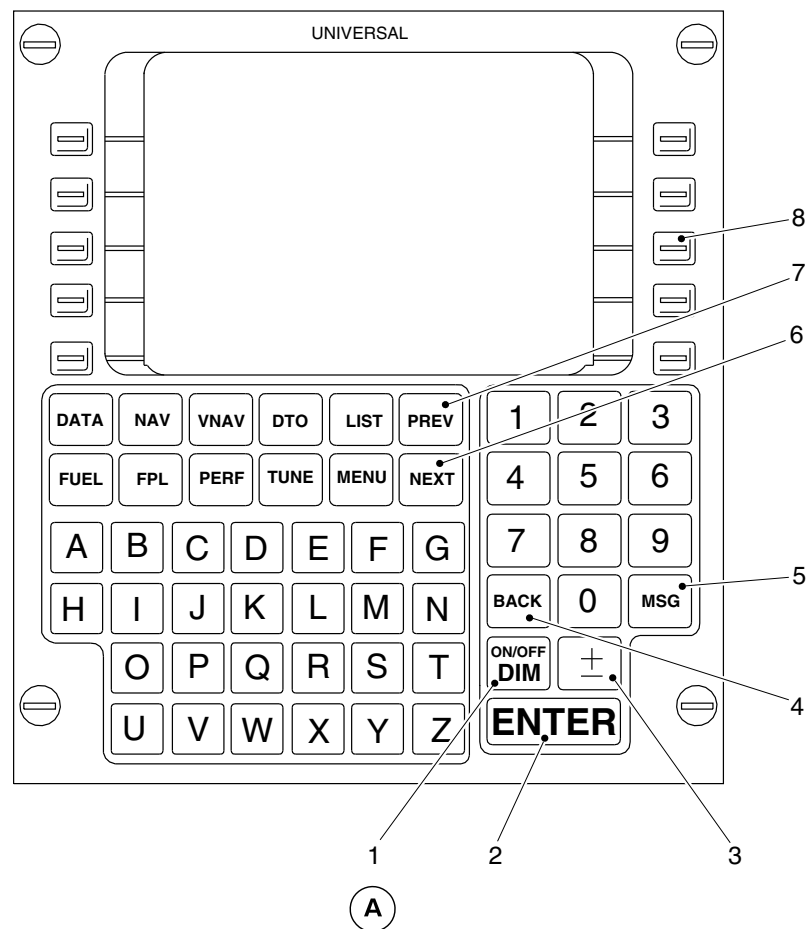
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LEGEND

1. ON/OFF – DIM key.
2. ENTER key.
3. $\pm$ key.
4. BACK key.
5. MSG key (message).
6. NEXT key.
7. PREV key (previous).
8. LSK key (line select key).



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UNS FMS MCDU – Control Keys
Figure 7

PSM 1-84-2A
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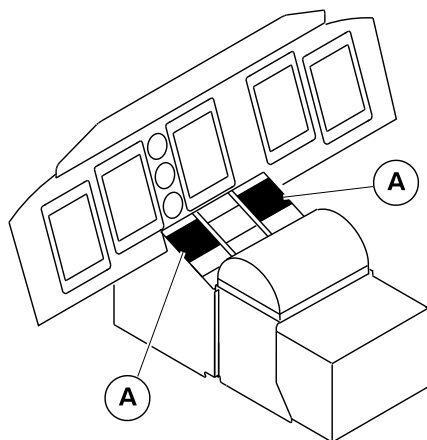
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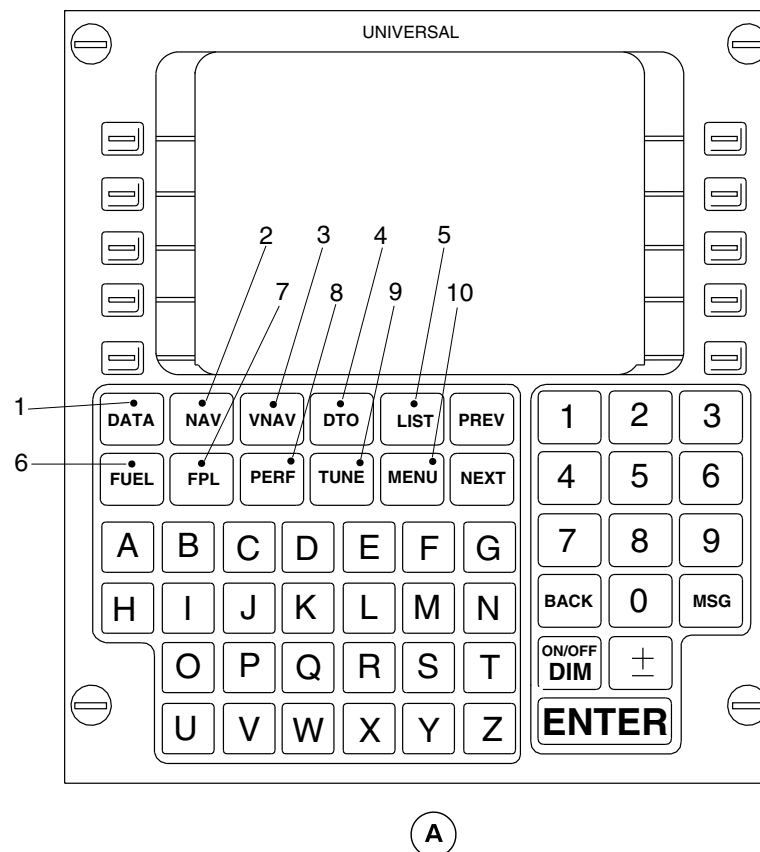
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LEGEND

1. DATA key.
2. NAV key (Navigation).
3. VNAV (Vertical Navigation).
4. DTO key (Direct To).
5. List key.
6. Fuel key.
7. FPL key (Flight Plan).
8. PERF key (Performance).
9. Tune key.
10. MENU key.



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UNS FMS MCDU – Function Keys
Figure 8

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